ETSI TS 129 002 V16.0.0 (2020-11)



Digital cellular telecommunications system (Phase 2+) (GSM);
Universal Mobile Telecommunications System (UMTS);
LTE;
5G;
Mobile Application Part (MAP) specification
(3GPP TS 29.002 version 16.0.0 Release 16)



Reference
RTS/TSGC-0429002vg00

Keywords
5G,GSM,LTE,UMTS

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Siret N° 348 623 562 00017 - NAF 742 C
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Sous-Préfecture de Grasse (06) N° 7803/88

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

It is necessary to transfer between entities of a Public Land Mobile Network (PLMN) information specific to the PLMN in order to deal with the specific behaviour of roaming Mobile Stations (MS)s. The Signalling System No. 7 specified by CCITT is used to transfer this information.

The present document describes the requirements for the signalling system and the procedures needed at the application level in order to fulfil these signalling needs.

Clauses 1 to 6 are related to general aspects such as terminology, mobile network configuration and other protocols required by MAP.

MAP consists of a set of MAP services that are provided to MAP service-users by a MAP service-provider.

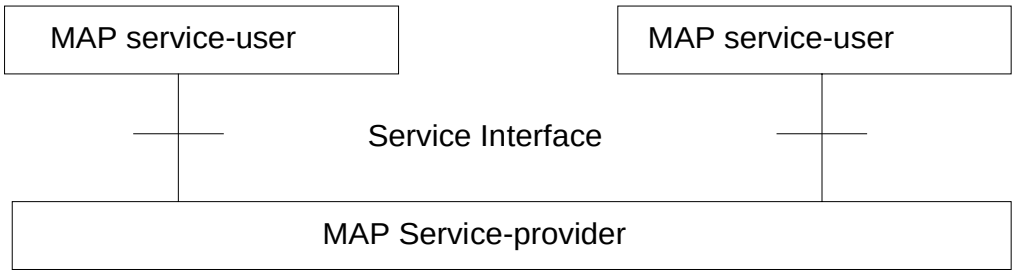


Figure 1.1/1: Modelling principles

Clauses 7 to 13A of the present document describe the MAP services.

Clauses 14 to 17 define the MAP protocol specification and the behaviour of service provider (protocol elements to be used to provide MAP services, mapping on to TC service primitives, abstract syntaxes, etc.).

Clauses 18 to 25 describe the MAP user procedures that make use of MAP services.

The present document specifies functions, procedures and information which apply to GERAN Iu mode. However, functionality related to GERAN Iu mode is neither maintained nor enhanced.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document in the same Release as the present document.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 22.001: "Digital cellular telecommunications system (Phase 2+); Principles of telecommunication services supported by a Public Land Mobile Network (PLMN)".

[3] 3GPP TS 22.002: "Bearer Services Supported by a Public Land Mobile Network (PLMN)".

[4] 3GPP TS 22.003: "Circuit Teleservices Supported by a Public Land Mobile Network (PLMN)".

[5] 3GPP TS 22.004: "General on Supplementary Services".

[6] 3GPP TS 42.009: "Digital cellular telecommunications system (Phase 2+); Security aspects".

[7] 3GPP TS 22.016: "International Mobile station Equipment Identities (IMEI)".

[8] 3GPP TS 22.041: "Operator Determined Barring".

[9] 3GPP TS 22.081: "Line identification supplementary services - Stage 1".

[10] 3GPP TS 22.082: "Call Forwarding (CF) supplementary services - Stage 1".

[11] 3GPP TS 22.083: "Call Waiting (CW) and Call Hold (HOLD) Supplementary Services - Stage 1".

[12] 3GPP TS 22.084: "Multi Party (MPTY) Supplementary Services - Stage 1".

[13] 3GPP TS 22.085: "Closed User Group (CUG) supplementary services - Stage 1".

[14] 3GPP TS 22.086: "Advice of charge (AoC) Supplementary Services - Stage 1".

[15] 3GPP TS 22.088: "Call Barring (CB) supplementary services - Stage 1".

[16] 3GPP TS 22.090: "Unstructured Supplementary Service Data (USSD); - Stage 1".

[17] 3GPP TS 23.003: "Numbering, addressing and identification".

[18] Void

[19] 3GPP TS 23.007: "Restoration procedures".

[20] 3GPP TS 23.008: "Organisation of subscriber data".

[21] 3GPP TS 23.009: "Handover procedures".

[22] 3GPP TS 23.011: "Technical realization of Supplementary Services - General Aspects".

[23] 3GPP TS 23.012: "Location management procedures".

[24] 3GPP TS 43.020: "Security related network functions".

[25] 3GPP TS 23.038: "Alphabets and language".

[25a] 3GPP TS 23.039: "Interface protocols for the connection of Short Message Service Centres (SMSCs) to Short Message Entities (SMEs)".

[26] 3GPP TS 23.040: "Technical realization of the Short Message Service (SMS) Point to Point (PP)".

[26a] 3GPP TS 23.271: "Functional stage2 description of LCS".

[27] 3GPP TS 23.081: "Line Identification Supplementary Services - Stage 2".

[28] 3GPP TS 23.082: "Call Forwarding (CF) Supplementary Services - Stage 2".

[29] 3GPP TS 23.083: "Call Waiting (CW) and Call Hold (HOLD) Supplementary Services - Stage 2".

[30] 3GPP TS 23.084: "Multi Party (MPTY) Supplementary Services - Stage 2".

[31] 3GPP TS 23.085: "Closed User Group (CUG) Supplementary Services - Stage 2".

[32] 3GPP TS 23.086: "Advice of Charge (AoC) Supplementary Services - Stage 2".

[33] 3GPP TS 23.088: "Call Barring (CB) Supplementary Services - Stage 2".

[34] 3GPP TS 23.090: "Unstructured Supplementary Services Data (USSD) - Stage 2".

[34a] 3GPP TS 33.204: "3G Security; Network domain security; TCAP user security".

[35] 3GPP TS 24.008: "Mobile Radio Interface Layer 3 specification; Core Network Protocols - Stage 3".

[36] 3GPP TS 24.010: "Mobile radio interface layer 3 Supplementary Services specification - General aspects".

[37] 3GPP TS 24.011: "Point-to-Point (PP) Short Message Service (SMS) support on mobile radio interface".

[37a] 3GPP TS 44.071: "Location Services (LCS) – stage 3".

[38] 3GPP TS 24.080: "Mobile radio interface layer 3 supplementary services specification - Formats and coding".

[39] 3GPP TS 24.081: "Line identification supplementary services - Stage 3".

[40] 3GPP TS 24.082: "Call Forwarding (CF) Supplementary Services - Stage 3".

[41] 3GPP TS 24.083: "Call Waiting (CW) and Call Hold (HOLD) supplementary services - Stage 3".

[42] 3GPP TS 24.084: "Multi Party (MPTY) Supplementary Services - Stage 3".

[43] 3GPP TS 24.085: "Closed User Group (CUG) Supplementary Services - Stage 3".

[44] 3GPP TS 24.086: "Advice of Charge (AoC) Supplementary Services - Stage 3".

[45] 3GPP TS 24.088: "Call Barring (CB) Supplementary Services - Stage 3".

[46] 3GPP TS 24.090: "Unstructured Supplementary Services Data - Stage 3".

[47] 3GPP TS 48.002: " Base Station System - Mobile-services Switching Centre (BSS - MSC) interface principles".

[48] 3GPP TS 48.006: "Signalling transport mechanism specification for the Base Station System - Mobile-services Switching Centre (BSS - MSC) interface".

[49] 3GPP TS 48.008: "Mobile Switching Centre - Base Station System (MSC - BSS) interface; Layer 3 specification".

[49a1] 3GPP TS 48.031: "Location Services (LCS); Serving Mobile Location Centre (SMLC) – Serving Mobile Location Centre (SMLC); SMLC Peer Protocol (SMLCPP)".

[49b] 3GPP TS 48.071: "Location Services (LCS); Serving Mobile Location Centre - Base Station System (SMLC - BSS) interface Layer 3 specification".

[50] 3GPP TS 49.001: "General network interworking scenarios".

[51] 3GPP TS 29.002: "Mobile Application Part (MAP) specification".

[52] Void

[53] Void

[54] Void

[55] 3GPP TS 29.006: "Interworking between a Public Land Mobile Network (PLMN) and a Packet Switched Public Data Network/Integrated Services Digital Network (PSPDN/ISDN) for the support of Packet Switched data transmission services".

[56] 3GPP TS 29.007: "General requirements on interworking between the Public Land Mobile Network (PLMN) and the Integrated Services Digital Network (ISDN) or Public Switched Telephone Network (PSTN)".

[57] 3GPP TS 29.008: "Application of the Base Station System Application Part (BSSAP) on the E-interface".

[58] 3GPP TS 29.010: "Information element mapping between Mobile Station - Base Station System and BSS - Mobile-services Switching Centre (MS - BSS - MSC) Signalling procedures and the Mobile Application Part (MAP)".

[59] 3GPP TS 29.011: "Signalling interworking for Supplementary Services".

[59a] 3GPP TS 49.031: "Digital cellular telecommunications system (Phase 2+); Location Services (LCS); Base Station System Application Part LCS Extension (BSSAP-LE)".

[60] Void

[61] 3GPP TS 52.008: "GSM Subscriber and Equipment Trace".

[62] ETS 300 102-1 (1990): "Integrated Services Digital Network (ISDN); User-network interface layer 3 specifications for basic call control".

[63] ETS 300 136 (1992): "Integrated Services Digital Network (ISDN); Closed User Group (CUG) supplementary service description".

[64] ETS 300 138 (1992): "Integrated Services Digital Network (ISDN); Closed User Group (CUG) supplementary service Digital Subscriber Signalling System No.one (DSS1) protocol".

[65] ETS 300 287: "Integrated Services Digital Network (ISDN); Signalling System No.7; Transaction Capabilities (TC) version 2".

[66] ETR 060: "Signalling Protocols and Switching (SPS); Guide-lines for using Abstract Syntax Notation One (ASN.1) in telecommunication application protocols".

[66b] ETR 091: "ETSI object identifier tree; Common domain Mobile domain"

[67] ITU-T Recommendation E.164: "The international public telecommunication numbering plan".

[68] ITU-T Recommendation E.212: "The international identification plan for mobile terminals and mobile users".

[69] ITU-T Recommendation E.213: "Telephone and ISDN numbering plan for land mobile stations in public land mobile networks (PLMN) ".

[70] ITU-T Recommendation E.214: "Structure of the land mobile global title for the signalling connection control part (SCCP) ".

[71] ITU-T Recommendation Q.699: "Interworking between ISDN access and non-ISDN access over ISDN User Part of Signalling System No. 7 ".

[72] ITU-T Recommendation Q.711: "Specifications of Signalling System No.7; Functional description of the Signalling Connection Control Part".

[73] ITU-T Recommendation Q.712: "Definition and function of SCCP messages".

[74] ITU-T Recommendation Q.713: "Specifications of Signalling System No.7; SCCP formats and codes".

[75] ITU-T Recommendation Q.714: "Specifications of Signalling System No.7; Signalling Connection Control Part procedures".

[76] ITU-T Recommendation Q.716: "Specifications of Signalling System No.7; Signalling connection control part (SCCP) performances".

[77] ITU-T Recommendation Q.721 (1988): "Specifications of Signalling System No.7; Functional description of the Signalling System No.7 Telephone user part".

[78] ITU-T Recommendation Q.722 (1988): "Specifications of Signalling System No.7; General function of Telephone messages and signals".

[79] ITU-T Recommendation Q.723 (1988): "Specifications of Signalling System No.7; Formats and codes".

[80] ITU-T Recommendation Q.724 (1988): "Specifications of Signalling System No.7; Signalling procedures".

[81] ITU-T Recommendation Q.725 (1988): "Specifications of Signalling System No.7; Signalling performance in the telephone application".

[82] ITU-T Recommendation Q.761 (1988): "Specifications of Signalling System No.7; Functional description of the ISDN user part of Signalling System No.7".

[83] ITU-T Recommendation Q.762 (1988): "Specifications of Signalling System No.7; General function of messages and signals".

[84] ITU-T Recommendation Q.763 (1988): "Specifications of Signalling System No.7; Formats and codes".

[85] ITU-T Recommendation Q.764 (1988): "Specifications of Signalling System No.7; Signalling procedures".

[86] ITU-T Recommendation Q.767: "Specifications of Signalling System No.7; Application of the ISDN user part of CCITT signalling System No.7 for international ISDN interconnections".

[87] ITU-T Recommendation Q.771: "Specifications of Signalling System No.7; Functional description of transaction capabilities".

[88] ITU-T Recommendation Q.772: "Specifications of Signalling System No.7; Transaction capabilities information element definitions".

[89] ITU-T Recommendation Q.773: "Specifications of Signalling System No.7; Transaction capabilities formats and encoding".

[90] ITU-T Recommendation Q.774: "Specifications of Signalling System No.7; Transaction capabilities procedures".

[91] ITU-T Recommendation Q.775: "Specifications of Signalling System No.7; Guide-lines for using transaction capabilities".

[92] ITU-T Recommendation X.200: "Reference Model of Open systems interconnection for CCITT Applications".

[93] ITU-T Recommendation X.680: "Information technology – Abstract Syntax Notation One (ASN.1): Specification of basic notation".

[93b] ITU-T Recommendation X.681: "Information technology – Abstract Syntax Notation One (ASN.1): Information object specification".

[94] ITU-T Recommendation X.690: "Information technology – ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER)".

[95] ITU-T Recommendation X.210: "Open systems interconnection layer service definition conventions".

[97] 3GPP TS 23.018: "Basic Call Handling".

[98] 3GPP TS 23.078: "Customised Applications for Mobile network Enhanced Logic (CAMEL) Phase 4 - Stage 2".

[99] 3GPP TS 23.079: "Support of Optimal Routeing (SOR) - Stage 2".

[100] 3GPP TS 43.068: "Voice Group Call Service (VGCS) - Stage 2".

[101] 3GPP TS 43.069: "Voice Broadcast service (VBS) - Stage 2".

[102] ANSI T1.113: "Signaling System No. 7 (SS7) - ISDN User Part".

[103] Void

[104] 3GPP TS 23.060: "General Packet Radio Service (GPRS) Service description; Stage 2".

[105] 3GPP TS 29.060: "General Packet Radio Service (GPRS); GPRS Tunnelling Protocol (GTP) across the Gn and Gp Interface".

[106] 3GPP TS 29.018: "General Packet Radio Service (GPRS); Serving GPRS Support Node (SGSN) - Visitors Location Register (VLR); Gs interface layer 3 specification".

[107] 3GPP TS 23.093: "Technical Realization of Completion of Calls to Busy Subscriber (CCBS); Stage 2".

[108] 3GPP TS 23.066: "Support of Mobile Number Portability (MNP); Technical Realisation Stage 2".

[109] ANSI T1.112 (1996): "Telecommunication – Signalling No. 7 - Signaling Connection Control Part (SCCP)".

[110] 3GPP TS 23.116: "Super-Charger Technical Realisation; Stage 2.".

[111] Void.

[112] Void

[113] Void

[114] Void

[115] Void

[116] ITU-T Recommendation Q.850 (May 1998): "Usage of cause and location in the Digital Subscriber Signalling System No. 1 and the Signalling System No. 7 ISDN User Part".

[117] 3GPP TS 22.135: "Multicall; Service description; Stage 1".

[118] 3GPP TS 23.135: "Multicall supplementary service; Stage 2".

[119] 3GPP TS 24.135: "Multicall supplementary service; Stage 3".

[120] 3GPP TS 25.413: "UTRAN Iu interface Radio Access Network Application Part (RANAP) signalling".

[121] 3GPP TS 29.202: "SS7 signalling transport in core network".

[122] 3GPP TS 23.032: "Universal Geographical Area Description (GAD)".

[123] 3GPP TS 22.071: "Location Services (LCS); Service description, Stage 1".

[124] ITU-T Recommendation X.880: "Data networks and open system communication - Open System Interconnection - Service definitions - Remote operations: Concepts, model and notation".

[125] 3GPP TS 23.278: "Customised Applications for Mobile Network Enhanced Logic (CAMEL) Phase 4 – Stage 2 IM CN Interworking (Rel-5)".

[126] 3GPP TS 23.172: "Technical realization of Circuit Switched (CS) multimedia service; UDI/RDI fallback and service modification".

[127] 3GPP TS 26.103: "Speech codec list for GSM and UMTS".

[128] 3GPP TS 23.141: "Presence Service; Architecture and Functional Description".

[129] 3GPP TS 23.094: "Follow Me (FM) – Stage 2".

[130] Void

[131] 3GPP TS 32.421: "Subscriber and equipment trace: Trace concepts and requirements".

[132] 3GPP TS 32.422: "Subscriber and equipment trace; Trace control and Configuration Management".

[133] 3GPP TS 23.236: "Intra-domain connection of Radio Access Network (RAN) nodes to multiple Core Network (CN) nodes".

[134] 3GPP TS 23.204: "Support of Short Message Service (SMS) over generic 3GPP Internet Protocol (IP) access".

[135] 3GPP TS 23.292: "IP Multimedia Subsystem (IMS) Centralized Services".

[136] 3GPP TS 23.067: "enhanced Multi-Level Precedence and Pre-emption service (eMLPP) - Stage 2".

[137] 3GPP TS 24.067: "Enhanced Multi-Level Precedence and Pre-emption service (eMLPP); Stage 3".

[138] 3GPP TS 22.011: "Service accessibility".

[139] IETF RFC 3588: "Diameter Base Protocol".

[140] Void

[141] 3GPP TS 29.173: "Locations Services; Diameter-based SLh interface for Control Plane LCS".

[142] Void

[143] 3GPP TS 23.272: "Circuit Switched (CS) fallback in Evolved Packet System (EPS); Stage 2".

[144] 3GPP TS 29.272: "Evolved Packet System (EPS); Mobility Management Entity (MME) and Service GPRS Support Node (SGSN) related interfaces based on Diameter protocol".

[145] 3GPP TS 23.401: "General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access".

[146] 3GPP TS 29.205: "Application of Q.1900 series to bearer independent Circuit Switched (CS) core network architecture; Stage 3".

[147] 3GPP TS 36.413: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP)".

[148] 3GPP TS 23.682: "Architecture enhancements to facilitate communications with packet data networks and applications".

[149] 3GPP TS 29.274: "3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunnelling Protocol for Control plane (GTPv2-C)".

[150] 3GPP TS 23.380: "IMS Restoration Procedures".

[151] 3GPP TS 29.273: "Evolved Packet System (EPS); 3GPP EPS AAA interfaces".

[152] 3GPP TS 29.118: "Mobility Management Entity (MME) - Visitor Location Register (VLR) SGs interface specification".

[153] 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)".

3 Abbreviations

ADD Automatic Device Detection
GANSS Galileo and Additional Navigation Satellite Systems

All other abbreviations used in the present document are listed in 3GPP TS 21.905.

4 Void

5 Overload and compatibility overview

5.1 Overload control

There is a requirement for an overload/congestion control for all entities of the Public Land Mobile Network and the underlying Signalling System No. 7.

5.1.1 Overload control for MSC (outside MAP)

For the entity MSC the following two procedures (outside MAP) may be applied to control the processor load:

- ISDN
  CCITT Recommendation Q.764 (Automatic Congestion Control), applicable to reduce the mobile terminating traffic;

- BSSAP
  3GPP TS 48.008 [49] (A-interface Flow Control), applicable to reduce the mobile originating traffic.

5.1.2 Overload control for MAP entities

For all MAP entities, especially the HLR, the following overload control method is applied.

If overload of a MAP entity is detected requests for certain MAP operations (see tables 5.1/1, 5.1/2, 5.1/3 and 5.1/4) may be ignored by the responder. The decision as to which MAP Operations may be ignored is made by the MAP service provider and is based upon the priority of the application context.

Since most of the affected MAP operations are supervised in the originating entity by TC timers (medium) an additional delay effect is achieved for the incoming traffic.

If overload levels are applicable in the Location Registers the MAP operations should be discarded taking into account the priority of their application context (see table 5.1/1 for HLR, table 5.1/2 for MSC/VLR, table 5.1/3 for the SGSN and table 5.1/4 for the SMLC; the lowest priority is discarded first).

The ranking of priorities given in the tables 5.1/1, 5.1/2, 5.1/3 and 5.1/4 is not normative. The tables can only be seen as a proposal that might be changed due to network operator/implementation matters.

Table 5.1/1: Priorities of Application Contexts for HLR as Responder

Responder = HLR	**Initiating Entity**
Priority high	
Mobility Management	
networkLocUp VLR (updateLocation), (restoreData/v2), (sendParameters/v1)	
gprsLocationUpdate (updateGPRSLocation/v3),	SGSN
infoRetrieval VLR/SGSN (sendAuthenticationInfo/v2/v3), (sendParameters/v1)	
istAlerting MSC (istAlert/v3) (purgeMS/v2/v3)	msPurging VLR
msPurging SGSN (purgeMS/v3)	
Short Message Service	
shortMsgGateway GMSC (sendRoutingInfoforSM), (reportSM-DeliveryStatus)	
mwdMngt VLR/SGSN (readyForSM/v2/v3), (noteSubscriberPresent/v1)	
Mobile Terminating Traffic	
locInfoRetrieval GMSC (sendRoutingInfo)	
anyTimeEnquiry gsmSCF (anyTimeInterrogation/v3)	
reporting VLR (statusReport)	
Location Services	
locationSvcGateway (sendRoutingInfoforLCS/v3)	GMLC
Subscriber Controlled Inputs (Supplementary Services)	
networkFunctionalSs (registerSS), (eraseSS), (activateSS), (deactivateSS), (interrogateSS), (registerPassword), (processUnstructuredSS-Data/v1), (beginSubscriberActivity/v1)	VLR
callCompletion VLR (registerCCEntry), (eraseCCEntry)	
networkUnstructuredSs (processUnstructuredSS-Request/v2)	VLR
imsiRetrieval VLR (sendIMSI/v2)	
gprsLocationInfoRetrieval (sendRoutingInfoForGprs/v3/v4)	GGSN/SGSN
failureReport GGSN/SGSN (failureReport/v3)	
authenticationFailureReport (authenticationFailureReport/v3)	VLR/SGSN
Priority low	

NOTE: The application context name is the last component but one of the object identifier.
Operation names are given in brackets for information with "/vn" appended to vn only operations.

Table 5.1/3: Priorities of Application Contexts for SGSN as Responder

Responder = SGSN	**Initiating Entity**
Priority high	
Mobility and Location Register Management	
locationCancel HLR (cancelLocation v3)	
reset HLR (reset)	
subscriberDataMngt (insertSubscriberData v3), (deleteSubscriberData v3)	HLR
tracing HLR (activateTraceMode), (deactivateTraceMode)	
Short Message Service	
shortMsgMT-Relay (MT-ForwardSM v3), (forwardSM v1/v2)	MSC
Location Services	
locationSvcEnquiry (provideSubscriberLocation v3)	GMLC
Network-Requested PDP context activation	
gprsNotify HLR (noteMsPresentForGprs v3),	
(*Subscriber Location & State retrieval)*	
subscriberInfoEnquiry (provideSubscriberInformation/v3)	HLR
Priority low	

NOTE: The application context name is the last component but one of the object identifier.
Operation names are given in brackets for information with "/vn" appended to vn.

Responder = MSC/VLR	Initiating Entity
Priority high	
Handover	
handoverControl MSC (prepareHandover/v2/v3), (performHandover/v1)	
Group call and Broadcast call	
groupCallControlMSC (prepareGroupCall/v3)	
groupCallInfoRetrieval (sendGroupCallInfo/v3)	MSC
Mobility and Location Register Management	
locationCancel HLR (cancelLocation)	
reset HLR (reset)	
immediateTermination (istCommand/v3)	HLR
interVlrInfoRetrieval (sendIdentification/v2/v3), (sendParameters/v1)	VLR
subscriberDataMngt (insertSubscriberData), (deleteSubscriberData)	HLR
tracing HLR (activateTraceMode), (deactivateTraceMode)	
Short Message Service	
shortMsgMO-Relay (MO-ForwardSM v3), (forwardSM v1/v2)	MSC/SGSN
shortMsgMT-Relay (MT-ForwardSM v3), (forwardSM v1/v2)	MSC
shortMsgAlert HLR (alertServiceCentre/v2), (alertServiceCentreWithoutResult/v1)	
Mobile Terminating Traffic	
resourceMngt GMSC (releaseResources)	
roamingNbEnquiry HLR (provideRoamingNumber)	
callControlTransfer (resumeCallHandling)	MSC
subscriberInfoEnquiry (provideSubscriberInformation/v3)	HLR
reporting HLR (remoteUserFree), (SetReportingState)	
Location Services	
locationSvcEnquiry (provideSubscriberLocation/v3)	GMLC
Network-Initiated USSD	
networkUnstructuredSs (unstructuredSS-Request/v2), (unstructuredSS-Notify/v2)	HLR
Priority low	

Table 5.1/2: Priorities of Application Contexts for MSC/VLR as Responder

NOTE: The application context name is the last component but one of the object identifier.
Operation names are given in brackets for information with "/vn" appended to vn only operations.

5.1.3 Congestion control for Signalling System No. 7

The requirements of SS7 Congestion control have to be taken into account as far as possible.

Means that could be applied to achieve the required traffic reductions are described in clauses 5.1.1 and 5.1.2.

5.2 Compatibility

5.2.1 General

The present document of the Mobile Application Part is designed in such a way that an implementation which conforms to it can also conform to the Mobile Application Part operational version 1 specifications, except on the MSC-VLR interface.

A version negotiation mechanism based on the use of an application-context-name is used to negotiate the protocol version used between two entities for supporting a MAP-user signalling procedure.

When starting a signalling procedure, the MAP-user supplies an application-context-name to the MAP-provider. This name refers to the set of application layer communication capabilities required for this dialogue. This refers to the required TC facilities (e.g. version 1 or 2) and the list of operation packages (i.e. set of operations) from which operations can be invoked during the dialogue.

A version one application-context-name may only be transferred to the peer user in a MAP-U-ABORT to an entity of version two or higher (i.e. to trigger a dialogue which involves only communication capabilities defined for MAP operational version 1).

If the proposed application-context-name can be supported by the responding entity the dialogue continues on this basis otherwise the dialogue is refused and the initiating user needs to start a new dialogue, which involves another application-context-name which requires less communication capabilities but provides similar functionality (if possible).

When a signalling procedure can be supported by several application contexts that differ by their version number, the MAP-User needs to select a name. It can either select the name that corresponds to the highest version it supports or follow a more specific strategy so that the number of protocol fallbacks due to version compatibility problems is minimised.

5.2.2 Strategy for selecting the Application Context (AC) version

A method should be used to minimise the number of protocol fall-backs which would occur sometimes if the highest supported AC-Name were always the one selected by GSM entities when initiating a dialogue. The following method is an example that can be used mainly at transitory phase stage when the network is one of mixed phase entities.

5.2.2.1 Proposed method

A table (table 1) may be set up by administrative action to define the highest application context (AC) version supported by each destination; a destination may be another node within the same or a different PLMN, or another PLMN considered as a single entity. The destination may be defined by an E.164 number or an E.214 number derived from an IMSI or in North America (World Zone 1) by an E.164 number or an IMSI (E.212 number). The table also includes the date when each destination is expected to be able to handle at least one AC of the latest version of the MAP protocol. When this date is reached, the application context supported by the node is marked as "unknown", which will trigger the use of table 2.

A second table (table 2) contains an entry for each destination that has an entry in table 1. For a given entity, the entry in table 2 may be a single application context version or a vector of different versions applying to different application contexts for that entity. Table 2 is managed as described in clause 5.2.2.2.

The data for each destination will go through the following states:

a) the version shown in table 1 is "version n-1", where 'n' is the highest version existing in this specification; table 2 is not used;

b) the version shown in table 1 is "unknown"; table 2 is used, and maintained as described in clause 5.2.2.2;

c) when the PLMN operator declares that an entity (single node or entire PLMN) has been upgraded to support all the MAP version n ACs defined for the relevant interface, the version shown in table 1 is set to "version n" by administrative action; table 2 is no longer used, and the storage space may be recovered.

5.2.2.2 Managing the version look-up table

WHEN it receives a MAP-OPEN ind the MAP-User determines the originating entity number either using the originating address parameter or the originating reference parameter or retrieving it from the subscriber data using the IMSI or the MSISDN.

IF the entity number is known:

THEN

It updates (if required) the associated list of highest supported ACs.

ELSE

It creates an entry for this entity and includes the received AC-name in the list of highest supported ACs.

WHEN starting a procedure, the originating MAP-user looks up its version control table.

IF the destination address is known and not timed-out.

THEN

It retrieves the appropriate AC-name and uses it

IF the dialogue is accepted by the peer

THEN

It does not modify the version control table

ELSE (this should never occur)

It starts a new dialogue with the common highest version supported (based on information implicitly or explicitly provided by the peer).

It replaces the old AC-name by the new one in the list of associated highest AC supported.

ELSE

It uses the AC-name that corresponds to the highest version it supports.

IF the dialogue is accepted by the peer.

THEN

It adds the destination node in its version control table and includes the AC-Name in the list of associated highest AC supported.

ELSE

It starts a new dialogue with the common highest version supported (based on information implicitly or explicitly provided by the peer).

IF the destination node was not known

THEN

It adds the destination node in its version control table and includes the new AC-Name in the list of associated highest AC supported.

ELSE

It replaces the old AC-name by the new one in the list of highest supported AC and reset the timer.

5.2.2.3 Optimising the method

A table look-up may be avoided in some cases if both the HLR and the VLR or both the HLR and the SGSN store for each subscriber the version of the AC-name used at location updating. Then:

- for procedures which make use of the same application-context, the same AC-name (thus the same version) can be selected (without any table look-up) when the procedure is triggered;
- for procedures which make use of a different application-context but which includes one of the packages used by the location updating AC, the same version can be selected (without any table look-up) when the procedure is triggered;

for HLR:

- Subscriber data modification (stand alone);

for VLR:

- Data Restoration.

6 Requirements concerning the use of SCCP and TC

6.1 Use of SCCP

The Mobile Application Part (MAP) makes use of the services offered by the Signalling Connection Control Part (SCCP).

MAP supports the following SCCP versions:

- Signalling Connection Control Part, Signalling System no. 7 CCITT ('Blue Book SCCP');
- Signalling Connection Control Part, Signalling System no. 7 ITU-T Recommendation (07/96) Q.711 to Q.716 ('White Book SCCP'). Support of White Book SCCP at the receiving side shall be mandated from 00:01hrs, 1st July 2002(UTC). However, for signalling over the MAP E-interface to support inter-MSC handover/relocation, the support of White Book SCCP shall be mandated with immediate effect.

A White Book SCCP message will fail if any signalling point used in the transfer of the message does not support White Book SCCP. Therefore it is recommended that the originator of the White Book SCCP message supports a drop back mechanism or route capability determination mechanism to interwork with signalling points that are beyond the control of GSM/UMTS network operators.

In North America (World Zone 1) the national version of SCCP is used as specified in ANSI T1.112. Interworking between a PLMN in North America and a PLMN outside North America will involve an STP to translate between ANSI SCCP and ITU-T/CCITT SCCP.

The SCCP is identified as an MTP3-user and the transport of SCCP messages between two entities shall be accomplished according to the 3GPP TS 29.202 [121].

6.1.1 SCCP Class

MAP will only make use of the connectionless classes (0 or 1) of the SCCP.

6.1.2 Sub-System Number (SSN)

The Application Entities (AEs) defined for MAP consist of several Application Service Elements (ASEs) and are addressed by sub-system numbers (SSNs). The SSNs for MAP are specified in 3GPP TS 23.003 [17].

When the SGSN emulates MSC behaviour for processing messages (MAP-MO-FORWARD-SHORT-MESSAGE, MAP_CHECK_IMEI, MAP_SUBSCRIBER_LOCATION_REPORT) towards entities which do not support interworking to SGSNs, it shall use the MSC SSN in the calling party address instead of the SGSN SSN.

When present in the network, the Presence Network Agent emulates the behaviour of the GSM Service Control Function (gsm SCF) for processing of messages (MAP-NOTE-MM-EVENT, MAP-ANY-TIME-INTERROGATION and MAP-ANY-TIME-MODIFICATION).

When a FFN (Follow Me Functional Node, see TS 23.094 [129]) is implemented in a network entity different from HLR, this network entity shall emulate HLR behaviour, i.e. it shall accept MAP-PROCESS-UNSTRUCTURED-SS-REQUEST messages addressed with SSN for HLR.

In an EPS, an Interworking Function (IWF) may be used to convert Diameter S6a messages to MAP Gr messages and vice versa; also an IWF may be used to convert Diameter S13 messages to MAP Gf messages and vice versa. An SSN value for the IWF does not exist. Instead the IWF shall use the SGSN SSN value when serving an MME and use the HLR SSN when serving an HSS. An IWF is said to serve an MME (or HSS) when Diameter messages are exchanged between the IWF and the MME (or HSS).

6.1.3 SCCP addressing

6.1.3.1 Introduction

Within the GSM System there will be a need to communicate between entities within the same PLMN and in different PLMNs. Using the Mobile Application Part (MAP) for this function implies the use of Transaction Capabilities (TC) and the Signalling Connection Control Part (SCCP) of CCITT Signalling System No. 7.

Only the entities that should be addressed are described below. If the CCITT or ITU-T SCCP is used, the format and coding of address parameters carried by the SCCP for that purpose shall comply with CCITT Recommendation Q.713 with the following restrictions:

1) Intra-PLMN addressing

   For communication between entities within the same PLMN, a MAP SSN shall always be included in the called and calling party addresses. All other aspects of SCCP addressing are network specific.

2) Inter-PLMN addressing

   a) Called Party Address

      - SSN indicator = 1 (MAP SSN always included);
      - Global title indicator = 0100 (Global title includes translation type, numbering plan, encoding scheme and nature of address indicator);
      - the translation type field will be coded "00000000" (Not used). For call related messages for non-optimal routed calls (as described in 3GPP TS 23.066 [108]) directed to another PLMN the translation type field may be coded "10000000" (CRMNP);
      - Routing indicator = 0 (Routing on global title);

   b) Calling Party Address

      - SSN indicator = 1 (MAP SSNs always included);
      - Point code indicator = 0;
      - Global title indicator = 0100 (Global title includes translation type, numbering plan, encoding scheme and nature of address indicator);

- Numbering Plan = 0001 (ISDN Numbering Plan, E.164; In Case of Inter-PLMN Signalling, the dialogue initiating entity and dialogue responding entity shall always include its own E.164 Global Title as Calling Party Address);
- the translation type field will be coded "00000000" (Not used);
- Routing indicator = 0 (Routing on Global Title).

If ANSI T1.112 SCCP is used, the format and coding of address parameters carried by the SCCP for that purpose shall comply with ANSI specification T1.112 with the following restrictions:

1) Intra-PLMN addressing

   For communication between entities within the same PLMN, a MAP SSN shall always be included in the called and calling party addresses. All other aspects of SCCP addressing are network specific.

2) Inter-PLMN addressing

   a) Called Party Address

   - SSN indicator = 1 (MAP SSN always included);
   - Global title indicator = 0010 (Global title includes translation type);
   - the Translation Type (TT) field will be coded as follows:

     TT = 9, if IMSI is included;

     TT = 14, if MSISDN is included;

     Or TT = 10, if Network Element is included. (If TT=10, then Number Portability GTT is not invoked, if TT=14, then Number Portability GTT may be invoked).

   - Routing indicator = 0 (Routing on global title);

   b) Calling Party Address

   - SSN indicator = 1 (MAP SSNs always included);
   - Point code indicator = 0;
   - Global Title indicator = 0010 (Global title includes translation type);

     TT = 9, if IMSI is included;

     TT = 14, if MSISDN is included;

     Or TT = 10, if Network Element is included. (If TT=10, then Number Portability GTT is not invoked, if TT=14, then Number Portability GTT may be invoked).

   Routing indicator = 0 (Routing on Global Title).

If a Global Title translation is required for obtaining routeing information, one of the numbering plans E.164, E.212 and E.214 is applicable.

- E.212 numbering plan.

  When CCITT or ITU-T SCCP is used, an E.212 number must not be included as Global Title in an SCCP UNITDATA message. The translation of an E.212 number into a Mobile Global Title is applicable in a dialogue initiating VLR, SGSN or GGSN if the routeing information towards the HLR is derived from the subscriber's IMSI. In World Zone 1 when ANSI SCCP is used, the IMSI (E.212 number) is used as a Global Title to address the HLR. When an MS moves from one VLR service area to another, the new VLR may derive the address of the previous VLR from the Location Area Identification provided by the MS in the location registration request. The PLMN where the previous VLR is located is identified by the E.212 numbering plan elements of the Location Area Identification, i.e. the Mobile Country Code (MCC) and the Mobile Network Code (MNC).

- E.214 and E.164 numbering plans.

  When CCITT or ITU-T SCCP is used, only address information belonging to either E.214 or E.164 numbering plan is allowed to be included as Global Title in the Called and Calling Party Address. In World Zone 1 when ANSI SCCP is used, the IMSI (E.212 number) is used as a Global Title to address the HLR.

  If the Calling Party Address associated with the dialogue initiating message contains a Global Title, the sending network entity shall include its E.164 entity number.

  When receiving an SCCP UNITDATA message, SCCP shall accept either of the valid numbering plans in the Called Party Address and in the Calling Party Address.

  When CCITT or ITU-T SCCP is used and an N-UNITDATA-REQUEST primitive from TC is received, SCCP shall accept an E.164 number or an E.214 number in the Called Address and in the Calling Address. In World Zone 1 when ANSI SCCP is used, the IMSI (E.212 number) is used instead of E.214 number.

The following clauses describe the method of SCCP addressing appropriate for each entity both for the simple intra-PLMN case and where an inter-PLMN communication is required. The following entities are considered:

- the Mobile-services Switching Centre (MSC);
- the Home location Register (HLR);
- the Visitor Location Register (VLR);
- the Gateway Mobile-services Switching Centre (GMSC);
- the GSM Service Control Function (gsmSCF);
- the Interworking Mobile-services Switching Centre (IWMSC);
- the Serving GPRS Support Node (SGSN);
- the Gateway GPRS Support Node (GGSN);
- the Gateway Mobile Location Centre (GMLC);
- the CSG Subscriber Server (CSS).

6.1.3.2 The Mobile-services Switching Centre (MSC)

There are several cases where it is necessary to address the MSC.

6.1.3.2.1 MSC interaction during handover or relocation

The address is derived from the target Cell id or from the target RNC id.

6.1.3.2.2 MSC for short message routing

When a short message has to be routed to an MS, the GMSC addresses the VMSC by an MSC identity received from the HLR that complies with E.164 rules.

For MS originating short message, the IWMSC address is derived from the Service Centre address.

6.1.3.2.3 MSC for location request routing

When a location request for a particular MS needs to be sent to the MS's VMSC, the GMLC addresses the VMSC using an E.164 address received from the MS's HLR.

6.1.3.2.4 MSC for LMU Control

When a control message has to be routed to an LMU from an SMLC, the SMLC addresses the serving MSC for the LMU using an E.164 address.

6.1.3.3 The Home Location Register (HLR)

There are several cases where the HLR has to be addressed.

6.1.3.3.1 During call set-up

When a call is initiated the HLR of the called mobile subscriber will be interrogated to discover the whereabouts of the MS. The addressing required by the SCCP will be derived from the MSISDN dialled by the calling subscriber. The dialled number will be translated into either an SPC, in the case of communications within a PLMN, or a Global Title if other networks are involved (i.e. if the communication is across a PLMN boundary).

If the calling subscriber is a fixed network subscriber, the interrogation can be initiated from the Gateway MSC of the home PLMN in the general case. If the topology of the network allows it, the interrogation could be initiated from any Signalling Point that has MAP capabilities, e.g. local exchange, outgoing International Switching Centre (ISC), etc.

6.1.3.3.2 Before location updating completion

When an MS registers for the first time in a VLR, the VLR has to initiate the update location dialogue with the MS's HLR and a preceding dialogue for authentication information retrieval if the authentication information must be retrieved from the HLR. When initiating either of these dialogues, the only data for addressing the HLR that the VLR has available is contained in the IMSI, and addressing information for SCCP must be derived from it. When continuing the established update location dialogue (as with any other dialogue), the VLR must derive the routeing information towards the HLR from the Calling Party Address received with the first responding CONTINUE message until the dialogue terminating message is received. This means that the VLR must be able to address the HLR based on:

- an E.214 Mobile Global Title originally derived by the VLR from the IMSI (when CCITT or ITU-T SCCP is used), or an E.212 number originally derived from IMSI (when ANSI SCCP is used, an IMSI); or
- an E.164 HLR address; or
- in the case of intra-PLMN signalling, an SPC.

When answering with Global Title to the VLR, the HLR shall insert its E.164 address in the Calling Party Address of the SCCP message containing the first responding CONTINUE message.

If the HLR is in the same PLMN as the VLR, local translation tables may exist to derive an SPC. For authentication information retrieval and location updating via the international PSTN/ISDN signalling network that requires the use of CCITT or ITU-T SCCP, the Global Title must be derived from the IMSI, using the principles contained in CCITT Recommendation E.214 and the Numbering Plan Indicator (NPI) value referenced by the SCCP Specifications. In World Zone 1 where the ANSI SCCP is used, IMSI (E.212 number) is used as Global Title. A summary of the translation from the IMSI (CCITT Recommendation E.212) to Mobile Global Title (described in CCITT Recommendation E.214) is shown below:

- E.212 Mobile Country Code translates to E.164 Country Code;
- E.212 Mobile Network Code translates to E.164 National Destination Code;
- E.212 Mobile Subscriber Identification Number (MSIN) is carried unchanged if within the E.164 number maximum length (15 digits). If the Mobile Global Title is more than 15 digits the number is truncated to 15 by deleting the least significant digits.

This translation will be done either at the application or at SCCP level in the VLR. The Mobile Global Title thus derived will be used to address the HLR.

If location updating is triggered by an MS that roams from one MSC Area into a different MSC Area served by the same VLR, the VLR shall address the HLR in the same way as if the MS registers for the first time in the VLR.

6.1.3.3.3 After location updating completion

In this case, the subscriber's basic MSISDN has been received from the HLR during the subscriber data retrieval procedure as well as the HLR number constituting a parameter of the MAP message indicating successful completion of the update location dialogue. From either of these E.164 numbers the address information for initiating dialogues with the roaming subscriber's HLR can be derived. Also the subscriber's IMSI may be used for establishing the routeing

information towards the HLR. This may apply in particular if the dialogue with the HLR is triggered by subscriber controlled input.

Thus the SCCP address of the roaming subscriber's HLR may be an SPC, or it may be a Global title consisting of the E.164 MSISDN or the E.164 number allocated to the HLR or either the E.214 Mobile Global Title derived from the IMSI if CCITT or ITU-T SCCP is used, or the IMSI if ANSI SCCP is used (ANSI SCCP is used in World Zone 1).

6.1.3.3.4 VLR restoration

If a roaming number is requested by the HLR for an IMSI that has no data record in the interrogated VLR, the VLR provides the roaming number in the dialogue terminating message. Subsequently the VLR must retrieve the authentication data from the MS's HLR, if required, and must then trigger the restore data procedure. For this purpose, the VLR has to initiate in succession two independent dialogues with the MS's HLR. The MTP and SCCP address information needed for routeing towards the HLR can be derived from the IMSI received as a parameter of the MAP message requesting the roaming number. In this case, the IMSI received from the HLR in the roaming number request shall be processed in the same way as the IMSI that is received from an MS that registers for the first time within a VLR. Alternatively to the IMSI, the Calling Party Address associated with the roaming number request may be used to obtain the routeing information towards the HLR.

6.1.3.3.5 During Network-Requested PDP Context Activation

When receiving a PDP PDU the GGSN may interrogate the HLR of the MS for information retrieval. When initiating such a dialogue, the only data for addressing the HLR that the GGSN has available is contained in the IMSI, and addressing information must be derived from it. The IMSI is obtained from the IP address or the X.25 address in the incoming IP message by means of a translation table. This means that the GGSN shall be able to address the HLR based on an E.214, (if CCITT or ITU-T SCCP is used), or E.212 (if ANSI SCCP is used), Mobile Global Title originally derived by the GGSN from the IMSI in the case of inter-PLMN signalling. In the case of intra-PLMN signalling, an SPC may also be used.

If the HLR is in the same PLMN as the GGSN, local translation tables may exist to derive an SPC. For information retrieval via the international PSTN/ISDN signalling network, the Global title must be derived from the IMSI, using the principles contained in CCITT Recommendation E.214 and the Numbering Plan Indicator (NPI) value referenced by the SCCP Specifications. A summary of the translation from the IMSI (CCITT Recommendation E.212) to Mobile Global Title (described in CCITT Recommendation E.214) is shown below:

- E.212 Mobile Country Code translates to E.164 Country Code;
- E.212 Mobile Network Code translates to E.164 National Destination Code;
- E.212 Mobile Subscriber Identification Number (MSIN) is carried unchanged if within the E.164 number maximum length (15 digits). If the Mobile Global Title is more than 15 digits the number is truncated to 15 by deleting the least significant digits.

This translation will be done either at the application or at SCCP level in the GGSN. The Mobile Global Title thus derived will be used to address the HLR.

6.1.3.3.6 Before GPRS location updating completion

When an MS registers for the first time in an SGSN, the SGSN has to initiate the update location dialogue with the MS's HLR and a preceding dialogue for authentication information retrieval if the authentication information must be retrieved from the HLR. When initiating either of these dialogues, the only data for addressing the HLR that the SGSN has available is contained in the IMSI, and addressing information for SCCP must be derived from it. When continuing the established update location dialogue (as with any other dialogue), the SGSN must derive the routeing information towards the HLR from the Calling Party Address received with the first responding CONTINUE message until the dialogue terminating message is received. This means that the SGSN must be able to address the HLR based on:

- an E.214 (if CCITT or ITU-T SCCP is used) or E.212 (if ANSI SCCP is used) Mobile Global Title originally derived by the SGSN from the IMSI; or
- an E.164 HLR address; or
- in the case of intra-PLMN signalling, an SPC.

If the HLR is in the same PLMN as the SGSN, local translation tables may exist to derive an SPC. For authentication information retrieval and location updating via the international PSTN/ISDN signalling network, the Global title must be derived from the IMSI, using the principles contained in CCITT Recommendation E.214 and the Numbering Plan Indicator (NPI) value referenced by the SCCP Specifications. A summary of the translation from the IMSI (CCITT Recommendation E.212) to Mobile Global Title (described in CCITT Recommendation E.214) is shown below:

- E.212 Mobile Country Code translates to E.164 Country Code;
- E.212 Mobile Network Code translates to E.164 National Destination Code;
- E.212 Mobile Subscriber Identification Number (MSIN) is carried unchanged if within the E.164 number maximum length (15 digits). If the Mobile Global Title is more than 15 digits the number is truncated to 15 by deleting the least significant digits.

This translation will be done either at the application or at SCCP level in the SGSN. The Mobile Global Title thus derived will be used to address the HLR.

6.1.3.3.7 After GPRS location updating completion

In this case, the subscriber's Basic MSISDN has been received from the HLR during the subscriber data retrieval procedure as well as the HLR number constituting a parameter of the MAP message indicating successful completion of the update location dialogue. From either of these E.164 numbers the address information for initiating dialogues with the roaming subscriber's HLR can be derived. Also the subscriber's IMSI may be used for establishing the routeing information towards the HLR.

Thus the SCCP address of the roaming subscriber's HLR may be an SPC, or it may be a Global title consisting of the E.164 MSISDN or the E.164 number allocated to the HLR or the E.214 Mobile Global Title derived from the IMSI.

6.1.3.3.8 Query for a Location Request

For a location request from an external client, the GMLC needs to address the home HLR of the target MS to obtain the address of the target MS's serving MSC. The GMLC uses either the international E.164 MSISDN, the international E.214 number (if CCITT or ITU-T SCCP is used) or the international E.212 number (if ANSI SCCP is used) of the MS as means to route a query to the HLR.

6.1.3.4 The Visitor Location Register (VLR)

6.1.3.4.0 General

There are several cases when the VLR needs to be addressed.

6.1.3.4.1 Inter-VLR information retrieval

When an MS moves from one VLR service area to another, the new VLR may request the IMSI and authentication sets from the previous VLR. The new VLR derives the address of the previous VLR from the Location Area Identification provided by the MS in the location registration request.

6.1.3.4.2 HLR request

The HLR will only request information from a VLR if it is aware that one of its subscribers is in the VLR service area. This means that a location updating dialogue initiated by the VLR has been successfully completed, i.e. the HLR has indicated successful completion of the update location procedure to the VLR.

When initiating dialogues towards the VLR after successful completion of location updating, the routeing information used by the HLR is derived from the E.164 VLR number received as a parameter of the MAP message initiating the update location dialogue. If the VLR is in the same PLMN as the HLR, the VLR may be addressed directly by an SPC derived from the E.164 VLR number. For dialogues via the international PSTN/ISDN signalling network, presence of the E.164 VLR number in the Called Party Address is required.

6.1.3.4.3 CSS request

The CSS will only request information from a VLR if it is aware that one of its subscribers is in the VLR service area. This means that a VCSG location updating dialogue initiated by the VLR has been successfully completed, i.e. the CSS has indicated successful completion of the update VCSG location procedure to the VLR.

When initiating dialogues towards the VLR after successful completion of VCSG location updating, the routeing information used by the CSS is derived from the E.164 VLR number received as a parameter of the MAP message initiating the update VCSG location dialogue. The VLR may be addressed either by the E.164 VLR number or directly by an SPC derived from the E.164 VLR number due to the VLR is in the same PLMN as the CSS.

6.1.3.5 The Interworking MSC (IWMSC) for Short Message Service

The IWMSC is the interface between the mobile network and the network to access to the Short Message Service Centre. This exchange has an E.164 address known in the SGSN or in the MSC.

6.1.3.6 The Equipment Identity Register (EIR)

The EIR address is either unique or could be derived from the IMEI. The type of address is not defined.

6.1.3.7 Void

6.1.3.8 The Serving GPRS Support Node (SGSN)

6.1.3.8.0 General

There are several cases when the SGSN needs to be addressed.

6.1.3.8.1 HLR request

The HLR will initiate dialogues towards the SGSN if it is aware that one of its subscribers is in the SGSN serving area. This means that a GPRS location updating has been successfully completed, i.e., the HLR has indicated successful completion of the GPRS location update to the SGSN. The routeing information used by the HLR is derived form the E.164 SGSN number received as parameter of the MAP message initiating the GPRS update location procedure. If the SGSN is in the same PLMN as the HLR, the SGSN may be addressed directly by an SPC derived from the E.164 SGSN number. For dialogues via the international PSTN/ISDN signalling network, the presence of the E.164 SGSN number in the Called Party Address is required.

6.1.3.8.2 GMSC request

When the GMSC initiates dialogues towards the SGSN the SGSN (MAP) SSN (See 3GPP TS 23.003 [17]) shall be included in the called party address. The routeing information used by the GMSC is derived from the E.164 SGSN number received as a parameter of the MAP message initiating the forward short message procedure. If the GMSC does not support the GPRS functionality the MSC (MAP) SSN value shall be included in the called party address.

NOTE: Every VMSC and SGSN shall have uniquely identifiable application using E.164 numbers, for the purpose of SMS over GPRS when the GMSC does not support the GPRS functionality.

6.1.3.8.3 CSS request

The CSS will initiate dialogues towards the SGSN if it is aware that one of its subscribers is in the SGSN serving area. This means that a VCSG location updating has been successfully completed, i.e., the CSS has indicated successful completion of the VCSG location update to the SGSN. The routeing information used by the CSS is derived from the E.164 SGSN number received as parameter of the MAP message initiating the update VCSG location procedure. The SGSN may be addressed either by the E.164 SGSN number or directly by an SPC derived from the E.164 SGSN number due to the SGSN is in the same PLMN as the CSS.

6.1.3.9 The Gateway GPRS Support Node (GGSN)

The GGSN provides interworking with external packet-switched networks, network screens and routing of the Network-Requested PDP Context activation. If a Network-Requested PDP Context activation fails, the HLR will alert

the GGSN when the subscriber becomes reachable. The HLR will use the E.164 GGSN number received as parameter of the MAP message reporting the failure.

6.1.3.10 The Gateway MSC (GMSC) for Short Message Service

The GMSC provides interworking with the network to access the Short Message Service Centre, the mobile network and routing of Send Routing Info For SM. The GMSC has on E.164 address known in the HLR, SGSN or MSC.

6.1.3.10A Void

6.1.3.10A.1 Void

6.1.3.10A.2 Void

6.1.3.10B The Gateway Mobile Location Centre (GMLC)

The GMLC initiates location requests on behalf of external clients. The E.164 address of the GMLC is provided to an HLR when the GMLC requests a serving MSC address or SGSN address from the HLR for a target MS. The E.164 address of the GMLC is also provided to a serving MSC or SGSN when the GMLC requests the location of a target MS served by this MSC or SGSN.

6.1.3.10C The CSG Subscriber Server (CSS)

The CSS address is either unique or could be derived from the IMSI. The type of address is not defined.

6.1.3.11 Summary table

The following tables summarise the SCCP address used for invoke operations. As a principle, within a PLMN either an SPC or a GT may be used (network operation option), whereas when addressing an entity outside the PLMN the GT must be used. The address type mentioned in the table (e.g. MSISDN) is used as GT or to derive the SPC.

For a response, the originating address passed in the invoke is used as SCCP Called Party Address. For extra-PLMN addressing the own E.164 entity address is used as SCCP Calling Party Address; for intra-PLMN addressing an SPC derived from the entity number may be used instead. When using an SPC, the SPC may be taken directly from MTP.

Table 6.1/1

to from	fixed net work	HLR	VLR	MSC	EIR	gsmSCF	SGSN	GGSN	CSS
fixed network	---	E:GT T:MSISDN	---	---	---	---	---	---	---
Home Location Register	---	---	I:SPC/GT E:GT T:VLR NUMBER	---	---	I:SPC/GT E:GT T:gsmSCF NUMBER	I:SPC/GT E:GT T:SGSN NUMBER	I:SPC/GT E:GT T:GGSN NUMBER	---
Visitor Location Register	---	I:SPC/GT E:GT T:MGT (outside World Zone 1)/MSISDN (World Zone 1/)HLR NUMBER (note)	I:SPC/GT E:GT T:VLR NUMBER	---	---	I:SPC/GT E:GT T:gsmSCF NUMBER	---	---	I: SPC/GT E:GT T:CSS NUMBER
mobile-services switching centre	---	I:SPC/GT E:GT T:MSISDN	I:SPC/GT E:GT T:VLR NUMBER	I:SPC/GT E:GT T:MSC NUMBER	I:SPC/GT E:GT T:EIR NUMBER	I:SPC/GT E:GT T:gsmSCF NUMBER	I:SPC/GT E:GT T:SGSN NUMBER	---	---
gsm Service Control Function	---	I:SPC/GT E:GT T:MSISDN	---	---	---	---	---	---	---
Serving GPRS Support Node	---	I:SPC/GT E:GT T:MGT/ MSISDN/HLR NUMBER	---	I:SPC/GT E:GT T:MSC NUMBER	I:SPC/GT E:GT T:EIR NUMBER	I:SPC/GT E:GT T:gsmSCF NUMBER	---	---	I:SPC/GT E:GT T:CSS NUMBER
Gateway GPRS Support Node	---	I:SPC/GT E:GT T:MGT	---	---	---	---	---	---	---
Gateway Mobile Location Centre	---	I:SPC/GT E:GT T:MSISDN, MGT (outside World Zone 1) or IMSI (World Zone 1) (note)	---	I:SPC/GT E:GT T:MSC NUMBER	---	---	I:SPC/GT E:GT T:SGSN NUMBER	---	---
CSG Subscriber Server	---	---	I:SPC/GT E:GT T:VLR NUMBER	---	---	---	I:SPC/GT E:GT T:SGSN NUMBER	---	---

I:	Intra-PLMN.	
E:	Extra (Inter)-PLMN.	
T:	Address Type.	
GT:	Global Title.	
MGT:	E.214 Mobile Global Title.	
SPC:	Signalling Point Code.	
NOTE:	For initiating the location updating procedure and an authentication information retrieval from the HLR preceding it, the VLR has to derive the HLR address from the IMSI of the MS. The result can be an SPC or an E.214 Mobile Global Title if CCITT or ITU-T SCCP is used, or IMSI itself if ANSI SCCP is used (ANSI SCCP is used in World Zone 1). When continuing the established update location dialogue (as with any other dialogue) the VLR must derive the routeing information towards the HLR from the Calling Party Address received with the first responding CONTINUE message until the dialogue terminating message is received. For transactions invoked by the VLR after update location completion, the VLR may derive the information for addressing the HLR from addresses received in the course of the update location procedure (MSISDN or HLR number) or from the IMSI. When invoking the Restore Data procedure and an authentication information retrieval from the HLR preceding it, the VLR must derive the information for addressing the HLR from the address information received in association with the roaming number request. This may be either the IMSI received as a parameter of the MAP message requesting the Roaming Number or the Calling Party Address associated with the MAP message requesting the Roaming Number. The gsmSCF shall be addressed using more than one Global Title number. The first Global Title number is used to address a gsmSCF for MAP. The second Global Title number is used to address a gsmSCF for CAP. For querying the HLR to obtain the VMSC address to support location services, the GMLC has to derive the HLR address from either the MSISDN or IMSI of the target MS. When using the IMSI, the result can be an SPC or an E.214 Mobile Global Title if CCITT or ITU-T SCCP is used, or IMSI itself if ANSI SCCP is used (ANSI SCCP is used in World Zone 1).	

Table 6.1/2

to **from**		**GMLC**
fixed network		---
Home Location Register		---
Visitor Location Register		---
Mobile-services Switching Centre		I:SPC/GT E:GT T:MLC Number
gsm Service Control Function		I:SPC/GT E:GT T:MSISDN
Serving GPRS Support Node		I:SPC/GT E:GT T:MLC Number
Gateway GPRS Support Node		---
Gateway Mobile Location Centre		

I:	Intra-PLMN.
E:	Extra (Inter)-PLMN.
T:	Address Type.
GT:	Global Title.
MGT:	E.214 Mobile Global Title.
SPC:	Signalling Point Code.

6.2 Use of TC

The Mobile Application part makes use of the services offered by the Transaction Capabilities (TC) of Signalling System No. 7. ETS 300 287, which is based on CCITT White Book Recommendations Q.771 to Q.775, should be consulted for the full specification of TC.

The MAP uses all the services provided by TC except the ones related to the unstructured dialogue facility.

From a modelling perspective, the MAP is viewed as a single Application Service Element. Further structuring of it is for further study.

Transaction Capabilities refers to a protocol structure above the network layer interface (i.e., the SCCP service interface) up to the application layer including common application service elements but not the specific application service elements using them.

TC is structured as a Component sub-layer above a Transaction sub-layer.

The Component sub-layer provides two types of application services: services for the control of end-to-end dialogues and services for Remote Operation handling. These services are accessed using the TC-Dialogue handling primitives and TC-Component handling primitives respectively.

Services for dialogue control include the ability to exchange information related to application-context negotiation as well as initialisation data.

Services for Remote Operation handling provide for the exchange of protocol data units invoking tasks (operations), and reporting their outcomes (results or errors) plus any non-application-specific protocol errors detected by the component sub-layer. The reporting of application-specific protocol errors by the TC user, as distinct from application process errors, is also provided. The Transaction sub-layer provides a simple end-to-end connection association service over which several related protocol data units (i.e. built by the Component Sub-Layer) can be exchanged. A Transaction termination can be prearranged (no indication provided to the TC user) or basic (indication provided).

7 General on MAP services

7.1 Terminology and definitions

The term service is used in clauses 7 to 12 as defined in CCITT Recommendation X.200. The service definition conventions of CCITT Recommendation X.210 are also used.

MAP services that are defined for use between HLR and SGSN are also used in an Evolved Packet System (EPS) between two IWFs and between HSS and IWF, where the IWF is an Interworking Function that converts MAP messages to Diameter messages and vice versa.

MAP services that are defined for use between SGSN and EIR are also used in an Evolved Packet System (EPS) between IWF and EIR.

IWFs may be connected via Diameter to MMEs and HSSs and they may be connected via MAP to HSSs, IWFs, and EIRs.

7.2 Modelling principles

MAP provides its users with a specified set of services and can be viewed by its users as a "black box" or abstract machine representing the MAP service-provider. The service interface can then be depicted as shown in figure 7.2/1.

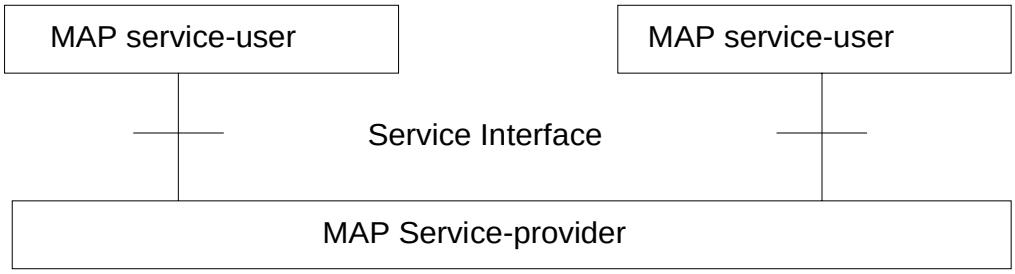


Figure 7.2/1: Modelling principles

The MAP service-users interact with the MAP service-provider by issuing or receiving MAP service-primitives at the service interface.

A MAP service-user may receive services from several instances of the MAP service-provider at the same time. In such cases the overall procedure is synchronised by the service-user.

The MAP service-primitives are named using the following notation:

MAP-ServicePrimitiveName **type**

where **type** can be any of: request (req), indication (ind), response (rsp) or confirm (cnf). (In the user arrow diagrams type is not indicated in the case of req/ind and indicated as "ack" in the case of rsp/cnf).

The services are further classified as unconfirmed-service, confirmed-service and provider-initiated-service where the first two categories refer to whether or not the service is confirmed by the service-provider. The confirmation may or may not correspond to a response provided by the other service-user.

MAP services are also classified as common MAP services that are available to all MAP service-users, and MAP service-user specific services, which are services available to one or several, but not all, MAP service-users.

A MAP dialogue is defined as an exchange of information between two MAP users in order to perform a common task. A MAP dialogue will consist of one or several MAP services.

7.3 Common MAP services

All MAP service-users require access to services for performing basic application layer functions:

- for establishing and clearing MAP dialogues between peer MAP service-users;
- for accessing functions supported by layers below the applications layer;
- for reporting abnormal situations;
- for handling of different MAP versions;
- for testing whether or not a persistent MAP dialogue is still active at each side.

For these purposes the following common services are defined:

- MAP-OPEN service;
- MAP-CLOSE service;
- MAP-DELIMITER service;
- MAP-U-ABORT service;
- MAP-P-ABORT service;
- MAP-NOTICE service.

In defining the service-primitives the following convention is used for categorising parameters:

M the inclusion of the parameter is mandatory. The M category can be used for any primitive type and specifies that the corresponding parameter must be present in the indicated primitive type;

O the inclusion of the parameter is a service-provider option. The O category can be used in indication and confirm type primitives and is used for parameters that may optionally be included by the service-provider;

U the inclusion of the parameter is a service-user option. The U category can be used in request and response type primitives. The inclusion of the corresponding parameter is the choice of the service-user;

C the inclusion of the parameter is conditional. The C category can be used for the following purposes:

- to indicate that if the parameter is received from another entity it must be included for the service being considered;
- to indicate that the service user must decide whether to include the parameter, based on the context on which the service is used;
- to indicate that one of a number of mutually exclusive parameters must be included (e.g. parameters indicating a positive result versus parameters indicating a negative result);
- to indicate that a service user optional parameter (marked "U") or a conditional parameter (marked "C") presented by the service user in a request or response type primitive is to be presented to the service user in the corresponding indication or confirm type primitive;

(=) when appended to one of the above, this symbol means that the parameter takes the same value as the parameter appearing immediately to its left;

blank the parameter is not present.

A primitive type may also be without parameters, i.e. no parameter is required with the primitive type; in this case the corresponding column of the table is empty.

7.3.1 MAP-OPEN service

This service is used for establishing a MAP dialogue between two MAP service-users. The service is a confirmed service with service primitives as shown in table 7.3/1.

Table 7.3/1: Service-primitives for the MAP-OPEN service

Parameters	Request	Indication	Response	Confirm
Application context name	M	M(=)	U	C(=)
Destination address	M	M(=)		
Destination reference	U	C(=)		
Originating address	U	O		
Originating reference	U	C(=)		
Specific information	U	C(=)	U	C(=)
Responding address			U	C(=)
Result			M	M(=)
Refuse-reason			C	C(=)
Provider error				O

Application context name:

This parameter identifies the type of application context being established. If the dialogue is accepted the received application context name shall be echoed. In case of refusal of dialogue this parameter shall indicate the highest version supported.

Destination address:

A valid SCCP address identifying the destination peer entity (see also clause 6). As an implementation option, this parameter may also, in the indication, be implicitly associated with the service access point at which the primitive is issued.

Destination-reference:

This parameter is a reference that refines the identification of the called process. It may be identical to Destination address but its value is to be carried at MAP level. Table 7.3/2 describes the MAP services using this parameter. Only these services are allowed to use it.

Table 7.3/2: Use of the destination reference

MAP service	Reference type	Use of the parameter
MAP-REGISTER-SS	IMSI	Subscriber identity
MAP-ERASE-SS	IMSI	Subscriber identity
MAP-ACTIVATE-SS	IMSI	Subscriber identity
MAP-DEACTIVATE-SS	IMSI	Subscriber identity
MAP-INTERROGATE-SS	IMSI	Subscriber identity
MAP-REGISTER-PASSWORD	IMSI	Subscriber identity
MAP-PROCESS-UNSTRUCTURED-SS-REQUEST	IMSI (note 1)	Subscriber identity
MAP-UNSTRUCTURED-SS-REQUEST	IMSI (note 2)	Subscriber identity
MAP-UNSTRUCTURED-SS-NOTIFY	IMSI (note 2)	Subscriber identity
MAP-FORWARD-SHORT-MESSAGE	IMSI (note 3)	Subscriber identity
MAP-REGISTER-CC-ENTRY	IMSI	Subscriber identity
MAP-ERASE-CC-ENTRY	IMSI	Subscriber identity

NOTE 1: On the HLR - HLR interface and on the HLR - gsmSCF interface the Destination reference shall be either IMSI or MSISDN.

NOTE 2: On the gsmSCF - HLR interface and on the HLR - HLR interface the Destination reference shall be either IMSI or MSISDN.

NOTE 3: Only when the IMSI and the LMSI are received together from the HLR in the mobile terminated short message transfer.

Originating address:

A valid SCCP address identifying the requestor of a MAP dialogue (see also clause 6). As an implementation option, this parameter may also, in the request, be implicitly associated with the service access point at which the primitive is issued.

Originating-reference:

This parameter is a reference that refines the identification of the calling process. It may be identical to the Originating address but its value is to be carried at MAP level. Table 7.3/3 describes the MAP services using the parameter. Only these services are allowed to use it. Processing of the Originating-reference shall be performed according to the supplementary service descriptions and other service descriptions, e.g. operator determined barring. Furthermore the receiving entity may be able to use the value of the Originating-reference to screen the service indication.

Table 7.3/3: Use of the originating reference

MAP service	Reference type	Use of the parameter
MAP-REGISTER-SS	ISDN-Address-String	Originated entity address
MAP-ERASE-SS	ISDN-Address-String	Originated entity address
MAP-ACTIVATE-SS	ISDN-Address-String	Originated entity address
MAP-DEACTIVATE-SS	ISDN-Address-String	Originated entity address
MAP-INTERROGATE-SS	ISDN-Address-String	Originated entity address
MAP-REGISTER-PASSWORD	ISDN-Address-String	Originated entity address
MAP-PROCESS-UNSTRUCTURED-SS-REQUEST	ISDN-Address-String	Originated entity address
MAP-UNSTRUCTURED-SS-REQUEST	ISDN-Address-String (note)	Originated entity address
MAP-UNSTRUCTURED-SS-NOTIFY	ISDN-Address-String (note)	Originated entity address
MAP-REGISTER-CC-ENTRY	ISDN-Address-String	Originated entity address
MAP-ERASE-CC-ENTRY	ISDN-Address-String	Originated entity address

NOTE: The Originating reference may be omitted.

Specific information:

This parameter may be used for passing any user specific information. Establishment and processing of the Specific information is not specified by GSM and shall be performed according to operator specific requirements.

Responding address:

An address identifying the responding entity. The responding address is included if required by the context (e.g. if it is different from the destination address).

Result:

This parameter indicates whether the peer accepts the dialogue.

Refuse reason:

This parameter is present only if the Result parameter indicates that the dialogue is refused. It takes one of the following values:

- Application-context-not-supported;
- Invalid-destination-reference;
- Invalid-originating-reference;
- No-reason-given;
- Remote node not reachable;
- Potential version incompatibility.

7.3.2 MAP-CLOSE service

This service is used for releasing a previously established MAP dialogue. The service may be invoked by either MAP service-user depending on rules defined within the service-user. The service is an unconfirmed service with parameters as shown in table 7.3/4.

Table 7.3/4: Service-primitives for the MAP-CLOSE service

Parameters	Request	Indication
Release method	M	
Specific Information	U	C(=)

Release method:

This parameter can take the following two values:

- normal release; in this case the primitive is mapped onto the protocol and sent to the peer;
- prearranged end; in this case the primitive is not mapped onto the protocol. Prearranged end is managed independently by the two users, i.e. only the request type primitive is required in this case.

Specific information:

This parameter may be used for passing any user specific information. Establishment and processing of the Specific information is not specified by GSM GSM and shall be performed according to operator specific requirements.

7.3.3 MAP-DELIMITER service

This service is used to explicitly request the transfer of the MAP protocol data units to the peer entities.

See also clause 7.4 and 7.5 for the detailed use of the MAP-DELIMITER service.

The service is an unconfirmed service with service-primitives as shown in table 7.3/5.

Table 7.3/5: Service-primitives for the MAP-DELIMITER service

Parameters	Request	Indication

7.3.4 MAP-U-ABORT service

This service enables the service-user to request the MAP dialogue to be aborted. The service is an unconfirmed service with service-primitives as shown in table 7.3/6.

Table 7.3/6: Service-primitives for the MAP-U-ABORT service

Parameters	Request	Indication
User reason	M	M(=)
Diagnostic information	U	C(=)
Specific information	U	C(=)

User reason:

This parameter can take the following values:

- resource limitation (congestion);

  the requested user resource is unavailable due to congestion;

- resource unavailable;

the requested user resource is unavailable for reasons other than congestion;

- application procedure cancellation;

  the procedure is cancelled for reasons detailed in the diagnostic information parameter;

- procedure error;

  processing of the procedure is terminated for procedural reasons.

Diagnostic information:

This parameter may be used to give additional information for some of the values of the user-reason parameter:

Table 7.3/7: User reason and diagnostic information

User reason	Diagnostic information
Resource limitation (congestion)	-
Resource unavailable	Short term/long term problem
Application procedure cancellation	Handover cancellation/ Radio Channel release/ Network path release/ Call release/ Associated procedure failure/ Tandem dialogue released/ Remote operations failure
Procedure error	-

Specific information:

This parameter may be used for passing any user specific information. Establishment and processing of the Specific information is not specified by GSM and shall be performed according to operator specific requirements.

7.3.5 MAP-P-ABORT service

This service enables the MAP service-provider to abort a MAP dialogue. The service is a provider-initiated service with service-primitives as shown in table 7.3/8.

Table 7.3/8: Service-primitives for the MAP-P-ABORT service

Parameters		Indication
Provider reason		M
Source		M

Provider reason:

This parameter indicates the reason for aborting the MAP dialogue:

- provider malfunction;
- supporting dialogue/transaction released;
- resource limitation;
- maintenance activity;
- version incompatibility;
- abnormal MAP dialogue.

Source:

This parameter indicates the source of the abort. For Transaction Capabilities (TC) applications the parameter may take the following values:

- MAP problem;
- TC problem;
- network service problem.

Table 7.3/9: Values of provider reason and source parameters and examples of corresponding events

Provider reason	Source	Corresponding event
Provider	MAP	Malfunction at MAP level at peer entity
malfunction	TC	"Unrecognised message type" or "Badly formatted transaction portion" or "Incorrect transaction portion" received in TC-P-ABORT "Abnormal dialogue"
	Network service	Malfunction at network service level at peer entity
Supporting dialogue/ transaction released	TC	"Unrecognised transaction ID" received in TC-ABORT
Resource	MAP	Congestion towards MAP peer service-user
limitation	TC	"Resource limitation" received in TC-P-ABORT
Maintenance	MAP	Maintenance at MAP peer service-user
activity	Network service	Maintenance at network peer service level
Abnormal MAP dialogue	MAP	MAP dialogue is not in accordance with specified application context
Version incompatibility	TC	A Provider Abort indicating "No common dialogue portion" is received in the dialogue initiated state

7.3.6 MAP-NOTICE service

This service is used to notify the MAP service-user about protocol problems related to a MAP dialogue not affecting the state of the protocol machines.

The service is a provider-initiated service with service-primitive as shown in table 7.3/10.

Table 7.3/10: Service-primitive for the MAP-NOTICE service

Parameters	Indication
Problem diagnostic	M

Problem diagnostic:

This parameter can take one of the following values:

- abnormal event detected by the peer;
- response rejected by the peer;
- abnormal event received from the peer;
- message cannot be delivered to the peer.

7.3.7 Void

7.3.8 Void

7.3.9 Void

7.3.10 Void

7.4 Sequencing of services

The sequencing of services is shown in figure 7.4/1 and is as follows:

Opening:

The MAP-OPEN service is invoked before any user specific service-primitive is accepted. The sequence may contain none, one or several user specific service-primitives. If no user specific service-primitive is contained between the MAP-OPEN and the MAP-DELIMITER primitives, then this will correspond to sending an empty Begin message in TC. If more than one user specific service-primitive is included, all are to be sent in the same Begin message. The sequence ends with a MAP-DELIMITER primitive.

Continuing:

This sequence may not be present in some MAP dialogues. If it is present, it ends with a MAP-DELIMITER primitive. If more than one user specific service-primitive is included, all are to be included in the same Continue message.

Closing:

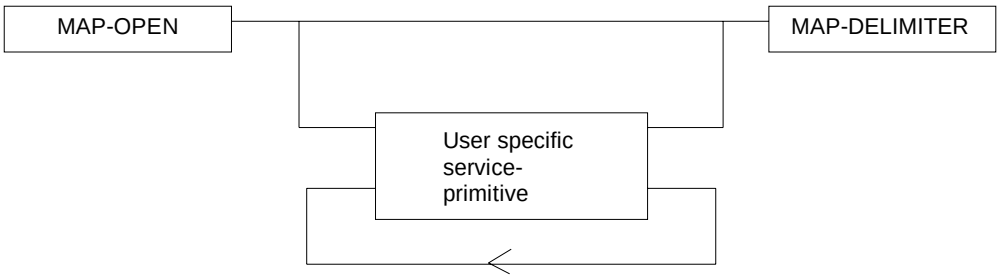
The sequence can only appear after an opening sequence or a continuing sequence. The sequence may contain none, one or several user specific service-primitives if the MAP-CLOSE primitive specifies normal release. If no user specific service-primitive is included, then this will correspond to sending an empty End message in TC. If more than one user specific service-primitive is included, all are to be sent in the same End message. If prearranged end is specified, the sequence cannot contain any user specific service-primitive. The MAP-CLOSE primitive must be sent after all user specific service-primitives have been delivered to the MAP service-provider.

Aborting:

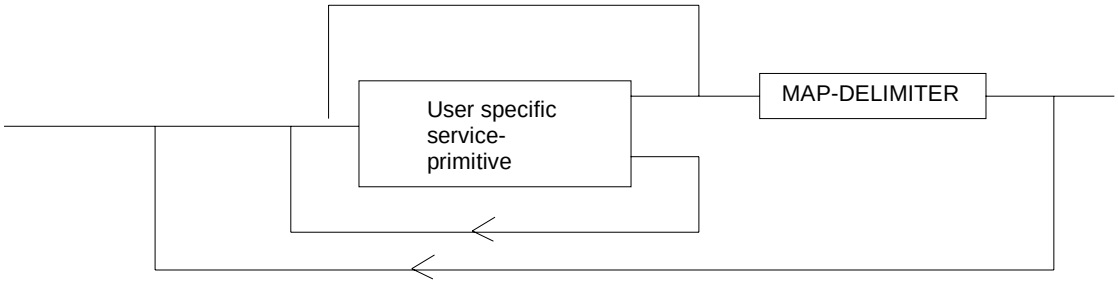
A MAP service-user can issue a MAP-U-ABORT primitive at any time after the MAP dialogue has been opened or as a response to an attempt to open a MAP dialogue.

The MAP service-provider may issue at any time a MAP-P-ABORT primitive towards a MAP service-user for which a MAP dialogue exists.

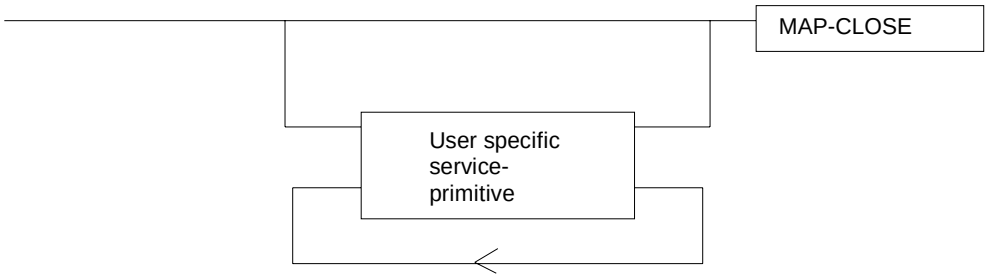
MAP-U-ABORT primitives and MAP-P-ABORT primitives terminate the MAP dialogue.



a) Opening



b) Continuing



c) Closing



d) Aborting

Figure 7.4/1: Sequencing of services

If the reason "resource unavailable (short term problem)" is indicated in the MAP-U-ABORT indication primitive, the MAP service-user may decide to attempt a new MAP dialogue establishment immediately.

Sequencing of user specific service-primitives is done by the MAP service-user and based on rules applicable for each MAP service-user instance.

A MAP-NOTICE indication primitive may be received at any time during the active period of a MAP dialogue.

7.5 General rules for mapping of services onto TC

7.5.1 Mapping of common services

Table 7.5/1 gives an overview of the mapping rules for mapping of common services onto TC-services. Table 7.5/2 gives the mapping rules for mapping of TC-services onto common services.

Protocol machine description is given in clauses 14 to 17.

Table 7.5/1: Mapping of common services onto TC services

MAP service-primitive	TC service-primitive
MAP-OPEN request (+ any user specific service primitives) + MAP-DELIMITER request	TC-BEGIN request (+ component handling primitives)
MAP-OPEN response (+ any user specific service primitives) + MAP-DELIMITER request	TC-CONTINUE request (note) (+ component handling primitives)
(any user specific service primitives) + MAP-DELIMITER request	TC-CONTINUE request (+ component handling primitives)
(any user specific service primitives) + MAP-CLOSE request	TC-END request (+ component handling primitives)

MAP-U-ABORT request	TC-U-ABORT request
NOTE: Or TC-END if the MAP-CLOSE request has been received before the MAP-DELIMITER request.	

Table 7.5/2: Mapping of TC services onto common service

TC service-primitive	MAP service-primitive
TC-BEGIN indication (+ component handling primitives)	MAP-OPEN indication (+ user specific service primitives) + MAP-DELIMITER indication (note 1)
TC-CONTINUE indication (+ component handling primitives)	First time: MAP-OPEN confirm (+ user specific service primitives) + MAP-DELIMITER indication (note 1) Subsequent times: (user specific service primitives) + MAP-DELIMITER indication (note 1)
TC-END indication (+ component handling primitives)	MAP-OPEN confirm (note 6) (user specific service primitives) + MAP-CLOSE indication
TC-U-ABORT indication	MAP-U-ABORT indication or MAP-P-ABORT indication (note 2) MAP-OPEN confirmation (note 3)
TC-P-ABORT indication	MAP-P-ABORT indication (note 4) MAP-OPEN confirmation (note 5)
NOTE 1: It may not be necessary to present this primitive to the user for MAP version 2 applications. NOTE 2: The mapping depends on whether the TC-U-ABORT indication primitive contains a MAP-abort-PDU from the remote MAP service-provider or a MAP-user-abort-PDU from the remote MAP service-user. NOTE 3: Only if the opening sequence is pending and if the "Abort Reason" in the TC-U-ABORT indication is set to "Application Context Not Supported". NOTE 4: If the "Abort Reason" in the TC-P-ABORT indication is set to a value different from "Incorrect Transaction Portion". NOTE 5: Only if the opening sequence is pending and if the "Abort Reason" in the TC-P-ABORT indication is set to "Incorrect Transaction Portion". NOTE 6: Only if opening sequence is pending.	

7.5.2 Mapping of user specific services

Table 7.5/3 gives the general mapping rules which apply to mapping of MAP user specific services onto TC services and table 7.5/4 gives the similar rules for mapping of TC services onto MAP user specific services. Detailed mapping is given in clauses 14 to 17.

Table 7.5/3: Mapping of MAP user specific services onto TC services

MAP service-primitive	TC-service-primitive
MAP-xx request	TC-INVOKE request
MAP-xx response (note 1)	TC-RESULT-L request TC-U-ERROR request TC-U-REJECT request TC-INVOKE request (note 2)

Table 7.5/4: Mapping of TC services onto MAP user specific services

TC-service-primitive	MAP service-primitive
TC-INVOKE indication	MAP-xx indication
TC-RESULT-L indication (note 4) TC-U-ERROR indication TC-INVOKE indication (note 2)	MAP-xx confirm

TC-L-CANCEL indication	
TC-U-REJECT indication TC-L-REJECT indication TC-R-REJECT indication	MAP-xx confirm or MAP-NOTICE indication (note 3)

Notes to tables 7.5/3 and 7.5/4:

NOTE 1: The mapping is determined by parameters contained in the MAP-xx response primitive.

NOTE 2: This applies only to TC class 4 operations where the operation is used to pass a result of another class 2 or class 4 operation.

NOTE 3: The detailed mapping rules are given in clause 16.

NOTE 4: If RESULT-NL components are present they are mapped onto the same MAP-xx confirm.

7.6 Definition of parameters

7.6.1 Common parameters

The following set of parameters is used in several MAP service-primitives.

7.6.1.1 Invoke Id

This parameter identifies corresponding service primitives. The parameter is supplied by the MAP service-user and must be unique over each service-user/service-provider interface.

7.6.1.2 Linked Id

This parameter is used for linked services and it takes the value of the invoke Id of the service linked to.

7.6.1.3 Provider error

This parameter is used to indicate a protocol related type of error:

- duplicated invoke Id;
- not supported service;
- mistyped parameter;
- resource limitation;
- initiating release, i.e. the peer has already initiated release of the dialogue and the service has to be released;
- unexpected response from the peer;
- service completion failure;
- no response from the peer;
- invalid response received.

7.6.1.4 User error

This parameter can take values as follows:

NOTE: The values are grouped in order to improve readability; the grouping has no other significance.

a) Generic error:

- system failure, i.e. a task cannot be performed because of a problem in the entity reporting the error or in another entity. The type of entity or network resource may be indicated by use of the network resource parameter or additional network resource parameter. If and only if the problem is in the entity reporting the error, a cause of failure (FailureCauseParam) shall be included;
- data missing, i.e. an optional parameter required by the context is missing;
- unexpected data value, i.e. the data type is formally correct but its value or presence is unexpected in the current context;
- resource limitation;
- initiating release, i.e. the receiving entity has started the release procedure;
- facility not supported, i.e. the requested facility is not supported by the PLMN with detailed reasons as follows:
  - Shape of location estimate not supported;
  - Needed LCS capability not supported in serving node;
- incompatible terminal, i.e. the requested facility is not supported by the terminal.

b) Identification or numbering problem:

- unknown subscriber, i.e. no such subscription exists;
- number changed, i.e. the subscription does not exist for that number any more;
- unknown MSC;
- unidentified subscriber, i.e. if the subscriber is not contained in the database and it has not or cannot be established whether or not a subscription exists;
- unallocated roaming number;
- unknown equipment;
- unknown location area.

c) Subscription problem:

- roaming not allowed, i.e. a location updating attempt is made in an area not covered by the subscription;
- illegal subscriber, i.e. illegality of the access has been established by use of authentication procedure;
- bearer service not provisioned;
- teleservice not provisioned;
- illegal equipment, i.e. the IMEI check procedure has shown that the IMEI is blacklisted or not whitelisted.

d) Handover problem:

- no handover number available, i.e. the VLR cannot allocate a number for handover or cannot allocate the required amount of numbers for relocation;
- subsequent handover failure, i.e. handover to a third MSC failed for some reason;
- target cell outside group call area.

e) Operation and maintenance problem:

- tracing buffer full, i.e. tracing cannot be performed because the tracing capacity is exceeded.

f) Call set-up problem:

- no roaming number available, i.e. a roaming number cannot be allocated because all available numbers are in use;
- absent subscriber, i.e. the subscriber has activated the detach service or the system detects the absence condition. This error may be qualified to indicate whether the subscriber was IMSI detached, in a restricted area or did not respond to paging;
- busy subscriber. This error may be qualified to indicate that the subscriber was busy due to CCBS and that CCBS is possible;
- no subscriber reply;
- forwarding violation, i.e. the call has already been forwarded the maximum number of times that is allowed;
- CUG reject, i.e. the call does not pass a CUG check; additional information may also be given in order to indicate rejection due to e.g. incoming call barred or non-CUG membership;
- call barred. Optionally, additional information may be included for indicating either that the call meets a barring condition set by the subscriber or that the call is barred for operator reasons. In the case of barring of Mobile Terminating Short Message, the additional information may indicate a barring condition due to "Unauthorised Message Originator"; if the call is rejected due to the application of the ACR supplementary service, the additional information shall indicate a barring condition due to "Anonymous Call Rejection";
- optimal routeing not allowed, i.e. the entity which sends the error does not support optimal routeing, or the HLR will not accept an optimal routeing interrogation from the GMSC, or the call cannot be optimally routed because it would contravene optimal routeing constraints;
- forwarding failed, i.e. the GMSC interrogated the HLR for forwarding information but the HLR returned an error.

g) Supplementary services problem:

- call barred;
- illegal SS operation;
- SS error status;
- SS not available;
- SS subscription violation;
- SS incompatibility;
- negative password check;
- password registration failure;
- Number of Password Attempts;
- USSD Busy;
- Unknown Alphabet;
- short term denial;
- long term denial.

For definition of these errors see 3GPP TS 24.080 [38].

h) Short message problem:

- SM delivery failure with detailed reason as follows:
  - memory capacity exceeded;
  - MS protocol error;

- MS not equipped;
- unknown service centre (SC);
- SC congestion;
- invalid SME address;
- subscriber is not an SC subscriber;
- and possibly detailed diagnostic information, coded as specified in 3GPP TS 23.040, under SMS-SUBMIT-REPORT and SMS-DELIVERY-REPORT. If the SM entity that returns the SM Delivery Failure error includes detailed diagnostic information, it shall be forwarded in the MAP_MO_FORWARD_SHORT_MESSAGE and in the MAP_MT_FORWARD_SHORT_MESSAGE response.

- message waiting list full, i.e. no further SC address can be added to the message waiting list.
- Subscriber busy for MT SMS, i.e. the mobile terminated short message transfer cannot be completed because:
  - another mobile terminated short message transfer is going on and the delivery node does not support message buffering; or
  - another mobile terminated short message transfer is going on and it is not possible to buffer the message for later delivery; or
  - the message was buffered but it is not possible to deliver the message before the expiry of the buffering time defined in 3GPP TS 23.040;
- Absent Subscriber SM, i.e. the mobile terminated short message transfer cannot be completed because the network cannot contact the subscriber. Diagnostic information regarding the reason for the subscriber's absence may be included with this error.

i) Location services problem:

- Unauthorised Requesting Network
- Unauthorised LCS Client with detailed reasons as follows:
  - NoAdditional Information
  - Client not in MS Privacy Exception List
  - Call to Client not setup
  - Disallowed by Local Regulatory Requirements
  - Unauthorised Privacy Class
  - Unauthorised Call/Session Unrelated External Client
  - Unauthorised Call/Session Related External Client
  - Privacy override not applicable
- Position method failure with detailed reasons as follows:
  - Congestion
  - Insufficient resources
  - Insufficient Measurement Data
  - Inconsistent Measurement Data
  - Location procedure not completed

- QoS not attainable
- Position Method Not Available in Network
- Position Method Not Available in Location Area
- Unknown or unreachable LCS Client.

7.6.1.5 All Information Sent

This parameter indicates to the receiving entity when the sending entity has sent all necessary information.

7.6.2 Numbering and identification parameters

7.6.2.1 IMSI

This parameter is the International Mobile Subscriber Identity defined in 3GPP TS 23.003 [17].

7.6.2.2 TMSI

This parameter is the Temporary Mobile Subscriber Identity defined in 3GPP TS 23.003 [17].

7.6.2.3 IMEI

This parameter is the International Mobile Equipment Identity defined in 3GPP TS 23.003 [17].

7.6.2.3a IMEISV

This parameter is the International Mobile Equipment Identity and Software Version Number defined in 3GPP TS 23.003 [17].

7.6.2.4 Previous location area Id

This parameter refers to the identity of the location area from which the subscriber has roamed.

7.6.2.5 Stored location area Id

This parameter refers to the location area where the subscriber is assumed to be located.

7.6.2.6 Current location area Id

This parameter is used to indicate the location area in which the subscriber is currently located.

7.6.2.7 Target location area Id

This parameter refers to the location area into which the subscriber intends to roam.

7.6.2.8 Target cell Id

This parameter refers to the identity of the cell to which a call has to be handed over.

7.6.2.8A Target RNC Id

This parameter refers to the identity of the RNC to which a call has to be relocated.

7.6.2.9 Void

7.6.2.10 Originating entity number

This parameter refers to an application layer identification of a system component in terms of its associated ISDN number.

7.6.2.11 MSC number

This parameter refers to the ISDN number of an MSC.

7.6.2.12 Target MSC number

This parameter refers to the ISDN number of an MSC to which a call has to be handed over.

7.6.2.13 HLR number

This parameter refers to the ISDN number of an HLR.

7.6.2.14 VLR number

This parameter refers to the ISDN number of a VLR.

7.6.2.15 HLR Id

This parameter refers to the identity of an HLR derived from the IMSI defined in CCITT Recommendation E.212.

7.6.2.16 LMSI

This parameter refers to a local identity allocated by the VLR to a given subscriber for internal management of data in the VLR. LMSI shall not be sent to the SGSN.

7.6.2.17 MS ISDN

This parameter refers to one of the ISDN numbers assigned to a mobile subscriber in accordance with CCITT Recommendation E.213.

7.6.2.17A Additional MSISDN

This parameter refers to an ISDN number assigned on top of the existing MSISDN, to a user with a connection to the PS domain (see 3GPP TS 23.003 [17]). If the Additional MSISDN is available its value shall be used as C-MSISDN on the Sv interface.

7.6.2.18 OMC Id

This parameter refers to the identity of an Operation and Maintenance Centre.

7.6.2.19 Roaming number

This parameter refers to the roaming number as defined in CCITT Recommendation E.213.

7.6.2.19A Relocation Number List

This parameter refers to the number(s) used for routing one call or several calls between MSCs during relocation.

7.6.2.20 Void

7.6.2.21 Handover number

This parameter refers to the number used for routing a call between MSCs during handover.

7.6.2.22 Forwarded-to number

This parameter refers to the address to which a call is to be forwarded. A subaddress may be appended. For subscribers having an originating CAMEL Phase 2 or higher subscription, this address need not be in E.164 international format.

7.6.2.22A Long forwarded-to number

This parameter refers to the address to which a call is to be forwarded. A subaddress may be appended. For subscribers having an originating CAMEL Phase 2 or higher subscription this address need not be in international format.

7.6.2.22B Long FTN Supported

This parameter indicates that the sending entity supports Long Forwarded-to Numbers.

7.6.2.23 Forwarded-to subaddress

This parameter refers to the sub-address attached to the address to which a call is to be forwarded.

7.6.2.24 Called number

This parameter refers to a called party number as defined in CCITT Recommendation Q.767.

7.6.2.25 Calling number

This parameter refers to a calling party number as defined in CCITT Recommendation Q.767.

7.6.2.26 Originally dialled number

This parameter refers to the number dialled by the calling party in order to reach a mobile subscriber.

7.6.2.27 Service centre address

This parameter represents the address of a Short Message Service Centre.

7.6.2.28 Zone Code

This parameter is used to define location areas into which the subscriber is allowed or not allowed to roam (regional subscription). With a complete list of Zone Codes the VLR or the SGSN or MME is able to determine for all its location areas, routing areas or tracking areas whether roaming is allowed or not.

7.6.2.29 MSIsdn-Alert

This parameter refers to the MSISDN stored in a Message Waiting Data File in the HLR. It is used to alert the Service Centre when the MS is again attainable.

7.6.2.30 Location Information

The VLR indicates in this parameter the location of the served subscriber as defined in 3GPP TS 23.018 [97].

7.6.2.30a Location Information for GPRS

The SGSN indicates in this parameter the location of the served subscriber as defined in 3GPP TS 23.078 [98].

7.6.2.30b Location Information for EPS

The MME (via an IWF) indicates in this parameter the location of the served subscriber.

7.6.2.31 GMSC Address

This parameter refers to the E.164 address of a GMSC.

7.6.2.32 VMSC Address

This parameter refers to the E.164 address of a VMSC.

7.6.2.33 Group Id

This parameter is used to describe groups a subscriber can be a member of. A subscriber can partake in all group calls (VBS/VGCS) where he subscribed to the respective groups.

7.6.2.34 North American Equal Access preferred Carrier Id

This parameter refers to the carrier identity preferred by the subscriber for calls requiring routing via an inter-exchange carrier. This identity is used at:

- outgoing calls: when the subscriber does not specify at call set-up a carrier identity;
- forwarded calls: when a call is forwarded by the subscriber;
- incoming calls: applicable to the roaming leg of the call.

7.6.2.35 Void

7.6.2.36 Void

7.6.2.37 Serving cell Id

This parameter indicates the cell currently being used by the served subscriber.

7.6.2.38 SGSN number

This parameter refers to the ISDN number of a SGSN.

7.6.2.39 SGSN address

This parameter refers to the IP-address of a SGSN. This parameter is defined in 3GPP TS 23.003 [17].

7.6.2.40 GGSN address

This parameter refers to the IP-address of a GGSN. This parameter is defined in 3GPP TS 23.003 [17].

7.6.2.41 GGSN number

This parameter refers to the ISDN number of a GGSN or the ISDN number of the protocol-converter if a protocol-converting GSN is used between the GGSN and the HLR.

7.6.2.42 APN

This parameter refers to the DNS name of a GGSN. This parameter is defined in 3GPP TS 23.060 [104].

7.6.2.43 Network Node number

This parameter refers to the ISDN number of an MT-SMS target node (MSC or MME, SGSN, or IP-SM-GW) or of an SMS Router.

7.6.2.43A Network Node Diameter Address

This parameter refers to the Diameter Name and Realm of the same MT-SMS target node or SMS Router of which the ISDN number is within the Network Node number parameter.

7.6.2.44 PDP-Type

This parameter indicates which type of protocol is used by the MS as defined in 3GPP TS 23.060 [104]. The allowed values are one of IPv4 encoded as HEX (21) or IPv6 encoded as HEX (57), or Non-IP encoded as HEX (02).

NOTE: To indicate both IPv4 and IPv6 PDP types are allowed, but not IPv4v6, two PDP contexts need to be present in the subscription for the same APN, one indicating IPv4 PDP type and one indicating IPv6 PDP type.

7.6.2.44A Extension PDP-Type

This parameter indicates the support of the dual-stack PDP-type (IPv4v6) encoded as HEX (8D) by a certain PDP, as defined in 3GPP TS 23.060 [104], and it is an extension to PDP-Type.

7.6.2.45 PDP-Address

This parameter indicates the address of the data protocol as defined in 3GPP TS 23.060 [104].

7.6.2.45A Extension PDP-Address

This parameter indicates an additional address of the data protocol, and it is included when the PDP supports dual-stack (IPv4v6).

It is an extension to PDP-Address and it is encoded in the same way. IPv4 or IPv6 address types can be used in this parameter but, when both parameters are present, each of them shall contain different address types.

7.6.2.46 Additional number

This parameter refers to the ISDN number of an additional MT-SMS target node (MSC or MME or SGSN) or of an SMS Router.

7.6.2.46A Additional Network Node Diameter Address

This parameter refers to an additional Diameter Name and Realm of the same MT-SMS target node or SMS Router of which the ISDN number is within the Additional number parameter.

7.6.2.46B Third Number

This parameter refers to the ISDN number of a third MT-SMS target node (MSC or MME or SGSN).

7.6.2.46C Third Network Node Diameter Address

This parameter refers to the third Diameter Name and Realm of the same MT-SMS target node of which the ISDN number is within the Third number parameter.

7.6.2.47 P-TMSI

This parameter is the Packet Temporary Mobile Subscriber Identity defined in 3GPP TS 23.003 [17].

7.6.2.48 B-subscriber number

This parameter refers to the number of the destination B dialled by the A user. This may include a subaddress.

7.6.2.49 B-subscriber subaddress

This parameter refers to the sub-address attached to the destination B dialled by the A user.

7.6.2.50 LMU Number

This parameter refers to a local number assigned to an LMU by an SMLC.

7.6.2.51 MLC Number

This parameter refers to the ISDN (E.164) number of an MLC.

7.6.2.52 Multicall Bearer Information

This parameter refers to the number of simultaneous bearers supported per user by the serving network.

7.6.2.53 Multiple Bearer Requested

This parameter indicates whether multiple bearers are requested for a relocation.

7.6.2.54 Multiple Bearer Not Supported

This parameter indicates whether multiple bearers are supported.

7.6.2.55 PDP-Charging Characteristics

This parameter indicates the charging characteristics associated with a specific PDP context as defined in 3GPP TS 32.215.

7.6.2.56 Selected RAB ID

The selected radio access bearer to be kept at subsequent inter-MSC handover from UMTS to GSM.

7.6.2.57 RAB ID

This parameter indicates the radio access bearer identifier as defined in 3GPP TS 25.413. This parameter is used to relate the radio resources with the radio access bearers.

7.6.2.58 gsmSCF Address

This parameter refers to the ISDN number assigned to the gsmSCF address. In an IP Multimedia Core Network, the gsmSCF-address shall contain the IM-SSF address when the IM-SSF takes the role of the gsmSCF.

7.6.2.59 V-GMLC Address

This parameter refers to the IP address of a V-GMLC.

7.6.2.60 Void

7.6.2.61 H-GMLC Address

This parameter refers to the IP address of a H-GMLC.

7.6.2.62 PPR Address

This parameter refers to the IP address of a Privacy Profile Register.

7.6.2.63 Routeing Number

This parameter refers to a number used for routeing purpose and identifying a network operator. See 3GPP TS 23.066 [108].

7.6.2.64 Additional V-GMLC Address

This parameter refers to the IP address of a V-GMLC.

7.6.2.65 MME Name

This parameter refers to the Diameter Identity of an MME as defined in 3GPP TS 23.003 [17].

7.6.2.66 3GPP AAA Server Name

This parameter refers to the Diameter Identity of a 3GPP AAA server as defined in 3GPP TS 29.273 [151].

7.6.2.67 CSS number

This parameter refers to the ISDN number of a CSS as defined in 3GPP TS 23.003[17].

7.6.2.68 SGSN Name

This parameter refers to the Diameter Identity of an SGSN as defined in 3GPP TS 23.003 [17].

7.6.2.69 SGSN Realm

This parameter refers to the Diameter Identity of an SGSN as defined in 3GPP TS 23.003 [17].

7.6.3 Subscriber management parameters

7.6.3.1 Category

This parameter refers to the calling party category as defined in CCITT Recommendation Q.767.

7.6.3.2 Equipment status

This parameter refers to the status of the mobile equipment as defined in 3GPP TS 22.016 [7].

7.6.3.2a BMUEF

This parameter refers to the Bit Map of UE Faults and corresponds to the UESBI-Iu parameter defined in 3GPP TS 25.413 [120].

7.6.3.3 Extensible Bearer service

This parameter may refer to a single bearer service, a set of bearer services or to all bearer services as defined in 3GPP TS 22.002 [3]. This parameter is used only for subscriber profile management. Extensible Bearer service values include all values defined for a Bearer service parameter (7.6.4.38).

7.6.3.4 Extensible Teleservice

This parameter may refer to a single teleservice, a set of teleservices or to all teleservices as defined in 3GPP TS 22.003 [4]. This parameter is used only for subscriber profile management. Extensible Teleservice values include all values defined for a Teleservice parameter (7.6.4.39).

7.6.3.5 Extensible Basic Service Group

This parameter refers to the Basic Service Group either as an extensible bearer service (see clause 7.6.3.3) or an extensible teleservice (see clause 7.6.3.4). This parameter is used only for subscriber profile management. The null value (i.e. neither extensible bearer service nor extensible teleservice) is used to denote the group containing all extensible bearer services and all extensible teleservices.

7.6.3.6 GSM bearer capability

This parameter refers to the GSM bearer capability information element defined in 3GPP TS 24.008 [35].

7.6.3.7 Subscriber Status

This parameter refers to the barring status of the subscriber:

- service granted;
- Operator Determined Barring.

7.6.3.8 CUG Outgoing Access indicator

This parameter represents the Outgoing Access as defined in ETS 300 136.

7.6.3.9 Operator Determined Barring General Data

This parameter refers to the set of subscriber features that the network operator or the service provider can regulate. This set only includes those limitations that can be

a) controlled in the VLR,
b) controlled in the SGSN or MME,
c) controlled in the SGSN applied for short message transfer only,
d) interrogated or modified by the gsmSCF:

ODB category	Controlled in the VLR	Controlled in the SGSN or MME	Controlled in the SGSN applied for short message transfer only	Interrogatable and modifyable by the gsmSCF
All outgoing calls barred	X		X	X
International outgoing calls barred	X		X	X
International outgoing calls except those to the home PLMN country barred	X		X	X
Interzonal outgoing calls barred	X		X	X
Interzonal outgoing calls except those to the	X		X	X

home PLMN country barred				
Interzonal outgoing calls AND international outgoing calls except those directed to the home PLMN country barred	X		X	X
Premium rate (information) outgoing calls barred	X			X
Premium rate (entertainment) outgoing calls barred	X			X
Supplementary service access barred	X			X
Invocation of call transfer barred	X			X
Invocation of chargeable call transfer barred	X			X
Invocation of internationally chargeable call transfer barred	X			X
Invocation of interzonally chargeable call transfer barred	X			X
Invocation of call transfer where both legs are chargeable barred	X			X
Invocation of call transfer if there is already an ongoing transferred call for the served subscriber in the serving MSC/VLR barred	X			X
All packet Oriented Services barred		X		X
Roamer Access to HPLMN-AP barred		X		X
Roamer Access to VPLMN-AP barred		X		X
Outgoing calls when roaming outside the home PLMN country				X
All incoming calls				X

Incoming calls when roaming outside the home PLMN country				X
Incoming calls when roaming outside the zone of the home PLMN country				X
Roaming outside the home PLMN				X
Roaming outside the home PLMN country				X
Registration of any call forwarded-to number				X
Registration of any international call forwarded-to number				X
Registration of any international call forwarded-to number except to a number within the HPLMN country				X
Registration of any inter-zone call forwarded-to number				X
Registration of any inter-zone call forwarded-to number except to a number within the HPLMN country				X

7.6.3.10 ODB HPLMN Specific Data

This parameter refers to the set of subscriber features that the network operator or the service provider can regulate only when the subscriber is registered in the HPLMN. This set only includes those limitations that can be controlled in the VLR or in the SGSN or MME:

- Operator Determined Barring Type 1;
- Operator Determined Barring Type 2;
- Operator Determined Barring Type 3;
- Operator Determined Barring Type 4.

7.6.3.11 Regional Subscription Data

This parameter defines the regional subscription area in which the subscriber is allowed to roam. It consists of a list of Zone Codes (see clause 7.6.2.28).

7.6.3.12 Regional Subscription Response

This parameter indicates either that the regional subscription data cannot be handled or that the current MSC or SGSN or MME area is entirely restricted because of regional subscription.

7.6.3.13 Roaming Restriction Due To Unsupported Feature

This parameter defines that a subscriber is not allowed to roam in the current MSC area. It may be used by the HLR if a feature or service is indicated as unsupported by the VLR.

7.6.3.14 Extensible SS-Info

This parameter refers to all the information related to a supplementary service and is a choice between:

- extensible forwarding information (see clause 7.6.3.15);
- extensible call barring information (see clause 7.6.3.20);
- CUG info (see clause 7.6.3.22);
- extensible SS-Data (see clause 7.6.3.29).

7.6.3.15 Extensible forwarding information

This parameter represents the information related to each call forwarding service:

- the SS-Code of the relevant call forwarding service (see clause 7.6.4.1);
- if required, a list of extensible forwarding feature parameters (see clause 7.6.3.16).

  The list may contain one item per Basic Service Group.

7.6.3.16 Extensible forwarding feature

This parameter applies to each combination of call forwarding service and Basic Service Group and contains the following information, as required:

- extensible Basic Service Group (see clause 7.6.3.5);
- extensible SS-Status (see clause 7.6.3.17);
- forwarded-to number (see clause 7.6.2.22);
- forwarded-to subaddress (see clause 7.6.2.23);
- extensible forwarding options (see clause 7.6.3.18);
- extensible no reply condition timer (see clause 7.6.4.19);
- long forwarded-to number (see clause 7.6.2.22A).

If a number is required to define the forwarded-to destination then:

- If the VLR supports Long Forwarded-to Numbers then the long forwarded-to number shall be present and the forwarded-to number shall be absent;
- If the VLR does not support Long Forwarded-to Numbers then the forwarded-to number shall be present and the long forwarded-to number shall be absent.

7.6.3.17 Extensible SS-Status

This parameter refers to the state information of individual supplementary services as defined in 3GPP TS 23.011 [22].

7.6.3.18 Extensible Forwarding Options

This parameter refers to a set of forwarding options attached to a supplementary service. It contains the following information:

- notification to forwarding party (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- redirection notification to the forwarded-to party (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- notification to calling party (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- redirecting presentation (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- forwarding reason (see 3GPP TS 22.082 [10] for the meaning of this parameter).

7.6.3.19 Extensible No reply condition timer

This parameter refers to the extensible no reply condition timer for call forwarding on no reply.

7.6.3.20 Extensible Call barring information

This parameter contains for each call barring service:

- SS-Code (see clause 7.6.4.1);
- a list of extensible call barring feature parameters (see clause 7.6.3.21).

  The list may contain one item per Basic Service Group.

7.6.3.21 Extensible Call barring feature

This parameter gives the status of call barring services as applicable to each Basic Service Group. The parameter contains the following information:

- Extensible Basic Service Group (see clause 7.6.3.5);
- provisioned SS-Status (see clause 7.6.3.17).

7.6.3.22 CUG info

This parameter refers to the overall information required for operation for each CUG:

- CUG subscriptionList;
- CUG featureList.

7.6.3.23 CUG subscription

This parameter refers to the set of basic information for each CUG defined in that subscription. The following information is stored:

- CUG index;
- CUG interlock;
- Intra CUG restrictions;
- Basic Service Group List.

7.6.3.24 CUG interlock

This parameter represents the CUG interlock code defined in ETS 300 138.

7.6.3.25 CUG index

This parameter represents the CUG index defined in ETS 300 138.

7.6.3.26 CUG feature

This parameter contains two parameters that are associated with the Basic Service Group. If the Basic Service Group Code is not present the feature applies to all Basic Services. The following parameters are included:

- Preferential CUG indicator:
  - indicates which CUG index is to be used at outgoing call set-up using the associated Basic Service Group;
- Inter CUG Option:
  - describes whether it for the associated Basic Service Group is allowed to make calls outside the CUG and whether incoming calls are allowed;
- Basic Service Group.

See 3GPP TS 22.085 [13] for meaning of this parameter.

7.6.3.27 Inter CUG options

This parameter indicates the subscribers' ability to make and receive calls outside a specific closed user group. It takes any of the following values:

- CUG only facility (only calls within CUG are allowed);
- CUG with outgoing access (calls outside CUG allowed);
- CUG with incoming access (calls from outside CUG into CUG allowed);
- CUG with both incoming and outgoing access (all calls allowed).

7.6.3.28 Intra CUG restrictions

This parameter describes whether or not the subscriber is allowed to originate calls to or to receive calls from within the CUG. It can take any of the following values:

- no CUG restrictions;
- CUG incoming calls barred;
- CUG outgoing calls barred.

7.6.3.29 Extensible SS-Data

This parameter refers to the necessary set of information required in order to characterise one supplementary service:

- SS-Code (see clause 7.6.4.1);
- Extensible SS-Status (if applicable) (see clause 7.6.3.17);
- Extensible Override subscription option (if applicable) (see clause 7.6.3.30);
- Extensible CLI Restriction (if applicable) (see clause 7.6.3.31);
- Extensible Basic Service Group Code (see clause 7.6.3.5).

7.6.3.30 Subscriber State

This parameter indicates the state of the MS as defined in 3GPP TS 23.018 [97].

7.6.3.31 Requested Info

This parameter indicates the subscriber information being requested as defined in 3GPP TS 23.018 [97] and 3GPP TS 23.078 [98].

7.6.3.31A Requested Domain

This parameter indicates the domain (circuit switched, i.e. from the MSC/VLR, or packet switched, i.e. from the SGSN) from which the requested information should be retrieved.

7.6.3.32 Suppression of Announcement

This parameter indicates if the announcement or tones shall be suppressed as defined in 3GPP TS 23.078 [98].

7.6.3.33 Suppress T-CSI

This parameter is used to suppress the invocation of terminating CAMEL services.

7.6.3.34 GMSC CAMEL Subscription Info

This parameter contains CAMEL subscription information, i.e. O-CSI and/or D-CSI and/or T-CSI, which indicates to the GMSC that originating and/or terminating CAMEL services shall be invoked for the incoming call.

7.6.3.35 VLR CAMEL Subscription Info

This parameter identifies the subscriber as having CAMEL services that are invoked in the MSC or VLR.

7.6.3.36 Supported CAMEL Phases in the VLR

This parameter indicates which phases of CAMEL are supported in the VLR.

7.6.3.36A Supported CAMEL Phases in the SGSN

This parameter indicates which phases of CAMEL are supported in the SGSN.

7.6.3.36B Offered CAMEL4 CSIs in the VLR

This parameter indicates which CSIs of CAMEL phase 4 are offered in the VLR as defined in 3GPP TS 23.078.

7.6.3.36C Offered CAMEL4 CSIs in the SGSN

This parameter indicates which CSIs of CAMEL phase 4 are offered in the SGSN as defined in 3GPP TS 23.078.

7.6.3.36D Offered CAMEL4 CSIs

This parameter indicates which CSIs of CAMEL phase 4 are offered as defined in 3GPP TS 23.078.

7.6.3.36E Offered CAMEL4 CSIs in interrogating node

This parameter indicates which CSIs of CAMEL phase 4 are offered in the GMSC or in the gsmSCF as defined in 3GPP TS 23.078.

7.6.3.36F Offered CAMEL4 CSIs in VMSC

This parameter indicates which CSIs of CAMEL phase 4 are offered in the VMSC as defined in 3GPP TS 23.078.

7.6.3.36G Offered CAMEL4 Functionalities

7.6.3.36H Supported CAMEL Phases

This parameter indicates which phases of CAMEL are supported as defined in 3GPP TS 23.078.

7.6.3.36I Supported CAMEL Phases in interrogating node

This parameter indicates which phases of CAMEL are supported as defined in 3GPP TS 23.078. The interrogating node may be a GMSC or a gsmSCF.

This parameter indicates which functionalities of CAMEL phase 4 are offered as defined in 3GPP TS 23.078.

7.6.3.37 CUG Subscription Flag

This parameter indicates that a subscriber with a T-CSI also has a CUG subscription. It is defined in 3GPP TS 23.078.

7.6.3.38 CAMEL Subscription Info Withdraw

This parameter indicates that CAMEL Subscription Info shall be deleted from the VLR or SGSN.

7.6.3.39 Voice Group Call Service (VGCS) Data

This parameter refers to one or more groups a subscriber may be a member of for voice group calls.

7.6.3.40 Voice Broadcast Service (VBS) Data

This parameter refers to one or more groups a subscriber may be a member of for the voice broadcast service. Per group it is further indicated whether the subscriber is only allowed to listen to respective group calls or whether he is in addition entitled to initiate respective voice broadcast calls.

7.6.3.41 ISDN bearer capability

This parameter refers to the ISDN bearer capability information element defined in 3GPP TS 29.007 [56].

7.6.3.42 Lower layer Compatibility

This parameter refers to the lower layer compatibility information element defined in 3GPP TS 24.008 [35].

7.6.3.43 High Layer Compatibility

This parameter refers to the high layer compatibility information element defined in 3GPP TS 24.008 [35].

7.6.3.44 Alerting Pattern

This parameter is an indication that can be used by the MS to alert the user in a specific manner in case of mobile terminating traffic (switched call or USSD). That indication can be an alerting level or an alerting category.

7.6.3.45 GPRS Subscription Data Withdraw

This parameter indicates that GPRS Subscription Data shall be deleted from the SGSN.

7.6.3.45A EPS Subscription Data Withdraw

This parameter indicates that EPS Subscription Data shall be deleted from the MME.

7.6.3.46 GPRS Subscription Data

This parameter refers to the list of PDP-Contexts the subscriber has subscribed to.

7.6.3.46A EPS Subscription Data

This parameter refers to the list of APN-Configurations the subscriber has subscribed to.

7.6.3.47 QoS-Subscribed

This parameter indicates the quality of service subscribed for a certain service. It is defined in 3GPP TS 23.060 [104].

7.6.3.48 VPLMN address allowed

This parameter specifies whether the MS is allowed to use a dynamic address allocated in the VPLMN. It is defined in 3GPP TS 23.060 [104].

7.6.3.49 Roaming Restricted In SGSN/MME Due To Unsupported Feature

This parameter defines that a subscriber is not allowed to roam in the current SGSN or MME area. It may be used by the HLR if a feature or service is indicated as unsupported by the SGSN or MME.

7.6.3.50 Network Access Mode

This parameter is defined in 3GPP TS 23.008 [20].

7.6.3.51 Mobile Not Reachable Reason

This parameter stores the reason for the MS being absent when an attempt to deliver a short message to an MS fails at the MSC, SGSN or both. It is defined in 3GPP TS 23.040.

7.6.3.52 Cancellation Type

This parameter indicates the reason of location cancellation. It is defined in 3GPP TS 23.060 [104]. The HLR shall not send Cancel Location with a Cancellation Type of "initialAttachProcedure" to the SGSN unless the SGSN has indicated support of this cancellation type within UpdateGprsLocation or the HLR has enough knowledge that the SGSN supports "initialAttachProcedure". If the HLR needs to send a cancellation type of "initialAttachProcedure" but cannot do so due to non-support by the SGSN, the HLR shall send Cancel Location with a Cancellation Type of "updateProcedure" and delete the stored SGSN-Number.

7.6.3.53 All GPRS Data

This parameter indicates to the SGSN that all GPRS Subscription Data shall be deleted for the subscriber.

7.6.3.54 Complete Data List Included

This parameter indicates to the SGSN or MME that the complete GPRS Subscription Data/EPS Subscription Data stored for the Subscriber shall be replaced with the GPRS Subscription Data/EPS Subscription Data received.

7.6.3.55 PDP Context Identifier

This parameter is used to identify a PDP context for the subscriber.

7.6.3.56 LSA Information

This parameter refers to one or more localised service areas a subscriber may be a member of, together with the priority, the preferential access indicator, the active mode support indicator and active mode indication of each localised service area. The access right outside these localised service areas is also indicated.

7.6.3.57 SoLSA support indicator

This parameter indicates that the VLR or the SGSN supports SoLSA subscription.

7.6.3.58 LSA Information Withdraw

This parameter indicates that LSA information shall be deleted from the VLR or the SGSN.

7.6.3.59 LMU Indicator

This parameter indicates the presence of an LMU.

7.6.3.60 LCS Information

This parameter defines the LCS related information for an MS subscriber and contains the following components:

- GMLC List (see clause 7.6.3.61).
- LCS Privacy Exception List (see clause 7.6.3.62).
- MO-LR List (see clause 7.6.3.65A).
- Additional LCS Privacy Exception List (see clause 7.6.3.62A).

7.6.3.61 GMLC List

This parameter contains the addresses of all GMLCs that are permitted to issue a call/session unrelated or call/session related MT-LR location request for this MS. Usage of this parameter is defined in 3GPP TS 23.271.

7.6.3.62 LCS Privacy Exception List

This parameter defines the classes of LCS Client that are allowed to locate any target MS. For each class, the following information is provided:

- SS-Code (see clause 7.6.4.1);
- a list of LCS privacy exception parameters (see clause 7.6.3.63).

7.6.3.62A Additional LCS Privacy Exception List

This parameter defines the classes of LCS Client that are allowed to locate any target MS. For each class, the following information is provided:

- SS-Code (see clause 7.6.4.1);
- a list of LCS privacy exception parameters (see clause 7.6.3.63).

The Additional LCS Privacy Exception List shall be present only if the LCS Privacy Exception List is present and contains LCS privacy exception parameters for 4 privacy exception classes.

7.6.3.63 LCS Privacy Exception Parameters

This parameter gives the status of each LCS privacy exception class and any additional parameters relevant to this class. The parameter contains the following information:

- provisioned SS-Status (see clause 7.6.3.17);
- privacy notification to MS user (see clause 7.6.3.65B);
- external client List (see clause 7.6.3.64);

- internal client List (see clause 7.6.3.65).
- service type List (see clause 7.6.3.65D);

7.6.3.64 External Client List

This parameter is only applicable to the call/session unrelated privacy class and call/session related privacy class, and gives the identities of the external clients that are allowed to locate a target MS for a MT-LR. Each identity is an international (e.g.E.164) address. For each identified external client, GMLC restrictions may be defined. It may also be indicated if the MS shall be notified of a non-restricted MT-LR from each identified LCS client and, if so, whether notification only or notification with privacy verification shall apply. Usage of this parameter is defined in 3GPP TS 23.271.

7.6.3.65 Internal Client List

This parameter is only applicable to the PLMN operator privacy class and gives the identities of the internal PLMN operator clients that are allowed to locate a target MS for an NI-LR or MT-LR. Usage of this parameter is defined in 3GPP TS 23.271.

7.6.3.65A MO-LR List

This parameter defines the classes of MO-LR for which a subscription exists for a particular MS. For each class, the following information is provided:

- SS-Code (see clause 7.6.4.1).

7.6.3.65B Privacy Notification to MS User

This parameter is applicable to the call/session unrelated privacy class and call/session related privacy class. For non-call/call related privacy class it indicates whether the MS user shall be notified for that class MT-LR from any value added LCS client when the MT-LR is restricted and be enabled to accept or override the restriction. Usage of this parameter is defined in 3GPP TS 23.271.

7.6.3.65C GMLC List Withdraw

This parameter indicates whether the subscriber's LCS GMLC list shall be deleted from the VLR or SGSN.

7.6.3.65D Service Type List

This parameter is only applicable to the Service type privacy class and gives the identities of the service type of the clients that are allowed to locate a target MS for an MT-LR. Usage of this parameter is defined in 3GPP TS 23.271.

7.6.3.66 IST Alert Timer

This parameter indicates the IST Alert Timer value that must be used in the MSC to inform the HLR about the call activities that the subscriber performs. Units are minutes.

7.6.3.67 Call Termination Indicator

This parameter indicates whether the MSC shall terminate a specific ongoing call, or all the call activities related to a specified subscriber.

7.6.3.68 IST Information Withdraw

This parameter indicates that IST information shall be deleted from the VMSC.

7.6.3.69 IST Support Indicator

This parameter indicates the degree of IST functionality supported by the MSC (Visited MSC or Gateway MSC). It can take one of the following values:

- Basic IST functionality;
- IST command service (in addition to the basic IST functionality and including the ability to terminate all calls being carried for the identified subscriber).

7.6.3.70 Super-Charger Supported In HLR

This parameter is used by the HLR to indicate support of the Super-Charger functionality and an indication of the age of the subscription data stored in the HLR.

7.6.3.71 Super-Charger Supported In Serving Network Entity

This parameter is used to indicate support of the Super-Charger functionality by the originating entity and to indicate either that subscription data is required or the date and time of the last know subscriber data modification.

7.6.3.72 Age Indicator

This parameter is used by the HLR to determine the validity of the subscription data retained by the serving network entity in a Super-Charged network.

7.6.3.73 GPRS enhancements support indicator

This parameter indicates to the HLR that the SGSN supports GPRS enhancements.

7.6.3.74 Extension QoS-Subscribed

This parameter indicates the enhanced QoS subscribed for a certain service. It is defined in 3GPP TS 23.060. This parameter is an extension to QoS-Subscribed.

7.6.3.75 SGSN CAMEL Subscription Info

This parameter identifies the subscriber as having CAMEL services that are invoked in the SGSN.

7.6.3.75A Extension-2 QoS-Subscribed

This parameter indicates the additional QoS information to the Extension QoS-subscribed parameter. It is a further extension to Extension QoS-Subscribed. This parameter shall be used when the maximum bit rate exceeds 8640 kbps. For more details, refer to 3GPP TS 24.008 [35].

7.6.3.75B Extension-3 QoS-Subscribed

This parameter indicates the additional QoS information to the Extension QoS-subscribed parameter. It is a further extension to Extension QoS-Subscribed. This parameter shall be used when the maximum/guaranteed bit rate for uplink exceeds 8640 kbps. For more details, refer to 3GPP TS 24.008 [35].

7.6.3.75C Extension-4 QoS-Subscribed

This parameter indicates the additional QoS information to the Extension QoS-subscribed parameter. It is a further extension to Extension QoS-Subscribed. This parameter shall be used to define the Evolved Allocation/Retention Priority parameter, which includes the Priority Level, the Preemption Capability value and the Preemption vulnerability value, as described in 3GPP TS 29.060 [105].

7.6.3.76 MO-SMS-CSI

This parameter identifies the subscriber as having mobile originating SMS CAMEL services as defined in 3GPP TS 23.078. For the CAMEL phase 3 the MO-SMS-CSI is the same as the SMS-CSI.

7.6.3.76a MT-SMS-CSI

This parameter identifies the subscriber as having mobile terminating SMS CAMEL services as defined in 3GPP TS 23.078.

7.6.3.77 GPRS-CSI

This parameter identifies the subscriber as having GPRS CAMEL services as defined in 3GPP TS 23.078.

7.6.3.78 CAMEL subscription info

This parameter indicates the CSI that can be controlled by CSE.

7.6.3.79 Extensible Call barring information for CSE

This parameter contains for each call barring service for CSE:

- SS-Code;
- a list of extensible call barring feature parameters.

  The list may contain one item per Basic Service Group.
- password;
- wrong password attempt counter;
- notification-to-CSE flag.

7.6.3.80 Extensible Forwarding information for CSE

This parameter represents the information for CSE related to each call forwarding service:

- the SS-Code of the relevant call forwarding service;
- if required, a list of extensible forwarding feature parameters;
- the list may contain one item per Basic Service Group;
- notification-to-CSE flag.

7.6.3.81 Modification Request for CSI

This parameter indicates the CAMEL subscription information to be modified by CSE.

7.6.3.81a Modification Request for ODB data

This parameter indicates the operator determined barring data to be modified by CSE.

7.6.3.82 Modification Request for SS Information

This parameter indicates the call forwarding, call barring, call hold, call waiting, explicit call transfer, calling line identification presentation and calling line identification restriction supplementary service data to be modified by CSE.

7.6.3.83 Call Barring Data

This parameter contains the extensible call barring feature list (see clause 7.6.3.21) and Notification to CSE flag.

7.6.3.84 Call Forwarding Data

This parameter contains the extensible call forwarding feature list (see clause 7.6.3.16) and Notification to CSE flag.

7.6.3.85 ODB Data

This parameter contains the ODB general data, ODB HPLMN specific data.

7.6.3.86 Requested Subscription Info

This parameter indicates the subscription information being requested.

7.6.3.87 CS Allocation/Retention priority

This parameter indicates the allocation/retention priority for Circuit Switched (CS). It corresponds to the allocation/retention priority that is defined in 3GPP TS 23.107.

7.6.3.88 ODB Info

This parameter contains the ODB data and Notification to CSE flag.

7.6.3.89 Suppress VT-CSI

This parameter is used to suppress the invocation of terminating CAMEL services at the VMSC.

7.6.3.90 Suppress Incoming Call Barring

This parameter is used to suppress the invocation of Incoming Call Barrings.

7.6.3.91 gsmSCF Initiated Call

This parameter is used to indicate that the call was initiated by the gsmSCF.

7.6.3.91a SuppressMTSS

This parameter is used to suppress the invocation of terminating supplementary services

7.6.3.92 Call barring support indicator

This parameter is used to indicate that the SGSN supports the call barring services for SMS.

7.6.3.93 MNP Info Result

This parameter refers to the Mobile Number Portability (MNP) information result (see 3GPP TS 23.078 [98] and 3GPP TS 23.066 [108]). This parameter may contain the following information:

- Routeing Number (see clause 7.6.2.63).
- IMSI (see 3GPP TS 23.078[98], see also clause 7.6.2.1).
- MSISDN (see clause 7.6.2.17).
- Number Portability Status (see clause 7.6.5.14).

7.6.3.94 Allowed Services

This parameter is used by the HLR to indicate which service is available for a call when two services have been requested, for the SCUDIF feature described in 3GPP TS 23.172 [126].

7.6.3.95 Unavailability Cause

This parameter is used to indicate the reason for the unavailability of one of the services as indicated by the Allowed Services IE (see 7.6.3.94) when two services have been requested, for the SCUDIF feature described in 3GPP TS 23.172 [126].

7.6.3.96 MNP Requested Info

This parameter indicates by its presence that Mobile Number Portability (MNP) information is requested for the subscriber, as defined in 3GPP TS 23.078 [98].

7.6.3.97 Access Restriction Data

This parameter refers to the radio access technologies that are possibly restricted to a subscriber via subscription data. For the use of the parameter, see 3GPP TS 23.012 [23] for the CS domain and 3GPP TS 23.060[104], 3GPP TS 29.060 [105] subclause 7.5.3 and 3GPP TS 29.274 [149] subclause 7.3.6 for the PS domain.

7.6.3.98 Supported RAT types indicator

This parameter indicates which RAT types are supported/served by the MSC/VLR or SGSN or MME

7.6.3.99 UE SRVCC Capability

This parameter indicates, if present, the support of SRVCC capability by the UE.

7.6.3.100 Temporary Empty CSG Subscription data Indicator

This parameter indicates that the CSS has currently no CSG subscription data for this roaming user but registers the VLR or SGSN, so to inform them if later changes in CSG subscription data occur.

7.6.3.101 WLAN-offloadability

This parameter refers to the WLAN offloadability for E-UTRAN or UTRAN. This parameter is defined in 3GPP TS 29.272 [144].

7.6.3.102 IMSI-Group-Id

This parameter refers to the IMSI-Group identifier. This parameter is defined in 3GPP TS 29.272 [144].

7.6.4 Supplementary services parameters

7.6.4.1 SS-Code

This parameter may refer to one supplementary service or a set of supplementary services as defined in 3GPP TS 22.004. For MAP this includes:

- Calling Line Identification Presentation service (CLIP);
- Calling Line Identification Restriction service (CLIR);
- Connected Line Identification Presentation service (COLP);
- Connected Line Identification Restriction service (COLR);
- Calling Name Presentation (CNAP);
- All Call Forwarding services, including Call Deflection;

- Call Waiting (CW);
- Call Hold (HOLD);
- Multi-Party service (MPTY);
- Closed User Group (CUG);
- All Charging services;
- All Call Restriction services;
- Explicit Call Transfer service (ECT);
- enhanced Multi-Level Precedence and Pre-emption service (eMLPP);
- Completion of Calls to Busy Subscriber, originating side (CCBS-A);
- Completion of Calls to Busy Subscriber, destination side (CCBS-B);
- All LCS privacy exceptions (see clause 7.6.4.44);
- Mobile Originating Location Request (MO-LR) (see clause 7.6.4.45);
- Multicall (MC).

7.6.4.1A SS-Code 2

This parameter is used to refer to one or a set of supplementary services (as 7.6.4.1 "SS-Code") related to Network Signal Info 2 for SCUDIF calls (see 3GPP TS 23.172 [126]).

7.6.4.2 SS-Status

This parameter refers to the state information of individual supplementary services as defined in 3GPP TS 23.011.

7.6.4.3 SS-Data

This parameter refers to the necessary set of information required in order to characterise one supplementary service:

- SS-Code (see clause 7.6.4.1);
- SS-Status (if applicable) (see clause 7.6.4.2);
- Override subscription option (see clause 7.6.4.4);
- CLI Restriction (see clause 7.6.4.5);
- Basic Service Group Code (see clause 7.6.4.40).

7.6.4.4 Override Category

This parameter refers to the subscription option Override Category attached to a supplementary service. It can take the following two values:

- Enabled;
- Disabled.

7.6.4.5 CLI Restriction Option

This parameter refers to the subscription option Restriction mode attached to the CLIR supplementary service. It can take the following three values:

- Permanent;

- Temporary (Default Restricted);
- Temporary (Default Allowed).

7.6.4.6 Forwarding Options

This parameter refers to a forwarding option attached to a supplementary service. It can take one of the following values:

- notification to forwarding party (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- notification to calling party (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- redirecting presentation (see 3GPP TS 22.082 [10] for the meaning of this parameter);
- Forwarding reason (see 3GPP TS 22.082 [10] for the meaning of this parameter).

7.6.4.7 No reply condition timer

This parameter refers to the no reply condition timer for call forwarding on no reply.

7.6.4.8 - 7.6.4.14 Void

7.6.4.15 Forwarding information

This parameter represents the information related to each call forwarding service:

- the SS-Code of the relevant call forwarding service (see clause 7.6.4.1);
- if required, a list of forwarding feature parameters (see clause 7.6.4.16).

  the list may contain one item per Basic Service Group.

7.6.4.16 Forwarding feature

This parameter applies to each combination of call forwarding service and Basic Service Group and contains the following information, as required:

- Basic Service Group (see clause 7.6.4.40);
- SS-Status (see clause 7.6.4.2);
- forwarded-to number (see clause 7.6.2.22);
- forwarded-to subaddress (see clause 7.6.2.23);
- forwarding options (see clause 7.6.4.6);
- no reply condition timer (see clause 7.6.4.7);
- long forwarded-to number (see clause 7.6.2.22A).

If a number is required to define the forwarded-to destination then:

- If the VLR supports Long Forwarded-to Numbers then the long forwarded-to number shall be present and the forwarded-to number shall be absent.
- If the VLR does not support Long Forwarded-to Numbers then the forwarded-to number shall be present and the long forwarded-to number shall be absent.

7.6.4.17 Void

7.6.4.18 Call barring information

This parameter contains for each call barring service:

- SS-Code (see clause 7.6.4.1);
- a list of call barring feature parameters (see clause 7.6.4.19).

  The list may contain one item per Basic Service Group.

7.6.4.19 Call barring feature

This parameter gives the status of call barring services as applicable to each Basic Service Group. The parameter contains the following information:

- Basic Service Group (see clause 7.6.4.40);
- SS-Status (see clause 7.6.4.2).

7.6.4.20 New password

This parameter refers to the password which the subscriber just registered in the network.

This parameter refers to a password used by the subscriber for supplementary service control.

7.6.4.21 Current password

This parameter refers to a password used by the subscriber for supplementary service control.

7.6.4.22 Guidance information

This parameter refers to guidance information given to a subscriber who is requested to provide a password. One of the following information may be given:

- "enter password";

  this information is used for checking of the old password;
- "enter new password";

  this information is used during password registration for the request of the first new password;
- "enter new password again";

  this information is used during password registration for the request of the new password again for verification.

7.6.4.23 Void

7.6.4.24 SS-Info

This parameter refers to all the information related to a supplementary service and is a choice between:

- forwarding information (see clause 7.6.4.15);
- call barring information (see clause 7.6.4.18);
- CUG info (see clause 7.6.4.8);
- SS-Data (see clause 7.6.4.3).
- eMLPP information (see clause 7.6.4.41).

7.6.4.25 - 7.6.4.35 Void

7.6.4.36 USSD Data Coding Scheme

This parameter contains the information of the alphabet and the language used for the unstructured information in an Unstructured Supplementary Service Data operation. The coding of this parameter is according to the Cell Broadcast Data Coding Scheme as specified in 3GPP TS 23.038 [25].

7.6.4.37 USSD String

This parameter contains a string of unstructured information in an Unstructured Supplementary Service Data operation. The string is sent either by the mobile user or the network. The contents of a string sent by the MS are interpreted by the network as specified in 3GPP TS 22.090 [16].

7.6.4.38 Bearer service

This parameter may refer to a single bearer service, a set of bearer services or to all bearer services as defined in 3GPP TS 22.002 [3]. This parameter is used only for supplementary service management.

7,6,4.38A Bearer Service 2

This parameter is used to indicate the bearer service or set of bearer services (as 7.6.4.38 "Bearer service") related to Network Signal Info 2 for SCUDIF calls (see 3GPP TS 23.172 [126]).

7.6.4.39 Teleservice

This parameter may refer to a single teleservice, a set of teleservices or to all teleservices as defined in 3GPP TS 22.003 [4]. This parameter is used only for supplementary service management.

7.6.4.39A Teleservice 2

This parameter is used to indicate the teleservice or set of teleservices (as 7.6.4.39 "Teleservice") related to Network Signal Info 2 for SCUDIF calls (see 3GPP TS 23.172 [126]).

7.6.4.40 Basic Service Group

This parameter refers to the Basic Service Group either as a bearer service (see clause 7.6.4.38) or a teleservice (see clause 7.6.4.39). This parameter is used only for supplementary service management. The null value (i.e. neither bearer service nor teleservice) is used to denote the group containing all bearer services and all teleservices.

7.6.4.41 eMLPP information

This parameter contains two parameters which are associated with the eMLPP service. The following two parameters are included:

- maximum entitled priority:
  - indicates the highest priority level the subscriber is allowed to apply for an outgoing call set-up;
- default priority:
  - defines the priority level which shall be assigned to a call if no explicit priority is indicated during call set-up.

7.6.4.42 SS-event

This parameter indicates the Supplementary Service for which an invocation notification is sent towards the gsmSCF. It can indicate one of the following services:

- Explicit Call Transfer (ECT)
- Call Deflection (CD)
- Multi-Party call (MPTY)

- Completion of Calls to Busy Subscriber (CCBS)

7.6.4.43 SS-event data

This parameter contains additional information related to Supplementary Service invocation. Depending on the service invoked it can contain the following information:

ECT A list with all Called Party Numbers involved.

CDThe called Party number involved.

7.6.4.44 LCS Privacy Exceptions

Distinct SS codes are assigned to the following classes of LCS client in a target MS subscriber's privacy exception list.

- Universal Class;
- Call/session related value added class;
- Call/session unrelated value added class;
- PLMN operator class.
- Service type class.

7.6.4.45 Mobile Originating Location Request (MO-LR)

Distinct SS codes are assigned to the following classes of MO-LR:

- Basic Self Location;
- Autonomous Self Location;
- Transfer to Third Party.

7.6.4.46 NbrUser

This parameter indicates the maximum number of parallel bearers that may be used as defined by the user at registration of the MC SS.

7.6.4.47 MC Subscription Data

This parameter contains two parameters which are associated with the MC service. The following two parameters are included:

- NbrUser:

  indicates the maximum number of parallel bearers that may be used as defined by the user at registration of the MC SS

- NbrSB:

  indicates the maximum number of parallel bearers that may be used as defined by the user's subscription.

7.6.4.48 MC Information

This parameter contains three parameters which are associated with the MC service. The following parameters are included:

- NbrSB;
- NbrUser;

- NbrSN.

Definitions of these parameters are provided in 3GPP TS 23.135.

7.6.4.49 CCBS Request State

This parameter indicates the current state of the CCBS request. It can take one of seven values:

- request;
- recall;
- active;
- completed;
- suspended;
- frozen;
- deleted.

7.6.4.50 Basic Service Group 2

This parameter refers to the Basic Service Group either as a bearer service (see clause 7.6.4.38) or a teleservice (see clause 7.6.4.39). This parameter is used only for supplementary service management.

7.6.5 Call parameters

7.6.5.1 Call reference number

This parameter refers to a call reference number allocated by a call control MSC.

7.6.5.2 Interrogation type

This parameter refers to the type of interrogation for routing information which is sent from a GMSC to an HLR. It can take either of two values:

- basic call (for information to route a call before the call has been extended to the VMSC of the called party);
- forwarding (for information to route the call to the forwarded-to destination after the VMSC of the forwarding party has requested the GMSC to resume handling of the call.

7.6.5.3 OR interrogation

This parameter indicates that the GMSC which interrogated the HLR for routeing information is not in the same PLMN as the HLR, and therefore that the call will potentially be optimally routed.

7.6.5.4 OR capability

This parameter indicates the phase of OR which the GMSC supports.

7.6.5.5 Forwarding reason

This parameter indicates the reason for which the call is to be forwarded. It can take one of three values:

- busy subscriber;
- mobile subscriber not reachable;
- no subscriber reply.

7.6.5.6 Forwarding interrogation required

This parameter indicates that if the VMSC of the forwarding subscriber requests the GMSC to resume handling of the call the GMSC shall interrogate the HLR for forwarding information.

7.6.5.7 O-CSI

This parameter identifies the subscriber as having originating CAMEL services as defined in 3GPP TS 23.078.

7.6.5.7A D-CSI

This parameter identifies the subscriber as having originating CAMEL dialled services as defined in 3GPP TS 23.078.

7.6.5.7B T-CSI

This parameter identifies the subscriber as having terminating CAMEL services in the GMSC, as defined in 3GPP TS 23.078.

7.6.5.7C VT-CSI

This parameter identifies the subscriber as having terminating CAMEL services in the VMSC, as defined in 3GPP TS 23.078.

7.6.5.7D O-IM-CSI

This parameter identifies the subscriber as having originating IP Multimedia Core Network CAMEL services as defined in 3GPP TS 23.278.

7.6.5.7E D-IM-CSI

This parameter identifies the subscriber as having originating IP Multimedia Core Network CAMEL dialled services as defined in 3GPP TS 23.278.

7.6.5.7F VT-IM-CSI

This parameter identifies the subscriber as having terminating IP Multimedia Core Network CAMEL services as defined in 3GPP TS 23.278.

7.6.5.8 Void

7.6.5.9 Void

7.6.5.10 Void

7.6.5.11 CCBS Feature

This parameter corresponds to the 'CCBS Description' parameter in 3GPP TS 23.093. It refers to the necessary set of information required in order to characterise a certain CCBS request. The parameter may contain the following information:

- CCBS Index (see 3GPP TS 23.093 for the use of this parameter);
- B-subscriber number (see clause 7.6.2.48);
- B-subscriber subaddress (see clause 7.6.2.49);
- Basic Service Group Code (see clause 7.6.4.40).

7.6.5.12 UU Data

This parameter includes User-To-User Data. It is defined in 3GPP TS 23.087.

7.6.5.13 UUS CF Interaction

This parameter indicates if the call forwarding or call deflection has been activated after UUS1 request has been accepted . It is defined in 3GPP TS 23.087.

7.6.5.14 Number Portability Status

This parameter indicates the number portability status of subscriber. See 3GPP TS 23.066 [108].

7.6.5.15 Pre-paging supported

This parameter indicates that the entity which sent it supports pre-paging.

7.6.5.16 MT Roaming Retry Supported

The parameter indicates that the entity which sent it supports MT Roaming Retry. When sent by the HLR, it further indicates that the GMSC also supports MT Roaming Retry.

7.6.5.17 MT Roaming Retry

The parameter indicates that the GMSC receiving the IE shall start MT roaming retry (see 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23]).

7.6.5.18 Paging Area

The parameter indicates the paging area where the MS is currently located (see 3GPP TS 23.012 [23] and 3GPP TS 23.018 [97]).

7.6.5.19 Call Priority

The parameter indicates the eMLPP priority of the call (see 3GPP TS 23.067 [136]).

7.6.5.20 MTRF Supported

The parameter indicates that the entity which sends it supports MT Roaming Forwarding.

7.6.5.21 LCLS Global Call Reference (LCLS GCR)

This parameter refers to a globally unique call identifier for the duration of the call (see 3GPP TS 29.205 [146]). This parameter is used to identify a call and to correlate the call legs of a call to determine if the call is a local call within the BSS.

7.6.5.22 LCLS-Negotiation

This parameter is used to request MSC-B to indicate LCLS, see 3GPP TS 29.205 [146] sub-clause B.2.1.4 LCLS Negotiation Request.

7.6.5.23 LCLS-Configuration-Preference

This parameter contains information to indicate the negotiated LCLS configuration preference, see 3GPP TS 29.205 [146] sub-clause B.2.1.10 LCLS Configuration Preference.

7.6.6 Radio parameters

7.6.6.1 - 7.6.6.3Void

7.6.6.4 GERAN Classmark

This information element is sent from one MSC to the other MSC in the signalling for inter MSC handover. It is used to convey information related to cell capabilities, as defined in 3GPP TS 48.008.

7.6.6.5 BSSMAP Service Handover

This parameter refers to the Service Handover information element defined in 3GPP TS 48.008

7.6.6.5A BSSMAP Service Handover List

This parameter refers to the list of Service Handover information elements defined in 3GPP TS 48.008. This parameter shall be used when there are multiple bearers and at least one of the bearers has an associated BSSMAP Service Handover parameter.

7.6.6.6 RANAP Service Handover

This parameter refers to the Service Handover information element defined in 3GPP TS 25.413.

7.6.6.7 HO-Number Not Required

This parameter indicates that no handover or relocation number allocation is necessary.

7.6.6.8 Integrity Protection Information

This parameter refers to the Integrity Protection Information element defined in 3GPP TS 25.413.

7.6.6.9 Encryption Information

This parameter refers to the Encryption Information element defined in 3GPP TS 25.413.

7.6.6.10 Radio Resource Information

This parameter refers to the Channel Type information element defined in 3GPP TS 48.008 [49].

7.6.6.10A Radio Resource List

This parameter refers to list of RAB-id's and their associated Channel Type information elements defined in 3GPP TS 48.008. This parameter shall be used when there are multiple bearers and at least one of the bearers has an associated Radio Resource Information parameter.

7.6.6.10B Chosen Radio Resource Information

This parameter refers to the Chosen Channel and Speech Version information elements defined in 3GPP TS 48.008.

7.6.6.11 Key Status

This parameter refers to the Key Status element defined in 3GPP TS 25.413.

7.6.6.12 Selected UMTS Algorithms

This parameters identifies the UMTS integrity and optionally encryption algorithms selected by MSC-B. Coding of this parameter is defined in 3GPP TS 25.413.

7.6.6.13 Allowed GSM Algorithms

This parameters identifies the allowed GSM algorithms in MSC-B. Coding of this parameter is defined in 3GPP TS 48.008.

7.6.6.14 Allowed UMTS Algorithms

This parameters identifies the allowed UMTS algorithms in MSC-B. Coding of this parameter is defined in 3GPP TS 25.413.

7.6.6.15 Selected GSM Algorithm

This parameter identifies the GSM algorithm selected by GSM BSC controlled by MSC-B. Coding of this parameter is defined in 3GPP TS 48.008.

7.6.6.16 Iu-Currently Used Codec

This parameter indicates the codec used at the Iu interface before handover.

7.6.6.17 Iu-Supported Codecs List

This parameter indicates the codecs supported by the UE and by MSC-A and the associated modes in priority order (the first entry being the highest priority codec). MSC-B uses this information to select the associated transcoder resources.

7.6.6.17A Iu-Available Codecs List

This parameter indicates the codecs available at the Iu interface in MSC-B and the associated modes. MSC-A uses this information to decide whether a change to a different codec at the Iu interface is possible.

7.6.6.18 Iu-Selected Codec

When sent by MSC-B, this parameter indicates the codec selected by MSC-B for the Iu interface. When sent by MSC-A, this parameter indicates the codec to be used by MSC-B at the Iu interface.

7.6.6.19 RAB Configuration Indicator

This parameter indicates by its presence that MSC-A (or MSC-B in case of subsequent handover) has generated the RAB parameters according to the preferred codec (first entry in the Iu-Supported Codecs List).

7.6.6.20 UESBI-Iu

This parameter refers to the UESBI-Iu (UE Specific Behaviour Information over the Iu interface) information element defined in 3GPP TS 25.413.

7.6.6.21 Alternative Channel Type

This parameter refers to the Channel Type information element defined in 3GPP TS 48.008 [49] for the alternative radio access bearer. This parameter is used for SCUDIF calls (see 3GPP TS 23.172 [126]).

7.6.6.22 AoIP-Supported Codecs List Anchor

This parameter is used for inter-MSC handover with AoIP access and the procedures and contents of the parameter are as defined in AoIP-Supported Codecs List (Anchor) in 3GPP TS 23.009 [21].

7.6.6.23 AoIP-Available Codecs List Map

This parameter is used for inter-MSC handover with AoIP access and the procedures and contents of the parameter are as defined in AoIP-Available Codecs List (Map) in 3GPP TS 23.009 [21].

7.6.6.24 AoIP-Selected Codec Target

This parameter is used for inter-MSC handover with AoIP access and the procedures and contents of the parameter are as defined in AoIP-Selected Codec (Target) in 3GPP TS 23.009 [21].

7.6.7 Authentication parameters

7.6.7.1 Authentication set list

This parameter represents a list of sets of authentication parameters for a given subscriber.

The list either contains Authentication Triplets (Rand, Sres, Kc) or Authentication Quintuplets (Rand, Xres, Ck, Ik, Autn). If the list contains Authentication Quintuplets, the order of sequence in this list is chronological, the first quintuplet in the list is the oldest one.

7.6.7.2 Rand

This parameter represents a random number used for authentication.

7.6.7.3 Sres

This parameter represents the response to an authentication request.

7.6.7.4 Kc

This parameter refers to a key used for ciphering purposes.

7.6.7.5 Xres

This parameter represents the response to an UMTS authentication request.

7.6.7.5A Ck

This parameter refers to a key used for UMTS ciphering purposes.

7.6.7.5B Ik

This parameter refers to the Integrity Key.

7.6.7.5C Autn

This parameter refers to the Authentication Token.

7.6.7.5D KASME

This parameter refers to the Key for the Access Security Management Entity.

7.6.7.6 Cksn

This parameter refers to a ciphering key sequence number.

7.6.7.6A Ksi

This parameter refers to a key set identifier.

7.6.7.6B Auts

This parameter refers to the resynchronisation token.

7.6.7.7 Ciphering mode

This parameter refers to the ciphering mode which is associated with a radio channel. It may take values as follows:

- no encryption;
- identification of specific ciphering algorithm.

7.6.7.8 Current Security Context

This parameter represents a list of security context parameters for a given subscriber.

The list either contains GSM Security Context data (Kc, Cksn) or UMTS Security Context Data (Ck, Ik, Ksi).

7.6.7.9 Failure cause

This parameter refers to an authentication failure which has occurred. It may take values as follows:

- wrong user response;
- wrong network signature.

7.6.7.10 Re-attempt

It indicates whether the failure ocurred in a normal authentication attempt or in an authentication reattempt (there was a previous unsuccessful authentication).

7.6.7.11 Access Type

It indicates whether the authentication procedure was initiated due to a call, an emergency call, a location updating, a supplementary service procedure, a short message transfer, a GPRS attach procedure, a routing area updating, a service request, a MS initiated Detach in GPRS, a PDP context activation or a PDP context deactivation procedure.

7.6.8 Short message parameters

7.6.8.1 SM-RP-DA

This parameter represents the destination address used by the short message service relay sub-layer protocol. It can be either of the following:

- IMSI (see clause 7.6.2.1);
- LMSI (see clause 7.6.2.16);
- MS-ISDN (see clause 7.6.2.17);
- roaming number (see clause 7.6.2.19);
- service centre address (see clause 7.6.2.27).

7.6.8.2 SM-RP-OA

This parameter refers to the originating address used by the short message service relay sub-layer protocol. It can be either of the following:

- MS-ISDN (see clause 7.6.2.17);
- service centre address (see clause 7.6.2.27).

7.6.8.3 MWD status

This parameter indicates whether or not the address of the originator service centre is already contained in the Message Waiting Data file. In addition, it contains the status of the Memory Capacity Exceeded Flag (MCEF), the status of the Mobile subscriber Not Reachable Flag (MNRF) and the status of the Mobile station Not Reachable for GPRS flag (MNRG).

7.6.8.4 SM-RP-UI

This parameter represents the user data field carried by the short message service relay sub-layer protocol.

7.6.8.5 SM-RP-PRI

This parameter is used to indicate whether or not delivery of the short message shall be attempted when a service centre address is already contained in the Message Waiting Data file.

7.6.8.6 SM Delivery Outcome

This parameter indicates the cause for setting the message waiting data. It can take one of the following values:

- Absent subscriber;
- MS memory capacity exceeded;
- Successful transfer.

7.6.8.7 More Messages To Send

This parameter is used to indicate whether or not the service centre has more short messages to send.

7.6.8.8 Alert Reason

This parameter is used to indicate the reason why the service centre is alerted. It can take one of the following values:

- MS present;
- Memory Available.

7.6.8.9 Absent Subscriber Diagnostic SM

This parameter is used to indicate the reason why the subscriber is absent. For the values for this parameter see 3GPP TS 23.040.

7.6.8.10 Alert Reason Indicator

This parameter indicates that the alert reason is sent to the HLR due to GPRS activity.

7.6.8.10A Additional Alert Reason Indicator

This parameter indicates that the alert reason is sent to the HLR due to IMS activity.

7.6.8.11 Additional SM Delivery Outcome

This parameter is used to indicate the GPRS delivery outcome in case a combination between delivery outcome for GPRS and non-GPRS are sent to the HLR.

7.6.8.12 Additional Absent Subscriber Diagnostic SM

This parameter indicates the reason of the additional SM Delivery Outcome.

7.6.8.13 Delivery Outcome Indicator

This parameter indicates that the delivery outcome sent to the HLR is for GPRS.

7.6.8.14 GPRS Node Indicator

This parameter indicates by its presence that the Network Node Number sent by the HLR, SMS-Router or IP-SM-GW is to be considered as the SGSN number (although it may actually be an SMS-Router Number or IP-SM-GW Number).

7.6.8.14A IMS Node Indicator

This parameter indicates by its presence that the Network Node Number sent by the HLR is an IP-SM-GW number.

7.6.8.15 GPRS Support Indicator

This parameter indicates that the SMS-GMSC supports GPRS specific procedure of combine delivery of Short Message via MSC and/or via the SGSN.

7.6.8.16 SM-RP-MTI

This parameter represents the RP-Message Type Indicator of the Short Message. It is used to distinguish a SM sent to the mobile station in order to acknowledge an MO-SM initiated by the mobile from a normal MT-SM. This parameter is formatted according to the formatting rules of address fields as described in 3GPP TS 23.040.

7.6.8.17 SM-RP-SMEA

This parameter represents the RP-Originating SME-address of the Short Message Entity that has originated the SM. This parameter is used by the short message service relay sub-layer protocol and is formatted according to the formatting rules of address fields as described in 3GPP TS 23.040.

7.6.8.18 IP-SM-GW SM Delivery Outcome

This parameter is used to indicate the delivery outcome for the IMS domain.

7.6.8.19 IP-SM-GW Absent Subscriber Diagnostic SM

This parameter indicates the reason of the IP-SM-GW SM Delivery Outcome.

7.6.8.20 IP-SM-GW Indicator

This parameter indicates by its presence that sm-deliveryOutcome is for delivery via IMS.

7.6.8.21 SM Delivery Timer

This parameter indicates the SM Delivery Timer value set in the SMS-GMSC to the IP-SM-GW, SGSN or MSC/VLR. It may be taken into account by the domain selection procedure in the IP-SM-GW. Units are in seconds.

7.6.8.22 SM Delivery Start Time

This parameter indicates the timestamp (in UTC) at which the SM Delivery Supervision Timer was started in the SMS-GMSC.

7.6.8.23 Maximum Retransmission Time

This parameter indicates the maximum retransmission time (in UTC) until which the SMS-GMSC is capable to retransmit the MT Short Message.

7.6.8.24 Requested Retransmission Time

This parameter indicates the retransmission time (in UTC) at which the SMS-GMSC is requested to retransmit the MT Short Message.

7.6.8.25 Maximum UE Availability Time

This parameter indicates the timestamp (in UTC) until which a UE using a power saving mechanism (such as extended idle mode DRX) is expected to be reachable for SM Delivery.

This information may be used by the SMS Center to prioritize the retransmission of Short Message to UEs using a power saving mechanism.

7.6.8.26 SMS-GMSC Alert Event

This parameter indicates the event that causes the MME (via an IWF) or the SGSN to alert the SMS-GMSC for retransmitting an MT Short Message.

7.6.8.27 SMS-GMSC Address

This parameter contains the E.164 number of the SMS-GMSC or SMS Router, in international number format as described in ITU-T Recommendation E.164 [67].

7.6.8.28 SMS-GMSC Diameter Address

This parameter contains the Diameter Identity of the SMS-GMSC or SMS Router.

7.6.8.29 New SGSN Number

This parameter contains the E.164 number of the new SGSN serving the MS.

7.6.8.30 New MME Number

This parameter contains the E.164 number of the new MME serving the MS.

7.6.8.31 New SGSN Diameter Address

This parameter contains the Diameter Identity of the new SGSN serving the MS.

7.6.8.32 New MME Diameter Address

This parameter contains the Diameter Identity of the new MME serving the MS.

7.6.8.33 New MSC Number

This parameter contains the E.164 number of the new MSC serving the MS.

7.6.9 Access and signalling system related parameters

7.6.9.1 AN-apdu

This parameter includes one or two concatenated complete 3GPP TS 25.413 or 3GPP TS 48.006 [48] messages, as described in 3GPP TS 23.009 and 3GPP TS 29.010. The access network protocol ID indicates that the message or messages are according to either 3GPP TS 48.006 [48] or 3GPP TS 25.413. For the coding of the messages see 3GPP TS 25.413, 3GPP TS 48.006 [48] and 3GPP TS 48.008 [49].

7.6.9.2 CM service type

This parameter identifies the service category being requested by the subscriber:

- mobile originating call;
- emergency call establishment;
- short message service;
- mobile originating call re-establishment;
- mobile terminating call;
- SS request;
- Voice group call set-up;
- Voice broadcast set-up.

7.6.9.3 Access connection status

This parameter represents the following access connection status information:

- RR-connection status (established/not established);
- ciphering mode (on/off);
- authentication status (authenticated/not authenticated).

7.6.9.4 External Signal Information

This parameter contains concatenated information elements (including tag and length) which are defined by a common protocol version, preceded by the associated protocol ID. It is used to transport information of the indicated protocol via MAP interfaces.

7.6.9.5 Access signalling information

This parameter refers to any set of information elements imported from 3GPP TS 24.008 [35].

7.6.9.6 Location update type

This parameter refers to the location update type (normal, periodic or IMSI attach) contained in the 3GPP TS 24.008 [35] LOCATION REGISTRATION REQUEST message.

7.6.9.7 Protocol ID

This parameter refers to the protocol to which the coding of the content of the associated External Signal Information conforms.

The following values are defined:

- 04.08;
- 08.06;
- ETS 300 102-1.

This value indicates the protocol defined by ETS 300 102-1 (EDSS1).

7.6.9.8 Network signal information

This parameter is transported as external signal information. The protocol ID shall be set to "ETS 300 102-1".

The network signal information may include the following information elements as defined in 3GPP TS 29.007 [56]:

- ISDN BC; the tag and length are defined by ETS 300 102-1.

For the content, see 3GPP TS 29.007 [56].

- HLC; the tag and length are defined by ETS 300 102-1.

  For the content, see 3GPP TS 29.007 [56].

- LLC; the tag and length are defined by ETS 300 102-1.

  For the content, see 3GPP TS 29.007 [56].

They are contained in the Signal Information parameter according to figure 7.6/1 (irrespective of the order):

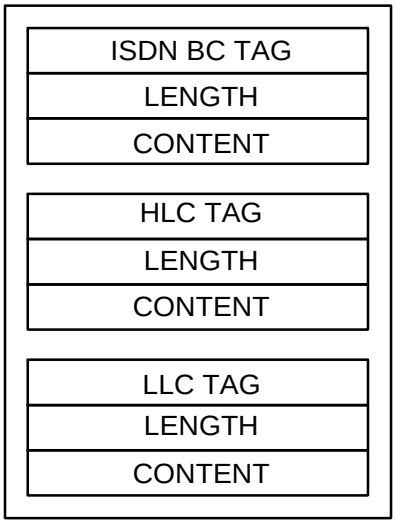


Figure 7.6/1: Network signal information parameter

7.6.9.8A Network signal information 2

This parameter is transported as additional external signal information for SCUDIF calls, described in 3GPP TS 23.172 [126]. The protocol ID and possibly included information elements are identical to Network Signal Information, defined in 7.6.9.8, "Network signal information".

7.6.9.9 Call Info

This parameter is transported as external signal information. The protocol ID shall be set to "3GPP TS 24.008 [35]".

The Call Info includes the set of information elements from the original SETUP message and is imported from 3GPP TS 24.008 [35].

7.6.9.10 Additional signal info

This parameter is transported as external signal information. The protocol ID shall be set to "ETS 300 356".

The additional signal information may include the following information elements:

- Calling Party Number as defined by ETS 300 356.
- Generic Number as defined by ETS 300 356.

They are contained in the Signal Information parameter according to figure 7.6/2 (irrespective of the order):

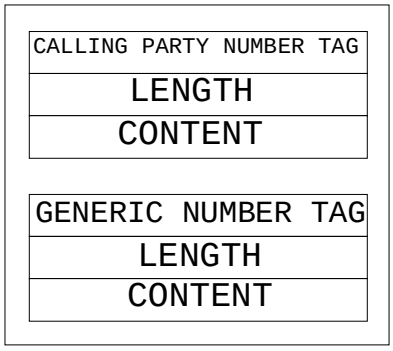


Figure 7.6/2: Additional signal information parameter

7.6.10 System operations parameters

7.6.10.1 Network resources

This parameter refers to a class or type of network resource:

- PLMN;
- HLR;
- VLR (current or previous);
- MSC (controlling or current);
- EIR;
- radio sub-system.

7.6.10.2 Trace reference

This parameter represents a reference associated with a GSM only tracing request as defined in 3GPP TS 52.008 [61]. The parameter is managed by OMC/EM.

7.6.10.2A Trace reference 2

This parameter represents a reference associated with a tracing request as defined in 3GPP TS 32.421 [131] and 3GPP TS 32.422 [132]. The parameter is managed by EM.

7.6.10.3 Trace type

This parameter identifies the type of trace for GSM only tracing request. Trace types are fully defined in 3GPP TS 52.008 [61]. If the activation of the tracing is requested only for UMTS, then this parameter shall contain value "No MSC Trace" for MSC Record Type and value "No BSS Trace" for BSS Record Type.

7.6.10.4 Additional network resources

This parameter refers to a class or type of network resource:

- SGSN;
- GGSN;
- GMLC;
- gsmSCF;
- NPLR;

- AuC.

7.6.10.5 Trace depth list

This parameter identifies the list of depths of trace per network element. See 3GPP TS 32.422 [132].

7.6.10.6 Trace NE type list

This parameter identifies the list of network elements to be traced. See 3GPP TS 32.422 [132].

7.6.10.7 Trace interface list

This parameter identifies the list of interfaces or protocols per network element to be traced. See 3GPP TS 32.422 [132].

7.6.10.8 Trace event list

This parameter identifies the list of events per network element, which trigger a Trace Recording Session. See 3GPP TS 32.422 [132].

7.6.10.9 Trace support indicator

This parameter indicates that UMTS trace parameters are supported in the VLR or in the SGSN.

7.6.10.10 Trace Propagation List

This parameter indicates UMTS trace propagation parameters sent from one MSC to the other MSC in the signalling for inter MSC handover/relocation. See 3GPP TS 32.422 [132].

7.6.10.11 MDT-Configuration

This parameter contains Minimization of Drive Test Configuration Data as defined in 3GPP TS 32.422 [132].

7.6.10.12 MDT User Consent

This parameter contains an indicator whether user consent for MDT activation is available or not as defined in 3GPP TS 32.422 [132].

7.6.11 Location Service Parameters

7.6.11.1 Age of Location Estimate

This parameter indicates how long ago the location estimate was obtained.

7.6.11.2 Deferred MT-LR Response Indicator

This parameter shows that this is a response to a deferred mt-lr request.

7.6.11.3 Deferred MT-LR Data

This parameter is used to report the deferred location event type, the location information and reason why the serving node aborted monitoring the event to the GMLC. The termination cause mt-lrRestart shall be used to trigger the GMLC to restart the location procedure in all the cases where the sending node detects that the location procedure cannot be successfully performed anymore by the sending node and that it could be successfully performed by another node (as for example when. Cancel Location or Send Identification has been received). The location information shall be included only if the termination cause is mt-lrRestart. The network node number contained in the location information refers to the node where the MS/UE has moved to and shall be included if available, like in case Send Identification has been received.

7.6.11.4 LCS Client ID

This parameter provides information related to the identity of an LCS client.

7.6.11.5 LCS Event

This parameter identifies an event associated with the triggering of a location estimate.

7.6.11.6 Void

7.6.11.7 LCS Priority

This parameter gives the priority of the location request.

7.6.11.8 LCS QoS

This parameter defines the Quality of Service (QoS) for any location request. It is composed of the following elements.

1) Response Time

   Indicates the category of response time – "low delay" or "delay tolerant".

2) Horizontal Accuracy

   Indicates the required horizontal accuracy of the location estimate.

3) Vertical Coordinate

   Indicates if a vertical coordinate is required (in addition to horizontal coordinates).

4) Vertical Accuracy

   Indicates the required vertical accuracy of the location estimate (inclusion is optional).

5) Velocity Request

   Indicates that velocity should be returned if available (inclusion is optional).

7.6.11.9 CS LCS Not Supported by UE

This parameter is used by the VLR to indicate to the HLR that the UE does not support neither UE Based nor UE Assisted positioning metheds for Circuit Switched Location Services. VLR defines the presence of this parameter on the basis of the Classmark 3 information.

7.6.11.10 PS LCS Not Supported by UE

This parameter is used by the SGSN to indicate to the HLR that the UE does not support neither UE Based nor UE Assisted positioning metheds for Packet Switched Location Services. SGSN defines the presence of this parameter on the basis of the UE capability information and the access technology supported by the SGSN.

7.6.11.11 Location Estimate

This parameter gives an estimate of the location of an MS in universal coordinates and the accuracy of the estimate. The estimate is expressed in terms of the geographical shapes defined by 3GPP TS 23.032. and is composed of the type of shape plus the encoding of the shape itself. Any type of shape defined in 3GPP TS 23.032 can be filled in in the Location Estimate parameter, but only the encoding of the following shapes shall be carried by Location Estimate:

- Ellipsoid point with uncertainty circle

- Ellipsoid point with uncertainty ellipse

- Ellipsoid point with altitude and uncertainty ellipsoid

- Ellipsoid arc

- Ellipsoid point

The encoding for the remaining types of shape, defined in the 3GPP TS 23.032, shall be filled in in the Additional Location Estimate parameter.

7.6.11.11A GERAN Positioning Data

This parameter provides positioning data associated with a successful or unsuccessful location attempt for a target MS described in 3GPP TS 49.031 [59a].

7.6.11.11B UTRAN Positioning Data

This parameter provides positioning data associated with a successful location attempt for a target MS as described in 3GPP TS 25.413 [120]. It contains the positioningDataDiscriminator and positioningDataSet parts of the RANAP PositionData element only.

7.6.11.11C GERAN GANSS Positioning Data

This parameter provides GANSS positioning data associated with a successful or unsuccessful location attempt for a target MS as described in 3GPP TS 49.031 [59a] if GANSS has been used.

7.6.11.11D UTRAN GANSS Positioning Data

This parameter provides GANSS positioning data associated with a successful location attempt for a target MS as described in 3GPP TS 25.413 [120] if GANSS has been used. It contains the GANSS-PositioningDataSet part of the RANAP PositionData element only.

7.6.11.11E UTRAN Additional Positioning Data

This parameter provides additional positioning data associated with a successful location attempt for a target MS as described in 3GPP TS 25.413 [120] if Additional Positioning has been used. It contains the Additional-PositioningDataSet part of the RANAP PositionData element only.

7.6.11.11F UTRAN Barometric Pressure Measurement

This parameter provides barometric pressure measurement associated with a successful location attempt for a target MS as described in 3GPP TS 25.413 [120].

7.6.11.11G UTRAN Civic Address

This parameter provides civic address associated with a successful location attempt for a target MS as described in 3GPP TS 25.413 [120].

7.6.11.12 Location Type

This parameter indicates the type of location estimate required by the LCS client. Possible location estimate types include:

- current location;

- current or last known location;

- initial location for an emergency services call;

- deferred location event type;

- notification verification only.

7.6.11.13 NA-ESRD

This parameter only applies to location for an emergency services call in North America and gives the North American Emergency Services Routing Digits.

7.6.11.14 NA-ESRK

This parameter only applies to location for an emergency services call in North America and gives the North American Emergency Services Routing Key.

7.6.11.15 LCS Service Type Id

This parameter defines the LCS Service Type of the current positioning request. The possible values are defined in 3GPP TS 22.071 [123]

7.6.11.16 Privacy Override

This parameter indicates if MS privacy is overridden by the LCS client when the GMLC and VMSC/SGSN for an MT-LR are in the same country.

7.6.11.17 Supported LCS Capability Sets

This parameter indicates which capability sets of LCS are supported in the VLR or SGSN.

7.6.11.18 LCS Codeword

This parameter contains the codeword associated to current positioning request as described in 3GPP TS 23.271 [26a].

7.6.11.19 NA-ESRK Request

This parameter allows the MSC to indicate that it requires the GMLC to allocate a NA-ESRK based on the target MS location estimate. This parameter only applies to emergency services calls in North America.

7.6.11.20 Supported GAD Shapes

This parameter indicates which of the shapes defined in 3GPP TS 23.032 are supported. If the parameter is not provided then the receiving node shall assume that the sending entity supports the following shapes:

- Ellipsoid point with uncertainty circle
- Ellipsoid point with uncertainty ellipse
- Ellipsoid point with altitude and uncertainty ellipsoid
- Ellipsoid arc
- Ellipsoid point

7.6.11.21 Additional Location Estimate

This parameter gives an estimate of the location of an MS/UE in universal coordinates and the accuracy of the estimate. This parameter allows the location estimate to be expressed in any of the geographical shapes defined in 3GPP TS 23.032

7.6.11.22 Cell Id Or SAI

For GERAN access, this parameter contains the Global Cell Identifier for the cell that the subscriber is currently attached to. For UTRAN access, this parameter contains the Service Area Identifier for the cell that the subscriber is currently attached to.

7.6.11.23 LCS-Reference Number

This parameter represents a reference between a request and a responce of a deferred mt-lr procedure as deccribed in 3GPP TS 23.271 [26a].

7.6.11.24 LCS Privacy Check

This parameter refers to the requested privacy check related actions (call/session unrelated and/or call/session related) from MSC or SGSN provided by H-GMLC. Possible requested actions are:

- positioning allowed without notifying the UE user;
- positioning allowed with notification to the UE user;
- positioning requires notification and verification by the UE user; positioning is allowed only if granted by the UE user or if there is no response to the notification;
- positioning requires notification and verification by the UE user; positioning is allowed only if granted by the UE user;
- positioning not allowed.

7.6.11.25 Additional LCS Capability Sets

This parameter indicates which capability sets of LCS are supported in the VLR or SGSN.

7.6.11.26 Area Event Info

This parameter defines the requested deferred MT-LR area event information. The parameter consists of area definition, type of area event, occurrence info and minimum interval time.

7.6.11.27 Velocity Estimate

This parameter gives an estimate of the velocity of an MS and the accuracy of the estimate. The estimate is expressed in terms of speed and bearing as defined by 3GPP TS 23.032 [122], and is composed of the velocity terms plus the encoding of the velocity itself. Only the encoding of the following velocity definitions shall be carried by the Velocity Estimate:

- Horizontal Velocity
- Horizontal with Vertical Velocity
- Horizontal Velocity with Uncertainty
- Horizontal with Vertical Velocity and Uncertainty

7.6.11.28 Accuracy Fulfilment Indicator

This parameter indicates the fulfilled accuracy of the positioning procedure. For details see 3GPP TS 23.271 [26a].

7.6.11.29 MO-LR Short Circuit Indicator

This parameter indicates whether MO-LR short circuit feature is permitted. For details see 3GPP TS 23.271 [26a].

7.6.11.30 Reporting PLMN List

This parameter provides a list of PLMNs in which subsequent periodic MO-LR TTTP requests will be made. For details see 3GPP TS 23.271 [26a].

7.6.11.31 Periodic LDR information

This parameter refers to the periodic reporting interval and reporting amount of the deferred periodic location. For details see 3GPP TS 23.271 [26a].

7.6.11.32 Sequence Number

This parameter refers to the number of the periodic location reports completed. The sequence number would be set to 1 in the first location report and increment by 1 for each new report. When the number reaches the reporting amount value, the H-GMLC (for a periodic MT-LR or a periodic MO-LR transfer to third party) will know the procedure is complete. For details see 3GPP TS 23.271 [26a].

7.6.12 Void

7.7 Representation of a list of a basic parameter in service-primitives

In some service-primitives several instances of a basic parameter of clause 7.6 are required. In the service descriptions such cases will be represented as

ParameterNameLIST

in the tables where ParameterName refers to one of the parameters defined in clause 7.6. This corresponds to the following construction rule:

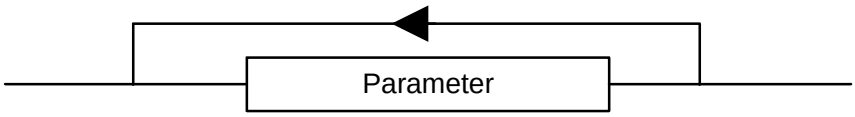


Figure 7.7/1: Construction of Lists

8 Mobility services

8.1 Location management services

8.1.1 Void

8.1.1.1 Void

8.1.1.2 Void

8.1.1.3 Void

8.1.2 MAP_UPDATE_LOCATION service

8.1.2.1 Definition

This service is used by the VLR to update the location information stored in the HLR.

This service is also used by an IWF that registers an MME as MSC for MT-SMS.

The MAP_UPDATE_LOCATION service is a confirmed service using the service primitives given in table 8.1/2.

8.1.2.2 Service primitives

Table 8.1/2: MAP_UPDATE_LOCATION

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
MSC Address	M	M(=)		
VLR number	M	M(=)		
LMSI	U	C(=)		
Supported CAMEL Phases	C	C(=)		
SoLSA Support Indicator	C	C(=)		
IST Support Indicator	C	C(=)		
Super-Charger Supported in Serving Network Entity	C	C(=)		
Long FTN Supported	C	C(=)		
Supported LCS Capability Sets	C	C(=)		
Offered CAMEL 4 CSIs	C	C(=)		
Inform Previous Network Entity	C	C(=)		
CS LCS Not Supported by UE	C	C(=)		
V-GMLC Address	U	C(=)		
IMEISV	C	C(=)		
Skip Subscriber Data Update	U	C(=)		
Supported RAT Types Indicator	U	C(=)		
Paging Area	U	C(=)		
Restoration Indicator	U	C(=)		
MTRF Supported	U	C(=)		
Equivalent PLMN List	C	C(=)		
MSISDN-less Operation Supported	C	C(=)		
MME-Diameter-Address-For MT-SMS	C	C(=)		
Reset-IDs Supported	C	C(=)		
ADD Capability			U	C(=)
Paging Area Capability			U	C(=)
HLR number			C	C(=)
User error			C	C(=)
Provider error				O

8.1.2.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

MSC Address

See definition for MSC number in clause 7.6.2. The MSC address is used for short message delivery only and for each incoming call set-up attempt the MSRN will be requested from the VLR.

VLR number

See definition in clause 7.6.2.

LMSI

See definition in clause 7.6.2. It is an operator option to provide the LMSI from the VLR; it is mandatory for the HLR to support the LMSI handling procedures.

Supported CAMEL Phases

This parameter indicates which phases of CAMEL are supported. Must be present if a CAMEL phase different from phase 1 is supported. Otherwise may be absent.

HLR number

See definition in clause 7.6.2. The presence of this parameter is mandatory in case of successful HLR updating.

SoLSA Support Indicator

This parameter is used by the VLR to indicate to the HLR in the Update Location indication that SoLSA is supported. If this parameter is not included in the Update Location indication and the Subscriber is marked as only allowed to roam in Subscribed LSAs, then the HLR shall reject the roaming and indicate to the VLR that roaming is not allowed to that Subscriber in the VLR.

This SoLSA Support Indicator shall be stored by the HLR per VLR where there are Subscribers roaming. If a Subscriber is marked as only allowed to roam in Subscribed LSAs while roaming in a VLR and no SoLSA Support indicator is stored for that VLR, the location status of that Subscriber shall be set to Restricted.

IST Support Indicator

This parameter is used to indicate to the HLR that the VMSC supports basic IST functionality, that is, the VMSC is able to terminate the Subscriber Call Activity that originated the IST Alert when it receives the IST alert response indicating that the call(s) shall be terminated. If this parameter is not included in the Update Location indication and the Subscriber is marked as an IST Subscriber, then the HLR may limit the service for the subscriber (by inducing an Operator Determined barring of Roaming, Incoming or Outgoing calls), or allow service assuming the associated risk of not having the basic IST mechanism available.

This parameter can also indicate that the VMSC supports the IST Command service, including the ability to terminate all calls being carried for the identified subscriber by using the IMSI as a key. If this additional capability is not included in the Update Location indication and the HLR supports the IST Command capability, then the HLR may limit the service for the subscriber (by inducing an Operator Determined barring of Roaming, Incoming or Outgoing calls), or allow service assuming the associated risk of not having the IST Command mechanism available.

Long FTN Supported

This parameter indicates that the VLR supports Long Forwarded-to Numbers.

Super-Charger Supported in Serving Network Entity

This parameter is used by the VLR to indicate to the HLR that the VLR supports the Super-Charger functionality and whether subscription data has been retained by the VLR. If subscription data has been retained by the VLR the age indicator shall be included. Otherwise the VLR shall indicate that subscriber data is required.

If this parameter is absent then the VLR does not support the Super-Charger functionality.

Supported LCS Capability Sets

This parameter indicates, if present, the capability sets of LCS which are supported. If the parameter is sent but no capability set is marked as supported then the VLR does not support LCS at all.

If this parameter is absent then the VLR may support at most LCS capability set 1, that is LCS Release98 or Release99 version.

Offered CAMEL 4 CSIs

This parameter indicates the CAMEL phase 4 CSIs offered in the VMSC/VLR (see clause 7.6.3.36D).

Inform Previous Network Entity

This parameter is used by the VLR to ask the HLR to inform the previous network entity about the update by sending the previous network entity a Cancel Location message. It is used if Super-Charger is supported in the network and either the serving network entity has not been able to inform the previous network entity that MS has moved (i.e. if it has not sent Send Identification to the previous serving entity) or the MTRF Supported flag is set in the MAP_UPDATE LOCATION request.

CS LCS Not Supported by UE

See definition in clause 7.6.11.

V-GMLC address

See definition in clause 7.6.2.

IMEISV

For definition of the parameter see clause 7.6.2. For the use of this parameter see 3GPP TS 23.012. IMEISV shall be present if ADD function is supported and a new IMEISV is to be notified to the HLR (The functional requirements for the presence of IMEISV due to ADD are described in 3GPP TS 22.101 clause 7.4).

Skip Subscriber Data Update

The presence of the parameter is optional and if present it indicates that the service is solely used to inform the HLR about change of IMEISV or Paging Area. The parameter is used to optimise signalling load during Location Update procedure.

Supported RAT Types Indicator

This parameter indicates, if present, which access technologies (e.g. GERAN and / or UTRAN) are served by the MSC/VLR (see clause 7.6.3)

Paging Area

This parameter indicates, if present, the paging area where the MS is currently located (see clause 7.6.5.18)

Restoration Indicator

This parameter indicates, if present, that the HLR shall send in the MAP-INSERT-SUBSCRIBER-DATA the MME Name if the subscriber is registered to EPS, or the SGSN Number if available and if the subscriber is registered to GPRS. The VLR may set this indicator during a CSFB mobile originated call if the VLR performs an implicit location update (see 3GPP TS 23.272 [143]).

MTRF Supported

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

Equivalent PLMN List

This parameter indicates the equivalent PLMN list of which the VLR requests the corresponding CSG Subscription data.

MSISDN-less Operation Supported

See subclause 3.6.1.5 of 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

MME-Diameter-Address-For-MT-SMS

This parameter may be sent by an IWF that registers an MME for MT-SMS. The MME-Diameter-Address-For-MT-SMS may be stored in the HLR and may be sent in SMS interrogation responses to SMS-GMSCs.

Reset-IDs Supported

This parameter indicates, if present, the support of Reset-IDs by the MSC.

ADD Capability

This parameter indicates, if present, the support of ADD function by the HLR.

Paging Area Capability

This parameter indicates, if present, the support of Paging Area function by the HLR. The HLR shall report the same capability for all subscribers.

User error

In case of unsuccessful updating, an error cause shall be returned by the HLR. The following error causes defined in clause 7.6.1 may be used, depending on the nature of the fault:

- unknown subscriber;
- roaming not allowed;

  This cause will be sent if the MS is not allowed to roam into the PLMN indicated by the VLR number. The cause is qualified by the roaming restriction reason "PLMN Not Allowed", "Supported RAT Types Not Allowed" or "Operator Determined Barring". If no qualification is received (HLR with MAP Version 1), "PLMN Not Allowed" is taken as default.

  This cause shall be used when the HLR rejects a MAP Update Location request received for an MSISDN-less subscription from a VLR not supporting MSISDN-less operation (see subclause 3.6.1.5 of 3GPP TS 23.012 [23]).
- system failure;
- unexpected data value.

Provider error

For definition of provider errors see clause 7.6.1.

8.1.3 MAP_CANCEL_LOCATION service

8.1.3.1 Definition

This service is used between HLR and VLR to delete a subscriber record from the VLR. It may be invoked automatically when an MS moves from one VLR area to another, to remove the subscriber record from the old VLR, or by the HLR operator to enforce a location updating from the VLR to the HLR, e.g. on withdrawal of a subscription.

Also this service is used between HLR and SGSN to delete a subscriber record from the SGSN. It may be invoked automatically when an MS moves from one SGSN area to another, to remove the subscriber record from the old SGSN, or by the HLR operator to enforce a location updating from the SGSN to the HLR. This service is also used to request the SGSN to indicate to the MS to initiate an immediate re-attach procedure.

In an EPS this service is used between HSS and IWF and between IWF and IWF to delete the subscriber record from the MME or SGSN or to release bearer resources without deleting the subscriber record. This service is also used to request the MME or SGSN to indicate to the UE to initiate an immediate re-attach procedure.

The MAP_CANCEL_LOCATION service is a confirmed service using the primitives defined in table 8.1/3.

8.1.3.2 Service primitives

Table 8.1/3: MAP_CANCEL_LOCATION

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
LMSI	C	C(=)		
Cancellation Type	C	C(=)		
MTRF Supported And Authorized	U	C(=)		
MTRF Supported And Not Authorized	U	C(=)		
New MSC Number	U	C(=)		
New VLR Number	U	C(=)		
New LMSI	U	C(=)		
Reattach Required	U	C(=)		
User error			C	C(=)
Provider error				O

8.1.3.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

LMSI

See definition in clause 7.6.2. The LMSI shall be included if it has been received from VLR. LMSI is not applicable between SGSN and HLR.

Value 0000 0000 can be used to indicate that the LMSI is not in use.

Cancellation Type

See definition in clause 7.6.3. The presence of this parameter is mandatory when the Cancel Location is sent to the SGSN or IWF. The parameter may also be sent during an inter-VLR location update If the VLR receives this parameter and does not understand it the VLR shall ignore it and should by default assume an Update procedure. If the SGSN receives this parameter indicating initial attach procedure, the SGSN shall do as specified in 3GPP TS 23.060 [104], and shall not delete the subscription data.

MTRF Supported And Authorized

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

MTRF Supported And Not Authorized

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

New MSC Number

This parameter refers to the E.164 address of the new VMSC. The use and conditions of presence of this parameter are specified in 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23]. This parameter shall be present if the MTRF Supported And Authorized flag is present.

New VLR Number

This parameter contains the new VLR Number. See definition in clause 7.6.2. The use and conditions of presence of this parameter are specified in 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23]. This parameter shall be present if the MTRF Supported And Authorized flag is present.

New LMSI

See definition in clause 7.6.2 for LMSI. This parameter shall be present if the MTRF Supported And Authorized flag is present and the HLR has received the LMSI in Update Location from the new VLR.

Reattach Required

When present and when the Cancellation Type indicates a subscription withdraw, this parameter indicates that the MME (informed via the IWF) or the SGSN shall delete the subscription data and request the UE or MS to initiate an immediate re-attach procedure as described in 3GPP TS 23.401 [145] and in 3GPP TS 23.060 [12].

User error

If the cancellation fails, an error cause is to be returned by the VLR or by the SGSN or IWF. One of the following error causes defined in clause 7.6.1 shall be used:

- unexpected data value;
- data missing.

Provider error

For definition of provider errors see clause 7.6.1.

8.1.4 MAP_SEND_IDENTIFICATION service

8.1.4.1 Definition

The MAP_SEND_IDENTIFICATION service is used between a VLR and a previous VLR to retrieve IMSI and authentication data for a subscriber registering afresh in that VLR.

It may also be used to send the MSC number from a VLR to a previous VLR.

The MAP_SEND_IDENTIFICATION service is a confirmed service using the service primitives defined in table 8.1/4.

8.1.4.2 Service primitives

Table 8.1/4: MAP_SEND_IDENTIFICATION

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
TMSI	M	M(=)		
Number of requested vectors	M	M(=)		
Segmentation prohibited indicator	C	C(=)		
MSC Number	U	C(=)		
Previous Location Area Id	U	C(=)		
Hop Counter	U	C (=)		
MTRF Supported	U	C(=)		
VLR Number	U	C(=)		
New LMSI	U	C(=)		
IMSI			C	C(=)
Authentication set			U	C(=)
Current Security Context			U	C(=)
MT call pending flag			U	C(=)
Last used LTE PLMN ID			U	C(=)
User error			C	C(=)
Provider error				O

8.1.4.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

TMSI

See definition in clause 7.6.2.

If multiple service requests are present in a dialogue then this parameter shall be present in every service request.

Number of requested vectors

A number indicating how many authentication vectors the new VLR is prepared to receive. The previous VLR shall not return more vectors than indicated by this parameter.

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one

Segmentation prohibited indicator

This parameter indicates if the new VLR or SGSN allows segmentation of the response at MAP user level.

This parameter may be present only in the first request of the dialogue.

IMSI

See definition in clause 7.6.2. The IMSI is to be returned if the service succeeds.

If multiple service requests are present in a dialogue and the service succeeds then this parameter shall not be present in any service response other than the first one

MSC Number

This is the ISDN number assigned to the MSC currently serving the MS. This parameter shall be present if the MTRF Supported flag is present.

Previous Location Area Id

See definition in clause 7.6.2. Together with the TMSI the Previous Location Area Id can be used to derive the IMSI.

Authentication set

See definition in clause 7.6.7. If the service succeeds a list of up to five authentication sets is returned, if there are any available.

Current Security Context

See definition in clause 7.6.7. If the service succeeds, a list of either GSM or UMTS Security Context parameters can be returned.

This parameter shall not be included if the Key Status associated to the current security context indicates this is a new keyset that has not been used yet. If this parameter is present in the message, the new VLR shall consider that the keyset has already been used (i.e. the key status is "old").

MT call pending flag

This flag indicates by its presence that there is a Mobile Terminating call pending in the old MSC/VLR. See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

Hop Counter

For the use of this parameter see 3GPP TS 23.012 [23].

MTRF Supported

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

VLR Number

This is the ISDN number assigned to the VLR currently serving the MS. See definition in clause 7.6.2. The use and conditions of presence of this parameter are specified in 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23]. This parameter shall be present if the MTRF Supported flag is present.

New LMSI

See definition in clause 7.6.2 for LMSI. This parameter may be present if the MTRF Supported flag is present.

Last used LTE PLMN ID

See 3GPP TS 23.272 [143] for the use of this parameter and the conditions for its presence.

User error

This parameter is mandatory if the service fails. The following error cause defined in clause 7.6.1 may be used, depending on the nature of the fault:

- unidentified subscriber.

Provider error

For definition of provider errors see clause 7.6.1.

8.1.5 Void

8.1.5.1 Void

8.1.5.2 Void

8.1.5.3 Void

8.1.6 MAP_PURGE_MS service

8.1.6.1 Definition

This service is used between the VLR and the HLR to cause the HLR to mark its data for an MS so that any request for routing information for a mobile terminated call or a mobile terminated short message will be treated as if the MS is not reachable. It is invoked when the subscriber record for the MS is to be deleted in the VLR, either by MMI interaction or automatically, e.g. because the MS has been inactive for several days. This service shall not be used if both the VLR and HLR support the Super-Charger functionality.

Also this service is used between the SGSN and the HLR to cause the HLR to mark its data for an MS so that any request for routing information for a mobile terminated short message or a network requested PDP-context activation will be treated as if the MS is not reachable. It is invoked when the subscriber record for the MS is to be deleted in the SGSN, either by MMI interaction or automatically, e.g. because the MS has been inactive for several days. This service shall not be used if both the SGSN and HLR support the Super-Charger functionality.

In an EPS this service is used between IWF and IWF and between IWF and HSS.

The MAP_PURGE_MS service is a confirmed service using the primitives defined in table 8.1/6.

8.1.6.2 Service primitives

Table 8.1/6: MAP_PURGE_MS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
VLR number	C	C(=)		
Freeze TMSI			C	C(=)
Freeze P-TMSI			C	C(=)
Freeze M-TMSI			C	C(=)
SGSN number	C	C(=)		
Last known location	C	C(=)		
User error			C	C(=)
Provider error				O

8.1.6.3 Parameter definitions and use

Invoke ID

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

VLR number

Shall be present if the sender is VLR. See definition in clause 7.6.2.

SGSN number

Shall be present if the sender is SGSN. See definition in clause 7.6.2.

In an EPS, this parameter may contain the IWF number.

Freeze TMSI

This parameter is sent to the VLR to indicate that the TMSI has to be frozen. It shall be present if the received VLR number matches the stored VLR number.

Freeze P-TMSI

This parameter is sent to the SGSN to indicate that the P-TMSI has to be frozen. It shall be present if the received SGSN number matches the stored SGSN number.

Freeze M-TMSI

This parameter is sent to the IWF to indicate that the M-TMSI has to be frozen. It shall be present if the received node number matches the stored IWF number.

Last known location

This parameter contains the last known location of the purged UE.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber.

Provider error

See definition of provider errors in clause 7.6.1.

8.1.7 MAP_UPDATE_GPRS_LOCATION service

8.1.7.1 Definition

This service is used by the SGSN to update the location information stored in the HLR.

In an EPS, this service is used between IWF and IWF and between IWF and HSS.

The MAP_UPDATE_GPRS_LOCATION service is a confirmed service using the service primitives given in table 8.1/7.

8.1.7.2 Service primitives

Table 8.1/7: MAP_UPDATE_GPRS_LOCATION

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
SGSN number	M	M(=)		
SGSN address	M	M(=)		
Supported CAMEL Phases	C	C(=)		
SoLSA Support Indicator	C	C(=)		
Super-Charger Supported in Serving Network Entity	C	C(=)		
GPRS enhancements support indicator	C	C(=)		
Supported LCS Capability Sets	C	C(=)		
Offered CAMEL 4 CSIs	C	C(=)		
Inform Previous Network Entity	C	C(=)		
PS LCS Not Supported by UE	C	C(=)		
V-GMLC Address	U	C(=)		
Call barring support indicator	C	C(=)		
IMEISV	C	C(=)		
Skip Subscriber Data Update	U	C(=)		

Parameter name	Request	Indication	Response	Confirm
Supported RAT Types Indicator	U	C(=)		
EPS Info	C	C(=)		
Serving Node Type Indicator	C	C(=)		
Supported Features	U	C(=)		
Used RAT Type	U	C(=)		
GPRS Subscription Data not needed Indicator	C	C(=)		
EPS Subscription Data Not Needed Indicator	C	C(=)		
Node-Type-Indicator	U	C(=)		
Area Restricted Indicator	C	C(=)		
UE Reachable Indicator	C	C(=)		
T-ADS Data Retrieval Support Indicator	C	C(=)		
Homogeneous Support Of IMS Voice Over PS Sessions	C	C(=)		
Update of Homogeneous Support Of IMS Voice Over PS Sessions	C	C(=)		
UE SRVCC Capability	C	C(=)		
Equivalent PLMN List	C	C(=)		
MME Number for MT SMS	C	C(=)		
SMS-Only	C	C(=)		
SMS Register Request	C	C(=)		
Removal of MME Registration for SMS	C	C(=)		
MSISDN-less Operation Supported	C	C(=)		
SGSN Name	C	C(=)		
SGSN Realm	C	C(=)		
Lgd Support Indicator	C	C(=)		
Adjacent-PLMNs	C	C(=)		
Reset-IDs Supported	C	C(=)		
ADD Capability			U	C(=)
SGSN-MME Separation Support Indicator			C	C(=)
HLR number			C	C(=)
MME Registered for SMS			C	C(=)
User error			C	C(=)
Provider error				O

8.1.7.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

SGSN number

See definition in clause 7.6.2.

In an EPS, this parameter is populated with an IWF number if received from an IWF.

SGSN address

See definition in clause 7.6.2.

In an EPS, this parameter is populated with an IWF address if received from an IWF.

Supported CAMEL Phases

This parameter indicates which phases of CAMEL are supported. The SGSN can only support CAMEL phase 3 or greater.

SoLSA Support Indicator

This parameter is used by the SGSN to indicate to the HLR in the Update GPRS Location indication that SoLSA is supported. If this parameter is not included in the Update GPRS Location indication and the Subscriber is marked as only allowed to roam in Subscribed LSAs, then the HLR shall reject the roaming and indicate to the SGSN that roaming is not allowed to that Subscriber in the SGSN.

This SoLSA Support Indicator shall be stored by the HLR per SGSN where there are Subscribers roaming. If a Subscriber is marked as only allowed to roam in Subscribed LSAs while roaming in a SGSN and no SoLSA Support indicator is stored for that SGSN, the location status of that Subscriber has to be set to Restricted.

Super-Charger Supported in Serving Network Entity

This parameter is used by the SGSN to indicate to the HLR that the SGSN supports the Super-Charger functionality and whether subscription data has been retained by the SGSN. If subscription data has been retained by the SGSN the age indicator shall be included. Otherwise the SGSN shall indicate that subscriber data is required.

If this parameter is absent then the SGSN does not support the Super-Charger functionality.

GPRS enhancements support indicator

This parameter is used by the SGSN to indicate to the HLR in the Update GPRS Location indication that GPRS enhancements are supported. If this parameter is included in the Update GPRS Location indication the HLR may send the extension QoS parameter in the PDP contexts to the SGSN. The HLR may send the extension-2 QoS, the extension-3 QoS and the extension-4 QoS parameters with the extension QoS parameter.

HLR number

See definition in clause 7.6.2. The presence of this parameter is mandatory in case of successful HLR updating.

Supported LCS Capability Sets

This parameter indicates, if present, the capability sets of LCS which are supported. If the parameter is sent but no capability set is marked as supported then the SGSN does not support LCS at all.

The SGSN is not allowed to indicate support for LCS capability set 1.

If this parameter is absent then the SGSN does not support LCS at all.

Offered CAMEL 4 CSIs

This parameter indicates the CAMEL phase 4 CSIs offered in the SGSN (see clause 7.6.3.36D).

Inform Previous Network Entity

This parameter is used by the SGSN to ask the HLR to inform the previous network entity about the update by sending the previous network entity a Cancel Location message. It is used in case Super-Charger is supported in the network and the serving network entity has not been able to inform the previous network entity that MS has moved, that is if it has not sent SGSN Context Request to the previous serving entity.

PS LCS Not Supported by UE

See definition in clause 7.6.11.

V-GMLC address

See definition in clause 7.6.2.

Call Barring support indicator

See definition in clause 7.6.3.92.

IMEISV

For definition of the parameter see clause 7.6.2. For the use of this parameter see 3GPP TS 23.060. IMEISV shall be present if ADD function is supported and the IMEISV is new in SGSN (The functional requirements for the presence of IMEISV due to ADD are described in 3GPP TS 22.101 clause 7.4).

Skip Subscriber Data Update

The presence of the parameter is optional and if present it indicates that subscriber data download during the updateGprsLocation procedure may be skipped by the HLR e.g. because the service is solely used to inform the HLR about change of IMEISV. The parameter is used to optimise signalling load during Location Update procedure.

Supported RAT Types Indicator

This parameter indicates, if present, which access technologies (e.g. GERAN and/or UTRAN and/or E-UTRAN) are served by the SGSN or MME (see clause 7.6.3)

EPS Info

This parameter may indicate that the MME or SGSN has selected a new PDN GW for an APN. If so, the HSS shall skip subscriber data update (insert subscriber data) and only note the new PDN GW.

Otherwise this parameter may indicate the appropriate instruction to be performed by the HSS which is one or more of

a) Update Location; i.e. send CancelLocation to the old MME and replace the stored MME id (if Serving Node Type Indicator is present and the stored MME id is different from the received MME id), or send CancelLocation to the old SGSN and replace the stored SGSN id (if Serving Node Type Indicator is absent and the stored SGSN id is different from the received SGSN id);

b) Cancel SGSN; i.e. send CancelLocation to the SGSN and delete the stored SGSN id.

c) Initial Attach; i.e. send CancelLocation to the MME (if Serving Node Type Indicator is absent) or to the SGSN (if Serving Node Type Indicator is present) with cancellation type set to "initial attach procedure"

Serving Node Type Indicator

This parameter indicates by its presence that the subscriber's serving node is an MME (which is either stand alone or combined with an SGSN) and it indicates by its absence that the subscriber's serving node is an SGSN (which is either stand alone or combined with an MME).

Supported Features

This parameter shall be used by an IWF to forward feature support indications as received from the MME or SGSN via S6a/S6d. It shall also be used by the SGSN to indicate support of the Dedicated Core Network functionality to the HLR.

Used RAT Type

This parameter may indicate the RAT type currently used by the serving node.

GPRS Subscription Data not needed Indicator

This parameter indicates by its presence that the SGSN (or MME/IWF) does not request GPRS Subscription Data in addition to EPS Subscription Data.

EPS Subscription Data Not Needed Indicator

This parameter indicates by its presence that the SGSN does not request EPS Subscription Data in addition to GPRS Subscription Data.

NOTE: The indicator is only applicable to an SGSN which only supports Gn and Gp interfaces and does not support S4 interface.

Node-Type Indicator

This parameter indicates by its presence that the requesting node is a combined MME/SGSN. Absence of this Indicator indicates that the requesting node is a single MME or SGSN.

When Node-Type Indicator is absent and Serving Node Type Indicator is present, the HSS may skip checking SMS/LCS supported features and skip the download of SMS/LCS-related subscription data to a standalone MME.

Area Restricted Indicator

This parameter indicates by its presence that the network node area is restricted due to regional subscription. This parameter is used by the IWF only.

UE-Reachable Indicator

This parameter indicates by its presence that the UE is reachable.

NOTE: In general any UpdateGPRS-Location request message (with or without UE-Reachable Indicator) implicitly conveys the information that the UE is now reachable.

This explicit indicator shall be set only when UpdateGPRS-Location is used for the only and no other purpose than indicating UE reachability. The HLR shall skip subscriber data downloading and any mobility management functionality other than reporting the UE's reachability to relevant core network entities.

T-ADS Data Retrieval Support Indicator

This parameter indicates by its presence that the SGSN supports retrieval of T-ADS data with the Provide-Subscriber-Info service.

Homogeneous Support Of IMS Voice Over PS Sessions

This parameter when present indicates that IMS voice over PS sessions is homogeneously supported in the complete SGSN area or that IMS voice over PS sessions is homogeneously not supported in the complete SGSN area.

Update of Homogeneous Support Of IMS Voice Over PS Sessions

This parameter when present indicates that Homogeneous Support of IMS Voice Over PS Sessions is updated. If the Homogeneous Support of IMS Voice Over PS Session is not present, the value of the Homogeneous Support of IMS Voice Over PS Sessions shall be updated as unknown to the serving node.

UE SRVCC Capability

See definition in clause 7.6.3.99.

Equivalent PLMN List

This parameter indicates the equivalent PLMN list of which the MME/SGSN requests the corresponding CSG Subscription data.

MME Number for MT SMS

This parameter contains the ISDN number of the MME allocated for MT SMS (see 3GPP TS 23.003 [17]). It is present when the MME requests to be registered for SMS.

SMS-Only

This parameter indicates to the HSS that the UE needs only PS domain services and SMS services.

SMS Register Request

This parameter indicates to the HSS that if the MME (via IWF) needs to be registered for SMS, prefers not to be registered for SMS or has no preference to be registered for SMS, see 3GPP TS 23.272 [143].

This parameter indicates to the HSS that if the SGSN needs to be registered for SMS, prefers not to be registered for SMS or has no preference to be registered for SMS, see 3GPP TS 23.060 [104].

Removal of MME Registration for SMS

This parameter indicates by its presence that the MME requests to remove its registration for SMS.

MSISDN-less Operation Supported

This parameter indicates by its presence that the SGSN supports MSISDN-less operation (see subclause 5.3.17 of 3GPP TS 23.060 [23]). An SGSN which supports MSISDN-less operation shall set this parameter.

SGSN Name

See definition in clause 7.6.2. This parameter is provided in a request when the serving node is an SGSN and the SGSN supports Lgd interface for LCS and/or Gdd interface for SMS.

SGSN Realm

See definition in clause 7.6.2. This parameter is provided in a request when the serving node is an SGSN and the SGSN supports Lgd interface for LCS and/or Gdd interface for SMS.

Lgd Support Indicator

This parameter, by its presence, indicates to the HSS that the SGSN supports Lgd interface for LCS. When absent the SGSN supports only Lg interface for LCS, if LCS is supported.

Adjacent PLMNs

This parameter indicates the list of PLMNs where an UE served by the SGSN is likely to make a handover from the PLMN where the SGSN is located. This list is statically configured by the operator in the SGSN, according to the geographical disposition of the different PLMNs in that area, the roaming agreements, etc...

Reset-IDs Supported

This parameter indicates, if present, the support of Reset-IDs by the SGSN.

ADD Capability

This parameter indicates, if present, the support of ADD function by the HLR.

SGSN-MME Separation Support Indicator

This parameter indicates by its presence that the HSS separately stores SGSN Id and MME Id. A combined MME/SGSN shall not send Update-GPRS-Location at intra node inter RAT routing area update if a Separation Indicator was not received previously.

MME Registered for SMS

This parameter indicates by its presence that the HSS has registered the MME for SMS.

User error

In case of unsuccessful updating, an error cause shall be returned by the HLR. The following error causes defined in clause 7.6.1 may be used, depending on the nature of the fault:

- unknown subscriber;
- roaming not allowed.

  This cause will be sent if the MS is not allowed to roam into the PLMN indicated by the SGSN number. The cause is qualified by the roaming restriction reason "PLMN Not Allowed", "Supported RAT Types Not Allowed" or "Operator Determined Barring".

  This cause shall be used when the HLR rejects a MAP Update Gprs Location request received for an MSISDN-less subscription from a SGSN not supporting MSISDN-less operation.
- system failure;
- unexpected data value.

The diagnostic in the Unknown Subscriber may indicate "Imsi Unknown" or "Gprs or EPS Subscription Unknown".

Provider error

For definition of provider errors see clause 7.6.1.

8.1.8 MAP-NOTE-MM-EVENT

8.1.8.1 Definition

This service is used between the VLR and the gsmSCF or between the SGSN and the gsmSCF when a mobility management event for a subscriber has been processed successfully, that subscriber is provisioned with M-CSI or MG-CSI and the relevant mobility management event is marked for reporting.

This service is also used between the VLR and the Presence Network Agent or between the SGSN and the Presence Network Agent to notify the Presence Network Agent when a mobility management event for a subscriber has been processed successfully, that subscriber is provisioned with M-CSI or MG-CSI and the relevant mobility management event is marked for reporting (see 3GPP TS 23.141 [128]).

8.1.8.2 Service primitives

The service primitives are shown in table 8.1/8.

Table 8.1/8: MAP_NOTE_MM_EVENT parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
Event Met	M	M(=)		
Service Key	M	M(=)		
IMSI	M	M(=)		
Basic MSISDN	M	M(=)		
Location Information for GPRS	C	C(=)		
Location Information	C	C(=)		
LSA Identity	C	C(=)		
Supported CAMEL Phases	M	M(=)		
Offered CAMEL 4 Functionalities	C	C(=)		
User error			C	C(=)
Provider error				O

8.1.8.3 Parameter use

Event Met
This parameter indicates the mobility management event that has lead to the notification. It shall have one of the following values for a mobility management event reported by the VLR:

- Location update in the same VLR service area;
- Location update to another VLR service area;
- IMSI attach;
- MS initiated IMSI detach (explicit detach);
- Network initiated IMSI detach (implicit detach).

It shall have one of the following values for a mobility management event reported by the SGSN:

- Routeing area update in the same SGSN service area;
- Routeing area update to another SGSN service area;
- GPRS attach;
- MS initiated GPRS detach;
- Network initiated GPRS detach;
- Network initiated transfer to the "not reachable for paging" state.

Service Key

See clause 7.6.x.

IMSI

See clause 7.6.x.

Basic MSISDN

See clause 7.6.x.

Location Information

See clause 7.6.2.30. This information shall be sent when the event is reported by a VLR, if available. If the event is reported as part of an SGs location update procedure, this information shall include the LAI and the Location Information for EPS if available.

Location Information for GPRS

See clause 7.6.2.30a. This information shall be sent when the event is reported by an SGSN, if available.

LSA Identity

See clause 7.6.x. This information shall be sent, if available.

Supported CAMEL Phases

See clause 7.6.x. This information shall always be sent.

Offered CAMEL 4 Functionalities

This parameter indicates the CAMEL phase 4 functionalities offered by the sending entity, VMSC/VLR or SGSN (see clause 7.6.3.36G).

User error

This parameter is sent by the receiving entity when an error is detected. It shall have one of the following values:

- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber;
- MM-EventNotSupported.

Provider error

This is defined in clause 7.6.1.

8.1.9 MAP_UPDATE_VCSG_LOCATION service

8.1.9.1 Definition

This procedure is used by the VLR or SGSN to register the MS in the CSS when

- the VPLMN supports Autonomous CSG Roaming
- and the HPLMN has enabled Autonomous CSG Roaming in the VPLMN
- and the MS has requested an initial attach or a location area procedure or a routing area procedure to a CSG cell
- and the VLR or SGSN has not yet registered the MS in the CSS.

The MAP_UPDATE_VCSG_LOCATION service is a confirmed service using the service primitives given in table 8.1/9.

8.1.9.2 Service primitives

Table 8.1/9: MAP_UPDATE_VCSG_LOCATION

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
VLR number	C	C(=)		
SGSN number	C	C(=)		
MSISDN	C	C(=)		
Temporary Empty CSG Subscription data Indicator			C	C(=)
User error			C	C(=)
Provider error				O

8.1.9.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

VLR number

See definition in clause 7.6.2. The presence of this parameter is mandatory when the service is used by the VLR.

SGSN number

See definition in clause 7.6.2. The presence of this parameter is mandatory when the service is used by the SGSN.

MSISDN

See definition in clause 7.6.2. Shall be present if this parameter is available.

Temporary Empty CSG Subscription data Indicator

See definition in clause 7.6.3.100. This parameter shall be present if the CSS accepts the request and there is no CSG Subscription data (empty CSG-ID list) for the roaming MS in the CSS.

User error

The following error causes defined in clause 7.6.1 may be used:

- unknown subscriber;
- system failure;
- unexpected data value.

Provider error

For definition of provider errors see clause 7.6.1

8.1.10 MAP_ CANCEL_VCSG_LOCATION service

8.1.10.1 Definition

This service is used between CSS and VLR to delete a roaming user record including the CSG subscription data and the CSS number from the VLR. The service is also used between CSS and SGSN to delete a roaming user record including the CSG subscription data and the CSS number from the SGSN. It may be invoked when there is removal of the CSG subscription data in CSS and of the MS registration including the case where a MS was registered in CSS but without CSG data.

The MAP_CANCEL_VCSG_LOCATION service is a confirmed service using the primitives defined in table 8.1/10.

8.1.10.2 Service primitives

Table 8.1/10: MAP_CANCEL_VCSG_LOCATION

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
User error			C	C(=)
Provider error				O

8.1.10.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

User error

If the cancellation fails, an error cause is to be returned by the VLR or by the SGSN. One of the following error causes defined in clause 7.6.1 shall be used:

- unexpected data value;
- data missing.

Provider error

For definition of provider errors see clause 7.6.1.

8.2 Paging and search

8.2.1 MAP_PAGE service

8.2.1.1 Definition

This service is used between VLR and MSC to initiate paging of an MS for mobile terminated short message or unstructured SS notification.

The MAP_PAGE service is a confirmed service using the primitives from table 8.2/1.

8.2.1.2 Service primitives

Table 8.2/1: MAP_PAGE

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
Stored location area Id	M	M(=)		
TMSI	U	C(=)		
User error			C	C(=)
Provider error				O

8.2.1.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2. The IMSI is used to define the paging subgroup. If the TMSI is not supplied, paging on the radio path uses the IMSI as an identifier.

Stored location area Id

See definition in clause 7.6.2.

TMSI

See definition in clause 7.6.2. The TMSI is included if paging on the radio channel is to use the TMSI as an identifier.

User error

The following error causes defined in clause 7.6.1 may be sent by the user in case of a paging error, depending on the failure reason:

- absent subscriber;
- unknown location area;
- busy subscriber;
- system failure;
- this corresponds to the case where there is no call associated with the MAP_PAGE service, i.e. if the call has been released but the dialogue to the VLR has not been aborted;
- unexpected data value.

Provider error

See definition in clause 7.6.1.

8.2.2 MAP_SEARCH_FOR_MS service

8.2.2.1 Definition

This service is used between VLR and MSC to initiate paging of an MS in all location areas of that VLR. It is used if the VLR does not hold location area information confirmed by radio contact.

The MAP_SEARCH_FOR_MS service is a confirmed service using the primitives from table 8.2/2.

8.2.2.2 Service primitives

Table 8.2/2: MAP_SEARCH_FOR_MS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
Current location area Id			C	C(=)
User error			C	C(=)
Provider error				O

8.2.2.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2. The IMSI is used to identify the subscriber when paging on the radio path.

Current location area Id

See definition in clause 7.6.2. In case of successful outcome of the service, i.e. if the MS responds to paging, the Location Area Id of the area in which the MS responded is given in the response.

User error

The following error causes defined in clause 7.6.1 shall be sent by the user if the search procedure fails, depending on the failure reason:

- absent subscriber;

  this error cause is returned by the MSC if the MS does not respond to the paging request;

- system failure;

- this corresponds to the case where there is no call associated with the MAP_SEARCH_FOR_MS service, i.e. if the call has been released but the dialogue to the VLR has not been aborted;

- busy subscriber;

- unexpected data value.

Provider error

See definition in clause 7.6.1.

8.3 Access management services

8.3.1 MAP_PROCESS_ACCESS_REQUEST service

8.3.1.1 Definition

This service is used between MSC and VLR to initiate processing of an MS access to the network, e.g. for mobile originated short message submission or after being paged by the network.

The MAP_PROCESS_ACCESS_REQUEST service is a confirmed service using the primitives from table 8.3/1.

8.3.1.2 Service primitives

Table 8.3/1: MAP_PROCESS_ACCESS_REQUEST

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
CM service type	M	M(=)		
Access connection status	M	M(=)		
Current Location Area Id	M	M(=)		
Serving cell Id	M	M(=)		
TMSI	C	C(=)		
Cksn	C	C(=)		
IMSI	C	C(=)	C	C(=)
IMEI	C	C(=)	C	C(=)
MSISDN			U	C(=)
User error			C	C(=)
Provider error				O

8.3.1.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

CM service type

See definition in clause 7.6.9.

Access connection status

See definition in clause 7.6.9.

Current Location Area Id

See definition in clause 7.6.2. This parameter is used to update the VLR in case of previous VLR failure.

Serving cell Id

See definition in clause 7.6.2.

TMSI

See definition in clause 7.6.2. Either TMSI or IMSI as received from the MS are included in the Request/Indication, but one shall be present. In case of CM Service Type "Emergency Call Establishment", the IMEI may replace IMSI/TMSI.

Cksn

See definition in clause 7.6.7. In case of access with TMSI, the Cksn shall be present.

IMSI

See definition in clause 7.6.2. Either TMSI or IMSI as received from the MS are included in the Request/Indication, but one shall be present. In case of CM Service Type "Emergency Call Establishment", the IMEI may replace IMSI/TMSI.

In the Response/Confirmation, the IMSI is to be sent in case of successful outcome of the service. In case of CM Service Type "Emergency Call Establishment", IMEI may replace IMSI.

IMEI

See definition in clause 7.6.2. The IMEI may replace IMSI/TMSI in the Request/Indication and IMSI in the Response/Confirmation only in case the CM Service Type indicates "Emergency Call Establishment".

MSISDN

See definition in clause 7.6.2. The MSISDN is included in case of successful outcome of the service as an operator option, e.g. if it is needed at the MSC for charging purposes in case of call forwarding.

User error

One of the following error causes defined in clause 7.6.1 shall be sent by the user if the access request fails, depending on the failure reason:

- unidentified subscriber;
- illegal subscriber;

  this error is sent if a correlated authentication procedure has not authenticated the subscriber;

- illegal equipment;

  this error is sent if an IMEI check failed, i.e. the IMEI is blacklisted or not white-listed;

- roaming not allowed;

- this cause is used after VLR restart if the subscriber has no subscription for the current location area, e.g. due to regional subscription. The cause will be qualified by "location area not allowed" or "national roaming not allowed", respectively;
- unknown location area;
- system failure;
- unexpected data value.

Provider error

For definition of provider errors see clause 7.6.1.

8.4 Handover services

It should be noted that the handover services used on the B-interface have not been updated for Release 99. The B-interface is not fully operational specified. It is strongly recommended not to implement the B-interface as an external interface.

8.4.1 MAP_PREPARE_HANDOVER service

8.4.1.1 Definition

This service is used between MSC-A and MSC-B (E-interface) when a call is to be handed over or relocated from MSC-A to MSC-B.

The MAP_PREPARE_HANDOVER service is a confirmed service using the primitives from table 8.4/1.

8.4.1.2 Service primitives

Table 8.4/1: MAP_PREPARE_HANDOVER

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Target Cell Id	C	C(=)		
Target RNC Id	C	C(=)		
HO-NumberNotRequired	C	C(=)		
IMSI	C	C(=)		
Integrity Protection Information	C	C(=)		
Encryption Information	C	C(=)		
Radio Resource Information	C	C(=)		
AN-APDU	C	C(=)	C	C(=)
Allowed GSM Algorithms	C	C(=)		
Allowed UMTS Algorithms	C	C(=)		
Radio Resource List	C	C(=)		
RAB ID	C	C(=)		
GERAN Classmark	C	C(=)		
BSSMAP Service Handover	C	C(=)		
BSSMAP Service Handover List	C	C(=)		
RANAP Service Handover	C	C(=)		
Iu-Currently Used Codec	C	C(=)		
Iu-Supported Codecs List	C	C(=)		
RAB Configuration Indicator	C	C(=)		
ASCI Call Reference	C	C(=)		
UESBI-Iu	C	C(=)		
IMEISV	C	C(=)		
Alternative Channel Type	C	C(=)		
Trace_Propagation_List	C	C(=)		
AoIP-Supported Codecs List Anchor	C	C(=)		

Regional Subscription Data	U	(C=)		
CSG Subscription Data	U	(C=)		
LCLS Global Call Reference	C	C(=)		
LCLS-Negotiation	C	C(=)		
LCLS-Configuration-Preference	C	C(=)		
Multiple Bearer Requested	C	C(=)		
Handover Number			C	C(=)
Relocation Number List			C	C(=)
Multicall Bearer Information			C	C(=)
Multiple Bearer Not Supported			C	C(=)
Selected UMTS Algorithms			C	C(=)
Chosen Radio Resource Information			C	C(=)
Iu-Selected Codec			C	C(=)
Iu-Available Codecs List			C	C(=)
AoIP-Selected Codec Target			C	C(=)
AoIP-Available Codecs List Map			C	C(=)
User error			C	C(=)
Provider error				O

8.4.1.3 Parameter use

Invoke Id

For definition of this parameter see clause 7.6.1.

Target Cell Id

For definition of this parameter see clause 7.6.2. This parameter is only included if the service is not in an ongoing transaction. This parameter shall also be excluded if the service is a part of the Inter-MSC SRNS Relocation procedure or the inter-system handover GSM to UMTS procedure described in 3GPP TS 23.009.

Target RNC Id

For definition of this parameter see clause 7.6.2. This parameter shall be included if the service is a part of the Inter-MSC SRNS Relocation procedure or the inter-system handover GSM to UMTS procedure described in 3GPP TS 23.009.

HO-Number Not Required

For definition of this parameter see clause 7.6.6.

IMSI

For definition of this parameter see clause 7.6.2. This UMTS parameter shall be included if:

- available and
- if the access network protocol is BSSAP and
- there is an indication that the MS also supports UMTS.

Integrity Protection Information

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if available and if the access network protocol is BSSAP.

Encryption Information

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if available and if the access network protocol is BSSAP.

Radio Resource Information

For definition of this parameter see clause 7.6.6. This GSM parameter shall be included if the access network protocol is RANAP and there is an indication that the UE also supports GSM. If the parameter Radio Resource List is sent , the parameter Radio Resource Information shall not be sent.

AN-APDU

For definition of this parameter see clause 7.6.9.

Allowed GSM Algorithms

For definition of this parameter see clause 7.6.6. This parameters includes allowed GSM algorithms. This GSM parameter shall be included if:

- the service is a part of the Inter-MSC SRNS Relocation procedure and
- Ciphering or Security Mode Setting procedure has been performed.and
- there is an indication that the UE also supports GSM.

Allowed UMTS Algorithms

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if all of the following conditions apply:

- access network protocol is BSSAP and
- Integrity Protection Information and Encryption Information are not available and
- Ciphering or Security Mode Setting procedure has been performed.

Radio Resource List

For definition of this parameter see clause 7.6.6. This parameter shall be included if the access network protocol is RANAP and there is an indication that the UE also supports GSM. This parameter shall be sent when MSC-A requests multiple bearers to MSC-B. If the parameter Radio Resource Information is sent , the parameter Radio Resource List shall not be sent.

RAB ID

For definition of this parameter see subclause 7.6.2. This parameter shall be included when MSC-A supports multiple bearers and access network protocol is BSSAP and the RAB ID has a value other than 1.

GERAN Classmark

For definition of this parameter see subclause 7.6.6 This parameter shall be included if available.

BSSMAP Service Handover

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the access network protocol is RANAP. If the parameter BSSMAP Service Handover List is sent, the parameter BSSMAP Service Handover shall not be sent.

BSSMAP Service Handover List

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the access network protocol is RANAP. This parameter shall be sent when MSC-A requests multiple bearers to MSC-B. If the parameter BSSMAP Service Handover is sent, the parameter BSSMAP Service Handover List shall not be sent.

RANAP Service Handover

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the access network protocol is BSSAP.

Iu-Currently Used Codec

For definition of this parameter see subclause 7.6.6. This parameter shall be included if the handover is requested for a speech bearer and the MS is in UMTS or GERAN Iu-mode access. This parameter shall not be included if the Iu-Supported Codecs List is not included.

Iu-Supported Codecs List

For definition of this parameter see subclause 7.6.6. This parameter shall be included by MSC-A, if the handover is requested for a speech bearer.

RAB Configuration Indicator

For definition of this parameter see subclause 7.6.6. This parameter may be included if the handover is requested for a speech bearer and MSC-A knows by means of configuration information that MSC-B supports the use of the Iu-Supported Codecs List parameter. This parameter shall not be included if the Iu-Supported Codecs List is not included.

ASCI Call Reference

This parameter contains either the broadcast call reference or group call reference. It shall be included if a subscriber is undergoing handover during a VGCS or VBS call, where MSC-B already has a Bearer established, so that MSC-B can determine the Group or Broadcast Call to which it shall attach the subscriber, see 3GPP TS 48.008 [49].

UESBI-Iu

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the access network protocol is BSSAP.

IMEISV

For definition of the parameter see clause 7.6.2. This parameter shall be present, if available. This is used e.g. for Management based Trace Activation (see 3GPP TS 32.422).

Alternative Channel Type

For definition of this parameter see clause 7.6.6 It shall be present for a SCUDIF call if the access network protocol is BSSAP.

Trace Propagation List

See definition in clause 7.6.10. This parameter shall be included when MSC-A requests trace invocation.

AoIP-Supported Codecs List Anchor

For definition of this parameter see subclause 7.6.6. This parameter may be included by MSC-A, if the handover is requested for a speech bearer and mobile terminal supports GSM codec types.

Regional Subscription Data

The list of subscribed Zone Codes as received from the HLR may be included by MSC-A at intra PLMN inter MSC handover and may be stored at MSC-B for use at subsequent intra MSC handover.

CSG Subscription Data

The subscribed CSG Subscription Information as received from the HLR may be included by MSC-A at intra PLMN inter MSC handover and at inter PLMN inter MSC handover when the target PLMN is an ePLMN, and may be stored at MSC-B for use at subsequent intra MSC handover.

LCLS Global Call Reference

For definition of this parameter see subclause 7.6.5.21. This parameter shall be included when MSC-A supports LCLS.

LCLS-Negotiation

For definition of this parameter see subclause 7.6.5.22. This parameter shall be included when MSC-A supports LCLS.

LCLS-Configurations-Preference

For definition of this parameter see subclause 7.6.5.23. This parameter shall be included when MSC-A supports LCLS.

Multiple Bearer Requested

For a definition of this parameter see clause 7.6.2. This parameter shall be sent when MSC-A requests multiple bearers to MSC-B.

Handover Number

For definition of this parameter see clause 7.6.2. This parameter shall be returned at handover, unless the parameter HO-NumberNotRequired is sent. If the parameter Handover Number is returned, the parameter Relocation Number List shall not be returned.

Relocation Number List

For definition of this parameter see clause 7.6.2. This parameter shall be returned at relocation, unless the parameter HO-NumberNotRequired is sent. If the parameter Relocation Number List is returned, the parameter Handover Number shall not be returned.

Multicall Bearer Information

For a definition of this parameter see clause 7.6.2. This parameter shall be returned at relocation in the case that MSC-B supports multiple bearers.

Multiple Bearer Not Supported

For a definition of this parameter see clause 7.6.2. This parameter shall be returned at relocation when MSC-B receives Multiple Bearer Requested parameter and MSC-B does not support multiple bearers.

Selected UMTS Algorithms

For definition of this parameter see clause 7.6.6. This parameters includes the UMTS integrity and optionally encryption algorithms selected by RNC under the control of MSC-B. This UMTS parameter shall be included if the service is a part of the inter MSC inter system handover from GSM to UMTS.

Chosen Radio Resource Information

For definition of this parameter see clause 7.6.6. This parameter shall be returned at relocation if the encapsulated PDU is RANAP RAB Assignment Response and MS is in GSM access.

Iu-Selected Codec

For definition of this parameter see subclause 7.6.6. This parameter shall be included if an Iu-Supported Codecs List was received in the service request and MSC-B supports the selection of codec based on the Iu-Supported Codecs List and the target radio access network is connected to MSC-B via the Iu interface, even if the Iu-Selected Codec is equal to the Iu-Currently Used Codec received in the service request. This parameter shall not be included if the Iu-Supported Codecs List was not received in the service request.

Iu-Available Codecs List

For definition of this parameter see subclause 7.6.6. This parameter shall be included by an MSC-B supporting TrFO, if the Iu-Supported Codecs List was included by MSC-A and the target radio access is UMTS or GERAN Iu-mode.

AoIP-Selected Codec Target

For definition of this parameter see subclause 7.6.6. This parameter may be included by an MSC-B supporting TrFO, if the AoIP-Supported Codecs List Anchor was included by MSC-A and if AoIP is used on the target A interface with transcoder inserted in the MGW.

AoIP-Available Codecs List Map

For definition of this parameter see subclause 7.6.6. This parameter may be included by an MSC-B supporting TrFO, if the AoIP-Supported Codecs List Anchor was included by MSC-A and if AoIP is used on the target A interface with transcoder inserted in the MGW.

User error

For definition of this parameter see clause 7.6.1. The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- No handover number available.
- Target cell outside group call area;
- System failure.
- Unexpected data value.
- Data Missing.

Provider error

See definition of provider errors in clause 7.6.1.

8.4.2 MAP_SEND_END_SIGNAL service

8.4.2.1 Definition

This service is used between MSC-B and MSC-A (E-interface) indicating that the radio path has been established by MSC-B to the MS. MSC-A retains then the main control of the call until it clears.

The response is used by MSC-A to inform MSC-B that all resources for the call can be released in MSC-B, either because the call has been released in MSC-A or because the call has been successfully handed over or relocated from MSC-B to another MSC.

The MAP_SEND_END_SIGNAL service is a confirmed service using the primitives from table 8.4/2.

8.4.2.2 Service primitives

Table 8.4/2: MAP_SEND_END_SIGNAL

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
AN-APDU	M	M(=)		
Provider error				O

8.4.2.3 Parameter use

Invoke Id

For definition of this parameter see clause 7.6.1.

AN-APDU

For definition of this parameter see clause 7.6.9.

Provider error

For definition of this parameter see clause 7.6.1.

8.4.3 MAP_PROCESS_ACCESS_SIGNALLING service

8.4.3.1 Definition

This service is used between MSC-B and MSC-A (E-interface) to pass information received on the A-interface or Iu-interface in MSC-B to MSC-A.

The MAP_PROCESS_ACCESS_SIGNALLING service is a non-confirmed service using the primitives from table 8.4/3.

8.4.3.2 Service primitives

Table 8.4/3: MAP_PROCESS_ACCESS_SIGNALLING

Parameter name	Request	Indication
Invoke Id	M	M(=)
AN-APDU	M	M(=)
Selected GSM Algorithm	C	C(=)
Selected UMTS Algorithms	C	C(=)
Chosen Radio Resource Information	C	C(=)
Selected RAB id	C	C(=)
Iu-Selected Codec	C	C(=)
Iu-Available Codecs List	C	C(=)
AoIP-Selected Codec Target	C	C(=)
AoIP-Available Codecs List Map	C	C(=)

8.4.3.3 Parameter use

Invoke Id

For definition of this parameter see clause 7.6.1.

AN-APDU

For definition of this parameter see clause 7.6.9.

Selected GSM algorithm

For definition of this parameter see clause 7.6.6. This parameter shall be present if the encapsulated PDU is Security Mode Complete and MS is in GSM access.

Selected UMTS Algorithms

For definition of this parameter see clause 7.6.6. This parameters includes the UMTS integrity and optionally encryption algorithms selected by RNC under the control of MSC-B. This UMTS parameter shall be included if the encapsulated PDU is BSSMAP Cipher Mode Complete and the MS is in UMTS, or an interystem handover to UMTS is performed in MSC-B, or in the case of intra MSC-B intra UMTS relocation.

Chosen Radio Resource Information

For definition of this parameter see clause 7.6.6. This parameter shall be sent if the encapsulated PDU is RANAP RAB Assignment Response and MS is in GSM access.

Selected RAB ID

The selected radio access bearer that was kept at subsequent intra-MSC handover from UMTS to GSM after multiple bearers were used.

Iu-Selected Codec

For definition of this parameter see subclause 7.6.6. This parameter shall be included

- if MSC-B changes the selected codec and the MS is in UMTS or GERAN Iu-mode access;
- if intersystem handover to UMTS or GERAN Iu-mode is performed in MSC-B; or
- if MSC-B received a Forward Access Signalling service request including an Iu-Supported Codecs List and the MS is in UMTS or GERAN Iu-mode access.

This parameter shall not be included if the Iu-Supported Codecs List was not received either in the Prepare Handover service request or in the Forward Access Signalling service request.

Iu-Available Codecs List

For definition of this parameter see subclause 7.6.6. This parameter shall be included by an MSC-B supporting TrFO

- if the Iu-Available Codecs List has changed in MSC-B;
- if intersystem handover to UMTS or GERAN Iu-mode is performed in MSC-B; or
- if MSC-B received a Forward Access Signalling service request including an Iu-Supported Codecs List and the MS is in UMTS or GERAN Iu-mode access.

AoIP-Selected Codec Target

For definition of this parameter see subclause 7.6.6. This parameter may be included

- if A interface codec is changed in MSC-B; or
- if intersystem handover to AoIP capable BSC is performed in MSC-B and if AoIP is used on the target A interface with transcoder inserted in the MGW; or
- if MSC-B received a Forward Access Signalling service request including an AoIP-Supported Codecs List and the MS is in AoIP capable GSM access where AoIP is used on the target A interface with transcoder inserted in the MGW.

This parameter shall not be included if the AoIP-Supported Codecs List Anchor was not received either in the Prepare Handover service request or in the Forward Access Signalling service request.

AoIP-Available Codecs List Map

For definition of this parameter see subclause 7.6.6. This parameter may be included by an MSC-B supporting TrFO

- if the AoIP-Available Codecs List has changed in MSC-B; or
- if intersystem handover to AoIP capable BSC is performed in MSC-B where AoIP is used on the target A interface with transcoder inserted in the MGW; or
- if MSC-B received a Forward Access Signalling service request including an AoIP-Supported Codecs List Anchor and the MS is in AoIP capable GSM access where AoIP is used on the target A interface with transcoder inserted in the MGW.

8.4.4 MAP_FORWARD_ACCESS_SIGNALLING service

8.4.4.1 Definition

This service is used between MSC-A and MSC-B (E-interface) to pass information to be forwarded to the A-interface or Iu-interface of MSC-B.

The MAP_FORWARD_ACCESS_SIGNALLING service is a non-confirmed service using the primitives from table 8.4/4.

8.4.4.2 Service primitives

Table 8.4/4: MAP_FORWARD_ACCESS_SIGNALLING

Parameter name	Request	Indication
Invoke Id	M	M(=)
Integrity Protection Information	C	C(=)
Encryption Information	C	C(=)
Key Status	C	C(=)
AN-APDU	M	M(=)
Allowed GSM Algorithms	C	C(=)

Allowed UMTS Algorithms	C	C(=)
Radio Resource Information	C	C(=)
Radio Resource List	C	C(=)
BSSMAP Service Handover	C	C(=)
BSSMAP Service Handover List	C	C(=)
RANAP Service Handover	C	C(=)
Iu-Currently Used Codec	C	C(=)
Iu-Supported Codecs List	C	C(=)
RAB Configuration Indicator	C	C(=)
Iu-Selected Codec	C	C(=)
Alternative Channel Type	C	C(=)
Trace Propagation List	C	C(=)
AoIP-Supported Codecs List Anchor	C	C(=)
AoIP-Selected Codec Target	C	C(=)
UESBI-Iu	C	C(=)
IMEISV	C	C(=)

8.4.4.3 Parameter use

For the definition and use of all parameters and errors, see clause 7.6.1.

Invoke Id

For definition of this parameter see clause 7.6.1.

Integrity Protection Information

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if available and if the encapsulated PDU is BSSMAP Cipher Mode Command.

Encryption Information

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if available and if the encapsulated PDU is BSSMAP Cipher Mode Command.

Key Status

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if available and if the encapsulated PDU is BSSMAP Cipher Mode Command.

AN-APDU

For definition of this parameter see clause 7.6.9.

Allowed GSM Algorithms

This parameters includes allowed GSM algorithms. This GSM parameter shall be included if the encapsulated PDU is RANAP Security Mode Command and there is an indication that the UE also supports GSM.

Allowed UMTS Algorithms

For definition of this parameter see clause 7.6.6. This UMTS parameter shall be included if Integrity Protection Information and Encryption Information are not available and the encapsulated PDU is BSSMAP Cipher Mode Command.

Radio Resource Information

For definition of this parameter see clause 7.6.6. This parameter shall be sent if the encapsulated PDU is RANAP RAB Assignment Request. If the parameter Radio Resource List is sent, the parameter Radio Resource Information shall not be sent.

Radio Resource List

For definition of this parameter see clause 7.6.6. This parameter shall be sent if the encapsulated PDU is RANAP RAB Assignment Request and MSC-A requests modification of multiple bearers. If the parameter Radio Resource Information is sent, the parameter Radio Resource List shall not be sent.

BSSMAP Service Handover

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the encapsulated PDU is RANAP RAB Assignment Request. If the parameter BSSMAP Service Handover List is sent, the parameter BSSMAP Service Handover shall not be sent.

BSSMAP Service Handover List

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the encapsulated PDU is RANAP RAB Assignment Request and MSC-A requests modification of multiple bearers. If the parameter BSSMAP Service Handover is sent, the parameter BSSMAP Service Handover List shall not be sent.

RANAP Service Handover

For definition of this parameter see clause 7.6.6.. It shall be present if it is available and the encapsulated PDU is BSSMAP Assignment Request.

Iu-Currently Used Codec

For definition of this parameter see subclause 7.6.6. This parameter shall be included if the encapsulated PDU is a RANAP RAB Assignment Request or BSSMAP Assignment Request for a speech bearer and the MS is in UMTS or GERAN Iu-mode access. This parameter shall not be included if the Iu-Supported Codecs List is not included.

Iu-Supported Codecs List

For definition of this parameter see subclause 7.6.6. This parameter shall be included if the encapsulated PDU is a RANAP RAB Assignment Request or BSSMAP Assignment Request and

- a new bearer is allocated for speech;
- an existing bearer is modified from data to speech; or
- for an existing speech bearer the order of priority in the Iu-Supported Codecs List needs to be modified.

This parameter shall not be included if the Iu-Selected Codec is included.

RAB Configuration Indicator

For definition of this parameter see subclause 7.6.6. This parameter may be included if the encapsulated PDU is a RANAP RAB Assignment Request for a speech bearer, and MSC-A knows by means of configuration information that MSC-B supports the use of the Iu-Supported Codecs List parameter. This parameter shall not be included if the Iu-Supported Codecs List is not included.

Iu-Selected Codec

For definition of this parameter see subclause 7.6.6. This parameter shall be included if

- the encapsulated PDU is a RANAP RAB Assignment Request or BSSMAP Assignment Request for an existing speech bearer; and
- the MS is in UMTS or GERAN Iu-mode access; and
- an Iu-Available Codecs List was received by MSC-A for this speech bearer before, either in the Prepare Handover service response or in the Process Access Signalling service request.

This parameter shall not be included if the Iu-Supported Codecs List is included.

Alternative Channel Type

For definition of this parameter see clause 7.6.6. This parameter shall be present for a SCUDIF call if the encapsulated PDU is BSSMAP Assignment Request.

Trace Propagation List

See definition in clause 7.6.10. This parameter shall be included when MSC-A requests trace invocation.

AoIP-Supported Codecs List Anchor

For definition of this parameter see subclause 7.6.6. This parameter may be included if the encapsulated PDU is a BSSMAP Assignment Request and

- a new bearer is allocated for speech;
- an existing bearer is modified from data to speech; or
- for an existing speech bearer the order of priority in the AoIP-Supported Codecs List needs to be modified.

This parameter shall not be included if the AoIP-Selected Codec Target is included.

AoIP-Selected Codec Target

For definition of this parameter see subclause 7.6.6. This parameter may be included if

- the encapsulated PDU is a BSSMAP Assignment Request for an existing speech bearer; and
- the MS is in AoIP capable GSM access where AoIP is used on the target A interface with transcoder inserted in the MGW; and
- an AoIP-Available Codecs List was received by MSC-A for this speech bearer before, either in the Prepare Handover service response or in the Process Access Signalling service request.

This parameter shall not be included if the AoIP-Supported Codecs List Anchor is included.

UESBI-Iu

For definition of this parameter see clause 7.6.6. It shall be present if it is available and the access network protocol is BSSAP and the parameter has not already been sent to the target MSC.

IMEISV

For definition of the parameter see clause 7.6.2. This parameter shall be present if available and if not already sent to the target MSC. This is used e.g. for Management based Trace Activation (see 3GPP TS 32.422).

8.4.5 MAP_PREPARE_SUBSEQUENT_HANDOVER service

8.4.5.1 Definition

This service is used between MSC-B and MSC-A (E-interface) to inform MSC-A that it has been decided that a handover or relocation to either MSC-A or a third MSC (MSC-B') is required.

The MAP_PREPARE_SUBSEQUENT_HANDOVER service is a confirmed service using the primitives from table 8.4/5.

8.4.5.2 Service primitives

Table 8.4/5: MAP_PREPARE_SUBSEQUENT_HANDOVER

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Target Cell Id	C	C(=)		
Target RNC Id	C	C(=)		
Target MSC Number	M	M(=)		
Selected RAB ID	C	C(=)		
GERAN Classmark	C	C(=)		
RAB Configuration Indicator	C	C(=)		
AN-APDU	M	M(=)	C	C(=)
User error			C	C(=)
Provider error				O

8.4.5.3 Parameter use

Invoke Id

For definition of this parameter see clause 7.6.1.

Target Cell Id

For definition of this parameter see clause 7.6.2. This parameter shall be excluded if the service is a part of the Inter-MSC SRNS Relocation procedure or the inter-system handover GSM to UMTS procedure described in 3GPP TS 23.009.

Target RNC Id

For definition of this parameter see clause 7.6.2. This parameter shall be included if the service is a part of the Inter-MSC SRNS Relocation procedure or the inter-system handover GSM to UMTS procedure described in 3GPP TS 23.009.

Target MSC Number

For definition of this parameter see clause 7.6.2.

Selected RAB ID

For definition of this parameter see clause 7.6.2.

GERAN Classmark

For definition of this parameter see subclause 7.6.6 This parameter shall be included if available.

RAB Configuration Indicator

For definition of this parameter see subclause 7.6.6. This parameter may be included if the call is a speech call and MSC-B knows by means of configuration information that MSC-B' (and MSC-A) supports the use of the Iu-Supported Codecs List parameter.

AN-APDU

For definition of this parameter see clause 7.6.9.

User error

For definition of this parameter see clause 7.6.1. The following error causes defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unknown MSC;
- Subsequent handover failure;

- Unexpected data value;
- Data Missing.

Provider error

For definition of this parameter see clause 7.6.1.

8.4.6 MAP_ALLOCATE_HANDOVER_NUMBER service

8.4.6.1 Definition

This service is used between MSC and VLR (B-interface) to request a handover number.

The MAP_ALLOCATE_HANDOVER_NUMBER service is a confirmed service using the primitives from table 8.4/6.

8.4.6.2 Service primitives

Table 8.4/6: MAP_ALLOCATE_HANDOVER_NUMBER

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
User error			C	C(=)
Provider error				O

8.4.6.3 Parameter use

Invoke Id

For definition of this parameter see clause 7.6.1.

User error

For definition of this parameter see clause 7.6.1. The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- No handover number available.

Provider error

For definition of this parameter see clause 7.6.1.

8.4.7 MAP_SEND_HANDOVER_REPORT service

8.4.7.1 Definition

This service is used between VLR and MSC-B (B-interface) to transfer the handover number to be forwarded to and used by MSC-A.

The MAP_SEND_HANDOVER_REPORT service is a confirmed service using the primitives from table 8.4/7.

8.4.7.2 Service primitives

Table 8.4/7: MAP_SEND_HANDOVER_REPORT

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Handover Number	M	M(=)		
Linked Id	M	M(=)		
Provider error				O

8.4.7.3 Parameter use

Invoke Id

For definition of this parameter see clause 7.6.1.

Handover Number

For definition of this parameter see clause 7.6.2.

Linked Id

For definition of this parameter see clause 7.6.1. This service is linked with MAP_ALLOCATE_HANDOVER_NUMBER.

Provider error

For definition of this parameter see clause 7.6.1.

8.5 Authentication management services

8.5.1 MAP_AUTHENTICATE service

The MAP_AUTHENTICATE service is used on the MAP B interface. This interface is not fully operational specified. It is strongly recommended not to implement the B-interface as an external interface.

8.5.1.1 Definition

This service is used between the VLR and the MSC when the VLR receives a MAP service indication from the MSC concerning a location registration, call set-up, operation on a supplementary service or a request from the MSC to initiate authentication.

The service is a confirmed service and consists of four service primitives.

8.5.1.2 Service primitives

The service primitives are shown in table 8.5/1.

Table 8.5/1: MAP_AUTHENTICATE parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
RAND	M	M(=)		
CKSN	M	M(=)		
SRES			M	M(=)
Provider error				O

8.5.1.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

RAND

See clause 7.6.7 for the use of this parameter.

CKSN

See clause 7.6.7 for the use of this parameter.

SRES

See clause 7.6.7 for the use of this parameter.

Provider error

See clause 7.6.1 for the use of this parameter.

8.5.2 MAP_SEND_AUTHENTICATION_INFO service

8.5.2.1 Definition

This service is used between the VLR and the HLR for the VLR to retrieve authentication information from the HLR. The VLR requests up to five authentication vectors.

Also this service is used between the SGSN and the HLR for the SGSN to retrieve authentication information and/or UE Usage Type from the HLR. The SGSN requests up to five authentication vectors.

Also this service is used between the BSF and the HLR for the BSF to retrieve authentication information from the HLR. The BSF shall only request one authentication vector at a time.

In an EPS, this service is used between IWF and IWF and between IWF and HSS.

If the requesting node type is different from "MME" and the user is a UMTS subscriber, the HLR shall return authentication quintuplets. If the requesting node type is different from MME and the user is a GSM subscriber, the HLR shall return authentication triplets.

If the requesting node type is "MME", the HSS shall return EPS authentication vectors.

If the requesting node type is a combined MME/SGSN, the HSS shall return requested authentication vectors for the actual RAT and may return additional authentication vectors for the other RAT.

If the HLR cannot provide the VLR, the SGSN or the BSF with triplets, an empty response is returned. The VLR, the SGSN, or the BSF may then re-use old authentication triplets, except where this is forbidden under the conditions specified in 3GPP TS 43.020 [24].

If the HLR cannot provide the VLR, the SGSN or the BSF with quintuplets, an empty response is returned. The VLR, the SGSN or the BSF shall not re-use old authentication quintuplets.

If the HSS cannot provide the IWF with EPS authentication vectors, an empty response is returned.

If the VLR or SGSN or IWF or BSF receives a MAP_SEND_AUTHENTICATION_INFO response containing a User Error parameter as part of the handling of an authentication procedure, the authentication procedure in the VLR or SGSN or MME or BSF shall fail.

Security related network functions are further described in 3GPP TS 43.020 [24] and 3GPP TS 33.200.

The service is a confirmed service and consists of four service primitives.

8.5.2.2 Service primitives

The service primitives are shown in table 8.5/2.

Table 8.5/2: MAP_SEND_AUTHENTICATION_INFO parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
Number of requested vectors	C	C(=)		
Requesting node type	C	C(=)		
Re-synchronisation Info	C	C(=)		
Segmentation prohibited indicator	C	C (=)		
Immediate response preferred indicator	U	C (=)		
Requesting PLMN ID	C	C(=)		
Number of additional requested vectors	C	C(=)		
Additional requested Vectors are for EPS	C	C(=)		
UE Usage Type Request Indication	C	C(=)		
AuthenticationSetList			C	C(=)
UE Usage Type			C	C(=)
User error			C	C(=)
Provider error				O

8.5.2.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

IMSI

See clause 7.6.2 for the use of this parameter.

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

Number of requested vectors

A number indicating how many authentication vectors the VLR, the SGSN, the MME or the BSF is prepared to receive. The HLR shall not return more vectors than indicated by this parameter.

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

Requesting node type

The type of the requesting node (SGSN, MME, combined MME/SGSN, VLR, or BSF).

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

Re-synchronisation Info

For definition and use of this parameter see 3GPP TS 33.200.

If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one..

Segmentation prohibited indicator

This parameter indicates if the VLR, the SGSN or the IWF allows segmentation of the response at MAP user level.

This parameter may be present only in the first request of the dialogue.

Immediate response preferred indicator

This parameter indicates that one of the requested authentication vectors is requested for immediate use in the VLR, the SGSN, the MME or the BSF. It may be used by the HLR together with the number of requested vectors and the number of vectors stored in the HLR to determine the number of vectors to be obtained from the AuC. It shall be ignored if the number of available vectors is greater than the number of requested vectors.

If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

Requesting PLMN ID

The PLMN-ID of the requesting node. See3GPP TS 23.003.

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

Number of additional requested vectors

A number indicating how many additional authentication vectors the combined MME/SGSN or IWF is prepared to receive. The HLR shall not return more vectors than indicated by this parameter. This parameter shall be present only if the requesting node type is a combined MME/SGSN. A combined MME/SGSN that wants to request only EPS-Vectors (only non-EPS-Vectors) shall set the requesting node type to "MME" ("SGSN").

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

Additional vectors are for EPS

This parameter shall be absent if Number of additional vectors is absent. The parameter indicates by its presence that additional vectors (i.e. not for immediate use) are for EPS.

This parameter shall be present in the first (or only) request of the dialogue. If multiple service requests are present in a dialogue then this parameter shall not be present in any service request other than the first one.

UE Usage Type Request Indication

This parameter indicates by its presence that the HLR (if it supports the Dedicated Core Network functionality) shall include the UE Usage Type in the response to the SGSN. This parameter is not applicable for VLRs.

AuthenticationSetList

A set of one to five authentication vectors are transferred from the HLR to the VLR, from the HLR to the SGSN or IWF or from the HLR to the BSF, if the outcome of the service was successful.

UE Usage Type

This parameter shall be present if UE Usage Type Request Indication was present in the request and the HLR supports the Dedicated Core Networks functionality (see 3GPP TS 23.060 [104]) and a UE Usage Type is available in the subscription data of the user. In this case, if the Immediate Response Preferred parameter is not set, the HLR may return no authentication vectors in the response.

User error

One of the following error causes defined in clause 7.6.1 shall be sent by the user in case of unsuccessful outcome of the service, depending on the respective failure reason:

- unknown subscriber;
- unexpected data value;
- system failure;
- data missing.

Provider error

See clause 7.6.1 for the use of this parameter.

8.5.3 MAP_AUTHENTICATION_FAILURE_REPORT service

8.5.3.1 Definition

This service is used between the VLR and the HLR or between the SGSN or HLR for reporting of authentication failures.

8.5.3.2 Service primitives

The service primitives are shown in table 8.5/3.

Table 8.5/3: MAP_AUTHENTICATION_FAILURE_REPORT parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
Failure cause	M	M(=)		
Re-attempt	M	M(=)		
Access Type	M	M(=)		
Rand	M	M(=)		
VLR number	C	C(=)		
SGSN number	C	C(=)		
User error			C	C(=)
Provider error				O

8.5.3.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

IMSI

See clause 7.6.2 for the use of this parameter.

Failure Cause

See clause 7.6.7 for use of this parameter.

Re-attempt

See clause 7.6.7 for use of this parameter.

Access Type

See clause 7.6.7 for use of this parameter.

Rand

This parameter identifies the specific AV that failed authentication.

See clause 7.6.7 for use of this parameter.

VLR number

Shall be present if the sender is VLR. See definition in clause 7.6.2.

SGSN number

Shall be present if the sender is SGSN. See definition in clause 7.6.2.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- Unknown Subscriber;
- System Failure;
- Unexpected Data Value.

Provider error

These are defined in clause 7.6.

8.6 Security management services

8.6.1 MAP_SET_CIPHERING_MODE service

8.6.1.1 Definitions

This service is used between the VLR and the MSC to set the ciphering mode and to start ciphering if applicable. It is called when another service requires that information is to be sent on the radio path in encrypted form.

The service is a non-confirmed service and consists of two service primitives.

8.6.1.2 Service primitives

The service primitives are shown in table 8.6/1.

Table 8.6/1: MAP_SET_CIPHERING_MODE parameters

Parameter name	Request	Indication
Invoke id	M	M(=)
Ciphering mode	M	M(=)
Kc	C	C(=)

8.6.1.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

Ciphering mode

See clause 7.6.7 for the use of this parameter.

Kc

The Kc parameter should be included when the ciphering mode parameter indicates that ciphering must be performed.

8.7 International mobile equipment identities management services

8.7.1 MAP_CHECK_IMEI service

8.7.1.1 Definition

This service is used between the VLR and the MSC, between the MSC and the EIR, between the SGSN and EIR, and between IWF and EIR to request check of IMEI. If the IMEI is not available in the MSC or in the SGSN, it is requested from the MS and transferred to the EIR in the service request.

This service may also be used to request the BMUEF from the EIR.

The service is a confirmed service and consists of four service primitives.

8.7.1.2 Service primitives

The service primitives are shown in table 8.7/1.

Table 8.7/1: MAP_CHECK_IMEI parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMEI	C	C(=)	C	C(=)
IMEISV	C	C(=)	C(=)	C(=)
Requested Equipment Info	M	M(=)		
Equipment status			C	C(=)
BMUEF			C	C(=)
User error			C	C(=)
Provider error				O

8.7.1.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

Requested Equipment Info

This parameter indicates whether Equipment Status or BMUEF or both is requested.

IMEI

See clause 7.6.2 for the use of this parameter. The parameter shall not be included in the service request between the VLR and the MSC, but one of IMEI and IMEISV is mandatory in the service request from the MSC to the EIR, from the SGSN to the EIR and from the IWF to the EIR. It is not included in the service response from the EIR to the MSC, the SGSN or the IWF, but one of IMEI and IMEISV is mandatory in the service response from the MSC to the VLR on successful outcome.

IMEISV

See clause 7.6.2 for the use of this parameter. IMEISV shall be present if BMUEF is requested.

Equipment status

See clause 7.6.3 for the use of this parameter. This parameter is sent by the responder in case of successful outcome of the service if Equipment status was requested.

BMUEF

See clause 7.6.4 for the use of this parameter. This parameter is sent by the responder in case of successful outcome of the service if BMUEF was requested.

User error

One of the following error causes defined in clause 7.6.1 shall be sent by the user in case of unsuccessful outcome of the service, depending on the respective failure reason:

- unknown equipment;

this error is returned by the responder when the IMEI is not known in the EIR;

- system failure;
- data missing.

Provider error

See clause 7.6.1 for the use of this parameter.

8.7.2 MAP_OBTAIN_IMEI service

8.7.2.1 Definition

This service is used between the VLR and the MSC to request the IMEI. If the IMEI is not available in the MSC, it is requested from the MS.

The service is a confirmed service and consists of four service primitives.

8.7.2.2 Service primitives

The service primitives are shown in table 8.7/2.

Table 8.7/2: MAP_OBTAIN_IMEI parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMEI			C	C(=)
User error			C	C(=)
Provider error				O

8.7.2.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

IMEI

See clause 7.6.2 for the use of this parameter. The parameter is included in the service response from the MSC to the VLR on successful outcome of the service.

User error

If the service fails, the VLR sends the user error System Failure (see clause 7.6.1) to the MSC.

Provider error

See clause 7.6.1 for the use of this parameter.

8.8 Subscriber management services

8.8.1 MAP-INSERT-SUBSCRIBER-DATA service

8.8.1.1 Definition

This service is used by an HLR to update a VLR with certain subscriber data in the following occasions:

- the operator has changed the subscription of one or more supplementary services, basic services or data of a subscriber. Note that in case of withdrawal of a Basic or Supplementary service this primitive shall not be used;

- the operator has applied, changed or removed Operator Determined Barring;
- the subscriber has changed data concerning one or more supplementary services by using a subscriber procedure;
- the HLR provides the VLR with subscriber parameters at location updating of a subscriber or at restoration. In this case, this service is used to indicate explicitly that a supplementary service is not provisioned, if the supplementary service specification requires it. The only supplementary services which have this requirement are the CLIR and COLR services. Network access mode is provided only in restoration. If the Super-Charger functionality is supported the HLR may not need to provide the VLR with subscriber parameters at location updating of a subscriber. See TS 23.116.

Also this service is used by an HLR to update an SGSN with certain subscriber data in the following occasions:

- if the GPRS subscription has changed;
- if the network access mode is changed;
- the operator has applied, changed or removed Operator Determined Barring;
- the subscriber has changed data concerning one or more supplementary services by using a subscriber procedure;
- the HLR provides the SGSN with subscriber parameters at GPRS location updating of a subscriber. If the Super-Charger functionality is supported the HLR may not need to provide the SGSN with subscriber parameters. See 3GPP TS 23.116.

Also this service is used by a CSS to update an SGSN or a VLR with VPLMN-CSG-Subscription data in the following occasions:

- if the VPLMN-CSG subscription data has changed;
- the CSS provides the VLR or SGSN with VPLMN-CSG subscription data at VCSG location updating of a subscriber.

In an EPS, this service is used by an HSS to update an MME via IWF with certain subscriber data in the following occasions:

- the EPS subscription has changed;
- the operator has applied, changed or removed Operator Determined Barring;
- the HSS provides the MME via IWF(MME) with subscriber parameters at EPS location updating of a subscriber unless an explicit indication to skip subscriber data update has been received.

In an EPS, this service is used by an IWF to indicate to the MME via IWF that the HSS has requested to be notified when the UE has become reachable.

It is a confirmed service and consists of the primitives shown in table 8.8/1.

8.8.1.2 Service primitives

Table 8.8/1: MAP-INSERT-SUBSCRIBER-DATA

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
MSISDN	C	C(=)		
Additional MSISDN	C	C(=)		
Category	C	C(=)		
Subscriber Status	C	C(=)		
Bearer service List	C	C(=)	C	C(=)
Teleservice List	C	C(=)	C	C(=)
Forwarding information List	C	C(=)		
Call barring information List	C	C(=)		
CUG information List	C	C(=)		

Parameter name	Request	Indication	Response	Confirm
SS-Data List	C	C(=)		
eMLPP Subscription Data	C	C(=)		
MC-Subscription Data	C	C(=)		
Operator Determined Barring General data	C	C(=)	C	C(=)
Operator Determined Barring HPLMN data	C	C(=)		
Roaming Restriction Due To Unsupported Feature	C	C(=)		
Regional Subscription Data	C	C(=)		
VLR CAMEL Subscription Info	C	C(=)		
Voice Broadcast Data	C	C(=)		
Voice Group Call Data	C	C(=)		
Network access mode	C	C(=)		
GPRS Subscription Data	C	C(=)		
EPS Subscription Data	C	C(=)		
VPLMN LIPA Allowed	C	C(=)		
Roaming Restricted In SGSN/MME Due To Unsupported Feature	C	C(=)		
North American Equal Access preferred Carrier Id List	U	C(=)		
SGSN CAMEL Subscription Info	C	C(=)		
LSA Information	C	C(=)		
IST Alert Timer	C	C(=)		
SS-Code List			C	C(=)
LMU Identifier	C	C(=)		
LCS Information	C	C(=)		
CS Allocation/Retention priority	C	C(=)		
Super-Charger Supported In HLR	C	C(=)		
Subscribed Charging Characteristics	C	C(=)		
Access Restriction Data	C	C(=)		
ICS Indicator	U	C(=)		
CSG Subscription Data	C	C(=)		
VPLMN CSG Subscription Data	C	C(=)		
UE Reachability Request Indicator	C	C(=)		
SGSN Number	C	C(=)		
MME-Name	C	C(=)		
Subscribed Periodic RAU-TAU Timer	C	C(=)		
Subscribed Periodic LAU Timer	C	C(=)		
MDT User Consent	C	C(=)		
PS and SMS-Only Service Provision	C	C(=)		
SMS in SGSN Allowed	C	C(=)		
CS-to-PS-SRVCC-Allowed-Indicator	C	C(=)		
P-CSCF Restoration Request	C	C(=)		
Adjacent Access Restriction Data	C	C(=)		
IMSI-Group-Id List	C	C(=)		
UE Usage Type	C	C(=)		
User Plane Integrity Protection Indicator	C	C(=)		
DL-Buffering Suggested Packet Count	C	C(=)		
Reset-IDs	C	C(=)		
eDRX-Cycle-Length List	C	C(=)		
IAB-Operation-Allowed-Indicator	C	C(==)		
Regional Subscription Response			C	C(=)
Supported CAMEL Phases			C	C (=)
Offered CAMEL 4 CSIs			C	C (=)
Supported Features			U	C (=)
User error			U	C(=)
Provider error				O

8.8.1.3 Parameter use

All parameters are described in clause 7.6. The following clarifications are applicable:

Network access mode

This parameter defines if the subscriber has access to MSC/VLR and/or to SGSN/MME. This parameter is used by SGSN/MME and MSC/VLR. In VLR, the parameter is used only as part of Restore Data Procedure and the parameter is not stored in the VLR. This parameter shall always be sent to the SGSN and viaIWF to the MME as part of the GPRS subscriber data at GPRS/MME location updating. It shall be sent to the SGSN and via IWF to the MME if it is changed as a result of administrative action.

This parameter shall not be used by the CSS.

IMSI

It is only included if the service is not used in an ongoing transaction (e.g. location updating). This parameter is used by the VLR and the SGSN and IWF.

MSISDN

For subscriptions with MSISDN:
It is included at location updating and when it is changed. The MSISDN sent shall be the basic MSISDN. This parameter is used by the VLR and the SGSN and IWF.

For a subscription without MSISDN:
The HLR shall not populate this parameter if the VLR or SGSN explicitly indicated support of MSISDN-less operation.

NOTE 1: See subclauses 8.1.2.3 and 8.1.7.3 for the case where the VLR or SGSN does not support MSISDN-less operation.

Additional MSISDN

If subscribed, the Additional MSISDN is included at location updating and when it is changed. This parameter is used by the SGSN and IWF. This parameter shall be ignored by the VLR if received.

If the SGSN does not indicate support of the feature the HSS shall not send the parameter.

Category

It is included either at location updating or when it is changed. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

Subscriber Status

It is included either at location updating or when it is changed.

To apply, remove or update Operator Determined Barring Categories the Subscriber Status is set to Operator Determined Barring. In this case ODB General Data shall also be present. If the Operator Determined Barring applies and the subscriber is registered in the HPLMN and HPLMN specific Operator Determined Barring applies then ODB HPLMN Specific Data shall also be present.

To remove all Operator Determined Barring Categories the Subscriber Status shall be set to "Service Granted". This parameter is used by the VLR and the SGSN and IWF.

This parameter shall not be used by the CSS.

Bearer service List

A list of Extensible Bearer service parameters (Extensible Bearer service is defined in clause 7.6). An Extensible Bearer service parameter must be the code for an individual Bearer service, except in the cases described below.

The codes for the Bearer service groups "allAlternateSpeech-DataCDA" and "allAlternateSpeech-DataCDS" shall, if applicable, be sent from the HLR to the VLR as a pair. The codes for the Bearer service groups "allSpeechFollowedByDataCDA" and "allSpeechFollowedByDataCDS" shall, if applicable, be sent from the HLR to the VLR as a pair.

If it is included in the Request/Indication, it includes either all Extensible Bearer services subscribed (at location updating or at restoration) or only the ones added (at subscriber data modification).

If the VLR receives an Indication containing any Extensible Bearer service parameters which it does not support/allocate it returns them in the response to the HLR and discards the unsupported Extensible Bearer services (no error is sent back), except in the cases described below.

If the VLR receives the codes for the Bearer service groups "allSpeechFollowedByDataCDA" and "allSpeechFollowedByDataCDS" and supports one or more of the circuit-switched synchronous or asynchronous data rates specified for simple data bearer services, it shall accept the bearer service codes, and not return them in the response to the HLR. If the VLR does not support any of the circuit-switched synchronous or asynchronous data rates specified for simple data bearer services, and receives the pair of codes for "allAlternateSpeech-DataCDA" and "allAlternateSpeech-DataCDS" or the pair of codes for "allSpeechFollowedByDataCDA" and "allSpeechFollowedByDataCDS", it shall reject the pair of codes by returning them in the response to the HLR. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Teleservice List

A list of Extensible Teleservice parameters (Extensible Teleservice is defined in clause 7.6). An Extensible Teleservice parameter must be the code for an individual Teleservice.

If it is included in the Request/Indication, it contains either all Extensible Teleservices subscribed (at location updating or at restoration) or the ones added (at subscriber data modification). Only the Extensible Teleservices that are relevant to the node at which the message is received should be included in the Teleservice List.

If the VLR or the SGSN or the IWF receives an Indication containing any Extensible Teleservice parameters which it does not support/allocate it returns them in the response to the HLR and discards the unsupported Extensible Teleservices (no error is sent back). This parameter is used by the VLR and the SGSN and the IWF.

This parameter shall not be used by the CSS.

Forwarding information List

A list of Extensible Forwarding information parameters (Extensible Forwarding information is defined in clause 7.6). It includes Call Forwarding services either at location updating or at restoration or when they are changed. Each Extensible Forwarding information parameter shall be treated independently of all other parameters in the primitive.

The Extensible Forwarding information shall include the SS-Code for an individual call forwarding supplementary service. The Extensible Forwarding information shall contain one or more Extensible Forwarding Features (Extensible Forwarding Feature is defined in clause 7.6).

The Extensible Forwarding Feature may include an Extensible Basic Service Group. This shall be interpreted according to the rules in clause 8.8.1.4.

The Extensible Forwarding Feature shall contain an Extensible SS-Status parameter.

If the Extensible SS-Status indicates that call forwarding is registered then (except for call forwarding unconditional) the Extensible Forwarding Feature shall contain a number to define the forwarded-to destination and, if available, the forwarded-to subaddress. In other states the forwarded-to number and, if applicable, the forwarded-to subaddress shall not be included. For call forwarding unconditional the forwarded-to number and, if applicable, the forwarded-to subaddress shall not be included. If the VLR does not receive a forwarded-to subaddress then it shall assume that a forwarded-to subaddress has not been registered.

The Extensible Forwarding Feature shall contain the extensible forwarding options (except for call forwarding unconditional where the extensible forwarding options shall not be included). Bits 3 and 4 of the extensible forwarding options shall be ignored by the VLR, and may be set to any value by the HLR.

For call forwarding on no reply: If the extensible SS-Status indicates that call forwarding is registered then the Extensible Forwarding Feature shall contain an extensible no reply condition timer. In other states the no reply condition timer shall not be included.

For call forwarding services other than call forwarding on no reply: The Extensible Forwarding Feature shall not contain a no reply condition timer.

If the VLR receives an Indication containing any Call Forwarding service codes which it does not support/allocate it returns them to the HLR in the parameter SS-Code List and discards the unsupported Call Forwarding service codes (no error is sent back). This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Call barring information List

A list of Extensible Call barring information parameters (Extensible Call barring information is defined in clause 7.6). It includes Call Barring services either at location updating or at restoration or when they are changed. Each Extensible Call barring information parameter shall be treated independently of all other parameters in the primitive.

The Extensible Call barring information shall include the SS-Code for an individual call barring supplementary service. The Extensible Call barring information shall contain one or more Extensible Call Barring Features (Extensible Call Barring Feature is defined in clause 7.6).

The Extensible Call Barring Feature may include an Extensible Basic Service Group. This shall be interpreted according to the rules in clause 8.8.1.4.

The Extensible Call Barring Feature shall contain an extensible SS-Status parameter.

If the VLR or the SGSN or the IWF receives an Indication containing any Extensible Call Barring service codes which it does not support/allocate it returns them to the HLR in the parameter SS-Code List and discards the unsupported Extensible Call Barring service codes (no error is sent back).

This parameter shall not be used by the CSS.

CUG information List

A list of CUG information list parameters (CUG information is defined in clause 7.6). It includes CUG information either at location updating or at restoration or when it is changed.

At location updating, restoration or when there is a change in CUG data, the HLR shall include the complete CUG-SubscriptionList and, if there are options per basic group, it shall also include the complete CUG-FeatureList. If there are not options per extensible basic service group the CUG-FeatureList shall not be included.

In any dialogue, the first insertSubscriberData message which contains CUG information shall include a non-empty CUG-SubscriptionList.

When the VLR receives CUG data it shall replace the stored CUG data with the received data set.

If CUG-FeatureList is omitted in the Insert Subscriber Data operation VLR shall interpret that no options per extensible basic service group exist, and then it shall apply the default values i.e. no outgoing access, no incoming access, no preferential CUG exists.

If CUG-Feature is received without preferential CUG, the VLR shall interpret that no preferential CUG applies.

If the VLR detects that there is overlapping in the information received within a dialogue, it shall send the error Unexpected Data Value.

Note that data consistency between CUG subscription data and CUG feature data is the responsibility of the HLR.

If the VLR does not support the CUG service it returns its code to the HLR in the parameter SS-Code List and discards the received information (no error is sent back). This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

SS-Data List

A list of Extensible SS-Data parameters (Extensible SS-Data is defined in clause 7.6). It is sent for any other supplementary service than Call Forwarding, Call Barring, CUG and eMLPP either at location updating or at restoration or when they are changed. Each SS-Data parameter shall be treated independently of all other parameters in the primitive.

The Extensible SS-Data shall include the SS-Code for an individual supplementary service.

The Extensible SS-Data shall contain an Extensible SS-Status parameter and any subscription options that are applicable to the service defined by the SS-Code.

The SS-Data may include a Basic Service Group List. This shall be interpreted according to the rules in clause 8.8.1.4.

If the VLR receives an Indication containing any supplementary service codes which it does not support/allocate it returns them to the HLR in the parameter SS-Code List and therefore discards the unsupported service codes received (no error is sent back).

This parameter is used by the SGSN only for LCS. If the SGSN receives an Indication containing any LCS related supplementary service codes which it does not support/allocate it returns them to the HLR in the parameter SS-Code List and therefore discards the unsupported service codes received (no error is sent back). SS-codes not related to the supported LCS capability set shall be discarded.

If the IWF receives an Indication containing any LCS related supplementary service codes, it returns them to the HSS in the parameter SS-Code List and therefore discards the service codes received (no error is sent back). SS-codes not related to the supported LCS capability set shall be discarded.

This parameter shall not be used by the CSS.

Operator Determined Barring General data

If it is included in a Request/Indication, it includes all the Operator Determined Barring categories that may be applied to a subscriber registered in any PLMN. This parameter is only included in a Request/Indication when the parameter Subscriber Status is set to the value Operator Determined Barring. Note that all General Operator Determined Barring Categories shall be set to their actual status.

If the VLR or the SGSN or IWF receives an Indication containing Operator Determined Barring General Data which shows that the subscriber is subject to barring not supported / not allocated by the VLR or by the SGSN, it returns Operator Determined Barring General Data in the response to the HLR to show the barring categories which are not supported / not allocated by the VLR or by the SGSN. This parameter is used by the VLR and the SGSN and IWF.

This parameter shall not be used by the CSS.

Operator Determined Barring HPLMN data

It includes all the Operator Determined Barring categories that may be applied only to a subscriber registered in the HPLMN. Therefore, it shall only be transferred to the VLR or to the SGSN or IWF when the subscriber is roaming into the HPLMN and when the parameter Subscriber Status is set to the value Operator Determined Barring. Note that all HPLMN Operator Determined Barring Categories shall be set to their actual status.

If Subscriber Status is set to the value Operator Determined Barring and no Operator Determined Barring HPLMN data is present then the VLR or the SGSN or IWF shall not apply any HPLMN specific ODB services to the subscriber. This parameter is used by the VLR and the SGSN and IWF.

This parameter shall not be used by the CSS.

eMLPP Subscription Data

If included in the Insert Subscriber Data request this parameter defines the priorities the subscriber might apply for a call (as defined in clause 7.6). It contains both subparameters of eMLPP.

If the VLR does not support the eMLPP service it returns its code to the HLR in the parameter SS-Code List and therefore discards the received information (no error is sent back).

eMLPP subscription data that have been stored previously in a subscriber data record in the VLR are completely replaced by the new eMLPP subscription data received in a MAP_INSERT_SUBSCRIBER_DATA during either an

Update Location or Restore Data procedure or a stand alone Insert Subscriber data procedure. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

MC Subscription Data

If included in the Insert Subscriber Data request, this parameter provides the MC Subscription Data as defined in clause 7.6.

If the VLR does not support the MC service, it returns its code to the HLR in the parameter SS-Code List and therefore discards the received information (no error is sent back).

MC subscription data that have been stored previously in a subscriber data record in the VLR are completely replaced by the new MC subscription data received in a MAP_INSERT_SUBSCRIBER_DATA during either an Update Location or Restore Data procedure or a stand alone Insert Subscriber data procedure. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Roaming Restriction Due To Unsupported Feature

The HLR may decide to include this parameter in the request if certain services or features are indicated as not supported by the MSC/VLR (e.g. Advice of Charge Charging Level).

If this parameter is sent to the VLR the MSC area is restricted by the HLR and the VLR. This parameter is used only by the VLR and if the SGSN or the IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Regional Subscription Data

If included in the Insert Subscriber Data request this parameter defines the subscriber's subscription area for the addressed VLR, for the addressed SGSN or for the addressed MME (as defined in clause 7.6). It contains the complete list of up to 10 Zone Codes that apply to a subscriber in the currently visited PLMN. The HLR shall send only those Zone Codes which are stored against the CC and NDC of the VLR, the SGSN or the MME to be updated.

NOTE 2: Support of this parameter is a network operator option and it will not be sent to networks which do not support Regional Subscription.

Regional subscription data that have been stored previously in a subscriber data record in the VLR, in the SGSN or in the MME are completely replaced by the regional subscription data received in an Insert Subscriber Data indication during either an Update Location or Restore Data procedure or a stand alone Insert Subscriber data procedure.

After the regional subscription data are inserted the VLR or the SGSN shall derive whether its location areas are allowed or not. If the whole MSC or SGSN area is restricted it will be reported to HLR by returning the Regional Subscription Response.

The VLR or the SGSN returns a Regional Subscription Response indicating that a problem with the Zone Code has been detected in one of the following cases:

- Too Many Zone Codes: more than 10 Zone Codes are to be stored in the VLR or in the SGSN.
- Regional Subscription Not Supported by the VLR or the SGSN.
- Zone Codes Conflict: the VLR or the SGSN detects that the zone codes indicate conflicting service permission for a location area.

Zone codes which have no mapping to location areas shall be ignored.

If a sequence of MAP_INSERT_SUBSCRIBER_DATA services is used during a dialogue, Regional Subscription Data shall be accepted only in one service. Regional Subscription Data received in a subsequent service shall be rejected with the error Unexpected Data Value.

If Regional Subscription Data are not included in any MAP_INSERT_SUBSCRIBER_DATA service, there is no restriction of roaming due to Regional Subscription. This parameter is used by the VLR, the SGSN and the IWF.

This parameter shall not be used by the CSS.

Voice Broadcast Data

This parameter contains a list of group id's a user might have subscribed to; (VBS-Data is defined in clause 7.6). It includes VBS information either at location updating or at restoration or when it is changed.

At location updating, restoration or when there is a change in VBS data, the HLR shall include the complete VBS-Data.

When the VLR receives VBS-Data within a dialogue it shall replace the stored VBS-data with the received data set. All subsequent VBS-data received within this dialogue shall be interpreted as add-on data.

If VBS-data is omitted in the Insert Subscriber Data operation the VLR shall keep the previously stored VBS data.

If the VLR detects that there is overlapping in the information received within a dialogue, it shall send the error Unexpected Data Value. This parameter is used only by the VLR and if the SGSN or the IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Voice Group Call Data

This parameter contains a list of group id's a user might have subscribed to; see clause 7.6.

At location updating, restoration or when there is a change in VGCS data, the HLR shall include the complete VGCS-Data.

When the VLR receives VGCS-Data within a dialogue it shall replace the stored VGCS-Data with the received data set. All VGCS-Data received within this dialogue shall be interpreted as add-on data.

If VBCS-Data is omitted in the Insert Subscriber Data operation the VLR shall keep the previously stored VGCS-Data.

If the VLR detects that there is overlapping in the information received within a dialogue, it shall send the error Unexpected Data Value. This parameter is used only by the VLR and if the SGSN or the IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

North American Equal Access preferred Carrier Id List

A list of the preferred carrier identity codes that are subscribed to.

When the VLR receives this parameter from the HLR, it shall replace the previously stored preferred carrier identity codes with the received ones. It is not possible to delete all the preferred carrier identity codes from the VLR using this service. To delete all the preferred carrier identity codes from the VLR, the HLR shall use the MAP_CANCEL_LOCATION service.

This parameter shall not be used by the CSS.

LSA Information

If included in the ISD request, this parameter contains a list of localised service area identities a user might have subscribed to together with the priority, the preferential access indicator, the active mode support indicator and active mode indication of each localised service area; see clause 7.6. The access right outside these localised service areas is also indicated. In all cases mentioned below, the LSA information shall only include LSA Data applicable to the VPLMN where the Subscriber is located. The VLR number, received in the MAP-UPDATE_LOCATION primitive, or the SGSN number, received in the MAP_UPDATE_GPRS_LOCATION primitive, can be used, alongside data stored in the HLR, to determine the LSA Data applicable to the VPLMN.

At restoration, location updating or GPRS location updating the HLR shall include the complete set of applicable LSA Information.

When there is a change in LSA data the HLR shall include at least the new and/or modified LSA data.

When there is a change in the access right outside the localised service areas the HLR shall include the LSA only access indicator.

When the SGSN or the VLR receives LSA information within a dialogue it shall check if the received data has to be considered as the entire LSA information. If so, it shall replace the stored LSA information with the received data set, otherwise it shall replace the data only for the modified LSA data (if any) and/or access right, and add the new LSA data (if any) to the stored LSA Information.

If the entire LSA information is received, it shall always include the LSA only access indicator value together with the LSA data applicable for the PLMN (if any).

If LSA Information is omitted in the Insert Subscriber Data operation the SGSN or the VLR shall keep the previously stored LSA Information.

If the SGSN or the VLR detects that there is overlapping in the information received within a dialogue, it shall send the error Unexpected Data Value. This parameter is used by the VLR and the SGSN, and if the IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

IST Alert Timer

This parameter contains the IST Alert timer value that must be used to inform the HLR about the call activities that the subscriber performs.

At Location Updating, restoration, or when there is a change in the IST data defined for the Subscriber, the HLR shall include the IST Alert timer.

This parameter shall not be used by the CSS.

LMU Identifier

This parameter indicates the presence of an LMU. This parameter is used only by the VLR and shall be ignored if received by an SGSN or an IWF.

This parameter shall not be used by the CSS.

LCS Information

This parameter provides the following LCS related information for an MS subscriber:

- list of GMLCs in the HPLMN;
- privacy exception list;
- MO-LR list.

At restoration and location updating, the HLR shall include the complete LCS data of the subscriber.

When there is a change in LCS subscriber data the HLR shall include at least the new and/or modified LCS data. LCS data that is not modified need not be included.

The VLR/SGSN shall keep any previously stored LCS Information that is not included in an Insert Subscriber Data operation.

If the VLR/SGSN detects that there is overlapping in the LCS information received within a dialogue, it shall send the error Unexpected Data Value. However, if the VLR receives the LCS code in both the LCS Information and the SS-Data List, then the VLR shall not interpret this as overlapping data. This parameter is used by the VLR and the SGSN and the IWF.

This parameter shall not be used by the CSS.

Super-Charger Supported In HLR

This parameter is used by the HLR to indicate support for the Super-Charger functionality. If this parameter is present it shall include an indication of the age of the subscription data stored in the HLR.

If this parameter is absent then the HLR does not support the Super-Charger functionality.

This parameter shall not be used by the CSS.

SS-Code List

The list of SS-Code parameters for the services that are provided to a subscriber but are not supported/allocated by the VLR/SGSN/IWF (SS-Code is defined in clause 7.6). The list can only include individual SS-Codes that were sent in the service request. For the VLR, this list can also include SS-Codes for the eMLPP and/or CUG services if the above mentioned conditions, as described in eMLPP Subscription Data and/or CUG information List, are met (that is, eMLPP Subscription Data and/or CUG information List are received).

This parameter shall not be used by the CSS.

ICS-Indicator

This optional flag indicates to the MSC Server enhanced for ICS (see 3GPP TS 23.292 [135]) whether the MSC Server shall attempt the IMS registration.

This parameter is used by the VLR and the SGSN.

This parameter shall not be used by the CSS.

CSG-Subscription Data

This parameter contains a list of CSG-Ids, the associated expiration dates (see 3GPP TS 22.011 [138]) and a list of corresponding APNs (see 3GPP TS 29.272 [144]. When the VLR or SGSN or MME receives CSG-Subscription Data from the HLR/HSS it shall replace the stored CSG-Subscription Data from the HLR/HSS (if any) with the received data. This parameter is used by the VLR and the SGSN and IWF, except the list of corresponding APNs is not applicable to the VLR, and the VLR shall ignore this list if it is received.

This parameter shall not be used by the CSS.

VPLMN CSG Subscription Data

This parameter contains a list of CSG-Ids, the associated expiration dates (see 3GPP TS 22.011 [138]). When the VLR or SGSN or MME receives VPLMN CSG Subscription Data from the CSS, it shall replace the stored VPLMN-CSG Subscription Data from the CSS (if any) with the received VPLMN CSG Subscription data. This parameter is used by the VLR, the SGSN and MME.

This parameter is not applicable for the HLR/HSS, and the VLR or SGSN or IWF shall ignore this parameter if it is received from the HLR/HSS.

CSG Subscription Data from the HLR/HSS and VPLMN CSG Subscription Data from the CSS are managed independently in the VLR or SGSN or MME. If the same CSG Id exists in both CSG subscription data from the CSS and CSG subscription data from the HLR/HSS, the CSG subscription data from the HLR/HSS shall take precedence over the CSG subscription data from the CSS in further use.

UE Reachability Request Indicator

This parameter indicates by its presence that the HSS is awaiting a Notification of UE Reachability. This parameter is used by the IWF only.

This parameter shall not be used by the CSS.

MME Name

This parameter contains the MME identity used over the SGs interface (see 3GPP TS 23.003 [17] subclause 19.4.2.4) when stored in the HSS. Otherwise this parameter contains the Diameter Identity of the MME (see 3GPP TS 23.003 [17]). If the subscriber is registered to EPS and the length of the MME Name does not exceed 55 octets, the HLR shall send the MME Name to the VLR during the data restoration procedure if the 'Restoration Indicator' is set in the MAP_RESTORE_DATA request, and during an Update Location procedure if the 'Restoration Indicator' is set in the MAP_UPDATE_LOCATION request. This parameter may be used by the MSC/VLR, e.g. to page the UE via SGs.

This parameter shall not be used by the CSS.

Subscribed Periodic RAU-TAU Timer

This parameter contains the subscribed periodic RAU/TAU timer (see 3GPP TS 23.401 [145] and 3GPP TS 23.060 [104]) and is used by the SGSN and MME (via IWF). The SGSN and MME shall handle the Subscribed Periodic RAU-TAU Timer as specified in subclause 5.2.1.1.2 of 3GPP TS 29.272 [144].

If the VLR receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Subscribed Periodic LAU Timer

This parameter contains the subscribed periodic LAU timer value (see 3GPP TS 23.012 [23]) and is used by the MSC/VLR. The MSC/VLR shall handle the Subscribed Periodic LAU Timer as specified in subclause 3.7.3 of 3GPP TS 23.012 [23].

If the SGSN receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

SGSN Number

This parameter contains the Identity of the SGSN (see 3GPP TS 23.003 [17]). If the subscriber is registered to GPRS, the HLR shall send the SGSN Number if available to the VLR during the data restoration procedure if the 'Restoration Indicator' is set in the MAP_RESTORE_DATA request, and during an Update Location procedure if the 'Restoration Indicator' is set in the MAP_UPDATE_LOCATION request. This parameter may be used by the MSC/VLR, e.g. to page the UE via Gs.

This parameter shall not be used by the CSS.

MDT User Consent

This parameter indicates the user consent availability for MDT activation, see 3GPP TS 32.422 [132]. This parameter is used by the VLR, the SGSN and the IWF.

This parameter shall not be used by the CSS.

PS and SMS-only Service Provision

This parameter indicates whether the subscription is for PS Only and permits CS service access only for SMS.

SMS in SGSN Allowed

This parameter indicates whether the HSS allows SMS to be provided by SGSN over NAS.

User Plane Integrity Protection Indicator

This parameter indicates by its presence that the SGSN may decide to activate integrity protection of the user plane when GERAN is used (see 3GPP TS 43.020 [24]).

If the VLR receives this parameter it shall ignore it.

DL-Buffering Suggested Packet Count

This parameter indicates a suggested DL-Buffering Packet Count. The MME (via IWF) and SGSN may take it into account in addition to local policies, to determine whether to invoke extended buffering of downlink packets at the SGW for High Latency Communication. Otherwise, the MME or SGSN shall make this determination based on local policies only (see 3GPP TS 29.272 [144]).

If the VLR receives this parameter it shall ignore it.

Reset-IDs

This parameter contains a list of subscribed Reset-IDs.

eDRX-Cycle-Length List

This list shall contain the subscribed eDRX cycle length, along with the RAT type to which it is applicable.

IAB-Operation-Allowed-Indicator

This parameter indicates by its presence that IAB operation is authorized for the UE. See 3GPP TS 401 [145].

Regional Subscription Response

If included in the response this parameter indicates one of:

- Network Node Area Restricted entirely because of regional subscription;
- Too Many Zone Codes to be inserted;
- Zone Codes Conflict;
- Regional Subscription not Supported by the VLR or by the SGSN or MME.

If the VLR determines after insertion of Regional Subscription Data that the entire MSC area is restricted, the VLR shall respond with a Regional Subscription Response indicating MSC Area Restricted. Otherwise MSC Area Restricted is not sent. The HLR shall check whether the current MSC area is no longer restricted.

If the SGSN determines after insertion of Regional Subscription Data that the entire SGSN area is restricted, the SGSN shall respond with a Regional Subscription Response indicating SGSN Area Restricted. Otherwise SGSN Area Restricted is not sent. The HLR shall check whether the current SGSN area is no longer restricted. This parameter is used by the VLR, the SGSN and the IWF.

This parameter shall not be used by the CSS.

VLR CAMEL Subscription Info

This parameter is sent for subscribers who have CAMEL services which are invoked in the MSC.

- In CAMEL phase 1, this parameter contains only the O-CSI.
- In CAMEL Phase 2, this parameter may contain O-CSI, SS-CSI and TIF-CSI. In CAMEL Phase 2 and onwards, TDP-Criteria for O-CSI may be associated with O-CSI.
- In CAMEL Phase 3, this parameter may contain O-CSI, D-CSI, SS-CSI, VT-CSI, MO-SMS-CSI, M-CSI and TIF-CSI. In CAMEL Phase 3 and onwards, TDP-Criteria for VT-CSI may be associated with VT-CSI.
- In CAMEL Phase 4, this parameter may contain O-CSI, D-CSI, SS-CSI, VT-CSI, MO-SMS-CSI, MT-SMS-CSI, M-CSI and TIF-CSI. In CAMEL Phase 4, TDP-Criteria for MT-SMS-CSI may be associated with MT-SMS-CSI.

The VLR CAMEL Subscription Info is sent at location updating or when any information in the applicable CAMEL Subscription Info in the HLR has been changed.

At location updating, the complete set of VLR CAMEL Subscription Info is sent in one dialogue.

When CAMEL Subscription Information is changed in the HLR and changed data have to be sent to the VLR, then:

- for CAMEL Phase 1 and CAMEL Phase 2, the complete set of VLR CAMEL Subscription Info is sent in one dialogue;
- for CAMEL Phase 3 or higher, one or more specific elements of VLR CAMEL Subscription Info are sent in one dialogue.

When the VLR receives a specific element of VLR CAMEL Subscription Info, it shall overwrite the corresponding specific element of VLR CAMEL Subscription Info (if any) which it has stored for that subscriber.

For CAMEL Phase 1 and CAMEL Phase 2 , the VLR CAMEL Subscription Info consists of any one or more of:

- O-CSI (irrespective of the value of the "CAMEL Capability Handling" inside O-CSI),TDP-Criteria for O-CSI,SS-CSI and TIF-CSI.

  (The complete set of above shall be sent even if only one CSI has changed in case of stand alone ISD. The omitted elements of above list will be withdrawn in the VLR.)

From CAMEL phase 3 onwards, the specific elements of VLR CAMEL Subscription Info which may be sent are:

O-CSI (irrespective of the value of the "CAMEL Capability Handling" inside O-CSI), TDP criteria for O-CSI, SS-CSI and TIF-CSI;

(The complete set of above shall be sent even if only one CSI has changed in case of stand alone ISD. The omitted elements of above list will be withdrawn in the VLR.)

- D-CSI;
- VT-CSI;
- TDP-Criteria for VT-CSI;
- MO-SMS-CSI;
- MT-SMS-CSI;
- TDP-Criteria for MT-SMS-CSI;
- M-CSI.

If the VLR CAMEL Subscription Info is omitted in the Insert Subscriber Data operation the VLR shall keep the previously stored VLR CAMEL Subscription Info. Within one dialogue subsequent received data are interpreted as add-on data. If the VLR detects that there is overlapping in the information received within a dialogue, it shall send the error Unexpected Data Value. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

The VLR CAMEL Subscription Info may contain the TIF-CSI (Translation Information Flag) for CAMEL Phase 2 and higher. See 3GPP TS 23.072 for the use of this parameter and the conditions for its presence.

This parameter shall not be used by the CSS.

Supported CAMEL Phases

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078. This parameter is used by the VLR and SGSN.

A VLR or SGSN not supporting any CAMEL Phase may omit this parameter. An IWF shall omit this parameter.

This parameter shall not be used by the CSS.

GPRS Subscription Data

This parameter contains a list of PDP-contexts a user has subscribed to; see clause 7.6.

At GPRS location updating the HLR shall include the complete GPRS Subscription Data.

When there is a change in GPRS subscriber data the HLR shall include only the new and/or modified PDP contexts.

When the SGSN receives GPRS Subscription Data within a dialogue it shall check if the received data has to be considered as the entire GPRS subscription data. If so, it shall replace the stored GPRS Subscription Data with the received data set, otherwise it shall replace the data only for the modified PDP contexts (if any) and add the new PDP contexts (if any) to the stored GPRS Subscription Data.

If GPRS Subscription Data is omitted in the Insert Subscriber Data operation the SGSN shall keep the previously stored GPRS Subscription Data.

If the SGSN detects that there is overlapping in the information received within a dialogue, it shall send the error Unexpected Data Value. This parameter is used only by the SGSN and if the VLR receives this parameter it shall ignore it.

The SGSN shall handle the SIPTO-Permission information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the SIPTO-Local-Network-Permission information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the LIPA Permission information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the Restoration-Priority information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

This parameter shall not be used by the CSS.

EPS Subscription Data

This parameter contains:

- the UE level APN-OI Replacement (see 3GPP TS 23.401 [145]), and
- the Subscriber Profile ID for RAT/Frequency Priority (RFSP-ID) (see 3GPP TS 23.401 [145], 3GPP TS 36.413 [147] and 3GPP TS 23.060 [104]), and
- the AMBR (see 3GPP TS 23.401 [145] and 3GPP TS 23.060 [104]), and
- a list of APN Configurations,
- a session transfer number for SRVCC (STN-SR) (see 3GPP TS 23.003 [17]).
- MPS CS Priority, which by its presence indicates the UE is subscribed to the eMLPP in the CS domain.
- MPS EPS Priority, which by its presence indicates the UE is subscribed to the MPS in the EPS domain.
- Subscribed vSRVCC (see 3GPP 29.272 [144]).

This parameter is used only by the MME via IWF and SGSN. If the VLR receives this parameter it shall ignore it.

The MPS CS Priority and MPS EPS Priority inside the parameter are used only by the MME via IWF. If the SGSN receives them it shall ignore them.

The SGSN shall handle the SIPTO-Permission information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the SIPTO-Local-Network-Permission information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the LIPA Permission information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the Restoration-Priority information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

The SGSN shall handle the WLAN-offloadability information as specified in 3GPP TS 29.272 [144] subclause 5.2.1.1.2.

This parameter shall not be used by the CSS.

VPLMN LIPA Allowed

This parameter by its presence indicates that the UE is allowed to use LIPA in the PLMN where the UE is attached (see 3GPP TS 23.401 [145] and 3GPP TS 23.060 [104]).

This parameter is used only by the IWF and SGSN. If the VLR receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

SGSN CAMEL Subscription Info

The SGSN CAMEL Subscription Info is sent at GPRS location updating or when any information in the applicable SGSN CAMEL Subscription Info in the HLR has been changed.

- In CAMEL Phase 3, this parameter may contain one or both of GPRS-CSI and MO-SMS-CSI.
- In CAMEL Phase 4, this parameter may contain GPRS-CSI, MO-SMS-CSI and MT-SMS-CSI and TDP-Criteria for MT-SMS-CSI.

At GPRS location updating the complete set of SGSN CAMEL Subscription Info is sent.

When CAMEL Subscription Information is changed in the HLR and changed data have to be sent to the SGSN, then one or more specific elements of SGSN CAMEL Subscription Info are sent in one dialogue.

When the SGSN receives a specific element of SGSN CAMEL Subscription Info, it shall overwrite the corresponding specific element of SGSN CAMEL Subscription Info (if any) which it has stored for that subscriber.

The specific elements of SGSN CAMEL Subscription Info which may be sent are:

- MO-SMS-CSI;
- MT-SMS-CSI;
- TDP-Criteria for MT-SMS-CSI;
- GPRS-CSI;
- MC-CSI.

This parameter is used only by the SGSN and if the VLR or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Roaming Restricted In SGSN/MME Due To Unsupported Feature

The HSS/HLR may decide to include this parameter in the request if certain services or features are indicated as not supported by the SGSN/IWF. This parameter is used only by the SGSN and IWFand if the VLR receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

CS Allocation/Retention priority

The CS Allocation/Retention priority is used only for Circuit Switched (CS). This parameter specifies relative importance to compare with other bearers about allocation and retention of bearer. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Offered CAMEL 4 CSIs

This parameter indicates the CAMEL phase 4 CSIs offered in the VMSC/VLR or SGSN (see clause 7.6.3.36D). An IWF shall omit this parameter.

This parameter shall not be used by the CSS.

Subscribed Charging Characteristics

This parameter refers to the Subscribed Charging Characteristics as defined in 3GPP TS 32.251.

For a detailed description of the use of the parameter, see 3GPP TS 32.251.

This parameter is used only by the SGSN and IWF and if the VLR receives this parameter it shall ignore it.

This parameter shall not be used by the CSS.

Access Restriction Data

This parameter indicates the allowed RAT for the PLMN where the UE is attached according to subscription data. (see clause 7.6.3.97)

If the VLR/SGSN/MME supports the Access Restriction feature but does not receive the Access Restriction Data parameter from the HSS/HLR at location updating or restoration, the VLR/SGSN/MME shall assume that the subscriber's profile does not have any restrictions enabled.

For a detailed description of the use of the parameter, see 3GPP TS 23.012 [23] for the CS domain and 3GPP TS 23.060[104], 3GPP TS 29.060 [105] subclause 7.5.3 and 3GPP TS 29.274 [149] subclause 7.3.6 for the PS domain.

This parameter shall not be used by the CSS.

Supported Features

This parameter shall be used by an IWF to forward feature support indications as received from the MME or SGSN via S6a/S6d.

This parameter shall not be used by the CSS.

CS-to-PS-SRVCC-Allowed-Indicator

This parameter indicates by its presence to the MSC Server enhanced for ICS (see 3GPP TS 23.292 [135]) that CS to PS SRVCC is subscribed.

This parameter is used by the VLR.

P-CSCF Restoration Request

This parameter indicates by its presence that the HSS requests to the SGSN or the MME (via the IWF) the execution of the HSS-based P-CSCF restoration procedures, as described in 3GPP TS 23.380 [150], clause 5.4.

This parameter shall not be used by the CSS.

Adjacent Access Restriction Data

This parameter indicates the allowed RAT in each one of the indicated PLMN IDs, according to subscription data.

This parameter shall not be used by the CSS.

IMSI-Group-Id List

A list of IMSI-Group-Id parameters each of which identifies an IMSI-Group the subscriber belongs to.

UE Usage Type

This parameter indicates the usage characteristics of the UE that enables the selection of a specific Dedicated Core Network . It shall not be sent to VLRs and shall not be sent to SGSNs that did not indicate support of the Dedicated Core Network functionality within GPRS-Location Update. When the Insert-Subscriber-Data operation is used within an Update-GPRS-Location Dialogue, the HLR shall include this parameter if the SGSN indicated support of the Dedicated Core Network functionality and a UE Usage Type is available in the subscription data of the user. Outside the Update-Gprs-Location Dialogue the HLR shall include this parameter towards the SGSN that supports the Dedicated Core Network functionality if the value changed.

User error

Only one of the following values is applicable:

- Unidentified subscriber;
- Data missing;
- Unexpected data value.

8.8.1.4 Basic service information related to supplementary services

A number of parameters that relate to supplementary services can be qualified by a Basic Service Group (or a Basic Service Group List). This clause explains how this information is to be interpreted. Supplementary service parameters to which this clause is applicable only apply to the basic service groups described in this clause, and only those basic service groups shall be overwritten at the VLR or the SGSN.

The Basic Service Group (or Basic Service Group List) is optional.

If present the Basic Service Group (or each element of the Basic Service Group List) shall be one of:

- an Elementary Basic Service Group for which the supplementary service is applicable to at least one basic service in the group and for which the subscriber has a subscription to at least one basic service in the group;
- the group "All Teleservices" provided that the service is applicable to at least one teleservice and that the subscriber has a subscription to at least one teleservice which is in the same Elementary Basic Service Group as a teleservice to which the service is applicable;
- the group "All Bearer Services" provided that the service is applicable to at least one bearer service and that the subscriber has a subscription to at least one bearer service which is in the same Elementary Basic Service Group as a basic service to which the service is applicable.

If the Basic Service Group (or Basic Service Group List) is not present then the parameter shall apply to all Basic Service Groups.

If the basic service information is not a single Elementary Basic Service Group then the parameter shall be taken as applying individually to all the Elementary Basic Service Groups for which:

- the supplementary service is applicable to at least one basic service in the Basic Service Group; and
- the subscriber has a subscription to at least one basic service in the Basic Service Group.

The VLR and the SGSN are not required to store supplementary services data for Basic Service Groups which are not supported at the VLR or the SGSN respectively.

8.8.2 MAP-DELETE-SUBSCRIBER-DATA service

8.8.2.1 Definition

This service is used by an HLR to remove certain subscriber data from a VLR or SGSN if the subscription of one or more supplementary services or basic services is withdrawn. Note that this service is not used in case of erasure or deactivation of supplementary services.

This service is also used by an HLR to remove GPRS subscription data from an SGSN.

This service is also used by an HSS via IWF to remove EPS subscription data from an MME.

This service is also used by a CSS to remove the CSG subscription data from an MME via IWF or a VLR/SGSN.

It is a confirmed service and consists of the primitives shown in table 8.8/2.

8.8.2.2 Service primitives

Table 8.8/2: MAP-DELETE-SUBSCRIBER-DATA

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
Basic service List	C	C(=)		
SS-Code List	C	C(=)		
Roaming Restriction Due To				
Unsupported Feature	C	C(=)		
Camel Subscription Info Withdraw	C	C(=)		
Specific CSI Withdraw	C	C(=)		
Regional Subscription Data	C	C(=)		
VBS Group Indication	C	C(=)		
VGCS Group Indication	C	C(=)		
GPRS Subscription Data Withdraw	C	C(=)		
EPS Subscription Data Withdraw	C	C(=)		
Roaming Restricted In SGSN/MME Due To Unsupported Feature	C	C(=)		

LSA Information Withdraw	C	C(=)		
IST Information Withdraw	C	C(=)		
Regional Subscription Response			C	C(=)
GMLC List Withdraw	C	C(=)		
Subscribed Charging Characteristics Withdraw	C	C(=)		
CSG Information Deleted	C	C(=)		
VPLMN CSG Information Deleted	C	C(=)		
APN-OI-Replacement Withdraw	C	C(=)		
STN-SR Withdraw	C	C(=)		
Subscribed vSRVCC Withdraw	C	C(=)		
Subscribed Periodic RAU-TAU Timer Withdraw	C	C(=)		
Subscribed Periodic LAU Timer Withdraw	C	C(=)		
Additional MSISDN Withdraw	C	C(=)		
CS-to-PS-SRVCC Withdraw	C	C(=)		
User Plane Integrity Protection Withdraw	C	C(=)		
DL-Buffering Suggested Packet Count Withdraw	C	C(=)		
UE-Usage-Type Withdraw	C	C(=)		
Reset-IDs Withdraw	C	C(=)		
IAB-Operation-Withdraw	C	C(=)		
User error			C	C(=)
Provider error				O

8.8.2.3 Parameter use

All parameters are described in clause 7.6. The following clarifications are applicable:

Basic service List

A list of Extensible Basic service parameters (Extensible Basic service is defined in clause 7.6). It is used when one, several or all basic services are to be withdrawn from the subscriber. If the VLR or the SGSN receives a value for an Extensible Basic Service which it does not support, it shall ignore that value. This parameter is used by the VLR and by the SGSN; if the IWF receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

SS-Code List

A list of SS-Code parameters (SS-Code is defined in clause 7.6). It is used when several or all supplementary services are to be withdrawn from the subscriber.

There are three possible options:

- deletion of basic service(s);

  The parameter Basic service List is only included.

- deletion of supplementary service(s);

  The parameter SS-Code List is only included.

- deletion of basic and supplementary services;

  Both Basic service List and SS-Code List are included.

This parameter is used by the VLR and SGSN and IWF for Call Barring and LCS. Otherwise, this parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Roaming Restriction Due To Unsupported Feature

This parameter is used if Roaming Restriction Due To Unsupported Feature is deleted from the subscriber data. This may occur if unsupported features or services are removed from the subscriber data in the HLR.

If this parameter is sent the VLR shall check if the current Location Area is possibly allowed now. This parameter is used only by the VLR and if the SGSN or IWF receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

CAMEL Subscription Info Withdraw

This parameter is used to indicate that CAMEL Subscription Info shall be deleted from the VLR or from the SGSN. All CAMEL Subscription Info for the subscriber shall be deleted. This parameter is used by the VLR and by the SGSN. This parameter should not be sent in the same message as the Specific CSI Withdraw parameter; if the IWF receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Specific CSI Withdraw

This parameter is used to indicate that one or more specific elements of CAMEL Subscription Info shall be deleted from the VLR or from the SGSN.

The specific elements of CAMEL Subscription Info which may be withdrawn are:

- O-CSI with TDP criteria for O-CSI;
- SS-CSI;
- TIF-CSI;
- D-CSI;
- VT-CSI with TDP criteria for VT-CSI;
- MO-SMS-CSI;
- MT-SMS-CSI with TDP-Criteria for MT-SMS-CSI;
- M-CSI;
- MG-CSI;
- GPRS-CSI.

This parameter is used by the VLR and by the SGSN; if the IWF receices this parameter it shall ignore it. It shall not be sent to VLRs that do not support CAMEL phase 3 or higher. This parameter should not be sent in the same message as the CAMEL Subscription Info Withdraw parameter.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Regional Subscription Identifier

Contains one single Zone Code (as defined in clause 7.6) and is used if all Zone Codes shall be deleted from the subscriber data. When all the Zone Codes are deleted, the VLR, the SGSN or the MME shall check for its location areas whether they are allowed or not. If the whole Network Node area is restricted, the VLR, the SGSN or the MME (via the IWF) will report it to HLR by returning the Regional Subscription Response "Network Node Area Restricted".

The binary coding of the Zone Code value received in a Delete Subscriber Data request shall not be checked by the VLR, the SGSN or the MME.

Note that support of this parameter is a network operator option and it shall not be sent to networks which do not support Regional Subscription.

If Regional Subscription is not supported by the VLR, the SGSN or the MME, the request for deletion of Zone Codes is refused by sending the Regional Subscription Response "Regional Subscription Not Supported" to the HLR.

If no Zone Codes are stored in the respective subscriber data record, the request for deleting all Zone Code information shall be ignored and no Regional Subscription Response shall be returned. This parameter is used by the VLR, the SGSN and the MME.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

VBS Group Indication

Contains an indication (flag) which is used if all Group Ids shall be deleted from the subscriber data for the Voice Broadcast teleservice.

If VBS is not supported in the VLR or no Group Ids are stored for VBS in the respective subscriber record, the request for deletion of all Group Ids shall be ignored. This parameter is used only by the VLR and if the SGSN or the IWF receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

VGCS Group Indication

Contains an indication (flag) which is used if all Group Id's shall be deleted from the subscriber data for the Voice Group Call teleservice. This parameter is used only by the VLR and if the SGSN receives this parameter it shall ignore it.

If VGCS is not supported in the VLR or no Group Ids are stored for VGCS in the respective subscriber record, the request for deletion of all Group Ids shall be ignored.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

GPRS Subscription Data Withdraw

This parameter is used to indicate whether all GPRS Subscription Data for the subscriber shall be deleted or if only a subset of the stored GPRS Subscription Data for the subscriber shall be deleted. In the latter case only those PDP contexts whose identifiers are included in the subsequent identifier list will be deleted. This parameter is used only by the SGSN and if the VLR receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

EPS Subscription Data Withdraw

This parameter is used to indicate whether all EPS Subscription Data for the subscriber shall be deleted or if only a subset of the stored EPS Subscription Data for the subscriber shall be deleted. In the latter case, only those APN Configurations whose identifiers are included in the subsequent identifier list will be deleted. This parameter is used only by the SGSN and the MME and if the VLR receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Roaming Restricted In SGSN/MME Due To Unsupported Feature

This parameter is used if Roaming Restricted In SGSN/MME Due To Unsupported Feature is deleted from the GPRS/EPS subscriber data. This may occur if unsupported features or services are removed from the GPRS/EPS subscriber data in the HLR.

If this parameter is sent the SGSN shall check if the current Location Area is possibly allowed now. This parameter is used only by the SGSN and the IWF and if the VLR receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

LSA Information Withdraw

This parameter is used to indicate whether all LSA Information for the subscriber shall be deleted or if only a subset of the stored LSA Information for the subscriber shall be deleted. In the latter case only the LSA data whose LSA identities are included in the subsequent LSA data list will be deleted. This parameter is used by the VLR and the SGSN.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

IST Information Withdraw

This parameter is used to indicate that the IST condition has been removed for the subscriber. See 3GPP TS 43.035 for the use of this parameter.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Regional Subscription Response

If included in the Delete Subscriber Data response this parameter indicates one of:

- Network Node Area Restricted;
- Regional Subscription Not Supported.

This parameter is used by the VLR, the SGSN and the IWF.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

GMLC List Withdraw

This parameter indicates that the subscriber's LCS GMLC List shall be deleted from the VLR or SGSN. This parameter is used by the VLR and the SGSN and IWF.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Subscribed Charging Characteristics Withdraw

This parameter indicates that the Subscribed Charging Characteristics shall be replaced with a local default value in the SGSN or in the MME (see 3GPP TS 32.251).
This parameter is used only by the SGSN and the IWF and if the VLR receives this parameter it shall ignore it.

This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

CSG Information Deleted

This parameter indicates that CSG Subscription Information received from the HLR/HSS shall be deleted from VLR, SGSN, or MME.
This parameter is used by the VLR, SGSN and the IWF. This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

VPLMN CSG Information Deleted

This parameter indicates that CSG Subscription Information received from the CSS shall be deleted from VLR, SGSN. This parameter is used by the VLR and SGSN. This parameter is not applicable for the HLR/HSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the HLR/HSS.

APN-OI-Replacement Withdraw

This parameter indicates that APN-OI-Replacement shall be deleted from the SGSN or the MME.
This parameter is used by the SGSN and the IWF and if the VLR receives this parameter it shall ignore it. This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

STN-SR Withdraw

This parameter indicates that STN-SR shall be deleted from the SGSN or the MME.
This parameter is used by the SGSN and the IWF and if the VLR receives this parameter it shall ignore it. This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Subscribed vSRVCC Withdraw

This parameter indicates that Subscribed vSRVCC shall be deleted from the MME.
This parameter is used by the MME and the IWF and if the SGSN or VLR receives this parameter it shall ignore it.
This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Subscribed Periodic RAU-TAU Timer Withdraw

This parameter indicates that Subscribed Periodic RAU-TAU Timer value shall be deleted from the SGSN or the MME.
This parameter is used by the SGSN and the IWF and if the VLR receives this parameter it shall ignore it. This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Subscribed Periodic LAU Timer Withdraw

This parameter indicates that Subscribed Periodic LAU Timer value shall be deleted from the VLR.
This parameter is used by the VLR and if the MME or SGSN receives this parameter it shall ignore it. This parameter is not applicable for the CSS and the VLR or SGSN or IWF shall ignore this parameter if it is received from the CSS.

Additional MSISDN Withdraw

This parameter indicates that Additional MSISDN shall be deleted from the SGSN or MME.
This parameter is used by the SGSN and the IWF.

CS-to-PS-SRVCC Withdraw

This parameter indicates by its presence that CS to PS SRVCC is no longer subscribed.

User Plane Integrity Protection Withdraw

This parameter indicates by its presence that User Plane Integrity Protection may no longer be required.

DL-Buffering Suggested Packet Count Withdraw

This parameter indicates by its presence that a suggested DL-Buffering Packet Count is no longer subscribed.

UE-Usage-Type Withdraw

This parameter indicates by its presence that a UE-Usage-Type is no longer subscribed.
This parameter is not applicable for VLRs.
The HLR shall include this parameter towards the SGSN or MME (via IWF) that supports the Dedicated Core Network functionality if the subscription to a UE-Usage-Type is removed.

Reset-IDs Withdraw

This parameter indicates by its presence that Reset-IDs are no longer subscribed.

IAB-Operation-Withdraw

This parameter indicates by its presence that IAB operation is no longer authorized for the UE.

User error

Only one of the following values is applicable:

- Unidentified subscriber;
- Data missing;

- Unexpected data value.

8.9 Identity management services

8.9.1 MAP-PROVIDE-IMSI service

8.9.1.1 Definition

This service is used by a VLR in order to get, via the MSC, the IMSI of a subscriber (e.g. when a subscriber has identified itself with a TMSI not allocated to any subscriber in the VLR).

It is a confirmed service and consists of the primitives shown in table 8.9/1.

8.9.1.2 Service primitives

Table 8.9/1: MAP-PROVIDE-IMSI

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI			C	C(=)
User error			C	C(=)
Provider error				O

8.9.1.3 Parameter use

All parameters are described in clause 7.6. The following clarifications are applicable:

IMSI

This parameter is received when the request is successfully carried out. It contains the requested IMSI.

User error

Only one of the following values is applicable:

- Absent subscriber.

8.9.2 MAP-FORWARD-NEW-TMSI service

8.9.2.1 Definition

This service is used by a VLR to allocate, via MSC, a new TMSI to a subscriber during an ongoing transaction (e.g. call set-up, location updating or supplementary services operation).

It is a confirmed service and consists of the primitives shown in table 8.9/2.

8.9.2.2 Service primitives

Table 8.9/2: MAP-FORWARD-NEW-TMSI

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
TMSI	M	M(=)		
Provider error				O

8.9.2.3 Parameter use

The parameter TMSI is described in clause 7.6.

8.10 Fault recovery services

8.10.1 MAP_RESET service

8.10.1.1 Definition

This service is used by the HSS/HLR or CSS, after a restart, to indicate to a list of VLRs, SGSNs or MMEs (via IWF) that a failure occurred.

This service may also be used by the HSS/HLR as part of operation and maintenance actions e.g. to allow planned HLR/HSS outage without service interruption, or to update subscription data shared by multiple subscribers.

The MAP_RESET service is a non-confirmed service using the service primitives defined in table 8.10/1.

8.10.1.2 Service primitives

Table 8.10/1: MAP_RESET

Parameter name	Request	Indication
Invoke Id	M	M(=)
Sending Node Number	M	M(=)
HLR Id LIST	U	C(=)
Reset-ID LIST	C	C(=)
Subscription Data	C	C(=)
Subscription Data Deletion	C	C(=)

8.10.1.3 Parameter definition and use

Invoke Id

See definition in clause 7.6.1.

SendingNode Number

For a restart of the HLR/HSS, this parameter shall contain the HLR number. See definition in clause 7.6.2.

For a restart of the CSS, this parameter shall contain the CSS number. See definition in clause 7.6.2.

HLR Id LIST

The HLR Id List is a list of HLR Ids. If the parameter is present in the indication, the VLR, the SGSN or the MME may base the retrieval of subscribers to be restored on their IMSI: the subscribers affected by the reset are those whose IMSI leading digits are equal to one of these numbers. If the parameter and the Reset-ID List is absent, subscribers to be restored are those for which the OriginatingEntityNumber received at location updating time matches the equivalent parameter of the Reset Indication.

This parameter shall only be applicable for a restart of the HLR/HSS. It shall not be present if Reset-ID LIST is present.

Reset-ID LIST

The Reset-ID LIST is a list of Reset-IDs. It shall not be present if Reset-IDs are not supported by the HLR/HSS and by theVLR or SGSN or MME (via IWF). If the parameter is present in the indication, the VLR, the SGSN or the MME may base the retrieval of affected subscribers (i.e. those impacted by the restoration or by the shared data update) on their subscribed Reset-IDs: The subscribers affected by the reset are those whose subscription contains at least one of these Reset-IDs.

Subscription Data

If the Reset Procedure is used to add/ modify subscription data shared by multiple subscribers, this Information Element shall contain the part of the subscription profile that either is to be added to the subscription profile stored in the VLR, MME or SGSN or combined MME/SGSN or is replacing a part of the subscription profiles of the impacted subscribers stored in the VLR, MME or SGSN.

Shall be absent if Subsciption Data Deletion is present.
Shall be absent if Reset-ID LIST is absent

Subscription Data Deletion

If the Reset Procedure is used to delete subscription data shared by multiple subscribers, this Information Element shall contain the identifications of the part of the subscription profile that is to be deleted from the subscription profiles of the impacted subscribers stored in the VLR, MME or SGSN.

Shall be absent if Subsciption Data is present.
Shall be absent if Reset-ID LIST is absent

8.10.2 MAP_FORWARD_CHECK_SS_INDICATION service

8.10.2.1 Definition

This service may be used by an HLR as an implementation option, to indicate to a mobile subscriber that supplementary services parameters may have been altered, e.g. due to a restart. If received from the HLR, the VLR shall forward this indication to the MSC, which in turn forwards it to the MS. The HLR only sends this indication after successful completion of the subscriber data retrieval from HLR to VLR that ran embedded in a MAP_UPDATE_LOCATION procedure.

The MAP_FORWARD_CHECK_SS_INDICATION service is a non-confirmed service using the service primitives defined in table 8.10/2.

8.10.2.2 Service primitives

Table 8.10/2: MAP_FORWARD_CHECK_SS_INDICATION

Parameter name	Request	Indication
Invoke Id	M	M(=)

8.10.2.3 Parameter definition and use

Invoke Id

See definition in clause 7.6.1.

8.10.3 MAP_RESTORE_DATA service

8.10.3.1 Definition

This service is invoked by the VLR on receipt of a MAP_PROVIDE_ROAMING_NUMBER indication for an unknown IMSI, or for a known IMSI with the indicator " Subscriber Data Confirmed by HLR" set to "Not confirmed". The service is used to update the LMSI in the HLR, if provided, and to request the HLR to send all data to the VLR that are to be stored in the subscriber's IMSI record.

This service may be invoked by the VLR on receipt of a "MAP-MT-FORWARD-SHORT-MESSAGE" message for an unknown IMSI, or for a known IMSI with the indicator "Subscriber Data Confirmed by HLR" set to "Not confirmed". The service is used to update the LMSI in the HLR, if provided, and to request the HLR to send all data to the VLR that are to be stored in the subscriber's IMSI record.

The HLR shall return the error "system failure" to the VLR if the subscriber is not registered on the VLR.

The MAP_RESTORE_DATA service is a confirmed service using the service primitives defined in table 8.10/3.

8.10.3.2 Service primitives

Table 8.10/3: MAP_RESTORE_DATA

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
LMSI	U	C(=)		
Supported CAMEL phases	C	C(=)		
SoLSA Support Indicator	C	C(=)		
IST Support Indicator	C	C(=)		
Super-Charger Supported in Serving Network Entity	C	C(=)		
Long FTN Supported	C	C(=)		
Supported LCS Capability Sets	C	C(=)		
Offered CAMEL 4 CSIs	C	C(=)		
Restoration Indicator	U	C(=)		
Supported RAT Types Indicator	U	C(=)		
MTRF Supported	U	C(=)		
MSISDN-less Operation Supported	C	C(=)		
HLR number			C	C(=)
MS Not Reachable Flag			C	C(=)
User error			C	C(=)
Provider error				O

8.10.3.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

LMSI

See definition in clause 7.6.2. It is an operator option to provide the LMSI from the VLR; it is mandatory for the HLR to support the LMSI handling procedures.

Supported CAMEL Phases

This parameter indicates which phases of CAMEL are supported. Must be present if a CAMEL phase different from phase 1 is supported. Otherwise may be absent.

SoLSA Support Indicator

This parameter is used by the VLR to indicate to the HLR in the Restore Data indication that SoLSA is supported. If this parameter is not included in the Restore Data indication then the HLR shall not perform any specific error handling.

This SoLSA Support Indicator shall be stored by the HLR per VLR where there are Subscribers roaming. If a Subscriber is marked as only allowed to roam in Subscribed LSAs while roaming in a VLR and no SoLSA Support indicator is stored for that VLR, the location status of that Subscriber shall be set to Restricted.

IST Support Indicator

This parameter is used to indicate to the HLR that the VMSC supports basic IST functionality, that is, the VMSC is able to terminate the Subscriber Call Activity that originated the IST Alert when it receives the IST alert response indicating that the call(s) shall be terminated. If this parameter is not included in the Restore Data indication and the Subscriber is marked as an IST Subscriber, then the HLR may limit the service for the subscriber (by inducing an Operator Determined barring of Outgoing calls), or allow service assuming the associated risk of not having the basic IST mechanism available.

This parameter can also indicate that the VMSC supports the IST Command service, including the ability to terminate all calls being carried for the identified subscriber by using the IMSI as a key. If this additional capability is not included in the Restore Data indication and the HLR supports the IST Command capability, then the HLR may limit the service for the subscriber (by inducing an Operator Determined barring of Outgoing calls), or allow service assuming the associated risk of not having the IST Command mechanism available.

Long FTN Supported

This parameter indicates that the VLR supports Long Forwarded-to Numbers.

Super-Charger Supported in Serving Network Entity

This parameter is used by the VLR to indicate to the HLR that the VLR supports the Super-Charger functionality and that subscriber data is required.

If this parameter is absent then the VLR does not support the Super-Charger functionality.

Supported LCS Capability Sets

This parameter indicates, if present, the capability sets of LCS which are supported. If the parameter is sent but no capability set is marked as supported then the VLR does not support LCS at all.

If this parameter is absent then the VLR may support at most LCS capability set 1, that is LCS Release98 or Release99 version.

Offered CAMEL 4 CSIs

This parameter indicates the CAMEL phase 4 CSIs offered in the VMSC/VLR (see clause 7.6.3.36D).

Restoration Indicator

This parameter indicates, if present, that the HLR shall send in the MAP-INSERT-SUBSCRIBER-DATA the MME Name if the subscriber is registered to EPS, or the SGSN Number if available and if the subscriber is registered to GPRS. The VLR may set this indicator if it supports Gs or SGs interfaces.

Supported RAT Types Indicator

This parameter indicates, if present, which access technologies (e.g. GERAN and / or UTRAN) are served by the MSC/VLR (see clause 7.6.3)

MTRF Supported

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

MSISDN-less Operation Supported

See subclause 3.6.1.5 of 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

HLR number

See definition in clause 7.6.2. The presence of this parameter is mandatory in case of successful outcome of the service.

MS Not Reachable Flag

See definition in clause 7.6.8. This parameter shall be present in case of successful outcome of the service, if the "MS Not Reachable flag" was set in the HLR.

User error

In case of unsuccessful outcome of the service, an error cause shall be returned by the HLR. The following error causes defined in clause 7.6.1 may be used, depending on the nature of the fault:

- unknown subscriber;
- system failure;
- unexpected data value;
- data missing.

Provider error

For definition of provider errors see clause 7.6.1.

8.11 Subscriber Information services

8.11.1 MAP-ANY-TIME-INTERROGATION service

8.11.1.1 Definition

This service is used by the gsmSCF, to request information (e.g. subscriber state and location) from the HLR or the GMLC at any time. This service may also be used by the gsmSCF to request the Mobile Number Portability (MNP) information from the NPLR.

This service is also used by the Presence Network Agent to request information, (e.g. subscriber state and location) about the subscriber (associated with a presentity) from the HLR at any time (see 3GPP TS 23.141 [128]).

When this service is used to the HLR, the subscriber state, location, Time Zone, or T-ADS data may be requested.

When this service is used to the GMLC, only the location may be requested.

When this service is used to the NPLR, only the MNP information may be requested.

The MAP-ANY-TIME-INTERROGATION service is a confirmed service using the service primitives defined in table 8.11/1.

8.11.1.2 Service primitives

Table 8.11/1: Any_Time_Interrogation

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
Requested Info	M	M(=)		
Requested domain	C	C(=)		
MNP Requested Info	C	C(=)		
gsmSCF-Address	M	M(=)		
IMSI	C	C(=)		

MSISDN	C	C(=)		
Location Information			C	C(=)
Location Information for GPRS			C	C(=)
Location Information for EPS			C	C(=)
Subscriber State			C	C(=)
PS Subscriber State			C	C(=)
EPS Subscriber State			C	C(=)
IMEI			C	C(=)
MS Classmark 2			C	C(=)
GPRS MS Class			C	C(=)
MNP info Result			C	C(=)
IMS Voice Over PS Sessions Support Indicator			C	C(=)
Last UE Activity Time			C	C(=)
Last RAT Type			C	C(=)
Time Zone			C	C(=)
Daylight Saving Time			C	C(=)
User error			C	C(=)
Provider error				O

8.11.1.3 Parameter definition and use

All parameters are described in clause 7.6. The use of these parameters and the requirements for their presence are specified in 3GPP TS 23.018 [97] and 3GPP TS 23.078 [98].

The HLR or GMLC may be able to use the value of the parameter gsmSCF-address to screen a MAP_Any_Time_Interrogation indication.

The use of the parameters and the requirements for their presence are specified in 3GPP TS 23.078 [98].

IMS Voice Over PS Sessions Support Indicator

This parameter indicates the most recent IMS-Voice-Over-PS-Sessions support (based on the Last UE Activity Time), as received from the serving nodes. This parameter shall be present if Requested Info indicates that T-ADS Data are requested.

Last UE Activity Time

This parameter indicates the most recent available point in time of the UE's last radio contact, as received from the serving nodes. This value may not represent the absolute last instant of radio activity of the UE, when any of the serving nodes has not answered to the T-ADS query. This parameter may be present if requested Info indicates that T-ADS Data are requested. This value may not be available when all the serving nodes have indicated an homogeneous support or an homogeneous non support of IMS Voice Over PS Sessions, since in that case, the serving nodes do not need to be explicitly asked for T-ADS Data.

Last RAT Type

This parameter indicates the most recent available RAT Type of the access (based on the Last UE Activity Time), as received from the serving nodes. This parameter shall be present if requested Info indicates that T-ADS Data are requested and the IMS Voice Over PS Sessions Support Indicator does not take the value "unknown". This value may not represent the absolute last RAT Type of the UE, when any of the serving nodes has not answered to the T-ADS query. This parameter may be present if requested Info indicates that T-ADS Data are requested. This value may not be available when all the serving nodes have indicated an homogeneous support or an homogeneous non support of IMS Voice Over PS Sessions, since in that case, the serving nodes do not need to be explicitly asked for T-ADS Data.

Time Zone

This parameter indicates the Time Zone of the location in the visited network where the UE is attached, including any adjustment for summertime (daylight saving time).

Daylight Saving Time

This parameter indicates the Daylight Saving Time (in steps of 1 hour) used to adjust for summertime the time zone of the location where the UE is attached in the visited network.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- System Failure;
- Any Time Interrogation Not Allowed;
- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber.

Provider error

These are defined in clause 7.6.1.

8.11.2 MAP-PROVIDE-SUBSCRIBER-INFO service

8.11.2.1 Definition

This service is used to request information (e.g. subscriber state and location) from the VLR, SGSN or MME (via an IWF) at any time.

The MAP-PROVIDE-SUBSCRIBER-INFO service is a confirmed service using the primitives defined in table 8.11/2.

8.11.2.2 Service primitives

Table 8.11/2: Provide_Subscriber_Information

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
Requested Info	M	M(=)		
IMSI	M	M(=)		
LMSI	U	O		
Call Priority	U	O		
Location Information			C	C(=)
Location Information for GPRS			C	C(=)
Subscriber State			C	C(=)
PS Subscriber State			C	C(=)
IMEI			C	C(=)
MS Classmark 2			C	C(=)
GPRS MS Class			C	C(=)
IMS Voice Over PS Sessions Support Indicator			C	C(=)
Last UE Activity Time			C	C(=)
Last RAT Type			C	C(=)
Location Information for EPS			C	C(=)
Time Zone			C	C(=)
Daylight Saving Time			C	C(=)
User error			C	C(=)
Provider error				O

8.11.2.3 Parameter definition and use

All parameters are defined in clause 7.6. The use of these parameters and the requirements for their presence are specified in 3GPP TS 23.018 [97] and 3GPP TS 23.078 [98].

Call Priority

This parameter indicates the eMLPP priority of the call (see 3GPP TS 24.067 [137]). This parameter should be present if the HLR supports this parameter and if the Call Priority was received in the MAP_SEND_ROUTING_INFORMATION request.

IMS Voice Over PS Sessions Support Indicator

This parameter indicates whether IMS Voice Over PS Sessions is supported at the UE's current Routing Area. This parameter shall be present if the UE's current Routing Area is known to the SGSN and the Requested Info indicates that T-ADS Data are requested; otherwise it shall be absent.

Last UE Activity Time

This parameter indicates the point in time of the UE's last radio contact. This parameter shall be present if requested Info indicates that T-ADS Data are request.

Last RAT Type

This parameter indicates the RAT Type of the access where the UE was present at the time of the last radio contact. This parameter shall be present if requested Info indicates that T-ADS Data are request.

Time Zone

This parameter indicates the Time Zone of the location in the visited network where the UE is attached, including any adjustment for summertime (daylight saving time).

Daylight Saving Time

This parameter indicates the Daylight Saving Time (in steps of 1 hour) used to adjust for summertime the time zone of the location where the UE is attached in the visited network.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Data Missing;
- Unexpected Data Value.

If the subscriber is not found on the VLR, SGSN or MME, this may be indicated to the requester with the "Unexpected Subscriber" value inside the Unexpected Data Value error

Provider error

These are defined in clause 7.6.1.

8.11.3 MAP-ANY-TIME-SUBSCRIPTION-INTERROGATION service

8.11.3.1 Definition

This service is used by the gsmSCF, to request subscription information (e.g. call forwarding supplementary service data or CSI) from the HLR at any time. In an IP Multimedia Core Network, an IM-SSF can take on the role of a gsmSCF for this service.

8.11.3.2 Service primitives

Table 8.11/3: Any_Time_Subscription_Interrogation

Parameter name	Request	Indication	Response	Confirm

Invoke id	M	M(=)	M(=)	M(=)
Requested Subscription Info	M	M(=)		
GsmSCF-Address	M	M(=)		
IMSI	C	C(=)		
MSISDN	C	C(=)		
Long FTN Supported	C	C(=)		
Call Forwarding Data			C	C(=)
Call Barring Data			C	C(=)
ODB Info			C	C(=)
CAMEL Subscription Info			C	C(=)
Supported CAMEL phases in VLR			C	C(=)
Supported CAMEL phases in SGSN			C	C(=)
Offered CAMEL 4 CSIs in VLR			C	C(=)
Offered CAMEL 4 CSIs in SGSN			C	C(=)
MSISDN-BS-List			C	C(=)
CSG Subscription Data			C	C(=)
Call Hold Data			C	C(=)
Call Waiting Data			C	C(=)
Explicit Call Transfer Data			C	C(=)
Calling Line Identification Presentation Data			C	C(=)
Calling Line Identification Restriction Data			C	C(=)
User error			C	C(=)
Provider error				O

8.11.3.3 Parameter definition and use

All parameters are described in clause 7.6.

The HLR may be able to use the value of the parameter gsmSCF-address to screen a MAP_Any_Time_Subscription_Interrogation indication. The gsmSCF-address shall contain the IM-SSF address when the IM-SSF takes the role of the gsmSCF.

The use of the parameters and the requirements for their presence are specified in 3GPP TS 23.078 [98] and 3GPP TS 23.278 [125].

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Unexpected Data Value;
- Unknown Subscriber;
- BearerServiceNotProvisioned;
- TeleserviceNotProvisioned;
- CallBarred;
- IllegalSS-Operation;
- SS-NotAvailable;
- InformationNotAvailable;
- Any Time Subscription Interrogation Not Allowed;
- Data Missing.

Provider error

These are defined in clause 7.6.1.

8.11.4 MAP-ANY-TIME-MODIFICATION service

8.11.4.1 Definition

This service is used by the gsmSCF, to modify information of the HLR at any time.

This service is also used by the Presence Network Agent to activate or deactivate reporting of mobility management events (associated with a presentity) from the VLR or SGSN (see 3GPP TS 23.141 [128]).

This service is also used by a Service Related Entity (e.g. the IP-SM-GW) to activate a one-time subscription of UE-reachability in the MME (see 3GPP TS 23.204 [134]) and SGSN (see 3GPP TS 23.060 [104]).

This service is also used by external Short Message Gateway (IP-SM-GW) for updating the IP-SM-GW Number stored in the HLR, and for retrieving SC Address from the HLR.

8.11.4.2 Service primitives

Table 8.11/4: Any_Time_Modification

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
gsmSCF-Address	M	M(=)		
Subscriber Identity	M	M(=)		
Modification request for ODB data	C	C(=)		
Modification request for SS information	C	C(=)		
Modification request for CSI	C	C(=)		
Modification request for CSG	C	C(=)		
Long FTN Supported	C	C(=)		
Modification request for IP-SM-GW data	C	C(=)		
Activation request for UE-Reachability	C	C(=)		
Ext Forwarding information-for-CSE			C	C(=)
Ext Call barring information-for-CSE			C	C(=)
ODB Info			C	C(=)
CAMEL subscription info			C	C(=)
Service Centre Address			C	C(=)
User error			C	C(=)
Provider error				O

8.11.4.3 Parameter definition and use

All parameters are described in clause 7.6.

The HLR may be able to use the value of the parameter gsmSCF-address to screen a MAP_Any_Time_Modification indication.

The use of parameters other than described below and the requirements for their presence are specified in 3GPP TS 23.078 [98] and 3GPP TS 23.278 [125].

gsmSCF-Address

This parameter indicates the address of the interrogating gsmSCF. The gsmSCF Address shall be in international E.164 format. If the service is used by IP-SM-GW, the parameter contains the address of the IP-SM-GW. See also 3GPP TS 23.078 [98] and 3GPP TS 23.278 [125].

Modification request for CSG

This parameter is used by the gsmSCF to request notification of modification of CSG subscription data.

Modification request for IP-SM-GW data

This parameter is used by the external IP-SM-GW for updating the IP-SM-GW Number and IP-SM-GW Diameter Address stored in the HLR. If this parameter is present then other modification requests shall not be present.

Activation request for UE Reachability

This parameter is used by the Service Related Entity (e.g. IP-SM-GW) to activate the one-time subscription for UE-Reachability. If this parameter is present then other modification requests shall not be present.

Service Centre Address

See definition in clause 7.6.2.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Any Time Modification Not Allowed;
- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber;
- Bearer service not provisioned;

  This error is returned only if not even a subset of the requested bearer service group has been subscribed to;
- Teleservice not provisioned;

  This error is returned only if not even a subset of the requested teleservice group has been subscribed to;
- Call Barred;
- Illegal SS operation;
- SS error status;
- SS incompatibility;
- SS subscription violation;
- Information Not Available.

Provider error

These are defined in clause 7.6.1.

8.11.5 MAP-NOTE-SUBSCRIBER-DATA-MODIFIED service

8.11.5.1 Definition

This service is used by the HLR to inform the gsmSCF that subscriber data have been modified. In an IP Multimedia Core Network, an IM-SSF can take on the role of a gsmSCF for this service.

This service is also used by the HLR to inform the Service Related Entity (e.g. IP-SM-GW) that the UE has become reachable (see 3GPP TS 23.204 [134]).

8.11.5.2 Service primitives

Table 8.11/5: Note_Subscriber_Data_Modified

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		

Parameter name	Request	Indication	Response	Confirm
MSISDN	M	M(=)		
Ext Forwarding information-for-CSE	C	C(=)		
Ext Call barring information-for-CSE	C	C(=)		
ODB Info	C	C(=)		
CAMEL subscription info	C	C(=)		
CSG Subscription Data	C	C		
CW info	C	C(=)		
CH info	C	C(=)		
CLIP Info	C	C(=)		
CLIR Info	C	C(=)		
ECT Info	C	C(=)		
All Information Sent	C	C(=)		
UE reachable	C	C(=)		
User error			C	C(=)
Provider error				O

8.11.5.3 Parameter definition and use

Invoke id

See clause 7.6.1 for the use of this parameter.

IMSI

See clause 7.6.2 for the use of this parameter.

MSISDN

See clause 7.6.2 for the use of this parameter. In an IP Multimedia Core Network, if no MSISDN is available, the HLR shall populate this parameter with the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]).

Ext Forwarding information-for-CSE

See clause 7.6.3 for the use of this parameter. The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98].

Ext Call barring information-for-CSE

See clause 7.6.3 for the use of this parameter. The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98].

ODB Info

See clause 7.6.3 for the use of this parameter. The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98].

CAMEL subscription info

See clause 7.6.3 for the use of this parameter. The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98] and 3GPP TS 23.278 [125].

CSG Subscription Data

This parameter contains a list of CSG-Ids and the associated expiration dates (see 3GPP TS 22.011 [138]). The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98].

CW Info

This parameter contains the status of the call waiting supplementary service. The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98]

CH Info

This parameter contains the status of the call hold supplementary service.The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98]

ECT Info

This parameter contains the status of the explicit call transfer supplementary service.The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98]

CLIP Info

This parameter contains the status of the calling line identification presentation supplementary service.The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98]

CLIR Info

This parameter contains the status of the calling line identification restriction supplementary service.The use of this parameter and the requirements for their presence are specified in 3GPP TS 23.078 [98]

All Information Sent

This parameter is set when the HLR has sent all information to gsmSCF.

UE Reachable

This parameter is used when the HLR indicates to the Service related entity (e.g. IP-SM-GW) that the UE is reachable again.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber.

Provider error

These are defined in clause 7.6.1.

The use of the parameters and the requirements for their presence are specified in 3GPP TS 23.078 [98] and 3GPP TS 23.278 [125].

9 Operation and maintenance services

9.1 Subscriber tracing services

9.1.1 MAP-ACTIVATE-TRACE-MODE service

9.1.1.1 Definition

This service is used between the HLR and the VLR to activate subscriber tracing in the VLR.

Also this service is used between the HLR and the SGSN to activate subscriber tracing in the SGSN.

The MAP-ACTIVATE-TRACE-MODE service is a confirmed service using the primitives from table 9.1/1.

9.1.1.2 Service primitives

Table 9.1/1: MAP-ACTIVATE-TRACE-MODE

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
Trace reference	M	M(=)		
Trace type	M	M(=)		
Trace reference 2	C	C(=)		
Trace depth list	C	C(=)		
Trace NE type list	C	C(=)		
Trace interface list	C	C(=)		
Trace event list	C	C(=)		
Trace support indicator			C	C(=)
OMC Id	U	C(=)		
MDT-Configuration	C	C(=)		
User error			C	C(=)
Provider error				O

9.1.1.3 Parameter use

Invoke id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2. The IMSI is a mandatory parameter in a stand-alone operation.

Trace reference

See definition in clause 7.6.10. This parameter contains trace reference for GSM only tracing request.

Trace type

See definition in clause 7.6.10. This parameter contains trace type for GSM only tracing request.

OMC Id

See definition in clause 7.6.2. The use of this parameter is an operator option.

Trace reference 2

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

Trace depth list

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

Trace NE type list

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

Trace interface list

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

Trace event list

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

Trace support indicator

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

MDT-Configuration

See definition in clause 7.6.10.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unidentified Subscriber;
- Facility Not Supported;
- Tracing Buffer Full;
- System Failure;
- Unexpected Data Value;
- Data missing.

Provider error

For definition of provider errors see clause 7.6.1.

9.1.2 MAP-DEACTIVATE-TRACE-MODE service

9.1.2.1 Definition

This service is used between the VLR and the HLR for deactivating subscriber tracing in the VLR.

Also this service is used between the SGSN and the HLR for deactivating subscriber tracing in the SGSN.

The MAP-DEACTIVATE-TRACE-MODE service is a confirmed service using the primitives from table 9.1/2.

9.1.2.2 Service primitives

Table 9.1/2: MAP-DEACTIVATE-TRACE-MODE

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
Trace reference	M	M(=)		
Trace reference 2	C	C(=)		
User error			C	C(=)
Provider error				O

9.1.2.3 Parameter use

Invoke id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2. The IMSI is a mandatory parameter in a stand-alone operation.

Trace reference

See definition in clause 7.6.10.

Trace reference 2

See definition in clause 7.6.10. This parameter shall be used for UMTS trace activation.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unidentified Subscriber;
- Facility Not Supported;
- System Failure;
- Unexpected Data Value;
- Data missing.

Provider error

For definition of provider errors see clause 7.6.1.

9.1.3 MAP-TRACE-SUBSCRIBER-ACTIVITY service

9.1.3.1 Definition

This service is used between the VLR and the MSC to activate the subscriber tracing in the MSC.

The MAP-TRACE-SUBSCRIBER-ACTIVITY service is a non-confirmed service using the primitives from table 9.1/3.

9.1.3.2 Service primitives

Table 9.1/3: MAP-TRACE-SUBSCRIBER-ACTIVITY

Parameter name	Request	Indication
Invoke id	M	M(=)
IMSI	C	C(=)
Trace reference	M	M(=)
Trace type	M	M(=)
OMC Id	U	C(=)

9.1.3.3 Parameter use

Invoke id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2. The controlling MSC shall provide either the IMSI or the IMEI to the servicing MSC.

Trace reference

See definition in clause 7.6.10.

Trace type

See definition in clause 7.6.10.

OMC Id

See definition in clause 7.6.2. The use of this parameter is an operator option.

9.2 Other operation and maintenance services

9.2.1 MAP-SEND-IMSI service

9.2.1.1 Definition

This service is used by a VLR in order to fetch the IMSI of a subscriber in case of some Operation & Maintenance procedure where subscriber data are needed in the Visited PLMN and MSISDN is the only subscriber's identity known.

It is a confirmed service and consists of the primitives shown in table 9.2/1.

9.2.1.2 Service primitives

Table 9.2/1: MAP-SEND-IMSI

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
MSISDN	M	M(=)		
IMSI			C	C(=)
User error			C	C(=)
Provider error				O

9.2.1.3 Parameter use

All parameters are described in clause 7.6. The following clarifications are applicable.

User error

Only one of the following values is applicable:

- Unknown subscriber;
- Unexpected data value;
- Data missing.

10 Call handling services

10.1 MAP_SEND_ROUTING_INFORMATION service

10.1.1 Definition

This service is used between the Gateway MSC and the HLR. The service is invoked by the Gateway MSC to perform the interrogation of the HLR in order to route a call towards the called MS.

This is a confirmed service using the primitives listed in table 10.1/1.

This service is also used between the GMSC and the NPLR and between the gsmSCF and the HLR.

10.1.2 Service primitives

Table 10.1/1: MAP_SEND_ROUTING_INFORMATION parameters

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Interrogation Type	M	M(=)		
GMSC or gsmSCF Address	M	M(=)		

Parameter name	Request	Indication	Response	Confirm
MSISDN	M	M(=)	C	C(=)
OR Interrogation	C	C(=)		
OR Capability	C	C(=)		
CUG Interlock	C	C(=)	C	C(=)
CUG Outgoing Access	C	C(=)	C	C(=)
Number of Forwarding	C	C(=)		
Network Signal Info	C	C(=)		
Supported CAMEL Phases	C	C(=)	C	C(=)
Suppress T-CSI	C	C(=)		
Offered CAMEL 4 CSIs	C	C(=)		
Suppression of Announcement	C	C(=)		
Call Reference Number	C	C(=)		
Forwarding Reason	C	C(=)		
Basic Service Group	C	C(=)		
Basic Service Group 2	C	C(=)		
Alerting Pattern	C	C(=)		
CCBS Call	C	C(=)		
Supported CCBS Phase	C	C(=)		
Additional Signal Info	C	C(=)		
IST Support Indicator	C	C(=)		
Pre-paging supported	C	C(=)		
Call Diversion Treatment Indicator	C	C(=)		
Long FTN Supported	C	C(=)		
Suppress VT-CSI	C	C(=)		
Suppress Incoming Call Barring	C	C(=)		
SuppressMTSS	C	C(=)		
gsmSCF Initiated Call	C	C(=)		
Network Signal Info 2	C	C(=)		
MT Roaming Retry Supported	U	C(=)		
Call Priority	U	C(=)		
IMSI			C	C(=)
MSRN			C	C(=)
Forwarding Data			C	C(=)
Forwarding Interrogation Required			C	C(=)
VMSC address			C	C(=)
ReleaseResourcesSupported			C	C(=)
GMSC Camel Subscription Info			C	C(=)
Location Information			C	C(=)
Subscriber State			C	C(=)
Basic Service Code			C	C(=)
CUG Subscription Flag			C	C(=)
North American Equal Access preferred Carrier Id			U	C(=)
User error			C	C(=)
SS-List			U	C(=)
CCBS Target			C	C(=)
Keep CCBS Call Indicator			C	C(=)
IST Alert Timer			C	C(=)
Number Portability Status			U	C(=)
Supported CAMEL Phases in VMSC			C	
Offered CAMEL 4 CSIs in VMSC			C	C(=)
MSRN 2			C	C(=)
Forwarding Data 2			C	C(=)
SS-List 2			C	C(=)
Basic Service Code 2			C	C(=)
Allowed Services			C	C(=)
Unavailability Cause			C	C(=)
Provider error				O
GSM Bearer Capability			U	C(=)

10.1.3 Parameter use

See clause 7.6 for a definition of the parameters used in addition to the following. Note that:

- a conditional parameter whose use is defined only in 3GPP TS 23.078 shall be absent if the sending entity does not support CAMEL;
- a conditional parameter whose use is defined only in 3GPP TS 23.079 [99] shall be absent if the sending entity does not support optimal routeing;
- a conditional parameter whose use is defined only in 3GPP TS 23.078 & 3GPP TS 23.079 [99] shall be absent if the sending entity supports neither CAMEL nor optimal routeing.

Interrogation Type

See 3GPP TS 23.079 [99] for the use of this parameter.

GMSC or gsmSCF address

The E.164 address of the GMSC or the gsmSCF. This parameter contains the gsmSCF address if the gsmSCF iniated call parameter is present, otherwise it is the GMSC address.

MSISDN

This is the Mobile Subscriber ISDN number assigned to the called subscriber. In the Request & Indication it is the number received by the GMSC in the ISUP IAM. If the call is to be forwarded and the HLR supports determination of the redirecting number, the HLR inserts the basic MSISDN in the Response.

See 3GPP TS 23.066 [108] for the use of this parameter and the conditions for its presence in the response.

OR Interrogation

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

OR Capability

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

CUG Interlock

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

CUG Outgoing Access

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

Number of Forwarding

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

Network Signal Info

See 3GPP TS 23.018 [97] for the conditions for the presence of the components of this parameter.

Supported CAMEL Phases

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

T-CSI Suppression

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

Offered CAMEL 4 CSIs

This parameter indicates the CAMEL phase 4 CSIs offered in the GMSC/VLR (see clause 7.6.3.36D).

Suppression Of Announcement

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

Call Reference Number

The use of this parameter and the conditions for its presence are specified in 3GPP TS 23.078 [98], 3GPP TS 23.079 [99] and 3GPP TS 23.018 [97].

Forwarding Reason

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

Basic Service Group

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

Basic Service Group 2

See 3GPP TS 23.079[99] for the use of this parameter and the conditions for its presence.

Alerting Pattern

See 3GPP TS 23.018 [97] and 3GPP TS 23.078 [98] for the use of this parameter and the conditions for its presence.

CCBS Call

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Supported CCBS Phase

This parameter indicates by its presence that CCBS is supported and the phase of CCBS which is supported.

Additional Signal Info

See 3GPP TS 23.081 [27] and 3GPP TS 23.088 [33] for the conditions for the presence of the components of this parameter.

IST Support Indicator

This parameter is used to indicate to the HLR that the GMSC supports basic IST functionality, that is, the GMSC is able to terminate the subscriber call activity that originated the IST Alert when it receives the IST Alert response indicating that the call(s) shall be terminated. If this parameter is not included in the Send Routing Information indication and the subscriber is marked as an IST subscriber, then the HLR may limit the service for the call (by barring the incoming call if it is not subject to forwarding, or suppressing Call Forwarding from the GMSC), or allow the call assuming the associated risk of not having the basic IST mechanism available.

This parameter can also indicate that the GMSC supports the IST Command, including the ability to terminate all calls being carried for the identified subscriber by using the IMSI as a key. If this additional capability is not included in the Send Routing Information indication and the subscriber is marked as an IST subscriber, then the HLR may limit the service for the subscriber (by barring the incoming calls if they are not subject to forwarding, or suppressing Call Forwarding from the GMSC), or allow the incoming calls assuming the associated risk of not having the IST Command mechanism available.

Pre-paging supported

See 3GPP TS 23.018 for the use of this parameter and the conditions for its presence.

Call Diversion Treatment Indicator

This parameter indicates whether or not call diversion is allowed.

Network Signal Info 2

See 3GPP TS 23.172 [126] for the conditions for the presence of the components of this parameter.

MT Roaming Retry Supported

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence.

Call Priority

This parameter indicates the eMLPP priority of the call (see 3GPP TS 24.067 [137]). This parameter should be present if the GMSC supports the eMLPP feature and if the call is an eMLPP call. The eMLPP priority levels A and B shall be mapped to the Call Priority level 0.

IMSI

See 3GPP TS 23.018 [97] and 3GPP TS 23.066 [108] for the use of this parameter and the conditions for its presence.

MSRN

See 3GPP TS 23.018 [97], 3GPP TS 23.066 [108] and 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence. If the NPLR returns only the MSISDN-number without Routeing Number to the GMSC, the MSISDN-number shall be returned as MSRN.

Forwarding Data

This parameter includes a number to define the forwarded-to destination, the forwarding reason and the forwarding options Notification to calling party and Redirecting presentation, and can include the forwarded-to subaddress. See 3GPP TS 23.018 [97] and 3GPP TS 23.079 [99] for the conditions for the presence of its components.

Forwarding Interrogation Required

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

Long FTN Supported

This parameter indicates that the GMSC supports Long Forwarded-to Numbers.

Suppress VT-CSI

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

Suppress Incoming Call Barring

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

gsmSCF Initiated Call

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

SuppressMTSS

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

VMSC address

See 3GPP TS 23.079 [99] and 3GPP TS 23.078 [98] for the use of this parameter and the conditions for its presence. In addition this parameter shall be present if the ReleaseResourcesSupported parameter is present.

Release Resources Supported

This parameter indicates by its presence that the MAP_RELEASE_RESOURCES service is supported at the VMSC. It shall be present if so indicated by the VMSC with MAP_PROVIDE_ROAMING_NUMBER confirm.

GMSC CAMEL Subscription Info

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

Location Information

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

Subscriber State

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

CUG Subscription Flag

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

North American Equal Access preferred Carrier Id

This parameter is returned to indicate the preferred carrier identity to be used to set-up the call (i.e. forwarding the call or establishing the roaming leg).

SS-List

This parameter includes SS-codes and will be returned as an operator option. The HLR shall not send PLMN-specific SS-codes across PLMN boundaries. However if the GMSC receives PLMN-specific SS-codes from a foreign PLMN's HLR the GMSC may ignore it. If the GMSC attempts to process the PLMN- specific SS- codes, this may lead to unpredictable behaviour but the GMSC shall continue call processing.

Basic Service Code

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

If the CAMEL service is not involved, this parameter includes the basic service code and will be returned as an operator option. The HLR shall not send a PLMN-specific Basic Service Code across PLMN boundaries. However if the GMSC receives a PLMN-specific Basic Service Code from a foreign PLMN's HLR the GMSC may ignore it. If the GMSC attempts to process the PLMN specific Basic Service codes, this may lead to unpredictable behaviour but the GMSC shall continue call processing.

CCBS Target

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Keep CCBS Call Indicator

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

IST Alert Timer

It includes the IST Alert timer value that must be used to inform the HLR about the call activities that the subscriber performs. This parameter is only sent to the GMSC in response to a Send Routing Information request which indicates the the GMSC supports IST.

Number Portability Status

This parameter indicates the number portability status of the subscriber. This parameter may be present if the sender of SRIack is NPLR.

Supported CAMEL Phases in VMSC

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078.

Offered CAMEL 4 CSIs in VMSC

This parameter is defined in clause 7.6.3.36F.

MSRN 2

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.172 [126].

Forwarding Data 2

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.172 [126].

SS-List 2

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.172 [126].

Basic Service Code 2

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.172 [126].

Allowed Services

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.172 [126].

Unavailability Cause

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.172 [126].

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Unknown Subscriber;

  The diagnostic for the Unknown Subscriber error may indicate "NPDB Mismatch".

- Number changed;

- Call Barred;

  This error will indicate that either incoming calls are barred for this MS or that calls are barred due to Operator Determined Barring (see 3GPP TS 22.041 [8] for a definition of this network feature);

- CUG Reject;

  The value of this error cause will indicate the reason for CUG Reject;

- Bearer Service Not Provisioned;

- Teleservice Not Provisioned;

  A subscription check has been performed and the call has not passed the check due to incompatibility with regard to the requested service. Depending on the nature of the incompatibility, either of these messages will be returned;

- Facility Not Supported;

- Absent Subscriber;

  This indicates that the location of the MS is not known (either the station is not registered and there is no location information available or the Provide Roaming Number procedure fails due to IMSI detached flag being set), or the GMSC requested forwarding information with a forwarding reason of not reachable, and the call forwarding on MS not reachable service is not active; this may also indicate that the MS has moved to a new MSC/VLR and that MT Roaming Retry is requested (see 3GPP TS 23.018 [97]);

- Busy Subscriber;

  This indicates that Call Forwarding on Busy was not active for the specified basic service group when the GMSC requested forwarding information with a forwarding reason of busy;

  The error may also indicate that the subscriber is busy due to an outstanding CCBS recall. In the error data it may then be specified that CCBS is possible for the busy encountered call;

- No Subscriber Reply;

  This indicates that Call Forwarding on No Reply was not active for the specified basic service group when the GMSC requested forwarding information with a forwarding reason of no reply;

- OR Not Allowed;

  This indicates that the HLR is not prepared to accept an OR interrogation from the GMSC, or that calls to the specified subscriber are not allowed to be optimally routed;

- Forwarding Violation;

- System Failure;

- Data Missing;

- Unexpected Data Value.

See clause 7.6.1.4 for a definition of these errors.

Provider error

These are defined in clause 7.6.

GSM Bearer Capability

This information is passed according to the rules specified in 3GPP TS 29.007 [56]. There may be two GSM Bearer Capabilities supplied.

10.2 MAP_PROVIDE_ROAMING_NUMBER service

10.2.1 Definition

This service is used between the HLR and VLR. The service is invoked by the HLR to request a VLR to send back a roaming number to enable the HLR to instruct the GMSC to route an incoming call to the called MS.

This service is also used between old VLR and new VLR during the MT Roaming Forwarding procedure. The service is invoked by the old VLR to request a roaming number from the new VLR to be able to route an incoming call to the called UE.

This is a confirmed service which uses the primitives described in table 10.2/1.

10.2.2 Service primitives

Table 10.2/1: MAP_PROVIDE_ROAMING_NUMBER parameters

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
MSC Number	M	M(=)		
MSISDN	U	C(=)		
LMSI	C	C(=)		
GSM Bearer Capability	C	C(=)		
Network Signal Info	C	C(=)		
Suppression Of Announcement	C	C(=)		
Call Reference Number	C	C(=)		
GMSC Address	C	C(=)		
OR Interrogation	C	C(=)		
OR Not Supported in GMSC	C	C(=)		
Alerting Pattern	C	C(=)		
CCBS Call	C	C(=)		
Supported CAMEL Phases in interrogating node	C	C(=)		
Additional Signal Info	C	C(=)		
Pre-paging supported	C	C(=)		
Long FTN Supported	C	C(=)		
Suppress VT-CSI	C	C(=)		
Offered CAMEL 4 CSIs in interrogating node	C	C(=)		
MT Roaming Retry Supported	U	C(=)		
Paging Area	U	C(=)		
Call Priority	U	C(=)		
MTRF Indicator	U	C(=)		
Old MSC Number	U	C (=)		
Last used LTE PLMN ID	U	C(=)		
Roaming Number			C	C(=)
VMSC address			U	C(=)
ReleaseResourcesSupported			U	C(=)
User error			C	C(=)

Parameter name	Request	Indication	Response	Confirm
Provider error				O

10.2.3 Parameter use

See clause 7.6 for a definition of the parameters used, in addition to the following. Note that:

- a conditional parameter whose use is defined only in 3GPP TS 23.078 [98] shall be absent if the sending entity does not support CAMEL;
- a conditional parameter whose use is defined only in 3GPP TS 23.079 [99] shall be absent if the sending entity does not support optimal routeing;
- a conditional parameter whose use is defined only in 3GPP TS 23.078 [98] & 3GPP TS 23.079 [99] shall be absent if the sending entity supports neither CAMEL nor optimal routeing.

IMSI

This is the IMSI of the called Subscriber.

MSC Number

This is the ISDN number assigned to the MSC currently serving the MS. When the service is used between HLR and VLR, the MSC number will have been stored in the HLR as provided at location updating. When used between old VLR and new VLR during an MT Roaming Forwarding procedure, the MSC number will have been provided at location cancelling or within Send Identification.

MSISDN

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

LMSI

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence. In addition, for the mobile terminating roaming forwarding procedure between the old VLR and the new VLR, this parameter shall be present if the MTRF Indicator is present and the old VLR has received the new LMSI in Cancel Location from the HLR or in Send Identification from the new VLR.

GSM Bearer Capability

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

This information is passed according to the rules specified in TS 3GPP TS 29.007 [56].

There may be two GSM Bearer Capabilities supplied.

Network Signal Info

See 3GPP TS 23.018 [97] for the conditions for the presence of the components of this parameter.

Suppression Of Announcement

The use of this parameter and the requirements for its presence are specified in 3GPP TS 23.078 [98].

Call Reference Number

The use of this parameter and the conditions for its presence are specified in 3GPP TS 23.078 [98], 3GPP TS 23.079 [99] and 3GPP TS 23.018 [97].

GMSC Address

The use of this parameter and the conditions for its presence are specified in 3GPP TS 23.078 [98], 3GPP TS 23.079 [99] and 3GPP TS 23.018 [97].

OR Interrogation

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

OR Not Supported in GMSC

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

Supported CAMEL Phases in interrogating node

This parameter is defined in clause 7.6.3.36I.Alerting Pattern

See 3GPP TS 23.078 [98] for the use of this parameter and the conditions for its presence.

CCBS Call

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Additional Signal Info

See 3GPP TS 23.081 [27] for the conditions for the presence of the components of this parameter.

Pre-paging supported

See 3GPP TS 23.018 for the use of this parameter and the conditions for its presence. This information element is not applicable for MTRF, and shall be ignored if received while the MTRF Indicator is present.

Long FTN supported

See 3GPP TS 23.082 for the use of this parameter and the conditions for its presence.

Suppress VT-CSI

See 3GPP TS 23.078 for the use of this parameter and the conditions for its presence.

Offered CAMEL 4 CSIs in interrogating node

This parameter is defined in clause 7.6.3.36E.

MT Roaming Retry Supported

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence. This information element is not applicable for MTRF, and shall be ignored if received while the MTRF Indicator is present.

Paging Area

See 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23] for the use of this parameter and the conditions for its presence. This information element is not applicable for MTRF, and shall be ignored if received while the MTRF Indicator is present.

Call Priority

This parameter indicates the eMLPP priority of the call (see 3GPP TS 24.067 [137]). This parameter should be present if the HLR supports this parameter and if the Call Priority was received in the MAP_SEND_ROUTING_INFORMATION request.

MTRF Indicator

This indicator indicates by its presence that the service is used between old VLR and new VLR during an MT Roaming Forwarding procedure. See 3GPP TS 23.018 [97].

Old MSC Number

This parameter refers to the E.164 address of the old MSC. The use of this parameter is specified in 3GPP TS 23.018 [97]. This information element is applicable only if the MTRF Indicator is set.

Last used LTE PLMN ID

See 3GPP TS 23.272 [143] for the use of this parameter and the conditions for its presence. This information element is applicable only if the MTRF Indicator is set.

Roaming Number

See 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence.

VMSC address

See 3GPP TS 23.079 [99], 3GPP TS 23.078 [98] and 3GPP TS 23.018 [97] for the use of this parameter and the conditions for its presence. This parameter shall be present during the Mobile Terminating Roaming Forwarding Call during Retrieval of Routeing Information procedure if an MSRN is allocated by the new MSC/VLR.

ReleaseResourcesSupported

This parameter indicates by its presence that the MAP_RELEASE_RESOURCES service is supported at the VMSC.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Absent Subscriber;

  This error will be returned if the IMSI detach flag is set.

- No Roaming Number Available;

- OR Not Allowed;

  This indicates that the MAP_PROVIDE_ROAMING_NUMBER indication included the OR interrogation indicator, but the VLR does not support optimal routeing.

- Facility Not Supported;

- System Failure;

- Data Missing;

- Unexpected Data Value.

See clause 7.6 for a definition of these reasons.

Provider error

These are defined in clause 7.6.

10.3 MAP_RESUME_CALL_HANDLING service

10.3.1 Definition

This service is used between the terminating VMSC and the GMSC. The service is invoked by the terminating VMSC to request the GMSC to resume handling the call and forward it to the specified destination.

This is a confirmed service which uses the Primitives listed in table 10.3/1.

10.3.2 Service primitives

Table 10.3/1: MAP_RESUME_CALL_HANDLING parameters

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Call Reference Number	C	C(=)		
Basic Service Group	C	C(=)		
Basic Service Group 2	C	C(=)		

IMSI	C	C(=)		
Forwarding Data	C	C(=)		
CUG Interlock	C	C(=)		
CUG Outgoing Access	C	C(=)		
O-CSI	C	C(=)		
D-CSI	C	C(=)		
CCBS Target	C	C(=)		
UU Data	C	C(=)		
UUS CF Interaction	C	C(=)		
All Information Sent	C	C(=)		
MSISDN	C	C(=)		
MT Roaming Retry	U	C(=)		
User error			C	C(=)
Provider error				O

10.3.3 Parameter use

Information received in subsequent segment of a segmented dialogue shall not overwrite information received in an earlier segment.

See clause 7.6 for a definition of the parameters used, in addition to the following.

Call Reference Number

See 3GPP TS 23.079 [99] for the use of this parameter. This parameter shall be present in the first segment of the dialogue.

Basic Service Group

See 3GPP TS 23.079 [99] for the use of this parameter. This parameter shall be present in the first segment of the dialogue.

Basic Service Group 2

See 3GPP TS 23.079[99] for the use of this parameter. If this parameter is present, it shall be in the first segment of the dialogue.

IMSI

This is the IMSI of the forwarding Subscriber. This parameter shall be present in the first segment of the dialogue.

Forwarding Data

This parameter includes a number to define the forwarded-to destination, the forwarding reason and the forwarding options Notification to calling party and Redirecting presentation, and can include the forwarded-to subaddress. See 3GPP TS 23.079 [99] for the conditions for the presence of its components. This parameter shall be present in a first segment of the dialogue.

CUG Interlock

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

CUG Outgoing Access

See 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence.

O-CSI

See 3GPP TS 23.078 for the use of this parameter and the conditions for its presence.

For CAMEL phases 1 & 2, the O-CSI shall contain only one set of O-BCSM TDP data.

D-CSI

The Dialled Services-CSI.

See 3GPP TS 23.078 for the use of this parameter and the conditions for its presence.

CCBS Target

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

UU Data

See 3GPP TS 23.087 for the use of this parameter and the conditions for its presence.

UUS CF Interaction

See 3GPP TS 23.087 for the use of this parameter and the conditions for its presence.

All Information Sent

This parameter is set when the VMSC has sent all information to GMSC.

MT Roaming Retry

See 3GPP TS 23.018 [97], 3GPP TS 23.012 [23] and 3GPP TS 23.079 [99] for the use of this parameter and the conditions for its presence. When this parameter is present, only the Call Reference Number and All Information Sent IEs shall be present; the other IEs shall be ignored by the GMSC if received.

MSISDN

This parameter is the basic MSISDN of the forwarding subscriber. It shall be present if the VMSC supports determination of the redirecting number.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Optimal Routeing not allowed;
- Forwarding failed;
- Unexpected Data Value;
- Data Missing.

Provider error

These are defined in clause 7.6.

10.4 MAP_PREPARE_GROUP_CALL service

10.4.1 Definition

This service is used by the Anchor_MSC to inform the Relay_MSC about a group call set-up.

The MAP_PREPARE_GROUP_CALL service is a confirmed service using the service primitives given in table 10.4/1.

10.4.2 Service primitives

Table 10.4/1: MAP_PREPARE_GROUP_CALL service

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Teleservice	M	M(=)		
ASCI Call Reference	M	M(=)		
Ciphering Algorithm	M	M(=)		
Group Key Number VK-Id	C	C(=)		
VSTK Key	C	C(=)		

VSTK-RAND	C	C(=)		
Priority	C	C(=)		
CODEC-Information	M	M(=)		
Uplink Free Indicator	M	M(=)		
Talker Channel Parameter	C	C(=)		
Uplink Reply Indicator	C	C(=)		
Group Call Number			M	M(=)
User Error			C	C(=)
Provider Error				O

10.4.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1.

Teleservice

Voice Broadcast Service or Voice Group Call Service.

ASCI Call Reference

Broadcast call reference or group call reference. This item is used to access the VBS-GCR or VGCS-GCR within the Relay_MSC.

Ciphering Algorithm

The ciphering algorithm to be used for the group call.

Group Key Number VK-Id

This Group Key Number has to be broadcast and is used by the mobile station to derive the key for ciphering on the radio interface (see 3GPP TS 43.020 [24]). Values 2 to 15 are reserved for future use.

Shall be present if the ciphering applies.

VSTK

The VGCS/VBS Short Term Key is used to derive the key for ciphering on the radio interface (see 3GPP TS 43.020 [24]).

Shall be present if the ciphering applies.

VSTK-RAND

This random number has to be broadcast and is used by the mobile station to derive the group key for ciphering on the radio interface (see 3GPP TS 43.020 [24]).

Shall be present if the ciphering applies.

Priority

Default priority level related to the call if eMLPP applies.

CODEC-Information

Information on the codecs allowed for this call.

Uplink Free Indicator

A flag indicating whether the call is initiated from a dispatcher.

Talker Channel Parameter

A flag indicating by its presence that a dedicated channel shall be established and maintained for the talking service subscriber.

Uplink Reply Indicator

A flag indicating by its presence that the uplink reply procedure is applicable for the voice group call or voice broadcast call.

Group Call Number

This temporary allocated E.164 number is used for routing the call from the Anchor MSC to the Relay MSC.

User Error

For definition of this parameter see clause 7.6.1 The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- No Group Call Number available;
- System Failure;
- Unexpected Data Value.

Provider Error

See definition of provider error in clause 7.6.1.

10.5 MAP_PROCESS_GROUP CALL_SIGNALLING service

10.5.1 Definitions

This service is used between Relay MSC and Anchor MSC for transmission of Group Call notifications.

The MAP_PROCESS_GROUP_CALL_SIGNALLING service is a non-confirmed service using the service primitives given in table 10.5/1.

10.5.2 Service primitives

Table 10.5/1: MAP_PROCESS_GROUP_CALL_SIGNALLING service

Parameter name	Request	Indication
Invoke Id	M	M(=)
Uplink Request	C	C(=)
Uplink Release Indication	C	C(=)
AN-APDU	C	C(=)
Release Group Call	C	C(=)
Talker Priority	C	C(=)
Additional Info	C	C(=)
Emergency Mode Reset Command Flag	C	C(=)

10.5.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1

Uplink Request

This information element indicates to the anchor MSC that a service subscriber roaming in the relay MSC area requests access to the uplink.

Uplink Release Indication

This information element if included by the Relay MSC indicates to the Anchor MSC that the uplink has become free.

AN-APDU

This parameter contains the Notification Data message as defined in3GPP TS 48.008 [49].

Release Group Call

This information element if included by the Relay MSC indicates to the Anchor MSC that the service subscriber who has initiated the call and who currently has access to the uplink terminates the call.

Talker Priority

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Additional Info

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Emergency Mode Reset Command Flag

For the definition and use of this parameter see 3GPP TS 43.068 [100]

10.6 MAP_FORWARD_GROUP_CALL_SIGNALLING service

10.6.1 Definitions

This service is used between Anchor MSC and Relay MSC for transmission of Group Call notifications.

The MAP_FORWARD_GROUP_CALL_SIGNALLING service is a non-confirmed service using the service primitives given in table 10.6/1.

10.6.2 Service primitives

Table 10.6/1: MAP_FORWARD_GROUP_CALL_SIGNALLING service

Parameter name	Request	Indication
Invoke Id	M	M(=)
IMSI	C	C(=)
Uplink Request Acknowledgement	C	C(=)
Uplink Release Indication	C	C(=)
Uplink Reject Command	C	C(=)
Uplink Seized Command	C	C(=)
Uplink Release Command	C	C(=)
AN-APDU	C	C(=)
State Attributes	C	C(=)
Talker Priority	C	C(=)
Additional Info	C	C(=)
Emergency Mode Reset Command Flag	C	C(=)
SM RP UI	C	C(=)

10.6.3 Parameter definitions and use

IMSI

Identity of the service subscriber who has established the call and who is allowed to terminate the call.

Invoke Id

See definition in clause 7.6.1.

Uplink Request Acknowledgement

This information element is used for positive acknowledgement of an uplink request.

Uplink Release Indication

This information element if included by the Anchor MSC indicates to the Relay MSC that the uplink has become free.

Uplink Reject Command

This information element is used for negative acknowledgement of an uplink request.

Uplink Seized Command

This information element if included by the Anchor MSC indicates to the Relay MSC that the uplink is no longer free.

Uplink Release Command

This information element if included by the Anchor MSC indicates to the Relay MSC that the uplink which is granted to a MS in the relay MSC area shall be released.

AN-APDU

This parameter contains the Notification Data message as defined in 3GPP TS 48.008 [49]

State Attributes

This information element is used to allow service logic running in an Anchor MSC to mute a VGCS talker even when the talker is served on a Relay MSC. The IE is used to build a GCC message that provides a mechanism to induce the VGCS talker terminal to mute/unmute the downlink at the Anchor MSC, as defined in 3GPP TS 44.068.

Talker Priority

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Additional Info

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Emergency Mode Reset Command Flag

For the definition and use of this parameter see 3GPP TS 43.068 [100]

SM RP UI

See definition in clause 7.6.8. The short message transfer protocol data unit received from the Service Centre is inserted in this parameter.

10.7 MAP_SEND_GROUP_CALL_END_SIGNAL service

10.7.1 Definitions

This service is used between the Relay MSC and the Anchor MSC. When the VGCS/ VBS calling service subscriber is in the Relay MSC area the MAP_SEND_GROUP_CALL_END_SIGNAL indicates that at least the downlink channel in the originating cell is established. For all other VGCS/ VBS call set-up scenarios (i.e. calling service subscriber in Anchor MSC area, calling service subscriber in other Relay MSC area, dispatcher originated call) the MAP_SEND_GROUP_CALL_END_SIGNAL indicates that at least the downlink channel in any one cell within the VGCS/ VBS call area in the Relay MSC is established. The response is used by the Anchor MSC to inform the Relay MSC that all resources for the call can be released in the Relay MSC because the call has been released in the Anchor MSC.

The MAP_SEND_GROUP_CALL_END_SIGNAL service is a confirmed service using the service primitives given in table 10.7/1.

10.7.2 Service primitives

Table 10.7/1: MAP_SEND_GROUP_CALL_END_SIGNAL service

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
Talker Priority	C	C(=)		
Additional Info	C	C(=)		
Provider Error				O

10.7.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1

IMSI

Identity of the service subscriber who has established the call and who is allowed to terminate the call.

Shall be present if the call was established by a service subscriber roaming in the relay MSC area.

Talker Priority

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Additional Info

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Provider Error

See definition of provider error in clause 7.6.1.

10.7A MAP_SEND_GROUP_CALL_INFO service

10.7A.1 Definitions

This service is used in a RANflex configuration (see 3GPP TS 23.236 [133]) between the subscriber's visited MSC and group call serving MSC of the subscriber's location area.

The MAP_SEND_GROUP_CALL_INFO service is a confirmed service using the service primitives given in table 10.7A/1.

10.7A.2 Service primitives

Table 10.7A/1: MAP_SEND_GROUP_CALL_INFO service

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Requested Info	M	M(=)		
Teleservice	M	M(=)		
Cell Id	C	C(=)		
Group Id	M	M(=)		
IMSI	C	C(=)	C	C(=)
Talker Priority	C	C(=)		
Additional Info	C	C(=)	C	C(=)
TMSI	C	C(=)		
CKSN	C	C(=)		
Anchor MSC Address			C	C(=)

ASCI Call Reference			C	C(=)
Additional Subscriptions			C	C(=)
Kc			C	C(=)
User Error			C	C(=)
Provider Error				O

10.7A.3 Parameter definitions and use

Invoke Id

See definition in clause 7.6.1

Requested Info

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Teleservice

Voice Broadcast Service or Voice Group Call Service.

Cell Id

Identity of the initiating service subscriber's current cell.

Group Id

For the definition and use of this parameter see 3GPP TS 43.068 [100] and 3GPP TS 43.069 [101].

If prefixes are used together with group IDs, the most significant digit of the Group Id contains the prefix.

IMSI

If sent in the request: Identity of the service subscriber who has established the call and who is allowed to terminate the call.
If sent in the response: Identity of the uplink requesting service subscriber.

Talker Priority

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Additional Info

For the definition and use of this parameter see 3GPP TS 43.068 [100]

TMSI

See definition in clause 7.6.2.

CKSN

See clause 7.6.7 for the use of this parameter.

Anchor MSC Address

For the definition and use of this parameter see 3GPP TS 43.068 [100] and 3GPP TS 43.069 [101]

ASCI Call Reference

For the definition and use of this parameter see 3GPP TS 43.068 [100] and 3GPP TS 43.069 [101]

Additional Subscriptions

For the definition and use of this parameter see 3GPP TS 43.068 [100]

Kc

See clause 7.6.7 for the use of this parameter.

User Error

For definition of this parameter see clause 7.6.1 The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- System Failure;
- Unexpected Data Value;
- Data Missing
- TeleserviceNotProvisioned;
- Unknown Subscriber;
- Ongoing Call.

Provider Error

See definition of provider error in clause 7.6.1.

10.8 Void

10.9 Void

10.10 MAP_SET_REPORTING_STATE service

10.10.1 Definition

This service is used between the HLR and the VLR to set the reporting state for a requested service. It is a confirmed service using the service primitives shown in table 10.10/1.

10.10.2 Service primitives

Table 10.10/1: MAP_SET_REPORTING_STATE parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
LMSI	C	C(=)		
CCBS Monitoring	C	C(=)		
CCBS Subscriber Status			C	C(=)
User error			C	C(=)
Provider error				O

10.10.3 Parameter use

See clause 7.6 for a definition of the parameters used, in addition to the following.

IMSI

The IMSI is a mandatory parameter if the service is used as the only one in a dialogue.

CCBS Monitoring

This parameter indicates whether monitoring for CCBS shall be started or stopped. If it indicates that monitoring shall be started this service corresponds to the message 'Start Reporting' in 3GPP TS 23.093 [107]; if it indicates that monitoring shall be stopped this service corresponds to the message 'Stop Reporting' in 3GPP TS 23.093 [107].

CCBS Subscriber Status

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- System Failure;
- Unidentified Subscriber;
- Unexpected Data Value;
- Data Missing;
- Resource Limitation;
- Facility Not Supported.

NOTE: This error is reserved for future use.

Provider error

These are defined in clause 7.6.

10.11 MAP_STATUS_REPORT service

10.11.1 Definition

This service is used by the VLR to report an event or call outcome to the HLR. It is a confirmed service using the service primitives shown in table 10.11/1.

10.11.2 Service primitives

Table 10.11/1: MAP_STATUS_REPORT parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
CCBS Subscriber Status	C	C(=)		
Monitoring Mode	C	C(=)		
Call Outcome	C	C(=)		
User error			C	C(=)
Provider error				O

10.11.3 Parameter use

See clause 7.6 for a definition of the parameters used, in addition to the following.

CCBS Subscriber Status

If this parameter is present without Monitoring Mode and Call Outcome this service corresponds to the message 'Event Report' in 3GPP TS 23.093 [107]. See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Monitoring Mode

If this parameter is present with CCBS Call Outcome this service corresponds to the message 'CCBS Call Report' in 3GPP TS 23.093 [107]. See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Call Outcome

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- Unknown Subscriber;
- System Failure;
- Unexpected Data Value;
- Data Missing.

Provider error

These are defined in clause 7.6.

10.12 MAP_REMOTE_USER_FREE service

10.12.1 Definition

This service is used between the HLR and the VLR to report that the B subscriber is now idle and that the A subscriber can be notified. It is a confirmed service using the service primitives shown in table 10.12/1.

10.12.2 Service primitives

Table 10.12/1: MAP_REMOTE_USER_FREE parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
Call Info	M	M(=)		
CCBS Feature	M	M(=)		
Translated B Number	M	M(=)		
Replace B Number	C	C(=)		
Alerting Pattern	C	C(=)		
RUF Outcome			C	C(=)
User error			C	C(=)
Provider error				O

10.12.3 Parameter use

See clause 7.6 for a definition of the parameters used, in addition to the following.

Call Info

See 3GPP TS 23.093 [107] for the use of this parameter.

CCBS Feature

See 3GPP TS 23.093 [107] for the conditions for the presence of the parameters included in the CCBS feature.

Translated B Number

See 3GPP TS 23.093 [107] for the use of this parameter.

Replace B Number

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Alerting Pattern

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

RUF Outcome

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- Unexpected Data Value;
- Data Missing;
- Incompatible Terminal;
- This error is returned by the responder when the terminal used for CCBS activation is not compatible with the terminal used for the CCBS recall. For details refer to 3GPP TS 24.008 [35];
- Absent Subscriber (IMSI Detach; Restricted Area; No Page Response);
- System Failure;
- Busy Subscriber (CCBS Busy).

Provider error

These are defined in clause 7.6.

10.13 MAP_IST_ALERT service

10.13.1 Definition

This service is used between the MSC (Visited MSC or Gateway MSC) and the HLR, to report that the IST timer running for a call for the Subscriber has expired. It is a confirmed service using the service primitives shown in table 10.13/1.

10.13.2 Service primitives

Table 10.13/1: MAP_IST_ALERT parameters

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
IST Alert Timer			C	C(=)
IST Information Withdraw			C	C(=)
Call termination Indicator			C	C(=)
User error			C	C(=)
Provider error				O

10.13.3 Parameter use

All parameters are described in clause 7.6. The following clarifications are applicable:

IST Alert Timer

If included in the IST Alert response, it includes the new IST Alert timer value that must be used to inform the HLR about the call activities that the subscriber performs.

IST Information Withdraw

If included in the IST Alert response, this parameter is used to indicate that the IST condition has been removed for the subscriber. When the MSC receives this parameter, IST control for that call shall be terminated.

Call termination Indicator

If included in the IST Alert response, this parameter is used to indicate whether the MSC shall terminate the call activity that had previously triggered the IST Alert procedure, or it shall also release all other call activities for the specified subscriber (outgoing call activities if the IST Alert is initiated by the VMSC, or incoming call activities if the IST Alert is initiated by the GMSC). Release of all other call activities is possible only if the MSC has the capability to link the call activities for the Subscriber by using the IMSI as key.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- System Failure;
- Unexpected Data Value;
- Resource Limitation;
- Facility Not Supported;
- Unknown Subscriber.

10.14 MAP_IST_COMMAND service

10.14.1 Definition

This service is used by the HLR to instruct the MSC (Visited MSC or Gateway MSC) to terminate ongoing call activities for a specific subscriber. It is a confirmed service using the service primitives shown in table 10.14/1.

10.14.2 Service primitives

Table 10.14/1: MAP_IST_COMMAND parameters

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
User error			C	C(=)
Provider error				O

10.14.3 Parameter use

All parameters are described in clause 7.6. The following clarifications are applicable:

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- System Failure;
- Unexpected Data Value;
- Resource Limitation;
- Facility Not Supported;

- Unknown Subscriber.

10.15 MAP_RELEASE_RESOURCES service

10.15.1 Definition

This service is used between the GMSC and the terminating VMSC. The service is invoked by the GMSC to request the VMSC to release the resources associated with the specified MSRN.

This is a confirmed service which uses the Primitives listed in table 10.15/1.

10.15.2 Service primitives

Table 10.15/1: MAP_RELEASE_RESOURCES parameters

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
MSRN	M	M(=)		
User error			C	C(=)
Provider error				O

10.15.3 Parameter use

MSRN

See 3GPP TS 23.018 [97] for the use of this parameter.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- System Failure;
- Unexpected Data Value;

Provider error

These are defined in clause 7.6.

11 Supplementary services related services

11.1 MAP_REGISTER_SS service

11.1.1 Definition

This service is used between the MSC and the VLR and between the VLR and the HLR to register data related to a supplementary service. The VLR will relay the message to the HLR.

The service is a confirmed service and uses the service primitives shown in table 11.1./1.

11.1.2 Service primitives

Table 11.1/1: MAP_REGISTER_SS parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)

SS-Code	M	M(=)		
Basic service	C	C(=)		
Forwarded-to number with subaddress	C	C(=)		
No reply condition time	C	C(=)		
EMLPP default priority	C	C(=)	C	C(=)
Long FTN Supported	C	C(=)		
NbrUser	C	C(=)	C	C(=)
Forwarding information			C	C(=)
User error			C	C(=)
Provider error				O

11.1.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

SS-Code

This parameter indicates the supplementary service which the mobile subscriber wants to register.

Basic service

This parameter indicates for which basic service group the supplementary service is to be registered. If it is not included, the registration request applies to all basic services.

Forwarded-to number with subaddress

This parameter is obligatory if the registration applies to one or more call forwarding supplementary services. It can optionally include a sub-address.

No reply condition time

This parameter is included if the registration applies to the Call Forwarding on No Reply supplementary service (or a superset of this service) and the mobile subscriber supplies a value for this time.

EMLPP default priority

This parameter is sent by the initiator to register the eMLPP default priority level and is returned by the responder at successful outcome of the service.

Long FTN Supported

This parameter indicates that the mobile station supports Long Forwarded-to Numbers.

NbrUser

This parameter is sent by the initiator to register the MC maximum number of user defined circuit switched bearers to be used.

Forwarding information

This parameter is returned by the responder at successful outcome of the service, if the registration request concerned one or a group of Call Forwarding supplementary services.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- System failure;
- Data missing;
- Unexpected data value;

- Call Barred;
- Bearer service not provisioned;
- This error is returned only if not even a subset of the requested bearer service group has been subscribed to;
- Teleservice not provisioned;

  This error is returned only if not even a subset of the requested teleservice group has been subscribed to;
- Illegal SS operation;
- SS error status;
- SS incompatibility.

Provider error

See clause 7.6.1 for the use of this parameter.

11.2 MAP_ERASE_SS service

11.2.1 Definition

This service is used between the MSC and the VLR and between the VLR and the HLR to erase data related to a supplementary service. The VLR will relay the message to the HLR.

The service is a confirmed service and uses the service primitives shown in table 11.2/1.

11.2.2 Service primitives

Table 11.2/1: MAP_ERASE_SS parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS-Code	M	M(=)		
Basic service	C	C(=)		
Forwarding information			C	C(=)
User error			C	C(=)
Provider error				O

11.2.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

SS-Code

This parameter indicates the supplementary service which the mobile subscriber wants to erase.

Basic service

This parameter indicates for which basic service group the supplementary service should be erased. If it is not included, the erasure request applies to all basic services.

Forwarding information

This parameter is returned by the responder at successful outcome of the service, if the erasure request concerned one or a group of Call Forwarding supplementary services.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data Missing;
- Unexpected data value;
- Bearer service not provisioned;

  This error is returned only if not even a subset of the requested bearer service group has been subscribed to;
- Teleservice not provisioned;

  This error is returned only if not even a subset of the requested teleservice group has been subscribed to;
- Call Barred;
- Illegal SS operation;
- SS error status.

Provider error

See clause 7.6.1 for the use of this parameter.

11.3 MAP_ACTIVATE_SS service

11.3.1 Definition

This service is used between the MSC and the VLR and between the VLR and the HLR to activate a supplementary service. The VLR will relay the message to the HLR.

The service is a confirmed service and uses the service primitives shown in table 11.3/1.

11.3.2 Service primitives

Table 11.3/1: MAP_ACTIVATE_SS parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS-Code	M	M(=)		
Long FTN Supported	C	C(=)		
Basic service	C	C(=)		
Forwarding information			C	C(=)
Call barring information			C	C(=)
SS-Data			C	C(=)
User error			C	C(=)
Provider error				O

11.3.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

SS-Code

This parameter indicates the supplementary service which the mobile subscriber wants to activate.

Basic service

This parameter indicates for which basic service groups the requested supplementary service(s) should be activated. If it is not included, the activation request applies to all basic services.

Forwarding information

This parameter is returned by the responder at successful outcome of the service, if the activation request concerned Call Forwarding.

Long FTN Supported

This parameter indicates that the mobile station supports Long Forwarded-to Numbers.

Call barring information

This parameter is returned by the responder at successful outcome of the service, if the activation request concerned Call Barring.

SS-Data

This parameter is returned by the responder at successful outcome of the service, if the activation request concerned for example Call Waiting.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data Missing;
- Unexpected data value;
- Bearer service not provisioned;
- This error is returned only if not even a subset of the requested bearer service group has been subscribed to.
- Teleservice not provisioned;
- This error is returned only if not even a subset of the requested teleservice group has been subscribed to.
- Call Barred;
- Illegal SS operation;
- SS error status;
- SS subscription violation;
- SS incompatibility;
- Negative PW check;
- Number Of PW Attempts Violation.

Provider error

See clause 7.6.1 for the use of this parameter.

11.4 MAP_DEACTIVATE_SS service

11.4.1 Definitions

This service is used between the MSC and the VLR and between the VLR and the HLR to deactivate a supplementary service. The VLR will relay the message to the HLR.

The service is a confirmed service and uses the service primitives shown in table 11.4/1.

11.4.2 Service primitives

Table 11.4/1: MAP_DEACTIVATE_SS parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS-Code	M	M(=)		
Basic service	C	C(=)		
Forwarding information			C	C(=)
Call barring information			C	C(=)
SS-Data			C	C(=)
User error			C	C(=)
Provider error				O

11.4.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

SS-Code

This parameter indicates the supplementary service which the mobile subscriber wants to deactivate.

Basic service

This parameter indicates for which basic service group the requested supplementary service(s) should be deactivated. If it is not included the deactivation request applies to all basic services.

Forwarding information

This parameter is returned by the responder at successful outcome of the service, if the deactivation request concerned one or a group of Call Forwarding supplementary services.

Call barring information

This parameter is returned by the responder at successful outcome of the service, if the activation request concerned one or a group of Call Barring supplementary services.

SS-Data

This parameter is returned by the responder at successful outcome of the service, for example if the deactivation request concerned the Call Waiting supplementary service.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data Missing;
- Unexpected data value;
- Bearer service not provisioned;

  This error is returned only if not even a subset of the requested bearer service group has been subscribed to;

- Teleservice not provisioned;

  This error is returned only if not even a subset of the requested teleservice group has been subscribed to;

- Call Barred;
- Illegal SS operation;
- SS error status;
- SS subscription violation;
- Negative PW check;
- Number Of PW Attempts Violation.

Provider error

See clause 7.6.1 for the use of this parameter.

11.5 MAP_INTERROGATE_SS service

11.5.1 Definitions

This service is used between the MSC and the VLR and between the VLR and the HLR to retrieve information related to a supplementary service. The VLR will relay the message to the HLR if necessary.

The service is a confirmed service and consists of four service primitives.

11.5.2 Service primitives

The service primitives are shown in table 11.5/1.

Table 11.5/1: MAP_INTERROGATE_SS parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS-Code	M	M(=)		
Basic service	C	C(=)		
Long FTN Supported	C	C(=)		
SS-Status			C	C(=)
Basic service Group LIST			C	C(=)
Forwarding feature LIST			C	C(=)
CLI restriction Info			C	C(=)
EMLPP Info			C	C(=)
MC Information			C	C(=)
CCBS Feature LIST			C	C(=)
User error			C	C(=)
Provider error				O

11.5.3 Parameter use

For additional information on parameter use refer to the GSM 04.8x and 04.9x-series of technical specifications.

Invoke id

See clause 7.6.1 for the use of this parameter.

SS-Code

The mobile subscriber can only interrogate a single supplementary service per service request.

Basic service

This parameter indicates for which basic service group the given supplementary service is interrogated. If it is not included, the interrogation request applies to all basic services.

SS-Status

This parameter is included by the responder if:

- the interrogated supplementary service can only be subscribed for all applicable basic services simultaneously; or
- the interrogated supplementary service is not active for any of the interrogated basic services, or
- the interrogation was for the CCBS supplementary service and no CCBS request is active or the service is not provisioned.

Basic service group LIST

This parameter LIST is used to include one or a series of basic service groups for which the interrogated supplementary service is active. If the interrogated supplementary service is not active for any of the interrogated (and provisioned) basic service groups, the SS-Status parameter is returned.

Long FTN Supported

This parameter indicates that the mobile station supports Long Forwarded-to Numbers.

Forwarding feature LIST

The forwarding feature parameter is described in clause 7.6.4. A list of one or more forwarding features is returned by the responder when the interrogation request applied to Call Forwarding supplementary service.

If no basic service code parameter is provided within this sequence, the forwarding feature parameter applies to all provisioned basic services.

CLI restriction Info

The CLI-RestrictionInfo parameter is returned by the responder when the interrogation request applies to the CLIR supplementary service.

EMLPP Info

The eMLPP info (maximum entitled priority and default priority) is returned by the responder if the interrogation request applies to the eMLPP supplementary service.

MC Information

The MC information (NbrSB, NbrUser and NbrSN) is returned by the responder if the interrogation request applies to the MC supplementary service. For a definition of these 3 components, refer to 3GPP TS 23.135 and 3GPP TS 24.135.

CCBS Feature LIST

The CCBS feature parameter is described in clause 7.6. A list of one or more CCBS features is returned by the responder when the interrogation request applied to the CCBS supplementary service. See 3GPP TS 23.093 [107] for the conditions for the presence of the parameters included in the CCBS feature.

User error

This error is sent by the responder upon unsuccessful outcome of the interrogation service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data Missing;
- Unexpected data value;
- Bearer Service not provisioned;

  This error is returned only if not even a subset of the interrogated bearer services are provided;

- Teleservice not provisioned;

This error is returned only if not even a subset of the interrogated teleservices are provided;

- Call Barred;
- Illegal SS operation;
- SS not available.

Provider error

See clause 7.6.1 for the use of this parameter.

11.6 Void

11.7 MAP_REGISTER_PASSWORD service

11.7.1 Definitions

This service is used between the MSC and the VLR and between the VLR and the HLR if the mobile subscriber requests to register a new password. The VLR will relay the message to the HLR.

The service is a confirmed service and consists of four service primitives.

11.7.2 Service primitives

The service primitives are shown in table 11.7/1.

Table 11.7/1: MAP_REGISTER_PASSWORD parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS-Code	M	M(=)		
New password			C	C(=)
User error			C	C(=)
Provider error				O

11.7.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

SS-Code

This parameter indicates for which supplementary service(s) the password should be registered.

New Password

See clause 7.6.4 for the use of this parameter.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data Missing;
- Unexpected data value;

- Call Barred;
- SS subscription violation;
- Password registration failure;
- Negative PW check;
- Number Of PW Attempts Violation.

Provider error

See clause 7.6.1 for the use of this parameter.

11.8 MAP_GET_PASSWORD service

11.8.1 Definitions

This service is used between the HLR and the VLR and between the VLR and the MSC when the HLR receives a request from the mobile subscriber for an operation on a supplementary service which requires a password from the subscriber. The VLR will relay the message to the MSC.

The service is a confirmed service and uses the service primitives shown in table 11.8/1.

11.8.2 Service primitives

Table 11.8/1: MAP_GET_PASSWORD parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
Linked id	C	C(=)		
Guidance info	M	M(=)		
Current password			M	M(=)
Provider error				O

11.8.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

Linked Id

See clause 7.6.1 for the use of this parameter. If the MAP_GET_PASSWORD service is used in conjunction with the MAP_REGISTER_PASSWORD service, this parameter must be present; otherwise it must be absent.

Guidance info

See clause 7.6.4 for the use of this parameter.

Current password

See clause 7.6.4 for the use of this parameter.

Provider error

See clause 7.6.1 for the use of this parameter.

11.9 MAP_PROCESS_UNSTRUCTURED_SS_REQUEST service

11.9.1 Definitions

This service is used between the MSC and the VLR, between the VLR and the HLR, between the HLR and gsmSCF and between the HLR and HLR to relay information in order to allow unstructured supplementary service operation.

The MAP_PROCESS_UNSTRUCTURED_SS_REQUEST service is a confirmed service using the primitives from table 11.9/1.

11.9.2 Service primitives

Table 11.9/1: MAP_PROCESS_UNSTRUCTURED_SS_REQUEST parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
USSD Data Coding Scheme	M	M(=)	C	C(=)
USSD String	M	M(=)	C	C(=)
MSISDN	C	C(=)		
User error			C	C(=)
Provider error				O

11.9.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

USSD Data Coding Scheme

See clause 7.6.4 for the use of this parameter. The presence of the parameter in the response is dependent on the unstructured supplementary service application. If this parameter is present, then the USSD String parameter has to be present.

USSD String

See clause 7.6.1 for the use of this parameter. The presence of the parameter in the response is dependent on the unstructured supplementary service application. If this parameter is present, then the USSD Data Coding Scheme parameter has to be present.

MSISDN

The subscriber's basic MSISDN.

See definition in clause 7.6.2. For Follow Me when the service request is sent from the HLR of the A subscriber, the parameter shall contain the MSISDN of the A subscriber, see 3GPP TS 23.094 [129]. For other purposes the MSISDN may be included as an operator option, e.g. to allow addressing the subscriber's data in the gsmSCF with the MSISDN.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- System failure;
- Data missing;
- Unexpected data value;

  This error is returned by the responder if it is not able to deal with the contents of the USSD string.

- Call Barred;
- Unknown Alphabet.

Provider error

See clause 7.6.1 for the use of this parameter.

11.10 MAP_UNSTRUCTURED_SS_REQUEST service

11.10.1 Definitions

This service is used between the gsmSCF and the HLR, the HLR and the VLR and between the VLR and the MSC when the invoking entity requires information from the mobile user, in connection with unstructured supplementary service handling.

The MAP_UNSTRUCTURED_SS_REQUEST service is a confirmed service using the primitives from table 11.10/1.

11.10.2 Service primitives

Table 11.10/1: MAP_UNSTRUCTURED_SS_REQUEST parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
USSD Data Coding Scheme	M	M(=)	C	C(=)
USSD String	M	M(=)	C	C(=)
Alerting Pattern	C	C(=)		
User error			C	C(=)
Provider error				O

11.10.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

USSD Data Coding Scheme

See clause 7.6.4 for the use of this parameter. The presence of the parameter in the response is dependent on the mobile user's MMI input. If this parameter is present, then the USSD String parameter has to be present.

USSD String

See clause 7.6.1 for the use of this parameter. The presence of the parameter in the response is dependent on the mobile user's MMI input. If this parameter is present, then the USSD Data Coding Scheme parameter has to be present.

Alerting Pattern

See clause 7.6.3 for the use of this parameter.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- System failure;
- Data missing;
- Unexpected data value;

  This error is returned by the responder if it is not able to deal with the contents of the USSD string;
- Absent Subscriber;
- Illegal Subscriber;

This error indicates that delivery of the unstructured supplementary service data failed because the MS failed authentication;

- Illegal Equipment;
- USSD Busy;
- Unknown Alphabet.

Provider error

See clause 7.6.1 for the use of this parameter.

11.11 MAP_UNSTRUCTURED_SS_NOTIFY service

11.11.1 Definitions

This service is used between the gsmSCF and the HLR, the HLR and the VLR and between the VLR and the MSC when the invoking entity requires a notification to be sent to the mobile user, in connection with unstructured supplementary services handling.

The MAP_UNSTRUCTURED_SS_NOTIFY service is a confirmed service using the primitives from table 11.11/1.

11.11.2 Service primitives

Table 11.11/1: MAP_UNSTRUCTURED_SS_NOTIFY parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
USSD Data Coding Scheme	M	M(=)		
USSD String	M	M(=)		
Alerting Pattern	C	C(=)		
User error			C	C(=)
Provider error				O

11.11.3 Parameter use

Invoke id

See clause 7.6.1 for the use of this parameter.

USSD Data Coding Scheme:

See clause 7.6.4 for the use of this parameter.

USSD String:

See clause 7.6.1 for the use of this parameter.

Alerting Pattern

See clause 7.6.3 for the use of this parameter.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values defined in clause 7.6.1:

- System failure;
- Data missing;

- Unexpected data value;

  This error is returned by the responder if it is not able to deal with the contents of the USSD string.

- Absent Subscriber;

- Illegal Subscriber;

  This error indicates that delivery of the unstructured supplementary service data failed because the MS failed authentication.

- Illegal Equipment;

- USSD Busy;

- Unknown Alphabet.

Provider error

See clause 7.6.1 for the use of this parameter.

11.12 MAP_SS_INVOCATION_NOTIFY

11.12.1 Definition

This service is used between the MSC and the gsmSCF when the subscriber invokes one of the following supplementary services; Call Deflection (CD), Explicit Call Transfer (ECT) or Multi Party (MPTY).

This service is used between the HLR and the gsmSCF when the subscriber invokes the CCBS supplementary service.

11.12.2 Service primitives

The service primitives are shown in table 11.12/1.

Table 11.12/1: SS_INVOCATION_NOTIFY parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
MSISDN	M	M(=)		
IMSI	M	M(=)		
SS- event	M	M(=)		
SS- event data	C	C(=)		
B-subscriber Number	C	C(=)		
CCBS Request State	C	C(=)		
User error			C	C(=)
Provider error				O

11.12.3 Parameter use

All parameters are described in clause 7.6. The use of these parameters and the requirements for their presence are specified in 3GPP TS 23.078.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Data Missing;

- Unexpected Data Value;

- Unknown Subscriber.

Provider error

This is defined in clause 7.6.1.

11.13 MAP_REGISTER_CC_ENTRY service

11.13.1 Definition

This service is used between the MSC and the VLR and between the VLR and the HLR to register data for a requested call completion supplementary service. The VLR will relay the message to the HLR.

The service is a confirmed service and uses the service primitives shown in table 11.13/1.

11.13.2 Service primitives

Table 11.13/1: MAP_REGISTER_CC_ENTRY parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS Code	M	M(=)		
CCBS Feature	C	C(=)	C	C(=)
Translated B number	C	C(=)		
Service Indicator	C	C(=)		
Call Info	C	C(=)		
Network Signal Info	C	C(=)		
User error			C	C(=)
Provider error				O

11.13.3 Parameter use

See clause 7.6 for a definition of the parameters used, in addition to the following.

SS-Code

This parameter indicates the call completion supplementary service for which the mobile subscriber wants to register an entry.

CCBS Feature

See 3GPP TS 23.093 [107] for the conditions for the presence of the parameters included in the CCBS feature.

Translated B Number

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Service Indicator

This parameter corresponds to the parameters 'Presentation Indicator' and 'CAMEL Invoked' in 3GPP TS 23.093 [107]. It indicates which services have been invoked for the original call (e.g. CLIR, CAMEL). See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Call Info

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

Network Signal Info

See 3GPP TS 23.093 [107] for the use of this parameter and the conditions for its presence.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data missing;
- Unexpected data value;
- Call Barred;
- Illegal SS operation;
- SS error status;
- SS incompatibility.
- Short Term Denial;
- Long Term Denial;
- Facility Not Supported;

NOTE: This error is reserved for future use.

Private Extensions shall not be sent with these user errors for this operation.

Provider error

See clause 7.6.1 for the use of this parameter.

11.14 MAP_ERASE_CC_ENTRY service

11.14.1 Definition

This service is used between the MSC and the VLR and between the VLR and the HLR to erase data related to a call completion supplementary service. The VLR will relay the message to the HLR.

The service is a confirmed service and uses the service primitives shown in table 11.14/1.

11.14.2 Service primitives

Table 11.14/1: MAP_ERASE_CC_ENTRY parameters

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
SS-Code	M	M(=)	C(=)	C(=)
CCBS Index	C	C(=)		
SS-Status			C	C(=)
User error			C	C(=)
Provider error				O

11.14.3 Parameter use

See clause 7.6 for a definition of the parameters used, in addition to the following.

SS-Code

This parameter indicates the call completion supplementary service for which the mobile subscriber wants to erase an entry/entries.

CCBS Index

See 3GPP TS 23.093 [107] for the use of this parameter and the condition for its presence.

SS-Status

Depending on the outcome of the service request this parameter may indicate either provisioned and active or not provisioned.

User error

This parameter is sent by the responder upon unsuccessful outcome of the service, and then takes one of the following values, defined in clause 7.6.1:

- System failure;
- Data Missing;
- Unexpected data value;
- Call Barred;
- Illegal SS operation;
- SS error status.

Private Extensions shall not be sent with these user errors for this operation.

Provider error

See clause 7.6.1 for the use of this parameter.

12 Short message service management services

12.1 MAP-SEND-ROUTING-INFO-FOR-SM service

12.1.1 Definition

This service is used between the gateway MSC and the HLR to retrieve the routing information needed for routing the short message to the servicing MSC or MME but not both, or SGSN, or (for T4-device triggering via the IMS) IP-SM-GW, or SMSF. This service is also used between the gateway MSC and SMS Router, and SMS Router and HLR in order to enforce routing of the SM delivery via the HPLMN of the receiving MS. This service is also used between HLR and IP-SM-GW, and between IP-SM-GW and HLR in order to allow MT-SM delivery (other than T4-device triggering) via the IMS.

This service is also used with an IWF interfacing the S6c interface.

The MAP-SEND-ROUTING-INFO-FOR-SM is a confirmed service using the primitives from table 12.1/1.

12.1.2 Service primitives

Table 12.1/1: MAP-SEND-ROUTING-INFO-FOR-SM

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
MSISDN	M	M(=)		
SM-RP-PRI	M	M(=)		
Service Centre Address	M	M(=)		
SM-RP-MTI	C	C(=)		
SM-RP-SMEA	C	C(=)		
GPRS Support Indicator	C	C(=)		
SM-Delivery Not Intended	U	C(=)		

IP-SM-GW Guidance Support Indicator	U	C(=)		
Single Attempt Delivery	C	C(=)		
IMSI	C	C(=)	C	C(=)
Correlation ID	C	C(=)		
T4 Trigger Indicator	C	C(=)		
SMSF Support Indicator	C	C(=)		
Network Node Number			C	C(=)
Network Node Diameter Address			C	C(=)
LMSI			C	C(=)
GPRS Node Indicator			C	C(=)
Additional Number			C	C(=)
Additional Network Node Diameter Address			C	C(=)
IP-SM-GW Guidance			U	C(=)
Third Number			C	C(=)
Third Network Node Diameter Address			C	C(=)
IMS Node Indicator			C	C(=)
SMSF 3GPP Number			C	C(=)
SMSF 3GPP Diameter Address			C	C(=)
SMSF Non-3GPP Number			C	C(=)
SMSF Non-3GPP Diameter Address			C	C(=)
SMSF 3GPP Address Indicator			C	C(=)
SMSF Non 3GPP Address Indicator			C	C(=)
User error			C	C(=)
Provider error				O

12.1.3 Parameter use

Invoke id

See definition in clause 7.6.1.

MSISDN

See definition in clause 7.6.2.

When SEND-ROUTING-INFO-FOR-SM is sent by the SMS-GMSC to the HLR following an T4 Submit Trigger (see 3GPP TS 23.682 [148]), MSISDN may not be available. In this case the UE shall be identified by the IMSI and the MSISDN shall take the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]).

When SEND-ROUTING-INFO-FOR-SM is sent by the SMS-GMSC to the HLR in a retry context of SMS for IMS UE to IMS UE without MSISDN (see 3GPP TS 23.204 [134]), MSISDN may not be available. In this case the UE shall be identified by a Correlation ID (SIP-URI-B) and the MSISDN shall take the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]).

SM-RP-PRI

See definition in clause 7.6.8.

Service Centre Address

See definition in clause 7.6.2.

SM-RP-MTI

See definition in clause 7.6.8. This parameter shall be present when the feature « SM filtering by the HPLMN » is supported by the SMS-GMSC and when the equivalent parameter is received from the short message service relay sub-layer protocol.

SM-RP-SMEA

See definition in clause 7.6.8. This parameter shall be present when the feature « SM filtering by the HPLMN » is supported by the SMS-GMSC and when the equivalent parameter is received from the short message service relay sub-layer protocol.

GPRS Support Indicator

See definition in clause 7.6.8. The presence of this parameter is mandatory if the SMS-GMSC supports receiving of the two numbers from the HLR.

SM-Delivery Not Intended

This parameter indicates by its presence that delivery of a short message is not intended. It further indicates whether only IMSI or only MCC+MNC are requested.

This parameter may be set by entities that request the service without intending to deliver a short message (e.g. MMS Relay/Server), and shall be evaluated by the SMS Router and may be evaluated by the HLR.

IP-SM-GW Guidance Support Indicator

This parameter indicates whether or not the SMS-GMSC is prepared to receive IP-SM-GW Guidance in the response.

Single Attempt Delivery

This parameter indicates the short message is only valid for delivering once, and the HLR/HSS does not need to add the received SC address into MWD list in the case there is no serving node available to provide SMS to the user.

IMSI

See definition in clause 7.6.2.

In Request and Indication:

IMSI shall be present if MSISDN is not available. When SEND-ROUTING-INFO-FOR-SM is sent by the SMS-GMSC to the HLR in a retry context of SMS for IMS UE to IMS UE without MSISDN (see 3GPP TS 23.204 [134]), IMSI may not be available. In this case the IMSI parameter shall be populated with the HLR-ID value.

In Response and Confirm: If enforcement of routing an SM via the HPLMN of the receiving MS is deployed, this parameter contains an MT Correlation ID instead of an IMSI when the service is used between SMS-GMSC and SMS Router (see 3GPP TS 23.040 [26] for more information). If the "SM-Delivery Not Intended" parameter was present in the Indication with a value of "only MCC+MNC requested", then this parameter may contain MCC+MNC+dummy MSIN.

The presence of this parameter is mandatory in a successful case.

T4 Trigger Indicator

This indicator indicates by its presence that the request is sent in the context of T4 device triggering (see 3GPP TS 23.682 [148]). When received, the HLR may return up to three serving node numbers and shall not forward the request to an IP-SM-GW or SMS Router.

SMSF Support Indicator

It indicates that the requesting node is capable of receiving ISDN numbers and/or Diameter addresses of the SMSF as target of MT-SMS.

Correlation ID

The Correlation ID parameter contains the SIP-URI-B identifying the (MSISDN-less) destination user. SIP-URI-A and HLR-ID shall be absent from this parameter.

The Correlation ID indicates by its presence that the request is sent in the context of MSISDN-less SMS delivery in IMS (see 3GPP TS 23.204 [134]). When received, the HLR shall return the IP-SM-GW number and shall not forward the request to an IP-SM-GW.

Network Node Number

See definition in clause 7.6.2. This parameter is provided in a successful response. If the "SM-Delivery Not Intended" parameter was present in the Indication a dummy address may be provided. See clause 12.1.4.

Network Node Diameter Address

See definition in clause 7.6.2. See clause 12.1.4.

LMSI

See definition in clause 7.6.2. It is an operator option to provide this parameter from the VLR; it is mandatory for the HLR to include the LMSI in a successful response, if the VLR has used the LMSI.

GPRS Node Indicator

See definition in clause 7.6.8.
Outside the context of T4 device triggering: The presence of this parameter is mandatory if only the SGSN number is sent in the Network Node Number (i.e. if the value within Network Node Number is to be considered as SGSN-Number and Additional Number is absent).
Within the context of T4 device triggering: The presence of this parameter is mandatory if the value within Network Node Number is to be considered as SGSN-Number and Third Number is absent.

Additional Number

See definition in clause 7.6.2. See clause 12.1.4.

Additional Network Node Diameter Address

See definition in clause 7.6.2. See clause 12.1.4.

IP-SM-GW Guidance

This parameter contains the recommended and the minimum timer values for supervision of MT-Forward-Short-Message response. Shall be absent if the IP-SM-GW-Guidance Support Indicator in the request is absent. This parameter is only used by IP-SM-GW and SMS-GMSC.

Third Number

See definition in clause 7.6.2. See clause 12.1.4.

Third Network Node Diameter Address

See definition in clause 7.6.2. See clause 12.1.4

IMS Node Indicator

See definition in clause 7.6.8.
Outside the context of T4 device triggering: The parameter is not applicable and shall be absent.
Within the context of T4 device triggering: The presence of this parameter is mandatory if the value within Network Node Number is to be considered as IP-SM-GW-Number and Third Number is absent.

SMSF 3GPP Number

This parameter contains the ISDN number of the SMSF target node for MT-SMS over 3GPP access.

SMSF 3GPP Diameter Address

This parameter contains the Diameter Name and Realm of the SMSF target node for MT-SMS over 3GPP access.

SMSF Non-3GPP Number

This parameter contains the ISDN number of the SMSF target node for MT-SMS over non-3GPP access.

SMSF Non-3GPP Diameter Address

This parameter contains the Diameter Name and Realm of the SMSF target node for MT-SMS over non-3GPP access.

SMSF 3GPP Address Indicator

It indicates that the parameter Network Node Number (and Network Node Diameter Address, if present) contains the address of an SMSF for 3GPP access.

SMSF Non-3GPP Address Indicator

It indicates that the parameter Network Node Number (and Network Node Diameter Address, if present) contains the address of an SMSF for non-3GPP access.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unknown subscriber;
- Call Barred;
- Teleservice Not Provisioned;
- Absent Subscriber_SM;
- Facility Not Supported;
- System failure;
- Unexpected Data Value;
- Data missing.

Provider error

For definition of provider errors see clause 7.6.1.

12.1.4 Identities of MT-SMS Target Nodes

In a successful MAP-Send-Routing-Info-For-SM response at least one MT-SMS Target Node identity or an SMS Router identity shall be present and this shall be an E.164 number within the Network Node Number parameter.

In addition, optionally a second Target Node identity or an SMS Router identity may be present as E.164 number within the Additional Number Parameter.

In T4 device trigger scenarios in addition to a second Target Node identity, a third Target Node Identity may be present as E.164 number within the Third Number parameter.

In addition to the E.164 identity of an MT-SMS Target Node or an SMSRouter, the presence of the Diameter Name/Realm of the corresponding target node or SMS Router follows the hereafter rules:

- If Network Node Number contains an MME number for SMS, Network Node Diameter Address shall be present and contain the Diameter address of the MME.
- If Network Node Number contains an MSC number, Network Node Diameter Address may be present and shall contain the Diameter address of the MME.
- If Network Node Number contains an SGSN number, Network Node Diameter Address shall be present only if the HSS has received the information that SGSN supports the Gdd interface.
- If Network Node Number contains an SMS Router number, Network Node Diameter Address may be present and shall contain the SMS Router Diameter address.
- If Network Node Number contains an IP-SM-GW number, Network Node Diameter Address may be present and shall contain the IP-SM-GW Diameter address.

Similar for Additional Number - Additional Network Node Diameter Address;

Similar for Third Number - Third Network Node Diameter Address.

In scenarios supporting interworking with 5G System, an E.164 Number and a Diameter Address of the SMSF may be present, for both 3GPP and non-3GPP accesses. In addition:

- If Network Node Number contains an SMSF 3GPP number, Network Node Diameter Address may be present and shall contain the SMSF 3GPP Diameter address.

- If Network Node Number contains an SMSF Non-3GPP number, Network Node Diameter Address may be present and shall contain the SMSF Non-3GPP Diameter address.

12.2 MAP-MO-FORWARD-SHORT-MESSAGE service

12.2.1 Definition

This service is used between the serving MSC or the SGSN or IP-SM-GW and the SMS Interworking MSC to forward mobile originated short messages.

The MAP-MO-FORWARD-SHORT-MESSAGE service is a confirmed service using the service primitives given in table 12.2/1.

12.2.2 Service primitives

Table 12.2/1: MAP-MO-FORWARD-SHORT-MESSAGE

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
SM RP DA	M	M(=)		
SM RP OA	M	M(=)		
SM RP UI	M	M(=)	C	C(=)
IMSI	C	C(=)		
Correlation ID	C	C(=)		
SM Delivery Outcome	C	C(=)		
User error			C	C(=)
Provider error				O

12.2.3 Parameter use

Invoke id

See definition in clause 7.6.1.

SM RP DA

See definition in clause 7.6.8.

In the mobile originated SM transfer this parameter contains the Service Centre address received from the mobile station.

SM RP OA

See definition in clause 7.6.8.

The MSISDN received from the VLR or from the SGSN is inserted in this parameter in the mobile originated SM transfer.

A Dummy MSISDN value is used for MSISDN-less SMS in IMS. In this case the originating user is identified by SIP-URI-A (see Correlation ID).

SM RP UI

See definition in clause 7.6.8. The short message transfer protocol data unit received from the Service Centre is inserted in this parameter.

IMSI

See definition in clause 7.6.2.1. The IMSI of the originating subscriber shall be inserted in this parameter in the mobile originated SM transfer.

Correlation ID

The Correlation ID is composed of an HLR-Id identifying the destination user's HLR, a SIP-URI-B identifying the MSISDN-less destination user, and a SIP-URI-A identifying the originating user.

The Correlation ID indicates by its presence that the request is sent in the context of MSISDN-less SMS delivery in IMS (see 3GPP TS 23.204 [134]), and that a Report-SM-Delivery status needs to be sent to the HLR to add the SC address to the MWD.

SM Delivery Outcome

See definition in clause 7.6.8. This parameter indicates the status of the mobile terminated SM delivery.

Shall be present if Correlation ID is present and shall take one of the unsuccessful outcome values.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Facility Not Supported;
- System Failure;
- SM Delivery Failure;
  - The reason of the SM Delivery Failure can be one of the following in the mobile originated SM:
    - unknown Service Centre address;
    - Service Centre congestion;
    - invalid Short Message Entity address;
    - subscriber not Service Centre subscriber;
    - protocol error.
- Unexpected Data Value

Provider error

For definition of provider errors see clause 7.6.1.

12.3 MAP-REPORT-SM-DELIVERY-STATUS service

12.3.1 Definition

This service is used between the gateway MSC and the HLR or the external Short Message Gateway (IP-SM-GW) and the HLR. The MAP-REPORT-SM-DELIVERY-STATUS service is used to set the Message Waiting Data into the HLR or to inform the HLR of successful SM transfer after polling. This service is invoked by the gateway MSC or the external Short Message Gateway (IP-SM-GW).

The MAP-REPORT-SM-DELIVERY-STATUS service is a confirmed service using the service primitives given in table 12.3/1.

12.3.2 Service primitives

Table 12.3/1: MAP-REPORT-SM-DELIVERY-STATUS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)

MSISDN	M	M(=)		
IMSI	C	C(=)		
Service Centre Address	M	M(=)		
SM Delivery Outcome	M	M(=)		
Absent Subscriber Diagnostic SM	C	C(=)		
GPRS Support Indicator	C	C(=)		
Delivery Outcome Indicator	C	C(=)		
Additional SM Delivery Outcome	C	C(=)		
Additional Absent Subscriber Diagnostic SM	C	C(=)		
IP-SM-GW-Indicator	C	C(=)		
IP-SM-GW SM Delivery Outcome	C	C(=)		
IP-SM-GW Absent Subscriber Diagnostic SM	C	C(=)		
Single Attempt Delivery	C	C(=)		
Correlation ID	C	C(=)		
SMSF 3GPP Delivery Outcome Indicator	C	C(=)		
SMSF 3GPP SM Delivery Outcome	C	C(=)		
SMSF 3GPP Absent Subscriber Diagnostic SM	C	C(=)		
SMSF non-3GPP Delivery Outcome Indicator	C	C(=)		
SMSF non-3GPP SM Delivery Outcome	C	C(=)		
SMSF non-3GPP Absent Subscriber Diagnostic SM	C	C(=)		
MSIsdn-Alert			C	C(=)
User error			C	C(=)
Provider error				O

12.3.3 Parameter use

Invoke id

See definition in clause 7.6.1.

MSISDN

See definition in clause 7.6.2.

When REPORT-SM-DELIVERY-STATUS is sent by the SMS-GMSC to the HLR following an T4 Submit Trigger (see 3GPP TS 23.682 [148]), MSISDN may not be available. In this case the UE shall be identified by the IMSI and the MSISDN shall take the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]).

When REPORT-SM-DELIVERY-STATUS is sent by the SMS-GMSC to the HLR in a retry context of SMS for IMS UE to IMS UE without MSISDN (see 3GPP TS 23.204 [134]), MSISDN may not be available. In this case the UE shall be identified by a Correlation ID (SIP-URI-B) and the MSISDN shall take the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]).

IMSI

See definition in clause 7.6.2. When REPORT-SM-DELIVERY-STATUS is sent by the SMS-GMSC to the HLR in a retry context of SMS for IMS UE to IMS UE without MSISDN (see 3GPP TS 23.204 [134]), IMSI may not be available. In this case the IMSI parameter shall be populated with an HLR-ID).

Service Centre Address

See definition in clause 7.6.2.

SM Delivery Outcome

See definition in clause 7.6.8. This parameter indicates the status of the mobile terminated SM delivery.

Absent Subscriber Diagnostic SM

See definition in clause 7.6.8.

GPRS Support Indicator

See definition in clause 7.6.8. The presence of this parameter is mandatory if the SMS-GMSC supports handling of two delivery outcomes.

Delivery Outcome Indicator

See definition in clause 7.6.8.

Additional SM Delivery Outcome

See definition in clause 7.6.8.

Additional Absent Subscriber Diagnostic SM

See definition in clause 7.6.8.

IP-SM-GW Indicator

See definition in clause 7.6.8.

IP-SM-GW SM Delivery Outcome

See definition in clause 7.6.8.

IP-SM-GW Absent Subscriber Diagnostic SM

See definition in clause 7.6.8.

Single Attempt Delivery

This parameter indicates the short message is only valid for delivering once, and the HLR/HSS does not need to add the received SC address into MWD list. It may only be present in the case the delivery of the short message failed due to absent subscriber or MS memory capacity exceeded.

Editor's Note: Description of the use of this parameter might be needed in 3GPP TS 23.040.

Correlation ID

The Correlation ID parameter contains the SIP-URI-B identifying the (MSISDN-less) destination user. SIP-URI-A and HLR-ID shall be absent from this parameter.

MSIsdn-Alert

See definition in clause 7.6.2. This parameter shall be present in case of unsuccessful delivery, when the MSISDN received in the operation is different from the stored MSIsdn-Alert; the stored MSIsdn-Alert is the value that is returned to the gateway MSC.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unknown Subscriber;
- Message Waiting List Full;
- Unexpected Data Value;
- Data missing.

Provider error

For definition of provider errors see clause 7.6.1.

12.4 MAP-READY-FOR-SM service

12.4.1 Definition

This service is used between the MSC and VLR as well as between the VLR and the HLR. The MSC initiates this service if a subscriber indicates memory available situation. The VLR uses the service to indicate this to the HLR.

The VLR initiates this service if a subscriber, whose message waiting flag is active in the VLR, has radio contact in the MSC.

Also this service is used between the SGSN and the HLR. The SGSN initiates this service if a subscriber indicates memory available situation. The SGSN uses the service to indicate this to the HLR.

Also this service is used between the HSS and the MME via an IWF. The MME initiates this service if a subscriber indicates memory available situation. The MME uses the service to indicate this to the HLR.

The SGSN initiates this service if a subscriber, whose message waiting flag is active in the SGSN, has radio contact in the GPRS.

The MME initiates this service if a subscriber, whose message waiting flag is active in the MME, has radio contact via LTE.

Also this service is used between the external Short Message Gateway (IP-SM-GW) and the HLR. The IP-SM-GW initiates this service if a subscriber indicates memory available situation. The IP-SM-GW uses the service to indicate this to the HLR.

The IP-SM-GW initiates this service if a subscriber, whose message waiting flag is active in the IP-SM-GW, is reachable in IMS.

The MAP-READY-FOR-SM service is a confirmed service using the primitives from table 12.4/1.

12.4.2 Service primitives

Table 12.4/1: MAP-READY-FOR-SM

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
IMSI	C	C(=)		
TMSI	C	C(=)		
Alert Reason	M	M(=)		
Alert Reason Indicator	C	C(=)		
Additional Alert Reason Indicator	C	C(=)		
Maximum UE Availability Time	C	C(=)		
User error			C	C(=)
Provider error				O

12.4.3 Parameter use

Invoke id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2. The IMSI is used always between the VLR and the HLR and between the SGSN and the HLR and between the HSS and the IWF. Between the MSC and the VLR the identification can be either IMSI or TMSI.

TMSI

See definition in clause 7.6.2. The identification can be either IMSI or TMSI between MSC and VLR.

Alert Reason

See definition in clause 7.6.8. This parameter indicates if the mobile subscriber is present or the MS has memory available.

Alert Reason Indicator

See definition in clause 7.6.8. This parameter by its presence indicates the message is sent from SGSN, and by its absence indicates the message is sent from VLR or MME via IWF.

Additional Alert Reason Indicator

See definition in clause 7.6.8.

Maximum UE Availability Time

See definition in clause 7.6.8. This information element may be included by the SGSN or MSC when notifying the HLR that the MS is reachable.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unknown Subscriber;
- Facility Not Supported;
- System Failure;
- Unexpected Data Value;
- Data missing.

Provider error

For definition of provider errors see clause 7.6.1.

12.5 MAP-ALERT-SERVICE-CENTRE service

12.5.1 Definition

This service is used between the HLR and the interworking MSC. The HLR initiates this service, if the HLR detects that a subscriber, whose MSISDN is in the Message Waiting Data file, is active or the MS has memory available.

This service is also used between an MME (via an IWF), SGSN or an MSC using Deployment Option 2 (see subclause 8.2.4a.1 of 3GPP TS 23.272 [143] and 3GPP TS 23.040 [6]) and the SMS-GMSC (possibly via an SMS Router), to indicate that a MS, for which one or more MT SMS have been requested to be retransmitted at a later time, is now available for MT SMS delivery or has moved under the coverage of another MME, SGSN or MSC. This procedure is used according to the call flows described in subclause 10.1 of 3GPP TS 23.040 [26].

The MAP-ALERT-SERVICE-CENTRE service is a confirmed service using the primitives from table 12.5/1.

12.5.2 Service primitives

Table 12.5/1: MAP-ALERT-SERVICE-CENTRE

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
MSIsdn-Alert	M	M(=)		
IMSI	C	C(=)		
Correlation ID	C	C(=)		
Service Centre Address	M	M(=)		
Maximum UE Availability Time	C	C(=)		
SMS-GMSC Alert Event	C	C(=)		
SMS-GMSC Diameter Address	C	C(=)		
New SGSN Number	C	C(=)		

New SGSN Diameter Address	C	C(=)		
New MME Number	C	C(=)		
New MME Diameter Address	C	C(=)		
New MSC Number	C	C(=)		
User error			C	C(=)
Provider error				O

12.5.3 Parameter use

Invoke id

See definition in clause 7.6.1.

MSIsdn-Alert

See definition in clause 7.6.2.

When the service is used between the HLR and the SMS-IWMSC, the provided MSISDN shall be the one which is stored in the Message Waiting Data file. If no MSISDN is available, the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]) shall be sent and an IMSI or Correlation ID (SIP-URI-B) shall be present.

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, the dummy MSISDN value (see clause 3 of 3GPP TS 23.003 [17]) shall be sent and an IMSI shall be present.

IMSI

When the service is used between the HLR and the SMS-IWMSC, the provided IMSI shall be the identifier which is stored in the Message Waiting Data file if no MSISDN is available in the context of T4 device triggering (see 3GPP TS 23.682 [148]).

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, this information element shall contain the IMSI in the request sent from the MME (via an IWF), SGSN or MSC, or the User Identifier Alert previously sent in the MT Forward Short Message response, when the request is sent from the SMS Router to the SMS-GMSC.

Correlation ID

When the service is used between the HLR and the SMS-IWMSC, the provided SIP-URI-B within the Correlation ID parameter shall be the identifier which is stored in the Message Waiting Data file if no MSISDN is available in a retry context of SMS for IMS UE to IMS UE without MSISDN (see 3GPP TS 23.204 [134]). HLR-ID and SIP-URI-A shall be absent.

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, this information element shall not be included.

Service Centre Address

See definition in clause 7.6.2.

When the service is used between the HLR and the SMS-IWMSC, this information element shall contain the E.164 number of the Service Center.

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, this information element shall contain the E.164 number of the SMS-GMSC (or SMS Router) previously received in the SMS-GMSC Address IE in the MT Forward Short Message Request.

Maximum UE Availability Time

See definition in clause 7.6.8.

When the service is used between the HLR and the SMS-IWMSC, this information element shall be included, if available.

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, this information element shall not be included.

SMS-GMSC Alert Event

See definition in clause 7.6.8.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, this information element shall either indicate that the MS is now available for MT SMS or that the MS has moved under the coverage of another MME, SGSN or MSC.

New SGSN Number

See definition in clause 7.6.8.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MME (via an IWF) or SGSN and the SMS-GMSC, this information element may be included if the SMS-GMSC Alert Event indicates that the MS has moved under the coverage of another SGSN. When present, it shall contain the E.164 number of the new SGSN serving the MS.

New MME Number

See definition in clause 7.6.8.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MME (via an IWF) or SGSN and the SMS-GMSC, this information element may be included if the SMS-GMSC Alert Event indicates that the MS has moved under the coverage of another MME. When present, it shall contain the E.164 number of the new MME serving the MS.

New SGSN Diameter Address

See definition in clause 7.6.8.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MME (via an IWF) or SGSN and the SMS-GMSC, this information element shall be included if available and if the SMS-GMSC Alert Event indicates that the MS has moved under the coverage of another SGSN. When present, it shall contain the Diameter Identity of the new SGSN serving the MS.

New MME Diameter Address

See definition in clause 7.6.8.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MME (via an IWF) or SGSN and the SMS-GMSC, this information element shall be included if available and if the SMS-GMSC Alert Event indicates that the MS has moved under the coverage of another MME. When present, it shall contain the Diameter Identity of the new MME serving the MS.

SMS-GMSC Diameter Address

See definition in clause 7.6.2.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MME (via an IWF), SGSN or MSC and the SMS-GMSC, this information element shall contain, if available, the Diameter Identity of the SMS-GMSC (or SMS Router) previously received in the SMS-GMSC Diameter Address IE in the MT Forward Short Message Request.

New MSC Number

See definition in clause 7.6.8.33.

When the service is used between the HLR and the SMS-IWMSC, this information element shall not be included.

When the service is used between an MSC and the SMS-GMSC, this information element may be included if the SMS-GMSC Alert Event indicates that the MS has moved under the coverage of another MSC. When present, it shall contain the E.164 number of the new MSC serving the MS.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- System Failure;
- Unexpected Data Value;
- Data missing.

Provider error

For definition of provider errors see clause 7.6.1.

12.6 MAP-INFORM-SERVICE-CENTRE service

12.6.1 Definition

This service is used between the HLR and the gateway MSC (transiting an SMS Router, if present) to inform the Service Centre which MSISDN number is stored in the Message Waiting Data file. If the stored MSISDN number is not the same as the one received from the gateway MSC in the MAP-SEND-ROUTING-INFO-FOR-SM service primitive the stored MSISDN number is included in the message.

Additionally the status of MCEF, MNRF and MNRG flags and the inclusion of the particular Service Centre address in the Message Waiting Data list is informed to the gateway MSC when appropriate.

If the HLR has stored a single MNRR, the value is included in the Absent Subscriber Diagnostic SM parameter.

If the HLR has stored a second MNRR, the value of the MNRR for the MSC is included in the Absent Subscriber Diagnostic SM parameter and the value of the MNRR for the SGSN is included in the Additional Absent Subscriber Diagnostic SM parameter.

The MAP-INFORM-SERVICE-CENTRE service is a non-confirmed service using the primitives from table 12.6/1.

12.6.2 Service primitives

Table 12.6/1: MAP-INFORM-SERVICE-CENTRE

Parameter name	Request	Indication
Invoke Id	M	M(=)
MSIsdn-Alert	C	C(=)
MWD Status	C	C(=)
Absent Subscriber Diagnostic SM	C	C(=)
Additional Absent Subscriber Diagnostic SM	C	C(=)

12.6.3 Parameter use

Invoke id

See definition in clause 7.6.1.

MSIsdn-Alert

See definition in clause 7.6.2. This parameter refers to the MSISDN stored in a Message Waiting Data file in the HLR.

MWD Status

See definition in clause 7.6.8. This parameter indicates the status of the MCEF, MNRF and MNRG flags and the status of the particular SC address presence in the Message Waiting Data list.

Absent Subscriber Diagnostic SM

See definition in clause 7.6.8.

Additional Absent Subscriber Diagnostic SM

See definition in clause 7.6.8.

12.7 MAP-SEND-INFO-FOR-MT-SMS service

12.7.1 Definition

This service is used between the MSC and the VLR. The service is invoked by the MSC receiving a mobile terminated short message to request subscriber related information from the VLR.

The MAP-SEND-INFO-FOR-MT-SMS service is a confirmed service using the primitives from table 12.7/1.

12.7.2 Service primitives

Table 12.7/1: MAP-SEND-INFO-FOR-MT-SMS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
SM RP DA	M	M(=)		
IMSI	C	C(=)		
MSISDN			C	C(=)
User error			C	C(=)
Provider error				O

12.7.3 Parameter use

Invoke id

See definition in clause 7.6.1.

SM RP DA

See definition in clause 7.6.8. This parameter shall contain either an IMSI or an LMSI.

IMSI

See definition in clause 7.6.2. This parameter shall be present if the SM RP DA parameter contains an LMSI; otherwise it shall be absent.

MSISDN

See definition in clause 7.6.2.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unknown subscriber;

- Unidentified Subscriber;
- Absent subscriber;
- Unexpected Data Value;
- Data Missing;
- Illegal subscriber;
- Illegal equipment;
- Subscriber busy for MT SMS;
- System Failure.

Provider error

For definition of provider errors see clause 7.6.1.

12.8 MAP-SEND-INFO-FOR-MO-SMS service

12.8.1 Definition

This service is used between the MSC and the VLR. The service is invoked by the MSC which has to handle a mobile originated short message request to request the subscriber related information from the VLR.

The MAP-SEND-INFO-FOR-MO-SMS service is a confirmed service using the primitives from table 12.8/1.

12.8.2 Service primitives

Table 12.8/1: MAP-SEND-INFO-FOR-MO-SMS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
Service Centre Address	M	M(=)		
MSISDN			C	C(=)
User error			C	C(=)
Provider error				O

12.8.3 Parameter use

Invoke id

See definition in clause 7.6.1.

Service Centre Address

See definition in clause 7.6.2.

MSISDN

See definition in clause 7.6.2.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Teleservice Not Provisioned;
- Call Barred;
- Unexpected Data Value;

- Data Missing.

Provider error

For definition of provider errors see clause 7.6.1.

12.9 MAP-MT-FORWARD-SHORT-MESSAGE service

12.9.1 Definition

This service is used between the gateway MSC and the serving MSC or the SGSN (transiting an SMS Router, if present) or the IP-SM-GW to forward mobile terminated short messages.

The MAP-MT-FORWARD-SHORT-MESSAGE service is a confirmed service using the service primitives given in table 12.9/1.

12.9.2 Service primitives

Table 12.9/1: MAP-MT-FORWARD-SHORT-MESSAGE

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
SM RP DA	M	M(=)		
SM RP OA	M	M(=)		
SM RP UI	M	M(=)	C	C(=)
More Messages To Send	C	C(=)		
SM Delivery Timer	C	C(=)		
SM Delivery Start Time	C	C(=)		
SMS Over IP Only Indicator	C	C(=)		
Correlation ID	C	C(=)		
Maximum Retransmission Time	C	C(=)		
SMS-GMSC Address	C	C(=)		
SMS-GMSC Diameter Address	C	C(=)		
Requested Retransmission Time			C	C(=)
User Identifier Alert			C	C(=)
User error			C	C(=)
Provider error				O

12.9.3 Parameter use

Invoke id

See definition in clause 7.6.1.

SM RP DA

See definition in clause 7.6.8. This parameter can contain either an IMSI or a LMSI. The use of the LMSI is an operator option. The LMSI can be provided if it is received from the HLR. The IMSI is used if the use of the LMSI is not available.

This parameter is omitted (i.e. is present and takes the value "noSM-RP-DA") in the mobile terminated subsequent SM transfers.

When a Correlation ID is present, the IMSI parameter within SM RP DA shall be populated with the HLR-ID and the destination user is identified by the SIP-URI-B within the Correlation ID.

SM RP OA

See definition in clause 7.6.8. The Service Centre address received from the originating Service Centre is inserted in this parameter.

This parameter is omitted in the mobile terminated subsequent SM transfers.

SM RP UI

See definition in clause 7.6.8. The short message transfer protocol data unit received from the Service Centre is inserted in this parameter. A short message transfer protocol data unit may also be inserted in this parameter in the message delivery acknowledgement from the MSC or from the SGSN to the Service Centre.

More Messages To Send

See definition in clause 7.6.8. The information from the MMS indication received from the Service Centre is inserted in this parameter.

SM Delivery Timer

See definition in clause 7.6.8.

SM Delivery Start Time

See definition in clause 7.6.8.

SMS Over IP Only Indicator

This indicator indicates by its presence that the IP-SM-GW shall try to deliver the short message via IMS without retrying to other domains. It shall be present in messages sent to the IP-SM-GW following a T4-Submit Trigger message (see 3GPP TS 23.682 [148]) but not in messages sent to MSC or SGSN (possibly transiting an SMS-Router).

The indicator also indicates to the IP-SM-GW by its presence that the IMSI within the message is a real IMSI and not a MT-Correlation ID allocated by the IP-SM-GW.

Correlation ID

The Correlation ID parameter contains the SIP-URI-B identifying the (MSISDN-less) destination user and the SIP-URI-A identifying the (MSISDN-less) originating user. HLR-ID shall be absent from this parameter.

Maximum Retransmission Time

See definition in clause 7.6.8.

SMS-GMSC Address

See definition in clause 7.6.8.

This information element shall be present if the Maximum Retransmission Time IE is present in the message. When present, it shall contain the E.164 number of the SMS-GMSC in the request sent by the SMS-GMSC or the E.164 number of the SMS Router in the request sent by the SMS Router.

SMS-GMSC Diameter Address

See definition in clause 7.6.8.

This information element shall be present if available and if the Maximum Retransmission Time IE is present in the message. When present, it shall contain the Diameter Identity of the SMS-GMSC in the request sent by the SMS-GMSC or the Diameter Identity of the SMS Router in the request sent by the SMS Router.

Requested Retransmission Time

See definition in clause 7.6.8. This information element may only be present if the MT Forward Short Message Response contains the User error set to Absent Subscriber_SM and if the Maximum Retransmission Time information element is present in the MT Forward Short Message Request. It may be included by an MSC using Deployment Option 2 (see subclause 8.2.4a.1 of 3GPP TS 23.272 [143] and 3GPP TS 23.040 [6]) or the SGSN if the UE is using a power saving mechanism (such as extended idle mode DRX) and the UE is currently not reachable.

The Requested Retransmission Time shall not exceed the Maximum Retransmission Time received from the SMS-GMSC.

User-Identifier-Alert

See definition in clause 7.6.8.

This information element shall be present in the message from the SMS Router to the SMS-GMSC, if the Requested Retransmission Time IE is present in the message. When present, this information shall contain an MT Correlation ID.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unidentified subscriber;
- Absent Subscriber_SM;
- Subscriber busy for MT SMS;
- Facility Not Supported;
- Illegal Subscriber indicates that delivery of the mobile terminated short message failed because the mobile station failed authentication;
- Illegal equipment indicates that delivery of the mobile terminated short message failed because an IMEI check failed, i.e. the IMEI was blacklisted or not white-listed;
- System Failure;
- SM Delivery Failure:
    - The reason of the SM Delivery Failure can be one of the following in the mobile terminated SM:
        - memory capacity exceeded in the mobile equipment;
        - protocol error;
        - mobile equipment does not support the mobile terminated short message service.
- Unexpected Data Value;
- Data Missing.

Provider error

For definition of provider errors see clause 7.6.1.

12.10 MAP-MT-FORWARD-SM-FOR-VGCS service

12.10.1 Definition

This service is used between the SMS gateway MSC and the Group Call Anchor MSC to forward mobile terminated short messages into an ongoing voice group call.

The MAP-MT-FORWARD-SM-FOR-VGCS service is a confirmed service using the service primitives given in table 12.10/1.

12.10.2 Service primitives

Table 12.10/1: MAP-MT-FORWARD-SM-VGCS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)

ASCI Call Reference	M	M(=)		
SM RP OA	M	M(=)		
SM RP UI	M	M(=)	C	C(=)
Dispatcher List			C	C(=)
Ongoing Call Indicator			C	C(=)
User error			C	C(=)
Provider error				O

12.10.3 Parameter use

Invoke id

See definition in clause 7.6.1.

ASCI Call Reference

Group call reference. This item is used to access the VGCS-GCR within the Anchor_MSC.

SM RP OA

See definition in clause 7.6.8. The Service Centre address received from the originating Service Centre is inserted in this parameter.

SM RP UI

See definition in clause 7.6.8. The short message transfer protocol data unit received from the Service Centre is inserted in this parameter. A short message transfer protocol data unit may also be inserted in this parameter in the message delivery acknowledgement from the MSC to the Service Centre.

Dispatcher List

A list of identities (international E.164 phone numbers) identifying the dispatchers of the VGCS call. It shall be present if received from the GCR; otherwise shall be absent.

Ongoing Call Indicator

Indicates by its presence that the VGCS call is ongoing.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- System Failure;
- Unexpected Data Value.

Provider error

For definition of provider errors see clause 7.6.1.

13 Network-Requested PDP Context Activation services

13.1 MAP_SEND_ROUTING_INFO_FOR_GPRS service

13.1.1 Definition

This service is used by the GGSN to request GPRS routing information from the HLR.

13.1.2 Service primitives

Table 13.1/1: MAP_SEND_ROUTING_INFO_FOR_GPRS

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
GGSN address	C	C(=)	C	C(=)
GGSN number	M	M(=)		
SGSN address			C	C(=)
Mobile Not Reachable Reason			C	C(=)
User error			C	C(=)
Provider error				O

13.1.3 Parameter definition and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

GGSN address

This parameter shall be present if the protocol-converting GSN is used between the GGSN and the HLR.

GGSN number

See definition in clause 7.6.2.

SGSN address

This parameter shall be present if the outcome of the Send Routing Info For GPRS request to the GPRS application process in the HLR is positive.

Mobile Not Reachable Reason

This parameter shall be present if the outcome of the Send Routing Info For GPRS request to the GPRS application process in the HLR is positive and the MNRG flag in the HLR is set. See definition in clause 7.6.3.51.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- Absent Subscriber;
- System Failure;
- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber.

The diagnostic in the Unknown Subscriber may indicate "Imsi Unknown" or "Gprs Subscription Unknown".

- Call Barred;

  This error will indicate that the received PDP PDUs in the GGSN shall be barred for this MS due to Operator Determined Barring. (The CallBarringCause must be the operatorBarring.)

Provider error

These are defined in clause 7.6.1.

13.2 MAP_FAILURE_REPORT service

13.2.1 Definition

This service is used by the GGSN to inform the HLR that network requested PDP-context activation has failed.

13.2.2 Service primitives

Table 13.2/1: MAP_FAILURE_REPORT

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
GGSN address	C	C(=)	C	C(=)
GGSN number	M	M(=)		
User error			C	C(=)
Provider error				O

13.2.3 Parameter definition and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

GGSN address

This parameter shall be present if the protocol-converting GSN is used between the GGSN and the HLR.

GGSN number

See definition in clause 7.6.2.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- System Failure;
- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber.

Provider error

These are defined in clause 7.6.1.

13.3 MAP_NOTE_MS_PRESENT_FOR_GPRS service

13.3.1 Definition

This service is used by the HLR to inform the GGSN that the MS is present for GPRS again.

13.3.2 Service primitives

Table 13.3/1: MAP_NOTE_MS_PRESENT_FOR_GPRS

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
IMSI	M	M(=)		
GGSN address	C	C(=)		
SGSN address	M	M(=)		
User error			C	C(=)
Provider error				O

13.3.3 Parameter definition and use

Invoke Id

See definition in clause 7.6.1.

IMSI

See definition in clause 7.6.2.

GGSN address

This parameter shall be present if the protocol-converting GSN is used between the GGSN and the HLR.

SGSN address

See definition in clause 7.6.2.

User error

This parameter is sent by the responder when an error is detected and if present, takes one of the following values:

- System Failure;
- Data Missing;
- Unexpected Data Value;
- Unknown Subscriber.

Provider error

These are defined in clause 7.6.1.

13A Location Service Management Services

13A.1 MAP-SEND-ROUTING-INFO-FOR-LCS Service

13A.1.1 Definition

This service is used between the GMLC and the HLR to retrieve the routing information needed for routing a location service request to the servicing VMSC, SGSN, MME or 3GPP AAA server. The MAP-SEND-ROUTING-INFO-FOR-LCS is a confirmed service using the primitives from table 13A.1/1.

13A.1.2 Service Primitives

Table 13A.1/1: MAP-SEND-ROUTING-INFO-FOR-LCS

Parameter name	Request	Indication	Response	Confirm
Invoke Id	M	M(=)	M(=)	M(=)
MLC Number	M	M(=)		
MSISDN	C	C(=)	C	C(=)
IMSI	C	C(=)	C	C(=)
LMSI			C	C(=)
Network Node Number			C	C(=)
GPRS Node Indicator			C	C(=)
Additional Number			C	C(=)
Supported LCS Capability Sets			C	C(=)
Additional LCS Capability Sets			C	C(=)
MME Name			C	C(=)
SGSN Name			C	C(=)
SGSN Realm			C	C(=)
AAA Server Name			C	C(=)
V-GMLC Address			U	C(=)
Additional V-GMLC Address			U	C(=)
H-GMLC Address			C	C(=)
PPR Address			U	C(=)
User error			C	C(=)
Provider error				O

13A.1.3 Parameter Use

Invoke id

See definition in clause 7.6.1.

MLC Number

See definition in clause 7.6.2.

MSISDN

See definition in clause 7.6.2. The request shall carry either the IMSI or MSISDN. The response shall carry whichever of these was not included in the request (see 3GPP TS 23.271 for details).

IMSI

See definition in clause 7.6.2.

LMSI

See definition in clause 7.6.2. It is an operator option to provide this parameter from the VLR; it is mandatory for the HLR to include the LMSI in a successful response, if the VLR has used the LMSI.

Network Node Number

See definition in clause 7.6.2. This parameter is provided in a successful response. If the Network Node Number and Additional Number are received in the GMLC, the Network Node Number is used in preference to the Additional Number. If the serving node has no ISDN number, the HLR shall populate the Network Node Number parameter with a dummy ISDN number of "0".

GPRS Node Indicator

See definition in clause 7.6.8. The presence of this parameter is mandatory only if the SGSN number is sent in the Network Node Number.

Additional Number

See definition in clause 7.6.2. This parameter is provided in a successful response. If the Network Node Number and Additional Number are received in the GMLC, the Network Node Number is used in preference to the Additional Number.

Supported LCS Capability Sets

See definition in clause 7.6.11. This parameter indicates the LCS capability of the serving node that is indicated by the Network Node Number. This parameter is provided only if LCS capability sets are available in HLR and Network Node Number is present in this message.

Additional LCS Capability Sets

See definition in clause 7.6.11. This parameter indicates the LCS capability of the serving node that is indicated by the Additional Number. This parameter is provided only if LCS capability sets are available in HLR and Additional Number is present in this message.

MME Name

See definition in clause 7.6.2. This parameter is provided in a successful response when the serving node is an MME.

SGSN Name

See definition in clause 7.6.2. This parameter is provided in a successful response when the serving node is an SGSN and the SGSN has indicated its support for Lgd interface.

SGSN Realm

See definition in clause 7.6.2. This parameter is provided in a successful response when the serving node is an SGSN and the SGSN has indicated its support for Lgd interface.

AAA Server Name

See definition in clause 7.6.2. This parameter is provided in a successful response when the serving node is a 3GPP AAA server.

V-GMLC address

See definition in clause 7.6.2. . This parameter indicates the V-GMLC address of the serving node that is indicated by the Network Node Number.

Additional V-GMLC address

See definition in clause 7.6.2. This parameter indicates the V-GMLC address of the serving node that is indicated by the Additional Number. This parameter is provided only if additional LCS capability sets are available in HLR and Additional Number is present in this message.

H-GMLC address

See definition in clause 7.6.2. The requirements for its presence are specified in 3GPP TS 23.271 [26a].

PPR address

See definition in clause 7.6.2.

User error

The following errors defined in clause 7.6.1 may be used, depending on the nature of the fault:

- Unknown subscriber;
- Absent Subscriber;
- Facility Not Supported;
- System failure;

- Unexpected Data Value;
- Data missing;
- Unauthorised requesting network.

Provider error

For definition of provider errors see clause 7.6.1.

13A.2 MAP-PROVIDE-SUBSCRIBER-LOCATION Service

13A.2.1 Definition

This service is used by a GMLC to request the location of a target MS from the visited MSC or SGSN at any time. This is a confirmed service using the primitives from table 13A.2/1.

13A.2.2 Service Primitives

Table 13A.2/1: Provide_Subscriber_Location

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
Location Type	M	M(=)		
MLC Number	M	M(=)		
LCS Client ID	M	M(=)		
Privacy Override	U	C(=)		
IMSI	C	C(=)		
MSISDN	C	C(=)		
LMSI	C	C(=)		
LCS Priority	C	C(=)		
LCS QoS	C	C(=)		
IMEI	U	C(=)		
Supported GAD Shapes	C	C(=)		
LCS-Reference Number	C	C(=)		
LCS Codeword	C	C(=)		
LCS Service Type Id	C	C(=)		
LCS Privacy Check	C	C(=)		
Area Event Info	C	C(=)		
H-GMLC Address	C	C(=)		
Reporting PLMN List	C	C(=)		
PeriodicLDRInfo	C	C(=)		
MO-LR Short Circuit Indicator	C	C(=)	C	C(=)
Location Estimate			M	M(=)
GERAN Positioning Data			C	C(=)
UTRAN Positioning Data			C	C(=)
GERAN GANSS Positioning Data			C	C(=)
UTRAN GANSS Positioning Data			C	C(=)
UTRAN Additional Positioning Data			C	C(=)
UTRAN Barometric Pressure Measurement			C	C(=)
UTRAN Civic Address			C	C(=)
Age of Location Estimate			C	C(=)
Additional Location Estimate			C	C(=)
Deferred MT-LR Response Indicator			C	C(=)
Cell Id Or SAI			C	C(=)

Accuracy Fulfilment Indicator			C	C(=)
Target Serving Node for Handover			C	C(=)
User error			C	C(=)
Provider error				O

13A.2.3 Parameter Definition and Use

All parameters are defined in clause 7.6. The use of these parameters and the requirements for their presence are specified in 3GPP TS 23.271 [26a].

Location Type

This parameter identifies the type of location information requested.

MLC Number

This is the E.164 number of the requesting GMLC.

LCS Client ID

This parameter provides information related to the identity of an LCS client.

Privacy Override

This parameter indicates if MS privacy is overridden by the LCS client when the GMLC and VMSC or SGSN for an MT-LR are in the same country.

IMSI

The IMSI is provided to identify the target MS. At least one of the IMSI or MSISDN is mandatory.

MSISDN

The MSISDN is provided to identify the target MS. At least one of the IMSI or MSISDN is mandatory.

LMSI

The LMSI shall be provided if previously supplied by the HLR. This parameter is only used in the case of the MT-LR for CS domain.

LCS Priority

This parameter indicates the priority of the location request.

LCS QoS

This parameter indicates the required quality of service in terms of response time and accuracy.

IMEI

The requirements for its presence are specified in 3GPP TS 23.271 [26a].

Supported GAD Shapes

This parameter indicates which of the shapes defined in 3GPP TS 23.032 [122] are supported.

LCS-Reference Number

This parameter shall be included if a deferred MT-LR procedure is performed for a UE available event, an area event or a periodic positioning event.

LCS Codeword

See definition in clause 7.6.11.18. The requirements for its presence are specified in 3GPP TS 23.271 [26a].

LCS Service Type Id

See definition in clause 7.6.11.15. The requirements for its presence are specified in 3GPP TS 23.271 [26a].

LCS Privacy Check

See definition in clause 7.6.11. The requirements for its and its components presence are specified in 3GPP TS 23.271 [26a].

Area Event Info

See definition in clause 7.6.11. The parameter shall be included if a deferred MT-LR procedure is performed for an area event.

H-GMLC address

See definition in clause 7.6.2. The parameter shall be included if a deferred MT-LR procedure is performed for a UE available event, an area event or a periodic positioning event.

Location Estimate

This parameter provides the location estimate if this is encoded in one of the supported geographical shapes. Otherwise this parameter shall consist of one octet, which shall be discarded by the receiving node.

GERAN Positioning Data

This parameter indicates the usage of each positioning method that was attempted to determine the location estimate either successfully or unsuccessfully. If Positioning Data received from the RAN contains no Positioning Methods, Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is GERAN, see 3GPP TS 23.271 [26a].

UTRAN Positioning Data

This parameter indicates the usage of each positioning method that was successfully attempted to determine the location estimate. If Position Data received from the RAN contains no Positioning Methods, UTRAN Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

GERAN GANSS Positioning Data

This parameter indicates the usage of each GANSS positioning method that was attempted to determine the location estimate either successfully or unsuccessfully. If GANSS Positioning Data received from the RAN contains no GANSS method, GERAN GANSS Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is GERAN, see 3GPP TS 23.271 [26a].

UTRAN GANSS Positioning Data

This parameter indicates the usage of each GANSS positioning method that was successfully attempted to determine the location estimate. If Position Data received from the RAN contains no GANSS Positioning Data Set, UTRAN GANSS Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

UTRAN Additional Positioning Data

This parameter indicates the usage of each Additional positioning method that was successfully attempted to determine the location estimate. If Position Data received from the RAN contains no Additional Positioning Data Set, UTRAN Additional Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

UTRAN Barometric Pressure Measurement

This parameter indicates the uncompensated barometric pressure measurement at the MS. The absence of this parameter implies that a barometric pressure measurement was not available or could not be successfully obtained. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

UTRAN Civic Address

This parameter indicates the civic address of the MS. The absence of this parameter implies that a civic address was not available or could not be successfully obtained. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

Age of Location Estimate

This parameter indicates how long ago the location estimate was obtained.

Additional Location Estimate

This parameter provides the location estimate when not provided by the Location Estimate parameter. It may be sent only if the parameter Supported GAD Shapes has been received in the Provide Subscriber Location indication and the shape to be included is supported by the GMLC.

Deferred MT-LR Response Indicator

See definition in clause 7.6.11.2.

Cell Id Or SAI

For GERAN access, this parameter indicates Global Cell Identifier of the cell that the served subscriber is currently attached to. For UTRAN access, this parameter contains the Service Area Identifier for the cell that the subscriber is currently attached to. This parameter is included only for North American Emergency Calls as described in 3GPP TS 23.271 [26a].

Accuracy Fulfilment Indicator

See definition in clause 7.6.11.28.

MO-LR Short Circuit Indicator

This parameter indicates whether MO-LR Short Circuit is permitted for periodic location.

Reporting PLMN List

This parameter indicates a list of PLMNs in which subsequent periodic MO-LR TTTP requests will be made.

Periodic LDR information

This parameter indicates the reporting amount and reporting interval of deferred periodic location.

Target Serving Node for Handover

This parameter provides the address of the target side serving node for handover of an IMS Emergency Call.

User error

This parameter is sent by the responder when the location request has failed or cannot proceed and if present, takes one of the following values defined in clause 7.6.1.

- System Failure;
- Data Missing;
- Unexpected Data Value;
- Facility Not Supported;
- Unidentified Subscriber;
- Illegal Subscriber;
- Illegal Equipment;
- Absent Subscriber (diagnostic information may also be provided);
- Unauthorised requesting network;

- Unauthorised LCS Client with detailed reason;
- Position method failure with detailed reason.

Provider error

These are defined in clause 7.6.1.

13A.3 MAP-SUBSCRIBER-LOCATION-REPORT Service

13A.3.1 Definition

This service is used by a VMSC or SGSN to provide the location of a target MS to a GMLC when a request for location is either implicitly administered or made at some earlier time. This is a confirmed service using the primitives from table 13A.3/1.

13A.3.2 Service Primitives

Table 13A.3/1: Subscriber_Location_Report

Parameter name	Request	Indication	Response	Confirm
Invoke id	M	M(=)	M(=)	M(=)
LCS Event	M	M(=)		
LCS Client ID	M	M(=)		
Network Node Number	M	M(=)		
IMSI	C	C(=)		
MSISDN	C	C(=)		
NA-ESRD	C	C(=)	C	C(=)
NA-ESRK	C	C(=)	C	C(=)
IMEI	U	C(=)		
Location Estimate	C	C(=)		
GERAN Positioning Data	C	C(=)		
UTRAN Positioning Data	C	C(=)		
GERAN GANSS Positioning Data	C	C(=)		
UTRAN GANSS Positioning Data	C	C(=)		
UTRAN Additional Positioning Data	C	C(=)		
UTRAN Barometric Pressure Measurement	C	C(=)		
UTRAN Civic Address	C	C(=)		
Age of Location Estimate	C	C(=)		
LMSI	U	C(=)		
GPRS Node Indicator	C	C(=)		
Additional Location Estimate	C	C(=)		
Deferred MT-LR Data	C	C(=)		
LCS-Reference Number	C	C(=)	C	C(=)
NA-ESRK Request	C	C(=)		
Cell Id Or SAI	C	C(=)		
H-GMLC Address	C	C(=)	C	C(=)
LCS Service Type Id	C	C(=)		
Pseudonym Indicator	C	C(=)		
Accuracy Fulfilment Indicator	C	C(=)		
Sequence Number	C	C(=)		
Periodic LDR Info	C	C(=)		
MO-LR Short Circuit Indicator	C	C(=)	C	C(=)
Target Serving Node for Handover	C	C(=)		
Reporting PLMN List			C	C(=)

User error			C	C(=)
Provider error				O

13A.3.3 Parameter Definition and Use

All parameters are defined in clause 7.6. The use of these parameters and the requirements for their presence are specified in. 3GPP TS 23.271 [26a].

LCS Event

This parameter indicates the event that triggered the Subscriber Location Report.

LCS Client ID

This parameter provides information related to the identity of the recipient LCS client.

Network Node Number

See definition in clause 7.6.2. This parameter provides the address of the sending node.

IMSI

The IMSI shall be provided if available to the VMSC or SGSN.

MSISDN

The MSISDN shall be provided if available to the VMSC or SGSN.

NA-ESRD

If the target MS has originated an emergency service call in North America, the NA-ESRD shall be provided by the VMSC if available.

If the target MS has originated an emergency service call in North America and NA-ESRK Request is included in Subscriber_Location_Report-Arg, an NA-ESRK or NA-ESRD, but not both, may also be included in the response to the MSC, see 3GPP TS 23.271 [26a].

NA-ESRK

If the target MS has originated an emergency service call in North America, the NA-ESRK shall be provided by the VMSC if assigned.

If the target MS has originated an emergency service call in North America and NA-ESRK Request is included in Subscriber_Location_Report-Arg, an NA-ESRK or NA-ESRD, but not both, may also be included in the response to the MSC, see 3GPP TS 23.271 [26a].

IMEI

The requirements for its presence are specified in 3GPP TS 23.271 [26a].

Location Estimate

This parameter provides the location estimate. The absence of this parameter implies that a location estimate was not available or could not be successfully obtained. If the obtained location estimate is not encoded in one of the supported geographical shapes then this parameter shall consist of one octet, which shall be discarded by the receiving node.

GERAN Positioning Data

This parameter indicates the usage of each positioning method that was attempted to determine the location estimate either successfully or unsuccessfully. If Positioning Data received from the RAN contains no Positioning Methods, Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is GERAN, see 3GPP TS 23.271 [26a].

UTRAN Positioning Data

This parameter indicates the usage of each positioning method that was successfullyattempted to determine the location estimate. If Position Data received from the RAN contains no Positioning Methods, UTRAN Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

GERAN GANSS Positioning Data

This parameter indicates the usage of each GANSS positioning method that was attempted to determine the location estimate either successfully or unsuccessfully. If GANSS Positioning Data received from the RAN contains no GANSS method, GERAN GANSS Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is GERAN, see 3GPP TS 23.271 [26a].

UTRAN GANSS Positioning Data

This parameter indicates the usage of each GANSS positioning method that was successfully attempted to determine the location estimate. If Position Data received from the RAN contains no GANSS Positioning Data Set, UTRAN GANSS Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

UTRAN Additional Positioning Data

This parameter indicates the usage of each Additional positioning method that was successfully attempted to determine the location estimate. If Position Data received from the RAN contains no Additional Positioning Data Set, UTRAN Additional Positioning Data is excluded from the MAP message. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

UTRAN Barometric Pressure Measurement

This parameter indicates the uncompensated barometric pressure measurement at the MS. The absence of this parameter implies that a barometric pressure measurement was not available or could not be successfully obtained. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

UTRAN Civic Address

This parameter indicates the civic address of the MS. The absence of this parameter implies that a civic address was not available or could not be successfully obtained. It may be included in the message only if the access network is UTRAN, see 3GPP TS 23.271 [26a].

Age of Location Estimate

This parameter indicates how long ago the location estimate was obtained.

LMSI

The LMSI may be provided if assigned by the VLR.

GPRS Node Indicator

See definition in clause 7.6.8. This presence of this parameter is mandatory only if the SGSN number is sent in the Network Node Number.

Additional Location Estimate

This parameter provides the location estimate when not provided by the Location Estimate parameter..

Deferred MT-LR Data

See definition in clause 7.6.11.3.

LCS-Reference Number

This parameter shall be included if the Subscriber Location Report is the response to a deferred MT location request.

NA-ESRK Request

If the target MS has originated an emergency service call in North America, NA-ESRK Request may be included to indicate that the MSC is able to accept NA-ESRK in the Response message, see section 7.6.11.19.

Cell Id Or SAI

For GERAN access, this parameter indicates Global Cell Identifier of the cell that the served subscriber is currently attached to. For UTRAN access, this parameter contains the Service Area Identifier for the cell that the subscriber is currently attached to. This parameter is included only for Emergency Calls as described in 3GPP TS 23.271 [26a].

H-GMLC address

See definition in clause 7.6.2. The parameter shall be included if the Subscriber Location Report is the response to a deferred MT location request for a UE available event, an area event or a periodic positioning event. This parameter shall be included in a Subscriber Location Report response if a deferred MO-LR TTTP procedure is initiated for a periodic positioning event.

LCS Service Type Id

See definition in clause 7.6.11.15. The requirements for its presence are specified in 3GPP TS 23.271 [26a].

Pseudonym Indicator

This parameter indicates by its presence that the pseudonym is required. Refer to 3GPP TS 23.271 [26a].

Accuracy Fulfilment Indicator

For a mobile terminated periodic LDR, this parameter indicates whether the obtained location estimate satisfies the requested accuracy or not, provided that this indication is obtained from RAN or the UE with the location estimate.

Periodic LDR Information

This parameter refers to the periodic reporting interval and reporting amount of the deferred periodic location.

MO-LR Short Circuit Indicator

This parameter indicates whether MO-LR Short Circuit is permitted for periodic location.

Reporting PLMN List

This parameter indicates a list of PLMNs in which subsequent periodic MO-LR TTTP requests will be made.

Sequence Number

This parameter refers to the number of the periodic location reports completed. The sequence number would be set to 1 in the first location report and increment by 1 for each new report. When the number reaches the reporting amount value, the H-GMLC (for a periodic MT-LR or a periodic MO-LR transfer to third party) will know the procedure is complete. For details see 3GPP TS 23.271 [26a].

Target Serving Node for Handover

This parameter provides the address of the target side serving node for handover of an IMS Emergency Call.

User error

This parameter is sent by the responder when the received message contains an error, cannot be forwarded or stored for an LCS client or cannot be accepted for some other reason and if present, takes one of the following values defined in clause 7.6.1.

- System Failure;
- Data Missing;
- Unexpected Data Value;
- Resource Limitation;
- Unknown Subscriber;
- Unauthorised requesting network;
- Unknown or unreachable LCS Client.

Provider error

These are defined in clause 7.6.1.

13A.4 Void

13A.4.1 Void

13A.4.2 Void

13A.4.3 Void

13A.5 Void

13A.5.1 Void

13A.5.2 Void

13A.5.3 Void

13A.6 Void

13A.6.1 Void

13A.6.2 Void

13A.6.3 Void

13A.7 Void

13A.7.1 Void

13A.7.2 Void

13A.7.3 Void

13A.8 Void

13A.8.1 Void

13A.8.2 Void

13A.8.3 Void

13A.9 Void

13A.9.1 Void

13A.9.2 Void

13A.9.3 Void

14 General

14.1 Overview

Clauses 14 to 17 specify the protocol elements to be used to provide the MAP services described in clause 7.

Clause 15 specifies the elements of procedures for the MAP protocol. Clause 16 specifies the mapping onto TC service primitives. Clause 17 specifies the application contexts, operation packages and abstract syntaxes for the MAP protocol as well as the encoding rules to be applied.

14.2 Underlying services

The MAP protocol relies on the services provided by the Transaction Capabilities (TC) of Signalling System Number No. 7, as referenced in clause 6.

14.3 Model

The MAP Protocol Machine (MAP PM) can be modelled as a collection of service state machines (SSMs) - one per MAP specific service invoked - coordinated by a MAP dialogue control function with its one state machine: MAP dialogue state machine (DSM). There are two types of Service State Machines: Requesting Service State Machines (RSM) and Performing Service State Machines (PSM).

A new invocation of a MAP PM is employed on the receipt of a MAP-OPEN request primitive or a TC-BEGIN indication primitive. Each invocation controls exactly one MAP dialogue. For each MAP specific service invoked during a dialogue, a MAP RSM is created at the requestor's side and a MAP PSM is created at the performer's side.

This modelling is used only to facilitate understanding and the MAP behaviour descriptions and is not intended to suggest any implementation. SDL descriptions are organised according to this model.

How the MAP-service-user and the MAP refer to a MAP dialogue (i.e. a MAP PM invocation) is a local implementation matter.

How TC dialogue identifiers are assigned to a MAP PM invocation is also a local implementation matter.

14.4 Conventions

The behaviour of the MAP PM depends on the application-context-name associated with the dialogue. One major difference is that the MAP requests the transfer of the application-context-name by TC only for those contexts which do not belong to the so-called "version one context set".

The "version one context set" is a set of application-contexts which model the behaviour of a MAP V1 implementation according to the latest phase 1 version of GSM 09.02. This set is defined in clause 15.

The procedures described in clause 15 are used when the application-context-name does not refer to a dialogue between an MSC and its VLR. When the application-context-name refers to a dialogue between an MSC and its VLR the MAP PM procedures are a local implementation matter.

15 Elements of procedure

15.1 Handling of unknown operations

Unknown operations (i.e. a standard operation introduced in a later version of the MAP specification, or a private operation) can be introduced into MAP in a backwards compatible way. This means that the receiver of an unknown operation shall, if the dialogue state allows it, send a TC-REJECT component to the sender of the operation indicating 'unrecognised operation' and continue with the processing of further components or messages exchanged within the dialogue as if the unknown operation had not been received.

The standardised structure of a MAP dialogue shall not be affected by the invocation of unknown operations, i.e. if a dialogue uses only a TC-BEGIN message which is acknowledged by a TC-END message, a TC-CONTINUE message shall not be used to invoke an unknown operation. However the standardised structure of a MAP dialogue may be affected by the rejection of unknown operations, i.e. if a dialogue uses only a TC-BEGIN message which is acknowledged by a TC-END message, a TC-CONTINUE message followed by a TC-END message may be used to carry the rejection of an unknown operation and the response to the standardised operation. The entity which initiated a dialogue whose standardised structure is a TC-BEGIN message which is acknowledged by a TC-END message shall not send any messages in that dialogue after the TC-BEGIN. Note that if the dialogue structure is affected as described in this paragraph the TC-CONTINUE shall include the dialogue portion required to confirm the acceptance of the dialogue.

Unknown operations may be invoked in the following types of message (there is no restriction as to how many unknown operations can be invoked in a message):

- TC-BEGIN: the component to invoke the unknown operation shall follow the component of the standard operation which is included in this message.
- TC-CONTINUE: the component to invoke the unknown operation may be transported as the only component in a stand-alone message or may be grouped with existing operations. In the latter case a specific sequencing of components is not required.
- TC-END: if the component to invoke the unknown operation is grouped with an existing operation a specific sequencing of components is not required

The TC-REJECT component may be sent in the following messages:

- TC-CONTINUE or TC-END: either as the only component of the message or grouped with an existing component. The choice is up to the MAP-Service User.

  If the received message contains only unknown operations the MAP-Service User shall send the TC-REJECT components in a TC-CONTINUE message to the peer entity, if the dialogue state allows it.

  If the received message contains unknown operations and standard operations and the standardised structure of the dialogue requires the response to the standard operation to be sent within a TC-END message, then the MAP-Service User may send the response to the standard operations and the TC-REJECT components for the unknown operations in a TC-CONTINUE message followed by a TC-END message. Neither a specific distribution of the components to the TC messages nor a specific sequencing of components is required.

Note that the SDL diagrams of clauses 19 - 25 do not show the report to the MAP-Service User about the reception of the unknown operation. This has been done for simplicity of description; the MAP PM may inform the MAP-Service User.

The sender of the unknown operation shall ensure that there is enough room in the used message for the unknown operation.

15.2 Dialogue establishment

The establishment of a MAP dialogue involves two MAP-service-users: the dialogue-initiator and the dialogue-responder.

This procedure is driven by the following signals:

- a MAP-OPEN request primitive from the dialogue-initiator;
- a TC-BEGIN indication primitive occurring at the responding side;
- a MAP-OPEN response primitive from the dialogue-responder;
- the first TC-CONTINUE indication primitive occurring at the initiating side;

and under specific conditions:

- a TC-END indication primitive occurring at the initiating side;
- a TC-U-ABORT indication primitive occurring at the initiating side;
- a TC-P-ABORT indication primitive occurring at the initiating side.

One instance of the MAP dialogue state machine runs at the initiating side, and one at the responding side.

15.2.1 Behaviour at the initiating side

The behaviour of the MAP dialogue state machine at the initiating side is defined in sheets 1 – 8 of the process MAP_DSM (figure 15.6/3).

Sheet 3: When the MAP dialogue state machine at the initiating side is waiting for a response from the responding side, a TC-END indication which echoes the AC name which was sent in the TC-BEGIN indicates acceptance of the dialogue.

Sheet 3: If the dialogue opening is accepted, any components included in the TC-END are processed and passed to the MAP-Service User. The dialogue is closed by sending a MAP-CLOSE to the MAP-Service User.

Sheet 3, sheet 4, sheet 5, sheet 6, sheet 7, sheet 8: when a dialogue is terminated, the MAP dialogue state machine terminates all instances of the Requesting_MAP_SSM which are active for this dialogue.

Sheet 4: A TC-P-ABORT with an abort parameter Incorrect_Transaction_Portion indicates that the responding side does not support a MAP version higher than 1. This triggers a MAP-OPEN confirm indicating that the dialogue is refused, with a refuse reason potential version incompatibility. The MAP-Service User may then decide to retry the dialogue at MAP version 1.

Sheet 8: When the MAP dialogue state machine at the initiating side is waiting for a response from the responding side, a TC-CONTINUE indication which echoes the AC name which was sent in the TC-BEGIN indicates acceptance of the dialogue.

Sheet 8: If the dialogue opening is accepted, any components included in the TC-CONTINUE are processed and passed to the MAP-Service User. The dialogue has then reached the established state.

15.2.2 Behaviour at the responding side

The behaviour of the MAP dialogue state machine at the responding side is defined in sheets 0 – 14 of the process MAP_DSM (figure 15.6/3).

Sheet 9: If no application context information is included in the TC-BEGIN indication, this implies a MAP version 1 dialogue. An explicit application context indicating version 1 is treated as abnormal behaviour.

Sheet 11: The v1 application context name which corresponds to a v1 operation is derived using the information in table 15.2/1.

Table 15.2/1: Mapping of V1 operation codes on to application-context-names

Operation	Application-context-name (note 1)
updateLocation	networkLocUpContext-v1
cancelLocation	locationCancellationContext-v1
provideRoamingNumber	roamingNumberEnquiryContext-v1
insertSubscriberData	subscriberDataMngtContext-v1
deleteSubscriberData	subscriberDataMngtContext-v1
sendParameters	infoRetrievalContext-v1
	networkLocUpContext-v1 (note 2)
beginSubscriberActivity	networkFunctionalSsContext-v1
sendRoutingInfo	locationInfoRetrievalContext-v1
performHandover	handoverControlContext-v1
reset	resetContext-v1
activateTraceMode	tracingContext-v1
deactivateTraceMode	tracingContext-v1
sendRoutingInfoForSM	shortMsgGatewayContext-v1
forwardSM	shortMsgRelayContext-v1
reportSM-deliveryStatus	shortMsgGatewayContext-v1
noteSubscriberPresent	mwdMngtContext-v1
alertServiceCentreWithoutResult	shortMsgAlertContext-v1
checkIMEI	EquipmentMngtContext-v1

NOTE 1: These symbolic names refer to the object identifier value defined in clause 17 and allocated to each application-context used for the MAP.
NOTE 2: The choice between the application contexts is based on the parameters received in the operation.

Sheet 12: If the dialogue is accepted, each component present in the TC-BEGIN is forwarded to an instance of a Performing_MAP_SSM, by executing the procedure Process_Components.

Sheet 13: If the MAP dialogue state machine receives a MAP-OPEN response with a result accepted, it waits for any MAP specific service request or response primitives or a MAP-DELIMITER request.

Sheet 13, sheet 14: When a dialogue is terminated, the MAP dialogue state machine terminates all instances of the Requesting_MAP_SSM or Performing_MAP_SSM which are active for this dialogue.

Sheet 14: A MAP-DELIMITER request triggers a TC-CONTINUE request to accept the dialogue. The dialogue has then reached the established state.

15.3 Dialogue continuation

Once established the dialogue is said to be in a continuation phase. The behaviour of the MAP dialogue state machine in this phase is defined in sheets 15 – 17 of the process MAP_DSM (figure 15.6/3).

Both MAP users can request the transfer of MAP APDUs until one of them requests the termination of the dialogue.

Normal closure of an established dialogue is shown on sheet 16; abnormal termination is shown on sheet 17.

15.4 Load control

If an entity which should respond to a MAP dialogue opening request is overloaded, it uses the AC of the request to determine whether to discard the request.

The priority level allocated to each application-context is described in clause 5, tables 5.1/1, 5.1/2, and 5.1/3.

15.5 Procedures for MAP specific services

This clause describes the MAP procedures for MAP specific services. These procedures are driven by the following types of event:

- a MAP specific request or a MAP specific response primitive;

- a component handling primitive from TC.

A Service State Machine is activated when of one of the following signals is received:

- a MAP request primitive, which activates a requesting SSM;
- a TC-INVOKE indication primitive without a linked identifier, which activates a performing SSM.

For component handling primitives there are two types of event:

- events which activate a Service State Machine or which can be related to an existing one;
- events which cannot be related to a Service State Machine.

15.5.1 Service invocation

The behaviour of the requesting SSM which handles a service is defined by the SDL for the process Requesting_MAP_SSM. The requesting SSM receives a MAP service request from the MAP-Service User via the MAP dialogue state machine and sends a TC-INVOKE request to TCAP. When a confirm is received from TCAP via the MAP dialogue state machine, the requesting SSM forwards a MAP service confirm to the MAP-Service User.

The response to a MAP service invocation may come in the form of a linked request. If the linked request corresponds to a class 4 operation, this is handled by the requesting SSM. If the linked request corresponds to a class 1, 2 or 3 operation, the MAP dialogue state machine sends a notification to the requesting SSM and creates an instance of a performing SSM to handle the linked request. The test "Linked_Operation_Allowed" on sheet 3 of the process Requesting_MAP_SSM takes the (TRUE) exit if the definition of the parent operation includes the received linked operation as a permitted linked operation; otherwise the test takes the (FALSE) exit.

The mapping of MAP specific services on to remote operations is given in table 16.2/1.

15.5.2 Void

15.5.3 Service invocation receipt

The behaviour of the performing SSM which handles a service is defined by the SDL for the process Performing_MAP_SSM. The performing SSM receives a TC-INVOKE component from TCAP via the MAP dialogue state machine and sends a MAP service indication to the MAP-Service User. When a MAP service response is received from the MAP-Service User via the MAP dialogue state machine, the performing SSM forwards a TC-RESULT or TC-U-ERROR component to TCAP.

15.5.4 Void

15.5.5 Handling of components received from TC

The procedure Process_Components shows the handling of components received in a TC-BEGIN, TC-CONTINUE or TC-END message.

Sheet 2: If a linked invoke component corresponds to a class 4 operation, the MAP dialogue state machine sends it to the requesting SSM instance identified by the linked invoke ID. If a linked invoke component corresponds to any other class of operation, the MAP dialogue state machine sends a notification to the requesting SSM instance identified by the linked invoke ID, creates an instance of a performing SSM and sends the invoke component to it.

15.6 SDL descriptions

The following SDL specification describes a system which includes three blocks: MAP-user, MAP-provider and TC.

Such a system resides in each network component supporting MAP and communicates with its peers via the lower layers of the signalling network which are part of the environment.

Only the MAP-provider is fully described in this clause. The various types of processes which form the MAP-User block and the TC block are described respectively in clauses 18 to 25 of the present document and in CCITT Recommendation Q.774.

The MAP-Provider block communicates with the MAP_USER via two channels U1 and U2. Via U1 the MAP-provider receives the MAP request and response primitives. Via U2 it sends the MAP indication and confirm primitives.

The MAP-Provider block communicates with TC via two channels P1 and P2. Via P1 the MAP-Provider sends all the TC request primitives. Via P2 it receives all the TC indication primitives.

The MAP-Provider block is composed of the four following types of process:

a) MAP_DSM: This type of process handles a dialogue for transport of MAP messages. There exists one process instance per MAP dialogue.

b) Load_Ctrl: This type of process is in charge of load control. There is only one instance of this process in each system.

c) Requesting_MAP_SSM: This type of process handles a MAP service requested during a dialogue. An instance of this process is created by the instance of the MAP_DSM process for each requested MAP service.

d) Performing_MAP_SSM: This type of process handles a MAP service performed during a dialogue. An instance of this process is created by the instance of the MAP_DSM process for each MAP service to be performed.

A process MAP_DSM exchanges external signals with other blocks as well as internal signals with the other processes of the MAP-Provider block. The external signals are either MAP service primitives or TC service primitives.

The signal routes used by the various processes are organised as follows:

a) A process MAP_DSM receives and sends events from/to the MAP_user via signal route User1/User2. These routes use channels U1 and U2 respectively.

b) A process MAP_DSM receives and sends events from/to the TCAP via signal route TC1/TC2. These routes use channels P1 and P2 respectively.

c) A process MAP_DSM receives and sends events from/to the LOAD_CTRL process via signal route Load1/Load2. These routes are internal.

d) A process MAP_DSM sends events to the Performing_MAP_SSM processes via signal route Intern1. This route is internal.

e) A process MAP_DSM sends events to the Requesting_MAP_SSM processes via signal route Intern2. This route is internal.

f) A process Performing_MAP_SSM sends events to the MAP_USER via signal route User3. This route uses channel U2.

g) A process Performing_MAP_SSM sends events to the TCAP via signal route TC3. This route uses channel P1.

h) A process Requesting_MAP_SSM sends events to the MAP_USER via signal route User4. This route uses channel U2.

i) A process Requesting_MAP_SSM sends events to the TCAP via signal route TC4. This route uses channel P1.

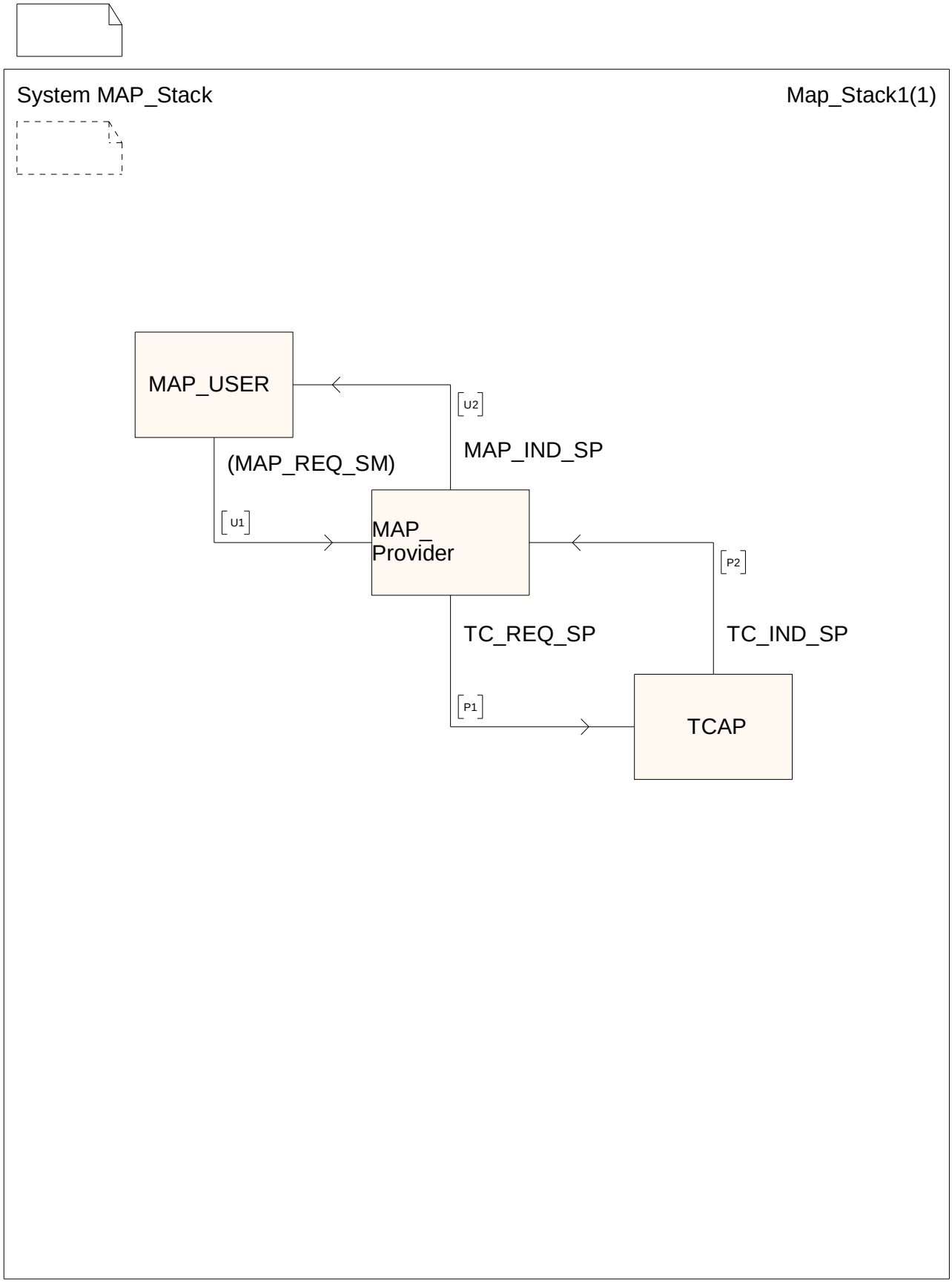


Figure 15.6/1: System MAP_Stack

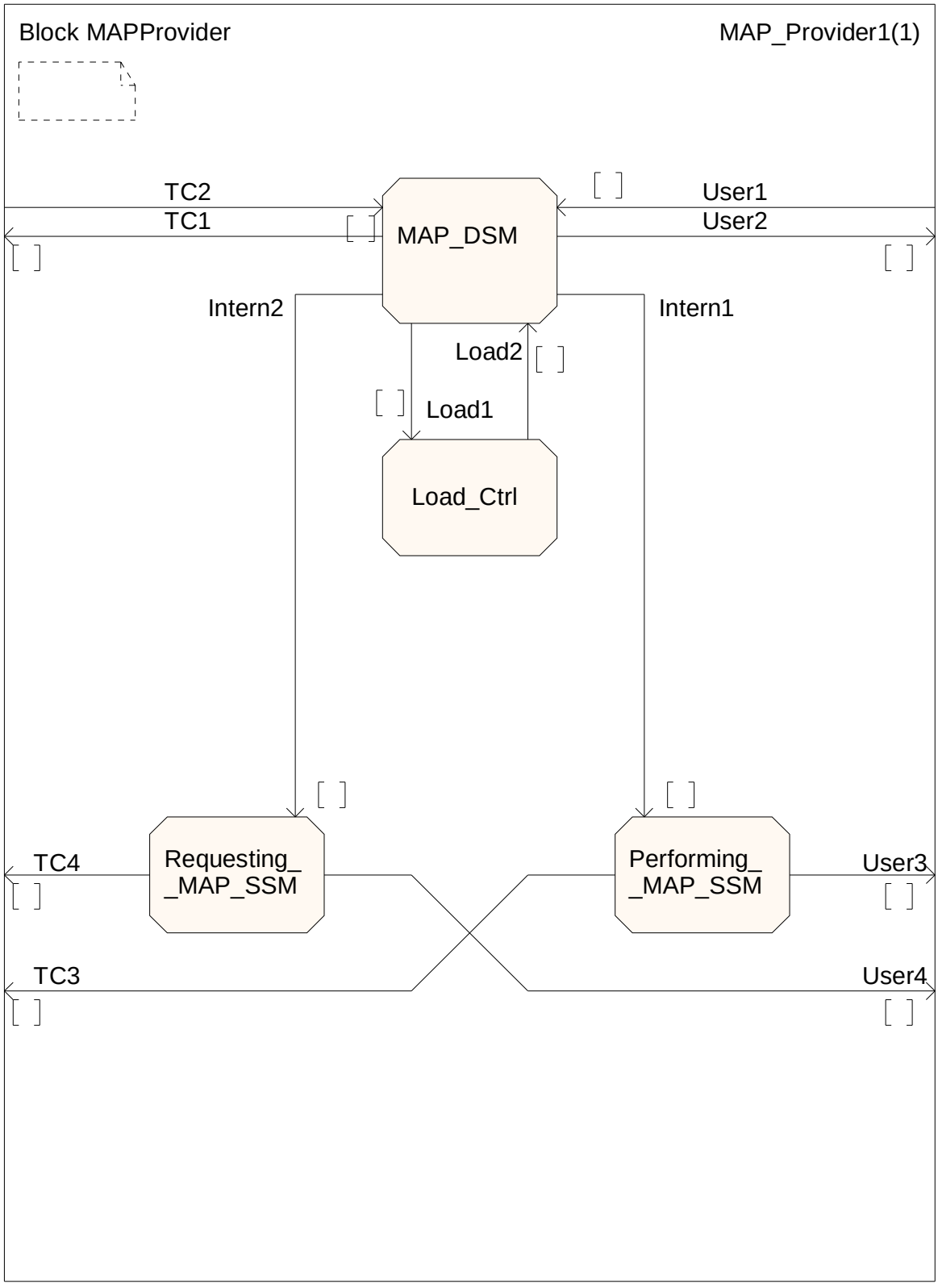


Figure 15.6/2: Block MAP_Provider

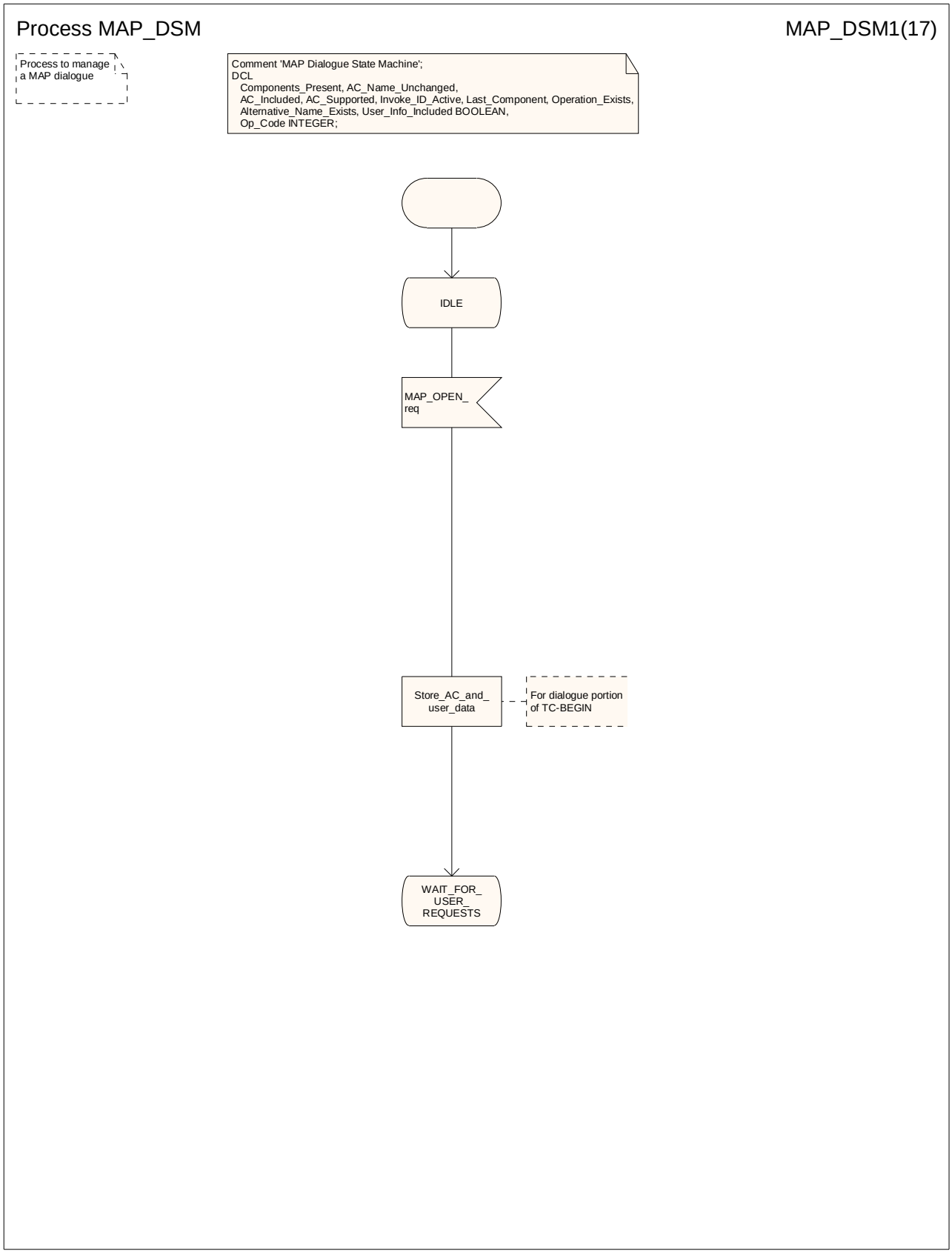


Figure 15.6/3a: Process MAP_DSM (sheet 1)

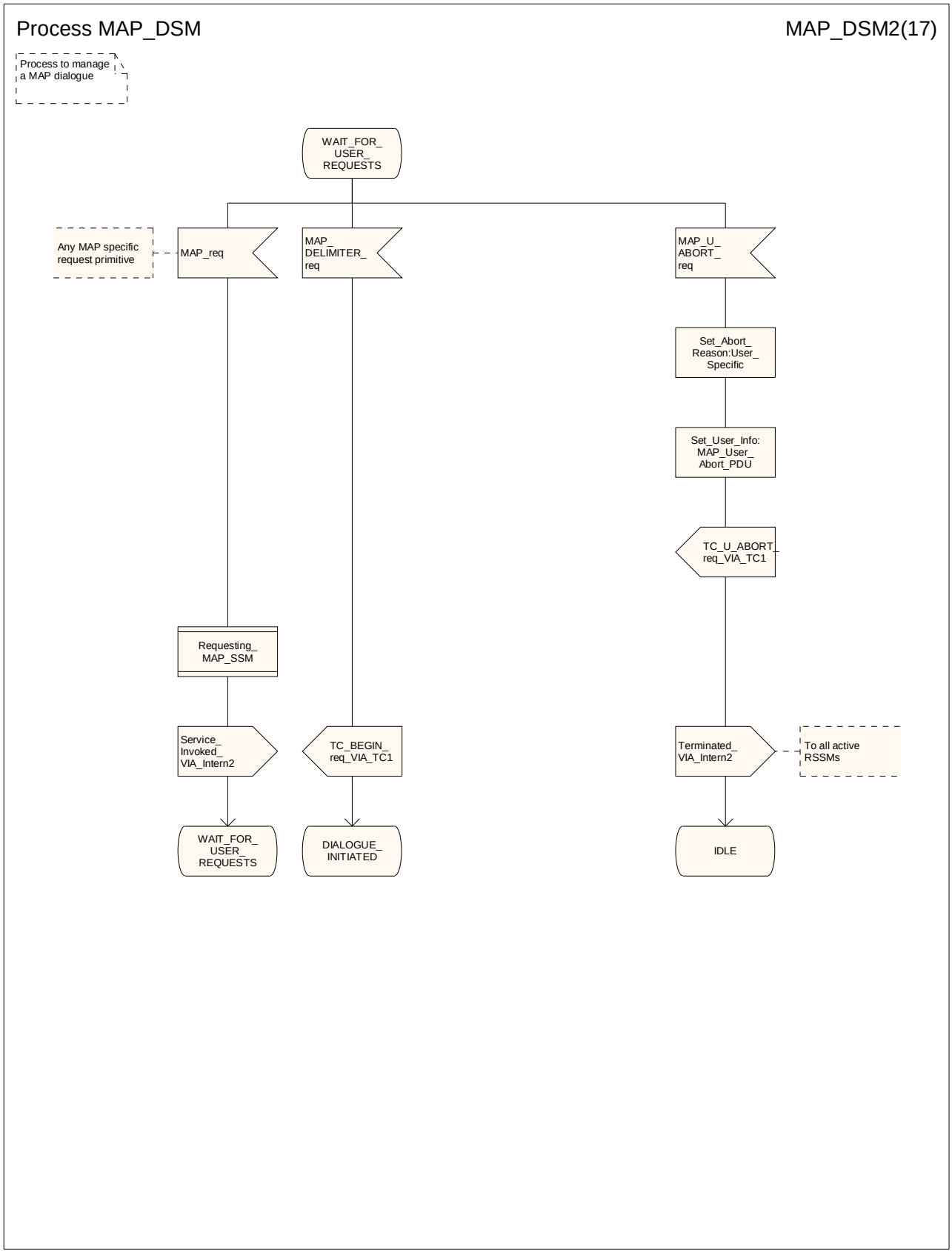


Figure 15.6/3b: Process MAP_DSM (sheet 2)

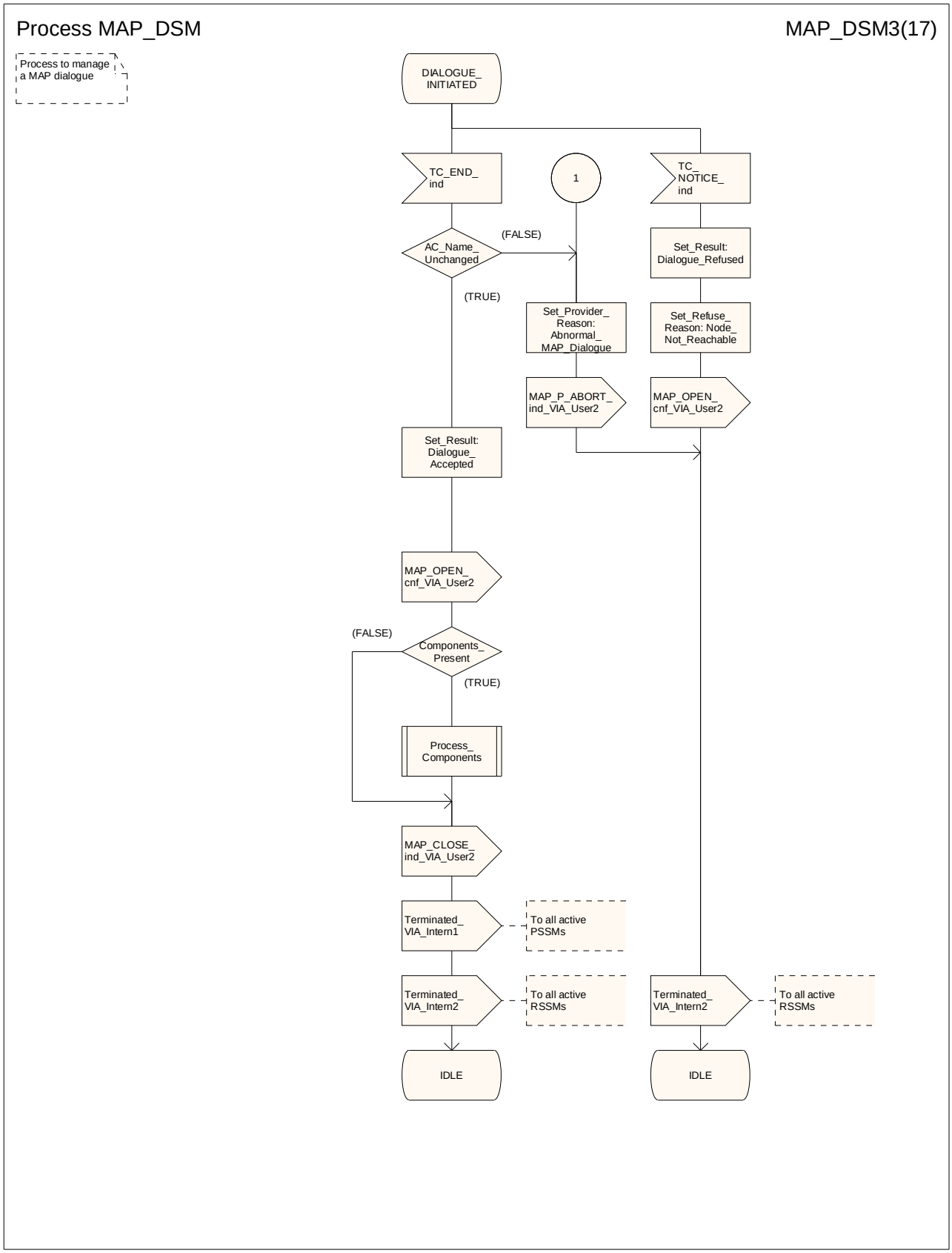


Figure 15.6/3c: Process MAP_DSM (sheet 3)

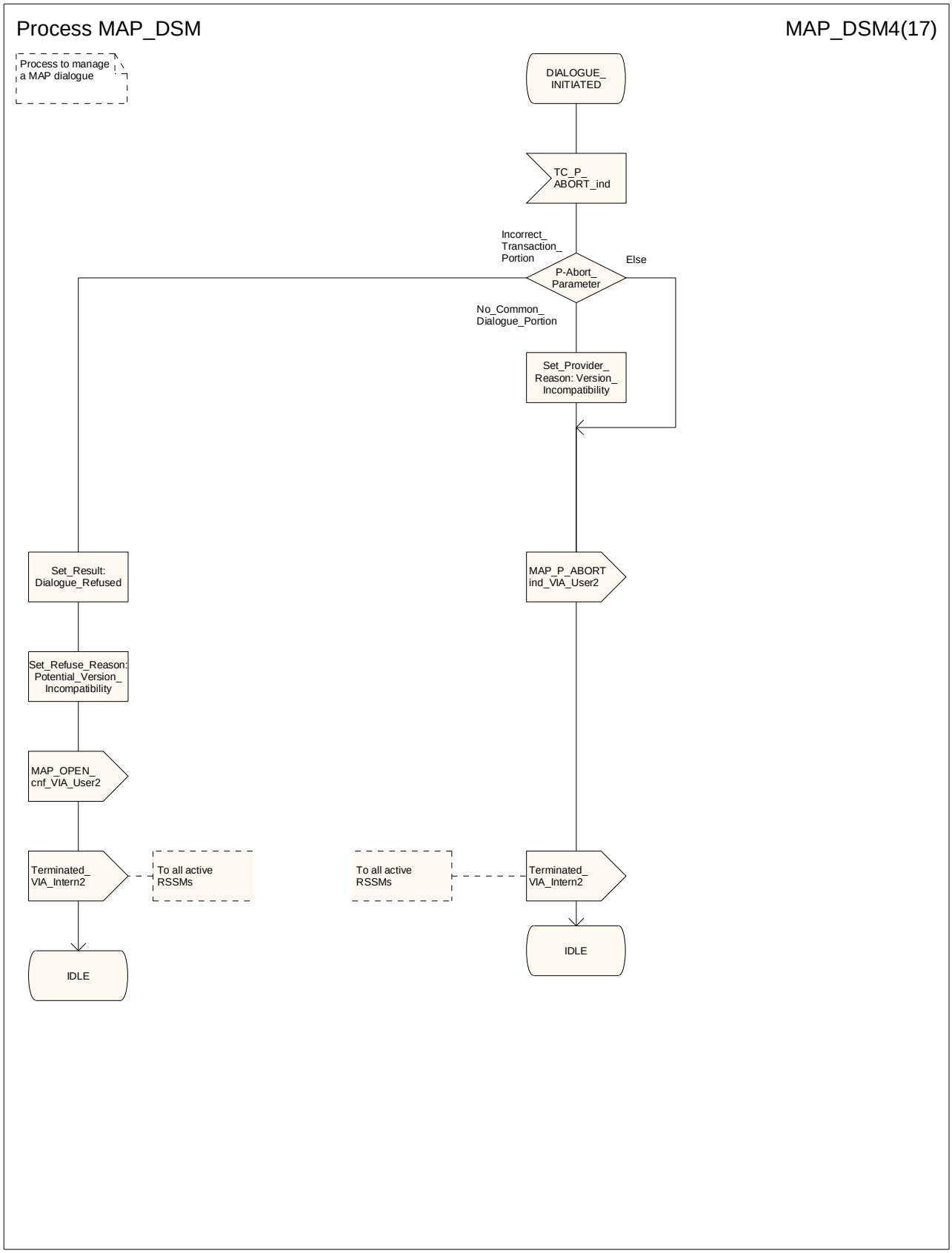


Figure 15.6/3d: Process MAP_DSM (sheet 4)

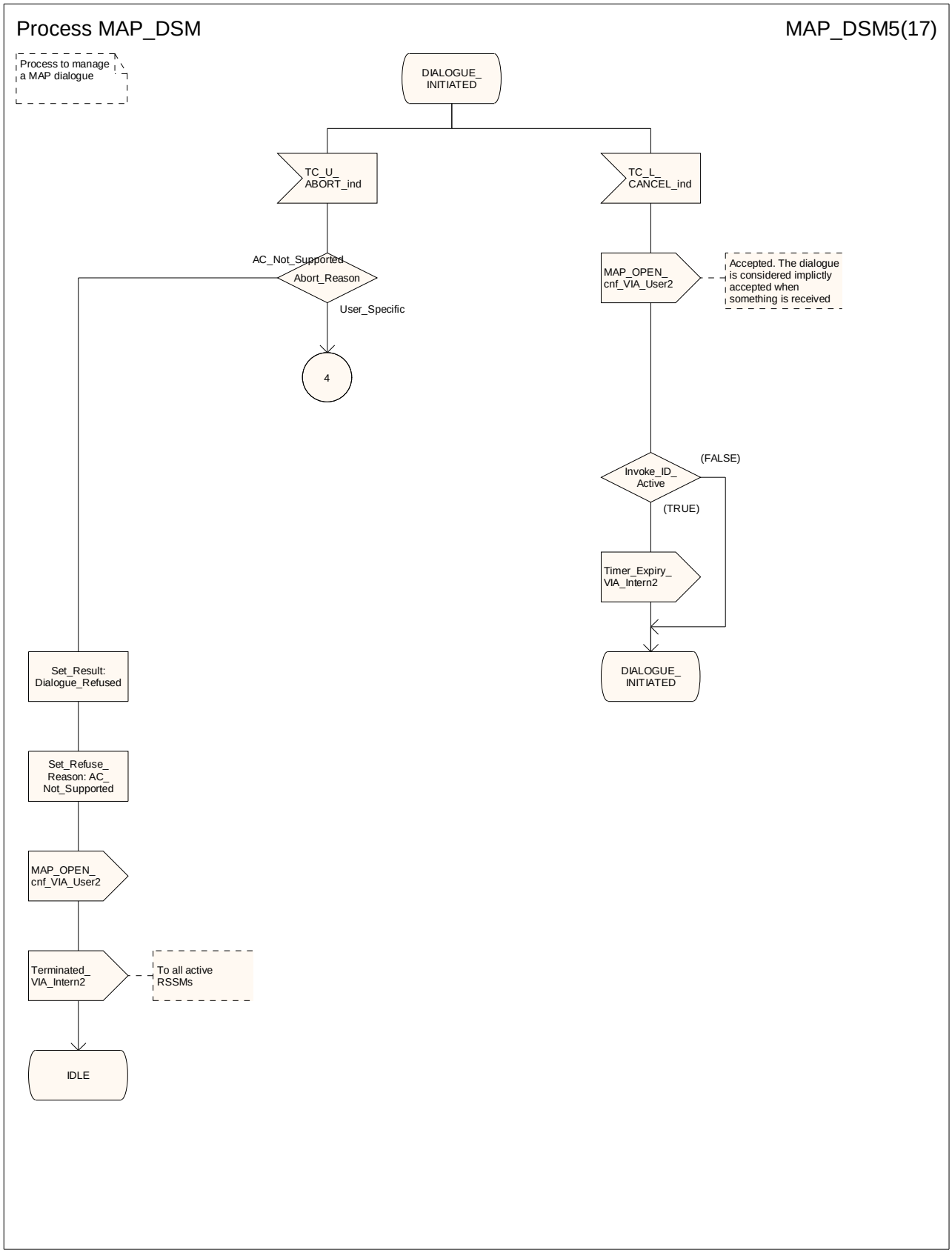


Figure 15.6/3e: Process MAP_DSM (sheet 5)

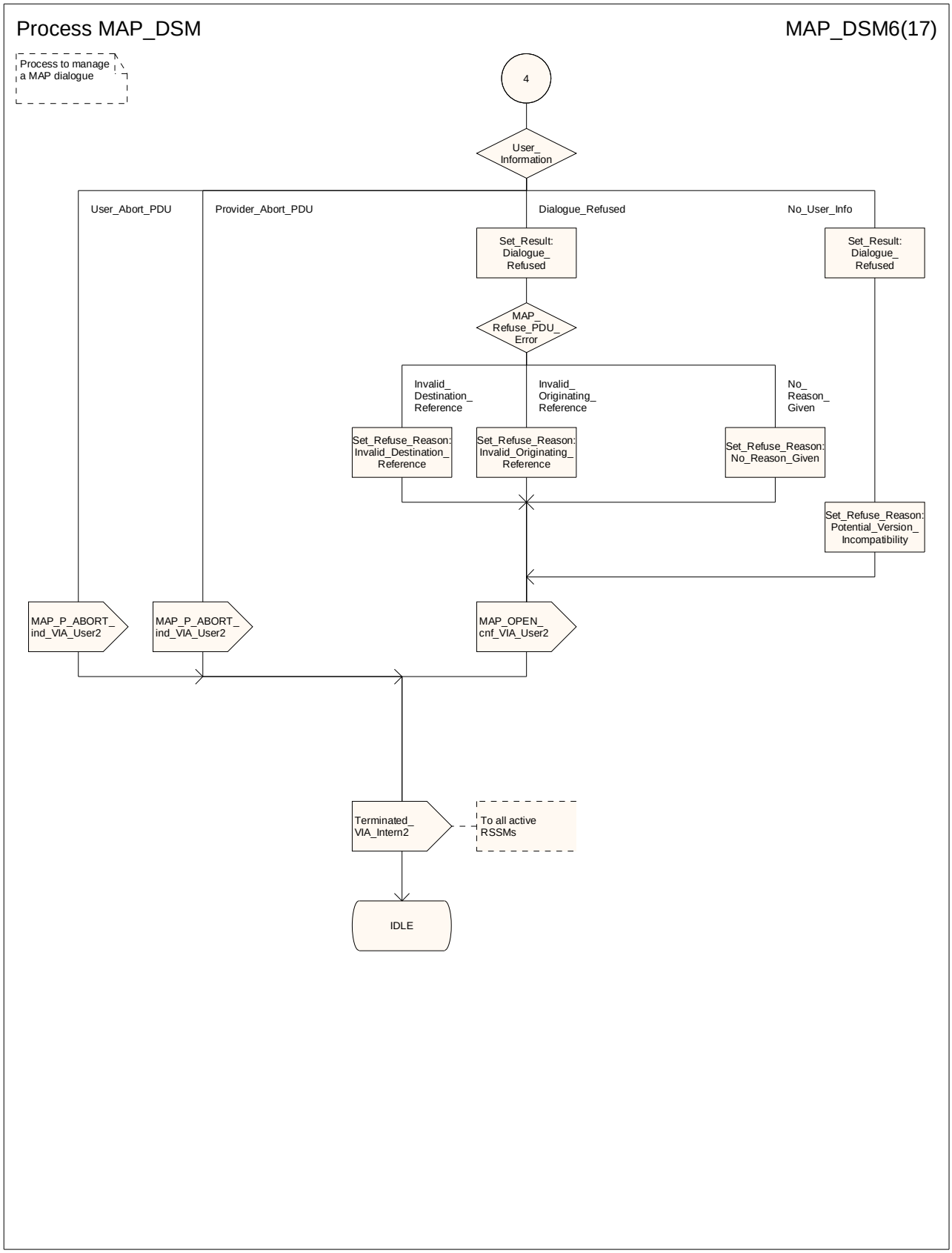


Figure 15.6/3f: Process MAP_DSM (sheet 6)

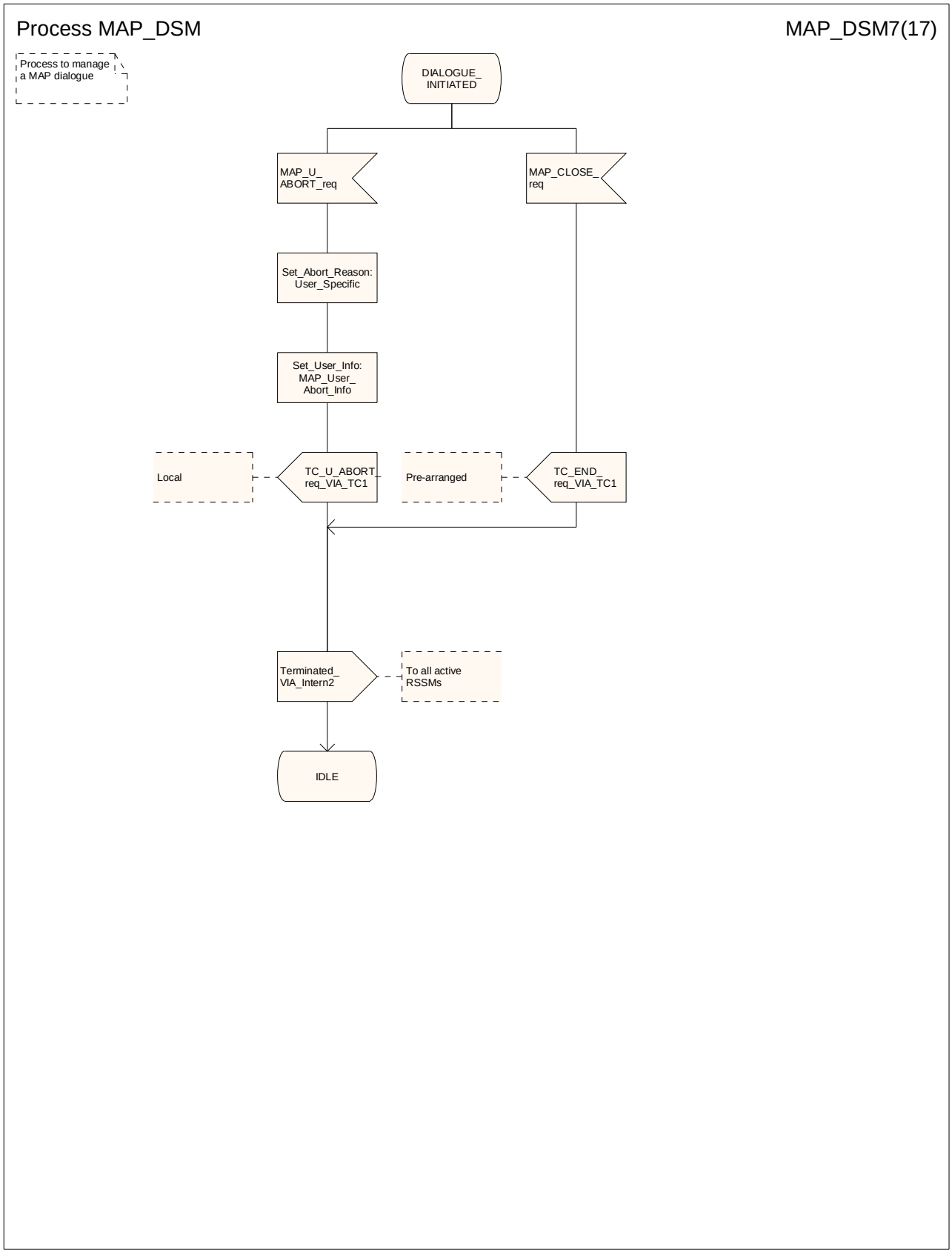


Figure 15.6/3g: Process MAP_DSM (sheet 7)

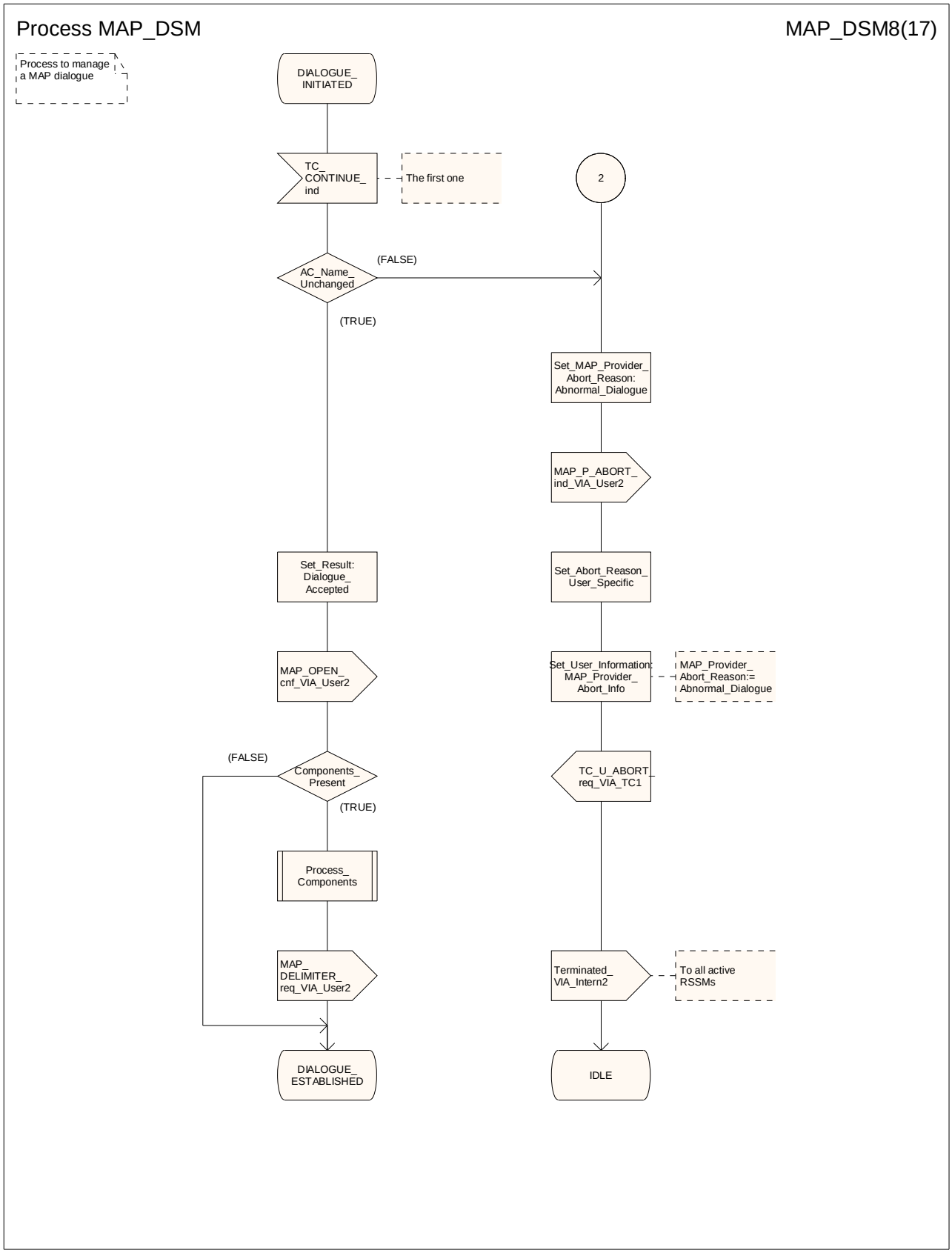


Figure 15.6/3h: Process MAP_DSM (sheet 8)

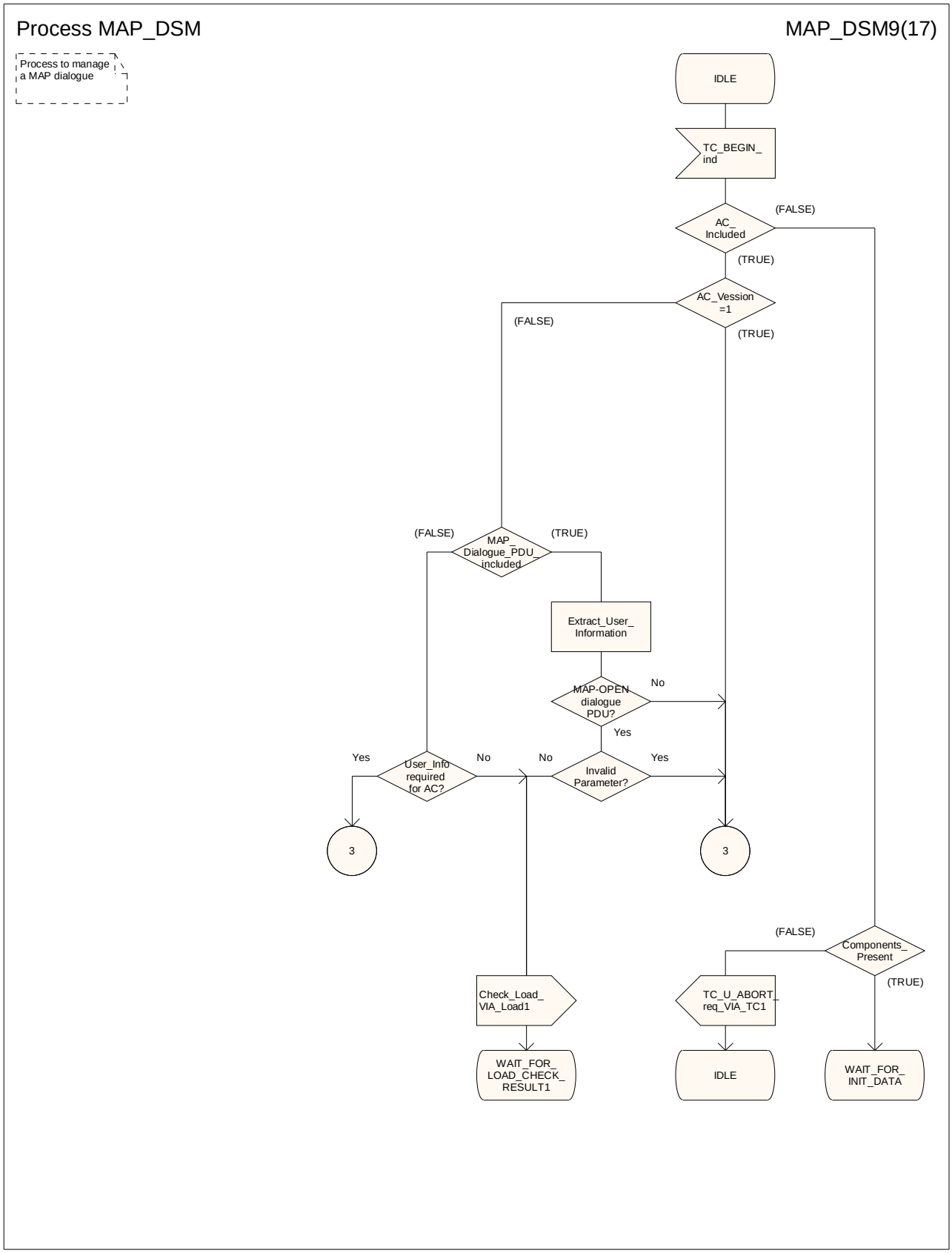


Figure 15.6/3i: Process MAP_DSM (sheet 9)

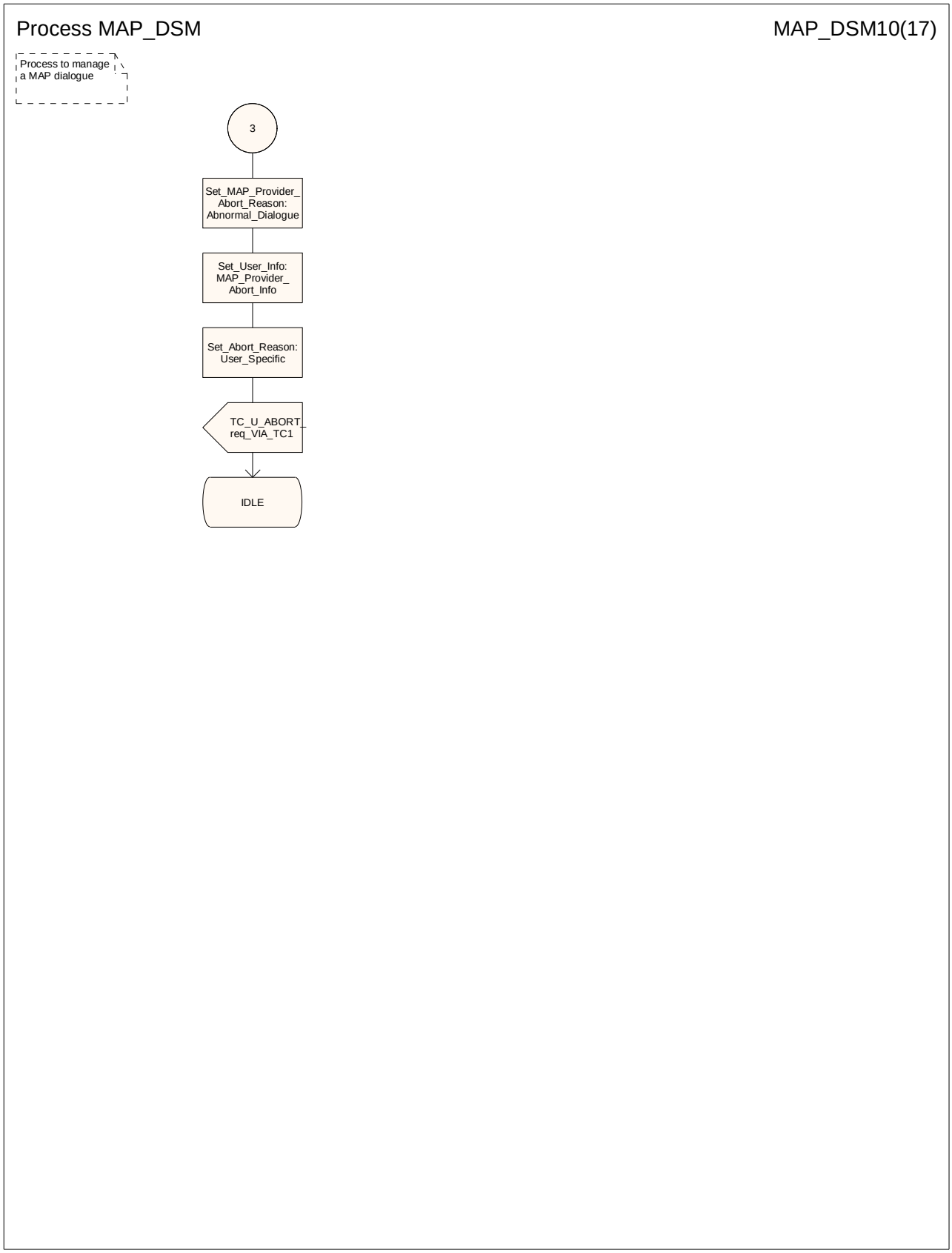


Figure 15.6/3j: Process MAP_DSM (sheet 10)

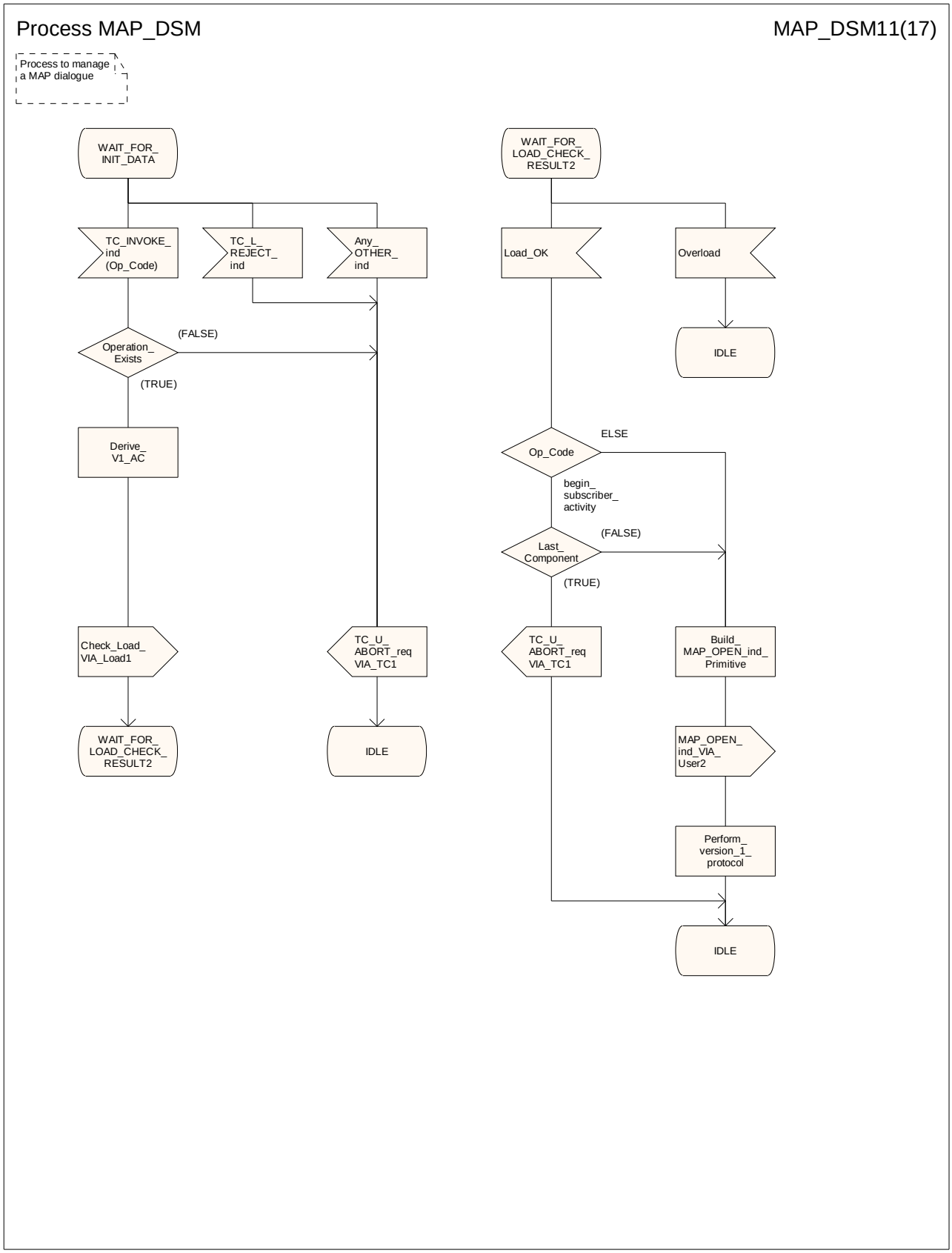


Figure 15.6/3k: Process MAP_DSM (sheet 11)

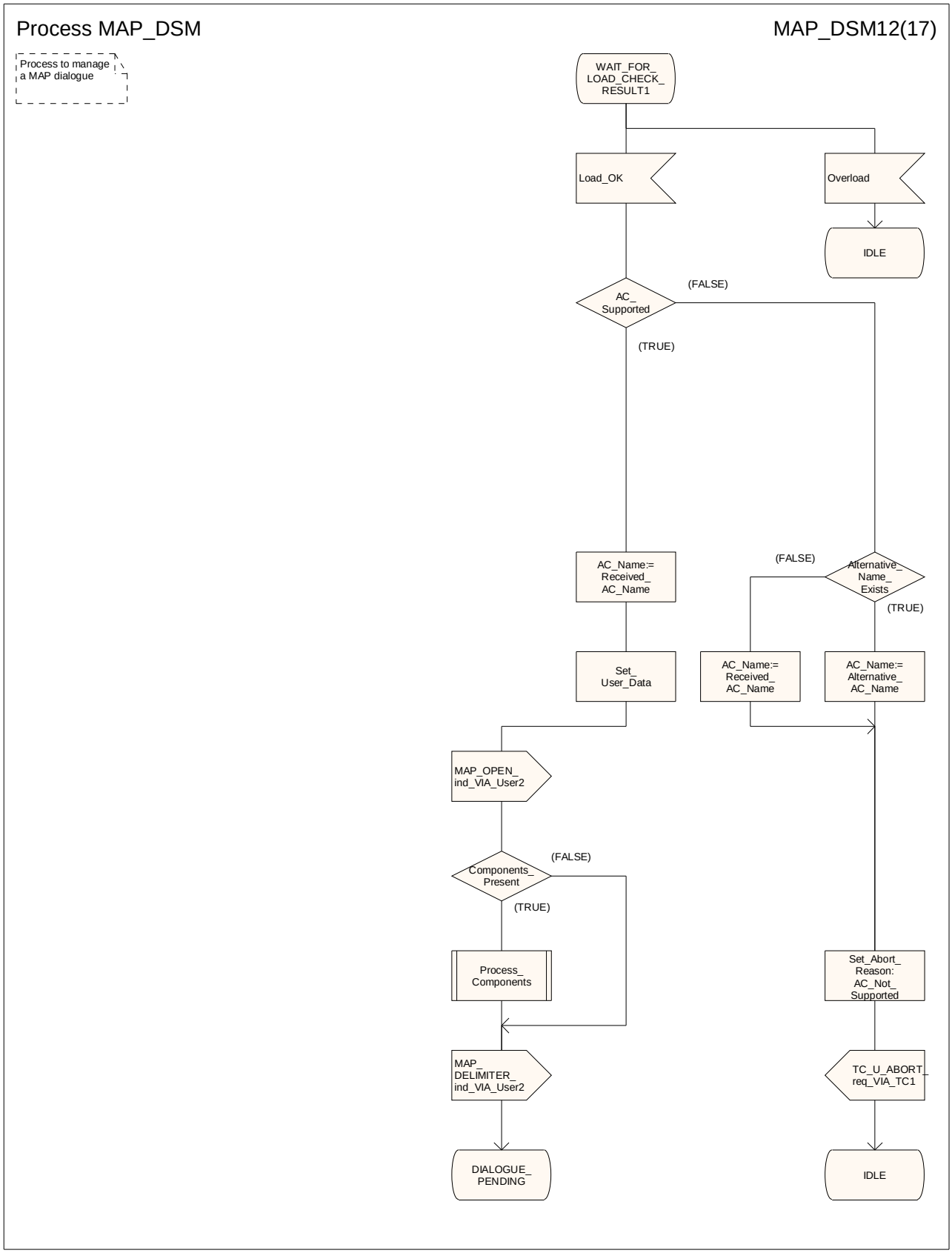


Figure 15.6/3l: Process MAP_DSM (sheet 12)

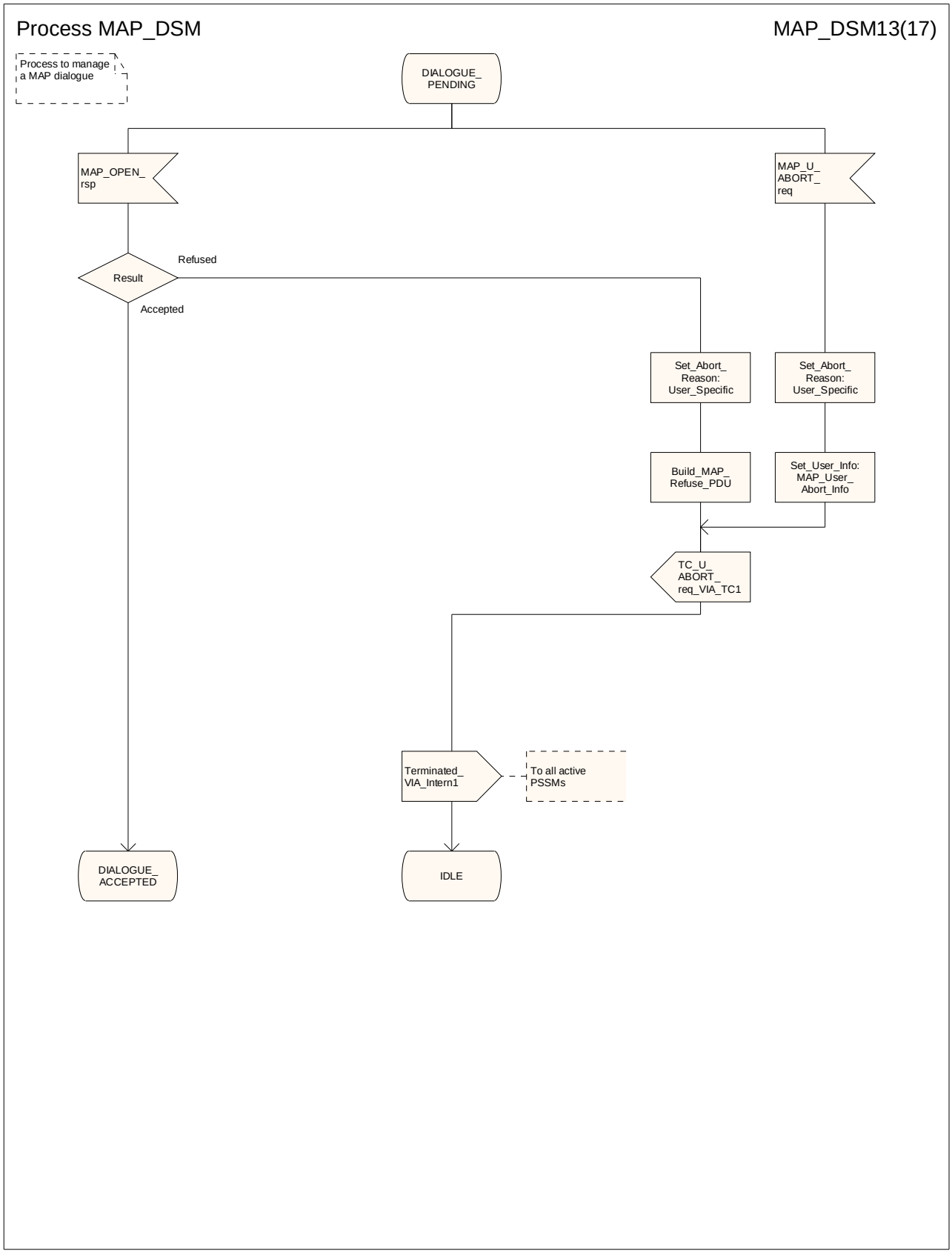


Figure 15.6/3m: Process MAP_DSM (sheet 13)

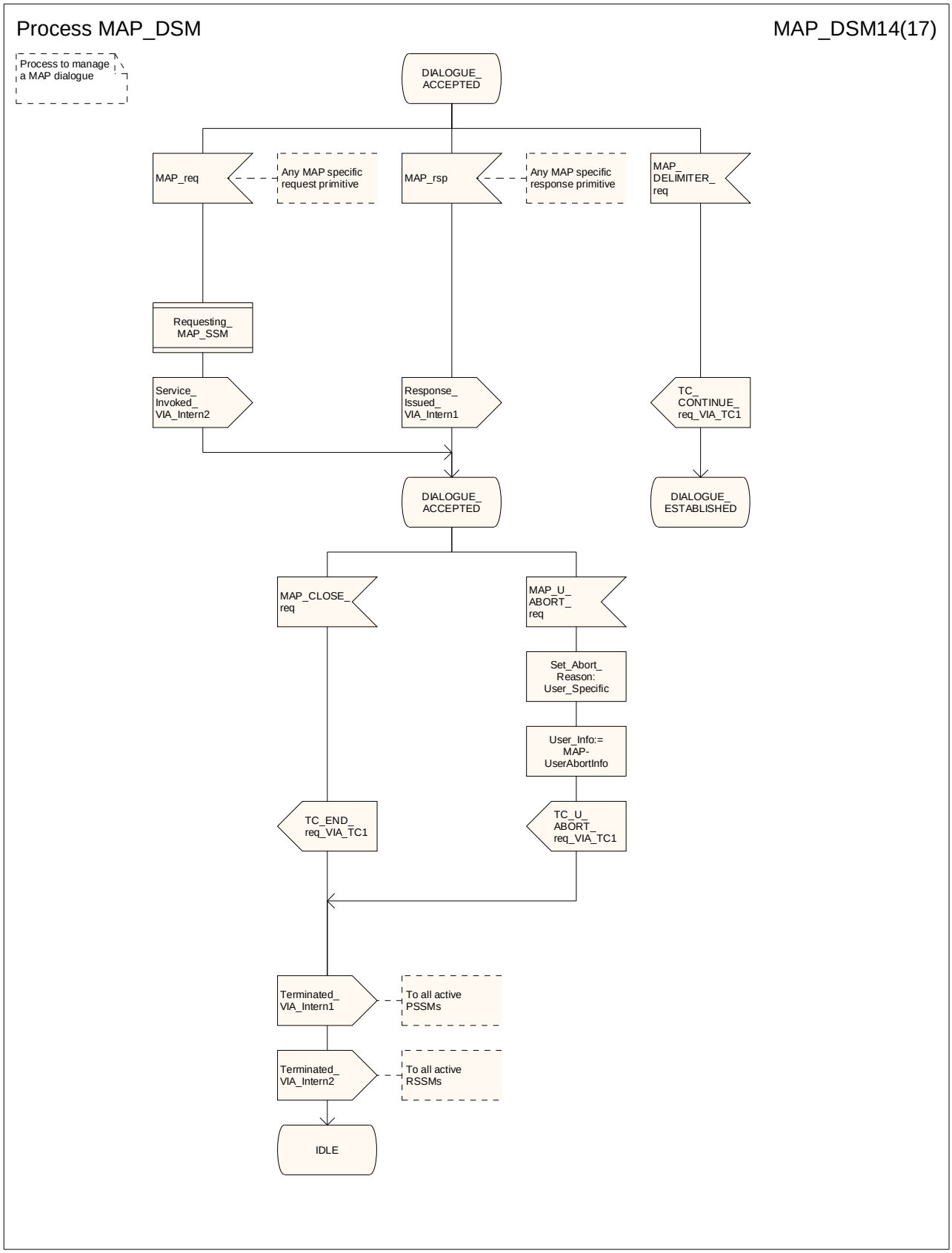


Figure 15.6/3n: Process MAP_DSM (sheet 14)

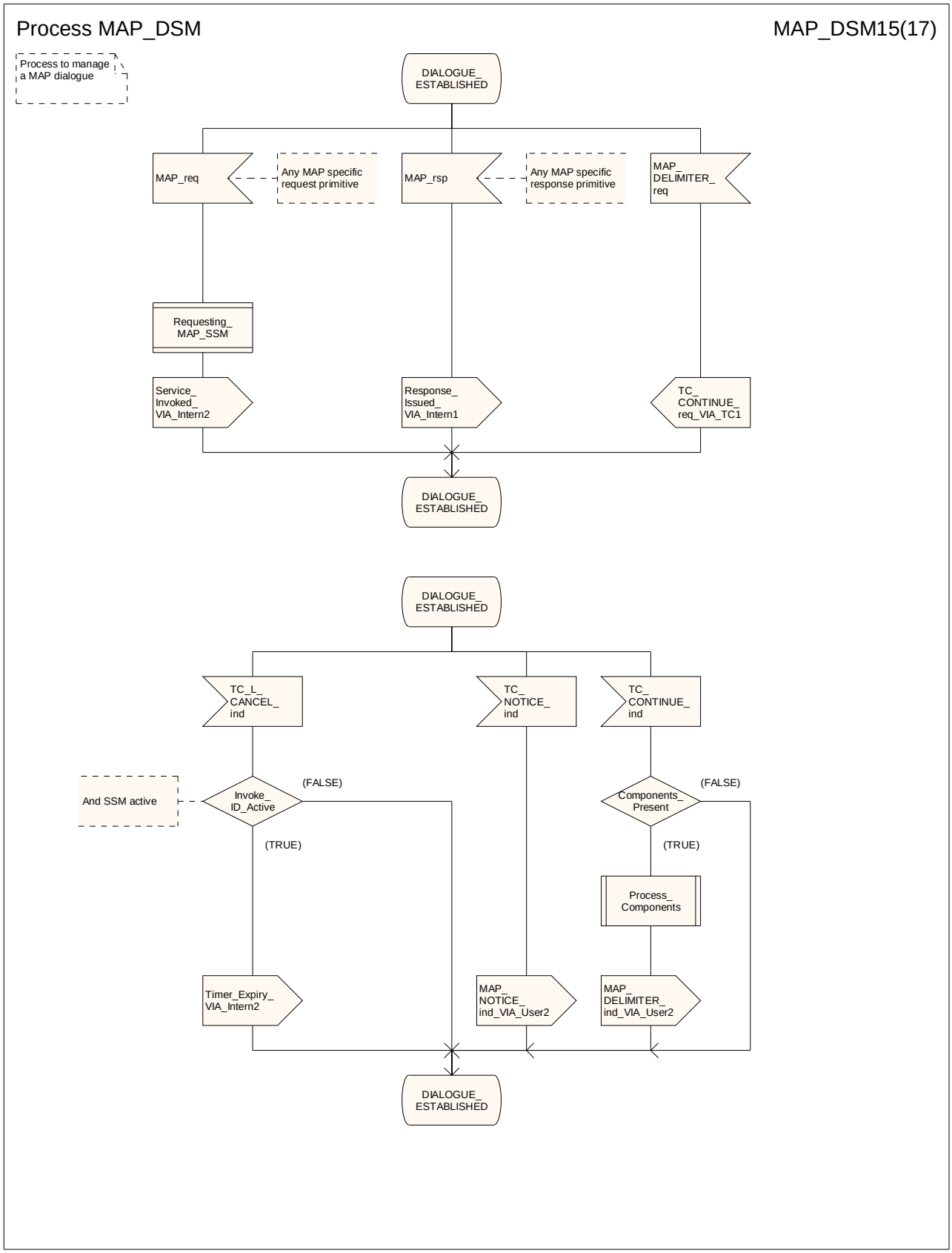


Figure 15.6/30: Process MAP_DSM (sheet 15)

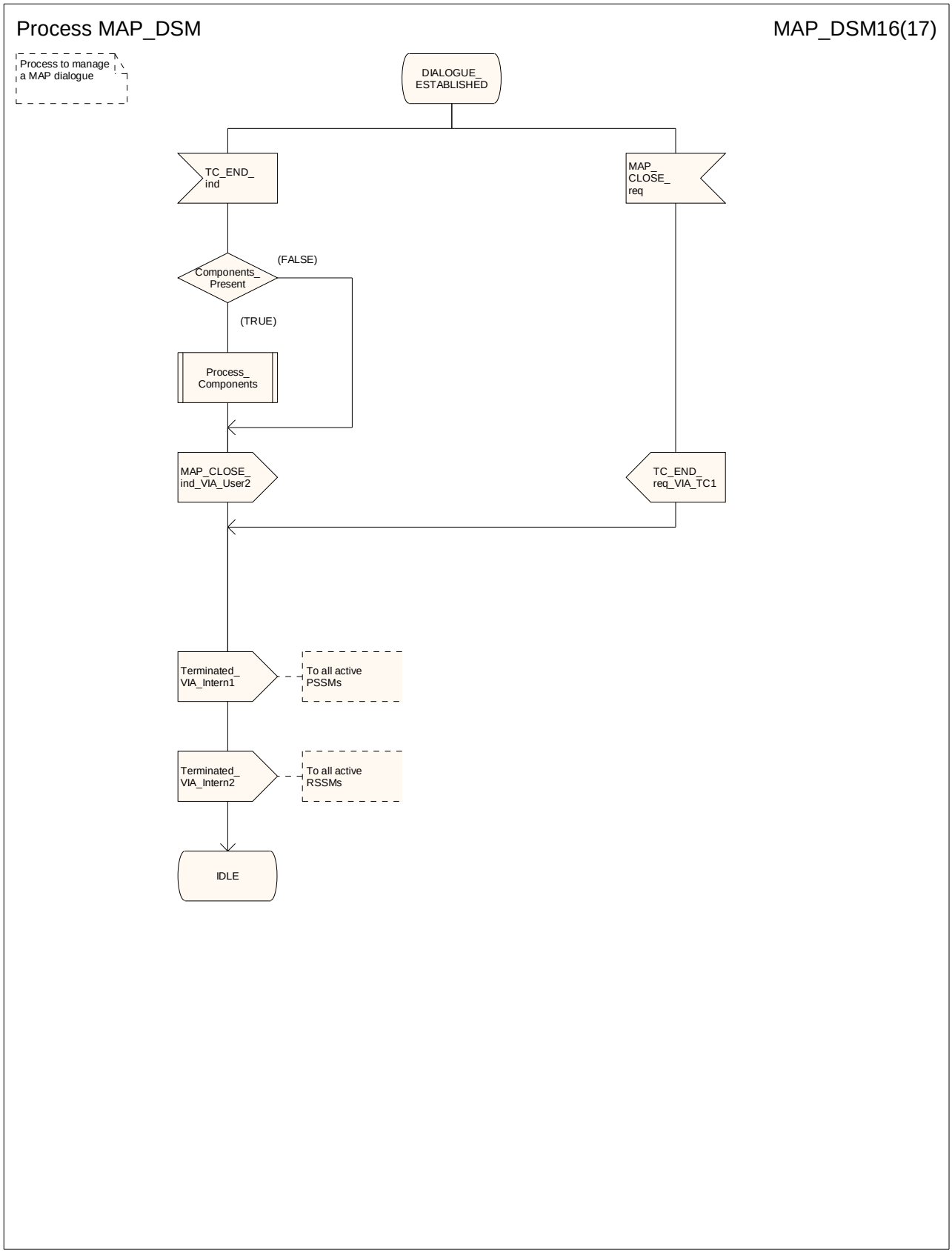


Figure 15.6/3p: Process MAP_DSM (sheet 16)

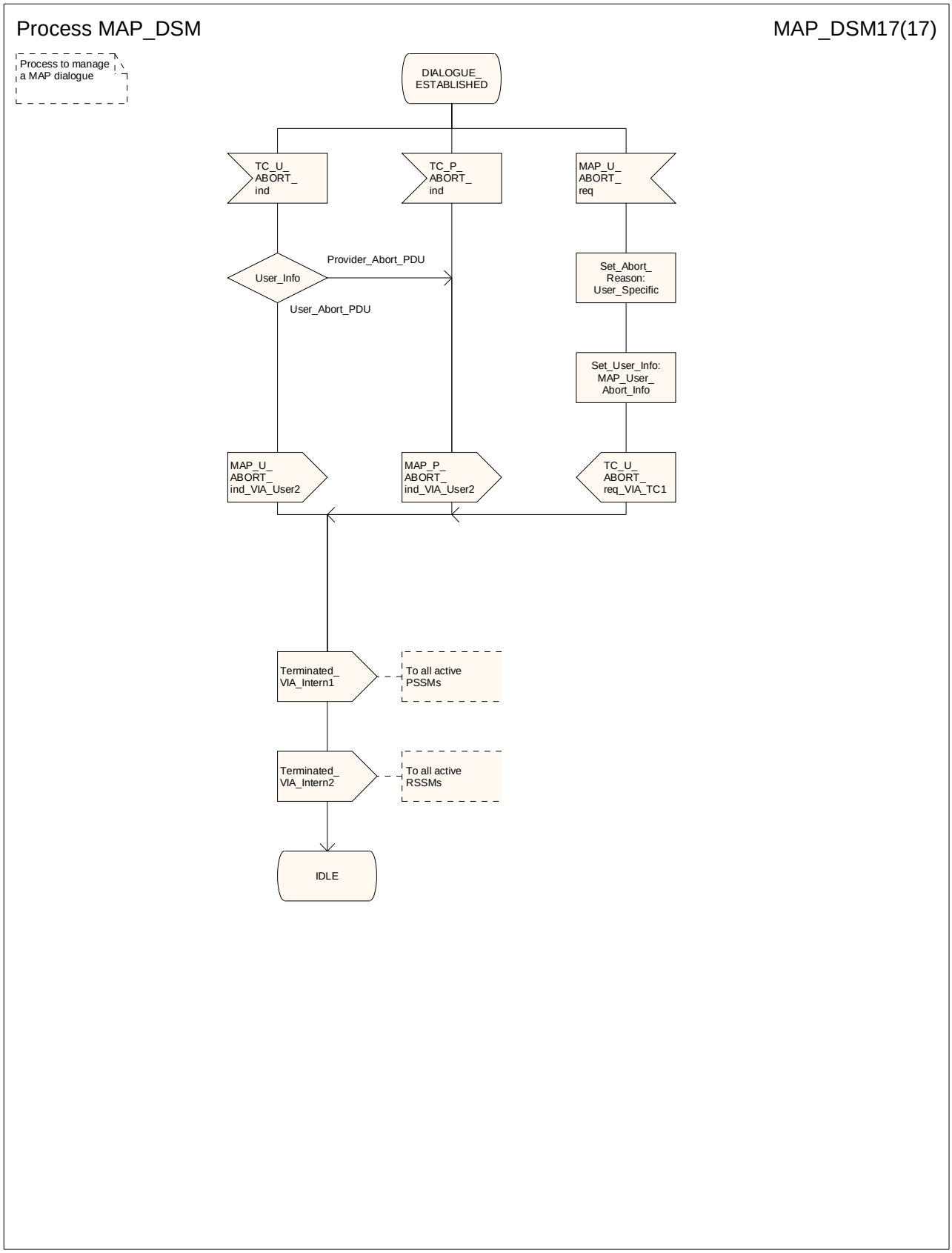


Figure 15.6/3q: Process MAP_DSM (sheet 17)

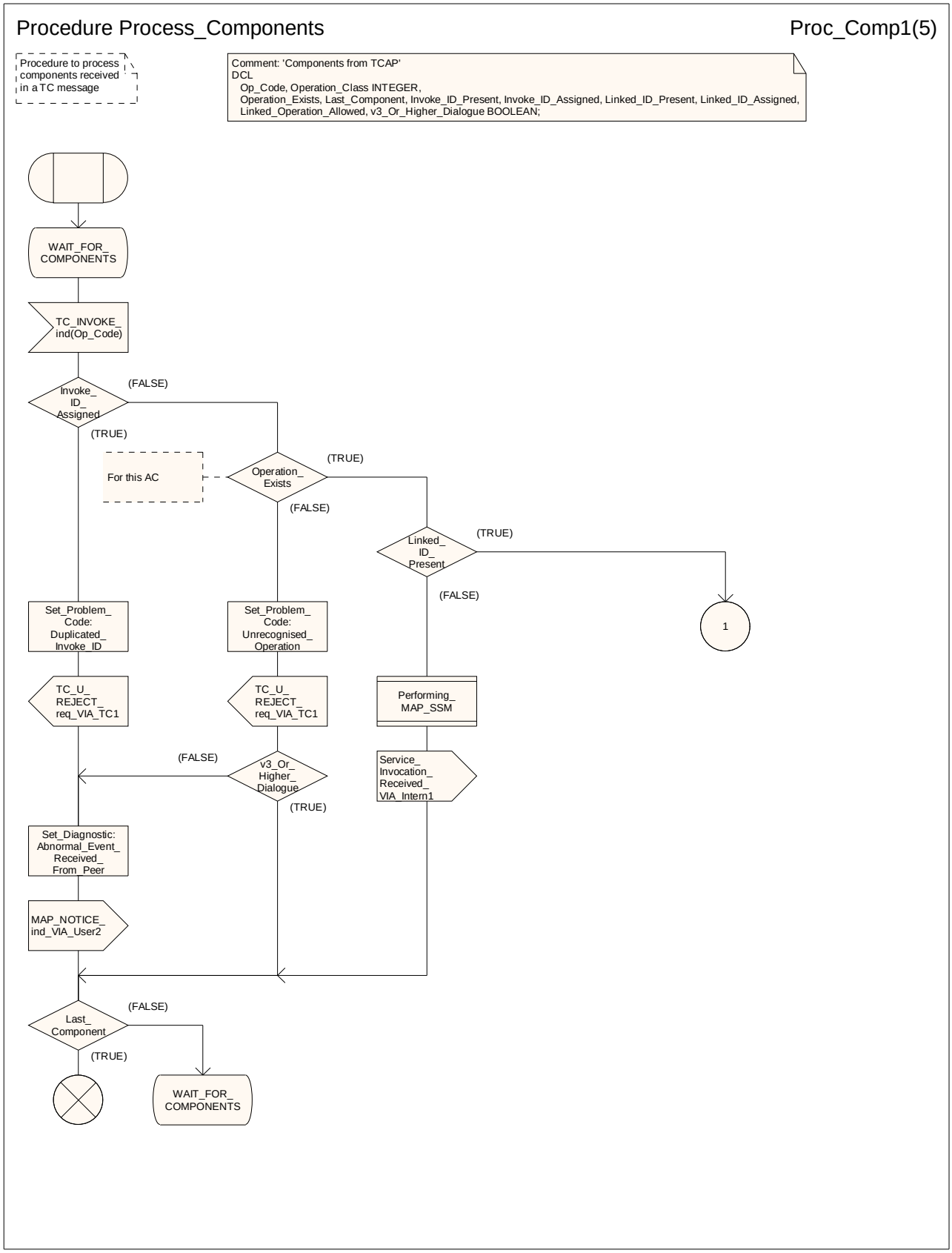


Figure 15.6/4a: Procedure Process_Components (sheet 1)

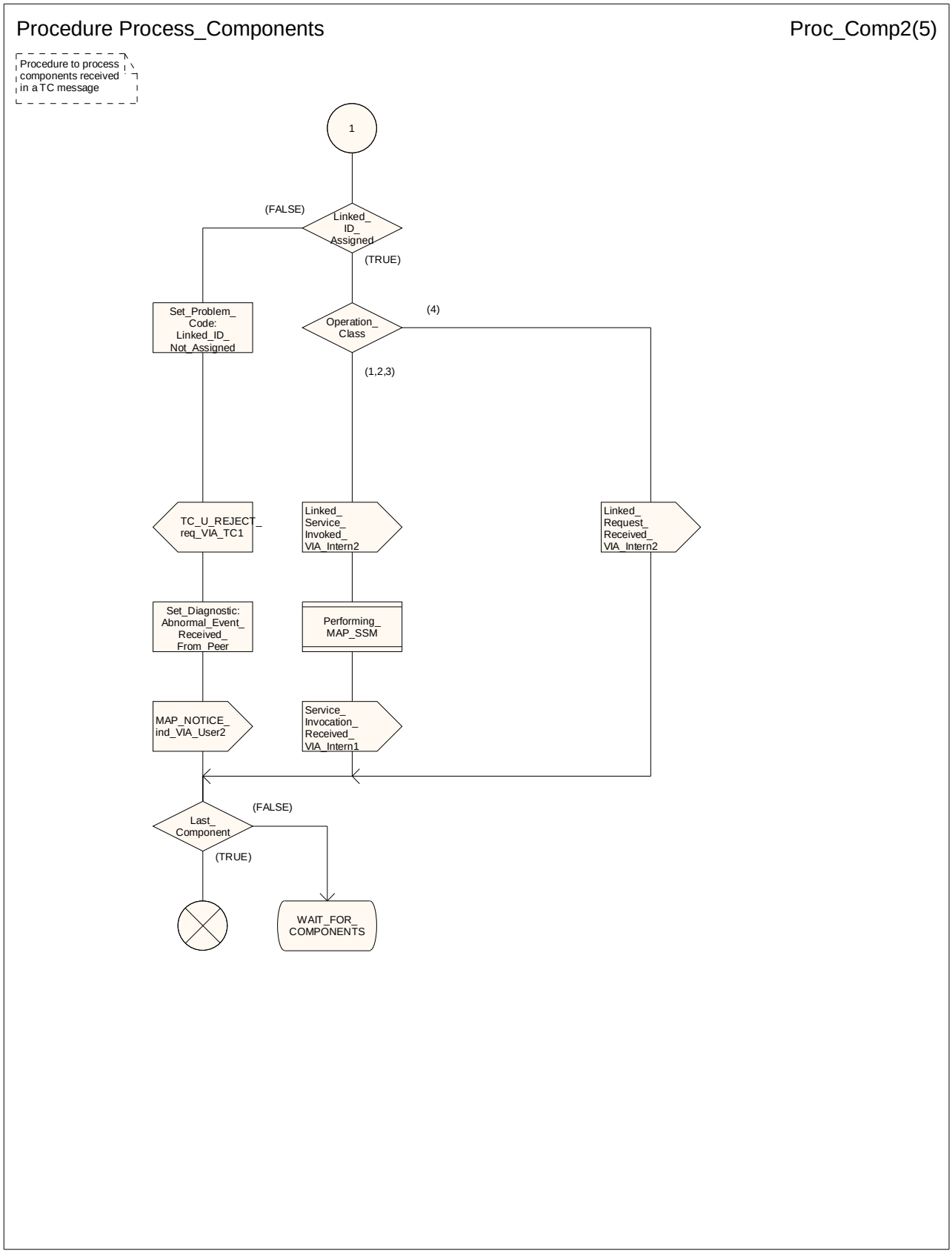


Figure 15.6/4b: Procedure Process_Components (sheet 2)

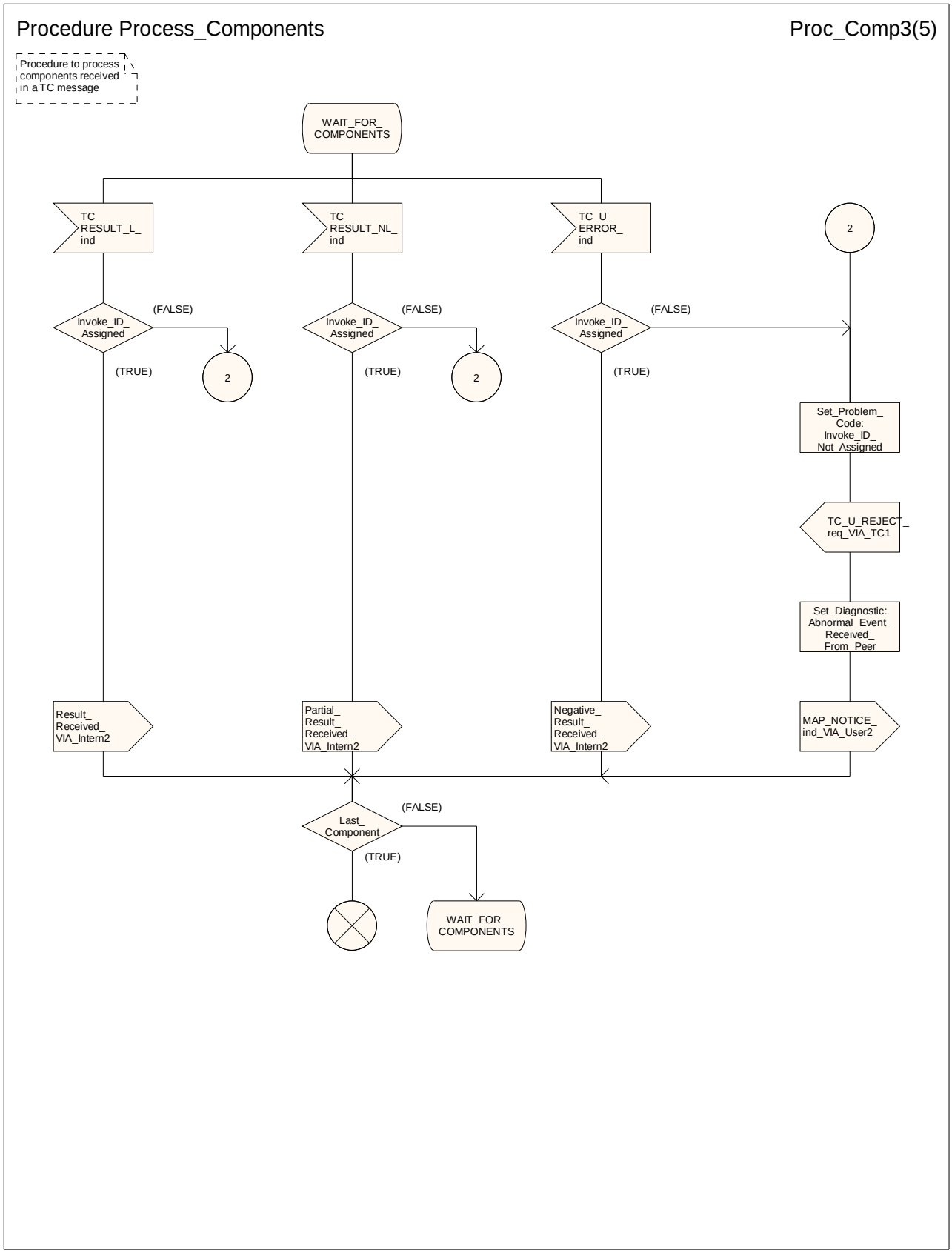


Figure 15.6/4c: Procedure Process_Components (sheet 3)

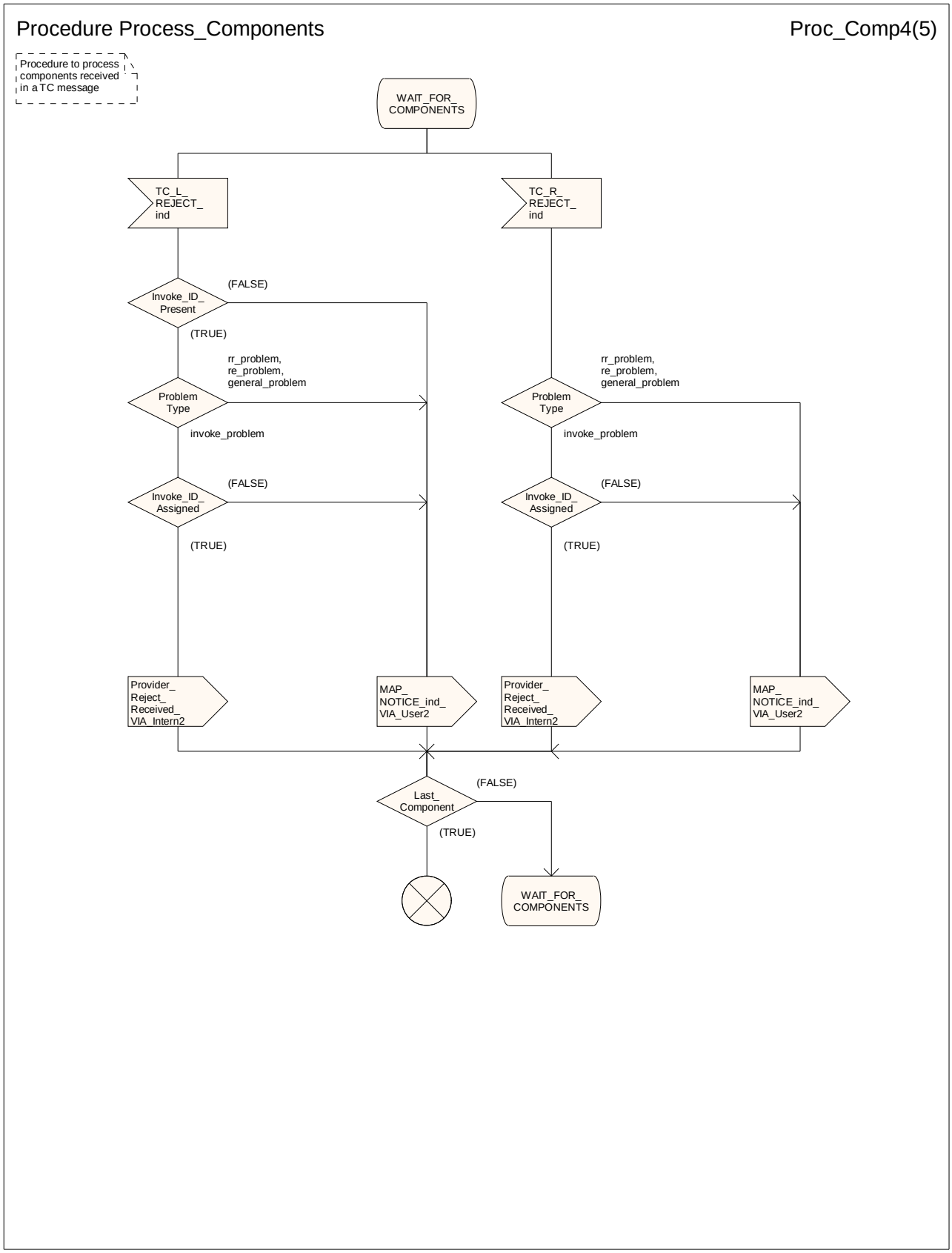


Figure 15.6/4d: Procedure Process_Components (sheet 4)

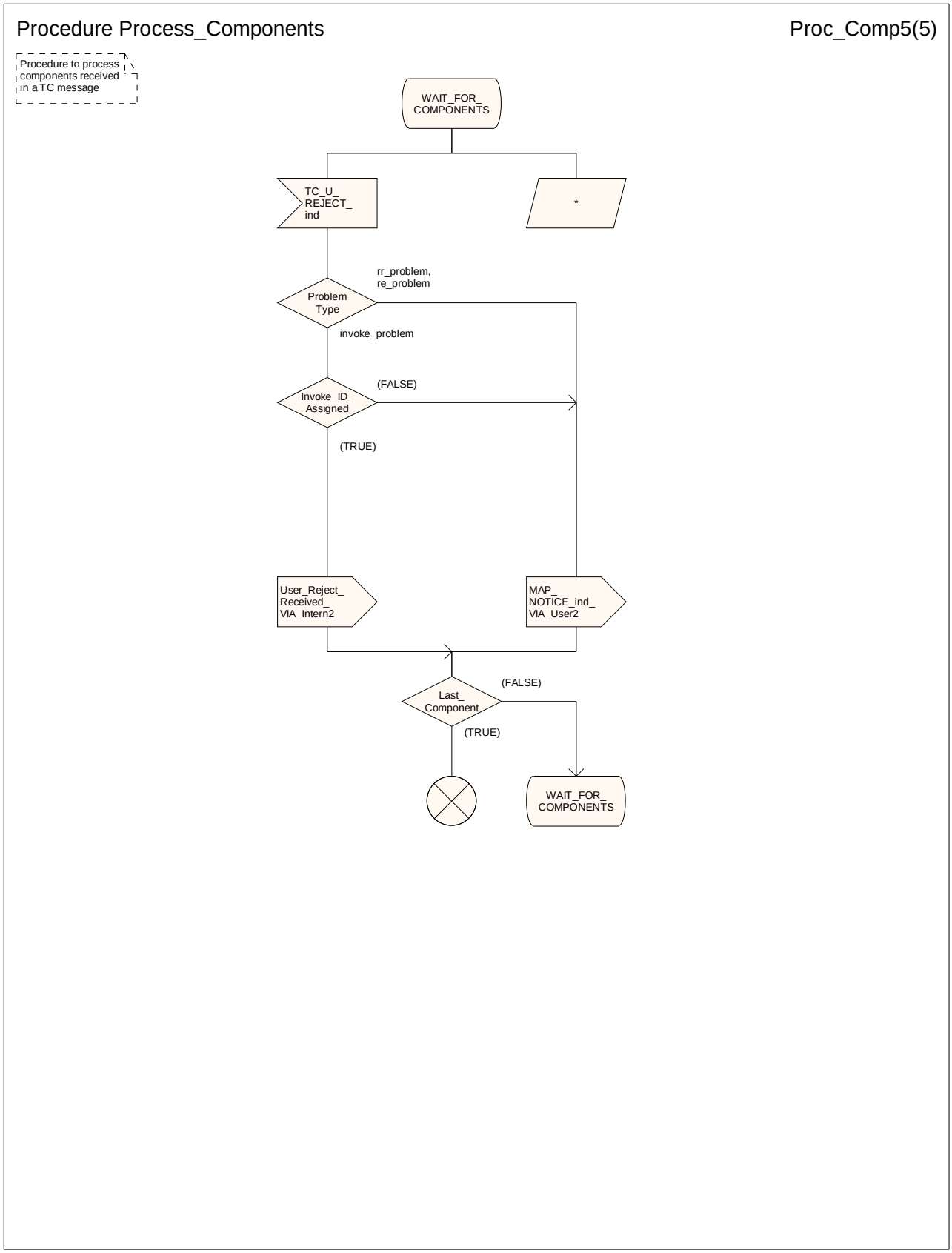


Figure 15.6/4e: Procedure Process_Components (sheet 5)

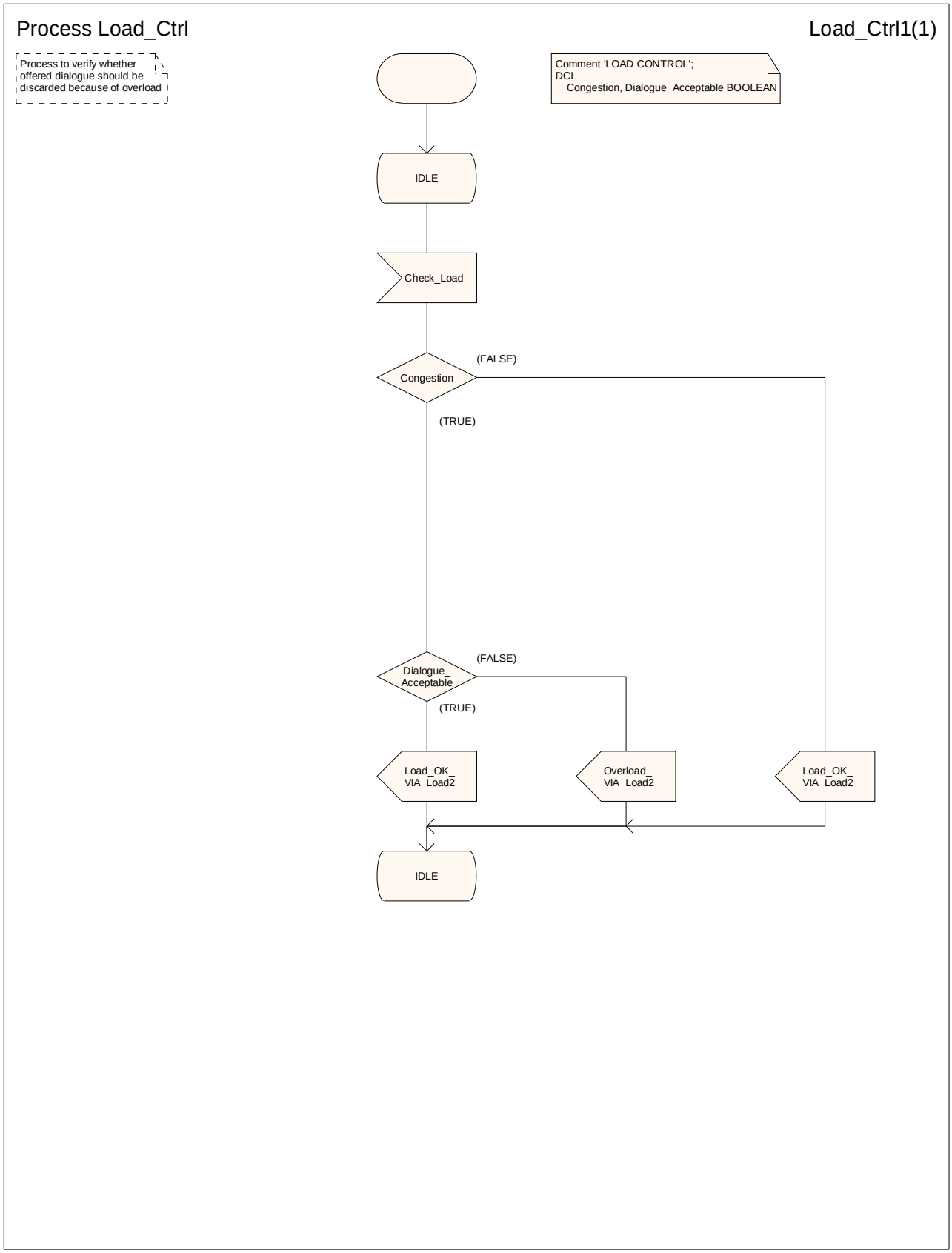


Figure 15.6/5: Process Load_Ctrl

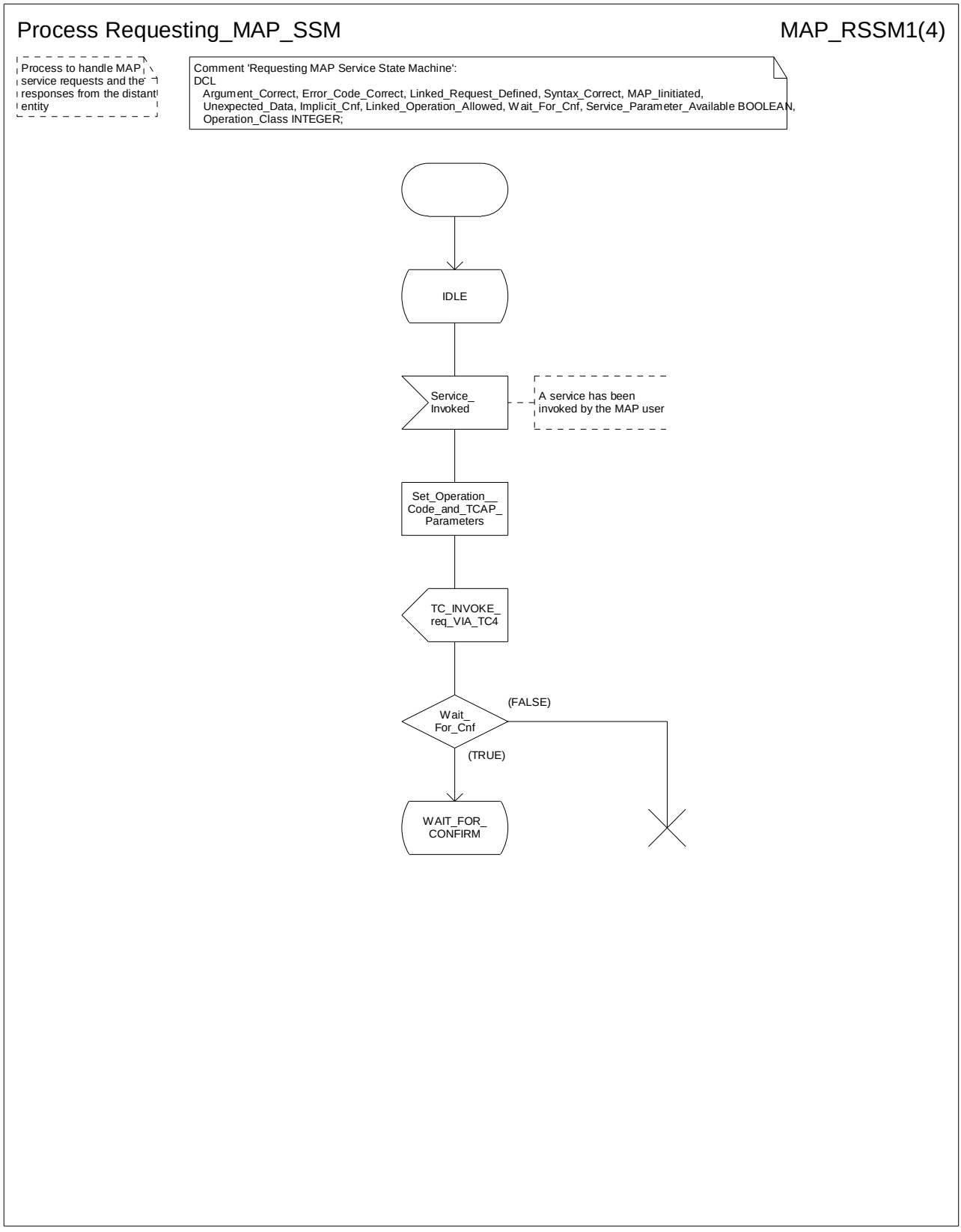


Figure 15.6/6a: Process Requesting_MAP_SSM (sheet 1)

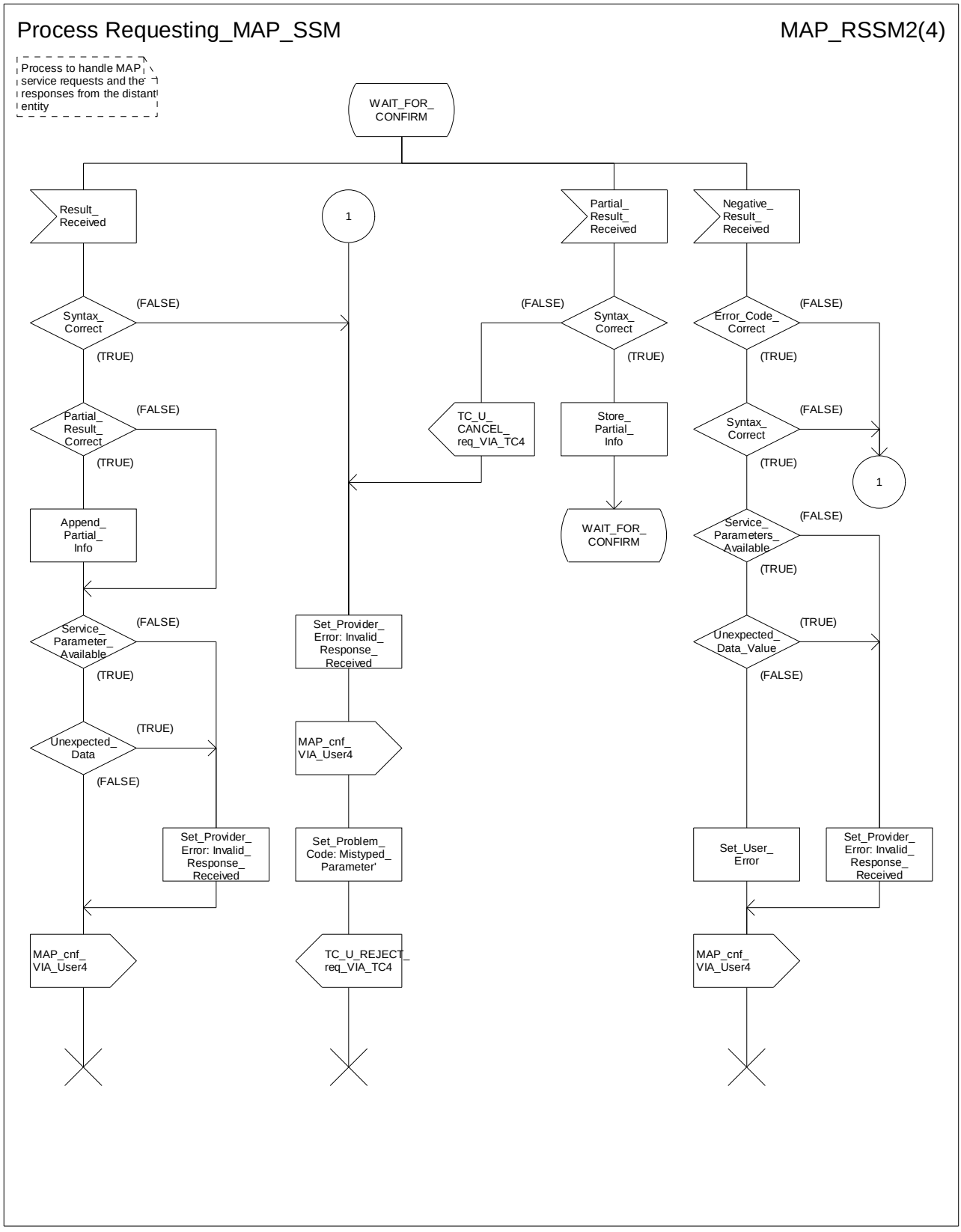


Figure 15.6/6b: Process Requesting_MAP_SSM (sheet 2)

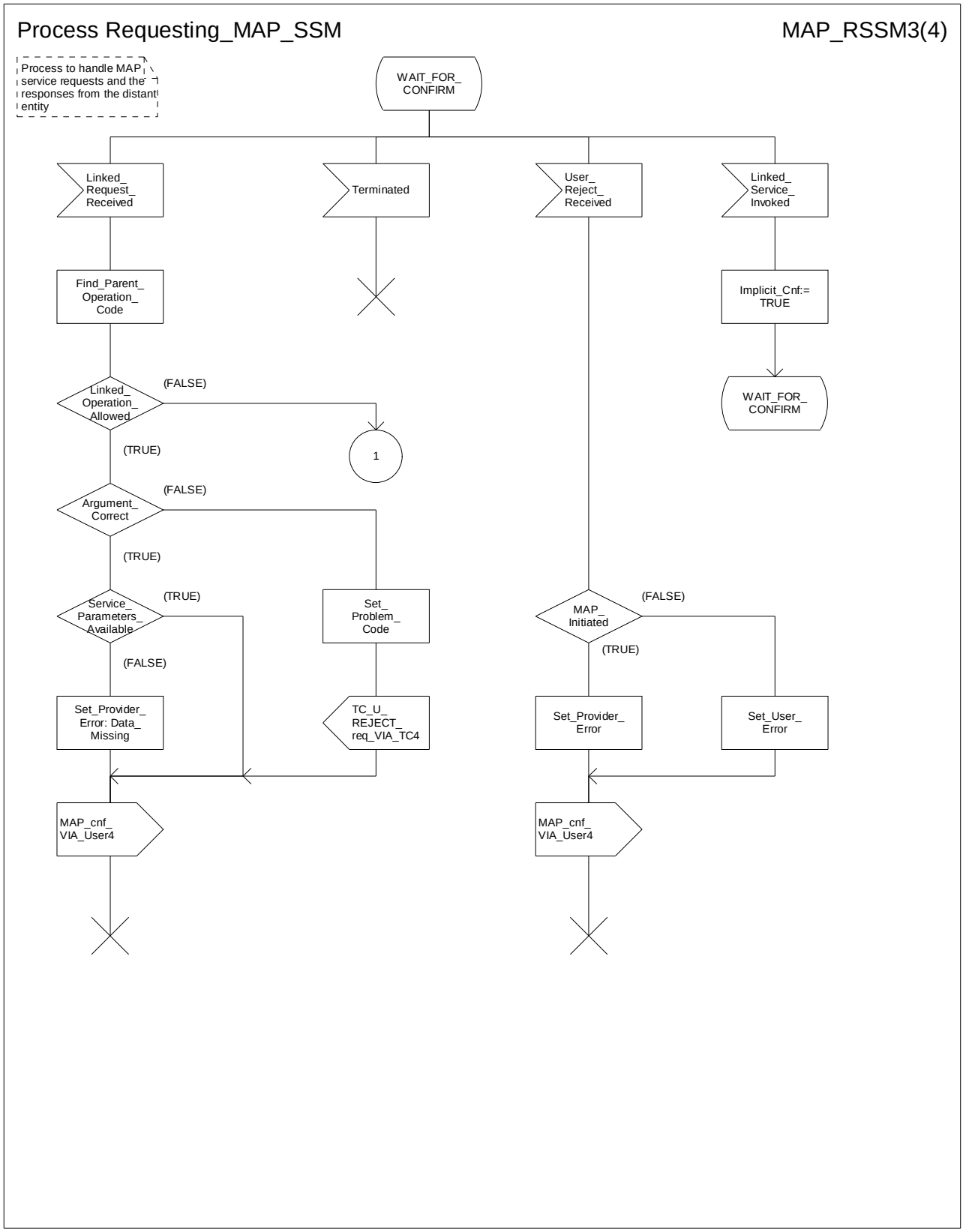


Figure 15.6/6c: Process Requesting_MAP_SSM (sheet 3)

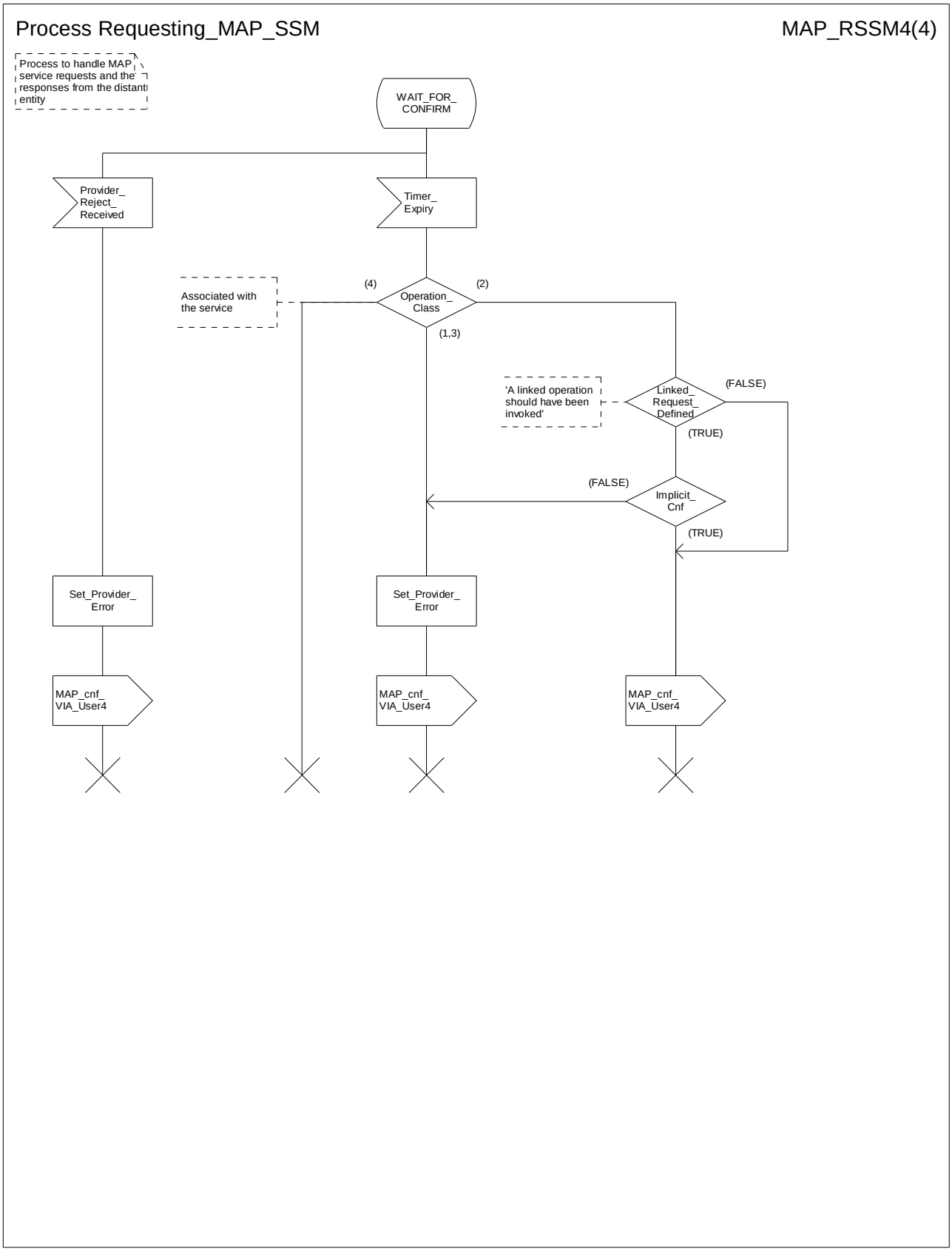


Figure 15.6/6d: Process Requesting_MAP_SSM (sheet 4)

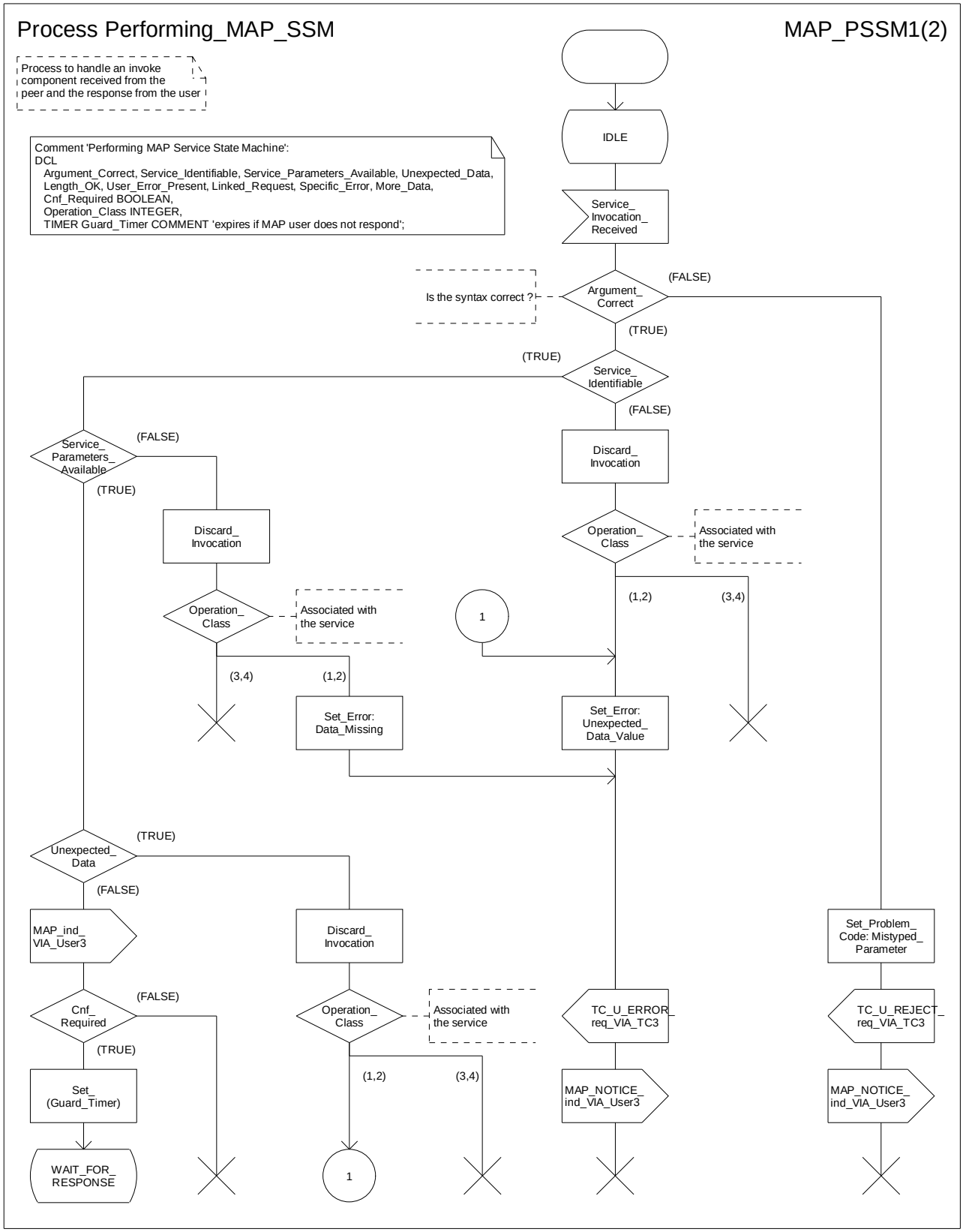

Process Performing_MAP_SSM
MAP_PSSM1(2)
Process to handle an invoke component received from the peer and the response from the user
Comment 'Performing MAP Service State Machine':
DCL
Argument_Correct, Service_Identifiable, Service_Parameters_Available, Unexpected_Data,
Length_OK, User_Error_Present, Linked_Request, Specific_Error, More_Data,
Cnf_Required BOOLEAN,
Operation_Class INTEGER,
TIMER Guard_Timer COMMENT 'expires if MAP user does not respond';
IDLE
Service_ Invocation_ Received
Is the syntax correct ?
Argument_ Correct
(FALSE)
(TRUE)
Service_ Identifiable
(TRUE)
(FALSE)
Discard_ Invocation
Operation_ Class
Associated with the service
(1,2)
(3,4)
Service_ Parameters_ Available
(FALSE)
(TRUE)
Discard_ Invocation
Operation_ Class
Associated with the service
1
(3,4)
(1,2)
Set_Error: Data_Missing
Set_Error: Unexpected_ Data_Value
Unexpected_ Data
(TRUE)
(FALSE)
MAP_ind_ VIA_User3
Discard_ Invocation
Set_Problem_ Code: Mistyped_ Parameter
Cnf_ Required
(FALSE)
(TRUE)
Operation_ Class
Associated with the service
TC_U_ERROR_ req_VIA_TC3
TC_U_REJECT_ req_VIA_TC3
Set_ (Guard_Timer)
(1,2)
(3,4)
MAP_NOTICE_ ind_VIA_User3
MAP_NOTICE_ ind_VIA_User3
WAIT_FOR_ RESPONSE
1

Figure 15.6/8a: Process Performing_MAP_SSM (sheet 1)

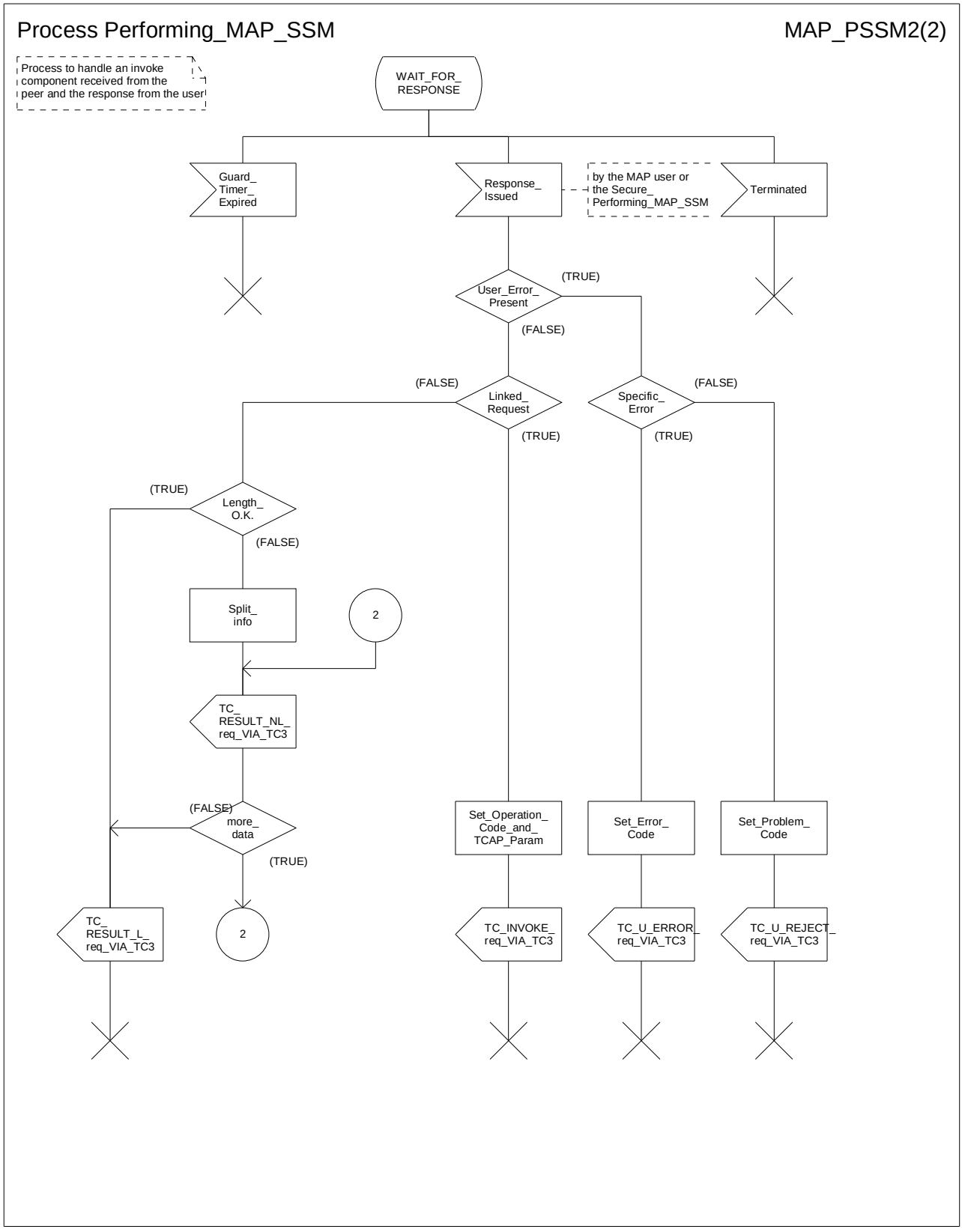


Figure 15.6/8b: Process Performing_MAP_SSM (sheet 2)

16 Mapping on to TC services

16.1 Dialogue control

Dialogue control services are mapped to TC dialogue handling services. The TC-UNI service is not used by the MAP PM.

16.1.1 Directly mapped parameters

The following parameters of the MAP-OPEN request and indication primitives are directly mapped on to the corresponding parameters of the TC-BEGIN primitives:

- destination address;
- originating address.

16.1.2 Use of other parameters of dialogue handling primitives

16.1.2.1 Dialogue Id

The value of this parameter is associated with the MAP PM invocation in an implementation dependent manner.

16.1.2.2 Application-context-name

The application-context-name parameter of a MAP primitive is mapped to the application-context-name parameter of TC dialogue handling primitives according to the rules described in clause 15.1.

16.1.2.3 User information

The user information parameter of TC dialogue primitives is used to carry the MAP dialogue APDUs.

16.1.2.4 Component present

This parameter is used by the MAP PM as described in CCITT Recommendation Q.771. It is not visible to the MAP user.

16.1.2.5 Termination

The value of this parameter of the TC-END request primitive is set by the MAP PM on the basis of the release method parameter of the MAP-CLOSE request primitive, except when the dialogue state machine is in the state DIALOGUE INITIATED, in which case the Termination parameter shall always indicate "pre-arranged end".

16.1.2.6 P-Abort-Cause

Values of the P-abort-cause parameter are mapped to the values of the provider-reason parameter of the MAP-P-ABORT indication primitive according to table 16.1/1, except in the dialogue initiated phase for the "incorrectTransactionPortion" and "noCommonDialoguePortion" values which are mapped to the "potential incompatibility problem" value of the refuse-reason parameter of the MAP-OPEN cnf primitive. The source parameter in the MAP-P-ABORT ind takes the value "TC problem".

16.1.2.7 Quality of service

The quality of service of TC request primitives is set by the MAP as shown below.

- Return option: "Return message on error" or "Discard message on error" as required by the network operator;
- Sequence control: "Sequence guaranteed" or "Sequence result not guaranteed" as required by the network operator;
- "Sequence guaranteed" shall be used when a segmented result is to be transferred (e.g. subscriber data in response to SendParameters). It may also be appropriate to use Sequence guaranteed when a series of InsertSubscriberData, ProcessAccessSignalling or ForwardAccessSignalling operations is used.

It is essential that the TC message which indicates acceptance of a dialogue opening request is received by the dialogue initiator before any subsequent message in that dialogue; otherwise the dialogue opening will fail. The dialogue responder shall ensure that this requirement is met by:

- Sending the dialogue acceptance message in a TC-END, if the dialogue structure requires it; or
- Using "Sequence guaranteed", if the dialogue acceptance message is sent in a TC-CONTINUE; or
- Waiting until the dialogue acceptance message has been acknowledged by the dialogue initiator before sending a subsequent message, if the dialogue acceptance message is sent in a TC-CONTINUE.

Table 16.1/1: Mapping of P-Abort cause in TC-P-ABORT indication on to provider-reason in MAP-P-ABORT indication

TC P-Abort cause	MAP provider-reason
unrecognised message type	provider malfunction
unrecognised transaction Id	supporting dialogue released
badlyFormattedTransactionPortion	provider malfunction
incorrectTransactionPortion	provider malfunction (note)
resourceLimitation	resource limitation
abnormalDialogue	provider malfunction
noCommonDialoguePortion	version incompatibility
NOTE: Or version incompatibility in the dialogue initiated phase.	

16.2 Service specific procedures

Specific services are mapped to TC component handling services.

16.2.1 Directly mapped parameters

The Invoke Id parameter of the MAP request and indication primitive is directly mapped on to the Invoke Id parameter of the component handling primitives.

16.2.2 Use of other parameters of component handling primitives

16.2.2.1 Dialogue Id

The value of this parameter is associated with the MAP PM invocation in an implementation dependent manner.

16.2.2.2 Class

The value of this parameter is set by the MAP PM according to the type of the operation to be invoked.

16.2.2.3 Linked Id

When a service response is mapped to a class 4 operation, the value of this parameter is set by the MAP PM and corresponds to the value assigned by the user to the initial service request (i.e. the value of the invoke ID parameter of the request primitive). Otherwise if such a parameter is included in MAP request/indication primitives it is directly mapped to the linked ID parameter of the associated TC-INVOKE request/indication primitives.

16.2.2.4 Operation

When mapping a request primitive on to a Remote Operations PDU (invoke), the MAP PM shall set the operation code according to the mapping described in table 16.2/1.

When mapping a response primitive on to a Remote Operations service, the MAP PM shall set the operation code of the TC-RESULT-L/NL primitive (if required) to the same value as the one received at invocation time.

Table 16.2/1: Mapping of MAP specific services on to MAP operations

MAP-SERVICE	operation
MAP-ACTIVATE-SS	activateSS
MAP-ACTIVATE-TRACE-MODE	activateTraceMode
MAP-ALERT-SERVICE-CENTRE	alertServiceCentre
MAP-ANY-TIME-INTERROGATION	anyTimeInterrogaton
MAP_AUTHENTICATION_FAILURE_REPORT	authenticationFailureReport
MAP-ANY-TIME-MODIFICATION	anyTimeModification
MAP-ANY-TIME-SUBSCRIPTION-INTERROGATION	anyTimeSubscriptionInterrogation
MAP-CANCEL-LOCATION	cancelLocation
MAP-CHECK-IMEI	checkIMEI
MAP-DEACTIVATE-SS	deactivateSS
MAP-DEACTIVATE-TRACE-MODE	deactivateTraceMode
MAP-DELETE-SUBSCRIBER-DATA	deleteSubscriberData
MAP-ERASE-CC-ENTRY	eraseCC-Entry
MAP-ERASE-SS	eraseSS
MAP-FAILURE-REPORT	failureReport
MAP-FORWARD-ACCESS-SIGNALLING	forwardAccessSignalling
MAP-FORWARD-CHECK-SS-INDICATION	forwardCheckSsIndication
MAP-FORWARD-GROUP-CALL-SIGNALLING	forwardGroupCallSignalling
MAP-MT-FORWARD-SHORT-MESSAGE	mt-forwardSM
MAP-MO-FORWARD-SHORT-MESSAGE	mo-forwardSM
MAP-GET-PASSWORD	getPassword
MAP-INFORM-SERVICE-CENTRE	informServiceCentre
MAP-INSERT-SUBSCRIBER-DATA	insertSubscriberData
MAP-INTERROGATE-SS	interrogateSs
MAP-IST-ALERT	istAlert
MAP-IST-COMMAND	istCommand
MAP-NOTE-MS-PRESENT-FOR-GPRS	noteMsPresentForGprs
MAP-NOTE-SUBSCRIBER-DATA-MODIFIED	noteSubscriberDataModified
MAP-PREPARE-GROUP-CALL	prepareGroupCall
MAP-PREPARE-HANDOVER	prepareHandover
MAP-PREPARE-SUBSEQUENT-HANDOVER	prepareSubsequentHandover
MAP-PROCESS-ACCESS-SIGNALLING	processAccessSignalling
MAP-PROCESS-GROUP-CALL-SIGNALLING	processGroupCallSignalling
MAP-PROCESS-UNSTRUCTURED-SS-REQUEST	processUnstructuredSS-Request
MAP-PROVIDE-ROAMING-NUMBER	provideRoamingNumber
MAP-PROVIDE-SUBSCRIBER-LOCATION	provideSubscriberLocation
MAP-PROVIDE-SUBSCRIBER-INFO	provideSubscriberInfo
MAP-PURGE-MS	purgeMS
MAP-READY-FOR-SM	readyForSM
MAP-REGISTER-CC-ENTRY	registerCC-Entry
MAP-REGISTER-PASSWORD	registerPassword
MAP-REGISTER-SS	registerSS
MAP-REMOTE-USER-FREE	remoteUserFree
MAP-REPORT-SM-DELIVERY-STATUS	reportSmDeliveryStatus
MAP-RESET	reset
MAP-RESTORE-DATA	restoreData
MAP-SEND_GROUP-CALL_END_SIGNAL	sendGroupCallEndSignal
MAP-SEND-GROUP-CALL-INFO	sendGroupCallInfo
MAP-SEND-END-SIGNAL	sendEndSignal
MAP-SEND-AUTHENTICATION-INFO	sendAuthenticationInfo
MAP-SEND-IMSI	sendIMSI
MAP-SEND-IDENTIFICATION	sendIdentification
MAP-SEND-ROUTING-INFO-FOR-SM	sendRoutingInfoForSM

MAP-SEND-ROUTING-INFO-FOR-GPRS	sendRoutingInfoForGprs
MAP-SEND-ROUTING-INFO-FOR-LCS	sendRoutingInfoForLCS
MAP-SEND-ROUTING-INFORMATION	sendRoutingInfo
MAP-SET-REPORTING-STATE	setReportingState
MAP-STATUS-REPORT	statusReport
MAP-SUBSCRIBER-LOCATION-REPORT	subscriberLocationReport
MAP-SUPPLEMENTARY-SERVICE-INVOCATION-NOTIFICATION	ss-Invocation-Notification
MAP-UNSTRUCTURED-SS-NOTIFY	unstructuredSS-Notify
MAP-UNSTRUCTURED-SS-REQUEST	unstructuredSS-Request
MAP-UPDATE-GPRS-LOCATION	updateGprsLocation
MAP-UPDATE-LOCATION	updateLocation
MAP-NOTE-MM-EVENT	NoteMM-Event
MAP-UPDATE-VCSG-LOCATION	updateVcsgLocation
MAP-CANCEL-VCSG-LOCATION	cancelVcsgLocation

16.2.2.5 Error

The error parameter in a TC-U-ERROR indication primitive is mapped to the user error parameter in the MAP confirm primitive of the service associated with the operation to which the error is attached.

The user error parameter in MAP response primitives is mapped to the error parameter of the TC-U-ERROR request primitive, except for "initiating-release" and "resource-limitation" which are mapped to the problem code parameter of the TC-U-REJECT request primitive.

16.2.2.6 Parameters

The parameters of MAP specific request and indication primitives are mapped to the argument parameter of TC-INVOKE primitives.

The parameters of MAP specific response and confirm primitives are mapped to the result parameter of TC-RESULT-L primitives, the parameter of TC-U-ERROR primitives or the argument of TC-INVOKE primitives when mapping on linked class 4 operations is used.

16.2.2.7 Time out

The value of this parameter is set by the MAP PM according to the type of operation invoked.

16.2.2.8 Last component

This parameter is used by the MAP PM as described in CCITT Recommendation Q.711. It is not visible from the MAP user.

16.2.2.9 Problem code

16.2.2.9.1 Mapping to MAP User Error

The following values of the user error parameter are mapped as follows to values of the TC problem code parameter. These values are generated by the MAP user. This mapping is valid from the TC-U-REJECT indication primitive to the MAP confirm service primitive and from the MAP response service primitive to the TC-U-REJECT request primitive.

Table 16.2/2: Mapping of MAP User Error parameter on to TC problem code in TC-U-REJECT primitives

MAP User Error	TC problem code
resource limitation	resource limitation
initiating release	initiating release

16.2.2.9.2 Mapping to MAP Provider Error parameter

The following values of the TC problem code parameter of the TC-U-REJECT indication primitive are mapped as follows to values of the MAP Provider Error parameter of the MAP confirm primitive.

Table 16.2/3: Mapping of TC problem code in TC-U-REJECT on to MAP Provider Error parameter

TC problem code	MAP Provider Error
duplicated invoke Id	duplicated invoke id
unrecognised operation	service not supported
mistyped parameter	mistyped parameter

The following values of the problem code parameters of the TC-L-REJECT primitive are mapped to values of the provider error parameter of the MAP confirm primitive as follows.

Table 16.2/4: Mapping of TC problem code in TC-L-REJECT on to MAP Provider Error parameter

TC problem code	MAP Provider Error
return result unexpected	unexpected response from the peer
return error unexpected	unexpected response from the peer

16.2.2.9.3 Mapping to diagnostic parameter

The following values of the problem code parameter of the TC-R-REJECT and TC-U-REJECT primitive are mapped to values of the diagnostic parameter of the MAP-NOTICE indication primitive as follows:

Table 16.2/5: Mapping of TC problem code of TC-R-REJECT and TC-U-REJECT on to diagnostic parameter

TC problem code	MAP diagnostic
General problem	- abnormal event detected by the peer
Invoke problem	
- unrecognised linked ID	- abnormal event detected by the peer
- linked response unexpected	- response rejected by the peer
- unexpected linked operation	- response rejected by the peer
Return result problem	
- unrecognised invoke ID	- response rejected by the peer
- return result unexpected	- response rejected by the peer
- mistyped parameter	- response rejected by the peer
Return error problem	
- unrecognised invoke ID	- response rejected by the peer
- return error unexpected	- response rejected by the peer
- unrecognised error	- response rejected by the peer
- unexpected error	- response rejected by the peer
- mistyped parameter	- response rejected by the peer

The following values of the problem code parameter of the TC-L-REJECT primitive are mapped to values of the diagnostic parameter of the MAP-NOTICE indication primitive as follows.

Table 16.2/6: Mapping of TC problem code of TC-L-REJECT on to diagnostic parameter

TC problem code	MAP diagnostic
General problems	- abnormal event received from the peer
Invoke problem	
- unrecognised linked ID	- abnormal event received from the peer
Return result problem	
- unrecognised invoke ID	- abnormal event received from the peer
Return error problem	
- unrecognised invoke ID	- abnormal event received from the peer

17 Abstract syntax of the MAP protocol

17.1 General

This clause specifies the Abstract Syntaxes for the Mobile Application Part as well as the associated set of Operations and Errors, using the Abstract Syntax Notation One (ASN.1), defined in ITU-T Recommendations X.680 and X.681 with additions as defined in clause 17.1.4 on Compatibility Considerations and the OPERATION and ERROR external information object classes, defined in ITU-T Recommendation X.880.

The Abstract Syntax is defined for all interfaces specified in clause 4.4 except for the A- and B-interfaces.

The Mobile Application Part protocol is defined by two Abstract Syntaxes:

- one Abstract Syntax which encompass all Operations and Errors identified by the various MAP subsystem numbers.

This Abstract Syntax represents the set of values each of which is a value of the ASN.1 type TCAPMessages. TCMessage as defined in ITU-T Recommendation Q.773 with the component relationconstraint sections resolved by the operation and error codes included in the ASN.1 modules MAP-*Operations and MAP-Errors. However, only the subset of this abstract syntax which is required by the procedures defined for an entity needs to be supported.

- one Abstract Syntax identified by the OBJECT IDENTIFIER value MAP-DialogueInformation.map-DialogueAS.

This Abstract Syntax represents the set of values each of which is a value of the ASN.1 type MAP-DialogueInformation.MAP-DialoguePDU. Such a value of the ASN.1 single-ASN.1-type element is contained within the user-information element of the TCAPMessages.DialoguePortion ASN.1 type. This Abstract Syntax name is to be used as a direct reference.

17.1.1 Encoding rules

The encoding rules which are applicable to the defined Abstract Syntaxes are the Basic Encoding Rules for Abstract Syntax Notation One, defined in ITU-T Recommendation X.690 with the same exceptions as in ITU-T Recommendation Q.773, clause 4 Message Representation.

When the definite form is used for length encoding, a data value of length less than 128 octets must have the length encoded in the short form.

When the long form is employed to code a length, the minimum number of octets shall be used to code the length field.

OCTET STRING values and BIT STRING values must be encoded in a primitive form.

There is no restriction to the use of empty constructors (e.g. an empty SEQUENCE type). That is, the encoding of the content of any data value shall consist of zero, one or more octets.

17.1.2 Use of TC

The mapping of OPERATION and ERROR to TC components is defined in ETS 300 287 (version 2) which is based on ITU-T Recommendation Q.773.

NOTE 1: The class of an operation is not stated explicitly but is specified as well in the ASN.1 operation definition.

Class 1: RESULT and ERROR appear in ASN.1 operation definition.

Class 2: only ERROR appears in ASN.1 operation definition.

Class 3: only RESULT appears in ASN.1 operation definition.

Class 4: both RESULT and ERROR do not appear in ASN.1 operation definition.

The field "ARGUMENT", "PARAMETER" or "RESULT" (for information objects of class OPERATION and ERROR) is always optional from a syntactic point of view. However, except when specifically mentioned with the

ASN.1 comment "-- optional" , the "parameter" part of a component has to be considered as mandatory from a semantic point of view.

When an optional element is missing in an invoke component or in an inner data structure while it is required by the context, an error component is returned if specified in the information object associated with the operation ; the associated type of error is "DataMissing". This holds also when the entire parameter of an invoke component is missing while it is required by the context.

NOTE 2: When a mandatory element is missing in the parameter or inner data structure of any component, a reject component is returned (if the dialogue still exists). The problem code to be used is "Mistyped parameter".

The Timer Values used in the operation definitions are indicated as ASN.1 comments. The Timer Value Ranges are:

s = from 3 seconds to 10 seconds;

m = from 15 seconds to 30 seconds;

ml = from 1 minute to 10 minutes;

l = from 28 hours to 38 hours.

17.1.2.1 Use of Global Operation and Error codes defined outside MAP

An entity supporting an application context greater than 2 shall be capable of receiving an operation or error code, within an application context defined in GSM 29.002, encoded as either an Object Identifier (as defined in ITU-T Recommendation X.690) or an integer value (as defined in clause 17.5). Related restrictions regarding the use of Object Identifiers are as follows:

- The length of the Object Identifier shall not exceed 16 octets and the number of components of the Object Identifier shall not exceed 16.
- Object Identifiers shall be used only for operations or errors defined outside of GSM 29.002.
- Global error codes may be sent only in response to a global operation. If a standard operation is received then a global error code shall not be sent in response.

Handling of an unknown operation codes by the receiving entity is defined in clause 15.1.1.

17.1.3 Use of information elements defined outside MAP

An information element or a set of information elements (messages) transparently carried in the Mobile Application Part but defined in other recommendations/technical specifications are handled in one of the following ways:

i) The contents of each information element (without the octets encoding the identifier and the length in the recommendation/technical specification where it is defined unless explicitly stated otherwise) is carried as the value of an ASN.1 type derived from the OCTET STRING data type. Additionally, the internal structure may be explained by means of comments. In case of misalignment the referred to recommendation/technical specification takes precedence.

ii) The complete information element (including the octets encoding the identifier and the length in the recommendation/technical specification where it is defined) or set of information elements and the identity of the associated protocol are carried as the value of the ExternalSignalInfo data type defined in the present document. Where more than one information element is carried, the information elements are sent contiguously with no filler octets between them.

17.1.4 Compatibility considerations

The following ASN.1 modules conform to ITU-T Recommendation X.680 and X.681 . An extension marker ("...") is used wherever future protocol extensions are foreseen.

The "..." construct applies only to SEQUENCE and ENUMERATED data types. An entity supporting a version greater than 1 shall not reject an unsupported extension following "..." of that SEQUENCE or ENUMERATED data type. The

Encoding Rules from clause 17.1.1 apply to every element of the whole Transfer Syntax especially to the ASN.1 type EXTERNAL.

The extension container "privateExtensionList" is defined in this specification in order to carry extensions which are defined outside this specification. Private extensions can be defined by, for example, network operators, manufacturers, and regional standardisation bodies.

Private extensions shall:

1) if included in operations of an AC of V2, follow the extension marker and be tagged using PRIVATE tags up to and including 29.

NOTE: This type of extension is in most cases used only within a PLMN.

2) if included in operations of an AC of V3 or higher: be included only in the Private Extension Container that is defined in the specification.

NOTE: This type of extension can be used between PLMNs.

Private extensions shall not be included in v2 supplementary service operations.

Private extensions shall not be included within user error for RegisterCCEntry and EraseCCEntry operations.

PCS extensions shall be included in the PCS Extension Container that is defined in this specification.

In order to improve extensibility, a few error parameters have been defined as a CHOICE between the version 2 description and a SEQUENCE including the version 2 description and an extension container. Operations used in a v2-application-context must consider only the first alternative while operations used in a vn-application-context (n>2) must consider only the second alternative.

17.1.5 Structure of the Abstract Syntax of MAP

For each MAP parameter which has to be transferred by a MAP Protocol Data Unit (MAP message), there is a PDU field (an ASN.1 type) which has the same name as the corresponding parameter, except for the differences required by the ASN.1 notation (blanks between words are removed or replaced by hyphen, the first letter of the first word is capital and the first letter of each of the following words ise capitalised, e.g. "no reply condition time" is mapped to "NoReplyConditionTime"). Additionally some words may be abbreviated as follows:

bs basic service

ch call handling

cug closed user group

ho handover

ic incoming call

id identity

info information

mm mobility management

lcs location services

ms mobile service

oc outgoing call

om operation & maintenance

pw Password

sm short message service

ss supplementary service

The MAP protocol is composed of several ASN.1 modules dealing with either operations, errors, data types, and, if applicable, split into those dealing with mobile services, call handling services, supplementary services and short message services. For operations and errors the code values are given as parameters, in order to allow use of the defined information objects also by other protocols (e.g. 3GPP TS 24.080 [38]). The ASN.1 source lines are preceded by line-numbers at the left margin in order to enable the usage of the cross-reference in annex A.

The module containing the definition of the operation packages for MAP is:

1. MAP-OperationPackages.

The module containing the definition of the application contexts for MAP is:

2. MAP-ApplicationContexts.

The module containing the data types for the Abstract Syntax to be used for TCAPMessages.DialoguePortion for MAP is:

3. MAP-DialogueInformation.

The module containing the supported operations is:

4. MAP-Protocol.

The modules containing all operation definitions for MAP are:

5. MAP-MobileServiceOperations;
6. MAP-OperationAndMaintenanceOperations;
7. MAP-CallHandlingOperations;
8. MAP-SupplementaryServiceOperations;
9. MAP-ShortMessageServiceOperations;
10. MAP-Group-Call-Operations;
11. MAP-LocationServiceOperations.

The module containing all error definitions for MAP is:

12. MAP-Errors.

Modules containing all data type definitions for MAP are:

13. MAP-MS-DataTypes;
14. MAP-OM-DataTypes;
15. MAP-CH-DataTypes;
16. MAP-SS-DataTypes;
17. MAP-SS-Code;
18. MAP-SM-DataTypes;
19. MAP-ER-DataTypes;
20. MAP-CommonDataTypes;
21. MAP-TS-Code;
22. MAP-BS-Code;
23. MAP-ExtensionDataTypes;
24. MAP-GR-DataTypes;

25. MAP-LCS-DataTypes.

References are made also to modules defined outside of the present document. They are defined in the technical specification Mobile Services Domain, technical specification Transaction Capability and ITU-T Recommendation X.880 respectively:

MobileDomainDefinitions;

TCAPMessages, DialoguePDUs ;

Remote-Operations-Information-Objects.

17.1.6 Application Contexts

The following informative table lists the latest versions of the Application Contexts used in this specification, with the operations used by them and, where applicable, whether or not the operation description is exactly the same as for previous versions. Information in 17.6 & 17.7 relates only to the ACs in this table.

AC Name	AC Version	Operations Used	Comments
locationCancellationContext	v3	cancelLocation	
equipmentMngtContext	V3	checkIMEI	
imsiRetrievalContext	v2	sendIMSI	
infoRetrievalContext	v3	sendAuthenticationInfo	
interVlrInfoRetrievalContext	v3	sendIdentification	
handoverControlContext	v3	prepareHandover forwardAccessSignalling sendEndSignal processAccessSignalling prepareSubsequentHandover	the syntax of this operation has been extended in comparison with release 98 version
mwdMngtContext	v3	readyForSM	
msPurgingContext	v3	purgeMS	
shortMsgAlertContext	v2	alertServiceCentre	
resetContext	v3	reset	
networkUnstructuredSsContext	v2	processUnstructuredSS-Request unstructuredSS-Request unstructuredSS-Notify	
tracingContext	v3	activateTraceMode deactivateTraceMode	
networkFunctionalSsContext	v2	registerSS eraseSS activateSS deactivateSS registerPassword interrogateSS getPassword	
shortMsgMO-RelayContext	v3	mo-forwardSM	
shortMsgMT-RelayContext	v3	mt-forwardSM	
shortMsgMT-VGCS-RelayContext	v3	mt-forwardSM-VGCS	
shortMsgGatewayContext	v3	sendRoutingInfoForSM reportSM-DeliveryStatus InformServiceCentre	the syntax of this operation has been extended in comparison with release 96 version
networkLocUpContext	v3	updateLocation forwardCheckSs-Indication restoreData insertSubscriberData activateTraceMode	the syntax is the same in v1 & v2
gprsLocationUpdateContext	v3	updateGprsLocation insertSubscriberData activateTraceMode	
subscriberDataMngtContext	v3	insertSubscriberData	

AC Name	AC Version	Operations Used	Comments
		deleteSubscriberData	
roamingNumberEnquiryContext	v3	provideRoamingNumber	
locationInfoRetrievalContext	v3	sendRoutingInfo	
gprsNotifyContext	v3	noteMsPresentForGprs	
gprsLocationInfoRetrievalContext	v4	sendRoutingInfoForGprs	
failureReportContext	v3	failureReport	
callControlTransferContext	v4	resumeCallHandling	
subscriberInfoEnquiryContext	v3	provideSubscriberInfo	
anyTimeEnquiryContext	v3	anyTimeInterrogation	
anyTimeInfoHandlingContext	v3	anyTimeSubscriptionInterrogation anyTimeModification	
ss-InvocationNotificationContext	v3	ss-InvocationNotification	
groupCallControlContext	v3	prepareGroupCall processGroupCallSignalling forwardGroupCallSignalling sendGroupCallEndSignal	
reportingContext	v3	setReportingState statusReport remoteUserFree	
callCompletionContext	v3	registerCC-Entry eraseCC-Entry	
• istAlertingContext	• v3	istAlert	•
• ServiceTerminationContext	• v3	istCommand	•
• locationSvcEnquiryContext	• v3	provideSubscriberLocation subscriberLocationReport	•
• locationSvcGatewayContext	• v3	sendRoutingInfoForLCS	•
• mm-EventReportingContext	• v3	noteMM-Event	•
subscriberDataModificationNotificationContext	v3	noteSubscriberDataModified	
authenticationFailureReportContext	v3	authenticationFailureReport	
resourceManagementContext	v3	releaseResources	
groupCallInfoRetievalContext	v3	sendGroupCallInfo	
vcsgLocationUpdateContext	v3	updateVcsgLocation insertSubscriberData	
vcsgLocationCancelContext	v3	cancelVcsgLocation	

NOTE (*): The syntax of the operations is not the same as in previous versions unless explicitly stated

17.2 Operation packages

17.2.1 General aspects

This clause describes the operation-packages which are used to build the application-contexts defined in clause 17.3.

Each operation-package is a specification of the roles of a pair of communicating objects (i.e. a pair of MAP-Providers), in terms of operations which they can invoke of each other.

The grouping of operations into one or several packages does not necessarily imply any grouping in terms of Application Service Elements.

The following ASN.1 information object class is used to describe operation-packages in this clause:

```
OPERATION-PACKAGE ::= CLASS {
     &Both     OPERATION                 OPTIONAL,
     &Consumer OPERATION                 OPTIONAL,
     &Supplier OPERATION                 OPTIONAL,
     &id       OBJECT IDENTIFIER         UNIQUE OPTIONAL }
WITH SYNTAX {
     [ OPERATIONS       &Both ]
     [ CONSUMER INVOKES &Supplier ]
     [ SUPPLIER INVOKES &Consumer ]
     [ ID               &id ] }
```

Since the application-context definitions provided in clause 17.3 use only an informal description technique, only the type notation is used in the following clauses to define operation-packages.

The following definitions are used throughout this clause (n>=2):

- v1-only operation: An operation which shall be used only in v1 application-contexts;
- vn-only operation: An operation which shall be used only in vn application-contexts;
- v(n-1)-operation: An operation whose specification has not been modified since the MAP v(n-1) specifications or if the modifications are considered as not affecting v(n-1) implementations;
- v(n-1)-equivalent operation: The version of an operation which excludes all the information elements and errors which have been added since the MAP v(n-1) specification;
- vn-only package: An operation package which contains only vn-only operations;
- v(n-1)-package: An operation package which contains only v(n-1)- operations.

The names of vn-packages are suffixed by "-vn" where n>=2.

For each operation package which is not vn-only (n>=2) and which does not include only v(n-1)-operations, there is a v(n-1)-equivalent package. Except when a definition is explicitly provided in the following clauses, the v(n-1)-equivalent package includes the v(n-1)-equivalent operations of the operations which belong to this package.

17.2.2 Packages specifications

17.2.2.1 Location updating

This operation package includes the operations required for location management procedures between HLR and VLR.

```
locationUpdatingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     updateLocation}
     SUPPLIER INVOKES {
     forwardCheckSs-Indication} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.2 Location cancellation

This operation package includes the operations required for location cancellation and MS purging procedures between HLR and VLR and between HLR and SGSN.

```
locationCancellationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR
     CONSUMER INVOKES {
     cancelLocation} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.3 Roaming number enquiry

This operation package includes the operations required for roaming number enquiry procedures between HLR or old VLR and VLR.

```
roamingNumberEnquiryPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR if Consumer is HLR or old VLR
     CONSUMER INVOKES {
     provideRoamingNumber} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.4 Information retrieval

This operation package includes the operation required for the authentication information retrieval procedure between HLR and VLR and between HLR and SGSN.

```
infoRetrievalPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is HLR if Consumer is SGSN
     CONSUMER INVOKES {
     sendAuthenticationInfo} }
```

The v2-equivalent package is defined as follows:

```
infoRetrievalPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is HLR if Consumer is SGSN
     CONSUMER INVOKES {
     sendAuthenticationInfo} }
```

The v1-equivalent package is defined as follows:

```
infoRetrievalPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is HLR or VLR if Consumer is VLR
     -- Supplier is HLR if Consumer is SGSN
     CONSUMER INVOKES {
     sendParameters} }
```

17.2.2.5 Inter-VLR information retrieval

This operation package includes the operations required for inter VLR information retrieval procedures.

```
interVlrInfoRetrievalPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR if Consumer is VLR
     CONSUMER INVOKES {
     sendIdentification} }
```

The v2-equivalent package is defined as follows:

```
interVlrInfoRetrievalPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is VLR if Consumer is VLR
     CONSUMER INVOKES {
     sendIdentification} }
```

The v1-equivalent package is : infoRetrievalPackage-v1.

17.2.2.6 IMSI retrieval

This operation package includes the operation required for the IMSI retrieval procedure between HLR and VLR.

```
imsiRetrievalPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     sendIMSI} }
```

This package is v2 only.

17.2.2.7 Call control transfer

This operation package includes the operation required for the call control transfer procedure between VMSC and GMSC.

```
callControlTransferPackage-v4  OPERATION-PACKAGE ::= {
     -- Supplier is GMSC if Consumer is VMSC
     CONSUMER INVOKES {
     resumeCallHandling} }
```

The v3-equivalent package can be determined according to the rules described in clause 17.2.1.

17.2.2.8 Void

17.2.2.9 Void

17.2.2.10 Interrogation

This operation package includes the operations required for interrogation procedures between MSC and HLR or NPLR or between HLR and gsmSCF.

```
interrogationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR or NPLR if Consumer is MSC
     -- Supplier is HLR if Consumer is gsmSCF
     CONSUMER INVOKES {
     sendRoutingInfo} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.11 Void

17.2.2.12 Handover Control

This operation package includes the operations required for handover procedures between MSCs.

```
handoverControlPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is MSCB if Consumer is MSCA
     CONSUMER INVOKES {
     prepareHandover |
     forwardAccessSignalling}
     SUPPLIER INVOKES {
     sendEndSignal |
     processAccessSignalling |
     prepareSubsequentHandover} }
```

The v2-equivalent package can be determined according to the rules described in clause 17.2.1.

The v1-equivalent package is defined as follows.

```
handoverControlPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is MSCB if Consumer is MSCA
     CONSUMER INVOKES {
     performHandover |
     forwardAccessSignalling |
     traceSubscriberActivity}
     SUPPLIER INVOKES {
     sendEndSignal |
     noteInternalHandover |
     processAccessSignalling |
     performSubsequentHandover} }
```

17.2.2.13 Subscriber Data management stand alone

This operation package includes the operations required for stand alone subscriber data management procedures between HLR and VLR or between HLR and SGSN. Also this operation package includes the operations required for stand alone subscriber data management procedures between CSS and VLR or between CSS and SGSN. For the CSS – VLR interface and CSS – SGSN interface only version 3 of this operation package is applicable.

```
subscriberDataMngtStandAlonePackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR or CSS
     CONSUMER INVOKES {
     insertSubscriberData |
     deleteSubscriberData} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.14 Equipment management

This operation package includes the operations required for equipment management procedures between EIR and MSC or between EIR and SGSN.

```
equipmentMngtPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is EIR if Consumer is MSC
     -- Supplier is EIR if Consumer is SGSN
     CONSUMER INVOKES {
     checkIMEI} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.15 Subscriber data management

This operation package includes the operations required for subscriber data management procedures between HLR and VLR or between HLR and SGSN. Also this operation package includes the operations required for subscriber data management procedures between CSS and VLR or between CSS and SGSN. For the CSS – VLR interface and CSS – SGSN interface only version 3 of this operation package is applicable.

```
subscriberDataMngtPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR or CSS
     CONSUMER INVOKES {
     insertSubscriberData} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.16 Location register restart

This operation package includes the operations required for location register restart procedures between HLR and VLR or between HLR and SGSN and also between CSS and VLR or between CSS and SGSN.

```
resetPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR or CSS
     CONSUMER INVOKES {
     reset} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

For CSS-VLR interface and CSS-SGSN interface, only version 3 of this operation package is applicable.

17.2.2.17 Tracing stand-alone

This operation package includes the operations required for stand alone tracing procedures between HLR and VLR or between HLR and SGSN.

```
tracingStandAlonePackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR
     CONSUMER INVOKES {
     activateTraceMode |
     deactivateTraceMode} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.18 Functional SS handling

This operation package includes the operations required for functional supplementary services procedures between VLR and HLR.

```
functionalSsPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     registerSS |
     eraseSS |
     activateSS |
     deactivateSS |
     registerPassword |
     interrogateSS}
     SUPPLIER INVOKES {
     getPassword} }
```

The v1-equivalent package can be determined according to the rules described in clause 17.2.1.

17.2.2.19 Tracing

This operation package includes the operations required for tracing procedures between HLR and VLR or between HLR and SGSN.

```
tracingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR
     CONSUMER INVOKES {
     activateTraceMode} }
```

The v1-equivalent and v2-equivalent packages can be determined according to the rules described in clause 17.2.1.

17.2.2.20 Binding

This operation package includes the operation required to initialise a supplementary service procedure between VLR and HLR or between gsmSCF and HLR.

```
bindingPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is gsmSCF if Consumer is HLR
     CONSUMER INVOKES {
     beginSubscriberActivity} }
```

This package is v1 only.

17.2.2.21 Unstructured SS handling

This operation package includes the operations required for unstructured supplementary services procedures between VLR and HLR, between the HLR and the gsmSCF, and between HLR and HLR.

```
unstructuredSsPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is gsmSCF or HLR if Consumer is HLR
     CONSUMER INVOKES {
     processUnstructuredSS-Request}
     SUPPLIER INVOKES {
     unstructuredSS-Request |
     unstructuredSS-Notify} }
```

The v1-equivalent package is defined as follows:

```
unstructuredSsPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is gsmSCF if Consumer is HLR
     CONSUMER INVOKES {
     processUnstructuredSS-Data} }
```

17.2.2.22 MO Short message relay services

This operation package includes the operations required for short message relay service procedures between IWMSC and VMSC or between GMSC and MSC or between SGSN and IWMSC.

```
mo-ShortMsgRelayPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is IWMSC if Consumer is MSC
     -- Supplier is IWMSC if Consumer is SGSN
     CONSUMER INVOKES {
     mo-forwardSM} }
```

The v2-equivalent package is defined as follows:

```
shortMsgRelayPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is IWMSC if Consumer is MSC
     -- Supplier is MSC or SGSN if Consumer is GMSC
     -- Supplier is IWMSC if Consumer is SGSN
     CONSUMER INVOKES {
     forwardSM} }
```

The v1-equivalent package can be determined according to the rules described in clause 17.2.1.

17.2.2.23 Short message gateway services

This operation package includes the operations required for short message service gateway procedures between MSC and HLR.

```
shortMsgGatewayPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is GMSC
     CONSUMER INVOKES {
     sendRoutingInfoForSM |
     reportSM-DeliveryStatus}
     SUPPLIER INVOKES {
     informServiceCentre} }
```

The v2-equivalent package can be determined according to the rules described in clause 17.2.1.

The v1-equivalent package is defined as follows:

```
shortMsgGatewayPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is GMSC
     CONSUMER INVOKES {
     sendRoutingInfoForSM |
     reportSMDeliveryStatus} }
```

17.2.2.24 MT Short message relay services

This operation package includes the operations required for short message relay service procedures between GMSC and MSC or between GMSC and SGSN.

```
mt-ShortMsgRelayPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is MSC or SGSN or SMS-Router or IP-SM-GW if Consumer is GMSC
     CONSUMER INVOKES {
     mt-forwardSM} }
```

The v2-equivalent package is: s**hortMsgRelayPackage-v2**

17.2.2.25 Void

17.2.2.26 Message waiting data management

This operation package includes the operations required for short message waiting data procedures between HLR and VLR, between HLR and SGSN.

```
mwdMngtPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is SGSN
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     readyForSM} }
```

The v2-equivalent package can be determined according to the rules described in clause 17.2.1.

The v1-equivalent package is defined as follows:

```
mwdMngtPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     noteSubscriberPresent} }
```

17.2.2.27 Alerting

This operation package includes the operations required for alerting between HLR and IWMSC.

```
alertingPackage-v2  OPERATION-PACKAGE ::= {
     -- Supplier is IWMSC if Consumer is HLR
     CONSUMER INVOKES {
     alertServiceCentre} }
```

The v1-equivalent package is defined as follows.

```
alertingPackage-v1  OPERATION-PACKAGE ::= {
     -- Supplier is IWMSC if Consumer is HLR
     CONSUMER INVOKES {
     alertServiceCentreWithoutResult} }
```

17.2.2.28 Data restoration

This operation package includes the operations required for VLR data restoration between HLR and VLR.

```
dataRestorationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     restoreData} }
```

The v2-equivalent package can be determined according to the rules described in clause 17.2.1.

The v1-equivalent package is: infoRetrievalPackage-v1

17.2.2.29 Purging

This operation package includes the operations required for purging between HLR and VLR or between HLR and SGSN.

```
purgingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is HLR if Consumer is SGSN
     CONSUMER INVOKES {
     purgeMS} }
```

The v2-equivalent package can be determined according to the rules described in clause 17.2.1.

17.2.2.30 Subscriber information enquiry

This operation package includes the operations required for subscriber information enquiry procedures between HLR and VLR or between HLR and SGSN.

```
subscriberInformationEnquiryPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is HLR
     CONSUMER INVOKES {
     provideSubscriberInfo} }
```

This package is v3 only.

17.2.2.31 Any time information enquiry

This operation package includes the operations required for any time information enquiry procedures between gsmSCF and HLR or between gsmSCF and GMLC or between gsmSCF and NPLR.

```
anyTimeInformationEnquiryPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR or GMLC or NPLR if Consumer is gsmSCF
     CONSUMER INVOKES {
     anyTimeInterrogation} }
```

This package is v3 only.

17.2.2.32 Group Call Control

This operation package includes the operations required for group call and broadcast call procedures between MSCs.

```
groupCallControlPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is relay MSC if Consumer is anchor MSC
     CONSUMER INVOKES {
     prepareGroupCall |
     forwardGroupCallSignalling}
     SUPPLIER INVOKES {
     sendGroupCallEndSignal |
     processGroupCallSignalling} }
```

This package is v3 only.

17.2.2.32A Group Call Info Retrieval

This operation package includes the operations required for group call and broadcast call info retrieval between MSCs.

```
groupCallInfoRetrievalPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is group call serving MSC if Consumer is visited MSC
     -- Supplier is visited MSC if Consumer is group call serving MSC
     CONSUMER INVOKES {
     sendGroupCallInfo} }
```

This package is v3 only.

17.2.2.33 Void

17.2.2.34 Void

17.2.2.35 Gprs location updating

This operation package includes the operations required for the gprs location management procedures between HLR and SGSN.

```
gprsLocationUpdatingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is SGSN
     CONSUMER INVOKES {
     updateGprsLocation} }
```

This package is v3 only.

17.2.2.36 Gprs Interrogation

This operation package includes the operations required for interrogation procedures between HLR and GGSN.

```
gprsInterrogationPackage-v4  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is GGSN
     CONSUMER INVOKES {
     sendRoutingInfoForGprs} }
```

The v3-equivalent package is defined as follows.

```
gprsInterrogationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is GGSN
     CONSUMER INVOKES {
     sendRoutingInfoForGprs} }
```

17.2.2.37 Failure reporting

This operation package includes the operations required for failure reporting between HLR and GGSN.

```
failureReportingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is GGSN
     CONSUMER INVOKES {
     failureReport} }
```

This package is v3 only.

17.2.2.38 GPRS notifying

This operation package includes the operations required for notifying that GPRS subscriber is present between HLR and GGSN.

```
gprsNotifyingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is GGSN if Consumer is HLR
     CONSUMER INVOKES {
     noteMsPresentForGprs} }
```

This package is v3 only.

17.2.2.39 Supplementary Service invocation notification

This operation package includes the operations required for Supplementary Service invocation notification procedures between the MSC and the gsmSCF and between the HLR and the gsmSCF.

```
ss-InvocationNotificationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is gsmSCF if Consumer is MSC
     -- Supplier is gsmSCF if Consumer is HLR
     CONSUMER INVOKES {
     ss-InvocationNotification} }
```

This package is v3 only.

17.2.2.40 Set Reporting State

This operation package includes the operation required for procedures between HLR and VLR to set the reporting state.

```
setReportingStatePackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR if Consumer is HLR
     CONSUMER INVOKES {
     setReportingState} }
```

This package is v3 only.

17.2.2.41 Status Report

This operation package includes the operation required for procedures between VLR and HLR to report call results and events.

```
statusReportPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     statusReport} }
```

This package is v3 only.

17.2.2.42 Remote User Free

This operation package includes the operation required by the HLR to indicate to the VLR that the remote user is free.

```
remoteUserFreePackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR if Consumer is HLR
     CONSUMER INVOKES {
     remoteUserFree} }
```

This package is v3 only.

17.2.2.43 Call Completion

This operation package includes the operations required for procedures between VLR and HLR for subscriber control of call completion services.

```
callCompletionPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     CONSUMER INVOKES {
     registerCC-Entry |
     eraseCC-Entry} }
```

This package is v3 only.

17.2.2.44 Location service gateway services

This operation package includes the operations required for location service gateway procedures between GMLC and HLR.

```
locationSvcGatewayPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is GMLC
     CONSUMER INVOKES {
     sendRoutingInfoForLCS} }
```

This package is v3 only.

17.2.2.45 Location service enquiry

This operation package includes the operations required for the location service enquiry procedures between GMLC and MSC and between GMLC and SGSN.

```
locationSvcEnquiryPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is MSC or SGSN if Consumer is GMLC
     CONSUMER INVOKES {
     provideSubscriberLocation} }
```

This package is v3 only.

17.2.2.45A Location service reporting

This operation package includes the operations required for the location service enquiry procedures between MSC and GMLC and between SGSN and GMLC.

```
locationSvcReportingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is GMLC if Consumer is MSC
     -- Supplier is GMLC if Consumer is SGSN
     CONSUMER INVOKES {
     subscriberLocationReport} }
```

17.2.2.46 Void

17.2.2.47 Void

17.2.2.48 Void

17.2.2.49 IST Alerting

This operation package includes the operation required for alerting procedures between the MSC (Visited MSC or Gateway MSC) and HLR.

```
ist-AlertingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VMSC
     -- Supplier is HLR if Consumer is GMSC
     CONSUMER INVOKES {
     istAlert} }
```

This package is v3 only.

17.2.2.50 Service Termination

This operation package includes the operation required for immediate service termination procedures between the HLR and the Visited MSC or between the HLR and the Gateway MSC.

```
serviceTerminationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VMSC or GMSC if Consumer is HLR
     CONSUMER INVOKES {
     istCommand} }
```

This package is v3 only.

17.2.2.51 Mobility Management event notification

This operation package includes the operations required for Mobility Management event notification procedures between VLR and gsmSCF.

```
mm-EventReportingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is gsmSCF if Consumer is VLR
     CONSUMER INVOKES {
     noteMM-Event} }
```

This package is v3 only.

17.2.2.52 Any time information handling

This operation package includes the operations required for any time information handling procedures between gsmSCF and HLR.

```
anyTimeInformationHandlingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is gsmSCF
     CONSUMER INVOKES {
     anyTimeSubscriptionInterrogation |
     anyTimeModification} }
```

This package is v3 only.

17.2.2.53 Subscriber Data modification notification

This operation package includes the operations required for Subscriber Data modification notification procedures between HLR and gsmSCF.

```
subscriberDataModificationNotificationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is gsmSCF if Consumer is HLR
     CONSUMER INVOKES {
     noteSubscriberDataModified} }
```

This package is v3 only.

17.2.2.54 Authentication Failure Report

This operation package includes the operation required for procedures between VLR and HLR or the SGSN and the HLR for reporting of authentication failures.

```
authenticationFailureReportPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is HLR if Consumer is VLR
     -- Supplier is HLR if Consumer is SGSN
     CONSUMER INVOKES {
     authenticationFailureReport} }
```

This package is v3 only.

17.2.2.55 Resource Management

This operation package includes the operation required for procedures between GMSC and VMSC for resource management purpose.

```
resourceManagementPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VMSC if Consumer is GMSC
     CONSUMER INVOKES {
     releaseResources} }
```

This package is v3 only.

17.2.2.56 MT Short message relay VGCS services

This operation package includes the operations required for short message relay service procedures between SMS GMSC and MSC.

```
mt-ShortMsgRelay-VGCS-Package-v3  OPERATION-PACKAGE ::= {
     -- Supplier is MSC if Consumer is GMSC
     CONSUMER INVOKES {
     mt-forwardSM-VGCS} }
```

This package is v3 only.

17.2.2.57 Vcsg location updating

This operation package includes the operations required for the vcsg location management procedures between CSS and VLR or between CSS and SGSN.

```
vcsgLocationUpdatingPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is CSS if Consumer is VLR or SGSN
     CONSUMER INVOKES {
     updateVcsgLocation} }
```

This operation package is v3 only.

17.2.2.58 Vcsg location cancellation

This operation package includes the operations required for the vcsg location cancellation procedures between CSS and VLR or between CSS and SGSN.

```
vcsgLocationCancellationPackage-v3  OPERATION-PACKAGE ::= {
     -- Supplier is VLR or SGSN if Consumer is CSS
     CONSUMER INVOKES {
     cancelVcsgLocation} }
```

This operation package is v3 only.

17.3 Application contexts

17.3.1 General aspects

An application-context is assigned for each dialogue established by a MAP-user. In the present document each application-context is assigned a name which is supplied in the MAP-OPEN Req primitive by the MAP-User and transmitted to the peer under certain circumstances.

The following ASN.1 information object class is used to describe the main aspects of application-contexts in the following clauses:

```
APPLICATION-CONTEXT ::= CLASS {
     &Symmetric           OPERATION-PACKAGE OPTIONAL,
     &InitiatorConsumerOf OPERATION-PACKAGE OPTIONAL,
     &ResponderConsumerOf OPERATION-PACKAGE OPTIONAL,
     &code                OBJECT IDENTIFIER }
WITH SYNTAX {
     [ OPERATIONS OF         &Symmetric ]
     [ INITIATOR CONSUMER OF &InitiatorConsumerOf
     RESPONDER CONSUMER OF &ResponderConsumerOf ]
     ID &code }
```

The following definitions are used throughout this clause:

- v1-application-context: An application-context which contains only v1-packages and uses only TC v1 facilities;
- v1 context set: the set of v1-application-contexts defined in the present document.
- vn-application-context (n>=2): An application-context which contains only vn-packages;

The names of v1-application-contexts are suffixed by "-v1" while other names are suffixed by "-vn" where n>=2.

Application-contexts which do not belong to the v1 context set use v2 TC facilities.

The last component of each application-context-name (i.e. the last component of the object identifier value) assigned to an application-context which belongs to the v1 context set indicates explicitly "version1".

For each application-context which does not belong to the "v1 context set" there is a v1-equivalent application context. This is a v1-application-context which includes the v1-equivalents of the packages included in the original context.

Each application-context uses the abstract-syntax associated with the operation-packages it includes and uses the transfer-syntax derived from it by applying the encoding rules defined in clause 17.1.1.

ACs which do not belong to the v1 context set require the support of the abstract-syntax identified by the object identifier value: MAP-DialogueInformation.map-Dialogue-AS defined in clause 17.4.

17.3.2 Application context definitions

17.3.2.1 Void

17.3.2.2 Location Updating

This application context is used between HLR and VLR for location updating procedures.

```
networkLocUpContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     locationUpdatingPackage-v3 |
     dataRestorationPackage-v3}
     RESPONDER CONSUMER OF {
     subscriberDataMngtPackage-v3 |
     tracingPackage-v3}
     ID   {map-ac networkLocUp(1) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac networkLocUp(1) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac networkLocUp(1) version1(1)}
```

17.3.2.3 Location Cancellation

This application context is used between HLR and VLR or between HLR and SGSN for location cancellation procedures. For the HLR - SGSN interface only version 3 of this application context is applicable.

```
locationCancellationContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR or SGSN if Initiator is HLR
     INITIATOR CONSUMER OF {
     locationCancellationPackage-v3}
     ID   {map-ac locationCancel(2) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   map-ac locationCancel(2) version2(2)
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   map-ac locationCancel(2) version1(1)
```

17.3.2.4 Roaming number enquiry

This application context is used between HLR and VLR for roaming number enquiry procedures.

```
roamingNumberEnquiryContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR if Initiator is HLR
     INITIATOR CONSUMER OF {
     roamingNumberEnquiryPackage-v3}
     ID   {map-ac roamingNbEnquiry(3) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac roamingNbEnquiry(3) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac roamingNbEnquiry(3) version1(1)}
```

17.3.2.5 Void

17.3.2.6 Location Information Retrieval

This application-context is used between GMSC and HLR or between GMSC and NPLR or between gsmSCF and HLR when retrieving location information. For the GMSC - NPLR interface version 1, version 2 and version 3 of this application context are applicable.

```
locationInfoRetrievalContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR or NPLR if Initiator is GMSC
     -- Responder is HLR if Initiator is gsmSCF
     INITIATOR CONSUMER OF {
     interrogationPackage-v3}
     ID   {map-ac locInfoRetrieval(5) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac locInfoRetrieval(5) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac locInfoRetrieval(5) version1(1)}
```

17.3.2.7 Call control transfer

This application context is used for the call control transfer procedure between the VMSC and the GMSC.

```
callControlTransferContext-v4 APPLICATION-CONTEXT ::= {
     -- Responder is GMSC if Initiator is VMSC
     INITIATOR CONSUMER OF {
     callControlTransferPackage-v4}
     ID   {map-ac callControlTransfer(6) version4(4)} }
```

The following application-context-name is assigned to the v3-equivalent application-context:

```
     ID   {map-ac callControlTransfer(6) version3(3)}
```

17.3.2.8 Void

17.3.2.9 Void

17.3.2.10 Void

17.3.2.11 Location registers restart

This application context is used between HLR and VLR or between HLR and SGSN for location register restart procedures or between CSS and VLR or between CSS and SGSN for CSG Subscriber Server restart procedures. For the HLR - VLR interface and for the HLR - SGSN interface version 1, version 2 and version 3 of this application context are applicable For the CSS - VLR interface and the CSS - SGSN interface version 3 of this application context is applicable.

```
resetContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR or SGSN if Initiator is HLR or CSS
     INITIATOR CONSUMER OF {
     resetPackage-v3}
     ID   {map-ac reset(10) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac reset(10) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac reset(10) version1(1)}
```

17.3.2.12 Handover control

This application context is used for handover procedures between MSCs.

```
handoverControlContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is MSCB if Initiator is MSCA
     INITIATOR CONSUMER OF {
     handoverControlPackage-v3}
     ID   {map-ac handoverControl(11) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac handoverControl(11) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac handoverControl(11) version1(1)}
```

17.3.2.13 IMSI Retrieval

This application context is used for IMSI retrieval between HLR and VLR.

```
imsiRetrievalContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     imsi-RetrievalPackage-v2}
     ID   {map-ac imsiRetrieval(26) version2(2)} }
```

This application-context is v2 only.

17.3.2.14 Equipment Management

This application context is used for equipment checking between MSC and EIR or between SGSN and EIR. For the SGSN - EIR interface version 1 and version 2 and version 3 of this application context are applicable:

```
equipmentMngtContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is EIR if Initiator is MSC
     -- Responder is EIR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     equipmentMngtPackage-v3}
     ID   {map-ac equipmentMngt(13) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
equipmentMngtContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is EIR if Initiator is MSC
     -- Responder is EIR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     equipmentMngtPackage-v2}
     ID   {map-ac equipmentMngt(13) version2(2)} }
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac equipmentMngt(13) version1(1)}
```

17.3.2.15 Information retrieval

This application context is used for authentication information retrieval between HLR and VLR or between HLR and SGSN. For the HLR - SGSN interface version 1 and version 2 and version 3 of this application context are applicable.

```
infoRetrievalContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     -- Responder is HLR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     infoRetrievalPackage-v3}
     ID   {map-ac infoRetrieval(14) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
infoRetrievalContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     -- Responder is HLR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     infoRetrievalPackage-v2}
     ID   {map-ac infoRetrieval(14) version2(2)} }
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac infoRetrieval(14) version1(1)}
```

17.3.2.16 Inter-VLR information retrieval

This application context is used for information retrieval between VLRs.

```
interVlrInfoRetrievalContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     interVlrInfoRetrievalPackage-v3}
     ID   {map-ac interVlrInfoRetrieval(15) version3(3)} }
```

The v2-equivalent application-context is:

```
interVlrInfoRetrievalContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is VLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     interVlrInfoRetrievalPackage-v2}
     ID   {map-ac interVlrInfoRetrieval(15) version2(2)} }
```

The v1-equivalent application-context is:

```
     ID   {map-ac infoRetrieval(14) version1(1)}
```

17.3.2.17 Stand Alone Subscriber Data Management

This application context is used for stand alone subscriber data management between HLR and VLR or between HLR and SGSN. For the HLR - SGSN interface only version 3 of this application context is applicable. Also this application context is used for stand alone subscriber data management between CSS and VLR or between CSS and SGSN. For the CSS – VLR interface and CSS – SGSN interface only version 3 of this application context is applicable:

```
subscriberDataMngtContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR or SGSN if Initiator is HLR or CSS
     INITIATOR CONSUMER OF {
     subscriberDataMngtStandAlonePackage-v3}
     ID   {map-ac subscriberDataMngt(16) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac subscriberDataMngt(16) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac subscriberDataMngt(16) version1(1)}
```

17.3.2.18 Tracing

This application context is used between HLR and VLR or between HLR and SGSN for stand alone tracing control procedures. For the HLR - SGSN interface version 1, version 2 and version 3 of this application context are applicable.

```
tracingContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR or SGSN if Initiator is HLR
     INITIATOR CONSUMER OF {
     tracingStandAlonePackage-v3}
     ID   {map-ac tracing(17) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac tracing(17) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac tracing(17) version1(1)}
```

17.3.2.19 Network functional SS handling

This application context is used for functional-like SS handling procedures between VLR and HLR.

```
networkFunctionalSsContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is HLR, Initiator is VLR
     INITIATOR CONSUMER OF {
     functionalSsPackage-v2}
     ID   {map-ac networkFunctionalSs(18) version2(2)} }
```

The v1-equivalent application-context is defined as follows:

```
networkFunctionalSsContext-v1 APPLICATION-CONTEXT ::= {
     -- Responder is HLR, Initiator is VLR
     INITIATOR CONSUMER OF {
     functionalSsPackage-v1 |
     unstructuredSsPackage-v1 |
     bindingPackage-v1}
     ID   {map-ac networkFunctionalSs(18) version1(1)} }
```

17.3.2.20 Network unstructured SS handling

This application context is used for handling stimuli-like procedures between HLR and VLR, between the HLR and gsmSCF, and between HLR and HLR.

```
networkUnstructuredSsContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is HLR, Initiator is VLR
     -- Responder is VLR, Initiator is HLR
     -- Responder is gsmSCF, Initiator is HLR
     -- Responder is HLR, Initiator is gsmSCF
     -- Responder is HLR, Initiator is HLR
     OPERATIONS OF {
     unstructuredSsPackage-v2}
     ID   {map-ac networkUnstructuredSs(19) version2(2)} }
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac networkFunctionalSs(18) version1(1)}
```

17.3.2.21 Short Message Gateway

This application context is used for short message gateway procedures.

```
shortMsgGatewayContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is GMSC
     INITIATOR CONSUMER OF {
     shortMsgGatewayPackage-v3}
     ID   {map-ac shortMsgGateway(20) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac shortMsgGateway(20) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac shortMsgGateway(20) version1(1)}
```

17.3.2.22 Mobile originating Short Message Relay

This application context is used between MSC and IWMSC or between SGSN and IWMSC for mobile originating short message relay procedures. For the SGSN - IWMSC interface version 1, version 2 and version 3 of this application context are applicable.

```
shortMsgMO-RelayContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is IWMSC if Initiator is MSC
     -- Responder is IWMSC if Initiator is SGSN
     INITIATOR CONSUMER OF {
     mo-ShortMsgRelayPackage-v3}
     ID   {map-ac shortMsgMO-Relay(21) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac shortMsgMO-Relay(21) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac shortMsg-Relay(21) version1(1)}
```

17.3.2.23 Void

17.3.2.24 Short message alert

This application context is used for short message alerting procedures.

```
shortMsgAlertContext-v2 APPLICATION-CONTEXT ::= {
     -- Responder is IWMSC if Initiator is HLR
     INITIATOR CONSUMER OF {
     alertingPackage-v2}
     ID   {map-ac shortMsgAlert(23) version2(2)} }
```

The following application-context-name is symbolically assigned to the v1-equivalent application-context:

```
     ID   {map-ac shortMsgAlert(23) version1(1)}
```

17.3.2.25 Short message waiting data management

This application context is used between VLR and HLR or between SGSN and HLR for short message waiting data management procedures. For the SGSN - HLR interface only version 3 of this application context is applicable.

```
mwdMngtContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is SGSN
     -- Responder is HLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     mwdMngtPackage-v3}
     ID   {map-ac mwdMngt(24) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac mwdMngt(24) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac mwdMngt(24) version1(1)}
```

17.3.2.26 Mobile terminating Short Message Relay

This application context is used between GMSC and MSC or between GMSC and SGSN for mobile terminating short message relay procedures. For the GMSC - SGSN interface version 2 and version 3 of this application context and the equivalent version 1 application context are applicable.

```
shortMsgMT-RelayContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is MSC or SGSN if Initiator is GMSC
     INITIATOR CONSUMER OF {
     mt-ShortMsgRelayPackage-v3}
     ID   {map-ac shortMsgMT-Relay(25) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac shortMsgMT-Relay(25) version2(2)}
```

The following application-context-name is assigned to the v1-equivalent application-context:

```
     ID   {map-ac shortMsg-Relay(21) version1(1)}
```

17.3.2.27 MS purging

This application context is used between HLR and VLR or between HLR and SGSN for MS purging procedures. For the SGSN - HLR interface only version 3 of this application context is applicable.

```
msPurgingContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     -- Responder is HLR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     purgingPackage-v3}
     ID   {map-ac msPurging(27) version3(3)} }
```

The following application-context-name is assigned to the v2-equivalent application-context:

```
     ID   {map-ac msPurging(27) version2(2)}
```

17.3.2.28 Subscriber information enquiry

This application context is used between HLR and VLR or between HLR and SGSN for subscriber information enquiry procedures.

```
subscriberInfoEnquiryContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR or SGSN if Initiator is HLR
     INITIATOR CONSUMER OF {
     subscriberInformationEnquiryPackage-v3}
     ID   {map-ac subscriberInfoEnquiry(28) version3(3)} }
```

This application-context is v3 only.

17.3.2.29 Any time information enquiry

This application context is used between gsmSCF and HLR or between gsmSCF and GMLC or between gsmSCF and NPLR for any time information enquiry procedures.

```
anyTimeInfoEnquiryContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR or GMLC or NPLR if Initiator is gsmSCF
     INITIATOR CONSUMER OF {
     anyTimeInformationEnquiryPackage-v3}
     ID   {map-ac anyTimeInfoEnquiry(29) version3(3)} }
```

This application-context is v3 only.

17.3.2.30 Group Call Control

This application context is used between anchor MSC and relay MSC for group call and broadcast call procedures.

```
groupCallControlContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is relay MSC if Initiator is anchor MSC
     INITIATOR CONSUMER OF {
     groupCallControlPackage-v3}
     ID   {map-ac groupCallControl(31) version3(3)} }
```

This application-context is v3 only.

17.3.2.30A Group Call Info Retrieval

This application context is used between group call serving MSC and visited MSC for group call and broadcast call procedures.

```
groupCallInfoRetControlContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is group call serving MSC if Initiator is visited MSC
     -- Responder is visited MSC if Initiator is group call serving MSC
     INITIATOR CONSUMER OF {
     groupCallInfoRetrievalPackage-v3}
     ID   {map-ac groupCallInfoRetrieval(45) version3(3)} }
```

This application-context is v3 only.

17.3.2.31 Void

17.3.2.32 Gprs Location Updating

This application context is used between HLR and SGSN for gprs location updating procedures.

```
gprsLocationUpdateContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     gprsLocationUpdatingPackage-v3}
     RESPONDER CONSUMER OF {
     subscriberDataMngtPackage-v3 |
     tracingPackage-v3}
     ID   {map-ac gprsLocationUpdate(32) version3(3)} }
```

This application-context is v3 only.

17.3.2.33 Gprs Location Information Retreival

This application context is used between HLR and GGSN when retrieving gprs location information.

```
gprsLocationInfoRetrievalContext-v4 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is GGSN
     INITIATOR CONSUMER OF {
     gprsInterrogationPackage-v4}
     ID   {map-ac gprsLocationInfoRetrieval(33) version4(4)} }
```

The following application-context-name is assigned to the v3-equivalent application-context:

```
     ID   {map-ac gprsLocationInfoRetrieval(33) version3(3)}
```

17.3.2.34 Failure Reporting

This application context is used between HLR and GGSN to inform that network requested PDP-context activation has failed.

```
failureReportContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is GGSN
     INITIATOR CONSUMER OF {
     failureReportingPackage-v3}
     ID   {map-ac failureReport(34) version3(3)} }
```

This application-context is v3 only.

17.3.2.35 GPRS Notifying

This application context is used between HLR and GGSN for notifying that GPRS subscriber is present again.

```
gprsNotifyContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is GGSN if Initiator is HLR
     INITIATOR CONSUMER OF {
     gprsNotifyingPackage-v3}
     ID   {map-ac gprsNotify(35) version3(3)} }
```

This application-context is v3 only.

17.3.2.36 Supplementary Service invocation notification

This application context is used between the MSC and the gsmSCF and between the HLR and the gsmSCF for Supplementary Service invocation notification procedures.

```
ss-InvocationNotificationContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is gsmSCF, Initiator is MSC
     -- Responder is gsmSCF, Initiator is HLR
     INITIATOR CONSUMER OF {
     ss-InvocationNotificationPackage-v3}
     ID   {map-ac ss-InvocationNotification(36) version3(3)} }
```

This application-context is v3 only.

17.3.2.37 Reporting

This application context is used between HLR and VLR for reporting procedures.

```
reportingContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR if Initiator is HLR
     -- Responder is HLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     setReportingStatePackage-v3 |
     statusReportPackage-v3 |
     remoteUserFreePackage-v3}
     RESPONDER CONSUMER OF {
     setReportingStatePackage-v3 |
     statusReportPackage-v3}
     ID   {map-ac reporting(7) version3(3)} }
```

This application-context is v3 only.

17.3.2.38 Call Completion

This application context is used between VLR and the HLR for subscriber control of call completion services.

```
callCompletionContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     INITIATOR CONSUMER OF {
     callCompletionPackage-v3}
     ID   {map-ac callCompletion(8) version3(3)} }
```

This application-context is v3 only.

17.3.2.39 Location Service Gateway

This application context is used for location service gateway procedures.

```
locationSvcGatewayContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is GMLC
     INITIATOR CONSUMER OF {
     locationSvcGatewayPackage-v3}
     ID   {map-ac locationSvcGateway(37) version3(3)} }
```

17.3.2.40 Location Service Enquiry

This application context is used for location service enquiry procedures.

```
locationSvcEnquiryContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is MSC or SGSN if Initiator is GMLC
     -- Responder is GMLC if Initiator is MSC
     -- Responder is GMLC if Initiator is SGSN
     INITIATOR CONSUMER OF {
     locationSvcEnquiryPackage-v3 |
     locationSvcReportingPackage-v3}
     ID   {map-ac locationSvcEnquiry(38) version3 (3)} }
```

17.3.2.41 Void

17.3.2.42 Void

17.3.2.43 Void

17.3.2.44 IST Alerting

This application context is used between MSC (Visited MSC or Gateway MSC) and HLR for alerting services within IST procedures.

```
istAlertingContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VMSC
     -- Responder is HLR if Initiator is GMSC
     INITIATOR CONSUMER OF {
     ist-AlertingPackage-v3}
     ID   {map-ac alerting(4) version3(3)} }
```

This application-context is v3 only.

17.3.2.45 Service Termination

This application context is used between HLR and MSC (Visited MSC or Gateway MSC) for service termination services within IST procedures.

```
serviceTerminationContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VMSC or GMSC if Initiator is HLR
     INITIATOR CONSUMER OF {
     serviceTerminationPackage-v3}
     ID   {map-ac serviceTermination(9) version3(3)} }
```

This application-context is v3 only.

17.3.2.46 Mobility Management event notification

This application context is used between VLR and gsmSCF for Mobility Management event notification procedures.

```
mm-EventReportingContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is gsmSCF, Initiator is VLR
     INITIATOR CONSUMER OF {
     mm-EventReportingPackage-v3}
     ID   {map-ac mm-EventReporting(42) version3(3)} }
```

This application-context is v3 only.

17.3.2.47 Any time information handling

This application context is used between gsmSCF and HLR for any time information handling procedures.

```
anyTimeInfohandlingContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is gsmSCF
     INITIATOR CONSUMER OF {
     anyTimeInformationHandlingPackage-v3}
     ID   {map-ac anyTimeInfoHandling(43) version3(3)} }
```

This application-context is v3 only.

17.3.2.48 Subscriber Data modification notification

This application context is used between HLR and gsmSCF for Subscriber Data modification notification procedures.

```
subscriberDataModificationNotificationContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is gsmSCF, Initiator is HLR
     INITIATOR CONSUMER OF {
     subscriberDataModificationNotificationPackage-v3}
     ID   {map-ac subscriberDataModificationNotification(22) version3(3)} }
```

This application-context is v3 only.

17.3.2.49 Authentication Failure Report

This application context is used between VLR and HLR or SGSN and HLR for reporting of authentication failures.

```
authenticationFailureReportContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is HLR if Initiator is VLR
     -- Responder is HLR if Initiator is SGSN
     INITIATOR CONSUMER OF {
     authenticationFailureReportPackage-v3 }
     ID   {map-ac authenticationFailureReport(39) version3(3)} }
```

This application-context is v3 only.

17.3.2.50 Resource Management

This application context is used between GMSC and VMSC for resource management purpose.

```
resourceManagementContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VMSC if Initiator is GMSC
     INITIATOR CONSUMER OF {
     resourceManagementPackage-v3 }
     ID   {map-ac resourceManagement(44) version3(3)} }
```

This application-context is v3 only.

17.3.2.51 Mobile terminating Short Message Relay VGCS

This application context is used between SMS-GMSC and MSC for mobile terminating short message relay procedures for VGCS.

```
shortMsgMT-Relay-VGCS-Context-v3 APPLICATION-CONTEXT ::= {
     -- Responder is MSC if Initiator is SMS-GMSC
     INITIATOR CONSUMER OF {
     mt-ShortMsgRelay-VGCS-Package-v3}
     ID   {map-ac shortMsgMT-Relay-VGCS(41) version3(3)} }
```

This application-context is v3 only.

17.3.2.52 Vcsg Location Updating

This application context is used between CSS and VLR or between CSS and SGSN for vcsg location updating procedures.

```
vcsgLocationUpdateContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is CSS if Initiator is VLR or SGSN
     INITIATOR CONSUMER OF {
     vcsgLocationUpdatingPackage-v3}
     RESPONDER CONSUMER OF {
     subscriberDataMngtPackage-v3}
     ID   {map-ac vcsgLocationUpdate(46) version3(3)} }
```

This application-context is v3 only.

17.3.2.53 Vcsg Location Cancellation

This application context is used between CSS and VLR or between CSS and SGSN for vcsg location cancellation procedures.

```
vcsgLocationCancellationContext-v3 APPLICATION-CONTEXT ::= {
     -- Responder is VLR or SGSN if Initiator is CSS
     INITIATOR CONSUMER OF {
     vcsgLocationCancellationPackage-v3}
     ID   {map-ac vcsgLocationCancel(47) version3(3)} }
```

This application-context is v3 only.

17.3.3 ASN.1 Module for application-context-names

The following ASN.1 module summarises the application-context-name assigned to MAP application-contexts.

```
MAP-ApplicationContexts {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ApplicationContexts (2) version18 (18)}

DEFINITIONS

::=

BEGIN


-- EXPORTS everything


IMPORTS
   gsm-NetworkId,
   ac-Id
FROM MobileDomainDefinitions {
   itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
   mobileDomainDefinitions (0) version1 (1)}
;
```

```
-- application-context-names

map-ac  OBJECT IDENTIFIER ::= {gsm-NetworkId ac-Id}

networkLocUpContext-v3  OBJECT IDENTIFIER ::=
     {map-ac networkLocUp(1) version3(3)}

locationCancellationContext-v3  OBJECT IDENTIFIER ::=
     {map-ac locationCancel(2) version3(3)}

roamingNumberEnquiryContext-v3  OBJECT IDENTIFIER ::=
     {map-ac roamingNbEnquiry(3) version3(3)}

authenticationFailureReportContext-v3  OBJECT IDENTIFIER ::=
     {map-ac authenticationFailureReport(39) version3(3)}

locationInfoRetrievalContext-v3  OBJECT IDENTIFIER ::=
     {map-ac locInfoRetrieval(5) version3(3)}

resetContext-v3  OBJECT IDENTIFIER ::=
     {map-ac reset(10) version3(3)}

handoverControlContext-v3  OBJECT IDENTIFIER ::=
     {map-ac handoverControl(11) version3(3)}

equipmentMngtContext-v3  OBJECT IDENTIFIER ::=
     {map-ac equipmentMngt(13) version3(3)}

infoRetrievalContext-v3  OBJECT IDENTIFIER ::=
     {map-ac infoRetrieval(14) version3(3)}

interVlrInfoRetrievalContext-v3  OBJECT IDENTIFIER ::=
     {map-ac interVlrInfoRetrieval(15) version3(3)}

subscriberDataMngtContext-v3  OBJECT IDENTIFIER ::=
     {map-ac subscriberDataMngt(16) version3(3)}

tracingContext-v3  OBJECT IDENTIFIER ::=
     {map-ac tracing(17) version3(3)}

networkFunctionalSsContext-v2  OBJECT IDENTIFIER ::=
     {map-ac networkFunctionalSs(18) version2(2)}

networkUnstructuredSsContext-v2  OBJECT IDENTIFIER ::=
     {map-ac networkUnstructuredSs(19) version2(2)}

shortMsgGatewayContext-v3  OBJECT IDENTIFIER ::=
     {map-ac shortMsgGateway(20) version3(3)}

shortMsgMO-RelayContext-v3  OBJECT IDENTIFIER ::=
     {map-ac shortMsgMO-Relay(21) version3(3)}

shortMsgAlertContext-v2  OBJECT IDENTIFIER ::=
     {map-ac shortMsgAlert(23) version2(2)}

mwdMngtContext-v3  OBJECT IDENTIFIER ::=
     {map-ac mwdMngt(24) version3(3)}

shortMsgMT-RelayContext-v3  OBJECT IDENTIFIER ::=
     {map-ac shortMsgMT-Relay(25) version3(3)}

shortMsgMT-Relay-VGCS-Context-v3  OBJECT IDENTIFIER ::=
     {map-ac shortMsgMT-Relay-VGCS(41) version3(3)}

imsiRetrievalContext-v2  OBJECT IDENTIFIER ::=
     {map-ac imsiRetrieval(26) version2(2)}

msPurgingContext-v3  OBJECT IDENTIFIER ::=
     {map-ac msPurging(27) version3(3)}

subscriberInfoEnquiryContext-v3  OBJECT IDENTIFIER ::=
     {map-ac subscriberInfoEnquiry(28) version3(3)}

anyTimeInfoEnquiryContext-v3  OBJECT IDENTIFIER ::=
     {map-ac anyTimeInfoEnquiry(29) version3(3)}
```

```
callControlTransferContext-v4  OBJECT IDENTIFIER ::=
     {map-ac callControlTransfer(6) version4(4)}

ss-InvocationNotificationContext-v3  OBJECT IDENTIFIER ::=
     {map-ac ss-InvocationNotification(36) version3(3)}

groupCallControlContext-v3  OBJECT IDENTIFIER ::=
     {map-ac groupCallControl(31) version3(3)}

groupCallInfoRetrievalContext-v3  OBJECT IDENTIFIER ::=
     {map-ac groupCallInfoRetrieval(45) version3(3)}

gprsLocationUpdateContext-v3  OBJECT IDENTIFIER ::=
     {map-ac gprsLocationUpdate(32) version3(3)}

gprsLocationInfoRetrievalContext-v4  OBJECT IDENTIFIER ::=
     {map-ac gprsLocationInfoRetrieval(33) version4(4)}

failureReportContext-v3  OBJECT IDENTIFIER ::=
     {map-ac failureReport(34) version3(3)}

gprsNotifyContext-v3  OBJECT IDENTIFIER ::=
     {map-ac gprsNotify(35) version3(3)}

reportingContext-v3  OBJECT IDENTIFIER ::=
     {map-ac reporting(7) version3(3)}

callCompletionContext-v3  OBJECT IDENTIFIER ::=
     {map-ac callCompletion(8) version3(3)}

istAlertingContext-v3  OBJECT IDENTIFIER ::=
     {map-ac istAlerting(4) version3(3)}

serviceTerminationContext-v3  OBJECT IDENTIFIER ::=
     {map-ac immediateTermination(9) version3(3)}

locationSvcGatewayContext-v3  OBJECT IDENTIFIER ::=
     {map-ac locationSvcGateway(37) version3(3)}

locationSvcEnquiryContext-v3  OBJECT IDENTIFIER ::=
     {map-ac locationSvcEnquiry(38) version3(3)}

mm-EventReportingContext-v3  OBJECT IDENTIFIER ::=
     {map-ac mm-EventReporting(42) version3(3)}

anyTimeInfoHandlingContext-v3  OBJECT IDENTIFIER ::=
     {map-ac anyTimeInfoHandling(43) version3(3)}

subscriberDataModificationNotificationContext-v3  OBJECT IDENTIFIER ::=
     {map-ac subscriberDataModificationNotification(22) version3(3)}

resourceManagementContext-v3  OBJECT IDENTIFIER ::=
     {map-ac resourceManagement(44) version3(3)}

vcsgLocationUpdateContext-v3  OBJECT IDENTIFIER ::=
     {map-ac vcsgLocationUpdate(46) version3(3)}

vcsgLocationCancellationContext-v3  OBJECT IDENTIFIER ::=
     {map-ac vcsgLocationCancel(47) version3(3)}


-- The following Object Identifiers are reserved for application-contexts
--  existing in previous versions of the protocol
```

-- AC Name & Version	Object Identifier	
--		
-- networkLocUpContext-v1	map-ac networkLocUp (1)	version1 (1)
-- networkLocUpContext-v2	map-ac networkLocUp (1)	version2 (2)
-- locationCancellationContext-v1	map-ac locationCancellation (2)	version1 (1)
-- locationCancellationContext-v2	map-ac locationCancellation (2)	version2 (2)
-- roamingNumberEnquiryContext-v1	map-ac roamingNumberEnquiry (3)	version1 (1)
-- roamingNumberEnquiryContext-v2	map-ac roamingNumberEnquiry (3)	version2 (2)
-- locationInfoRetrievalContext-v1	map-ac locationInfoRetrieval (5)	version1 (1)
-- locationInfoRetrievalContext-v2	map-ac locationInfoRetrieval (5)	version2 (2)
-- resetContext-v1	map-ac reset (10)	version1 (1)
-- resetContext-v2	map-ac reset (10)	version2 (2)
-- handoverControlContext-v1	map-ac handoverControl (11)	version1 (1)
-- handoverControlContext-v2	map-ac handoverControl (11)	version2 (2)
-- **sIWFSAllocationContext**-v3	map-ac sIWFSAllocation **(12)**	version3 (3)
-- equipmentMngtContext-v1	map-ac equipmentMngt (13)	version1 (1)
-- equipmentMngtContext-v2	map-ac equipmentMngt (13)	version2 (2)
-- infoRetrievalContext-v1	map-ac infoRetrieval (14)	version1 (1)
-- infoRetrievalContext-v2	map-ac infoRetrieval (14)	version2 (2)
-- interVlrInfoRetrievalContext-v2	map-ac interVlrInfoRetrieval (15)	version2 (2)
-- subscriberDataMngtContext-v1	map-ac subscriberDataMngt (16)	version1 (1)
-- subscriberDataMngtContext-v2	map-ac subscriberDataMngt (16)	version2 (2)
-- tracingContext-v1	map-ac tracing (17)	version1 (1)
-- tracingContext-v2	map-ac tracing (17)	version2 (2)
-- *networkFunctionalSsContext-v1*	*map-ac networkFunctionalSs (18)*	*version1 (1)*
-- shortMsgGatewayContext-v1	map-ac shortMsgGateway (20)	version1 (1)
-- shortMsgGatewayContext-v2	map-ac shortMsgGateway (20)	version2 (2)
-- shortMsgRelayContext-v1	map-ac shortMsgRelay (21)	version1 (1)
-- shortMsgAlertContext-v1	map-ac shortMsgAlert (23)	version1 (1)
-- *mwdMngtContext-v1*	*map-ac mwdMngt (24)*	*version1 (1)*
-- mwdMngtContext-v2	map-ac mwdMngt (24)	version2 (2)
-- shortMsgMT-RelayContext-v2	map-ac shortMsgMT-Relay (25)	version2 (2)
-- msPurgingContext-v2	map-ac msPurging (27)	version2 (2)
-- callControlTransferContext-v3	map-ac callControlTransferContext (6)	version3 (3)
-- gprsLocationInfoRetrievalContext-v3	map-ac gprsLocationInfoRetrievalContext (33)	version3 (3)

```
END
```

17.4 MAP Dialogue Information

```
MAP-DialogueInformation {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-DialogueInformation (3) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   map-DialogueAS,
   MAP-DialoguePDU

;

IMPORTS
   gsm-NetworkId,
   as-Id
FROM MobileDomainDefinitions {
   itu-t (0) identified-organization (4) etsi (0) mobileDomain (0)
   mobileDomainDefinitions (0) version1 (1)}

   AddressString
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network(1) modules (3) map-CommonDataTypes (18) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}
```

```
;

-- abstract syntax name for MAP-DialoguePDU

map-DialogueAS  OBJECT IDENTIFIER ::=
     {gsm-NetworkId as-Id map-DialoguePDU (1) version1 (1)}

MAP-DialoguePDU ::= CHOICE {
     map-open  [0] MAP-OpenInfo,
     map-accept                          [1] MAP-AcceptInfo,
     map-close [2] MAP-CloseInfo,
     map-refuse                          [3] MAP-RefuseInfo,
     map-userAbort                       [4] MAP-UserAbortInfo,
     map-providerAbort                   [5] MAP-ProviderAbortInfo}

MAP-OpenInfo ::= SEQUENCE {
     destinationReference                [0] AddressString               OPTIONAL,
     originationReference                [1] AddressString               OPTIONAL,
     ...,
     extensionContainer                  ExtensionContainer              OPTIONAL
     -- extensionContainer must not be used in version 2
     }

MAP-AcceptInfo ::= SEQUENCE {
     ...,
     extensionContainer                  ExtensionContainer              OPTIONAL
     -- extensionContainer must not be used in version 2
     }

MAP-CloseInfo ::= SEQUENCE {
     ...,
     extensionContainer                  ExtensionContainer              OPTIONAL
     -- extensionContainer must not be used in version 2
     }

MAP-RefuseInfo ::= SEQUENCE {
     reason    Reason,
     ...,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     -- extensionContainer must not be used in version 2
     alternativeApplicationContext       OBJECT IDENTIFIER               OPTIONAL
     -- alternativeApplicationContext must not be used in version 2
     }

Reason ::= ENUMERATED {
     noReasonGiven                       (0),
     invalidDestinationReference         (1),
     invalidOriginatingReference         (2)}

MAP-UserAbortInfo ::= SEQUENCE {
     map-UserAbortChoice                 MAP-UserAbortChoice,
     ...,
     extensionContainer                  ExtensionContainer              OPTIONAL
     -- extensionContainer must not be used in version 2
     }

MAP-UserAbortChoice ::= CHOICE {
     userSpecificReason                  [0] NULL,
     userResourceLimitation              [1] NULL,
     resourceUnavailable                 [2] ResourceUnavailableReason,
     applicationProcedureCancellation    [3] ProcedureCancellationReason}

ResourceUnavailableReason ::= ENUMERATED {
     shortTermResourceLimitation  (0),
     longTermResourceLimitation  (1)}

ProcedureCancellationReason ::= ENUMERATED {
     handoverCancellation  (0),
     radioChannelRelease  (1),
     networkPathRelease  (2),
     callRelease  (3),
     associatedProcedureFailure  (4),
     tandemDialogueRelease  (5),
     remoteOperationsFailure  (6)}
```

```
MAP-ProviderAbortInfo ::= SEQUENCE {
     map-ProviderAbortReason             MAP-ProviderAbortReason,
     ...,
     extensionContainer                  ExtensionContainer                OPTIONAL
     -- extensionContainer must not be used in version 2
     }
```

```
MAP-ProviderAbortReason ::= ENUMERATED {
     abnormalDialogue  (0),
     invalidPDU  (1)}
```

```
END
```

17.5 MAP operation and error codes

```
MAP-Protocol {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Protocol (4) version18 (18)}

DEFINITIONS

::=

BEGIN

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4) informationObjects(5) version1(0)}

   updateLocation,
   cancelLocation,
   cancelVcsgLocation,
   purgeMS,
   sendIdentification,
   updateGprsLocation,
   updateVcsgLocation,
   prepareHandover,
   sendEndSignal,
   processAccessSignalling,
   forwardAccessSignalling,
   prepareSubsequentHandover,
   sendAuthenticationInfo,
   authenticationFailureReport,
   checkIMEI,
   insertSubscriberData,
   deleteSubscriberData,
   reset,
   forwardCheckSS-Indication,
   restoreData,
   provideSubscriberInfo,
   anyTimeInterrogation,
   anyTimeSubscriptionInterrogation,
   anyTimeModification,
   sendRoutingInfoForGprs,
   failureReport,
   noteMsPresentForGprs,
   noteMM-Event,
   noteSubscriberDataModified


FROM MAP-MobileServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MobileServiceOperations (5)
   version18 (18)}

   activateTraceMode,
   deactivateTraceMode,
   sendIMSI
FROM MAP-OperationAndMaintenanceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-OperationAndMaintenanceOperations (6)
   version18 (18)}

   sendRoutingInfo,
   provideRoamingNumber,
   resumeCallHandling,
   setReportingState,
```

```
   statusReport,
   remoteUserFree,
   ist-Alert,
   ist-Command,
   releaseResources
FROM MAP-CallHandlingOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CallHandlingOperations (7)
   version18 (18)}

   registerSS,
   eraseSS,
   activateSS,
   deactivateSS,
   interrogateSS,
   processUnstructuredSS-Request,
   unstructuredSS-Request,
   unstructuredSS-Notify,
   registerPassword,
   getPassword,
   ss-InvocationNotification,
   registerCC-Entry,
   eraseCC-Entry
FROM MAP-SupplementaryServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SupplementaryServiceOperations (8)
   version18 (18)}

   sendRoutingInfoForSM,
   mo-ForwardSM,
   mt-ForwardSM,
   reportSM-DeliveryStatus,
   alertServiceCentre,
   informServiceCentre,
   readyForSM,
   mt-ForwardSM-VGCS
FROM MAP-ShortMessageServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ShortMessageServiceOperations (9)
   version18 (18)}

   prepareGroupCall,
   processGroupCallSignalling,
   forwardGroupCallSignalling,
   sendGroupCallEndSignal,
   sendGroupCallInfo
FROM MAP-Group-Call-Operations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Group-Call-Operations (22)
   version18 (18)}

   provideSubscriberLocation,
   sendRoutingInfoForLCS,
   subscriberLocationReport
FROM MAP-LocationServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-LocationServiceOperations (24)
   version18 (18)}


;
Supported-MAP-Operations OPERATION ::= {updateLocation | cancelLocation | cancelVcsgLocation |
   purgeMS |
   sendIdentification | updateGprsLocation | updateVcsgLocation | prepareHandover | sendEndSignal |
   processAccessSignalling | forwardAccessSignalling | prepareSubsequentHandover |
   sendAuthenticationInfo | authenticationFailureReport | checkIMEI | insertSubscriberData |
   deleteSubscriberData | reset | forwardCheckSS-Indication | restoreData | provideSubscriberInfo |
   anyTimeInterrogation | anyTimeSubscriptionInterrogation | anyTimeModification |
   sendRoutingInfoForGprs | failureReport |noteMsPresentForGprs | noteMM-Event |
   noteSubscriberDataModified | activateTraceMode | deactivateTraceMode | sendIMSI |
   sendRoutingInfo | provideRoamingNumber | resumeCallHandling | setReportingState | statusReport |
   remoteUserFree | ist-Alert |
   ist-Command | registerSS | eraseSS | activateSS | deactivateSS | interrogateSS |
   processUnstructuredSS-Request | unstructuredSS-Request | unstructuredSS-Notify |
   registerPassword | getPassword | ss-InvocationNotification | registerCC-Entry | eraseCC-Entry |
   sendRoutingInfoForSM | mo-ForwardSM | mt-ForwardSM | reportSM-DeliveryStatus |
   alertServiceCentre | informServiceCentre | readyForSM | prepareGroupCall |
   processGroupCallSignalling | forwardGroupCallSignalling | sendGroupCallEndSignal |
   provideSubscriberLocation | sendRoutingInfoForLCS | subscriberLocationReport |
```

```
   releaseResources | mt-ForwardSM-VGCS | sendGroupCallInfo }


-- The following operation codes are reserved for operations
-- existing in previous versions of the protocol
```

-- Operation Name	AC used	Oper. Code
--		
-- sendParameters	map-ac infoRetrieval (14) version1 (1)	local:9
-- processUnstructuredSS-Data	map-ac networkFunctionalSs (18) version1 (1)	local:19
-- performHandover	map-ac handoverControl (11) version1 (1)	local:28
-- performSubsequentHandover	map-ac handoverControl (11) version1 (1)	local:30
-- provideSIWFSNumber	map-ac sIWFSAllocation (12) version3 (3)	local:31
-- siwfs-SignallingModify	map-ac sIWFSAllocation (12) version3 (3)	local:32
-- noteInternalHandover	map-ac handoverControl (11) version1 (1)	local:35
-- noteSubscriberPresent	map-ac mwdMngt (24) version1 (1)	local:48
-- alertServiceCentreWithoutResult	map-ac shortMsgAlert (23) version1 (1)	local:49
-- traceSubscriberActivity	map-ac handoverControl (11) version1 (1)	local:52
-- beginSubscriberActivity	map-ac networkFunctionalSs (18) version1 (1)	local:54

```
-- The following error codes are reserved for errors
-- existing in previous versions of the protocol
```

-- Error Name	AC used	Error Code
--		
-- unknownBaseStation	map-ac handoverControl (11) version1 (1)	local:2
-- invalidTargetBaseStation	map-ac handoverControl (11) version1 (1)	local:23
-- noRadioResourceAvailable	map-ac handoverControl (11) version1 (1)	local:24

```
END
```

17.6 MAP operations and errors

17.6.1 Mobile Service Operations

```
MAP-MobileServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MobileServiceOperations (5)
   version18 (18)}

DEFINITIONS

::=

BEGIN

EXPORTS

   -- location registration operations
   updateLocation,
   cancelLocation,
   purgeMS,
   sendIdentification,

   -- gprs location registration operations
   updateGprsLocation,

   -- vcsg location registration operations
   updateVcsgLocation,
   cancelVcsgLocation,

   -- subscriber information enquiry operations
   provideSubscriberInfo,

   -- any time information enquiry operations
   anyTimeInterrogation,

   -- any time information handling operations
   anyTimeSubscriptionInterrogation,
   anyTimeModification,

   -- subscriber data modification notification operations
   noteSubscriberDataModified,
```

```
   -- handover operations
   prepareHandover,
   sendEndSignal,
   processAccessSignalling,
   forwardAccessSignalling,
   prepareSubsequentHandover,

   -- authentication management operations
   sendAuthenticationInfo,
   authenticationFailureReport,

   -- IMEI management operations
   checkIMEI,

   -- subscriber management operations
   insertSubscriberData,
   deleteSubscriberData,

   -- fault recovery operations
   reset,
   forwardCheckSS-Indication,
   restoreData,

-- gprs location information retrieval operations
   sendRoutingInfoForGprs,

   -- failure reporting operations
   failureReport,

   -- gprs notification operations
   noteMsPresentForGprs,

   -- Mobility Management operations
   noteMM-Event

;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
   dataMissing,
   unexpectedDataValue,
   unknownSubscriber,
   unknownMSC,
   unidentifiedSubscriber,
   unknownEquipment,
   roamingNotAllowed,
   ati-NotAllowed,
   noHandoverNumberAvailable,
   subsequentHandoverFailure,
   absentSubscriber,
   mm-EventNotSupported,
   atsi-NotAllowed,
   atm-NotAllowed,
   bearerServiceNotProvisioned,
   teleserviceNotProvisioned,
   callBarred,
   illegalSS-Operation,
   ss-ErrorStatus,
   ss-NotAvailable,
   ss-Incompatibility,
   ss-SubscriptionViolation,
   informationNotAvailable,
   targetCellOutsideGroupCallArea


FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

   UpdateLocationArg,
   UpdateLocationRes,
   CancelLocationArg,
   CancelLocationRes,
```

```
   PurgeMS-Arg,
   PurgeMS-Res,
   SendIdentificationArg,
   SendIdentificationRes,
   UpdateGprsLocationArg,
   UpdateGprsLocationRes,
   UpdateVcsgLocationArg,
   UpdateVcsgLocationRes,
   CancelVcsgLocationArg,
   CancelVcsgLocationRes,
   PrepareHO-Arg,
   PrepareHO-Res,
   ForwardAccessSignalling-Arg,
   ProcessAccessSignalling-Arg,
   SendEndSignal-Arg,
   SendEndSignal-Res,
   PrepareSubsequentHO-Res,
   PrepareSubsequentHO-Arg,
   SendAuthenticationInfoArg,
   SendAuthenticationInfoRes,
   AuthenticationFailureReportArg,
   AuthenticationFailureReportRes,
   CheckIMEI-Arg,
   CheckIMEI-Res,
   InsertSubscriberDataArg,
   InsertSubscriberDataRes,
   DeleteSubscriberDataArg,
   DeleteSubscriberDataRes,
   ResetArg,
   RestoreDataArg,
   RestoreDataRes,
   ProvideSubscriberInfoArg,
   ProvideSubscriberInfoRes,
   AnyTimeSubscriptionInterrogationArg,
   AnyTimeSubscriptionInterrogationRes,
   AnyTimeModificationArg,
   AnyTimeModificationRes,
   NoteSubscriberDataModifiedArg,
   NoteSubscriberDataModifiedRes,
   AnyTimeInterrogationArg,
   AnyTimeInterrogationRes,
   SendRoutingInfoForGprsArg,
   SendRoutingInfoForGprsRes,
   FailureReportArg,
   FailureReportRes,
   NoteMsPresentForGprsArg,
   NoteMsPresentForGprsRes,
   NoteMM-EventArg,
   NoteMM-EventRes


FROM MAP-MS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MS-DataTypes (11) version18 (18)}

;


-- location registration operations

updateLocation  OPERATION ::= {               --Timer m
     ARGUMENT
     UpdateLocationArg
     RESULT
     UpdateLocationRes
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber |
     roamingNotAllowed}
     CODE local:2 }
```

```
cancelLocation  OPERATION ::= {              --Timer m
     ARGUMENT
     CancelLocationArg
     RESULT
     CancelLocationRes
          -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue}
     CODE local:3 }
```

```
purgeMS  OPERATION ::= {                     --Timer m
     ARGUMENT
     PurgeMS-Arg
     RESULT
     PurgeMS-Res
          -- optional
     ERRORS{
     dataMissing |
     unexpectedDataValue|
     unknownSubscriber}
     CODE local:67 }
```

```
sendIdentification  OPERATION ::= {          --Timer s
     ARGUMENT
     SendIdentificationArg
     RESULT
     SendIdentificationRes
     ERRORS {
     dataMissing |
     unidentifiedSubscriber}
     CODE local:55 }
```

-- gprs location registration operations

```
updateGprsLocation  OPERATION ::= {          --Timer m
     ARGUMENT
     UpdateGprsLocationArg
     RESULT
     UpdateGprsLocationRes
     ERRORS {
     systemFailure |
     unexpectedDataValue |
     unknownSubscriber |
     roamingNotAllowed}
     CODE local:23 }
```

-- subscriber information enquiry operations

```
provideSubscriberInfo  OPERATION ::= {       --Timer m
     ARGUMENT
     ProvideSubscriberInfoArg
     RESULT
     ProvideSubscriberInfoRes
     ERRORS {
     dataMissing |
     unexpectedDataValue}
     CODE local:70 }
```

-- any time information enquiry operations

```
anyTimeInterrogation  OPERATION ::= {        --Timer m
     ARGUMENT
     AnyTimeInterrogationArg
     RESULT
     AnyTimeInterrogationRes
     ERRORS {
     systemFailure |
     ati-NotAllowed |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:71 }
```

-- any time information handling operations

```
anyTimeSubscriptionInterrogation  OPERATION ::= {                        --Timer m
     ARGUMENT
     AnyTimeSubscriptionInterrogationArg
     RESULT
     AnyTimeSubscriptionInterrogationRes
     ERRORS {
     atsi-NotAllowed |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-NotAvailable |
     informationNotAvailable}
     CODE local:62 }
```

```
anyTimeModification  OPERATION ::= {          --Timer m
     ARGUMENT
     AnyTimeModificationArg
     RESULT
     AnyTimeModificationRes
     ERRORS {
     atm-NotAllowed |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-SubscriptionViolation |
     ss-ErrorStatus |
     ss-Incompatibility |
     informationNotAvailable}
     CODE local:65 }
```

-- subscriber data modification notification operations

```
noteSubscriberDataModified  OPERATION ::= {        --Timer m
     ARGUMENT
     NoteSubscriberDataModifiedArg
     RESULT
     NoteSubscriberDataModifiedRes
          -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:5 }
```

-- handover operations

```
prepareHandover  OPERATION ::= {              --Timer m
     ARGUMENT
     PrepareHO-Arg
     RESULT
     PrepareHO-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     noHandoverNumberAvailable |
     targetCellOutsideGroupCallArea }
     CODE local:68 }
```

```
sendEndSignal  OPERATION ::= {                --Timer l
     ARGUMENT
     SendEndSignal-Arg
     RESULT
     SendEndSignal-Res
     CODE local:29 }
```

```
processAccessSignalling  OPERATION ::= {      --Timer s
     ARGUMENT
     ProcessAccessSignalling-Arg
     CODE local:33 }
```

```
forwardAccessSignalling  OPERATION ::= {      --Timer s
     ARGUMENT
     ForwardAccessSignalling-Arg
     CODE local:34 }
```

```
prepareSubsequentHandover  OPERATION ::= {         --Timer m
     ARGUMENT
     PrepareSubsequentHO-Arg
     RESULT
     PrepareSubsequentHO-Res
     ERRORS {
     unexpectedDataValue |
     dataMissing |
     unknownMSC |
     subsequentHandoverFailure}
     CODE local:69 }
```

```
-- authentication management operations
```

```
sendAuthenticationInfo  OPERATION ::= {       --Timer m
     ARGUMENT
     SendAuthenticationInfoArg
     -- optional
     -- within a dialogue sendAuthenticationInfoArg shall not be present in
     -- subsequent invoke components. If received in a subsequent invoke component
     -- it shall be discarded.

     RESULT
     SendAuthenticationInfoRes
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:56 }
```

```
authenticationFailureReport  OPERATION ::= {       --Timer m
     ARGUMENT
     AuthenticationFailureReportArg
     RESULT
     AuthenticationFailureReportRes
     -- optional
     ERRORS {
     systemFailure |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:15 }
```

```
-- IMEI management operations
```

```
checkIMEI  OPERATION ::= {                    --Timer m
     ARGUMENT
     CheckIMEI-Arg
     RESULT
     CheckIMEI-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unknownEquipment}
     CODE local:43 }
```

```
-- subscriber management operations
```

```
insertSubscriberData  OPERATION ::= {          --Timer m
     ARGUMENT
     InsertSubscriberDataArg
     RESULT
     InsertSubscriberDataRes
     -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unidentifiedSubscriber}
     CODE local:7 }
```

```
deleteSubscriberData  OPERATION ::= {          --Timer m
     ARGUMENT
     DeleteSubscriberDataArg
     RESULT
     DeleteSubscriberDataRes
     -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unidentifiedSubscriber}
     CODE local:8 }
```

-- fault recovery operations

```
reset  OPERATION ::= {                         --Timer m
     ARGUMENT
     ResetArg
     CODE local:37 }
```

```
forwardCheckSS-Indication  OPERATION ::= {          --Timer s
     CODE local:38 }
```

```
restoreData  OPERATION ::= {                   --Timer m
     ARGUMENT
     RestoreDataArg
     RESULT
     RestoreDataRes
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:57 }
```

-- gprs location information retrieval operations

```
sendRoutingInfoForGprs  OPERATION ::= {        --Timer m
     ARGUMENT
     SendRoutingInfoForGprsArg
     RESULT
     SendRoutingInfoForGprsRes
     ERRORS {
     absentSubscriber |
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber |
     callBarred }
     CODE local:24 }
```

-- failure reporting operations

```
failureReport  OPERATION ::= {                 --Timer m
     ARGUMENT
     FailureReportArg
     RESULT
     FailureReportRes
          -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:25 }
```

-- gprs notification operations

```
noteMsPresentForGprs  OPERATION ::= {          --Timer m
     ARGUMENT
     NoteMsPresentForGprsArg
     RESULT
     NoteMsPresentForGprsRes
          -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:26 }
```

```
noteMM-Event  OPERATION ::= {                  --Timer m
     ARGUMENT
     NoteMM-EventArg
     RESULT
     NoteMM-EventRes
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber |
     mm-EventNotSupported}
     CODE local:89 }
```

```
-- vcsg location registration operations
```

```
updateVcsgLocation  OPERATION ::= {            --Timer m
     ARGUMENT
     UpdateVcsgLocationArg
     RESULT
     UpdateVcsgLocationRes
     ERRORS {
     systemFailure |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:53 }
```

```
cancelVcsgLocation  OPERATION ::= {            --Timer m
     ARGUMENT
     CancelVcsgLocationArg
     RESULT
     CancelVcsgLocationRes
          -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue}
     CODE local:36 }
```

```
END
```

17.6.2 Operation and Maintenance Operations

```
MAP-OperationAndMaintenanceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-OperationAndMaintenanceOperations (6)
   version18 (18)}

DEFINITIONS

::=

BEGIN

EXPORTS
   activateTraceMode,
   deactivateTraceMode,
   sendIMSI
;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
```

```
   dataMissing,
   unexpectedDataValue,
   facilityNotSupported,
   unknownSubscriber,
   unidentifiedSubscriber,
   tracingBufferFull
FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

   ActivateTraceModeArg,
   ActivateTraceModeRes,
   DeactivateTraceModeArg,
   DeactivateTraceModeRes
FROM MAP-OM-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-OM-DataTypes (12) version18 (18)}

   ISDN-AddressString,
   IMSI
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}
;
```

```
activateTraceMode  OPERATION ::= {            --Timer m
     ARGUMENT
     ActivateTraceModeArg
     RESULT
     ActivateTraceModeRes
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unidentifiedSubscriber |
     tracingBufferFull}
     CODE local:50 }
```

```
deactivateTraceMode  OPERATION ::= {          --Timer m
     ARGUMENT
     DeactivateTraceModeArg
     RESULT
     DeactivateTraceModeRes
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unidentifiedSubscriber}
     CODE local:51 }
```

```
sendIMSI  OPERATION ::= {                     --Timer m
     ARGUMENT
     ISDN-AddressString
     RESULT
     IMSI
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:58 }
```

```
END
```

17.6.3 Call Handling Operations

```
MAP-CallHandlingOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CallHandlingOperations (7)
   version18 (18)}

DEFINITIONS

::=
```

```
BEGIN

EXPORTS
   sendRoutingInfo,
   provideRoamingNumber,
   resumeCallHandling,
   setReportingState,
   statusReport,
   remoteUserFree,
   ist-Alert,
   ist-Command,
   releaseResources
;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
   dataMissing,
   unexpectedDataValue,
   facilityNotSupported,
   or-NotAllowed,
   unknownSubscriber,
   numberChanged,
   bearerServiceNotProvisioned,
   teleserviceNotProvisioned,
   noRoamingNumberAvailable,
   absentSubscriber,
   busySubscriber,
   noSubscriberReply,
   callBarred,
   forwardingViolation,
   forwardingFailed,
   cug-Reject,
   resourceLimitation,
   incompatibleTerminal,
   unidentifiedSubscriber

FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}
   SendRoutingInfoArg,
   SendRoutingInfoRes,
   ProvideRoamingNumberArg,
   ProvideRoamingNumberRes,
   ResumeCallHandlingArg,
   ResumeCallHandlingRes,
   SetReportingStateArg,
   SetReportingStateRes,
   StatusReportArg,
   StatusReportRes,
   RemoteUserFreeArg,
   RemoteUserFreeRes,
   IST-AlertArg,
   IST-AlertRes,
   IST-CommandArg,
   IST-CommandRes,
   ReleaseResourcesArg,
   ReleaseResourcesRes
FROM MAP-CH-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CH-DataTypes (13) version18 (18)}

;
```

```
sendRoutingInfo  OPERATION ::= {              --Timer m
-- The timer is set to the upper limit of the range if the GMSC supports pre-paging.
     ARGUMENT
     SendRoutingInfoArg
     RESULT
     SendRoutingInfoRes
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     or-NotAllowed |
     unknownSubscriber |
     numberChanged |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     absentSubscriber |
     busySubscriber |
     noSubscriberReply |
     callBarred |
     cug-Reject |
     forwardingViolation}
     CODE local:22 }
```

```
provideRoamingNumber  OPERATION ::= {         --Timer m
-- The timer is set to the upper limit of the range if the HLR supports pre-paging.
     ARGUMENT
     ProvideRoamingNumberArg
     RESULT
     ProvideRoamingNumberRes
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     or-NotAllowed |
     absentSubscriber |
     noRoamingNumberAvailable}
     CODE local:4 }
```

```
resumeCallHandling  OPERATION ::= {           --Timer m
     ARGUMENT
     ResumeCallHandlingArg
     RESULT
     ResumeCallHandlingRes
     -- optional
     ERRORS {
     forwardingFailed |
     or-NotAllowed |
     unexpectedDataValue |
     dataMissing }
     CODE local:6 }
```

```
setReportingState  OPERATION ::= {            --Timer m
     ARGUMENT
     SetReportingStateArg
     RESULT
     SetReportingStateRes
     -- optional
     ERRORS {
     systemFailure |
     unidentifiedSubscriber |
     unexpectedDataValue |
     dataMissing |
     resourceLimitation |
     facilityNotSupported}
     CODE local:73 }
```

```
statusReport  OPERATION ::= {                --Timer m
     ARGUMENT
     StatusReportArg
     RESULT
     StatusReportRes
     -- optional
     ERRORS {
     unknownSubscriber |
     systemFailure |
     unexpectedDataValue |
     dataMissing}
     CODE local:74 }
```

```
remoteUserFree  OPERATION ::= {              --Timer ml
     ARGUMENT
     RemoteUserFreeArg
     RESULT
     RemoteUserFreeRes
     ERRORS {
     unexpectedDataValue |
     dataMissing |
     incompatibleTerminal |
     absentSubscriber |
     systemFailure |
     busySubscriber}
     CODE local:75 }
```

```
ist-Alert  OPERATION ::= {                   --Timer m
     ARGUMENT
     IST-AlertArg
     RESULT
     IST-AlertRes
     -- optional
     ERRORS {
     unexpectedDataValue |
     resourceLimitation |
     unknownSubscriber |
     systemFailure |
     facilityNotSupported}
     CODE local:87 }
```

```
ist-Command  OPERATION::= {                  --Timer m
     ARGUMENT
     IST-CommandArg
     RESULT
     IST-CommandRes
     -- optional
     ERRORS {
     unexpectedDataValue |
     resourceLimitation |
     unknownSubscriber |
     systemFailure |
     facilityNotSupported}
     CODE local:88 }
```

```
releaseResources  OPERATION::= {             --Timer m
     ARGUMENT
     ReleaseResourcesArg
     RESULT
     ReleaseResourcesRes
     -- optional
     ERRORS {
     unexpectedDataValue |
     systemFailure }
     CODE local:20 }
```

```
END
```

17.6.4 Supplementary service operations

```
MAP-SupplementaryServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SupplementaryServiceOperations (8)
   version18 (18)}

DEFINITIONS

::=
```

```
BEGIN

EXPORTS
   registerSS,
   eraseSS,
   activateSS,
   deactivateSS,
   interrogateSS,
   processUnstructuredSS-Request,
   unstructuredSS-Request,
   unstructuredSS-Notify,
   registerPassword,
   getPassword,
   ss-InvocationNotification,
   registerCC-Entry,
   eraseCC-Entry
;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
   dataMissing,
   unexpectedDataValue,
   unknownSubscriber,
   bearerServiceNotProvisioned,
   teleserviceNotProvisioned,
   callBarred,
   illegalSS-Operation,
   ss-ErrorStatus,
   ss-NotAvailable,
   ss-SubscriptionViolation,
   ss-Incompatibility,
   pw-RegistrationFailure,
   negativePW-Check,
   numberOfPW-AttemptsViolation,
   unknownAlphabet,
   ussd-Busy,
   absentSubscriber,
   illegalSubscriber,
   illegalEquipment,
   shortTermDenial,
   longTermDenial,
   facilityNotSupported
FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

   RegisterSS-Arg,
   SS-Info,
   SS-ForBS-Code,
   InterrogateSS-Res,
   USSD-Arg,
   USSD-Res,
   Password,
   GuidanceInfo,
   SS-InvocationNotificationArg,
   SS-InvocationNotificationRes,
   RegisterCC-EntryArg,
   RegisterCC-EntryRes,
   EraseCC-EntryArg,
   EraseCC-EntryRes
FROM MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-DataTypes (14) version18 (18)}

   SS-Code
FROM MAP-SS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-Code (15) version18 (18)}
;


-- supplementary service handling operations
```

```
registerSS  OPERATION ::= {                    --Timer m
     ARGUMENT
     RegisterSS-Arg
     RESULT
     SS-Info
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-ErrorStatus |
     ss-Incompatibility}
     CODE local:10 }
```

```
eraseSS  OPERATION ::= {                       --Timer m
     ARGUMENT
     SS-ForBS-Code
     RESULT
     SS-Info
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-ErrorStatus
     }
     CODE local:11 }
```

```
activateSS  OPERATION ::= {                    --Timer m
     ARGUMENT
     SS-ForBS-Code
     RESULT
     SS-Info
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-ErrorStatus |
     ss-SubscriptionViolation |
     ss-Incompatibility |
     negativePW-Check |
     numberOfPW-AttemptsViolation}
     CODE local:12 }
```

```
deactivateSS  OPERATION ::= {                  --Timer m
     ARGUMENT
     SS-ForBS-Code
     RESULT
     SS-Info
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-ErrorStatus |
     ss-SubscriptionViolation |
     negativePW-Check |
     numberOfPW-AttemptsViolation}
     CODE local:13 }
```

```
interrogateSS  OPERATION ::= {                --Timer m
     ARGUMENT
     SS-ForBS-Code
     RESULT
     InterrogateSS-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     bearerServiceNotProvisioned |
     teleserviceNotProvisioned |
     callBarred |
     illegalSS-Operation |
     ss-NotAvailable}
     CODE local:14 }
```

```
processUnstructuredSS-Request  OPERATION ::= {     --Timer 10 minutes
     ARGUMENT
     USSD-Arg
     RESULT
     USSD-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     unknownAlphabet |
     callBarred}
     CODE local:59 }
```

```
unstructuredSS-Request  OPERATION ::= {       --Timer ml
     ARGUMENT
     USSD-Arg
     RESULT
     USSD-Res
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     absentSubscriber |
     illegalSubscriber |
     illegalEquipment |
     unknownAlphabet |
     ussd-Busy}
     CODE local:60 }
```

```
unstructuredSS-Notify  OPERATION ::= {        --Timer ml
     ARGUMENT
     USSD-Arg
     RETURN RESULT TRUE
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     absentSubscriber |
     illegalSubscriber |
     illegalEquipment |
     unknownAlphabet |
     ussd-Busy}
     CODE local:61 }
```

```
registerPassword  OPERATION ::= {             --Timer ml
     ARGUMENT
     SS-Code
     RESULT
     Password
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     callBarred |
     ss-SubscriptionViolation |
     pw-RegistrationFailure |
     negativePW-Check |
     numberOfPW-AttemptsViolation}
     LINKED {
     getPassword}
     CODE local:17 }
```

```
getPassword  OPERATION ::= {                  --Timer m
     ARGUMENT
     GuidanceInfo
     RESULT
     Password
     CODE local:18 }
```

```
ss-InvocationNotification  OPERATION ::= {         --Timer m
     ARGUMENT
     SS-InvocationNotificationArg
     RESULT
     SS-InvocationNotificationRes
     -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber}
     CODE local:72 }
```

```
registerCC-Entry  OPERATION ::= {             --Timer m
     ARGUMENT
     RegisterCC-EntryArg
     RESULT
     RegisterCC-EntryRes
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     callBarred |
     illegalSS-Operation |
     ss-ErrorStatus |
     ss-Incompatibility |
     shortTermDenial |
     longTermDenial |
     facilityNotSupported}
     CODE local:76 }
```

```
eraseCC-Entry  OPERATION ::= {                --Timer m
     ARGUMENT
     EraseCC-EntryArg
     RESULT
     EraseCC-EntryRes
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     callBarred |
     illegalSS-Operation |
     ss-ErrorStatus}
     CODE local:77 }
```

```
END
```

17.6.5 Short message service operations

```
MAP-ShortMessageServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ShortMessageServiceOperations (9)
   version18 (18)}
```

```
DEFINITIONS

::=

BEGIN

EXPORTS
   sendRoutingInfoForSM,
   mo-ForwardSM,
   mt-ForwardSM,
   reportSM-DeliveryStatus,
   alertServiceCentre,
   informServiceCentre,
   readyForSM,
   mt-ForwardSM-VGCS
;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
   dataMissing,
   unexpectedDataValue,
   facilityNotSupported,
   unknownSubscriber,
   unidentifiedSubscriber,
   illegalSubscriber,
   illegalEquipment,
   teleserviceNotProvisioned,
   callBarred,
   subscriberBusyForMT-SMS,
   sm-DeliveryFailure,
   messageWaitingListFull,
   absentSubscriberSM
FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

   RoutingInfoForSM-Arg,
   RoutingInfoForSM-Res,
   MO-ForwardSM-Arg,
   MO-ForwardSM-Res,
   MT-ForwardSM-Arg,
   MT-ForwardSM-Res,
   ReportSM-DeliveryStatusArg,
   ReportSM-DeliveryStatusRes,
   AlertServiceCentreArg,
   InformServiceCentreArg,
   ReadyForSM-Arg,
   ReadyForSM-Res,
   MT-ForwardSM-VGCS-Arg,
   MT-ForwardSM-VGCS-Res
FROM MAP-SM-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SM-DataTypes (16) version18 (18)}

;
```

```
sendRoutingInfoForSM  OPERATION ::= {         --Timer m
     ARGUMENT
     RoutingInfoForSM-Arg
     RESULT
     RoutingInfoForSM-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unknownSubscriber |
     teleserviceNotProvisioned |
     callBarred |
     absentSubscriberSM}
     CODE local:45 }
```

```
mo-ForwardSM  OPERATION ::= {                --Timer ml
     ARGUMENT
     MO-ForwardSM-Arg
     RESULT
     MO-ForwardSM-Res
     -- optional
     ERRORS {
     systemFailure |
     unexpectedDataValue |
     facilityNotSupported |
     sm-DeliveryFailure}
     CODE local:46 }
```

```
mt-ForwardSM  OPERATION ::= {                --Timer ml
     -- the timer value may be subject to negotiation between GMSC and IP-SM-GW
     ARGUMENT
     MT-ForwardSM-Arg
     RESULT
     MT-ForwardSM-Res
     -- optional
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unidentifiedSubscriber |
     illegalSubscriber |
     illegalEquipment |
     subscriberBusyForMT-SMS |
     sm-DeliveryFailure |
     absentSubscriberSM}
     CODE local:44 }
```

```
reportSM-DeliveryStatus  OPERATION ::= {      --Timer s
     ARGUMENT
     ReportSM-DeliveryStatusArg
     RESULT
     ReportSM-DeliveryStatusRes
     -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     unknownSubscriber |
     messageWaitingListFull}
     CODE local:47 }
```

```
alertServiceCentre  OPERATION ::= {          --Timer s
     ARGUMENT
     AlertServiceCentreArg
     RETURN RESULT TRUE
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue}
     CODE local:64 }
```

```
informServiceCentre  OPERATION ::= {         --Timer s
     ARGUMENT
     InformServiceCentreArg
     CODE local:63 }
```

```
readyForSM  OPERATION ::= {                  --Timer m
     ARGUMENT
     ReadyForSM-Arg
     RESULT
     ReadyForSM-Res
     -- optional
     ERRORS {
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unknownSubscriber}
     CODE local:66 }
```

```
mt-ForwardSM-VGCS  OPERATION ::= {            --Timer ml
     ARGUMENT
     MT-ForwardSM-VGCS-Arg
     RESULT
     MT-ForwardSM-VGCS-Res
     -- optional
     ERRORS {
     systemFailure |
     unexpectedDataValue }
     CODE local:21 }
```

```
END
```

17.6.6 Errors

```
MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

DEFINITIONS

::=

BEGIN

EXPORTS

   -- generic errors
   systemFailure,
   dataMissing,
   unexpectedDataValue,
   facilityNotSupported,
   incompatibleTerminal,
   resourceLimitation,

   -- identification and numbering errors
   unknownSubscriber,
   numberChanged,
   unknownMSC,
   unidentifiedSubscriber,
   unknownEquipment,

   -- subscription errors
   roamingNotAllowed,
   illegalSubscriber,
   illegalEquipment,
   bearerServiceNotProvisioned,
   teleserviceNotProvisioned,

   -- handover errors
   noHandoverNumberAvailable,
   subsequentHandoverFailure,
   targetCellOutsideGroupCallArea,

   -- operation and maintenance errors
   tracingBufferFull,

   -- call handling errors
   or-NotAllowed,
   noRoamingNumberAvailable,
   busySubscriber,
   noSubscriberReply,
   absentSubscriber,
   callBarred,
   forwardingViolation,
   forwardingFailed,
   cug-Reject,

   -- any time interrogation errors
   ati-NotAllowed,

   -- any time information handling errors
   atsi-NotAllowed,
   atm-NotAllowed,
   informationNotAvailable,
```

```
   -- supplementary service errors
   illegalSS-Operation,
   ss-ErrorStatus,
   ss-NotAvailable,
   ss-SubscriptionViolation,
   ss-Incompatibility,
   unknownAlphabet,
   ussd-Busy,
   pw-RegistrationFailure,
   negativePW-Check,
   numberOfPW-AttemptsViolation,
   shortTermDenial,
   longTermDenial,

   -- short message service errors
   subscriberBusyForMT-SMS,
   sm-DeliveryFailure,
   messageWaitingListFull,
   absentSubscriberSM,

   -- Group Call errors
   noGroupCallNumberAvailable,
   ongoingGroupCall,

   -- location service errors
   unauthorizedRequestingNetwork,
   unauthorizedLCSClient,
   positionMethodFailure,
   unknownOrUnreachableLCSClient,

   -- Mobility Management errors
   mm-EventNotSupported


;

IMPORTS
   ERROR
FROM Remote-Operations-Information-Objects {joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0) }

   SS-Status
FROM MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-DataTypes (14) version18 (18)}

   SS-IncompatibilityCause,
   PW-RegistrationFailureCause,
   SM-DeliveryFailureCause,
   SystemFailureParam,
   DataMissingParam,
   UnexpectedDataParam,
   FacilityNotSupParam,
   UnknownSubscriberParam,
   NumberChangedParam,
   UnidentifiedSubParam,
   RoamingNotAllowedParam,
   IllegalSubscriberParam,
   IllegalEquipmentParam,
   BearerServNotProvParam,
   TeleservNotProvParam,
   TracingBufferFullParam,
   NoRoamingNbParam,
   OR-NotAllowedParam,
   AbsentSubscriberParam,
   BusySubscriberParam,
   NoSubscriberReplyParam,
   CallBarredParam,
   ForwardingViolationParam,
   ForwardingFailedParam,
   CUG-RejectParam,
   ATI-NotAllowedParam,
   SubBusyForMT-SMS-Param,
   MessageWaitListFullParam,
   AbsentSubscriberSM-Param,
   ResourceLimitationParam,
   NoGroupCallNbParam,
   IncompatibleTerminalParam,
   ShortTermDenialParam,
```

```
   LongTermDenialParam,
   UnauthorizedRequestingNetwork-Param,
   UnauthorizedLCSClient-Param,
   PositionMethodFailure-Param,
   UnknownOrUnreachableLCSClient-Param,
   MM-EventNotSupported-Param,
   ATSI-NotAllowedParam,
   ATM-NotAllowedParam,
   IllegalSS-OperationParam,
   SS-NotAvailableParam,
   SS-SubscriptionViolationParam,
   InformationNotAvailableParam,
   TargetCellOutsideGCA-Param,
   OngoingGroupCallParam
FROM MAP-ER-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ER-DataTypes (17) version18 (18)}
;

-- generic errors

systemFailure  ERROR ::= {
     PARAMETER
     SystemFailureParam
     -- optional
     CODE local:34 }

dataMissing  ERROR ::= {
     PARAMETER
     DataMissingParam
     -- optional
     -- DataMissingParam must not be used in version <3
     CODE local:35 }

unexpectedDataValue  ERROR ::= {
     PARAMETER
     UnexpectedDataParam
     -- optional
     -- UnexpectedDataParam must not be used in version <3
     CODE local:36 }

facilityNotSupported  ERROR ::= {
     PARAMETER
     FacilityNotSupParam
     -- optional
     -- FacilityNotSupParam must not be used in version <3
     CODE local:21 }

incompatibleTerminal  ERROR ::= {
     PARAMETER
     IncompatibleTerminalParam
     -- optional
     CODE local:28 }

resourceLimitation  ERROR ::= {
     PARAMETER
     ResourceLimitationParam
     -- optional
     CODE local:51 }

-- identification and numbering errors

unknownSubscriber  ERROR ::= {
     PARAMETER
     UnknownSubscriberParam
     -- optional
     -- UnknownSubscriberParam must not be used in version <3
     CODE local:1 }

numberChanged  ERROR ::= {
     PARAMETER
     NumberChangedParam
     -- optional
     CODE local:44 }

unknownMSC  ERROR ::= {
     CODE local:3 }
```

```
unidentifiedSubscriber  ERROR ::= {
     PARAMETER
     UnidentifiedSubParam
     -- optional
     -- UunidentifiedSubParam must not be used in version <3
     CODE local:5 }
```

```
unknownEquipment  ERROR ::= {
     CODE local:7 }
```

```
-- subscription errors
```

```
roamingNotAllowed  ERROR ::= {
     PARAMETER
     RoamingNotAllowedParam
     CODE local:8 }
```

```
illegalSubscriber  ERROR ::= {
     PARAMETER
     IllegalSubscriberParam
     -- optional
     -- IllegalSubscriberParam must not be used in version <3
     CODE local:9 }
```

```
illegalEquipment  ERROR ::= {
     PARAMETER
     IllegalEquipmentParam
     -- optional
     -- IllegalEquipmentParam must not be used in version <3
     CODE local:12 }
```

```
bearerServiceNotProvisioned  ERROR ::= {
     PARAMETER
     BearerServNotProvParam
     -- optional
     -- BearerServNotProvParam must not be used in version <3
     CODE local:10 }
```

```
teleserviceNotProvisioned  ERROR ::= {
     PARAMETER
     TeleservNotProvParam
     -- optional
     -- TeleservNotProvParam must not be used in version <3
     CODE local:11 }
```

```
-- handover errors
```

```
noHandoverNumberAvailable  ERROR ::= {
     CODE local:25 }
```

```
subsequentHandoverFailure  ERROR ::= {
     CODE local:26 }
```

```
targetCellOutsideGroupCallArea  ERROR ::= {
     PARAMETER
     TargetCellOutsideGCA-Param
     -- optional
     CODE local:42 }
```

```
-- operation and maintenance errors
```

```
tracingBufferFull  ERROR ::= {
     PARAMETER
     TracingBufferFullParam
     -- optional
     CODE local: 40 }
```

```
-- call handling errors
```

```
noRoamingNumberAvailable  ERROR ::= {
     PARAMETER
     NoRoamingNbParam
     -- optional
     CODE local:39 }
```

```
absentSubscriber  ERROR ::= {
     PARAMETER
     AbsentSubscriberParam
     -- optional
     -- AbsentSubscriberParam must not be used in version <3
     CODE local:27 }
```

```
busySubscriber  ERROR ::= {
     PARAMETER
     BusySubscriberParam
     -- optional
     CODE local:45 }
```

```
noSubscriberReply  ERROR ::= {
     PARAMETER
     NoSubscriberReplyParam
     -- optional
     CODE local:46 }
```

```
callBarred  ERROR ::= {
     PARAMETER
     CallBarredParam
     -- optional
     CODE local:13 }
```

```
forwardingViolation  ERROR ::= {
     PARAMETER
     ForwardingViolationParam
     -- optional
     CODE local:14 }
```

```
forwardingFailed  ERROR ::= {
     PARAMETER
     ForwardingFailedParam
     -- optional
     CODE local:47 }
```

```
cug-Reject  ERROR ::= {
     PARAMETER
     CUG-RejectParam
     -- optional
     CODE local:15 }
```

```
or-NotAllowed  ERROR ::= {
     PARAMETER
     OR-NotAllowedParam
     -- optional
     CODE local:48 }
```

```
-- any time interrogation errors
ati-NotAllowed  ERROR ::= {
     PARAMETER
     ATI-NotAllowedParam
     -- optional
     CODE local:49 }
```

```
-- any time information handling errors
atsi-NotAllowed  ERROR ::= {
     PARAMETER
     ATSI-NotAllowedParam
     -- optional
     CODE local:60 }
```

```
atm-NotAllowed  ERROR ::= {
     PARAMETER
     ATM-NotAllowedParam
     -- optional
     CODE local:61 }
```

```
informationNotAvailable  ERROR ::= {
     PARAMETER
     InformationNotAvailableParam
     -- optional
     CODE local:62 }
```

```
-- supplementary service errors
```

```
illegalSS-Operation  ERROR ::= {
     PARAMETER
     IllegalSS-OperationParam
     -- optional
     -- IllegalSS-OperationParam must not be used in version <3
     CODE local:16 }

ss-ErrorStatus  ERROR ::= {
     PARAMETER
     SS-Status
     -- optional
     CODE local:17 }

ss-NotAvailable  ERROR ::= {
     PARAMETER
     SS-NotAvailableParam
     -- optional
     -- SS-NotAvailableParam must not be used in version <3
     CODE local:18 }

ss-SubscriptionViolation  ERROR ::= {
     PARAMETER
     SS-SubscriptionViolationParam
     -- optional
     -- SS-SubscriptionViolationParam must not be used in version <3
     CODE local:19 }

ss-Incompatibility  ERROR ::= {
     PARAMETER
     SS-IncompatibilityCause
     -- optional
     CODE local:20 }

unknownAlphabet  ERROR ::= {
     CODE local:71 }

ussd-Busy  ERROR ::= {
     CODE local:72 }

pw-RegistrationFailure  ERROR ::= {
     PARAMETER
     PW-RegistrationFailureCause
     CODE local:37 }

negativePW-Check  ERROR ::= {
     CODE local:38 }

numberOfPW-AttemptsViolation  ERROR ::= {
     CODE local:43 }

shortTermDenial  ERROR ::= {
     PARAMETER
     ShortTermDenialParam
     -- optional
     CODE local:29 }

longTermDenial  ERROR ::= {
     PARAMETER
     LongTermDenialParam
     -- optional
     CODE local:30 }

-- short message service errors

subscriberBusyForMT-SMS  ERROR ::= {
     PARAMETER
     SubBusyForMT-SMS-Param
     -- optional
     CODE local:31 }

sm-DeliveryFailure  ERROR ::= {
     PARAMETER
     SM-DeliveryFailureCause
     CODE local:32 }
```

```
messageWaitingListFull  ERROR ::= {
     PARAMETER
     MessageWaitListFullParam
     -- optional
     CODE local:33 }
```

```
absentSubscriberSM  ERROR ::= {
     PARAMETER
     AbsentSubscriberSM-Param
     -- optional
     CODE local:6 }
```

```
-- Group Call errors
```

```
noGroupCallNumberAvailable  ERROR ::= {
     PARAMETER
     NoGroupCallNbParam
     -- optional
     CODE local:50 }
```

```
ongoingGroupCall  ERROR ::= {
     PARAMETER
     OngoingGroupCallParam
     -- optional
     CODE local:22 }
```

```
-- location service errors
```

```
unauthorizedRequestingNetwork  ERROR ::= {
     PARAMETER
     UnauthorizedRequestingNetwork-Param
     -- optional
     CODE local:52 }
```

```
unauthorizedLCSClient  ERROR ::= {
     PARAMETER
     UnauthorizedLCSClient-Param
     -- optional
     CODE local:53 }
```

```
positionMethodFailure  ERROR ::= {
     PARAMETER
     PositionMethodFailure-Param
     -- optional
     CODE local:54 }
```

```
unknownOrUnreachableLCSClient  ERROR ::= {
     PARAMETER
     UnknownOrUnreachableLCSClient-Param
     -- optional
     CODE local:58 }
```

```
mm-EventNotSupported  ERROR ::= {
     PARAMETER
     MM-EventNotSupported-Param
     -- optional
     CODE local:59 }
```

```
END
```

17.6.7 Group Call operations

```
MAP-Group-Call-Operations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Group-Call-Operations (22)
   version18 (18)}

DEFINITIONS

::=

BEGIN

EXPORTS
   prepareGroupCall,
   sendGroupCallEndSignal,
```

```
   forwardGroupCallSignalling,
   processGroupCallSignalling,
   sendGroupCallInfo
;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
   unexpectedDataValue,
   noGroupCallNumberAvailable,
   ongoingGroupCall,
   unknownSubscriber,
   teleserviceNotProvisioned,
   dataMissing
FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

   PrepareGroupCallArg,
   PrepareGroupCallRes,
   SendGroupCallEndSignalArg,
   SendGroupCallEndSignalRes,
   ForwardGroupCallSignallingArg,
   ProcessGroupCallSignallingArg,
   SendGroupCallInfoArg,
   SendGroupCallInfoRes
FROM MAP-GR-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-GR-DataTypes (23) version18 (18)}

;
```

```
prepareGroupCall  OPERATION ::= {            --Timer m
     ARGUMENT
     PrepareGroupCallArg
     RESULT
     PrepareGroupCallRes
     ERRORS {
     systemFailure |
     noGroupCallNumberAvailable |
     unexpectedDataValue}
     CODE local:39 }
```

```
sendGroupCallEndSignal  OPERATION ::= {      --Timer l
     ARGUMENT
     SendGroupCallEndSignalArg
     RESULT
     SendGroupCallEndSignalRes
     CODE local:40 }
```

```
processGroupCallSignalling  OPERATION ::= {       --Timer s
     ARGUMENT
     ProcessGroupCallSignallingArg
     CODE local:41 }
```

```
forwardGroupCallSignalling  OPERATION ::= {       --Timer s
     ARGUMENT
     ForwardGroupCallSignallingArg
     CODE local:42 }
```

```
sendGroupCallInfo  OPERATION ::= {           --Timer m
     ARGUMENT
     SendGroupCallInfoArg
     RESULT
     SendGroupCallInfoRes
     ERRORS {
     systemFailure |
     ongoingGroupCall |
     unexpectedDataValue |
     dataMissing |
     teleserviceNotProvisioned |
     unknownSubscriber}
     CODE local:84 }
```

```
END
```

17.6.8 Location service operations

```
MAP-LocationServiceOperations {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-LocationServiceOperations (24)
   version18 (18)}

DEFINITIONS

::=

BEGIN

EXPORTS
   provideSubscriberLocation,
   sendRoutingInfoForLCS,
   subscriberLocationReport
;

IMPORTS
   OPERATION
FROM Remote-Operations-Information-Objects {
   joint-iso-itu-t remote-operations(4)
  informationObjects(5) version1(0)}

   systemFailure,
   dataMissing,
   unexpectedDataValue,
   facilityNotSupported,
   unknownSubscriber,
   absentSubscriber,
   unauthorizedRequestingNetwork,
   unauthorizedLCSClient,
   positionMethodFailure,
   resourceLimitation,
   unknownOrUnreachableLCSClient,
   unidentifiedSubscriber,
   illegalEquipment,
   illegalSubscriber
FROM MAP-Errors {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-Errors (10) version18 (18)}

   RoutingInfoForLCS-Arg,
   RoutingInfoForLCS-Res,
   ProvideSubscriberLocation-Arg,
   ProvideSubscriberLocation-Res,
   SubscriberLocationReport-Arg,
   SubscriberLocationReport-Res
FROM MAP-LCS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-LCS-DataTypes (25) version18 (18)}
;

sendRoutingInfoForLCS  OPERATION ::= {        --Timer m
     ARGUMENT
     RoutingInfoForLCS-Arg
     RESULT
     RoutingInfoForLCS-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unknownSubscriber |
     absentSubscriber |
     unauthorizedRequestingNetwork }
     CODE local:85 }
```

```
provideSubscriberLocation  OPERATION ::= {      --Timer ml
     ARGUMENT
     ProvideSubscriberLocation-Arg
     RESULT
     ProvideSubscriberLocation-Res
     ERRORS {
     systemFailure |
     dataMissing |
     unexpectedDataValue |
     facilityNotSupported |
     unidentifiedSubscriber |
     illegalSubscriber |
     illegalEquipment |
     absentSubscriber |
     unauthorizedRequestingNetwork |
     unauthorizedLCSClient |
     positionMethodFailure }
     CODE local:83 }

subscriberLocationReport  OPERATION ::= {     --Timer m
     ARGUMENT
     SubscriberLocationReport-Arg
     RESULT
     SubscriberLocationReport-Res
     ERRORS {
     systemFailure |
     dataMissing |
     resourceLimitation |
     unexpectedDataValue |
     unknownSubscriber |
     unauthorizedRequestingNetwork |
     unknownOrUnreachableLCSClient}
     CODE local:86 }


END
```

17.6.9 Void

17.7 MAP constants and data types

17.7.1 Mobile Service data types

```
MAP-MS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MS-DataTypes (11) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS

   -- location registration types
   UpdateLocationArg,
   UpdateLocationRes,
   CancelLocationArg,
   CancelLocationRes,
   PurgeMS-Arg,
   PurgeMS-Res,
   SendIdentificationArg,
   SendIdentificationRes,
   UpdateGprsLocationArg,
   UpdateGprsLocationRes,
   IST-SupportIndicator,
   SupportedLCS-CapabilitySets,
   UpdateVcsgLocationArg,
```

```
    UpdateVcsgLocationRes,
    CancelVcsgLocationArg,
    CancelVcsgLocationRes,


    -- handover types
    ForwardAccessSignalling-Arg,
    PrepareHO-Arg,
    PrepareHO-Res,
    PrepareSubsequentHO-Arg,
    PrepareSubsequentHO-Res,
    ProcessAccessSignalling-Arg,
    SendEndSignal-Arg,
    SendEndSignal-Res,

    -- authentication management types
    SendAuthenticationInfoArg,
    SendAuthenticationInfoRes,
    AuthenticationFailureReportArg,
    AuthenticationFailureReportRes,

    -- security management types
    Kc,
    Cksn,

    -- equipment management types
    CheckIMEI-Arg,
    CheckIMEI-Res,

    -- subscriber management types
    InsertSubscriberDataArg,
    InsertSubscriberDataRes,
    LSAIdentity,
    DeleteSubscriberDataArg,
    DeleteSubscriberDataRes,
    Ext-QoS-Subscribed,
    Ext2-QoS-Subscribed,
    Ext3-QoS-Subscribed,
    Ext4-QoS-Subscribed,
    SubscriberData,
    ODB-Data,
    SubscriberStatus,
    ZoneCodeList,
    maxNumOfZoneCodes,
    O-CSI,
    D-CSI,
    O-BcsmCamelTDPCriteriaList,
    T-BCSM-CAMEL-TDP-CriteriaList,
    SS-CSI,
    ServiceKey,
    DefaultCallHandling,
    DefaultSMS-Handling,
    DefaultGPRS-Handling,
    CamelCapabilityHandling,
    BasicServiceCriteria,
    SupportedCamelPhases,
    OfferedCamel4CSIs,
    OfferedCamel4Functionalities,
    maxNumOfCamelTDPData,
    CUG-Index,
    CUG-Info,
    CUG-Interlock,
    InterCUG-Restrictions,
    IntraCUG-Options,
    NotificationToMSUser,
    QoS-Subscribed,
    IST-AlertTimerValue,
    T-CSI,
    T-BcsmTriggerDetectionPoint,
    APN,
    AdditionalInfo,

    -- fault recovery types
    ResetArg,
    RestoreDataArg,
    RestoreDataRes,
```

```
-- provide subscriber info types
   GeographicalInformation,
   MS-Classmark2,
   GPRSMSClass,

   -- subscriber information enquiry types
   ProvideSubscriberInfoArg,
   ProvideSubscriberInfoRes,
   SubscriberInfo,
   LocationInformation,
   LocationInformationGPRS,
   SubscriberState,
   GPRSChargingID,
   MNPInfoRes,
   RouteingNumber,

   -- any time information enquiry types
   AnyTimeInterrogationArg,
   AnyTimeInterrogationRes,

   -- any time information handling types
   AnyTimeSubscriptionInterrogationArg,
   AnyTimeSubscriptionInterrogationRes,
   AnyTimeModificationArg,
   AnyTimeModificationRes,

   -- subscriber data modification notification types
   NoteSubscriberDataModifiedArg,
   NoteSubscriberDataModifiedRes,

   -- gprs location information retrieval types
   SendRoutingInfoForGprsArg,
   SendRoutingInfoForGprsRes,

   -- failure reporting types
   FailureReportArg,
   FailureReportRes,

   -- gprs notification types
   NoteMsPresentForGprsArg,
   NoteMsPresentForGprsRes,

   -- Mobility Management types
   NoteMM-EventArg,
   NoteMM-EventRes,
   NumberPortabilityStatus,
   PagingArea,

   -- VGCS / VBS types types
   GroupId,
   Long-GroupId,
   AdditionalSubscriptions

;

IMPORTS
   maxNumOfSS,
   SS-SubscriptionOption,
   SS-List,
   SS-ForBS-Code,
   Password,
   OverrideCategory,
   CliRestrictionOption
FROM MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-DataTypes (14) version18 (18)}

   SS-Code
FROM MAP-SS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-Code (15) version18 (18)}

   Ext-BearerServiceCode
FROM MAP-BS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-BS-Code (20) version18 (18)}

   Ext-TeleserviceCode
FROM MAP-TS-Code {
```

```
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-TS-Code (19) version18 (18)}

   AddressString,
   ISDN-AddressString,
   ISDN-SubaddressString,
   FTN-AddressString,
   AccessNetworkSignalInfo,
   IMSI,
   IMEI,
   TMSI,
   HLR-List,
   LMSI,
   Identity,
   GlobalCellId,
   CellGlobalIdOrServiceAreaIdOrLAI,
   Ext-BasicServiceCode,
   NAEA-PreferredCI,
   EMLPP-Info,
   MC-SS-Info,
   SubscriberIdentity,
   AgeOfLocationInformation,
   LCSClientExternalID,
   LCSClientInternalID,
   Ext-SS-Status,
   LCSServiceTypeID,
   ASCI-CallReference,
   TBCD-STRING,
   LAIFixedLength,
   PLMN-Id,
   EMLPP-Priority,
   GSN-Address,
   DiameterIdentity,
   Time,
   E-UTRAN-CGI,
   NR-CGI,
   TA-Id,
   RAIdentity,
   NetworkNodeDiameterAddress
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}

   AbsentSubscriberDiagnosticSM
FROM MAP-ER-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ER-DataTypes (17) version18 (18)}

   TracePropagationList
FROM MAP-OM-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-OM-DataTypes (12) version18 (18)}

;

-- location registration types
```

```
UpdateLocationArg ::= SEQUENCE {
    imsi     IMSI,
    msc-Number                          [1] ISDN-AddressString,
    vlr-Number                          ISDN-AddressString,
    lmsi     [10] LMSI                  OPTIONAL,
    extensionContainer                  ExtensionContainer              OPTIONAL,
    ... ,
    vlr-Capability                      [6] VLR-Capability              OPTIONAL,
    informPreviousNetworkEntity         [11] NULL OPTIONAL,
    cs-LCS-NotSupportedByUE             [12] NULL OPTIONAL,
    v-gmlc-Address                      [2]  GSN-Address                OPTIONAL,
    add-info  [13] ADD-Info             OPTIONAL,
    pagingArea                          [14] PagingArea                 OPTIONAL,
    skipSubscriberDataUpdate            [15] NULL OPTIONAL,
    -- The skipSubscriberDataUpdate parameter in the UpdateLocationArg and the ADD-Info
    -- structures carry the same semantic.
    restorationIndicator                [16] NULL OPTIONAL,
    eplmn-List                          [3] EPLMN-List                  OPTIONAL,
    mme-DiameterAddress                 [4] NetworkNodeDiameterAddress  OPTIONAL
    }
```

```
VLR-Capability ::= SEQUENCE{
    supportedCamelPhases                [0] SupportedCamelPhases        OPTIONAL,
    extensionContainer                  ExtensionContainer              OPTIONAL,
    ... ,
    solsaSupportIndicator               [2] NULL  OPTIONAL,
    istSupportIndicator                 [1] IST-SupportIndicator        OPTIONAL,
    superChargerSupportedInServingNetworkEntity   [3] SuperChargerInfo  OPTIONAL,
    longFTN-Supported                   [4]  NULL OPTIONAL,
    supportedLCS-CapabilitySets         [5]  SupportedLCS-CapabilitySets  OPTIONAL,
    offeredCamel4CSIs                   [6] OfferedCamel4CSIs           OPTIONAL,
    supportedRAT-TypesIndicator         [7]  SupportedRAT-Types         OPTIONAL,
    longGroupID-Supported               [8]  NULL OPTIONAL,
    mtRoamingForwardingSupported        [9]  NULL OPTIONAL,
    msisdn-lessOperation-Supported      [10] NULL OPTIONAL,
    reset-ids-Supported                 [11] NULL OPTIONAL }
```

```
SupportedRAT-Types::= BIT STRING {
    utran  (0),
    geran  (1),
    gan    (2),
    i-hspa-evolution (3),
    e-utran  (4),
    nb-iot   (5)} (SIZE (2..8))
    -- exception handling: bits 6 to 7 shall be ignored if received and not understood
```

```
SuperChargerInfo ::= CHOICE {
    sendSubscriberData                  [0] NULL,
    subscriberDataStored                [1] AgeIndicator }
```

```
AgeIndicator ::= OCTET STRING (SIZE (1..6))
    -- The internal structure of this parameter is implementation specific.
```

```
IST-SupportIndicator ::=  ENUMERATED {
    basicISTSupported                   (0),
    istCommandSupported                 (1),
    ...}
-- exception handling:
-- reception of values > 1 shall be mapped to ' istCommandSupported '
```

```
SupportedLCS-CapabilitySets ::= BIT STRING {
     lcsCapabilitySet1 (0),
     lcsCapabilitySet2 (1),
     lcsCapabilitySet3 (2),
     lcsCapabilitySet4 (3) ,
     lcsCapabilitySet5 (4) } (SIZE (2..16))
-- Core network signalling capability set1 indicates LCS Release98 or Release99 version.
-- Core network signalling capability set2 indicates LCS Release4.
-- Core network signalling capability set3 indicates LCS Release5.
-- Core network signalling capability set4 indicates LCS Release6.
-- Core network signalling capability set5 indicates LCS Release7 or later version.
-- A node shall mark in the BIT STRING all LCS capability sets it supports.
-- If no bit is set then the sending node does not support LCS.
-- If the parameter is not sent by an VLR then the VLR may support at most capability set1.
-- If the parameter is not sent by an SGSN then no support for LCS is assumed.
-- An SGSN is not allowed to indicate support of capability set1.
-- Other bits than listed above shall be discarded.
```

```
UpdateLocationRes ::= SEQUENCE {
     hlr-Number                          ISDN-AddressString,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     add-Capability                      NULL     OPTIONAL,
     pagingArea-Capability               [0]NULL                         OPTIONAL }
```

```
ADD-Info ::= SEQUENCE {
     imeisv    [0] IMEI,
     skipSubscriberDataUpdate            [1] NULL  OPTIONAL,
     -- The skipSubscriberDataUpdate parameter in the UpdateLocationArg and the ADD-Info
     -- structures carry the same semantic.
     ...}
```

```
PagingArea ::= SEQUENCE SIZE (1..5) OF LocationArea
```

```
LocationArea ::= CHOICE {
     laiFixedLength                      [0] LAIFixedLength,
     lac       [1] LAC}
```

```
LAC ::= OCTET STRING (SIZE (2))
     -- Refers to Location Area Code of the Location Area Identification defined in
      -- 3GPP TS 23.003 [17].
     -- Location Area Code according to 3GPP TS 24.008 [35]
```

```
CancelLocationArg ::= [3] SEQUENCE {
     identity  Identity,
     cancellationType                    CancellationType                OPTIONAL,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     typeOfUpdate                        [0] TypeOfUpdate                OPTIONAL,
     mtrf-SupportedAndAuthorized         [1] NULL  OPTIONAL,
     mtrf-SupportedAndNotAuthorized      [2] NULL  OPTIONAL,
     newMSC-Number                       [3] ISDN-AddressString          OPTIONAL,
     newVLR-Number                       [4] ISDN-AddressString          OPTIONAL,
     new-lmsi  [5] LMSI                  OPTIONAL,
     reattach-Required                   [6] NULL  OPTIONAL
     }
     --mtrf-SupportedAndAuthorized and mtrf-SupportedAndNotAuthorized shall not
     -- both be present
```

```
TypeOfUpdate ::= ENUMERATED {
     sgsn-change (0),
     mme-change (1),
     ...}
     -- TypeOfUpdate shall be absent if CancellationType is different from updateProcedure
     -- and initialAttachProcedure
```

```
CancellationType ::= ENUMERATED {
     updateProcedure                     (0),
     subscriptionWithdraw                (1),
     ...,
     initialAttachProcedure               (2)}
     -- The HLR shall not send values other than listed above
```

```
CancelLocationRes ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
PurgeMS-Arg ::= [3] SEQUENCE {
     imsi      IMSI,
     vlr-Number                          [0] ISDN-AddressString            OPTIONAL,
     sgsn-Number                         [1]  ISDN-AddressString           OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     locationInformation                 [2] LocationInformation           OPTIONAL,
     locationInformationGPRS             [3] LocationInformationGPRS       OPTIONAL,
     locationInformationEPS              [4] LocationInformationEPS        OPTIONAL }
```

```
PurgeMS-Res ::= SEQUENCE {
     freezeTMSI                          [0]  NULL OPTIONAL,
     freezeP-TMSI                        [1]  NULL OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     freezeM-TMSI                        [2]  NULL OPTIONAL }
```

```
SendIdentificationArg ::= SEQUENCE {
     tmsi      TMSI,
     numberOfRequestedVectors            NumberOfRequestedVectors          OPTIONAL,
     -- within a dialogue numberOfRequestedVectors shall be present in
     -- the first service request and shall not be present in subsequent service requests.
     -- If received in a subsequent service request it shall be discarded.
     segmentationProhibited              NULL      OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     msc-Number                          ISDN-AddressString                OPTIONAL,
     previous-LAI                        [0] LAIFixedLength                OPTIONAL,
     hopCounter                          [1] HopCounter                    OPTIONAL,
     mtRoamingForwardingSupported        [2] NULL  OPTIONAL,
     newVLR-Number                       [3] ISDN-AddressString            OPTIONAL,
     new-lmsi  [4] LMSI                  OPTIONAL }
```

```
HopCounter ::= INTEGER (0..3)
```

```
SendIdentificationRes ::= [3] SEQUENCE {
     imsi      IMSI                              OPTIONAL,
     -- IMSI shall be present in the first (or only) service response of a dialogue.
     -- If multiple service requests are present in a dialogue then IMSI
     -- shall not be present in any service response other than the first one.
     authenticationSetList               AuthenticationSetList             OPTIONAL,
     currentSecurityContext              [2]CurrentSecurityContext         OPTIONAL,
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     ...,
     lastUsedLtePLMN-Id                  [4] PLMN-Id                       OPTIONAL,
     mtCallPendingFlag                   [5]  NULL OPTIONAL }
```

```
-- authentication management types
```

```
AuthenticationSetList ::= CHOICE {
     tripletList                         [0] TripletList,
     quintupletList                      [1] QuintupletList }
```

```
TripletList ::= SEQUENCE SIZE (1..5) OF
          AuthenticationTriplet
```

```
QuintupletList ::= SEQUENCE SIZE (1..5) OF
          AuthenticationQuintuplet
```

```
AuthenticationTriplet ::= SEQUENCE {
     rand      RAND,
     sres      SRES,
     kc        Kc,
     ...}
```

```
AuthenticationQuintuplet ::= SEQUENCE {
     rand      RAND,
     xres      XRES,
     ck        CK,
     ik        IK,
     autn      AUTN,
     ...}
```

```
CurrentSecurityContext ::= CHOICE {
     gsm-SecurityContextData              [0] GSM-SecurityContextData,
     umts-SecurityContextData             [1] UMTS-SecurityContextData }
```

```
GSM-SecurityContextData ::= SEQUENCE {
     kc        Kc,
     cksn      Cksn,
     ... }
```

```
UMTS-SecurityContextData ::= SEQUENCE {
     ck        CK,
     ik        IK,
     ksi       KSI,
     ... }
```

```
RAND ::= OCTET STRING (SIZE (16))
```

```
SRES ::= OCTET STRING (SIZE (4))
```

```
Kc ::= OCTET STRING (SIZE (8))
```

```
XRES ::= OCTET STRING (SIZE (4..16))
```

```
CK ::= OCTET STRING (SIZE (16))
```

```
IK ::= OCTET STRING (SIZE (16))
```

```
AUTN ::= OCTET STRING (SIZE (16))
```

```
AUTS ::= OCTET STRING (SIZE (14))
```

```
Cksn ::= OCTET STRING (SIZE (1))
     -- The internal structure is defined in 3GPP TS 24.008
```

```
KSI ::= OCTET STRING (SIZE (1))
     -- The internal structure is defined in 3GPP TS 24.008
```

```
AuthenticationFailureReportArg ::= SEQUENCE {
     imsi      IMSI,
     failureCause                         FailureCause,
     extensionContainer                   ExtensionContainer              OPTIONAL,
     ... ,
     re-attempt                           BOOLEAN   OPTIONAL,
     accessType                           AccessType                      OPTIONAL,
     rand      RAND                            OPTIONAL,
     vlr-Number                           [0] ISDN-AddressString          OPTIONAL,
     sgsn-Number                          [1] ISDN-AddressString          OPTIONAL }
```

```
AccessType ::= ENUMERATED {
     call (0),
     emergencyCall (1),
     locationUpdating (2),
     supplementaryService (3),
     shortMessage (4),
     gprsAttach (5),
     routingAreaUpdating (6),
     serviceRequest (7),
     pdpContextActivation (8),
     pdpContextDeactivation (9),
     ...,
     gprsDetach (10)}
     -- exception handling:
     -- received values greater than 10 shall be ignored.
```

```
AuthenticationFailureReportRes ::= SEQUENCE {
     extensionContainer                   ExtensionContainer              OPTIONAL,
     ...}
```

```
FailureCause ::= ENUMERATED {
     wrongUserResponse  (0),
     wrongNetworkSignature  (1)}
```

```
-- gprs location registration types
```

```
UpdateGprsLocationArg ::= SEQUENCE {
    imsi     IMSI,
    sgsn-Number                         ISDN-AddressString,
    sgsn-Address                        GSN-Address,
    extensionContainer                  ExtensionContainer              OPTIONAL,
    ... ,
    sgsn-Capability                     [0] SGSN-Capability             OPTIONAL,
    informPreviousNetworkEntity         [1]  NULL OPTIONAL,
    ps-LCS-NotSupportedByUE             [2]  NULL OPTIONAL,
    v-gmlc-Address                      [3]  GSN-Address                OPTIONAL,
    add-info [4]  ADD-Info              OPTIONAL,
    eps-info [5]                        EPS-Info OPTIONAL,
    servingNodeTypeIndicator            [6]  NULL OPTIONAL,
    skipSubscriberDataUpdate            [7] NULL  OPTIONAL,
    usedRAT-Type                        [8] Used-RAT-Type               OPTIONAL,
    gprsSubscriptionDataNotNeeded       [9] NULL  OPTIONAL,
    nodeTypeIndicator                   [10] NULL OPTIONAL,
    areaRestricted                      [11] NULL OPTIONAL,
    ue-reachableIndicator               [12] NULL OPTIONAL,
    epsSubscriptionDataNotNeeded        [13] NULL OPTIONAL,
    ue-srvcc-Capability                 [14] UE-SRVCC-Capability        OPTIONAL,
    eplmn-List                          [15] EPLMN-List                 OPTIONAL,
    mmeNumberforMTSMS                   [16] ISDN-AddressString         OPTIONAL,
    smsRegisterRequest                  [17] SMSRegisterRequest         OPTIONAL,
    sms-Only [18] NULL                  OPTIONAL,
    removalofMMERegistrationforSMS      [22] NULL OPTIONAL,
    sgsn-Name [19] DiameterIdentity     OPTIONAL,
    sgsn-Realm                          [20] DiameterIdentity           OPTIONAL,
    lgd-supportIndicator                [21] NULL OPTIONAL,
    adjacentPLMN-List                   [23] AdjacentPLMN-List          OPTIONAL }
```

```
SMSRegisterRequest::= ENUMERATED {
    sms-registration-required  (0),
    sms-registration-not-preferred  (1),
    no-preference  (2),
    ...}
```

```
Used-RAT-Type::= ENUMERATED {
    utran  (0),
    geran  (1),
    gan    (2),
    i-hspa-evolution (3),
    e-utran  (4),
    ...,
    nb-iot   (5)}
    -- The value e-utran indicates wide-band e-utran
```

```
EPS-Info ::= CHOICE{
    pdn-gw-update                       [0] PDN-GW-Update,
    isr-Information                     [1] ISR-Information }
```

```
PDN-GW-Update ::= SEQUENCE{
    apn      [0] APN                    OPTIONAL,
    pdn-gw-Identity                     [1] PDN-GW-Identity             OPTIONAL,
    contextId [2] ContextId                   OPTIONAL,
    extensionContainer                  [3] ExtensionContainer          OPTIONAL,
    ... }
--  The pdn-gw-update IE shall include the pdn-gw-Identity, and the apn or/and the contextID.
--  The HSS shall ignore the eps-info IE if it includes a pdn-gw-update IE which does not
--  include pdn-gw-Identity.
--  The pdn-gw-Identity is defined as OPTIONAL for backward compatility reason with
--  outdated earlier versions of this specification.
```

```
ISR-Information::= BIT STRING {
    updateLocation  (0),
    cancelSGSN  (1),
    initialAttachIndicator  (2)} (SIZE (3..8))
    -- exception handling: reception of unknown bit assignments in the
    -- ISR-Information data type shall be discarded by the receiver
```

```
SGSN-Capability ::= SEQUENCE{
     solsaSupportIndicator                NULL     OPTIONAL,
     extensionContainer                   [1] ExtensionContainer            OPTIONAL,
     ... ,
     superChargerSupportedInServingNetworkEntity   [2] SuperChargerInfo     OPTIONAL ,
     gprsEnhancementsSupportIndicator     [3] NULL  OPTIONAL,
     supportedCamelPhases                 [4] SupportedCamelPhases          OPTIONAL,
     supportedLCS-CapabilitySets          [5]  SupportedLCS-CapabilitySets  OPTIONAL,
     offeredCamel4CSIs                    [6] OfferedCamel4CSIs             OPTIONAL,
     smsCallBarringSupportIndicator       [7]  NULL OPTIONAL,               supportedRAT-
TypesIndicator[8]                         SupportedRAT-Types                OPTIONAL,
     supportedFeatures                    [9] SupportedFeatures             OPTIONAL,
     t-adsDataRetrieval                   [10] NULL OPTIONAL,
     homogeneousSupportOfIMSVoiceOverPSSessions [11] BOOLEAN                OPTIONAL,
     --   "true" indicates homogeneous support, "false" indicates homogeneous non-support
     --   in the complete SGSN or MME area
     cancellationTypeInitialAttach        [12] NULL OPTIONAL,
     msisdn-lessOperation-Supported       [14] NULL OPTIONAL,
     updateofHomogeneousSupportOfIMSVoiceOverPSSessions [15] NULL           OPTIONAL,
     reset-ids-Supported                  [16] NULL OPTIONAL,
     ext-SupportedFeatures                [17] Ext-SupportedFeatures        OPTIONAL
     }
   --     the supportedFeatures , t-adsDataRetrieval,
     --   homogeneousSupportOfIMSVoiceOverPSSessions
     --   /updateofHomogeneousSupportOfIMSVoiceOverPSSessions and
     --   ext-SupportedFeatures are also applied to the MME/IWF
```

```
SupportedFeatures::= BIT STRING {
     odb-all-apn (0),
     odb-HPLMN-APN (1),
     odb-VPLMN-APN (2),
     odb-all-og (3),
     odb-all-international-og (4),
     odb-all-int-og-not-to-HPLMN-country (5),
     odb-all-interzonal-og (6),
     odb-all-interzonal-og-not-to-HPLMN-country (7),
     odb-all-interzonal-og-and-internat-og-not-to-HPLMN-country (8),
     regSub (9),
     trace (10),
     lcs-all-PrivExcep (11),
     lcs-universal (12),
     lcs-CallSessionRelated (13),
     lcs-CallSessionUnrelated (14),
     lcs-PLMN-operator (15),
     lcs-ServiceType (16),
     lcs-all-MOLR-SS (17),
     lcs-basicSelfLocation (18),
     lcs-autonomousSelfLocation (19),
     lcs-transferToThirdParty (20),
     sm-mo-pp (21),
     barring-OutgoingCalls (22),
     baoc (23),
     boic (24),
     boicExHC (25),
     localTimeZoneRetrieval (26),
     additionalMsisdn (27),
     smsInMME (28),
     smsInSGSN (29),
     ue-Reachability-Notification (30),
     state-Location-Information-Retrieval (31),
     partialPurge (32),
     gddInSGSN (33),
     sgsnCAMELCapability (34),
     pcscf-Restoration (35),
     dedicatedCoreNetworks (36),
     non-IP-PDN-Type-APNs (37),
     non-IP-PDP-Type-APNs (38),
     nrAsSecondaryRAT (39) } (SIZE (26..40))
     --   for the definition and usage of the above features see the 3GPP TS 29.272 [144].
     --   Additional supported features are encoded in Ext-SupportedFeatures bit string.
```

```
Ext-SupportedFeatures ::= BIT STRING {
     unlicensedSpectrumAsSecondaryRAT (0) } (SIZE (1..40))
```

```
UE-SRVCC-Capability::= ENUMERATED {
     ue-srvcc-not-supported  (0),
     ue-srvcc-supported  (1),
     ...}
```

```
UpdateGprsLocationRes ::= SEQUENCE {
     hlr-Number                          ISDN-AddressString,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     add-Capability                      NULL     OPTIONAL,
     sgsn-mmeSeparationSupported         [0] NULL  OPTIONAL,
     mmeRegisteredforSMS                 [1] NULL  OPTIONAL }
```

```
EPLMN-List ::= SEQUENCE SIZE (1..50) OF
          PLMN-Id
```

```
AdjacentPLMN-List ::= SEQUENCE SIZE (1..50) OF
          PLMN-Id
```

-- handover types

```
ForwardAccessSignalling-Arg ::= [3] SEQUENCE {
     an-APDU  AccessNetworkSignalInfo,
     integrityProtectionInfo             [0] IntegrityProtectionInformation OPTIONAL,
     encryptionInfo                      [1] EncryptionInformation          OPTIONAL,
     keyStatus [2]                       KeyStatus OPTIONAL,
     allowedGSM-Algorithms               [4]  AllowedGSM-Algorithms         OPTIONAL,
     allowedUMTS-Algorithms              [5]  AllowedUMTS-Algorithms        OPTIONAL,
     radioResourceInformation            [6] RadioResourceInformation       OPTIONAL,
     extensionContainer                  [3]  ExtensionContainer            OPTIONAL,
     ...,
     radioResourceList                   [7]  RadioResourceList             OPTIONAL,
     bssmap-ServiceHandover              [9]  BSSMAP-ServiceHandover        OPTIONAL,
     ranap-ServiceHandover               [8]  RANAP-ServiceHandover         OPTIONAL,
     bssmap-ServiceHandoverList          [10] BSSMAP-ServiceHandoverList    OPTIONAL,
     currentlyUsedCodec                  [11] Codec                         OPTIONAL,
     iuSupportedCodecsList               [12] SupportedCodecsList           OPTIONAL,
     rab-ConfigurationIndicator          [13] NULL OPTIONAL,
     iuSelectedCodec                     [14] Codec                         OPTIONAL,
     alternativeChannelType              [15] RadioResourceInformation      OPTIONAL,
     tracePropagationList                [17] TracePropagationList          OPTIONAL,
     aoipSupportedCodecsListAnchor       [18] AoIPCodecsList                OPTIONAL,
     aoipSelectedCodecTarget             [19] AoIPCodec                     OPTIONAL,
     uesbi-Iu  [20]                      UESBI-Iu  OPTIONAL,
     imeisv    [21]                      IMEI OPTIONAL }
```

```
AllowedGSM-Algorithms ::= OCTET STRING (SIZE (1))
     -- internal structure is coded as Algorithm identifier octet from
     -- Permitted Algorithms defined in 3GPP TS 48.008
     -- A node shall mark all GSM algorithms that are allowed in MSC-B
```

```
AllowedUMTS-Algorithms ::= SEQUENCE {
     integrityProtectionAlgorithms       [0]  PermittedIntegrityProtectionAlgorithms
     OPTIONAL,
     encryptionAlgorithms                [1]  PermittedEncryptionAlgorithms OPTIONAL,
     extensionContainer                  [2]  ExtensionContainer            OPTIONAL,
     ...}
```

```
PermittedIntegrityProtectionAlgorithms ::=
     OCTET STRING (SIZE (1..maxPermittedIntegrityProtectionAlgorithmsLength))
     -- Octets contain a complete PermittedIntegrityProtectionAlgorithms data type
     -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
     -- mandated by 3GPP TS 25.413.
     -- Padding bits are included, if needed, in the least significant bits of the
     -- last octet of the octet string.
```

```
maxPermittedIntegrityProtectionAlgorithmsLength INTEGER ::= 9
```

```
PermittedEncryptionAlgorithms ::=
     OCTET STRING (SIZE (1..maxPermittedEncryptionAlgorithmsLength))
     -- Octets contain a complete PermittedEncryptionAlgorithms data type
     -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
     -- mandated by 3GPP TS 25.413
     -- Padding bits are included, if needed, in the least significant bits of the
     -- last octet of the octet string.
```

```
maxPermittedEncryptionAlgorithmsLength INTEGER ::= 9
```

```
KeyStatus ::= ENUMERATED {
     old  (0),
     new  (1),
     ...}
     -- exception handling:
     -- received values in range 2-31 shall be treated as "old"
     -- received values greater than 31 shall be treated as "new"
```

```
PrepareHO-Arg ::= [3] SEQUENCE {
     targetCellId                        [0] GlobalCellId                  OPTIONAL,
     ho-NumberNotRequired                NULL      OPTIONAL,
     targetRNCId                         [1] RNCId OPTIONAL,
     an-APDU   [2] AccessNetworkSignalInfo OPTIONAL,
     multipleBearerRequested             [3] NULL  OPTIONAL,
     imsi      [4] IMSI                  OPTIONAL,
     integrityProtectionInfo             [5] IntegrityProtectionInformation OPTIONAL,
     encryptionInfo                      [6] EncryptionInformation         OPTIONAL,
     radioResourceInformation            [7] RadioResourceInformation      OPTIONAL,
     allowedGSM-Algorithms               [9]  AllowedGSM-Algorithms        OPTIONAL,
     allowedUMTS-Algorithms              [10] AllowedUMTS-Algorithms       OPTIONAL,
     radioResourceList                   [11] RadioResourceList            OPTIONAL,
     extensionContainer                  [8] ExtensionContainer            OPTIONAL,
     ... ,
     rab-Id    [12] RAB-Id               OPTIONAL,
     bssmap-ServiceHandover              [13] BSSMAP-ServiceHandover       OPTIONAL,
     ranap-ServiceHandover               [14] RANAP-ServiceHandover        OPTIONAL,
     bssmap-ServiceHandoverList          [15] BSSMAP-ServiceHandoverList   OPTIONAL,
     asciCallReference                   [20] ASCI-CallReference           OPTIONAL,
     geran-classmark                     [16] GERAN-Classmark              OPTIONAL,
     iuCurrentlyUsedCodec                [17] Codec                        OPTIONAL,
     iuSupportedCodecsList               [18] SupportedCodecsList          OPTIONAL,
     rab-ConfigurationIndicator          [19] NULL OPTIONAL,
     uesbi-Iu  [21]                      UESBI-Iu  OPTIONAL,
     imeisv    [22]                      IMEI OPTIONAL,
     alternativeChannelType              [23] RadioResourceInformation     OPTIONAL,
     tracePropagationList                [25] TracePropagationList         OPTIONAL,
     aoipSupportedCodecsListAnchor       [26] AoIPCodecsList               OPTIONAL,
     regionalSubscriptionData            [27] ZoneCodeList                 OPTIONAL,
     lclsGlobalCallReference             [28] LCLS-GlobalCallReference     OPTIONAL,
     lcls-Negotiation                    [29] LCLS-Negotiation             OPTIONAL,
     lcls-Configuration-Preference       [30] LCLS-ConfigurationPreference OPTIONAL,
     csg-SubscriptionDataList            [31] CSG-SubscriptionDataList     OPTIONAL
     }
```

```
LCLS-GlobalCallReference ::= OCTET STRING (SIZE (13..15))
     -- Octets are coded as specified in 3GPP TS 29.205 [146]
```

```
LCLS-Negotiation::= BIT STRING {
     permission-indicator-not-allowed-bit (0),
     permission-indicator-spare-bit       (1)} (SIZE (2..8))
     --for definition and allowed combination of bits 0 and 1 see 3GPP TS 29.205
     -- exception handling: bits 2 to 7 shall be ignored if received and not understood
```

```
LCLS-ConfigurationPreference::= BIT STRING {
     forward-data-sending-indicator       (0),
     backward-data-sending-indicator      (1),
     forward-data-reception-indicator     (2),
     backward-data-reception-indicator    (3)} (SIZE (4..8))
     -- exception handling: bits 4 to 7 shall be ignored if received and not understood
```

```
BSSMAP-ServiceHandoverList ::= SEQUENCE SIZE (1.. maxNumOfServiceHandovers) OF
          BSSMAP-ServiceHandoverInfo
```

```
BSSMAP-ServiceHandoverInfo ::= SEQUENCE {
    bssmap-ServiceHandover              BSSMAP-ServiceHandover,
    rab-Id    RAB-Id,
    -- RAB Identity is needed to relate the service handovers with the radio access bearers.
    ...}
```

```
maxNumOfServiceHandovers  INTEGER ::= 7
```

```
BSSMAP-ServiceHandover ::= OCTET STRING (SIZE (1))
    -- Octets are coded according the Service Handover information element in
    -- 3GPP TS 48.008.
```

```
RANAP-ServiceHandover ::= OCTET STRING (SIZE (1))
    -- Octet contains a complete Service-Handover data type
    -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
    -- mandated by 3GPP TS 25.413
    -- Padding bits are included in the least significant bits.
```

```
RadioResourceList ::= SEQUENCE SIZE (1.. maxNumOfRadioResources) OF
         RadioResource
```

```
RadioResource ::= SEQUENCE {
    radioResourceInformation            RadioResourceInformation,
    rab-Id    RAB-Id,
    -- RAB Identity is needed to relate the radio resources with the radio access bearers.
    ...}
```

```
maxNumOfRadioResources  INTEGER ::= 7
```

```
PrepareHO-Res ::= [3] SEQUENCE {
    handoverNumber                      [0] ISDN-AddressString              OPTIONAL,
    relocationNumberList                [1]  RelocationNumberList           OPTIONAL,
    an-APDU  [2]                        AccessNetworkSignalInfo             OPTIONAL,
    multicallBearerInfo                 [3]  MulticallBearerInfo            OPTIONAL,
    multipleBearerNotSupported          NULL      OPTIONAL,
    selectedUMTS-Algorithms             [5]  SelectedUMTS-Algorithms        OPTIONAL,
    chosenRadioResourceInformation      [6] ChosenRadioResourceInformation OPTIONAL,
    extensionContainer                  [4]  ExtensionContainer             OPTIONAL,
    ...,
    iuSelectedCodec                     [7] Codec OPTIONAL,
    iuAvailableCodecsList               [8] CodecList                       OPTIONAL,
    aoipSelectedCodecTarget             [9] AoIPCodec                       OPTIONAL,
    aoipAvailableCodecsListMap          [10] AoIPCodecsList                 OPTIONAL }
```

```
SelectedUMTS-Algorithms ::= SEQUENCE {
    integrityProtectionAlgorithm        [0]  ChosenIntegrityProtectionAlgorithm   OPTIONAL,
    encryptionAlgorithm                 [1]  ChosenEncryptionAlgorithm     OPTIONAL,
    extensionContainer                  [2]  ExtensionContainer            OPTIONAL,
    ...}
```

```
ChosenIntegrityProtectionAlgorithm ::= OCTET STRING (SIZE (1))
    -- Octet contains a complete IntegrityProtectionAlgorithm data type
    -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
    -- mandated by 3GPP TS 25.413
    -- Padding bits are included in the least significant bits.
```

```
ChosenEncryptionAlgorithm ::= OCTET STRING (SIZE (1))
    -- Octet contains a complete EncryptionAlgorithm data type
    -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
    -- mandated by 3GPP TS 25.413
    -- Padding bits are included in the least significant bits.
```

```
ChosenRadioResourceInformation ::= SEQUENCE {
    chosenChannelInfo                   [0] ChosenChannelInfo               OPTIONAL,
    chosenSpeechVersion                 [1] ChosenSpeechVersion             OPTIONAL,
    ...}
```

```
ChosenChannelInfo ::= OCTET STRING (SIZE (1))
    -- Octets are coded according the Chosen Channel information element in 3GPP TS 48.008
```

```
ChosenSpeechVersion ::= OCTET STRING (SIZE (1))
    -- Octets are coded according the Speech Version (chosen) information element in 3GPP TS
    -- 48.008
```

```
PrepareSubsequentHO-Arg ::= [3] SEQUENCE {
    targetCellId                    [0] GlobalCellId                OPTIONAL,
    targetMSC-Number                [1] ISDN-AddressString,
    targetRNCId                     [2] RNCId OPTIONAL,
    an-APDU  [3]                    AccessNetworkSignalInfo         OPTIONAL,
    selectedRab-Id                  [4]  RAB-Id                     OPTIONAL,
    extensionContainer              [5]  ExtensionContainer         OPTIONAL,
    ...,
    geran-classmark                 [6] GERAN-Classmark             OPTIONAL,
    rab-ConfigurationIndicator      [7] NULL  OPTIONAL }
```

```
PrepareSubsequentHO-Res ::= [3] SEQUENCE {
    an-APDU  AccessNetworkSignalInfo,
    extensionContainer              [0]  ExtensionContainer         OPTIONAL,
    ...}
```

```
ProcessAccessSignalling-Arg ::= [3] SEQUENCE {
    an-APDU  AccessNetworkSignalInfo,
    selectedUMTS-Algorithms         [1]  SelectedUMTS-Algorithms      OPTIONAL,
    selectedGSM-Algorithm           [2]  SelectedGSM-Algorithm        OPTIONAL,
    chosenRadioResourceInformation  [3] ChosenRadioResourceInformation OPTIONAL,
    selectedRab-Id                  [4] RAB-Id                        OPTIONAL,
    extensionContainer              [0]  ExtensionContainer           OPTIONAL,
    ...,
    iUSelectedCodec                 [5] Codec OPTIONAL,
    iuAvailableCodecsList           [6] CodecList                     OPTIONAL,
    aoipSelectedCodecTarget         [7] AoIPCodec                     OPTIONAL,
    aoipAvailableCodecsListMap      [8] AoIPCodecsList                OPTIONAL }
```

```
AoIPCodecsList ::= SEQUENCE {
    codec1   [1] AoIPCodec,
    codec2   [2] AoIPCodec              OPTIONAL,
    codec3   [3] AoIPCodec              OPTIONAL,
    codec4   [4] AoIPCodec              OPTIONAL,
    codec5   [5] AoIPCodec              OPTIONAL,
    codec6   [6] AoIPCodec              OPTIONAL,
    codec7   [7] AoIPCodec              OPTIONAL,
    codec8   [8] AoIPCodec              OPTIONAL,
    extensionContainer              [9] ExtensionContainer          OPTIONAL,
    ...}
    -- Codecs are sent in priority order where codec1 has highest priority
```

```
AoIPCodec ::= OCTET STRING (SIZE (1..3))

    -- The internal structure is defined as follows:
    -- octet 1                      Coded as Speech Codec Elements in 3GPP TS 48.008
    --       with the exception that FI, PI, PT and TF bits shall
    --       be set to 0
    -- octets 2,3                   Optional; in case of AMR codec types it defines
    --       the supported codec configurations as defined in
    --       3GPP TS 48.008
```

```
SupportedCodecsList ::= SEQUENCE {
    utranCodecList                  [0] CodecList                   OPTIONAL,
    geranCodecList                  [1] CodecList                   OPTIONAL,
    extensionContainer              [2] ExtensionContainer          OPTIONAL,
    ...}
```

```
CodecList ::= SEQUENCE {
    codec1   [1] Codec,
    codec2   [2] Codec                  OPTIONAL,
    codec3   [3] Codec                  OPTIONAL,
    codec4   [4] Codec                  OPTIONAL,
    codec5   [5] Codec                  OPTIONAL,
    codec6   [6] Codec                  OPTIONAL,
    codec7   [7] Codec                  OPTIONAL,
    codec8   [8] Codec                  OPTIONAL,
    extensionContainer              [9] ExtensionContainer          OPTIONAL,
    ...}
    -- Codecs are sent in priority order where codec1 has highest priority
```

```
Codec ::= OCTET STRING (SIZE (1..4))

    -- The internal structure is defined as follows:
    -- octet 1                          Coded as Codec Identification code in 3GPP TS 26.103
    -- octets 2,3,4                     Parameters for the Codec as defined in 3GPP TS
    --       26.103, if available, length depending on the codec
```

```
GERAN-Classmark ::= OCTET STRING (SIZE (2..87))
    -- Octets are coded according the GERAN Classmark information element in 3GPP TS 48.008
```

```
SelectedGSM-Algorithm ::= OCTET STRING (SIZE (1))
    -- internal structure is coded as Algorithm identifier octet from Chosen Encryption
    -- Algorithm defined in 3GPP TS 48.008
    -- A node shall mark only the selected GSM algorithm
```

```
SendEndSignal-Arg ::= [3] SEQUENCE {
    an-APDU   AccessNetworkSignalInfo,
    extensionContainer                  [0]  ExtensionContainer           OPTIONAL,
    ...}
```

```
SendEndSignal-Res ::= SEQUENCE {
    extensionContainer                  [0]  ExtensionContainer           OPTIONAL,
    ...}
```

```
RNCId ::= OCTET STRING (SIZE (7))
    -- The internal structure is defined as follows:
    -- octet 1 bits 4321                Mobile Country Code 1st digit
    --         bits 8765                Mobile Country Code 2nd digit
    -- octet 2 bits 4321                Mobile Country Code 3rd digit
    --         bits 8765                Mobile Network Code 3rd digit
    --       or filler (1111) for 2 digit MNCs
    -- octet 3 bits 4321                Mobile Network Code 1st digit
    --         bits 8765                Mobile Network Code 2nd digit
    -- octets 4 and 5                   Location Area Code according to 3GPP TS 24.008
    -- octets 6 and 7                   RNC Id or Extended RNC Id value according to
    --       3GPP TS 25.413
```

```
RelocationNumberList ::= SEQUENCE SIZE (1..maxNumOfRelocationNumber) OF
         RelocationNumber
```

```
MulticallBearerInfo ::= INTEGER (1..maxNumOfRelocationNumber)
```

```
RelocationNumber ::= SEQUENCE {
    handoverNumber                      ISDN-AddressString,
    rab-Id    RAB-Id,
    -- RAB Identity is needed to relate the calls with the radio access bearers.
    ...}
```

```
RAB-Id ::= INTEGER (1..maxNrOfRABs)
```

```
maxNrOfRABs INTEGER ::= 255
```

```
maxNumOfRelocationNumber  INTEGER ::= 7
```

```
RadioResourceInformation ::= OCTET STRING (SIZE (3..13))
    -- Octets are coded according the Channel Type information element in 3GPP TS 48.008
```

```
IntegrityProtectionInformation ::= OCTET STRING (SIZE (18..maxNumOfIntegrityInfo))
    -- Octets contain a complete IntegrityProtectionInformation data type
    -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
    -- mandated by 3GPP TS 25.413
    -- Padding bits are included, if needed, in the least significant bits of the
    -- last octet of the octet string.
```

```
maxNumOfIntegrityInfo INTEGER ::= 100
```

```
EncryptionInformation ::= OCTET STRING (SIZE (18..maxNumOfEncryptionInfo))
    -- Octets contain a complete EncryptionInformation data type
    -- as defined in 3GPP TS 25.413, encoded according to the encoding scheme
    -- mandated by 3GPP TS 25.413
    -- Padding bits are included, if needed, in the least significant bits of the
    -- last octet of the octet string.
```

```
maxNumOfEncryptionInfo INTEGER ::= 100
```

```
-- authentication management types

SendAuthenticationInfoArg ::= SEQUENCE {
     imsi     [0] IMSI,
     numberOfRequestedVectors             NumberOfRequestedVectors,
     segmentationProhibited               NULL     OPTIONAL,
     immediateResponsePreferred           [1] NULL                          OPTIONAL,
     re-synchronisationInfo               Re-synchronisationInfo            OPTIONAL,
     extensionContainer                   [2] ExtensionContainer            OPTIONAL,
     ...,
     requestingNodeType                   [3] RequestingNodeType            OPTIONAL,
     requestingPLMN-Id                    [4] PLMN-Id                       OPTIONAL,
     numberOfRequestedAdditional-Vectors  [5] NumberOfRequestedVectors      OPTIONAL,
     additionalVectorsAreForEPS           [6] NULL  OPTIONAL,
     ueUsageTypeRequestIndication         [7] NULL  OPTIONAL }

NumberOfRequestedVectors ::= INTEGER (1..5)

Re-synchronisationInfo ::= SEQUENCE {
     rand      RAND,
     auts      AUTS,
     ...}

SendAuthenticationInfoRes ::= [3] SEQUENCE {
     authenticationSetList                AuthenticationSetList             OPTIONAL,
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ...,
     eps-AuthenticationSetList            [2] EPS-AuthenticationSetList     OPTIONAL,
     ueUsageType                          [3] UE-UsageType                  OPTIONAL }

EPS-AuthenticationSetList ::= SEQUENCE SIZE (1..5) OF
          EPC-AV

UE-UsageType::= OCTET STRING (SIZE (4))
     -- octets are coded as defined in 3GPP TS 29.272 [144]

EPC-AV ::= SEQUENCE {
     rand      RAND,
     xres      XRES,
     autn      AUTN,
     kasme     KASME,
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ...}

KASME ::= OCTET STRING (SIZE (32))

RequestingNodeType ::= ENUMERATED {
     vlr  (0),
     sgsn  (1),
     ...,
     s-cscf  (2),
     bsf  (3),
     gan-aaa-server  (4),
     wlan-aaa-server  (5),
     mme  (16),
     mme-sgsn (17)
     }
     -- the values 2, 3, 4 and 5 shall not be used on the MAP-D or Gr interfaces
     -- exception handling:
     -- received values in the range (6-15) shall be treated as "vlr"
     -- received values greater than 17 shall be treated as "sgsn"

-- equipment management types

CheckIMEI-Arg ::= SEQUENCE {
     imei      IMEI,
     requestedEquipmentInfo               RequestedEquipmentInfo,
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ...}

CheckIMEI-Res ::= SEQUENCE {
     equipmentStatus                      EquipmentStatus                   OPTIONAL,
     bmuef     UESBI-Iu                   OPTIONAL,
     extensionContainer                   [0] ExtensionContainer            OPTIONAL,
     ...}
```

```
RequestedEquipmentInfo::= BIT STRING {
    equipmentStatus  (0),
    bmuef  (1)} (SIZE (2..8))
    -- exception handling: reception of unknown bit assignments in the
    -- RequestedEquipmentInfo data type shall be discarded by the receiver
```

```
UESBI-Iu ::= SEQUENCE {
    uesbi-IuA [0] UESBI-IuA                 OPTIONAL,
    uesbi-IuB [1] UESBI-IuB                 OPTIONAL,
    ...}
```

```
UESBI-IuA     ::= BIT STRING (SIZE(1..128))
-- See 3GPP TS 25.413
```

```
UESBI-IuB     ::= BIT STRING (SIZE(1..128))
-- See 3GPP TS 25.413
```

```
EquipmentStatus ::= ENUMERATED {
    whiteListed  (0),
    blackListed  (1),
    greyListed  (2)}
```

-- subscriber management types

```
InsertSubscriberDataArg ::= SEQUENCE {
    imsi     [0] IMSI                   OPTIONAL,
    COMPONENTS OF                       SubscriberData,
    extensionContainer                  [14] ExtensionContainer           OPTIONAL,
    ... ,
    naea-PreferredCI                    [15] NAEA-PreferredCI             OPTIONAL,
    -- naea-PreferredCI is included at the discretion of the HLR operator.
    gprsSubscriptionData                [16] GPRSSubscriptionData         OPTIONAL,
    roamingRestrictedInSgsnDueToUnsupportedFeature [23]                   NULL
                                        OPTIONAL,
    networkAccessMode                   [24] NetworkAccessMode            OPTIONAL,
    lsaInformation                      [25] LSAInformation               OPTIONAL,
    lmu-Indicator                       [21] NULL OPTIONAL,
    lcsInformation                      [22] LCSInformation               OPTIONAL,
    istAlertTimer                       [26] IST-AlertTimerValue          OPTIONAL,
    superChargerSupportedInHLR          [27] AgeIndicator                 OPTIONAL,
    mc-SS-Info                          [28] MC-SS-Info                   OPTIONAL,
    cs-AllocationRetentionPriority      [29] CS-AllocationRetentionPriority OPTIONAL,
    sgsn-CAMEL-SubscriptionInfo         [17] SGSN-CAMEL-SubscriptionInfo  OPTIONAL,
    chargingCharacteristics             [18] ChargingCharacteristics      OPTIONAL,
    accessRestrictionData               [19] AccessRestrictionData        OPTIONAL,
    ics-Indicator                       [20] BOOLEAN                      OPTIONAL,
    eps-SubscriptionData                [31] EPS-SubscriptionData         OPTIONAL,
    csg-SubscriptionDataList            [32] CSG-SubscriptionDataList     OPTIONAL,
    ue-ReachabilityRequestIndicator     [33] NULL OPTIONAL,
    sgsn-Number                         [34] ISDN-AddressString           OPTIONAL,
    mme-Name [35]                       DiameterIdentity                  OPTIONAL,
    subscribedPeriodicRAUTAUtimer       [36] SubscribedPeriodicRAUTAUtimer OPTIONAL,
    vplmnLIPAAllowed                    [37] NULL OPTIONAL,
    mdtUserConsent                      [38] BOOLEAN                      OPTIONAL,
    subscribedPeriodicLAUtimer          [39] SubscribedPeriodicLAUtimer   OPTIONAL,
    vplmn-Csg-SubscriptionDataList      [40] VPLMN-CSG-SubscriptionDataList OPTIONAL,
    additionalMSISDN                    [41] ISDN-AddressString           OPTIONAL,
    psAndSMS-OnlyServiceProvision       [42] NULL OPTIONAL,
    smsInSGSNAllowed                    [43] NULL OPTIONAL,
    cs-to-ps-SRVCC-Allowed-Indicator    [44] NULL OPTIONAL,
    pcscf-Restoration-Request           [45] NULL OPTIONAL,
    adjacentAccessRestrictionDataList   [46] AdjacentAccessRestrictionDataList OPTIONAL,
    imsi-Group-Id-List                  [47] IMSI-GroupIdList             OPTIONAL,
    ueUsageType                         [48] UE-UsageType                 OPTIONAL,
    userPlaneIntegrityProtectionIndicator    [49] NULL                    OPTIONAL,
    dl-Buffering-Suggested-Packet-Count [50] DL-Buffering-Suggested-Packet-Count  OPTIONAL,
    reset-Id-List                       [51] Reset-Id-List                OPTIONAL,
    eDRX-Cycle-Length-List              [52] EDRX-Cycle-Length-List       OPTIONAL,
    ext-AccessRestrictionData           [53] Ext-AccessRestrictionData    OPTIONAL,
    iab-Operation-Allowed-Indicator     [54] NULL OPTIONAL }
    -- If the Network Access Mode parameter is sent, it shall be present only in
    -- the first sequence if seqmentation is used
```

```
EDRX-Cycle-Length-List ::= SEQUENCE SIZE (1..8) OF
        EDRX-Cycle-Length
```

```
EDRX-Cycle-Length ::= SEQUENCE {
        rat-Type                        [0]  Used-RAT-Type,
```

```
        eDRX-Cycle-Length-Value         [1]  EDRX-Cycle-Length-Value,
        ...}
    -- The eDRX-Cycle-Length contains the subscribed eDRX-Cycle-Length applicable to a
    -- a specific RAT Type.
```

```
EDRX-Cycle-Length-Value ::= OCTET STRING (SIZE (1))
    -- The EDRX-Cycle-Length-Value shall be encoded as specified in subclause 7.3.216 of
    -- 3GPP TS 29.272 [144].
```

```
Reset-Id-List ::= SEQUENCE SIZE (1..50) OF
        Reset-Id
```

```
Reset-Id ::= OCTET STRING (SIZE (1..4))
    -- Reset-Ids shall be unique within the HPLMN.
```

```
DL-Buffering-Suggested-Packet-Count ::= INTEGER (-1..2147483647)
    -- values are defined in 3GPP TS 29.272 [144]
```

```
Group-Service-ID ::= INTEGER (0..4294967295)
    -- values are defined in 3GPP TS 29.272 [144]
```

```
Local-GroupID ::= OCTET STRING (SIZE (1..10))
    -- Refers to Local group ID defined by an operator identified by the PLMN-ID.
    --   for details see 3GPP TS 29.272 [144]
```

```
IMSI-GroupIdList ::= SEQUENCE SIZE (1..50) OF
        IMSI-GroupId
```

```
IMSI-GroupId ::= SEQUENCE {
        group-Service-Id                [0]  Group-Service-ID,
        plmnId                          [1]  PLMN-Id,
        local-Group-ID                  [2]  Local-GroupID,
        ...}
```

```
SubscribedPeriodicRAUTAUtimer ::= INTEGER (0..4294967295)
    -- This parameter carries the subscribed periodic TAU/RAU timer value in seconds as
    -- specified in 3GPP TS 24.008 [35].
```

```
SubscribedPeriodicLAUtimer ::= INTEGER (0..4294967295)
    -- This parameter carries the subscribed periodic LAU timer value in seconds as
    -- specified in 3GPP TS 24.008 [35].
```

```
CSG-SubscriptionDataList ::= SEQUENCE SIZE (1..50) OF
        CSG-SubscriptionData
```

```
CSG-SubscriptionData ::= SEQUENCE {
    csg-Id                              CSG-Id,
    expirationDate                      Time OPTIONAL,
    extensionContainer                  ExtensionContainer              OPTIONAL,
    ...,
    lipa-AllowedAPNList                 [0] LIPA-AllowedAPNList         OPTIONAL,
    plmn-Id                             [1] PLMN-Id                     OPTIONAL
}
```

```
VPLMN-CSG-SubscriptionDataList ::= SEQUENCE SIZE (1..50) OF
        CSG-SubscriptionData
```

```
CSG-Id ::= BIT STRING (SIZE (27))
    -- coded according to 3GPP TS 23.003 [17].
```

```
LIPA-AllowedAPNList ::= SEQUENCE SIZE (1..maxNumOfLIPAAllowedAPN) OF
        APN
```

```
maxNumOfLIPAAllowedAPN  INTEGER ::= 50
```

```
EPS-SubscriptionData ::= SEQUENCE {
    apn-oi-Replacement                  [0]  APN-OI-Replacement           OPTIONAL,
    -- this apn-oi-Replacement refers to the UE level apn-oi-Replacement.
    rfsp-id   [2]                       RFSP-ID   OPTIONAL,
    ambr      [3]                       AMBR OPTIONAL,
    apn-ConfigurationProfile            [4]  APN-ConfigurationProfile     OPTIONAL,
    stn-sr    [6]                       ISDN-AddressString                OPTIONAL,
    extensionContainer                  [5]  ExtensionContainer           OPTIONAL,
    ...,
    mps-CSPriority                      [7]  NULL OPTIONAL,
    mps-EPSPriority                     [8]  NULL OPTIONAL,
    subscribed-vsrvcc                   [9]  NULL OPTIONAL }
    -- mps-CSPriority by its presence indicates that the UE is subscribed to the eMLPP in
    -- the CS domain, referring to the 3GPP TS 29.272 [144] for details.
    -- mps-EPSPriority by its presence indicates that the UE is subscribed to the MPS in
    -- the EPS domain, referring to the 3GPP TS 29.272 [144] for details.
    --
    -- subscribed-vsrvcc by its presence indicates that the UE is subscribed to the vSRVCC in
    -- the EPS domain, referring to the 3GPP TS 29.272 [144] for details.
```

```
APN-OI-Replacement ::=  OCTET STRING (SIZE (9..100))
    -- Octets are coded as APN Operator Identifier according to TS 3GPP TS 23.003 [17]
```

```
RFSP-ID ::=  INTEGER (1..256)
```

```
APN-ConfigurationProfile ::= SEQUENCE {
    defaultContext                      ContextId,
    completeDataListIncluded            NULL      OPTIONAL,
    -- If segmentation is used, completeDataListIncluded may only be present in the
    -- first segment of APN-ConfigurationProfile.
    epsDataList                         [1]  EPS-DataList,
    extensionContainer                  [2]  ExtensionContainer           OPTIONAL,
    ... ,
    additionalDefaultContext            [3]  ContextId                    OPTIONAL
    --   for details see the 3GPP TS 29.272 [144].
 }
```

```
EPS-DataList ::= SEQUENCE SIZE (1..maxNumOfAPN-Configurations) OF
        APN-Configuration
```

```
maxNumOfAPN-Configurations  INTEGER ::= 50
```

```
APN-Configuration ::= SEQUENCE {
    contextId [0] ContextId,
    pdn-Type  [1] PDN-Type,
    servedPartyIP-IPv4-Address          [2] PDP-Address                   OPTIONAL,
    apn       [3] APN,
    eps-qos-Subscribed                  [4] EPS-QoS-Subscribed,
    pdn-gw-Identity                     [5] PDN-GW-Identity               OPTIONAL,
    pdn-gw-AllocationType               [6] PDN-GW-AllocationType         OPTIONAL,
    vplmnAddressAllowed                 [7] NULL  OPTIONAL,
    chargingCharacteristics             [8] ChargingCharacteristics       OPTIONAL,
    ambr      [9] AMBR                  OPTIONAL,
    specificAPNInfoList                 [10] SpecificAPNInfoList          OPTIONAL,
    extensionContainer                  [11] ExtensionContainer           OPTIONAL,
    servedPartyIP-IPv6-Address          [12] PDP-Address                  OPTIONAL,
    ...,
    apn-oi-Replacement                  [13] APN-OI-Replacement           OPTIONAL,
    -- this apn-oi-Replacement refers to the APN level apn-oi-Replacement.
    sipto-Permission                    [14] SIPTO-Permission             OPTIONAL,
    lipa-Permission                     [15] LIPA-Permission              OPTIONAL,
    restoration-Priority                [16] Restoration-Priority         OPTIONAL,
    sipto-local-network-Permission      [17] SIPTO-Local-Network-Permission OPTIONAL,
    wlan-offloadability                 [18] WLAN-Offloadability          OPTIONAL,
    non-IP-PDN-Type-Indicator           [19] NULL OPTIONAL,
    nIDD-Mechanism                      [20] NIDD-Mechanism               OPTIONAL,
    sCEF-ID   [21]                      FQDN OPTIONAL,
    pdn-ConnectionContinuity            [22] PDN-ConnectionContinuity     OPTIONAL
    -- absence of pdn-ConnectionContinuity indicates that the handling is left to
    -- local VPLMN policy
 }
```

```
PDN-ConnectionContinuity ::= ENUMERATED {
    maintainPDN-Connection                    (0),
    disconnectPDN-ConnectionWithReactivationRequest                    (1),
    disconnectPDN-ConnectionWithoutReactivationRequest                 (2)
 }
```

```
NIDD-Mechanism ::= ENUMERATED {
    sGi-based-data-delivery             (0),
    sCEF-based-data-delivery            (1)
    -- The default value, when this information element is not present, is
    -- sGi-based-data-delivery (0)
 }
```

```
PDN-Type ::= OCTET STRING (SIZE (1))
    -- Octet is coded  as follows:
    --   Bits
    --   3 2 1
    --   0 0 1 IPv4
    --   0 1 0 IPv6
    --   0 1 1 IPv4v6
    --   1 0 0 IPv4_or_IPv6
    --   Bits 8-4 shall be coded as zero.
    --   for details see 3GPP TS 29.272 [144]
```

```
EPS-QoS-Subscribed ::= SEQUENCE {
    qos-Class-Identifier                [0] QoS-Class-Identifier,
    allocation-Retention-Priority       [1] Allocation-Retention-Priority,
    extensionContainer                  [2] ExtensionContainer            OPTIONAL,
    ... }
```

```
AMBR ::= SEQUENCE {
    max-RequestedBandwidth-UL           [0] Bandwidth,
    max-RequestedBandwidth-DL           [1] Bandwidth,
    extensionContainer                  [2] ExtensionContainer            OPTIONAL,
    ...,
    extended-Max-RequestedBandwidth-UL  [3] BandwidthExt                  OPTIONAL,
    extended-Max-RequestedBandwidth-DL  [4] BandwidthExt                  OPTIONAL
    -- extended-Max-RequestedBandwidth-UL/DL shall be populated according to the
    -- description of the corresponding parameters in 3GPP TS 29.272 [144]
}
```

```
SpecificAPNInfoList ::= SEQUENCE SIZE (1..maxNumOfSpecificAPNInfos) OF
         SpecificAPNInfo
```

```
maxNumOfSpecificAPNInfos  INTEGER ::= 50
```

```
SpecificAPNInfo ::= SEQUENCE {
    apn      [0] APN,
    pdn-gw-Identity                     [1] PDN-GW-Identity,
    extensionContainer                  [2] ExtensionContainer            OPTIONAL,
    ... }
```

```
Bandwidth ::= INTEGER
    -- bits per second
```

```
BandwidthExt ::= INTEGER
    -- kilobits per second
```

```
QoS-Class-Identifier ::= INTEGER (1..9)
    -- values are defined in  3GPP TS 29.212
```

```
Allocation-Retention-Priority ::= SEQUENCE {
    priority-level                      [0] INTEGER,
    pre-emption-capability              [1] BOOLEAN                       OPTIONAL,
    pre-emption-vulnerability           [2] BOOLEAN                       OPTIONAL,
    extensionContainer                  [3] ExtensionContainer            OPTIONAL,
    ... }
```

```
PDN-GW-Identity ::= SEQUENCE {
    pdn-gw-ipv4-Address                 [0] PDP-Address             OPTIONAL,
    pdn-gw-ipv6-Address                 [1] PDP-Address             OPTIONAL,
    pdn-gw-name                         [2] FQDN  OPTIONAL,
    extensionContainer                  [3] ExtensionContainer      OPTIONAL,
    ... }
```

```
FQDN ::=  OCTET STRING (SIZE (9..255))
```

```
PDN-GW-AllocationType ::= ENUMERATED {
    static   (0),
    dynamic  (1)}
```

```
WLAN-Offloadability ::= SEQUENCE {
    wlan-offloadability-EUTRAN          [0] WLAN-Offloadability-Indication  OPTIONAL,
    wlan-offloadability-UTRAN           [1] WLAN-Offloadability-Indication   OPTIONAL,
    extensionContainer                  [2] ExtensionContainer     OPTIONAL,
    ... }
```

```
WLAN-Offloadability-Indication ::= ENUMERATED {
    notAllowed   (0),
    allowed      (1)}
```

```
AccessRestrictionData ::= BIT STRING {
    utranNotAllowed (0),
    geranNotAllowed (1),
    ganNotAllowed   (2),
    i-hspa-evolutionNotAllowed (3),
    wb-e-utranNotAllowed (4),
    ho-toNon3GPP-AccessNotAllowed (5),
    nb-iotNotAllowed (6),
    enhancedCoverageNotAllowed (7) } (SIZE (2..8))
    -- exception handling:
    -- The VLR shall ignore the access restriction data related to an access type not
    -- supported by the node.
    -- The handling of the access restriction data by the SGSN is described in subclause
    -- 5.3.19 of TS 23.060, in subclause 7.5.3 of TS 29.060 and subclause 7.3.6 of TS 29.274.
    -- Additional access restrictions are encoded in Ext-AccessRestrictionData bit string.
```

```
Ext-AccessRestrictionData ::= BIT STRING {
    nrAsSecondaryRATNotAllowed (0),
    unlicensedSpectrumAsSecondaryRATNotAllowed (1) } (SIZE (1..32))
```

```
AdjacentAccessRestrictionDataList ::= SEQUENCE SIZE (1..50) OF
        AdjacentAccessRestrictionData
```

```
AdjacentAccessRestrictionData ::= SEQUENCE {
        plmnId                          [0]  PLMN-Id,
        accessRestrictionData           [1]  AccessRestrictionData,
        ... ,
        ext-AccessRestrictionData       [2]  Ext-AccessRestrictionData     OPTIONAL }
```

```
CS-AllocationRetentionPriority ::= OCTET STRING (SIZE (1))
    -- This data type encodes each priority level defined in TS 23.107 as the binary value
    -- of the priority level.
```

```
IST-AlertTimerValue ::= INTEGER (15..255)
```

```
LCSInformation ::= SEQUENCE {
    gmlc-List [0]                       GMLC-List OPTIONAL,
    lcs-PrivacyExceptionList            [1]  LCS-PrivacyExceptionList      OPTIONAL,
    molr-List [2]                       MOLR-List OPTIONAL,
    ...,
    add-lcs-PrivacyExceptionList        [3]  LCS-PrivacyExceptionList      OPTIONAL }
    -- add-lcs-PrivacyExceptionList may be sent only if lcs-PrivacyExceptionList is
    -- present and contains four instances of LCS-PrivacyClass. If the mentioned condition
    -- is not satisfied the receiving node shall discard add-lcs-PrivacyExceptionList.
    -- If an LCS-PrivacyClass is received both in lcs-PrivacyExceptionList and in
    -- add-lcs-PrivacyExceptionList with the same SS-Code, then the error unexpected
    -- data value shall be returned.
```

```
GMLC-List ::= SEQUENCE SIZE (1..maxNumOfGMLC) OF
        ISDN-AddressString
    -- if segmentation is used, the complete GMLC-List shall be sent in one segment
```

```
maxNumOfGMLC  INTEGER ::= 5
```

```
NetworkAccessMode ::= ENUMERATED {
    packetAndCircuit                    (0),
    onlyCircuit                         (1),
    onlyPacket                          (2),
    ...}
    -- if unknown values are received in NetworkAccessMode
    -- they shall be discarded.
```

```
GPRSDataList ::= SEQUENCE SIZE (1..maxNumOfPDP-Contexts) OF
        PDP-Context
```

```
maxNumOfPDP-Contexts  INTEGER ::= 50
```

```
PDP-Context ::= SEQUENCE {
    pdp-ContextId                       ContextId,
    pdp-Type  [16] PDP-Type,
    pdp-Address                         [17] PDP-Address                    OPTIONAL,
    qos-Subscribed                      [18] QoS-Subscribed,
    vplmnAddressAllowed                 [19] NULL OPTIONAL,
    apn      [20] APN,
    extensionContainer                  [21] ExtensionContainer             OPTIONAL,
    ... ,
    ext-QoS-Subscribed                  [0] Ext-QoS-Subscribed              OPTIONAL,
    pdp-ChargingCharacteristics         [1] ChargingCharacteristics         OPTIONAL,
    ext2-QoS-Subscribed                 [2] Ext2-QoS-Subscribed             OPTIONAL,
    -- ext2-QoS-Subscribed may be present only if ext-QoS-Subscribed is present.
    ext3-QoS-Subscribed                 [3] Ext3-QoS-Subscribed             OPTIONAL,
    -- ext3-QoS-Subscribed may be present only if ext2-QoS-Subscribed is present.
    ext4-QoS-Subscribed                 [4] Ext4-QoS-Subscribed             OPTIONAL,
    -- ext4-QoS-Subscribed may be present only if ext3-QoS-Subscribed is present.
    apn-oi-Replacement                  [5]  APN-OI-Replacement             OPTIONAL,
    -- this apn-oi-Replacement refers to the APN level apn-oi-Replacement and has
    -- higher priority than UE level apn-oi-Replacement.
    ext-pdp-Type                        [6] Ext-PDP-Type                    OPTIONAL,
    -- contains the value IPv4v6 defined in 3GPP TS 29.060 [105], if the PDP can be
    -- accessed by dual-stack UEs
    ext-pdp-Address                     [7] PDP-Address                     OPTIONAL,
    -- contains an additional IP address in case of dual-stack static IP address assignment
    -- for the UE.
    -- it may contain an IPv4 or an IPv6 address/prefix, and it may be present
    -- only if pdp-Address is present; if both are present, each parameter shall
    -- contain a different type of address (IPv4 or IPv6).
    ambr     [10] AMBR                  OPTIONAL,
    -- this ambr contains the AMBR associated to the APN included in the
    -- PDP-Context (APN-AMBR).
    sipto-Permission                    [8] SIPTO-Permission                OPTIONAL,
    lipa-Permission                     [9] LIPA-Permission                 OPTIONAL,
    restoration-Priority                [11] Restoration-Priority           OPTIONAL,
    sipto-local-network-Permission      [12] SIPTO-Local-Network-Permission OPTIONAL,
    nIDD-Mechanism                      [13] NIDD-Mechanism                 OPTIONAL,
    sCEF-ID   [14]                      FQDN OPTIONAL
    }
```

```
Restoration-Priority ::= OCTET STRING (SIZE (1))
    -- Octet 1:
    --  Restoration Priority. This octet encodes the Restoration Priority,
    --  as the binary value of the Restoration-Priority described in 3GPP TS 29.272 [144].
```

```
SIPTO-Permission ::= ENUMERATED {
    siptoAboveRanAllowed  (0),
    siptoAboveRanNotAllowed  (1)
    }
```

```
SIPTO-Local-Network-Permission ::= ENUMERATED {
    siptoAtLocalNetworkAllowed  (0),
    siptoAtLocalNetworkNotAllowed  (1)
    }
```

```
LIPA-Permission ::= ENUMERATED {
     lipaProhibited  (0),
     lipaOnly  (1),
     lipaConditional  (2)
     }
```

```
ContextId ::= INTEGER (1..maxNumOfPDP-Contexts)
```

```
GPRSSubscriptionData ::= SEQUENCE {
     completeDataListIncluded            NULL     OPTIONAL,
     -- If segmentation is used, completeDataListIncluded may only be present in the
     -- first segment of GPRSSubscriptionData.
     gprsDataList                        [1]  GPRSDataList,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...,
     apn-oi-Replacement                  [3]  APN-OI-Replacement           OPTIONAL
     -- this apn-oi-Replacement refers to the UE level apn-oi-Replacement.
 }
```

```
SGSN-CAMEL-SubscriptionInfo ::= SEQUENCE {
     gprs-CSI [0]                        GPRS-CSI  OPTIONAL,
     mo-sms-CSI                          [1]  SMS-CSI                      OPTIONAL,
     extensionContainer                  [2]  ExtensionContainer           OPTIONAL,
     ...,
     mt-sms-CSI                          [3]  SMS-CSI                      OPTIONAL,
     mt-smsCAMELTDP-CriteriaList         [4]  MT-smsCAMELTDP-CriteriaList  OPTIONAL,
     mg-csi    [5]                       MG-CSI    OPTIONAL
     }
```

```
GPRS-CSI ::= SEQUENCE {
     gprs-CamelTDPDataList               [0] GPRS-CamelTDPDataList         OPTIONAL,
     camelCapabilityHandling             [1] CamelCapabilityHandling       OPTIONAL,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     notificationToCSE                   [3]  NULL OPTIONAL,
     csi-Active                          [4]  NULL OPTIONAL,
     ...}
--   notificationToCSE and csi-Active shall not be present when GPRS-CSI is sent to SGSN.
--   They may only be included in ATSI/ATM ack/NSDC message.
--   GPRS-CamelTDPData and  camelCapabilityHandling shall be present in
--   the GPRS-CSI sequence.
--   If GPRS-CSI is segmented, gprs-CamelTDPDataList and camelCapabilityHandling shall be
--   present in the first segment
```

```
GPRS-CamelTDPDataList ::= SEQUENCE SIZE (1..maxNumOfCamelTDPData) OF
     GPRS-CamelTDPData
--   GPRS-CamelTDPDataList shall not contain more than one instance of
--   GPRS-CamelTDPData containing the same value for gprs-TriggerDetectionPoint.
```

```
GPRS-CamelTDPData ::= SEQUENCE {
     gprs-TriggerDetectionPoint          [0] GPRS-TriggerDetectionPoint,
     serviceKey                          [1] ServiceKey,
     gsmSCF-Address                      [2] ISDN-AddressString,
     defaultSessionHandling              [3] DefaultGPRS-Handling,
     extensionContainer                  [4] ExtensionContainer            OPTIONAL,
     ...
     }
```

```
DefaultGPRS-Handling ::= ENUMERATED {
     continueTransaction (0) ,
     releaseTransaction (1) ,
     ...}
-- exception handling:
-- reception of values in range 2-31 shall be treated as "continueTransaction"
-- reception of values greater than 31 shall be treated as "releaseTransaction"
```

```
GPRS-TriggerDetectionPoint ::= ENUMERATED {
     attach                              (1),
     attachChangeOfPosition              (2),
     pdp-ContextEstablishment            (11),
     pdp-ContextEstablishmentAcknowledgement  (12),
     pdp-ContextChangeOfPosition         (14),
     ... }
-- exception handling:
-- For GPRS-CamelTDPData sequences containing this parameter with any
-- other value than the ones listed the receiver shall ignore the whole
-- GPRS-CamelTDPDatasequence.
```

```
APN ::=  OCTET STRING (SIZE (2..63))
     -- Octets are coded according to TS 3GPP TS 23.003 [17]
```

```
PDP-Type ::= OCTET STRING (SIZE (2))
     -- Octets are coded according to TS 3GPP TS 29.060 [105]
     -- Only the values PPP, IPv4, IPv6 and Non-IP are allowed for this parameter.
```

```
Ext-PDP-Type ::= OCTET STRING (SIZE (2))
     -- Octets are coded, similarly to PDP-Type, according to TS 3GPP TS 29.060 [105].
     -- Only the value IPv4v6 is allowed for this parameter.
```

```
PDP-Address ::= OCTET STRING (SIZE (1..16))
     -- Octets are coded according to TS 3GPP TS 29.060 [105]

     -- The possible size values are:
     -- 1-7 octets  X.25 address type
     --  4  octets  IPv4 address type
     -- 16  octets  Ipv6 address type
```

```
QoS-Subscribed ::= OCTET STRING (SIZE (3))
     -- Octets are coded according to TS 3GPP TS 24.008 [35] Quality of Service Octets
     -- 3-5.
```

```
Ext-QoS-Subscribed ::= OCTET STRING (SIZE (1..9))
     -- OCTET 1:
     --  Allocation/Retention Priority (This octet encodes each priority level defined in
     --     23.107 as the binary value of the priority level, declaration in 29.060)
     -- Octets 2-9 are coded according to 3GPP TS 24.008 [35] Quality of Service Octets
     -- 6-13.
```

```
Ext2-QoS-Subscribed ::= OCTET STRING (SIZE (1..3))
     -- Octets 1-3 are coded according to 3GPP TS 24.008 [35] Quality of Service Octets 14-16.
     -- If Quality of Service information is structured with 14 octet length, then
     -- Octet 1 is coded according to 3GPP TS 24.008 [35] Quality of Service Octet 14.
```

```
Ext3-QoS-Subscribed ::= OCTET STRING (SIZE (1..2))
     -- Octets 1-2 are coded according to 3GPP TS 24.008 [35] Quality of Service Octets 17-18.
```

```
Ext4-QoS-Subscribed ::= OCTET STRING (SIZE (1))
     -- Octet 1:
     --  Evolved Allocation/Retention Priority. This octet encodes the Priority Level (PL),
     --  the Preemption Capability (PCI) and Preemption Vulnerability  (PVI) values, as
     --  described in 3GPP TS 29.060 [105].
```

```
ChargingCharacteristics ::= OCTET STRING (SIZE (2))
     -- Octets are coded according to 3GPP TS 32.215.
```

```
LSAOnlyAccessIndicator ::= ENUMERATED {
     accessOutsideLSAsAllowed  (0),
     accessOutsideLSAsRestricted (1)}
```

```
LSADataList ::= SEQUENCE SIZE (1..maxNumOfLSAs) OF
          LSAData
```

```
maxNumOfLSAs  INTEGER ::= 20
```

```
LSAData ::= SEQUENCE {
     lsaIdentity                         [0] LSAIdentity,
     lsaAttributes                       [1] LSAAttributes,
     lsaActiveModeIndicator              [2] NULL  OPTIONAL,
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     ...}
```

```
LSAInformation ::= SEQUENCE {
     completeDataListIncluded            NULL      OPTIONAL,

     -- If segmentation is used, completeDataListIncluded may only be present in the
     -- first segment.
     lsaOnlyAccessIndicator              [1]  LSAOnlyAccessIndicator       OPTIONAL,
     lsaDataList                         [2]  LSADataList                  OPTIONAL,
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     ...}
```

```
LSAIdentity ::= OCTET STRING (SIZE (3))
     -- Octets are coded according to TS 3GPP TS 23.003 [17]
```

```
LSAAttributes ::= OCTET STRING (SIZE (1))
     -- Octets are coded according to TS 3GPP TS 48.008 [49]
```

```
SubscriberData ::= SEQUENCE {
     msisdn    [1] ISDN-AddressString     OPTIONAL,
     category  [2] Category               OPTIONAL,
     subscriberStatus                     [3] SubscriberStatus              OPTIONAL,
     bearerServiceList                    [4] BearerServiceList             OPTIONAL,
     -- The exception handling for reception of unsupported / not allocated
     -- bearerServiceCodes is defined in section 8.8.1
     teleserviceList                      [6] TeleserviceList               OPTIONAL,
     -- The exception handling for reception of unsupported / not allocated
     -- teleserviceCodes is defined in section 8.8.1
     provisionedSS                        [7] Ext-SS-InfoList               OPTIONAL,
     odb-Data  [8] ODB-Data               OPTIONAL,
     roamingRestrictionDueToUnsupportedFeature  [9] NULL                    OPTIONAL,
     regionalSubscriptionData             [10] ZoneCodeList                 OPTIONAL,
     vbsSubscriptionData                  [11] VBSDataList                  OPTIONAL,
     vgcsSubscriptionData                 [12] VGCSDataList                 OPTIONAL,
     vlrCamelSubscriptionInfo             [13] VlrCamelSubscriptionInfo     OPTIONAL
     }
```

```
Category ::= OCTET STRING (SIZE (1))
     -- The internal structure is defined in ITU-T Rec Q.763.
```

```
SubscriberStatus ::= ENUMERATED {
     serviceGranted  (0),
     operatorDeterminedBarring  (1)}
```

```
BearerServiceList ::= SEQUENCE SIZE (1..maxNumOfBearerServices) OF
          Ext-BearerServiceCode
```

```
maxNumOfBearerServices  INTEGER ::= 50
```

```
TeleserviceList ::= SEQUENCE SIZE (1..maxNumOfTeleservices) OF
          Ext-TeleserviceCode
```

```
maxNumOfTeleservices  INTEGER ::= 20
```

```
ODB-Data ::= SEQUENCE {
     odb-GeneralData                      ODB-GeneralData,
     odb-HPLMN-Data                       ODB-HPLMN-Data                    OPTIONAL,
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ...}
```

```
ODB-GeneralData ::= BIT STRING {
     allOG-CallsBarred  (0),
     internationalOGCallsBarred  (1),
     internationalOGCallsNotToHPLMN-CountryBarred  (2),
     interzonalOGCallsBarred (6),
     interzonalOGCallsNotToHPLMN-CountryBarred (7),
     interzonalOGCallsAndInternationalOGCallsNotToHPLMN-CountryBarred (8),
     premiumRateInformationOGCallsBarred  (3),
     premiumRateEntertainementOGCallsBarred  (4),
     ss-AccessBarred  (5),
     allECT-Barred (9),
     chargeableECT-Barred (10),
     internationalECT-Barred (11),
     interzonalECT-Barred (12),
     doublyChargeableECT-Barred (13),
     multipleECT-Barred (14),
     allPacketOrientedServicesBarred (15),
     roamerAccessToHPLMN-AP-Barred  (16),
     roamerAccessToVPLMN-AP-Barred  (17),
     roamingOutsidePLMNOG-CallsBarred  (18),
     allIC-CallsBarred  (19),
     roamingOutsidePLMNIC-CallsBarred  (20),
     roamingOutsidePLMNICountryIC-CallsBarred  (21),
     roamingOutsidePLMN-Barred  (22),
     roamingOutsidePLMN-CountryBarred  (23),
     registrationAllCF-Barred  (24),
     registrationCFNotToHPLMN-Barred  (25),
     registrationInterzonalCF-Barred  (26),
     registrationInterzonalCFNotToHPLMN-Barred  (27),
     registrationInternationalCF-Barred  (28)} (SIZE (15..32))
     -- exception handling: reception of unknown bit assignments in the
     -- ODB-GeneralData type shall be treated like unsupported ODB-GeneralData
     -- When the ODB-GeneralData type is removed from the HLR for a given subscriber,
     -- in NoteSubscriberDataModified operation sent toward the gsmSCF
     -- all bits shall be set to "0".
```

```
ODB-HPLMN-Data ::= BIT STRING {
     plmn-SpecificBarringType1  (0),
     plmn-SpecificBarringType2  (1),
     plmn-SpecificBarringType3  (2),
     plmn-SpecificBarringType4  (3)} (SIZE (4..32))
     -- exception handling: reception of unknown bit assignments in the
     -- ODB-HPLMN-Data type shall be treated like unsupported ODB-HPLMN-Data
     -- When the ODB-HPLMN-Data type is removed from the HLR for a given subscriber,
     -- in NoteSubscriberDataModified operation sent toward the gsmSCF
     -- all bits shall be set to "0".
```

```
Ext-SS-InfoList ::= SEQUENCE SIZE (1..maxNumOfSS) OF
         Ext-SS-Info
```

```
Ext-SS-Info ::= CHOICE {
     forwardingInfo                      [0] Ext-ForwInfo,
     callBarringInfo                     [1] Ext-CallBarInfo,
     cug-Info  [2] CUG-Info,
     ss-Data   [3] Ext-SS-Data,
     emlpp-Info                          [4] EMLPP-Info}
```

```
Ext-ForwInfo ::= SEQUENCE {
     ss-Code   SS-Code,
     forwardingFeatureList               Ext-ForwFeatureList,
     extensionContainer                  [0] ExtensionContainer           OPTIONAL,
     ...}
```

```
Ext-ForwFeatureList ::= SEQUENCE SIZE (1..maxNumOfExt-BasicServiceGroups) OF
         Ext-ForwFeature
```

```
Ext-ForwFeature ::= SEQUENCE {
    basicService                    Ext-BasicServiceCode            OPTIONAL,
    ss-Status [4] Ext-SS-Status,
    forwardedToNumber               [5] ISDN-AddressString          OPTIONAL,
    -- When this data type is sent from an HLR which supports CAMEL Phase 2
    -- to a VLR that supports CAMEL Phase 2 the VLR shall not check the
    -- format of the number
    forwardedToSubaddress           [8] ISDN-SubaddressString       OPTIONAL,
    forwardingOptions               [6] Ext-ForwOptions             OPTIONAL,
    noReplyConditionTime            [7] Ext-NoRepCondTime           OPTIONAL,
    extensionContainer              [9] ExtensionContainer          OPTIONAL,
    ...,
    longForwardedToNumber           [10] FTN-AddressString          OPTIONAL }
```

```
Ext-ForwOptions ::= OCTET STRING (SIZE (1..5))

    -- OCTET 1:

    --  bit 8: notification to forwarding party
    --   0  no notification
    --   1  notification

    --  bit 7: redirecting presentation
    --   0 no presentation
    --   1  presentation

    --  bit 6: notification to calling party
    --   0  no notification
    --   1  notification

    --  bit 5: 0 (unused)

    --  bits 43: forwarding reason
    --   00  ms not reachable
    --   01  ms busy
    --   10  no reply
    --   11  unconditional

    -- bits 21: 00 (unused)

    -- OCTETS 2-5: reserved for future use. They shall be discarded if
    -- received and not understood.
```

```
Ext-NoRepCondTime ::= INTEGER (1..100)
    -- Only values 5-30 are used.
    -- Values in the ranges 1-4 and 31-100 are reserved for future use
    -- If received:
    --   values 1-4 shall be mapped on to value 5
    --   values 31-100 shall be mapped on to value 30
```

```
Ext-CallBarInfo ::= SEQUENCE {
    ss-Code  SS-Code,
    callBarringFeatureList          Ext-CallBarFeatureList,
    extensionContainer              ExtensionContainer              OPTIONAL,
    ...}
```

```
Ext-CallBarFeatureList ::= SEQUENCE SIZE (1..maxNumOfExt-BasicServiceGroups) OF
        Ext-CallBarringFeature
```

```
Ext-CallBarringFeature ::= SEQUENCE {
    basicService                    Ext-BasicServiceCode            OPTIONAL,
    ss-Status [4] Ext-SS-Status,
    extensionContainer              ExtensionContainer              OPTIONAL,
    ...}
```

```
CUG-Info ::= SEQUENCE {
    cug-SubscriptionList            CUG-SubscriptionList,
    cug-FeatureList                 CUG-FeatureList                 OPTIONAL,
    extensionContainer              [0] ExtensionContainer          OPTIONAL,
    ...}
```

```
CUG-SubscriptionList ::= SEQUENCE SIZE (0..maxNumOfCUG) OF
        CUG-Subscription
```

```
CUG-Subscription ::= SEQUENCE {
    cug-Index CUG-Index,
    cug-Interlock                   CUG-Interlock,
    intraCUG-Options                IntraCUG-Options,
    basicServiceGroupList           Ext-BasicServiceGroupList       OPTIONAL,
    extensionContainer              [0] ExtensionContainer          OPTIONAL,
    ...}
```

```
CUG-Index ::= INTEGER (0..32767)
    -- The internal structure is defined in ETS 300 138.
```

```
CUG-Interlock ::= OCTET STRING (SIZE (4))
```

```
IntraCUG-Options ::= ENUMERATED {
    noCUG-Restrictions  (0),
    cugIC-CallBarred  (1),
    cugOG-CallBarred  (2)}
```

```
maxNumOfCUG  INTEGER ::= 10
```

```
CUG-FeatureList ::= SEQUENCE SIZE (1..maxNumOfExt-BasicServiceGroups) OF
        CUG-Feature
```

```
Ext-BasicServiceGroupList ::= SEQUENCE SIZE (1..maxNumOfExt-BasicServiceGroups) OF
        Ext-BasicServiceCode
```

```
maxNumOfExt-BasicServiceGroups  INTEGER ::= 32
```

```
CUG-Feature ::= SEQUENCE {
    basicService                    Ext-BasicServiceCode            OPTIONAL,
    preferentialCUG-Indicator       CUG-Index OPTIONAL,
    interCUG-Restrictions           InterCUG-Restrictions,
    extensionContainer              ExtensionContainer              OPTIONAL,
    ...}
```

```
InterCUG-Restrictions ::= OCTET STRING (SIZE (1))

    -- bits 876543: 000000 (unused)
    -- Exception handling:
    -- bits 876543 shall be ignored if received and not understood

    -- bits 21
    --   00  CUG only facilities
    --   01  CUG with outgoing access
    --   10  CUG with incoming access
    --   11  CUG with both outgoing and incoming access
```

```
Ext-SS-Data ::= SEQUENCE {
    ss-Code  SS-Code,
    ss-Status [4] Ext-SS-Status,
    ss-SubscriptionOption           SS-SubscriptionOption           OPTIONAL,
    basicServiceGroupList           Ext-BasicServiceGroupList       OPTIONAL,
    extensionContainer              [5] ExtensionContainer          OPTIONAL,
    ...}
```

```
LCS-PrivacyExceptionList ::= SEQUENCE SIZE (1..maxNumOfPrivacyClass) OF
        LCS-PrivacyClass
```

```
maxNumOfPrivacyClass  INTEGER ::= 4
```

```
LCS-PrivacyClass ::= SEQUENCE {
     ss-Code   SS-Code,
     ss-Status Ext-SS-Status,
     notificationToMSUser                [0] NotificationToMSUser          OPTIONAL,
     -- notificationToMSUser may be sent only for SS-codes callSessionRelated
     -- and callSessionUnrelated. If not received for SS-codes callSessionRelated
     -- and callSessionUnrelated,
     -- the default values according to 3GPP TS 23.271 shall be assumed.
     externalClientList                  [1] ExternalClientList            OPTIONAL,
     -- externalClientList may be sent only for SS-code callSessionUnrelated to a
     -- visited node that does not support LCS Release 4 or later versions.
     -- externalClientList may be sent only for SS-codes callSessionUnrelated and
     -- callSessionRelated to a visited node that supports LCS Release 4 or later versions.
     plmnClientList                      [2] PLMNClientList                OPTIONAL,
     -- plmnClientList may be sent only for SS-code plmnoperator.
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     ...,
     ext-externalClientList              [4] Ext-ExternalClientList        OPTIONAL,
     -- Ext-externalClientList may be sent only if the visited node supports LCS Release 4 or
     -- later versions, the user did specify more than 5 clients, and White Book SCCP is used.
     serviceTypeList                     [5]  ServiceTypeList              OPTIONAL
     -- serviceTypeList may be sent only for SS-code serviceType and if the visited node
     -- supports LCS Release 5 or later versions.
     --
     -- if segmentation is used, the complete LCS-PrivacyClass shall be sent in one segment
}
```

```
ExternalClientList ::= SEQUENCE SIZE (0..maxNumOfExternalClient) OF
          ExternalClient
```

```
maxNumOfExternalClient  INTEGER ::= 5
```

```
PLMNClientList ::= SEQUENCE SIZE (1..maxNumOfPLMNClient) OF
          LCSClientInternalID
```

```
maxNumOfPLMNClient  INTEGER ::= 5
```

```
Ext-ExternalClientList ::= SEQUENCE SIZE (1..maxNumOfExt-ExternalClient) OF
          ExternalClient
```

```
maxNumOfExt-ExternalClient  INTEGER ::= 35
```

```
ExternalClient ::= SEQUENCE {
     clientIdentity                      LCSClientExternalID,
     gmlc-Restriction                    [0] GMLC-Restriction              OPTIONAL,
     notificationToMSUser                [1] NotificationToMSUser          OPTIONAL,
     -- If notificationToMSUser is not received, the default value according to
     -- 3GPP TS 23.271 shall be assumed.
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ... }
```

```
GMLC-Restriction ::= ENUMERATED {
     gmlc-List (0),
     home-Country                        (1) ,
     ... }
-- exception handling:
-- At reception of any other value than the ones listed the receiver shall ignore
-- GMLC-Restriction.
```

```
NotificationToMSUser ::= ENUMERATED {
     notifyLocationAllowed               (0),
     notifyAndVerify-LocationAllowedIfNoResponse   (1),
     notifyAndVerify-LocationNotAllowedIfNoResponse(2),
     ...,
     locationNotAllowed (3) }
-- exception handling:
-- At reception of any other value than the ones listed the receiver shall ignore
-- NotificationToMSUser.
```

```
ServiceTypeList ::= SEQUENCE SIZE (1..maxNumOfServiceType) OF
          ServiceType
```

```
maxNumOfServiceType  INTEGER ::= 32
```

```
ServiceType ::= SEQUENCE {
     serviceTypeIdentity                LCSServiceTypeID,
     gmlc-Restriction                   [0] GMLC-Restriction              OPTIONAL,
     notificationToMSUser               [1] NotificationToMSUser          OPTIONAL,
     -- If notificationToMSUser is not received, the default value according to
     -- 3GPP TS 23.271 shall be assumed.
     extensionContainer                 [2] ExtensionContainer            OPTIONAL,
     ... }
```

```
MOLR-List ::= SEQUENCE SIZE (1..maxNumOfMOLR-Class) OF
          MOLR-Class
```

```
maxNumOfMOLR-Class  INTEGER ::= 3
```

```
MOLR-Class ::= SEQUENCE {
     ss-Code   SS-Code,
     ss-Status Ext-SS-Status,
     extensionContainer                 [0] ExtensionContainer            OPTIONAL,
     ...}
```

```
ZoneCodeList ::= SEQUENCE SIZE (1..maxNumOfZoneCodes)
          OF ZoneCode
```

```
ZoneCode ::= OCTET STRING (SIZE (2))
     -- internal structure is defined in TS 3GPP TS 23.003 [17]
```

```
maxNumOfZoneCodes  INTEGER ::= 10
```

```
InsertSubscriberDataRes ::= SEQUENCE {
     teleserviceList                    [1] TeleserviceList               OPTIONAL,
     bearerServiceList                  [2] BearerServiceList             OPTIONAL,
     ss-List   [3] SS-List              OPTIONAL,
     odb-GeneralData                    [4] ODB-GeneralData               OPTIONAL,
     regionalSubscriptionResponse       [5] RegionalSubscriptionResponse  OPTIONAL,
     supportedCamelPhases               [6] SupportedCamelPhases          OPTIONAL,
     extensionContainer                 [7] ExtensionContainer            OPTIONAL,
     ... ,
     offeredCamel4CSIs                  [8] OfferedCamel4CSIs             OPTIONAL,
     supportedFeatures                  [9] SupportedFeatures             OPTIONAL,
     ext-SupportedFeatures              [10] Ext-SupportedFeatures        OPTIONAL }
```

```
RegionalSubscriptionResponse ::= ENUMERATED {
     networkNode-AreaRestricted         (0),
     tooManyZoneCodes                   (1),
     zoneCodesConflict                  (2),
     regionalSubscNotSupported          (3)}
```

```
DeleteSubscriberDataArg ::= SEQUENCE {
     imsi      [0] IMSI,
     basicServiceList                   [1] BasicServiceList              OPTIONAL,
     -- The exception handling for reception of unsupported/not allocated
     -- basicServiceCodes is defined in section 6.8.2
     ss-List   [2] SS-List              OPTIONAL,
     roamingRestrictionDueToUnsupportedFeature [4] NULL                   OPTIONAL,
     regionalSubscriptionIdentifier     [5] ZoneCode                      OPTIONAL,
     vbsGroupIndication                 [7] NULL  OPTIONAL,
     vgcsGroupIndication                [8] NULL  OPTIONAL,
     camelSubscriptionInfoWithdraw      [9] NULL  OPTIONAL,
     extensionContainer                 [6] ExtensionContainer            OPTIONAL,
     ...,
     gprsSubscriptionDataWithdraw       [10] GPRSSubscriptionDataWithdraw OPTIONAL,
     roamingRestrictedInSgsnDueToUnsuppportedFeature [11] NULL            OPTIONAL,
     lsaInformationWithdraw             [12] LSAInformationWithdraw       OPTIONAL,
     gmlc-ListWithdraw                  [13] NULL OPTIONAL,
     istInformationWithdraw             [14] NULL OPTIONAL,
     specificCSI-Withdraw               [15] SpecificCSI-Withdraw         OPTIONAL,
     chargingCharacteristicsWithdraw    [16] NULL OPTIONAL,
     stn-srWithdraw                     [17] NULL OPTIONAL,
     epsSubscriptionDataWithdraw        [18] EPS-SubscriptionDataWithdraw   OPTIONAL,
     apn-oi-replacementWithdraw         [19] NULL OPTIONAL,
     csg-SubscriptionDeleted            [20] NULL OPTIONAL,
     subscribedPeriodicTAU-RAU-TimerWithdraw  [22] NULL                   OPTIONAL,
     subscribedPeriodicLAU-TimerWithdraw [23] NULL OPTIONAL,
     subscribed-vsrvccWithdraw          [21] NULL OPTIONAL,
     vplmn-Csg-SubscriptionDeleted      [24] NULL OPTIONAL,
     additionalMSISDN-Withdraw          [25] NULL OPTIONAL,
     cs-to-ps-SRVCC-Withdraw            [26] NULL OPTIONAL,
```

```
    imsiGroupIdList-Withdraw             [27] NULL OPTIONAL,
    userPlaneIntegrityProtectionWithdraw [28] NULL OPTIONAL,
    dl-Buffering-Suggested-Packet-Count-Withdraw  [29] NULL              OPTIONAL,
    ue-UsageTypeWithdraw                 [30] NULL OPTIONAL,
    reset-idsWithdraw                    [31] NULL OPTIONAL,
    iab-OperationWithdraw                [32] NULL OPTIONAL }
```

```
SpecificCSI-Withdraw ::= BIT STRING {
    o-csi (0),
    ss-csi (1),
    tif-csi (2),
    d-csi (3),
    vt-csi (4),
    mo-sms-csi (5),
    m-csi (6),
    gprs-csi (7),
    t-csi (8),
    mt-sms-csi (9),
    mg-csi (10),
    o-IM-CSI (11),
    d-IM-CSI (12),
    vt-IM-CSI (13) } (SIZE(8..32))
-- exception handling:
-- bits 11 to 31 shall be ignored if received by a non-IP Multimedia Core Network entity.
-- bits 0-10 and 14-31 shall be ignored if received by an IP Multimedia Core Network entity.
-- bits 11-13 are only applicable in an IP Multimedia Core Network.
-- Bit 8 and bits 11-13 are only applicable for the NoteSubscriberDataModified operation.
```

```
GPRSSubscriptionDataWithdraw ::= CHOICE {
    allGPRSData                          NULL,
    contextIdList                        ContextIdList}
```

```
EPS-SubscriptionDataWithdraw ::= CHOICE {
    allEPS-Data                          NULL,
    contextIdList                        ContextIdList}
```

```
ContextIdList ::= SEQUENCE SIZE (1..maxNumOfPDP-Contexts) OF
        ContextId
```

```
LSAInformationWithdraw ::= CHOICE {
    allLSAData                           NULL,
    lsaIdentityList                      LSAIdentityList }
```

```
LSAIdentityList ::= SEQUENCE SIZE (1..maxNumOfLSAs) OF
        LSAIdentity
```

```
BasicServiceList ::= SEQUENCE SIZE (1..maxNumOfBasicServices) OF
        Ext-BasicServiceCode
```

```
maxNumOfBasicServices  INTEGER ::= 70
```

```
DeleteSubscriberDataRes ::= SEQUENCE {
    regionalSubscriptionResponse         [0] RegionalSubscriptionResponse  OPTIONAL,
    extensionContainer                   ExtensionContainer                OPTIONAL,
    ...}
```

```
VlrCamelSubscriptionInfo ::= SEQUENCE {
    o-CSI     [0] O-CSI                  OPTIONAL,
    extensionContainer                   [1] ExtensionContainer            OPTIONAL,
    ...,
    ss-CSI    [2] SS-CSI                 OPTIONAL,
    o-BcsmCamelTDP-CriteriaList          [4] O-BcsmCamelTDPCriteriaList    OPTIONAL,
    tif-CSI   [3] NULL                   OPTIONAL,
    m-CSI     [5] M-CSI                  OPTIONAL,
    mo-sms-CSI                           [6] SMS-CSI                       OPTIONAL,
    vt-CSI    [7] T-CSI                  OPTIONAL,
    t-BCSM-CAMEL-TDP-CriteriaList        [8] T-BCSM-CAMEL-TDP-CriteriaList OPTIONAL,
    d-CSI     [9] D-CSI                  OPTIONAL,
    mt-sms-CSI                           [10] SMS-CSI                      OPTIONAL,
    mt-smsCAMELTDP-CriteriaList          [11] MT-smsCAMELTDP-CriteriaList  OPTIONAL
    }
```

```
MT-smsCAMELTDP-CriteriaList ::= SEQUENCE SIZE (1.. maxNumOfCamelTDPData) OF
    MT-smsCAMELTDP-Criteria
```

```
MT-smsCAMELTDP-Criteria ::= SEQUENCE {
     sms-TriggerDetectionPoint           SMS-TriggerDetectionPoint,
     tpdu-TypeCriterion                  [0]  TPDU-TypeCriterion          OPTIONAL,
     ... }
```

```
TPDU-TypeCriterion ::= SEQUENCE SIZE (1..maxNumOfTPDUTypes) OF
     MT-SMS-TPDU-Type
```

```
maxNumOfTPDUTypes INTEGER ::= 5
```

```
MT-SMS-TPDU-Type ::= ENUMERATED {
     sms-DELIVER                         (0),
     sms-SUBMIT-REPORT                   (1),
     sms-STATUS-REPORT                   (2),
     ... }

--   exception handling:
--   For TPDU-TypeCriterion sequences containing this parameter with any
--   other value than the ones listed above the receiver shall ignore
--   the whole TPDU-TypeCriterion sequence.
--   In CAMEL phase 4, sms-SUBMIT-REPORT shall not be used and a received TPDU-TypeCriterion
--   sequence containing sms-SUBMIT-REPORT shall be wholly ignored.
```

```
D-CSI ::= SEQUENCE {
     dp-AnalysedInfoCriteriaList         [0] DP-AnalysedInfoCriteriaList   OPTIONAL,
     camelCapabilityHandling             [1] CamelCapabilityHandling       OPTIONAL,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     notificationToCSE                   [3]  NULL OPTIONAL,
     csi-Active                          [4]  NULL OPTIONAL,
     ...}
--   notificationToCSE and csi-Active shall not be present when D-CSI is sent to VLR/GMSC.
--   They may only be included in ATSI/ATM ack/NSDC message.
--   DP-AnalysedInfoCriteria and  camelCapabilityHandling shall be present in
--   the D-CSI sequence.
--   If D-CSI is segmented, then the first segment shall contain dp-AnalysedInfoCriteriaList
--   and camelCapabilityHandling. Subsequent segments shall not contain
--   camelCapabilityHandling, but may contain dp-AnalysedInfoCriteriaList.
```

```
DP-AnalysedInfoCriteriaList  ::= SEQUENCE SIZE (1..maxNumOfDP-AnalysedInfoCriteria) OF
         DP-AnalysedInfoCriterium
```

```
maxNumOfDP-AnalysedInfoCriteria INTEGER ::= 10
```

```
DP-AnalysedInfoCriterium ::= SEQUENCE {
     dialledNumber                       ISDN-AddressString,
     serviceKey                          ServiceKey,
     gsmSCF-Address                      ISDN-AddressString,
     defaultCallHandling                 DefaultCallHandling,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
SS-CSI ::= SEQUENCE {
     ss-CamelData                        SS-CamelData,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     notificationToCSE                   [0]  NULL OPTIONAL,
     csi-Active                          [1]  NULL OPTIONAL
--   notificationToCSE and csi-Active shall not be present when SS-CSI is sent to VLR.
--   They may only be included in ATSI/ATM ack/NSDC message.
}
```

```
SS-CamelData  ::= SEQUENCE {
     ss-EventList                        SS-EventList,
     gsmSCF-Address                      ISDN-AddressString,
     extensionContainer                  [0] ExtensionContainer            OPTIONAL,
     ...}
```

```
SS-EventList  ::= SEQUENCE SIZE (1..maxNumOfCamelSSEvents) OF SS-Code
    -- Actions for the following SS-Code values are defined in CAMEL Phase 3:
    -- ect    SS-Code ::= '00110001'B
    -- multiPTY                        SS-Code ::= '01010001'B
    -- cd     SS-Code ::= '00100100'B
    -- ccbs   SS-Code ::= '01000100'B
    -- all other SS codes shall be ignored
    -- When SS-CSI is sent to the VLR, it shall not contain a marking for ccbs.
    -- If the VLR receives SS-CSI containing a marking for ccbs, the VLR shall discard the
    -- ccbs marking in SS-CSI.
```

```
maxNumOfCamelSSEvents INTEGER ::= 10
```

```
O-CSI ::= SEQUENCE {
    o-BcsmCamelTDPDataList              O-BcsmCamelTDPDataList,
    extensionContainer                  ExtensionContainer              OPTIONAL,
    ...,
    camelCapabilityHandling             [0] CamelCapabilityHandling     OPTIONAL,
    notificationToCSE                   [1]  NULL OPTIONAL,
    csiActive [2]                       NULL OPTIONAL}
--  notificationtoCSE and csiActive shall not be present when O-CSI is sent to VLR/GMSC.
--  They may only be included in ATSI/ATM ack/NSDC message.
--  O-CSI shall not be segmented.
```

```
O-BcsmCamelTDPDataList ::= SEQUENCE SIZE (1..maxNumOfCamelTDPData) OF
    O-BcsmCamelTDPData
    -- O-BcsmCamelTDPDataList shall not contain more than one instance of
    -- O-BcsmCamelTDPData containing the same value for o-BcsmTriggerDetectionPoint.
    -- For CAMEL Phase 2, this means that only one instance of O-BcsmCamelTDPData is allowed
    -- with o-BcsmTriggerDetectionPoint being equal to DP2.
```

```
maxNumOfCamelTDPData  INTEGER ::= 10
```

```
O-BcsmCamelTDPData ::= SEQUENCE {
    o-BcsmTriggerDetectionPoint         O-BcsmTriggerDetectionPoint,
    serviceKey                          ServiceKey,
    gsmSCF-Address                      [0] ISDN-AddressString,
    defaultCallHandling                 [1] DefaultCallHandling,
    extensionContainer                  [2] ExtensionContainer          OPTIONAL,
    ...
    }
```

```
ServiceKey ::= INTEGER (0..2147483647)
```

```
O-BcsmTriggerDetectionPoint ::= ENUMERATED {
    collectedInfo (2),
    ...,
    routeSelectFailure (4) }
    -- exception handling:
    -- For O-BcsmCamelTDPData sequences containing this parameter with any
    -- other value than the ones listed the receiver shall ignore the whole
    -- O-BcsmCamelTDPDatasequence.
    -- For O-BcsmCamelTDP-Criteria sequences containing this parameter with any
    -- other value than the ones listed the receiver shall ignore the whole
    -- O-BcsmCamelTDP-Criteria sequence.
```

```
O-BcsmCamelTDPCriteriaList ::= SEQUENCE SIZE (1..maxNumOfCamelTDPData) OF
    O-BcsmCamelTDP-Criteria
```

```
T-BCSM-CAMEL-TDP-CriteriaList ::= SEQUENCE SIZE (1..maxNumOfCamelTDPData) OF
    T-BCSM-CAMEL-TDP-Criteria
```

```
O-BcsmCamelTDP-Criteria ::= SEQUENCE {
    o-BcsmTriggerDetectionPoint         O-BcsmTriggerDetectionPoint,
    destinationNumberCriteria           [0] DestinationNumberCriteria   OPTIONAL,
    basicServiceCriteria                [1] BasicServiceCriteria        OPTIONAL,
    callTypeCriteria                    [2] CallTypeCriteria            OPTIONAL,
    ...,
    o-CauseValueCriteria                [3] O-CauseValueCriteria        OPTIONAL,
    extensionContainer                  [4] ExtensionContainer          OPTIONAL }
```

```
T-BCSM-CAMEL-TDP-Criteria ::= SEQUENCE {
    t-BCSM-TriggerDetectionPoint        T-BcsmTriggerDetectionPoint,
    basicServiceCriteria                [0] BasicServiceCriteria        OPTIONAL,
    t-CauseValueCriteria                [1] T-CauseValueCriteria        OPTIONAL,
    ... }
```

```
DestinationNumberCriteria  ::= SEQUENCE {
    matchType [0] MatchType,
    destinationNumberList               [1] DestinationNumberList         OPTIONAL,
    destinationNumberLengthList         [2] DestinationNumberLengthList   OPTIONAL,
    -- one or both of destinationNumberList and destinationNumberLengthList
    -- shall be present
    ...}
```

```
DestinationNumberList  ::= SEQUENCE SIZE   (1..maxNumOfCamelDestinationNumbers) OF
        ISDN-AddressString
    -- The receiving entity shall not check the format of a number in
    -- the dialled number list
```

```
DestinationNumberLengthList  ::= SEQUENCE SIZE (1..maxNumOfCamelDestinationNumberLengths)
OF
             INTEGER(1..maxNumOfISDN-AddressDigits)
```

```
BasicServiceCriteria   ::= SEQUENCE SIZE(1..maxNumOfCamelBasicServiceCriteria) OF
    Ext-BasicServiceCode
```

```
maxNumOfISDN-AddressDigits  INTEGER ::= 15
```

```
maxNumOfCamelDestinationNumbers  INTEGER ::= 10
```

```
maxNumOfCamelDestinationNumberLengths  INTEGER ::= 3
```

```
maxNumOfCamelBasicServiceCriteria  INTEGER ::= 5
```

```
CallTypeCriteria       ::= ENUMERATED {
    forwarded (0),
    notForwarded                        (1)}
```

```
MatchType       ::= ENUMERATED {
    inhibiting                          (0),
    enabling  (1)}
```

```
O-CauseValueCriteria   ::= SEQUENCE SIZE(1..maxNumOfCAMEL-O-CauseValueCriteria) OF
    CauseValue
```

```
T-CauseValueCriteria   ::= SEQUENCE SIZE(1..maxNumOfCAMEL-T-CauseValueCriteria) OF
    CauseValue
```

```
maxNumOfCAMEL-O-CauseValueCriteria  INTEGER ::= 5
```

```
maxNumOfCAMEL-T-CauseValueCriteria  INTEGER ::= 5
```

```
CauseValue ::= OCTET STRING (SIZE(1))
-- Type extracted from Cause parameter in ITU-T Recommendation Q.763.
-- For the use of cause value refer to ITU-T Recommendation Q.850.
```

```
DefaultCallHandling ::= ENUMERATED {
    continueCall (0) ,
    releaseCall (1) ,
    ...}
    -- exception handling:
    -- reception of values in range 2-31 shall be treated as "continueCall"
    -- reception of values greater than 31 shall be treated as "releaseCall"
```

```
CamelCapabilityHandling ::= INTEGER(1..16)
    -- value 1 = CAMEL phase 1,
    -- value 2 = CAMEL phase 2,
    -- value 3 = CAMEL Phase 3,
    -- value 4 = CAMEL phase 4:
    -- reception of values greater than 4 shall be treated as CAMEL phase 4.
```

```
SupportedCamelPhases ::= BIT STRING {
    phase1 (0),
    phase2 (1),
    phase3 (2),
    phase4 (3)} (SIZE (1..16))
-- A node shall mark in the BIT STRING all CAMEL Phases it supports.
-- Other values than listed above shall be discarded.
```

```
OfferedCamel4CSIs ::= BIT STRING {
     o-csi    (0),
     d-csi    (1),
     vt-csi   (2),
     t-csi    (3),
     mt-sms-csi                          (4),
     mg-csi   (5),
     psi-enhancements                    (6)
} (SIZE (7..16))
-- A node supporting Camel phase 4 shall mark in the BIT STRING all Camel4 CSIs
-- it offers.
-- Other values than listed above shall be discarded.
```

```
OfferedCamel4Functionalities ::= BIT STRING {
     initiateCallAttempt                 (0),
     splitLeg (1),
     moveLeg  (2),
     disconnectLeg                       (3),
     entityReleased                      (4),
     dfc-WithArgument                    (5),
     playTone (6),
     dtmf-MidCall                        (7),
     chargingIndicator                   (8),
     alertingDP                          (9),
     locationAtAlerting                  (10),
     changeOfPositionDP                  (11),
     or-Interactions                     (12),
     warningToneEnhancements             (13),
     cf-Enhancements                     (14),
     subscribedEnhancedDialledServices   (15),
     servingNetworkEnhancedDialledServices (16),
     criteriaForChangeOfPositionDP       (17),
     serviceChangeDP                     (18),
     collectInformation                  (19)
} (SIZE (15..64))
-- A node supporting Camel phase 4 shall mark in the BIT STRING all CAMEL4
-- functionalities it offers.
-- Other values than listed above shall be discarded.
```

```
SMS-CSI ::= SEQUENCE {
     sms-CAMEL-TDP-DataList              [0] SMS-CAMEL-TDP-DataList        OPTIONAL,
     camelCapabilityHandling             [1] CamelCapabilityHandling       OPTIONAL,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     notificationToCSE                   [3] NULL  OPTIONAL,
     csi-Active                          [4] NULL  OPTIONAL,
     ...}
--   notificationToCSE and csi-Active shall not be present
--   when MO-SMS-CSI or MT-SMS-CSI is sent to VLR or SGSN.
--   They may only be included in ATSI/ATM ack/NSDC message.
--   SMS-CAMEL-TDP-Data and  camelCapabilityHandling shall be present in
--   the SMS-CSI sequence.
--   If SMS-CSI is segmented, sms-CAMEL-TDP-DataList and camelCapabilityHandling shall be
--   present in the first segment
```

```
SMS-CAMEL-TDP-DataList ::= SEQUENCE SIZE (1..maxNumOfCamelTDPData) OF
     SMS-CAMEL-TDP-Data
--   SMS-CAMEL-TDP-DataList shall not contain more than one instance of
--   SMS-CAMEL-TDP-Data containing the same value for sms-TriggerDetectionPoint.
```

```
SMS-CAMEL-TDP-Data ::= SEQUENCE {
     sms-TriggerDetectionPoint           [0] SMS-TriggerDetectionPoint,
     serviceKey                          [1] ServiceKey,
     gsmSCF-Address                      [2] ISDN-AddressString,
     defaultSMS-Handling                 [3] DefaultSMS-Handling,
     extensionContainer                  [4] ExtensionContainer            OPTIONAL,
     ...
     }
```

```
SMS-TriggerDetectionPoint ::= ENUMERATED {
     sms-CollectedInfo (1),
     ...,
     sms-DeliveryRequest (2)
     }
--   exception handling:
--   For SMS-CAMEL-TDP-Data and MT-smsCAMELTDP-Criteria sequences containing this
--   parameter with any other value than the ones listed the receiver shall ignore
--   the whole sequence.
--
--   If this parameter is received with any other value than sms-CollectedInfo
--   in an SMS-CAMEL-TDP-Data sequence contained in mo-sms-CSI, then the receiver shall
--   ignore the whole SMS-CAMEL-TDP-Data sequence.
--
--   If this parameter is received with any other value than sms-DeliveryRequest
--   in an SMS-CAMEL-TDP-Data sequence contained in mt-sms-CSI then the receiver shall
--   ignore the whole SMS-CAMEL-TDP-Data sequence.
--
--   If this parameter is received with any other value than sms-DeliveryRequest
--   in an MT-smsCAMELTDP-Criteria sequence then the receiver shall
--   ignore the whole MT-smsCAMELTDP-Criteria sequence.
```

```
DefaultSMS-Handling ::= ENUMERATED {
     continueTransaction (0) ,
     releaseTransaction (1) ,
     ...}
--   exception handling:
--   reception of values in range 2-31 shall be treated as "continueTransaction"
--   reception of values greater than 31 shall be treated as "releaseTransaction"
```

```
M-CSI ::= SEQUENCE {
     mobilityTriggers                    MobilityTriggers,
     serviceKey                          ServiceKey,
     gsmSCF-Address                      [0]  ISDN-AddressString,
     extensionContainer                  [1]  ExtensionContainer           OPTIONAL,
     notificationToCSE                   [2] NULL  OPTIONAL,
     csi-Active                          [3] NULL  OPTIONAL,
     ...}
--   notificationToCSE and csi-Active shall not be present when M-CSI is sent to VLR.
--   They may only be included in ATSI/ATM ack/NSDC message.
```

```
MG-CSI ::= SEQUENCE {
     mobilityTriggers                    MobilityTriggers,
     serviceKey                          ServiceKey,
     gsmSCF-Address                      [0]  ISDN-AddressString,
     extensionContainer                  [1]  ExtensionContainer           OPTIONAL,
     notificationToCSE                   [2] NULL  OPTIONAL,
     csi-Active                          [3] NULL  OPTIONAL,
     ...}
--   notificationToCSE and csi-Active shall not be present when MG-CSI is sent to SGSN.
--   They may only be included in ATSI/ATM ack/NSDC message.
```

```
MobilityTriggers  ::= SEQUENCE SIZE (1..maxNumOfMobilityTriggers) OF
     MM-Code
```

```
maxNumOfMobilityTriggers INTEGER ::= 10
```

```
MM-Code ::= OCTET STRING (SIZE (1))
--   This type is used to indicate a Mobility Management event.
--   Actions for the following MM-Code values are defined in CAMEL Phase 4:
--
--   CS domain MM events:
--   Location-update-in-same-VLR          MM-Code ::= '00000000'B
--   Location-update-to-other-VLR         MM-Code ::= '00000001'B
--   IMSI-Attach                          MM-Code ::= '00000010'B
--   MS-initiated-IMSI-Detach             MM-Code ::= '00000011'B
--   Network-initiated-IMSI-Detach        MM-Code ::= '00000100'B
--
--   PS domain MM events:
--   Routeing-Area-update-in-same-SGSN    MM-Code ::= '10000000'B
--   Routeing-Area-update-to-other-SGSN-update-from-new-SGSN
--        MM-Code ::= '10000001'B
--   Routeing-Area-update-to-other-SGSN-disconnect-by-detach
--        MM-Code ::= '10000010'B
--   GPRS-Attach                          MM-Code ::= '10000011'B
--   MS-initiated-GPRS-Detach             MM-Code ::= '10000100'B
--   Network-initiated-GPRS-Detach        MM-Code ::= '10000101'B
--   Network-initiated-transfer-to-MS-not-reachable-for-paging
--        MM-Code ::= '10000110'B
--
--   If the MSC receives any other MM-code than the ones listed above for the
--   CS domain, then the MSC shall ignore that MM-code.
--   If the SGSN receives any other MM-code than the ones listed above for the
--   PS domain, then the SGSN shall ignore that MM-code.
```

```
T-CSI ::= SEQUENCE {
     t-BcsmCamelTDPDataList              T-BcsmCamelTDPDataList,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     camelCapabilityHandling             [0] CamelCapabilityHandling       OPTIONAL,
     notificationToCSE                   [1] NULL  OPTIONAL,
     csi-Active                          [2] NULL  OPTIONAL}
--   notificationToCSE and csi-Active shall not be present when VT-CSI/T-CSI is sent
--   to VLR/GMSC.
--   They may only be included in ATSI/ATM ack/NSDC message.
--   T-CSI shall not be segmented.
```

```
T-BcsmCamelTDPDataList ::= SEQUENCE SIZE (1..maxNumOfCamelTDPData) OF
     T-BcsmCamelTDPData
     --- T-BcsmCamelTDPDataList shall not contain more than one instance of
     --- T-BcsmCamelTDPData containing the same value for t-BcsmTriggerDetectionPoint.
     --- For CAMEL Phase 2, this means that only one instance of T-BcsmCamelTDPData is
allowed
     --- with t-BcsmTriggerDetectionPoint being equal to DP12.
     --- For CAMEL Phase 3, more TDP's are allowed.
```

```
T-BcsmCamelTDPData ::= SEQUENCE {
     t-BcsmTriggerDetectionPoint         T-BcsmTriggerDetectionPoint,
     serviceKey                          ServiceKey,
     gsmSCF-Address                      [0] ISDN-AddressString,
     defaultCallHandling                 [1] DefaultCallHandling,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...}
```

```
T-BcsmTriggerDetectionPoint ::= ENUMERATED {
     termAttemptAuthorized (12),
     ... ,
     tBusy (13),
     tNoAnswer (14)}
     -- exception handling:
     -- For T-BcsmCamelTDPData sequences containing this parameter with any other
     -- value than the ones listed above, the receiver shall ignore the whole
     -- T-BcsmCamelTDPData sequence.
```

```
-- gprs location information retrieval types
```

```
SendRoutingInfoForGprsArg ::= SEQUENCE {
     imsi     [0] IMSI,
     ggsn-Address                        [1] GSN-Address                   OPTIONAL,
     ggsn-Number                         [2]  ISDN-AddressString,
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     ...}
```

```
SendRoutingInfoForGprsRes ::= SEQUENCE {
     sgsn-Address                        [0] GSN-Address,
     ggsn-Address                        [1]  GSN-Address                    OPTIONAL,
     mobileNotReachableReason            [2]  AbsentSubscriberDiagnosticSM  OPTIONAL,
     extensionContainer                  [3] ExtensionContainer              OPTIONAL,
     ...}
```

-- failure report types

```
FailureReportArg ::= SEQUENCE {
     imsi     [0] IMSI,
     ggsn-Number                         [1] ISDN-AddressString              ,
     ggsn-Address                        [2] GSN-Address                     OPTIONAL,
     extensionContainer                  [3] ExtensionContainer              OPTIONAL,
     ...}
```

```
FailureReportRes ::= SEQUENCE {
     ggsn-Address                        [0] GSN-Address                     OPTIONAL,
     extensionContainer                  [1] ExtensionContainer              OPTIONAL,
     ...}
```

-- gprs notification types

```
NoteMsPresentForGprsArg ::= SEQUENCE {
     imsi     [0] IMSI,
     sgsn-Address                        [1] GSN-Address,
     ggsn-Address                        [2] GSN-Address                     OPTIONAL,
     extensionContainer                  [3] ExtensionContainer              OPTIONAL,
     ...}
```

```
NoteMsPresentForGprsRes ::= SEQUENCE {
     extensionContainer                  [0] ExtensionContainer              OPTIONAL,
     ...}
```

-- fault recovery types

```
ResetArg ::= SEQUENCE {
     sendingNodenumber                   SendingNode-Number,
     hlr-List  HLR-List                  OPTIONAL,
     -- The hlr-List parameter shall only be applicable for a restart of the HSS/HLR.
     extensionContainer                  [0] ExtensionContainer              OPTIONAL,
     ...,
     reset-Id-List                       [1]  Reset-Id-List                  OPTIONAL,
     subscriptionData                    [2]  InsertSubscriberDataArg        OPTIONAL,
     subscriptionDataDeletion            [3]  DeleteSubscriberDataArg        OPTIONAL}
```

```
SendingNode-Number ::= CHOICE {
     hlr-Number                          ISDN-AddressString,
     css-Number                          [1] ISDN-AddressString}
```

```
RestoreDataArg ::= SEQUENCE {
     imsi     IMSI,
     lmsi     LMSI                            OPTIONAL,
     extensionContainer                  ExtensionContainer                  OPTIONAL,
     ... ,
     vlr-Capability                      [6] VLR-Capability                  OPTIONAL,
     restorationIndicator                [7]  NULL OPTIONAL
 }
```

```
RestoreDataRes ::= SEQUENCE {
     hlr-Number                          ISDN-AddressString,
     msNotReachable                      NULL     OPTIONAL,
     extensionContainer                  ExtensionContainer                  OPTIONAL,
     ...}
```

-- VBS/VGCS types

```
VBSDataList ::= SEQUENCE SIZE (1..maxNumOfVBSGroupIds) OF
          VoiceBroadcastData
```

```
VGCSDataList ::= SEQUENCE SIZE (1..maxNumOfVGCSGroupIds) OF
          VoiceGroupCallData
```

```
maxNumOfVBSGroupIds  INTEGER ::= 50
```

```
maxNumOfVGCSGroupIds  INTEGER ::= 50
```

```
VoiceGroupCallData  ::= SEQUENCE {
     groupId  GroupId,
     -- groupId shall be filled with six TBCD fillers (1111)if the longGroupId is present
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     additionalSubscriptions             AdditionalSubscriptions           OPTIONAL,
     additionalInfo                      [0] AdditionalInfo                OPTIONAL,
     longGroupId                         [1] Long-GroupId                  OPTIONAL }

     -- VoiceGroupCallData containing a longGroupId shall not be sent to VLRs that did not
     -- indicate support of long Group IDs within the Update Location or Restore Data
     -- request message
```

```
AdditionalInfo ::= BIT STRING (SIZE (1..136))
--   Refers to Additional Info as specified in 3GPP TS 43.068
```

```
AdditionalSubscriptions ::= BIT STRING {
     privilegedUplinkRequest (0),
     emergencyUplinkRequest (1),
     emergencyReset (2)} (SIZE (3..8))
-- Other bits than listed above shall be discarded.
```

```
VoiceBroadcastData ::= SEQUENCE {
     groupid  GroupId,
     -- groupId shall be filled with six TBCD fillers (1111)if the longGroupId is present
     broadcastInitEntitlement            NULL     OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     longGroupId                         [0] Long-GroupId                  OPTIONAL }

     -- VoiceBroadcastData containing a longGroupId shall not be sent to VLRs that did not
     -- indicate support of long Group IDs within the Update Location or Restore Data
     -- request message
```

```
GroupId  ::= TBCD-STRING (SIZE (3))
     -- When Group-Id is less than six characters in length, the TBCD filler (1111)
     -- is used to fill unused half octets.
     -- Refers to the Group Identification as specified in 3GPP TS 23.003
     -- and 3GPP TS 43.068/ 43.069
```

```
Long-GroupId  ::= TBCD-STRING (SIZE (4))
     -- When Long-Group-Id is less than eight characters in length, the TBCD filler (1111)
     -- is used to fill unused half octets.
     -- Refers to the Group Identification as specified in 3GPP TS 23.003
     -- and 3GPP TS 43.068/ 43.069
```

-- provide subscriber info types

```
ProvideSubscriberInfoArg ::= SEQUENCE {
     imsi [0] IMSI,
     lmsi [1] LMSI                       OPTIONAL,
     requestedInfo                       [2] RequestedInfo,
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     ...,
     callPriority                        [4]  EMLPP-Priority               OPTIONAL
     }
```

```
ProvideSubscriberInfoRes ::= SEQUENCE {
     subscriberInfo                      SubscriberInfo,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
SubscriberInfo ::= SEQUENCE {
     locationInformation                 [0] LocationInformation           OPTIONAL,
     subscriberState                     [1] SubscriberState               OPTIONAL,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ... ,
     locationInformationGPRS             [3] LocationInformationGPRS       OPTIONAL,
     ps-SubscriberState                  [4] PS-SubscriberState            OPTIONAL,
     imei     [5] IMEI                   OPTIONAL,
     ms-Classmark2                       [6] MS-Classmark2                 OPTIONAL,
     gprs-MS-Class                       [7] GPRSMSClass                   OPTIONAL,
     mnpInfoRes                          [8] MNPInfoRes                    OPTIONAL,
     imsVoiceOverPS-SessionsIndication   [9] IMS-VoiceOverPS-SessionsInd   OPTIONAL,
     lastUE-ActivityTime                 [10] Time OPTIONAL,
     lastRAT-Type                        [11] Used-RAT-Type                OPTIONAL,
     eps-SubscriberState                 [12] PS-SubscriberState           OPTIONAL,
     locationInformationEPS              [13] LocationInformationEPS       OPTIONAL,
     timeZone [14] TimeZone              OPTIONAL,
     daylightSavingTime                  [15] DaylightSavingTime           OPTIONAL,
     locationInformation5GS              [16] LocationInformation5GS       OPTIONAL }

--   If the HLR receives locationInformation, subscriberState or ms-Classmark2 from an SGSN or
--   MME (via an IWF), it shall discard them.
--   If the HLR receives locationInformationGPRS, ps-SubscriberState, gprs-MS-Class or
--   locationInformationEPS (outside the locationInformation IE) from a VLR, it shall
--   discard them.
--   If the HLR receives parameters which it has not requested, it shall discard them.
--   The locationInformation5GS IE should be absent if UE did not access via 5GS and IM-SSF.
```

```
IMS-VoiceOverPS-SessionsInd ::= ENUMERATED {
     imsVoiceOverPS-SessionsNotSupported  (0),
     imsVoiceOverPS-SessionsSupported     (1),
     unknown   (2)
     }
--   "unknown" shall not be used within ProvideSubscriberInfoRes
```

```
TimeZone ::= OCTET STRING (SIZE (2..3))
--   Refer to the 3GPP TS 29.272 [144] for details.
```

```
DaylightSavingTime ::= ENUMERATED {
     noAdjustment                        (0),
     plusOneHourAdjustment               (1),
     plusTwoHoursAdjustment              (2)
     }
--   Refer to the 3GPP TS 29.272 [144] for details.
```

```
MNPInfoRes ::= SEQUENCE {
     routeingNumber                      [0] RouteingNumber                OPTIONAL,
     imsi     [1] IMSI                   OPTIONAL,
     msisdn   [2] ISDN-AddressString     OPTIONAL,
     numberPortabilityStatus             [3] NumberPortabilityStatus       OPTIONAL,
     extensionContainer                  [4] ExtensionContainer            OPTIONAL,
     ... }
--   The IMSI parameter contains a generic IMSI, i.e. it is not tied necessarily to the
--   Subscriber. MCC and MNC values in this IMSI shall point to the Subscription Network of
--   the Subscriber. See 3GPP TS 23.066 [108].
```

```
RouteingNumber ::= TBCD-STRING (SIZE (1..5))
```

```
NumberPortabilityStatus ::= ENUMERATED {
     notKnownToBePorted                  (0),
     ownNumberPortedOut                  (1),
     foreignNumberPortedToForeignNetwork (2),
     ...,
     ownNumberNotPortedOut               (4),
     foreignNumberPortedIn               (5)
     }
     --   exception handling:
     --  reception of other values than the ones listed the receiver shall ignore the
     --  whole NumberPortabilityStatus;
     --  ownNumberNotPortedOut or foreignNumberPortedIn may only be included in Any Time
     --  Interrogation message.
```

```
MS-Classmark2 ::= OCTET STRING (SIZE (3))
     -- This parameter carries the value part of the MS Classmark 2 IE defined in
     -- 3GPP TS 24.008 [35].
```

```
GPRSMSClass ::= SEQUENCE {
     mSNetworkCapability                [0] MSNetworkCapability,
     mSRadioAccessCapability            [1] MSRadioAccessCapability       OPTIONAL
     }
```

```
MSNetworkCapability ::= OCTET STRING (SIZE (1..8))
     -- This parameter carries the value part of the MS Network Capability IE defined in
     -- 3GPP TS 24.008 [35].
```

```
MSRadioAccessCapability ::= OCTET STRING (SIZE (1..50))
     -- This parameter carries the value part of the MS Radio Access Capability IE defined in
     -- 3GPP TS 24.008 [35].
```

```
RequestedInfo ::= SEQUENCE {
     locationInformation                [0] NULL  OPTIONAL,
     subscriberState                    [1] NULL  OPTIONAL,
     extensionContainer                 [2] ExtensionContainer            OPTIONAL,
     ...,
     currentLocation                    [3] NULL  OPTIONAL,
     requestedDomain                    [4] DomainType                    OPTIONAL,
     imei      [6] NULL                 OPTIONAL,
     ms-classmark                       [5] NULL  OPTIONAL,
     mnpRequestedInfo                   [7] NULL  OPTIONAL,
     locationInformationEPS-Supported   [11] NULL OPTIONAL,
     t-adsData [8] NULL                 OPTIONAL,
     requestedNodes                     [9] RequestedNodes                OPTIONAL,
     servingNodeIndication              [10] NULL OPTIONAL,
     localTimeZoneRequest               [12] NULL OPTIONAL
 }

--   currentLocation and locationInformationEPS-Supported shall be absent if
--   locationInformation is absent
--   t-adsData shall be absent in messages sent to the VLR
--   requestedNodes shall be absent if requestedDomain is "cs-Domain"
--   servingNodeIndication shall be absent if locationInformation is absent;
--   servingNodeIndication shall be absent if current location is present;
--   servingNodeIndication indicates by its presence that only the serving node's
--   address (MME-Name or SGSN-Number or VLR-Number) is requested.
```

```
DomainType ::=  ENUMERATED {
     cs-Domain (0),
     ps-Domain (1),
     ...}
-- exception handling:
-- reception of values > 1 shall be mapped to 'cs-Domain'
```

```
RequestedNodes ::= BIT STRING {
     mme  (0),
     sgsn (1)} (SIZE (1..8))
-- Other bits than listed above shall be discarded.
```

```
LocationInformation ::= SEQUENCE {
     ageOfLocationInformation            AgeOfLocationInformation          OPTIONAL,
     geographicalInformation             [0] GeographicalInformation       OPTIONAL,
     vlr-number                          [1] ISDN-AddressString            OPTIONAL,
     locationNumber                      [2] LocationNumber                OPTIONAL,
     cellGlobalIdOrServiceAreaIdOrLAI    [3] CellGlobalIdOrServiceAreaIdOrLAI   OPTIONAL,
     extensionContainer                  [4] ExtensionContainer            OPTIONAL,
     ... ,
     selectedLSA-Id                      [5] LSAIdentity                   OPTIONAL,
     msc-Number                          [6] ISDN-AddressString            OPTIONAL,
     geodeticInformation                 [7] GeodeticInformation           OPTIONAL,
     currentLocationRetrieved            [8] NULL  OPTIONAL,
     sai-Present                         [9] NULL  OPTIONAL,
     locationInformationEPS              [10] LocationInformationEPS       OPTIONAL,
     userCSGInformation                  [11] UserCSGInformation           OPTIONAL }
-- sai-Present indicates that the cellGlobalIdOrServiceAreaIdOrLAI parameter contains
-- a Service Area Identity.
-- currentLocationRetrieved shall be present
-- if the location information were retrieved after a successfull paging.
-- if the locationinformationEPS IE is present then the cellGlobalIdOrServiceAreaIdOrLAI IE,
-- the ageOfLocationInformation IE, the geographicalInformation IE, the geodeticInformation IE
-- and the currentLocationRetrieved IE (outside the locationInformationEPS IE) shall be
-- absent. As an exception, both the cellGlobalIdOrServiceAreaIdOrLAI IE including an LAI and
-- the locationinformationEPS IE may be present in a MAP-NOTE-MM-EVENT.
-- UserCSGInformation contains the CSG ID, Access mode, and the CSG Membership Indication in
-- the case the Access mode is Hybrid Mode.
-- The locationInformationEPS IE should be absent if locationInformationEPS-Supported was not
-- received in the RequestedInfo IE.
```

```
LocationInformationEPS ::= SEQUENCE {
     e-utranCellGlobalIdentity           [0] E-UTRAN-CGI                   OPTIONAL,
     trackingAreaIdentity                [1] TA-Id OPTIONAL,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     geographicalInformation             [3] GeographicalInformation       OPTIONAL,
     geodeticInformation                 [4] GeodeticInformation           OPTIONAL,
     currentLocationRetrieved            [5] NULL                          OPTIONAL,
     ageOfLocationInformation            [6] AgeOfLocationInformation      OPTIONAL,
     ...,
     mme-Name [7] DiameterIdentity       OPTIONAL }
-- currentLocationRetrieved shall be present if the location information
-- was retrieved after successful paging.
```

```
LocationInformationGPRS ::= SEQUENCE {
     cellGlobalIdOrServiceAreaIdOrLAI    [0] CellGlobalIdOrServiceAreaIdOrLAI OPTIONAL,
     routeingAreaIdentity                [1] RAIdentity                    OPTIONAL,
     geographicalInformation             [2] GeographicalInformation       OPTIONAL,
     sgsn-Number                         [3] ISDN-AddressString            OPTIONAL,
     selectedLSAIdentity                 [4] LSAIdentity                   OPTIONAL,
     extensionContainer                  [5] ExtensionContainer            OPTIONAL,
     ...,
     sai-Present                         [6] NULL  OPTIONAL,
     geodeticInformation                 [7] GeodeticInformation           OPTIONAL,
     currentLocationRetrieved            [8] NULL  OPTIONAL,
     ageOfLocationInformation            [9] AgeOfLocationInformation      OPTIONAL,
     userCSGInformation                  [10] UserCSGInformation           OPTIONAL }
-- sai-Present indicates that the cellGlobalIdOrServiceAreaIdOrLAI parameter contains
-- a Service Area Identity.
-- currentLocationRetrieved shall be present if the location information
-- was retrieved after successful paging.
-- UserCSGInformation contains the CSG ID, Access mode, and the CSG Membership Indication in
-- the case the Access mode is Hybrid Mode.
```

```
LocationInformation5GS ::= SEQUENCE {
     nrCellGlobalIdentity                [0] NR-CGI                          OPTIONAL,
     e-utranCellGlobalIdentity           [1] E-UTRAN-CGI                     OPTIONAL,
     geographicalInformation             [2] GeographicalInformation         OPTIONAL,
     geodeticInformation                 [3] GeodeticInformation             OPTIONAL,
     amf-address                         [4] FQDN                            OPTIONAL,
     trackingAreaIdentity                [5] TA-Id                           OPTIONAL,
     currentLocationRetrieved            [6] NULL                            OPTIONAL,
     ageOfLocationInformation            [7] AgeOfLocationInformation        OPTIONAL,
     vplmnId   [8] PLMN-Id               OPTIONAL,
     localtimeZone                       [9] TimeZone                        OPTIONAL,
     rat-Type  [10] Used-RAT-Type        OPTIONAL,
     extensionContainer                  [11] ExtensionContainer             OPTIONAL,
     ...}
-- currentLocationRetrieved shall be present if the location information
-- was retrieved after successful paging.
```

```
UserCSGInformation ::= SEQUENCE {
     csg-Id    [0] CSG-Id,
     extensionContainer                  [1] ExtensionContainer              OPTIONAL,
     ...,
     accessMode                          [2] OCTET STRING (SIZE(1))          OPTIONAL,
     cmi       [3] OCTET STRING (SIZE(1)) OPTIONAL }
-- The encoding of the accessMode and cmi parameters are as defined in 3GPP TS 29.060 [105].
```

```
GeographicalInformation ::= OCTET STRING (SIZE (8))
--   Refers to geographical Information defined in 3GPP TS 23.032.
--   Only the description of an ellipsoid point with uncertainty circle
--   as specified in 3GPP TS 23.032 is allowed to be used
--   The internal structure according to 3GPP TS 23.032 is as follows:
--   Type of shape (ellipsoid point with uncertainty circle)              1 octet
--   Degrees of Latitude                      3 octets
--   Degrees of Longitude                     3 octets
--   Uncertainty code                         1 octet
```

```
GeodeticInformation ::= OCTET STRING (SIZE (10))
--   Refers to Calling Geodetic Location defined in Q.763 (1999).
--   Only the description of an ellipsoid point with uncertainty circle
--   as specified in Q.763 (1999) is allowed to be used
--   The internal structure according to Q.763 (1999) is as follows:
--   Screening and presentation indicators    1 octet
--   Type of shape (ellipsoid point with uncertainty circle)              1 octet
--   Degrees of Latitude                      3 octets
--   Degrees of Longitude                     3 octets
--   Uncertainty code                         1 octet
--   Confidence                               1 octet
```

```
LocationNumber ::= OCTET STRING (SIZE (2..10))
     -- the internal structure is defined in ITU-T Rec Q.763
```

```
SubscriberState ::= CHOICE {
     assumedIdle                         [0] NULL,
     camelBusy [1] NULL,
     netDetNotReachable                  NotReachableReason,
     notProvidedFromVLR                  [2] NULL}
```

```
PS-SubscriberState ::= CHOICE {
     notProvidedFromSGSNorMME            [0] NULL,
     ps-Detached                         [1] NULL,
     ps-AttachedNotReachableForPaging    [2] NULL,
     ps-AttachedReachableForPaging       [3] NULL,
     ps-PDP-ActiveNotReachableForPaging  [4] PDP-ContextInfoList,
     ps-PDP-ActiveReachableForPaging     [5] PDP-ContextInfoList,
     netDetNotReachable                  NotReachableReason }
```

```
PDP-ContextInfoList ::= SEQUENCE SIZE (1..maxNumOfPDP-Contexts) OF
          PDP-ContextInfo
```

```
PDP-ContextInfo ::= SEQUENCE {
    pdp-ContextIdentifier               [0] ContextId,
    pdp-ContextActive                   [1] NULL  OPTIONAL,
    pdp-Type  [2] PDP-Type,
    pdp-Address                         [3] PDP-Address                 OPTIONAL,
    apn-Subscribed                      [4] APN   OPTIONAL,
    apn-InUse [5] APN                   OPTIONAL,
    nsapi     [6] NSAPI                 OPTIONAL,
    transactionId                       [7] TransactionId               OPTIONAL,
    teid-ForGnAndGp                     [8] TEID  OPTIONAL,
    teid-ForIu                          [9] TEID  OPTIONAL,
    ggsn-Address                        [10] GSN-Address                OPTIONAL,
    qos-Subscribed                      [11] Ext-QoS-Subscribed         OPTIONAL,
    qos-Requested                       [12] Ext-QoS-Subscribed         OPTIONAL,
    qos-Negotiated                      [13] Ext-QoS-Subscribed         OPTIONAL,
    chargingId                          [14] GPRSChargingID             OPTIONAL,
    chargingCharacteristics             [15] ChargingCharacteristics    OPTIONAL,
    rnc-Address                         [16] GSN-Address                OPTIONAL,
    extensionContainer                  [17] ExtensionContainer         OPTIONAL,
    ...,
    qos2-Subscribed                     [18] Ext2-QoS-Subscribed        OPTIONAL,
    -- qos2-Subscribed may be present only if qos-Subscribed is present.
    qos2-Requested                      [19] Ext2-QoS-Subscribed        OPTIONAL,
    -- qos2-Requested may be present only if qos-Requested is present.
    qos2-Negotiated                     [20] Ext2-QoS-Subscribed        OPTIONAL,
    -- qos2-Negotiated may be present only if qos-Negotiated is present.
    qos3-Subscribed                     [21] Ext3-QoS-Subscribed        OPTIONAL,
    -- qos3-Subscribed may be present only if qos2-Subscribed is present.
    qos3-Requested                      [22] Ext3-QoS-Subscribed        OPTIONAL,
    -- qos3-Requested may be present only if qos2-Requested is present.
    qos3-Negotiated                     [23] Ext3-QoS-Subscribed        OPTIONAL,
    -- qos3-Negotiated may be present only if qos2-Negotiated is present.
    qos4-Subscribed                     [25] Ext4-QoS-Subscribed        OPTIONAL,
    -- qos4-Subscribed may be present only if qos3-Subscribed is present.
    qos4-Requested                      [26] Ext4-QoS-Subscribed        OPTIONAL,
    -- qos4-Requested may be present only if qos3-Requested is present.
    qos4-Negotiated                     [27] Ext4-QoS-Subscribed        OPTIONAL,
    -- qos4-Negotiated may be present only if qos3-Negotiated is present.
    ext-pdp-Type                        [28] Ext-PDP-Type               OPTIONAL,
    -- contains the value IPv4v6 defined in 3GPP TS 29.060 [105], if the PDP can be
    -- accessed by dual-stack UEs.
    ext-pdp-Address                     [29] PDP-Address                OPTIONAL
    -- contains an additional IP address in case of dual-stack static IP address assignment
    -- for the UE.
    -- it may contain an IPv4 or an IPv6 address/prefix, and it may be present
    -- only if pdp-Address is present; if both are present, each parameter shall
    -- contain a different type of address (IPv4 or IPv6).

}
```

```
NSAPI ::= INTEGER (0..15)
--   This type is used to indicate the Network layer Service Access Point
```

```
TransactionId ::= OCTET STRING (SIZE (1..2))
--   This type carries the value part of the transaction identifier which is used in the
--   session management messages on the access interface. The encoding is defined in
--   3GPP TS 24.008
```

```
TEID ::= OCTET STRING (SIZE (4))
--   This type carries the value part of the Tunnel Endpoint Identifier which is used to
--   distinguish between different tunnels between the same pair of entities which communicate
--   using the GPRS Tunnelling Protocol The encoding is defined in 3GPP TS 29.060.
```

```
GPRSChargingID ::= OCTET STRING (SIZE (4))
--   The Charging ID is a unique four octet value generated by the GGSN when
--   a PDP Context is activated. A Charging ID is generated for each activated context.
--   The encoding is defined in 3GPP TS 29.060.
```

```
NotReachableReason ::= ENUMERATED {
    msPurged (0),
    imsiDetached (1),
    restrictedArea (2),
    notRegistered (3)}
```

```
-- any time interrogation info types
```

```
AnyTimeInterrogationArg ::= SEQUENCE {
     subscriberIdentity                  [0] SubscriberIdentity,
     requestedInfo                       [1] RequestedInfo,
     gsmSCF-Address                      [3] ISDN-AddressString,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...}
```

```
AnyTimeInterrogationRes ::= SEQUENCE {
     subscriberInfo                      SubscriberInfo,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

-- any time information handling types

```
AnyTimeSubscriptionInterrogationArg ::= SEQUENCE {
     subscriberIdentity                  [0] SubscriberIdentity,
     requestedSubscriptionInfo           [1] RequestedSubscriptionInfo,
     gsmSCF-Address                      [2] ISDN-AddressString,
     extensionContainer                  [3] ExtensionContainer            OPTIONAL,
     longFTN-Supported                   [4]  NULL OPTIONAL,
     ...}
```

```
AnyTimeSubscriptionInterrogationRes ::= SEQUENCE {
     callForwardingData                  [1] CallForwardingData            OPTIONAL,
     callBarringData                     [2] CallBarringData               OPTIONAL,
     odb-Info  [3] ODB-Info              OPTIONAL,
     camel-SubscriptionInfo              [4] CAMEL-SubscriptionInfo        OPTIONAL,
     supportedVLR-CAMEL-Phases           [5] SupportedCamelPhases          OPTIONAL,
     supportedSGSN-CAMEL-Phases          [6] SupportedCamelPhases          OPTIONAL,
     extensionContainer                  [7] ExtensionContainer            OPTIONAL,
     ... ,
     offeredCamel4CSIsInVLR              [8] OfferedCamel4CSIs             OPTIONAL,
     offeredCamel4CSIsInSGSN             [9] OfferedCamel4CSIs             OPTIONAL,
     msisdn-BS-List                      [10] MSISDN-BS-List               OPTIONAL,
     csg-SubscriptionDataList            [11] CSG-SubscriptionDataList     OPTIONAL,
     cw-Data   [12]                      CallWaitingData                   OPTIONAL,
     ch-Data   [13]                      CallHoldData                      OPTIONAL,
     clip-Data [14] ClipData             OPTIONAL,
     clir-Data [15]                      ClirData  OPTIONAL,
     ect-data  [16] EctData              OPTIONAL }
```

```
CallWaitingData ::= SEQUENCE {
     cwFeatureList                       [1] Ext-CwFeatureList,
     notificationToCSE                   [2] NULL  OPTIONAL,
     ... }
```

```
Ext-CwFeatureList ::= SEQUENCE SIZE (1..maxNumOfExt-BasicServiceGroups) OF
          Ext-CwFeature
```

```
Ext-CwFeature ::= SEQUENCE {
     basicService                        [1] Ext-BasicServiceCode,
     ss-Status [2] Ext-SS-Status,
     ... }
```

```
ClipData ::= SEQUENCE {
     ss-Status [1] Ext-SS-Status,
     overrideCategory                    [2] OverrideCategory,
     notificationToCSE                   [3] NULL  OPTIONAL,
     ... }
```

```
ClirData ::= SEQUENCE {
     ss-Status [1] Ext-SS-Status,
     cliRestrictionOption                [2] CliRestrictionOption          OPTIONAL,
     notificationToCSE                   [3] NULL  OPTIONAL,
     ... }
```

```
CallHoldData ::= SEQUENCE {
     ss-Status [1] Ext-SS-Status,
     notificationToCSE                   [2] NULL  OPTIONAL,
     ... }
```

```
EctData ::= SEQUENCE {
     ss-Status [1] Ext-SS-Status,
     notificationToCSE                   [2] NULL  OPTIONAL,
     ... }
```

```
RequestedSubscriptionInfo ::= SEQUENCE {
    requestedSS-Info                    [1] SS-ForBS-Code                 OPTIONAL,
    odb      [2] NULL                   OPTIONAL,
    requestedCAMEL-SubscriptionInfo     [3] RequestedCAMEL-SubscriptionInfo OPTIONAL,
    supportedVLR-CAMEL-Phases           [4] NULL  OPTIONAL,
    supportedSGSN-CAMEL-Phases          [5] NULL  OPTIONAL,
    extensionContainer                  [6] ExtensionContainer            OPTIONAL,
    ...,
    additionalRequestedCAMEL-SubscriptionInfo
         [7] AdditionalRequestedCAMEL-SubscriptionInfo
                                        OPTIONAL,
    msisdn-BS-List                      [8] NULL  OPTIONAL,
    csg-SubscriptionDataRequested       [9] NULL  OPTIONAL,
    cw-Info  [10]                       NULL OPTIONAL,
    clip-Info [11] NULL                 OPTIONAL,
    clir-Info [12] NULL                 OPTIONAL,
    hold-Info [13] NULL                 OPTIONAL,
    ect-Info  [14] NULL                 OPTIONAL }
```

```
MSISDN-BS-List ::= SEQUENCE SIZE (1..maxNumOfMSISDN) OF
        MSISDN-BS
```

```
maxNumOfMSISDN  INTEGER ::= 50
```

```
MSISDN-BS ::= SEQUENCE {
    msisdn                              ISDN-AddressString,
    basicServiceList                    [0]  BasicServiceList             OPTIONAL,
    extensionContainer                  [1]  ExtensionContainer           OPTIONAL,
    ...}
```

```
RequestedCAMEL-SubscriptionInfo ::= ENUMERATED {
    o-CSI    (0),
    t-CSI    (1),
    vt-CSI   (2),
    tif-CSI  (3),
    gprs-CSI (4),
    mo-sms-CSI                          (5),
    ss-CSI   (6),
    m-CSI    (7),
    d-csi    (8)}
```

```
AdditionalRequestedCAMEL-SubscriptionInfo ::= ENUMERATED {
    mt-sms-CSI                          (0),
    mg-csi   (1),
    o-IM-CSI (2),
    d-IM-CSI (3),
    vt-IM-CSI(4),
    ...}
--  exception handling: unknown values shall be discarded by the receiver.
```

```
CallForwardingData ::= SEQUENCE {
    forwardingFeatureList               Ext-ForwFeatureList,
    notificationToCSE                   NULL     OPTIONAL,
    extensionContainer                  [0] ExtensionContainer            OPTIONAL,
    ...}
```

```
CallBarringData ::= SEQUENCE {
    callBarringFeatureList              Ext-CallBarFeatureList,
    password  Password                  OPTIONAL,
    wrongPasswordAttemptsCounter        WrongPasswordAttemptsCounter      OPTIONAL,
    notificationToCSE                   NULL     OPTIONAL,
    extensionContainer                  ExtensionContainer                OPTIONAL,
    ...}
```

```
WrongPasswordAttemptsCounter ::= INTEGER (0..4)
```

```
ODB-Info ::= SEQUENCE {
    odb-Data  ODB-Data,
    notificationToCSE                   NULL     OPTIONAL,
    extensionContainer                  ExtensionContainer                OPTIONAL,
    ...}
```

```
CAMEL-SubscriptionInfo ::= SEQUENCE {
    o-CSI     [0]                          O-CSI     OPTIONAL,
    o-BcsmCamelTDP-CriteriaList            [1]  O-BcsmCamelTDPCriteriaList    OPTIONAL,
    d-CSI     [2]                          D-CSI     OPTIONAL,
    t-CSI     [3]                          T-CSI     OPTIONAL,
    t-BCSM-CAMEL-TDP-CriteriaList          [4]  T-BCSM-CAMEL-TDP-CriteriaList OPTIONAL,
    vt-CSI    [5]                          T-CSI     OPTIONAL,
    vt-BCSM-CAMEL-TDP-CriteriaList         [6]  T-BCSM-CAMEL-TDP-CriteriaList OPTIONAL,
    tif-CSI   [7]                          NULL OPTIONAL,
    tif-CSI-NotificationToCSE              [8]  NULL OPTIONAL,
    gprs-CSI  [9]                          GPRS-CSI  OPTIONAL,
    mo-sms-CSI                             [10] SMS-CSI                       OPTIONAL,
    ss-CSI    [11]                         SS-CSI    OPTIONAL,
    m-CSI     [12]                         M-CSI     OPTIONAL,
    extensionContainer                     [13] ExtensionContainer            OPTIONAL,
    ...,
    specificCSIDeletedList                 [14] SpecificCSI-Withdraw          OPTIONAL,
    mt-sms-CSI                             [15] SMS-CSI                       OPTIONAL,
    mt-smsCAMELTDP-CriteriaList            [16] MT-smsCAMELTDP-CriteriaList   OPTIONAL,
    mg-csi    [17]                         MG-CSI    OPTIONAL,
    o-IM-CSI  [18] O-CSI                   OPTIONAL,
    o-IM-BcsmCamelTDP-CriteriaList         [19] O-BcsmCamelTDPCriteriaList    OPTIONAL,
    d-IM-CSI  [20] D-CSI                   OPTIONAL,
    vt-IM-CSI [21] T-CSI                   OPTIONAL,
    vt-IM-BCSM-CAMEL-TDP-CriteriaList      [22] T-BCSM-CAMEL-TDP-CriteriaList OPTIONAL
    }
```

```
AnyTimeModificationArg ::= SEQUENCE {
    subscriberIdentity                     [0]  SubscriberIdentity,
    gsmSCF-Address                         [1]  ISDN-AddressString,
    modificationRequestFor-CF-Info         [2]  ModificationRequestFor-CF-Info OPTIONAL,
    modificationRequestFor-CB-Info         [3]  ModificationRequestFor-CB-Info OPTIONAL,
    modificationRequestFor-CSI             [4]  ModificationRequestFor-CSI    OPTIONAL,
    extensionContainer                     [5]  ExtensionContainer            OPTIONAL,
    longFTN-Supported                      [6]  NULL OPTIONAL,
    ...,
    modificationRequestFor-ODB-data        [7]  ModificationRequestFor-ODB-data OPTIONAL,
    modificationRequestFor-IP-SM-GW-Data [8]  ModificationRequestFor-IP-SM-GW-Data OPTIONAL,
    activationRequestForUE-reachability  [9]  RequestedServingNode          OPTIONAL,
    modificationRequestFor-CSG             [10] ModificationRequestFor-CSG    OPTIONAL,
    modificationRequestFor-CW-Data         [11] ModificationRequestFor-CW-Info OPTIONAL,
    modificationRequestFor-CLIP-Data       [12] ModificationRequestFor-CLIP-Info  OPTIONAL,
    modificationRequestFor-CLIR-Data       [13] ModificationRequestFor-CLIR-Info  OPTIONAL,
    modificationRequestFor-HOLD-Data       [14] ModificationRequestFor-CH-Info OPTIONAL,
    modificationRequestFor-ECT-Data        [15] ModificationRequestFor-ECT-Info   OPTIONAL }
```

```
ModificationRequestFor-CW-Info ::= SEQUENCE {
    basicService                           [0]  Ext-BasicServiceCode          OPTIONAL,
    ss-Status [1]                          Ext-SS-Status                      OPTIONAL,
    modifyNotificationToCSE                [2]  ModificationInstruction       OPTIONAL,
    extensionContainer                     [3]  ExtensionContainer            OPTIONAL,
    ...}
```

```
ModificationRequestFor-CH-Info ::= SEQUENCE {
    ss-Status [0]                          Ext-SS-Status                      OPTIONAL,
    modifyNotificationToCSE                [1]  ModificationInstruction       OPTIONAL,
    extensionContainer                     [2]  ExtensionContainer            OPTIONAL,
    ...}
```

```
ModificationRequestFor-ECT-Info ::= SEQUENCE {
    ss-Status [0]                          Ext-SS-Status                      OPTIONAL,
    modifyNotificationToCSE                [1]  ModificationInstruction       OPTIONAL,
    extensionContainer                     [2]  ExtensionContainer            OPTIONAL,
    ...}
```

```
ModificationRequestFor-CLIR-Info ::= SEQUENCE {
    ss-Status [0]                          Ext-SS-Status                      OPTIONAL,
    cliRestrictionOption                   [1]  CliRestrictionOption          OPTIONAL,
    modifyNotificationToCSE                [2]  ModificationInstruction       OPTIONAL,
    extensionContainer                     [3]  ExtensionContainer            OPTIONAL,
    ...}
```

```
ModificationRequestFor-CLIP-Info ::= SEQUENCE {
     ss-Status [0]                          Ext-SS-Status                    OPTIONAL,
     overrideCategory                       [1]  OverrideCategory            OPTIONAL,
     modifyNotificationToCSE                [2]  ModificationInstruction     OPTIONAL,
     extensionContainer                     [3]  ExtensionContainer          OPTIONAL,
     ...}
```

```
ModificationRequestFor-CSG ::= SEQUENCE {
     modifyNotificationToCSE                [0]  ModificationInstruction     OPTIONAL,
     extensionContainer                     [1]  ExtensionContainer          OPTIONAL,
     ...}
```

```
RequestedServingNode ::= BIT STRING {
     mmeAndSgsn  (0)} (SIZE (1..8))
```

```
ServingNode ::= BIT STRING {
     mme (0),
     sgsn (1)} (SIZE (2..8))
-- Other bits than listed above shall be discarded.
```

```
AnyTimeModificationRes ::= SEQUENCE {
     ss-InfoFor-CSE                         [0]  Ext-SS-InfoFor-CSE          OPTIONAL,
     camel-SubscriptionInfo                 [1]  CAMEL-SubscriptionInfo      OPTIONAL,
     extensionContainer                     [2]  ExtensionContainer          OPTIONAL,
     ...,
     odb-Info  [3]                          ODB-Info  OPTIONAL,
     cw-Data   [4]                          CallWaitingData                  OPTIONAL,
     ch-Data   [5]                          CallHoldData                     OPTIONAL,
     clip-Data [6] ClipData                 OPTIONAL,
     clir-Data [7]                          ClirData  OPTIONAL,
     ect-data  [8] EctData                  OPTIONAL,
     serviceCentreAddress                   [9] AddressString                OPTIONAL
 }
```

```
ModificationRequestFor-CF-Info ::= SEQUENCE {
     ss-Code   [0]                          SS-Code,
     basicService                           [1]  Ext-BasicServiceCode        OPTIONAL,
     ss-Status [2]                          Ext-SS-Status                    OPTIONAL,
     forwardedToNumber                      [3]  AddressString               OPTIONAL,
     forwardedToSubaddress                  [4]  ISDN-SubaddressString       OPTIONAL,
     noReplyConditionTime                   [5]  Ext-NoRepCondTime           OPTIONAL,
     modifyNotificationToCSE                [6]  ModificationInstruction     OPTIONAL,
     extensionContainer                     [7]  ExtensionContainer          OPTIONAL,
     ...}
```

```
ModificationRequestFor-CB-Info ::= SEQUENCE {
     ss-Code   [0]                          SS-Code,
     basicService                           [1]  Ext-BasicServiceCode        OPTIONAL,
     ss-Status [2]                          Ext-SS-Status                    OPTIONAL,
     password  [3]                          Password  OPTIONAL,
     wrongPasswordAttemptsCounter           [4]  WrongPasswordAttemptsCounter  OPTIONAL,
     modifyNotificationToCSE                [5]  ModificationInstruction     OPTIONAL,
     extensionContainer                     [6]  ExtensionContainer          OPTIONAL,
     ...}
```

```
ModificationRequestFor-ODB-data ::= SEQUENCE {
     odb-data  [0]                          ODB-Data  OPTIONAL,
     modifyNotificationToCSE                [1]  ModificationInstruction     OPTIONAL,
     extensionContainer                     [2]  ExtensionContainer          OPTIONAL,
     ...}
```

```
ModificationRequestFor-CSI ::= SEQUENCE {
     requestedCamel-SubscriptionInfo        [0]  RequestedCAMEL-SubscriptionInfo,
     modifyNotificationToCSE                [1]  ModificationInstruction     OPTIONAL,
     modifyCSI-State                        [2]  ModificationInstruction     OPTIONAL,
     extensionContainer                     [3]  ExtensionContainer          OPTIONAL,
     ...,
     additionalRequestedCAMEL-SubscriptionInfo
          [4] AdditionalRequestedCAMEL-SubscriptionInfo
                                            OPTIONAL }
-- requestedCamel-SubscriptionInfo shall be discarded if
-- additionalRequestedCAMEL-SubscriptionInfo is received
```

```
ModificationRequestFor-IP-SM-GW-Data ::= SEQUENCE {
    modifyRegistrationStatus            [0]  ModificationInstruction      OPTIONAL,
    extensionContainer                  [1]  ExtensionContainer           OPTIONAL,
    ...,
    ip-sm-gw-DiameterAddress            [2]  NetworkNodeDiameterAddress   OPTIONAL
    -- ip-sm-gw-DiameterAddress may be present when ModificationInstruction is "activate"
    }
```

```
ModificationInstruction ::= ENUMERATED {
    deactivate                          (0),
    activate  (1)}
```

-- subscriber data modification notification types

```
NoteSubscriberDataModifiedArg ::= SEQUENCE {
    imsi     IMSI,
    msisdn   ISDN-AddressString,
    forwardingInfoFor-CSE               [0] Ext-ForwardingInfoFor-CSE     OPTIONAL,
    callBarringInfoFor-CSE              [1] Ext-CallBarringInfoFor-CSE    OPTIONAL,
    odb-Info  [2] ODB-Info              OPTIONAL,
    camel-SubscriptionInfo              [3] CAMEL-SubscriptionInfo        OPTIONAL,
    allInformationSent                  [4] NULL  OPTIONAL,
    extensionContainer                  ExtensionContainer                OPTIONAL,
    ...,
    ue-reachable                        [5] ServingNode                   OPTIONAL,
    csg-SubscriptionDataList            [6] CSG-SubscriptionDataList      OPTIONAL,
    cw-Data   [7]                       CallWaitingData                   OPTIONAL,
    ch-Data   [8]                       CallHoldData                      OPTIONAL,
    clip-Data [9] ClipData              OPTIONAL,
    clir-Data [10]                      ClirData  OPTIONAL,
    ect-data  [11] EctData              OPTIONAL }
```

```
NoteSubscriberDataModifiedRes ::= SEQUENCE {
    extensionContainer                  ExtensionContainer                OPTIONAL,
    ...}
```

```
-- mobility management event notificatioon info types
```

```
NoteMM-EventArg::= SEQUENCE {
    serviceKey                          ServiceKey,
    eventMet  [0]                       MM-Code,
    imsi      [1]                       IMSI,
    msisdn    [2]                       ISDN-AddressString,
    locationInformation                 [3]  LocationInformation          OPTIONAL,
    supportedCAMELPhases                [5]  SupportedCamelPhases         OPTIONAL,
    extensionContainer                  [6]  ExtensionContainer           OPTIONAL,
    ...,
    locationInformationGPRS             [7]  LocationInformationGPRS      OPTIONAL,
    offeredCamel4Functionalities        [8] OfferedCamel4Functionalities  OPTIONAL
}
```

```
NoteMM-EventRes ::= SEQUENCE {
    extensionContainer                  ExtensionContainer                OPTIONAL,
    ...}
```

```
Ext-SS-InfoFor-CSE ::= CHOICE {
    forwardingInfoFor-CSE               [0] Ext-ForwardingInfoFor-CSE,
    callBarringInfoFor-CSE              [1] Ext-CallBarringInfoFor-CSE
    }
```

```
Ext-ForwardingInfoFor-CSE ::= SEQUENCE {
    ss-Code   [0]                       SS-Code,
    forwardingFeatureList               [1]  Ext-ForwFeatureList,
    notificationToCSE                   [2]  NULL OPTIONAL,
    extensionContainer                  [3]  ExtensionContainer           OPTIONAL,
    ...}
```

```
Ext-CallBarringInfoFor-CSE ::= SEQUENCE {
    ss-Code   [0]                       SS-Code,
    callBarringFeatureList              [1]  Ext-CallBarFeatureList,
    password  [2]                       Password  OPTIONAL,
    wrongPasswordAttemptsCounter        [3]  WrongPasswordAttemptsCounter OPTIONAL,
    notificationToCSE                   [4]  NULL OPTIONAL,
    extensionContainer                  [5]  ExtensionContainer           OPTIONAL,
    ...}
```

```
-- vcsg location registration types

UpdateVcsgLocationArg ::= SEQUENCE {
     imsi      IMSI,
     msisdn    [2] ISDN-AddressString     OPTIONAL,
     vlr-Number                           [0] ISDN-AddressString            OPTIONAL,
     sgsn-Number                          [1] ISDN-AddressString            OPTIONAL,
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ... }

UpdateVcsgLocationRes ::= SEQUENCE {
     temporaryEmptySubscriptiondataIndicator  NULL OPTIONAL,
     extensionContainer                       ExtensionContainer           OPTIONAL,
     ... }

CancelVcsgLocationArg ::= SEQUENCE {
     identity  Identity,
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ...
     }

CancelVcsgLocationRes ::= SEQUENCE {
     extensionContainer                       ExtensionContainer           OPTIONAL,
     ... }


END
```

17.7.2 Operation and maintenance data types

```
MAP-OM-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-OM-DataTypes (12) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   ActivateTraceModeArg,
   ActivateTraceModeRes,
   DeactivateTraceModeArg,
   DeactivateTraceModeRes,
   TracePropagationList
;

IMPORTS
   AddressString,
   IMSI,
   GSN-Address,
   GlobalCellId,
   E-UTRAN-CGI,
   TA-Id,
   RAIdentity,
   LAIFixedLength,
   PLMN-Id
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}

;
```

```
ActivateTraceModeArg ::= SEQUENCE {
    imsi      [0] IMSI                  OPTIONAL,
    traceReference                      [1] TraceReference,
    traceType [2] TraceType,
    omc-Id    [3] AddressString         OPTIONAL,
    extensionContainer                  [4] ExtensionContainer          OPTIONAL,
    ...,
    traceReference2                     [5] TraceReference2             OPTIONAL,
    traceDepthList                      [6] TraceDepthList              OPTIONAL,
    traceNE-TypeList                    [7] TraceNE-TypeList            OPTIONAL,
    traceInterfaceList                  [8] TraceInterfaceList          OPTIONAL,
    traceEventList                      [9] TraceEventList              OPTIONAL,
    traceCollectionEntity               [10] GSN-Address                OPTIONAL,
    mdt-Configuration                   [11] MDT-Configuration          OPTIONAL
    }
```

```
MDT-Configuration ::= SEQUENCE {
    jobType   JobType,
    areaScope AreaScope                 OPTIONAL,
    listOfMeasurements                  ListOfMeasurements              OPTIONAL,
    reportingTrigger                    [0] ReportingTrigger            OPTIONAL,
    reportInterval                      ReportInterval                  OPTIONAL,
    reportAmount                        [1] ReportAmount                OPTIONAL,
    eventThresholdRSRP                  EventThresholdRSRP              OPTIONAL,
    eventThresholdRSRQ                  [2] EventThresholdRSRQ          OPTIONAL,
    loggingInterval                     [3] LoggingInterval             OPTIONAL,
    loggingDuration                     [4] LoggingDuration             OPTIONAL,
    extensionContainer                  [5] ExtensionContainer          OPTIONAL,
    ...,
    measurementPeriodUMTS               [6] PeriodUMTS                  OPTIONAL,
    measurementPeriodLTE                [7] PeriodLTE                   OPTIONAL,
    collectionPeriodRRM-UMTS            [8] PeriodUMTS                  OPTIONAL,
    collectionPeriodRRM-LTE             [9] PeriodLTE                   OPTIONAL,
    positioningMethod                   [10] PositioningMethod          OPTIONAL,
    measurementQuantity                 [11] MeasurementQuantity        OPTIONAL,
    eventThreshold1F                    [12] EventThreshold1F           OPTIONAL,
    eventThreshold1I                    [13] EventThreshold1I           OPTIONAL,
    mdt-Allowed-PLMN-List               [14] MDT-Allowed-PLMNId-List    OPTIONAL }
```

```
MDT-Allowed-PLMNId-List ::= SEQUENCE SIZE (1..16) OF
        PLMN-Id
```

```
PeriodUMTS ::= ENUMERATED {
    d250ms (0),
    d500ms (1),
    d1000ms (2),
    d2000ms (3),
    d3000ms (4),
    d4000ms (5),
    d6000ms (6),
    d8000ms (7),
    d12000ms (8),
    d16000ms (9),
    d20000ms (10),
    d24000ms (11),
    d28000ms (12),
    d32000ms (13),
    d64000ms (14)}
```

```
PeriodLTE ::= ENUMERATED {
    d1024ms (0),
    d1280ms (1),
    d2048ms (2),
    d2560ms (3),
    d5120ms (4),
    d10240ms (5),
    d1min (6)}
```

```
PositioningMethod ::= OCTET STRING (SIZE (1))
    -- Octet is coded as described in 3GPP TS 32.422 [132].
```

```
MeasurementQuantity ::= OCTET STRING (SIZE (1))
    -- Octet is coded as described in 3GPP TS 32.422 [132].
```

```
EventThreshold1F ::= INTEGER
    (-120..165)
```

```
EventThreshold1I ::= INTEGER
     (-120..-25)
```

```
JobType ::= ENUMERATED {
     immediate-MDT-only (0),
     logged-MDT-only (1),
     trace-only (2),
     immediate-MDT-and-trace (3)}
```

```
AreaScope ::= SEQUENCE {
     cgi-List  [0] CGI-List               OPTIONAL,
     e-utran-cgi-List                     [1] E-UTRAN-CGI-List             OPTIONAL,
     routingAreaId-List                   [2] RoutingAreaId-List           OPTIONAL,
     locationAreaId-List                  [3] LocationAreaId-List          OPTIONAL,
     trackingAreaId-List                  [4] TrackingAreaId-List          OPTIONAL,
     extensionContainer                   [5] ExtensionContainer           OPTIONAL,
     ... }
```

```
CGI-List ::= SEQUENCE SIZE (1..32) OF
          GlobalCellId
```

```
E-UTRAN-CGI-List ::= SEQUENCE SIZE (1..32) OF
          E-UTRAN-CGI
```

```
RoutingAreaId-List ::= SEQUENCE SIZE (1..8) OF
          RAIdentity
```

```
LocationAreaId-List ::= SEQUENCE SIZE (1..8) OF
          LAIFixedLength
```

```
TrackingAreaId-List ::= SEQUENCE SIZE (1..8) OF
          TA-Id
```

```
ListOfMeasurements ::= OCTET STRING (SIZE (4))
     -- Octets are coded as described in 3GPP TS 32.422.
```

```
ReportingTrigger ::= OCTET STRING (SIZE (1))
     -- Octet is coded as described in 3GPP TS 32.422.
```

```
ReportInterval ::= ENUMERATED {
     umts250ms (0),
     umts500ms (1),
     umts1000ms (2),
     umts2000ms (3),
     umts3000ms (4),
     umts4000ms (5),
     umts6000ms (6),
     umts8000ms (7),
     umts12000ms (8),
     umts16000ms (9),
     umts20000ms (10),
     umts24000ms (11),
     umts28000ms (12),
     umts32000ms (13),
     umts64000ms (14),
     lte120ms (15),
     lte240ms (16),
     lte480ms (17),
     lte640ms (18),
     lte1024ms (19),
     lte2048ms (20),
     lte5120ms (21),
     lte10240ms (22),
     lte1min (23),
     lte6min (24),
     lte12min (25),
     lte30min (26),
     lte60min (27)}
```

```
ReportAmount ::= ENUMERATED {
    d1 (0),
    d2 (1),
    d4 (2),
    d8 (3),
    d16 (4),
    d32 (5),
    d64 (6),
    infinity (7)}
```

```
EventThresholdRSRP ::= INTEGER
    (0..97)
```

```
EventThresholdRSRQ ::= INTEGER
    (0..34)
```

```
LoggingInterval ::= ENUMERATED {
    d1dot28 (0),
    d2dot56 (1),
    d5dot12 (2),
    d10dot24 (3),
    d20dot48 (4),
    d30dot72 (5),
    d40dot96 (6),
    d61dot44 (7)}
```

```
LoggingDuration ::= ENUMERATED {
    d600sec (0),
    d1200sec (1),
    d2400sec (2),
    d3600sec (3),
    d5400sec (4),
    d7200sec (5)}
```

```
TraceReference ::= OCTET STRING (SIZE (1..2))
```

```
TraceReference2 ::= OCTET STRING (SIZE (3))
```

```
TraceRecordingSessionReference ::= OCTET STRING (SIZE (2))
```

```
TraceType ::= INTEGER
    (0..255)
    -- Trace types are fully defined in  3GPP TS 52.008. [61]
```

```
TraceDepthList ::= SEQUENCE {
    msc-s-TraceDepth                [0] TraceDepth                  OPTIONAL,
    mgw-TraceDepth                  [1] TraceDepth                  OPTIONAL,
    sgsn-TraceDepth                 [2] TraceDepth                  OPTIONAL,
    ggsn-TraceDepth                 [3] TraceDepth                  OPTIONAL,
    rnc-TraceDepth                  [4] TraceDepth                  OPTIONAL,
    bmsc-TraceDepth                 [5] TraceDepth                  OPTIONAL,
    ... ,
    mme-TraceDepth                  [6] TraceDepth                  OPTIONAL,
    sgw-TraceDepth                  [7] TraceDepth                  OPTIONAL,
    pgw-TraceDepth                  [8] TraceDepth                  OPTIONAL,
    eNB-TraceDepth                  [9] TraceDepth                  OPTIONAL,
    msc-s-TraceDepthExtension       [10] TraceDepthExtension        OPTIONAL,
    mgw-TraceDepthExtension         [11] TraceDepthExtension        OPTIONAL,
    sgsn-TraceDepthExtension        [12] TraceDepthExtension        OPTIONAL,
    ggsn-TraceDepthExtension        [13] TraceDepthExtension        OPTIONAL,
    rnc-TraceDepthExtension         [14] TraceDepthExtension        OPTIONAL,
    bmsc-TraceDepthExtension        [15] TraceDepthExtension        OPTIONAL,
    mme-TraceDepthExtension         [16] TraceDepthExtension        OPTIONAL,
    sgw-TraceDepthExtension         [17] TraceDepthExtension        OPTIONAL,
    pgw-TraceDepthExtension         [18] TraceDepthExtension        OPTIONAL,
    eNB-TraceDepthExtension         [19] TraceDepthExtension        OPTIONAL }

-- If one of the TraceDepthExtension types is sent, the corresponding TraceDepth type
-- shall also be sent with the same enumeration value to allow the receiver not supporting
-- the Extension to fall back to the non extended type.
-- If one of the TraceDepthExtension types is received and supported, the corresponding
-- TraceDepth type shall be ignored.
```

```
TraceDepth ::= ENUMERATED {
     minimum (0),
     medium (1),
     maximum (2),
     ...}
-- The value medium is applicable only for RNC. For other network elements, if value medium
-- is received, value minimum shall be applied.
```

```
TraceDepthExtension ::= ENUMERATED {
     minimumWithoutVendorSpecificExtension (0),
     mediumWithoutVendorSpecificExtension (1),
     maximumWithoutVendorSpecificExtension (2),
     ...}
-- The value mediumWithoutVendorSpecificExtension is applicable only for RNC. For other
-- network elements, if value mediumWithoutVendorSpecificExtension is received, value
-- minimumWithoutVendorSpecificExtension shall be applied.
```

```
TraceNE-TypeList ::= BIT STRING {
     msc-s (0),
     mgw (1),
     sgsn (2),
     ggsn (3),
     rnc (4),
     bm-sc (5) ,
     mme (6),
     sgw (7),
     pgw (8),
     eNB (9)} (SIZE (6..16))
-- Other bits than listed above shall be discarded.
```

```
TraceInterfaceList ::= SEQUENCE {
     msc-s-List                          [0] MSC-S-InterfaceList          OPTIONAL,
     mgw-List  [1] MGW-InterfaceList     OPTIONAL,
     sgsn-List [2] SGSN-InterfaceList    OPTIONAL,
     ggsn-List [3] GGSN-InterfaceList    OPTIONAL,
     rnc-List  [4] RNC-InterfaceList     OPTIONAL,
     bmsc-List [5] BMSC-InterfaceList    OPTIONAL,
     ...,
     mme-List  [6] MME-InterfaceList     OPTIONAL,
     sgw-List  [7] SGW-InterfaceList     OPTIONAL,
     pgw-List  [8] PGW-InterfaceList     OPTIONAL,
     eNB-List  [9] ENB-InterfaceList     OPTIONAL}
```

```
MSC-S-InterfaceList ::= BIT STRING {
     a (0),
     iu (1),
     mc (2),
     map-g (3),
     map-b (4),
     map-e (5),
     map-f (6),
     cap (7),
     map-d (8),
     map-c (9)} (SIZE (10..16))
-- Other bits than listed above shall be discarded.
```

```
MGW-InterfaceList ::= BIT STRING {
     mc (0),
     nb-up (1),
     iu-up (2)} (SIZE (3..8))
-- Other bits than listed above shall be discarded.
```

```
SGSN-InterfaceList ::= BIT STRING {
     gb (0),
     iu (1),
     gn (2),
     map-gr (3),
     map-gd (4),
     map-gf (5),
     gs (6),
     ge (7),
     s3 (8),
     s4 (9),
     s6d (10)} (SIZE (8..16))
-- Other bits than listed above shall be discarded.
```

```
GGSN-InterfaceList ::= BIT STRING {
     gn (0),
     gi (1),
     gmb (2)} (SIZE (3..8))
-- Other bits than listed above shall be discarded.
```

```
RNC-InterfaceList ::= BIT STRING {
     iu (0),
     iur (1),
     iub (2),
     uu (3)} (SIZE (4..8))
-- Other bits than listed above shall be discarded.
```

```
BMSC-InterfaceList ::= BIT STRING {
     gmb (0)} (SIZE (1..8))
-- Other bits than listed above shall be discarded.
```

```
MME-InterfaceList ::= BIT STRING {
     s1-mme (0),
     s3 (1),
     s6a (2),
     s10 (3),
     s11 (4)} (SIZE (5..8))
-- Other bits than listed above shall be discarded.
```

```
SGW-InterfaceList ::= BIT STRING {
     s4 (0),
     s5 (1),
     s8b (2),
     s11 (3),
     gxc (4)} (SIZE (5..8))
-- Other bits than listed above shall be discarded.
```

```
PGW-InterfaceList ::= BIT STRING {
     s2a (0),
     s2b (1),
     s2c (2),
     s5 (3),
     s6b (4),
     gx (5),
     s8b (6),
     sgi (7)} (SIZE (8..16))
-- Other bits than listed above shall be discarded.
```

```
ENB-InterfaceList ::= BIT STRING {
     s1-mme (0),
     x2 (1),
     uu (2)} (SIZE (3..8))
-- Other bits than listed above shall be discarded.
```

```
TraceEventList ::= SEQUENCE {
     msc-s-List                      [0] MSC-S-EventList             OPTIONAL,
     mgw-List  [1] MGW-EventList     OPTIONAL,
     sgsn-List [2] SGSN-EventList    OPTIONAL,
     ggsn-List [3] GGSN-EventList    OPTIONAL,
     bmsc-List [4] BMSC-EventList    OPTIONAL,
     ...,
     mme-List  [5] MME-EventList     OPTIONAL,
     sgw-List  [6] SGW-EventList     OPTIONAL,
     pgw-List  [7] PGW-EventList     OPTIONAL}
```

```
MSC-S-EventList ::= BIT STRING {
     mo-mtCall (0),
     mo-mt-sms (1),
     lu-imsiAttach-imsiDetach (2),
     handovers (3),
     ss (4)} (SIZE (5..16))
-- Other bits than listed above shall be discarded.
```

```
MGW-EventList ::= BIT STRING {
     context (0)} (SIZE (1..8))
-- Other bits than listed above shall be discarded.
```

```
SGSN-EventList ::= BIT STRING {
     pdpContext (0),
     mo-mt-sms (1),
     rau-gprsAttach-gprsDetach (2),
     mbmsContext (3)} (SIZE (4..16))
-- Other bits than listed above shall be discarded.
```

```
GGSN-EventList ::= BIT STRING {
     pdpContext (0),
     mbmsContext (1)} (SIZE (2..8))
-- Other bits than listed above shall be discarded.
```

```
BMSC-EventList ::= BIT STRING {
     mbmsMulticastServiceActivation (0)} (SIZE (1..8))
-- Other bits than listed above shall be discarded.
```

```
MME-EventList ::= BIT STRING {
     ue-initiatedPDNconectivityRequest (0),
     serviceRequestts (1),
     initialAttachTrackingAreaUpdateDetach (2),
     ue-initiatedPDNdisconnection (3),
     bearerActivationModificationDeletion (4),
     handover (5)} (SIZE (6..8))
-- Other bits than listed above shall be discarded.
```

```
SGW-EventList ::= BIT STRING {
     pdn-connectionCreation (0),
     pdn-connectionTermination (1),
     bearerActivationModificationDeletion (2)} (SIZE (3..8))
-- Other bits than listed above shall be discarded.
```

```
PGW-EventList ::= BIT STRING {
     pdn-connectionCreation (0),
     pdn-connectionTermination (1),
     bearerActivationModificationDeletion (2)} (SIZE (3..8))
-- Other bits than listed above shall be discarded.
```

```
TracePropagationList ::= SEQUENCE {
     traceReference                      [0] TraceReference                OPTIONAL,
     traceType [1] TraceType             OPTIONAL,
     traceReference2                     [2] TraceReference2               OPTIONAL,
     traceRecordingSessionReference      [3] TraceRecordingSessionReference OPTIONAL,
     rnc-TraceDepth                      [4] TraceDepth                    OPTIONAL,
     rnc-InterfaceList                   [5] RNC-InterfaceList             OPTIONAL,
     msc-s-TraceDepth                    [6] TraceDepth                    OPTIONAL,
     msc-s-InterfaceList                 [7] MSC-S-InterfaceList           OPTIONAL,
     msc-s-EventList                     [8] MSC-S-EventList               OPTIONAL,
     mgw-TraceDepth                      [9] TraceDepth                    OPTIONAL,
     mgw-InterfaceList                   [10] MGW-InterfaceList            OPTIONAL,
     mgw-EventList                       [11] MGW-EventList                OPTIONAL,
     ...,
     rnc-TraceDepthExtension             [12] TraceDepthExtension          OPTIONAL,
     msc-s-TraceDepthExtension           [13] TraceDepthExtension          OPTIONAL,
     mgw-TraceDepthExtension             [14] TraceDepthExtension          OPTIONAL
}
-- If one of the TraceDepthExtension types is sent, the corresponding TraceDepth type
-- shall also be sent with the same enumeration value to allow the receiver not supporting
-- the Extension to fall back to the non extended type.
-- If one of the TraceDepthExtension types is received and supported, the corresponding
-- TraceDepth type shall be ignored.
```

```
ActivateTraceModeRes ::= SEQUENCE {
     extensionContainer                  [0] ExtensionContainer            OPTIONAL,
     ...,
     traceSupportIndicator               [1]  NULL OPTIONAL
     }
```

```
DeactivateTraceModeArg ::= SEQUENCE {
     imsi     [0] IMSI                   OPTIONAL,
     traceReference                      [1] TraceReference,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...,
     traceReference2                     [3] TraceReference2               OPTIONAL
     }
```

```
DeactivateTraceModeRes ::= SEQUENCE {
     extensionContainer                  [0] ExtensionContainer            OPTIONAL,
     ...}
```

```
END
```

17.7.3 Call handling data types

```
MAP-CH-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CH-DataTypes (13) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   SendRoutingInfoArg,
   SendRoutingInfoRes,
   ProvideRoamingNumberArg,
   ProvideRoamingNumberRes,
   ResumeCallHandlingArg,
   ResumeCallHandlingRes,
   NumberOfForwarding,
   SuppressionOfAnnouncement,
   CallReferenceNumber,
   SetReportingStateArg,
   SetReportingStateRes,
   StatusReportArg,
   StatusReportRes,
   RemoteUserFreeArg,
   RemoteUserFreeRes,
   IST-AlertArg,
   IST-AlertRes,
   IST-CommandArg,
   IST-CommandRes,
   UU-Data,
   ReleaseResourcesArg,
   ReleaseResourcesRes
;

IMPORTS
   SubscriberInfo,
   SupportedCamelPhases,
   OfferedCamel4CSIs,
   CUG-Interlock,
   O-CSI,
   D-CSI,
   O-BcsmCamelTDPCriteriaList,
   T-BCSM-CAMEL-TDP-CriteriaList,
   IST-SupportIndicator,
   IST-AlertTimerValue,
   T-CSI,
   NumberPortabilityStatus,
   PagingArea
FROM MAP-MS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MS-DataTypes (11) version18 (18)}

   ForwardingOptions,
   SS-List,
   CCBS-Feature
FROM MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-DataTypes (14) version18 (18)}

   ISDN-AddressString,
   ISDN-SubaddressString,
   FTN-AddressString,
   ExternalSignalInfo,
   Ext-ExternalSignalInfo,
   IMSI,
   LMSI,
   Ext-BasicServiceCode,
```

```
   AlertingPattern,
   NAEA-PreferredCI,
   EMLPP-Priority,
   PLMN-Id
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}
;
```

```
CUG-CheckInfo ::= SEQUENCE {
     cug-Interlock                    CUG-Interlock,
     cug-OutgoingAccess               NULL     OPTIONAL,
     extensionContainer               ExtensionContainer              OPTIONAL,
     ...}
```

```
NumberOfForwarding ::= INTEGER (1..5)
```

```
SendRoutingInfoArg ::= SEQUENCE {
     msisdn    [0] ISDN-AddressString,
     cug-CheckInfo                    [1] CUG-CheckInfo               OPTIONAL,
     numberOfForwarding               [2] NumberOfForwarding          OPTIONAL,
     interrogationType                [3] InterrogationType,
     or-Interrogation                 [4] NULL  OPTIONAL,
     or-Capability                    [5] OR-Phase                    OPTIONAL,
     gmsc-OrGsmSCF-Address            [6] ISDN-AddressString,
     callReferenceNumber              [7] CallReferenceNumber         OPTIONAL,
     forwardingReason                 [8] ForwardingReason            OPTIONAL,
     basicServiceGroup                [9] Ext-BasicServiceCode        OPTIONAL,
     networkSignalInfo                [10] ExternalSignalInfo         OPTIONAL,
     camelInfo [11] CamelInfo         OPTIONAL,
     suppressionOfAnnouncement        [12] SuppressionOfAnnouncement  OPTIONAL,
     extensionContainer               [13] ExtensionContainer         OPTIONAL,
     ...,
     alertingPattern                  [14] AlertingPattern            OPTIONAL,
     ccbs-Call [15] NULL                   OPTIONAL,
     supportedCCBS-Phase              [16] SupportedCCBS-Phase        OPTIONAL,
     additionalSignalInfo             [17] Ext-ExternalSignalInfo     OPTIONAL,
     istSupportIndicator              [18] IST-SupportIndicator       OPTIONAL,
     pre-pagingSupported              [19] NULL OPTIONAL,
     callDiversionTreatmentIndicator  [20] CallDiversionTreatmentIndicator   OPTIONAL,
     longFTN-Supported                [21] NULL OPTIONAL,
     suppress-VT-CSI                  [22] NULL OPTIONAL,
     suppressIncomingCallBarring      [23] NULL OPTIONAL,
     gsmSCF-InitiatedCall             [24] NULL OPTIONAL,
     basicServiceGroup2               [25] Ext-BasicServiceCode       OPTIONAL,
     networkSignalInfo2               [26] ExternalSignalInfo         OPTIONAL,
     suppressMTSS                     [27] SuppressMTSS               OPTIONAL,
     mtRoamingRetrySupported          [28] NULL OPTIONAL,
     callPriority                     [29] EMLPP-Priority             OPTIONAL
     }
```

```
SuppressionOfAnnouncement ::= NULL
```

```
SuppressMTSS ::= BIT STRING {
     suppressCUG                      (0),
     suppressCCBS                     (1) } (SIZE (2..16))
     --   Other bits than listed above shall be discarded
```

```
InterrogationType ::= ENUMERATED {
     basicCall  (0),
     forwarding  (1)}
```

```
OR-Phase ::= INTEGER (1..127)
```

```
CallReferenceNumber ::= OCTET STRING (SIZE (1..8))
```

```
ForwardingReason ::= ENUMERATED {
     notReachable  (0),
     busy  (1),
     noReply  (2)}
```

```
SupportedCCBS-Phase ::= INTEGER (1..127)
-- exception handling:
-- Only value 1 is used.
-- Values in the ranges 2-127 are reserved for future use.
-- If received values 2-127 shall be mapped on to value 1.
```

```
CallDiversionTreatmentIndicator ::= OCTET STRING (SIZE(1))
--  callDiversionAllowed (xxxx xx01)
--  callDiversionNotAllowed (xxxx xx10)
--  network default is call diversion allowed
```

```
SendRoutingInfoRes ::= [3] SEQUENCE {
    imsi     [9] IMSI                   OPTIONAL,
    -- IMSI must be present if SendRoutingInfoRes is not segmented.
    -- If the TC-Result-NL segmentation option is taken the IMSI must be
    -- present in one segmented transmission of SendRoutingInfoRes.
    extendedRoutingInfo                 ExtendedRoutingInfo              OPTIONAL,
    cug-CheckInfo                       [3] CUG-CheckInfo                OPTIONAL,
    cugSubscriptionFlag                 [6] NULL  OPTIONAL,
    subscriberInfo                      [7] SubscriberInfo               OPTIONAL,
    ss-List   [1] SS-List               OPTIONAL,
    basicService                        [5] Ext-BasicServiceCode         OPTIONAL,
    forwardingInterrogationRequired     [4] NULL  OPTIONAL,
    vmsc-Address                        [2] ISDN-AddressString           OPTIONAL,
    extensionContainer                  [0] ExtensionContainer           OPTIONAL,
    ... ,
    naea-PreferredCI                    [10] NAEA-PreferredCI            OPTIONAL,
    -- naea-PreferredCI is included at the discretion of the HLR operator.
    ccbs-Indicators                     [11] CCBS-Indicators             OPTIONAL,
    msisdn    [12] ISDN-AddressString   OPTIONAL,
    numberPortabilityStatus             [13] NumberPortabilityStatus     OPTIONAL,
    istAlertTimer                       [14] IST-AlertTimerValue         OPTIONAL,
    supportedCamelPhasesInVMSC          [15] SupportedCamelPhases        OPTIONAL,
    offeredCamel4CSIsInVMSC             [16] OfferedCamel4CSIs           OPTIONAL,
    routingInfo2                        [17] RoutingInfo                 OPTIONAL,
    ss-List2  [18] SS-List              OPTIONAL,
    basicService2                       [19] Ext-BasicServiceCode        OPTIONAL,
    allowedServices                     [20] AllowedServices             OPTIONAL,
    unavailabilityCause                 [21] UnavailabilityCause         OPTIONAL,
    releaseResourcesSupported           [22] NULL OPTIONAL,
    gsm-BearerCapability                [23] ExternalSignalInfo          OPTIONAL
    }
```

```
AllowedServices ::= BIT STRING {
    firstServiceAllowed                 (0),
    secondServiceAllowed                (1) } (SIZE (2..8))
    --   firstService is the service indicated in the networkSignalInfo
    --   secondService is the service indicated in the networkSignalInfo2
    --   Other bits than listed above shall be discarded
```

```
UnavailabilityCause ::= ENUMERATED {
    bearerServiceNotProvisioned         (1),
    teleserviceNotProvisioned           (2),
    absentSubscriber                    (3),
    busySubscriber                      (4),
    callBarred                          (5),
    cug-Reject                          (6),
    ...}
    --   exception handling:
    --   Reception of other values than the ones listed shall result in the service
    --   being unavailable for that call.
```

```
CCBS-Indicators ::= SEQUENCE {
    ccbs-Possible                       [0]  NULL OPTIONAL,
    keepCCBS-CallIndicator              [1]  NULL OPTIONAL,
    extensionContainer                  [2]  ExtensionContainer          OPTIONAL,
    ...}
```

```
RoutingInfo ::= CHOICE {
    roamingNumber                       ISDN-AddressString,
    forwardingData                      ForwardingData}
```

```
ForwardingData ::= SEQUENCE {
    forwardedToNumber                   [5] ISDN-AddressString          OPTIONAL,
    -- When this datatype is sent from an HLR which supports CAMEL Phase 2
    -- to a GMSC which supports CAMEL Phase 2 the GMSC shall not check the
    -- format of the number
    forwardedToSubaddress               [4] ISDN-SubaddressString       OPTIONAL,
    forwardingOptions                   [6] ForwardingOptions           OPTIONAL,
    extensionContainer                  [7] ExtensionContainer          OPTIONAL,
    ...,
    longForwardedToNumber               [8] FTN-AddressString           OPTIONAL}
```

```
ProvideRoamingNumberArg ::= SEQUENCE {
    imsi     [0] IMSI,
    msc-Number                          [1] ISDN-AddressString,
    msisdn    [2] ISDN-AddressString    OPTIONAL,
    lmsi     [4] LMSI                   OPTIONAL,
    gsm-BearerCapability                [5] ExternalSignalInfo          OPTIONAL,
    networkSignalInfo                   [6] ExternalSignalInfo          OPTIONAL,
    suppressionOfAnnouncement           [7] SuppressionOfAnnouncement   OPTIONAL,
    gmsc-Address                        [8] ISDN-AddressString          OPTIONAL,
    callReferenceNumber                 [9] CallReferenceNumber         OPTIONAL,
    or-Interrogation                    [10] NULL OPTIONAL,
    extensionContainer                  [11] ExtensionContainer         OPTIONAL,
    ... ,
    alertingPattern                     [12] AlertingPattern            OPTIONAL,
    ccbs-Call [13] NULL                 OPTIONAL,
    supportedCamelPhasesInInterrogatingNode  [15] SupportedCamelPhases  OPTIONAL,
    additionalSignalInfo                [14] Ext-ExternalSignalInfo     OPTIONAL,
    orNotSupportedInGMSC                [16] NULL OPTIONAL,
    pre-pagingSupported                 [17] NULL OPTIONAL,
    longFTN-Supported                   [18] NULL OPTIONAL,
    suppress-VT-CSI                     [19] NULL OPTIONAL,
    offeredCamel4CSIsInInterrogatingNode [20] OfferedCamel4CSIs         OPTIONAL,
    mtRoamingRetrySupported             [21] NULL OPTIONAL,
    pagingArea                          [22] PagingArea                 OPTIONAL,
    callPriority                        [23] EMLPP-Priority             OPTIONAL,
    mtrf-Indicator                      [24] NULL OPTIONAL,
    oldMSC-Number                       [25] ISDN-AddressString         OPTIONAL,
    lastUsedLtePLMN-Id                  [26] PLMN-Id                    OPTIONAL
    }
```

```
ProvideRoamingNumberRes ::= SEQUENCE {
    roamingNumber                       ISDN-AddressString,
    extensionContainer                  ExtensionContainer              OPTIONAL,
    ...,
    releaseResourcesSupported           NULL     OPTIONAL,
    vmsc-Address                        ISDN-AddressString              OPTIONAL }
```

```
ResumeCallHandlingArg ::= SEQUENCE {
    callReferenceNumber                 [0] CallReferenceNumber         OPTIONAL,
    basicServiceGroup                   [1] Ext-BasicServiceCode        OPTIONAL,
    forwardingData                      [2] ForwardingData              OPTIONAL,
    imsi     [3] IMSI                   OPTIONAL,
    cug-CheckInfo                       [4] CUG-CheckInfo               OPTIONAL,
    o-CSI    [5] O-CSI                  OPTIONAL,
    extensionContainer                  [7] ExtensionContainer          OPTIONAL,
    ccbs-Possible                       [8]  NULL OPTIONAL,
    msisdn   [9]                        ISDN-AddressString              OPTIONAL,
    uu-Data  [10] UU-Data               OPTIONAL,
    allInformationSent                  [11] NULL OPTIONAL,
    ...,
    d-csi    [12]                       D-CSI    OPTIONAL,
    o-BcsmCamelTDPCriteriaList          [13] O-BcsmCamelTDPCriteriaList OPTIONAL,
    basicServiceGroup2                  [14] Ext-BasicServiceCode       OPTIONAL,
    mtRoamingRetry                      [15] NULL OPTIONAL
    }
```

```
UU-Data ::= SEQUENCE {
    uuIndicator                         [0] UUIndicator                 OPTIONAL,
    uui      [1] UUI                    OPTIONAL,
    uusCFInteraction                    [2] NULL  OPTIONAL,
    extensionContainer                  [3] ExtensionContainer          OPTIONAL,
    ...}
```

```
UUIndicator ::= OCTET STRING (SIZE (1))
    -- Octets are coded according to ETS 300 356
```

```
UUI  ::= OCTET STRING (SIZE (1..131))
```

```
    -- Octets are coded according to ETS 300 356
```

```
ResumeCallHandlingRes ::= SEQUENCE {
    extensionContainer                  ExtensionContainer                  OPTIONAL,
    ...}
```

```
CamelInfo ::= SEQUENCE {
    supportedCamelPhases                SupportedCamelPhases,
    suppress-T-CSI                      NULL     OPTIONAL,
    extensionContainer                  ExtensionContainer                  OPTIONAL,
    ... ,
    offeredCamel4CSIs                   [0] OfferedCamel4CSIs               OPTIONAL }
```

```
ExtendedRoutingInfo ::= CHOICE {
    routingInfo                         RoutingInfo,
    camelRoutingInfo                    [8] CamelRoutingInfo}
```

```
CamelRoutingInfo ::= SEQUENCE {
    forwardingData                      ForwardingData                      OPTIONAL,
    gmscCamelSubscriptionInfo           [0] GmscCamelSubscriptionInfo,
    extensionContainer                  [1] ExtensionContainer              OPTIONAL,
    ...}
```

```
GmscCamelSubscriptionInfo ::= SEQUENCE {
    t-CSI     [0] T-CSI                 OPTIONAL,
    o-CSI     [1] O-CSI                 OPTIONAL,
    extensionContainer                  [2] ExtensionContainer              OPTIONAL,
    ...,
    o-BcsmCamelTDP-CriteriaList         [3] O-BcsmCamelTDPCriteriaList      OPTIONAL,
    t-BCSM-CAMEL-TDP-CriteriaList       [4]  T-BCSM-CAMEL-TDP-CriteriaList OPTIONAL,
    d-csi     [5]                       D-CSI     OPTIONAL}
```

```
SetReportingStateArg ::= SEQUENCE {
    imsi      [0]                       IMSI OPTIONAL,
    lmsi      [1]                       LMSI OPTIONAL,
    ccbs-Monitoring                     [2]  ReportingState                 OPTIONAL,
    extensionContainer                  [3]  ExtensionContainer             OPTIONAL,
    ...}
```

```
ReportingState ::= ENUMERATED {
    stopMonitoring                      (0),
    startMonitoring                     (1),
    ...}
    -- exception handling:
    -- reception of values 2-10 shall be mapped to 'stopMonitoring'
    -- reception of values > 10 shall be mapped to 'startMonitoring'
```

```
SetReportingStateRes ::= SEQUENCE{
    ccbs-SubscriberStatus               [0]  CCBS-SubscriberStatus          OPTIONAL,
    extensionContainer                  [1]  ExtensionContainer             OPTIONAL,
    ...}
```

```
CCBS-SubscriberStatus ::= ENUMERATED {
    ccbsNotIdle                         (0),
    ccbsIdle (1),
    ccbsNotReachable                    (2),
    ...}
    --  exception handling:
    --  reception of values 3-10 shall be mapped to 'ccbsNotIdle'
    --  reception of values 11-20 shall be mapped to 'ccbsIdle'
    --  reception of values > 20 shall be mapped to 'ccbsNotReachable'
```

```
StatusReportArg ::= SEQUENCE{
    imsi [0]  IMSI,
    eventReportData                     [1]  EventReportData                OPTIONAL,
    callReportdata                      [2]  CallReportData                 OPTIONAL,
    extensionContainer                  [3]  ExtensionContainer             OPTIONAL,
    ...}
```

```
EventReportData ::= SEQUENCE{
    ccbs-SubscriberStatus               [0]  CCBS-SubscriberStatus          OPTIONAL,
    extensionContainer                  [1]  ExtensionContainer             OPTIONAL,
    ...}
```

```
CallReportData ::= SEQUENCE{
     monitoringMode                  [0]  MonitoringMode              OPTIONAL,
     callOutcome                     [1]  CallOutcome                 OPTIONAL,
     extensionContainer              [2]  ExtensionContainer          OPTIONAL,
     ...}
```

```
MonitoringMode ::= ENUMERATED {
     a-side   (0),
     b-side   (1),
     ...}
     --   exception handling:
     --  reception of values 2-10 shall be mapped 'a-side'
     --  reception of values > 10 shall be mapped to 'b-side'
```

```
CallOutcome ::= ENUMERATED {
     success  (0),
     failure  (1),
     busy     (2),
     ...}
     --   exception handling:
     --  reception of values 3-10 shall be mapped to 'success'
     --  reception of values 11-20 shall be mapped to 'failure'
     --  reception of values > 20 shall be mapped to 'busy'
```

```
StatusReportRes ::= SEQUENCE {
     extensionContainer              [0]  ExtensionContainer          OPTIONAL,
     ...}
```

```
RemoteUserFreeArg ::= SEQUENCE{
     imsi     [0]                    IMSI,
     callInfo [1]                    ExternalSignalInfo,
     ccbs-Feature                    [2]  CCBS-Feature,
     translatedB-Number              [3]  ISDN-AddressString,
     replaceB-Number                 [4]  NULL OPTIONAL,
     alertingPattern                 [5]  AlertingPattern             OPTIONAL,
     extensionContainer              [6]  ExtensionContainer          OPTIONAL,
     ...}
```

```
RemoteUserFreeRes ::= SEQUENCE{
     ruf-Outcome                     [0]  RUF-Outcome,
     extensionContainer              [1]  ExtensionContainer          OPTIONAL,
     ...}
```

```
RUF-Outcome ::= ENUMERATED{
     accepted (0),
     rejected (1),
     noResponseFromFreeMS (2), -- T4 Expiry
     noResponseFromBusyMS (3), -- T10 Expiry
     ududFromFreeMS (4),
     ududFromBusyMS (5),
     ...}
     -- exception handling:
     -- reception of values 6-20 shall be mapped to 'accepted'
     -- reception of values 21-30 shall be mapped to 'rejected'
     -- reception of values 31-40 shall be mapped to 'noResponseFromFreeMS'
     -- reception of values 41-50 shall be mapped to 'noResponseFromBusyMS'
     -- reception of values 51-60 shall be mapped to 'ududFromFreeMS'
     -- reception of values > 60 shall be mapped to 'ududFromBusyMS'
```

```
IST-AlertArg ::= SEQUENCE{
     imsi     [0]                    IMSI,
     extensionContainer              [1]  ExtensionContainer          OPTIONAL,
     ...}
```

```
IST-AlertRes ::= SEQUENCE{
     istAlertTimer                   [0]  IST-AlertTimerValue         OPTIONAL,
     istInformationWithdraw          [1]  NULL OPTIONAL,
     callTerminationIndicator        [2]  CallTerminationIndicator    OPTIONAL,
     extensionContainer              [3]  ExtensionContainer          OPTIONAL,
     ...}
```

```
IST-CommandArg ::= SEQUENCE{
     imsi     [0]                    IMSI,
     extensionContainer              [1]  ExtensionContainer          OPTIONAL,
     ...}
```

```
IST-CommandRes ::= SEQUENCE{
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
CallTerminationIndicator ::= ENUMERATED {
     terminateCallActivityReferred       (0),
     terminateAllCallActivities          (1),
     ...}
     -- exception handling:
     -- reception of values 2-10 shall be mapped to ' terminateCallActivityReferred '
     -- reception of values > 10 shall be mapped to ' terminateAllCallActivities '

     -- In MSCs not supporting linkage of all call activities, any value received shall
     -- be interpreted as ' terminateCallActivityReferred '
```

```
ReleaseResourcesArg ::= SEQUENCE{
     msrn     ISDN-AddressString,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
ReleaseResourcesRes ::= SEQUENCE{
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
END
```

17.7.4 Supplementary service data types

```
MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-DataTypes (14) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   RegisterSS-Arg,
   SS-Info,
   SS-Status,
   SS-SubscriptionOption,
   SS-ForBS-Code,
   InterrogateSS-Res,
   USSD-Arg,
   USSD-Res,
   USSD-DataCodingScheme,
   USSD-String,
   Password,
   GuidanceInfo,
   SS-List,
   SS-InfoList,
   OverrideCategory,
   CliRestrictionOption,
   NoReplyConditionTime,
   ForwardingOptions,
   maxNumOfSS,
   SS-Data,
   SS-InvocationNotificationArg,
   SS-InvocationNotificationRes,
   CCBS-Feature,
   RegisterCC-EntryArg,
   RegisterCC-EntryRes,
   EraseCC-EntryArg,
   EraseCC-EntryRes
;

IMPORTS
   AddressString,
   ISDN-AddressString,
   ISDN-SubaddressString,
   FTN-AddressString,
   IMSI,
   BasicServiceCode,
```

```
   AlertingPattern,
   EMLPP-Priority,
   MaxMC-Bearers,
   MC-Bearers,
   ExternalSignalInfo
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}

   SS-Code
FROM MAP-SS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-Code (15) version18 (18)}
;
```

```
RegisterSS-Arg ::= SEQUENCE {
     ss-Code  SS-Code,
     basicService                    BasicServiceCode                OPTIONAL,
     forwardedToNumber               [4] AddressString               OPTIONAL,
     forwardedToSubaddress           [6] ISDN-SubaddressString       OPTIONAL,
     noReplyConditionTime            [5] NoReplyConditionTime        OPTIONAL,
     ...,
     defaultPriority                 [7] EMLPP-Priority              OPTIONAL,
     nbrUser  [8] MC-Bearers         OPTIONAL,
     longFTN-Supported               [9]  NULL OPTIONAL }
```

```
NoReplyConditionTime ::= INTEGER (5..30)
```

```
SS-Info ::= CHOICE {
     forwardingInfo                  [0] ForwardingInfo,
     callBarringInfo                 [1] CallBarringInfo,
     ss-Data  [3] SS-Data}
```

```
ForwardingInfo ::= SEQUENCE {
     ss-Code  SS-Code                OPTIONAL,
     forwardingFeatureList           ForwardingFeatureList,
     ...}
```

```
ForwardingFeatureList ::=
     SEQUENCE SIZE (1..maxNumOfBasicServiceGroups) OF
          ForwardingFeature
```

```
ForwardingFeature ::= SEQUENCE {
     basicService                    BasicServiceCode                OPTIONAL,
     ss-Status [4] SS-Status         OPTIONAL,
     forwardedToNumber               [5] ISDN-AddressString          OPTIONAL,
     forwardedToSubaddress           [8] ISDN-SubaddressString       OPTIONAL,
     forwardingOptions               [6] ForwardingOptions           OPTIONAL,
     noReplyConditionTime            [7] NoReplyConditionTime        OPTIONAL,
     ...,
     longForwardedToNumber           [9] FTN-AddressString           OPTIONAL }
```

```
SS-Status ::= OCTET STRING (SIZE (1))

     -- bits 8765: 0000 (unused)
     -- bits 4321: Used to convey the "P bit","R bit","A bit" and "Q bit",
     --     representing supplementary service state information
     --     as defined in TS 3GPP TS 23.011 [22]

     -- bit 4: "Q bit"

     -- bit 3: "P bit"

     -- bit 2: "R bit"

     -- bit 1: "A bit"
```

```
ForwardingOptions ::= OCTET STRING (SIZE (1))

     -- bit 8: notification to forwarding party
     --   0  no notification
     --   1  notification

     -- bit 7: redirecting presentation
     --   0 no presentation
     --   1  presentation

     -- bit 6: notification to calling party
     --   0  no notification
     --   1  notification

     -- bit 5: 0 (unused)

     -- bits 43: forwarding reason
     --   00  ms not reachable
     --   01  ms busy
     --   10  no reply
     --   11  unconditional when used in a SRI Result,
     --     or call deflection when used in a RCH Argument
     -- bits 21: 00 (unused)
```

```
CallBarringInfo ::= SEQUENCE {
     ss-Code   SS-Code                    OPTIONAL,
     callBarringFeatureList               CallBarringFeatureList,
     ...}
```

```
CallBarringFeatureList ::= SEQUENCE SIZE (1..maxNumOfBasicServiceGroups) OF
          CallBarringFeature
```

```
CallBarringFeature ::= SEQUENCE {
     basicService                         BasicServiceCode                 OPTIONAL,
     ss-Status [4] SS-Status              OPTIONAL,
     ...}
```

```
SS-Data ::= SEQUENCE {
     ss-Code   SS-Code                    OPTIONAL,
     ss-Status [4] SS-Status              OPTIONAL,
     ss-SubscriptionOption                SS-SubscriptionOption            OPTIONAL,
     basicServiceGroupList                BasicServiceGroupList            OPTIONAL,
     ...,
     defaultPriority                      EMLPP-Priority                   OPTIONAL,
     nbrUser   [5] MC-Bearers             OPTIONAL
     }
```

```
SS-SubscriptionOption ::= CHOICE {
     cliRestrictionOption                 [2] CliRestrictionOption,
     overrideCategory                     [1] OverrideCategory}
```

```
CliRestrictionOption ::= ENUMERATED {
     permanent  (0),
     temporaryDefaultRestricted  (1),
     temporaryDefaultAllowed  (2)}
```

```
OverrideCategory ::= ENUMERATED {
     overrideEnabled  (0),
     overrideDisabled  (1)}
```

```
SS-ForBS-Code ::= SEQUENCE {
     ss-Code   SS-Code,
     basicService                         BasicServiceCode                 OPTIONAL,
     ...,
     longFTN-Supported                    [4]  NULL OPTIONAL }
```

```
GenericServiceInfo ::= SEQUENCE {
     ss-Status SS-Status,
     cliRestrictionOption                 CliRestrictionOption             OPTIONAL,
     ...,
     maximumEntitledPriority              [0] EMLPP-Priority               OPTIONAL,
     defaultPriority                      [1] EMLPP-Priority               OPTIONAL,
     ccbs-FeatureList                     [2] CCBS-FeatureList             OPTIONAL,
     nbrSB     [3] MaxMC-Bearers          OPTIONAL,
     nbrUser   [4] MC-Bearers             OPTIONAL,
     nbrSN     [5] MC-Bearers             OPTIONAL }
```

```
CCBS-FeatureList ::= SEQUENCE SIZE (1..maxNumOfCCBS-Requests) OF
         CCBS-Feature
```

```
maxNumOfCCBS-Requests  INTEGER ::= 5
```

```
CCBS-Feature ::= SEQUENCE {
     ccbs-Index                          [0] CCBS-Index                    OPTIONAL,
     b-subscriberNumber                  [1] ISDN-AddressString            OPTIONAL,
     b-subscriberSubaddress              [2] ISDN-SubaddressString         OPTIONAL,
     basicServiceGroup                   [3] BasicServiceCode              OPTIONAL,
     ...}
```

```
CCBS-Index  ::= INTEGER (1..maxNumOfCCBS-Requests)
```

```
InterrogateSS-Res ::= CHOICE {
     ss-Status [0] SS-Status,
     basicServiceGroupList               [2] BasicServiceGroupList,
     forwardingFeatureList               [3] ForwardingFeatureList,
     genericServiceInfo                  [4]  GenericServiceInfo }
```

```
USSD-Arg ::= SEQUENCE {
     ussd-DataCodingScheme               USSD-DataCodingScheme,
     ussd-String                         USSD-String,
     ... ,
     alertingPattern                     AlertingPattern                   OPTIONAL,
     msisdn    [0] ISDN-AddressString    OPTIONAL }
```

```
USSD-Res ::= SEQUENCE {
     ussd-DataCodingScheme               USSD-DataCodingScheme,
     ussd-String                         USSD-String,
     ...}
```

```
USSD-DataCodingScheme ::= OCTET STRING (SIZE (1))
     -- The structure of the USSD-DataCodingScheme is defined by
     -- the Cell Broadcast Data Coding Scheme as described in
     -- TS 3GPP TS 23.038 [25]
```

```
USSD-String ::= OCTET STRING (SIZE (1..maxUSSD-StringLength))
     -- The structure of the contents of the USSD-String is dependent
     -- on the USSD-DataCodingScheme as described in TS 3GPP TS 23.038 [25].
```

```
maxUSSD-StringLength  INTEGER ::= 160
```

```
Password ::= NumericString
     (FROM ("0"|"1"|"2"|"3"|"4"|"5"|"6"|"7"|"8"|"9"))
     (SIZE (4))
```

```
GuidanceInfo ::= ENUMERATED {
     enterPW  (0),
     enterNewPW  (1),
     enterNewPW-Again  (2)}
     -- How this information is really delivered to the subscriber
     -- (display, announcement, ...) is not part of this
     -- specification.
```

```
SS-List ::= SEQUENCE SIZE (1..maxNumOfSS) OF
         SS-Code
```

```
maxNumOfSS  INTEGER ::= 30
```

```
SS-InfoList ::= SEQUENCE SIZE (1..maxNumOfSS) OF
         SS-Info
```

```
BasicServiceGroupList ::= SEQUENCE SIZE (1..maxNumOfBasicServiceGroups) OF
         BasicServiceCode
```

```
maxNumOfBasicServiceGroups  INTEGER ::= 13
```

```
SS-InvocationNotificationArg ::= SEQUENCE {
    imsi     [0] IMSI,
    msisdn   [1] ISDN-AddressString,
    ss-Event [2] SS-Code,
    -- The following SS-Code values are allowed :
    -- ect    SS-Code ::= '00110001'B
    -- multiPTY                        SS-Code ::= '01010001'B
    -- cd     SS-Code ::= '00100100'B
    -- ccbs   SS-Code ::= '01000100'B
    ss-EventSpecification              [3] SS-EventSpecification         OPTIONAL,
    extensionContainer                 [4] ExtensionContainer            OPTIONAL,
    ...,
    b-subscriberNumber                 [5]  ISDN-AddressString           OPTIONAL,
    ccbs-RequestState                  [6]  CCBS-RequestState            OPTIONAL
    }
```

```
CCBS-RequestState ::= ENUMERATED {
    request   (0),
    recall    (1),
    active    (2),
    completed (3),
    suspended (4),
    frozen    (5),
    deleted   (6)
    }
```

```
SS-InvocationNotificationRes ::= SEQUENCE {
    extensionContainer                 ExtensionContainer                OPTIONAL,
    ...
    }
```

```
SS-EventSpecification ::= SEQUENCE SIZE (1..maxEventSpecification) OF
        AddressString
```

```
maxEventSpecification  INTEGER ::= 2
```

```
RegisterCC-EntryArg ::= SEQUENCE {
    ss-Code   [0]                      SS-Code,
    ccbs-Data [1]                      CCBS-Data OPTIONAL,
    ...}
```

```
CCBS-Data ::= SEQUENCE {
    ccbs-Feature                       [0]  CCBS-Feature,
    translatedB-Number                 [1]  ISDN-AddressString,
    serviceIndicator                   [2]  ServiceIndicator             OPTIONAL,
    callInfo  [3]                      ExternalSignalInfo,
    networkSignalInfo                  [4]  ExternalSignalInfo,
    ...}
```

```
ServiceIndicator ::= BIT STRING {
    clir-invoked (0),
    camel-invoked (1)} (SIZE(2..32))
    -- exception handling:
    -- bits 2 to 31 shall be ignored if received and not understood
```

```
RegisterCC-EntryRes ::= SEQUENCE {
    ccbs-Feature                       [0] CCBS-Feature                  OPTIONAL,
    ...}
```

```
EraseCC-EntryArg ::= SEQUENCE {
    ss-Code   [0]                      SS-Code,
    ccbs-Index                         [1]  CCBS-Index                   OPTIONAL,
    ...}
```

```
EraseCC-EntryRes ::= SEQUENCE {
    ss-Code   [0]                      SS-Code,
    ss-Status [1] SS-Status            OPTIONAL,
    ...}
```

```
END
```

17.7.5 Supplementary service codes

```
MAP-SS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-Code (15) version18 (18)}
```

```
DEFINITIONS

::=

BEGIN

SS-Code ::= OCTET STRING (SIZE (1))
     -- This type is used to represent the code identifying a single
     -- supplementary service, a group of supplementary services, or
     -- all supplementary services. The services and abbreviations
     -- used are defined in TS 3GPP TS 22.004 [5]. The internal structure is
     -- defined as follows:
     --
     -- bits 87654321: group (bits 8765), and specific service
     -- (bits 4321)

allSS          SS-Code ::= '00000000'B
     -- reserved for possible future use
     -- all SS

allLineIdentificationSS                  SS-Code ::= '00010000'B
     -- reserved for possible future use
     -- all line identification SS
clip      SS-Code ::= '00010001'B
     -- calling line identification presentation
clir      SS-Code ::= '00010010'B
     -- calling line identification restriction
colp      SS-Code ::= '00010011'B
     -- connected line identification presentation
colr      SS-Code ::= '00010100'B
     -- connected line identification restriction
mci       SS-Code ::= '00010101'B
     -- reserved for possible future use
     -- malicious call identification

allNameIdentificationSS                  SS-Code ::= '00011000'B
     -- all name identification SS
cnap      SS-Code ::= '00011001'B
     -- calling name presentation

     -- SS-Codes '00011010'B to '00011111'B are reserved for future
     -- NameIdentification Supplementary Service use.

allForwardingSS                          SS-Code ::= '00100000'B
     -- all forwarding SS
cfu       SS-Code ::= '00100001'B
     -- call forwarding unconditional
allCondForwardingSS                      SS-Code ::= '00101000'B
     -- all conditional forwarding SS
cfb       SS-Code ::= '00101001'B
     -- call forwarding on mobile subscriber busy
cfnry          SS-Code ::= '00101010'B
     -- call forwarding on no reply
cfnrc          SS-Code ::= '00101011'B
     -- call forwarding on mobile subscriber not reachable
cd        SS-Code ::= '00100100'B
     -- call deflection

allCallOfferingSS                        SS-Code ::= '00110000'B
     -- reserved for possible future use
     -- all call offering SS includes also all forwarding SS
ect       SS-Code ::= '00110001'B
     -- explicit call transfer
mah       SS-Code ::= '00110010'B
     -- reserved for possible future use
     -- mobile access hunting
```

```
allCallCompletionSS                     SS-Code ::= '01000000'B
     -- reserved for possible future use
     -- all Call completion SS
cw        SS-Code ::= '01000001'B
     -- call waiting
hold      SS-Code ::= '01000010'B
     -- call hold
ccbs-A         SS-Code ::= '01000011'B
     -- completion of call to busy subscribers, originating side
     -- this SS-Code is used only in InsertSubscriberData, DeleteSubscriberData
     -- and InterrogateSS
ccbs-B         SS-Code ::= '01000100'B
     -- completion of call to busy subscribers, destination side
     -- this SS-Code is used only in InsertSubscriberData and DeleteSubscriberData
mc        SS-Code ::= '01000101'B
     -- multicall
```

```
allMultiPartySS                         SS-Code ::= '01010000'B
     -- reserved for possible future use
     -- all multiparty SS
multiPTY       SS-Code ::= '01010001'B
     -- multiparty
```

```
allCommunityOfInterest-SS               SS-Code ::= '01100000'B
     -- reserved for possible future use
     -- all community of interest SS
cug       SS-Code ::= '01100001'B
     -- closed user group
```

```
allChargingSS SS-Code ::= '01110000'B
     -- reserved for possible future use
     -- all charging SS
aoci      SS-Code ::= '01110001'B
     -- advice of charge information
aocc      SS-Code ::= '01110010'B
     -- advice of charge charging
```

```
allAdditionalInfoTransferSS             SS-Code ::= '10000000'B
     -- reserved for possible future use
     -- all additional information transfer SS
uus1      SS-Code ::= '10000001'B
     -- UUS1 user-to-user signalling
uus2      SS-Code ::= '10000010'B
     -- UUS2 user-to-user signalling
uus3      SS-Code ::= '10000011'B
     -- UUS3 user-to-user signalling
```

```
allBarringSS  SS-Code ::= '10010000'B
     -- all barring SS
barringOfOutgoingCalls                  SS-Code ::= '10010001'B
     -- barring of outgoing calls
baoc      SS-Code ::= '10010010'B
     -- barring of all outgoing calls
boic      SS-Code ::= '10010011'B
     -- barring of outgoing international calls
boicExHC       SS-Code ::= '10010100'B
     -- barring of outgoing international calls except those directed
     -- to the home PLMN Country
barringOfIncomingCalls                  SS-Code ::= '10011001'B
     -- barring of incoming calls
baic      SS-Code ::= '10011010'B
     -- barring of all incoming calls
bicRoam        SS-Code ::= '10011011'B
     -- barring of incoming calls when roaming outside home PLMN
     -- Country
```

```
allPLMN-specificSS                  SS-Code ::= '11110000'B
plmn-specificSS-1                   SS-Code ::= '11110001'B
plmn-specificSS-2                   SS-Code ::= '11110010'B
plmn-specificSS-3                   SS-Code ::= '11110011'B
plmn-specificSS-4                   SS-Code ::= '11110100'B
plmn-specificSS-5                   SS-Code ::= '11110101'B
plmn-specificSS-6                   SS-Code ::= '11110110'B
plmn-specificSS-7                   SS-Code ::= '11110111'B
plmn-specificSS-8                   SS-Code ::= '11111000'B
plmn-specificSS-9                   SS-Code ::= '11111001'B
plmn-specificSS-A                   SS-Code ::= '11111010'B
plmn-specificSS-B                   SS-Code ::= '11111011'B
plmn-specificSS-C                   SS-Code ::= '11111100'B
plmn-specificSS-D                   SS-Code ::= '11111101'B
plmn-specificSS-E                   SS-Code ::= '11111110'B
plmn-specificSS-F                   SS-Code ::= '11111111'B
```

```
allCallPrioritySS                   SS-Code ::= '10100000'B
     -- reserved for possible future use
     -- all call priority SS
emlpp          SS-Code ::= '10100001'B
     -- enhanced Multilevel Precedence Pre-emption (EMLPP) service
```

```
allLCSPrivacyException              SS-Code ::= '10110000'B
     -- all LCS Privacy Exception Classes
universal      SS-Code ::= '10110001'B
     -- allow location by any LCS client
callSessionRelated                  SS-Code ::= '10110010'B
     -- allow location by any value added LCS client to which a call/session
     -- is established from the target MS
callSessionUnrelated                SS-Code ::= '10110011'B
     -- allow location by designated external value added LCS clients
plmnoperator   SS-Code ::= '10110100'B
     -- allow location by designated PLMN operator LCS clients
serviceType    SS-Code ::= '10110101'B
     -- allow location by LCS clients of a designated LCS service type
```

```
allMOLR-SS     SS-Code ::= '11000000'B
     -- all Mobile Originating Location Request Classes
basicSelfLocation                   SS-Code ::= '11000001'B
     -- allow an MS to request its own location
autonomousSelfLocation              SS-Code ::= '11000010'B
     -- allow an MS to perform self location without interaction
     -- with the PLMN for a predetermined period of time
transferToThirdParty                SS-Code ::= '11000011'B
     -- allow an MS to request transfer of its location to another LCS client
```

```
END
```

17.7.6 Short message data types

```
MAP-SM-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SM-DataTypes (16) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   RoutingInfoForSM-Arg,
   RoutingInfoForSM-Res,
   MO-ForwardSM-Arg,
   MO-ForwardSM-Res,
   MT-ForwardSM-Arg,
   MT-ForwardSM-Res,
   ReportSM-DeliveryStatusArg,
   ReportSM-DeliveryStatusRes,
   AlertServiceCentreArg,
   InformServiceCentreArg,
   ReadyForSM-Arg,
   ReadyForSM-Res,
   SM-DeliveryOutcome,
   AlertReason,
```

```
   Additional-Number,
   MT-ForwardSM-VGCS-Arg,
   MT-ForwardSM-VGCS-Res
;

IMPORTS
   AddressString,
   ISDN-AddressString,
   SignalInfo,
   IMSI,
   LMSI,
   ASCI-CallReference,
   Time,
   NetworkNodeDiameterAddress,
   HLR-Id

FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   AbsentSubscriberDiagnosticSM
FROM MAP-ER-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ER-DataTypes (17) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}
;
```

```
RoutingInfoForSM-Arg ::= SEQUENCE {
     msisdn    [0] ISDN-AddressString,
     sm-RP-PRI [1] BOOLEAN,
     serviceCentreAddress                [2] AddressString,
     extensionContainer                  [6] ExtensionContainer            OPTIONAL,
     ... ,
     gprsSupportIndicator                [7]  NULL OPTIONAL,
     -- gprsSupportIndicator is set only if the SMS-GMSC supports
     -- receiving of two numbers from the HLR
     sm-RP-MTI [8] SM-RP-MTI             OPTIONAL,
     sm-RP-SMEA                          [9] SM-RP-SMEA              OPTIONAL,
     sm-deliveryNotIntended              [10] SM-DeliveryNotIntended       OPTIONAL,
     ip-sm-gwGuidanceIndicator           [11] NULL OPTIONAL,
     imsi      [12] IMSI                 OPTIONAL,
     t4-Trigger-Indicator                [14] NULL OPTIONAL,
     singleAttemptDelivery               [13] NULL OPTIONAL,
     correlationID                       [15] CorrelationID                OPTIONAL,
     smsf-supportIndicator               [16] NULL                         OPTIONAL }
```

```
SM-DeliveryNotIntended ::= ENUMERATED {
     onlyIMSI-requested  (0),
     onlyMCC-MNC-requested  (1),
     ...}
```

```
SM-RP-MTI ::= INTEGER (0..10)
     -- 0 SMS Deliver
     -- 1 SMS Status Report
     -- other values are reserved for future use and shall be discarded if
     -- received
```

```
SM-RP-SMEA ::= OCTET STRING (SIZE (1..12))
     -- this parameter contains an address field which is encoded
     -- as defined in 3GPP TS 23.040. An address field contains 3 elements :
     --   address-length
     --   type-of-address
     --   address-value
```

```
RoutingInfoForSM-Res ::= SEQUENCE {
     imsi      IMSI,
     locationInfoWithLMSI                [0] LocationInfoWithLMSI,
     extensionContainer                  [4] ExtensionContainer            OPTIONAL,
     ...,
     ip-sm-gwGuidance                    [5] IP-SM-GW-Guidance             OPTIONAL }
```

```
IP-SM-GW-Guidance ::= SEQUENCE {
     minimumDeliveryTimeValue            SM-DeliveryTimerValue,
     recommendedDeliveryTimeValue        SM-DeliveryTimerValue,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
LocationInfoWithLMSI ::= SEQUENCE {
     networkNode-Number                  [1] ISDN-AddressString,
     lmsi      LMSI                          OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     gprsNodeIndicator                   [5]  NULL OPTIONAL,
     -- gprsNodeIndicator is set only if the SGSN number is sent as the
     -- Network Node Number
     additional-Number                   [6] Additional-Number             OPTIONAL,
     networkNodeDiameterAddress          [7] NetworkNodeDiameterAddress    OPTIONAL,
     additionalNetworkNodeDiameterAddress [8] NetworkNodeDiameterAddress   OPTIONAL,
     thirdNumber                         [9] Additional-Number             OPTIONAL,
     thirdNetworkNodeDiameterAddress     [10] NetworkNodeDiameterAddress   OPTIONAL,
     imsNodeIndicator                    [11] NULL OPTIONAL,
     -- gprsNodeIndicator and imsNodeIndicator shall not both be present.
     -- additionalNumber and thirdNumber shall not both contain the same type of number.
     smsf-3gpp-Number                    [12] ISDN-AddressString           OPTIONAL,
     smsf-3gpp-DiameterAddress           [13] NetworkNodeDiameterAddress   OPTIONAL,
     smsf-non-3gpp-Number                [14] ISDN-AddressString           OPTIONAL,
     smsf-non-3gpp-DiameterAddress       [15] NetworkNodeDiameterAddress   OPTIONAL,
     smsf-3gpp-address-indicator         [16] NULL                         OPTIONAL,
     smsf-non-3gpp-address-indicator     [17] NULL                         OPTIONAL
     --
     -- If smsf-supportIndicator was not included in the request, in RoutingInfoForSM-Arg,
     -- then smsf-3gpp Number/DiameterAddress, smsf-non-3gpp Number/DiameterAddress and
     -- smsf-address-indicator and smsf-non-3gpp-address-indicator shall be absent.
     --
     -- If smsf-3gpp-address-indicator is present, it indicates that the networkNode-Number
     -- (and networkNodeDiameterAddress, if present) contains the address of an SMSF for
     -- 3GPP access.
     --
     -- If smsf-non-3gpp-address-indicator is present, it indicates that the
     -- networkNode-Number (and networkNodeDiameterAddress, if present) contains the
     -- address of an SMSF for non 3GPP access.
     --
     -- At most one of gprsNodeIndicator, imsNodeIndicator, smsf-3gpp-address-indicator
     -- and smsf-non-3gpp-address-indicator shall be present. Absence of all these
     -- indicators indicate that the networkNode-Number (and networkNodeDiameterAddress,
     -- if present) contains the address of an MSC/MME.

     }
```

```
Additional-Number ::= CHOICE {
     msc-Number                          [0] ISDN-AddressString,
     sgsn-Number                         [1] ISDN-AddressString}
     -- msc-number can be the MSC number or
     -- the SMS Router number or the MME number for MT SMS
     -- sgsn-number can be the SGSN number or the SMS Router number
```

```
MO-ForwardSM-Arg ::= SEQUENCE {
     sm-RP-DA  SM-RP-DA,
     sm-RP-OA  SM-RP-OA,
     sm-RP-UI  SignalInfo,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ... ,
     imsi      IMSI                      OPTIONAL,
     correlationID                       [0] CorrelationID                 OPTIONAL,
     sm-DeliveryOutcome                  [1] SM-DeliveryOutcome            OPTIONAL
 }
```

```
MO-ForwardSM-Res ::= SEQUENCE {
     sm-RP-UI  SignalInfo                OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
MT-ForwardSM-Arg ::= SEQUENCE {
     sm-RP-DA  SM-RP-DA,
     sm-RP-OA  SM-RP-OA,
     sm-RP-UI  SignalInfo,
     moreMessagesToSend                  NULL     OPTIONAL,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     smDeliveryTimer                     SM-DeliveryTimerValue           OPTIONAL,
     smDeliveryStartTime                 Time     OPTIONAL,
     smsOverIP-OnlyIndicator             [0] NULL  OPTIONAL,
     correlationID                       [1] CorrelationID               OPTIONAL,
     maximumRetransmissionTime           [2] Time  OPTIONAL,
     smsGmscAddress                      [3] ISDN-AddressString          OPTIONAL,
     smsGmscDiameterAddress              [4] NetworkNodeDiameterAddress  OPTIONAL }
     -- SM-DeliveryTimerValue contains the value used by the SMS-GMSC
```

```
CorrelationID ::= SEQUENCE {
     hlr-id    [0] HLR-Id                OPTIONAL,
     sip-uri-A [1] SIP-URI               OPTIONAL,
     sip-uri-B [2] SIP-URI}
```

```
SIP-URI ::= OCTET STRING
-- octets are coded as defined in IETF RFC 3261
```

```
MT-ForwardSM-Res ::= SEQUENCE {
     sm-RP-UI  SignalInfo                OPTIONAL,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ... }
```

```
SM-RP-DA ::= CHOICE {
     imsi      [0] IMSI,
     lmsi      [1] LMSI,
     serviceCentreAddressDA              [4] AddressString,
     noSM-RP-DA                          [5] NULL}
```

```
SM-RP-OA ::= CHOICE {
     msisdn    [2] ISDN-AddressString,
     serviceCentreAddressOA              [4] AddressString,
     noSM-RP-OA                          [5] NULL}
```

```
SM-DeliveryTimerValue ::= INTEGER (30..600)
```

```
ReportSM-DeliveryStatusArg ::= SEQUENCE {
     msisdn    ISDN-AddressString,
     serviceCentreAddress                AddressString,
     sm-DeliveryOutcome                  SM-DeliveryOutcome,
     absentSubscriberDiagnosticSM        [0] AbsentSubscriberDiagnosticSM
                                         OPTIONAL,
     extensionContainer                  [1] ExtensionContainer            OPTIONAL,
     ...,
     gprsSupportIndicator                [2]  NULL OPTIONAL,
     -- gprsSupportIndicator is set only if the SMS-GMSC supports
     -- handling of two delivery outcomes
     deliveryOutcomeIndicator            [3]  NULL OPTIONAL,
     -- DeliveryOutcomeIndicator is set when the SM-DeliveryOutcome
     -- is for GPRS
     additionalSM-DeliveryOutcome        [4]  SM-DeliveryOutcome           OPTIONAL,
     -- If received, additionalSM-DeliveryOutcome is for GPRS
     -- If DeliveryOutcomeIndicator is set, then AdditionalSM-DeliveryOutcome shall be absent
     additionalAbsentSubscriberDiagnosticSM   [5]  AbsentSubscriberDiagnosticSM OPTIONAL,
     -- If received additionalAbsentSubscriberDiagnosticSM is for GPRS
     -- If DeliveryOutcomeIndicator is set, then AdditionalAbsentSubscriberDiagnosticSM
     -- shall be absent
     ip-sm-gw-Indicator                  [6]  NULL OPTIONAL,
     -- the ip-sm-gw indicator indicates by its presence that sm-deliveryOutcome
     -- is for delivery via IMS
     -- If present, deliveryOutcomeIndicator shall be absent.
     ip-sm-gw-sm-deliveryOutcome         [7]  SM-DeliveryOutcome           OPTIONAL,
     -- If received ip-sm-gw-sm-deliveryOutcome is for delivery via IMS
     -- If ip-sm-gw-Indicator is set, then ip-sm-gw-sm-deliveryOutcome shall be absent
     ip-sm-gw-absentSubscriberDiagnosticSM    [8]  AbsentSubscriberDiagnosticSM OPTIONAL,
     -- If received ip-sm-gw-sm-absentSubscriberDiagnosticSM is for delivery via IMS
     -- If ip-sm-gw-Indicator is set, then ip-sm-gw-sm-absentSubscriberDiagnosticSM
     -- shall be absent
     imsi     [9] IMSI                   OPTIONAL,
     singleAttemptDelivery               [10] NULL OPTIONAL,
     correlationID                       [11] CorrelationID                OPTIONAL,
     smsf-3gpp-deliveryOutcomeIndicator  [12] NULL                         OPTIONAL,
     -- smsf-3gpp-deliveryOutcome is set when the SM-DeliveryOutcome
     -- is for 3GPP-SMSF
     smsf-3gpp-deliveryOutcome           [13] SM-DeliveryOutcome           OPTIONAL,
     -- If smsf-3gpp-deliveryOutcomeIndicator is set, then smsf-3gpp-deliveryOutcome
     -- shall be absent
     smsf-3gpp-absentSubscriberDiagSM    [14] AbsentSubscriberDiagnosticSM  OPTIONAL,
     -- If smsf-3gpp-deliveryOutcomeIndicator is set, then
     -- smsf-3gpp-absentSubscriberDiagSM shall be absent
     smsf-non-3gpp-deliveryOutcomeIndicator   [15] NULL                    OPTIONAL,
     -- smsf-non-3gpp-deliveryOutcomeIndicator is set when the SM-DeliveryOutcome
     -- is for non-3GPP-SMSF
     smsf-non-3gpp-deliveryOutcome       [16] SM-DeliveryOutcome           OPTIONAL,
     -- If smsf-non-3gpp-deliveryOutcomeIndicator is set, then smsf-non-3gpp-deliveryOutcome
     -- shall be absent
     smsf-non-3gpp-absentSubscriberDiagSM [17] AbsentSubscriberDiagnosticSM  OPTIONAL
     -- If smsf-non-3gpp-deliveryOutcomeIndicator is set, then
     -- smsf-non-3gpp-absentSubscriberDiagSM shall be absent

}
```

```
SM-DeliveryOutcome ::= ENUMERATED {
     memoryCapacityExceeded  (0),
     absentSubscriber  (1),
     successfulTransfer  (2)}
```

```
ReportSM-DeliveryStatusRes ::= SEQUENCE {
     storedMSISDN                        ISDN-AddressString                OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
AlertServiceCentreArg ::= SEQUENCE {
     msisdn    ISDN-AddressString,
     serviceCentreAddress                AddressString,
     ...,
     imsi      IMSI                           OPTIONAL,
     correlationID                       CorrelationID                     OPTIONAL,
     maximumUeAvailabilityTime           [0] Time                          OPTIONAL,
     smsGmscAlertEvent                   [1] SmsGmsc-Alert-Event           OPTIONAL,
     smsGmscDiameterAddress              [2] NetworkNodeDiameterAddress    OPTIONAL,
     newSGSNNumber                       [3] ISDN-AddressString            OPTIONAL,
     newSGSNDiameterAddress              [4] NetworkNodeDiameterAddress    OPTIONAL,
     newMMENumber                        [5] ISDN-AddressString            OPTIONAL,
     newMMEDiameterAddress               [6] NetworkNodeDiameterAddress    OPTIONAL,
     newMSCNumber                        [7] ISDN-AddressString            OPTIONAL }
```

```
SmsGmsc-Alert-Event ::= ENUMERATED {
     msAvailableForMtSms  (0),
     msUnderNewServingNode  (1)  }
```

```
InformServiceCentreArg ::= SEQUENCE {
     storedMSISDN                        ISDN-AddressString                OPTIONAL,
     mw-Status MW-Status                 OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ... ,
     absentSubscriberDiagnosticSM        AbsentSubscriberDiagnosticSM      OPTIONAL,
     additionalAbsentSubscriberDiagnosticSM    [0]  AbsentSubscriberDiagnosticSM OPTIONAL }
     -- additionalAbsentSubscriberDiagnosticSM may be present only if
     -- absentSubscriberDiagnosticSM is present.
     -- if included, additionalAbsentSubscriberDiagnosticSM is for GPRS and
     -- absentSubscriberDiagnosticSM is for non-GPRS
```

```
MW-Status ::= BIT STRING {
     sc-AddressNotIncluded  (0),
     mnrf-Set  (1),
     mcef-Set  (2) ,
     mnrg-Set  (3)} (SIZE (6..16))
     -- exception handling:
     -- bits 4 to 15 shall be ignored if received and not understood
```

```
ReadyForSM-Arg ::= SEQUENCE {
     imsi      [0] IMSI,
     alertReason                         AlertReason,
     alertReasonIndicator                NULL      OPTIONAL,
     -- alertReasonIndicator is set only when the alertReason
     -- sent to HLR is for GPRS
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     additionalAlertReasonIndicator      [1] NULL  OPTIONAL,
     -- additionalAlertReasonIndicator is set only when the alertReason
     -- sent to HLR is for IP-SM-GW
     maximumUeAvailabilityTime           Time      OPTIONAL }
```

```
ReadyForSM-Res ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
AlertReason ::= ENUMERATED {
     ms-Present  (0),
     memoryAvailable  (1)}
```

```
MT-ForwardSM-VGCS-Arg ::= SEQUENCE {
     asciCallReference                   ASCI-CallReference,
     sm-RP-OA  SM-RP-OA,
     sm-RP-UI  SignalInfo,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
MT-ForwardSM-VGCS-Res ::= SEQUENCE {
     sm-RP-UI  [0] SignalInfo            OPTIONAL,
     dispatcherList                      [1] DispatcherList                OPTIONAL,
     ongoingCall                         NULL      OPTIONAL,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...,
     additionalDispatcherList            [3] AdditionalDispatcherList      OPTIONAL }
     -- additionalDispatcherList shall be absent if dispatcherList is absent or
     -- contains less than 5 ISDN-AddressStrings
```

```
DispatcherList ::=
     SEQUENCE SIZE (1..maxNumOfDispatchers) OF
          ISDN-AddressString
```

```
maxNumOfDispatchers  INTEGER ::= 5
```

```
AdditionalDispatcherList ::=
     SEQUENCE SIZE (1..maxNumOfAdditionalDispatchers) OF
          ISDN-AddressString
```

```
maxNumOfAdditionalDispatchers  INTEGER ::= 15
```

```
END
```

17.7.7 Error data types

```
MAP-ER-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ER-DataTypes (17) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   RoamingNotAllowedParam,
   CallBarredParam,
   CUG-RejectParam,
   SS-IncompatibilityCause,
   PW-RegistrationFailureCause,
   SM-DeliveryFailureCause,
   SystemFailureParam,
   DataMissingParam,
   UnexpectedDataParam,
   FacilityNotSupParam,
   OR-NotAllowedParam,
   UnknownSubscriberParam,
   NumberChangedParam,
   UnidentifiedSubParam,
   IllegalSubscriberParam,
   IllegalEquipmentParam,
   BearerServNotProvParam,
   TeleservNotProvParam,
   TracingBufferFullParam,
   NoRoamingNbParam,
   AbsentSubscriberParam,
   BusySubscriberParam,
   NoSubscriberReplyParam,
   ForwardingViolationParam,
   ForwardingFailedParam,
   ATI-NotAllowedParam,
   SubBusyForMT-SMS-Param,
   MessageWaitListFullParam,
   AbsentSubscriberSM-Param,
   AbsentSubscriberDiagnosticSM,
   ResourceLimitationParam,
   NoGroupCallNbParam,
   IncompatibleTerminalParam,
   ShortTermDenialParam,
   LongTermDenialParam,
   UnauthorizedRequestingNetwork-Param,
   UnauthorizedLCSClient-Param,
   PositionMethodFailure-Param,
   UnknownOrUnreachableLCSClient-Param,
   MM-EventNotSupported-Param,
   ATSI-NotAllowedParam,
   ATM-NotAllowedParam,
   IllegalSS-OperationParam,
   SS-NotAvailableParam,
   SS-SubscriptionViolationParam,
   InformationNotAvailableParam,
   TargetCellOutsideGCA-Param,
   OngoingGroupCallParam,
```

```
   PositionMethodFailure-Diagnostic,
   UnauthorizedLCSClient-Diagnostic

;

IMPORTS
   SS-Status
FROM MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-DataTypes (14) version18 (18)}

   SignalInfo,
   BasicServiceCode,
   NetworkResource,
   AdditionalNetworkResource,
   IMSI,
   Time
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}


   SS-Code
FROM MAP-SS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-Code (15) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}
;
```

```
RoamingNotAllowedParam ::= SEQUENCE {
     roamingNotAllowedCause              RoamingNotAllowedCause,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     additionalRoamingNotAllowedCause    [0] AdditionalRoamingNotAllowedCause OPTIONAL }

--   if the additionalRoamingNotallowedCause is received by the MSC/VLR or SGSN then the
--   roamingNotAllowedCause shall be discarded.
```

```
AdditionalRoamingNotAllowedCause ::= ENUMERATED {
     supportedRAT-TypesNotAllowed (0),
     ...}
```

```
RoamingNotAllowedCause ::= ENUMERATED {
     plmnRoamingNotAllowed  (0),
     operatorDeterminedBarring  (3)}
```

```
CallBarredParam ::= CHOICE {
     callBarringCause                    CallBarringCause,
     -- call BarringCause must not be used in version 3 and higher
     extensibleCallBarredParam           ExtensibleCallBarredParam
     -- extensibleCallBarredParam must not be used in version <3
     }
```

```
CallBarringCause ::= ENUMERATED {
     barringServiceActive  (0),
     operatorBarring  (1)}
```

```
ExtensibleCallBarredParam ::= SEQUENCE {
     callBarringCause                    CallBarringCause                  OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ... ,
     unauthorisedMessageOriginator       [1] NULL  OPTIONAL,
     anonymousCallRejection              [2] NULL  OPTIONAL }

-- unauthorisedMessageOriginator and anonymousCallRejection shall be mutually exclusive.
```

```
CUG-RejectParam ::= SEQUENCE {
     cug-RejectCause                     CUG-RejectCause                   OPTIONAL,
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
CUG-RejectCause ::= ENUMERATED {
     incomingCallsBarredWithinCUG  (0),
     subscriberNotMemberOfCUG  (1),
     requestedBasicServiceViolatesCUG-Constraints  (5),
     calledPartySS-InteractionViolation  (7)}
```

```
SS-IncompatibilityCause ::= SEQUENCE {
     ss-Code   [1] SS-Code                OPTIONAL,
     basicService                         BasicServiceCode                 OPTIONAL,
     ss-Status [4] SS-Status              OPTIONAL,
     ...}
```

```
PW-RegistrationFailureCause ::= ENUMERATED {
     undetermined  (0),
     invalidFormat  (1),
     newPasswordsMismatch  (2)}
```

```
SM-EnumeratedDeliveryFailureCause ::= ENUMERATED {
     memoryCapacityExceeded  (0),
     equipmentProtocolError  (1),
     equipmentNotSM-Equipped  (2),
     unknownServiceCentre  (3),
     sc-Congestion  (4),
     invalidSME-Address  (5),
     subscriberNotSC-Subscriber  (6)}
```

```
SM-DeliveryFailureCause ::= SEQUENCE {
     sm-EnumeratedDeliveryFailureCause    SM-EnumeratedDeliveryFailureCause,
     diagnosticInfo                       SignalInfo                       OPTIONAL,
     extensionContainer                   ExtensionContainer               OPTIONAL,
     ...}
```

```
AbsentSubscriberSM-Param ::= SEQUENCE {
     absentSubscriberDiagnosticSM         AbsentSubscriberDiagnosticSM      OPTIONAL,
     -- AbsentSubscriberDiagnosticSM can be either for non-GPRS
     -- or for GPRS
     extensionContainer                   ExtensionContainer                OPTIONAL,
     ...,
     additionalAbsentSubscriberDiagnosticSM   [0]  AbsentSubscriberDiagnosticSM OPTIONAL,
     -- if received, additionalAbsentSubscriberDiagnosticSM
     -- is for GPRS and absentSubscriberDiagnosticSM is
     -- for non-GPRS
     imsi     [1] IMSI                    OPTIONAL,
     -- when sent from HLR to IP-SM-GW, IMSI shall be present if UNRI is not set
     -- to indicate that the absent condition is met for CS and PS but not for IMS.
     requestedRetransmissionTime          [2] Time  OPTIONAL,
     userIdentifierAlert                  [3] IMSI  OPTIONAL }
```

```
AbsentSubscriberDiagnosticSM ::= INTEGER (0..255)
     -- AbsentSubscriberDiagnosticSM values are defined in 3GPP TS 23.040
```

```
SystemFailureParam ::= CHOICE {
     networkResource                      NetworkResource,
     -- networkResource must not be used in version 3
     extensibleSystemFailureParam         ExtensibleSystemFailureParam
     -- extensibleSystemFailureParam must not be used in version <3
     }
```

```
ExtensibleSystemFailureParam ::= SEQUENCE {
     networkResource                      NetworkResource                  OPTIONAL,
     extensionContainer                   ExtensionContainer               OPTIONAL,
     ...,
     additionalNetworkResource            [0] AdditionalNetworkResource    OPTIONAL,
     failureCauseParam                    [1] FailureCauseParam            OPTIONAL }
```

```
FailureCauseParam ::= ENUMERATED {
     limitReachedOnNumberOfConcurrentLocationRequests (0),
     ... }
     -- if unknown value is received in FailureCauseParam it shall be ignored
```

```
DataMissingParam ::= SEQUENCE {
     extensionContainer                   ExtensionContainer               OPTIONAL,
     ...}
```

```
UnexpectedDataParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     unexpectedSubscriber [0]            NULL      OPTIONAL}
-- the unexpectedSubscriber indication in the unexpectedDataValue error shall not be used
-- for operations that allow the unidentifiedSubscriber error.
```

```
FacilityNotSupParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     shapeOfLocationEstimateNotSupported [0]  NULL OPTIONAL,
     neededLcsCapabilityNotSupportedInServingNode [1] NULL                 OPTIONAL }
```

```
OR-NotAllowedParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
UnknownSubscriberParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     unknownSubscriberDiagnostic         UnknownSubscriberDiagnostic       OPTIONAL}
```

```
UnknownSubscriberDiagnostic ::= ENUMERATED {
     imsiUnknown  (0),
     gprs-eps-SubscriptionUnknown  (1),
     ...,
     npdbMismatch  (2)}
     -- if unknown values are received in
     -- UnknownSubscriberDiagnostic they shall be discarded
```

```
NumberChangedParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
UnidentifiedSubParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
IllegalSubscriberParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
IllegalEquipmentParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
BearerServNotProvParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
TeleservNotProvParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
TracingBufferFullParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
NoRoamingNbParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
AbsentSubscriberParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     absentSubscriberReason              [0] AbsentSubscriberReason        OPTIONAL}
```

```
AbsentSubscriberReason ::= ENUMERATED {
     imsiDetach (0),
     restrictedArea (1),
     noPageResponse (2),
     ... ,
     purgedMS (3),
     mtRoamingRetry (4),
     busySubscriber (5)}
-- exception handling: at reception of other values than the ones listed the
-- AbsentSubscriberReason shall be ignored.
-- The AbsentSubscriberReason: purgedMS is defined for the Super-Charger feature
-- (see TS 23.116). If this value is received in a Provide Roaming Number response
-- it shall be mapped to the AbsentSubscriberReason: imsiDetach in the Send Routeing
-- Information response
-- The AbsentSubscriberReason: mtRoamingRetry is used during MT Roaming Retry,
-- see 3GPP TS 23.018[97].
-- The AbsentSubscriberReason: busySubscriber is used during MT Roaming Forwarding,
-- see 3GPP TS 23.018[97].
```

```
BusySubscriberParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     ccbs-Possible                       [0] NULL  OPTIONAL,
     ccbs-Busy [1] NULL                  OPTIONAL}
```

```
NoSubscriberReplyParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ForwardingViolationParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ForwardingFailedParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ATI-NotAllowedParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ATSI-NotAllowedParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ATM-NotAllowedParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
IllegalSS-OperationParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
SS-NotAvailableParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
SS-SubscriptionViolationParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
InformationNotAvailableParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
SubBusyForMT-SMS-Param ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ... ,
     gprsConnectionSuspended             NULL     OPTIONAL }
     -- If GprsConnectionSuspended is not understood it shall
     -- be discarded
```

```
MessageWaitListFullParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ResourceLimitationParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
NoGroupCallNbParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
IncompatibleTerminalParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
ShortTermDenialParam ::= SEQUENCE {
     ...}
```

```
LongTermDenialParam ::= SEQUENCE {
     ...}
```

```
UnauthorizedRequestingNetwork-Param ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
UnauthorizedLCSClient-Param ::= SEQUENCE {
     unauthorizedLCSClient-Diagnostic    [0] UnauthorizedLCSClient-Diagnostic   OPTIONAL,
     extensionContainer                  [1] ExtensionContainer                 OPTIONAL,
     ... }
```

```
UnauthorizedLCSClient-Diagnostic ::= ENUMERATED {
     noAdditionalInformation (0),
     clientNotInMSPrivacyExceptionList (1),
     callToClientNotSetup (2),
     privacyOverrideNotApplicable (3),
     disallowedByLocalRegulatoryRequirements (4),
     ...,
     unauthorizedPrivacyClass (5),
     unauthorizedCallSessionUnrelatedExternalClient (6),
     unauthorizedCallSessionRelatedExternalClient (7) }
--   exception handling:
--   any unrecognized value shall be ignored
```

```
PositionMethodFailure-Param ::= SEQUENCE {
     positionMethodFailure-Diagnostic    [0] PositionMethodFailure-Diagnostic   OPTIONAL,
     extensionContainer                  [1] ExtensionContainer                 OPTIONAL,
     ... }
```

```
PositionMethodFailure-Diagnostic ::= ENUMERATED {
     congestion  (0),
     insufficientResources  (1),
     insufficientMeasurementData  (2),
     inconsistentMeasurementData  (3),
     locationProcedureNotCompleted  (4),
     locationProcedureNotSupportedByTargetMS  (5),
     qoSNotAttainable  (6),
     positionMethodNotAvailableInNetwork  (7),
     positionMethodNotAvailableInLocationArea (8),
     ... }
--   exception handling:
--   any unrecognized value shall be ignored
```

```
UnknownOrUnreachableLCSClient-Param ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
MM-EventNotSupported-Param ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
TargetCellOutsideGCA-Param ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
OngoingGroupCallParam ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...}
```

```
END
```

17.7.8 Common data types

```
MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS

   -- general data types and values
   AddressString,
   ISDN-AddressString,
   maxISDN-AddressLength,
   FTN-AddressString,
   ISDN-SubaddressString,
   ExternalSignalInfo,
   Ext-ExternalSignalInfo,
   AccessNetworkSignalInfo,
   SignalInfo,
   maxSignalInfoLength,
   AlertingPattern,
   TBCD-STRING,
   DiameterIdentity,
   Time,
   HLR-Id,

   -- data types for numbering and identification
   IMSI,
   TMSI,
   Identity,
   SubscriberId,
   IMEI,
   HLR-List,
   LMSI,
   GlobalCellId,
   NetworkResource,
   AdditionalNetworkResource,
   NAEA-PreferredCI,
   NAEA-CIC,
   ASCI-CallReference,
   SubscriberIdentity,
   PLMN-Id,
   E-UTRAN-CGI,
   NR-CGI,
   TA-Id,
   RAIdentity,
   NetworkNodeDiameterAddress,

   -- data types for CAMEL
   CellGlobalIdOrServiceAreaIdOrLAI,
   CellGlobalIdOrServiceAreaIdFixedLength,
   LAIFixedLength,

   -- data types for subscriber management
   BasicServiceCode,
   Ext-BasicServiceCode,
   EMLPP-Info,
   EMLPP-Priority,
   MC-SS-Info,
   MaxMC-Bearers,
   MC-Bearers,
   Ext-SS-Status,

   -- data types for geographic location
   AgeOfLocationInformation,
   LCSClientExternalID,
   LCSClientInternalID,
   LCSServiceTypeID,
```

```
   -- gprs location registration types
   GSN-Address

;

IMPORTS
   TeleserviceCode,
   Ext-TeleserviceCode
FROM MAP-TS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-TS-Code (19) version18 (18)}

   BearerServiceCode,
   Ext-BearerServiceCode
FROM MAP-BS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-BS-Code (20) version18 (18)}

   SS-Code
FROM MAP-SS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SS-Code (15) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}
;


-- general data types
```

```
TBCD-STRING ::= OCTET STRING
     -- This type (Telephony Binary Coded Decimal String) is used to
     -- represent several digits from 0 through 9, *, #, a, b, c, two
     -- digits per octet, each digit encoded 0000 to 1001 (0 to 9),
     -- 1010 (*), 1011 (#), 1100 (a), 1101 (b) or 1110 (c); 1111 used
     -- as filler when there is an odd number of digits.

     -- bits 8765 of octet n encoding digit 2n
     -- bits 4321 of octet n encoding digit 2(n-1) +1
```

```
DiameterIdentity ::= OCTET STRING (SIZE(9..255))
-- content of DiameterIdentity is defined in IETF RFC 3588 [139]
```

```
AddressString ::= OCTET STRING (SIZE (1..maxAddressLength))
    -- This type is used to represent a number for addressing
    -- purposes. It is composed of
    --   a)   one octet for nature of address, and numbering plan
    --   indicator.
    --   b)   digits of an address encoded as TBCD-String.

    -- a)     The first octet includes a one bit extension indicator, a
    --   3 bits nature of address indicator and a 4 bits numbering
    --   plan indicator, encoded as follows:

    -- bit 8: 1  (no extension)

    -- bits 765: nature of address indicator
    --   000  unknown
    --   001  international number
    --   010  national significant number
    --   011  network specific number
    --   100  subscriber number
    --   101  reserved
    --   110  abbreviated number
    --   111  reserved for extension

    -- bits 4321: numbering plan indicator
    --   0000  unknown
    --   0001  ISDN/Telephony Numbering Plan (Rec ITU-T E.164)
    --   0010  spare
    --   0011  data numbering plan (ITU-T Rec X.121)
    --   0100  telex numbering plan (ITU-T Rec F.69)
    --   0101  spare
    --   0110  land mobile numbering plan (ITU-T Rec E.212)
    --   0111  spare
    --   1000  national numbering plan
    --   1001  private numbering plan
    --   1111  reserved for extension

    --   all other values are reserved.

    -- b)     The following octets representing digits of an address
    --   encoded as a TBCD-STRING.
```

```
maxAddressLength  INTEGER ::= 20
```

```
ISDN-AddressString ::=
         AddressString (SIZE (1..maxISDN-AddressLength))
    -- This type is used to represent ISDN numbers.
```

```
maxISDN-AddressLength  INTEGER ::= 9
```

```
FTN-AddressString ::=
         AddressString (SIZE (1..maxFTN-AddressLength))
    -- This type is used to represent forwarded-to numbers.
    -- If NAI = international the first digits represent the country code (CC)
    -- and the network destination code (NDC) as for E.164.
```

```
maxFTN-AddressLength  INTEGER ::= 15
```

```
ISDN-SubaddressString ::=
        OCTET STRING (SIZE (1..maxISDN-SubaddressLength))
    -- This type is used to represent ISDN subaddresses.
    -- It is composed of
    --   a)   one octet for type of subaddress and odd/even indicator.
    --   b)   20 octets for subaddress information.

    --   a)   The first octet includes a one bit extension indicator, a
    --   3 bits type of subaddress and a one bit odd/even indicator,
    --   encoded as follows:

    --   bit 8: 1  (no extension)

    --   bits 765: type of subaddress
    --   000  NSAP (X.213/ISO 8348 AD2)
    --   010  User Specified
    --   All other values are reserved

    --   bit 4: odd/even indicator
    --   0  even number of address signals
    --   1  odd number of address signals
    --   The odd/even indicator is used when the type of subaddress
    --   is "user specified" and the coding is BCD.

    --   bits 321: 000 (unused)

    --   b) Subaddress information.
    --   The NSAP X.213/ISO8348AD2 address shall be formatted as specified
    --   by octet 4 which contains the Authority and Format Identifier
    --   (AFI). The encoding is made according to the "preferred binary
    --   encoding" as defined in X.213/ISO834AD2. For the definition
    --   of this type of subaddress, see ITU-T Rec I.334.

    --   For User-specific subaddress, this field is encoded according
    --   to the user specification, subject to a maximum length of 20
    --   octets. When interworking with X.25 networks BCD coding should
    --   be applied.
```

```
maxISDN-SubaddressLength  INTEGER ::= 21
```

```
ExternalSignalInfo ::= SEQUENCE {
    protocolId                      ProtocolId,
    signalInfo                      SignalInfo,
    -- Information about the internal structure is given in
    -- clause 7.6.9.
    extensionContainer              ExtensionContainer              OPTIONAL,
    -- extensionContainer must not be used in version 2
    ...}
```

```
SignalInfo ::= OCTET STRING (SIZE (1..maxSignalInfoLength))
```

```
maxSignalInfoLength  INTEGER ::= 200
    -- This NamedValue represents the theoretical maximum number of octets which is
    -- available to carry a single instance of the SignalInfo data type,
    -- without requiring segmentation to cope with the network layer service.
    -- However, the actual maximum size available for an instance of the data
    -- type may be lower, especially when other information elements
    -- have to be included in the same component.
```

```
ProtocolId ::= ENUMERATED {
    gsm-0408  (1),
    gsm-0806  (2),
    gsm-BSSMAP  (3),
    -- Value 3 is reserved and must not be used
    ets-300102-1  (4)}
```

```
Ext-ExternalSignalInfo ::= SEQUENCE {
    ext-ProtocolId                  Ext-ProtocolId,
    signalInfo                      SignalInfo,
    -- Information about the internal structure is given in
    -- clause 7.6.9.10
    extensionContainer              ExtensionContainer              OPTIONAL,
    ...}
```

```
Ext-ProtocolId ::= ENUMERATED {
     ets-300356  (1),
     ...
     }
-- exception handling:
-- For Ext-ExternalSignalInfo sequences containing this parameter with any
-- other value than the ones listed the receiver shall ignore the whole
-- Ext-ExternalSignalInfo sequence.
```

```
AccessNetworkSignalInfo ::= SEQUENCE {
     accessNetworkProtocolId             AccessNetworkProtocolId,
     signalInfo                          LongSignalInfo,
     -- Information about the internal structure is given in clause 7.6.9.1

     extensionContainer                  ExtensionContainer               OPTIONAL,
     ...}
```

```
LongSignalInfo ::= OCTET STRING (SIZE (1..maxLongSignalInfoLength))
```

```
maxLongSignalInfoLength  INTEGER ::= 2560
     -- This Named Value represents the maximum number of octets which is available
     -- to carry a single instance of the LongSignalInfo data type using
     -- White Book SCCP with the maximum number of segments.
     -- It takes account of the octets used by the lower layers of the protocol, and
     -- other information elements which may be included in the same component.
```

```
AccessNetworkProtocolId ::= ENUMERATED {
     ts3G-48006   (1),
     ts3G-25413 (2),
     ...}
     -- exception handling:
     -- For AccessNetworkSignalInfo sequences containing this parameter with any
     -- other value than the ones listed the receiver shall ignore the whole
     -- AccessNetworkSignalInfo sequence.
```

```
AlertingPattern ::= OCTET STRING (SIZE (1) )
     -- This type is used to represent Alerting Pattern

     --   bits 8765 : 0000 (unused)

     --   bits 43 : type of Pattern
     --   00 level
     --   01 category
     --   10 category
     --   all other values are reserved.

     --   bits 21 : type of alerting

alertingLevel-0   AlertingPattern ::= '00000000'B
alertingLevel-1   AlertingPattern ::= '00000001'B
alertingLevel-2   AlertingPattern ::= '00000010'B
     -- all other values of Alerting level are reserved
     -- Alerting Levels are defined in GSM 02.07

alertingCategory-1   AlertingPattern ::= '00000100'B
alertingCategory-2   AlertingPattern ::= '00000101'B
alertingCategory-3   AlertingPattern ::= '00000110'B
alertingCategory-4   AlertingPattern ::= '00000111'B
alertingCategory-5   AlertingPattern ::= '00001000'B
     -- all other values of Alerting Category are reserved
     -- Alerting categories are defined in GSM 02.07
```

```
GSN-Address ::= OCTET STRING (SIZE (5..17))
     -- Octets are coded according to TS 3GPP TS 23.003 [17]
```

```
Time ::= OCTET STRING (SIZE (4))
     -- Octets are coded according to IETF RFC 3588 [139]
```

```
-- data types for numbering and identification
```

```
IMSI ::= TBCD-STRING (SIZE (3..8))
     -- digits of MCC, MNC, MSIN are concatenated in this order.
```

```
Identity ::= CHOICE {
     imsi      IMSI,
     imsi-WithLMSI                         IMSI-WithLMSI}
```

```
IMSI-WithLMSI ::= SEQUENCE {
     imsi      IMSI,
     lmsi      LMSI,
     -- a special value 00000000 indicates that the LMSI is not in use
     ...}
```

```
ASCI-CallReference ::= TBCD-STRING (SIZE (1..8))
     -- digits of VGCS/VBS-area,Group-ID are concatenated in this order if there is a
     -- VGCS/VBS-area.
```

```
TMSI ::= OCTET STRING (SIZE (1..4))
```

```
SubscriberId ::= CHOICE {
     imsi      [0] IMSI,
     tmsi      [1] TMSI}
```

```
IMEI ::= TBCD-STRING (SIZE (8))
     --   Refers to International Mobile Station Equipment Identity
     --   and Software Version Number (SVN) defined in TS 3GPP TS 23.003 [17].
     --   If the SVN is not present the last octet shall contain the
     --   digit 0 and a filler.
     --   If present the SVN shall be included in the last octet.
```

```
HLR-Id ::= IMSI
     -- leading digits of IMSI, i.e. (MCC, MNC, leading digits of
     -- MSIN) forming HLR Id defined in TS 3GPP TS 23.003 [17].
```

```
HLR-List ::= SEQUENCE SIZE (1..maxNumOfHLR-Id) OF
          HLR-Id
```

```
maxNumOfHLR-Id  INTEGER ::= 50
```

```
LMSI ::= OCTET STRING (SIZE (4))
```

```
GlobalCellId ::= OCTET STRING (SIZE (5..7))
     -- Refers to Cell Global Identification defined in TS 3GPP TS 23.003 [17].
     -- The internal structure is defined as follows:
     -- octet 1 bits 4321                Mobile Country Code 1st digit
     --         bits 8765                Mobile Country Code 2nd digit
     -- octet 2 bits 4321                Mobile Country Code 3rd digit
     --         bits 8765                Mobile Network Code 3rd digit
     --       or filler (1111) for 2 digit MNCs
     -- octet 3 bits 4321                Mobile Network Code 1st digit
     --         bits 8765                Mobile Network Code 2nd digit
     -- octets 4 and 5                   Location Area Code according to TS 3GPP TS 24.008
[35]
     -- octets 6 and 7                   Cell Identity (CI) according to TS 3GPP TS 24.008
[35]
```

```
NetworkResource ::= ENUMERATED {
     plmn  (0),
     hlr  (1),
     vlr  (2),
     pvlr  (3),
     controllingMSC  (4),
     vmsc  (5),
     eir  (6),
     rss  (7)}
```

```
AdditionalNetworkResource ::= ENUMERATED {
     sgsn (0),
     ggsn (1),
     gmlc (2),
     gsmSCF (3),
     nplr (4),
     auc (5),
     ... ,
     ue (6),
     mme (7)}
     -- if unknown value is received in AdditionalNetworkResource
     -- it shall be ignored.
```

```
NAEA-PreferredCI ::= SEQUENCE {
    naea-PreferredCIC                   [0] NAEA-CIC,
    extensionContainer                  [1] ExtensionContainer          OPTIONAL,
    ...}
```

```
NAEA-CIC ::= OCTET STRING (SIZE (3))
    -- The internal structure is defined by the Carrier Identification
    -- parameter in ANSI T1.113.3. Carrier codes between "000" and "999" may
    -- be encoded as 3 digits using "000" to "999" or as 4 digits using
    -- "0000" to "0999". Carrier codes between "1000" and "9999" are encoded
    -- using 4 digits.
```

```
SubscriberIdentity ::= CHOICE {
    imsi     [0] IMSI,
    msisdn   [1] ISDN-AddressString
    }
```

```
LCSClientExternalID ::= SEQUENCE {
    externalAddress                     [0] ISDN-AddressString          OPTIONAL,
    extensionContainer                  [1] ExtensionContainer          OPTIONAL,
    ... }
```

```
LCSClientInternalID ::= ENUMERATED {
    broadcastService                    (0),
    o-andM-HPLMN                        (1),
    o-andM-VPLMN                        (2),
    anonymousLocation                   (3),
    targetMSsubscribedService           (4),
    ... }
-- for a CAMEL phase 3 PLMN operator client, the value targetMSsubscribedService shall be used
```

```
LCSServiceTypeID ::= INTEGER (0..127)
    -- the integer values 0-63 are reserved for Standard LCS service types
    -- the integer values 64-127 are reserved for Non Standard LCS service types
```

```
-- Standard LCS Service Types
emergencyServices                       LCSServiceTypeID ::= 0
emergencyAlertServices                  LCSServiceTypeID ::= 1
personTracking                          LCSServiceTypeID ::= 2
fleetManagement                         LCSServiceTypeID ::= 3
assetManagement                         LCSServiceTypeID ::= 4
trafficCongestionReporting              LCSServiceTypeID ::= 5
roadsideAssistance                      LCSServiceTypeID ::= 6
routingToNearestCommercialEnterprise    LCSServiceTypeID ::= 7
navigation                              LCSServiceTypeID ::= 8
    --this service type is reserved for use in previous releases
citySightseeing                         LCSServiceTypeID ::= 9
localizedAdvertising                    LCSServiceTypeID ::= 10
mobileYellowPages                       LCSServiceTypeID ::= 11
trafficAndPublicTransportationInfo      LCSServiceTypeID ::= 12
weather       LCSServiceTypeID ::= 13
assetAndServiceFinding                  LCSServiceTypeID ::= 14
gaming        LCSServiceTypeID ::= 15
findYourFriend                          LCSServiceTypeID ::= 16
dating        LCSServiceTypeID ::= 17
chatting      LCSServiceTypeID ::= 18
routeFinding                            LCSServiceTypeID ::= 19
whereAmI      LCSServiceTypeID ::= 20

-- The values of LCSServiceTypeID are defined according to 3GPP TS 22.071.
```

```
-- Non Standard LCS Service Types
serv64        LCSServiceTypeID ::= 64
serv65        LCSServiceTypeID ::= 65
serv66        LCSServiceTypeID ::= 66
serv67        LCSServiceTypeID ::= 67
serv68        LCSServiceTypeID ::= 68
serv69        LCSServiceTypeID ::= 69
serv70        LCSServiceTypeID ::= 70
serv71        LCSServiceTypeID ::= 71
serv72        LCSServiceTypeID ::= 72
serv73        LCSServiceTypeID ::= 73
serv74        LCSServiceTypeID ::= 74
serv75        LCSServiceTypeID ::= 75
serv76        LCSServiceTypeID ::= 76
serv77        LCSServiceTypeID ::= 77
serv78        LCSServiceTypeID ::= 78
serv79        LCSServiceTypeID ::= 79
serv80        LCSServiceTypeID ::= 80
serv81        LCSServiceTypeID ::= 81
serv82        LCSServiceTypeID ::= 82
serv83        LCSServiceTypeID ::= 83
serv84        LCSServiceTypeID ::= 84
serv85        LCSServiceTypeID ::= 85
serv86        LCSServiceTypeID ::= 86
serv87        LCSServiceTypeID ::= 87
serv88        LCSServiceTypeID ::= 88
serv89        LCSServiceTypeID ::= 89
serv90        LCSServiceTypeID ::= 90
serv91        LCSServiceTypeID ::= 91
serv92        LCSServiceTypeID ::= 92
serv93        LCSServiceTypeID ::= 93
serv94        LCSServiceTypeID ::= 94
serv95        LCSServiceTypeID ::= 95
serv96        LCSServiceTypeID ::= 96
serv97        LCSServiceTypeID ::= 97
serv98        LCSServiceTypeID ::= 98
serv99        LCSServiceTypeID ::= 99
serv100       LCSServiceTypeID ::= 100
serv101       LCSServiceTypeID ::= 101
serv102       LCSServiceTypeID ::= 102
serv103       LCSServiceTypeID ::= 103
serv104       LCSServiceTypeID ::= 104
serv105       LCSServiceTypeID ::= 105
serv106       LCSServiceTypeID ::= 106
serv107       LCSServiceTypeID ::= 107
serv108       LCSServiceTypeID ::= 108
serv109       LCSServiceTypeID ::= 109
serv110       LCSServiceTypeID ::= 110
serv111       LCSServiceTypeID ::= 111
serv112       LCSServiceTypeID ::= 112
serv113       LCSServiceTypeID ::= 113
serv114       LCSServiceTypeID ::= 114
serv115       LCSServiceTypeID ::= 115
serv116       LCSServiceTypeID ::= 116
serv117       LCSServiceTypeID ::= 117
serv118       LCSServiceTypeID ::= 118
serv119       LCSServiceTypeID ::= 119
serv120       LCSServiceTypeID ::= 120
serv121       LCSServiceTypeID ::= 121
serv122       LCSServiceTypeID ::= 122
serv123       LCSServiceTypeID ::= 123
serv124       LCSServiceTypeID ::= 124
serv125       LCSServiceTypeID ::= 125
serv126       LCSServiceTypeID ::= 126
serv127       LCSServiceTypeID ::= 127
```

```
PLMN-Id ::= OCTET STRING (SIZE (3))
    -- The internal structure is defined as follows:
    -- octet 1 bits 4321                Mobile Country Code 1st digit
    --         bits 8765                Mobile Country Code 2nd digit
    -- octet 2 bits 4321                Mobile Country Code 3rd digit
    --         bits 8765                Mobile Network Code 3rd digit
    --       or filler (1111) for 2 digit MNCs
    -- octet 3 bits 4321                Mobile Network Code 1st digit
    --         bits 8765                Mobile Network Code 2nd digit
```

```
E-UTRAN-CGI ::= OCTET STRING (SIZE (7))
     -- Octets are coded as described in 3GPP TS 29.118 [152].
```

```
NR-CGI ::= OCTET STRING (SIZE (8))
     -- Octets are coded as described in 3GPP TS 38.413 [153].
```

```
TA-Id ::= OCTET STRING (SIZE (5))
     -- Octets are coded as described in 3GPP TS 29.118 [152].
```

```
RAIdentity ::= OCTET STRING (SIZE (6))
-- Routing Area Identity is coded in accordance with 3GPP TS 29.060 [105].
-- It shall contain the value part defined in 3GPP TS 29.060 only. I.e. the 3GPP TS 29.060
-- type identifier octet shall not be included.
```

```
NetworkNodeDiameterAddress::= SEQUENCE {
     diameter-Name                       [0] DiameterIdentity,
     diameter-Realm                      [1] DiameterIdentity }
```

-- data types for CAMEL

```
CellGlobalIdOrServiceAreaIdOrLAI ::= CHOICE {
     cellGlobalIdOrServiceAreaIdFixedLength   [0] CellGlobalIdOrServiceAreaIdFixedLength,
     laiFixedLength                      [1] LAIFixedLength}
```

```
CellGlobalIdOrServiceAreaIdFixedLength ::= OCTET STRING (SIZE (7))
     -- Refers to Cell Global Identification or Service Are Identification
     -- defined in 3GPP TS 23.003.
     -- The internal structure is defined as follows:
     -- octet 1 bits 4321                Mobile Country Code 1st digit
     --         bits 8765                Mobile Country Code 2nd digit
     -- octet 2 bits 4321                Mobile Country Code 3rd digit
     --         bits 8765                Mobile Network Code 3rd digit
     --       or filler (1111) for 2 digit MNCs
     -- octet 3 bits 4321                Mobile Network Code 1st digit
     --         bits 8765                Mobile Network Code 2nd digit
     -- octets 4 and 5                   Location Area Code according to 3GPP TS 24.008
     -- octets 6 and 7                   Cell Identity (CI) value or
     --       Service Area Code (SAC) value
     --       according to 3GPP TS 23.003
```

```
LAIFixedLength ::= OCTET STRING (SIZE (5))
     -- Refers to Location Area Identification defined in 3GPP TS 23.003 [17].
     -- The internal structure is defined as follows:
     -- octet 1 bits 4321                Mobile Country Code 1st digit
     --         bits 8765                Mobile Country Code 2nd digit
     -- octet 2 bits 4321                Mobile Country Code 3rd digit
     --         bits 8765                Mobile Network Code 3rd digit
     --       or filler (1111) for 2 digit MNCs
     -- octet 3 bits 4321                Mobile Network Code 1st digit
     --         bits 8765                Mobile Network Code 2nd digit
     -- octets 4 and 5                   Location Area Code according to 3GPP TS 24.008 [35]
```

-- data types for subscriber management

```
BasicServiceCode ::= CHOICE {
     bearerService                       [2] BearerServiceCode,
     teleservice                         [3] TeleserviceCode}
```

```
Ext-BasicServiceCode ::= CHOICE {
     ext-BearerService                   [2] Ext-BearerServiceCode,
     ext-Teleservice                     [3] Ext-TeleserviceCode}
```

```
EMLPP-Info ::= SEQUENCE {
     maximumentitledPriority             EMLPP-Priority,
     defaultPriority                     EMLPP-Priority,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
EMLPP-Priority ::= INTEGER (0..15)
     -- The mapping from the values A,B,0,1,2,3,4 to the integer-value is
     -- specified as follows where A is the highest and 4 is the lowest
     -- priority level
     -- the integer values 7-15 are spare and shall be mapped to value 4
```

```
priorityLevelAEMLPP-Priority ::= 6
priorityLevelBEMLPP-Priority ::= 5
priorityLevel0EMLPP-Priority ::= 0
priorityLevel1EMLPP-Priority ::= 1
priorityLevel2EMLPP-Priority ::= 2
priorityLevel3EMLPP-Priority ::= 3
priorityLevel4EMLPP-Priority ::= 4
```

```
MC-SS-Info ::= SEQUENCE {
     ss-Code   [0] SS-Code,
     ss-Status [1] Ext-SS-Status,
     nbrSB     [2] MaxMC-Bearers,
     nbrUser   [3] MC-Bearers,
     extensionContainer                  [4] ExtensionContainer            OPTIONAL,
     ...}
```

```
MaxMC-Bearers ::= INTEGER (2..maxNumOfMC-Bearers)
```

```
MC-Bearers ::= INTEGER (1..maxNumOfMC-Bearers)
```

```
maxNumOfMC-Bearers  INTEGER ::= 7
```

```
Ext-SS-Status ::= OCTET STRING (SIZE (1..5))

     -- OCTET 1:
     --
     -- bits 8765: 0000 (unused)
     -- bits 4321: Used to convey the "P bit","R bit","A bit" and "Q bit",
     --     representing supplementary service state information
     --     as defined in TS 3GPP TS 23.011 [22]

     -- bit 4: "Q bit"

     -- bit 3: "P bit"

     -- bit 2: "R bit"

     -- bit 1: "A bit"

     -- OCTETS 2-5: reserved for future use. They shall be discarded if
     -- received and not understood.
```

```
   -- data types for geographic location
```

```
AgeOfLocationInformation ::= INTEGER (0..32767)
-- the value represents the elapsed time in minutes since the last
-- network contact of the mobile station (i.e. the actuality of the
-- location information).
-- value "0" indicates that the MS is currently in contact with the
--           network
-- value "32767" indicates that the location information is at least
--               32767 minutes old
```

```
END
```

17.7.9 Teleservice Codes

```
MAP-TS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-TS-Code (19) version18 (18)}

DEFINITIONS

::=

BEGIN
```

```
TeleserviceCode ::= OCTET STRING (SIZE (1))
     -- This type is used to represent the code identifying a single
     -- teleservice, a group of teleservices, or all teleservices. The
     -- services are defined in TS GSM 22.003 [4].
     -- The internal structure is defined as follows:

     -- bits 87654321: group (bits 8765) and specific service
     -- (bits 4321)
```

```
Ext-TeleserviceCode ::= OCTET STRING (SIZE (1..5))
     -- This type is used to represent the code identifying a single
     -- teleservice, a group of teleservices, or all teleservices. The
     -- services are defined in TS GSM 22.003 [4].
     -- The internal structure is defined as follows:

     -- OCTET 1:
     -- bits 87654321: group (bits 8765) and specific service
     -- (bits 4321)

     -- OCTETS 2-5: reserved for future use. If received the
    -- Ext-TeleserviceCode shall be
     -- treated according to the exception handling defined for the
     -- operation that uses this type.

     -- Ext-TeleserviceCode includes all values defined for TeleserviceCode.
```

```
allTeleservices                          TeleserviceCode ::= '00000000'B
```

```
allSpeechTransmissionServices            TeleserviceCode ::= '00010000'B
telephony     TeleserviceCode ::= '00010001'B
emergencyCallsTeleserviceCode ::= '00010010'B
```

```
allShortMessageServices                  TeleserviceCode ::= '00100000'B
shortMessageMT-PP                        TeleserviceCode ::= '00100001'B
shortMessageMO-PP                        TeleserviceCode ::= '00100010'B
```

```
allFacsimileTransmissionServices         TeleserviceCode ::= '01100000'B
facsimileGroup3AndAlterSpeech            TeleserviceCode ::= '01100001'B
automaticFacsimileGroup3                 TeleserviceCode ::= '01100010'B
facsimileGroup4                          TeleserviceCode ::= '01100011'B
```

```
-- The following non-hierarchical Compound Teleservice Groups
-- are defined in TS 3GPP TS 22.030:
allDataTeleservices                      TeleserviceCode ::= '01110000'B
     -- covers Teleservice Groups 'allFacsimileTransmissionServices'
     -- and 'allShortMessageServices'
allTeleservices-ExeptSMS                 TeleserviceCode ::= '10000000'B
     -- covers Teleservice Groups 'allSpeechTransmissionServices' and
     -- 'allFacsimileTransmissionServices'
--
-- Compound Teleservice Group Codes are only used in call
-- independent supplementary service operations, i.e. they
-- are not used in InsertSubscriberData or in
-- DeleteSubscriberData messages.
```

```
allVoiceGroupCallServices                TeleserviceCode ::= '10010000'B
voiceGroupCallTeleserviceCode ::= '10010001'B
voiceBroadcastCall                       TeleserviceCode ::= '10010010'B
```

```
allPLMN-specificTS                       TeleserviceCode ::= '11010000'B
plmn-specificTS-1                        TeleserviceCode ::= '11010001'B
plmn-specificTS-2                        TeleserviceCode ::= '11010010'B
plmn-specificTS-3                        TeleserviceCode ::= '11010011'B
plmn-specificTS-4                        TeleserviceCode ::= '11010100'B
plmn-specificTS-5                        TeleserviceCode ::= '11010101'B
plmn-specificTS-6                        TeleserviceCode ::= '11010110'B
plmn-specificTS-7                        TeleserviceCode ::= '11010111'B
plmn-specificTS-8                        TeleserviceCode ::= '11011000'B
plmn-specificTS-9                        TeleserviceCode ::= '11011001'B
plmn-specificTS-A                        TeleserviceCode ::= '11011010'B
plmn-specificTS-B                        TeleserviceCode ::= '11011011'B
plmn-specificTS-C                        TeleserviceCode ::= '11011100'B
plmn-specificTS-D                        TeleserviceCode ::= '11011101'B
plmn-specificTS-E                        TeleserviceCode ::= '11011110'B
plmn-specificTS-F                        TeleserviceCode ::= '11011111'B
```

```
END
```

17.7.10 Bearer Service Codes

```
MAP-BS-Code {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-BS-Code (20) version18 (18)}

DEFINITIONS
```

```
::=

BEGIN

BearerServiceCode ::= OCTET STRING (SIZE (1))
     -- This type is used to represent the code identifying a single
     -- bearer service, a group of bearer services, or all bearer
     -- services. The services are defined in TS 3GPP TS 22.002 [3].
     -- The internal structure is defined as follows:
     --
     -- plmn-specific bearer services:
     -- bits 87654321: defined by the HPLMN operator

     -- rest of bearer services:
     -- bit 8: 0 (unused)
     -- bits 7654321: group (bits 7654), and rate, if applicable
     -- (bits 321)

Ext-BearerServiceCode ::= OCTET STRING (SIZE (1..5))
     -- This type is used to represent the code identifying a single
     -- bearer service, a group of bearer services, or all bearer
     -- services. The services are defined in TS 3GPP TS 22.002 [3].
     -- The internal structure is defined as follows:
     --
     -- OCTET 1:
     -- plmn-specific bearer services:
     -- bits 87654321: defined by the HPLMN operator
     --
     -- rest of bearer services:
     -- bit 8: 0 (unused)
     -- bits 7654321: group (bits 7654), and rate, if applicable
     -- (bits 321)

     -- OCTETS 2-5: reserved for future use. If received the
    -- Ext-TeleserviceCode shall be
     -- treated according to the exception handling defined for the
     -- operation that uses this type.


     -- Ext-BearerServiceCode includes all values defined for BearerServiceCode.

allBearerServices                       BearerServiceCode ::= '00000000'B

allDataCDA-Services                     BearerServiceCode ::= '00010000'B
dataCDA-300bpsBearerServiceCode ::= '00010001'B
dataCDA-1200bps                         BearerServiceCode ::= '00010010'B
dataCDA-1200-75bps                      BearerServiceCode ::= '00010011'B
dataCDA-2400bps                         BearerServiceCode ::= '00010100'B
dataCDA-4800bps                         BearerServiceCode ::= '00010101'B
dataCDA-9600bps                         BearerServiceCode ::= '00010110'B
general-dataCDA                         BearerServiceCode ::= '00010111'B

allDataCDS-Services                     BearerServiceCode ::= '00011000'B
dataCDS-1200bps                         BearerServiceCode ::= '00011010'B
dataCDS-2400bps                         BearerServiceCode ::= '00011100'B
dataCDS-4800bps                         BearerServiceCode ::= '00011101'B
dataCDS-9600bps                         BearerServiceCode ::= '00011110'B
general-dataCDS                         BearerServiceCode ::= '00011111'B

allPadAccessCA-Services                 BearerServiceCode ::= '00100000'B
padAccessCA-300bps                      BearerServiceCode ::= '00100001'B
padAccessCA-1200bps                     BearerServiceCode ::= '00100010'B
padAccessCA-1200-75bps                  BearerServiceCode ::= '00100011'B
padAccessCA-2400bps                     BearerServiceCode ::= '00100100'B
padAccessCA-4800bps                     BearerServiceCode ::= '00100101'B
padAccessCA-9600bps                     BearerServiceCode ::= '00100110'B
general-padAccessCA                     BearerServiceCode ::= '00100111'B

allDataPDS-Services                     BearerServiceCode ::= '00101000'B
dataPDS-2400bps                         BearerServiceCode ::= '00101100'B
dataPDS-4800bps                         BearerServiceCode ::= '00101101'B
dataPDS-9600bps                         BearerServiceCode ::= '00101110'B
general-dataPDS                         BearerServiceCode ::= '00101111'B

allAlternateSpeech-DataCDA              BearerServiceCode ::= '00110000'B

allAlternateSpeech-DataCDS              BearerServiceCode ::= '00111000'B
```

```
allSpeechFollowedByDataCDA                BearerServiceCode ::= '01000000'B
```

```
allSpeechFollowedByDataCDS                BearerServiceCode ::= '01001000'B
```

```
-- The following non-hierarchical Compound Bearer Service
-- Groups are defined in TS 3GPP TS 22.030:
allDataCircuitAsynchronous                BearerServiceCode ::= '01010000'B
     -- covers "allDataCDA-Services", "allAlternateSpeech-DataCDA" and
     -- "allSpeechFollowedByDataCDA"
allAsynchronousServices                   BearerServiceCode ::= '01100000'B
     -- covers "allDataCDA-Services", "allAlternateSpeech-DataCDA",
     -- "allSpeechFollowedByDataCDA" and "allPadAccessCDA-Services"
allDataCircuitSynchronous                 BearerServiceCode ::= '01011000'B
     -- covers "allDataCDS-Services", "allAlternateSpeech-DataCDS" and
     -- "allSpeechFollowedByDataCDS"
allSynchronousServices                    BearerServiceCode ::= '01101000'B
     -- covers "allDataCDS-Services", "allAlternateSpeech-DataCDS",
     -- "allSpeechFollowedByDataCDS" and "allDataPDS-Services"
--
-- Compound Bearer Service Group Codes are only used in call
-- independent supplementary service operations, i.e. they
-- are not used in InsertSubscriberData or in
-- DeleteSubscriberData messages.
```

```
allPLMN-specificBS                        BearerServiceCode ::= '11010000'B
plmn-specificBS-1                         BearerServiceCode ::= '11010001'B
plmn-specificBS-2                         BearerServiceCode ::= '11010010'B
plmn-specificBS-3                         BearerServiceCode ::= '11010011'B
plmn-specificBS-4                         BearerServiceCode ::= '11010100'B
plmn-specificBS-5                         BearerServiceCode ::= '11010101'B
plmn-specificBS-6                         BearerServiceCode ::= '11010110'B
plmn-specificBS-7                         BearerServiceCode ::= '11010111'B
plmn-specificBS-8                         BearerServiceCode ::= '11011000'B
plmn-specificBS-9                         BearerServiceCode ::= '11011001'B
plmn-specificBS-A                         BearerServiceCode ::= '11011010'B
plmn-specificBS-B                         BearerServiceCode ::= '11011011'B
plmn-specificBS-C                         BearerServiceCode ::= '11011100'B
plmn-specificBS-D                         BearerServiceCode ::= '11011101'B
plmn-specificBS-E                         BearerServiceCode ::= '11011110'B
plmn-specificBS-F                         BearerServiceCode ::= '11011111'B
```

```
END
```

17.7.11 Extension data types

```
MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS

   PrivateExtension,
   ExtensionContainer,
   SLR-ArgExtensionContainer;


-- IOC for private MAP extensions
```

```
MAP-EXTENSION  ::= CLASS {
    &ExtensionType                    OPTIONAL,
    &extensionId                      OBJECT IDENTIFIER }
    -- The length of the Object Identifier shall not exceed 16 octets and the
    -- number of components of the Object Identifier shall not exceed 16
```

```
-- data types
```

```
ExtensionContainer ::= SEQUENCE {
    privateExtensionList                [0]PrivateExtensionList           OPTIONAL,
    pcs-Extensions                      [1]PCS-Extensions                 OPTIONAL,
    ...}
```

```
SLR-ArgExtensionContainer ::= SEQUENCE {
    privateExtensionList                [0]PrivateExtensionList           OPTIONAL,
    slr-Arg-PCS-Extensions              [1]SLR-Arg-PCS-Extensions         OPTIONAL,
    ...}
```

```
PrivateExtensionList ::= SEQUENCE SIZE (1..maxNumOfPrivateExtensions) OF
        PrivateExtension
```

```
PrivateExtension ::= SEQUENCE {
    extId    MAP-EXTENSION.&extensionId
        ({ExtensionSet}),
    extType  MAP-EXTENSION.&ExtensionType
        ({ExtensionSet}{@extId})        OPTIONAL}
```

```
maxNumOfPrivateExtensions  INTEGER ::= 10
```

```
ExtensionSet MAP-EXTENSION ::=
    {...
    -- ExtensionSet is the set of all defined private extensions
    }
    -- Unsupported private extensions shall be discarded if received.
```

```
PCS-Extensions ::= SEQUENCE {
    ...}
```

```
SLR-Arg-PCS-Extensions ::= SEQUENCE {
    ...,
    na-ESRK-Request                     [0]  NULL OPTIONAL }
```

```
END
```

17.7.12 Group Call data types

```
MAP-GR-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-GR-DataTypes (23) version18 (18)}

DEFINITIONS

IMPLICIT TAGS

::=

BEGIN

EXPORTS
   PrepareGroupCallArg,
   PrepareGroupCallRes,
   SendGroupCallEndSignalArg,
   SendGroupCallEndSignalRes,
   ForwardGroupCallSignallingArg,
   ProcessGroupCallSignallingArg,
   SendGroupCallInfoArg,
   SendGroupCallInfoRes
;

IMPORTS
   ISDN-AddressString,
   IMSI,
   TMSI,
   EMLPP-Priority,
   ASCI-CallReference,
   SignalInfo,
   GlobalCellId,
   AccessNetworkSignalInfo
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   Ext-TeleserviceCode
FROM MAP-TS-Code {
```

```
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-TS-Code (19) version18 (18)}

   Kc,
   AdditionalInfo,
   GroupId,
   Long-GroupId,
   AdditionalSubscriptions,
   Cksn
FROM MAP-MS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MS-DataTypes (11) version18 (18)}

   ExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}
;
```

```
PrepareGroupCallArg ::= SEQUENCE {
     teleservice                         Ext-TeleserviceCode,
     asciCallReference                   ASCI-CallReference,
     codec-Info                          CODEC-Info,
     cipheringAlgorithm                  CipheringAlgorithm,
     groupKeyNumber-Vk-Id                [0] GroupKeyNumber              OPTIONAL,
     groupKey  [1] Kc                    OPTIONAL,
     -- this parameter shall not be sent and shall be discarded if received
     priority  [2] EMLPP-Priority        OPTIONAL,
     uplinkFree                          [3] NULL  OPTIONAL,
     extensionContainer                  [4] ExtensionContainer          OPTIONAL,
     ...,
     vstk      [5] VSTK                  OPTIONAL,
     vstk-rand [6] VSTK-RAND             OPTIONAL,
     talkerChannelParameter              [7] NULL  OPTIONAL,
     uplinkReplyIndicator                [8] NULL  OPTIONAL}
```

```
VSTK ::= OCTET STRING (SIZE (16))
```

```
VSTK-RAND ::= OCTET STRING (SIZE (5))
     -- The 36 bit value is carried in bit 7 of octet 1 to bit 4 of octet 5
     -- bits 3, 2, 1, and 0 of octet 5 are padded with zeros.
```

```
PrepareGroupCallRes ::= SEQUENCE {
     groupCallNumber                     ISDN-AddressString,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
SendGroupCallEndSignalArg ::= SEQUENCE {
     imsi      IMSI                           OPTIONAL,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     talkerPriority                      [0]TalkerPriority               OPTIONAL,
     additionalInfo                      [1]AdditionalInfo               OPTIONAL }
```

```
TalkerPriority ::= ENUMERATED {
     normal  (0),
     privileged  (1),
     emergency  (2)}
```

```
SendGroupCallEndSignalRes ::= SEQUENCE {
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...}
```

```
ForwardGroupCallSignallingArg ::= SEQUENCE {
     imsi     IMSI                              OPTIONAL,
     uplinkRequestAck                    [0] NULL  OPTIONAL,
     uplinkReleaseIndication             [1] NULL  OPTIONAL,
     uplinkRejectCommand                 [2] NULL  OPTIONAL,
     uplinkSeizedCommand                 [3] NULL  OPTIONAL,
     uplinkReleaseCommand                [4] NULL  OPTIONAL,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     stateAttributes                     [5] StateAttributes             OPTIONAL,
     talkerPriority                      [6] TalkerPriority              OPTIONAL,
     additionalInfo                      [7] AdditionalInfo              OPTIONAL,
     emergencyModeResetCommandFlag       [8] NULL  OPTIONAL,
     sm-RP-UI  [9] SignalInfo            OPTIONAL,
     an-APDU   [10] AccessNetworkSignalInfo    OPTIONAL
 }
```

```
ProcessGroupCallSignallingArg ::= SEQUENCE {
     uplinkRequest                       [0] NULL  OPTIONAL,
     uplinkReleaseIndication             [1] NULL  OPTIONAL,
     releaseGroupCall                    [2] NULL  OPTIONAL,
     extensionContainer                  ExtensionContainer              OPTIONAL,
     ...,
     talkerPriority                      [3] TalkerPriority              OPTIONAL,
     additionalInfo                      [4] AdditionalInfo              OPTIONAL,
     emergencyModeResetCommandFlag       [5] NULL  OPTIONAL,
     an-APDU   [6] AccessNetworkSignalInfoOPTIONAL }
```

```
GroupKeyNumber ::= INTEGER (0..15)
```

```
CODEC-Info ::= OCTET STRING (SIZE (5..10))
     -- Refers to channel type
     -- coded according to 3GPP TS 48.008 [49] and including Element identifier and Length
```

```
CipheringAlgorithm ::= OCTET STRING (SIZE (1))
     -- Refers to 'permitted algorithms' in 'encryption information'
     -- coded according to 3GPP TS 48.008 [49]:

     -- Bits 8-1
     -- 8765 4321
     -- 0000 0001                        No encryption
     -- 0000 0010                        GSM A5/1
     -- 0000 0100                        GSM A5/2
     -- 0000 1000                        GSM A5/3
     -- 0001 0000                        GSM A5/4
     -- 0010 0000                        GSM A5/5
     -- 0100 0000                        GSM A5/6
     -- 1000 0000                        GSM A5/7
```

```
StateAttributes ::= SEQUENCE {
     downlinkAttached                         [5] NULL                    OPTIONAL,
     uplinkAttached                           [6] NULL                    OPTIONAL,
     dualCommunication                        [7] NULL                    OPTIONAL,
     callOriginator                           [8] NULL                    OPTIONAL }

     -- Refers to 3GPP TS 44.068 for definitions of StateAttributes fields.
```

```
SendGroupCallInfoArg ::= SEQUENCE {
     requestedInfo                       RequestedInfo,
     groupId  Long-GroupId,
     teleservice                         Ext-TeleserviceCode,
     cellId    [0] GlobalCellId          OPTIONAL,
     imsi      [1] IMSI                  OPTIONAL,
     tmsi      [2] TMSI                  OPTIONAL,
     additionalInfo                      [3] AdditionalInfo              OPTIONAL,
     talkerPriority                      [4] TalkerPriority              OPTIONAL,
     cksn      [5] Cksn                  OPTIONAL,
     extensionContainer                  [6] ExtensionContainer          OPTIONAL,
     ... }
```

```
RequestedInfo ::= ENUMERATED {
     anchorMSC-AddressAndASCI-CallReference        (0),
     imsiAndAdditionalInfoAndAdditionalSubscription(1),
     ... }
--   exception handling:
--   an unrecognized value shall be rejected by the receiver with a return error cause of
--   unexpected data value
```

```
SendGroupCallInfoRes ::= SEQUENCE {
     anchorMSC-Address                  [0] ISDN-AddressString           OPTIONAL,
     asciCallReference                  [1] ASCI-CallReference           OPTIONAL,
     imsi      [2] IMSI                 OPTIONAL,
     additionalInfo                     [3] AdditionalInfo               OPTIONAL,
     additionalSubscriptions            [4] AdditionalSubscriptions      OPTIONAL,
     kc        [5] Kc                   OPTIONAL,
     extensionContainer                 [6] ExtensionContainer           OPTIONAL,
     ... }
```

```
END
```

17.7.13 Location service data types

```
MAP-LCS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-LCS-DataTypes (25) version18 (18)}

DEFINITIONS
IMPLICIT TAGS
::=
BEGIN

EXPORTS
   RoutingInfoForLCS-Arg,
   RoutingInfoForLCS-Res,
   ProvideSubscriberLocation-Arg,
   ProvideSubscriberLocation-Res,
   SubscriberLocationReport-Arg,
   SubscriberLocationReport-Res,
   LocationType,
   DeferredLocationEventType,
   LCSClientName,
   LCS-QoS,
   Horizontal-Accuracy,
   ResponseTime,
   Ext-GeographicalInformation,
   VelocityEstimate,
   SupportedGADShapes,
   Add-GeographicalInformation,
   LCSRequestorID,
   LCS-ReferenceNumber,
   LCSCodeword,
   AreaEventInfo,
   ReportingPLMNList,
   PeriodicLDRInfo,
   SequenceNumber,
   LCSClientType,
   LCS-Priority,
   OccurrenceInfo,
   IntervalTime
;

IMPORTS
   AddressString,
   ISDN-AddressString,
   IMEI,
   IMSI,
   LMSI,
   SubscriberIdentity,
   AgeOfLocationInformation,
   LCSClientExternalID,
   LCSClientInternalID,
   LCSServiceTypeID,
   CellGlobalIdOrServiceAreaIdOrLAI,
   PLMN-Id,
   GSN-Address,
   DiameterIdentity
```

```
FROM MAP-CommonDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-CommonDataTypes (18) version18 (18)}

   ExtensionContainer,
   SLR-ArgExtensionContainer
FROM MAP-ExtensionDataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-ExtensionDataTypes (21) version18 (18)}

   USSD-DataCodingScheme,
   USSD-String
FROM MAP-SS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0) gsm-Network (1) modules (3)
   map-SS-DataTypes (14) version18 (18)}

   APN,
   SupportedLCS-CapabilitySets
FROM MAP-MS-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-MS-DataTypes (11) version18 (18)}

   Additional-Number
FROM MAP-SM-DataTypes {
   itu-t identified-organization (4) etsi (0) mobileDomain (0)
   gsm-Network (1) modules (3) map-SM-DataTypes (16) version18 (18)}
;
```

```
RoutingInfoForLCS-Arg ::= SEQUENCE {
     mlcNumber [0] ISDN-AddressString,
     targetMS  [1] SubscriberIdentity,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...}
```

```
RoutingInfoForLCS-Res ::= SEQUENCE {
     targetMS  [0] SubscriberIdentity,
     lcsLocationInfo                     [1] LCSLocationInfo,
     extensionContainer                  [2] ExtensionContainer            OPTIONAL,
     ...,
     v-gmlc-Address                      [3]  GSN-Address                  OPTIONAL,
     h-gmlc-Address                      [4]  GSN-Address                  OPTIONAL,
     ppr-Address                         [5]  GSN-Address                  OPTIONAL,
     additional-v-gmlc-Address           [6]  GSN-Address                  OPTIONAL }
```

```
LCSLocationInfo ::= SEQUENCE {
     networkNode-Number                  ISDN-AddressString,
     -- NetworkNode-number can be msc-number, sgsn-number or a dummy value of "0"
     lmsi      [0] LMSI                  OPTIONAL,
     extensionContainer                  [1] ExtensionContainer            OPTIONAL,
     ... ,
     gprsNodeIndicator                   [2] NULL  OPTIONAL,
     -- gprsNodeIndicator is set only if the SGSN number is sent as the Network Node Number
     additional-Number                   [3] Additional-Number             OPTIONAL,
     supportedLCS-CapabilitySets         [4]  SupportedLCS-CapabilitySets  OPTIONAL,
     additional-LCS-CapabilitySets       [5]  SupportedLCS-CapabilitySets  OPTIONAL,
     mme-Name  [6]                       DiameterIdentity                  OPTIONAL,
     aaa-Server-Name                     [8]  DiameterIdentity             OPTIONAL,
     sgsn-Name [9]                       DiameterIdentity                  OPTIONAL,
     sgsn-Realm                          [10] DiameterIdentity             OPTIONAL
     }
```

```
ProvideSubscriberLocation-Arg ::= SEQUENCE {
     locationType                         LocationType,
     mlc-Number                           ISDN-AddressString,
     lcs-ClientID                         [0] LCS-ClientID                 OPTIONAL,
     privacyOverride                      [1] NULL  OPTIONAL,
     imsi      [2] IMSI                   OPTIONAL,
     msisdn    [3] ISDN-AddressString     OPTIONAL,
     lmsi      [4] LMSI                   OPTIONAL,
     imei      [5] IMEI                   OPTIONAL,
     lcs-Priority                         [6] LCS-Priority                 OPTIONAL,
     lcs-QoS   [7] LCS-QoS                OPTIONAL,
     extensionContainer                   [8] ExtensionContainer           OPTIONAL,
     ... ,
     supportedGADShapes                   [9]  SupportedGADShapes          OPTIONAL,
     lcs-ReferenceNumber                  [10] LCS-ReferenceNumber         OPTIONAL,
     lcsServiceTypeID                     [11] LCSServiceTypeID            OPTIONAL,
     lcsCodeword                          [12] LCSCodeword                 OPTIONAL,
     lcs-PrivacyCheck                     [13] LCS-PrivacyCheck            OPTIONAL,
     areaEventInfo                        [14] AreaEventInfo               OPTIONAL,
     h-gmlc-Address                       [15] GSN-Address                 OPTIONAL,
     mo-lrShortCircuitIndicator           [16] NULL OPTIONAL,
     periodicLDRInfo                      [17] PeriodicLDRInfo             OPTIONAL,
     reportingPLMNList                    [18] ReportingPLMNList           OPTIONAL }

     -- one of imsi or msisdn is mandatory
     -- If a location estimate type indicates activate deferred location or cancel deferred
     -- location, a lcs-Reference number shall be included.
```

```
LocationType ::= SEQUENCE {
     locationEstimateType                 [0] LocationEstimateType,
     ...,
     deferredLocationEventType            [1] DeferredLocationEventType    OPTIONAL }
```

```
LocationEstimateType ::= ENUMERATED {
     currentLocation                      (0),
     currentOrLastKnownLocation           (1),
     initialLocation                      (2),
     ...,
     activateDeferredLocation             (3),
     cancelDeferredLocation               (4) ,
     notificationVerificationOnly         (5) }
--   exception handling:
--   a ProvideSubscriberLocation-Arg containing an unrecognized LocationEstimateType
--   shall be rejected by the receiver with a return error cause of unexpected data value
```

```
DeferredLocationEventType ::= BIT STRING {
     msAvailable                          (0) ,
     enteringIntoArea                     (1),
     leavingFromArea                      (2),
     beingInsideArea                      (3) ,
     periodicLDR                          (4)  } (SIZE (1..16))
-- beingInsideArea is always treated as oneTimeEvent regardless of the possible value
-- of occurrenceInfo inside areaEventInfo.
-- exception handling:
-- a ProvideSubscriberLocation-Arg containing other values than listed above in
-- DeferredLocationEventType shall be rejected by the receiver with a return error cause of
-- unexpected data value.
```

```
LCS-ClientID ::= SEQUENCE {
     lcsClientType                        [0] LCSClientType,
     lcsClientExternalID                  [1] LCSClientExternalID          OPTIONAL,
     lcsClientDialedByMS                  [2] AddressString                OPTIONAL,
     lcsClientInternalID                  [3] LCSClientInternalID          OPTIONAL,
     lcsClientName                        [4] LCSClientName                OPTIONAL,
     ...,
     lcsAPN    [5] APN                    OPTIONAL,
     lcsRequestorID                       [6] LCSRequestorID               OPTIONAL }
```

```
LCSClientType ::= ENUMERATED {
     emergencyServices                  (0),
     valueAddedServices                 (1),
     plmnOperatorServices               (2),
     lawfulInterceptServices            (3),
     ... }
     --   exception handling:
     --   unrecognized values may be ignored if the LCS client uses the privacy override
     --   otherwise, an unrecognized value shall be treated as unexpected data by a receiver
     --   a return error shall then be returned if received in a MAP invoke
```

```
LCSClientName ::= SEQUENCE {
     dataCodingScheme                   [0] USSD-DataCodingScheme,
     nameString                         [2] NameString,
     ...,
     lcs-FormatIndicator                [3] LCS-FormatIndicator          OPTIONAL }

-- The USSD-DataCodingScheme shall indicate use of the default alphabet through the
-- following encoding
--   bit  7 6 5 4 3 2 1 0
--   0 0 0 0 1 1 1 1
```

```
NameString ::= USSD-String (SIZE (1..maxNameStringLength))
```

```
maxNameStringLength  INTEGER ::= 63
```

```
LCSRequestorID ::= SEQUENCE {
     dataCodingScheme                   [0] USSD-DataCodingScheme,
     requestorIDString                  [1] RequestorIDString,
     ...,
     lcs-FormatIndicator                [2] LCS-FormatIndicator          OPTIONAL }
```

```
RequestorIDString ::= USSD-String (SIZE (1..maxRequestorIDStringLength))
```

```
maxRequestorIDStringLength  INTEGER ::= 63
```

```
LCS-FormatIndicator ::= ENUMERATED {
     logicalName                        (0),
     e-mailAddress                      (1),
     msisdn    (2),
     url       (3),
     sipUrl    (4),
     ... }
```

```
LCS-Priority ::= OCTET STRING (SIZE (1))
     -- 0 = highest priority
     -- 1 = normal priority
     -- all other values treated as 1
```

```
LCS-QoS ::= SEQUENCE {
     horizontal-accuracy                [0] Horizontal-Accuracy          OPTIONAL,
     verticalCoordinateRequest          [1] NULL  OPTIONAL,
     vertical-accuracy                  [2] Vertical-Accuracy            OPTIONAL,
     responseTime                       [3] ResponseTime                 OPTIONAL,
     extensionContainer                 [4] ExtensionContainer           OPTIONAL,
     ...,
     velocityRequest                    [5] NULL  OPTIONAL
}
```

```
Horizontal-Accuracy ::= OCTET STRING (SIZE (1))
     -- bit 8 = 0
     -- bits 7-1 = 7 bit Uncertainty Code defined in 3GPP TS 23.032. The horizontal location
     -- error should be less than the error indicated by the uncertainty code with 67%
     -- confidence.
```

```
Vertical-Accuracy ::= OCTET STRING (SIZE (1))
     -- bit 8 = 0
     -- bits 7-1 = 7 bit Vertical Uncertainty Code defined in 3GPP TS 23.032.
     -- The vertical location error should be less than the error indicated
     -- by the uncertainty code with 67% confidence.
```

```
ResponseTime ::= SEQUENCE {
     responseTimeCategory               ResponseTimeCategory,
     ...}
--   note: an expandable SEQUENCE simplifies later addition of a numeric response time.
```

```
ResponseTimeCategory ::= ENUMERATED {
     lowdelay  (0),
     delaytolerant  (1),
     ... }
--   exception handling:
--   an unrecognized value shall be treated the same as value 1 (delaytolerant)
```

```
SupportedGADShapes ::= BIT STRING {
     ellipsoidPoint  (0),
     ellipsoidPointWithUncertaintyCircle (1),
     ellipsoidPointWithUncertaintyEllipse (2),
     polygon (3),
     ellipsoidPointWithAltitude (4),
     ellipsoidPointWithAltitudeAndUncertaintyElipsoid (5),
     ellipsoidArc  (6) } (SIZE (7..16))
-- A node shall mark in the BIT STRING all Shapes defined in 3GPP TS 23.032 it supports.
-- exception handling: bits 7 to 15 shall be ignored if received.
```

```
LCS-ReferenceNumber::= OCTET STRING (SIZE(1))
```

```
LCSCodeword ::= SEQUENCE {
     dataCodingScheme                    [0] USSD-DataCodingScheme,
     lcsCodewordString                   [1] LCSCodewordString,
     ...}
```

```
LCSCodewordString ::= USSD-String (SIZE (1..maxLCSCodewordStringLength))
```

```
maxLCSCodewordStringLength  INTEGER ::= 20
```

```
LCS-PrivacyCheck ::= SEQUENCE {
     callSessionUnrelated                [0] PrivacyCheckRelatedAction,
     callSessionRelated                  [1] PrivacyCheckRelatedAction     OPTIONAL,
     ...}
```

```
PrivacyCheckRelatedAction ::= ENUMERATED {
     allowedWithoutNotification (0),
     allowedWithNotification (1),
     allowedIfNoResponse (2),
     restrictedIfNoResponse (3),
     notAllowed (4),
     ...}
--   exception handling:
--   a ProvideSubscriberLocation-Arg containing an unrecognized PrivacyCheckRelatedAction
--   shall be rejected by the receiver with a return error cause of unexpected data value
```

```
AreaEventInfo ::= SEQUENCE {
     areaDefinition                      [0]  AreaDefinition,
     occurrenceInfo                      [1]  OccurrenceInfo               OPTIONAL,
     intervalTime                        [2]  IntervalTime                 OPTIONAL,
     ...}
```

```
AreaDefinition ::= SEQUENCE {
     areaList [0]                        AreaList,
     ...}
```

```
AreaList ::= SEQUENCE SIZE (1..maxNumOfAreas) OF Area
```

```
maxNumOfAreas  INTEGER ::= 10
```

```
Area ::= SEQUENCE {
     areaType [0]                        AreaType,
     areaIdentification                  [1]  AreaIdentification,
     ...}
```

```
AreaType ::= ENUMERATED {
     countryCode                         (0),
     plmnId   (1),
     locationAreaId                      (2),
     routingAreaId                       (3),
     cellGlobalId                        (4),
     ...,
     utranCellId                         (5) }
```

```
AreaIdentification ::= OCTET STRING (SIZE (2..7))
    -- The internal structure is defined as follows:
    -- octet 1 bits 4321                Mobile Country Code 1st digit
    --         bits 8765                Mobile Country Code 2nd digit
    -- octet 2 bits 4321                Mobile Country Code 3rd digit
    --         bits 8765                Mobile Network Code 3rd digit if 3 digit MNC included
    --       or filler (1111)
    -- octet 3 bits 4321                Mobile Network Code 1st digit
    --         bits 8765                Mobile Network Code 2nd digit
    -- octets 4 and 5                   Location Area Code (LAC) for Local Area Id,
    --       Routing Area Id and Cell Global Id
    -- octet 6                          Routing Area Code (RAC) for Routing Area Id
    -- octets 6 and 7                   Cell Identity (CI) for Cell Global Id
    -- octets 4 until 7                 Utran Cell Identity (UC-Id) for Utran Cell Id
```

```
OccurrenceInfo ::= ENUMERATED {
    oneTimeEvent                        (0),
    multipleTimeEvent                   (1),
    ...}
```

```
IntervalTime ::= INTEGER (1..32767)
    -- minimum interval time between area reports in seconds
```

```
PeriodicLDRInfo ::= SEQUENCE {
    reportingAmount ReportingAmount,
    reportingInterval   ReportingInterval,
    ...}
-- reportingInterval x reportingAmount shall not exceed 8639999 (99 days, 23 hours,
-- 59 minutes and 59 seconds) for compatibility with OMA MLP and RLP
```

```
ReportingAmount ::= INTEGER (1..maxReportingAmount)
```

```
maxReportingAmount INTEGER ::= 8639999
```

```
ReportingInterval ::= INTEGER (1..maxReportingInterval)
-- ReportingInterval is in seconds
```

```
maxReportingInterval INTEGER ::= 8639999
```

```
ReportingPLMNList::= SEQUENCE {
    plmn-ListPrioritized        [0] NULL            OPTIONAL,
    plmn-List           [1] PLMNList,
    ...}
```

```
PLMNList::= SEQUENCE SIZE (1..maxNumOfReportingPLMN) OF
         ReportingPLMN
```

```
maxNumOfReportingPLMN INTEGER ::= 20
```

```
ReportingPLMN::= SEQUENCE {
    plmn-Id         [0] PLMN-Id,
    ran-Technology          [1] RAN-Technology      OPTIONAL,
    ran-PeriodicLocationSupport [2] NULL            OPTIONAL,
    ...}
```

```
RAN-Technology ::= ENUMERATED {
    gsm     (0),
    umts    (1),
    ...}
```

```
ProvideSubscriberLocation-Res ::= SEQUENCE {
     locationEstimate                    Ext-GeographicalInformation,
     ageOfLocationEstimate               [0] AgeOfLocationInformation      OPTIONAL,
     extensionContainer                  [1] ExtensionContainer            OPTIONAL,
     ... ,
     add-LocationEstimate                [2] Add-GeographicalInformation   OPTIONAL,
     deferredmt-lrResponseIndicator      [3] NULL  OPTIONAL,
     geranPositioningData                [4] PositioningDataInformation    OPTIONAL,
     utranPositioningData                [5] UtranPositioningDataInfo      OPTIONAL,
     cellIdOrSai                         [6] CellGlobalIdOrServiceAreaIdOrLAI   OPTIONAL,
     sai-Present                         [7] NULL  OPTIONAL,
     accuracyFulfilmentIndicator         [8] AccuracyFulfilmentIndicator   OPTIONAL,
     velocityEstimate                    [9] VelocityEstimate              OPTIONAL,
     mo-lrShortCircuitIndicator          [10] NULL OPTIONAL,
     geranGANSSpositioningData           [11] GeranGANSSpositioningData    OPTIONAL,
     utranGANSSpositioningData           [12] UtranGANSSpositioningData    OPTIONAL,
     targetServingNodeForHandover        [13] ServingNodeAddress           OPTIONAL,
     utranAdditionalPositioningData      [14] UtranAdditionalPositioningData OPTIONAL,
     utranBaroPressureMeas               [15] UtranBaroPressureMeas        OPTIONAL,
     utranCivicAddress                   [16] UtranCivicAddress            OPTIONAL }

--   if deferredmt-lrResponseIndicator is set, locationEstimate is ignored.

-- the add-LocationEstimate parameter shall not be sent to a node that did not indicate the
-- geographic shapes supported in the ProvideSubscriberLocation-Arg
-- The locationEstimate and the add-locationEstimate parameters shall not be sent if
-- the supportedGADShapes parameter has been received in ProvideSubscriberLocation-Arg
-- and the shape encoded in locationEstimate or add-LocationEstimate is not marked
-- as supported in supportedGADShapes. In such a case ProvideSubscriberLocation
-- shall be rejected with error FacilityNotSupported with additional indication
-- shapeOfLocationEstimateNotSupported.
-- sai-Present indicates that the cellIdOrSai parameter contains a Service Area Identity.
```

```
AccuracyFulfilmentIndicator ::= ENUMERATED {
     requestedAccuracyFulfilled  (0),
     requestedAccuracyNotFulfilled  (1),
     ... }
```

```
Ext-GeographicalInformation ::= OCTET STRING (SIZE (1..maxExt-GeographicalInformation))
    -- Refers to geographical Information defined in 3GPP TS 23.032.
    -- This is composed of 1 or more octets with an internal structure according to
    -- 3GPP TS 23.032
    -- Octet 1: Type of shape, only the following shapes in 3GPP TS 23.032 are allowed:
    --   (a) Ellipsoid point with uncertainty circle
    --   (b) Ellipsoid point with uncertainty ellipse
    --   (c) Ellipsoid point with altitude and uncertainty ellipsoid
    --   (d) Ellipsoid Arc
    --   (e) Ellipsoid Point
    -- Any other value in octet 1 shall be treated as invalid
    -- Octets 2 to 8 for case (a) – Ellipsoid point with uncertainty circle
    --   Degrees of Latitude                3 octets
    --   Degrees of Longitude               3 octets
    --   Uncertainty code                   1 octet
    -- Octets 2 to 11 for case (b) – Ellipsoid point with uncertainty ellipse:
    --   Degrees of Latitude                3 octets
    --   Degrees of Longitude               3 octets
    --   Uncertainty semi-major axis        1 octet
    --   Uncertainty semi-minor axis        1 octet
    --   Angle of major axis                1 octet
    --   Confidence                         1 octet
    -- Octets 2 to 14 for case (c) – Ellipsoid point with altitude and uncertainty ellipsoid
    --   Degrees of Latitude                3 octets
    --   Degrees of Longitude               3 octets
    --   Altitude                           2 octets
    --   Uncertainty semi-major axis        1 octet
    --   Uncertainty semi-minor axis        1 octet
    --   Angle of major axis                1 octet
    --   Uncertainty altitude               1 octet
    --   Confidence                         1 octet
    -- Octets 2 to 13 for case (d) – Ellipsoid Arc
    --   Degrees of Latitude                3 octets
    --   Degrees of Longitude               3 octets
    --   Inner radius                       2 octets
    --   Uncertainty radius                 1 octet
    --   Offset angle                       1 octet
    --   Included angle                     1 octet
    --   Confidence                         1 octet
    -- Octets 2 to 7 for case (e) – Ellipsoid Point
    --   Degrees of Latitude                3 octets
    --   Degrees of Longitude               3 octets

    --
    -- An Ext-GeographicalInformation parameter comprising more than one octet and
    -- containing any other shape or an incorrect number of octets or coding according
    -- to 3GPP TS 23.032 shall be treated as invalid data by a receiver.
    --
    -- An Ext-GeographicalInformation parameter comprising one octet shall be discarded
    -- by the receiver if an Add-GeographicalInformation parameter is received
    -- in the same message.
    --
    -- An Ext-GeographicalInformation parameter comprising one octet shall be treated as
    -- invalid data by the receiver if an Add-GeographicalInformation parameter is not
    -- received in the same message.
```

```
maxExt-GeographicalInformation  INTEGER ::= 20
    -- the maximum length allows for further shapes in 3GPP TS 23.032 to be included in later
    -- versions of 3GPP TS 29.002
```

```
VelocityEstimate ::= OCTET STRING (SIZE (4..7))
    -- Refers to Velocity description defined in 3GPP TS 23.032.
    -- This is composed of 4 or more octets with an internal structure according to
    -- 3GPP TS 23.032
    -- Octet 1: Type of velocity, only the following types in 3GPP TS 23.032 are allowed:
    --   (a) Horizontal Velocity
    --   (b) Horizontal with Vertical Velocity
    --   (c) Horizontal Velocity with Uncertainty
    --   (d) Horizontal with Vertical Velocity and Uncertainty
    -- For types Horizontal with Vertical Velocity and Horizontal with Vertical Velocity
    -- and Uncertainty, the direction of the Vertical Speed is also included in Octet 1
    -- Any other value in octet 1 shall be treated as invalid
    -- Octets 2 to 4 for case (a) Horizontal velocity:
    --   Bearing                          1 octet
    --   Horizontal Speed                 2 octets
    -- Octets 2 to 5 for case (b) – Horizontal with Vertical Velocity:
    --   Bearing                          1 octet
    --   Horizontal Speed                 2 octets
    --   Vertical Speed                   1 octet
    -- Octets 2 to 5 for case (c) – Horizontal velocity with Uncertainty:
    --   Bearing                          1 octet
    --   Horizontal Speed                 2 octets
    --   Uncertainty Speed                1 octet
    -- Octets 2 to 7 for case (d) – Horizontal with Vertical Velocity and Uncertainty:
    --   Bearing                          1 octet
    --   Horizontal Speed                 2 octets
    --   Vertical Speed                   1 octet
    --   Horizontal Uncertainty Speed     1 octet
    --   Vertical Uncertainty Speed       1 octet
```

```
PositioningDataInformation ::= OCTET STRING (SIZE (2..maxPositioningDataInformation))
    -- Refers to the Positioning Data defined in 3GPP TS 49.031.
    -- This is composed of 2 or more octets with an internal structure according to
    -- 3GPP TS 49.031.
```

```
maxPositioningDataInformation INTEGER ::= 10
    --
```

```
UtranPositioningDataInfo ::= OCTET STRING (SIZE (3..maxUtranPositioningDataInfo))
    -- Refers to the Position Data defined in 3GPP TS 25.413.
    -- This is composed of the positioningDataDiscriminator and the positioningDataSet
    -- included in positionData as defined in 3GPP TS 25.413.
```

```
maxUtranPositioningDataInfo INTEGER ::= 11
    --
```

```
GeranGANSSpositioningData ::= OCTET STRING (SIZE (2..maxGeranGANSSpositioningData))
    -- Refers to the GANSS Positioning Data defined in 3GPP TS 49.031.
    -- This is composed of 2 or more octets with an internal structure according to
    -- 3GPP TS 49.031.
```

```
maxGeranGANSSpositioningData INTEGER ::= 10
    --
```

```
UtranGANSSpositioningData ::= OCTET STRING (SIZE (1..maxUtranGANSSpositioningData))
    -- Refers to the Position Data defined in 3GPP TS 25.413.
    -- This is composed of the GANSS-PositioningDataSet only, included in PositionData
    -- as defined in 3GPP TS 25.413.
```

```
maxUtranGANSSpositioningData INTEGER ::= 9
    --
```

```
UtranAdditionalPositioningData ::= OCTET STRING (SIZE (1..maxUtranAdditionalPositioningData))
    -- Refers to the Position Data defined in 3GPP TS 25.413.
    -- This is composed of the Additional-PositioningDataSet only, included in PositionData
    -- as defined in 3GPP TS 25.413.
```

```
maxUtranAdditionalPositioningData INTEGER ::= 8
    --
```

```
UtranBaroPressureMeas ::= INTEGER (30000..115000)
    -- Refers to the barometric pressure measurement defined in 3GPP TS 25.413.
    -- This is composed of the BarometricPressureMeasurement only as defined in 3GPP TS
    -- 25.413.
```

```
UtranCivicAddress ::= OCTET STRING
    -- Refers to the civic address defined in 3GPP TS 25.413.
    -- This is composed of the CivicAddress only as defined in 3GPP TS 25.413.
```

```
Add-GeographicalInformation ::= OCTET STRING (SIZE (1..maxAdd-GeographicalInformation))
    -- Refers to geographical Information defined in 3GPP TS 23.032.
    -- This is composed of 1 or more octets with an internal structure according to
    -- 3GPP TS 23.032
    -- Octet 1: Type of shape, all the shapes defined in 3GPP TS 23.032 are allowed:
    -- Octets 2 to n (where n is the total number of octets necessary to encode the shape
    -- according to 3GPP TS 23.032) are used to encode the shape itself in accordance with
the
    -- encoding defined in 3GPP TS 23.032
    --
    -- An Add-GeographicalInformation parameter, whether valid or invalid, received
    -- together with a valid Ext-GeographicalInformation parameter in the same message
    -- shall be discarded.
    --
    -- An Add-GeographicalInformation parameter containing any shape not defined in
    -- 3GPP TS 23.032 or an incorrect number of octets or coding according to
    -- 3GPP TS 23.032 shall be treated as invalid data by a receiver if not received
    -- together with a valid Ext-GeographicalInformation parameter in the same message.
```

```
maxAdd-GeographicalInformation  INTEGER ::= 91
    -- the maximum length allows support for all the shapes currently defined in 3GPP TS
23.032
```

```
SubscriberLocationReport-Arg ::= SEQUENCE {
    lcs-Event LCS-Event,
    lcs-ClientID                        LCS-ClientID,
    lcsLocationInfo                     LCSLocationInfo,
    msisdn    [0] ISDN-AddressString    OPTIONAL,
    imsi      [1] IMSI                  OPTIONAL,
    imei      [2] IMEI                  OPTIONAL,
    na-ESRD   [3] ISDN-AddressString    OPTIONAL,
    na-ESRK   [4] ISDN-AddressString    OPTIONAL,
    locationEstimate                    [5] Ext-GeographicalInformation   OPTIONAL,
    ageOfLocationEstimate               [6] AgeOfLocationInformation      OPTIONAL,
    slr-ArgExtensionContainer           [7] SLR-ArgExtensionContainer     OPTIONAL,
    ... ,
    add-LocationEstimate                [8] Add-GeographicalInformation   OPTIONAL,
    deferredmt-lrData                   [9] Deferredmt-lrData             OPTIONAL,
    lcs-ReferenceNumber                 [10] LCS-ReferenceNumber          OPTIONAL,
    geranPositioningData                [11] PositioningDataInformation   OPTIONAL,
    utranPositioningData                [12] UtranPositioningDataInfo     OPTIONAL,
    cellIdOrSai                         [13] CellGlobalIdOrServiceAreaIdOrLAI  OPTIONAL,
    h-gmlc-Address                      [14] GSN-Address                  OPTIONAL,
    lcsServiceTypeID                    [15] LCSServiceTypeID             OPTIONAL,
    sai-Present                         [17] NULL OPTIONAL,
    pseudonymIndicator                  [18] NULL OPTIONAL,
    accuracyFulfilmentIndicator         [19] AccuracyFulfilmentIndicator  OPTIONAL,
    velocityEstimate                    [20] VelocityEstimate             OPTIONAL,
    sequenceNumber                      [21] SequenceNumber               OPTIONAL,
    periodicLDRInfo                     [22] PeriodicLDRInfo              OPTIONAL,
    mo-lrShortCircuitIndicator          [23] NULL OPTIONAL,
    geranGANSSpositioningData           [24] GeranGANSSpositioningData    OPTIONAL,
    utranGANSSpositioningData           [25] UtranGANSSpositioningData    OPTIONAL,
    targetServingNodeForHandover        [26] ServingNodeAddress           OPTIONAL,
    utranAdditionalPositioningData      [27] UtranAdditionalPositioningData OPTIONAL,
    utranBaroPressureMeas               [28] UtranBaroPressureMeas        OPTIONAL,
    utranCivicAddress                   [29] UtranCivicAddress            OPTIONAL }

    -- one of msisdn or imsi is mandatory
    -- a location estimate that is valid for the locationEstimate parameter should
    -- be transferred in this parameter in preference to the add-LocationEstimate.
    -- the deferredmt-lrData parameter shall be included if and only if the lcs-Event
    -- indicates a deferredmt-lrResponse.
    -- if the lcs-Event indicates a deferredmt-lrResponse then the locationEstimate
    -- and the add-locationEstimate parameters shall not be sent if the
    -- supportedGADShapes parameter had been received in ProvideSubscriberLocation-Arg
    -- and the shape encoded in locationEstimate or add-LocationEstimate was not marked
    -- as supported in supportedGADShapes. In such a case terminationCause
    -- in deferredmt-lrData shall be present with value
    -- shapeOfLocationEstimateNotSupported.
    -- If a lcs event indicates deferred mt-lr response, the lcs-Reference number shall be
    -- included.
    -- sai-Present indicates that the cellIdOrSai parameter contains a Service Area Identity.
```

```
Deferredmt-lrData ::= SEQUENCE {
     deferredLocationEventType           DeferredLocationEventType,
     terminationCause                    [0] TerminationCause              OPTIONAL,
     lcsLocationInfo                     [1] LCSLocationInfo               OPTIONAL,
     ...}
     -- lcsLocationInfo may be included only if a terminationCause is present
     -- indicating mt-lrRestart.
```

```
LCS-Event ::= ENUMERATED {
     emergencyCallOrigination  (0),
     emergencyCallRelease  (1),
     mo-lr  (2),
     ...,
     deferredmt-lrResponse  (3) ,
     deferredmo-lrTTTPInitiation  (4),
     emergencyCallHandover (5)  }
     --   deferredmt-lrResponse is applicable to the delivery of a location estimate
     --   for an LDR initiated earlier by either the network (via an MT-LR activate deferred
     --   location) or the UE (via a deferred MO-LR TTTP initiation)
     --   exception handling:
     --   a SubscriberLocationReport-Arg containing an unrecognized LCS-Event
     --   shall be rejected by a receiver with a return error cause of unexpected data value
```

```
TerminationCause ::= ENUMERATED {
     normal  (0),
     errorundefined  (1),
     internalTimeout  (2),
     congestion  (3),
     mt-lrRestart  (4),
     privacyViolation  (5),
     ...,
     shapeOfLocationEstimateNotSupported (6) ,
     subscriberTermination (7),
     uETermination (8),
     networkTermination (9)  }
-- mt-lrRestart shall be used to trigger the GMLC to restart the location procedure,
-- either because the sending node knows that the terminal has moved under coverage
-- of another MSC or SGSN (e.g. Send Identification received), or because the subscriber
-- has been deregistered due to a Cancel Location received from HLR.
--
-- exception handling
-- an unrecognized value shall be treated the same as value 1 (errorundefined)
```

```
SequenceNumber ::= INTEGER (1..maxReportingAmount)
```

```
ServingNodeAddress ::= CHOICE {
     msc-Number                          [0] ISDN-AddressString,
     sgsn-Number                         [1] ISDN-AddressString,
     mme-Number                          [2] DiameterIdentity }
```

```
SubscriberLocationReport-Res ::= SEQUENCE {
     extensionContainer                  ExtensionContainer                OPTIONAL,
     ...,
     na-ESRK   [0] ISDN-AddressString    OPTIONAL,
     na-ESRD   [1] ISDN-AddressString    OPTIONAL,
     h-gmlc-Address                      [2]  GSN-Address                  OPTIONAL,
     mo-lrShortCircuitIndicator          [3] NULL  OPTIONAL,
     reportingPLMNList                   [4] ReportingPLMNList             OPTIONAL,
     lcs-ReferenceNumber                 [5]  LCS-ReferenceNumber          OPTIONAL }

-- na-ESRK and na-ESRD are mutually exclusive
--
-- exception handling
-- receipt of both na-ESRK and na-ESRD shall be treated the same as a return error
```

```
END
```

17.7.14 Void

18 General on MAP user procedures

18.1 Introduction

Clauses 18 to 25 describe the use of MAP services for GSM signalling procedures. GSM signalling procedures may involve one or several interfaces running one or several application protocols. The present document addresses only the signalling procedures which require at least the use of one MAP service.

When a signalling procedure takes place in the network, an application process invocation is created in each system component involved. Part of the application process invocation acts as a MAP user and handles one or several MAP dialogues. For each dialogue it employs an instance of the MAP service provider. It may also use other communication services to exchange information on other interfaces, but detailed description of these aspects is outside the scope of the present document.

18.2 Common aspects of user procedure descriptions

18.2.1 General conventions

For each signalling procedure the present document provides a brief textual overview accompanied by a flow diagram which represent the functional interactions between system components. Functional interactions are labelled using the MAP service name when the interaction results from a service request or by this service name followed by the symbol "ack" when this interaction results from a service response.

For each of the system components involved, the present document also provides a detailed textual description of the application process behaviour as well as an SDL diagram. SDL diagrams describe the sequence of events, as seen by the MAP-User, which occurs at MAP service provider boundaries as well as external events which occur at other interfaces and which impact on the previous sequence.

External events do not necessarily correspond to the messages of other protocols used in the system component. The MAP-user procedures are described as if a set of interworking functions (IWF) between the MAP-user and the other protocol entities was implemented (see figure 18.2/1). Such interworking functions are assumed to perform either an identity mapping or some processing or translation as required to eliminate information irrelevant to the MAP-user.

The mapping of service primitives on to protocol elements is described in clauses 14 to 17.

GSM signalling procedures are built from one or more sub-procedures (e.g. authentication, ciphering, ...).
Sub-procedures from which signalling procedures are built are represented using SDL MACRO descriptions.

In case of any discrepancy between the textual descriptions and the SDL descriptions, the latter take precedence.

18.2.2 Naming conventions

Events related to MAP are represented by MAP service primitives. The signal names used in the SDL diagrams are derived from the service primitive names defined in clauses 7 to 12, with some lexical transformations for readability and parsability purposes (blanks between words are replaced by underscores, the first letter of each word is capitalised).

Events received and sent on other interfaces are named by appending the message or signal name to a symbol representing the interface type, with some lexical transformations for readability and parsability purposes (blanks between words are replaced by underscores, the first letter of each word is capitalised).

The following symbols are used to represent the interface types:

"I": For interfaces to the fixed network. "I" stands for ISUP interface.

"A": For interfaces between the MSC and the BSS (i.e. A-interfaces);

"Gb": For interfaces between the SGSN and the BSS (i.e. Gb-interfaces);

"OM": For network management interfaces (communication with OMC, MML interface, ...);

"SC": For interfaces to a Service Centre;

"HO_CA": For internal interfaces to the Handover Control Application.

"US": For a local USSD application.

These naming conventions can be summarised by the following BNF description:

<Event_Name> ::= <MAP_Primitive> | <External_Event>

<MAP_Primitive> ::= <MAP_Open> | <MAP_Close> | <MAP_U_Abort> | <MAP_P_Abort> |
<MAP_Specific> | <MAP_Notice>

<MAP_Open> ::= MAP_Open_Req | MAP_Open_Ind | MAP_Open_Rsp | MAP_Open_Cnf

<MAP_Close> ::= MAP_Close_Req | MAP_Close_Ind

<MAP_U_Abort> ::= MAP_U_Abort_Req | MAP_U_Abort_Ind

<MAP_P_Abort> ::= MAP_P_Abort_Ind

<MAP_Notice> ::= MAP_Notice_Ind

<MAP_Specific>::= <MAP_Req> | <MAP_Ind> | <MAP_Rsp> | <MAP_Cnf>

<MAP_Req> ::= MAP_<Service_Name>_Req

<MAP_Ind> ::= MAP_<Service_Name>_Ind

<MAP_Rsp> ::= MAP_<Service_Name>_Rsp

<MAP_Cnf> ::= MAP_<Service_Name>_Cnf

<External_Event> ::= <Interface_Type>_<External_Signal>

<Interface_Type> ::= I | A | Gb | OM | SC | HO AC | US

<External_Signal> ::= <Lexical_Unit>

<Service_Name> ::= <Lexical_Unit>

<Lexical_Unit> ::= <Lexical_Component> | <Lexical_Unit>_ <Lexical_Component>

<Lexical_Component> ::= <Upper_Case_Letter><Letter_Or_Digit_List>

<Letter_Or_Digit_List> ::= <Letter_Or_Digit> | <Letter_Or_Digit_List><Letter_Or_Digit>

<Letter_Or_Digit> ::= <Letter> | <Digit>

<Letter> ::= <Lower_Case_Letter> | <Upper_Case_Letter>

<Upper_Case_Letter> ::= A|B|C|D|E|F|G|H|I|J|K|L|M|N|O|P|Q|R|S|T|U|V|W|X|Y|Z

<Lower_Case_Letter> ::= a|b|c|d|e|f|g|h|i|j|k|l|m|n|o|p|q|r|s|t|u|v|w|x|y|z

<Digit> ::= 1|2|3|4|5|6|7|8|9|0

Figure 18.2/1: Interfaces applicable to the MAP-User

18.2.3 Convention on primitives parameters

18.2.3.1 Open service

When the originating and destination reference parameters shall be included in the MAP-OPEN request primitive, their value are indicated as a comment to the signal which represents this primitive.

18.2.3.2 Close service

When a pre-arranged released is requested, a comment is attached to the signal which represents the MAP-CLOSE request primitive. In the absence of comment, a normal release is assumed.

18.2.4 Version handling at dialogue establishment

Unless explicitly indicated in subsequent clauses, the following principles regarding version handling procedures at dialogue establishment are applied by the MAP-user.

18.2.4.1 Behaviour at the initiating side

When a MAP user signalling procedure has to be executed, the MAP-user issues a MAP-OPEN request primitive with an appropriate application-context-name. If several names are supported (i.e. several versions) a suitable one is selected using the procedures described in clause 5.

If version n is selected (where 1 < n <= highest existing version) and a MAP-OPEN Confirm primitive is received in response to the MAP-OPEN request with a result parameter set to "refused" and a diagnostic parameter indicating "application context not supported" or "potential version incompatibility problem", the MAP-User issues a new MAP-OPEN request primitive with the equivalent version y context (where 1 <= y < n). This is informally represented in the SDL diagrams by task symbols indicating 'Perform Vr procedure".

18.2.4.2 Behaviour at the responding side

On receipt of a MAP-OPEN indication primitive, the MAP-User analyses the application-context-name and executes the procedure associated with the requested version context. For example,if it refers to a version one context, the associated V1 procedure is executed; if it refers to a version two context, the associated V2 procedure is executed;etc.

18.2.5 Abort Handling

Unless explicitly indicated in subsequent clauses, the following principles are applied by the MAP-user regarding abort handling procedures:

On receipt of a MAP-P-ABORT indication or MAP-U-ABORT Indication primitive from any MAP-provider invocation, the MAP-User issues a MAP-U-ABORT Request primitive to each MAP-provider invocation associated with the same user procedure.

If applicable a decision is made to decide if the affected user procedure has to be retried or not.

18.2.6 SDL conventions

The MAP SDLs make use of a number of SDL concepts and conventions, where not all of them may be widely known. Therefore, this clause outlines the use of a few concepts and conventions to improve understanding of the MAP SDLs.

The MAP User SDLs make use of SDL Processes, Procedures and Macros. Processes are independent from each other even if one process starts another one: The actions of both of them have no ordering in time. SDL Procedures and Macros are just used to ease writing of the specification: They contain parts of a behaviour used in several places, and the corresponding Procedure/Macro definition has to be expanded at the position of the Procedure/Macro call.

All Processes are started at system initialisation and live forever, unless process creation/termination is indicated explicitly (i.e. a process is created by some other process).

The direction of Input/Output Signals in the SDL graphs is used to indicate the entity to which/from which communication is directed. If a process A communicates in parallel with processes B and C, all Inputs/Outputs to/from B are directed to one side, whereas communication with C is directed to the other side. However, there has been no

formal convention used that communication to a certain entity (e.g. a HLR) will always be directed to a certain side (e.g. right).

In each state all those Input Signals are listed, which result in an action and/or state change. If an Input Signal is not listed in a state, receipt of this input should lead to an implicit consumption without any action or state change (according to the SDL rules). This implicit consumption is mainly used for receipt of the MAP DELIMITER indication and for receipt of a MAP CLOSE indication, except for a premature MAP CLOSE.

18.3 Interaction between MAP Provider and MAP Users

Each MAP User is defined by at least one SDL process. On the dialogue initiating side, the MAP User will create a new instance of a MAP Provider implicit by issuing a MAP-OPEN request. This instance corresponds to a TC Dialogue and lives as long as the dialogue exists (see also clause 14.3). There is a fixed relation between MAP User and this Provider instance, i.e. all MAP service primitives from the MAP User for this dialogue are sent to this instance and all TC components received by this MAP Provider are mapped onto service primitives sent to this MAP User.

On the receiving side a MAP Provider instance is created implicit by receipt of a TC BEGIN indication. The corresponding MAP User is determined by the Application Context name included in this primitive, i.e. each Application Context is associated with one and only one MAP User. An instance of this User will be created implicitly by receiving a MAP-OPEN indication. Note that in some cases there exist several SDL Processes for one MAP User (Application Context), e.g. the processes Register_SS_HLR, Erase_SS_HLR, Activate_SS_HLR, Deactivate_SS_HLR, Interrogate_SS_HLR, and Register_Password for the AC Network_Functional_SS_Handling. In these cases, a coordinator process is introduced acting as a MAP User, which in turn starts a sub-process depending on the first MAP service primitive received.

19 Mobility procedures

19.1 Location management Procedures

The signalling procedures in this subclause support:

- Interworking between the VLR and the HLR and between the VLR and the previous VLR (PVLR) when a non-GPRS subscriber performs a location update to a new VLR service area;
- Interworking between the SGSN, the HLR and the VLR when a subscriber with both GPRS and non-GPRS subscriptions performs a routeing area update in an SGSN and the Gs interface is implemented;
- Interworking between the SGSN and the VLR when a GPRS subscriber performs a routeing area update to a new SGSN service area;
- Interworking between the HLR and the VLR and between the HLR and the SGSN to delete a subscriber record from the VLR or the SGSN;
- Interworking between the VLR and the HLR and between the SGSN and the HLR to report to the HLR that a subscriber record has been purged from the VLR or the SGSN.

The MAP co-ordinating process in the HLR to handle a dialogue opened with the network location updating context is shown in figure 19.1/1. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see clause 25.1.1.

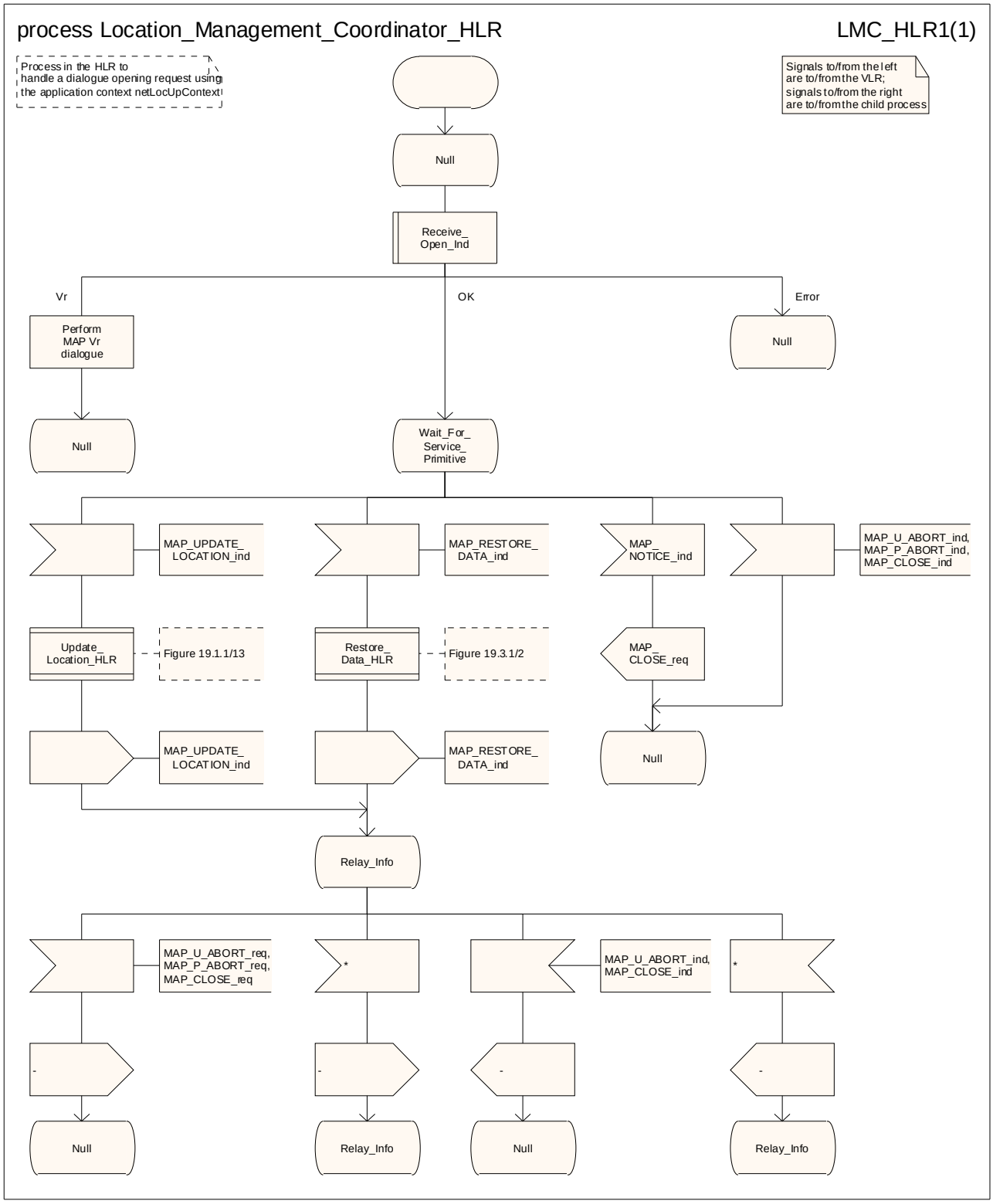


Figure 19.1/1: Process Location_Management_Coordinator_HLR

19.1.1 Location updating

19.1.1.1 General

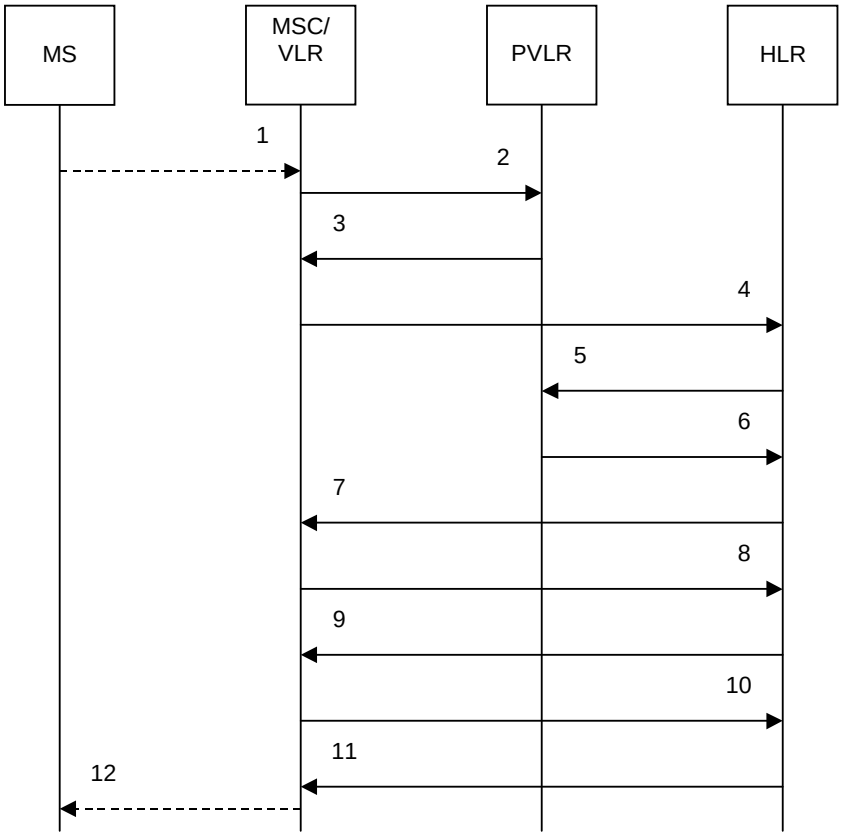
The stage 2 specification for GPRS is in 3GPP TS 23.060 [104]. The interworking between the MAP signalling procedures and the GPRS procedures in the SGSN and the HLR is shown by the transfer of signals between these procedures.

The message flow for successful inter-VLR location updating when the IMSI can be retrieved from the PVLR is shown in figure 19.1.1/2.

The message flow for successful inter-VLR location updating when the IMSI cannot be retrieved from the PVLR is shown in figure 19.1.1/3.

The message flow for successful GPRS Attach/RA update procedure (Gs interface not installed) is shown in figure 19.1.1/4.

The message flow for successful GPRS Attach/RA update procedure combined with a successful VLR location updating (Gs interface installed) is shown in figure 19.1.1/5.

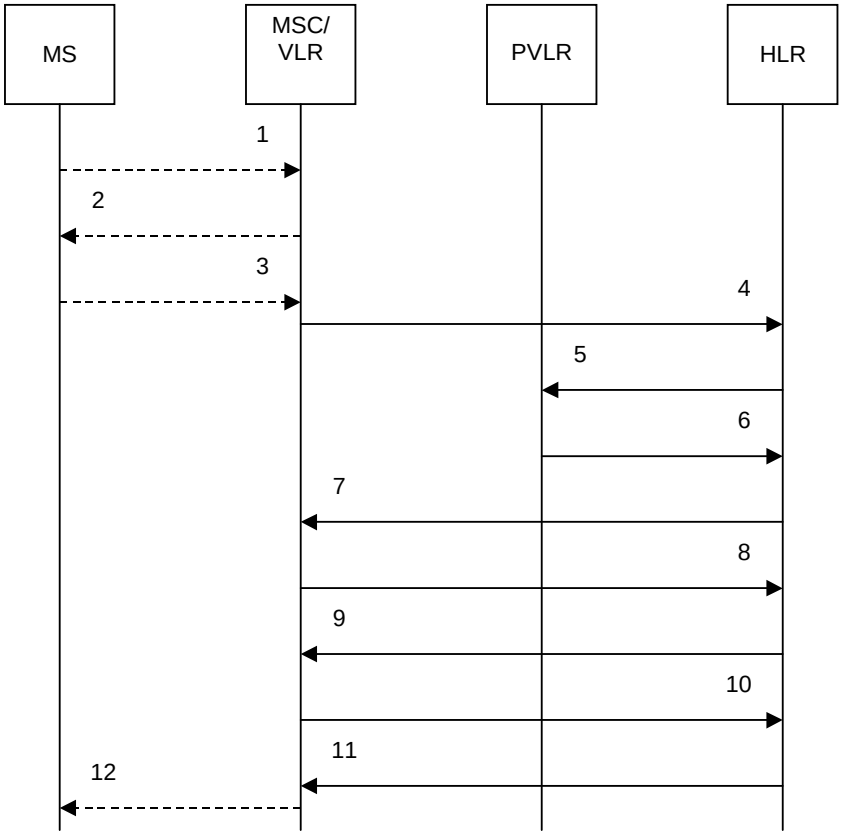


PVLR = Previous VLR

1) A_LU_REQUEST (Note 1)
2) MAP_SEND_IDENTIFICATION_req/ind
3) MAP_SEND_IDENTIFICATION_rsp/cnf
4) MAP_UPDATE_LOCATION_req/ind
5) MAP_CANCEL_LOCATION_req/ind
6) MAP_CANCEL_LOCATION_rsp/cnf
7) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 2)
8) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 2)
9) MAP_INSERT_SUBSCRIBER_DATA_req/ind
10) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
11) MAP_UPDATE_LOCATION_rsp/cnf
12) A_LU_CONFIRM (Note 1)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35]. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: Services printed in *italics* are optional.

Figure 19.1.1/2: Message flow for location updating to a new VLR area, when the IMSI can be retrieved from the previous VLR

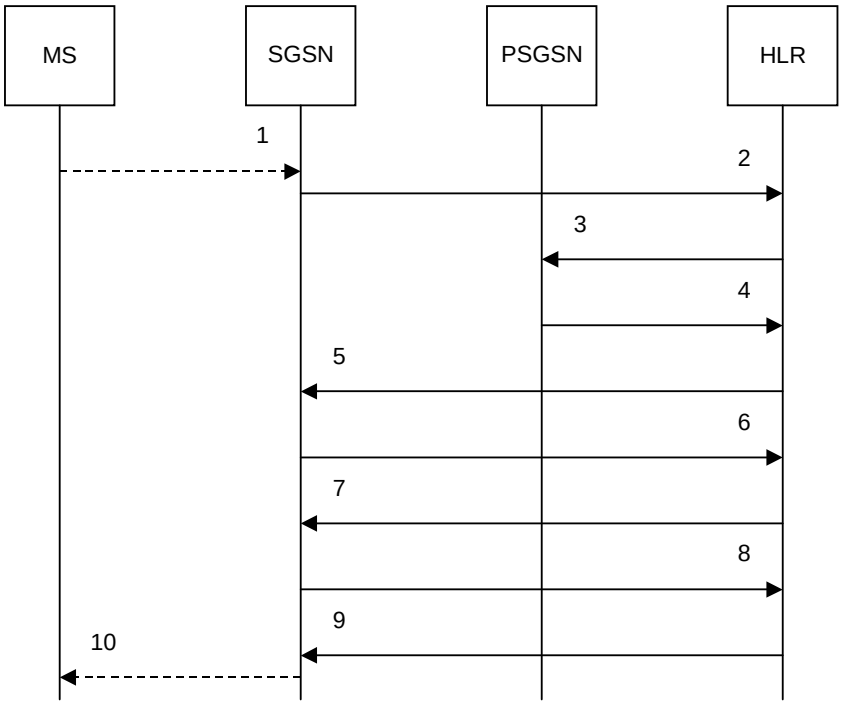


PVLR = Previous VLR

1) A_LU_REQUEST (Note 1)
2) A_IDENTITY_REQUEST (Note 1)
3) A_IDENTITY_RESPONSE (Note 1)
4) MAP_UPDATE_LOCATION_req/ind
5) MAP_CANCEL_LOCATION_req/ind
6) MAP_CANCEL_LOCATION_rsp/cnf
7) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 2)
8) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 2)
9) MAP_INSERT_SUBSCRIBER_DATA_req/ind
10) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
11) MAP_UPDATE_LOCATION_rsp/cnf
12) A_LU_CONFIRM (Note 1)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35]. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: Services printed in *italics* are optional.

Figure 19.1.1/3: Message flow for location updating to a new VLR area, when the IMSI cannot be retrieved from the previous VLR

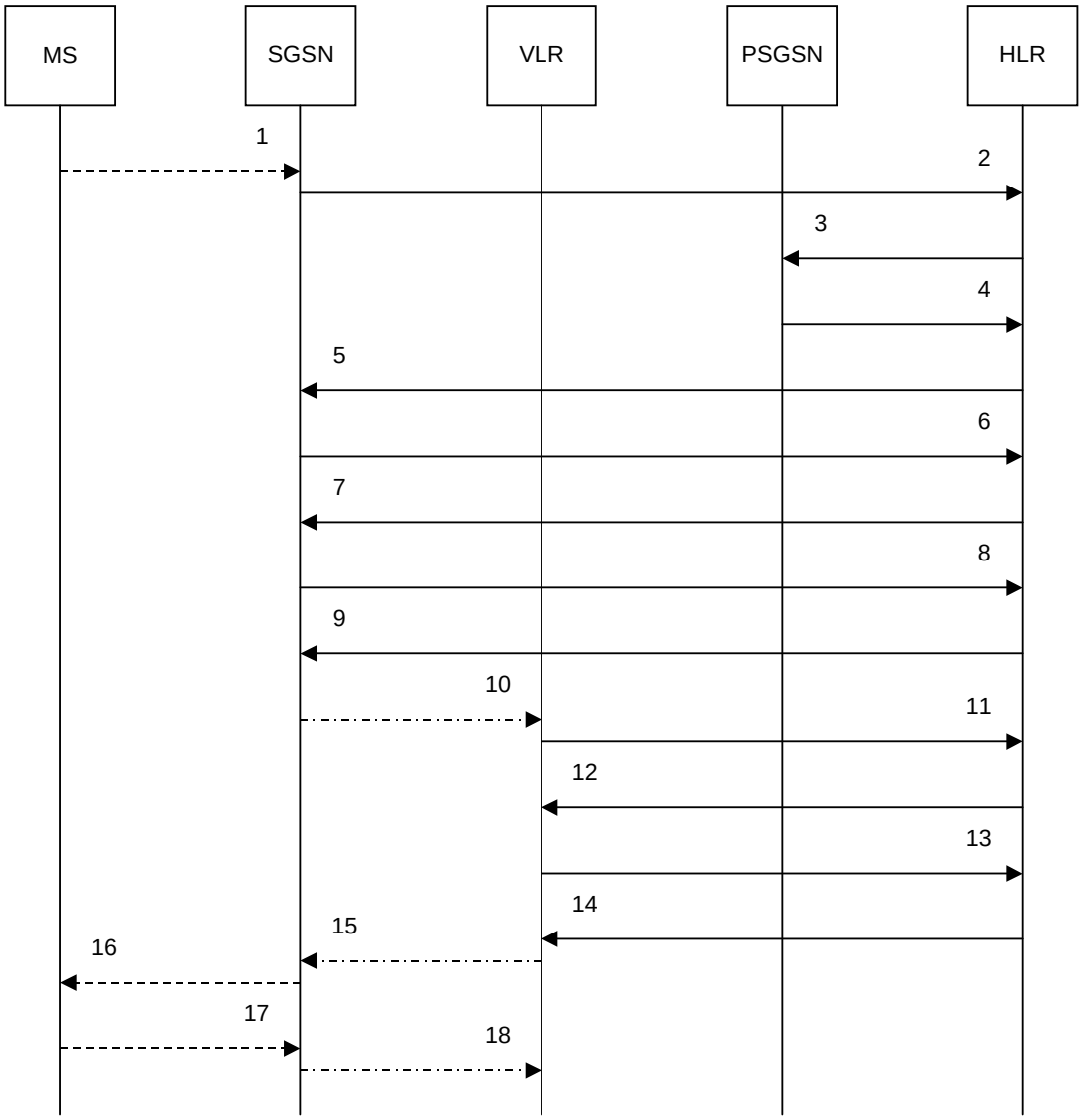


PSGSN = Previous SGSN

1) Gb_ATTACH_REQUEST or RA_UPDATE_REQUEST (Note 1, note 2)
2) MAP_UPDATE_GPRS_LOCATION_req/ind
3) MAP_CANCEL_LOCATION_req/ind
4) MAP_CANCEL_LOCATION_rsp/cnf
5) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 3)
6) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 3)
7) MAP_INSERT_SUBSCRIBER_DATA_req/ind
8) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
9) MAP_UPDATE_GPRS_LOCATION_rsp/cnf
10) Gb_ATTACH_ACCEPT or RA_UPDATE_ACCEPT (Note 1)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35]. The services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For security functions (authentication, ciphering, IMEI check) triggering refer to 3GPP TS 23.060 [104]. The MAP signalling invoked for these functions is described in clause 25 of the present document.
NOTE 3: Services printed in *italics* are optional.
NOTE 4: Refer to 3GPP TS 23.060 [104] for termination of the procedure and triggering of the signalling on the interface between the BSS and the SGSN.

Figure 19.1.1/4: Message flow for GPRS location updating (Gs interface not installed)



1) Gb_ATTACH_REQUEST or RA_UPDATE_REQUEST (Note 1, note 2)
2) MAP_UPDATE_GPRS_LOCATION_req/ind
3) MAP_CANCEL_LOCATION_req/ind
4) MAP_CANCEL_LOCATION_rsp/cnf
5) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 3)
6) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 3)
7) MAP_INSERT_SUBSCRIBER_DATA_req/ind
8) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
9) MAP_UPDATE_GPRS_LOCATION_rsp/cnf
10) Gs_LOCATION_UPDATE_REQUEST (Note 4)
11) MAP_UPDATE_LOCATION_req/ind (Note 5)
12) MAP_INSERT_SUBSCRIBER_DATA_req/ind
13) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
14) MAP_UPDATE_LOCATION_rsp/cnf
15) Gs_LOCATION_UPDATE_ACCEPT (Note 4)
16) Gb_ATTACH_ACCEPT or RA_UPDATE_ACCEPT (Note 1)
17) Gb_TMSI_REALLOCATION_COMPLETE (Note 1)
18) Gs_TMSI_REALLOCATION_COMPLETE (Note 4)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35]. The services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For security functions (authentication, ciphering, IMEI check) triggering refer to 3GPP TS 23.060 [104]. MAP processes invoked for those procedures are described in subclause 25.5.
NOTE 3: Services printed in *italics* are optional.

NOTE 5: For details of the procedure on the path between the SGSN and the VLR, see 3GPP TS 29.018 [106]. The services shown in chain lines indicate the trigger provided by the signalling on the path between the SGSN and the VLR, and the signalling triggered on the path between the SGSN and the VLR.
NOTE 4: Refer to 3GPP TS 23.060 [104] for termination of the procedure and triggering of the signalling on the interface between the BSS and the SGSN.
NOTE 5: For simplicity, the Location Cancellation procedure towards the previous VLR and optional tracing activation towards the new VLR are not shown in this figure.

Figure 19.1.1/5: Message flow for GPRS location updating (Gs interface installed)

19.1.1.2 Procedures in the VLR

The MAP process in the VLR for location updating for a non-GPRS subscriber is shown in figure 19.1.1/6. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The MAP process in the VLR to retrieve the IMSI of a subscriber from the previous VLR (PVLR) is shown in figure 19.1.1/7. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The process in the VLR for location updating for a GPRS subscriber when the Gs interface is installed is shown in figure 19.1.1/8.

The macro GPRS_Location_Update_Completion_VLR is shown in figure 19.1.1/9. The macro invokes a process not defined in this clause; the definition of this process can be found as follows:

Subscriber_Present_VLR see subclause 25.10.1.

The macro GPRS_Update_HLR_VLR is shown in figure 19.1.1/10. The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Insert_Subs_Data_VLR see subclause 25.7.1;

Activate_Tracing_VLR see subclause 25.9.4.

19.1.1.3 Procedure in the PVLR

The MAP process in the PVLR to handle a request for the IMSI of a subscriber from the new VLR is shown in figure 19.1.1/11. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

19.1.1.4 Procedure in the SGSN

The MAP process in the SGSN for location updating for a GPRS subscriber is shown in figure 19.1.1/12. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Insert_Subs_Data_SGSN see subclause 25.7.2;

Activate_Tracing_SGSN see subclause 25.9.5.

Sheet 2: The procedure Check_User_Error_In_Serving_Network_Entity is specific to Super-Charger; it is specified in 3GPP TS 23.116 [110].

19.1.1.5 Procedures in the HLR

The MAP process in the HLR to handle a location updating request from a VLR is shown in figure 19.1.1/13. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Confirmation see subclause 25.2.2.

The MAP process in the HLR to handle a location updating request from an SGSN is shown in figure 19.1.1/14. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2;

Control_Tracing_With_SGSN_HLR see subclause 25.9.7.

Sheet 2: The procedure Super_Charged_Cancel_Location_HLR is specific to Super-Charger; it is specified in 3GPP TS 23.116 [110]. If the HLR does not support the Super-Charger functionality, processing continues from the "No" exit of the test "Result=Pass?".

Sheet 2: The procedure Super_Charged_Location_Updating_HLR is specific to Super-Charger; it is specified in 3GPP TS 23.116 [110]. If the HLR does not support the Super-Charger functionality, processing continues from the "No" exit of the test "Result=Pass?".

Sheet 2: If the HLR supports the Administrative Restriction of Subscribers' Access feature and roaming is allowed in the VPLMN then the HLR may check the "Supported RAT Types" received from the VLR against the access restriction parameters. If this check fails then the decision box "Roaming allowed in this PLMN" shall take the exit "No".

The MAP process in the HLR to notify Short Message Service Centres that a subscriber is now reachable is shown in figure 19.1.1/15. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Alert_Service_Centre_HLR see subclause 25.10.3.

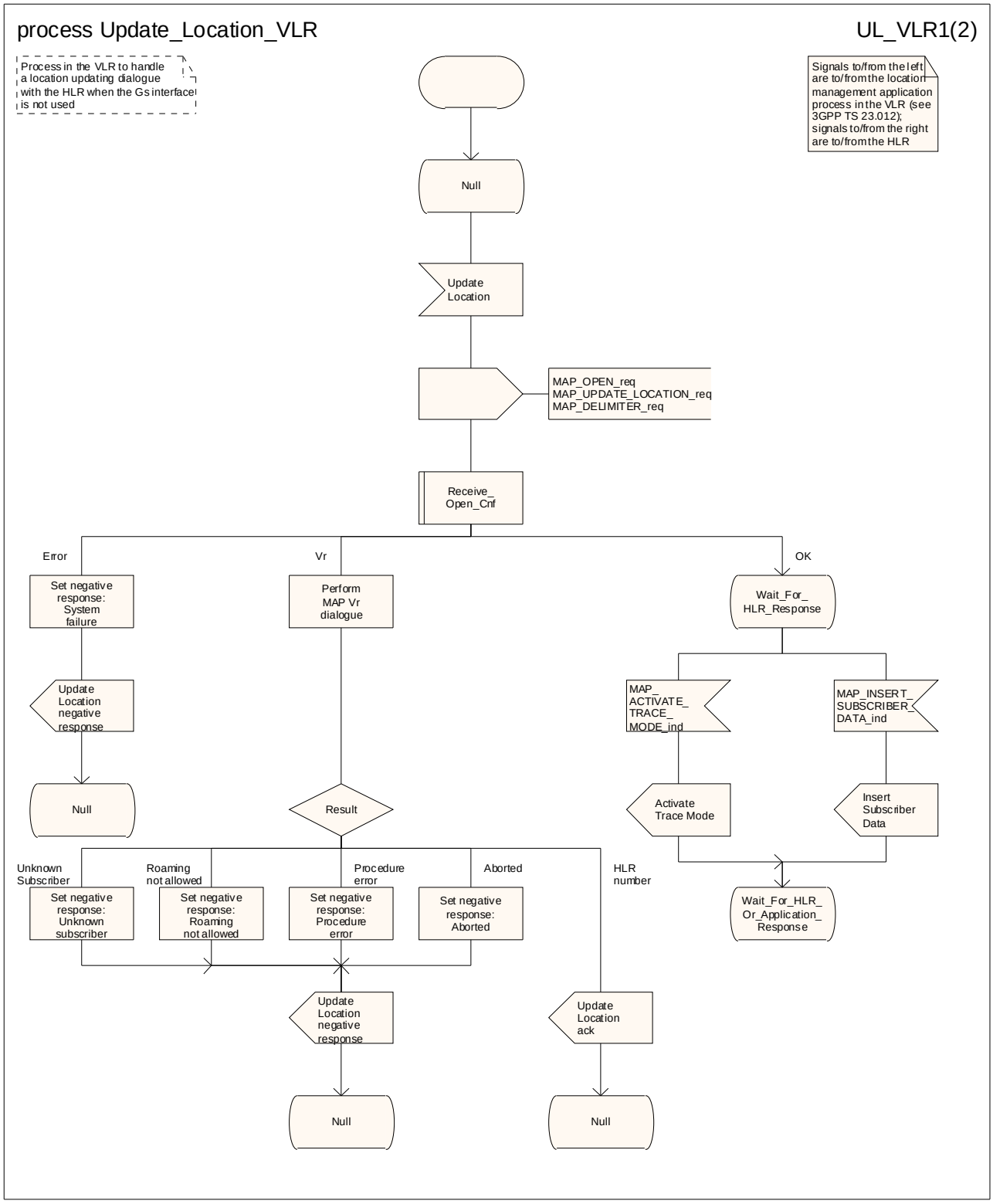


Figure 19.1.1/6 (sheet 1 of 2): Process Update_Location_VLR

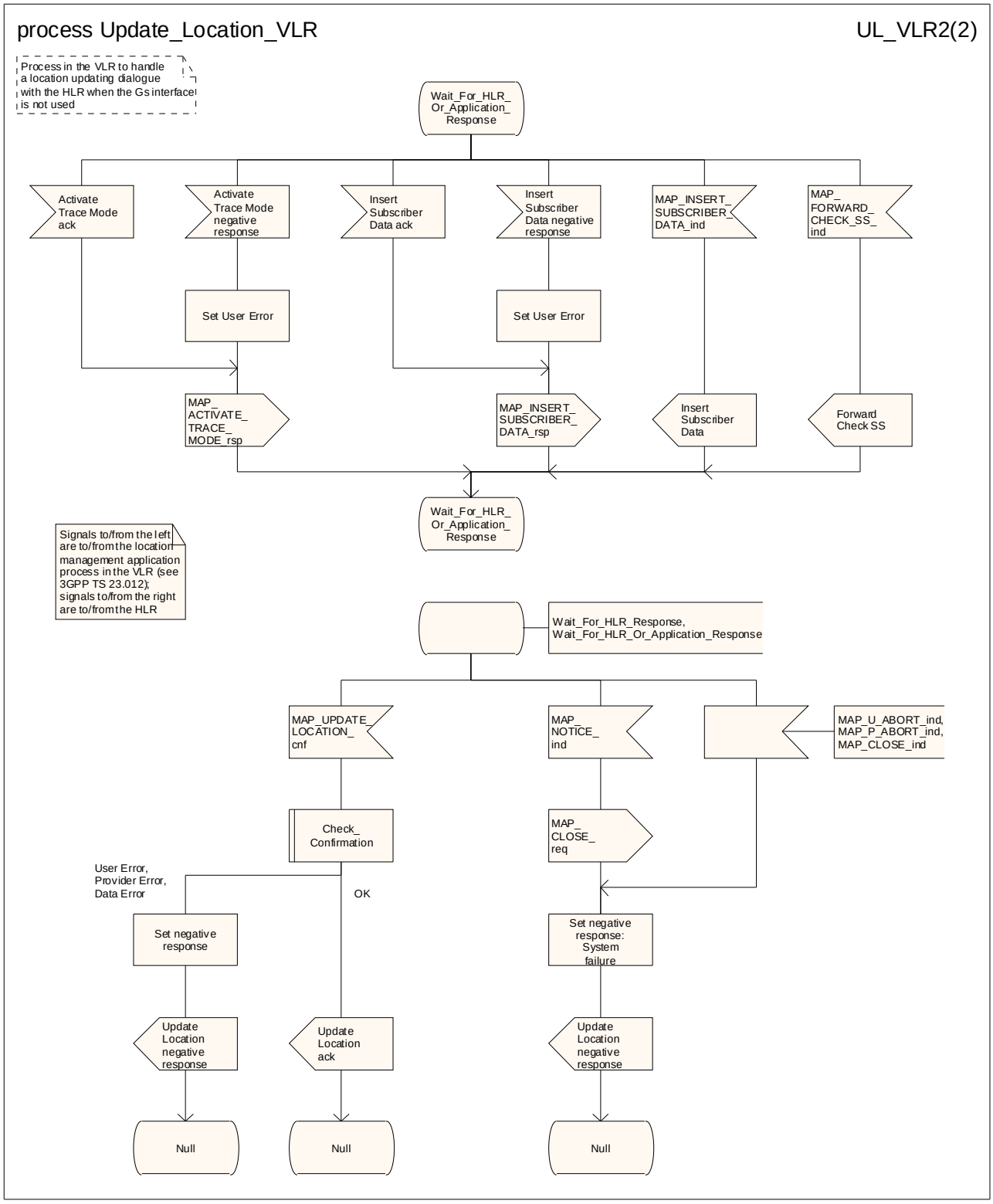


Figure 19.1.1/6 (sheet 2 of 2): Process Update_Location_VLR

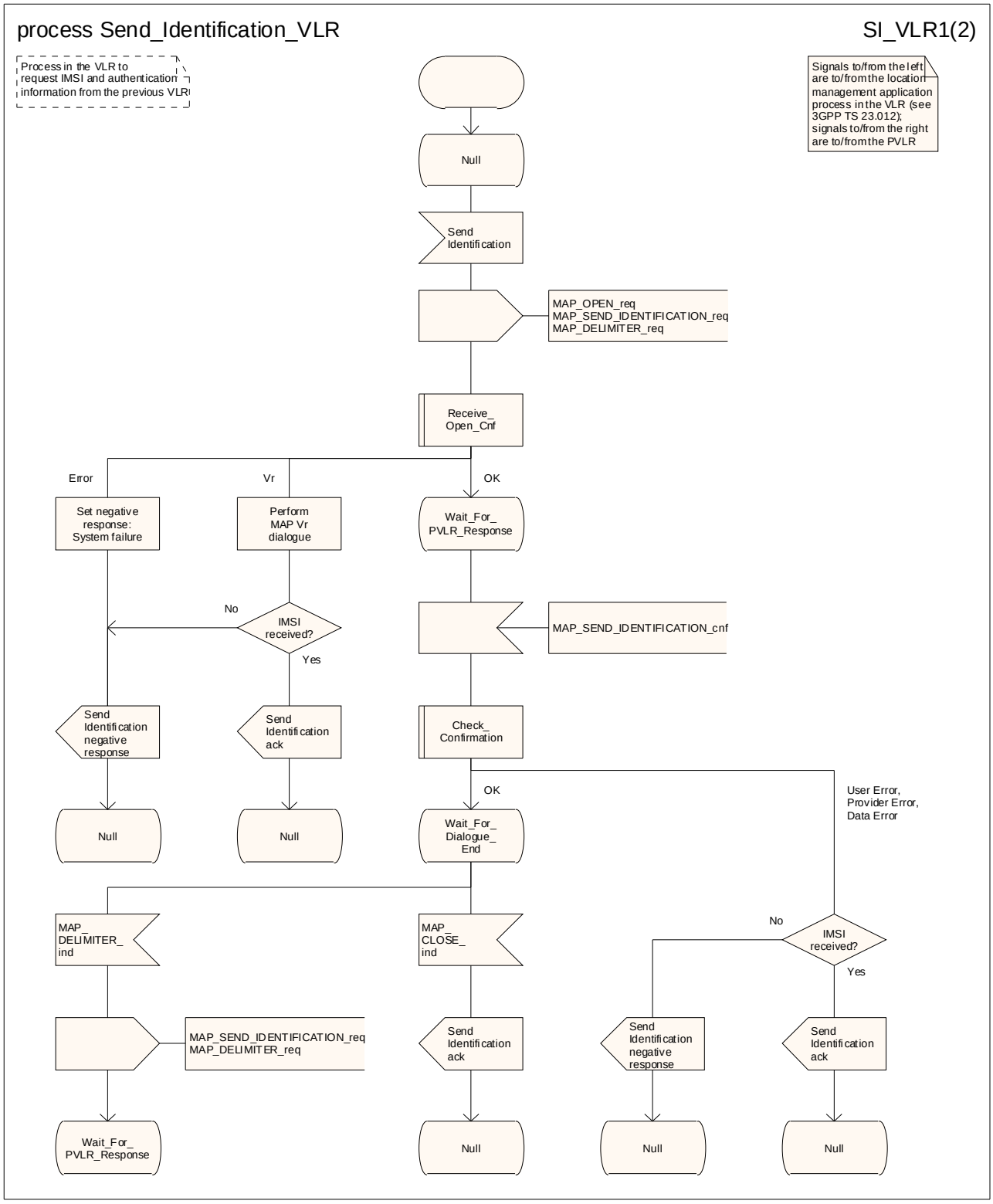


Figure 19.1.1/7 (sheet 1 of 2): Process Send_Identification_VLR

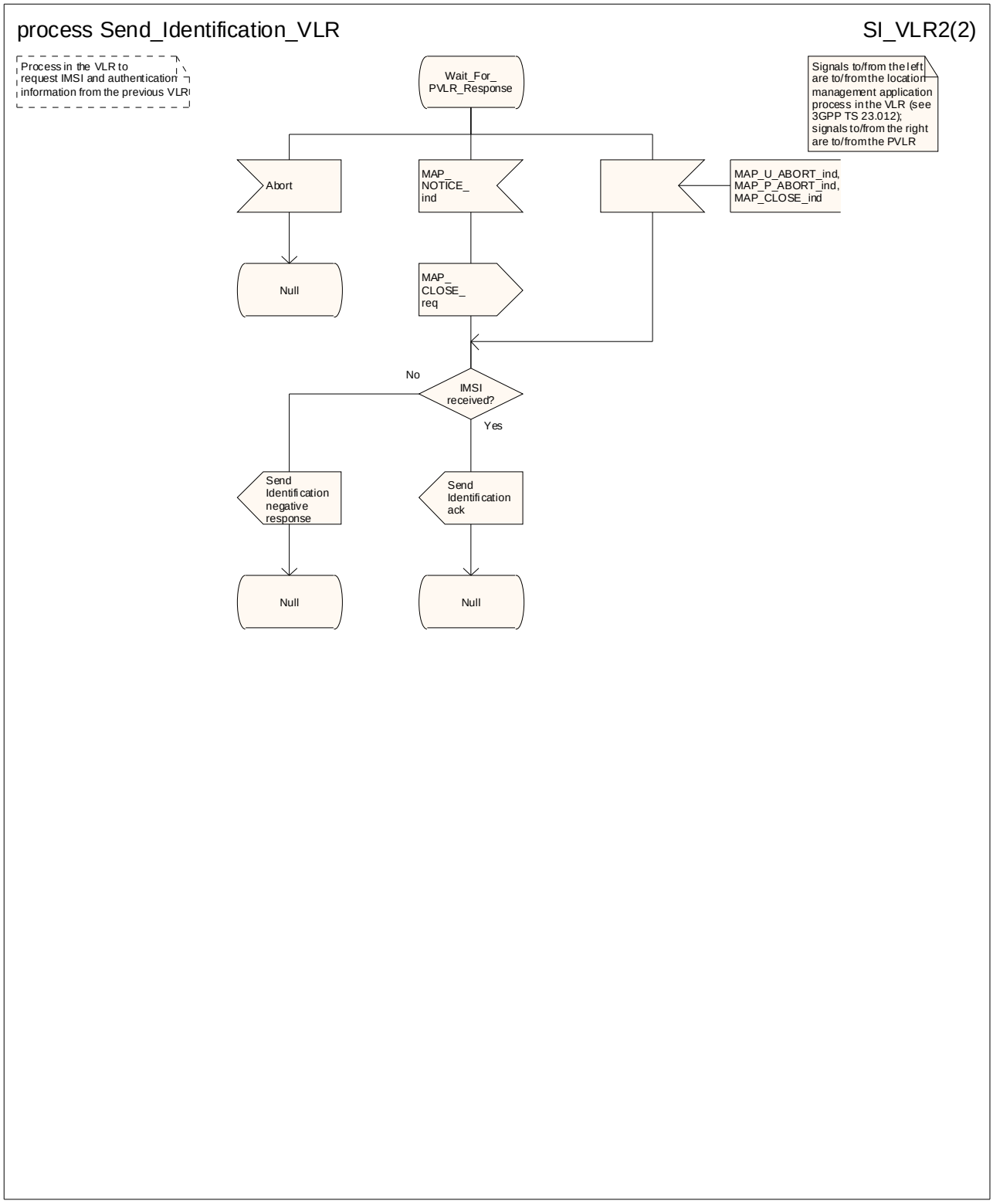


Figure 19.1.1/7 (sheet 2 of 2): Process Send_Identification_VLR

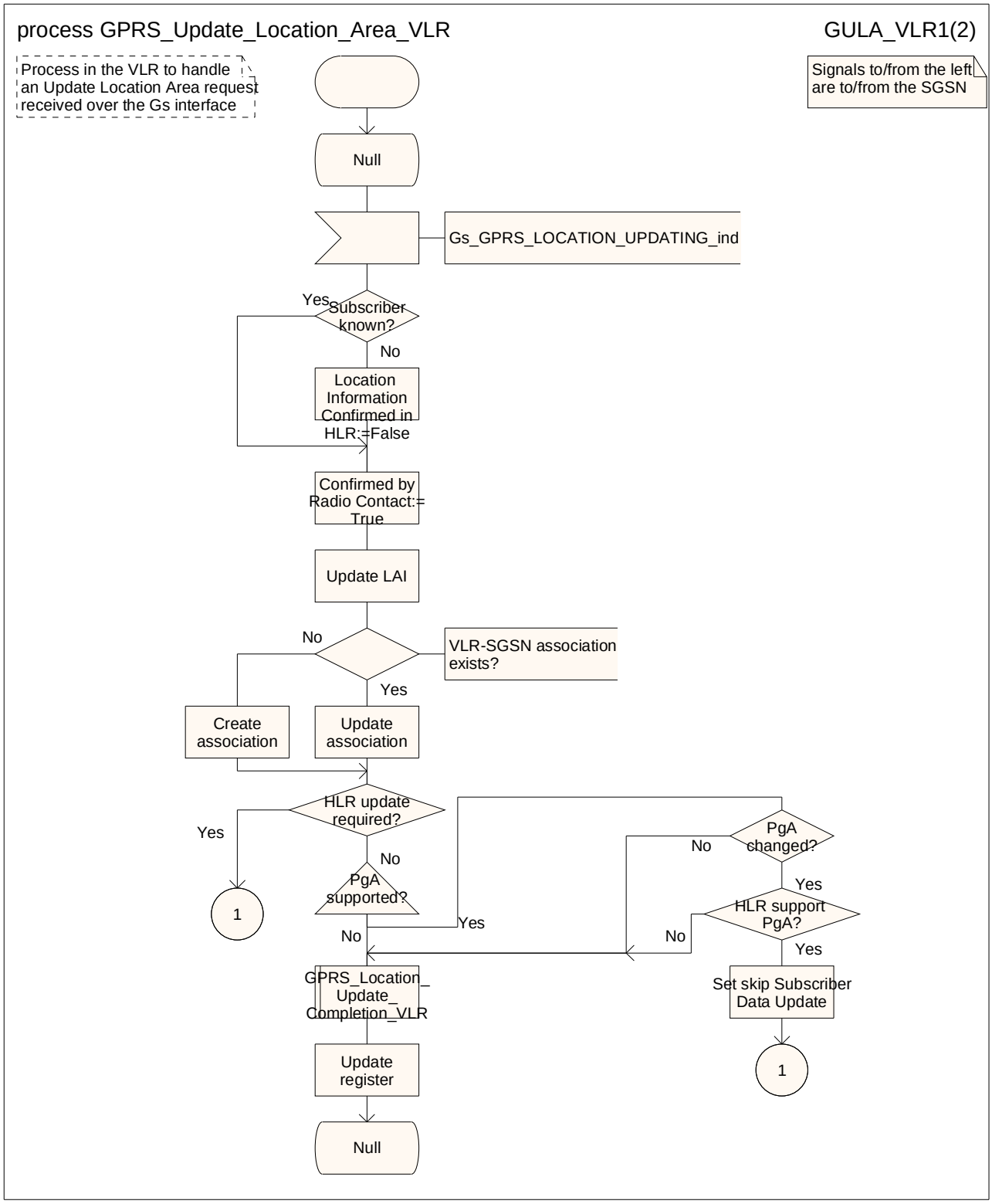


Figure 19.1.1/8 (sheet 1 of 2): Process GPRS_Update_Location_Area_VLR

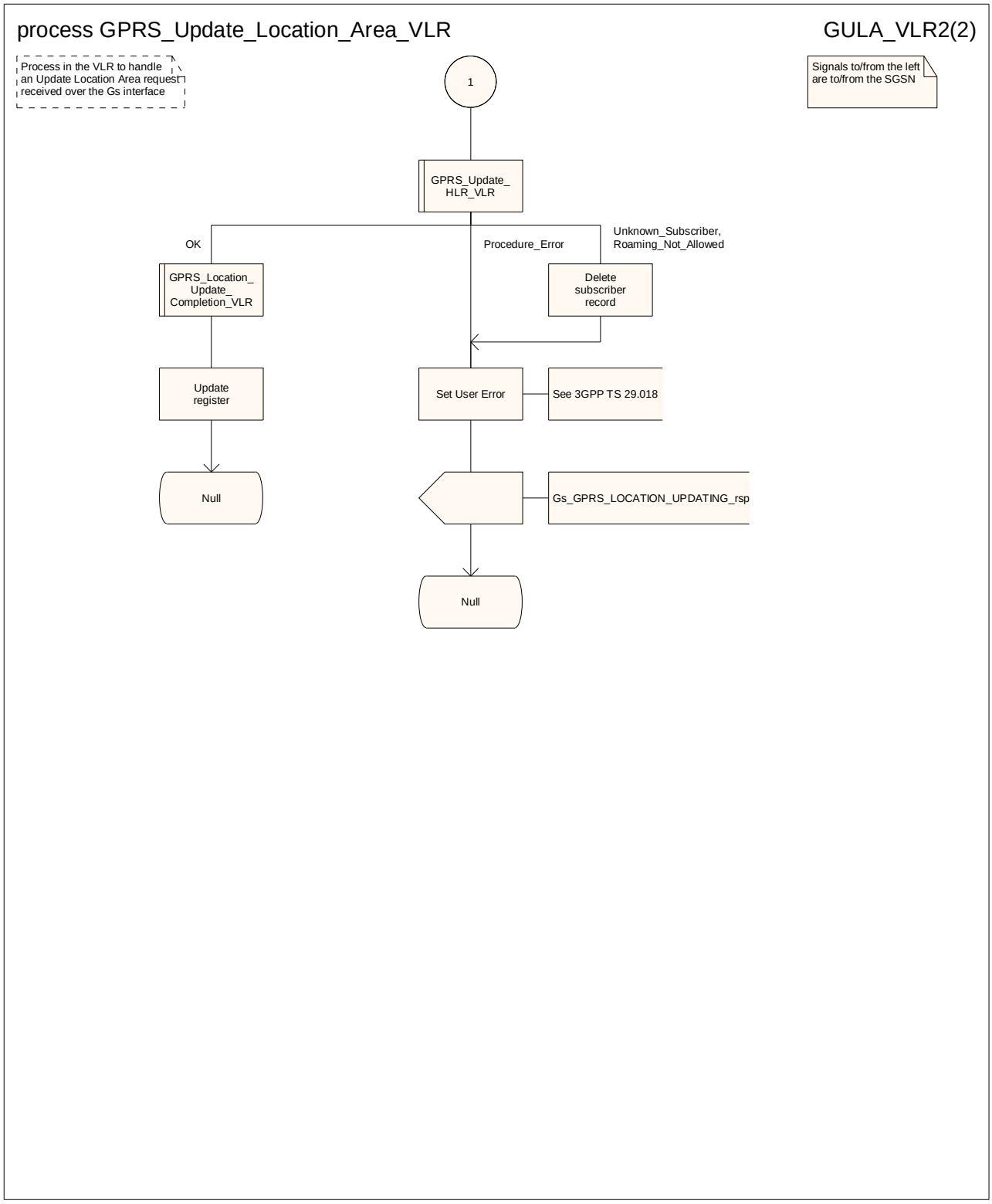


Figure 19.1.1/8 (sheet 2 of 2): Process GPRS_Update_Location_Area_VLR

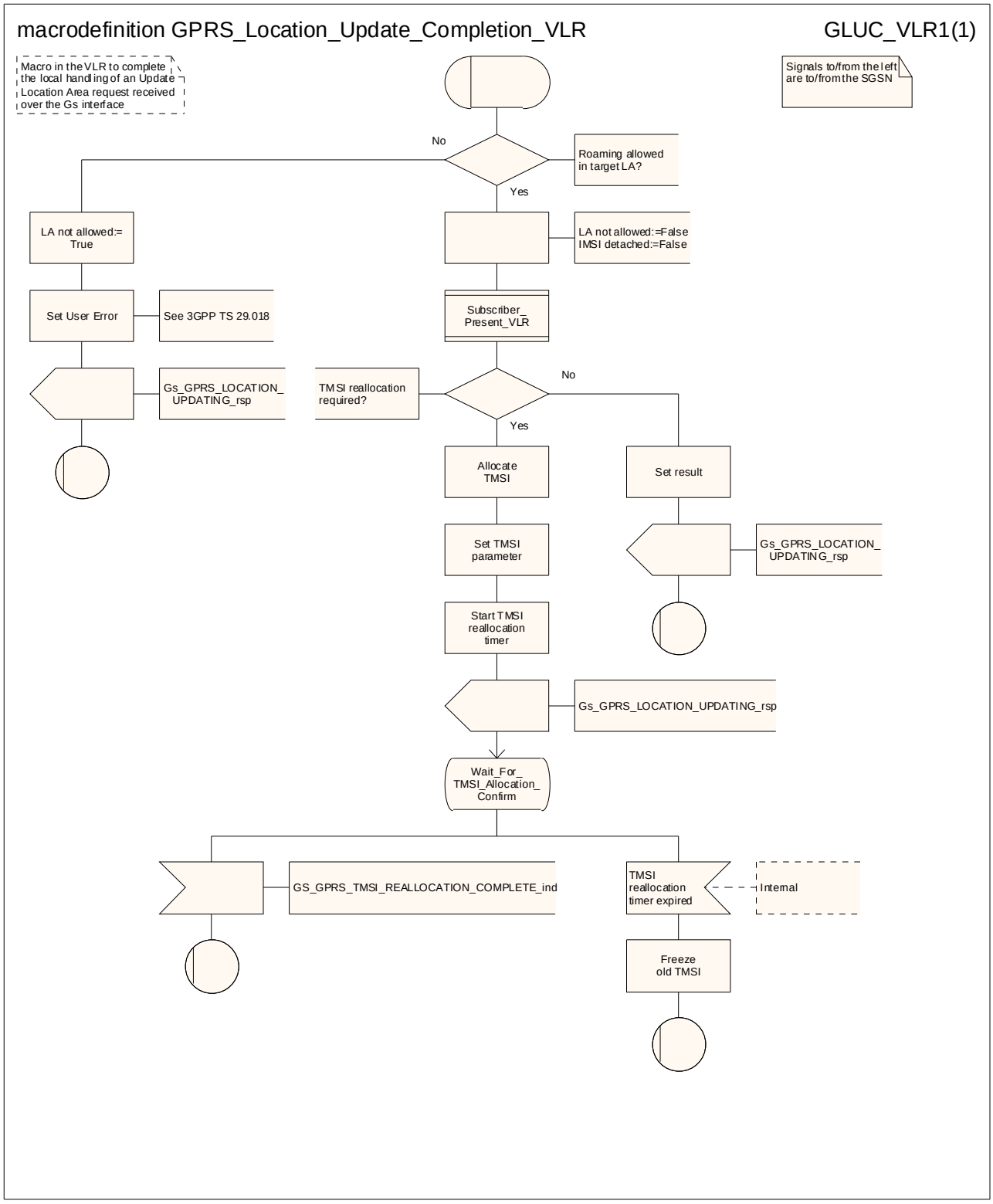


Figure 19.1.1/9: Macro GPRS_Location_Update_Completion_VLR

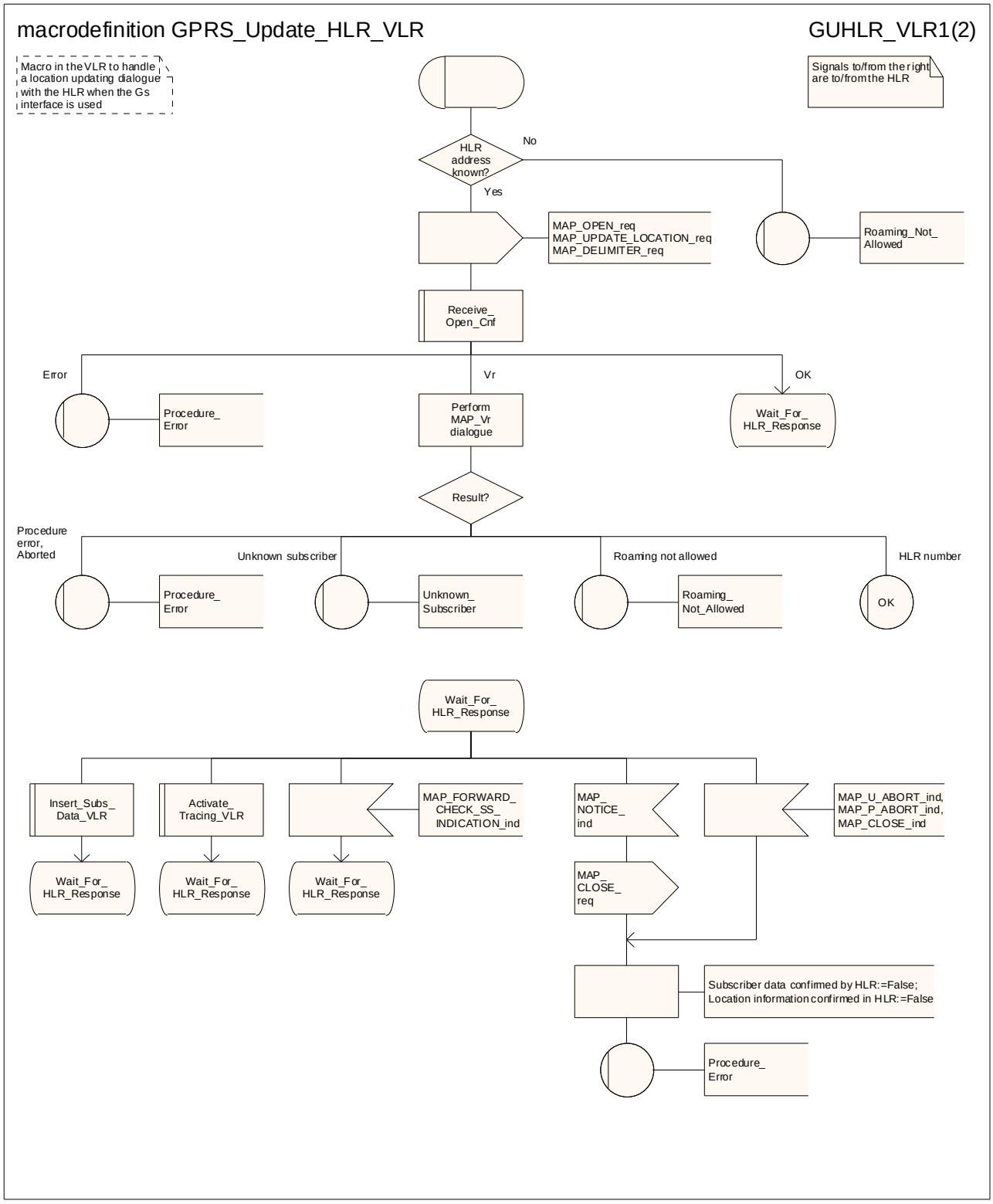


Figure 19.1.1/10 (sheet 1 of 2): Macro GPRS_Update_HLR_VLR

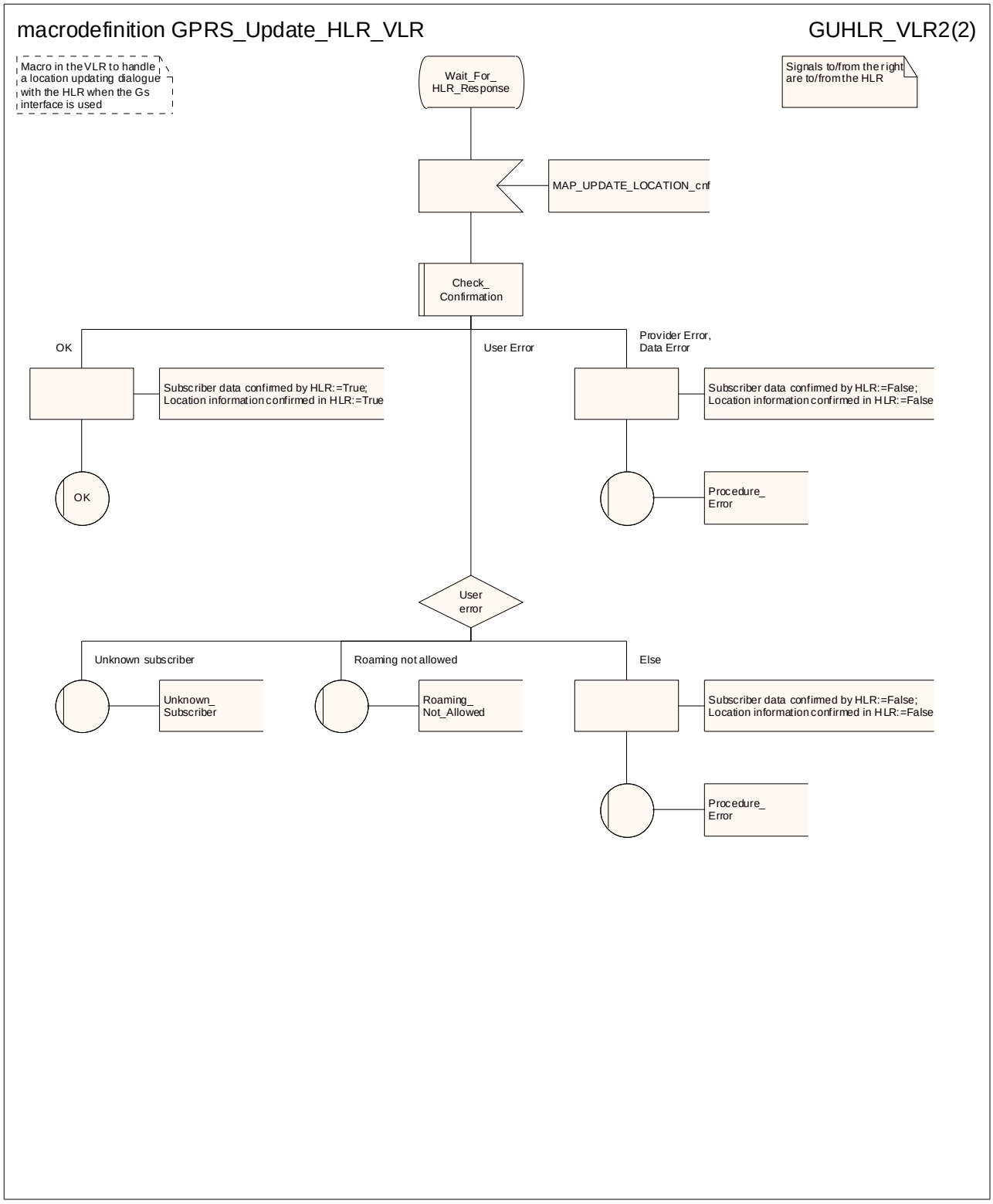


Figure 19.1.1/10 (sheet 2 of 2): Macro GPRS_Update_HLR_VLR

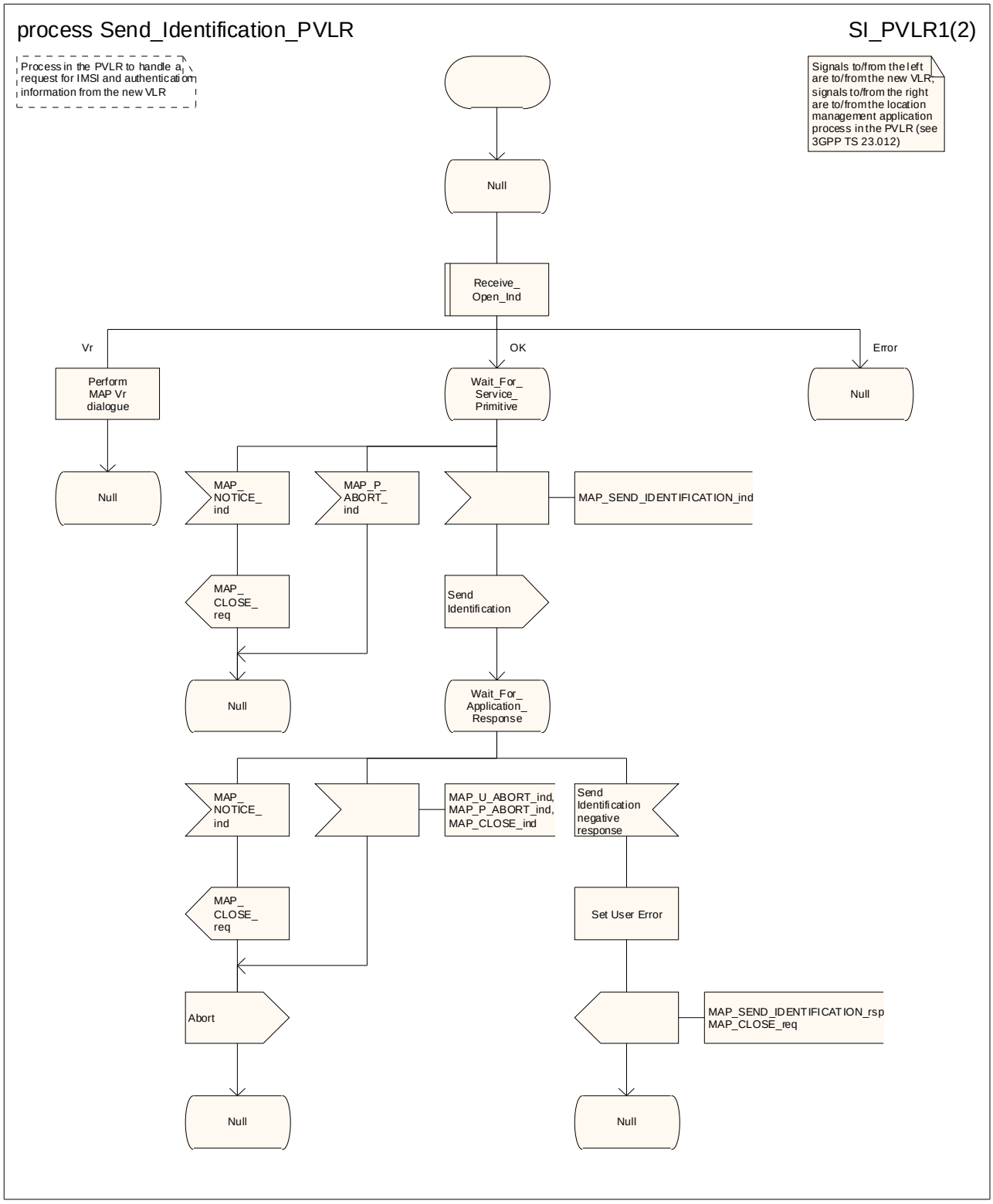


Figure 19.1.1/11 (sheet 1 of 2): Process Send_Identification_PVLR

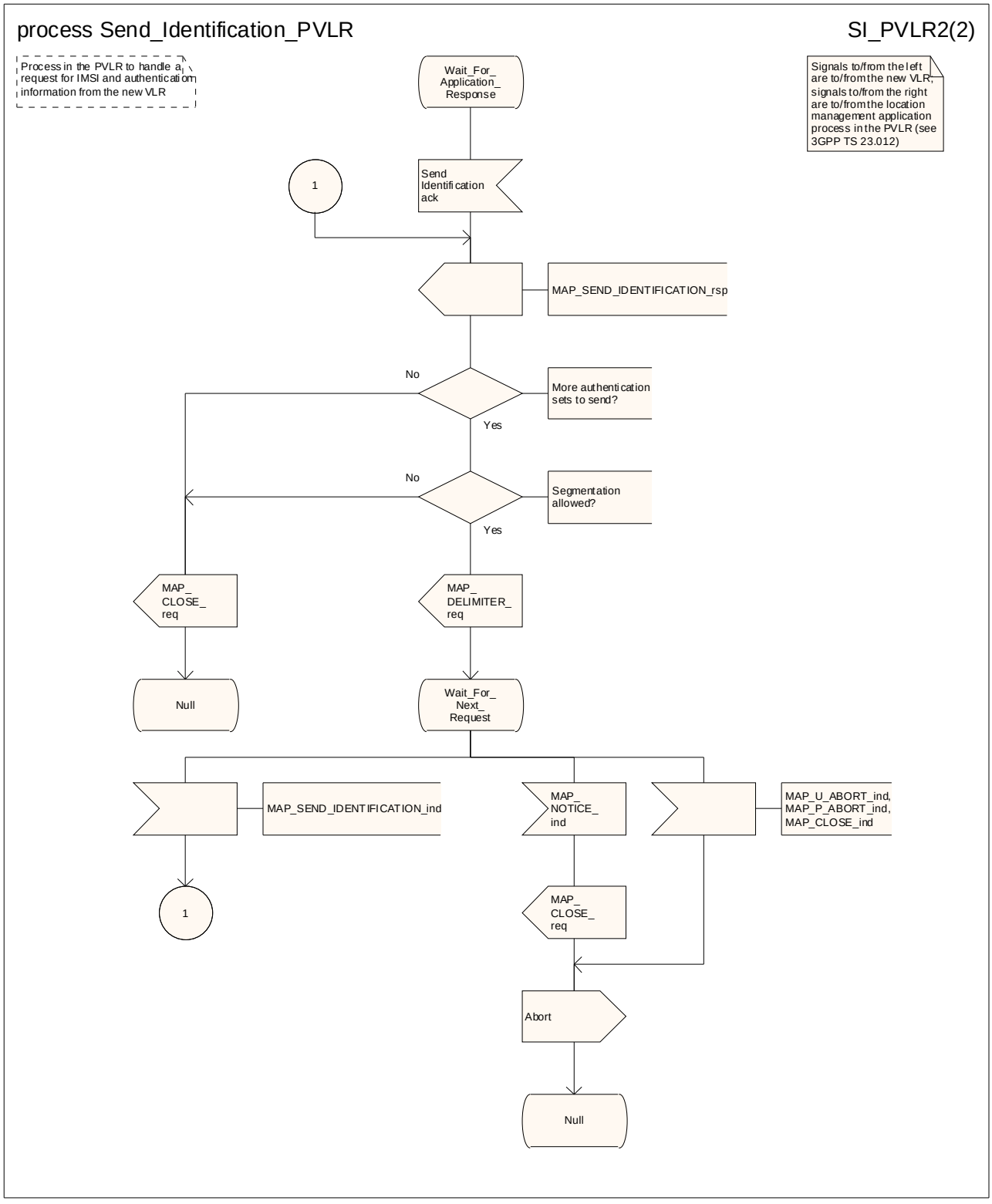


Figure 19.1.1/11 (sheet 2 of 2): Process Send_Identification_PVLR

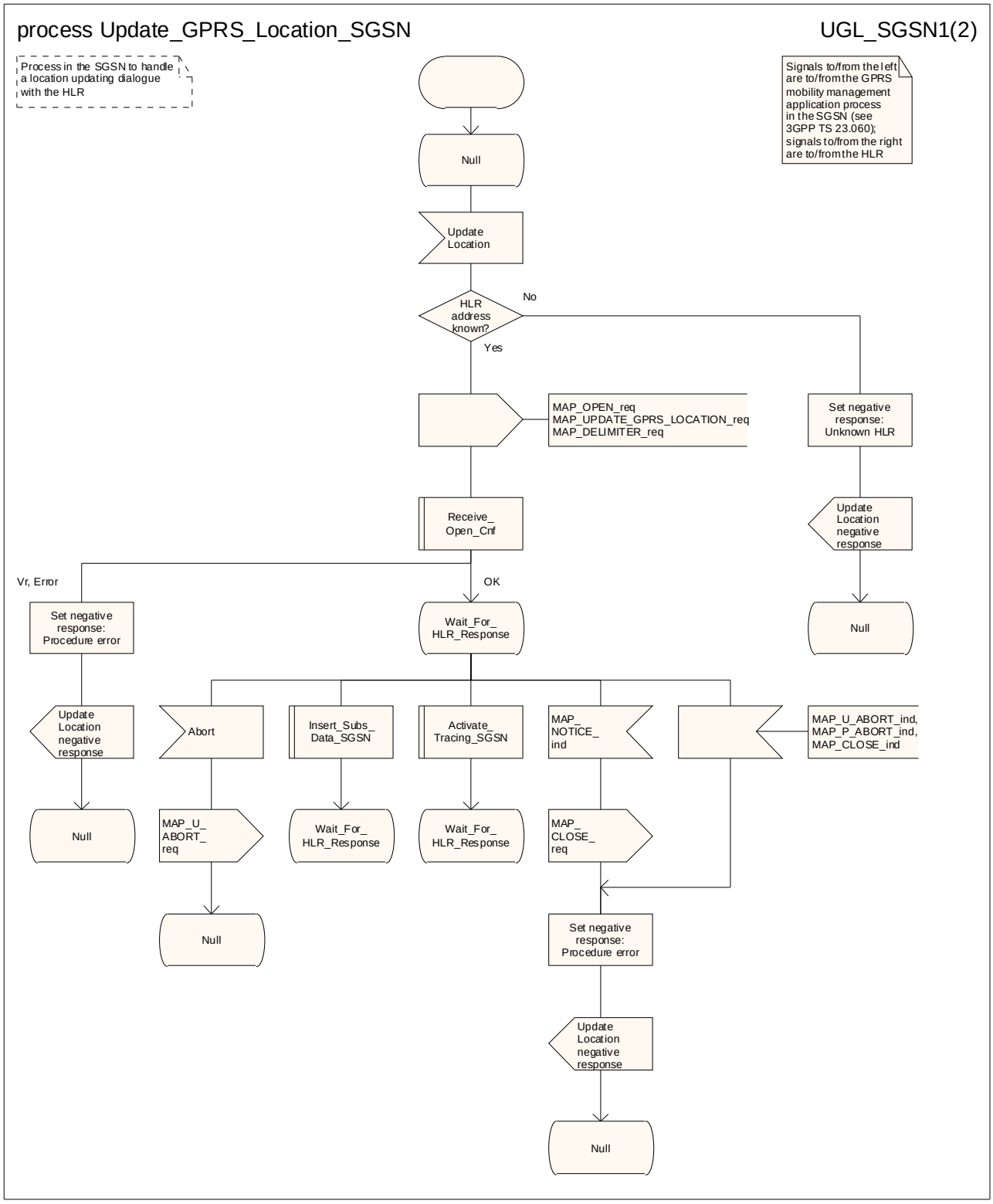


Figure 19.1.1/12 (sheet 1 of 2): Process Update_GPRS_Location_SGSN

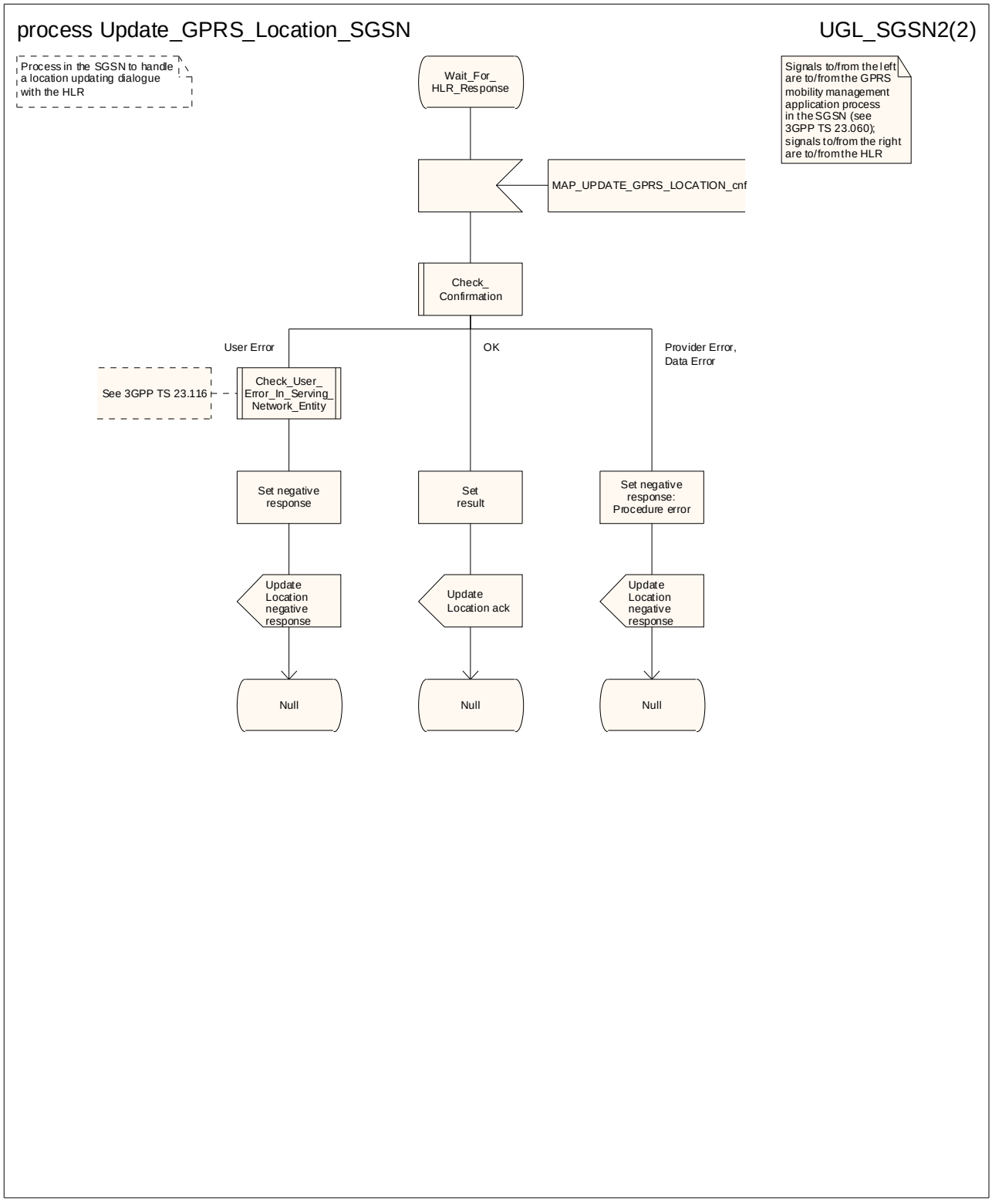


Figure 19.1.1/12 (sheet 2 of 2): Process Update_GPRS_Location_SGSN

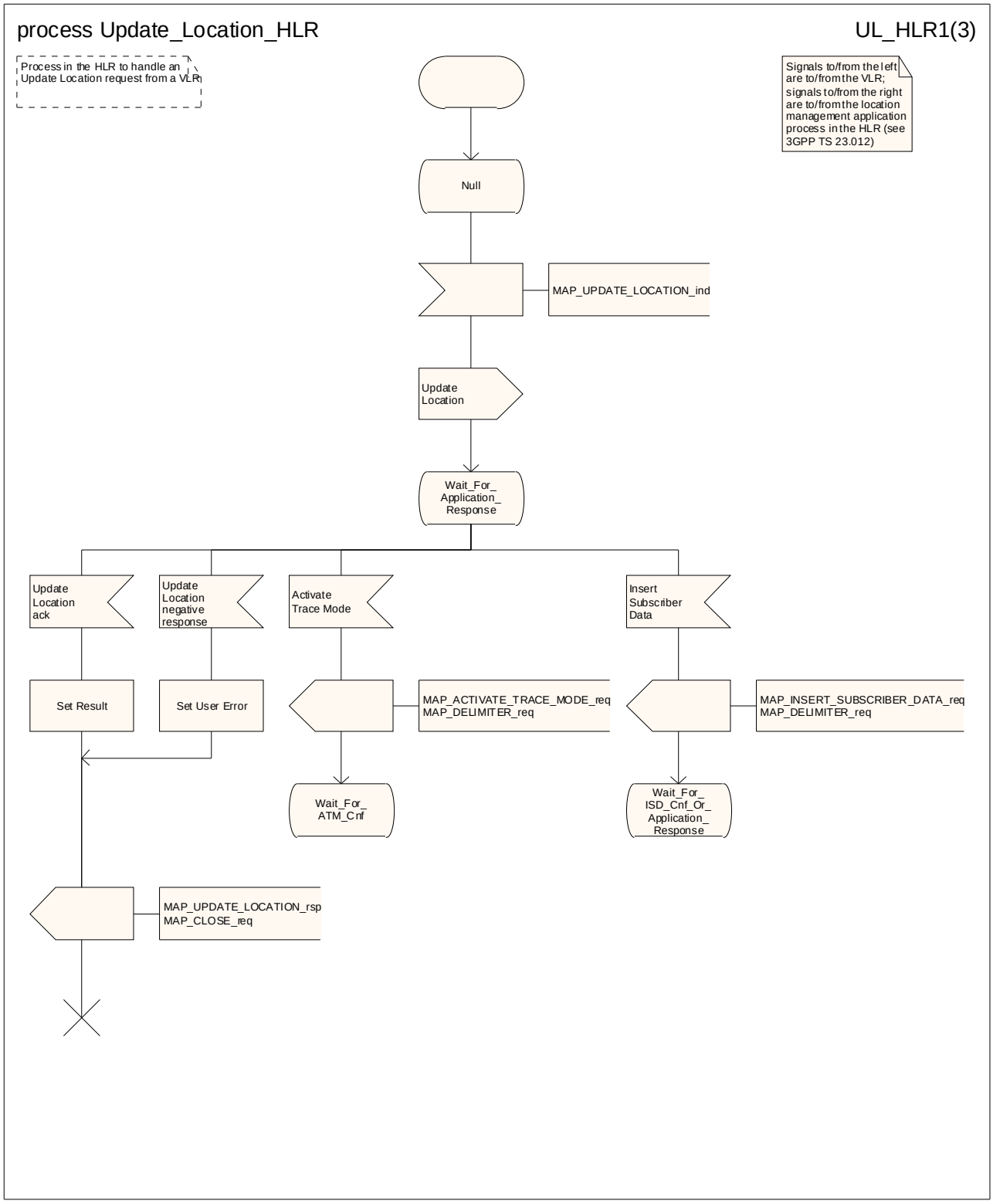


Figure 19.1.1/13 (sheet 1 of 3): Process Update_Location_HLR

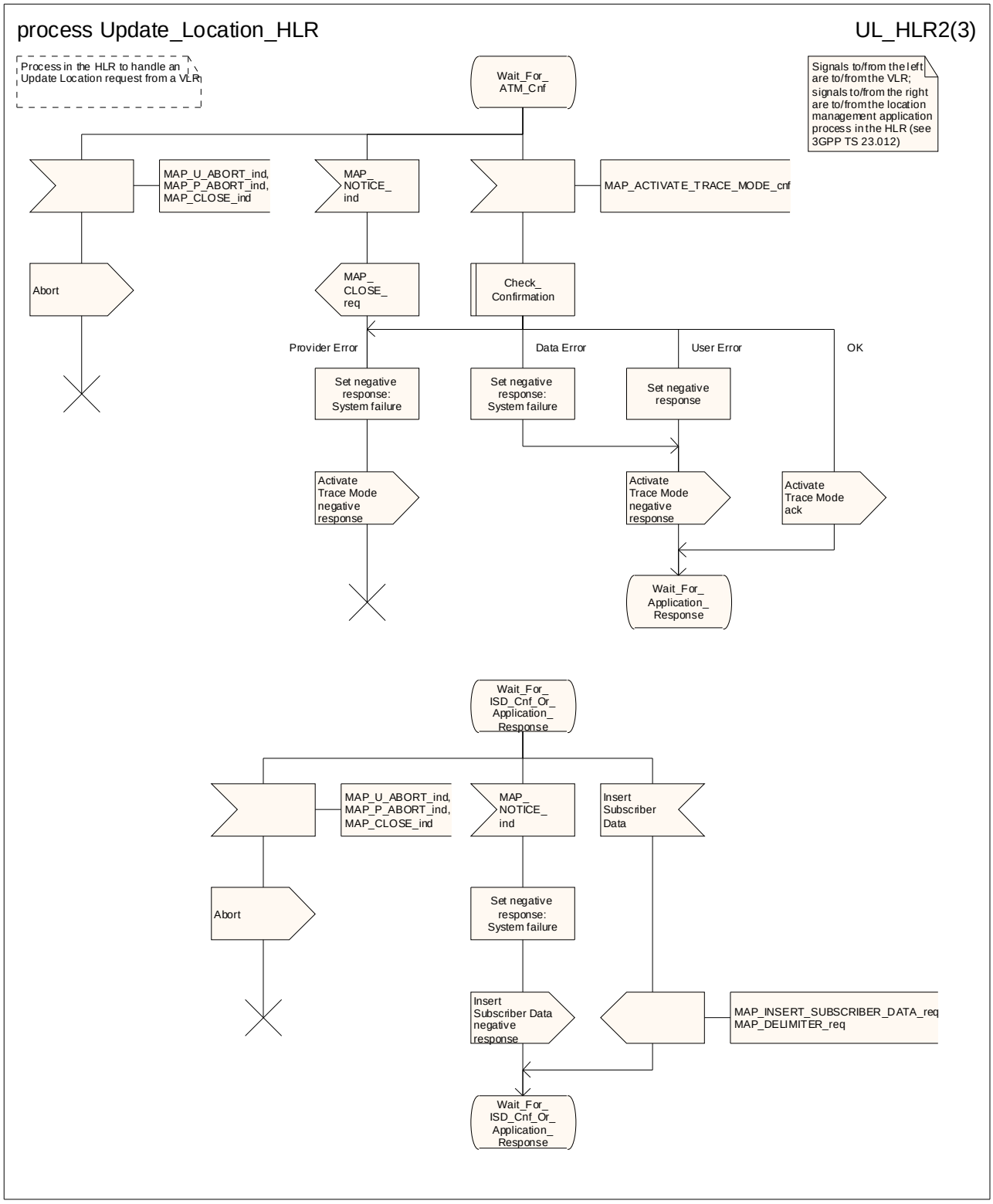


Figure 19.1.1/13 (sheet 2 of 3): Process Update_Location_HLR

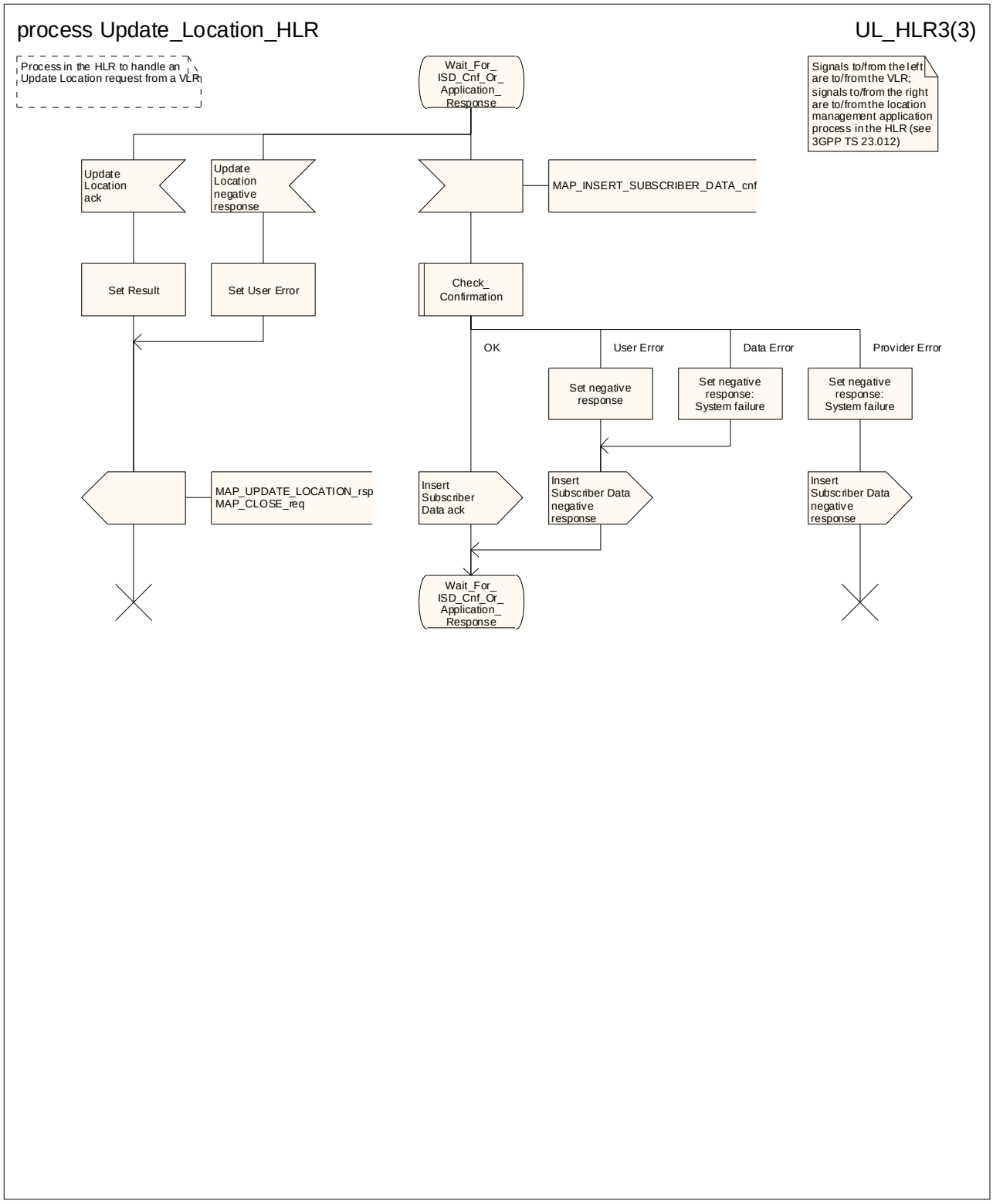


Figure 19.1.1/13 (sheet 3 of 3): Process Update_Location_HLR

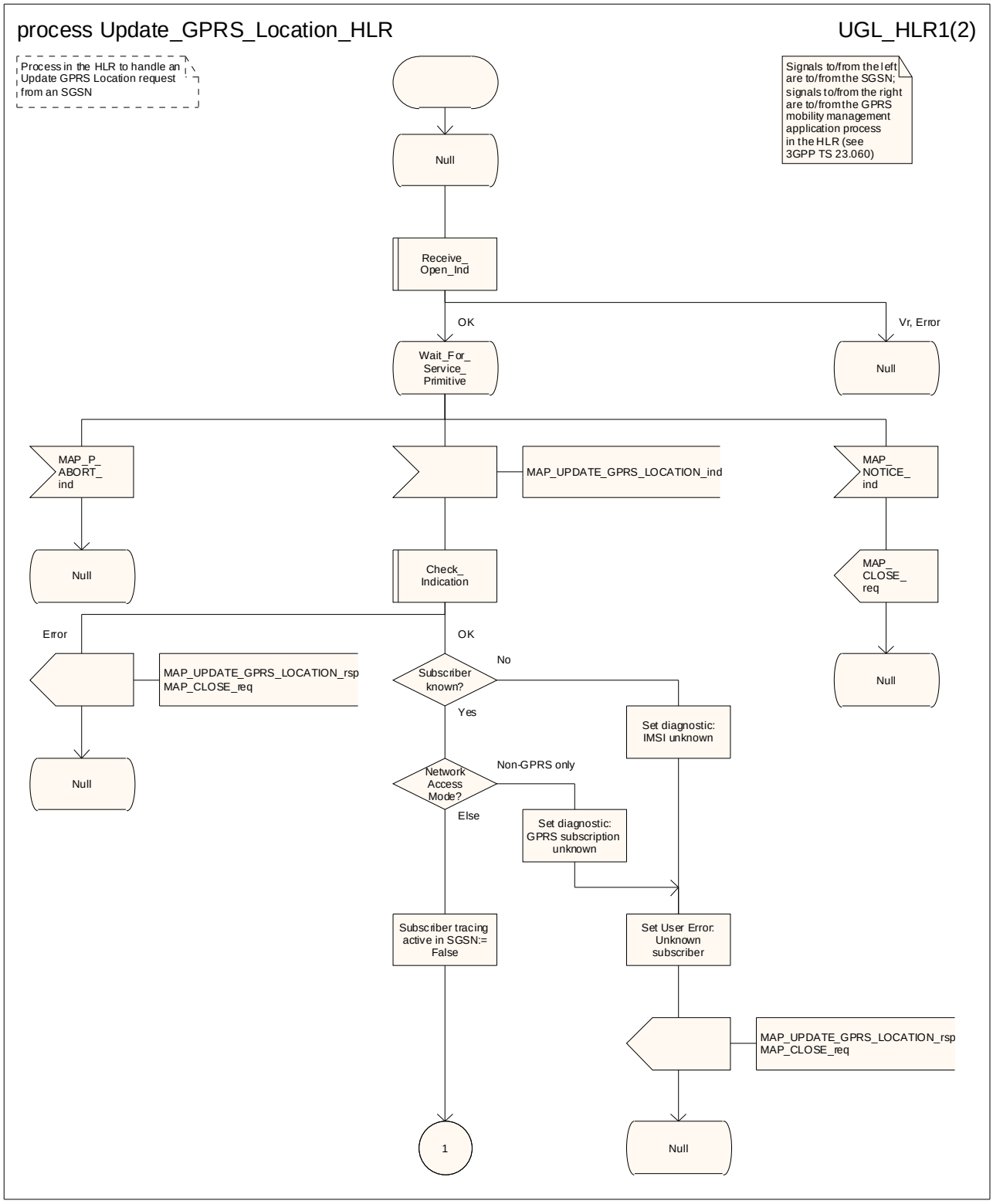


Figure 19.1.1/14 (sheet 1 of 2): Process Update_GPRS_Location_HLR

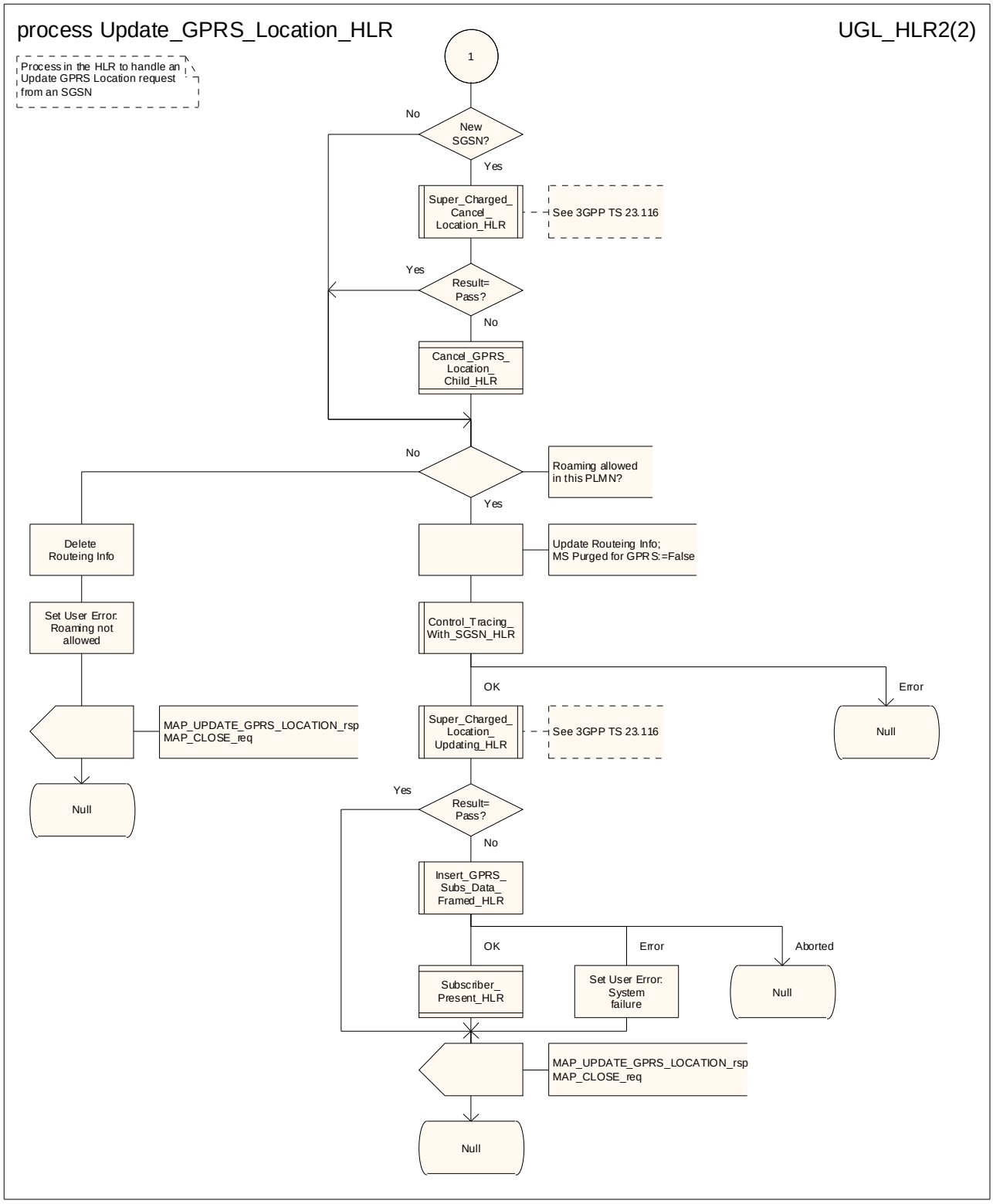


Figure 19.1.1/14 (sheet 2 of 2): Process Update_GPRS_Location_HLR

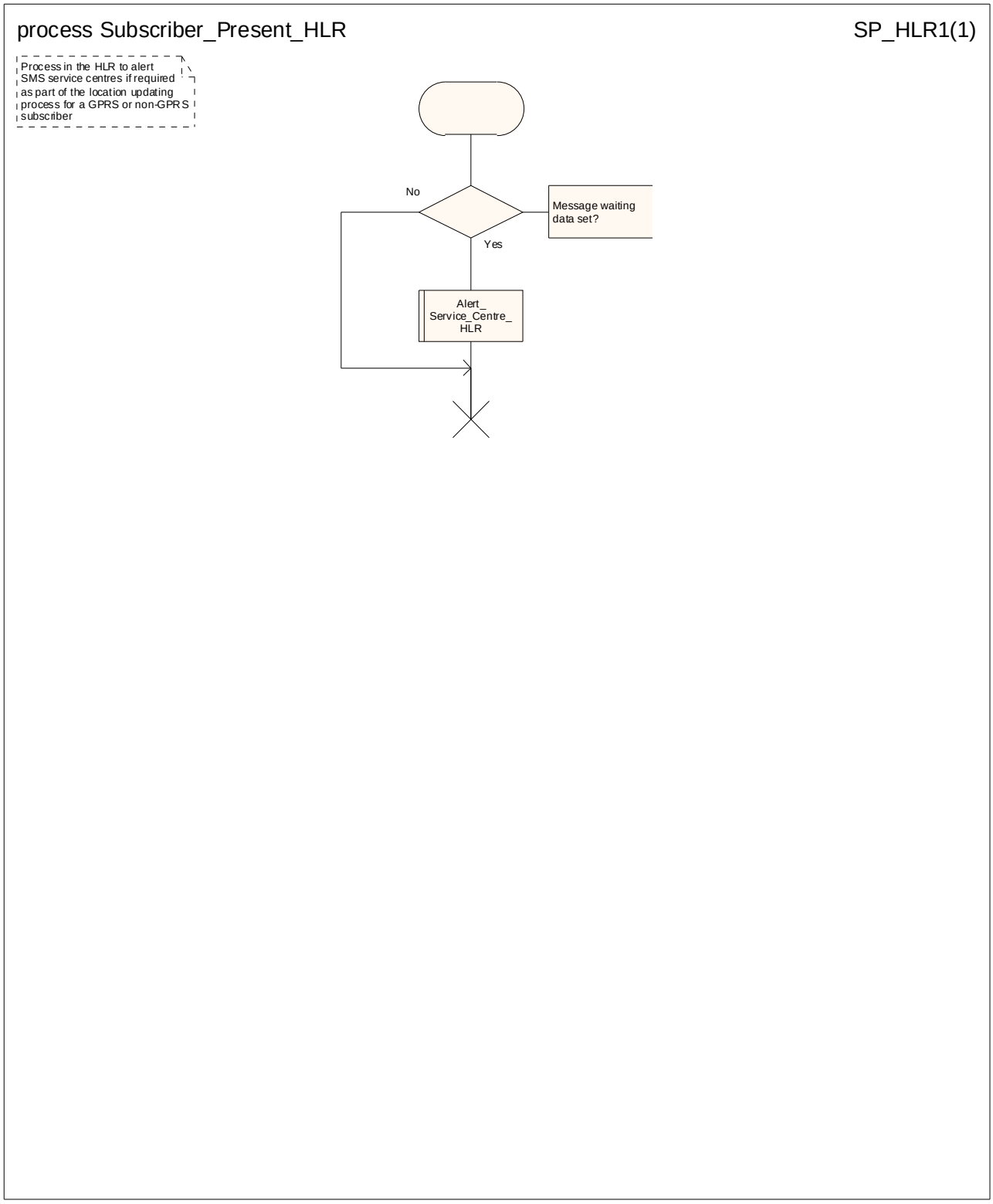
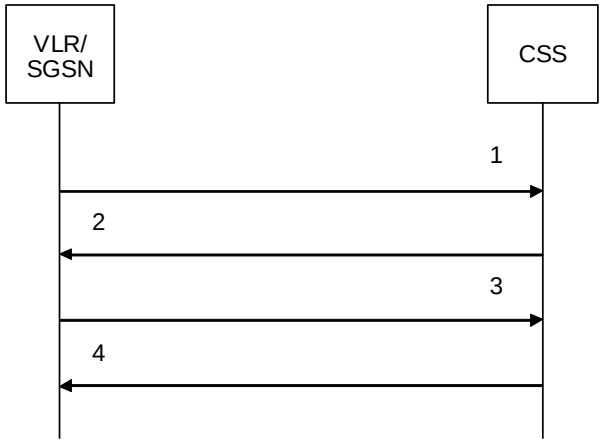


Figure 19.1.1/15: Process Subscriber_Present_HLR

19.1.1A Location updating for VCSG

19.1.1A.1 General

The message flow for successful VCSG location updating between VLR and CSS or between SGSN and CSS is shown in figure 19.1.1A/1.



1) MAP_UPDATE_VCSG_LOCATION_req/ind
2) MAP_INSERT_SUBSCRIBER_DATA_req/ind
3) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
4) MAP_UPDATE_VCSG_LOCATION_rsp/cnf

Figure 19.1.1A/1: Message flow for VCSG location updating

19.1.1A.2 Procedures in the VLR

The MAP process in the VLR for VCSG location updating for a roaming subscriber is shown in figure 19.1.1A/2.

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Insert_Subs_Data_VLR see subclause 25.7.2.

19.1.1A.3 Procedures in the SGSN

The MAP process in the SGSN for VCSG location updating for a roaming subscriber is shown in figure 19.1.1A/3.

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Insert_Subs_Data_SGSN see subclause 25.7.2.

19.1.1A.4 Procedures in the CSS

The MAP process in the CSS to handle a VCSG location updating request from a VLR or SGSN is shown in figure 19.1.1A/4. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2;

Insert_VCSG_Subs_Data_Framed_CSS see subclause 19.5A.1.

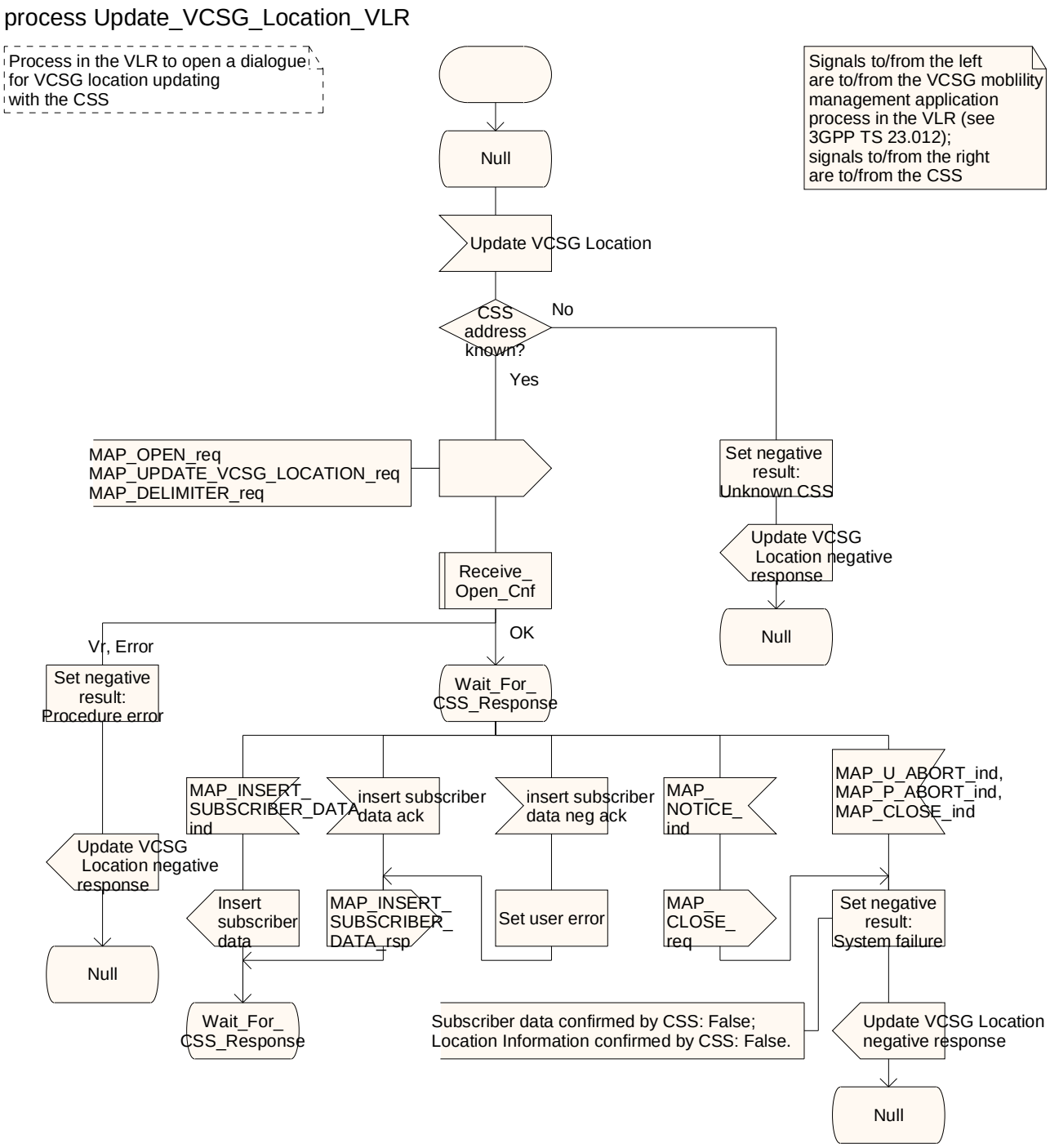


Figure 19.1.1A/2 (sheet 1 of 2): Process Update_VCSG_Location_VLR

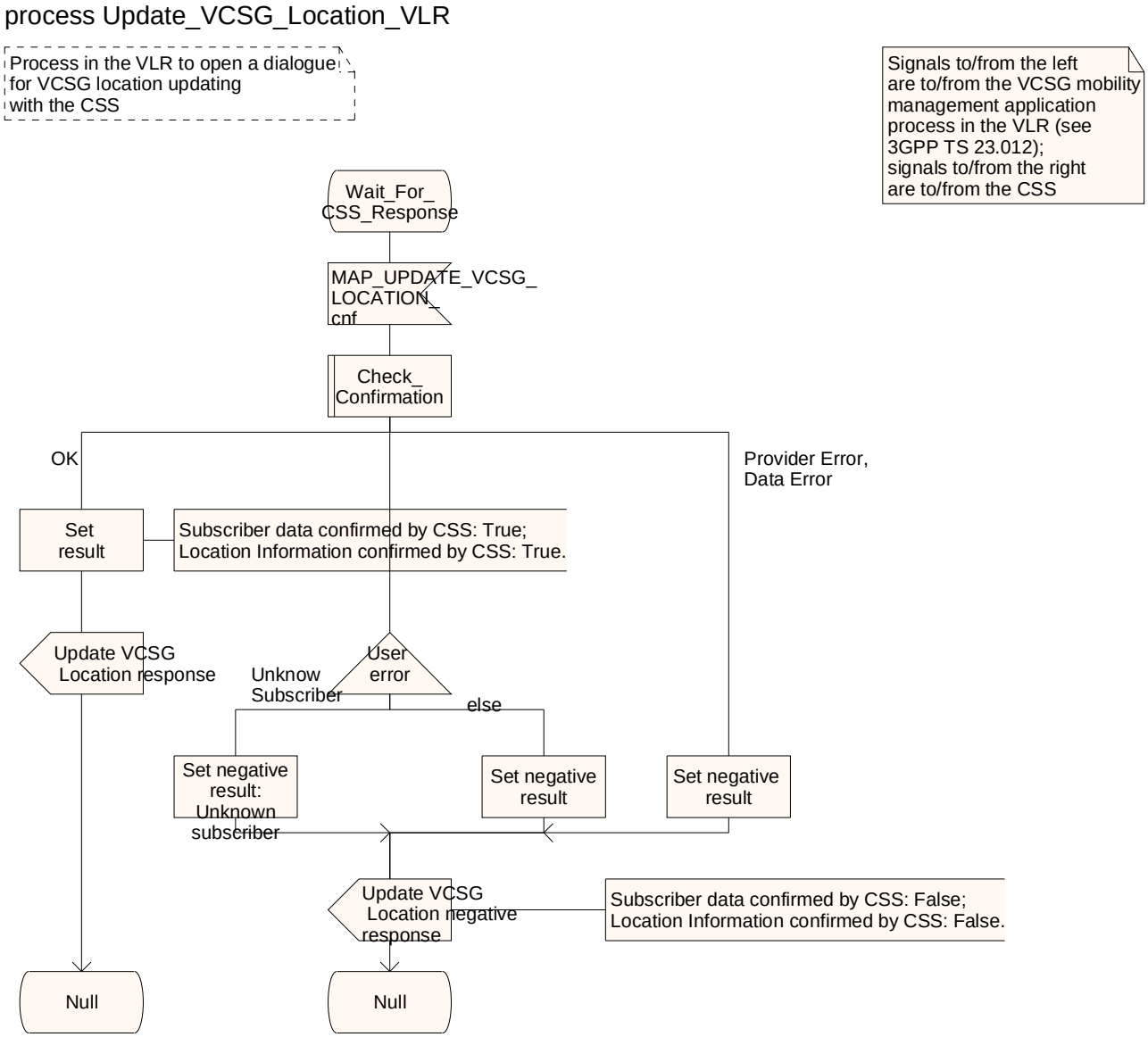


Figure 19.1.1A/2 (sheet 2 of 2): Process Update_VCSG_Location_VLR

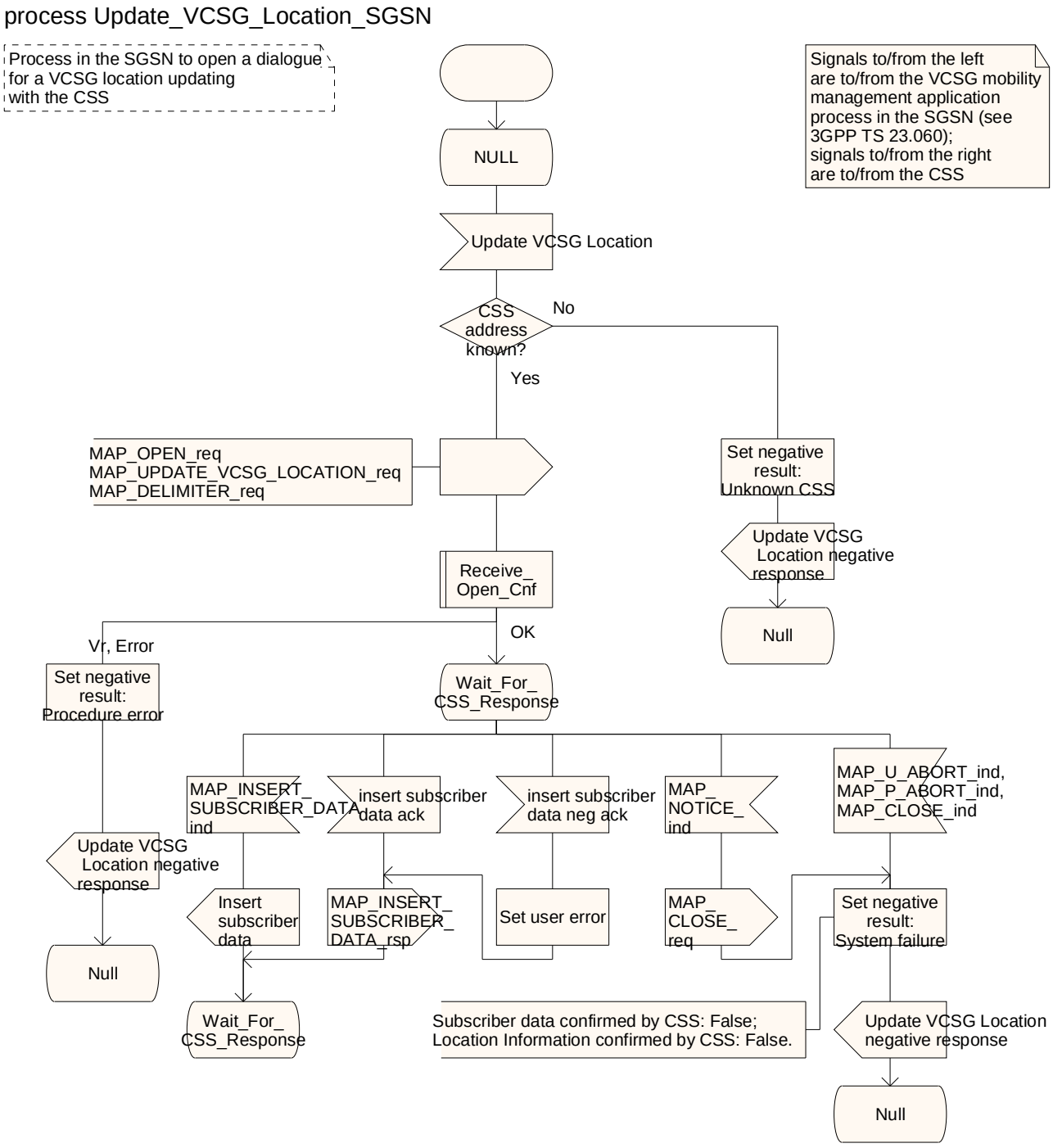


Figure 19.1.1A/3 (sheet 1 of 2): Process Update_VCSG_Location_SGSN

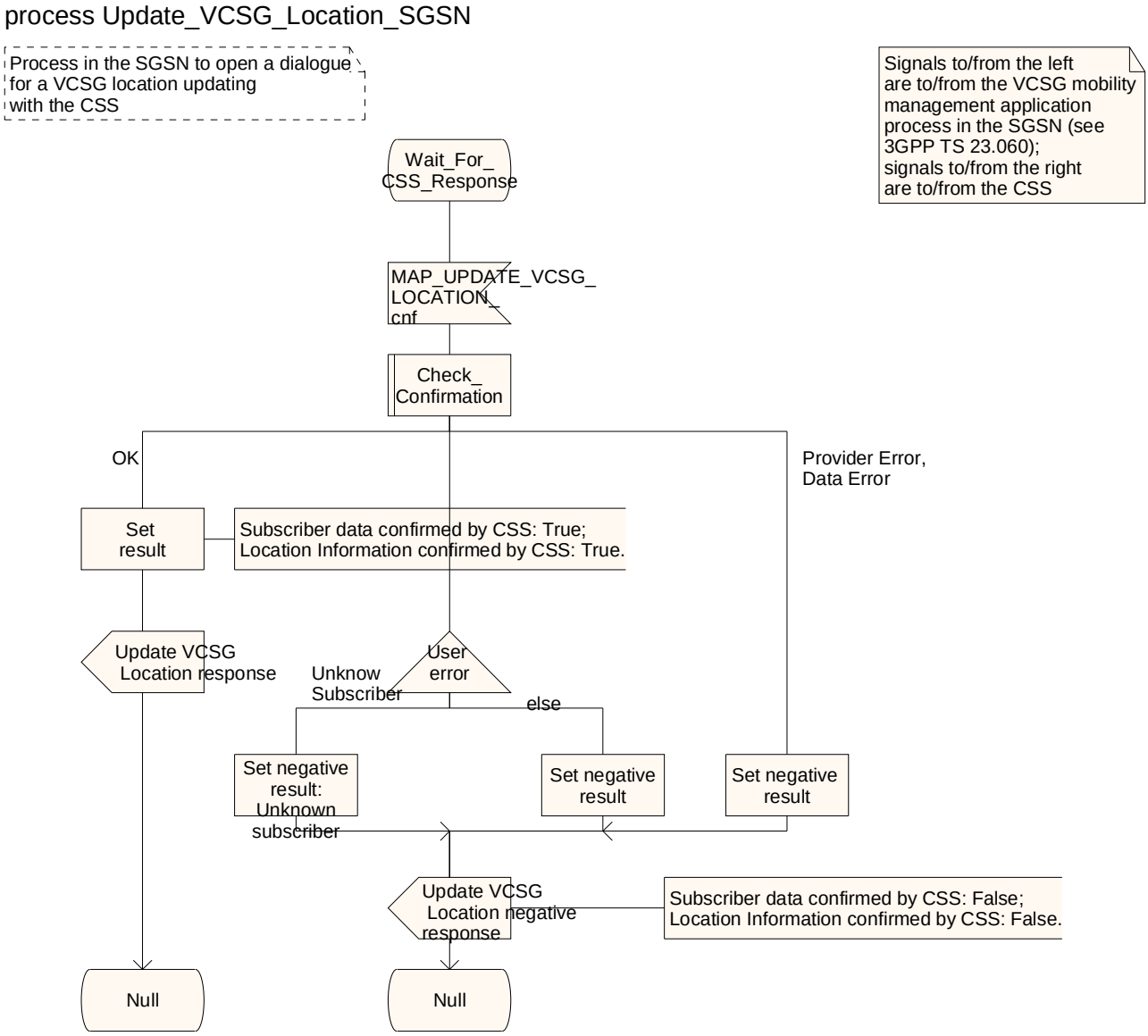


Figure 19.1.1A/3 (sheet 2 of 2): Process Update_VCSG_Location_SGSN

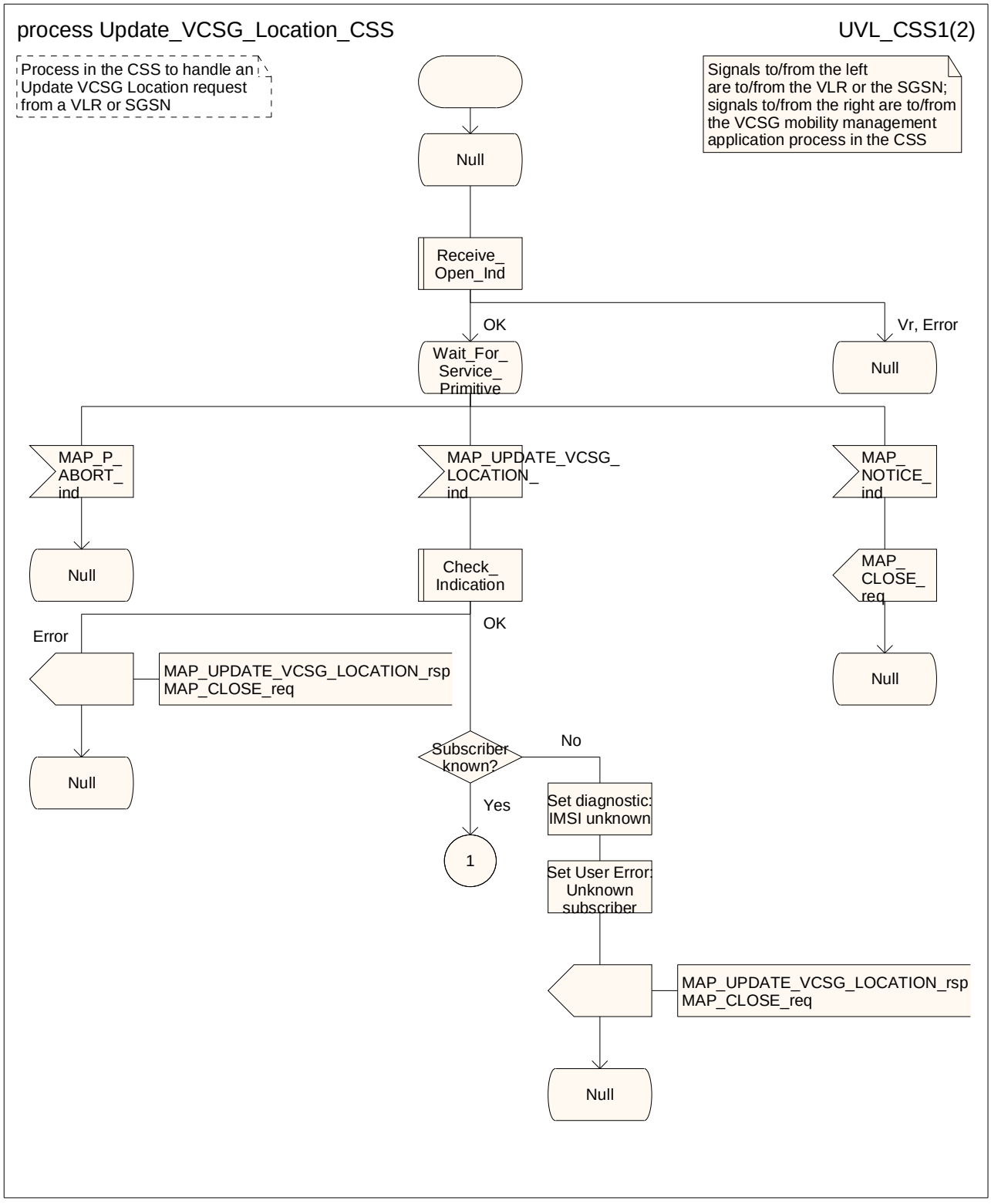


Figure 19.1.1A/4 (sheet 1 of 2): Process Update_VCSG_Location_CSS

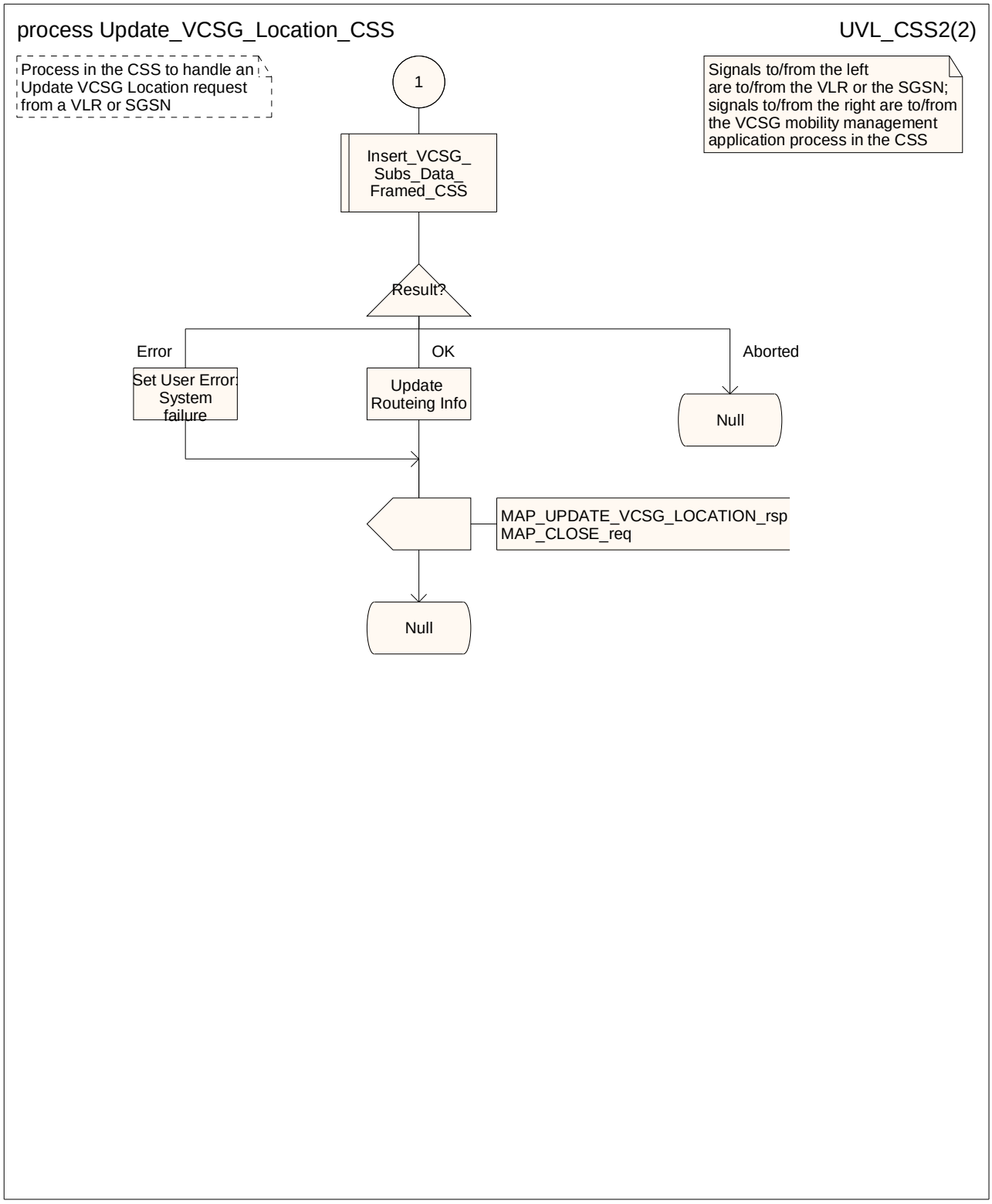


Figure 19.1.1A/4 (sheet 2 of 2): Process Update_VCSG_Location_CSS

19.1.2 Location Cancellation

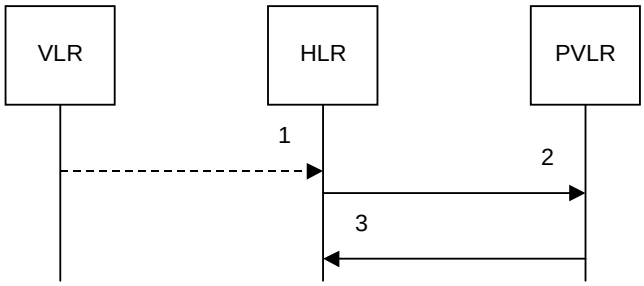
19.1.2.1 General

Location cancellation is used to delete a subscriber record from the serving node (VLR or SGSN). The procedure is invoked:

- because the subscriber has registered with a new serving node, or
- because the HPLMN operator has decided to delete the subscriber record from the serving node, e.g. because the subscription has been withdrawn, or because roaming restrictions have been imposed. Location cancellation can be used to force location updating including updating of subscriber data in the serving node at the next subscriber access.

The message flow for location cancellation for a non-GPRS subscriber is shown in figure 19.1.2/1.

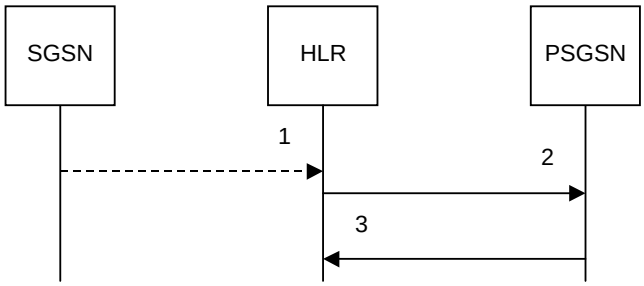
The message flow for location cancellation for a GPRS subscriber is shown in figure 19.1.2/2.



1) MAP_UPDATE_LOCATION_req/ind
2) MAP_CANCEL_LOCATION_req/ind
3) MAP_CANCEL_LOCATION_rsp/cnf

NOTE: The service shown in dotted lines indicates the trigger provided by other MAP signalling.

Figure 19.1.2/1: Message flow for Location Cancellation (non-GPRS)



1) MAP_UPDATE_GPRS_LOCATION_req/ind
2) MAP_CANCEL_LOCATION_req/ind
3) MAP_CANCEL_LOCATION_rsp/cnf

NOTE: The service shown in dotted lines indicates the trigger provided by other MAP signalling.

Figure 19.1.2/2: Message flow for Location Cancellation (GPRS)

19.1.2.2 Procedure in the HLR

The MAP process in the HLR to cancel the location information in a VLR is shown in figure 19.1.2/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The MAP process in the HLR to cancel the location information in a VLR as an independent process invoked from another process is shown in figure 19.1.2/4.

The MAP process in the HLR to cancel the location information in an SGSN is shown in figure 19.1.2/5. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The MAP process in the HLR to cancel the location information in an SGSN as an independent process invoked from another process is shown in figure 19.1.2/6.

19.1.2.3 Procedure in the VLR

The MAP process in the VLR to handle a location cancellation request is shown in figure 19.1.2/7. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

19.1.2.4 Procedure in the SGSN

The MAP process in the SGSN to handle a location cancellation request is shown in figure 19.1.2/8. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

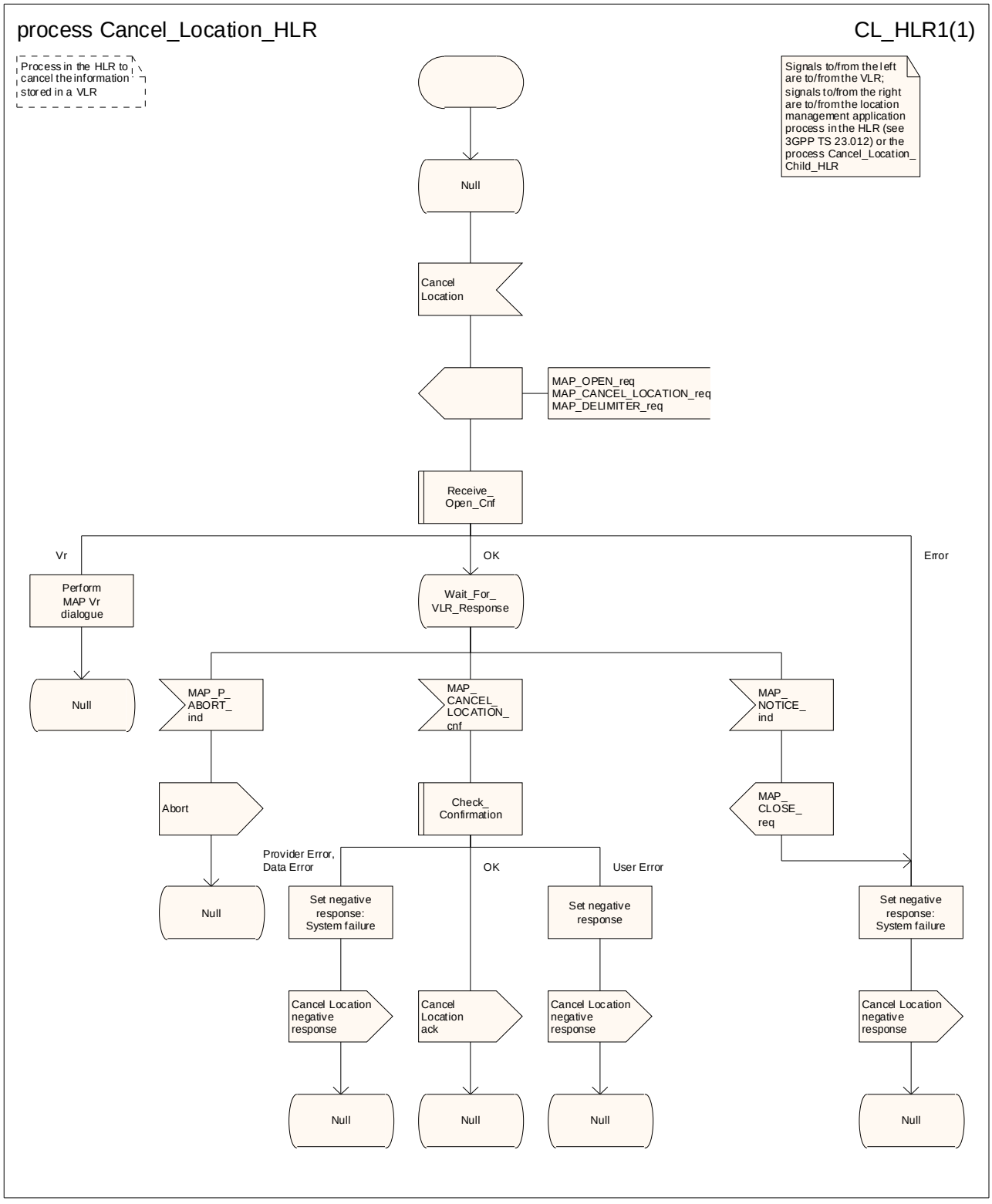


Figure 19.1.2/3: Process Cancel_Location_HLR

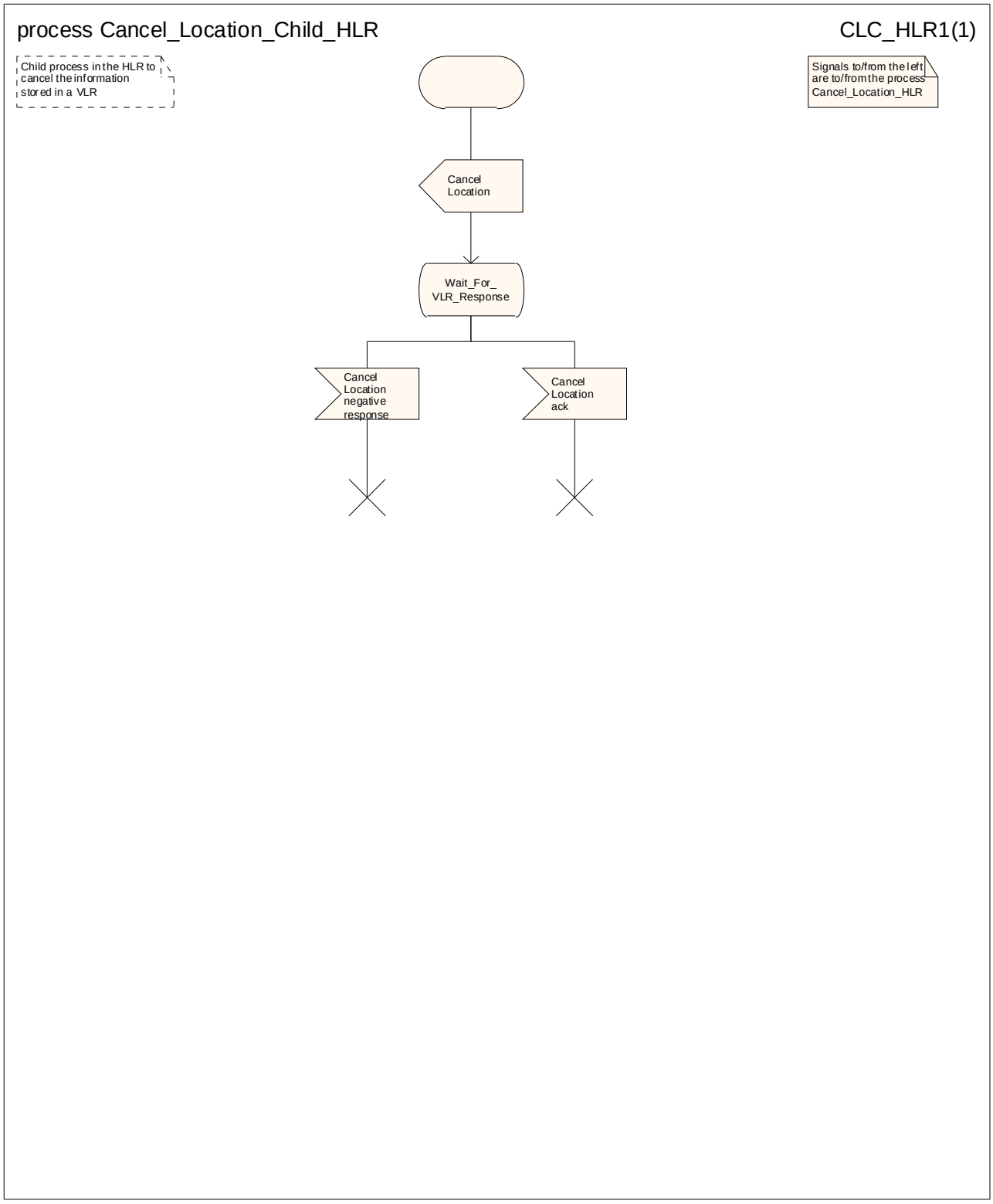


Figure 19.1.2/4: Process Cancel_Location_Child_HLR

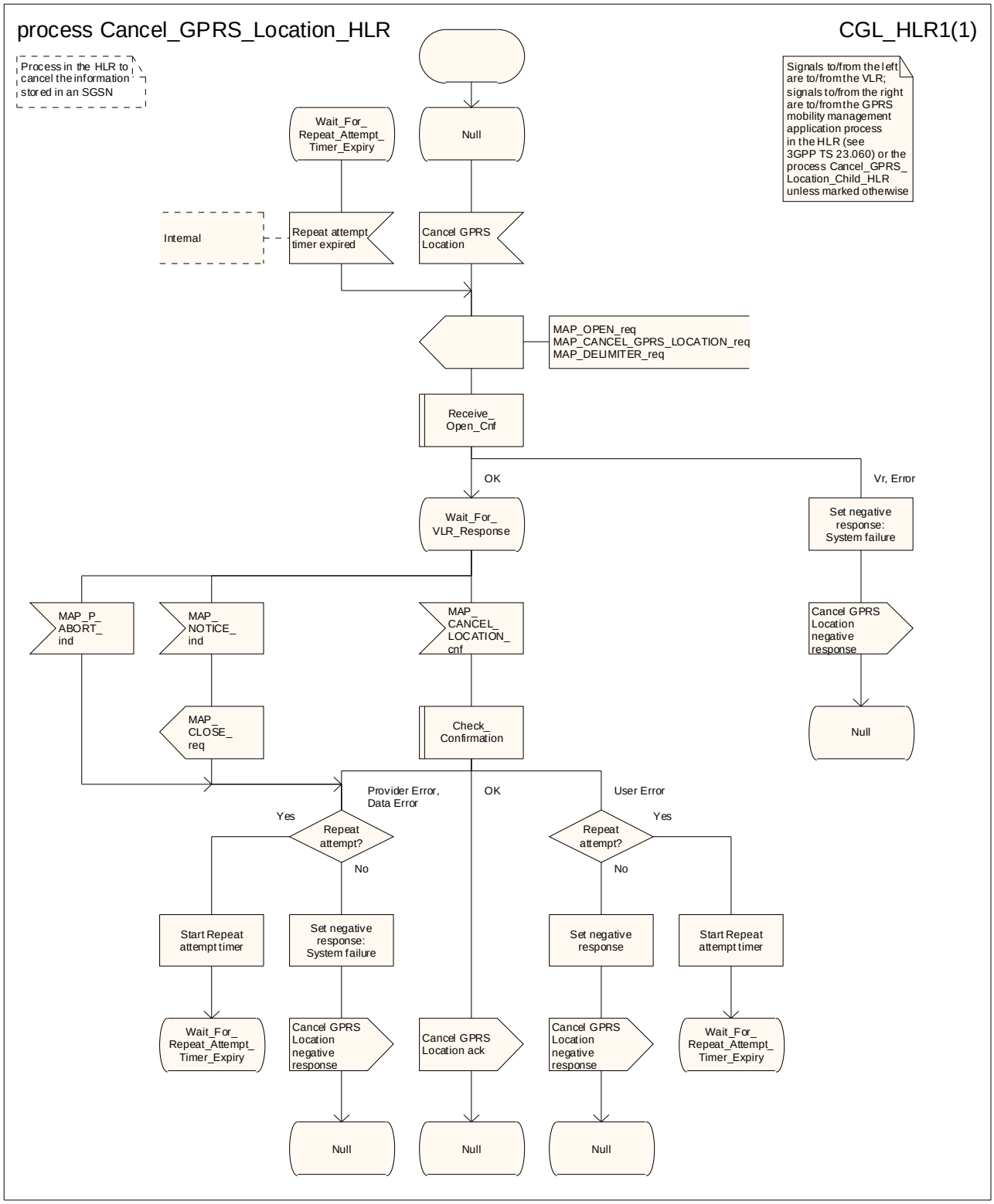


Figure 19.1.2/5: Process Cancel_GPRS_Location_HLR

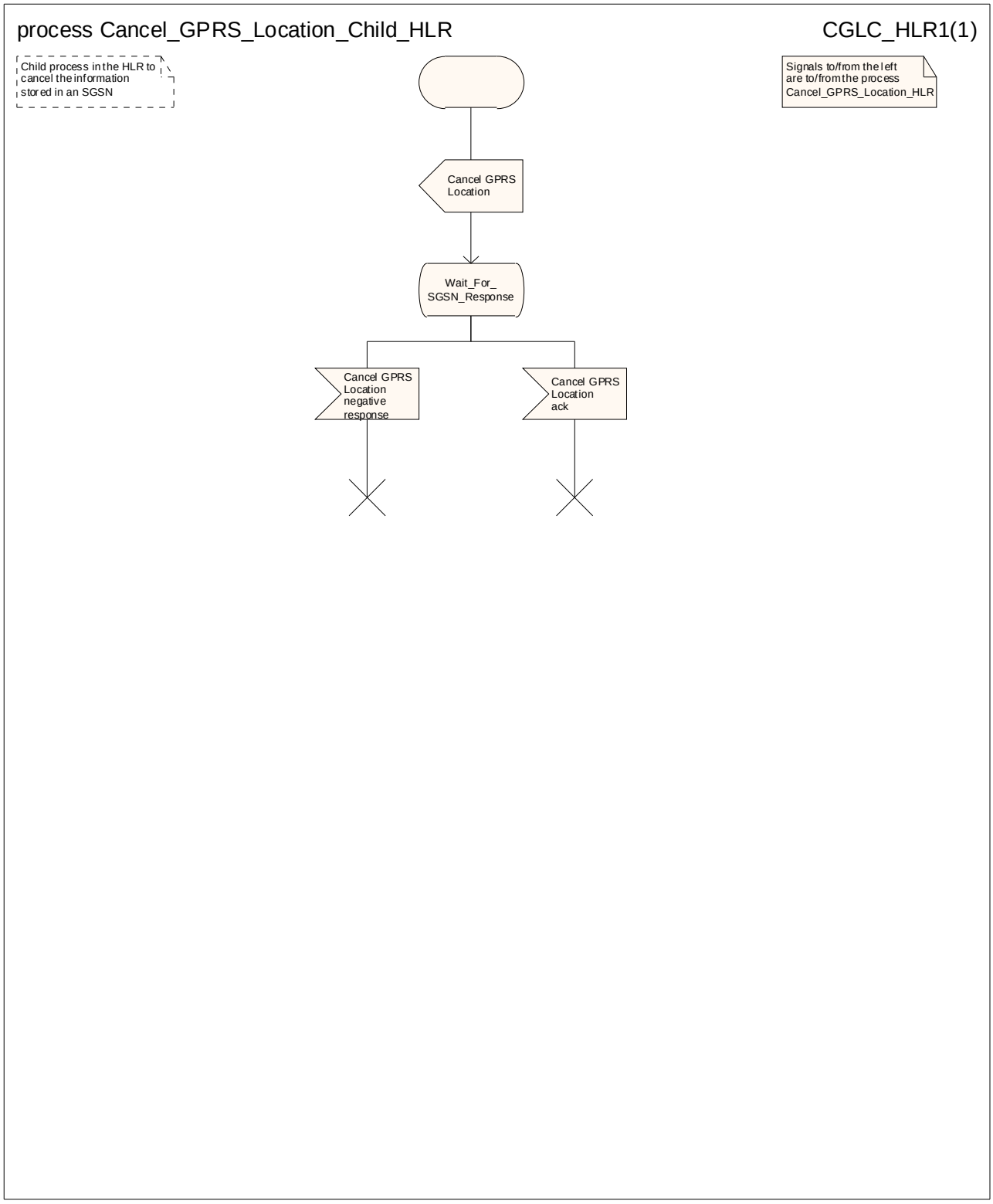


Figure 19.1.2/6: Process Cancel_GPRS_Location_Child_HLR

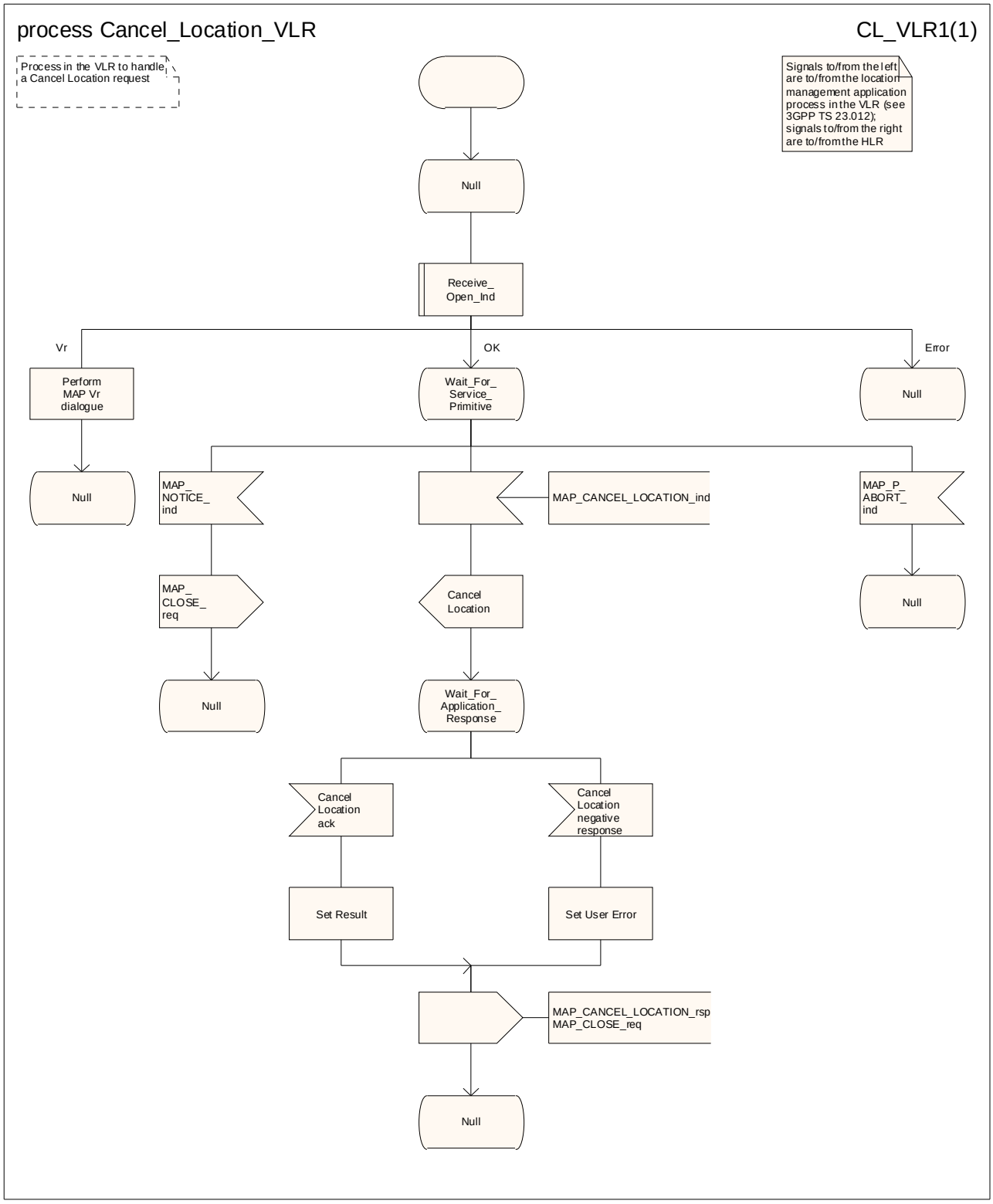


Figure 19.1.2/7: Process Cancel_Location_VLR

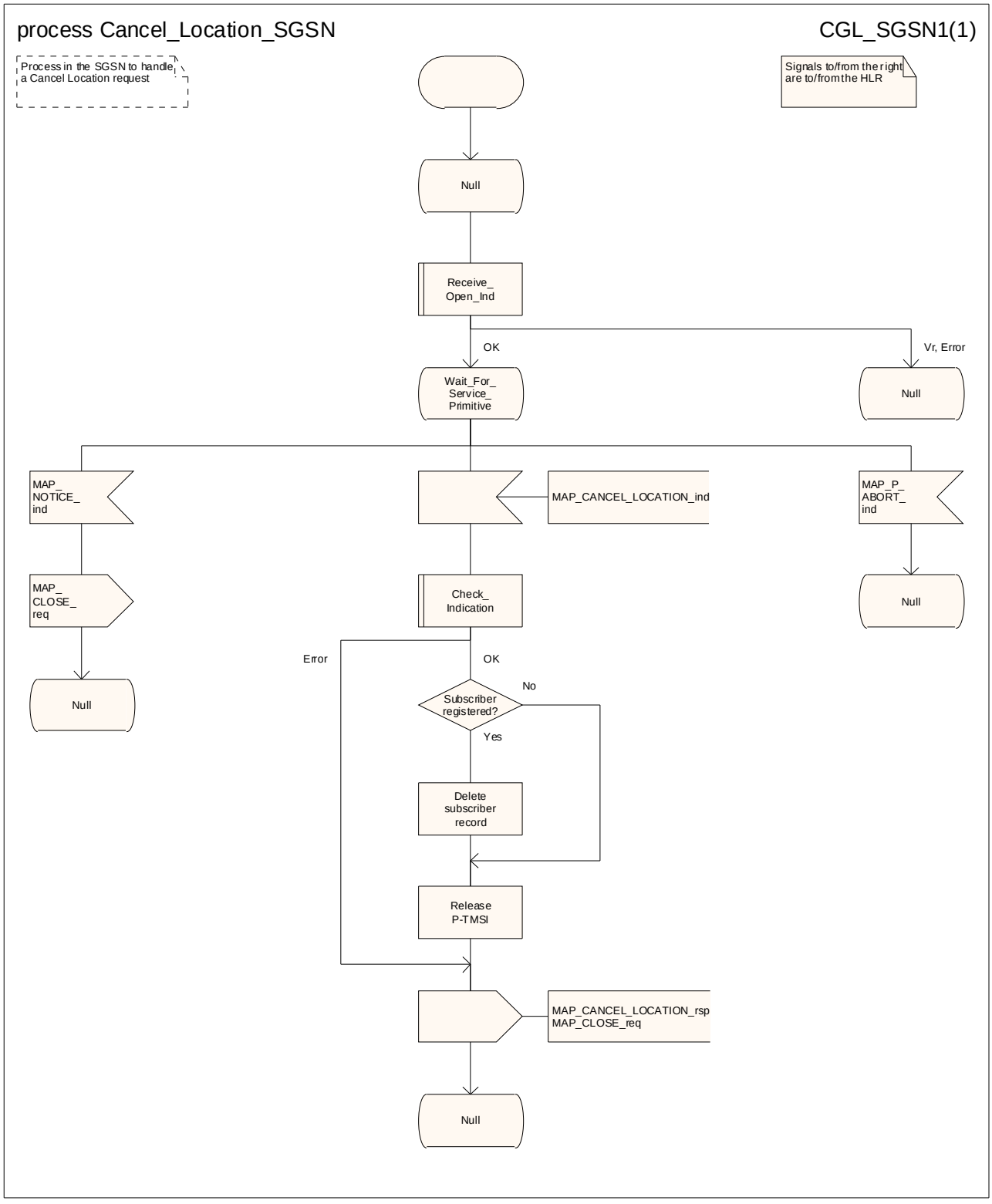


Figure 19.1.2/8: Process Cancel_Location_SGSN

19.1.2A Location Cancellation for VCSG

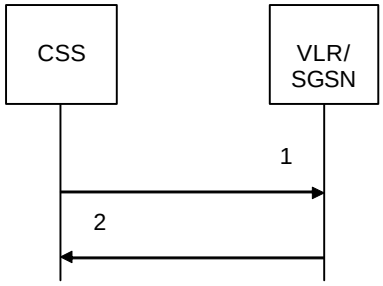
19.1.2A.1 General

Location cancellation for VCSG is used to delete a subscriber record from the serving node (VLR or SGSN). The procedure is invoked:

- because there is a removal of the CSG subscription data in the CSS and of the MS registration
- because the CSS has decided to cancel the registration of the MS which does not have CSG subscription data in the CSS.

NOTE: How the CSS determines when to cancel the registration of the MS is implementation dependent.

The message flow for VCSG location cancellation for a subscriber is shown in figure 19.1.2A/1.



1) MAP_CANCEL_VCSG_LOCATION_req/ind
2) MAP_CANCEL_VCSG_LOCATION_rsp/cnf

Figure 19.1.2A/1: Message flow for VCSG Location Cancellation

19.1.2A.2 Procedure in the CSS

The MAP process in the CSS to cancel the VCSG location information in a VLR is shown in figure 19.1.2A/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The MAP process in the CSS to cancel the VCSG location information in a SGSN is shown in figure 19.1.2A/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

19.1.2A.3 Procedure in the VLR

The MAP process in the VLR to handle a VCSG location cancellation request is shown in figure 19.1.2A/4. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

19.1.2A.4 Procedure in the SGSN

The MAP process in the SGSN to handle a VCSG location cancellation request is shown in figure 19.1.2A/5. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind	see subclause 25.1.1.

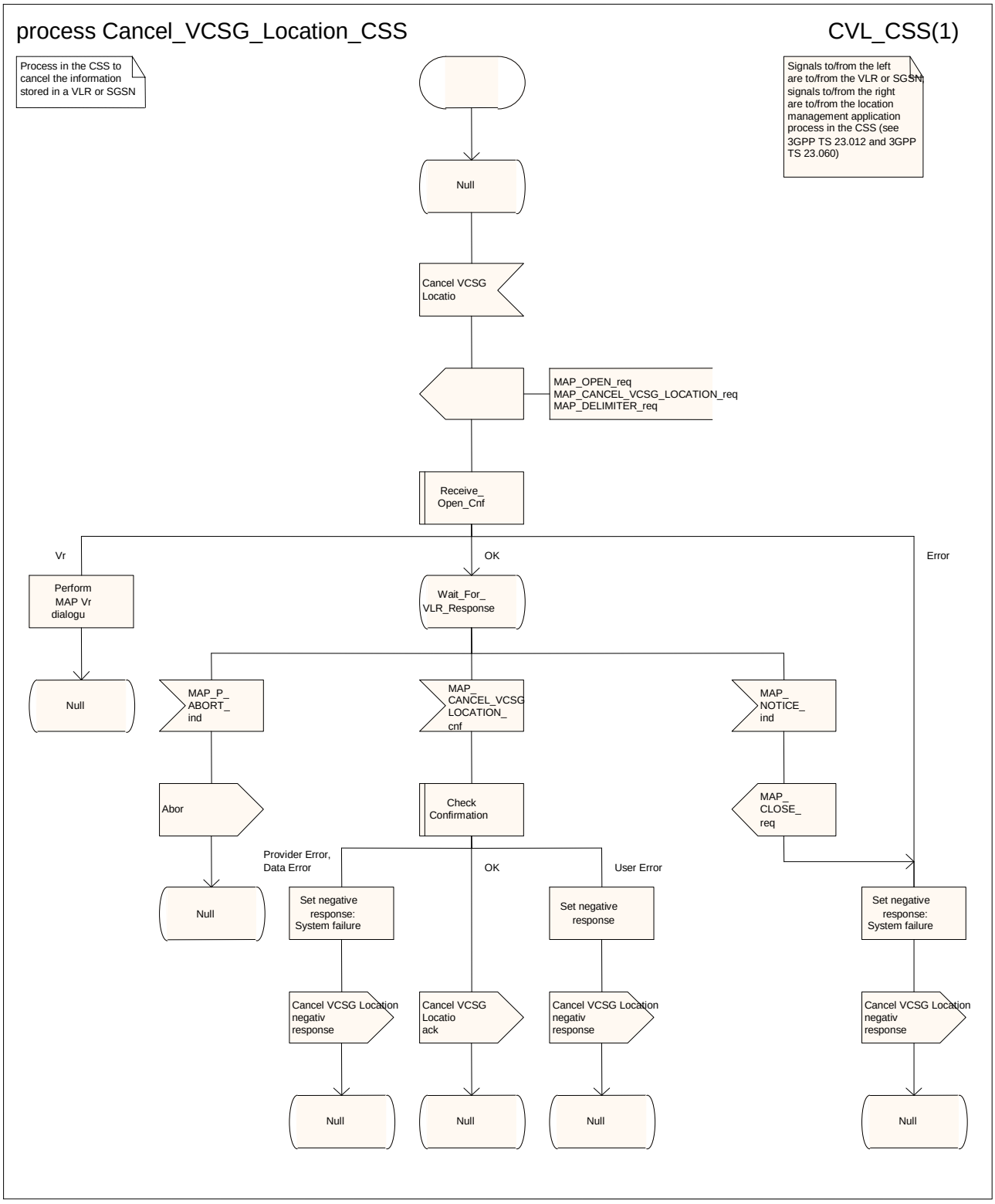


Figure 19.1.2A/2: Process Cancel_VCSG_Location_CSS

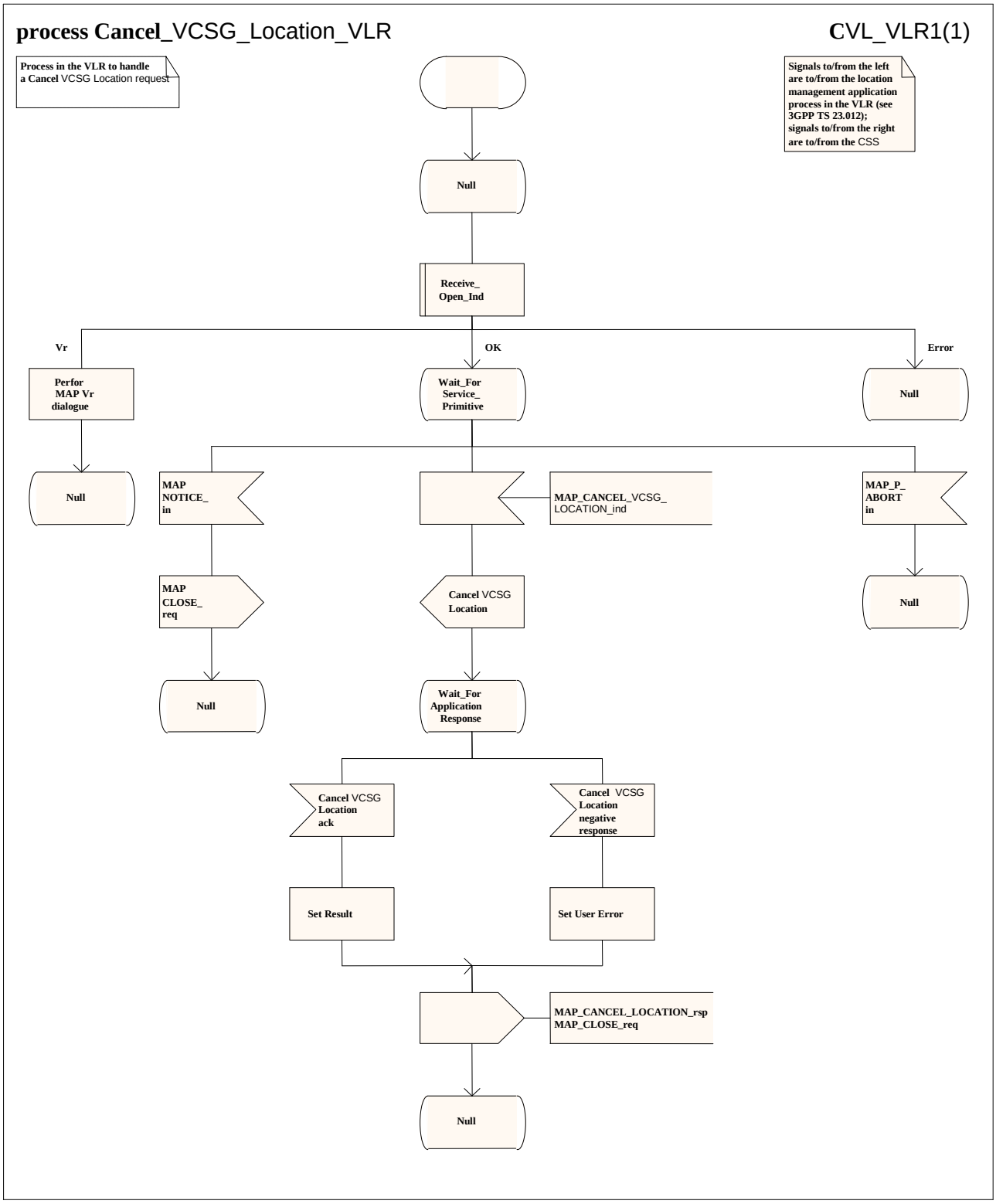


Figure 19.1.2A/3: Process Cancel_VCSG_Location_VLR

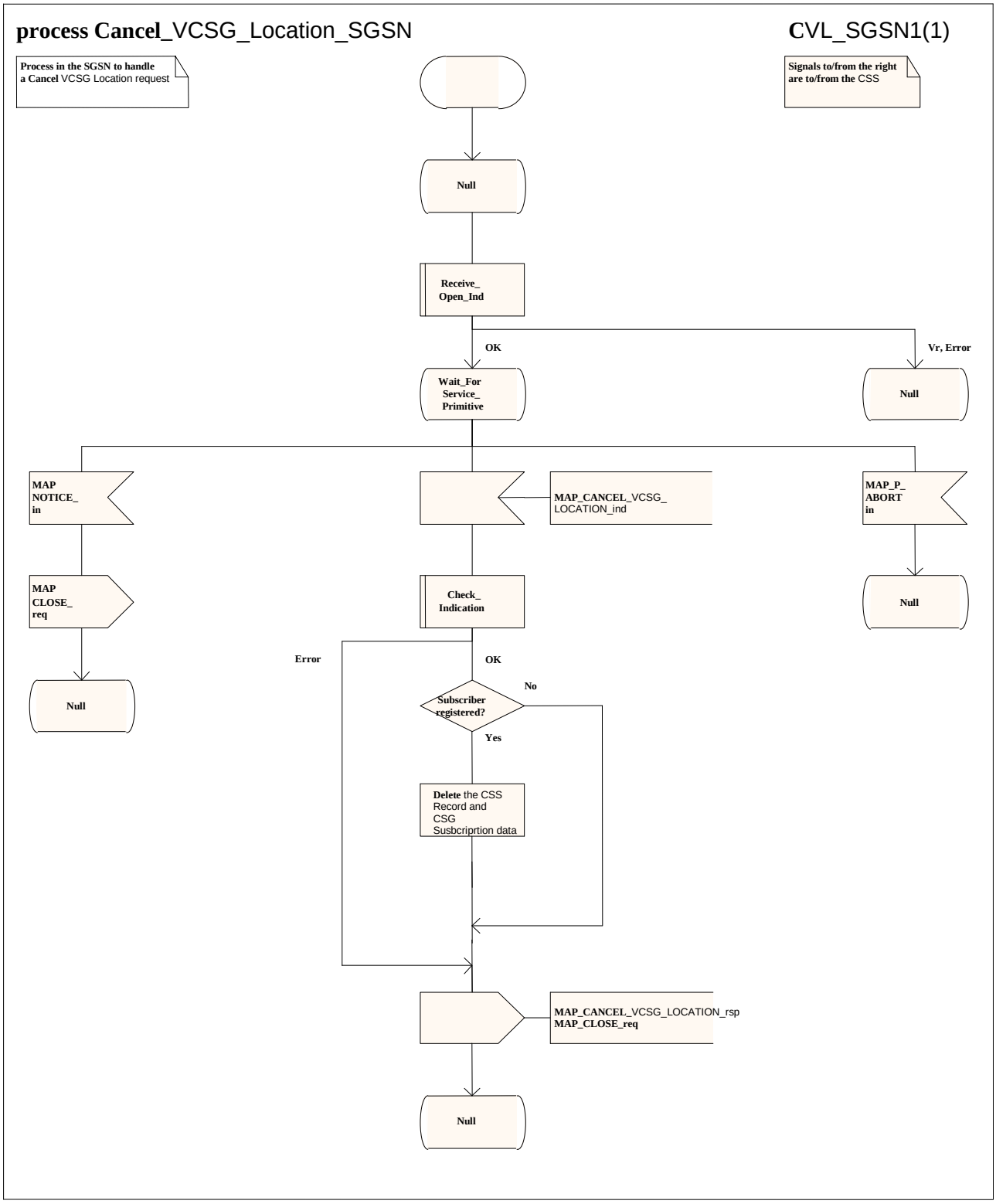


Figure 19.1.2A/4: Process Cancel_VCSG_Location_SGSN

19.1.3 Void

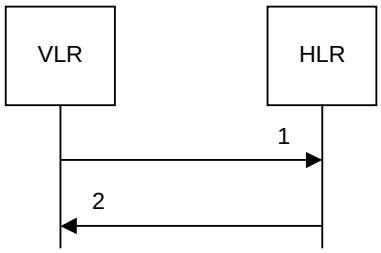
19.1.4 MS Purging

19.1.4.1 General

O&M procedures in the VLR or SGSN can trigger MS purging either because of administrative action or because the MS has been inactive for an extended period. The O&M process in the VLR or in the SGSN should ensure that during the MS purging procedure any other attempt to access the MS record is blocked, to maintain consistency of data.

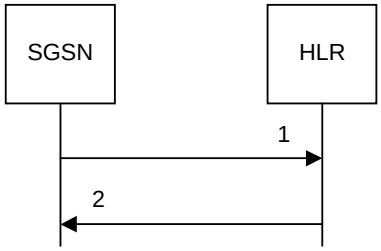
The message flow for a VLR to report MS purging to the HLR is shown in figure 19.1.4/1.

The message flow for an SGSN to report MS purging to the HLR is shown in figure 19.1.4/2.



1) MAP_PURGE_MS_req/ind
2) MAP_PURGE_MS_rsp/cnf

Figure 19.1.4/1: Message flow for MS purging (non-GPRS)



1) MAP_PURGE_MS_req/ind
2) MAP_PURGE_MS_rsp/cnf

Figure 19.1.4/2: Message flow for MS purging (GPRS)

19.1.4.2 Procedure in the VLR

The MAP process in the VLR to report MS purging to the HLR is shown in figure 19.1.4/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

19.1.4.3 Procedure in the SGSN

The MAP process in the SGSN to report MS purging to the HLR is shown in figure 19.1.4/4. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 1: The procedure Purge_MS_In_Serving_Network_Entity is specific to Super-Charger; it is specified in 3GPP TS 23.116 [110]. If the HLR does not support the Super-Charger functionality, processing continues from the "No" exit of the test "Result=Pass?".

19.1.4.4 Procedure in the HLR

The MAP process in the HLR to handle a notification from a VLR or an SGSN that an MS record has been purged is shown in figure 19.1.4/5. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

If the notification was received from a VLR, the MAP process communicates with the location management application process specified in 3GPP TS 23.012 [23]; if the notification was received from an SGSN, the MAP process communicates with the GPRS mobility management application process specified in 3GPP TS 23.060 [104].

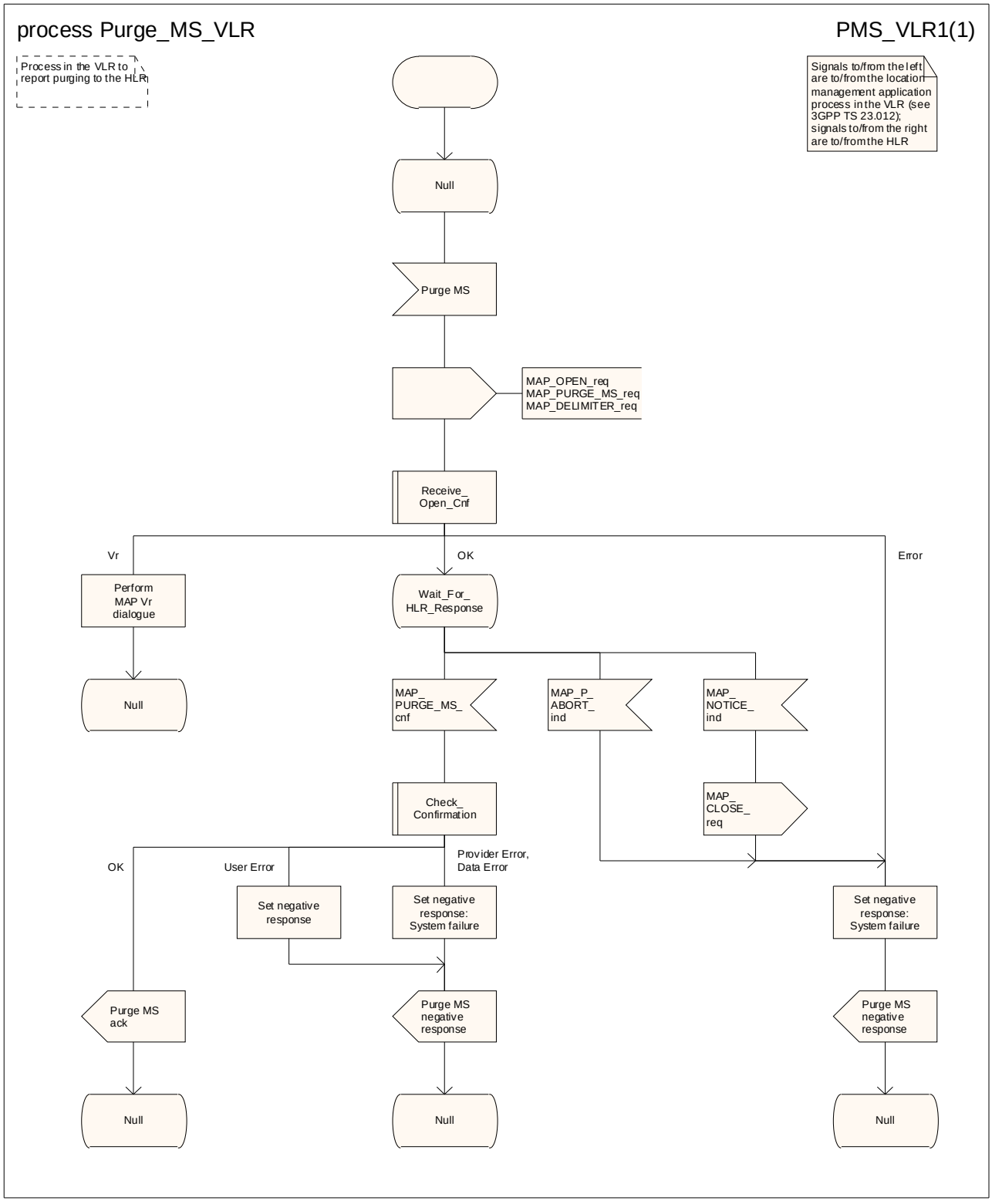


Figure 19.1.4/3: Process Purge_MS_VLR

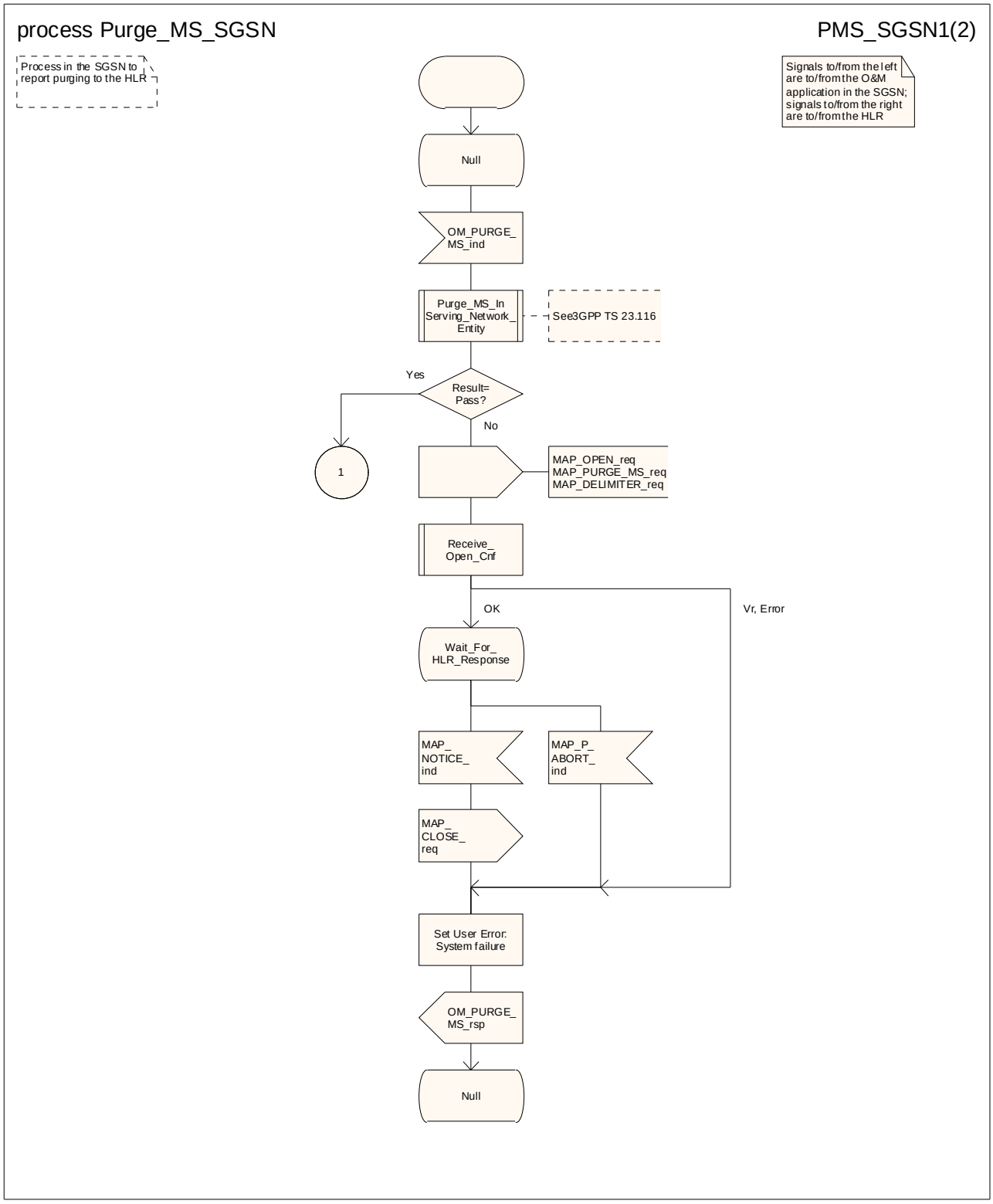


Figure 19.1.4/4 (sheet 1 of 2): Process Purge_MS_SGSN

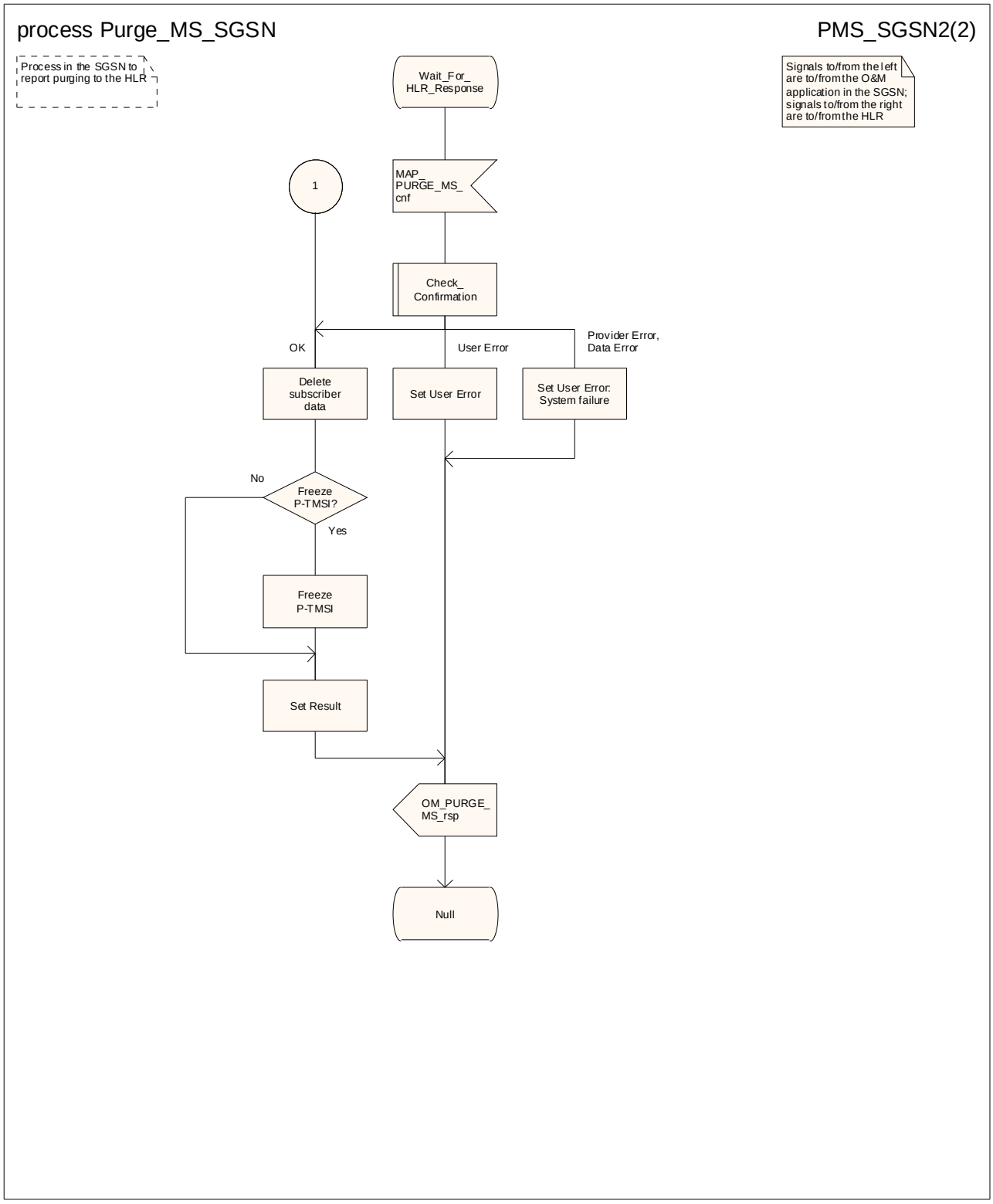


Figure 19.1.4/4 (sheet 2 of 2): Process Purge_MS_SGSN

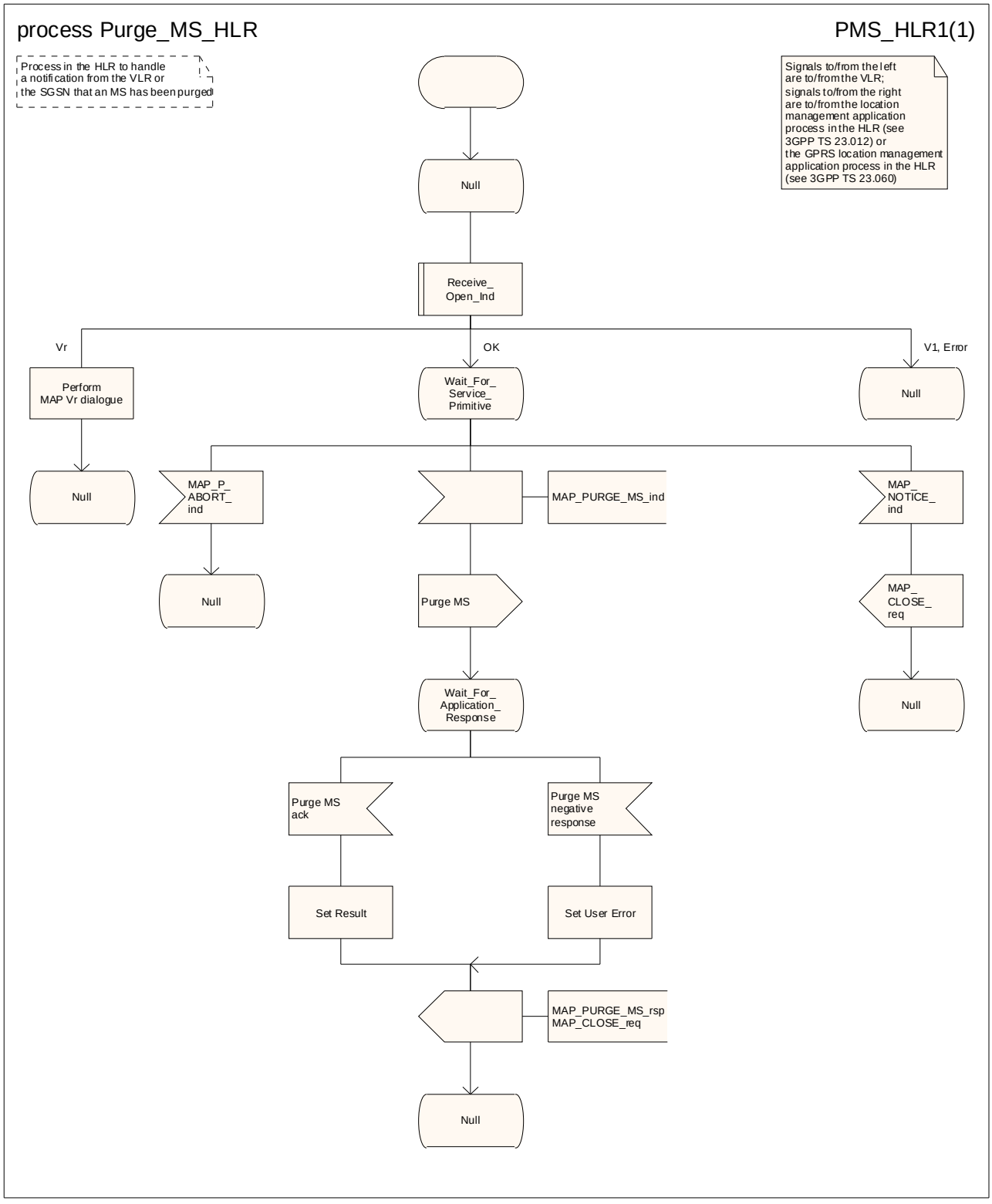


Figure 19.1.4/5: Process Purge_MS_HLR

19.2 Handover procedures

19.2.1 General

In this subclause, the term "Inter-MSC handover" is used to denote handover or relocation between different MSCs.

The interfaces involved for Inter-MSC handover are shown in figure 19.2/1. There are two Inter-MSC handover procedures:

1) Basic Inter-MSC handover:

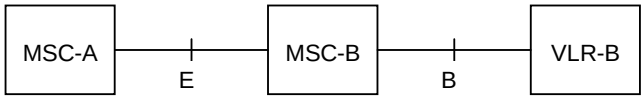
   The call is handed over from the controlling MSC(MSC—A) to another MSC(MSC—B) (figure 19.2/1a).

   Figure 19.2/2 shows the message flow for a successful handover from MSC-A to MSC—B, including a request for handover number allocation from MSC-B to VLR-B.

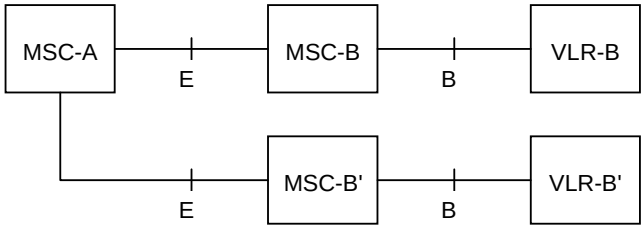
2) Subsequent Inter-MSC handover:

   After the call has been handed over from MSC-A to MSC-B, a further handover either to MSC-A (figure 19.2/1a) or to a third MSC (MSC-B') (figure 19.2/1b) may be necessary in order to continue the call.

   Figure 19.2/3 shows the message flow for a successful subsequent handover to MSC-B'. For a successful subsequent handover to MSC-A, the messages to and from MSC-B' and VLR-B' are omitted..

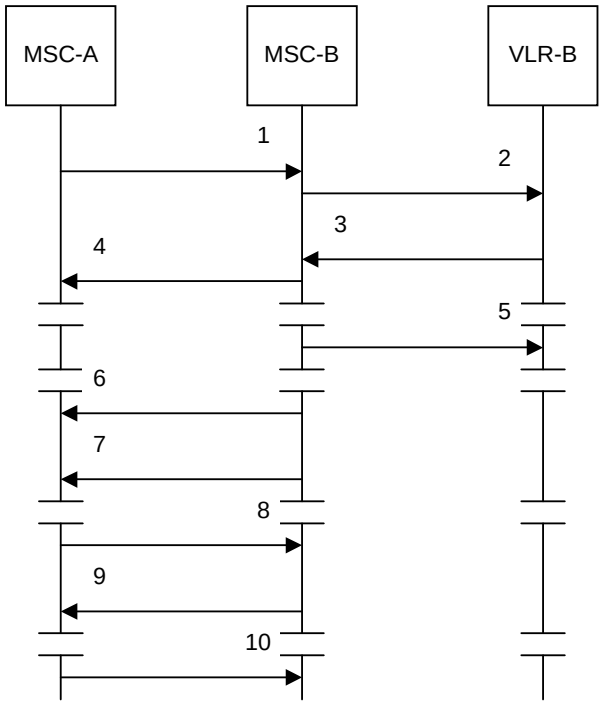


a) Basic handover procedure MSC-A to MSC-B and subsequent handover procedure MSC-B to MSC-A.



b) Subsequent handover procedure MSC-B to MSC-B'.

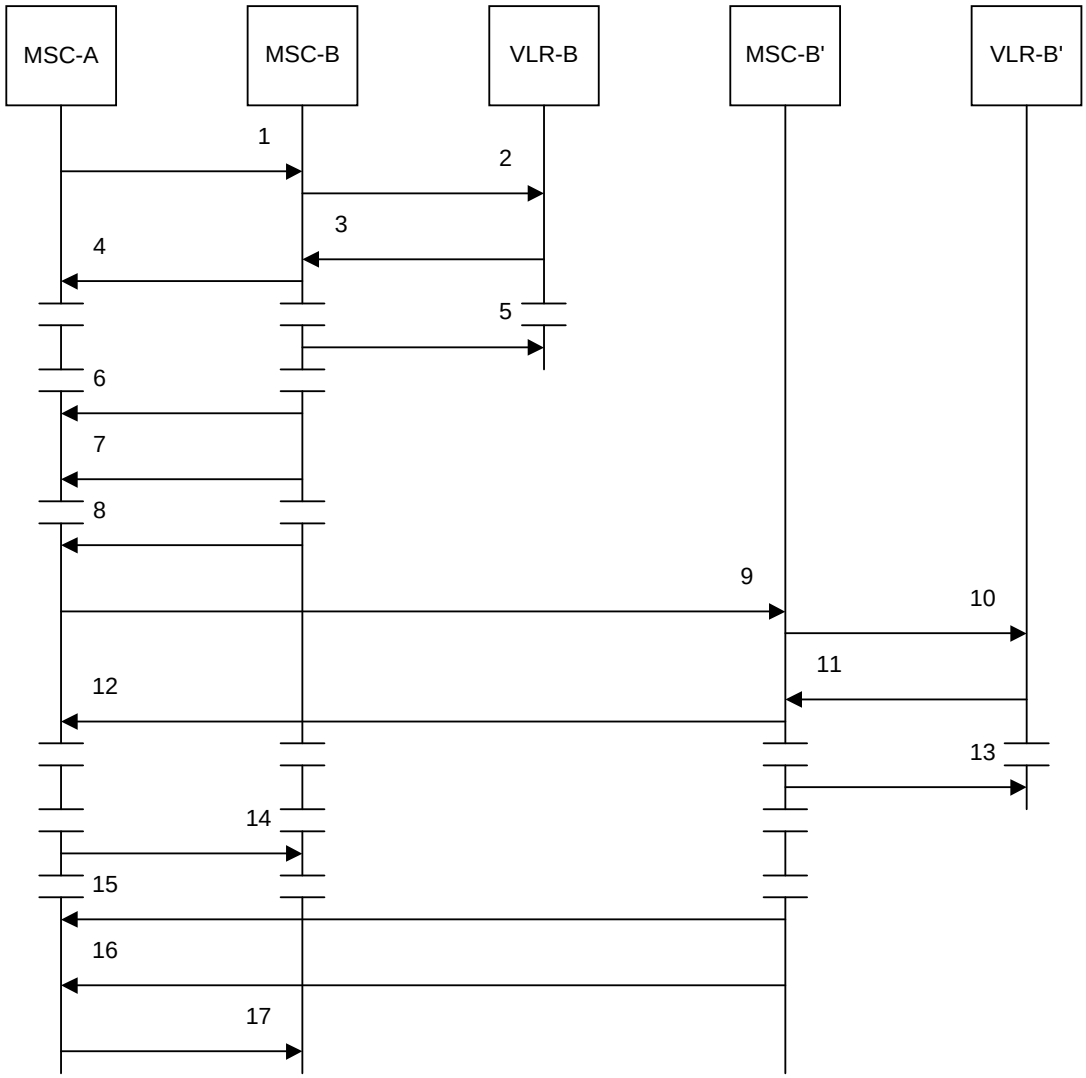
Figure 19.2/1: Interface structure for handover



1) MAP_PREPARE_HANDOVER_req/ind
2) MAP_ALLOCATE_HANDOVER_NUMBER_req/ind
3) MAP_SEND_HANDOVER_REPORT_req/ind
4) MAP_PREPARE_HANDOVER_rsp/cnf
5) MAP_SEND_HANDOVER_REPORT_rsp/cnf (Note)
6) MAP_PROCESS_ACCESS_SIGNALLING_req/ind
7) MAP_SEND_END_SIGNAL_req/ind
8) MAP_FORWARD_ACCESS_SIGNALLING_req/ind
9) MAP_PROCESS_ACCESS_SIGNALLING_req/ind
10) MAP_SEND_END_SIGNAL_rsp/cnf

NOTE: This can be sent at any time after the connection between MSC-A and MSC-B is established.

Figure 19.2/2: Example of a successful basic handover procedure to MSC-B



1) MAP_PREPARE_HANDOVER_req/ind
2) MAP_ALLOCATE_HANDOVER_NUMBER_req/ind
3) MAP_SEND_HANDOVER_REPORT_req/ind
4) MAP_PREPARE_HANDOVER_rsp/cnf
5) MAP_SEND_HANDOVER_REPORT_rsp/cnf (Note 1)
6) MAP_PROCESS_ACCESS_SIGNALLING_req/ind
7) MAP_SEND_END_SIGNAL_req/ind
8) MAP_PREPARE_SUBSEQUENT_HANDOVER_req/ind
9) MAP_PREPARE_HANDOVER_req/ind
10) MAP_ALLOCATE_HANDOVER_NUMBER_req/ind
11) MAP_SEND_HANDOVER_REPORT_req/ind
12) MAP_PREPARE_HANDOVER_rsp/cnf
13) MAP_SEND_HANDOVER_REPORT_rsp/cnf (Note 2)
14) MAP_PREPARE_SUBSEQUENT_HANDOVER_rsp/cnf
15) MAP_PROCESS_ACCESS_SIGNALLING_req/ind
16) MAP_SEND_END_SIGNAL_req/ind
17) MAP_SEND_END_SIGNAL_rsp/cnf (Note 3)

NOTE 1: This can be sent at any time after the connection between MSC-A and MSC-B is established.
NOTE 2: This can be sent at any time after the connection between MSC-A and MSC-B' is established.
NOTE 3: At this stage, the subsequent handover is complete. Any further interworking between MSC-A and MSC-B' is the same as the interworking between MSC-A and MSC-B after basic handover

Figure 19.2/3: Example of a successful subsequent handover to a third MSC

The MAP signalling procedures for inter-MSC handover support the allocation of a handover number or one or more relocation numbers and the transfer of encapsulated BSSAP or RANAP messages.

The minimum application context version for the MAP handover application context shall be:

- version 3 for inter-MSC UTRAN to UTRAN handover;
- version 3 for inter-MSC intersystem handover from GSM BSS to UTRAN;
- version 2 for inter-MSC intersystem handover from UTRAN to GSM BSS.

NOTE: If the MAP handover application context version 2 is used, subsequent handover to UTRAN is not possible.

The minimum application context version for the MAP handover application context should be version 2 for inter-MSC handover from GSM BSS to GSM BSS.

NOTE: If the MAP handover application context version 2 or lower is used, subsequent handover to UTRAN is not possible.

The BSSAP or RANAP messages encapsulated in MAP messages are processed by the Handover Control Application in each MSC. The information in the encapsulated BSSAP or RANAP messages is passed from the Handover Control Application to the MAP process at the sending end; the notation used in the SDL diagrams for the MAP processes is "HO_CA_MESSAGE_ind(Message transfer)". The information in the encapsulated BSSAP or RANAP messages is passed from the MAP process to the Handover Control Application at the sending end; the notation used in the SDL diagrams for the MAP processes is "HO_CA_MESSAGE_req(Message transfer)".

For details of the interworking between the A-interface and MAP procedures or the Iu-interface and MAP procedures, see 3GPP TS 23.009 [21] and 3GPP TS 29.010 [58].

19.2.2 Procedure in MSC-A

This subclause describes the inter-MSC handover procedure in MSC-A; it covers basic inter-MSC handover to another MSC (MSC-B) and subsequent inter-MSC handover to a third MSC (MSC-B') or back to the controlling MSC (MSC-A).

The MAP process in MSC-A to handle inter-MSC handover is shown in figure 19.2/4. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Indication see subclause 25.2.1.

Check_Confirmation see subclause 25.2.2.

Communication between the MAP handover process and the Handover Control application is represented by the HO_CA_MESSAGE service. For a detailed description of the interworking between the Handover Control applications in different MSCs for the inter-MSC handover procedure, see 3GPP TS 23.009 [21].

19.2.2.1 Basic handover

The handling in MSC-A for basic inter-MSC handover is shown in sheets 1 to 6 of figure 19.2/4.

Sheet 1: The MAP_PREPARE_HANDOVER request may contain:

- an indication that handover number allocation is not required;
- the target Cell ID, for compatibility for handover to GSM;
- the target RNC ID, for SRNS relocation or inter-system handover from GSM to UMTS;
- the IMSI;
- UMTS encryption information and UMTS integrity protection information, which are necessary for inter-system handover from GSM to UMTS;

- GSM radio resource information (channel type);
- the LCLS Global Call Reference;
- the LCLS-Negotiation;
- the LCLS-Configuration-Preference.

The conditions for the presence of these parameters and the processing in MSC-B (3G_MSC-B) are described in detail in 3GPP TS 29.010 [58], 3GPP TS 23.009 [21] and 3GPP TS 29.205 [146].

Sheet 2: The MAP_PREPARE_HANDOVER confirmation contains one of:

- no handover number, if the MAP_PREPARE_HANDOVER request included an indication that handover number allocation is not required;
- a handover number;
- one or more relocation numbers.

Sheet 2: The MAP_PREPARE_HANDOVER confirmation contains BSSAP or RANAP signalling information, which is passed to the Handover Control application in MSC-A.

Sheet 2: If the MAP_PREPARE_HANDOVER confirmation contains an indication that MSC-B does not support multiple bearers, the Handover Control application in MSC-A may request handover of one bearer to the same cell in MSC-B.

Sheet 5: If the original MAP_PREPARE_HANDOVER request included a parameter indicating that handover number allocation is not required, the Handover Control application in MSC-A may request a handover number (or one or more relocation numbers); this triggers a further MAP_PREPARE_HANDOVER request towards MSC-B.

19.2.2.2 Handling of access signalling

The Handover Control application in MSC-A may forward access signalling to any of the MS, RNS-B or BSS-B using the MAP_FORWARD_ACCESS_SIGNALLING service; any of the MS, RNS-B or BSS-B may forward access signalling to the Handover Control application in MSC-A using the MAP_PROCESS_ACCESS_SIGNALLING service. These are non-confirmed services.

19.2.2.3 Subsequent handover

The handling in MSC-A for subsequent inter-MSC handover is shown in sheets 7 & 8 of figure 19.2/4. If the Handover Control Application determines that the call is to be handed over to a third MSC (MSC-B') it triggers another instance of the MAP process to handle the basic handover to MSC-B', and reports the result of the subsequent handover to the instance of the MAP process which handles the dialogue with MSC-B.

Sheet 8: While the MAP process in MSC-A is waiting for the completion of subsequent handover, it relays access signalling between the Handover Control application and the MS, RNS-B or BSS-B as described in subclause 19.2.2.2.

19.2.3 Procedure in MSC-B

This subclause describes the handover or relocation procedure in MSC-B; it covers basic handover or relocation from the controlling MSC (MSC-A) and subsequent handover or relocation.

The MAP process in MSC-B to handle handover or relocation is shown in figure 19.2/5. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind	see subclause 25.1.1;
Receive_Open_Cnf	see subclause 25.1.2;
Check_Indication	see subclause 25.2.1.
Check_Confirmation	see subclause 25.2.2.

Communication between the MAP handover process and the Handover Control application is represented by the HO_CA_MESSAGE service. For a detailed description of the interworking between the Handover Control applications in different MSCs for the inter-MSC handover procedure, see 3GPP TS 23.009 [21].

The ordering of allocation of handover number and radio resources shown in the SDL diagrams is not mandatory.

19.2.3.1 Basic handover

The handling in MSC-B for basic inter-MSC handover is shown in sheets 1 to 7 of figure 19.2/5.

Sheet 2: If the MAP_PREPARE_HANDOVER indication included a parameter requesting multiple bearers but MSC-B does not support multiple bearers, MSC-B sends a MAP_PREPARE_HANDOVER response indicating that multiple bearers are not supported, and waits for a possible MAP_PREPARE_HANDOVER indication requesting handover of a single bearer.

Sheet 6: If the original MAP_PREPARE_HANDOVER indication included a parameter indicating that handover number allocation is not required, MSC-A may send a further MAP_PREPARE_HANDOVER request to request the allocation of a handover number (or one or more relocation numbers).

19.2.3.2 Handling of access signalling

The Handover Control application in MSC-A may forward access signalling to any of the MS, RNS-B or BSS-B using the MAP_FORWARD_ACCESS_SIGNALLING service; any of the MS, RNS-B or BSS-B may forward access signalling to the Handover Control application in MSC-A using the MAP_PROCESS_ACCESS_SIGNALLING service. These are non-confirmed services. Signals to or from any of the MS, RNS-B or BSS-B are routed through the Handover Control application in MSC-B.

19.2.3.3 Subsequent handover

The handling in MSC-B for subsequent inter-MSC handover is shown in sheet 8 of figure 19.2/5.

While the MAP process in MSC-B is waiting for the completion of subsequent handover, it relays access signalling between MSC-A and the MS, RNS-B or BSS-B through the Handover Control application as described in subclause 19.2.3.2.

19.2.4 Macro Receive_Error_From_HO_CA

This macro is used by the handover processes in MSC-A and MSC-B to receive errors from the Handover Control Application at any state of a handover process.

19.2.5 Procedure in VLR-B

The process in VLR-B to handle a request for a handover number is shown in figure 19.2/7. The process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind	see subclause 25.1.1;
Check_Indication	see subclause 25.2.1.

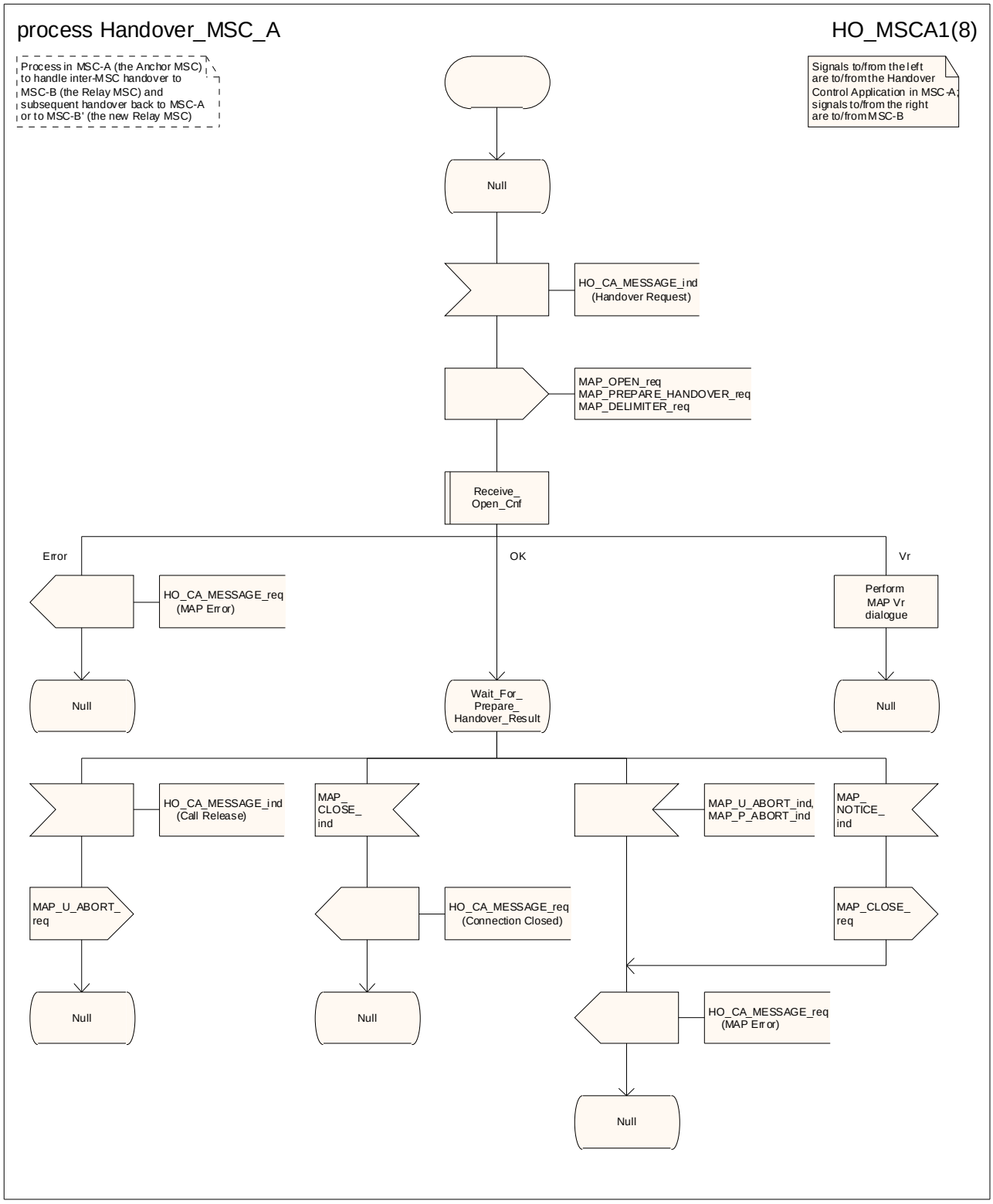


Figure 19.2/4 (sheet 1 of 8): Process HO_MSC_A

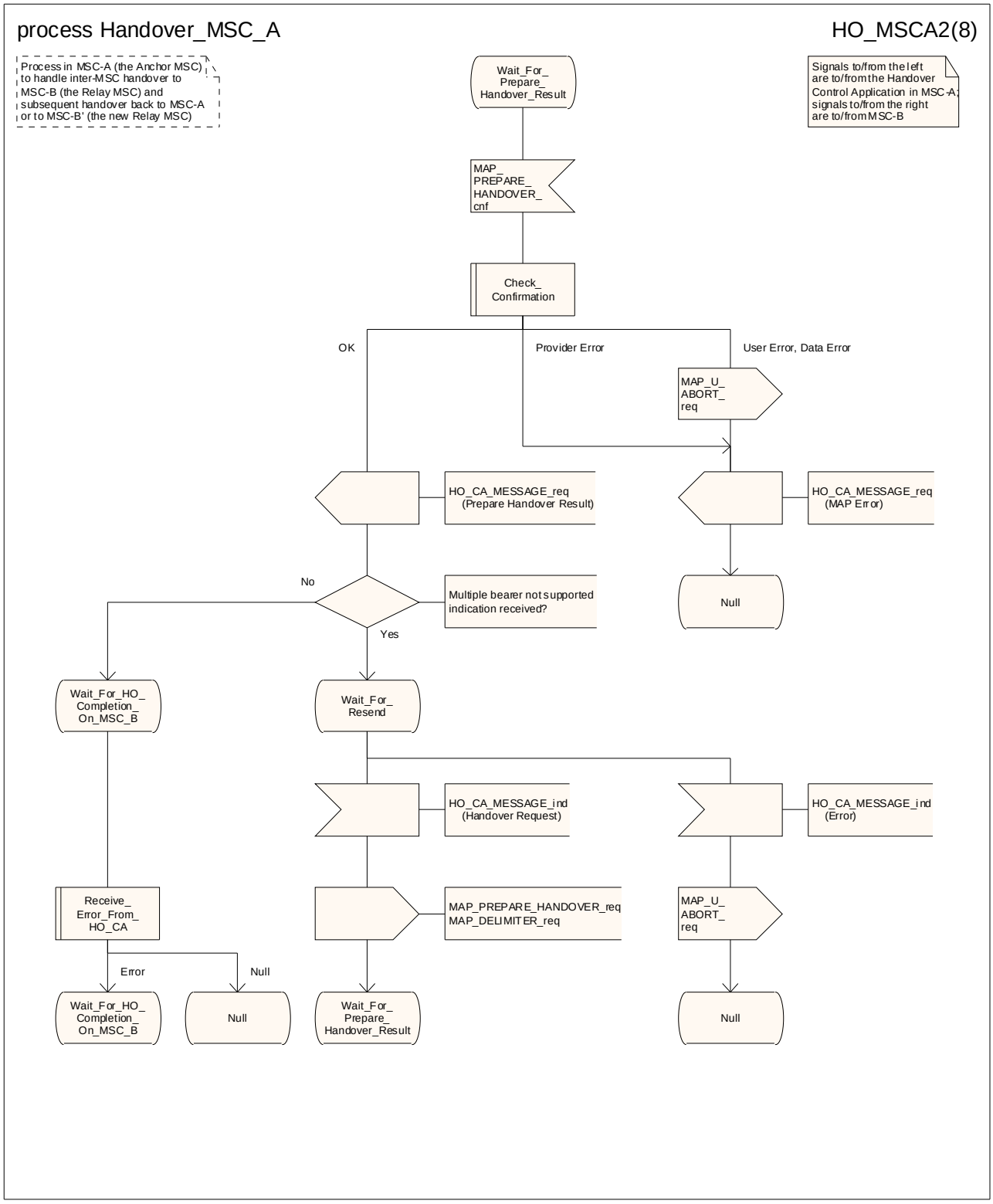


Figure 19.2/4 (sheet 2 of 8): Process HO_MSC_A

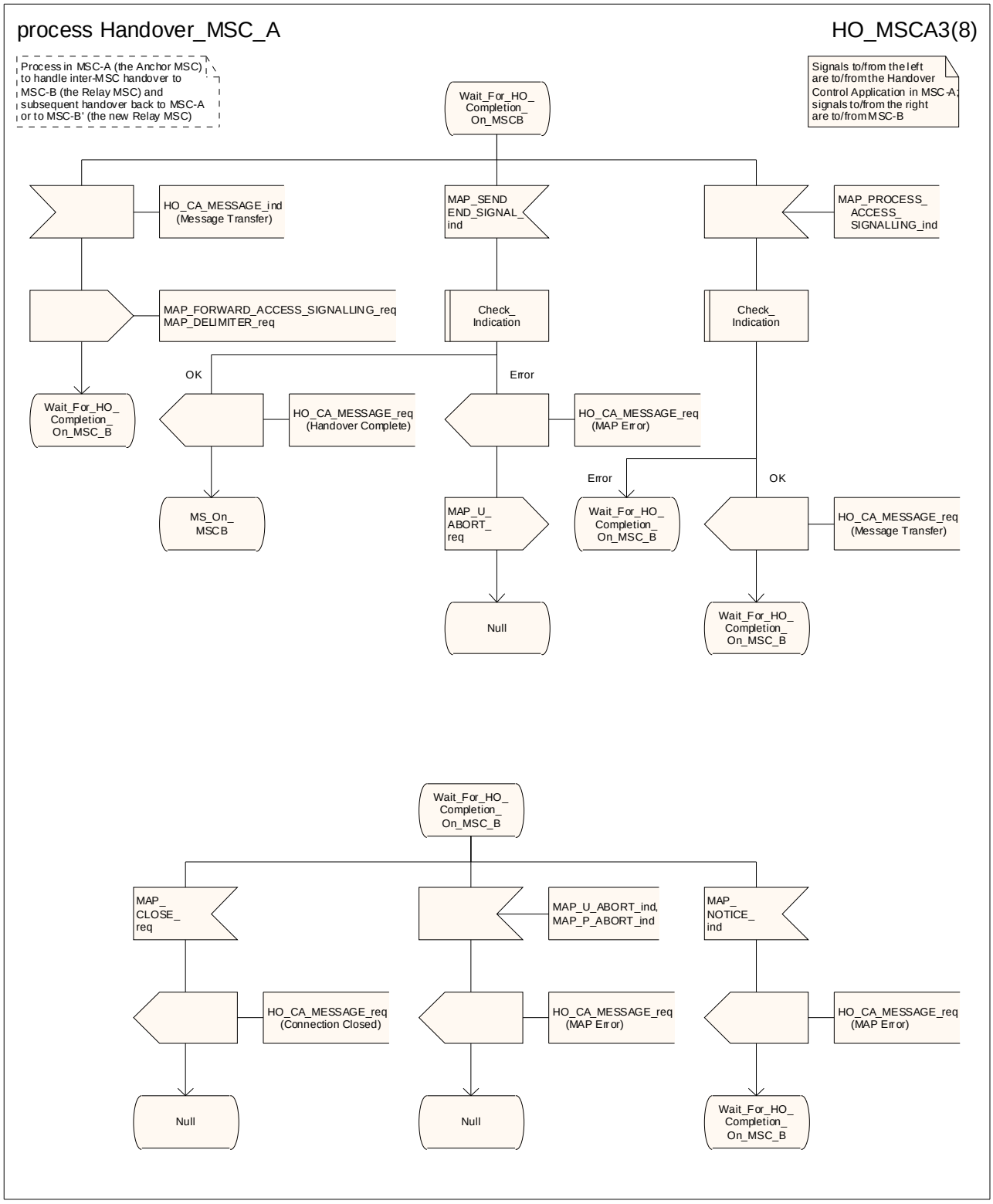


Figure 19.2/4 (sheet 3 of 8): Process HO_MSC_A

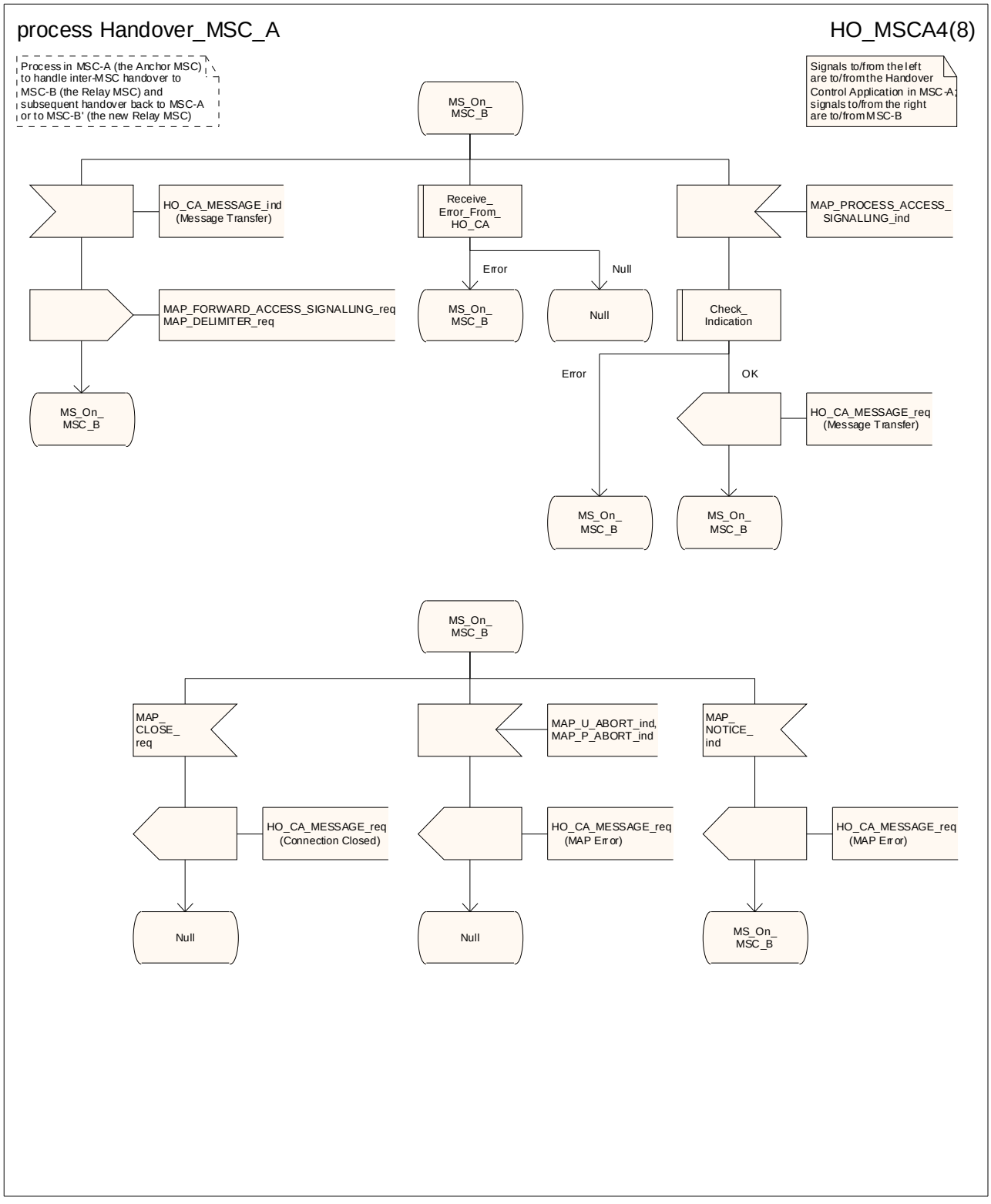


Figure 19.2/4 (sheet 4 of 8): Process HO_MSC_A

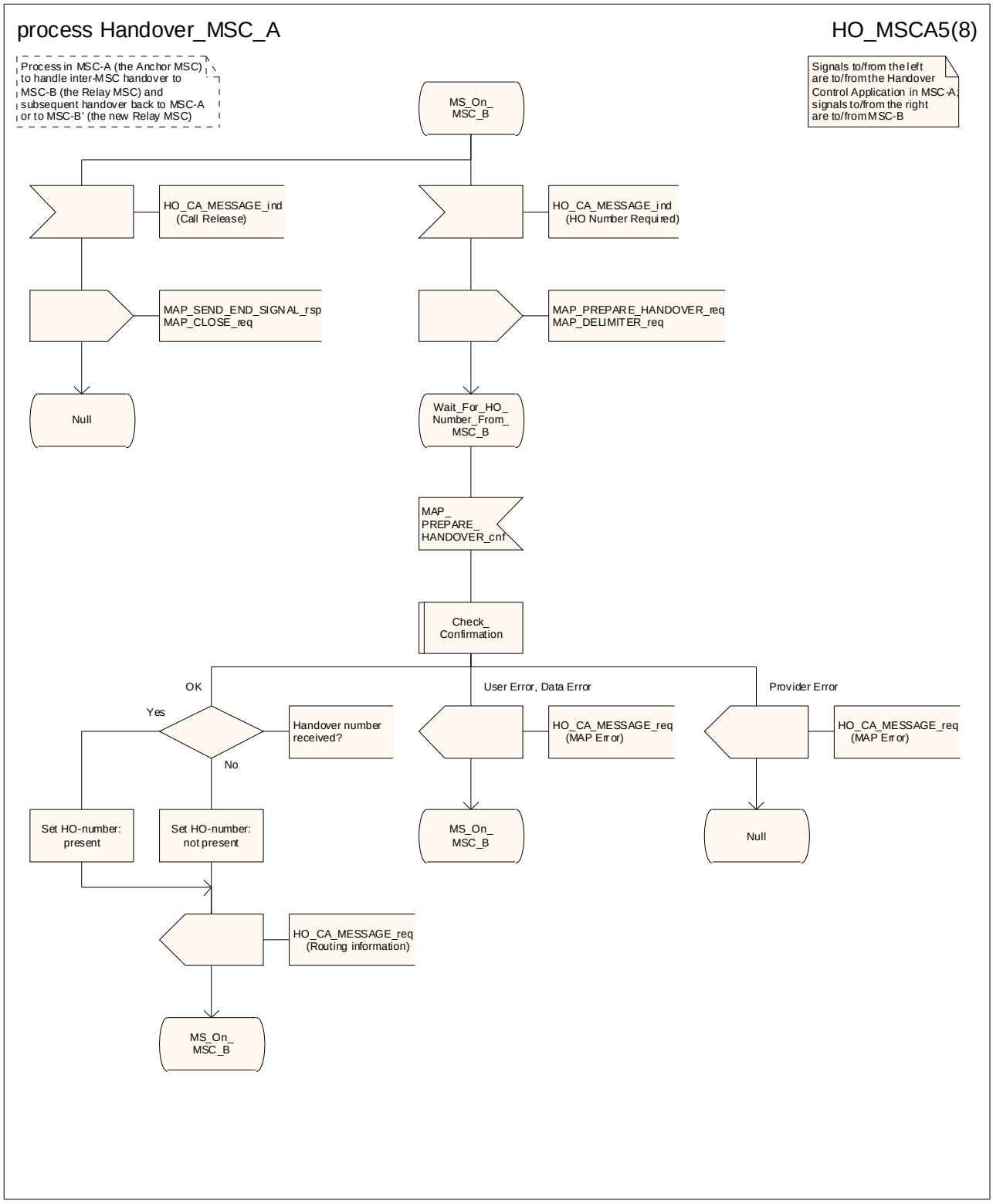


Figure 19.2/4 (sheet 5 of 8): Process HO_MSC_A

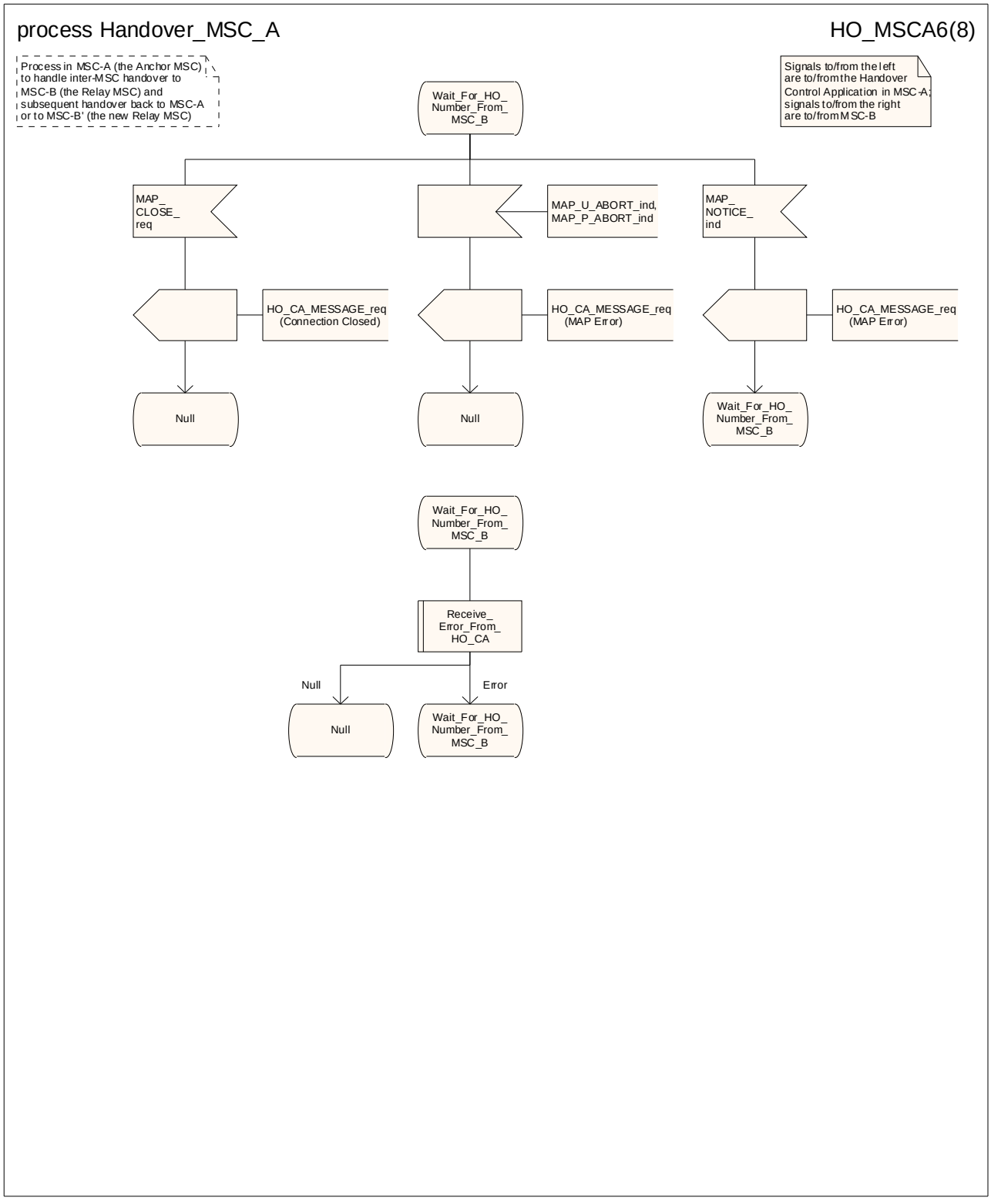


Figure 19.2/4 (sheet 6 of 8): Process HO_MSC_A

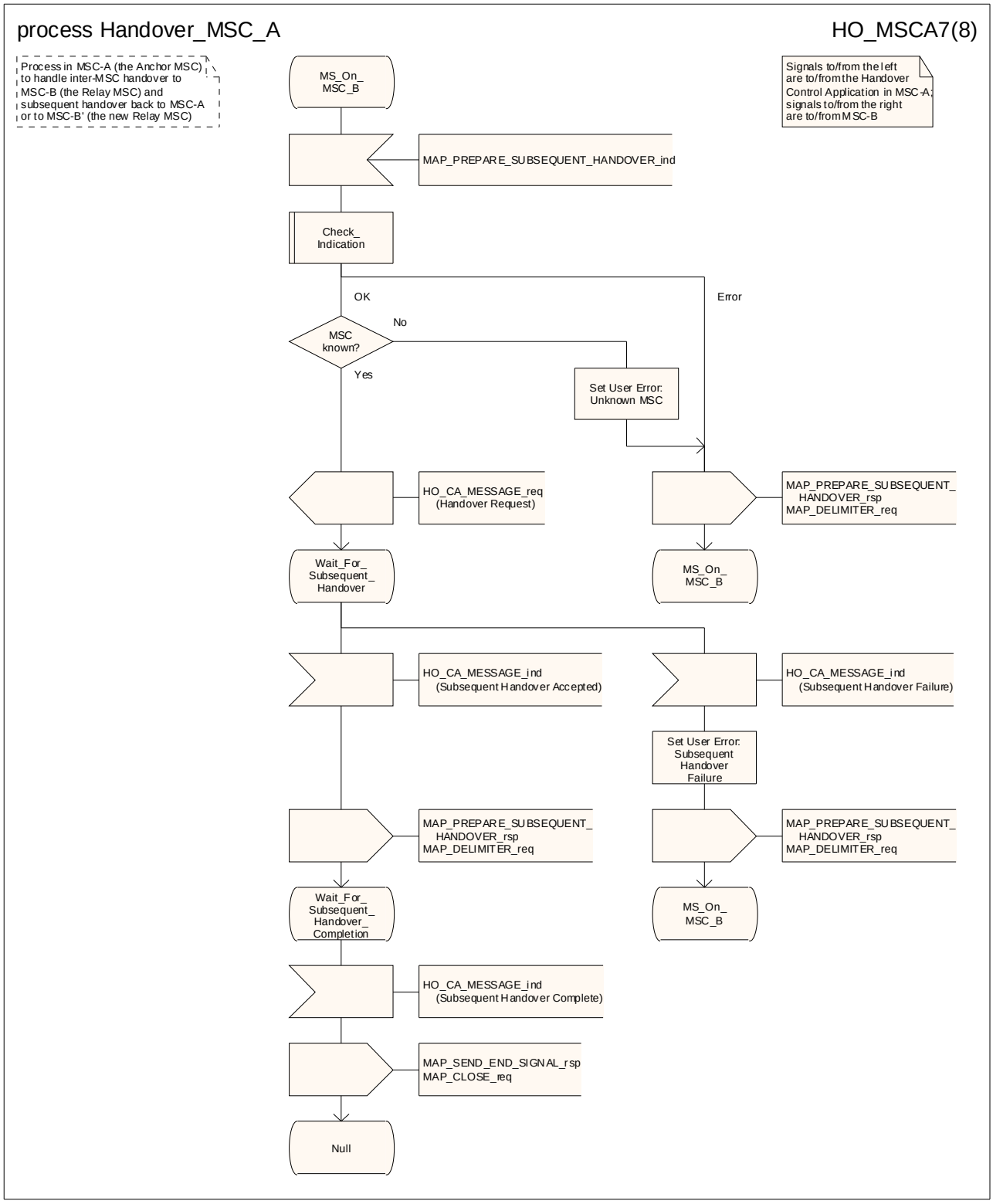


Figure 19.2/4 (sheet 7 of 8): Process HO_MSC_A

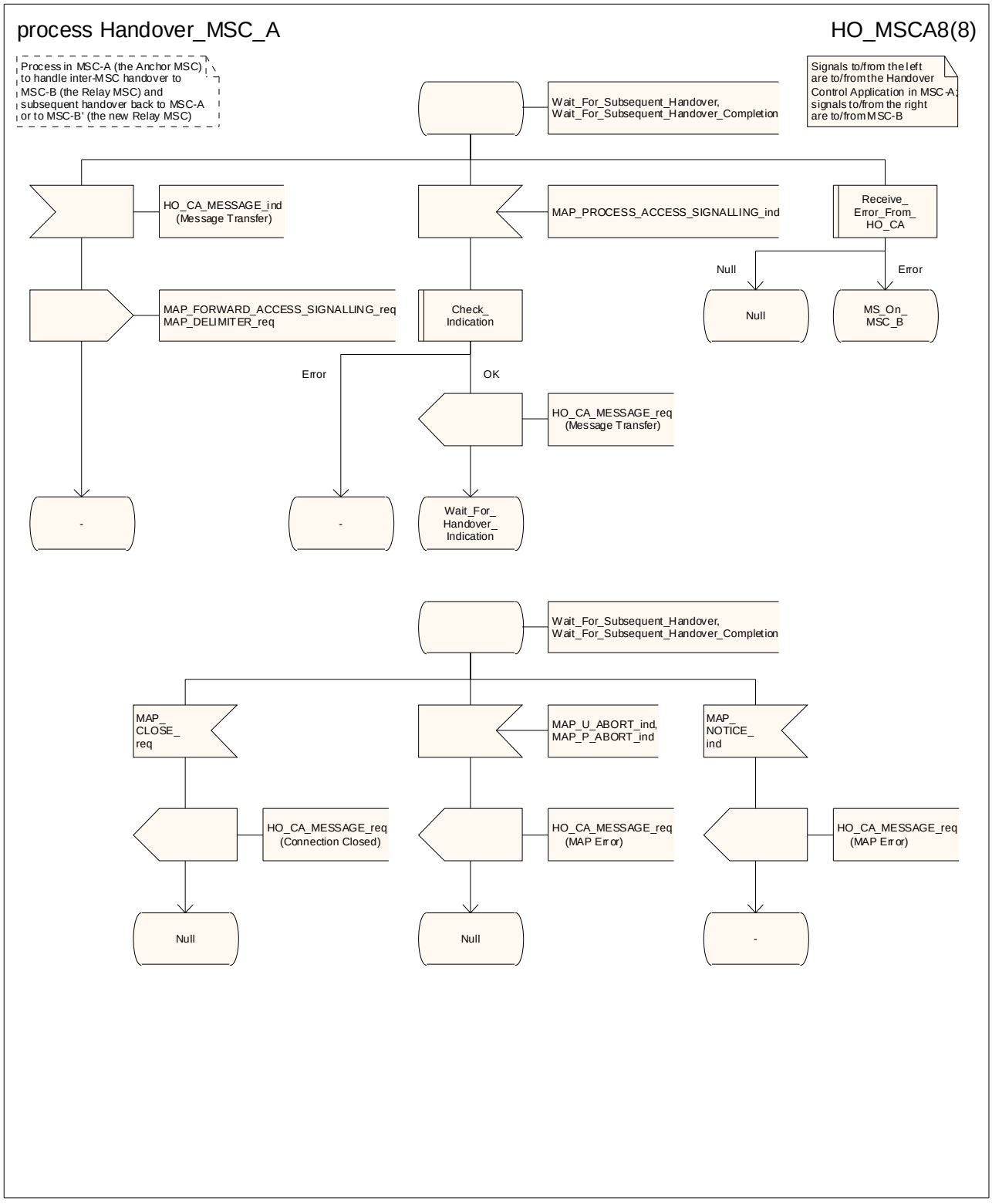


Figure 19.2/4 (sheet 8 of 8): Process HO_MSC_A

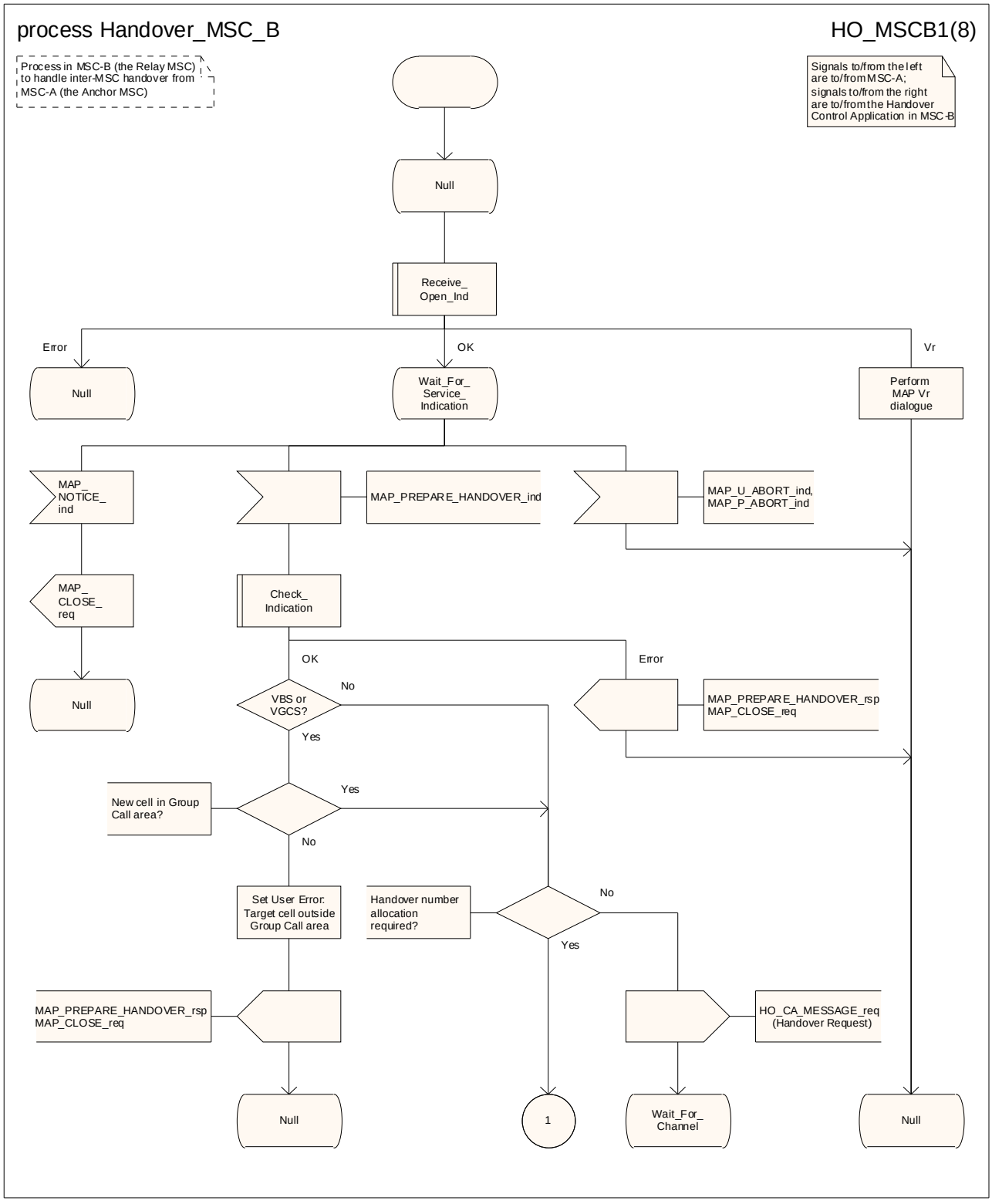


Figure 19.2/5 (sheet 1 of 8): Process HO_MSC_B

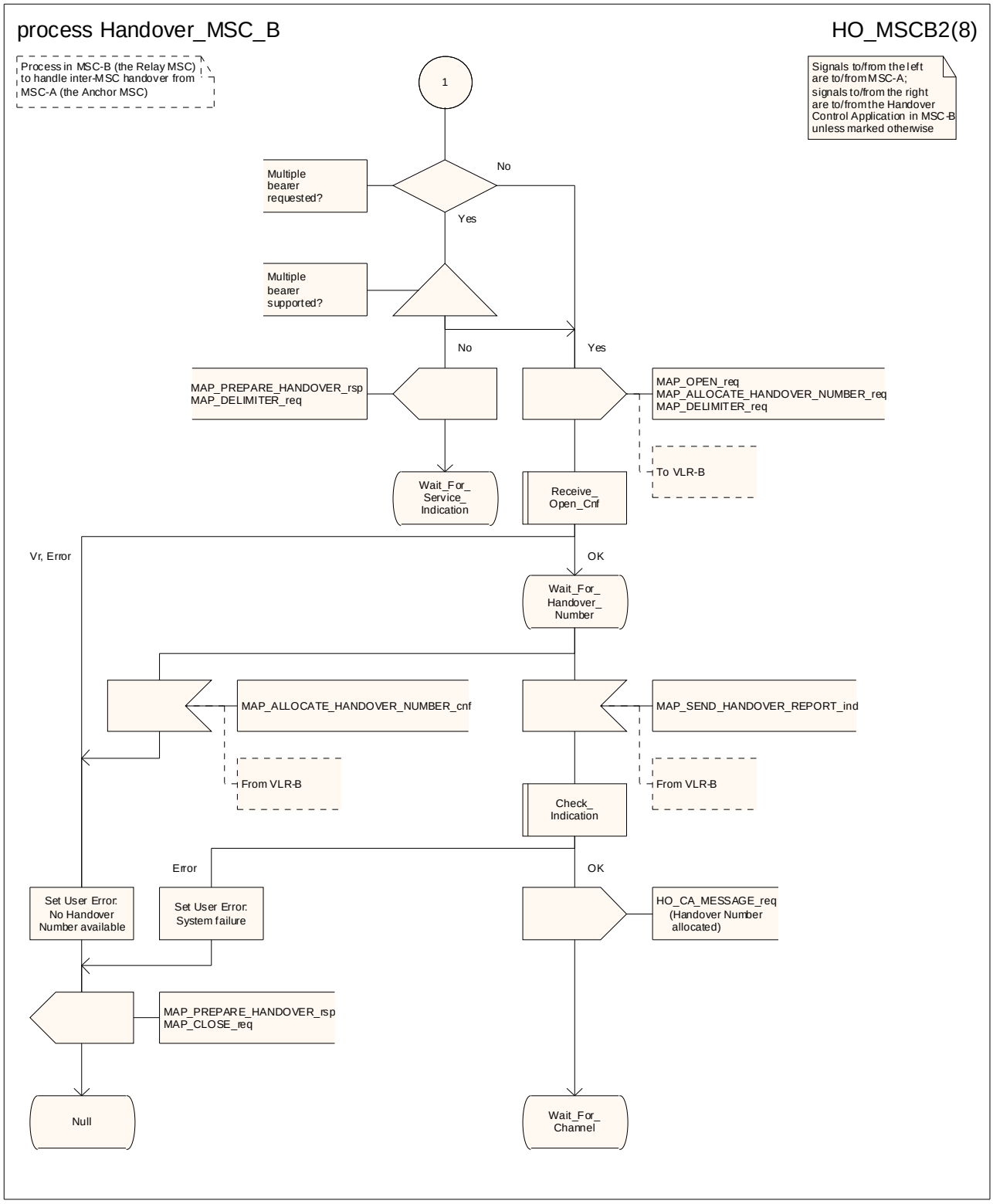


Figure 19.2/5 (sheet 2 of 8): Process HO_MSC_B

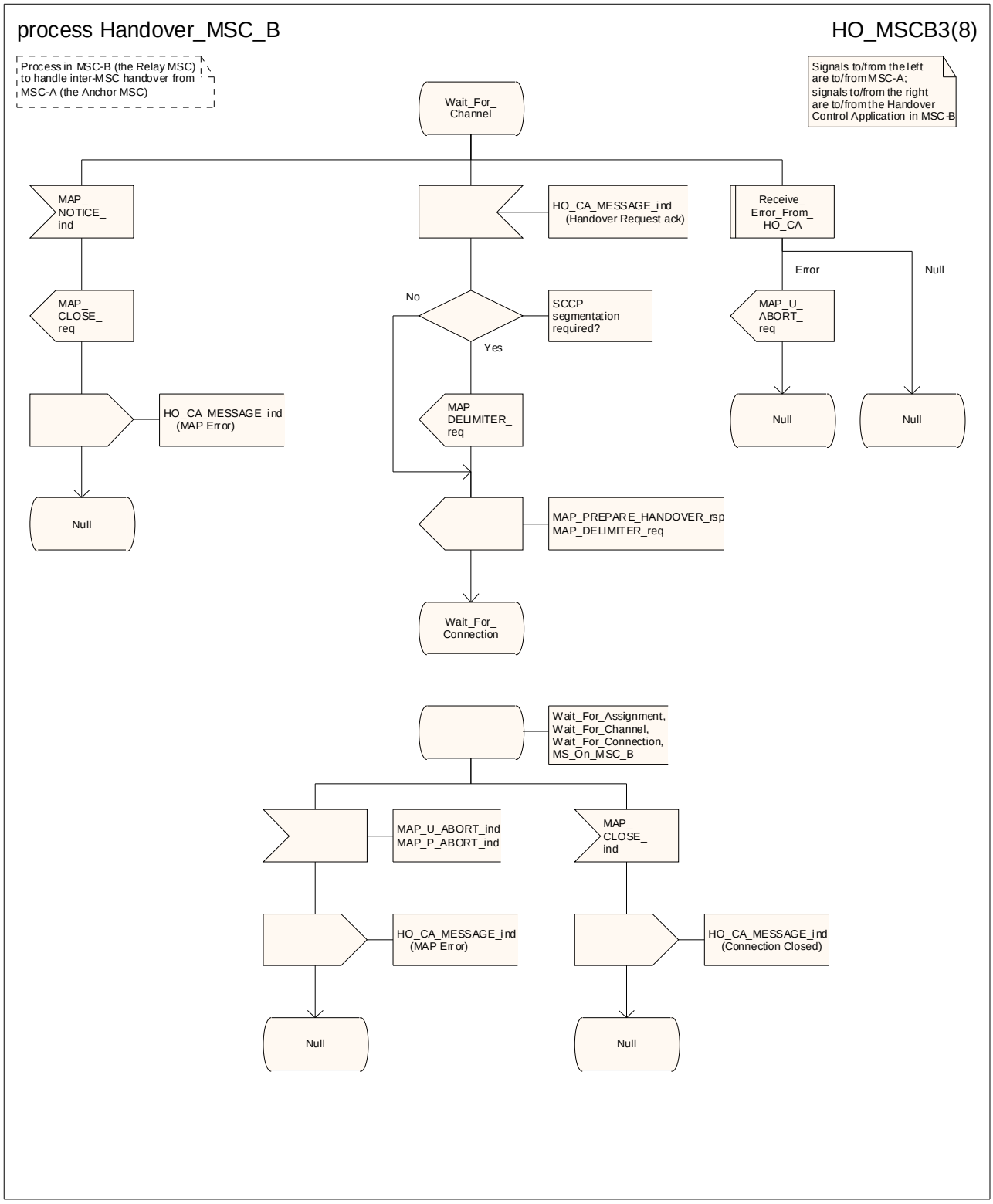


Figure 19.2/5 (sheet 3 of 8): Process HO_MSC_B

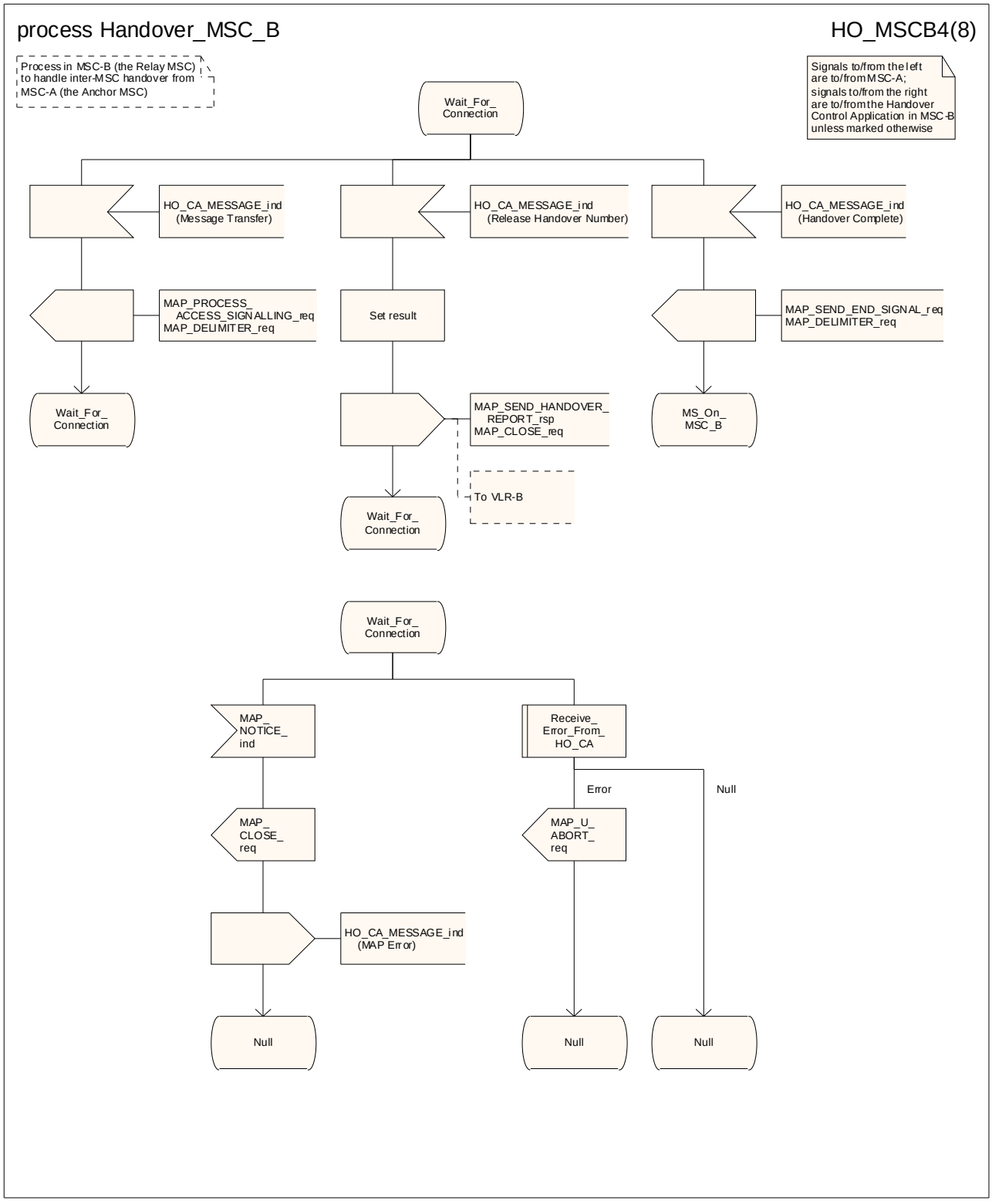


Figure 19.2/5 (sheet 4 of 8): Process HO_MSC_B

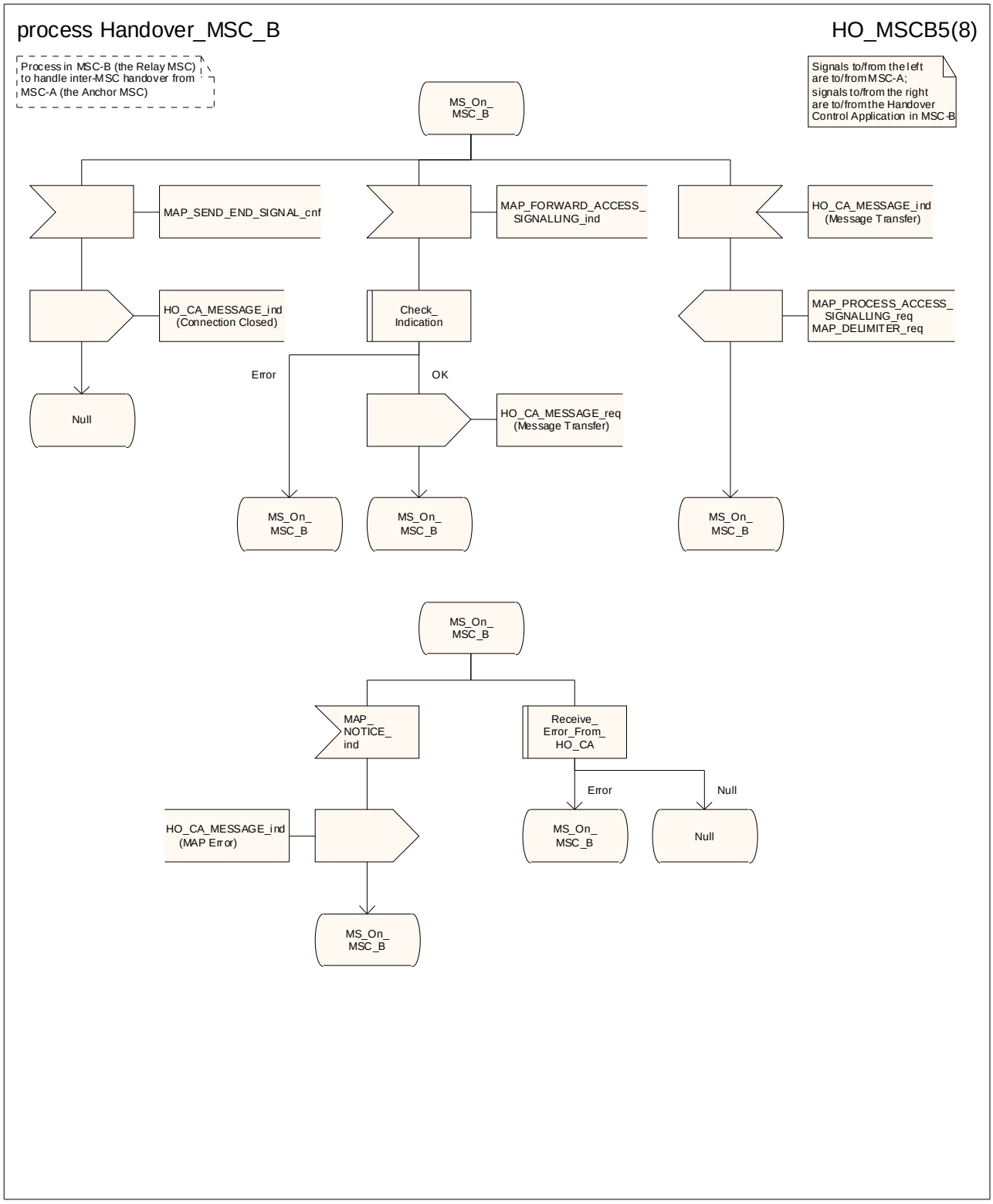


Figure 19.2/5 (sheet 5 of 8): Process HO_MSC_B

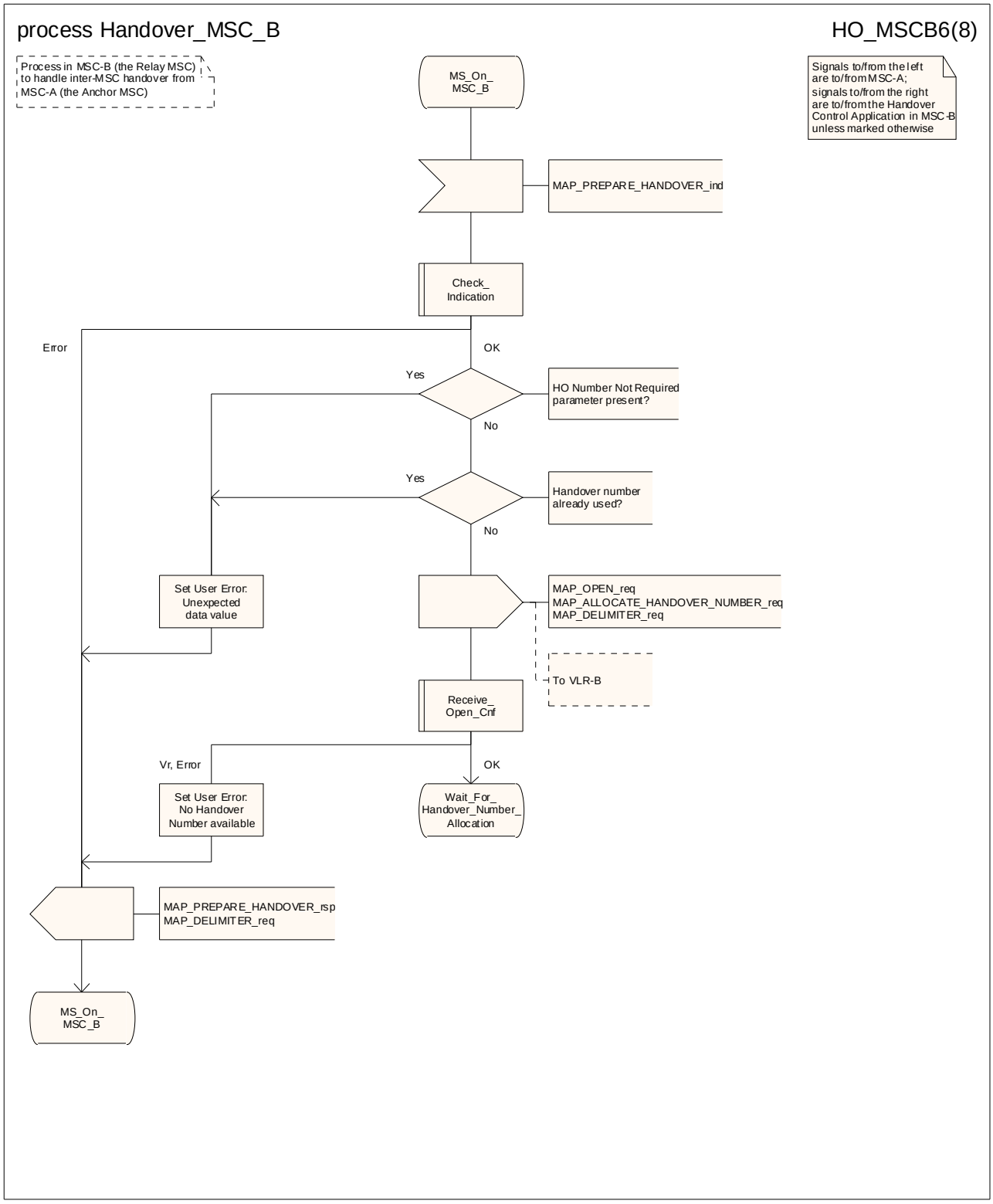


Figure 19.2/5 (sheet 6 of 8): Process HO_MSC_B

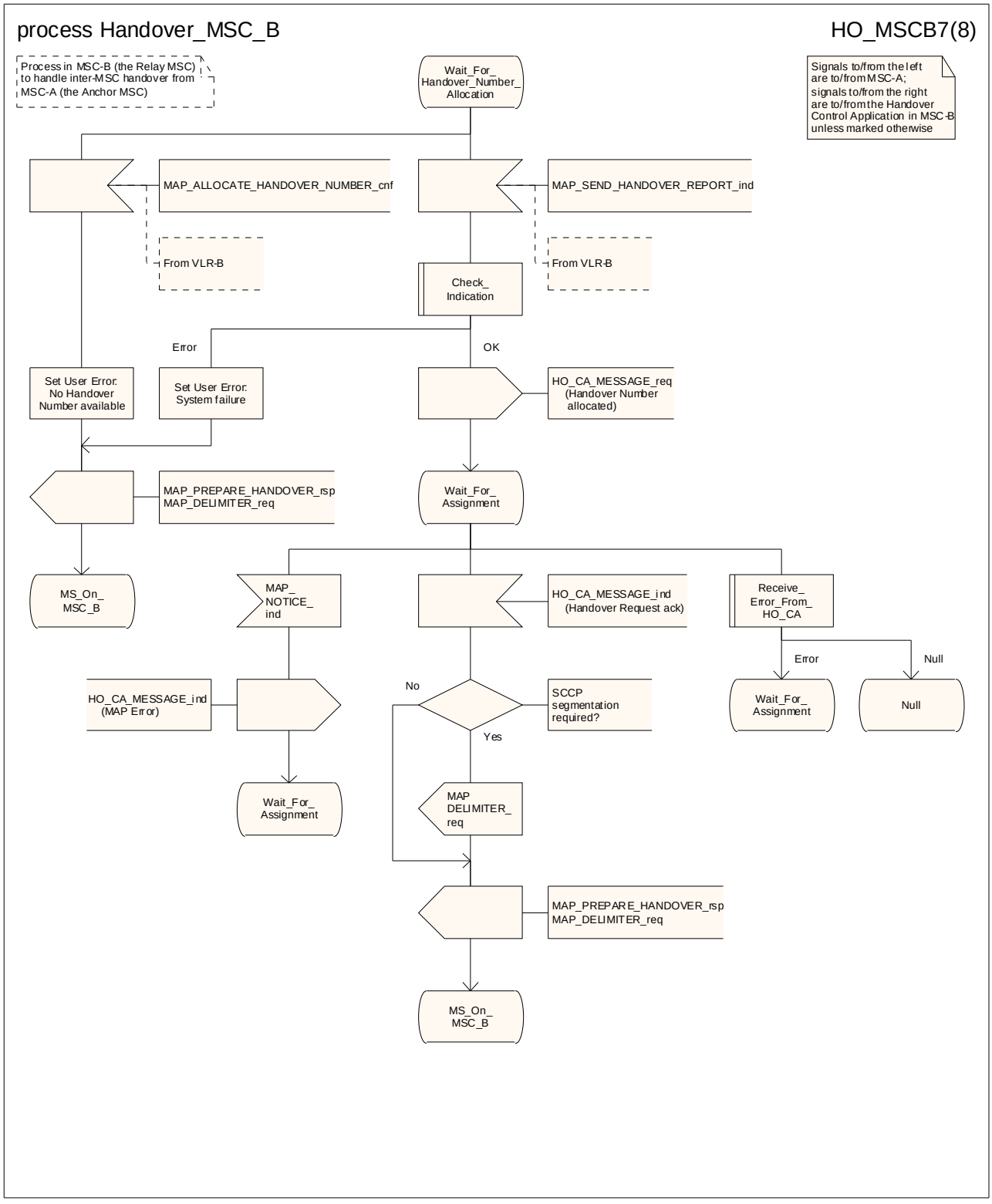


Figure 19.2/5 (sheet 7 of 8): Process HO_MSC_B

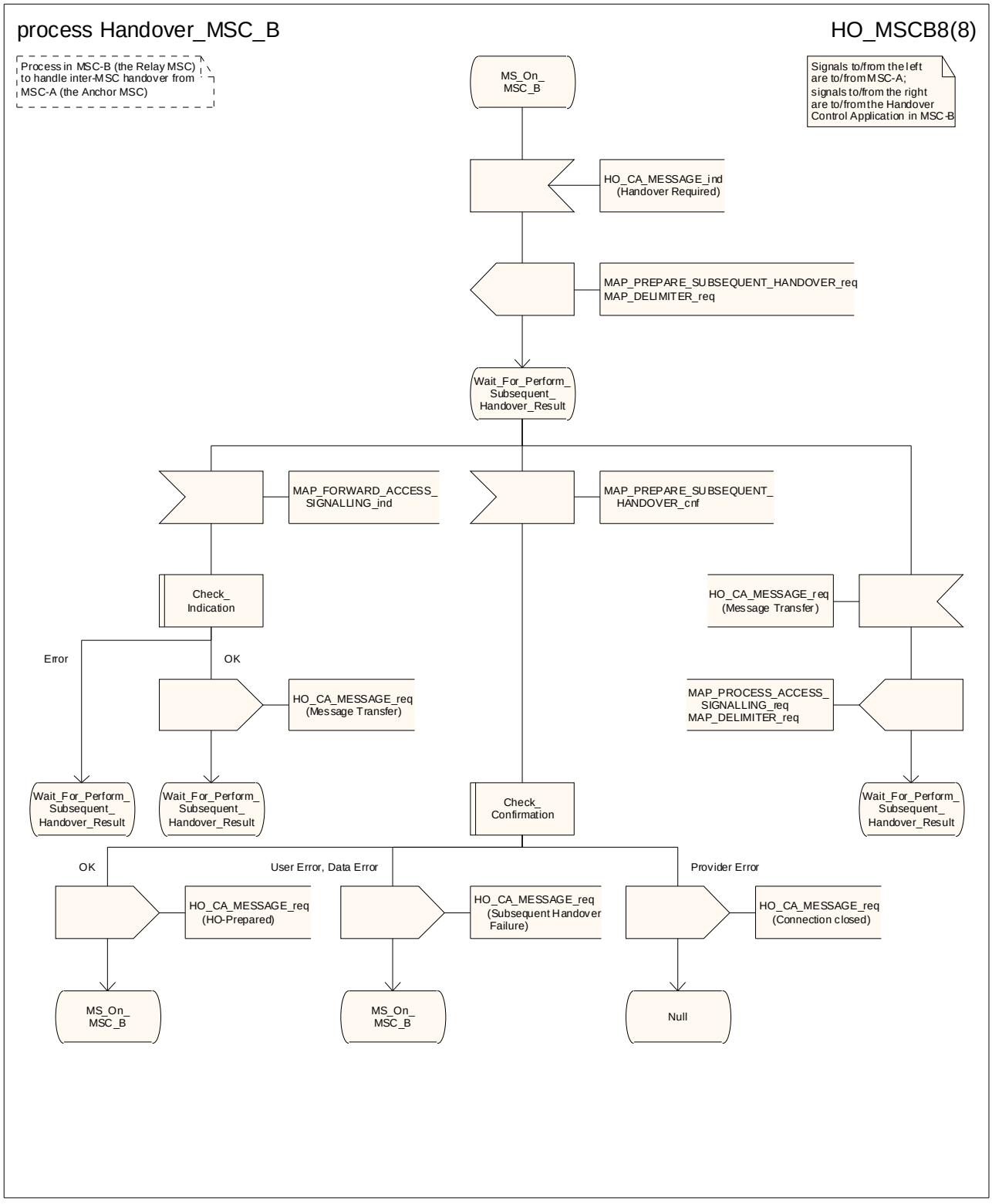


Figure 19.2/5 (sheet 8 of 8): Process HO_MSC_B

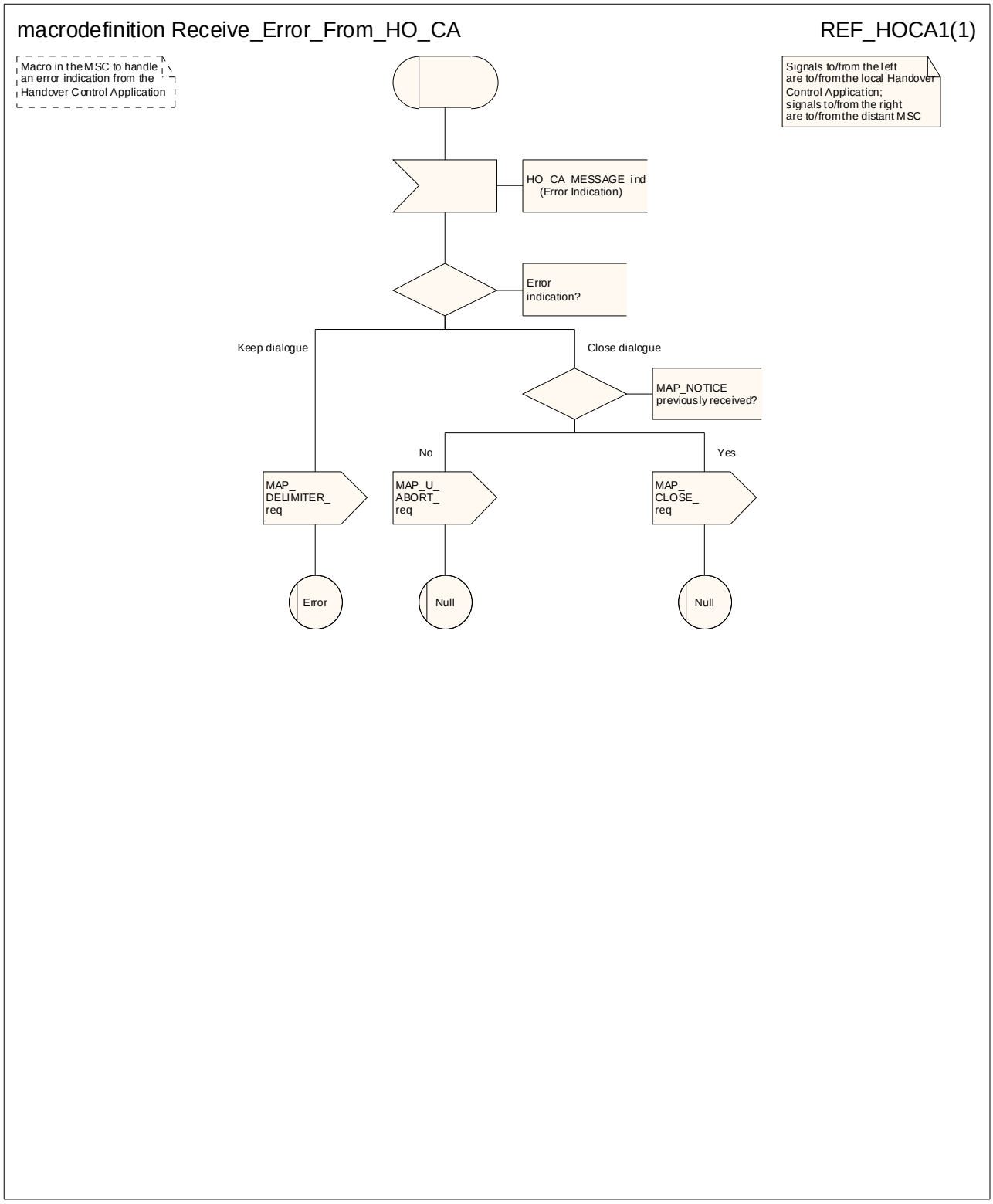


Figure 19.2/6: Macro Receive_error_from_HO_CA

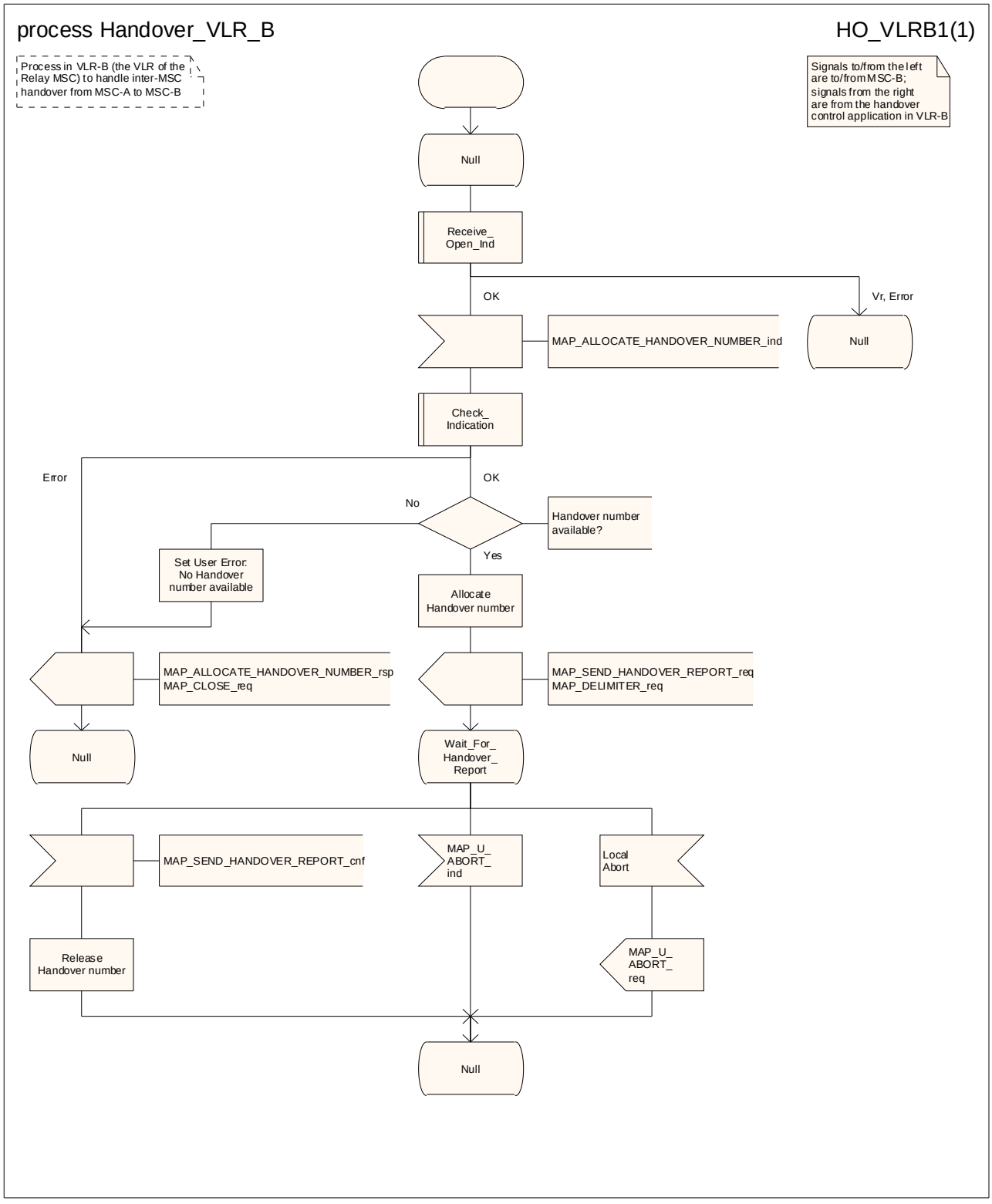


Figure 19.2/7: Process HO_VLR_B

19.3 Fault recovery procedures

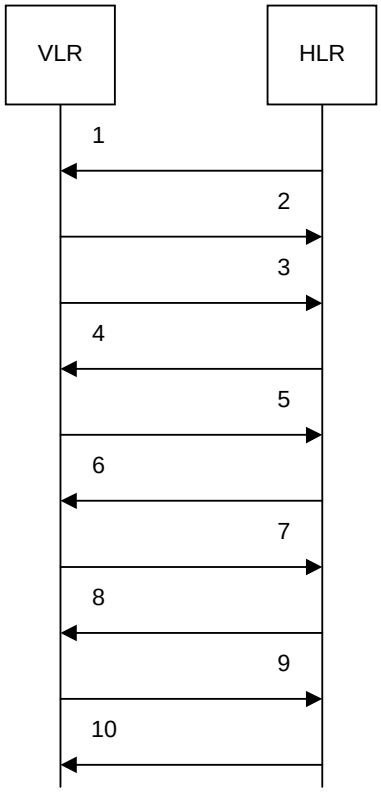
When a location register has restarted after a fault, the fault recovery procedures ensure that the subscriber data in the VLR or in the SGSN become consistent with the subscriber data that are stored in the HLR for the MS concerned and that the location information in the HLR , the VLR and the SGSN reflect accurately the current location of the MS.

The stage 2 specification of fault recovery procedures in location registers is 3GPP TS 23.007 [19].

19.3.1 VLR fault recovery procedures

19.3.1.1 General

Restoration of an IMSI record in a VLR can be triggered by a location registration request from the MS or by a request from the HLR for a roaming number to route a mobile terminated call to the MS. If the restoration is triggered by a location registration request from the MS, the VLR performs the location updating procedure described in 3GPP TS 23.012 [23] and subclause 19.1.1 of the present document. If the restoration is triggered by a request for a roaming number, the VLR provides the roaming number and triggers an independent dialogue to restore the subscriber data as described in 3GPP TS 23.018 [97]. The message flow for data restoration triggered by a request for a roaming number is shown in figure 19.3.1/1.



1) MAP_PROVIDE_ROAMING_NUMBER_req/ind
2) MAP_PROVIDE_ROAMING_NUMBER_rsp/cnf
3) *MAP_SEND_AUTHENTICATION_INFO_req/ind* (Note 1, note 2)
4) *MAP_SEND_AUTHENTICATION_INFO_rsp/cnf* (Note 1, note 2)
5) MAP_RESTORE_DATA_req/ind
6) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 1, note 3)
7) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 1, note 3)
8) MAP_INSERT_SUBSCRIBER_DATA_req/ind
9) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
10) MAP_RESTORE_DATA_rsp/cnf

NOTE 1: Services printed in *italics* are optional.
NOTE 2: If authentication is required.

NOTE 3: If subscriber tracing is active in the HLR.

Figure 19.3/1: Message flow for VLR restoration at mobile terminated call set-up

19.3.1.2 Procedure in the VLR

The procedure in the VLR to handle a dialogue for subscriber data restoration is defined in subclause 21.2.6 of the present document.

19.3.1.3 Procedure in the HLR

The MAP process in the HLR to handle a request for data restoration in the VLR is shown in figure 19.3.1/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Indication see subclause 25.2.1;

Control_Tracing_With_VLR_HLRsee subclause 25.9.6.

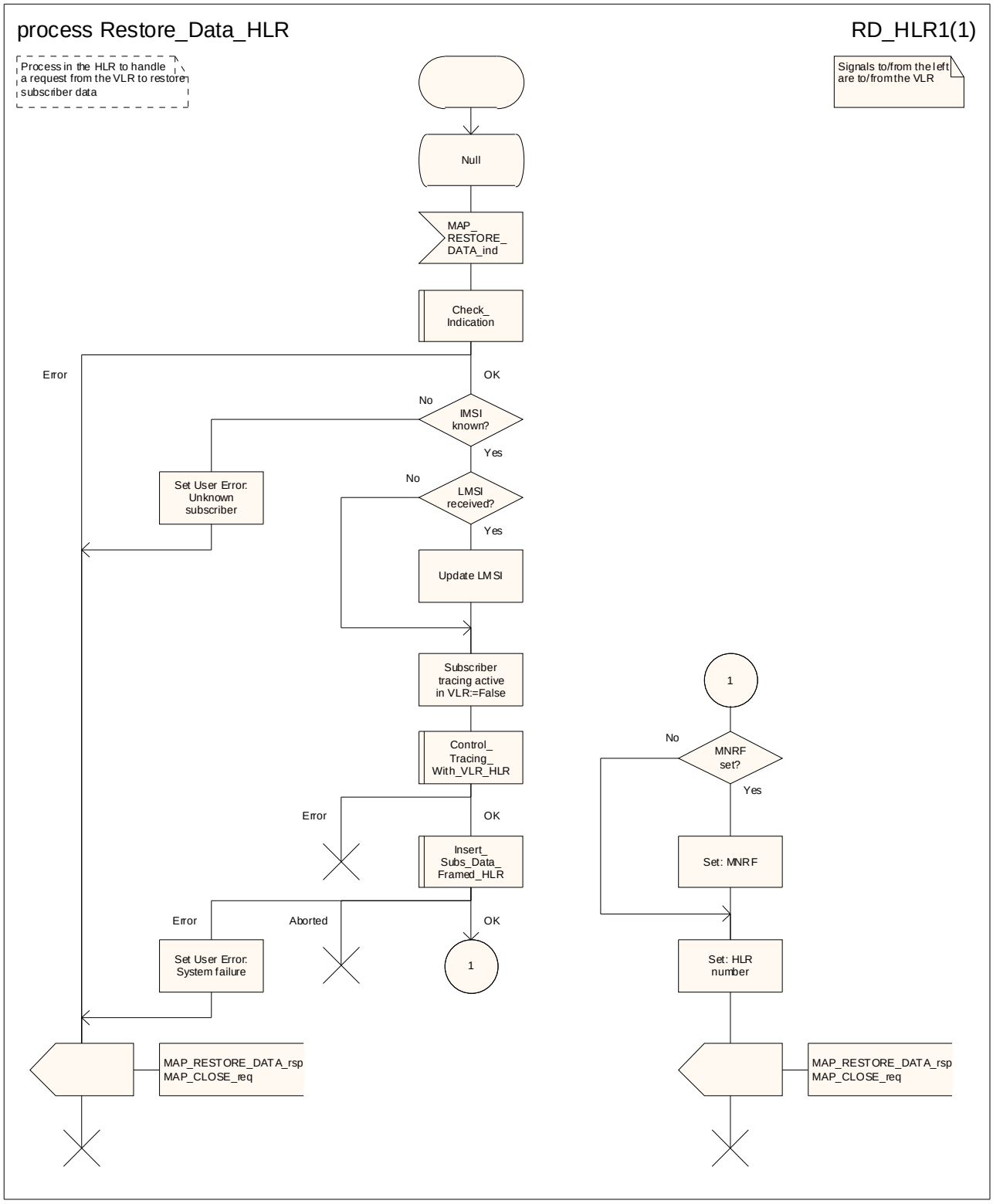


Figure 19.3.1/2: Process Restore_Data_HLR

19.3.2 HLR fault recovery procedures

19.3.2.1 General

For the HLR, periodic back-up of data to non-volatile memory is mandatory.

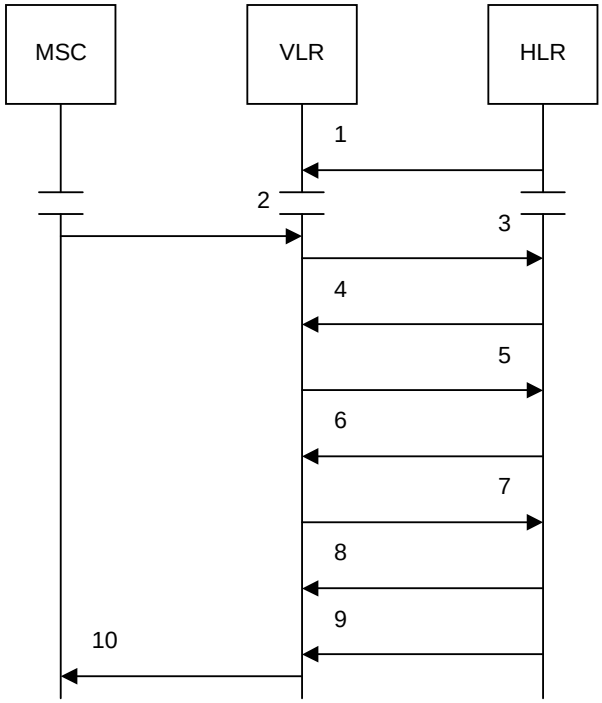
Data that have been changed after the last back-up and before the restart of the HLR cannot be recovered by reload from the non-volatile memory. Therefore, a restoration procedure is triggered for each IMSI record that has been affected by the HLR fault at the first authenticated radio contact with the MS concerned.

As an implementation option, a notification can be forwarded to the MS to alert the subscriber to check the parameters for supplementary services that allow subscriber controlled input (MAP_FORWARD_CHECK_SS_INDICATION service). If the VLR receives this notification from the HLR it shall forward the notification to the MS. If the Gs-interface is implemented the VLR shall not forward this notification.

A restoration procedure may also be triggered for IMSI records that shares subscription data with other IMSI records when the shared subscription data is modified, added or deleted. This option presumes the support of Reset-IDs.

The message flow for HLR restoration for a non-GPRS subscriber is shown in figure 19.3.2/1.

The message flow for HLR restoration for a GPRS subscriber is shown in figure 19.3.2/2.

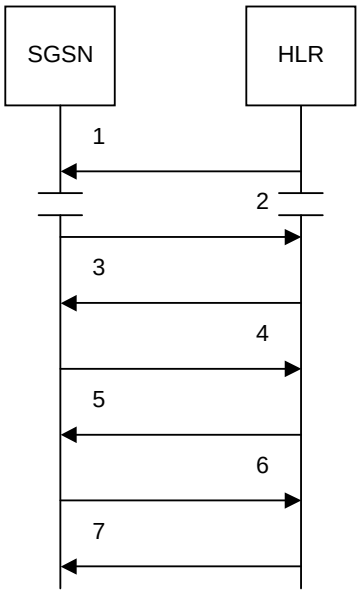


1) MAP_RESET_req/ind
2) MAP_PROCESS_ACCESS_REQUEST_req/ind
3) MAP_UPDATE_LOCATION_req/ind
4) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 1, Note 2)
5) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 1, Note 2)
6) MAP_INSERT_SUBSCRIBER_DATA_req/ind
7) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
8) MAP_UPDATE_LOCATION_rsp/cnf
9) *MAP_FORWARD_CHECK_SS_INDICATION_req/ind* (Note 1)
10) *MAP_FORWARD_CHECK_SS_INDICATION_req/ind* (Note 1)

NOTE 1: Services printed in *italics* are optional.
NOTE 2: If subscriber tracing is active in the HLR.

Steps 2 to 10 may be skipped if the procedure is used to update shared subscription data.

Figure 19.3.2/1: Message flow for HLR restoration (non-GPRS)



1) MAP_RESET_req/ind
2) MAP_UPDATE_GPRS_LOCATION_req/ind
3) *MAP_ACTIVATE_TRACE_MODE_req/ind* (Note 1, Note 2)
4) *MAP_ACTIVATE_TRACE_MODE_rsp/cnf* (Note 1, Note 2)
5) MAP_INSERT_SUBSCRIBER_DATA_req/ind
6) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
7) MAP_UPDATE_GPRS_LOCATION_rsp/cnf

NOTE 1: Services printed in *italics* are optional.
NOTE 2: If subscriber tracing is active in the HLR.

Steps 2 to 7 may be skipped if the procedure is used to update shared subscription data.

Figure 19.3.2/2: Message flow for HLR restoration (GPRS)

19.3.2.2 Procedure in the HLR

The MAP process in the HLR to notify the relevant serving nodes that the HLR has restarted is shown in figure 19.3.2/3.

The SGSN address list includes one instance of the address of each SGSN in which (according to the HLR data retrieved from the non-volatile memory) there is at least one subscriber registered who is affected by the HLR restart.

The VLR address list includes one instance of the address of each VLR in which (according to the HLR data retrieved from the non-volatile memory) there is at least one subscriber registered who is affected by the HLR restart.

The MAP process in the HLR to notify a VLR that the HLR has restarted is shown in figure 19.3.2/4. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Cnf see subclause 25.1.2.

The MAP process in the HLR to notify an SGSN that the HLR has restarted is shown in figure 19.3.2/5. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Cnf see subclause 25.1.2.

19.3.2.3 Procedure in the VLR

The MAP process in the VLR to handle a notification that an HLR has restarted is shown in figure 19.3.2/6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

When Reset-IDs are not supported or not present in the MAP_RESET indication, the VLR uses the HLR number or the HLR identity list included in the MAP_RESET indication to identify the IMSI records which are affected by the HLR restart.
When Reset-IDs are supported and present in the MAP_RESET indication, the VLR uses the Reset-IDs included in the MAP_RESET indication to identify the affected IMSI records.

The task "Update Data" includes any required action to let the update come into effect.
If the update of shared subscription data requires only local updates in the VLR (i.e., the update of the profile does not imply to initiate any signalling interaction towards other network nodes), the updates should be performed immediately (e.g. deleting an Operator Determined Barring).
If the update of shared subscription data implies initiating a signalling interaction towards other nodes, the signalling towards other nodes should be deferred to the next authenticated radio contact with that UE.

NOTE: The rational for the recommendation to defer signalling towards other nodes until the next authenticated radio contact is to consider impacts to the network only when the updates are required, and to spread the signalling towards other nodes over some time, based on user's activity.

To avoid high processing/signalling load resulting from shared subscription data update, processing/signalling actions resulting from data updates in the VLR may take a maximum operator configuration-depending time.

19.3.2.4 Procedure in the SGSN

The MAP process in the SGSN to handle a notification that an HLR has restarted is shown in figure 19.3.2/7. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

When Reset-IDs are not supported or not present in the MAP_RESET indication, the SGSN uses the HLR number or the HLR identity list included in the MAP_RESET indication to identify the IMSI records which are affected by the HLR restart.
When Reset-IDs are supported and present in the MAP_RESET indication, the SGSN uses the Reset-IDs included in the MAP_RESET indication to identify the affected IMSI records.
The task "Update Data" includes any required action to let the update come into effect.

If the update of shared subscription data requires only local updates in the SGSN (i.e., the update of the profile does not imply to initiate any signalling interaction towards other network nodes), the updates should be performed immediately (e.g. deleting an Operator Determined Barring).
If the update of shared subscription data implies initiating a signalling interaction towards other nodes, the signalling towards other nodes should be deferred to the next authenticated radio contact with that UE.

NOTE: The rational for the recommendation to defer signalling towards other nodes until the next authenticated radio contact is to consider impacts to the network only when the updates are required, and to spread the signalling towards other nodes over some time, based on user's activity.

To avoid high processing/signalling load resulting from shared subscription data update, processing/signalling actions resulting from data updates in the SGSN may take a maximum operator configuration-depending time.

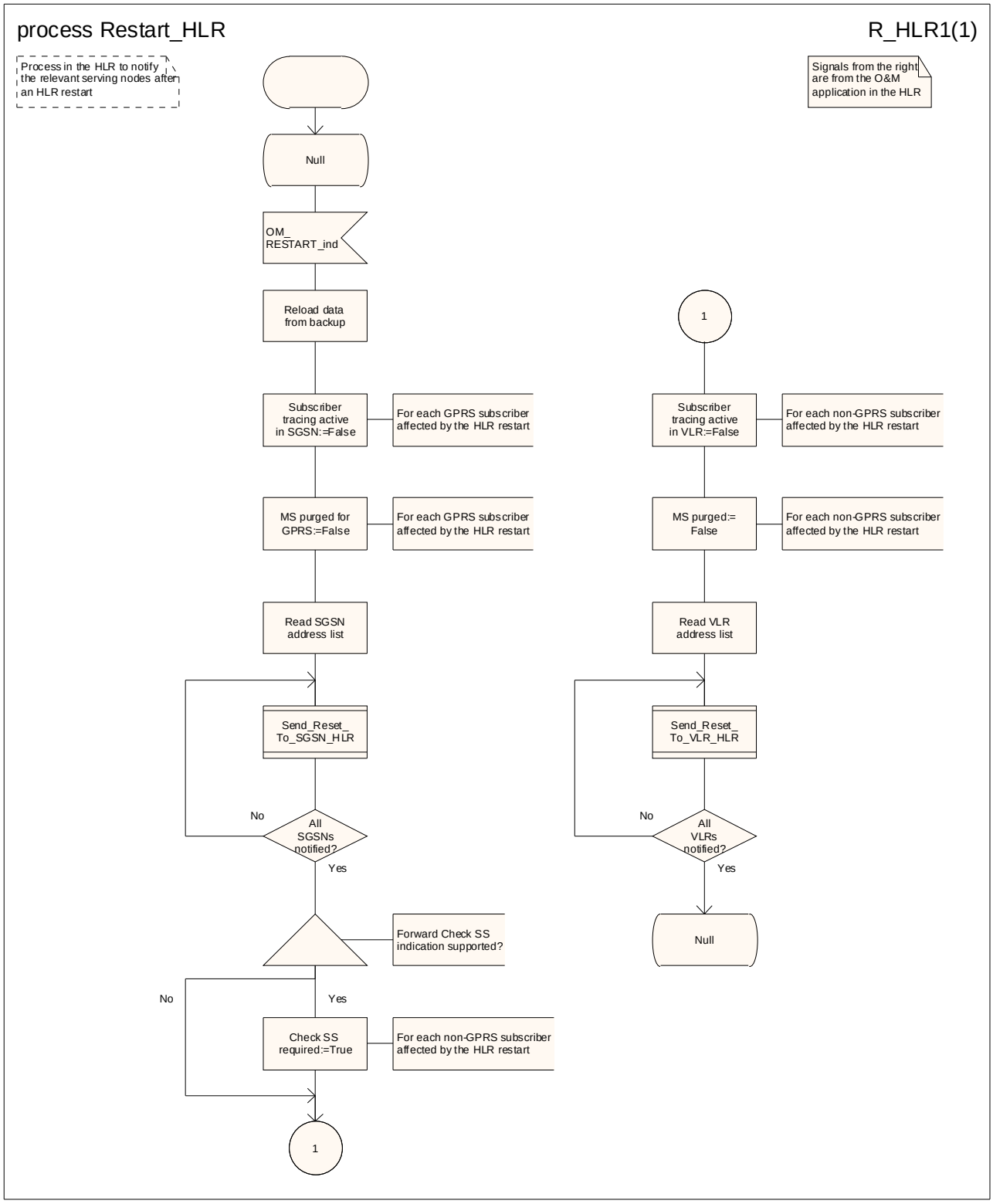


Figure 19.3.2/3: Process Restart_HLR

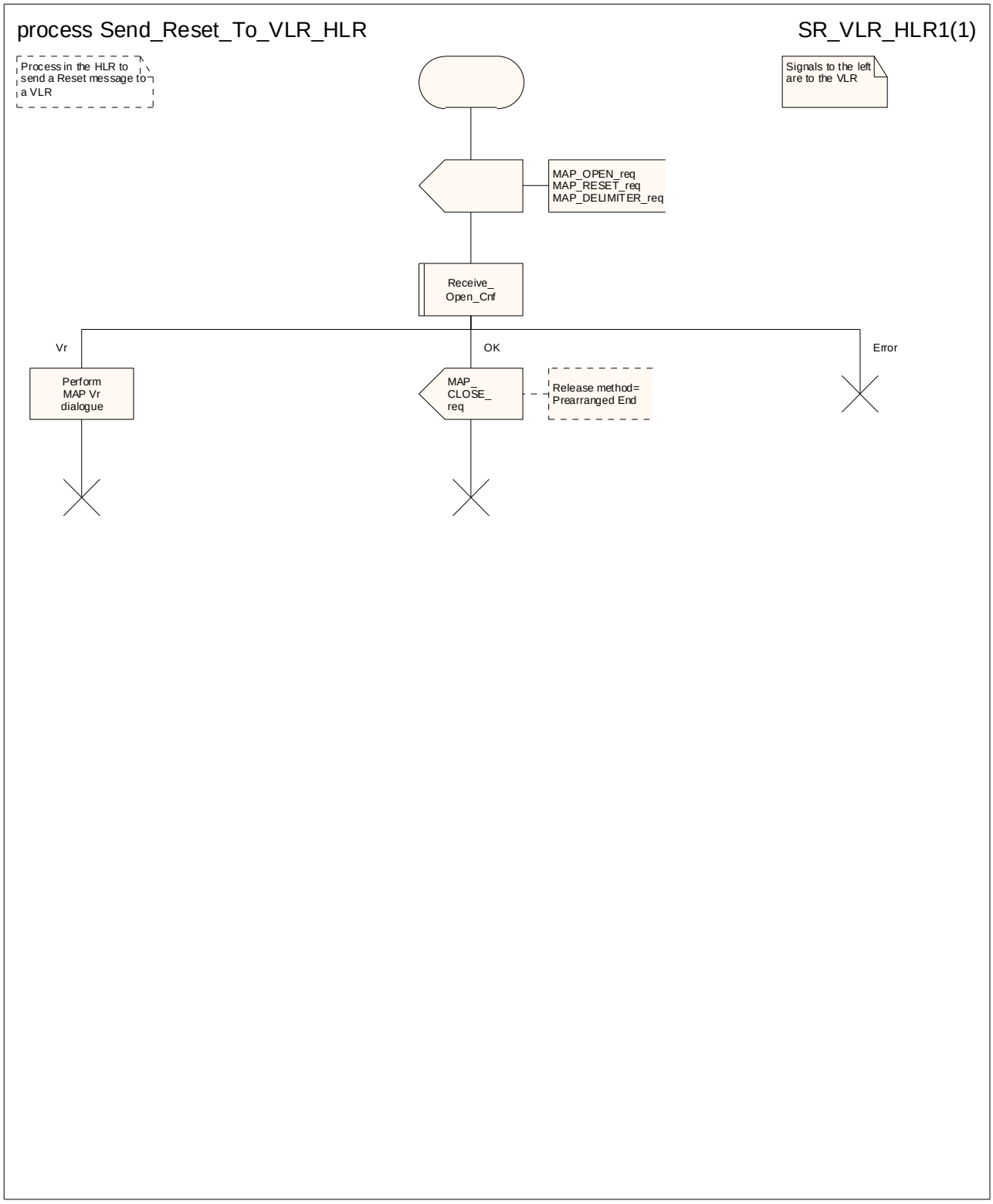


Figure 19.3.2/4: Process Send_Reset_To_VLR_HLR

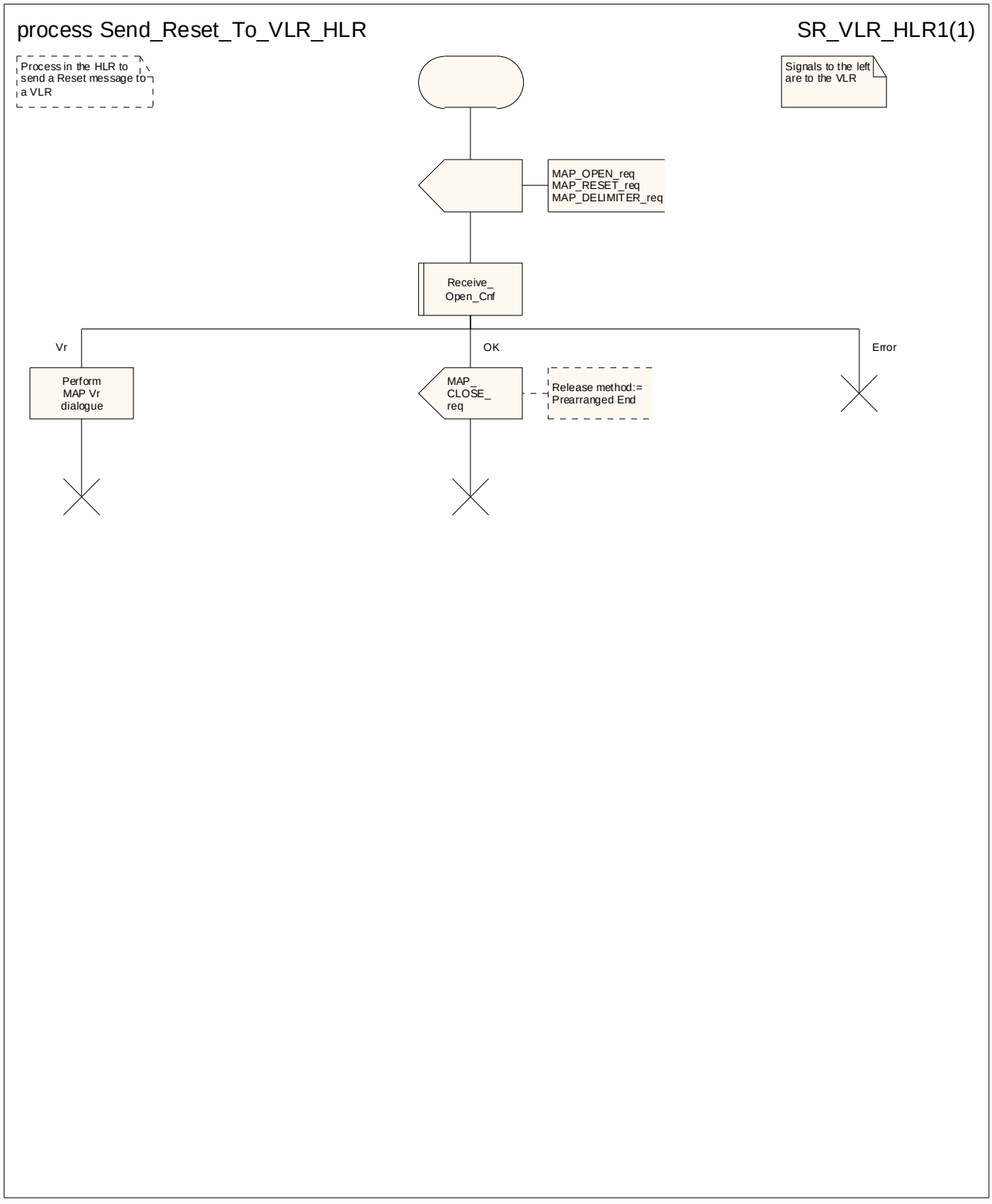


Figure 19.3.2/5: Process Send_Reset_To_SGSN_HLR

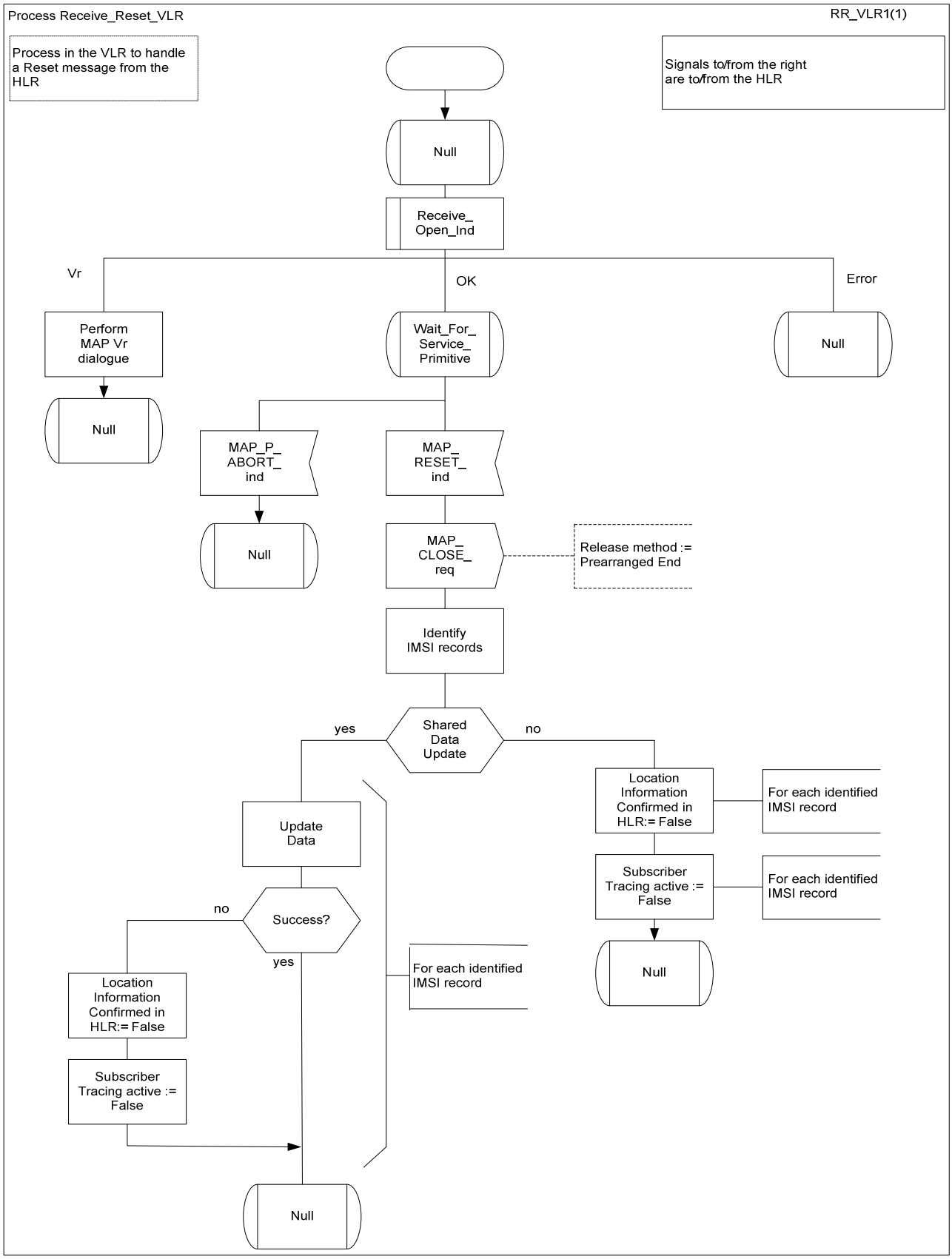


Figure 19.3.2/6: Process Receive_Reset_VLR

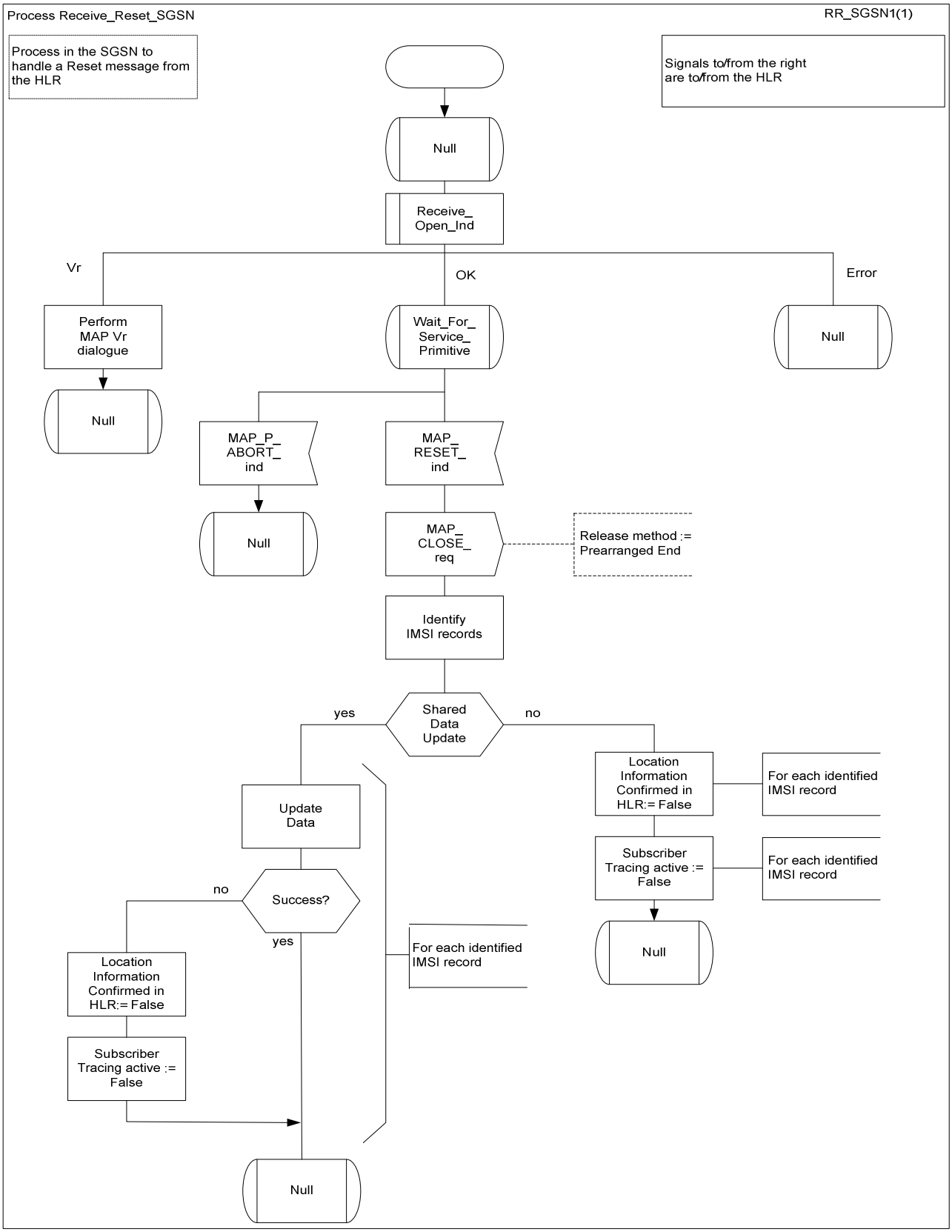


Figure 19.3.2/7: Process Receive_Reset_SGSN

19.3.3 CSS fault recovery procedures

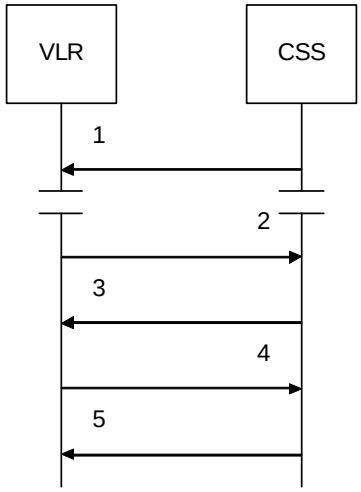
19.3.3.1 General

For the CSS, periodic back-up of data to non-volatile memory is mandatory.

Serving node numbers that have been changed after the last back-up and before the restart of the CSS cannot be recovered by reload from the non-volatile memory. Therefore, a restoration procedure is triggered for each IMSI record that has been affected by the CSS fault at the first authenticated radio contact with the MS concerned.

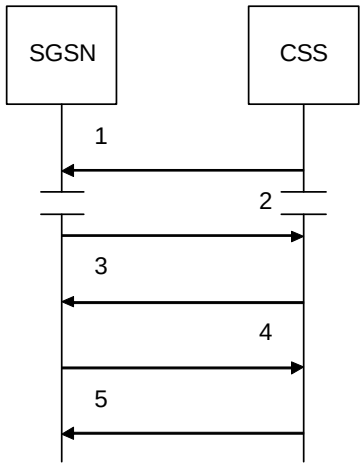
The message flow for CSS restoration for a non-GPRS subscriber is shown in figure 19.3.3/1.

The message flow for CSS restoration for a GPRS subscriber is shown in figure 19.3.3/2.



1) MAP_RESET_req/ind
2) MAP_UPDATE_VCSG_LOCATION_req/ind
3) MAP_INSERT_SUBSCRIBER_DATA_req/ind
4) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
5) MAP_UPDATE_VCSG_LOCATION_rsp/cnf

Figure 19.3.3/1: Message flow for CSS restoration (non-GPRS)



1) MAP_RESET_req/ind
2) MAP_UPDATE_VCSG_LOCATION_req/ind
3) MAP_INSERT_SUBSCRIBER_DATA_req/ind

4) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf
5) MAP_UPDATE_VCSG_LOCATION_rsp/cnf

Figure 19.3.3/2: Message flow for CSS restoration (GPRS)

19.3.3.2 Procedure in the CSS

The MAP process in the CSS to notify the relevant serving nodes that the CSS has restarted is shown in figure 19.3.3/3.

The SGSN address list includes one instance of the address of each SGSN in which (according to the CSS data retrieved from the non-volatile memory) there is at least one subscriber registered who is affected by the CSS restart.

The VLR address list includes one instance of the address of each VLR in which (according to the CSS data retrieved from the non-volatile memory) there is at least one subscriber registered who is affected by the CSS restart.

The MAP process in the CSS to notify a VLR that the CSS has restarted is shown in figure 19.3.3/4. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Cnf see subclause 25.1.2.

The MAP process in the CSS to notify an SGSN that the CSS has restarted is shown in figure 19.3.3/5. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Cnf see subclause 25.1.2.

19.3.3.3 Procedure in the VLR

The MAP process in the VLR to handle a notification that a CSS has restarted is shown in figure 19.3.3/6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

The VLR uses the CSS number (filled in Sending Node number parameter) included in the MAP_RESET indication to identify the user's IMSI records which are affected by the CSS restart.

19.3.3.4 Procedure in the SGSN

The MAP process in the SGSN to handle a notification that a CSS has restarted is shown in figure 19.3.3/6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

The SGSN uses the CSS number (filled in Sending Node number parameter) included in the MAP_RESET indication to identify the user's IMSI records which are affected by the CSS restart.

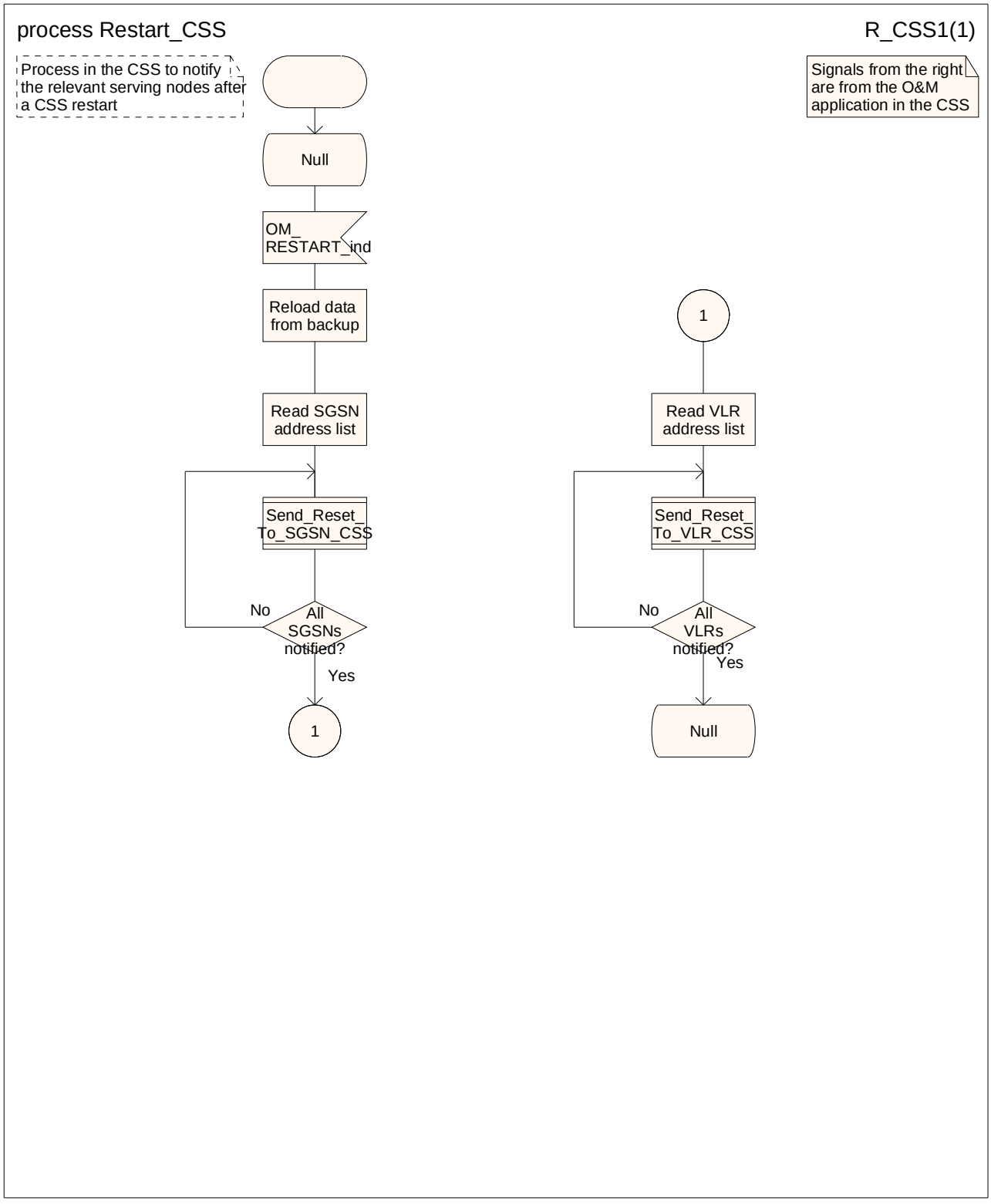


Figure 19.3.3/3: Process Restart_CSS

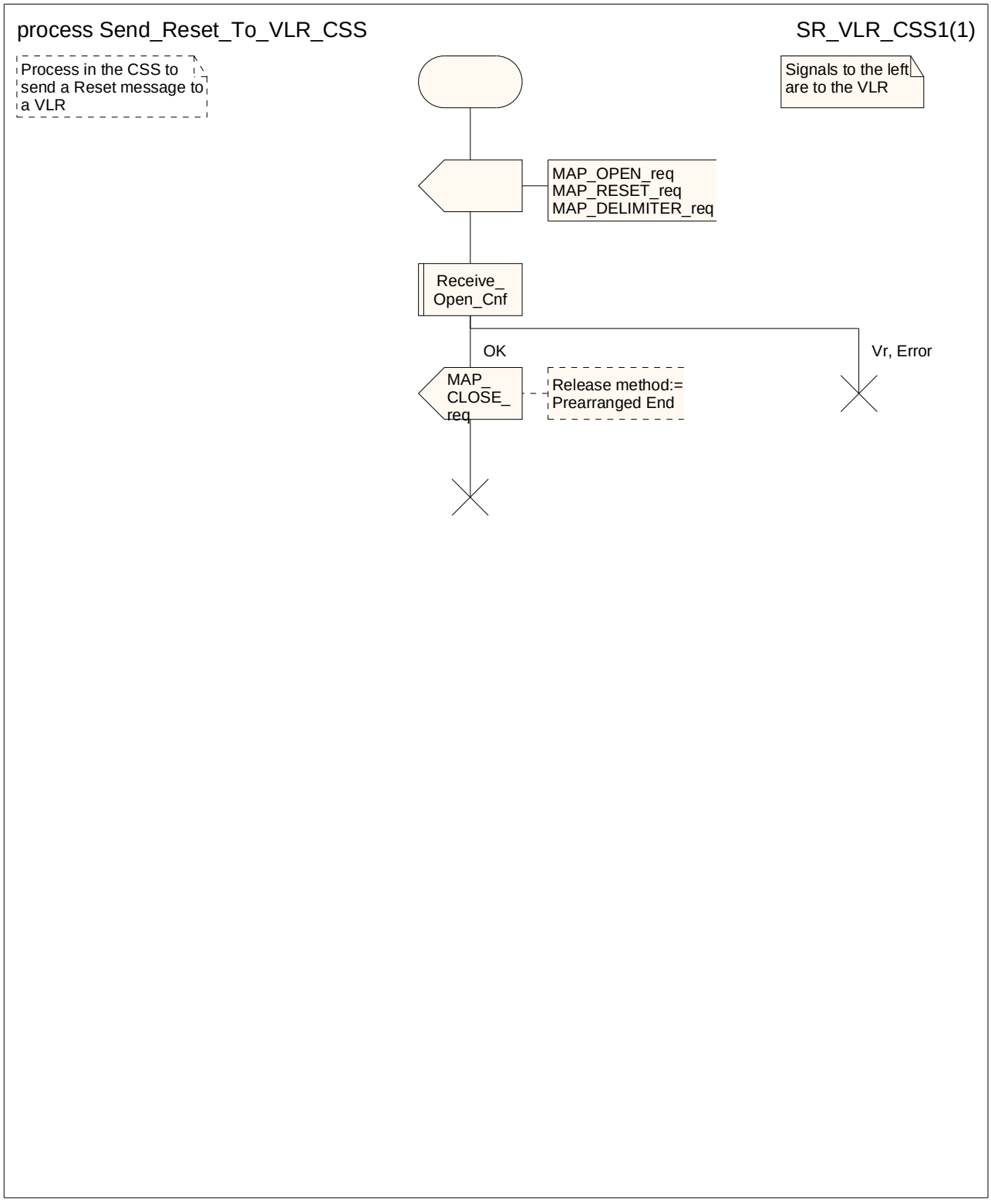


Figure 19.3.3/4: Process Send_Reset_To_VLR_CSS

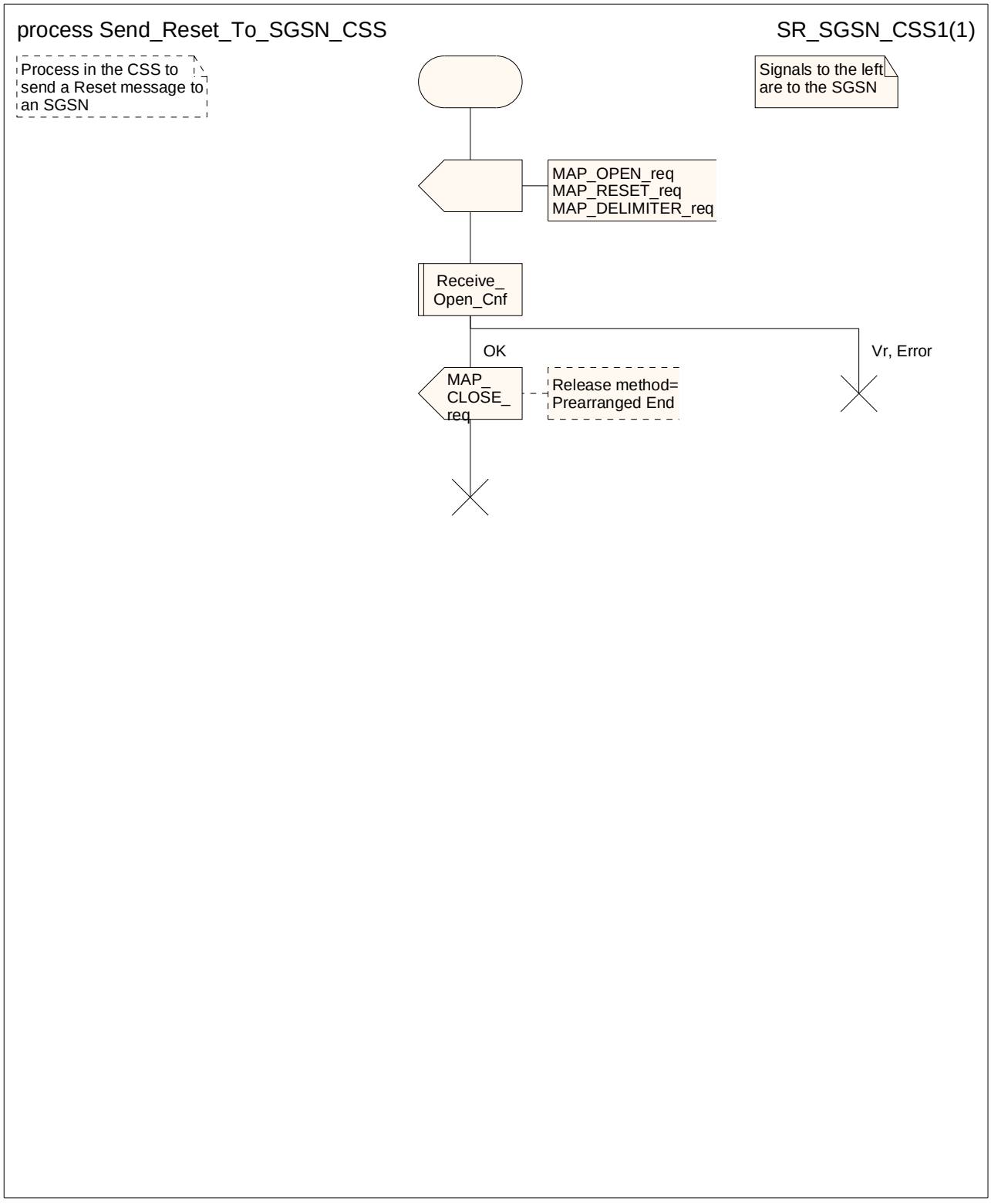


Figure 19.3.3/5: Process Send_Reset_To_SGSN_CSS

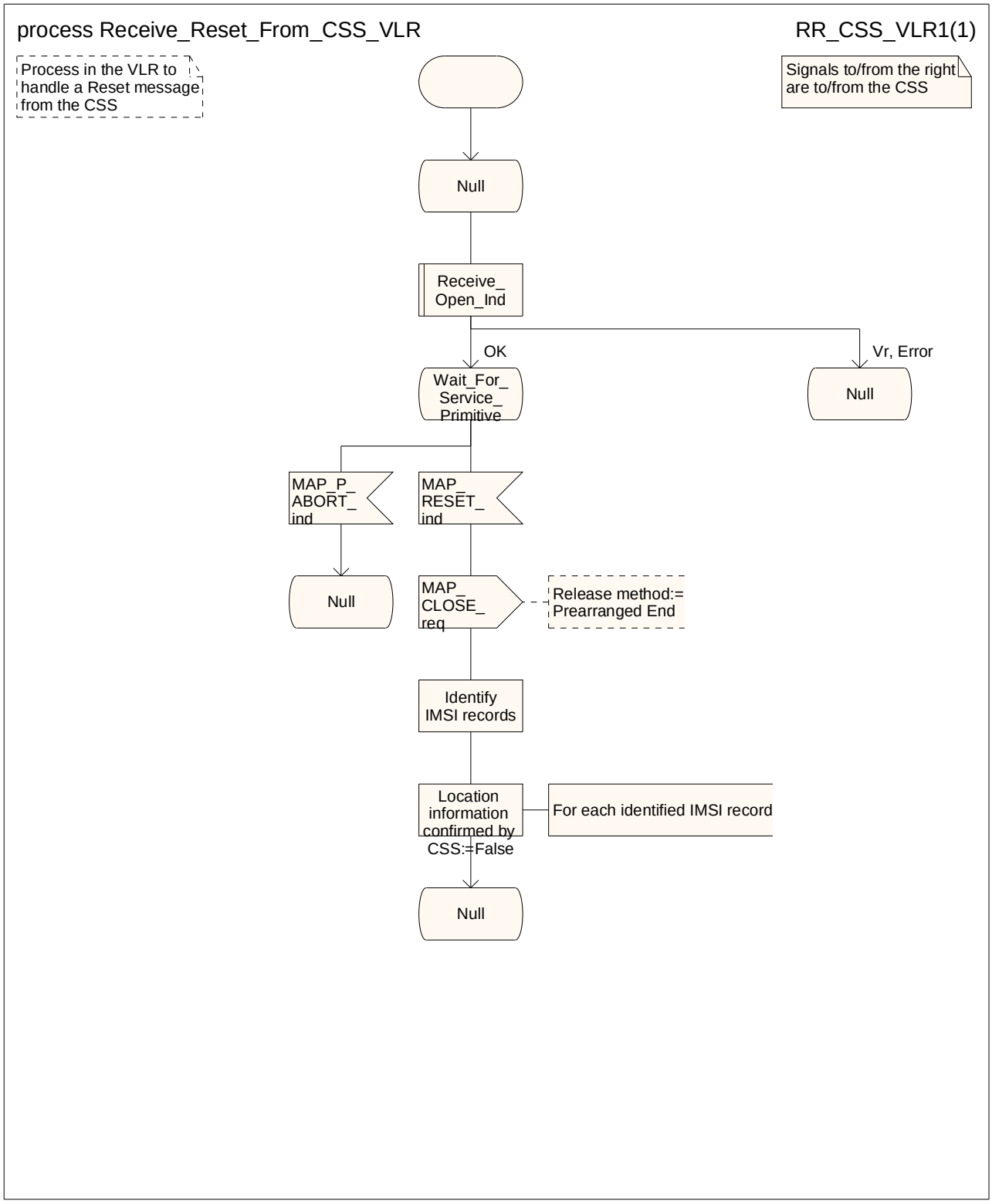


Figure 19.3.3/6: Process Receive_Reset_From_CSS_VLR

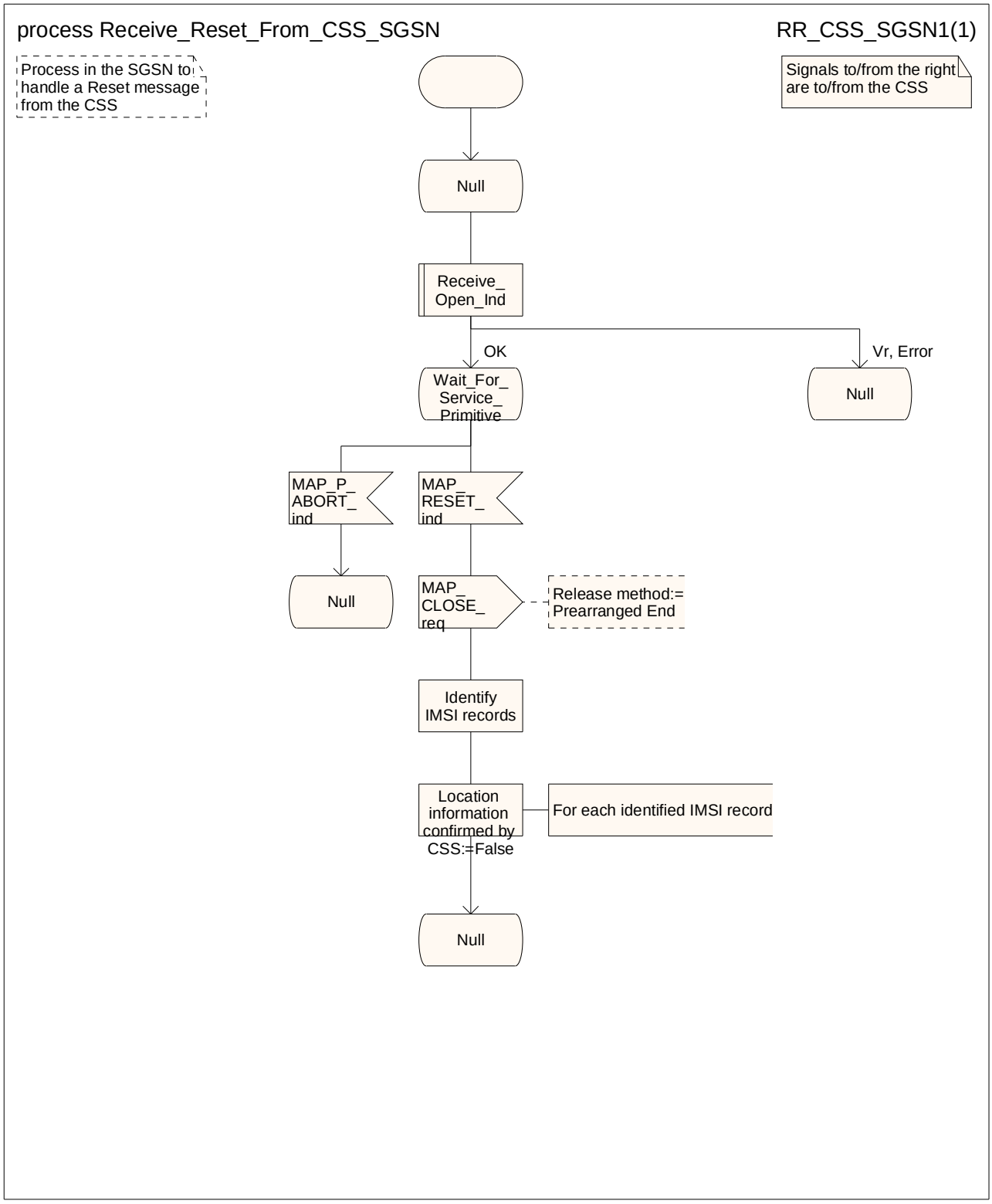


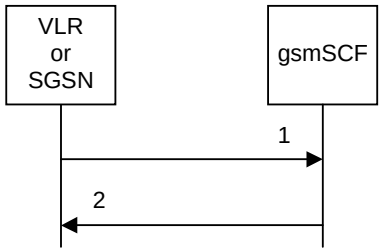
Figure 19.3.3/7: Process Receive_Reset_From_CSS_SGSN

19.4 Mobility Management event notification procedure

19.4.1 General

The Mobility Management event notification procedure is used to notify a gsmSCF about the successful completion of a Mobility Management event.

The message flow for Mobility Management event notification is shown in figure 19.4/1.



1) MAP_REPORT_MM_EVENT_req/ind
2) MAP_REPORT_MM_EVENT_rsp/cnf

Figure 19.5/1: Message flow for Mobility Management event notification

19.4.2 Procedure in the VLR or SGSN

The MAP process in the VLR or the SGSN to report a Mobility Management event to the gsmSCF is shown in figure 19.4/2.The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation: see subclause 25.2.2.

19.4.3 Procedure in the gsmSCF

The MAP process in the gsmSCF to handle the report of a Mobility Management event is shown in figure 19.4/3.The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

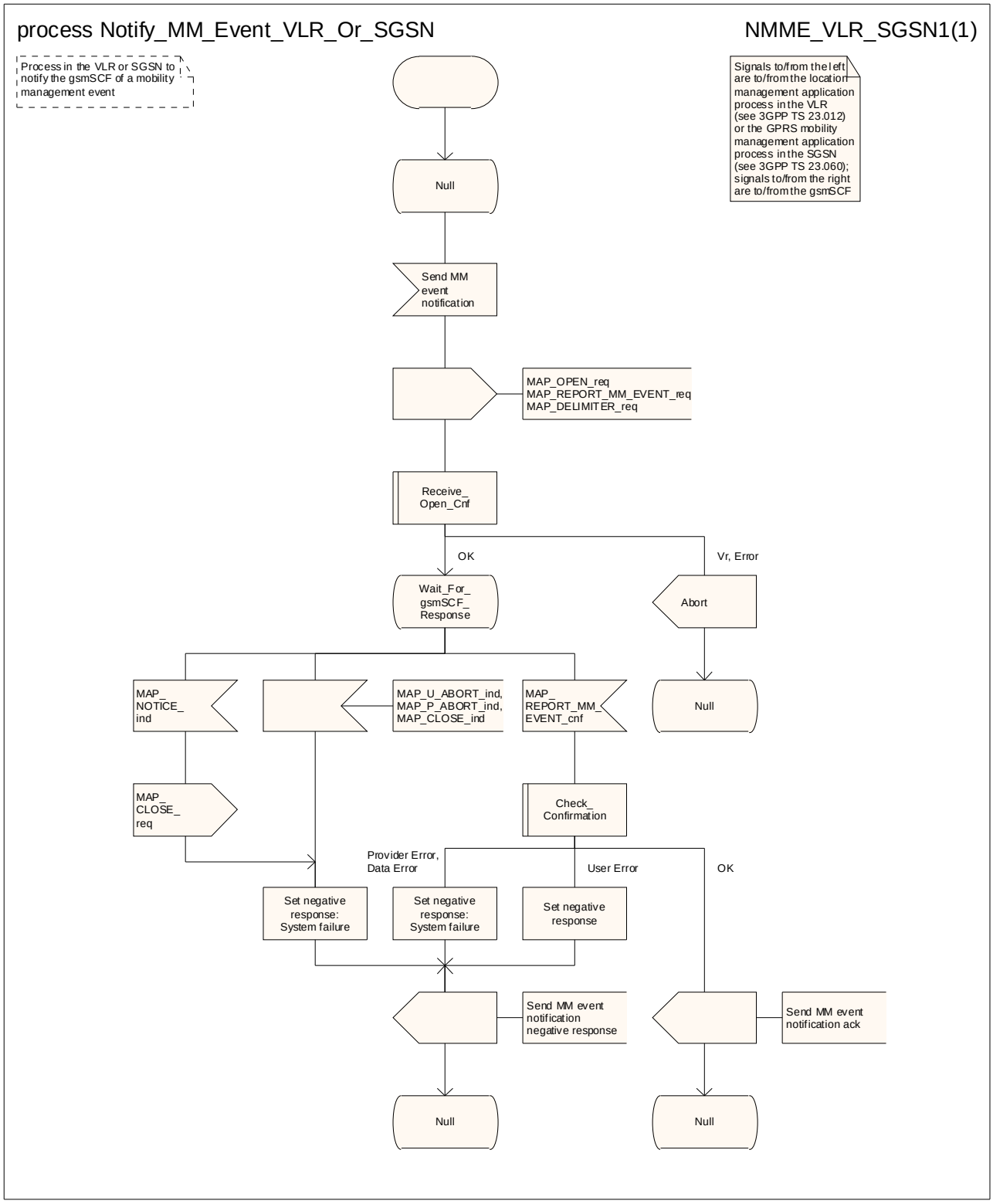


Figure 19.4/2: Process Notify_MM_Event_VLR_Or_SGSN

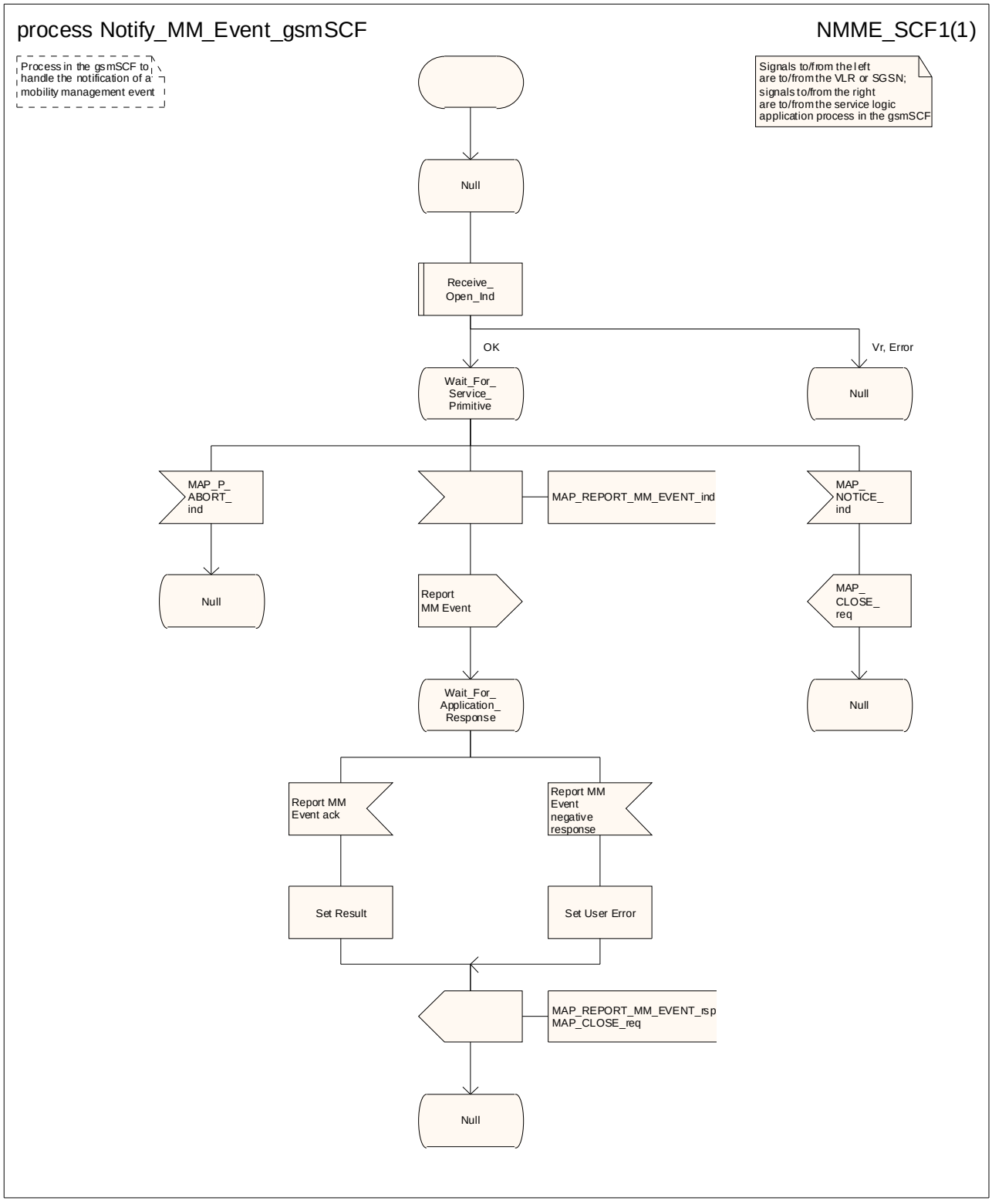


Figure 19.4/3: Process Notify_MM_Event_gsmSCF

19.5 HLR Insert Subscriber Data macros

19.5.1 Macro Insert_Subs_Data_Framed_HLR

This macro is used to transfer subscriber data to the VLR as part of an existing dialogue for location updating or data restoration. The macro invokes a macro and a process not defined in this clause; the definitions of the macro and the process can be found as follows:

Wait_For_Insert_Subs_Data_Cnf see subclause 25.7.5;

Send_Insert_Subs_Data_HLR: see subclause 25.7.7.

The HLR may wait for each MAP_INSERT_SUBSCRIBER_DATA request to be acknowledged before it sends the next request, or it may handle the requests and the confirmations in parallel.

If the VLR has indicated that it does not support a service or feature (e.g. Closed User Group or Advice Of Charge Charging Level) which the HLR operator regards as essential for the subscriber, the macro Wait_for_Insert_Subs_Data_Cnf takes the Replace_Service exit; the HLR sets the Roaming Restriction Due To Unsupported Feature flag to roaming restricted and sends Roaming Restriction Due To Unsupported Feature in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

If the HLR operator does not regard the unsupported service or feature as essential for the subscriber but the macro Wait_for_Insert_Subs_Data_Cnf takes the Replace_Service exit, the HLR sends the data for a replacement service in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

If subscriber data for CAMEL Phase 2 or later services are sent to a VLR which does not support the appropriate phase of CAMEL, the service behaviour may be unpredictable or incorrect. The HLR should therefore ensure that at the conclusion of a stand alone Insert Subscriber data procedure the data in the VLR do not require a capability that the VLR does not have. Possible mechanisms to ensure this are described in 3GPP TS 23.078 [98].

The HLR should send a Forwarded-to number which is not in E.164 international format to the VLR only when the HLR has ascertained that the VLR supports CAMEL Phase 2 or later. Thus, the ISD message containing the Forwarded-to number which is not in E.164 international format shall be sent to the VLR only if the HLR previously received confirmation from the VLR at Location Update that CAMEL Phase 2 or later is supported.

19.5.2 Macro Insert_GPRS_Subs_Data_Framed_HLR

This macro is used to transfer subscriber data to the SGSN as part of an existing dialogue for location updating. The macro invokes a macro and a process not defined in this clause; the definitions of the macro and the process can be found as follows:

Wait_For_Insert_GPRS_Subs_Data_Cnf see subclause 25.7.5;

Send_Insert_Subs_Data_HLR: see subclause 25.7.7.

The HLR may wait for each MAP_INSERT_SUBSCRIBER_DATA request to be acknowledged before it sends the next request, or it may handle the requests and the confirmations in parallel.

If the SGSN has indicated that it does not support a service or feature which the HLR operator regards as essential for the subscriber, the macro Wait_for_Insert_GPRS_Subs_Data_Cnf takes the Replace_Service exit; the HLR sets the Roaming Restricted In SGSN Due To Unsupported Feature flag to roaming restricted and sends Roaming Restricted In SGSN Due To Unsupported Feature in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

If the HLR operator does not regard the unsupported service or feature as essential for the subscriber but the macro Wait_for_Insert_GPRS_Subs_Data_Cnf takes the Replace_Service exit, the HLR sends the data for a replacement service in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

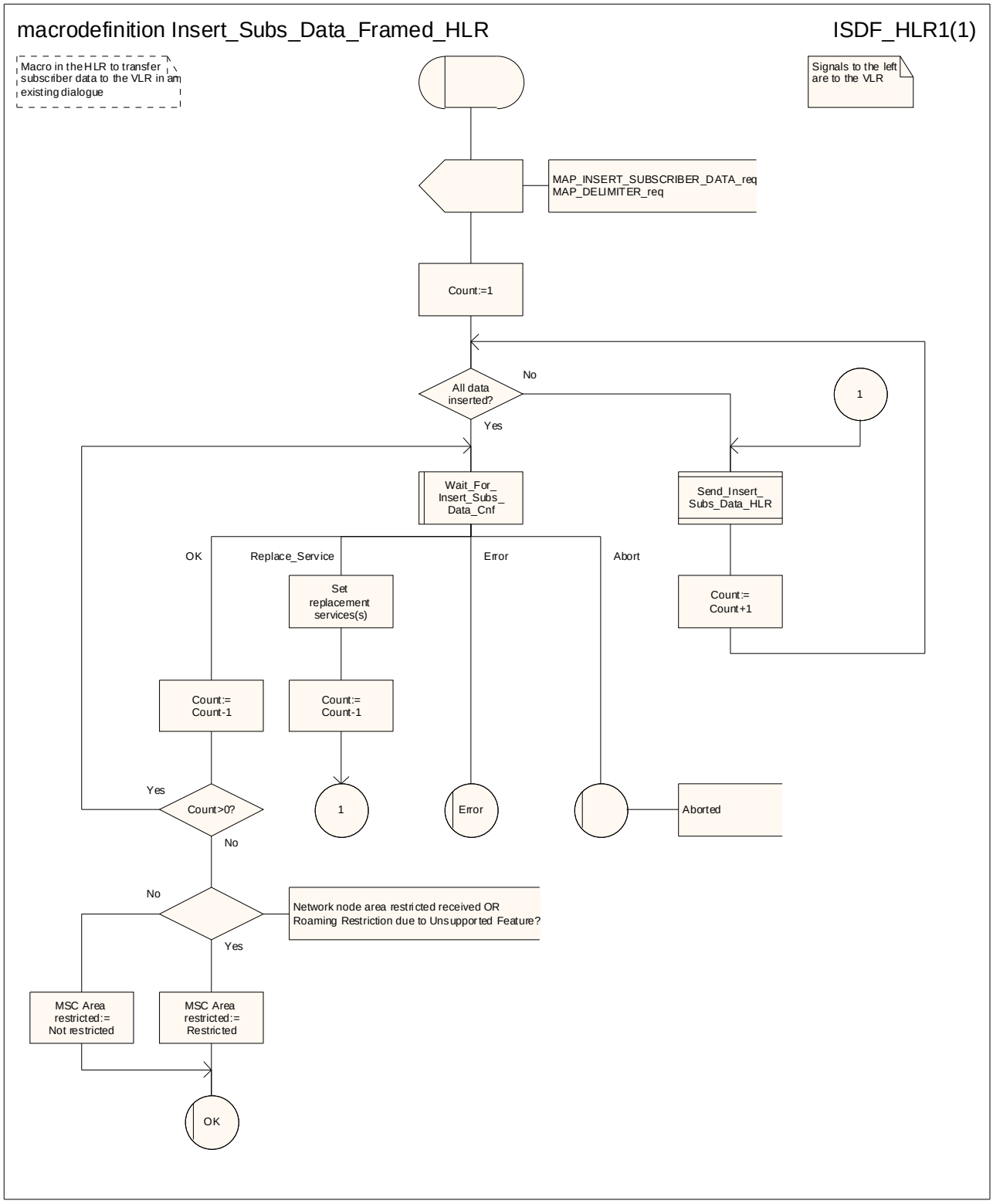


Figure 19.5/1: Macro Insert_Subs_Data_Framed_HLR

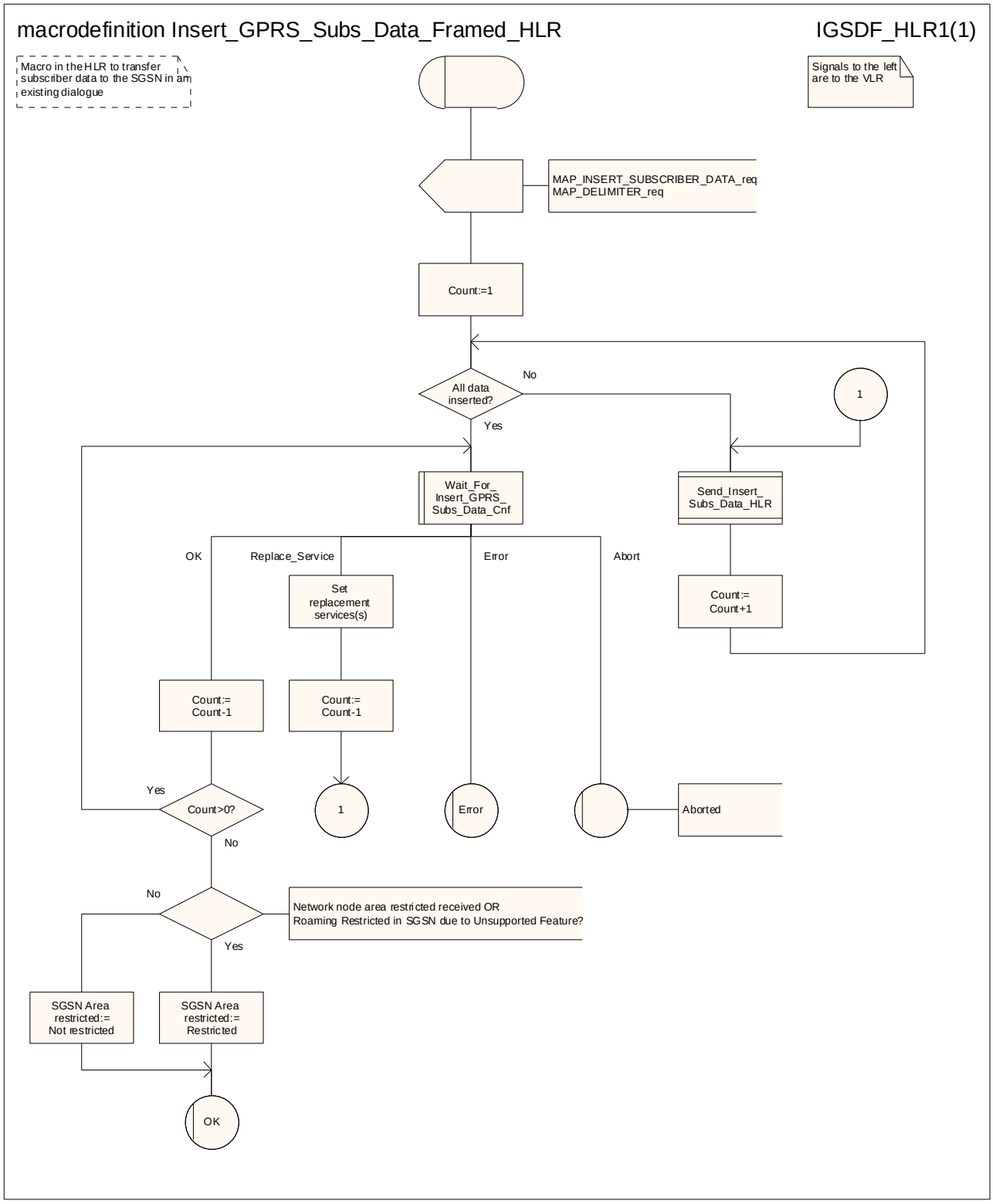


Figure 19.5/2: Macro Insert_GPRS_Subs_Data_Framed_HLR

19.5A CSS Insert Subscriber Data macros

19.5A.1 Macro Insert_VCSG_Subs_Data_Framed_CSS

This macro is used to transfer CSG subscriber data from the CSS to the VLR or the SGSN as part of an existing dialogue for VCSG location updating. The macro invokes a macro and a process not defined in this clause; the definitions of the macro and the process can be found as follows:

Wait_For_Insert_VCSG_Subs_Data_Cnf see subclause 25.7.9;

Send_Insert_VCSG_Subs_Data_CSS: see subclause 25.7.10.

The CSS may wait for each MAP_INSERT_SUBSCRIBER_DATA request to be acknowledged before it sends the next request, or it may handle the requests and the confirmations in parallel.

If the VLR or the SGSN has indicated that it does not support a service or feature which the CSS operator regards as essential for the subscriber, the macro Wait_for_Insert_VCSG_Subs_Data_Cnf takes the Replace_Service exit.

If the CSS operator does not regard the unsupported service or feature as essential for the subscriber but the macro Wait_for_Insert_VCSG_Subs_Data_Cnf takes the Replace_Service exit, the CSS sends the data for a replacement service in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

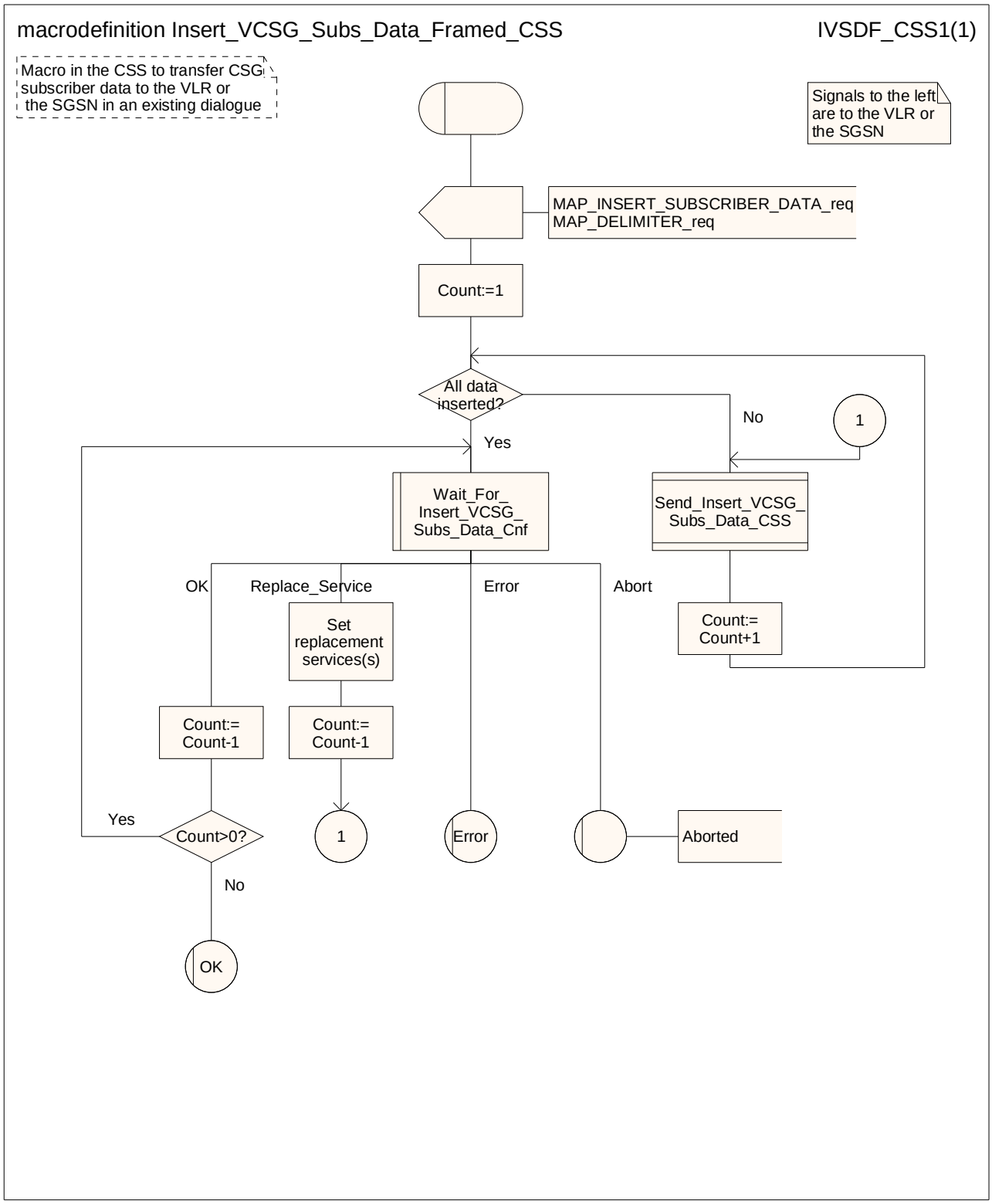


Figure 19.5A/1: Macro Insert_VCSG_Subs_Data_Framed_CSS

20 Operation and maintenance procedures

20.1 General

The Operation and Maintenance procedures are used to support operation and maintenance of the network.

The following procedures exist for operation and maintenance purposes:

i) Tracing procedures;

ii) Subscriber Data Management procedures;

iii) Subscriber Identity procedure.

The following application contexts refer to complex MAP Users consisting of several processes:

- subscriberDataManagementContext;
- tracingContext.

Each of these two application contexts needs a co-ordinating process in the VLR or in the SGSN as described in the following subclauses.

20.1.1 Tracing Co-ordinator for the VLR

The Tracing Co-ordinator process in the VLR is shown the figure 20.1/1. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

20.1.2 Tracing Co-ordinator for the SGSN

The Tracing Co-ordinator process in the SGSN is shown in figure 20.1/2. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

20.1.3 Subscriber Data Management Co-ordinator for the VLR

The Subscriber Data Management Co-ordinator process in the VLR is shown in figure 20.1/3 and figure 20.1/5. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

20.1.4 Subscriber Data Management Co-ordinator for the SGSN

The Subscriber Data Management Co-ordinator process in the SGSN is shown in figure 20.1/4 and figure 20.1/6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

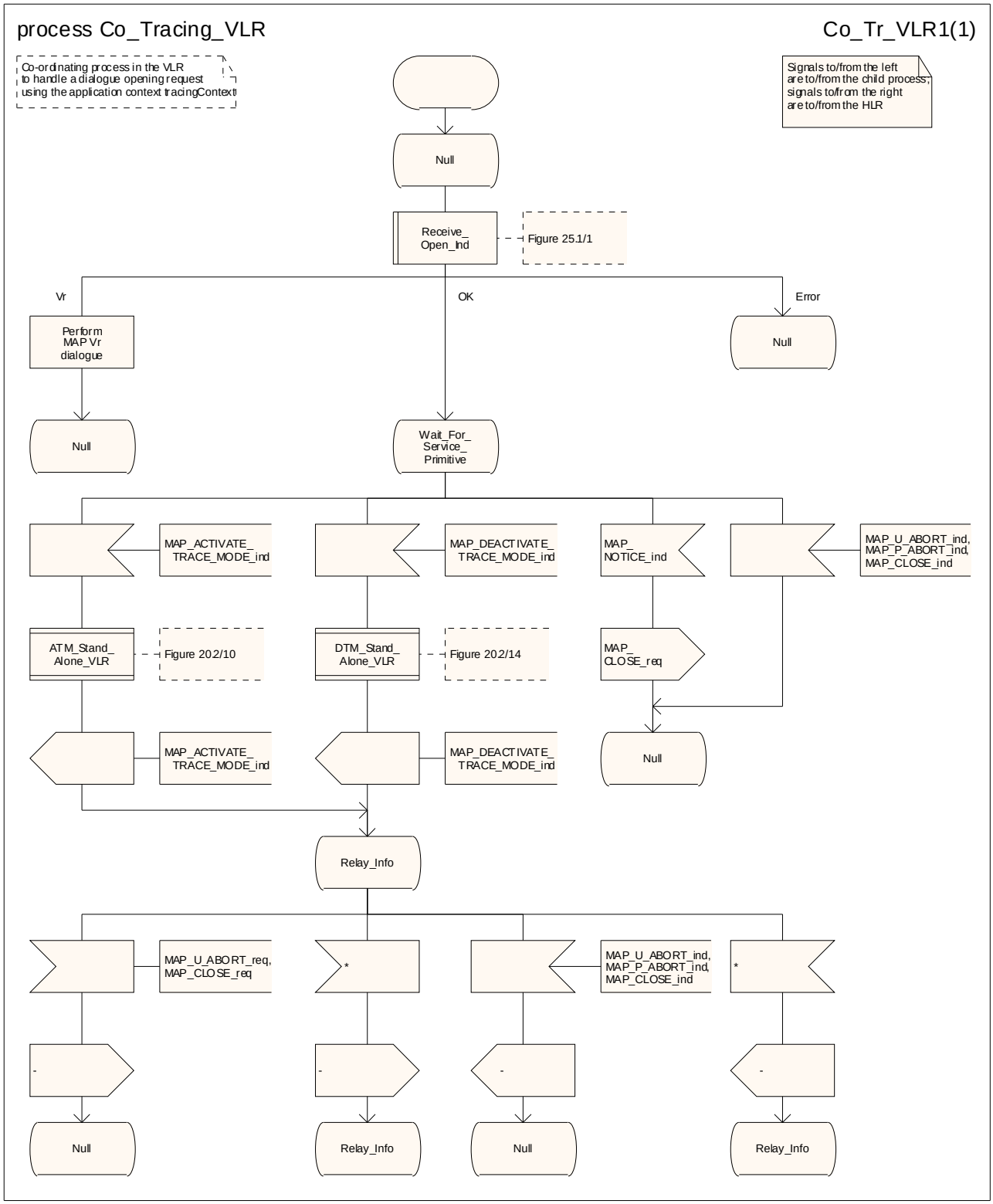


Figure 20.1/1: Process Co_Tracing_VLR

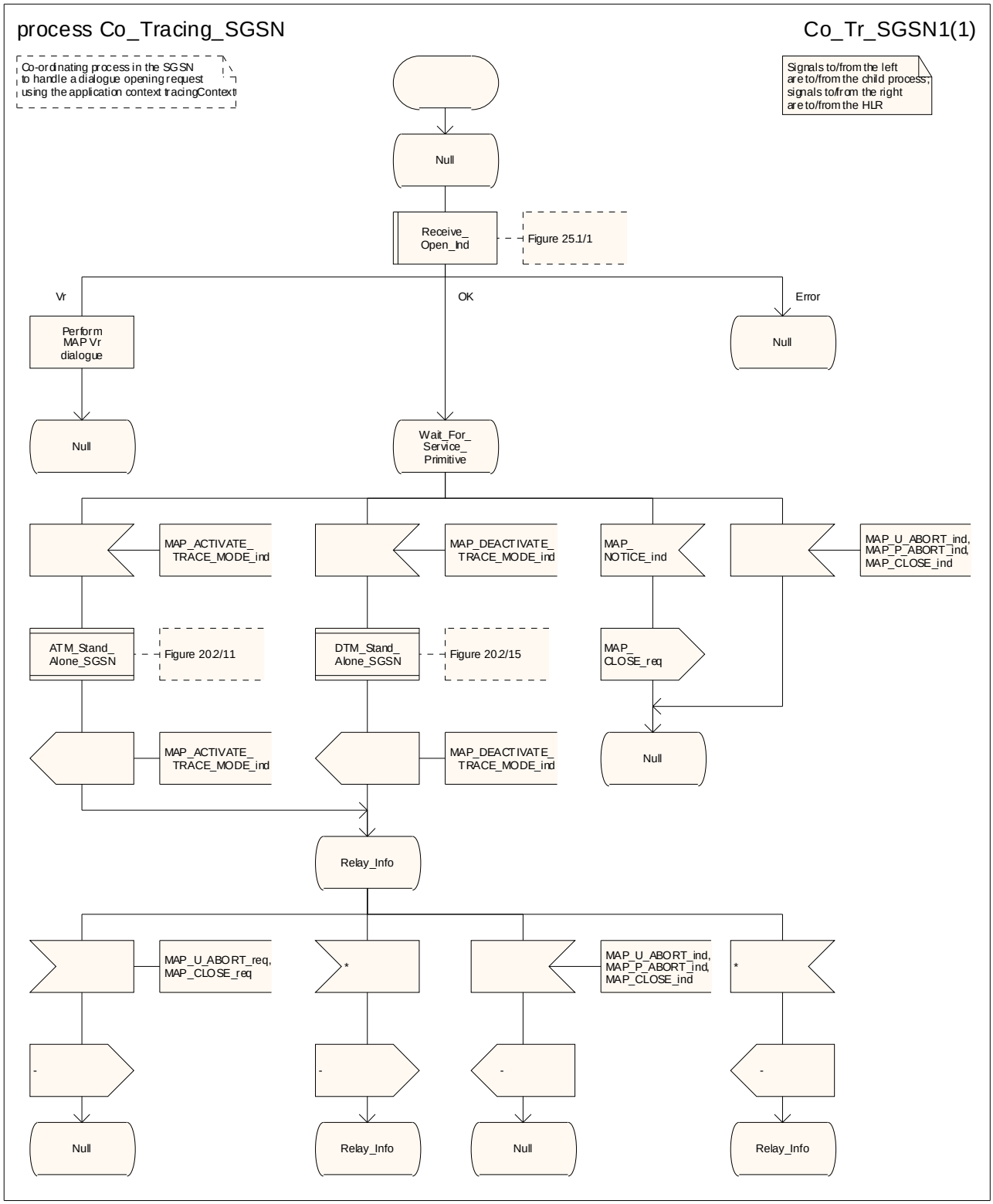


Figure 20.1/2: Process Co_Tracing_SGSN

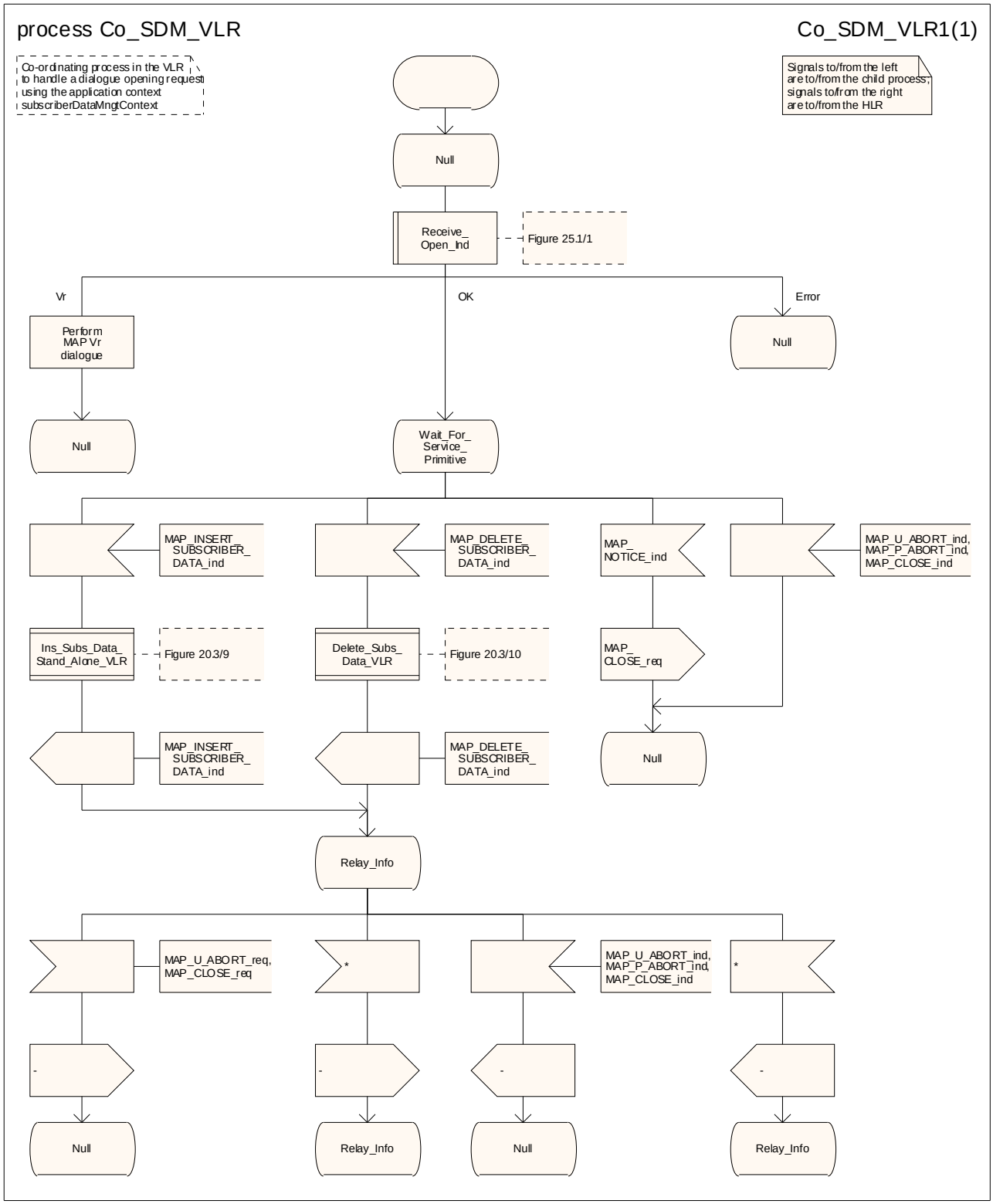


Figure 20.1/3: Process Co_SDM_VLR

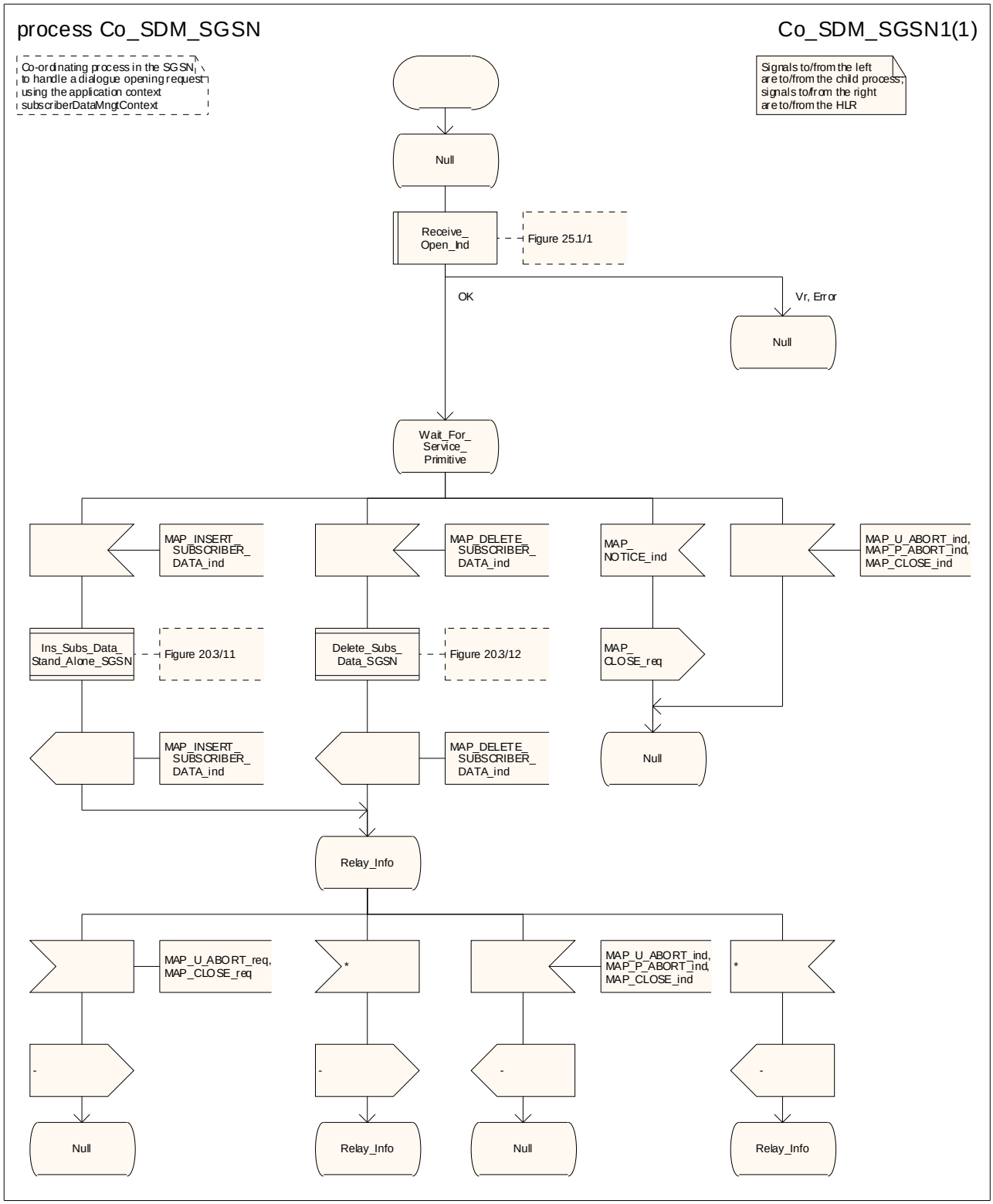


Figure 20.1/4: Process Co_SDM_SGSN

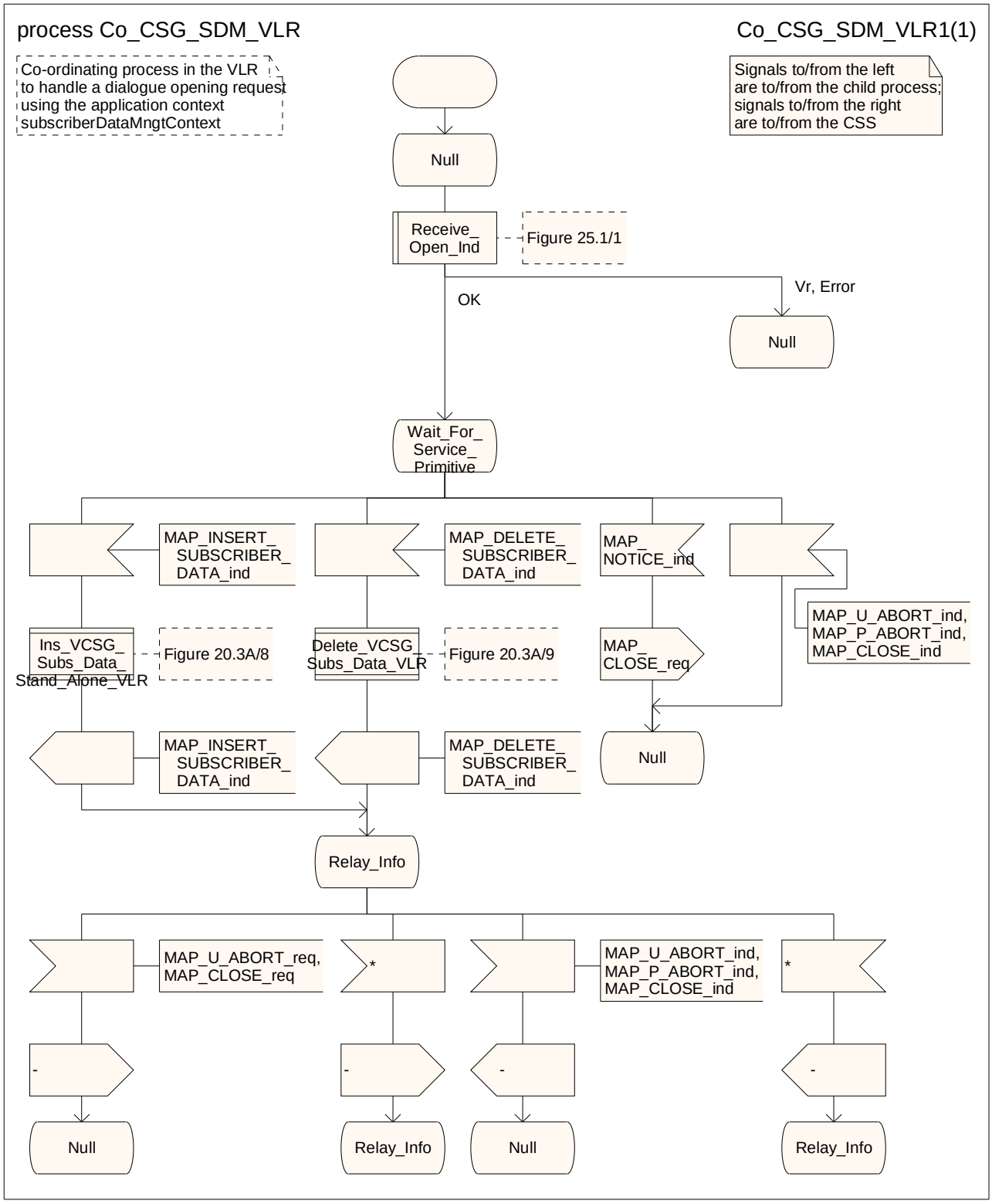


Figure 20.1/5: Process Co_CSG_SDM_VLR

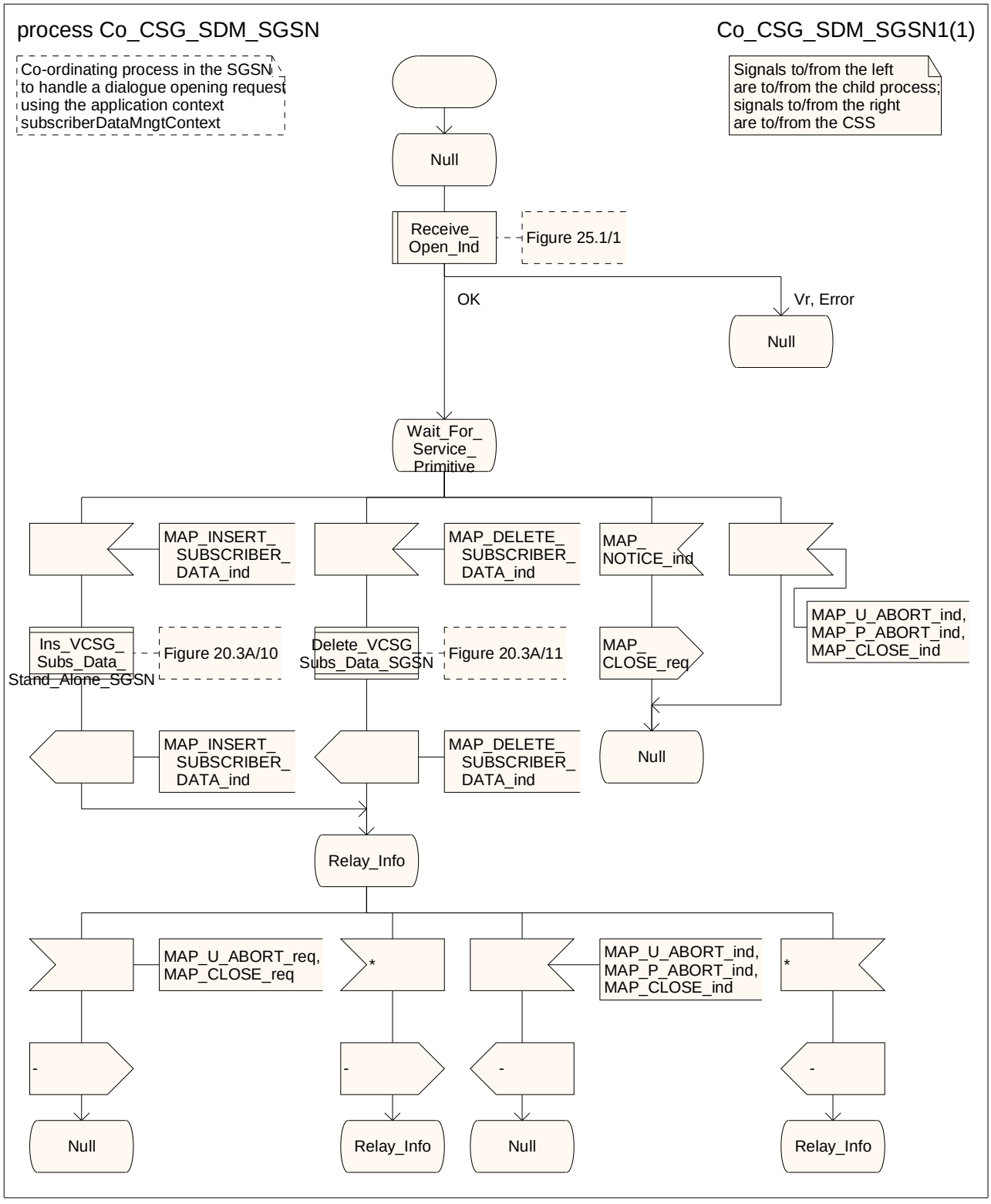


Figure 20.1/6: Process Co_CSG_SDM_SGSN

20.2 Tracing procedures

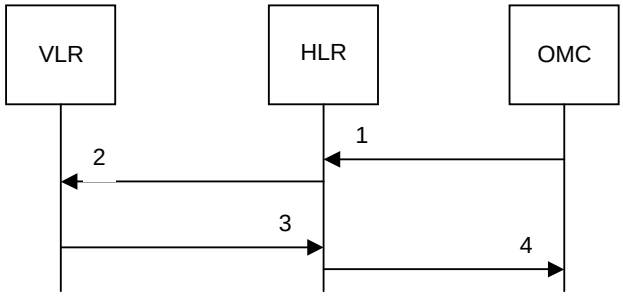
Three types of tracing procedures exist:

i) Subscriber tracing management procedures;

ii) Subscriber tracing procedures;

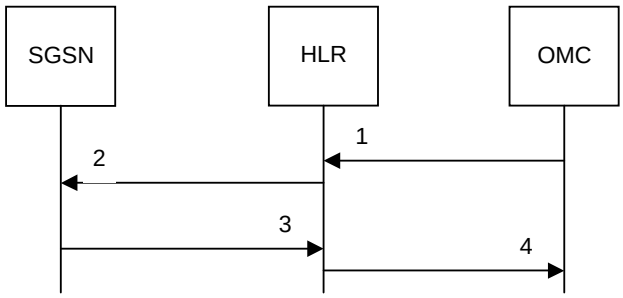
iii) Event tracing procedures.

The subscriber tracing management procedures are used to manage the status and the type of the tracing. The subscriber tracing activation procedure is used at location updating or data restoration when the trace mode of a subscriber is set active in the HLR or, as a stand alone procedure, when the subscriber is already registered and the trace mode becomes active in the HLR. The procedures to activate tracing in the VLR are shown in figures 20.2/1 and 20.2/3. The procedures to activate tracing in the SGSN are shown in figures 20.2/2 and 20.2/4.



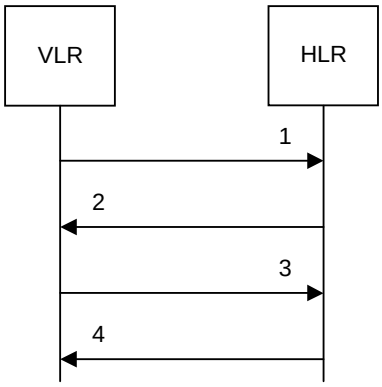
1) Subscriber Tracing Activation
2) MAP_ACTIVATE_TRACE_MODE_req/ind
3) MAP_ACTIVATE_TRACE_MODE_rsp/cnf
4) Subscriber Tracing Activation Accepted

Figure 20.2/1: Stand-alone subscriber tracing activation procedure for non-GPRS



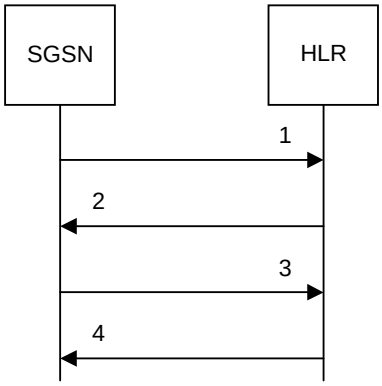
1) Subscriber Tracing Activation
2) MAP_ACTIVATE_TRACE_MODE_req/ind
3) MAP_ACTIVATE_TRACE_MODE_rsp/cnf
4) Subscriber Tracing Activation Accepted

Figure 20.2/2: Stand-alone subscriber tracing activation procedure for GPRS



1) MAP_UPDATE_LOCATION or MAP_RESTORE_DATA_req/ind
2) MAP_ACTIVATE_TRACE_MODE_req/ind
3) MAP_ACTIVATE_TRACE_MODE_rsp/cnf
4) MAP_UPDATE_LOCATION_rsp/cnf or MAP_RESTORE_DATA_rsp/cnf

Figure 20.2/3: Subscriber tracing activation procedure at location updating or data restoration

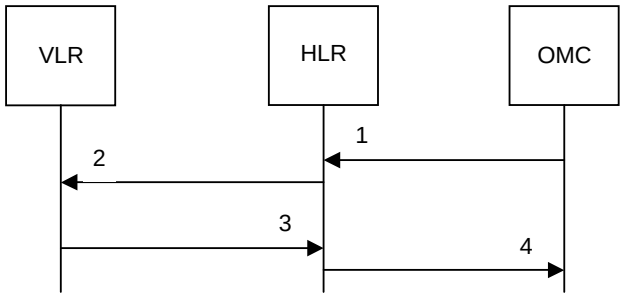


1) MAP_UPDATE_GPRS_LOCATION_req/ind
2) MAP_ACTIVATE_TRACE_MODE_req/ind
3) MAP_ACTIVATE_TRACE_MODE_rsp/cnf
4) MAP_UPDATE_GPRS_LOCATION_rsp/cnf

Figure 20.2/4: Subscriber tracing activation procedure at GPRS location updating

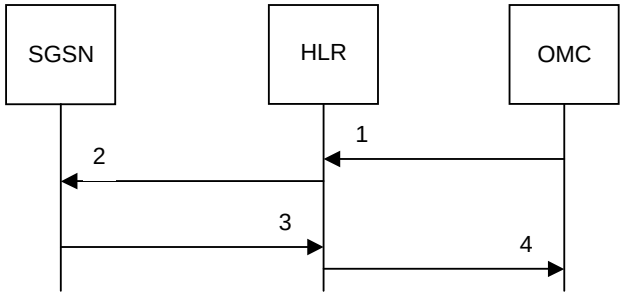
The MAP_ACTIVATE_TRACE_MODE request includes the IMSI, trace reference, trace type and identity of the OMC.

The subscriber tracing deactivation procedure is used when tracing of a subscriber in the VLR or in the SGSN is no longer required. The procedures are shown in figures 20.2/5 and 20.2/6.



1) Subscriber Tracing Deactivation
2) MAP_DEACTIVATE_TRACE_MODE_req/ind
3) MAP_DEACTIVATE_TRACE_MODE_rsp/cnf
4) Subscriber Tracing Deactivation Accepted

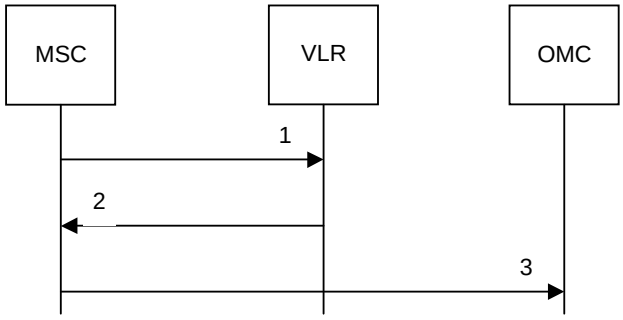
Figure 20.2/5: Subscriber tracing deactivation procedure for non-GPRS



1) Subscriber Tracing Deactivation
2) MAP_DEACTIVATE_TRACE_MODE_req/ind
3) MAP_DEACTIVATE_TRACE_MODE_rsp/cnf
4) Subscriber Tracing Deactivation Accepted

Figure 20.2/6: Subscriber tracing deactivation procedure for GPRS

The subscriber tracing procedures are used when the VLR detects any subscriber related activity for which the trace mode is activated, e.g. the VLR receives a MAP_PROCESS_ACCESS_REQUEST indication. The procedure is shown in figure 20.2/7.



1) MAP_PROCESS_ACCESS_REQUEST_req/ind
2) MAP_TRACE_SUBSCRIBER_ACTIVITY_req/ind
3) Subscriber tracing information

Figure 20.2/7: Subscriber tracing procedure in the serving MSC

20.2.1 Subscriber tracing activation procedure

20.2.1.1 Procedures in the HLR

A subscriber tracing activation request from the OMC starts the appropriate process in the HLR: ATM_With_VLR_HLR if tracing is required in the MSC/VLR, ATM_With_SGSN_HLR if tracing is required in the SGSN.

The process in the HLR to activate tracing in the VLR is shown in figure 20.2/8. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 1: If the Repeat attempt counter has reached its limit, the test "Repeat Attempt" takes the "No" exit; otherwise the test takes the "Yes" exit. The number of repeat attempts and the interval between successive repeat attempts are operator options.

The process in the HLR to activate tracing in the SGSN is shown in figure 20.2/9. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 1: If the Repeat attempt counter has reached its limit, the test "Repeat Attempt" takes the "No" exit; otherwise the test takes the "Yes" exit. The number of repeat attempts and the interval between successive repeat attempts are operator options.

20.2.1.2 Procedure in the VLR

The process in the VLR to activate tracing in a stand-alone dialogue is shown in figure 20.2/10. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

20.2.1.3 Procedure in the SGSN

The process in the SGSN to activate tracing in a stand-alone dialogue is shown in figure 20.2/11. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

20.2.2 Subscriber tracing deactivation procedure

20.2.2.1 Procedures in the HLR

A subscriber tracing deactivation request from the OMC starts the appropriate process in the HLR: DTM_HLR_With_VLR if tracing is no longer required in the MSC/VLR, DTM_HLR_With_SGSN if tracing is no longer required in the SGSN.

The process in the HLR to deactivate tracing in the VLR is shown in figure 20.2/12. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 1: If the Repeat attempt counter has reached its limit, the test "Repeat Attempt" takes the "No" exit; otherwise the test takes the "Yes" exit. The number of repeat attempts and the interval between successive repeat attempts are operator options.

The process in the HLR to deactivate tracing in the SGSN is shown in figure 20.2/13. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 1: If the Repeat attempt counter has reached its limit, the test "Repeat Attempt" takes the "No" exit; otherwise the test takes the "Yes" exit. The number of repeat attempts and the interval between successive repeat attempts are operator options.

20.2.2.2 Procedure in the VLR

The process in the VLR to deactivate tracing is shown in figure 20.2/14. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

20.2.2.3 Procedure in the SGSN

The process in the SGSN to deactivate tracing is shown in figure 20.2/15. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

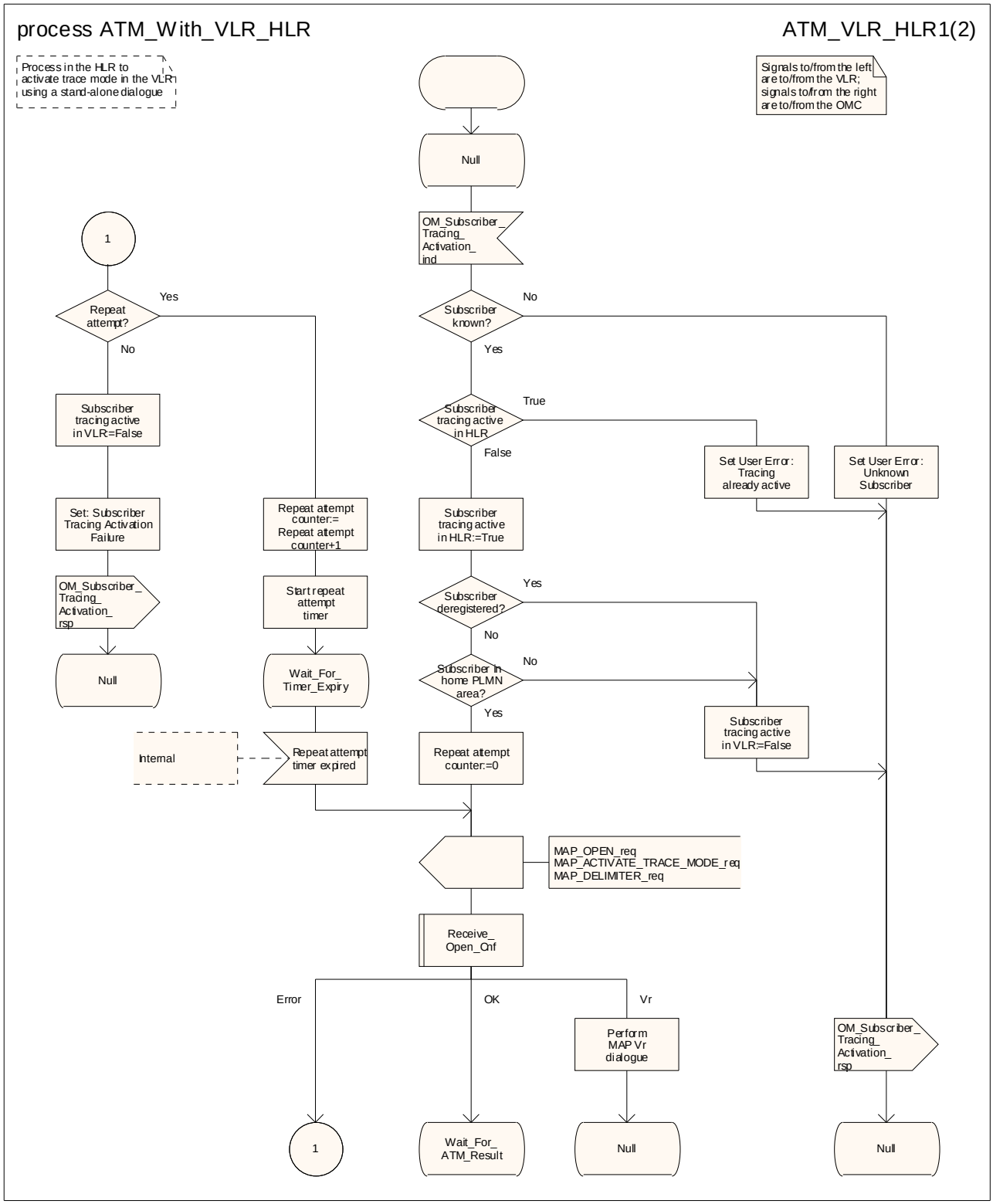


Figure 20.2/8 (sheet 1 of 2): Process ATM_With_VLR_HLR

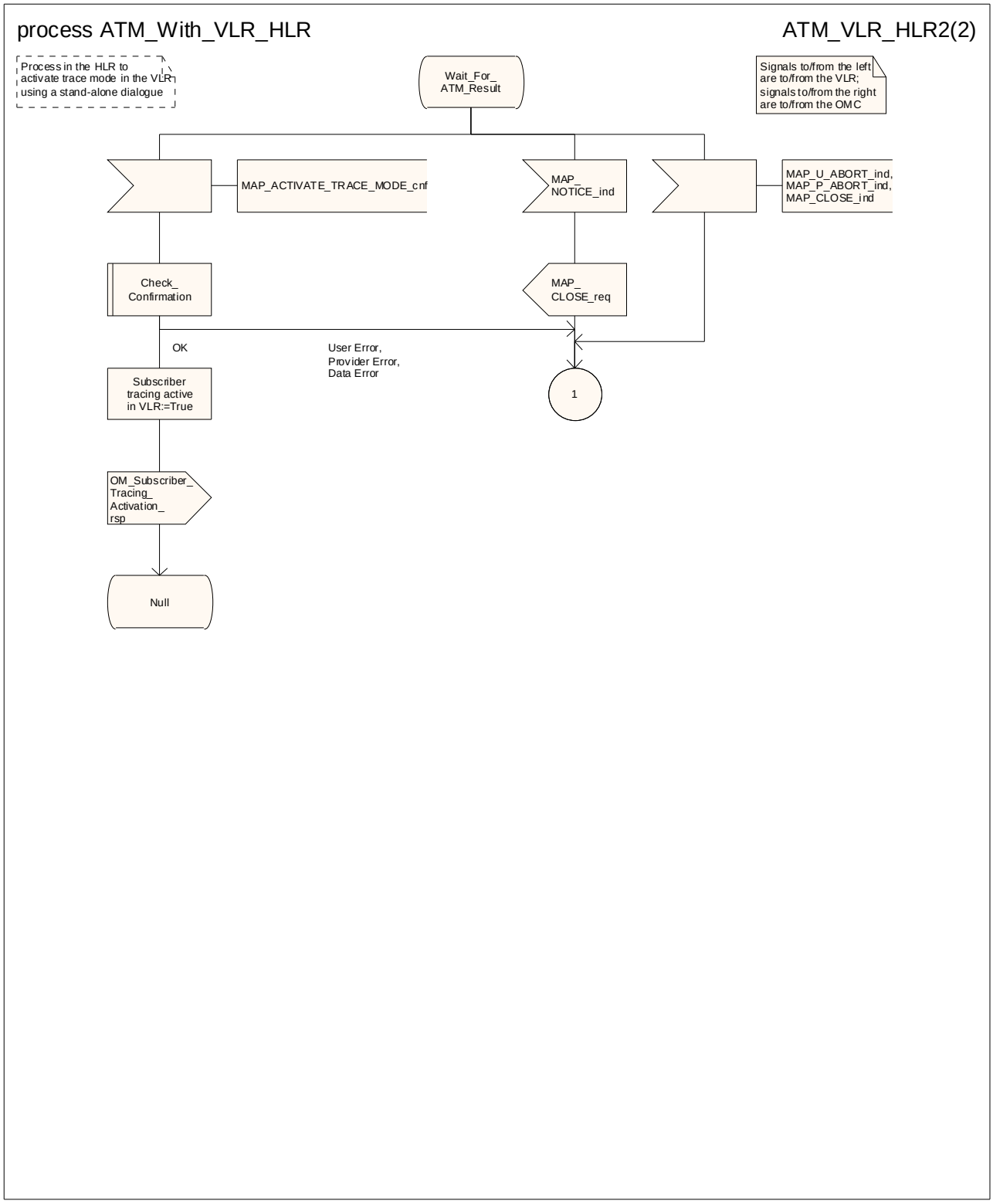


Figure 20.2/8 (sheet 2 of 2): Process ATM_With_VLR_HLR

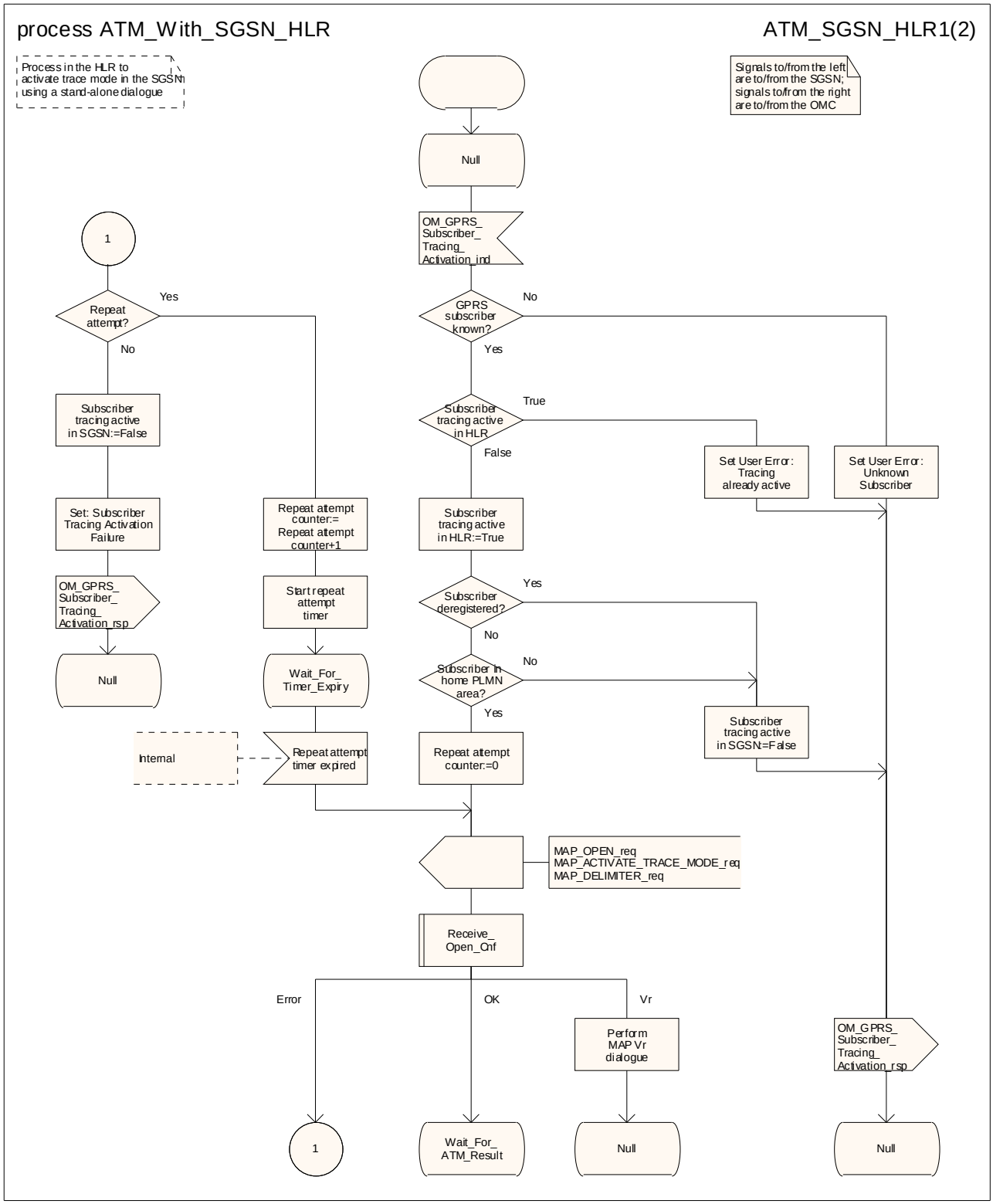


Figure 20.2/9 (sheet 1 of 2): Process ATM_with_SGSN_HLR

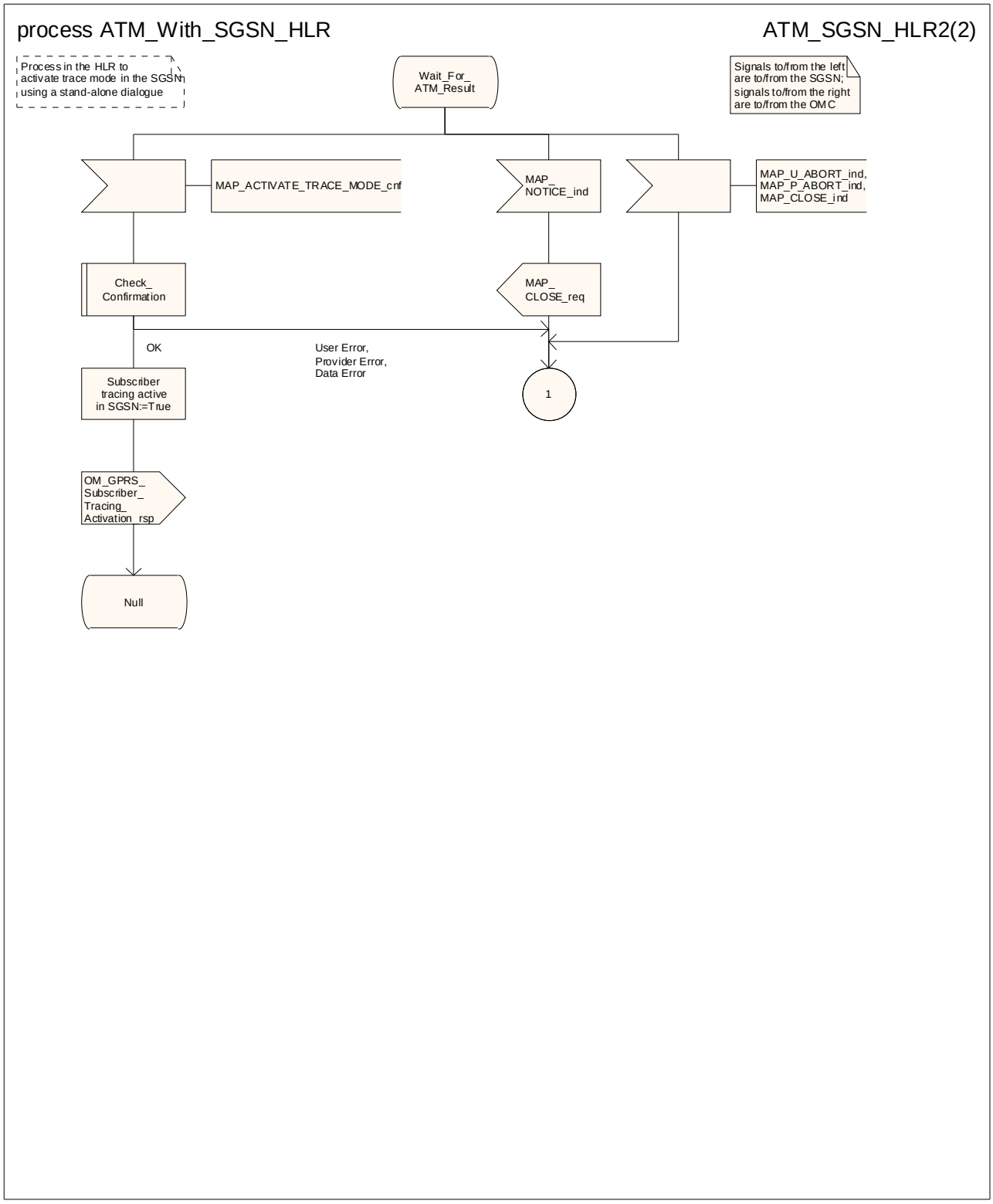


Figure 20.2/9 (sheet 2 of 2): Process ATM_with_SGSN_HLR

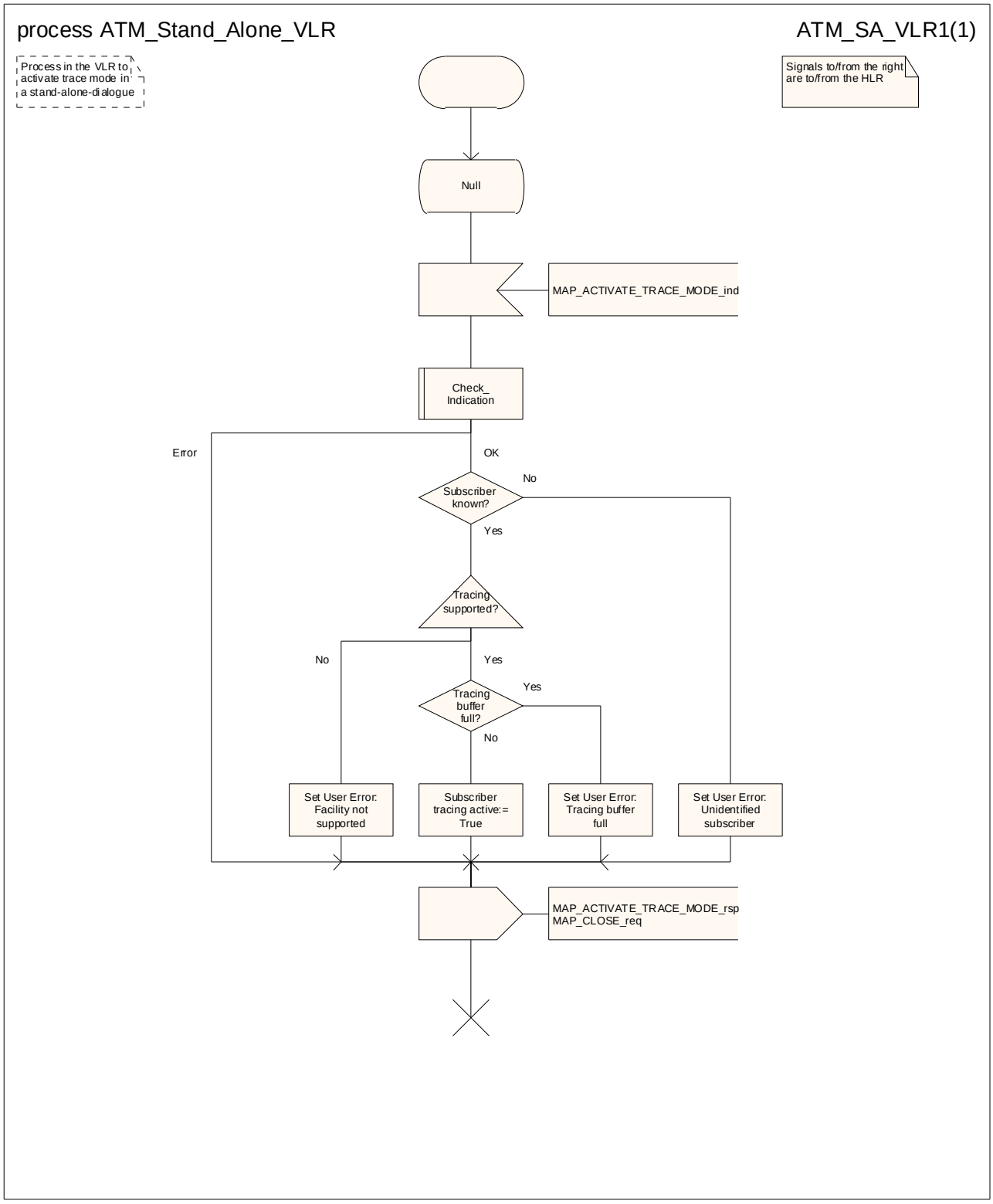


Figure 20.2/10: Process ATM_Standalone_VLR

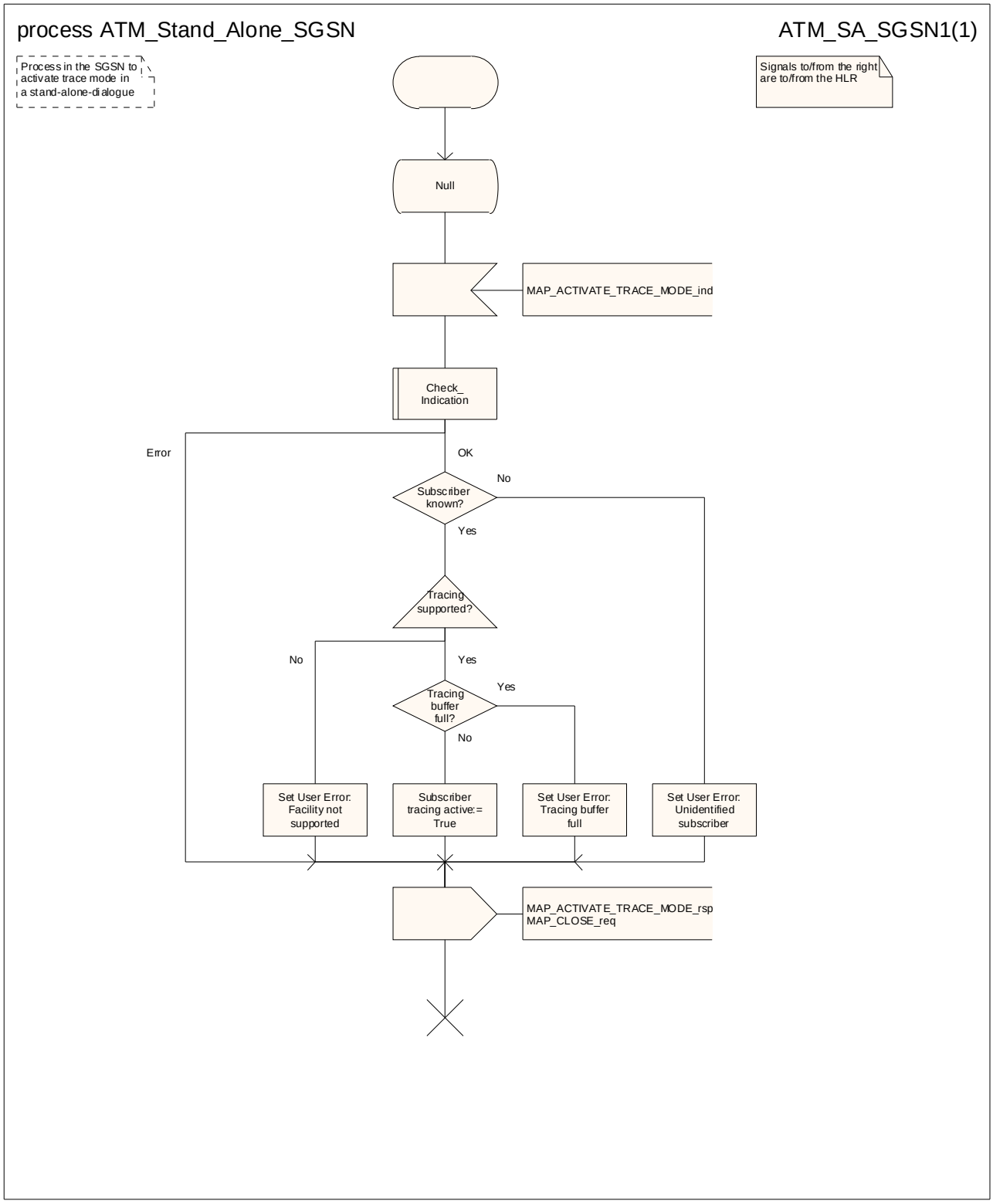


Figure 20.2/11: Process ATM_Standalone_SGSN

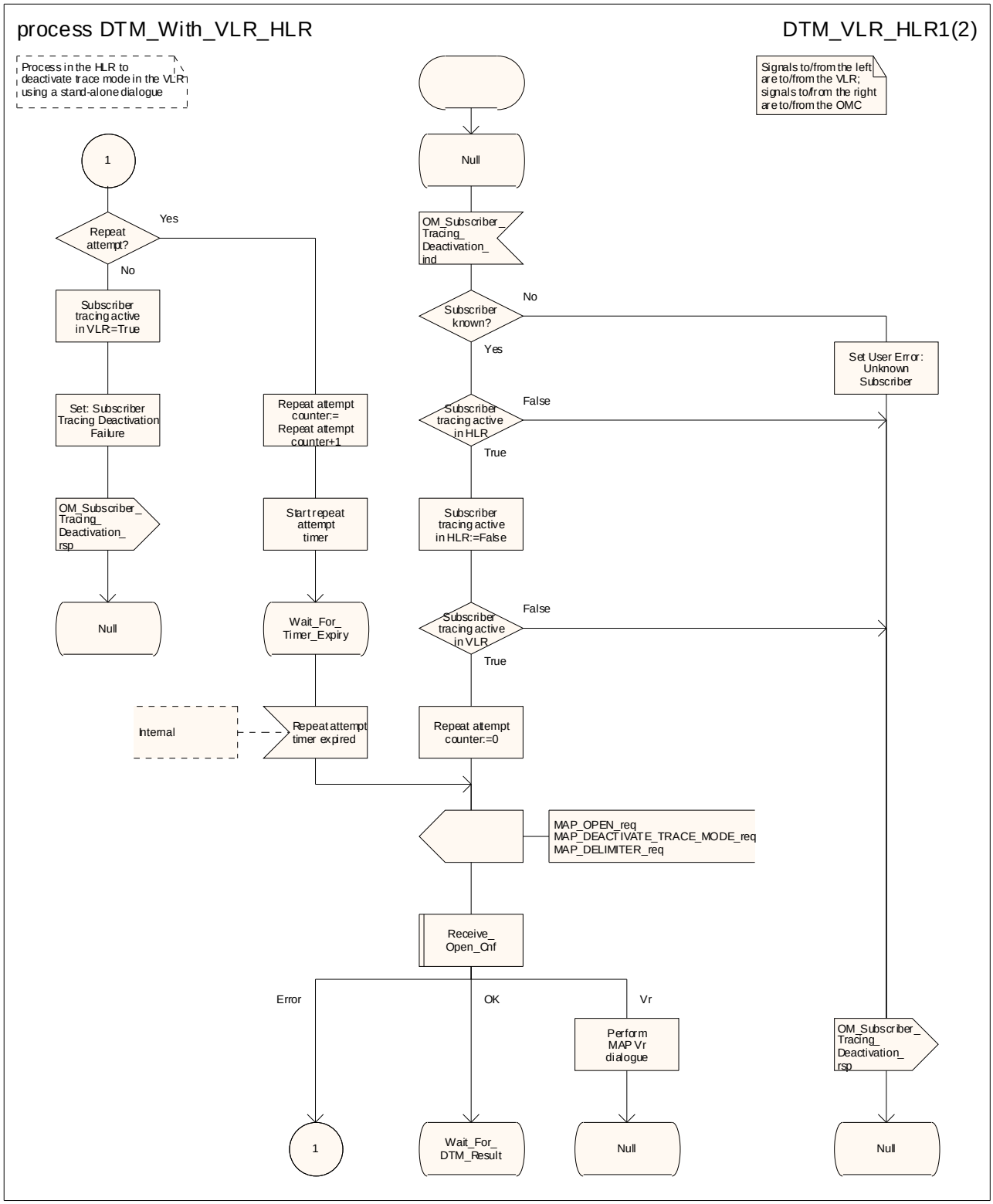


Figure 20.2/12 (sheet 1 of 2): Process DTM_with_VLR_HLR

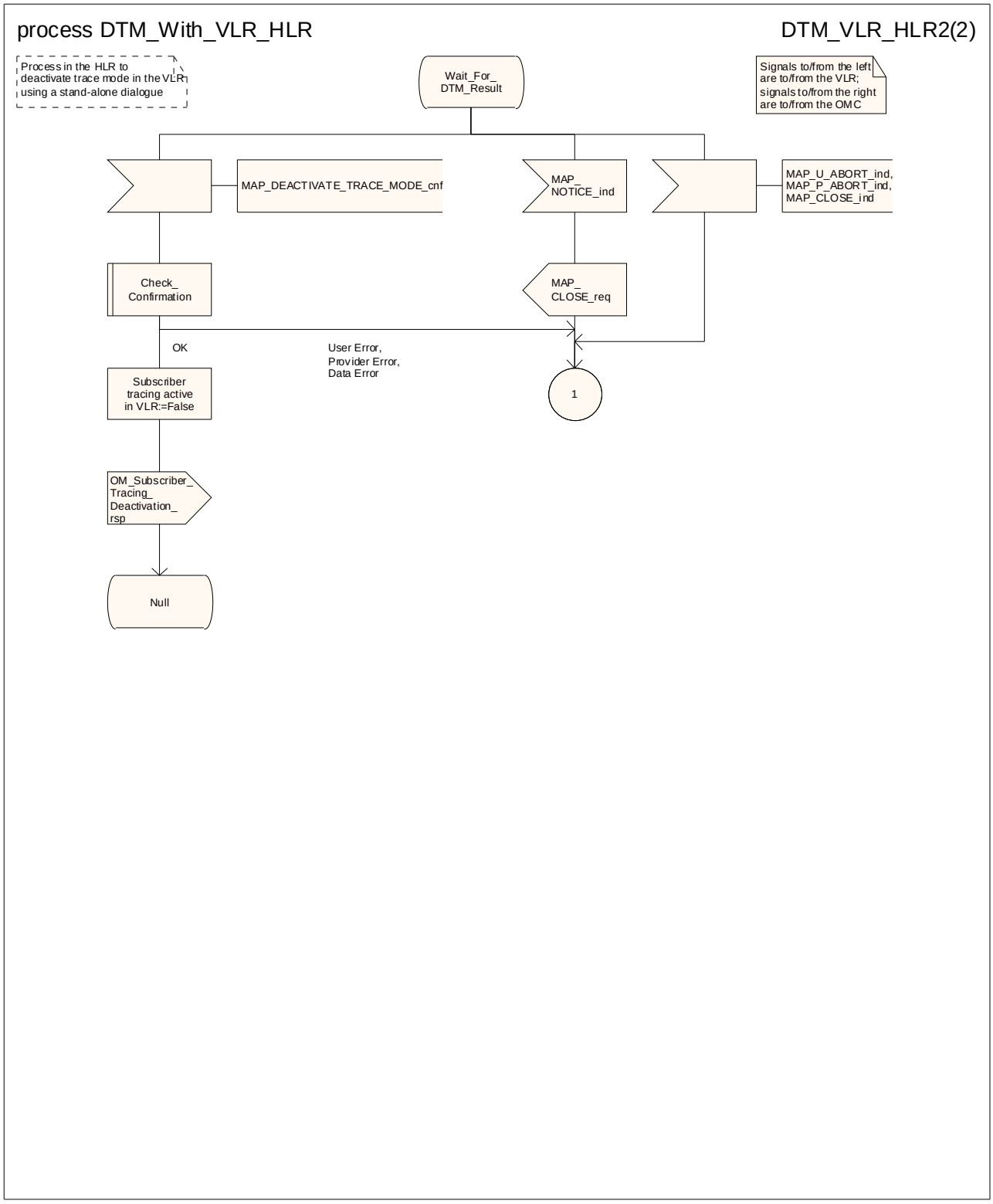


Figure 20.2/12 (sheet 2 of 2): Process DTM_with_VLR_HLR

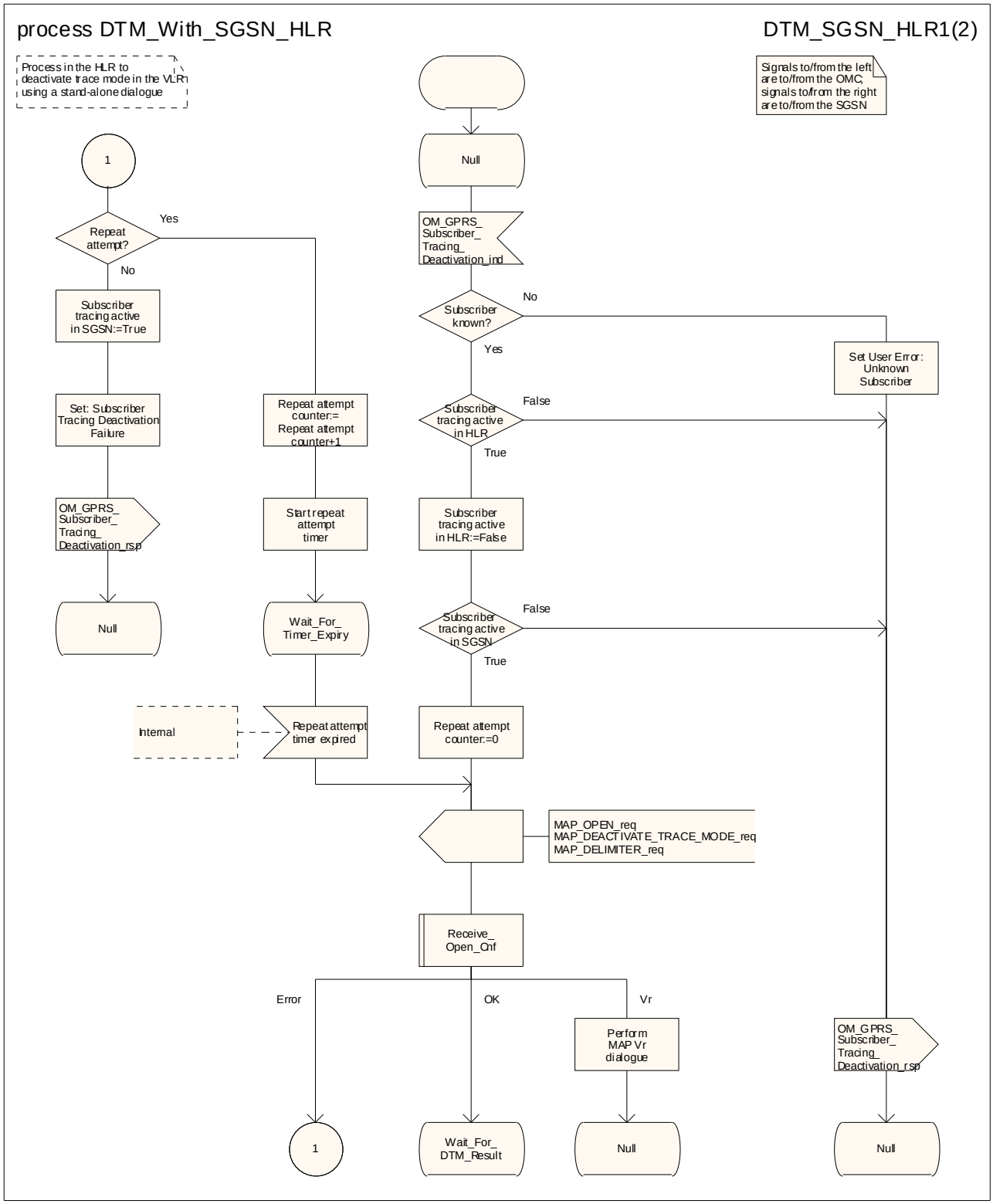


Figure 20.2/13 (sheet 1 of 2): Process DTM_with_SGSN_HLR

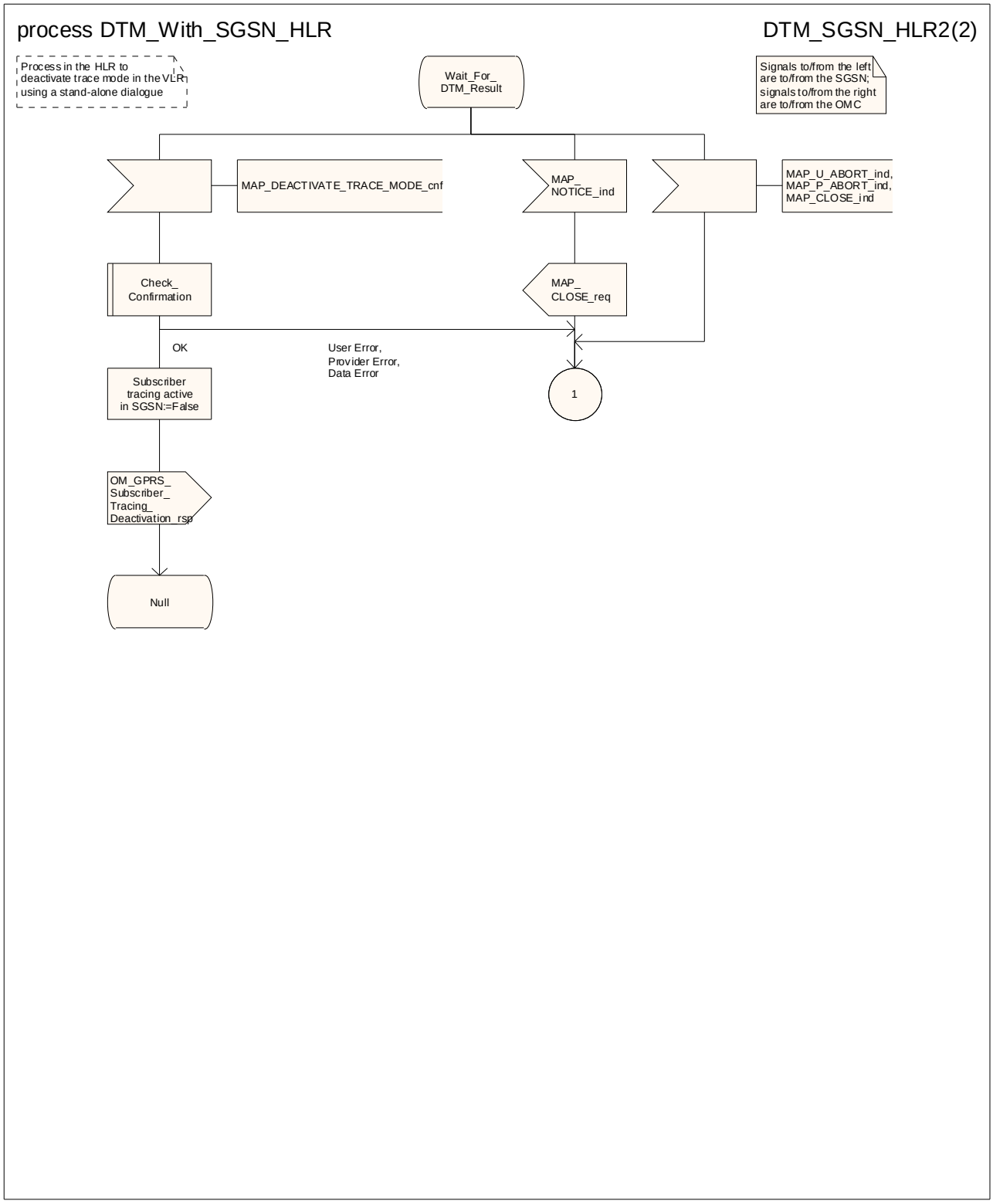


Figure 20.2/13 (sheet 2 of 2): Process DTM_with_SGSN_HLR

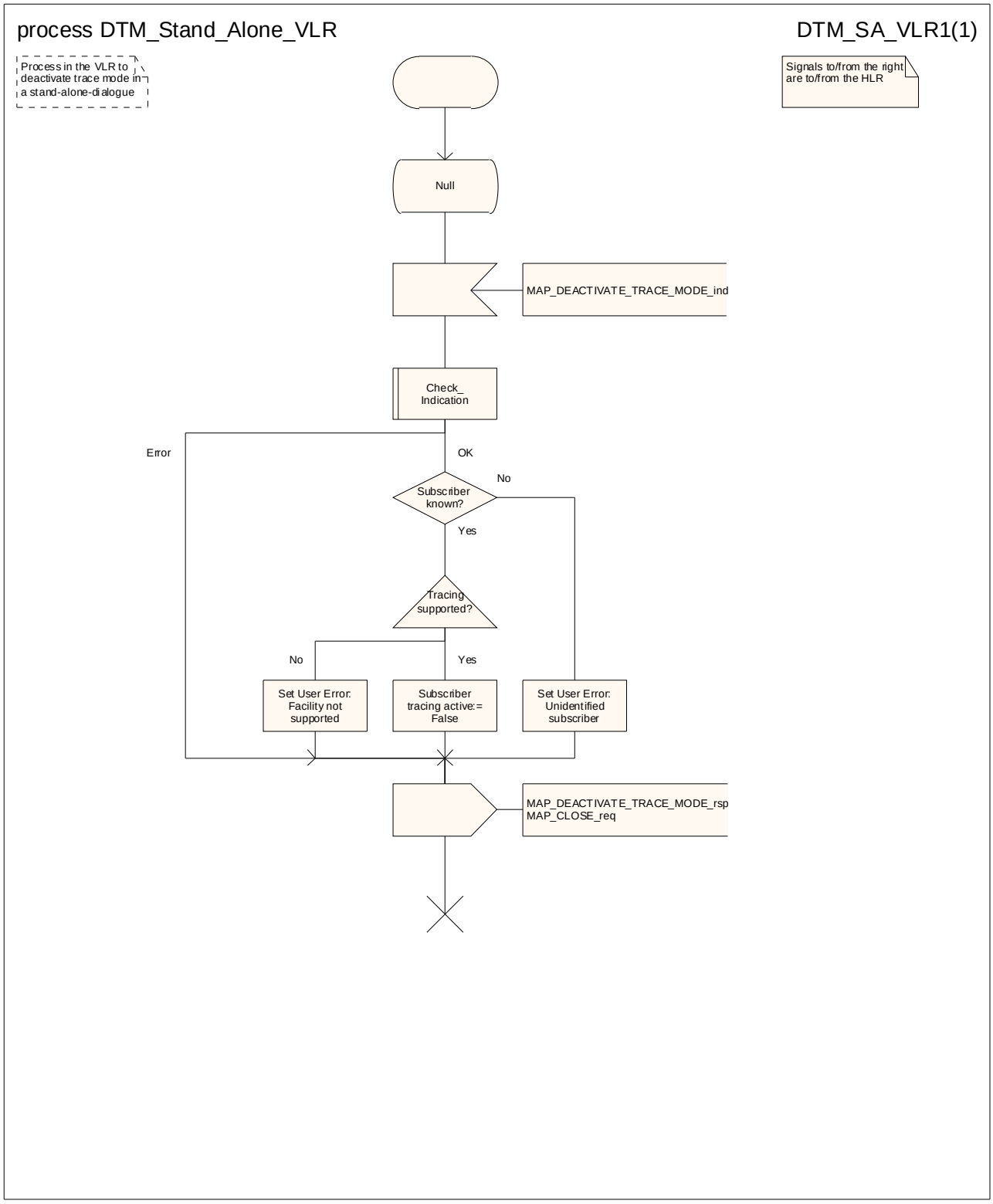


Figure 20.2/14: Process DTM_Standalone_VLR

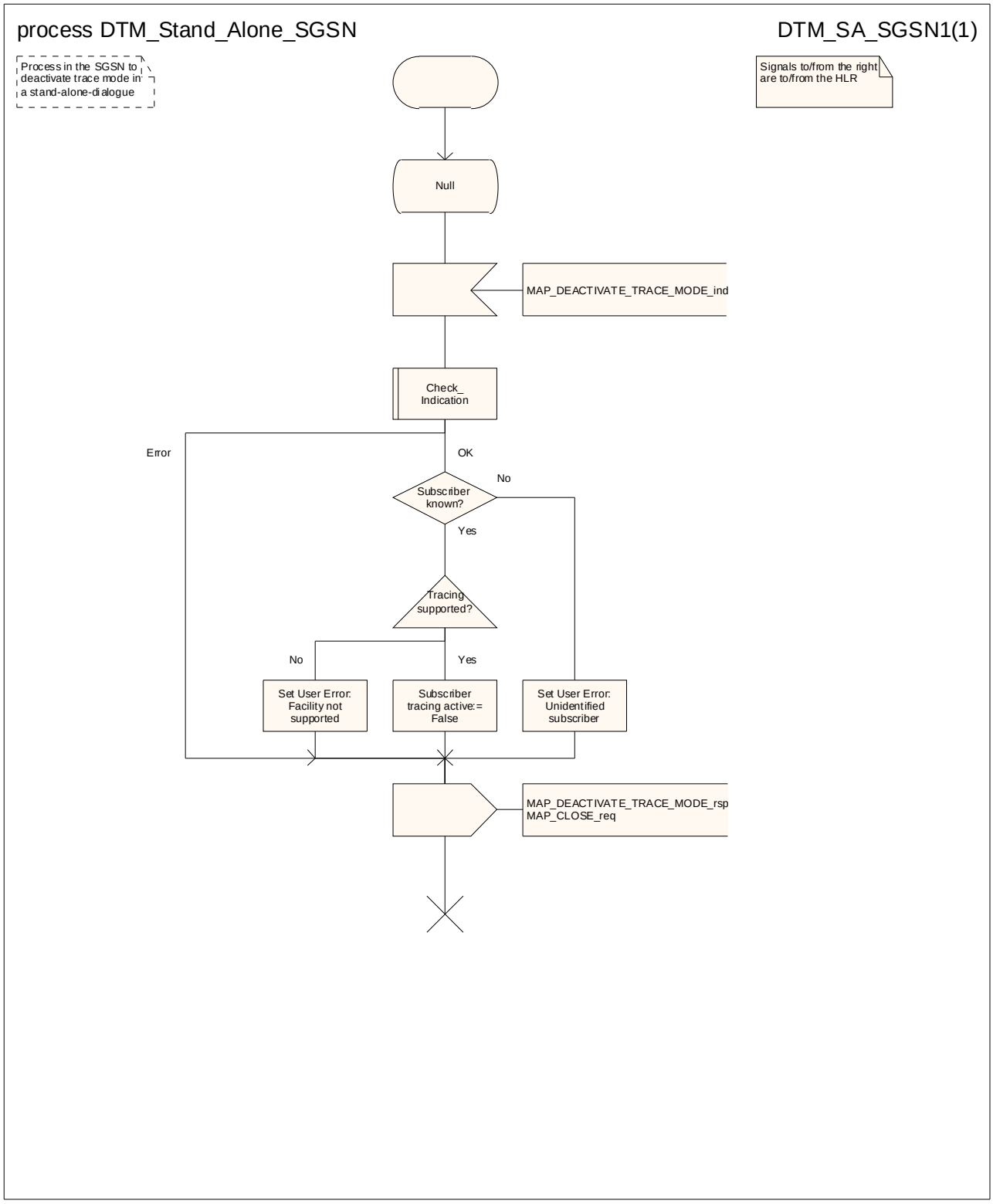


Figure 20.2/15: Process DTM_Standalone_SGSN

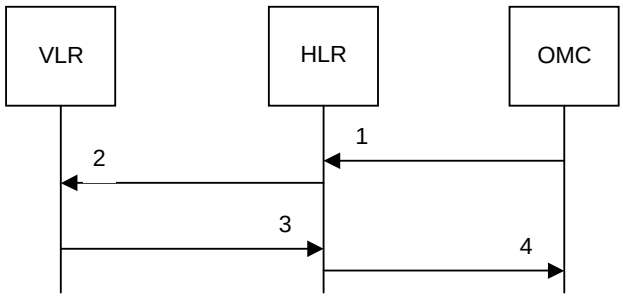
20.3 Subscriber data management procedures for HLR

Two types of subscriber data management procedures exist:

1) Subscriber Deletion;

2) Subscriber Data Modification.

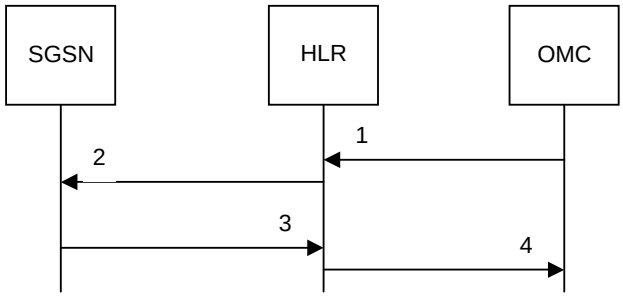
The subscriber deletion and subscriber data modification procedures are initiated by the OMC (see figures 20.3/1 , 20.3/2, 20.3/3 and 20.3/4).



1) Delete Subscriber
2) MAP_CANCEL_LOCATION_req/ind
3) MAP_CANCEL_LOCATION_rsp/cnf
4) Subscriber Deleted

Figure 20.3/1: Subscriber deletion procedure for non-GPRS

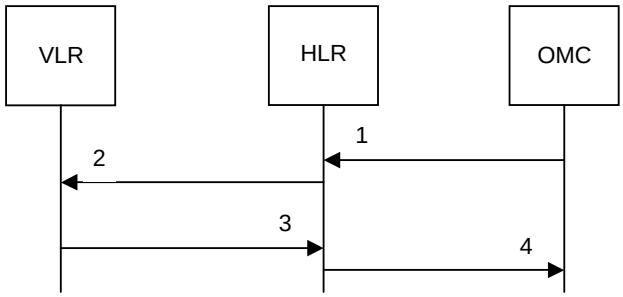
In the subscriber deletion procedure for a non-GPRS subscriber the subscriber data are removed from the VLR and the HLR. The HLR uses the MAP_CANCEL_LOCATION service.



1) Delete GPRS Subscriber
2) MAP_CANCEL_LOCATION_req/ind
3) MAP_CANCEL_LOCATION_rsp/cnf
4) GPRS Subscriber Deleted

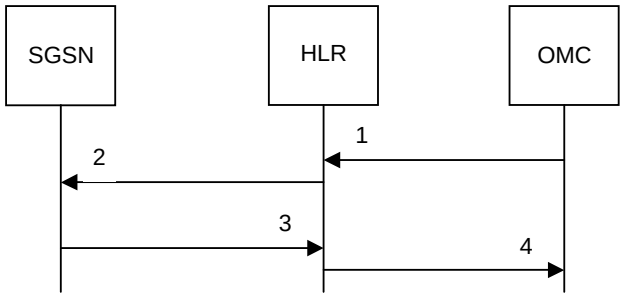
Figure 20.3/2: Subscriber deletion procedure for GPRS

In the subscriber deletion procedure for a GPRS subscriber the subscriber data are removed from the SGSN and the HLR. The HLR uses the MAP_CANCEL_LOCATION service.



1) Modify Subscriber Data
2) MAP_CANCEL_LOCATION_req/ind, MAP_INSERT_SUBSCRIBER_DATA_req/ind or MAP_DELETE_SUBSCRIBER_DATA_req/ind
3) MAP_CANCEL_LOCATION_rsp/cnf, MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf or MAP_DELETE_SUBSCRIBER_DATA_rsp/cnf
4) Subscriber Data Modified

Figure 20.3/3: Subscriber data modification procedure for non-GPRS



1) Modify GPRS Subscriber Data
2) MAP_CANCEL_LOCATION_req/ind, MAP_INSERT_SUBSCRIBER_DATA_req/ind or MAP_DELETE_SUBSCRIBER_DATA_req/ind
3) MAP_CANCEL_LOCATION_rsp/cnf, MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf or MAP_DELETE_SUBSCRIBER_DATA_rsp/cnf
4) GPRS Subscriber Data Modified

Figure 20.3/4: Subscriber data modification procedure for GPRS

In the subscriber data modification procedure the subscriber data are modified in the HLR and when necessary also in the VLR or in the SGSN. The HLR initiates one of the MAP_INSERT_SUBSCRIBER_DATA, MAP_DELETE_SUBSCRIBER_DATA or MAP_CANCEL_LOCATION services depending on the modified data.

20.3.1 Subscriber deletion procedure

20.3.1.1 Procedure in the HLR

The subscriber deletion process in the HLR is shown in figure 20.3/5. The MAP process invokes processes not defined in this clause; the definitions of these processes can be found as follows:

Cancel_GPRS_Location_Child_HLR see subclause 19.1.2.2;

Cancel_Location_Child_HLR see subclause 19.1.2.2.

20.3.1.2 Procedure in the VLR

The subscriber deletion procedure in the VLR is described in subclause 19.1.2.3 of the present document.

20.3.1.3 Procedure in the SGSN

The subscriber deletion procedure in the SGSN is described in subclause 19.1. 2.4 of the present document.

20.3.2 Subscriber data modification procedure

20.3.2.1 Procedure in the HLR

The OMC can modify the subscriber data in several different ways. The modifications can be categorised in the following groups:

1) data shall be modified in the HLR; no effect in the VLR;
2) data shall be modified in both the HLR and the VLR;
3) withdrawal of a basic service or a supplementary service requiring change to VLR data;
4) modification affects the roaming permission for the subscriber and the subscriber record shall be removed from the VLR data base;
5) withdrawal of non-GPRS Subscription caused by a change of Network Access Mode;
6) data shall be modified in the HLR; no effect in the SGSN;
7) data shall be modified in both the HLR and the SGSN;
8) withdrawal of GPRS subscription data or a basic service or a supplementary service requiring change to SGSN data;
9) modification affects the roaming permission for the subscriber and the subscriber record shall be removed from the SGSN data base;
10) withdrawal of GPRS Subscription caused by a change of Network Access Mode;
11) authentication algorithm or authentication key of the subscriber is modified.

In cases 2 and 7 the HLR uses the MAP_INSERT_SUBSCRIBER_DATA service.

In cases 3 and 8 the HLR uses the MAP_DELETE_SUBSCRIBER_DATA service.

In cases 4, 5, 9, 10 and 11 the HLR uses the MAP_CANCEL_LOCATION service.

If the deletion of subscriber data fails, the HLR may repeat the request; the number of repeat attempts and the time in between are HLR operator options, depending on the error returned by the VLR or the SGSN.

The subscriber data modification process in the HLR is shown in figure 20.3/6. The MAP process invokes processes not defined in this clause; the definitions of these processes can be found as follows:

Insert_Subs_Data_Stand_Alone_HLR see subclause 25.7.3;

Cancel_Location_Child_HLR see subclause 19.1.2.2;

Insert_GPRS_Subs_Data_Stand_Alone_HLR see subclause 25.7.4;

Cancel_GPRS_Location_Child_HLR see subclause 19.1.2.2.

The macro Delete_Subscriber_Data_HLR is shown in figure 20.3/7. The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The macro Delete_GPRS_Subscriber_Data_HLR is shown in figure 20.3/8. The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

20.3.2.2 Procedures in the VLR

The process in the VLR to update subscriber data in a stand-alone dialogue is shown in figure 20.3/9. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Indication see subclause 25.2.1;

Insert_Subs_Data_VLR see subclause 25.7.1.

The process in the VLR to delete subscriber data is shown in figure 20.3/10. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

20.3.2.3 Procedures in the SGSN

The process in the SGSN to update subscriber data in a stand-alone dialogue is shown in figure 20.3/11. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Indication see subclause 25.2.1;

Insert_Subs_Data_SGSN see subclause 25.7.2.

The process in the SGSN to delete subscriber data is shown in figure 20.3/12. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

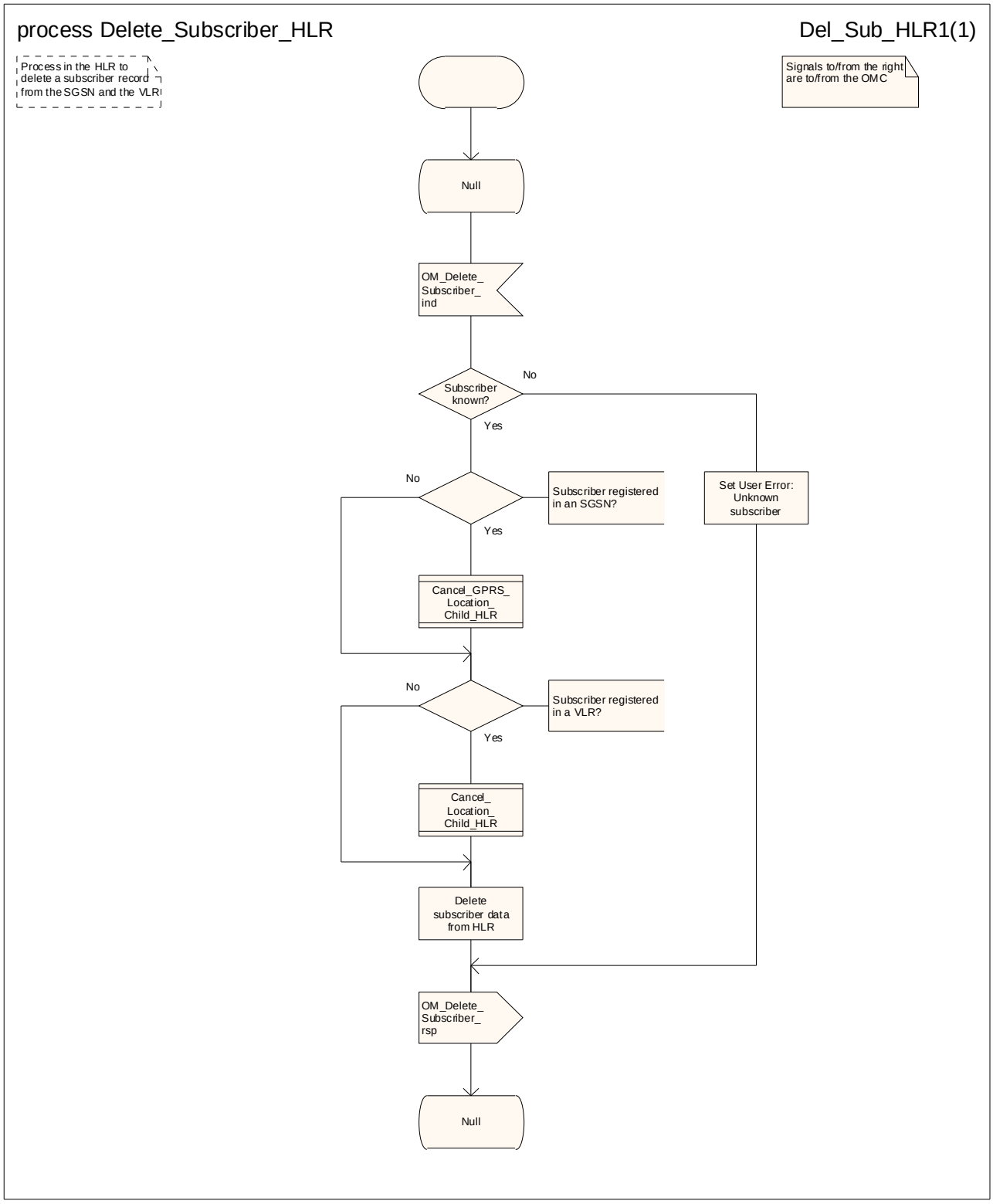


Figure 20.3/5: Process Delete_Subscriber_HLR

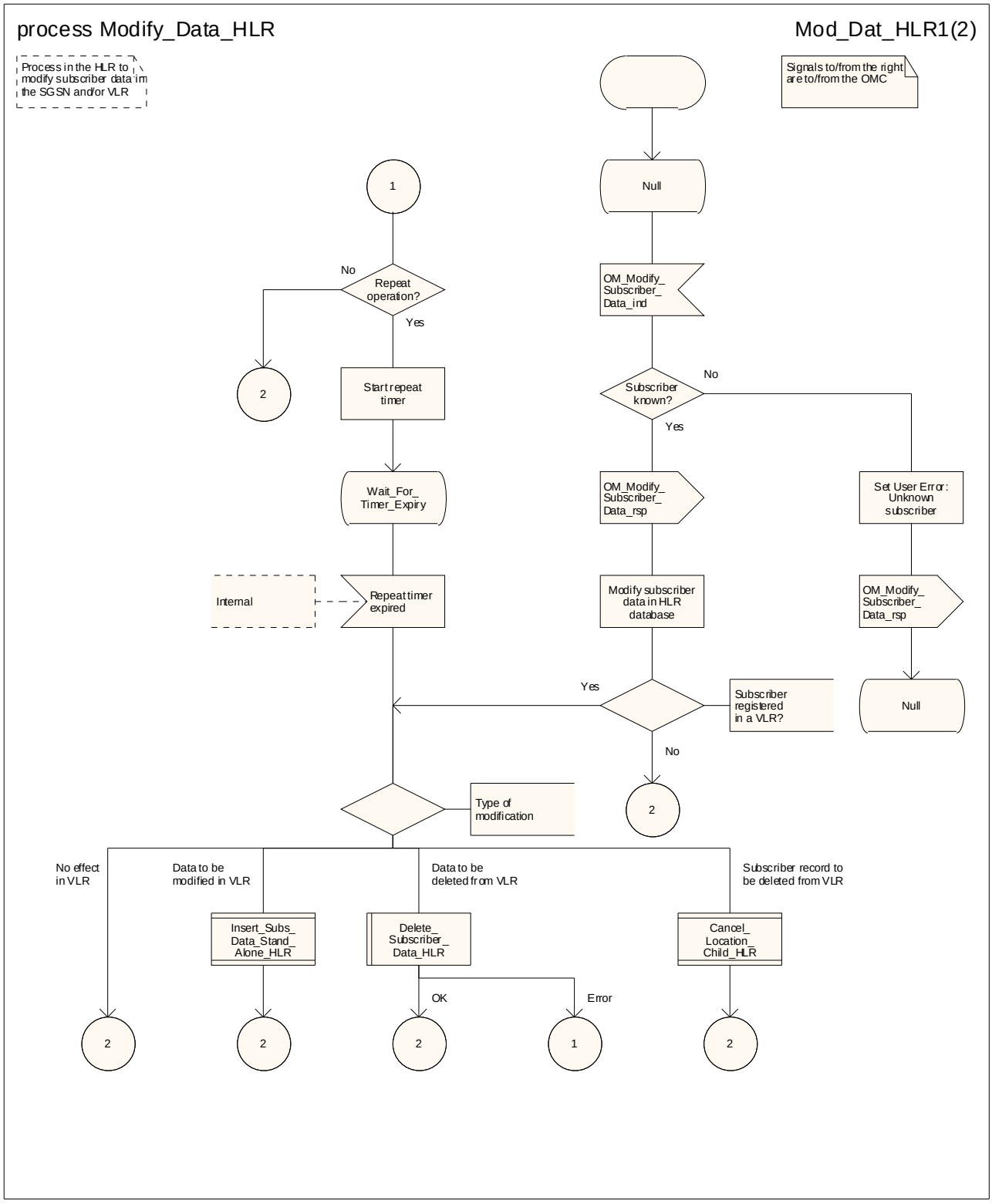


Figure 20.3/6 (sheet 1 of 2): Process Modify_Data_HLR

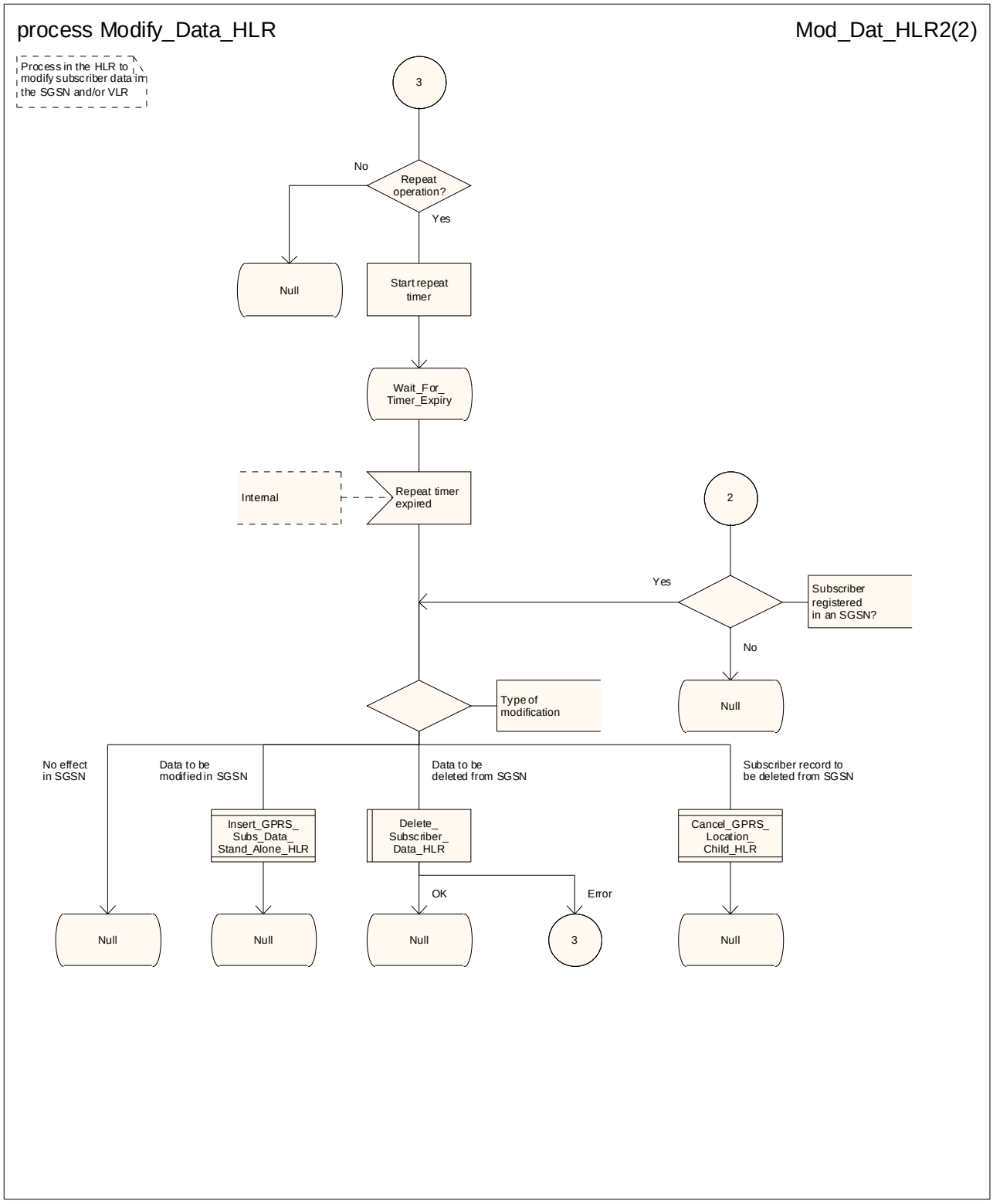


Figure 20.3/6 (sheet 2 of 2): Process Modify_Data_HLR

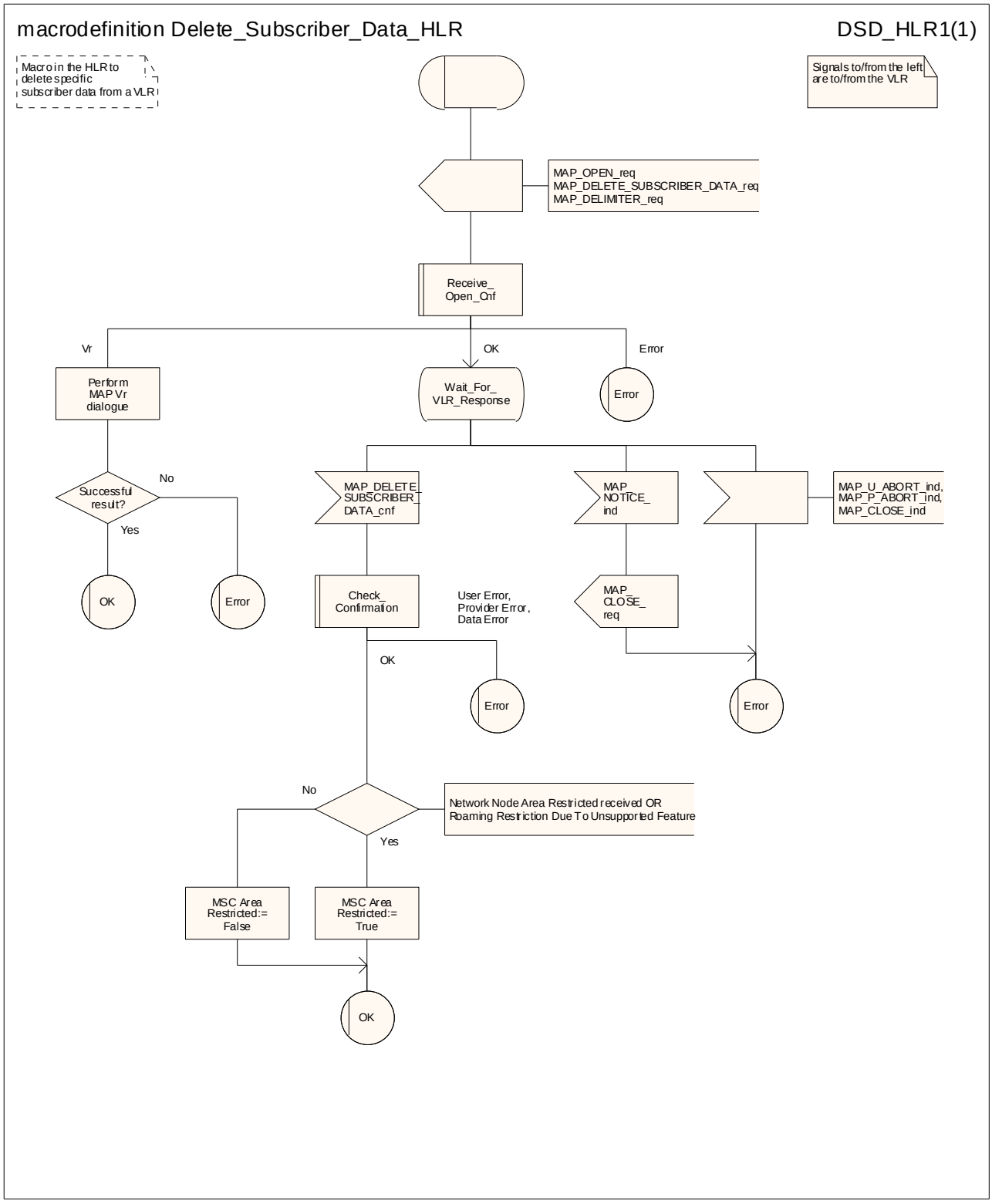


Figure 20.3/7: Macro Delete_Subscriber_Data_HLR

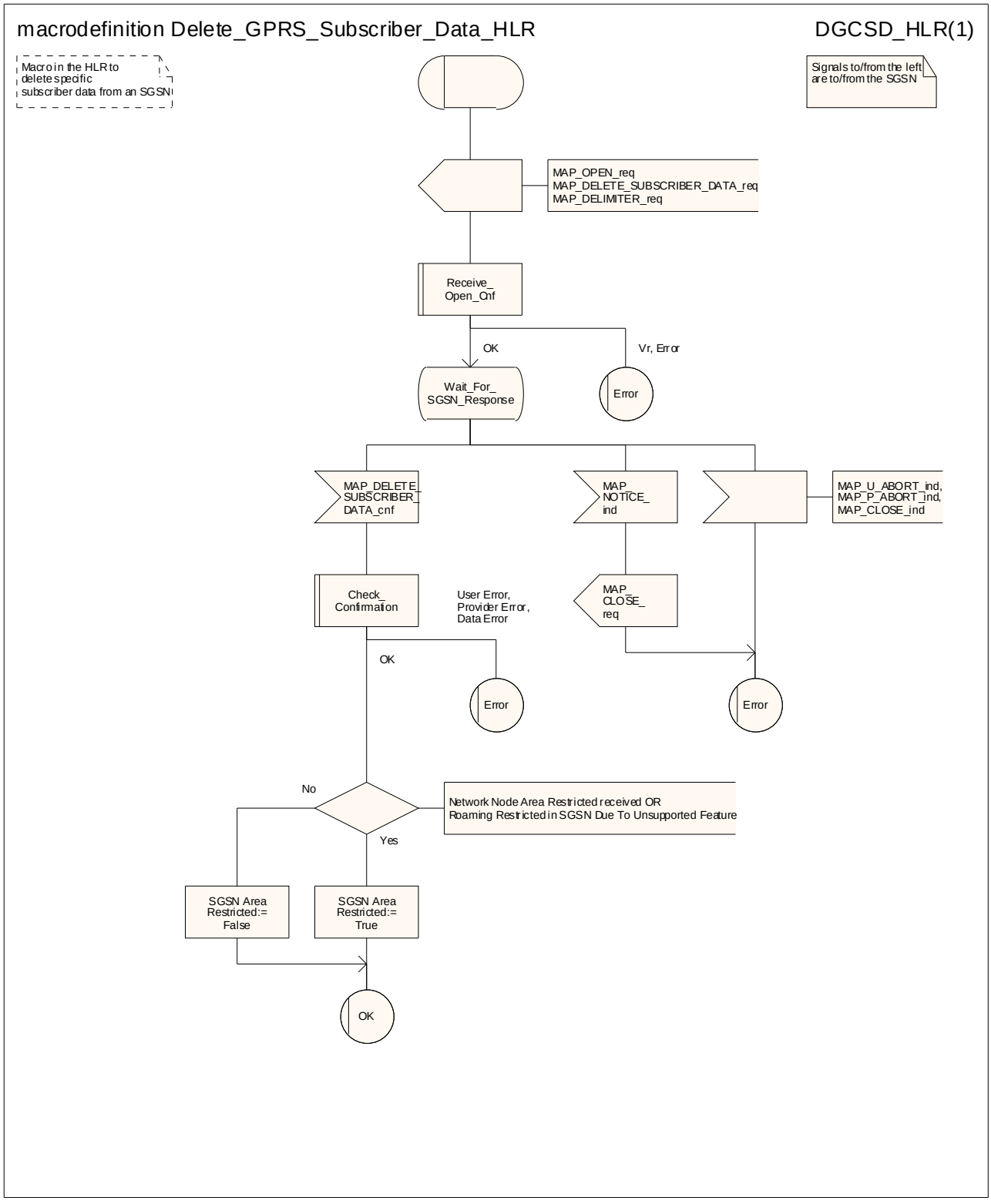


Figure 20.3/8: Macro Delete_GPRS_Subscriber_Data_HLR

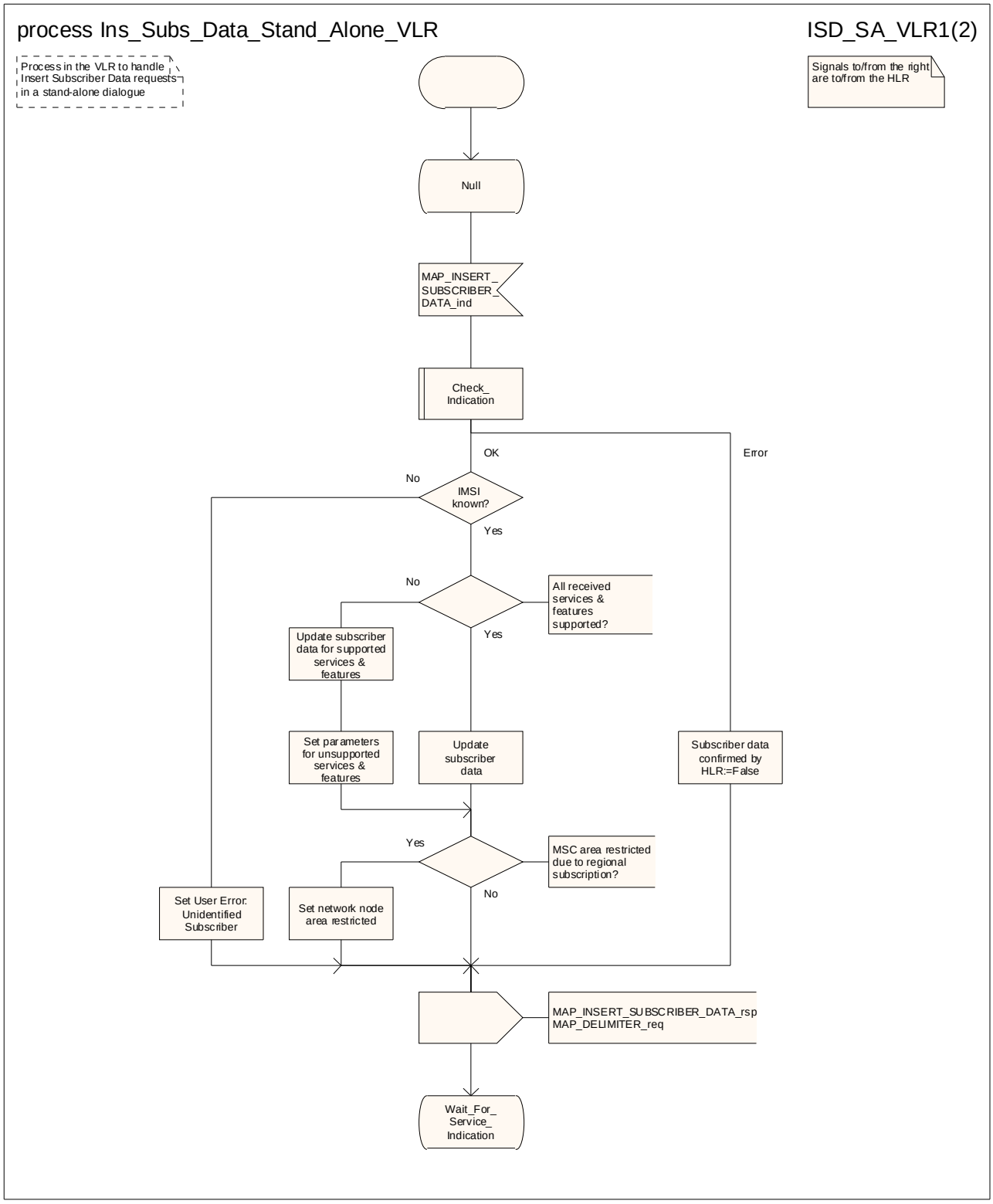


Figure 20.3/9 (sheet 1 of 2): Process Ins_Subs_Data_Stand_Alone_VLR

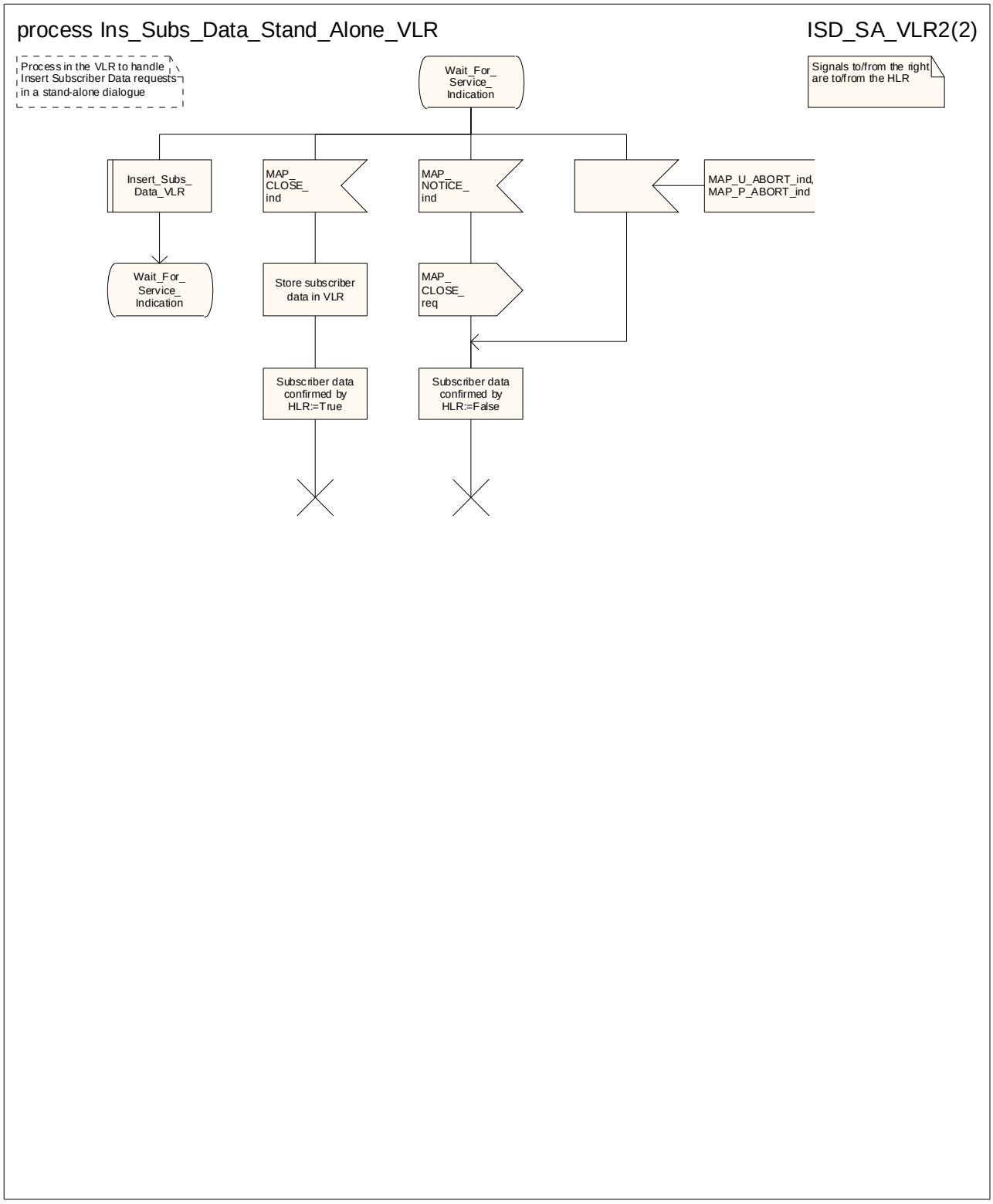


Figure 20.3/9 (sheet 2 of 2): Process Ins_Subs_Data_Stand_Alone_VLR

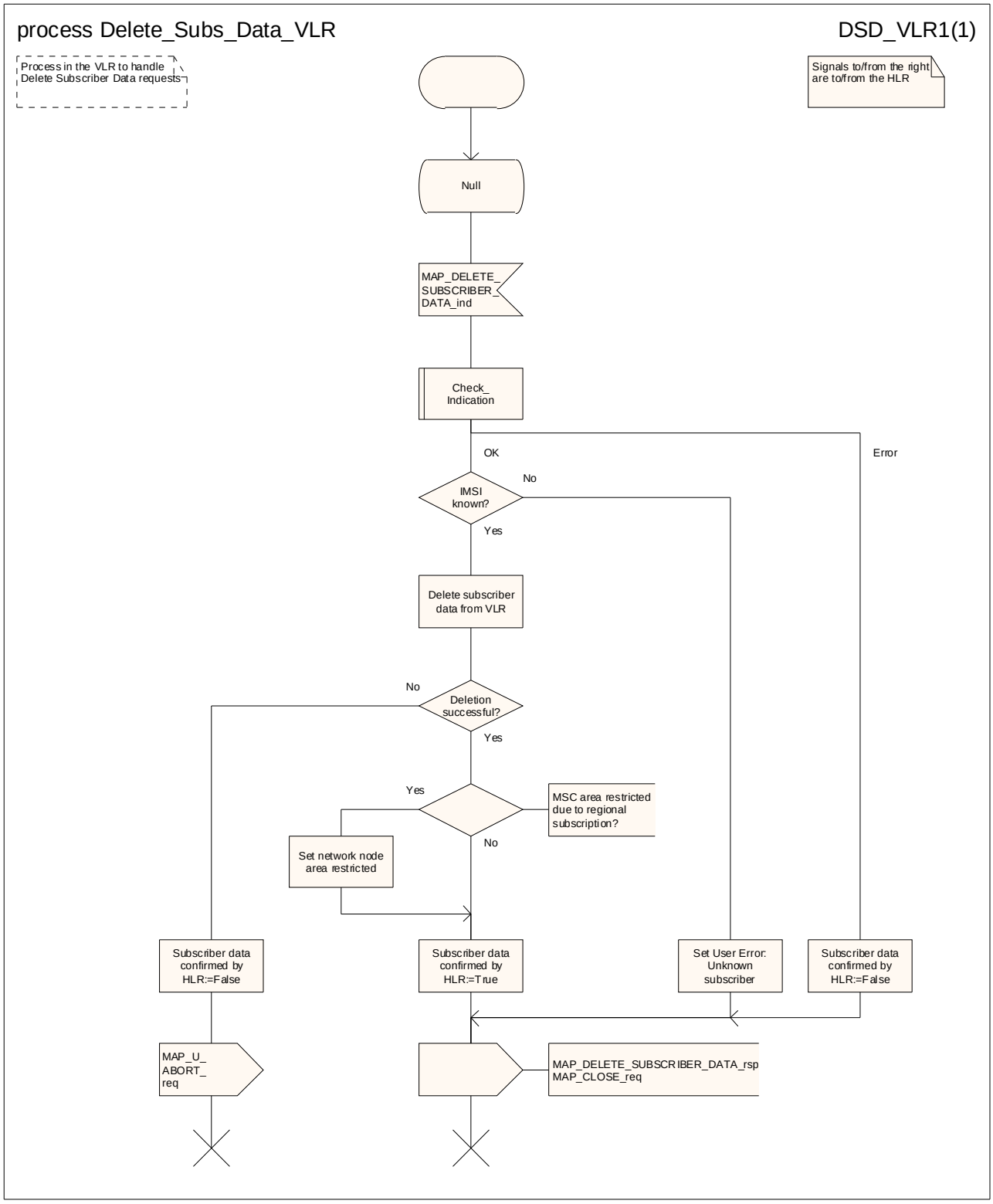


Figure 20.3/10: Process Delete_Subs_Data_VLR

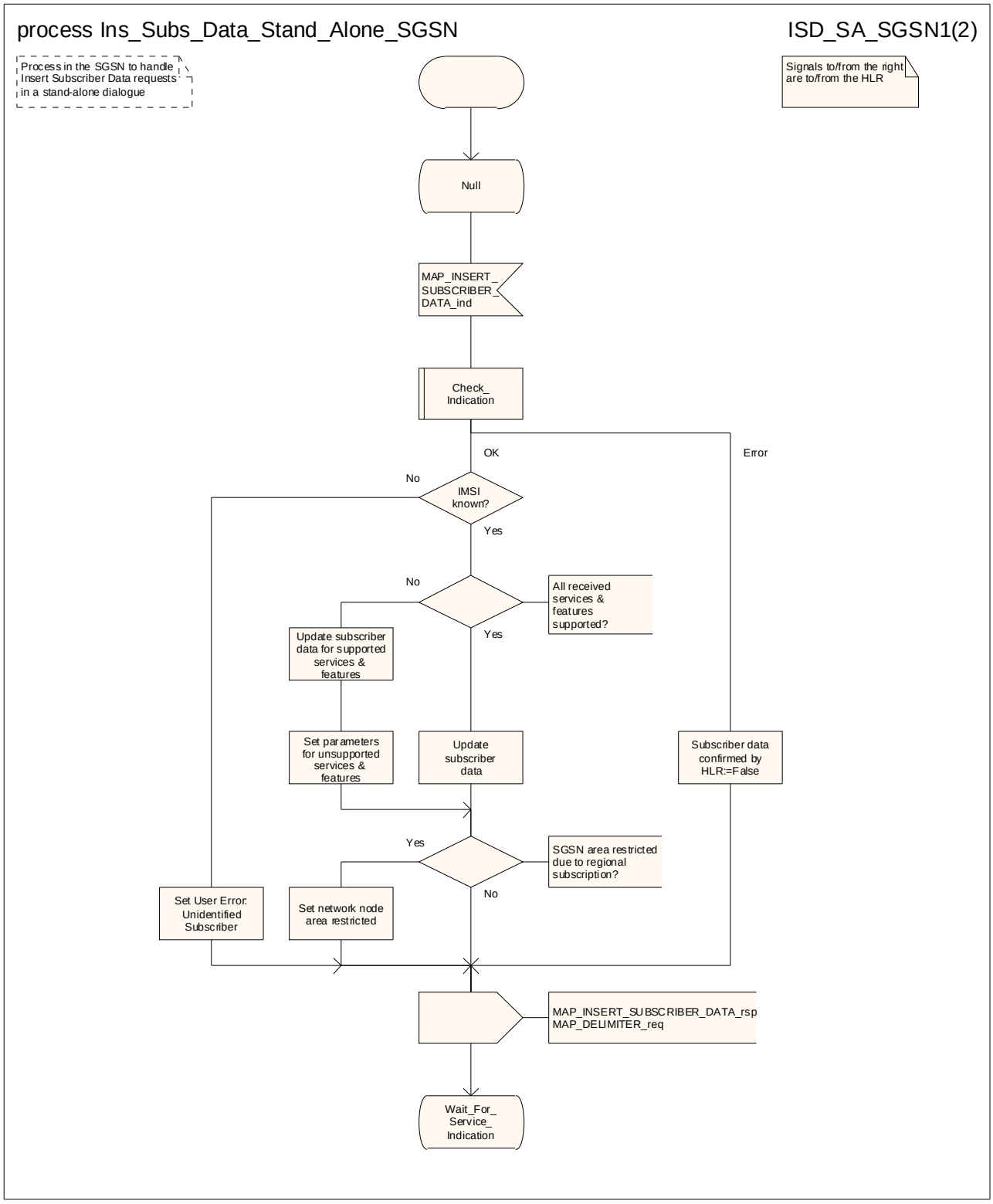


Figure 20.3/11 (sheet 1 of 2): Process Ins_Subs_Data_Stand_Alone_SGSN

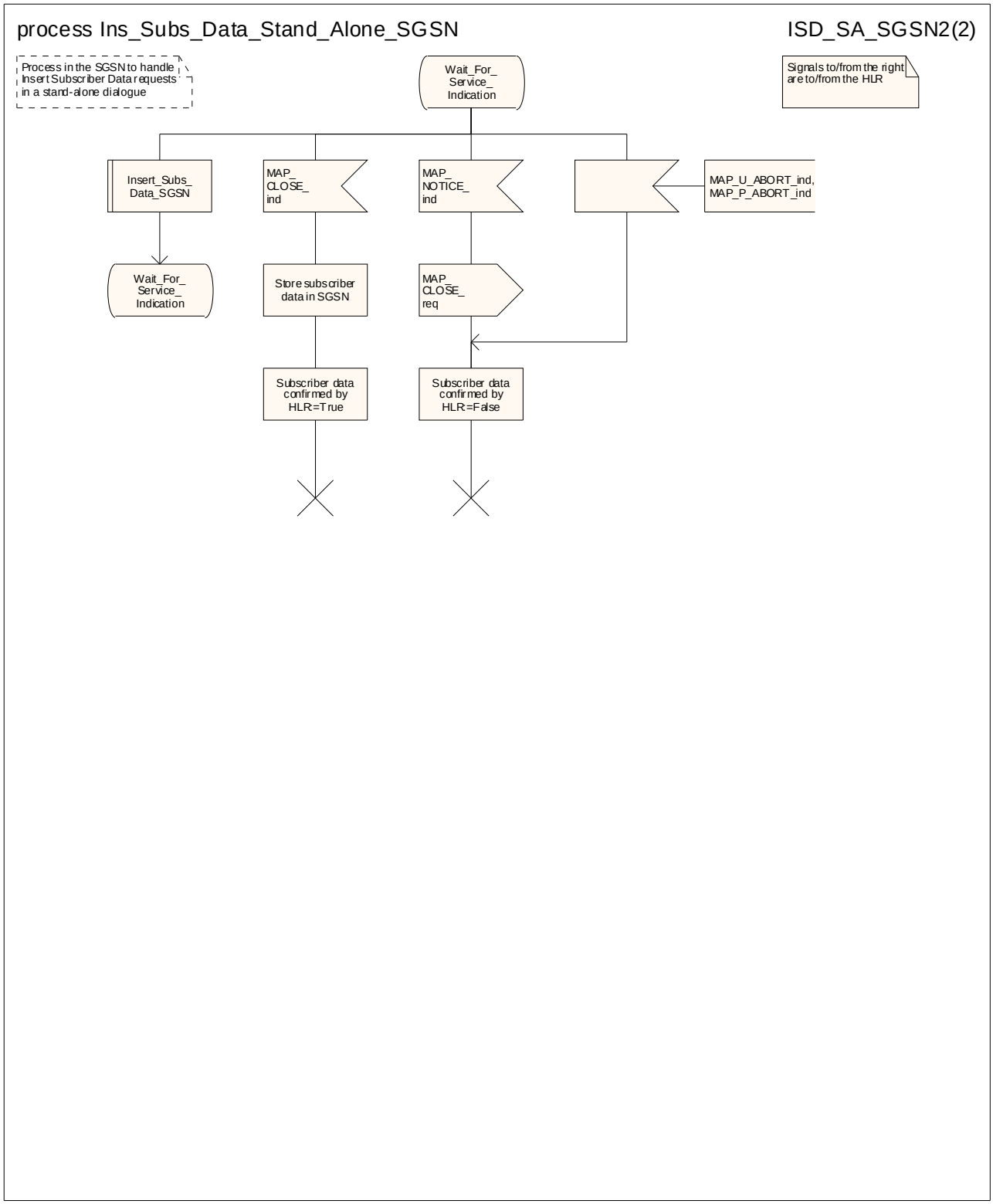


Figure 20.3/11 (sheet 2 of 2): Process Ins_Subs_Data_Stand_Alone_SGSN

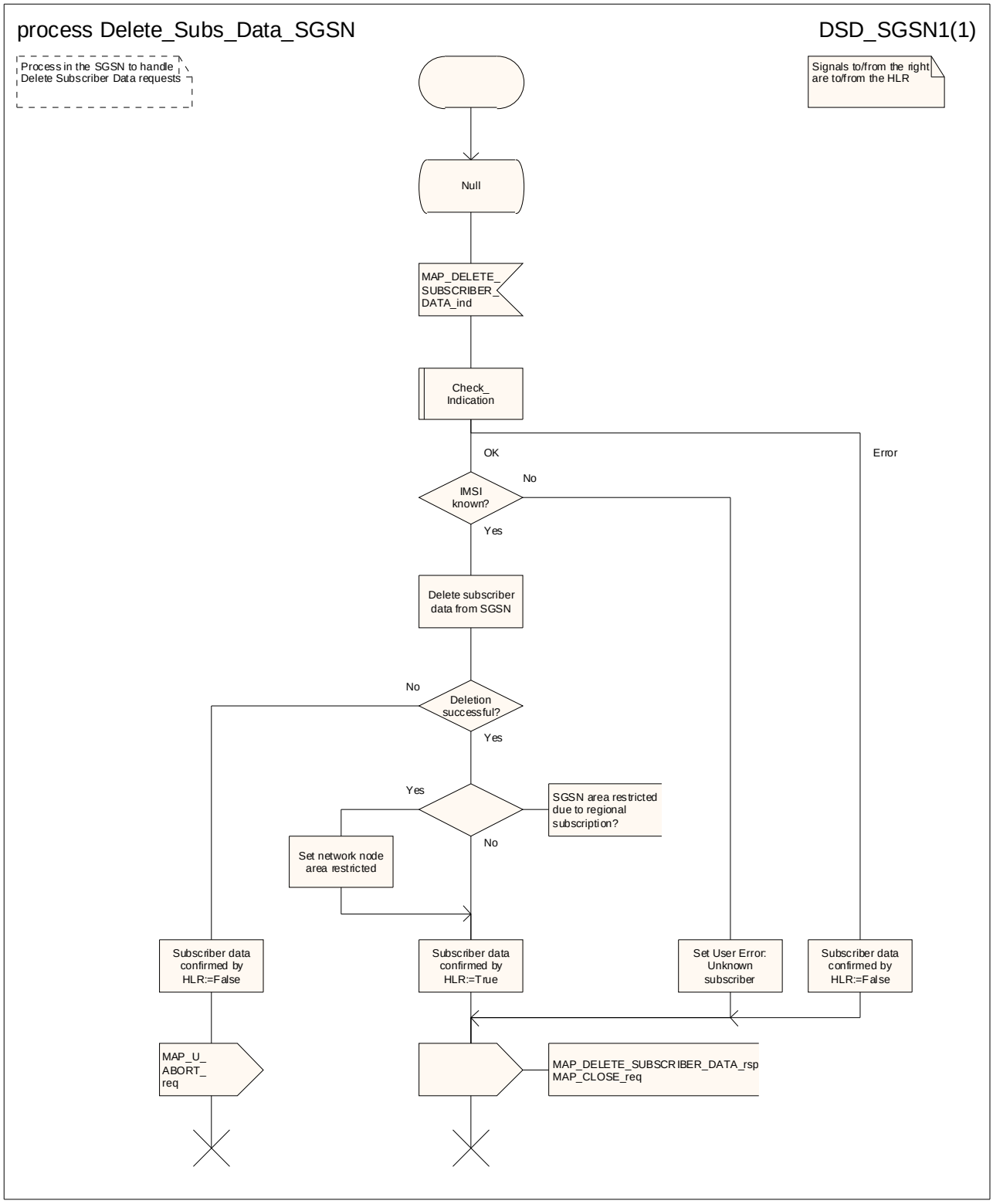


Figure 20.3/12: Process Delete_Subs_Data_SGSN

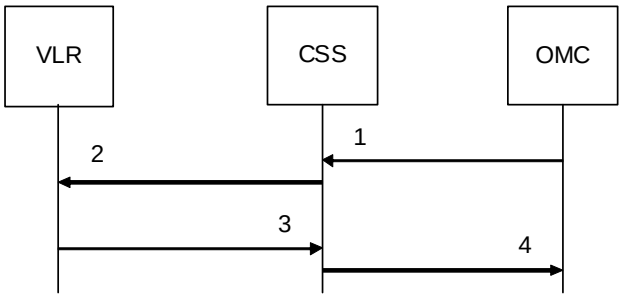
20.3A Subscriber Data Management procedures for CSS

Two types of subscriber data management procedures exist:

1) Subscriber Deletion;

2) Subscriber Data Modification.

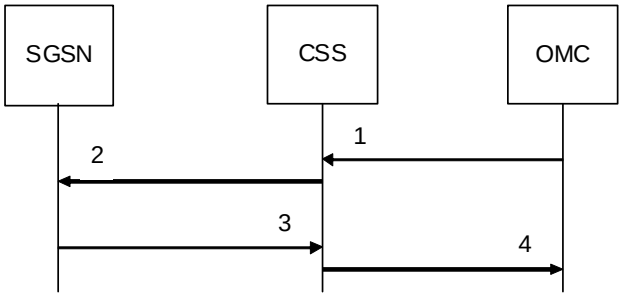
The subscriber deletion and subscriber data modification procedures are initiated by the OMC (see figures 20.3A/1, 20.3A/2, 20.3A/3 and 20.3A/4).



1) Delete Subscriber
2) MAP_DELETE_SUBSCRIBER_DATA_req/ind
3) MAP_DELETE_SUBSCRIBER_DATA_rsp/cnf
4) Subscriber Deleted

Figure 20.3A/1: Subscriber deletion procedure for non-GPRS

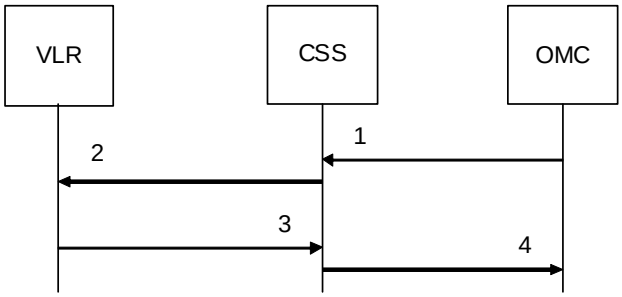
In the subscriber deletion procedure for a non-GPRS subscriber the CSG subscription data for the subscriber in the VPLMN are removed from the VLR and the CSS. The CSS uses the MAP_DELETE_SUBSCRIBER_DATA service.



1) Delete GPRS Subscriber
2) MAP_DELETE_SUBSCRIBER_DATA_req/ind
3) MAP_DELETE_SUBSCRIBER_DATA_rsp/cnf
4) GPRS Subscriber Deleted

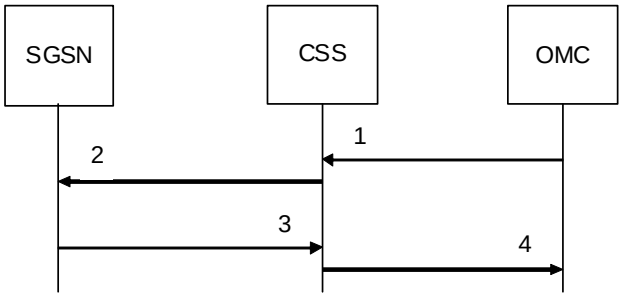
Figure 20.3A/2: Subscriber deletion procedure for GPRS

In the subscriber deletion procedure for a GPRS subscriber the CSG subscription data for the GPRS subscriber in the VPLMN are removed from the SGSN and the CSS. The CSS uses the MAP_DELETE_SUBSCRIBER_DATA service.



1) Modify Subscriber Data
2) MAP_INSERT_SUBSCRIBER_DATA_req/ind or MAP_DELETE_SUBSCRIBER_DATA_req/ind
3) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf or MAP_DELETE_SUBSCRIBER_DATA_rsp/cnf
4) Subscriber Data Modified

Figure 20.3A/3: Subscriber data modification procedure for non-GPRS



1) Modify GPRS Subscriber Data
2) MAP_INSERT_SUBSCRIBER_DATA_req/ind or MAP_DELETE_SUBSCRIBER_DATA_req/ind
3) MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf or MAP_DELETE_SUBSCRIBER_DATA_rsp/cnf
4) GPRS Subscriber Data Modified

Figure 20.3A/4: Subscriber data modification procedure for GPRS

In the subscriber data modification procedure the CSG subscription data in the VPLMN for the subscriber data are modified in the CSS and when necessary also in the VLR or in the SGSN. The CSS initiates one of the MAP_INSERT_SUBSCRIBER_DATA or MAP_DELETE_SUBSCRIBER_DATA services depending on the modified data.

20.3A.1 Subscriber deletion procedure

20.3A.1.1 Procedure in the CSS

The process in the CSS to delete subscriber is shown in figure 20.3A/5. In this case the CSS uses the MAP_DELETE_SUBSCRIBER_DATA service.

20.3A.1.2 Procedure in the VLR

The process in the VLR for the CSG subscriber deletion procedure is shown in figure 20.3A/9.

20.3A.1.3 Procedure in the SGSN

The process in the SGSN for the CSG subscriber deletion procedure is shown in figure 20.3A/11.

20.3A.2 Subscriber data modification procedure

20.3A.2.1 Procedure in the CSS

The OMC can modify the CSG subscriber data in several different ways. The modifications can be categorised in the following groups:

1) CSG subsctiption data shall be modified in the CSS; no effect in the VLR;
2) CSG subsctiption data shall be modified in both the CSS and the VLR;
3) withdrawal of CSG subscription data requiring change to VLR data.
4) CSG subsctiption data shall be modified in the CSS; no effect in the SGSN;
5) CSG subsctiption data shall be modified in both the CSS and the SGSN;
6) withdrawal of CSG subscription data requiring change to SGSN data.

In cases 2 and 5 the CSS uses the MAP_INSERT_SUBSCRIBER_DATA service.

In cases 3 and 6 the CSS uses the MAP_DELETE_SUBSCRIBER_DATA service.

If the deletion of CSG subscriber data fails, the CSS may repeat the request; the number of repeat attempts and the time in between are CSS operator options, depending on the error returned by the VLR or the SGSN. The CSS removes the routeting information after the completion of CSG subscriber data deletion procedure.

The CSG subscriber data modification process in the CSS is shown in figure 20.3A/6. The MAP process invokes processes not defined in this clause; the definitions of these processes can be found as follows:

Insert_VCSG_Subs_Data_Stand_Alone_CSS see subclause 25.7.8;

The macro Delete_VCSG_Subs_Data_CSS is shown in figure 20.3A/7. The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

20.3A.2.2 Procedures in the VLR

The process in the VLR to update CSG subscription data in the VPLMN for the subscriber in a stand-alone dialogue is shown in figure 20.3A/8. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Indication see subclause 25.2.1;

Insert_Subs_Data_VLR see subclause 25.7.1.

The process in the VLR to delete CSG subscriber data is shown in figure 20.3A/9. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

20.3A.2.3 Procedures in the SGSN

The process in the SGSN to update CSG subscription data in the VPLMN for the GPRS subscriber in a stand-alone dialogue is shown in figure 20.3A/10. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Indication see subclause 25.2.1;

Insert_Subs_Data_SGSN see subclause 25.7.2.

The process in the SGSN to delete subscriber data is shown in figure 20.3A/11. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

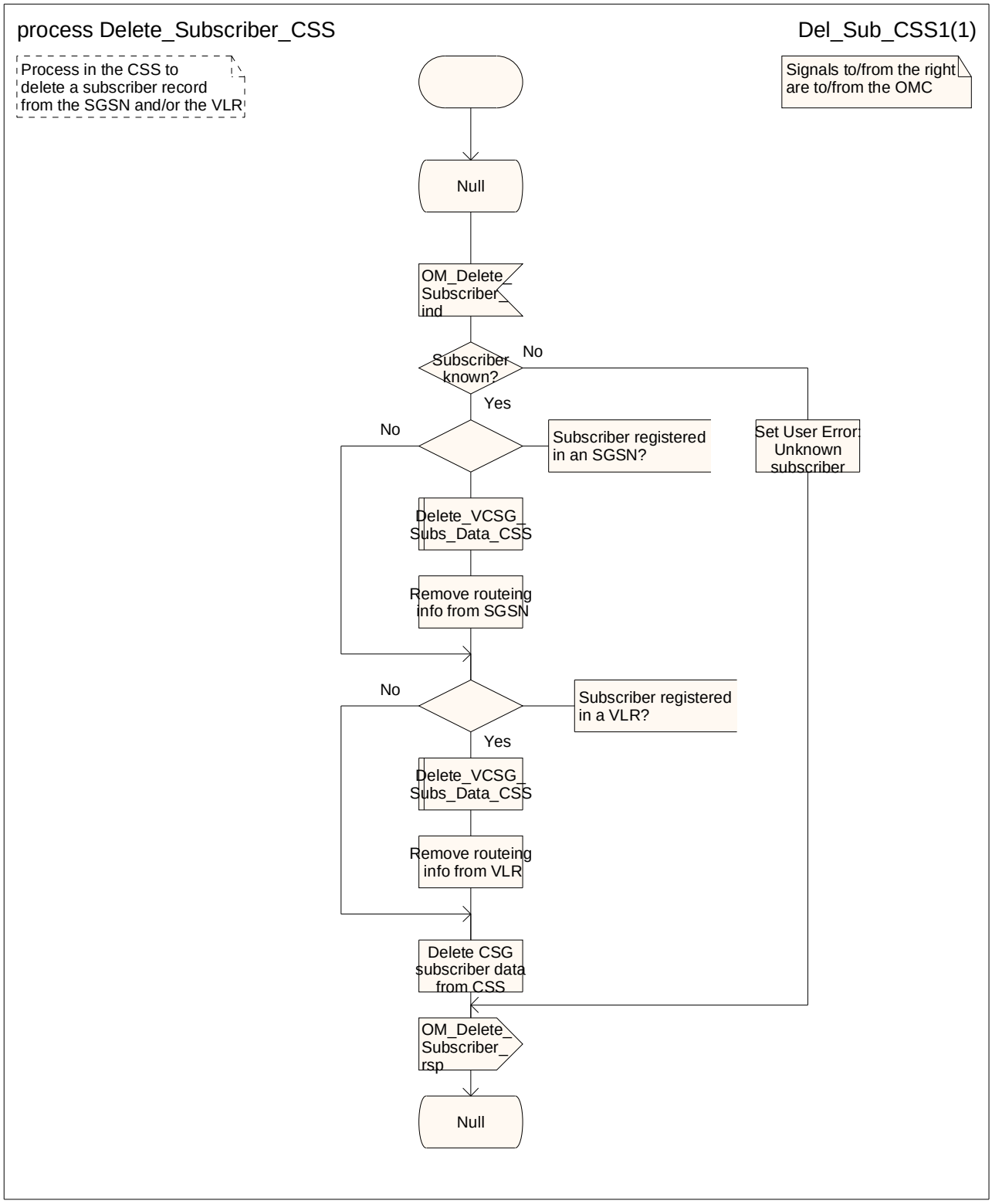


Figure 20.3A/5: Process Delete_Subscriber_CSS

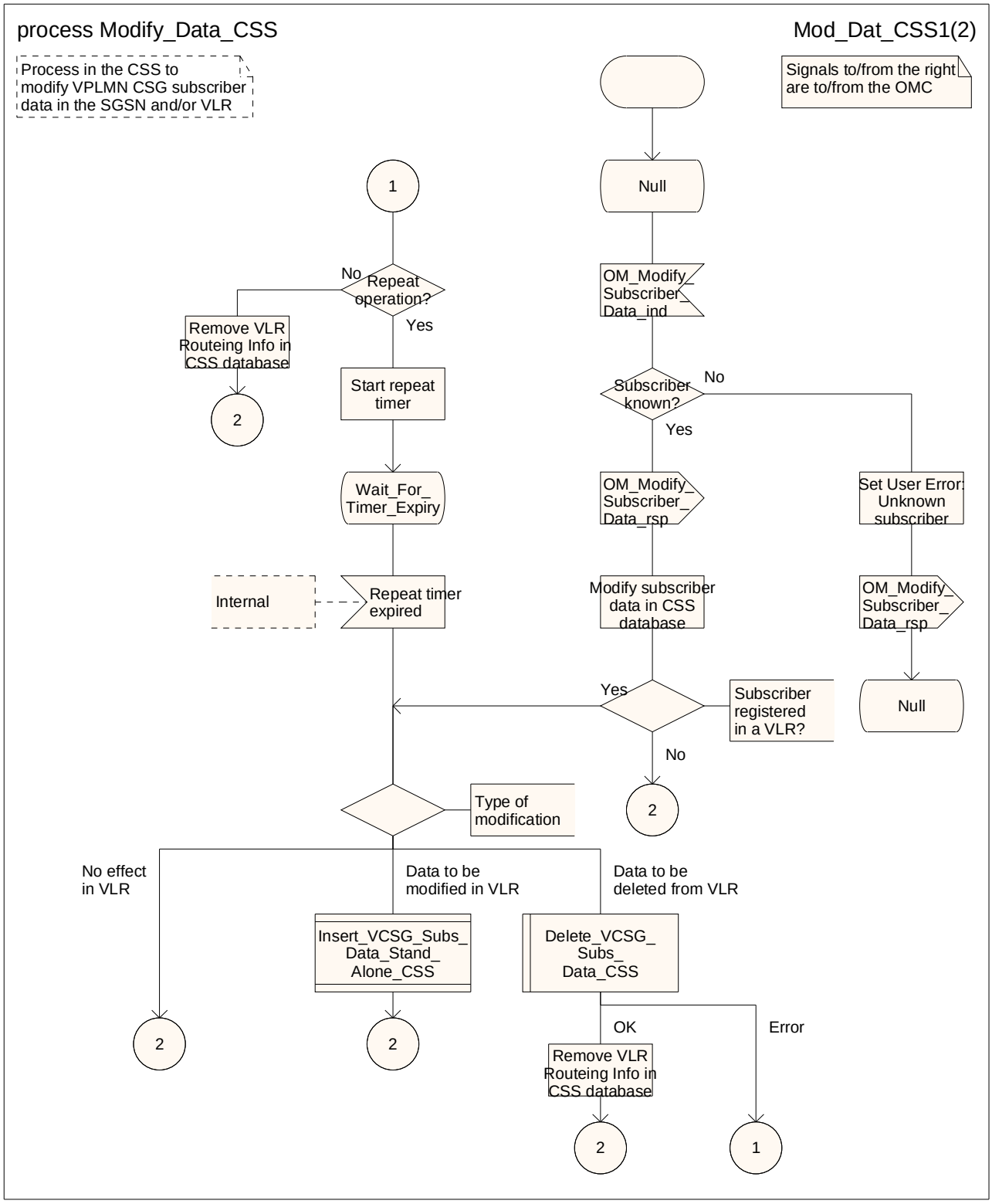


Figure 20.3A/6 (sheet 1 of 2): Process Modify_Data_CSS

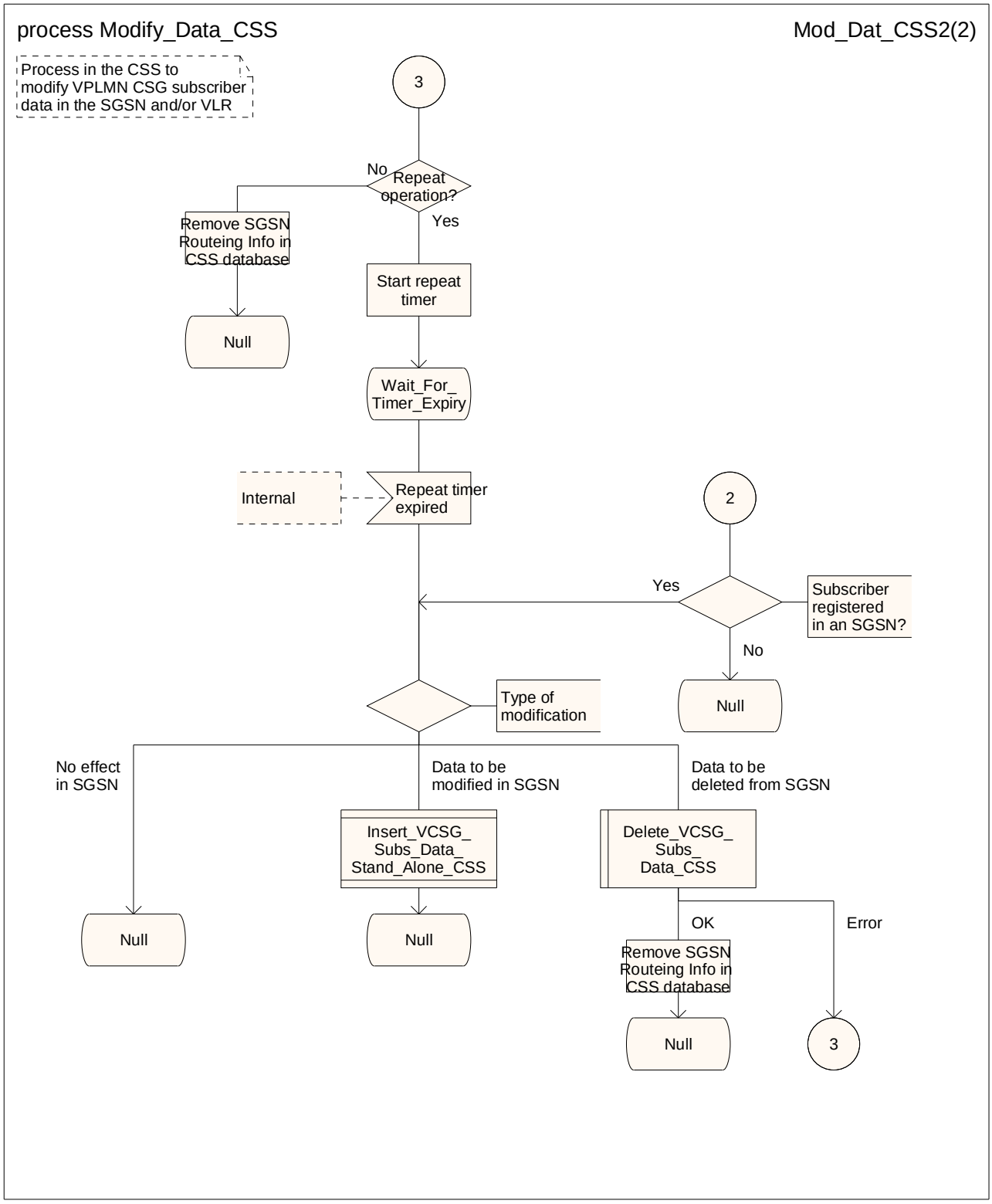


Figure 20.3A/6 (sheet 2 of 2): Process Modify_Data_CSS

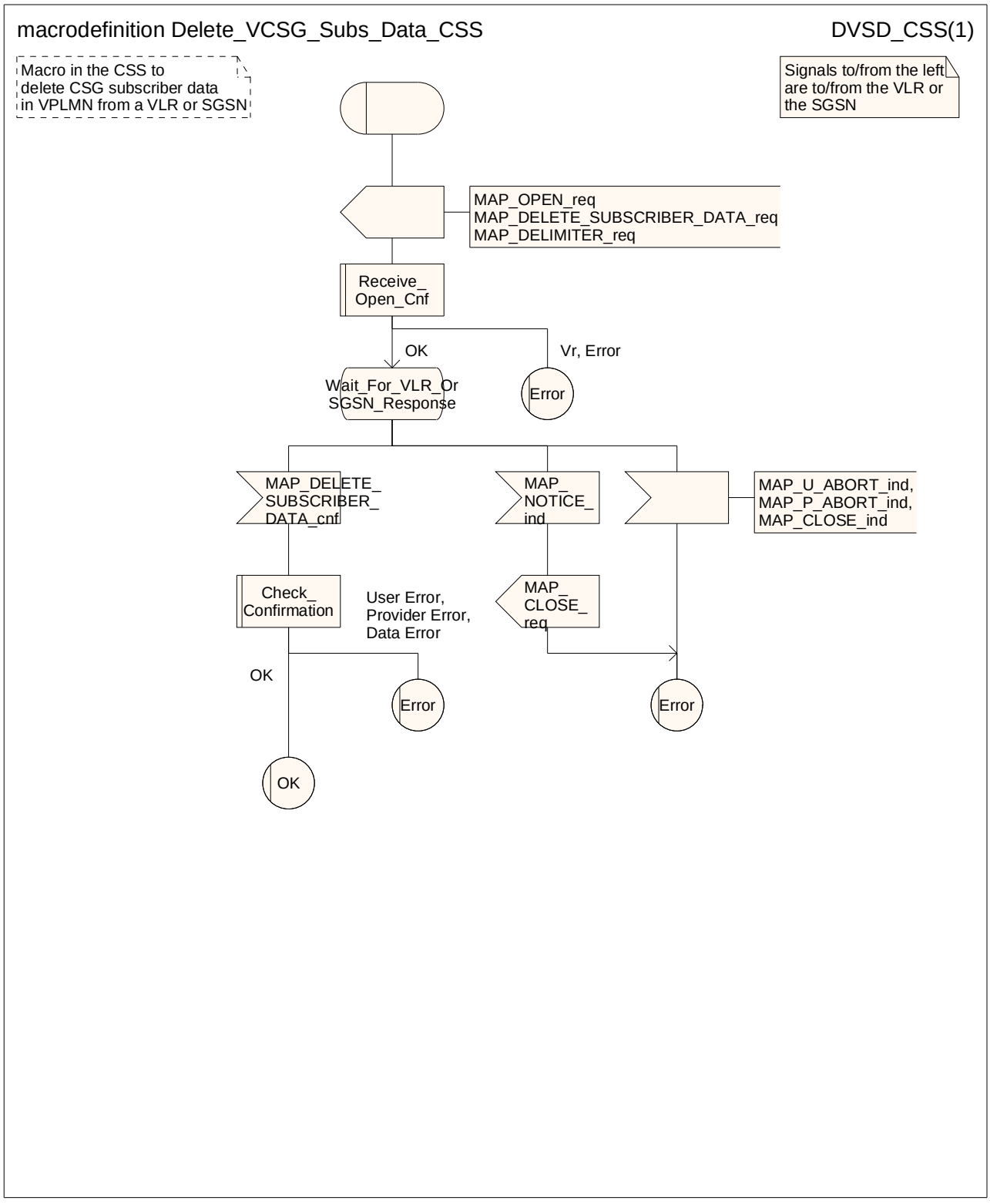


Figure 20.3A/7: Macro Delete_VCSG_Subs_Data_CSS

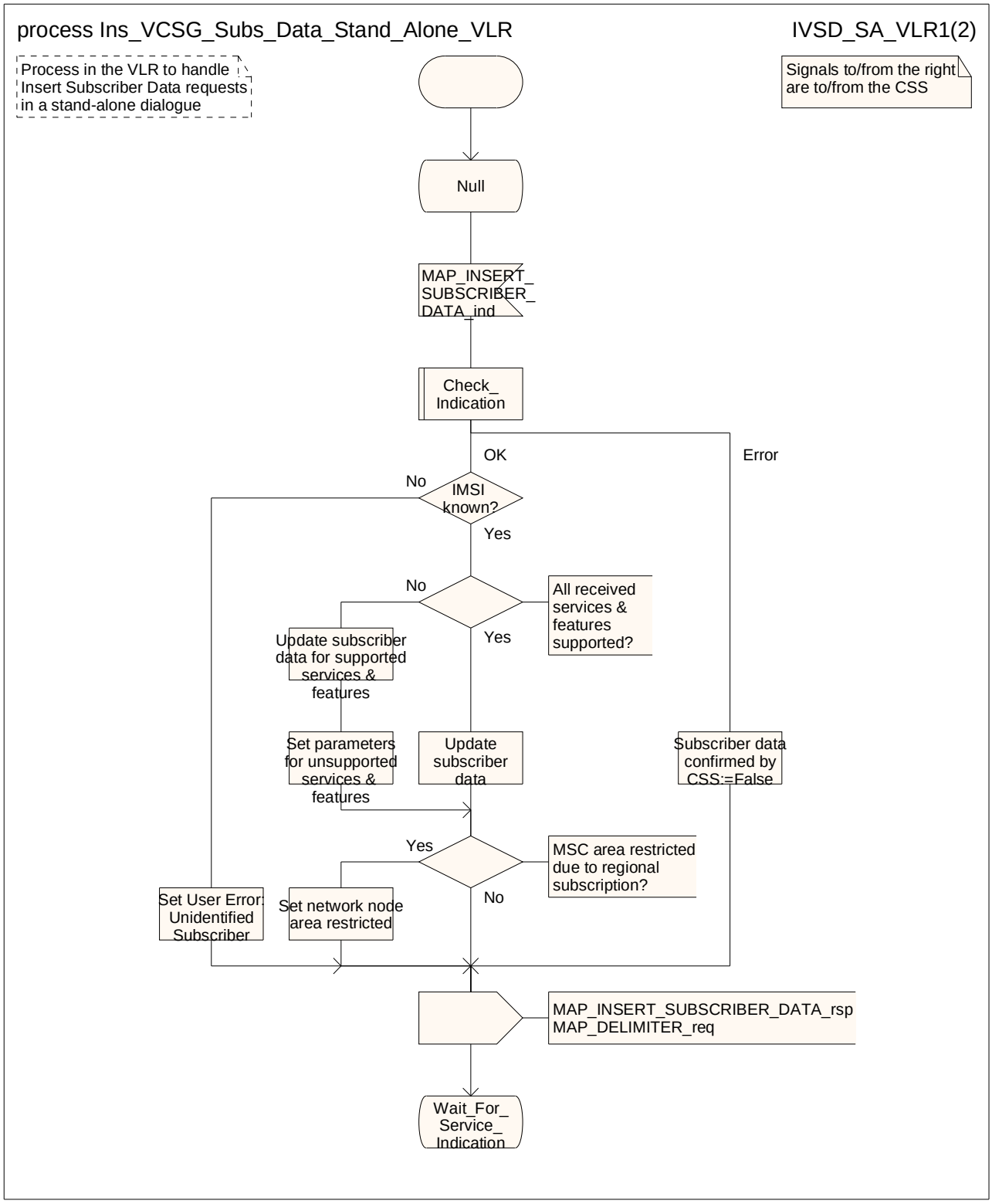


Figure 20.3A/8 (sheet 1 of 2): Process Ins_VCSG_Subs_Data_Stand_Alone_VLR

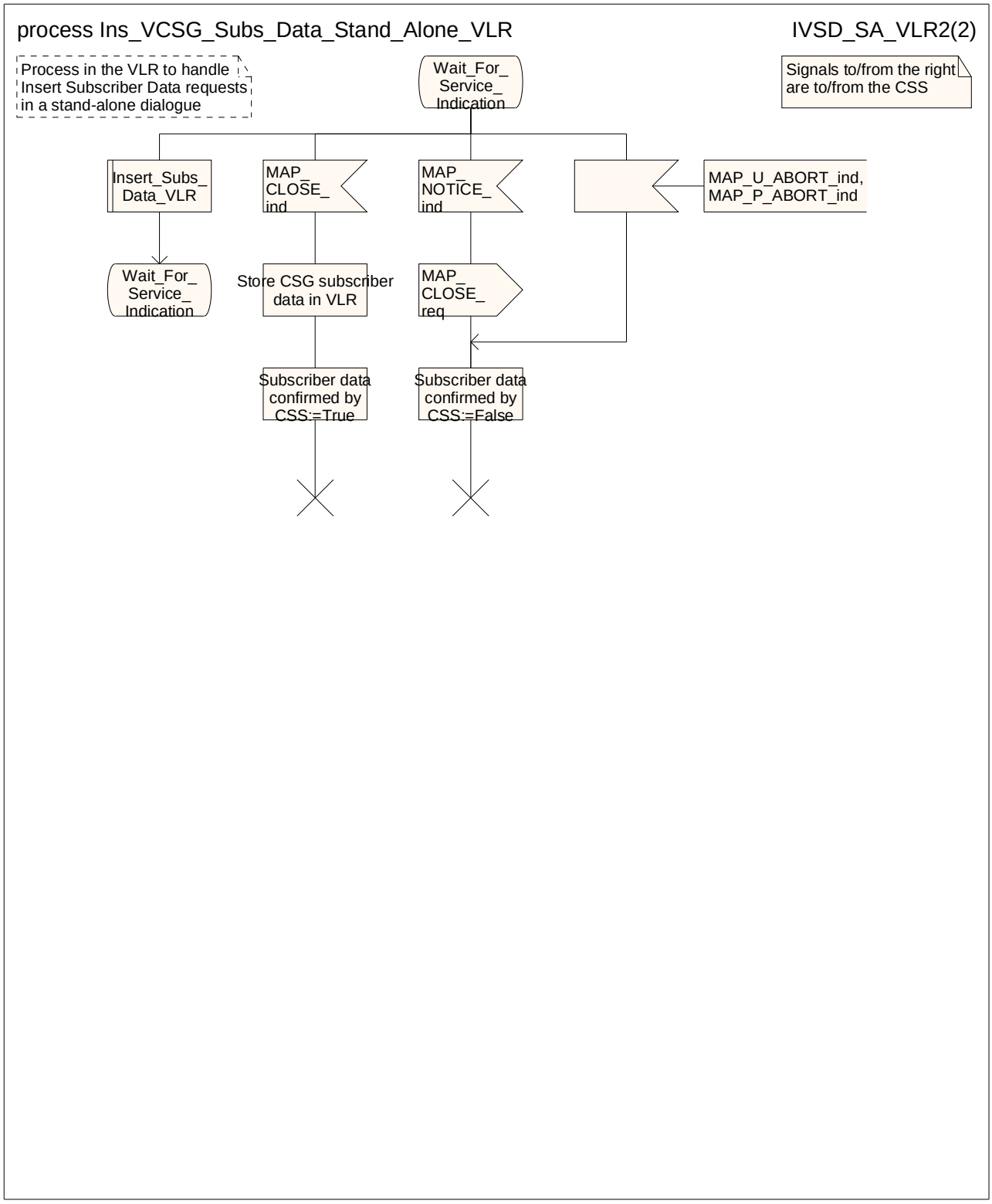


Figure 20.3A/8 (sheet 2 of 2): Process Ins_VCSG_Subs_Data_Stand_Alone_VLR

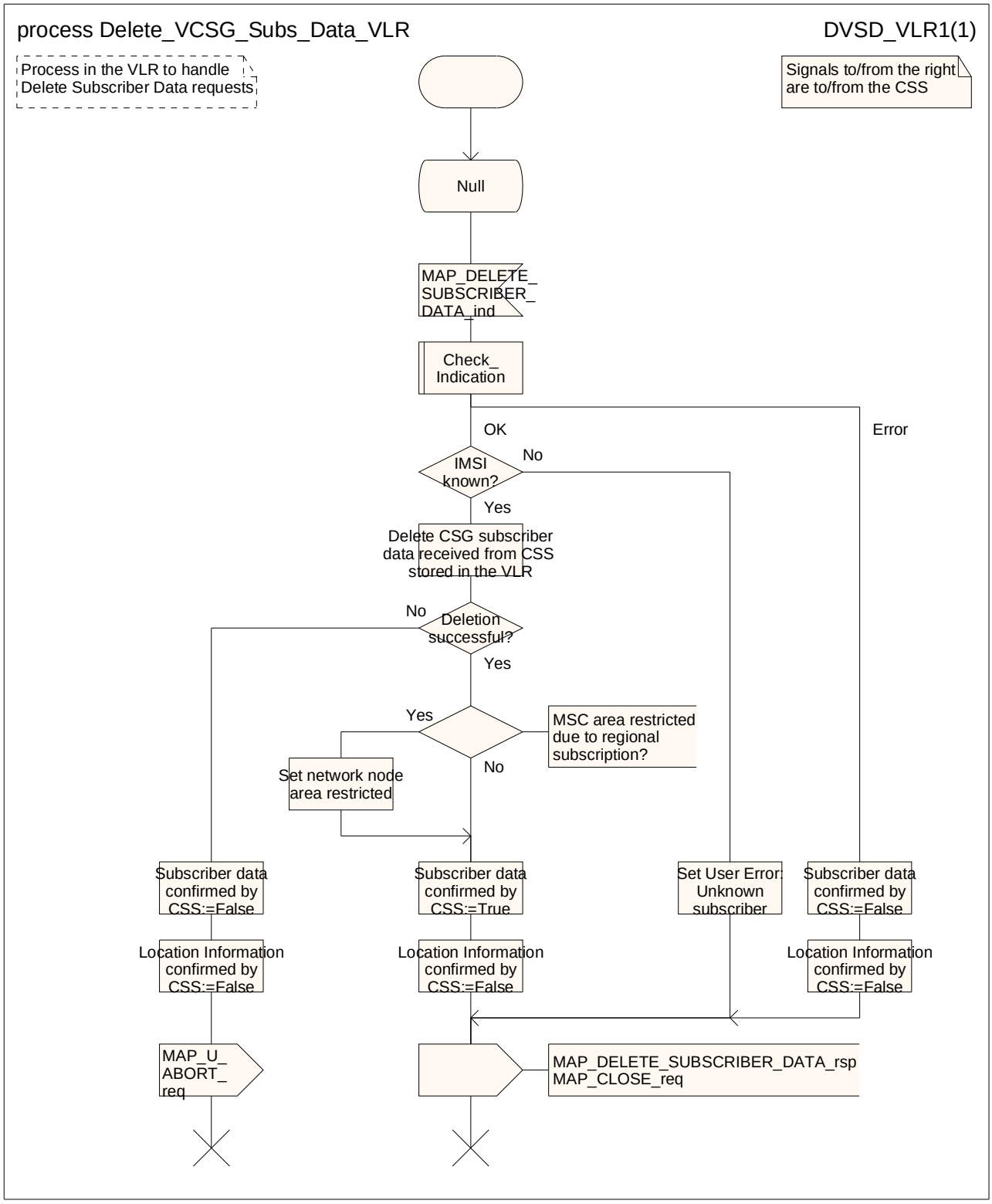


Figure 20.3A/9: Process Delete_VCSG_Subs_Data_VLR

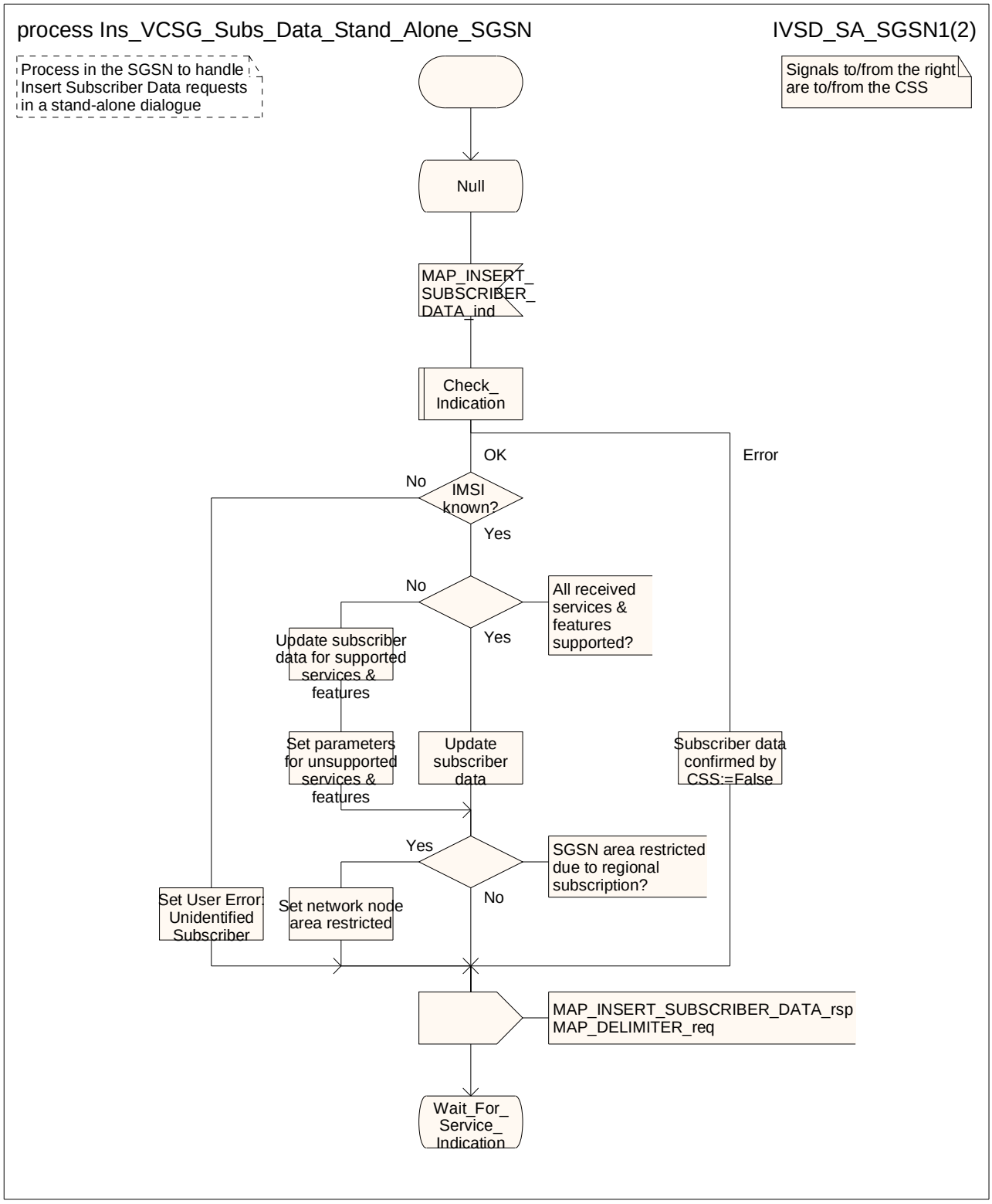


Figure 20.3A/10 (sheet 1 of 2): Process Ins_VCSG_Subs_Data_Stand_Alone_SGSN

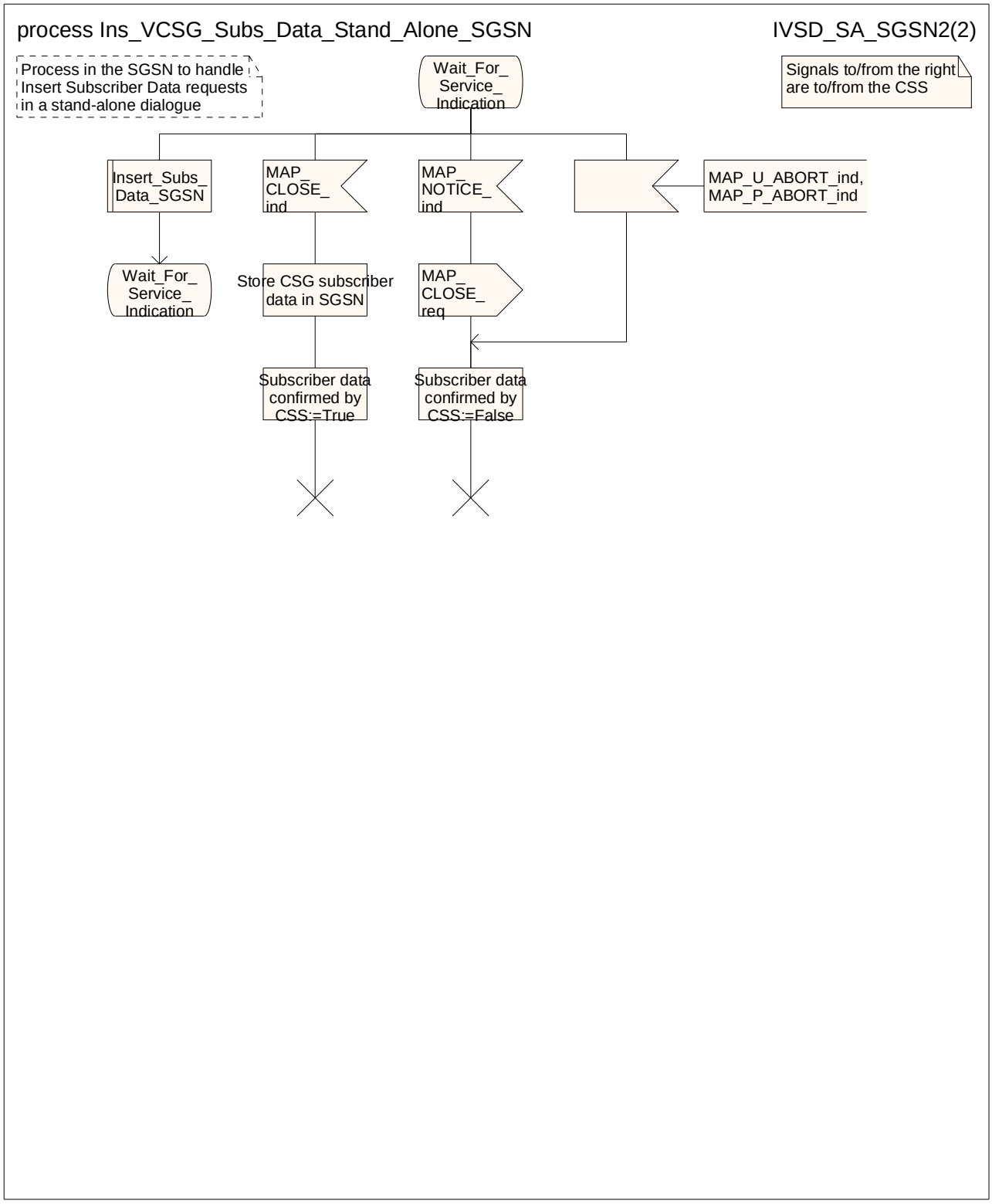


Figure 20.3A/10 (sheet 2 of 2): Process Ins_VCSG_Subs_Data_Stand_Alone_SGSN

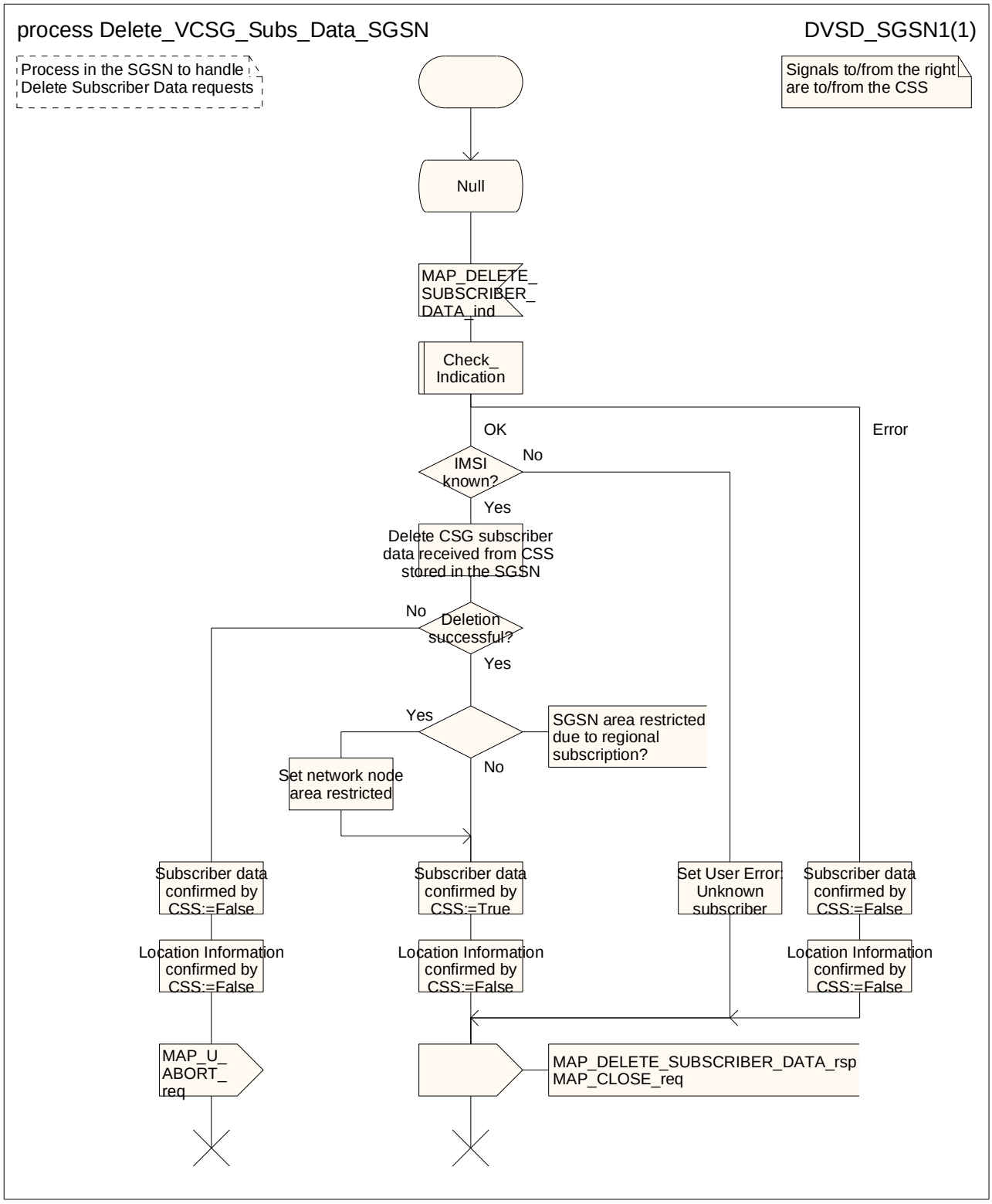
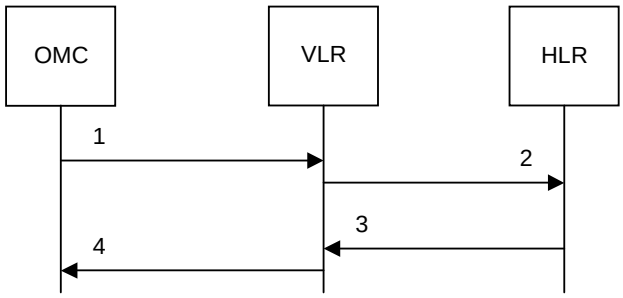


Figure 20.3A/11: Process Delete_VCSG_Subs_Data_SGSN

20.4 Subscriber Identity procedure

In the subscriber identity procedure the IMSI of the subscriber is retrieved from the HLR. The procedure is shown in figure 20.4/1.



1) Identity request
2) MAP_SEND_IMSI_req/ind
3) MAP_SEND_IMSI_rsp/cnf
4) Identity confirm

Figure 20.4/1: The subscriber identity procedure

20.4.1 Procedure in the VLR

The subscriber identity process in the VLR is shown in figure 20.4/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf	see subclause 25.1.2;
Check_Confirmation	see subclause 25.2.2.

20.4.2 Procedure in the HLR

The subscriber identity process in the HLR is shown in figure 20.4/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind	see subclause 25.1.1;
Check_Indication	see subclause 25.2.1.

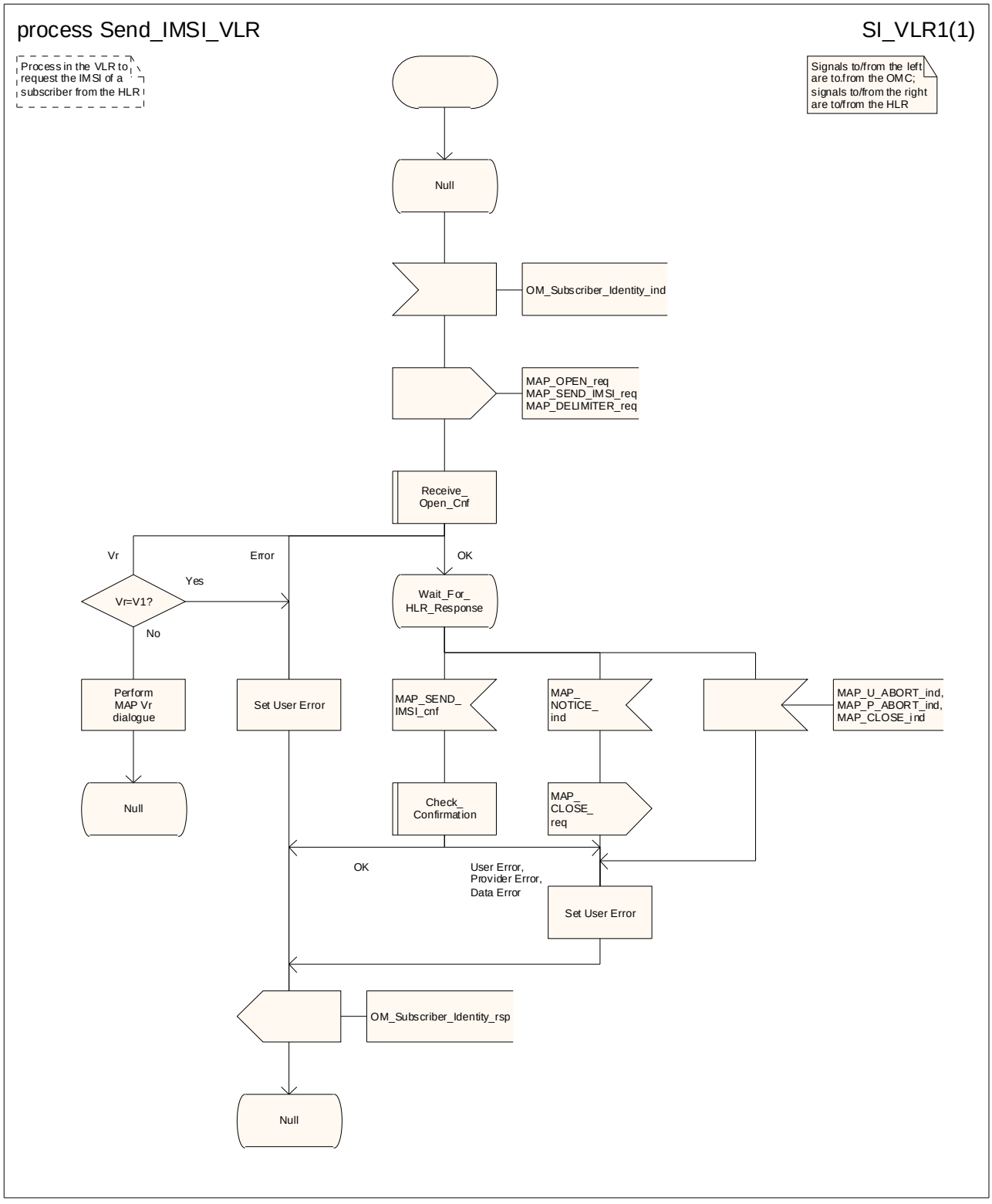


Figure 20.4/2: Process Send_IMSI_VLR

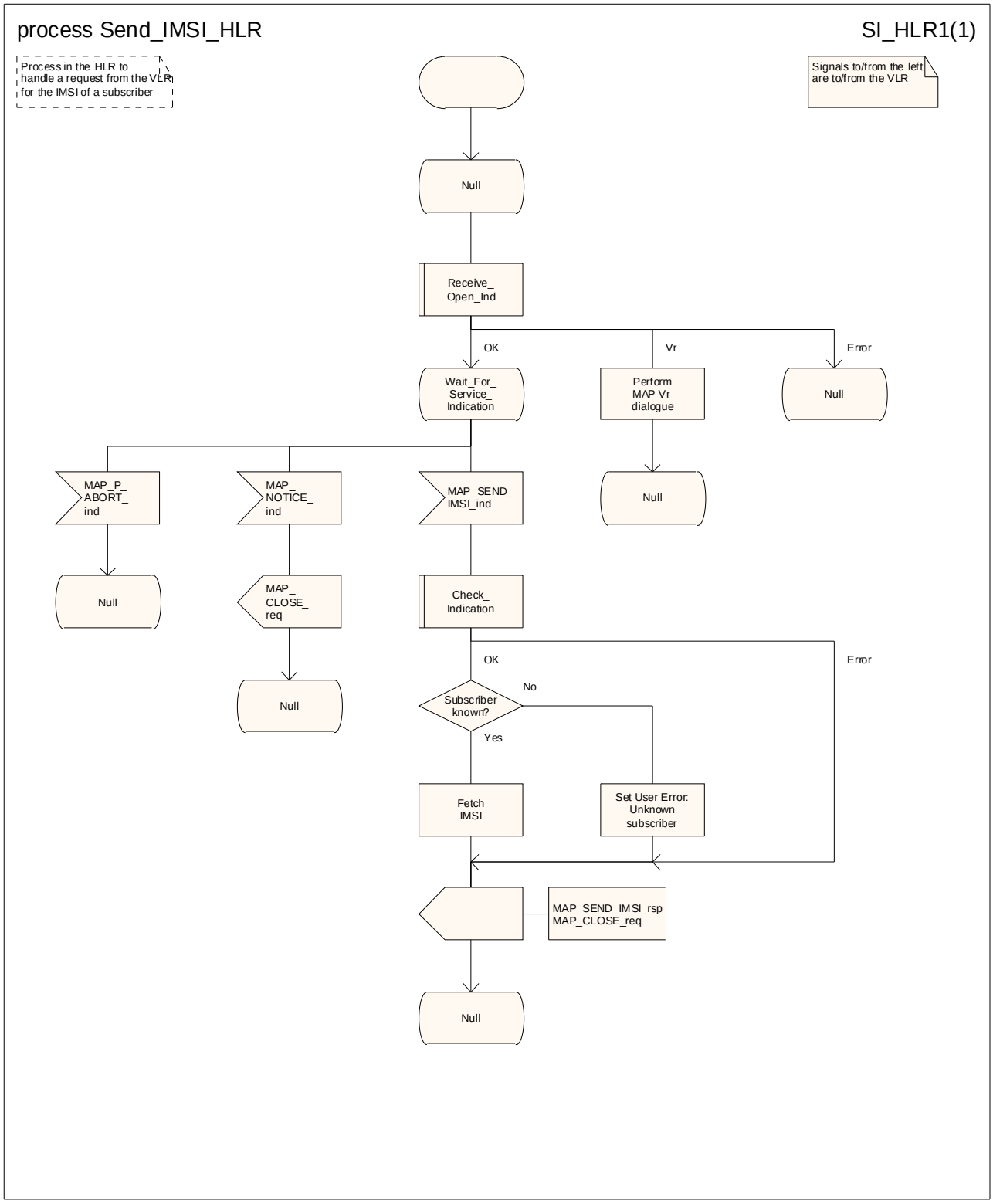


Figure 20.4/3: Process Send_IMSI_HLR

21 Call handling procedures

21.1 General

The MAP call handling procedures are used:

- to retrieve routeing information to handle a mobile terminating call;
- to transfer control of a call back to the GMSC if the call is to be forwarded;
- to retrieve and transfer information between anchor MSC and relay MSC for inter MSC group calls / broadcast calls;
- to handle the reporting of MS status for call completion services;
- to handle the notification of remote user free for CCBS;
- to handle the alerting and termination of ongoing call activities for a specific subscriber;
- to handle early release of no longer needed resources;
- to relay a mobile terminating call from the old to the new MSC during the Mobile Terminating Roaming Forwarding procedure.

The procedures to handle a mobile originating call and a mobile terminating call after the call has arrived at the destination MSC do not require any signalling over a MAP interface. These procedures are specified in 3GPP TS 23.018 [97].

The stage 2 specification for the retrieval of routeing information to handle a mobile terminating call is in 3GPP TS 23.018 [97]; modifications to this procedure for CAMEL are specified in 3GPP TS 23.078 [98], for optimal routeing of a basic mobile-to-mobile call in 3GPP TS 23.079 [99] and for CCBS in 3GPP TS 23.093 [107]. The interworking between the MAP signalling procedures and the call handling procedures for each entity (GMSC, HLR and VLR) is shown by the transfer of signals between these procedures.

The stage 2 specification for the transfer of control of a call back to the GMSC if the call is to be forwarded is in 3GPP TS 23.079 [99]. The interworking between the MAP signalling procedures and the call handling procedures for each entity (VMSC and GMSC) is shown by the transfer of signals between these procedures.

The stage 2 specifications for inter MSC group calls / broadcast calls are in 3GPP TS 43.068 [100] and 3GPP TS 43.069 [101]. The interworking between the MAP signalling procedures and the group call /broadcast call procedures for each entity (Anchor MSC and Relay MSC) is shown by the transfer of signals between these procedures.

The interworking between the call handling procedures and signalling protocols other than MAP are shown in 3GPP TS 23.018, 3GPP TS 23.078 and 3GPP TS 23.079 [99].

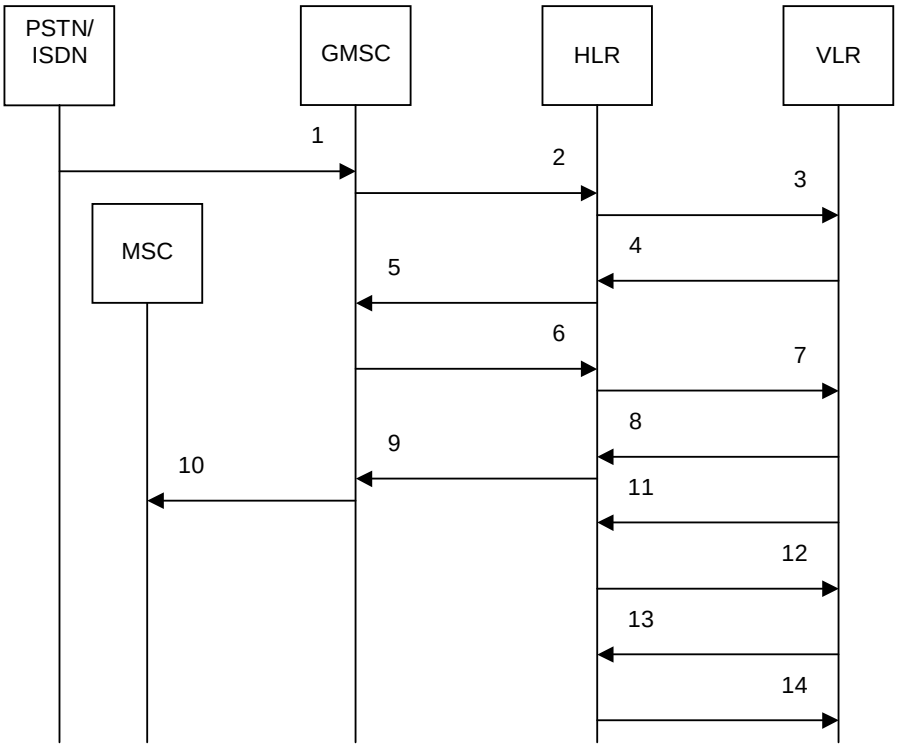
The stage 2 specification for the handling of reporting of MS status for call completion services and notification of remote user free for CCBS is in 3GPP TS 23.093 [107].

The stage 2 specification for the Mobile Terminating Roaming Forwarding procedure is in 3GPP TS 23.018 [97] and 3GPP TS 23.012 [23].

21.2 Retrieval of routing information

21.2.1 General

The message flows for successful retrieval of routeing information for a mobile terminating call are shown in figure 21.2/1 (mobile terminating call which has not been optimally routed) and 21.2/2 (mobile-to-mobile call which has been optimally routed). The message flow for successful retrieval of routeing information for a gsmSCF initiated call is shown in figure 21.2/3. The message flow for a successful Mobile Terminating Roaming Forwarding procedure is shown in figure 21.2/4.

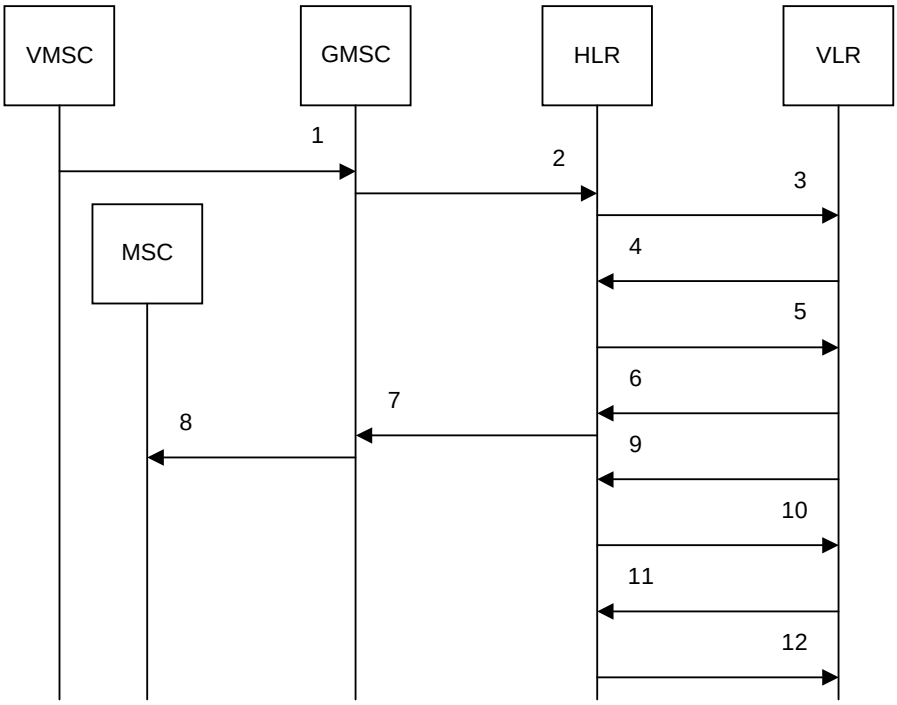


1) I_IAM (Note 1)
2) MAP_SEND_ROUTING_INFORMATION_req/ind (Note 2)
3) *MAP_PROVIDE_SUBSCRIBER_INFO_req/ind* (Note 3, Note 4)
4) *MAP_PROVIDE_SUBSCRIBER_INFO_rsp/cnf* (Note 4)
5) *MAP_SEND_ROUTING_INFORMATION_rsp/cnf* (Note 4)
6) *MAP_SEND_ROUTING_INFORMATION_req/ind* (Note 4)
7) MAP_PROVIDE_ROAMING_NUMBER_req/ind
8) MAP_PROVIDE_ROAMING_NUMBER_rsp/cnf
9) MAP_SEND_ROUTING_INFORMATION_rsp/cnf
10) I_IAM (Note 1)
11) *MAP_RESTORE_DATA_req/ind* (Note 4)
12) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 4)
13) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 4)
12) *MAP_RESTORE_DATA_rsp/cnf* (Note 4)

NOTE 1: TUP or ISUP may be used in signalling between MSCs, depending on the network type between the MSCs. For further details on the TUP and ISUP procedures refer to the following ITU-T Recommendations and ETSI specification:
- Q.721-725 - Telephone User Part (TUP);
- ETS 300 356-1 - Integrated Services Digital Network (ISDN); Signalling System No.7; ISDN User Part (ISUP) version 2 for the international interface; Part 1: Basic services.

NOTE 2: This service may also be used by an ISDN exchange for obtaining routing information from the HLR.
NOTE 3: As a network operator option, the HLR sends MAP_PROVIDE_SUBSCRIBER_INFORMATION to the VLR. For further details on the CAMEL procedures refer to 3GPP TS 23.078 [98].
NOTE 4: Services printed in *italics* are optional.

Figure 21.2/1: Message flow for retrieval of routeing information (non-optimally routed call)



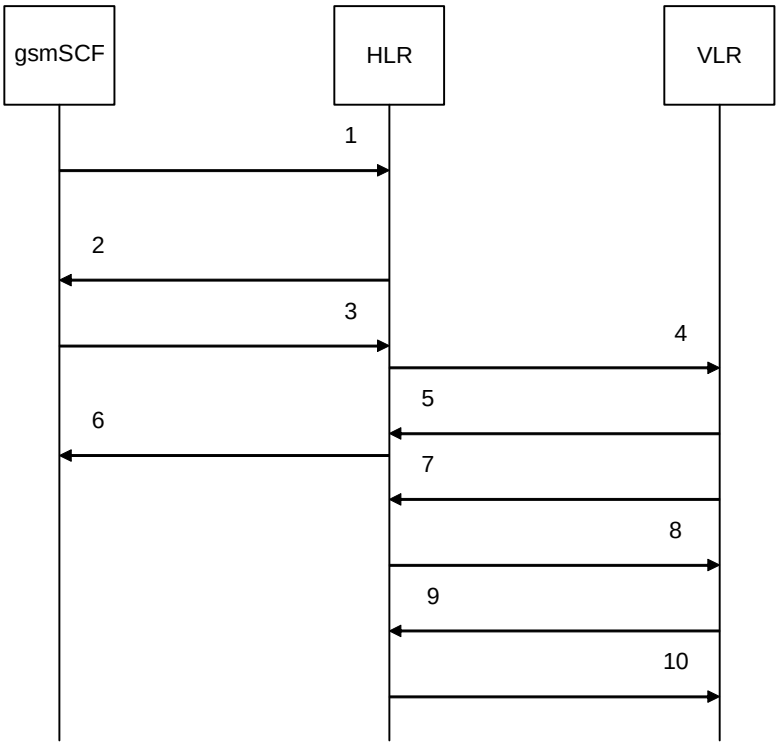
1) I_IAM (Note 1)
2) MAP_SEND_ROUTING_INFORMATION_req/ind
3) MAP_PROVIDE_SUBSCRIBER_INFO_req/ind (Note 2)
4) MAP_PROVIDE_SUBSCRIBER_INFO_rsp/cnf (Note 2)
5) MAP_PROVIDE_ROAMING_NUMBER_req/ind (Note 2)
6) MAP_PROVIDE_ROAMING_NUMBER_rsp/cnf (Note 2)
7) MAP_SEND_ROUTING_INFORMATION_rsp/cnf
8) I_IAM (Note 1)
9) *MAP_RESTORE_DATA_req/ind* (Note 3)
10) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 3)
11) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 3)
12) *MAP_RESTORE_DATA_rsp/cnf* (Note 3)

NOTE 1: TUP or ISUP may be used in signalling between MSCs, depending on the network type between the MSCs. For further details on the TUP and ISUP procedures refer to the following ITU-T Recommendations & ETSI specification:
- Q.721-725 - Telephone User Part (TUP);
- ETS 300 356-1 - Integrated Services Digital Network (ISDN); Signalling System No.7; ISDN User Part (ISUP) version 2 for the international interface; Part 1: Basic services.

NOTE 2: For Optimal Routeing phase 1, only one of the information flows for Provide Subscriber Info and Provide Roaming Number is used.

NOTE 3: Services printed in *italics* are optional.

Figure 21.2/2: Message flow for retrieval of routeing information (optimally routed call)



1) MAP_SEND_ROUTING_INFORMATION_req/ind
2) *MAP_SEND_ROUTING_INFORMATION_rsp/cnf* (Note 1)
3) *MAP_SEND_ROUTING_INFORMATION_req/ind* (Note 1)
4) MAP_PROVIDE_ROAMING_NUMBER_req/ind
5) MAP_PROVIDE_ROAMING_NUMBER_rsp/cnf
6) MAP_SEND_ROUTING_INFORMATION_rsp/cnf
7) *MAP_RESTORE_DATA_req/ind* (Note 1)
8) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 1)
9) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 1)
10) *MAP_RESTORE_DATA_rsp/cnf* (Note 1)

NOTE 1: Services printed in *italics* are optional.

Figure 21.2/3: Message flow for retrieval of routeing information for a gsmSCF initiated call

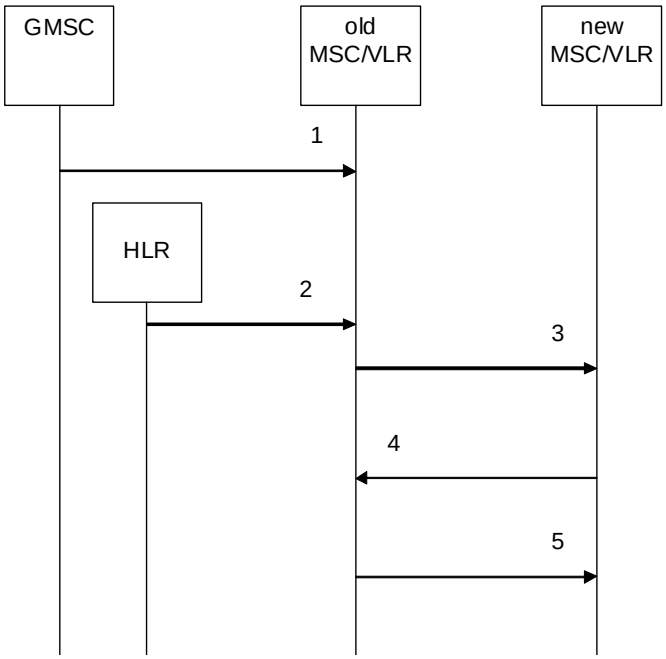
The following MAP services are used to retrieve routing information:

MAP_SEND_ROUTING_INFORMATION see subclause 10.1;

MAP_PROVIDE_ROAMING_NUMBER see subclause 10.2;

MAP_PROVIDE_SUBSCRIBER_INFO see subclause 8.11.2;

MAP_RESTORE_DATA see subclause 8.10.3.



1) I_IAM
2) MAP_CANCEL_LOCATION_req/ind
3) MAP_PROVIDE_ROAMING_NUMBER_req/ind
4) MAP_ PROVIDE_ROAMING_NUMBER_rsp/cnf
5) I_IAM

Figure 21.2/4: Message flow for Mobile Terminating Roaming Forwarding

The following MAP services are used for the Mobile Terminating Roaming Forwarding procedure:

MAP_PROVIDE_ ROAMING_NUMBER see subclause 10.2;

21.2.2 Procedure in the GMSC

The MAP process in the GMSC to retrieve routeing information for a mobile terminating call is shown in figure 21.2/6. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 1: if the MAP_SEND_ROUTING_INFORMATION request included the OR Interrogation parameter, the test "OR interrogation?" takes the "Yes" exit; otherwise the test takes the "No" exit.

21.2.9 Process in the gsmSCF

For the purposes of retrieving routeing information from the HLR, the gsmSCF takes the role of the GMSC and follows the process specified in subclause 21.2.2.

21.2.4 Procedure in the HLR

The MAP process in the HLR to retrieve routeing information for a mobile terminating call is shown in figure 21.2/7. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 3: if the MAP_PROVIDE_ROAMING_NUMBER request included the OR Interrogation parameter, the test "OR interrogation?" takes the "Yes" exit; otherwise the test takes the "No" exit.

21.2.5 Procedure in the VLR to provide a roaming number

The MAP process in the VLR to provide a roaming number for a mobile terminating call is shown in figure 21.2/8. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

21.2.6 Procedure in the VLR to restore subscriber data

The MAP process in the HLR to restore subscriber data is shown in figure 21.2/9. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Insert_Subs_Data_VLR see subclause 25.7.1;

Activate_Tracing_VLR see subclause 25.9.4.

21.2.7 Procedure in the VLR to provide subscriber information

The MAP process in the VLR to provide subscriber information for a mobile terminating call subject to CAMEL invocation is shown in figure 21.2/9. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

21.2.8 Procedure in the old VLR to request a Roaming Number (MTRF)

The MAP process in the old VLR for Mobile Terminating Roaming Forwarding is shown in figure 21.2/11. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Cnf see subclause 25.1.2

Check_Confirmation see subclause 25.2.2

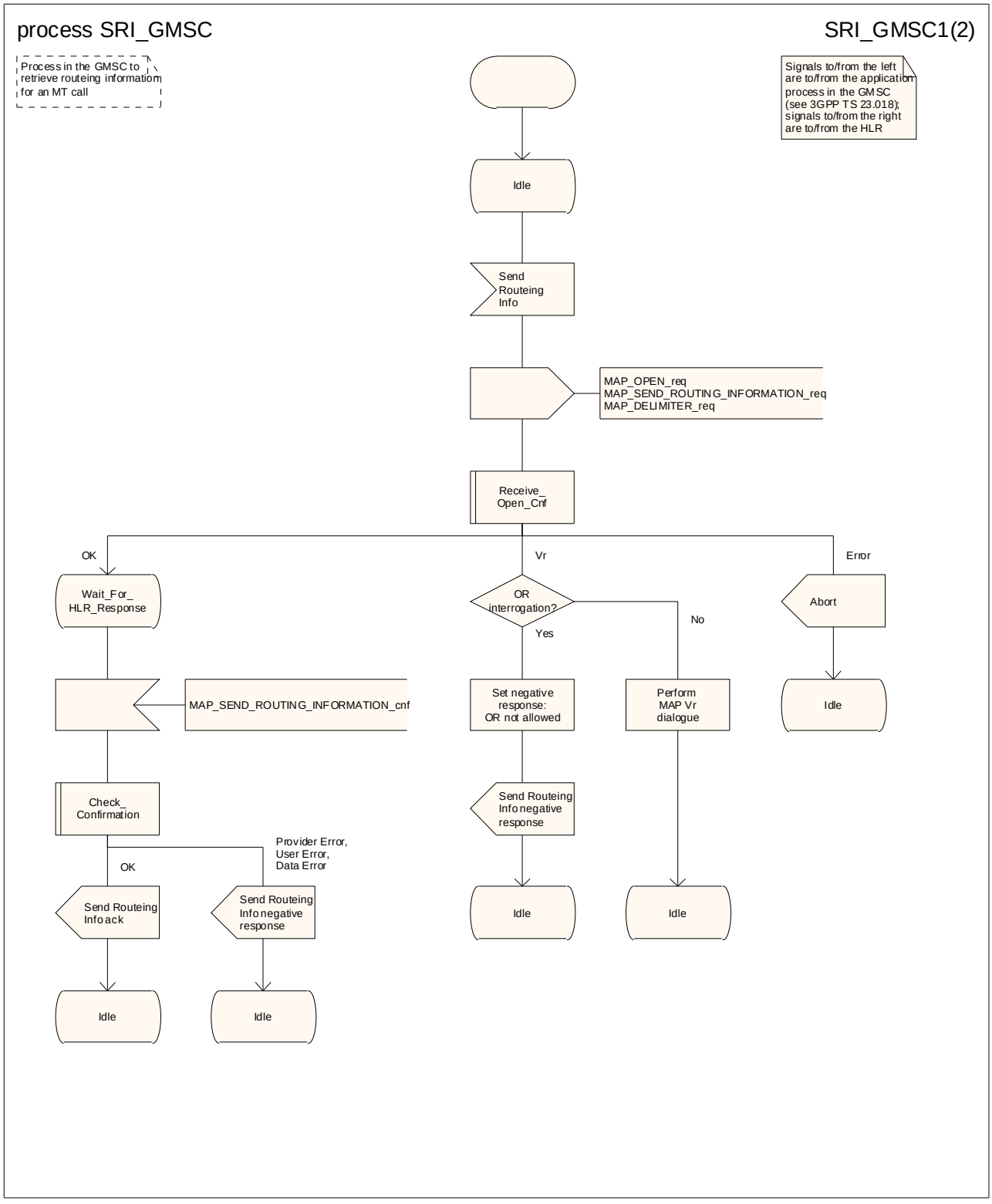


Figure 21.2/6 (sheet 1 of 2): Process SRI_GMSC

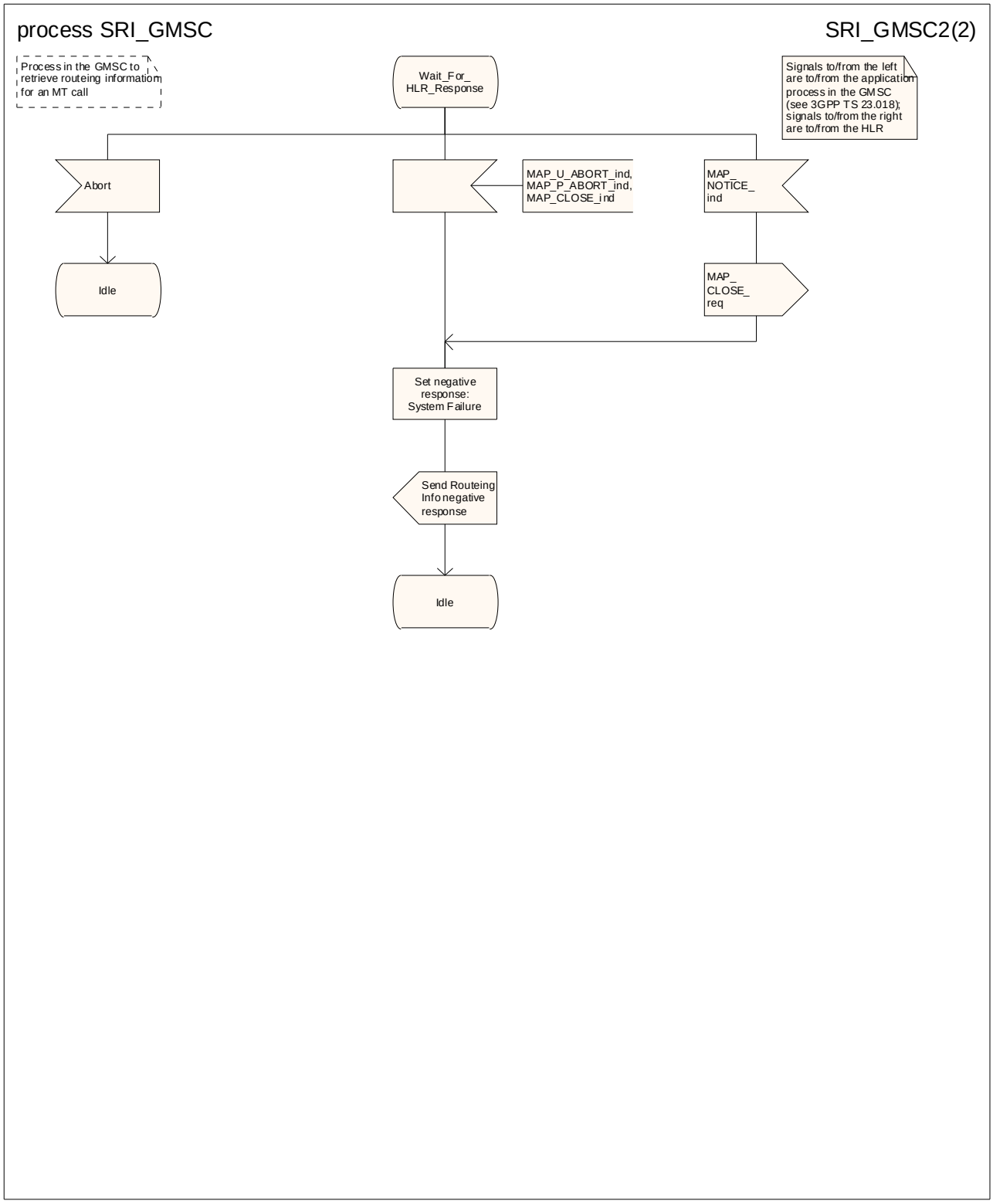


Figure 21.2/6 (sheet 2 of 2): Process SRI_GMSC

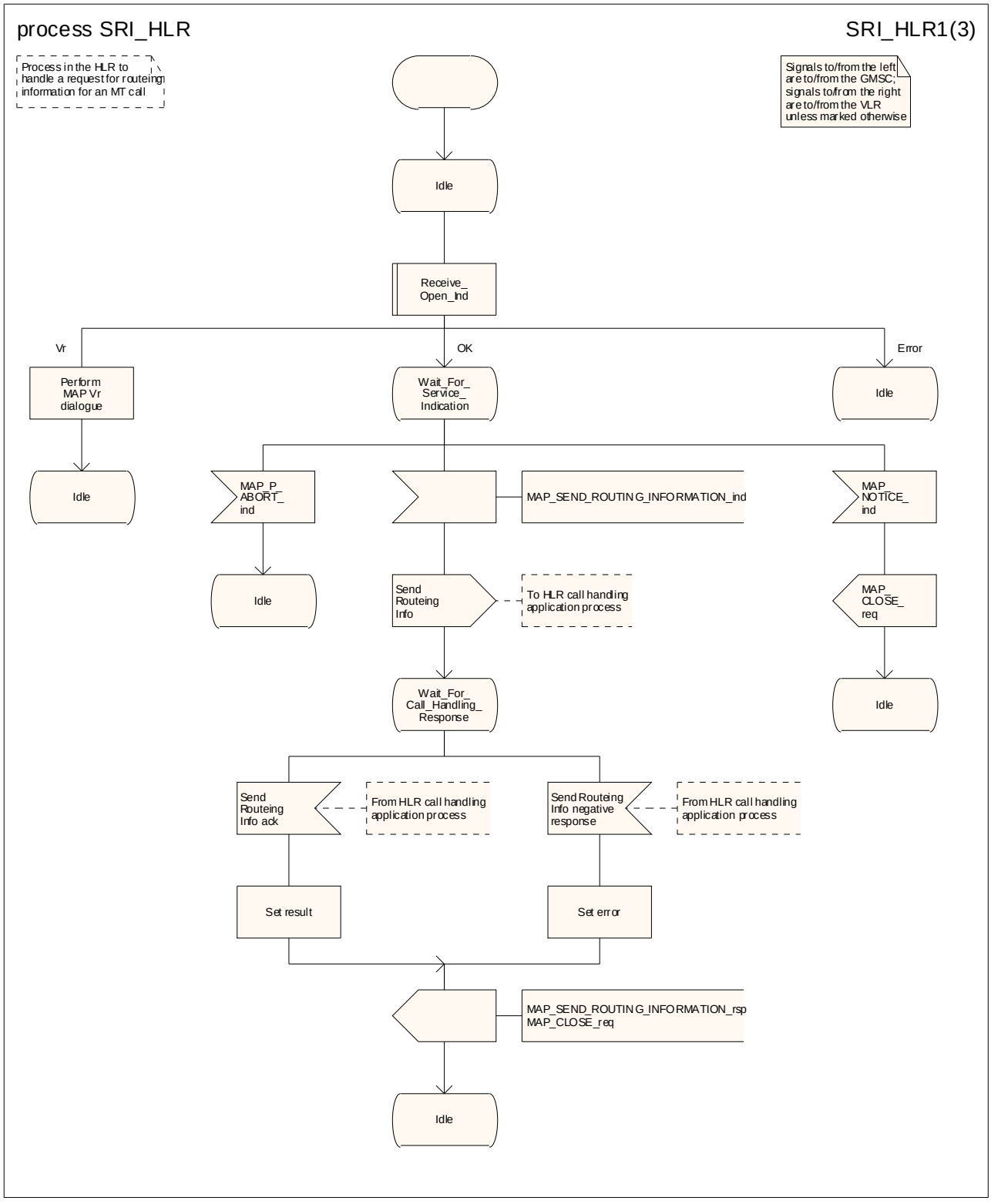


Figure 21.2/7 (sheet 1 of 3): Process SRI_HLR

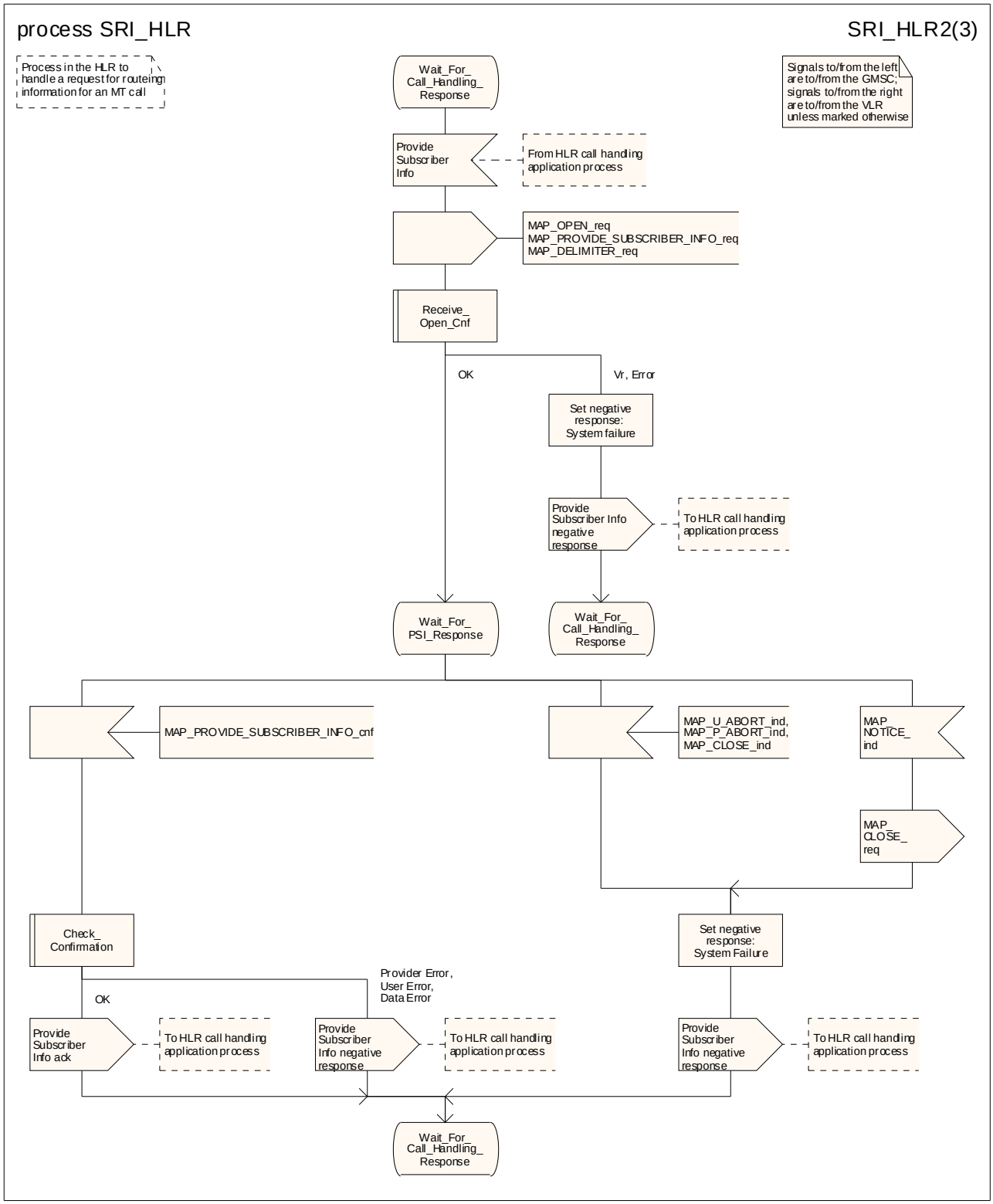


Figure 21.2/7 (sheet 2 of 3): Process SRI_HLR

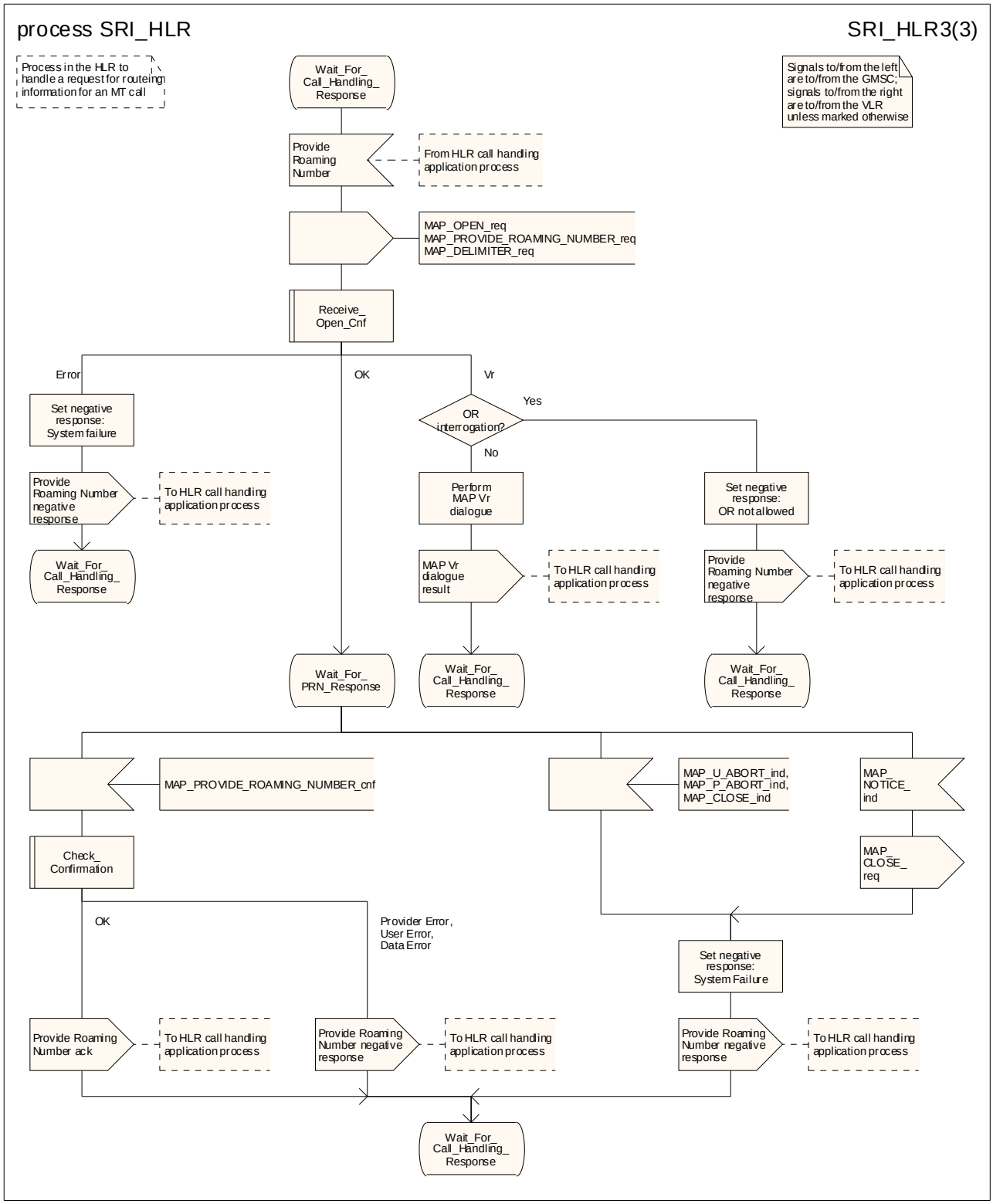


Figure 21.2/7 (sheet 3 of 3): Process SRI_HLR

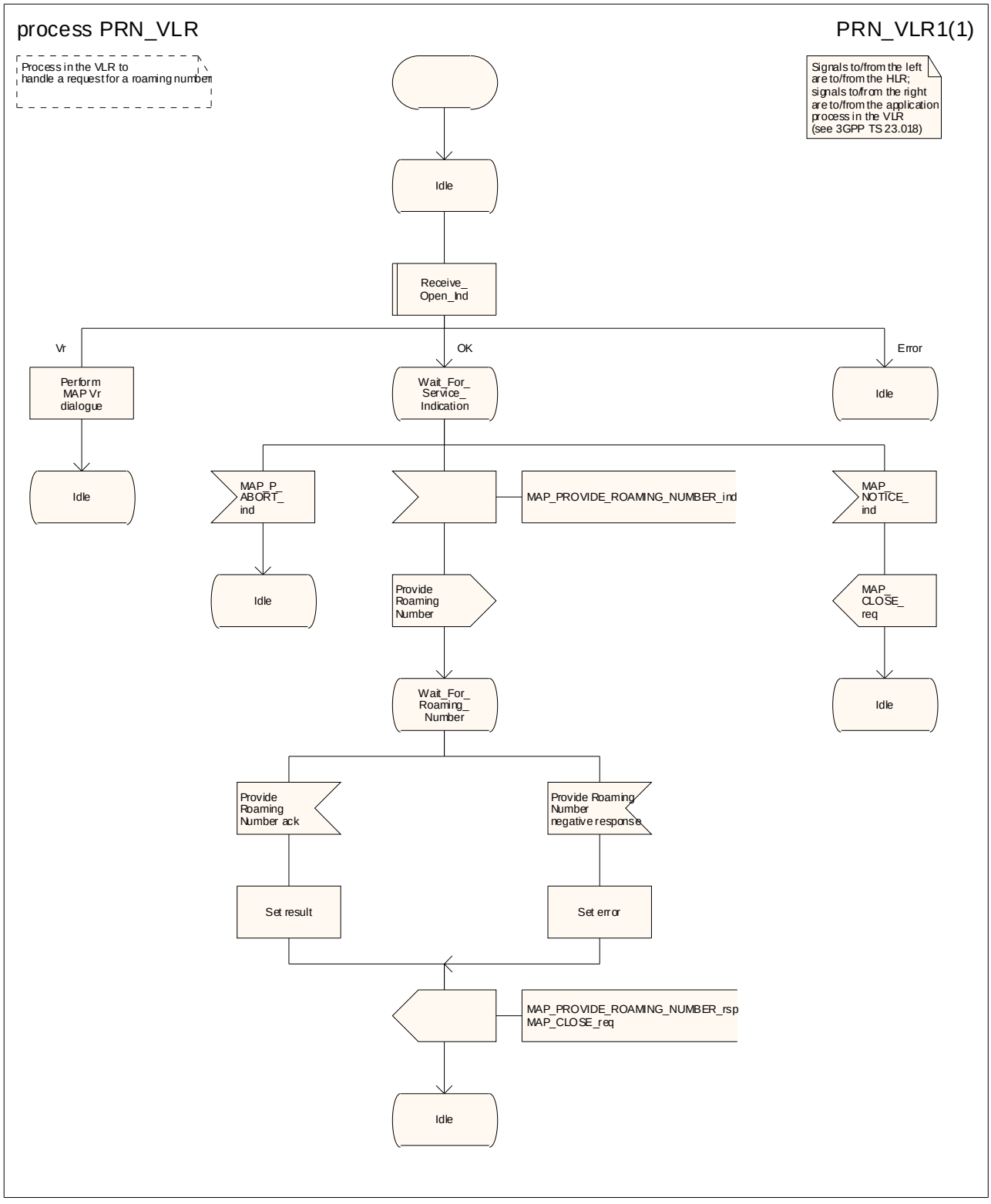


Figure 21.2/8: Process PRN_VLR

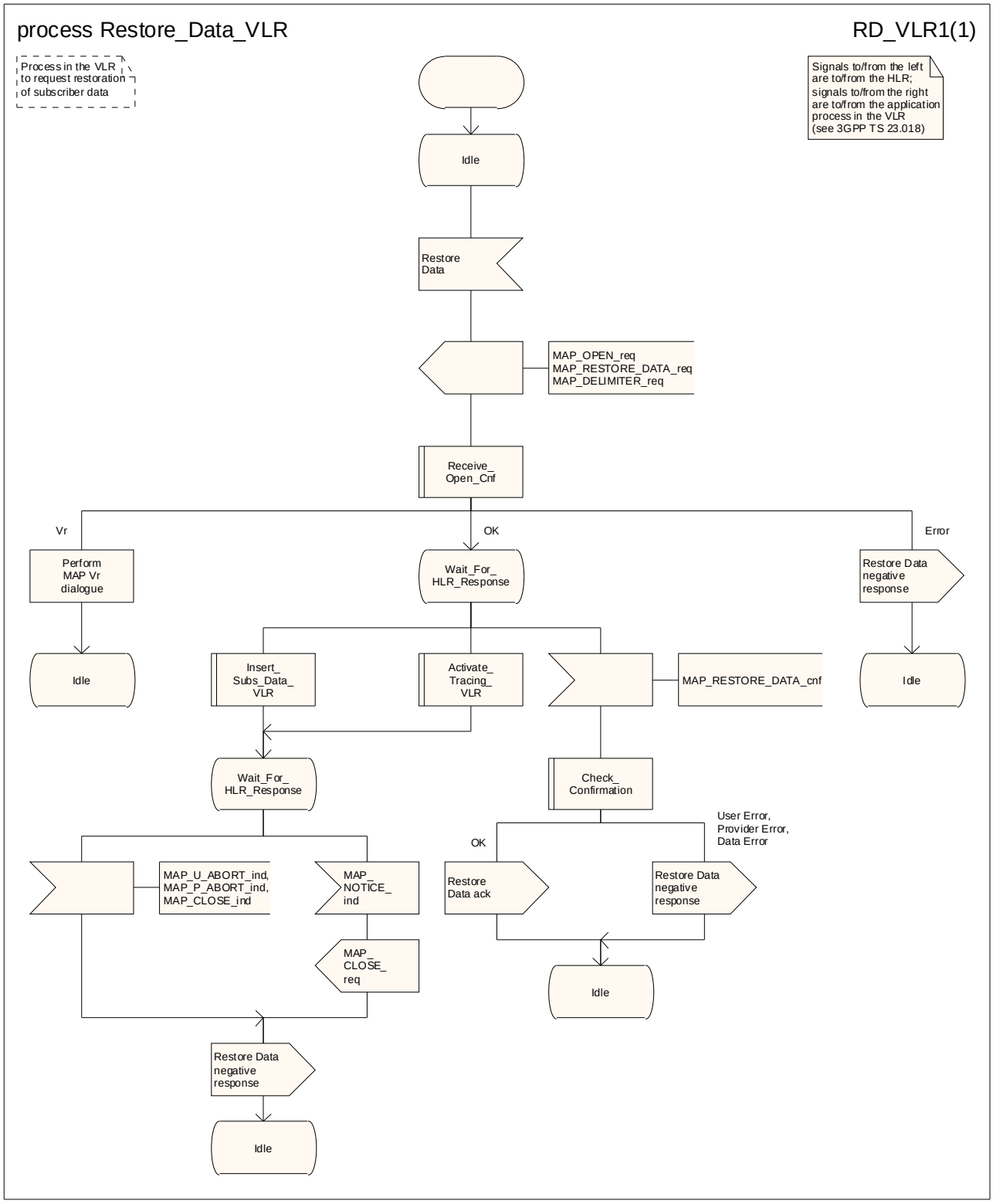


Figure 21.2/9: Process Restore_Data_VLR

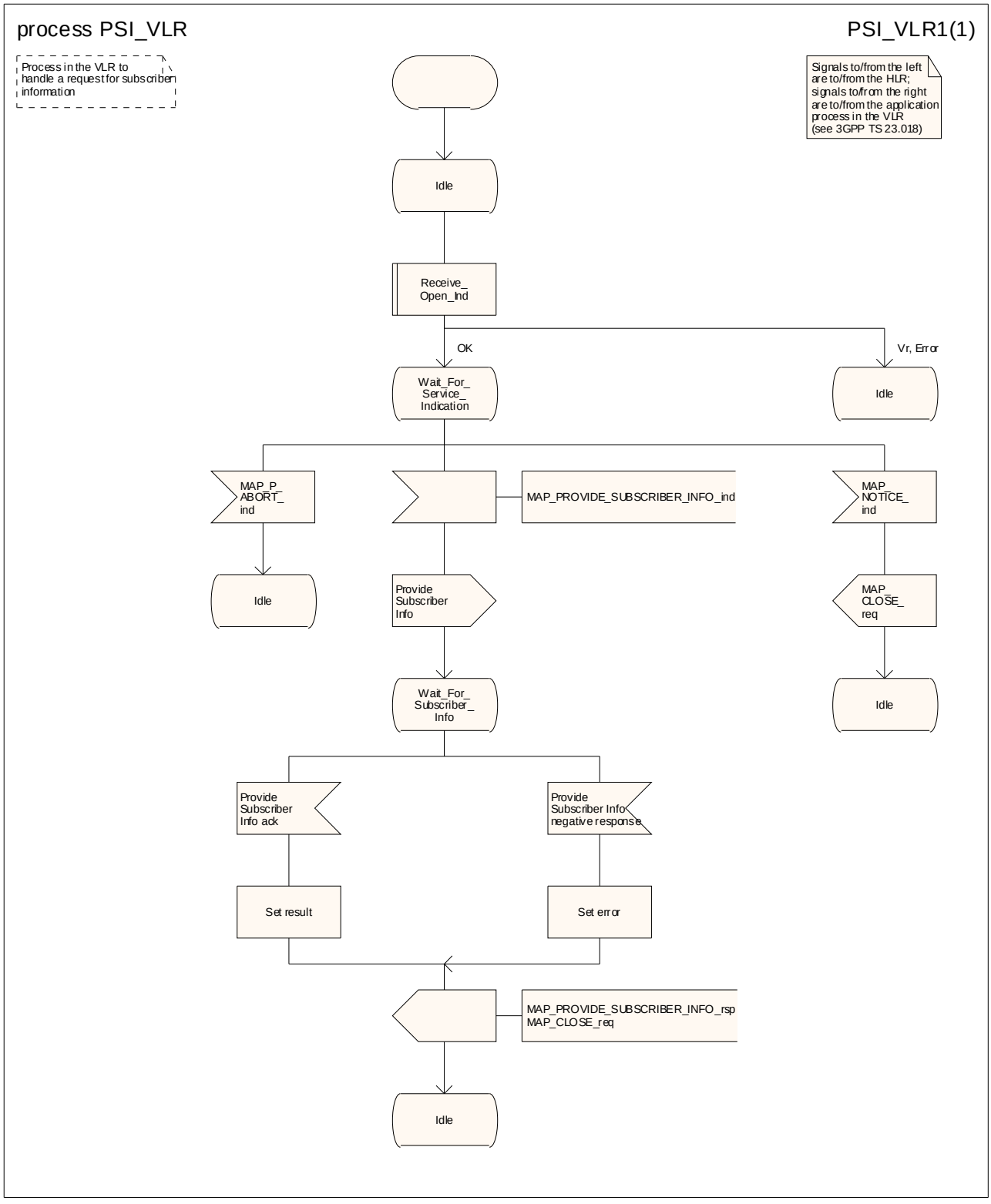


Figure 21.2/10: Process PSI_VLR

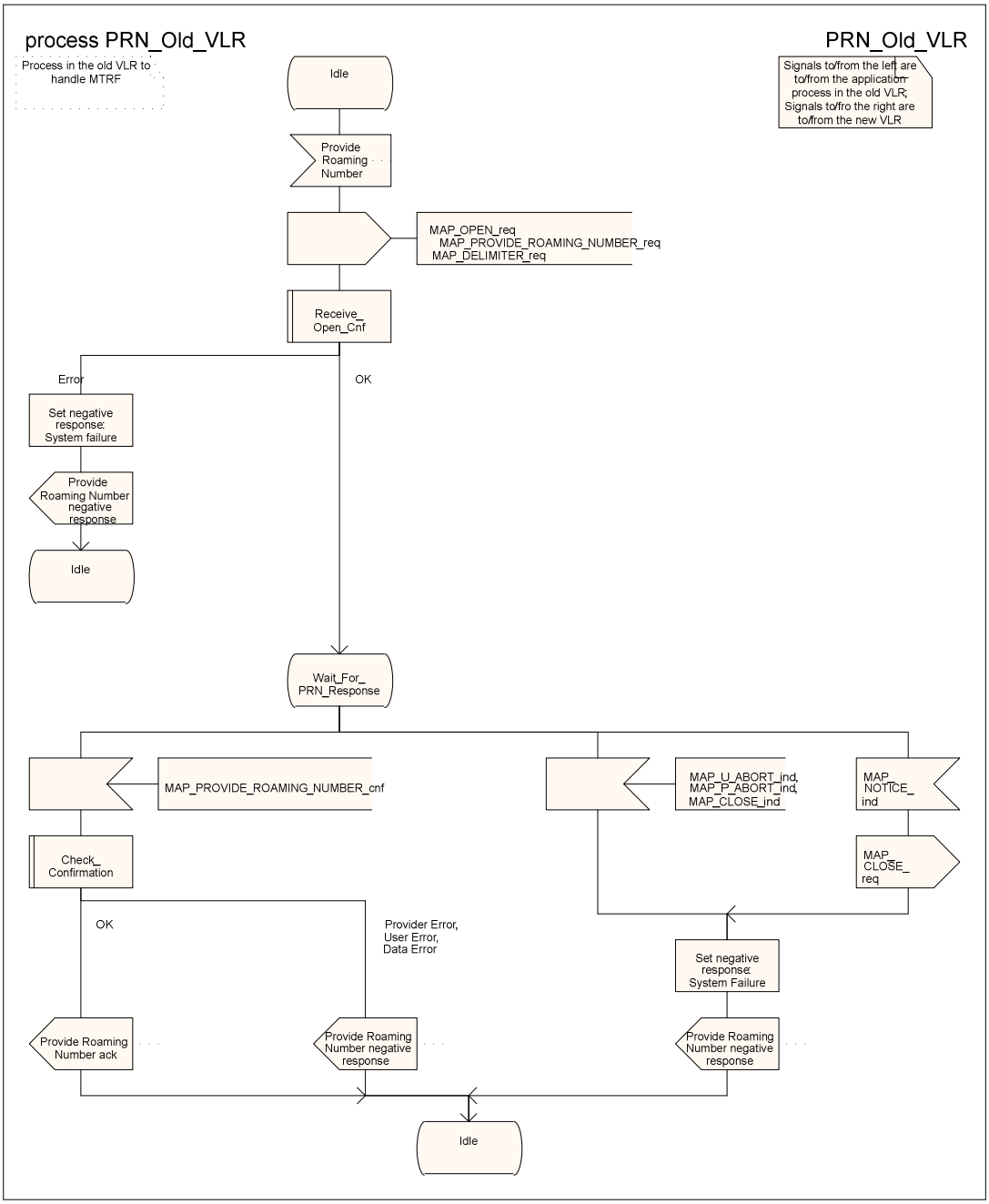
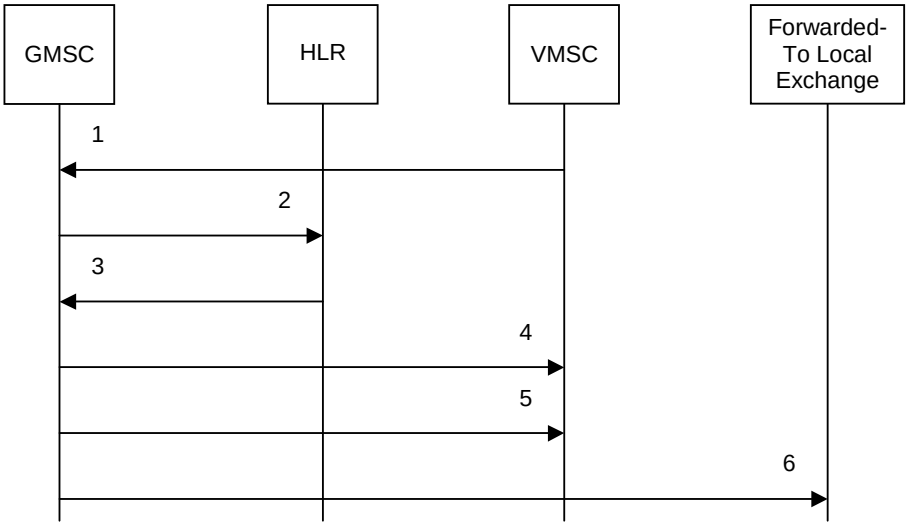


Figure 21.2/11: Process PRN_Old_VLR

21.3 Transfer of call handling

21.3.1 General

The message flow for successful transfer of call handling to forward a call is shown in figure 21.3/1.



1) MAP_RESUME_CALL_HANDLING_req/ind
2) *MAP_SEND_ROUTING_INFORMATION_req/ind* (Note 2)
3) *MAP_SEND_ROUTING_INFORMATION_rsp/cnf* (Note 2)
4) MAP_RESUME_CALL_HANDLING_rsp/cnf
5) I_REL (Note 1)
6) I_IAM (Note 1)

NOTE 1: TUP or ISUP may be used in signalling between MSCs, depending on the network type between the MSCs. For further details on the TUP and ISUP procedures refer to the following ITU-T Recommendations & ETSI specification:
- Q.721-725 - Telephone User Part (TUP);
- ETS 300 356-1 - Integrated Services Digital Network (ISDN); Signalling System No.7; ISDN User Part (ISUP) version 2 for the international interface; Part 1: Basic services.

NOTE 2: Services printed in *italics* are optional.

Figure 21.3/1: Message flow for transfer of call handling

If the HLR indicated in the response to the original request for routeing information that forwarding interrogation is required, the GMSC executes the Send Routeing Information procedure with the HLR to obtain forwarding information; otherwise the GMSC uses the forwarding data which were sent in the MAP_RESUME_CALL_HANDLING req/ind.

21.3.2 Process in the VMSC

The MAP process in the VMSC to retrieve routeing information for a mobile terminating call is shown in figure 21.3/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

If the capacity of a message signal unit in the lower layers of the protocol is enough to carry all the information which has to be sent to the GMSC, the test "Segmentation needed?" takes the "No" exit; otherwise the test takes the "Yes" exit.

21.3.3 Process in the GMSC

The MAP process in the GMSC to handle a request for the GMSC to resume call handling is shown in figure 21.3/3. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

If the parameter All Information Sent was present in the MAP_RESUME_CALL_HANDLING indication, the test "All Information Sent" takes the "Yes" exit; otherwise the test takes the "No" exit.

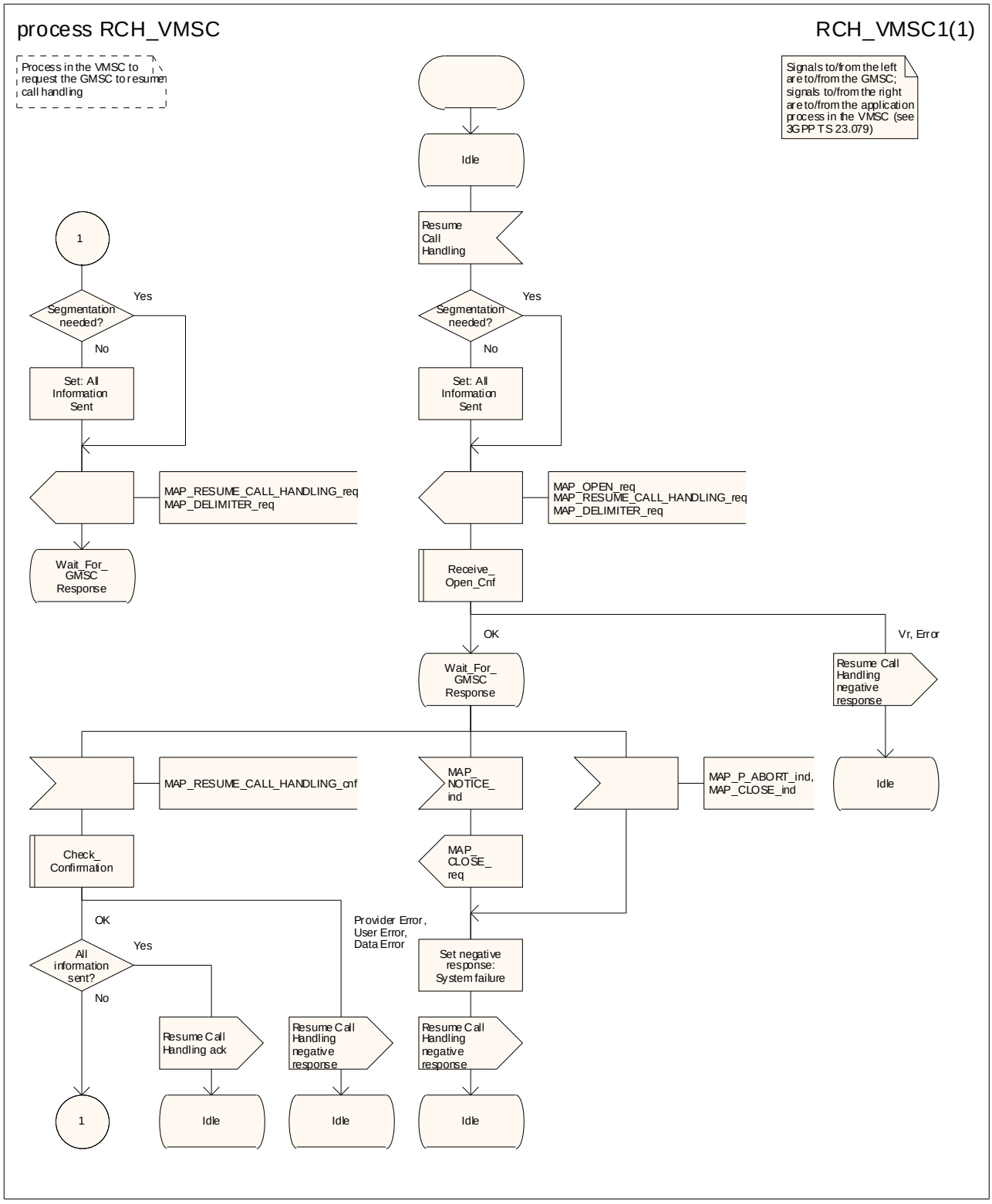


Figure 21.3/2: Process RCH_VMSC

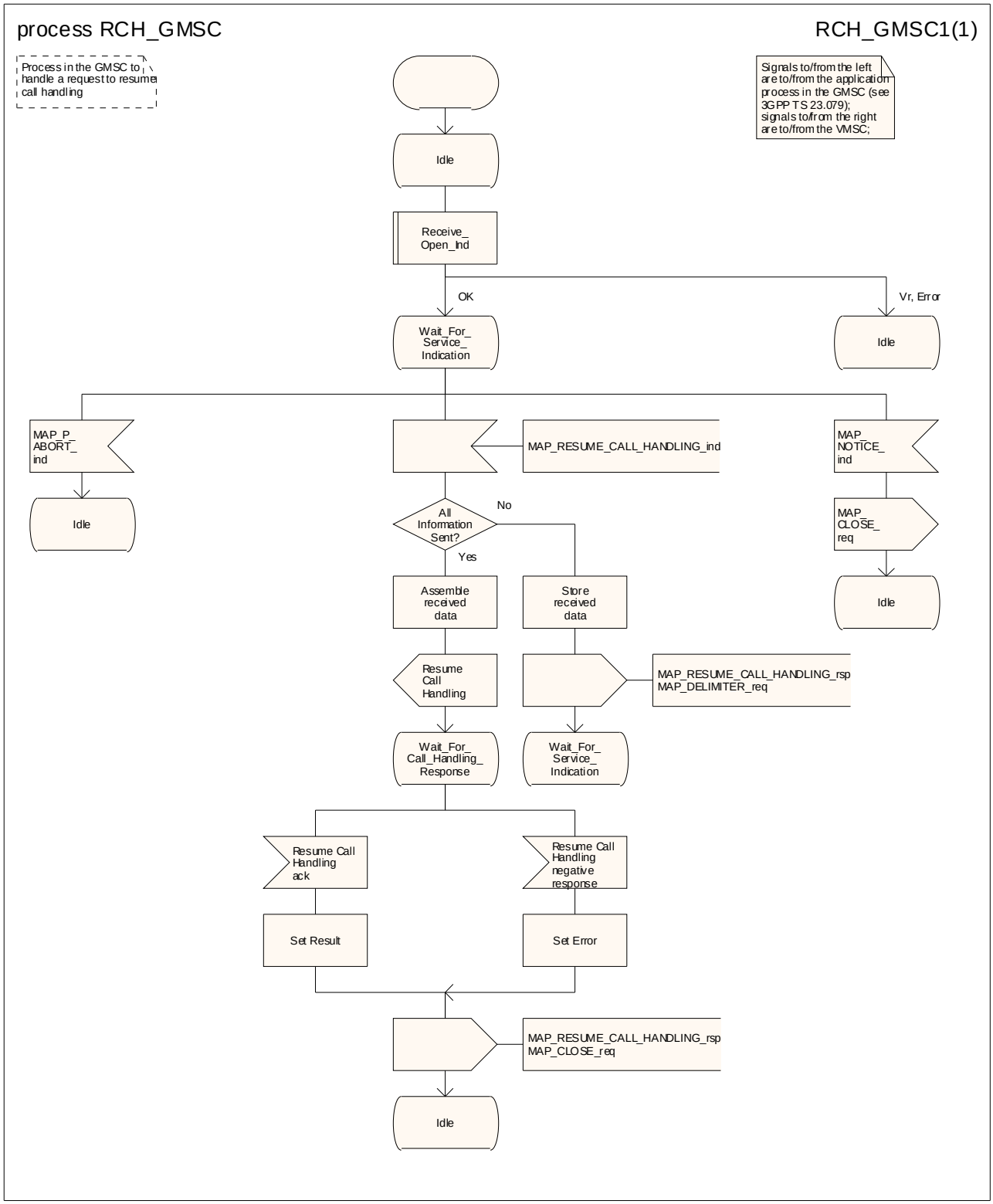
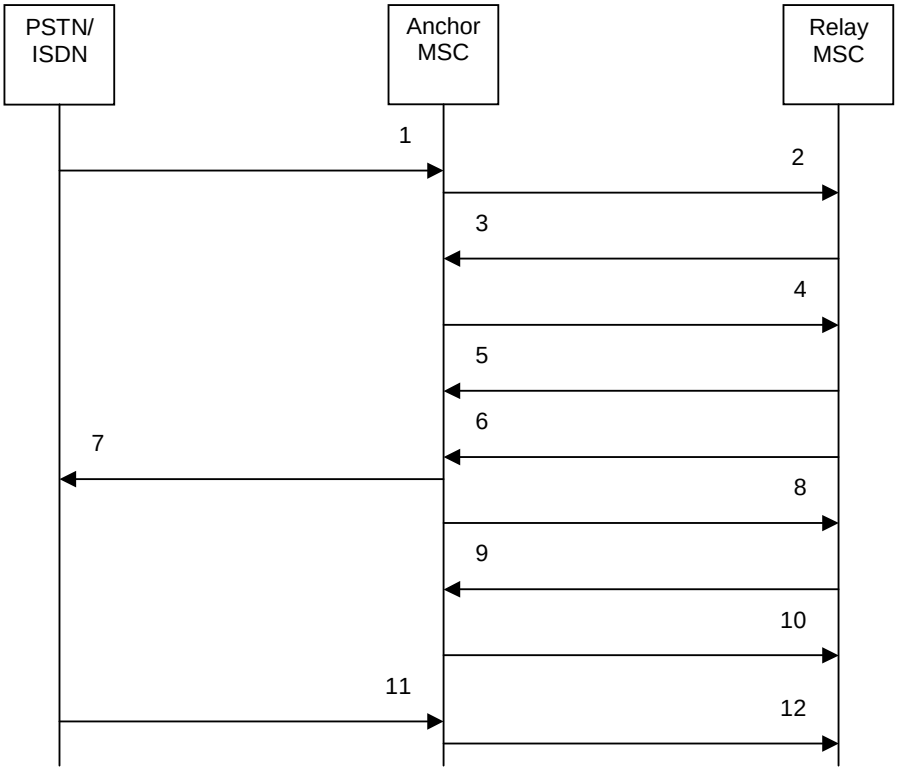


Figure 21.3/3: Process RCH_GMSC

21.4 Inter MSC Group Call Procedures

21.4.1 General

The message flow for successful inter MSC group call / broadcast call set-up is shown in figure 21.4/1.



1) I_IAM (Note 1)
2) MAP_PREPARE_GROUP_CALL_req/ind
3) MAP_PREPARE_GROUP_CALL_rsp/cnf
4) I_IAM (Note 1)
5) MAP_SEND_GROUP_CALL_END_SIGNAL_req/ind
6) I_ACM (Note 1)
7) I_ACM (Note 1)
8) MAP_FORWARD_GROUP_CALL_SIGNALLING_req/ind (Note 2)
9) MAP_PROCESS_GROUP_CALL_SIGNALLING_req/ind (Note 2)
10) MAP_SEND_GROUP_CALL_END_SIGNAL_rsp/cnf
11) I_REL (Note 3)
12) I_REL (Note 3)

NOTE 1: TUP or ISUP may be used in signalling between MSCs, depending on the network type between the MSCs. For further details on the TUP and ISUP procedures refer to the following ITU-T Recommendations and ETSI specification:
- Q.721-725 - Telephone User Part (TUP);
- ETS 300 356-1 - Integrated Services Digital Network (ISDN); Signalling System No.7; ISDN User Part (ISUP) version 2 for the international interface; Part 1: Basic services.

NOTE 2: The MAP_FORWARD_GROUP_CALL_SIGNALLING and MAP_PROCESS_GROUP_CALL_SIGNALLING services are not applicable for voice broadcast calls.

NOTE 3: The call can be released from the PSTN/ISDN or the Relay MSC

Figure 21.4/1: Message flow for inter MSC group call / broadcast call

21.4.2 Process in the Anchor MSC

The MAP process in the Anchor MSC to retrieve and transfer information from / to the Relay MSC for VBS and VGCS calls is shown in figure 21.4/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2.

21.4.3 Process in the Relay MSC

The MAP process in the Relay MSC to receive and transfer information from / to the Anchor MSC for VBS and VGCS calls is shown in figure 21.4/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.2;

Check_Indication see subclause 25.2.1.

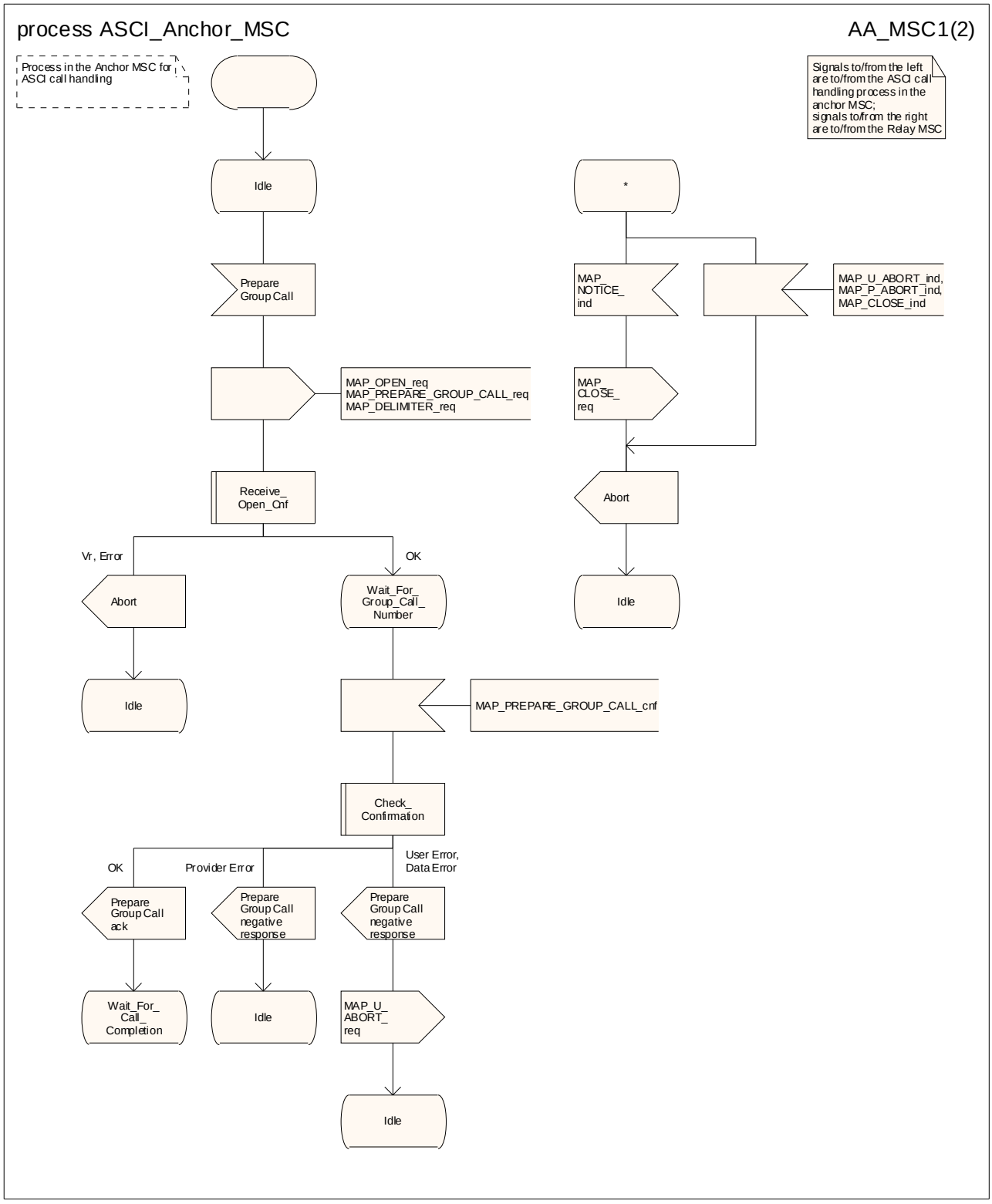


Figure 21.4/2 (sheet 1 of 2): Process ASCI_Anchor_MSC

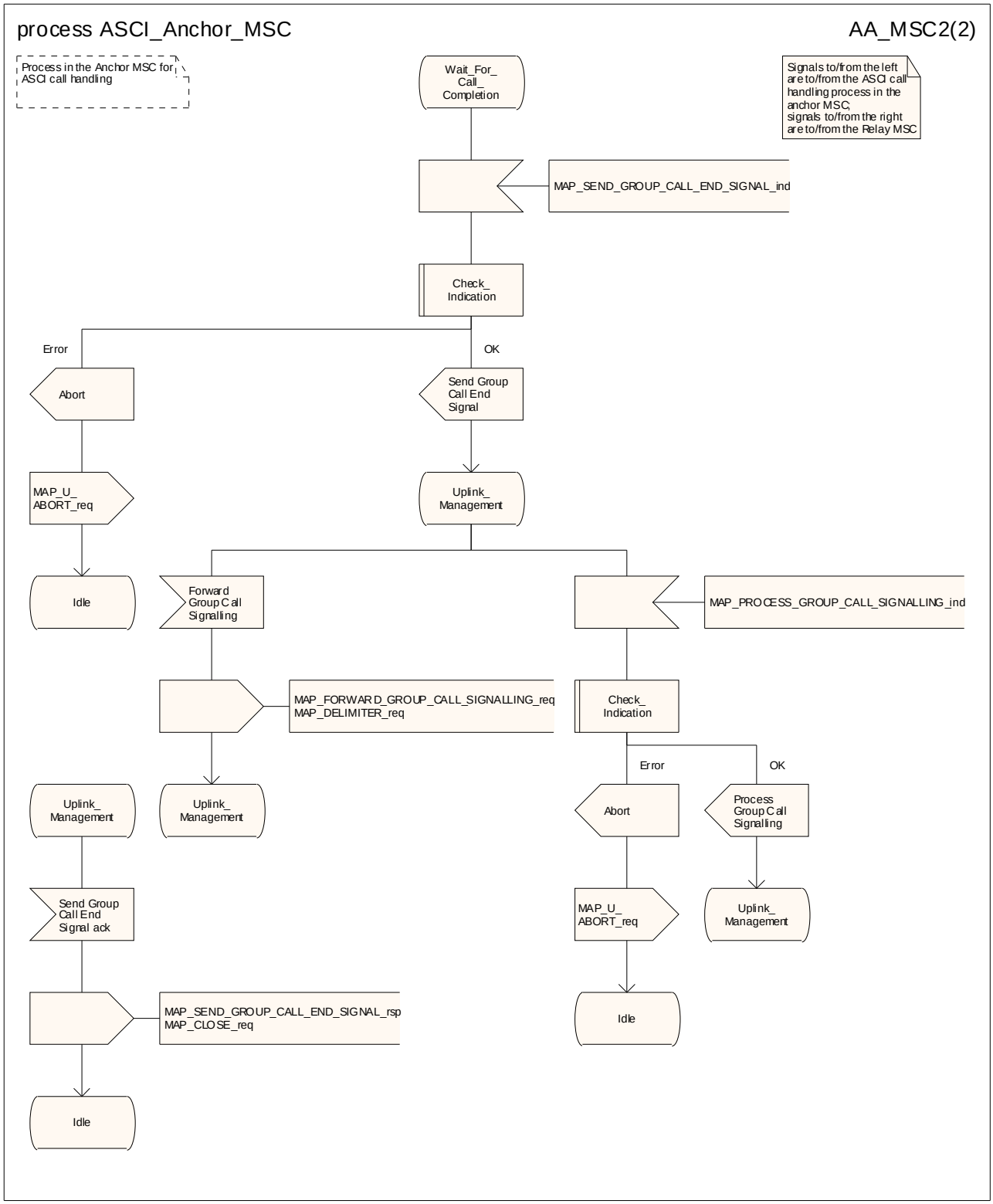


Figure 21.4/2 (sheet 2 of 2): Process ASCI_Anchor_MSC

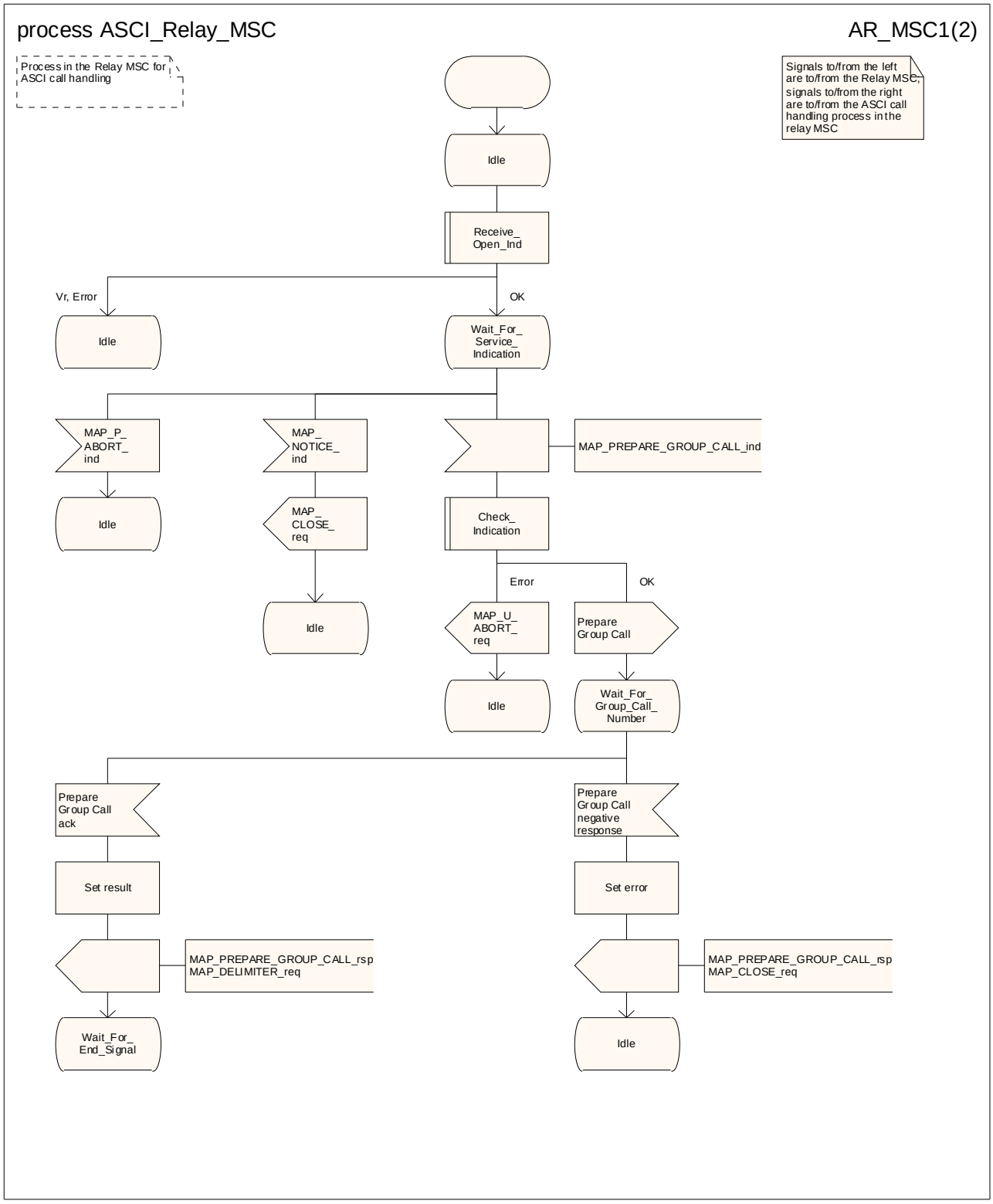


Figure 21.4/3 (sheet 1 of 2): Process ASCI_Relay_MSC

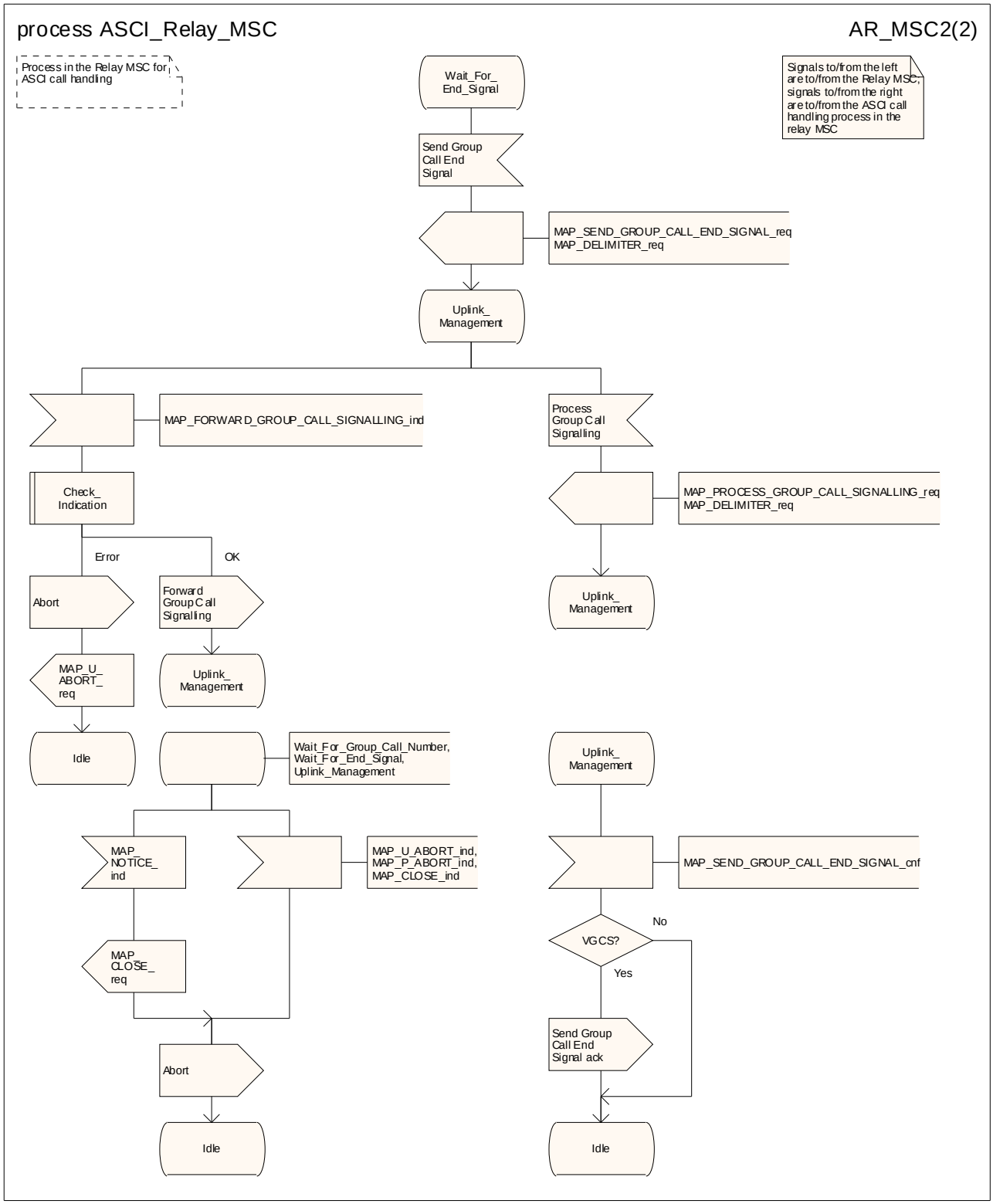


Figure 21.4/3 (sheet 2 of 2): Process ASCI_Relay_MSC

21.4A Inter MSC Group Call Info Retrieval

21.4A.1 General

The message flow for successful inter MSC group call info retrieval is shown in figure 21.4A/1.

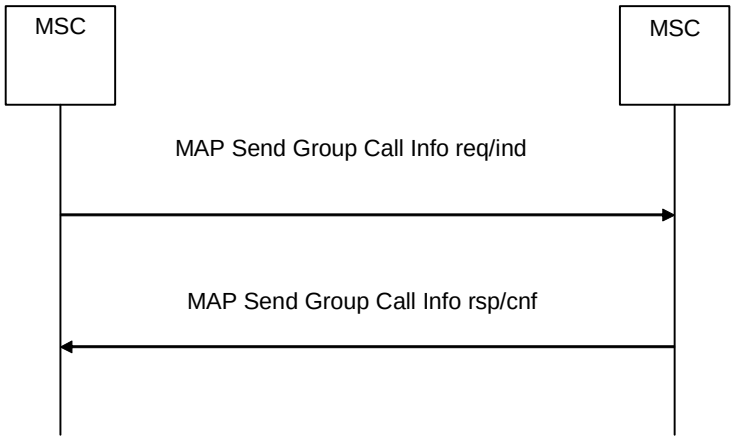


Figure 21.4A/1: Message flow for inter MSC group call info retrieval

21.4A.2 Process in the MSC

The MAP process in the MSC to retrieve and group call information is shown in figure 21.4A/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Receive_Open_Ind see subclause 25.1.2;

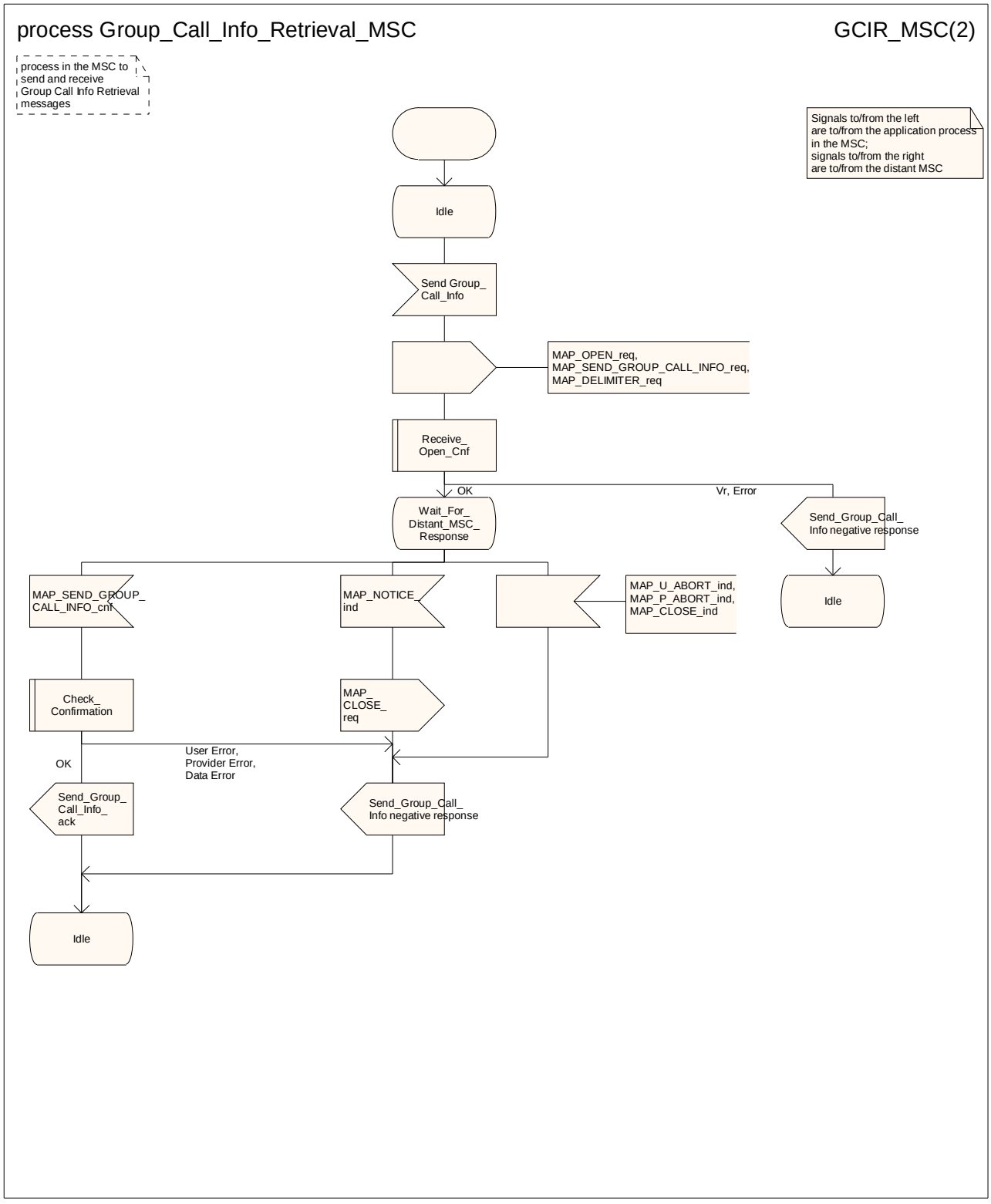


Figure 21.4A/2 (sheet 1 of 2): Process Group_Call_Info_Retrieval_MSC

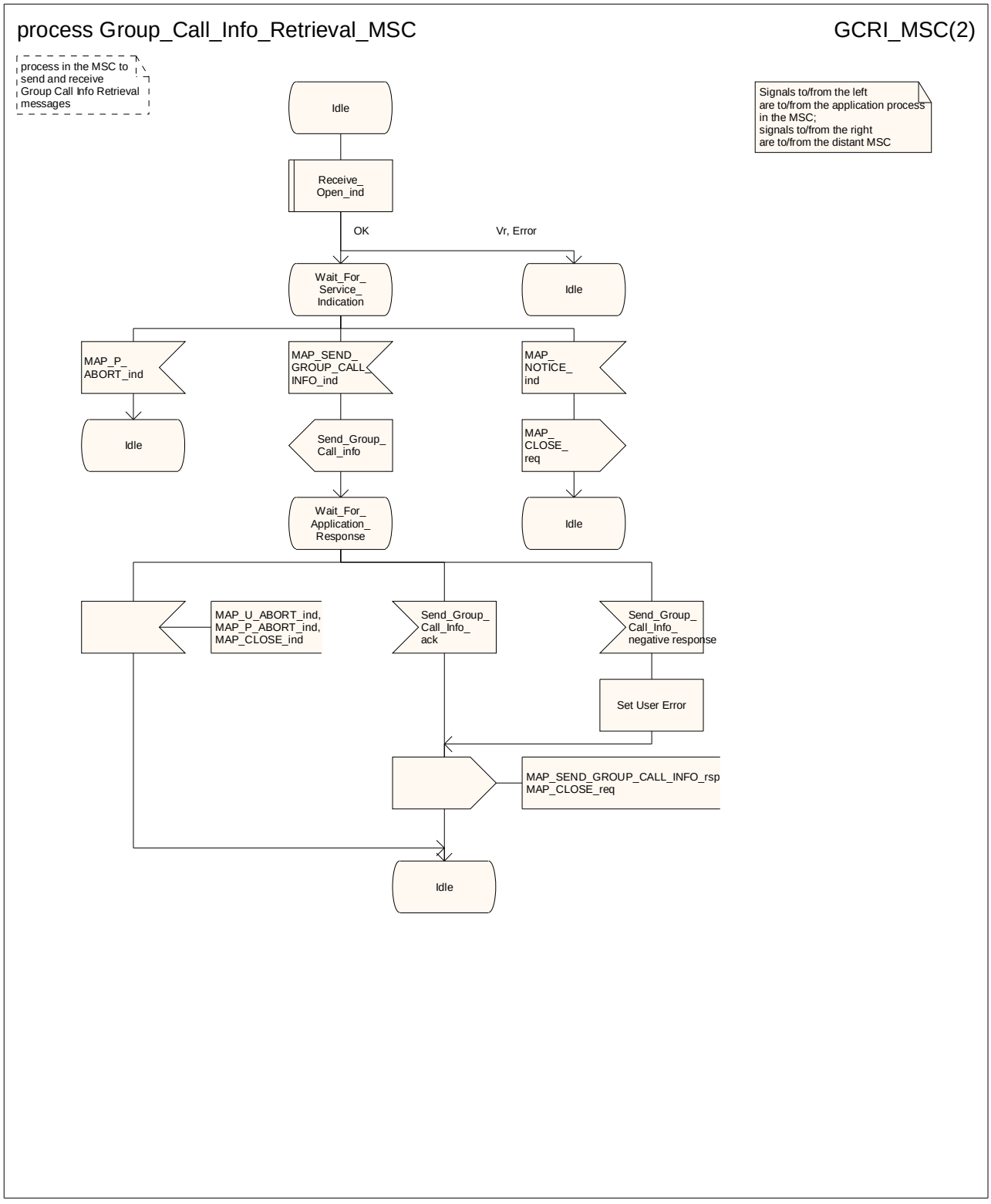


Figure 21.4A/2 (sheet 2 of 2): Process Group_Call_Info_Retrieval_MSC

21.5 Void

21.6 CCBS: monitoring and reporting the status of the subscriber

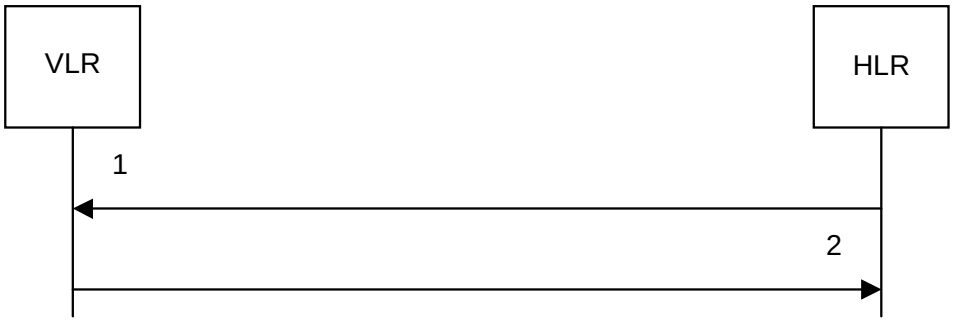
21.6.1 Reporting co-ordinator process in the VLR

The MAP co-ordinating process in the VLR to handle a dialogue opened with the reporting application context is shown in figure 21.6/6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

21.6.2 Setting the reporting state – stand-alone

The message flow for setting the reporting state in a stand-alone dialogue is shown in figure 21.6/1.



1) MAP_SET_REPORTING_STATE_req/ind
2) MAP_SET_REPORTING_STATE_rsp/cnf

Figure 21.6/1: Message flow for setting the reporting state – stand-alone dialogue

21.6.2.1 Process in the HLR

The MAP process in the HLR to set the reporting state in the VLR in a stand-alone dialogue is shown in figure 21.6/7. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The result of a request to stop reporting is not reported to the CCBS application in the HLR.

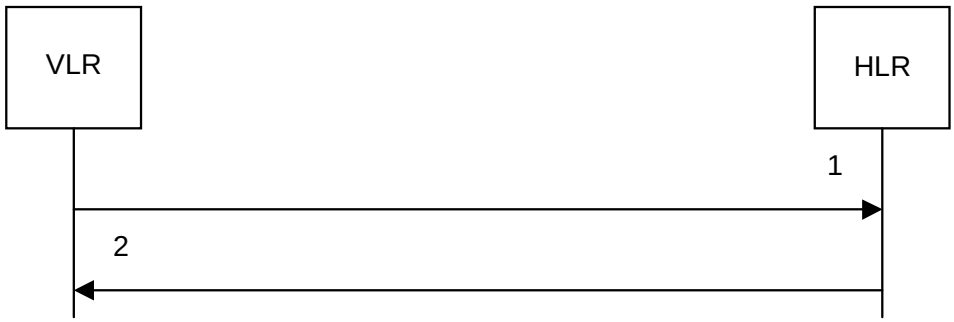
21.6.2.2 Process in the VLR

The MAP process in the VLR to set the reporting state is shown in figure 21.6/8.

The macro Set_Reporting_State_VLR is shown in figure 21.6/9.

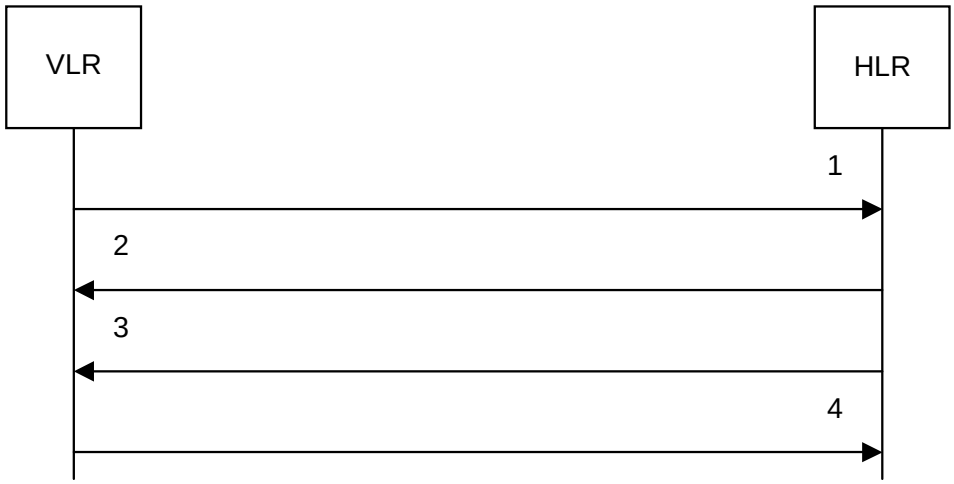
21.6.3 Status Reporting

The message flows for reporting the status of a subscriber are shown in figures 21.6/2 and 21.6/3.



1) MAP_STATUS_REPORT_req/ind
2) MAP_STATUS_REPORT_rsp/cnf

Figure 21.6/2: Message flow for status reporting, when monitoring continues in the VLR



1) MAP_STATUS_REPORT_req/ind
2) MAP_STATUS_REPORT_rsp/cnf
3) MAP_SET_REPORTING_STATE_req/ind
4) MAP_SET_REPORTING_STATE_rsp/cnf

Figure 21.6/3: Message flow for status reporting, when monitoring stops

The MAP_SET_REPORTING_STATE request is used to stop monitoring in the VLR. If the HLR requires the VLR to continue monitoring, it closes the dialogue without sending a MAP_SET_REPORTING_STATE request.

21.6.3.1 Process in the VLR

The MAP process in the VLR to send a status report to the HLR is shown in figure 21.6/10. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

This process can be used to report:

- an event, such as the user becoming free, or
- the result of a CCBS call attempt

to the HLR

21.6.3.2 Process in the HLR

The MAP process in the HLR to handle a status report is shown in figure 21.6/11. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

It is an implementation option whether to send the MAP_DELIMITER request before invoking the macro Set_Reporting_State_HLR.

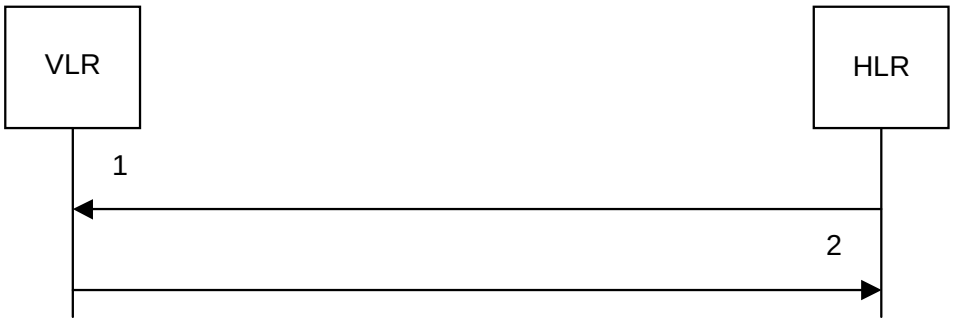
The macro Receive_Status_Report_HLR is shown in figure 21.6/12.

The macro Set_Reporting_State_HLR is shown in figure 21.6/13. The macro invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Confirmation see subclause 25.2.2.

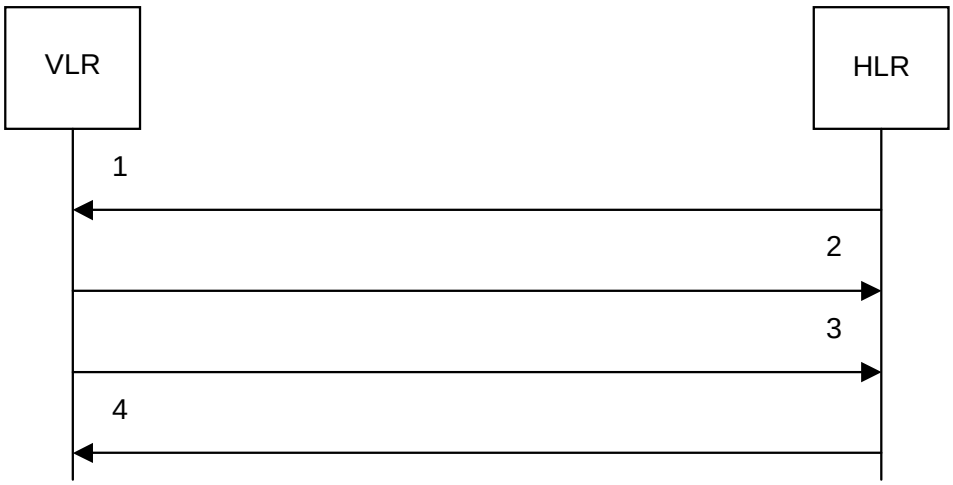
21.6.4 CCBS: Remote User Free

The message flows for handling remote user free are shown in figures 21.6/4 and 21.6/5.



1) MAP_REMOTE_USER_FREE_req/ind
2) MAP_REMOTE_USER_FREE_rsp/cnf

Figure 21.6/4: Remote User Free: recall not accepted



1) MAP_REMOTE_USER_FREE_req/ind
2) MAP_REMOTE_USER_FREE_rsp/cnf
3) MAP_STATUS_REPORT_req/ind
4) MAP_STATUS_REPORT_rsp/cnf

Figure 21.6/5: Remote User Free: recall accepted

21.6.4.1 Process in the HLR

The MAP process in the HLR to handle Remote User Free is shown in figure 21.6/14. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

21.6.3.2 Process in the VLR

The MAP process in the VLR to handle Remote User Free is shown in figure 21.6/15. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Confirmation see subclause 25.2.2.

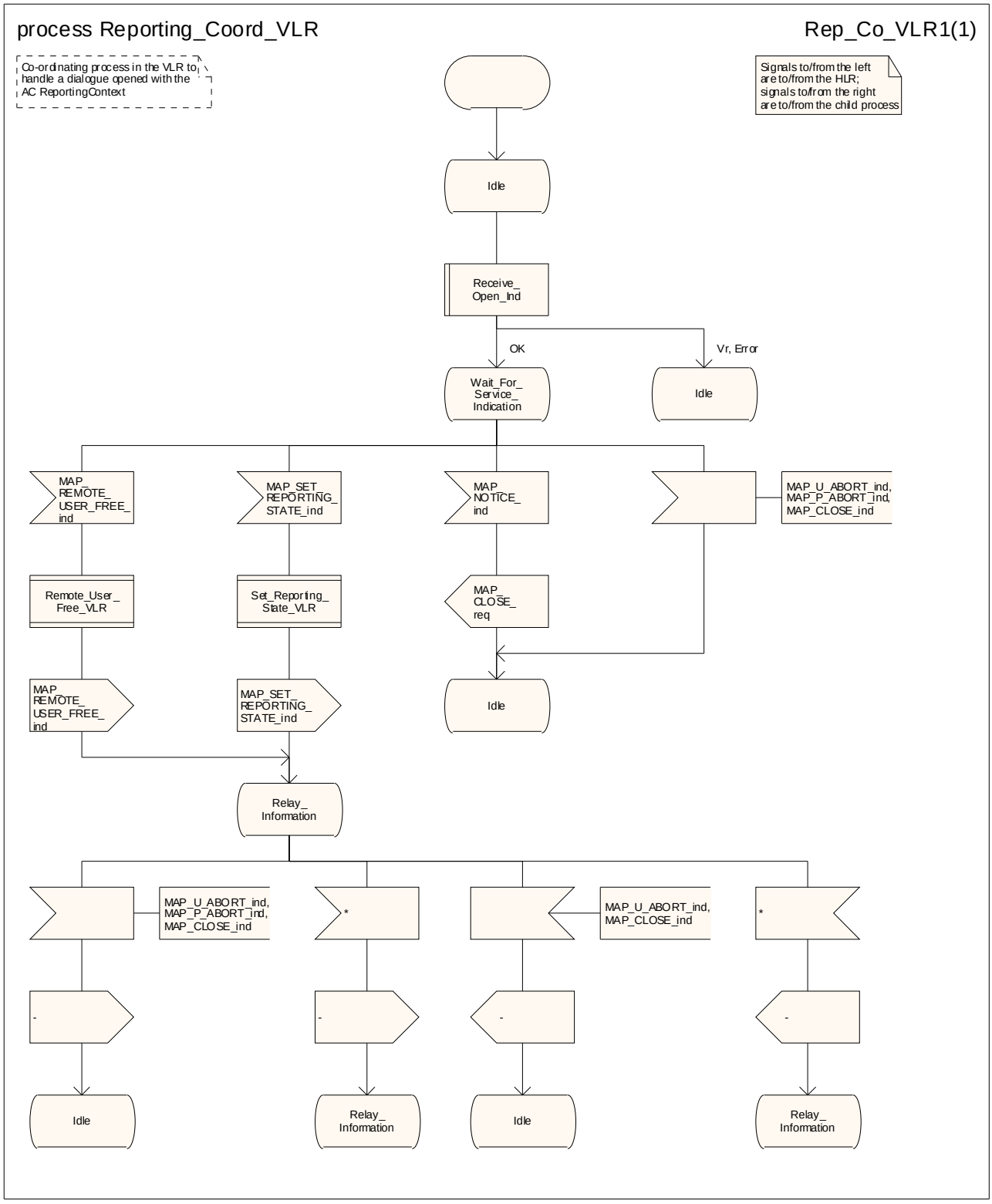


Figure 21.6/6: Process Reporting_Coord_VLR

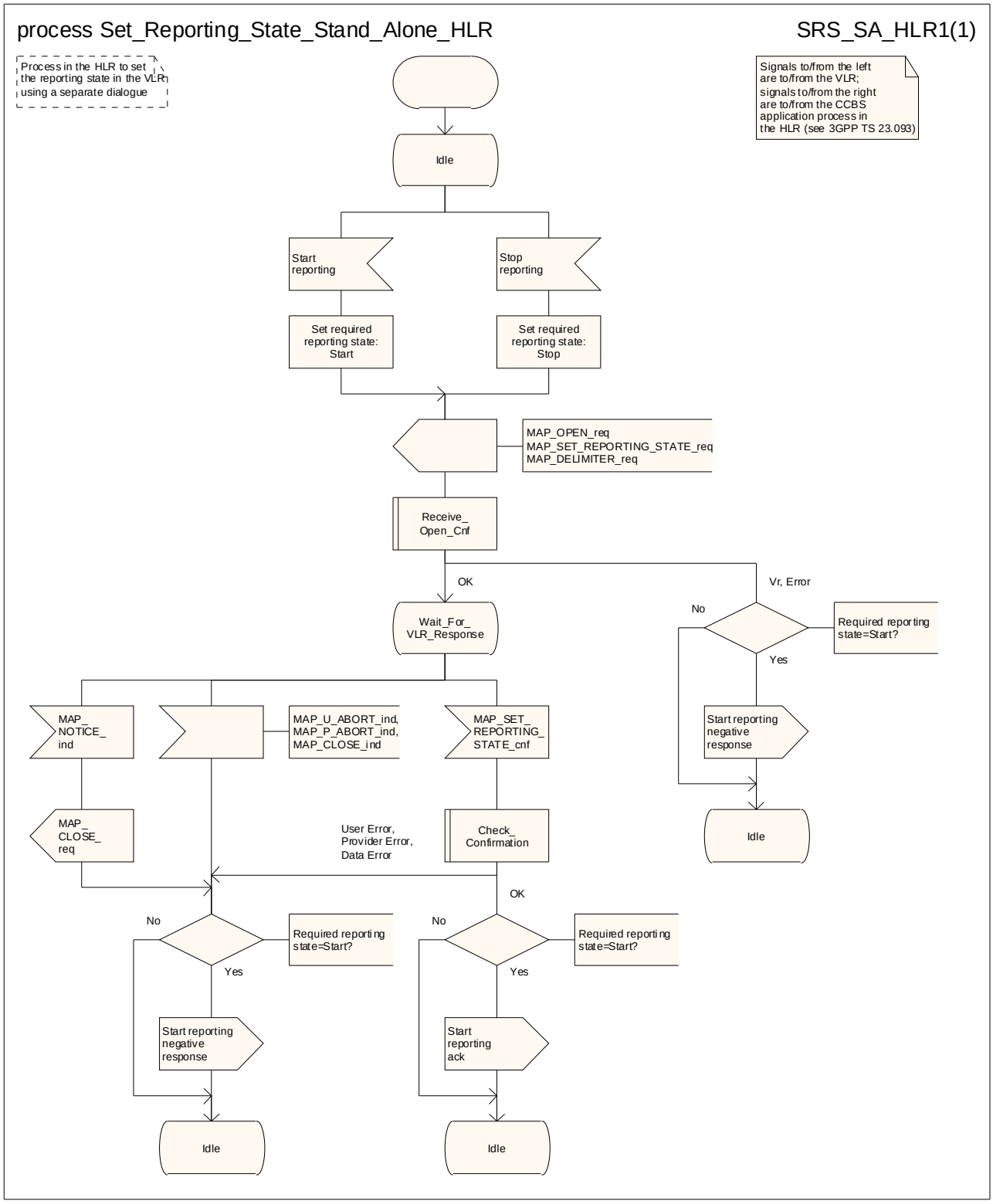


Figure 21.6/7: Process Set_Reporting_State_Stand_Alone_HLR

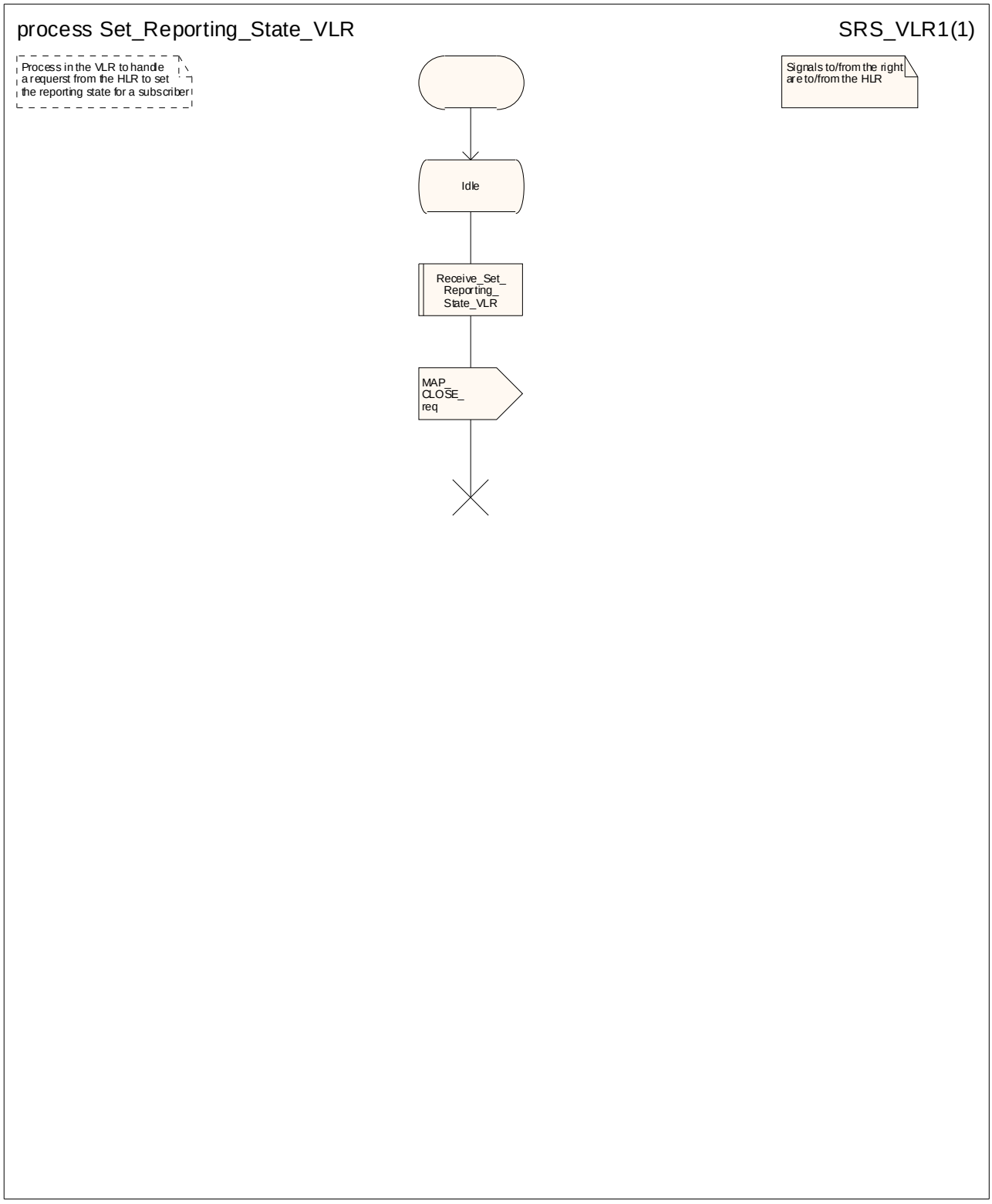


Figure 21.6/8: Process Set_Reporting_State_VLR

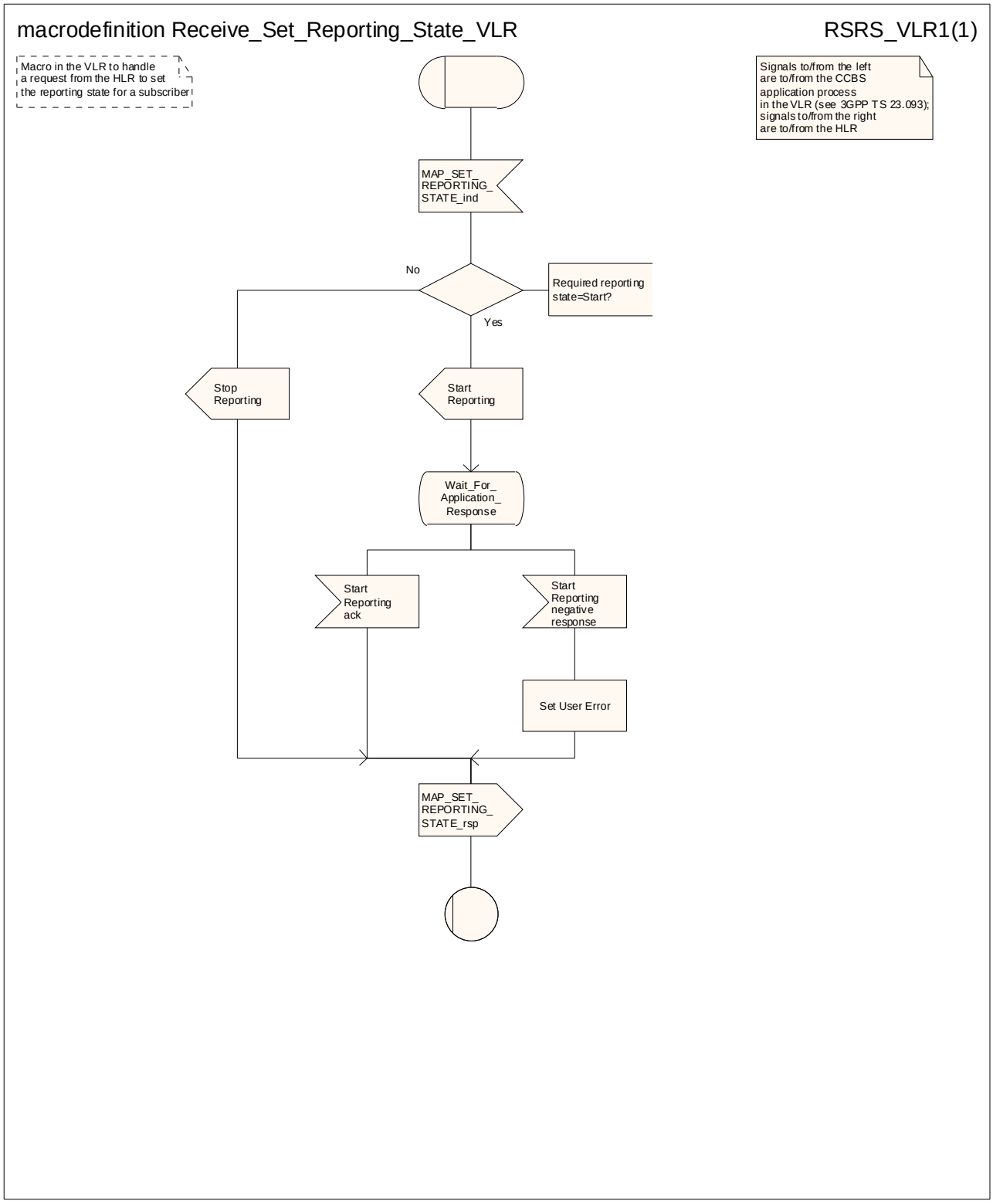


Figure 21.6/9: Macro Receive_Set_Reporting_State_VLR

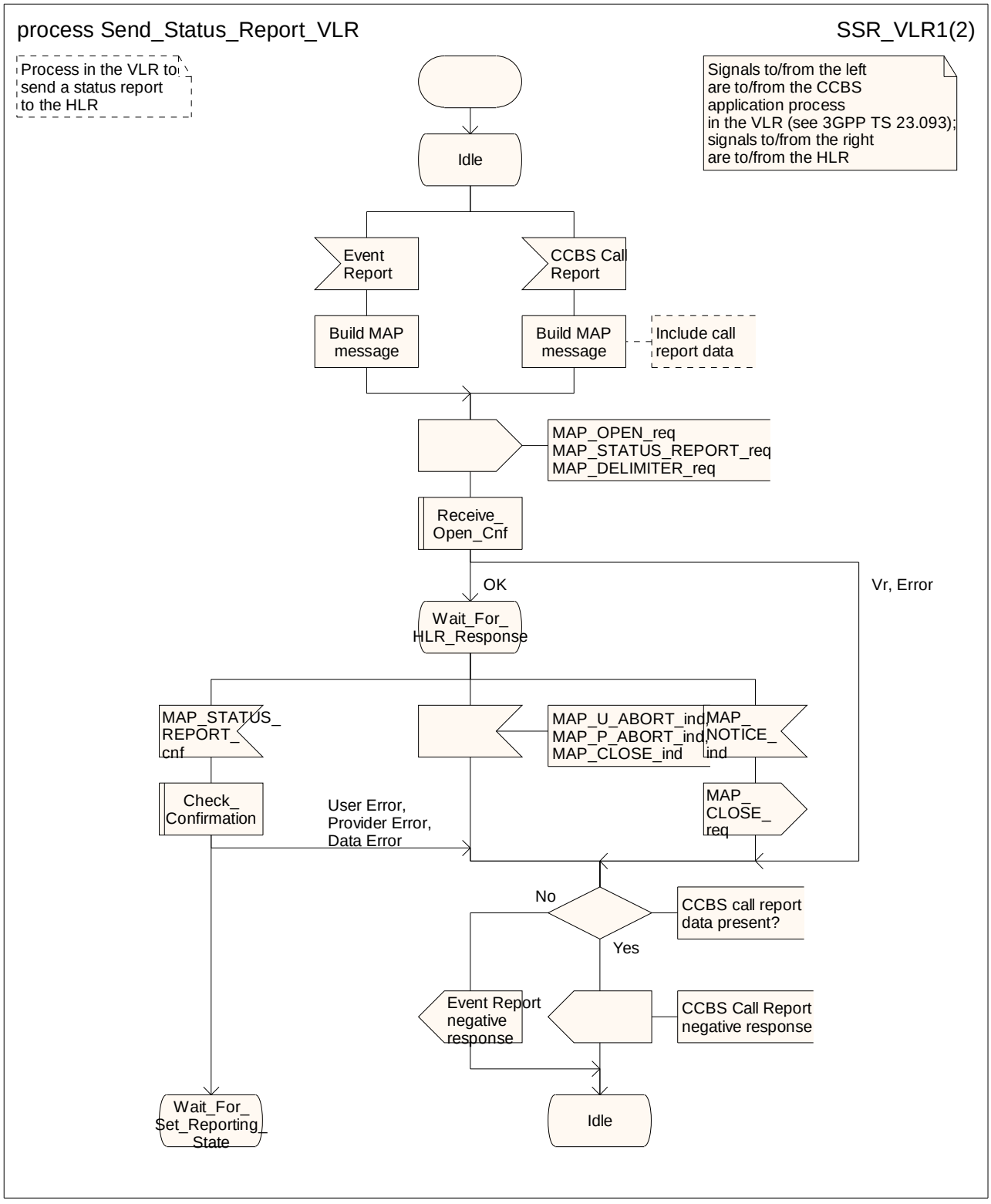


Figure 21.6/10 (sheet 1 of 2): Process Send_Status_Report_VLR

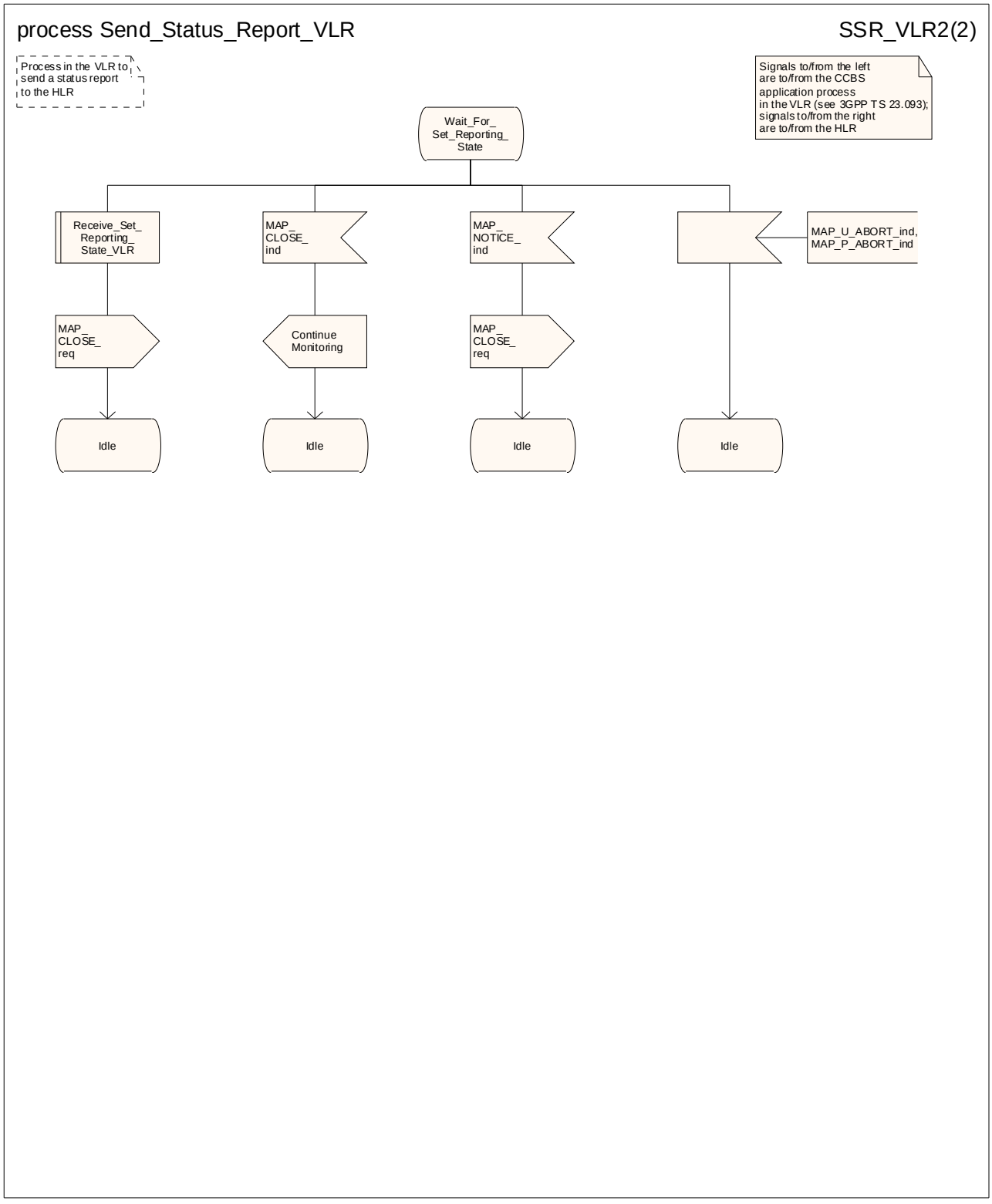


Figure 21.6/10 (sheet 2 of 2): Process Send_Status_Report_VLR

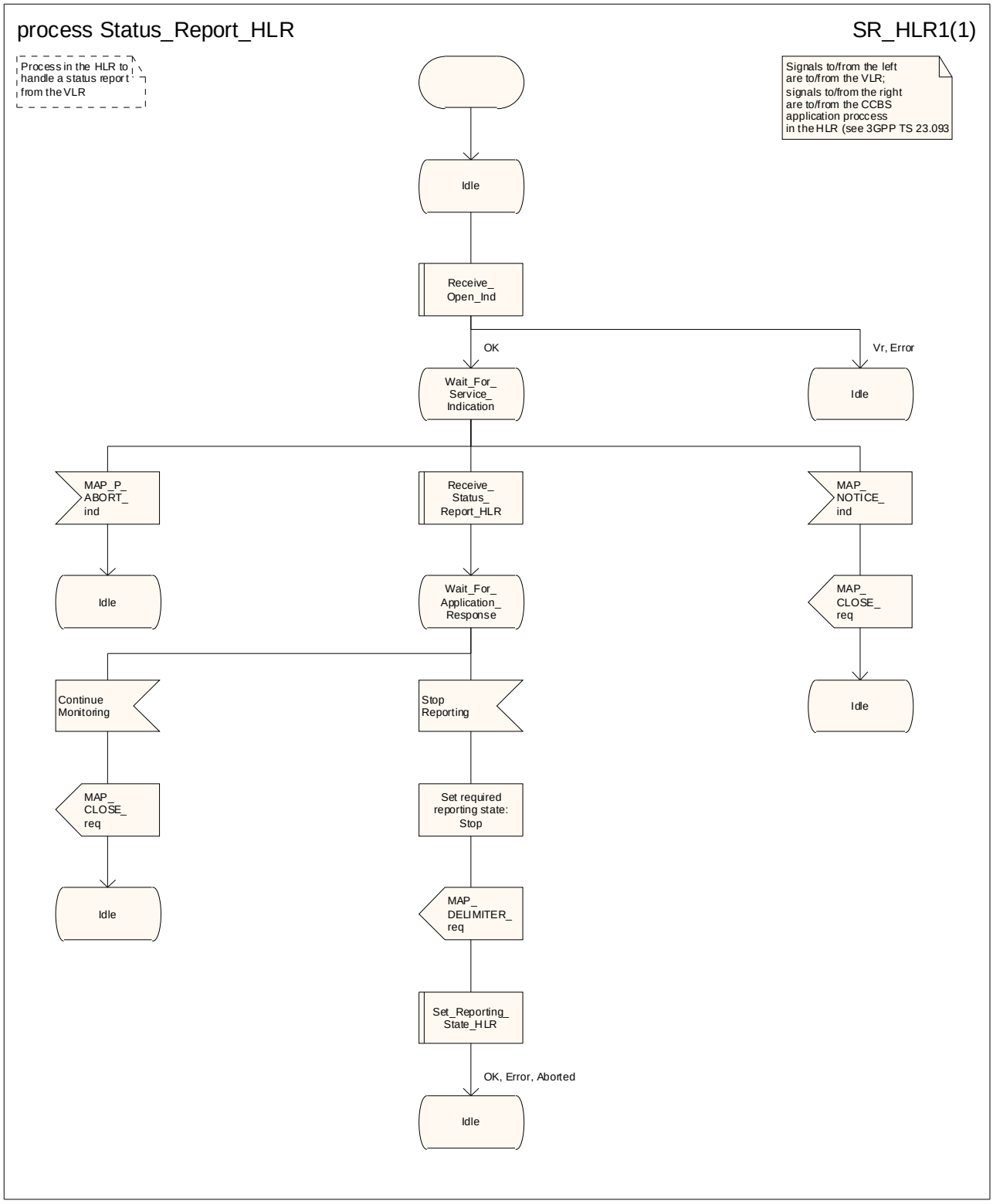


Figure 21.6/11: Process Status Report_HLR

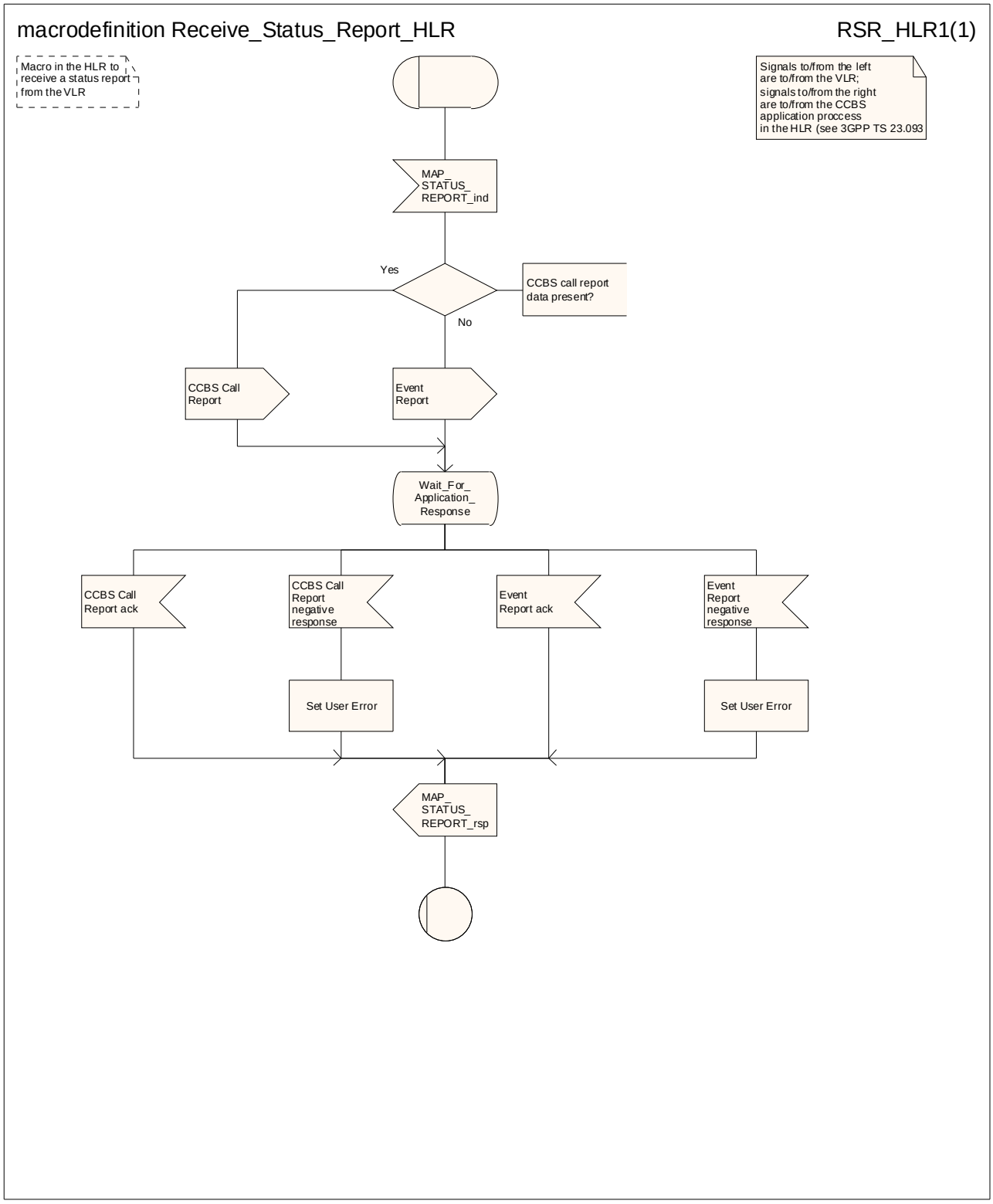


Figure 21.6/12: Macro Receive_Status_Report_HLR

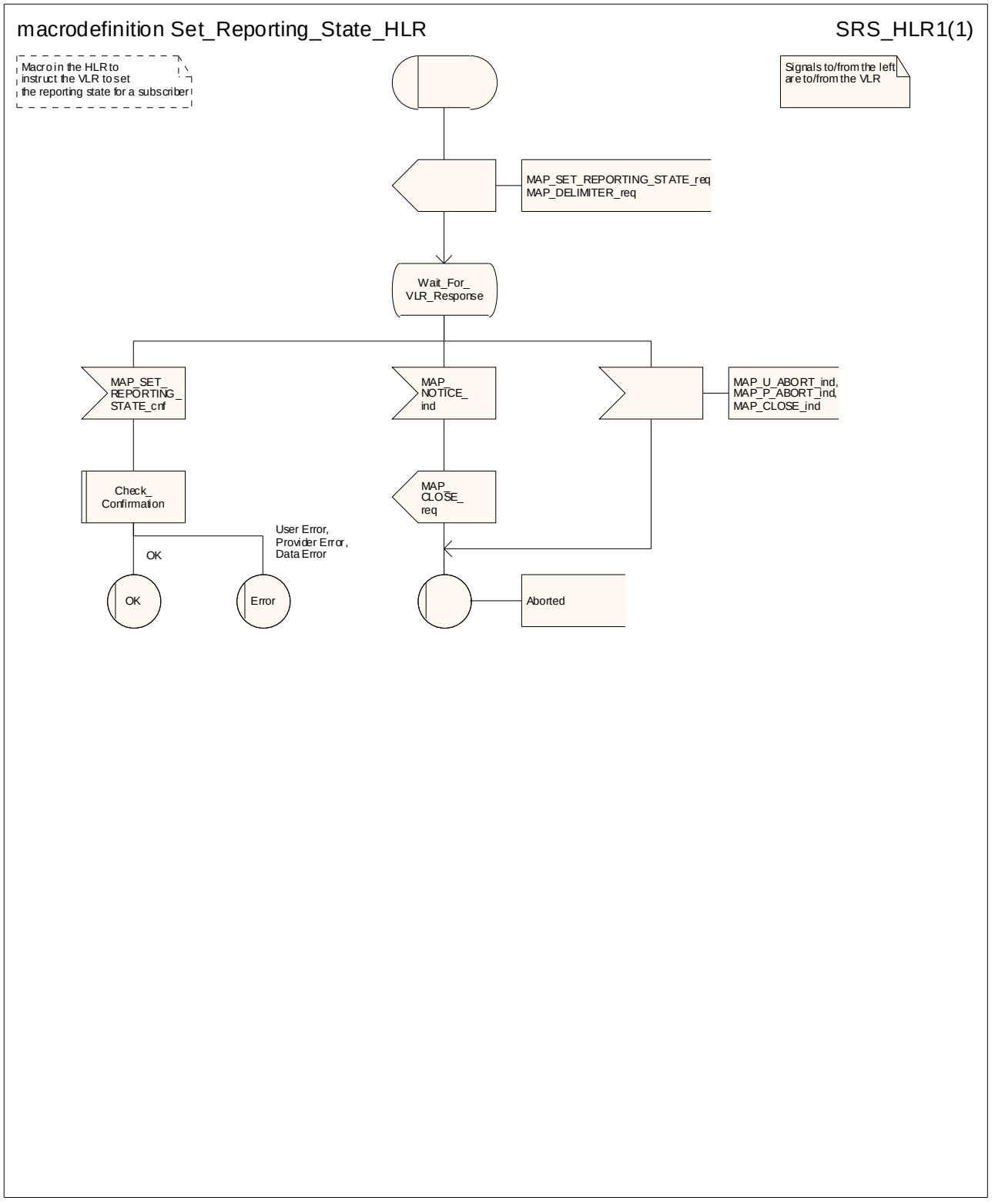


Figure 21.6/13: Macro Set_Reporting_State_HLR

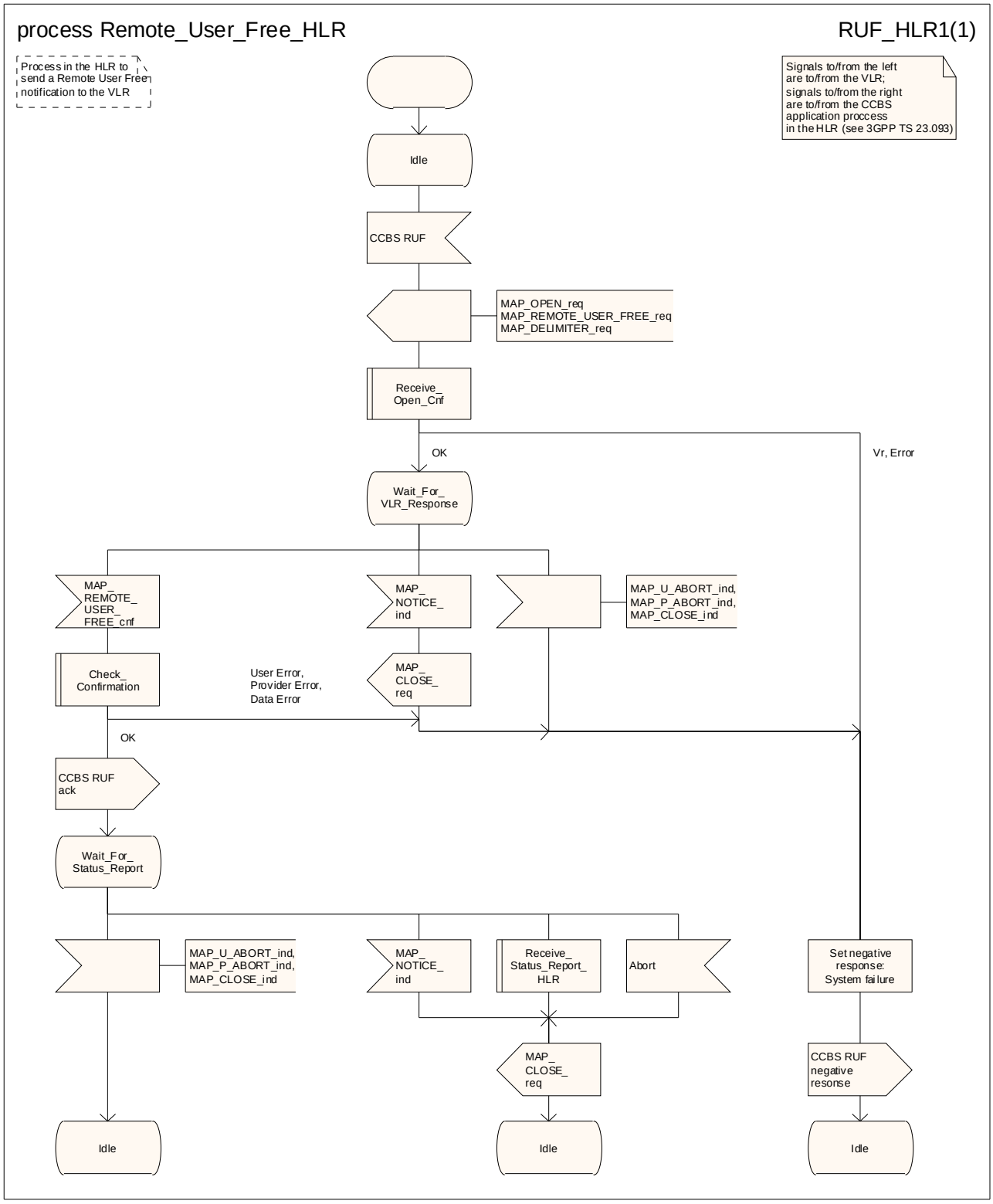


Figure 21.6/14: Process Remote_User_Free_HLR

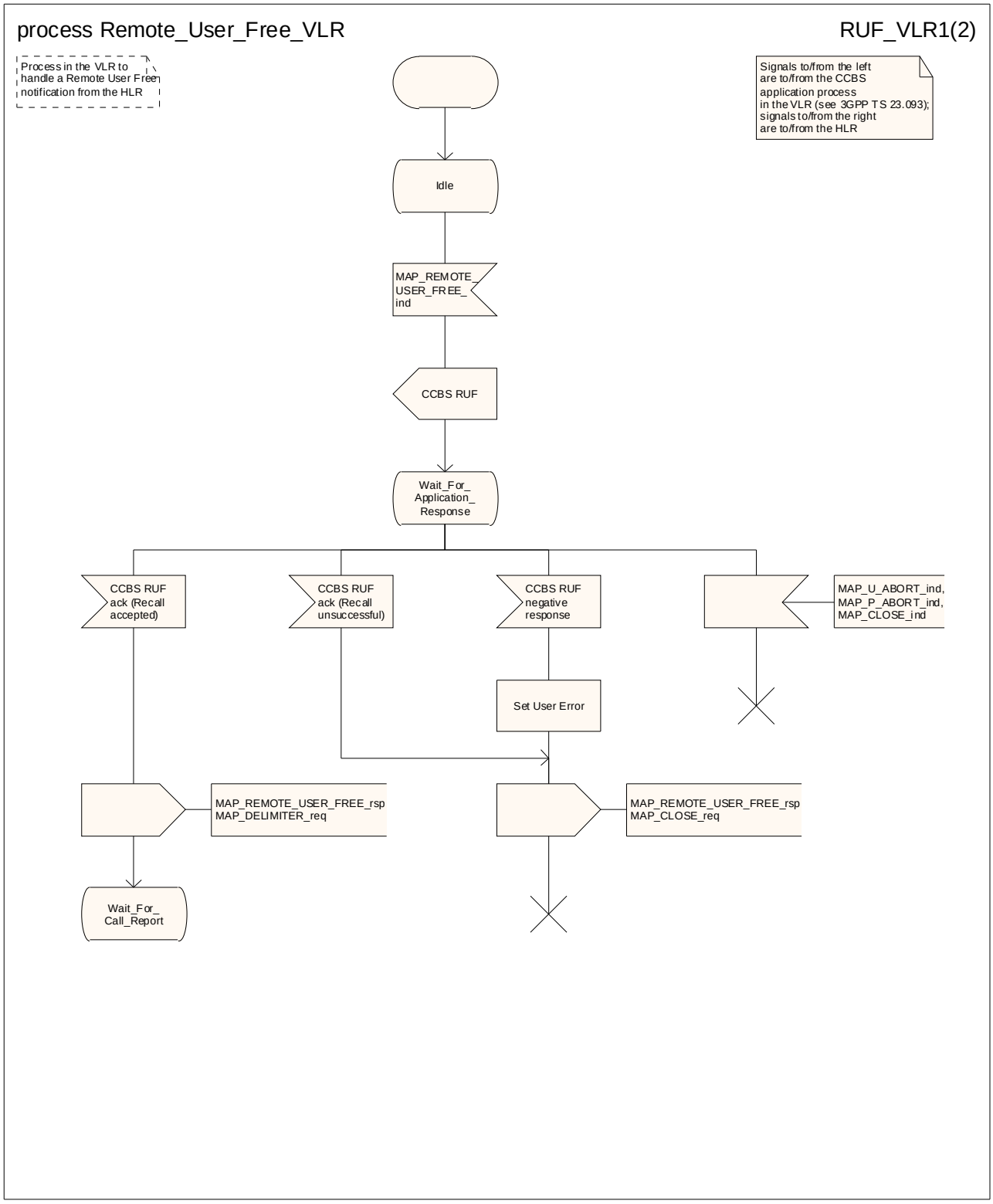


Figure 21.6/15 (sheet 1 of 2): Process Remote_User_Free_VLR

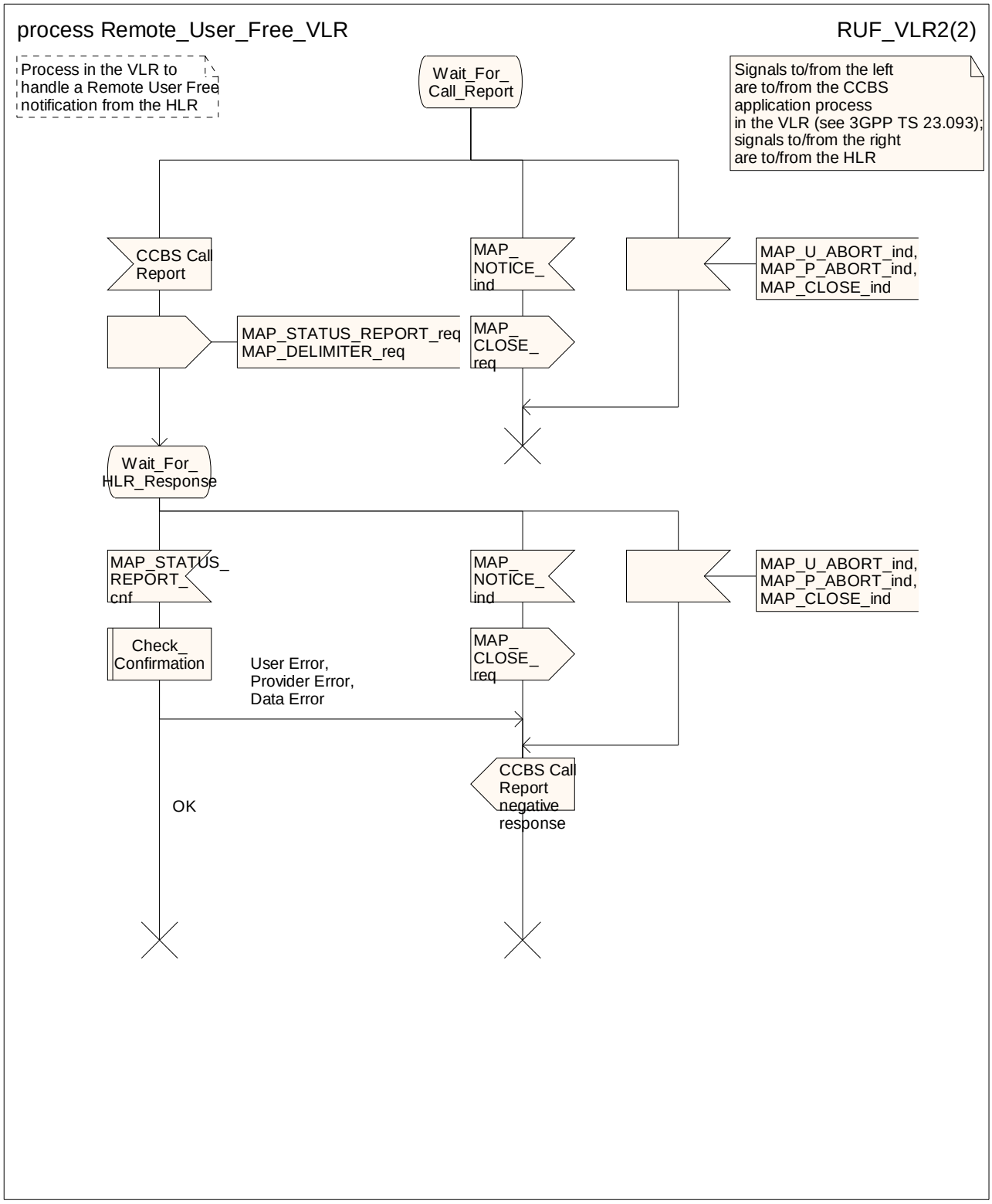


Figure 21.6/15 (sheet 2 of 2): Process Remote_User_Free_VLR

21.7 Void

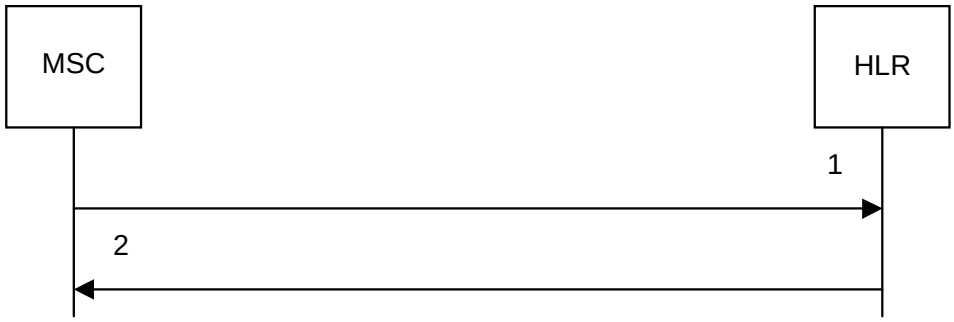
21.8 Void

21.9 Immediate Service Termination (IST)

21.9.1 IST Alert

The Immediate Service Termination Alert procedure is used to keep track of the call activities performed by subscribers who are marked as being subject to IST monitoring and, possibly, to terminate the call activities for which the alert was sent, or all the call activities related to the subscriber for whom the alert was sent.

The message flow for alerting is shown in figure 21.9/1; the MSC may be a Visited MSC or a Gateway MSC.



1) MAP_IST_ALERT_req/ind
2) MAP_IST_ALERT_rsp/cnf

Figure 21.9/1: Message flow for IST Alert

21.9.1.1 Procedure in the MSC

The MAP process in the MSC (Visited MSC or Gateway MSC) is shown in figure 21.9/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

21.9.1.2 Procedure in the HLR

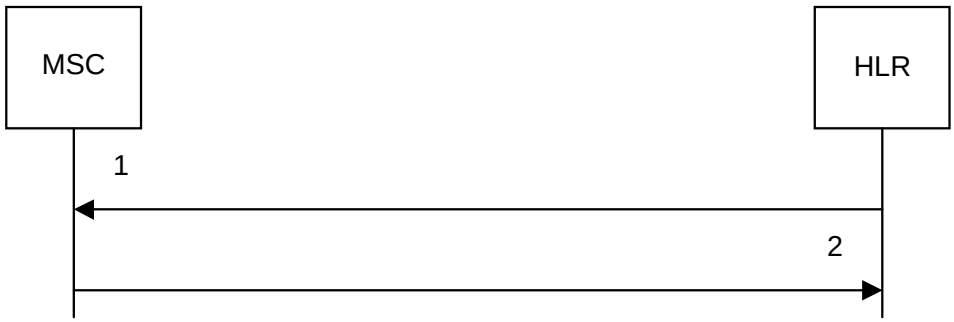
The MAP process in the HLR is shown in figure 21.9/4. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

21.9.2 IST Command

The Immediate Service Termination Command procedure is used to terminate the call activities related to a subscriber.

The message flow for the IST Command procedure is shown in figure 21.9/2; the MSC may be a Visited MSC or a Gateway MSC.



1) MAP_IST_COMMAND_req/ind
2) MAP_IST_COMMAND_rsp/cnf

Figure 21.9/2: Message flow for IST Command

21.9.2.1 Procedure in the HLR

The MAP process in the HLR is shown in figure 21.9/5. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

21.9.2.2 Procedure in the MSC

The MAP process in the MSC is shown in figure 21.9.6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

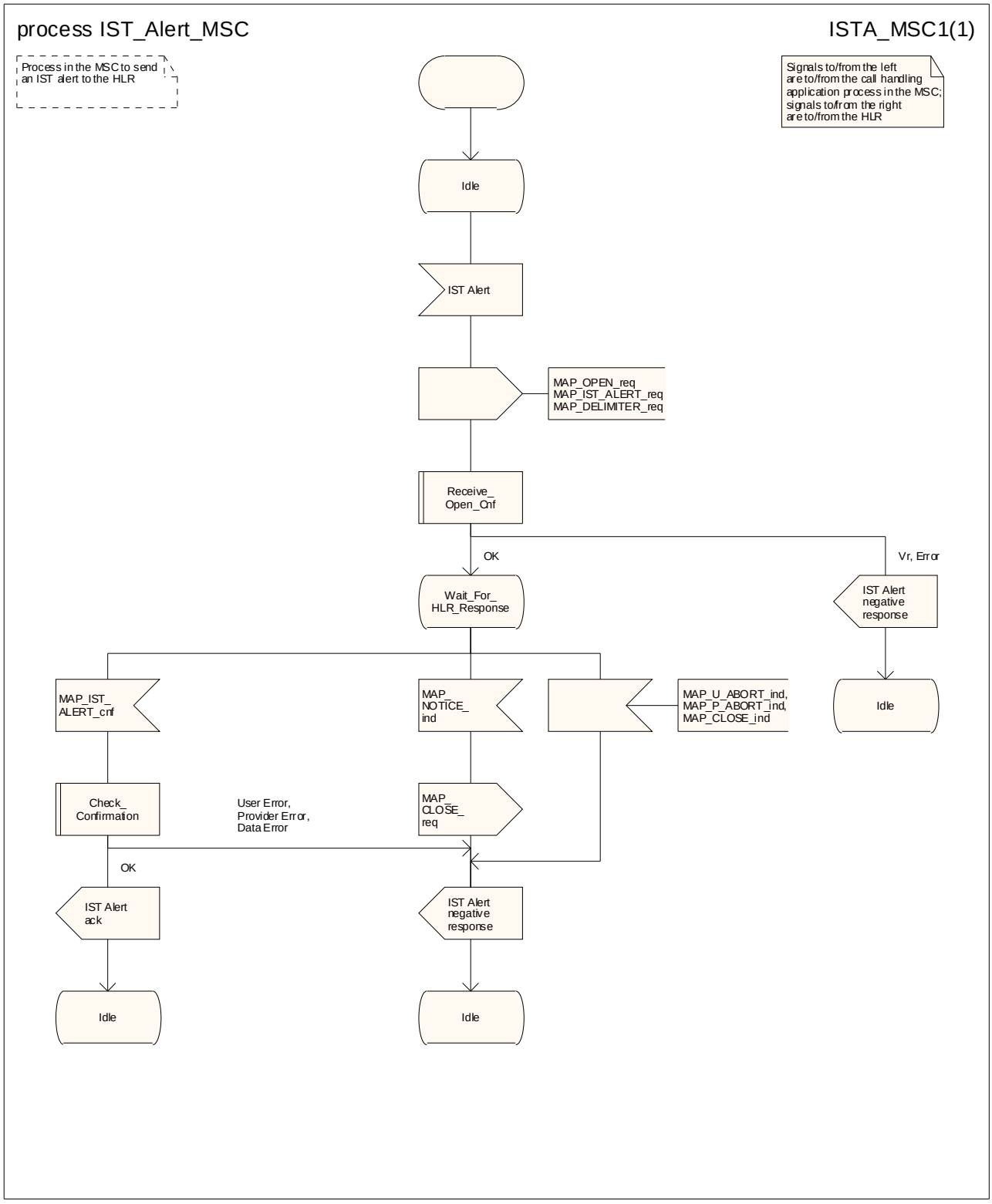


Figure 21.9/3: Process IST_Alert_MSC

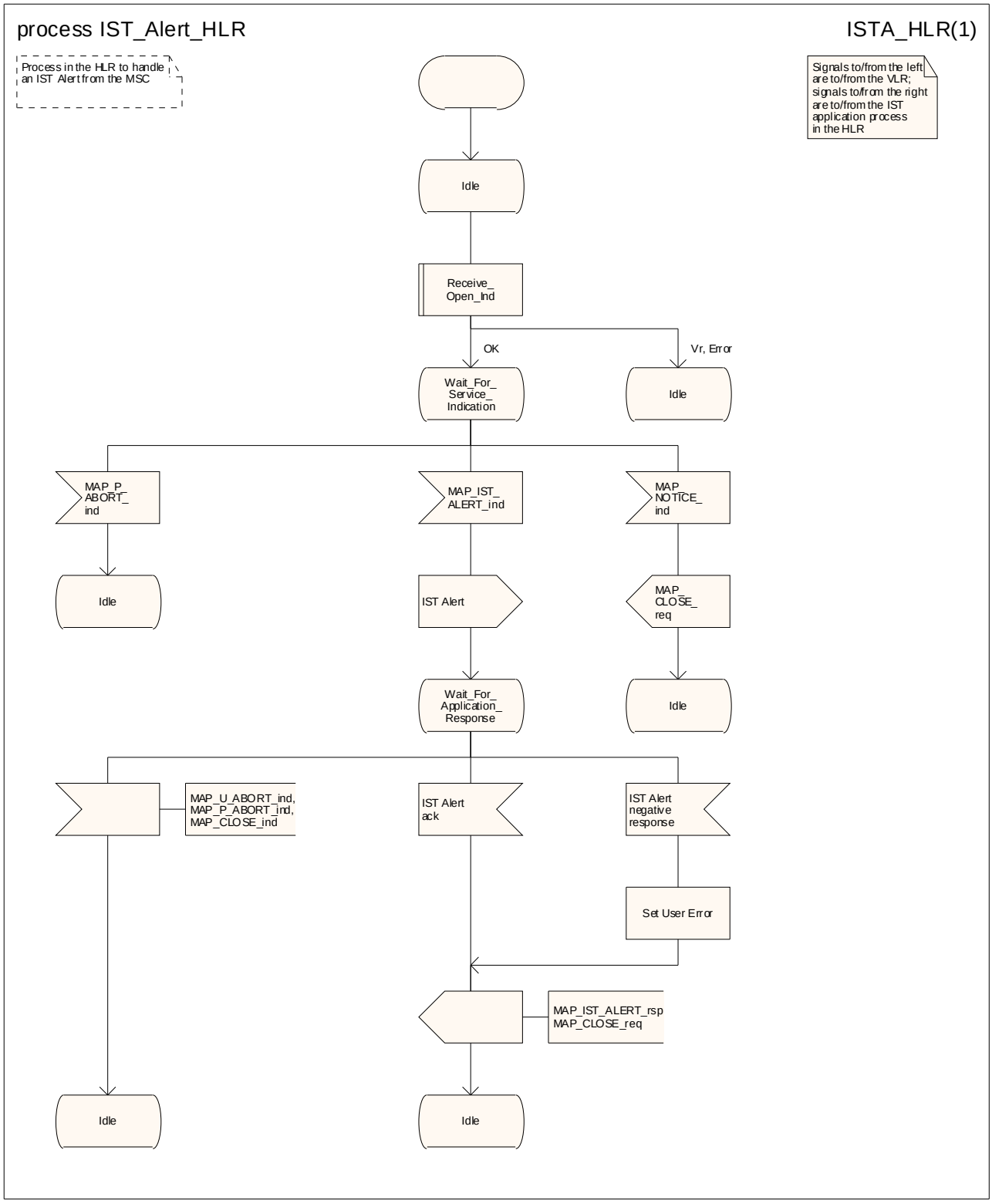


Figure 21.9/4: Process IST_Alert_HLR

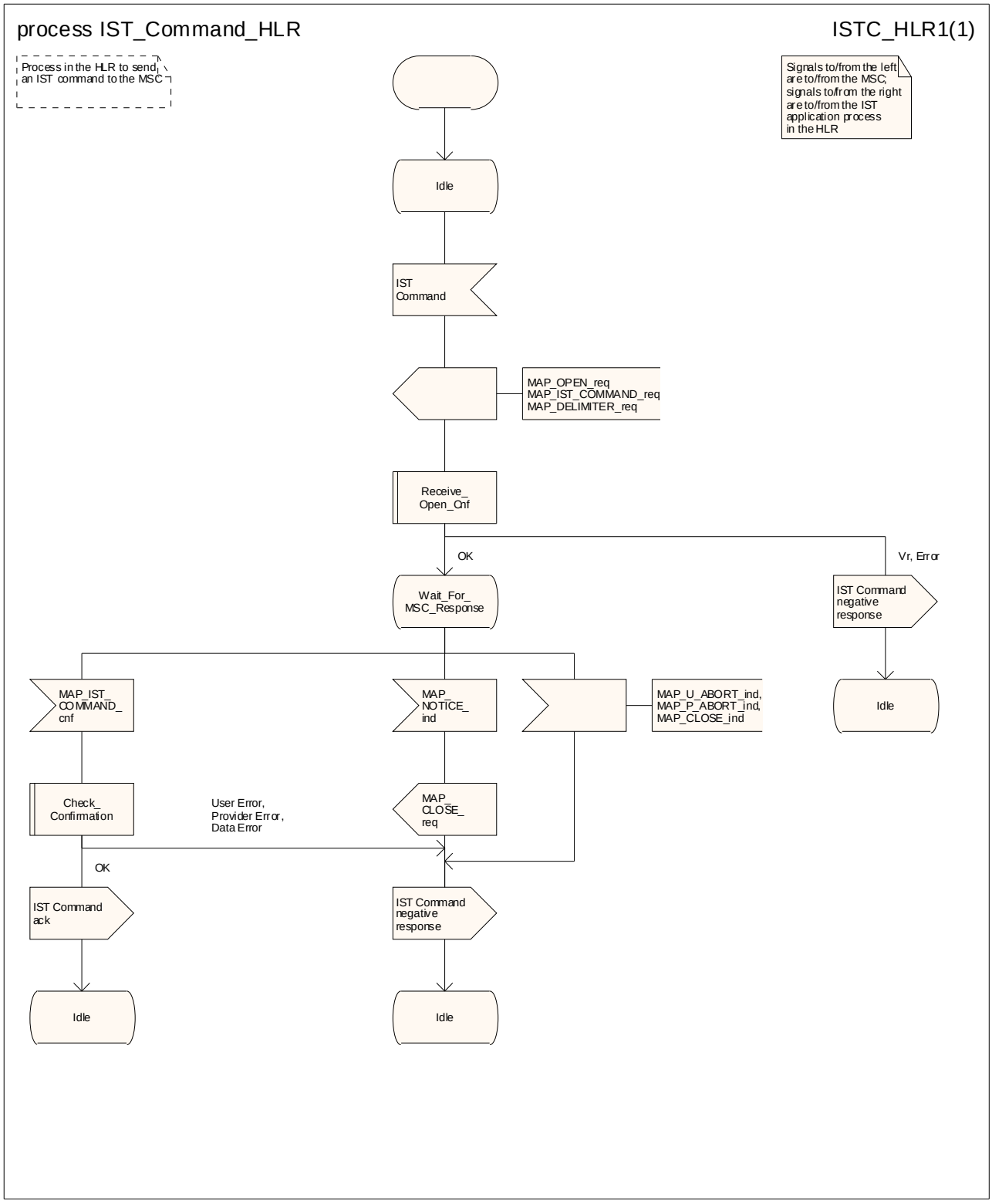


Figure 21.9/5: Process IST_Command_HLR

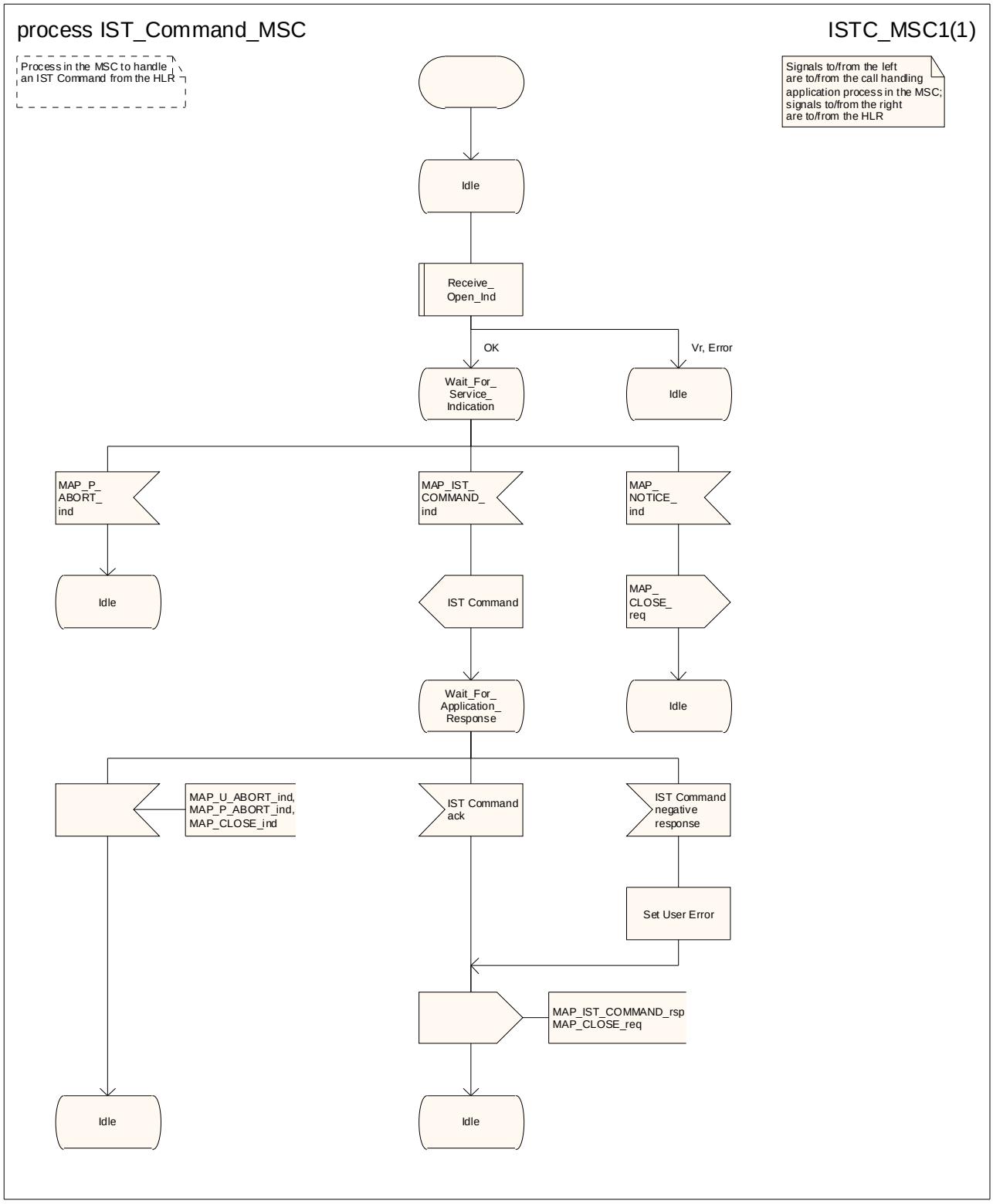
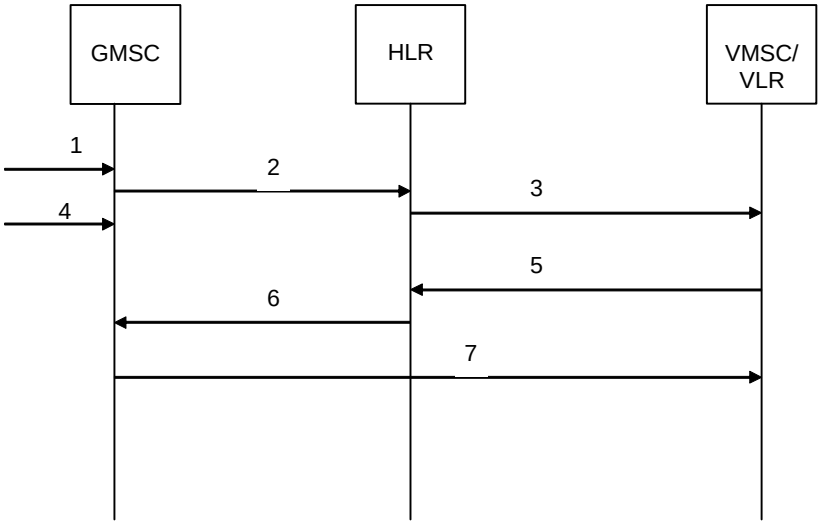


Figure 21.9/6: Process IST_Command_MSC

21.10 Resource Management

21.10.1 General

The message flow for successful release of resources is shown in figure 21.10/1.



1) I_IAM (Note 1)
2) MAP_SEND_ROUTING_INFORMATION_req/ind
3) MAP_PROVIDE_ROAMING_NUMBER_req/ind
4) I_REL (Note 1)
5) MAP_PROVIDE_ROAMING_NUMBER_rsp/cnf
6) MAP_SEND_ROUTING_INFORMATION_rsp/cnf
7) MAP_RELEASE_RESOURCES (Note 2)

NOTE 1: TUP or ISUP may be used in signalling between MSCs, depending on the network type between the MSCs. For further details on the TUP and ISUP procedures refer to the following ITU-T Recommendations & ETSI specification:
- Q.721-725 - Telephone User Part (TUP);
- ETS 300 356-1 - Integrated Services Digital Network (ISDN); Signalling System No.7; ISDN User Part (ISUP) version 2 for the international interface; Part 1: Basic services.

NOTE 2: Services printed in *italics* are optional.

Figure 21.10/1: Message flow for early release of resources

21.3.2 Process in the GMSC

The MAP process in the GMSC to release resources is shown in figure 21.10/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

21.3.3 Process in the VMSC

The MAP process in the VMSC to handle a request for the GMSC to release resources is shown in figure 21.10/3. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

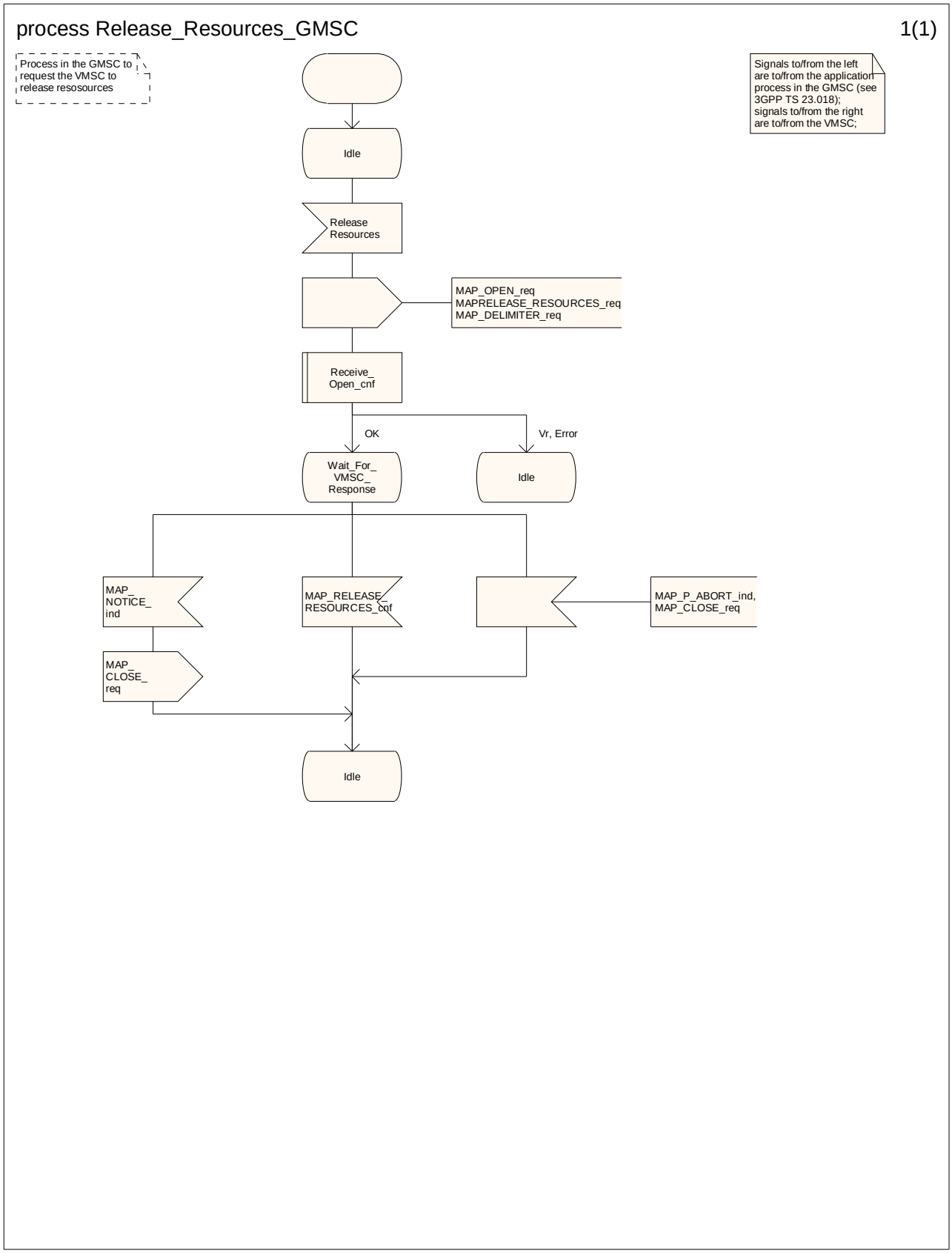


Figure 21.10/2: Process Release Resources_GMSC

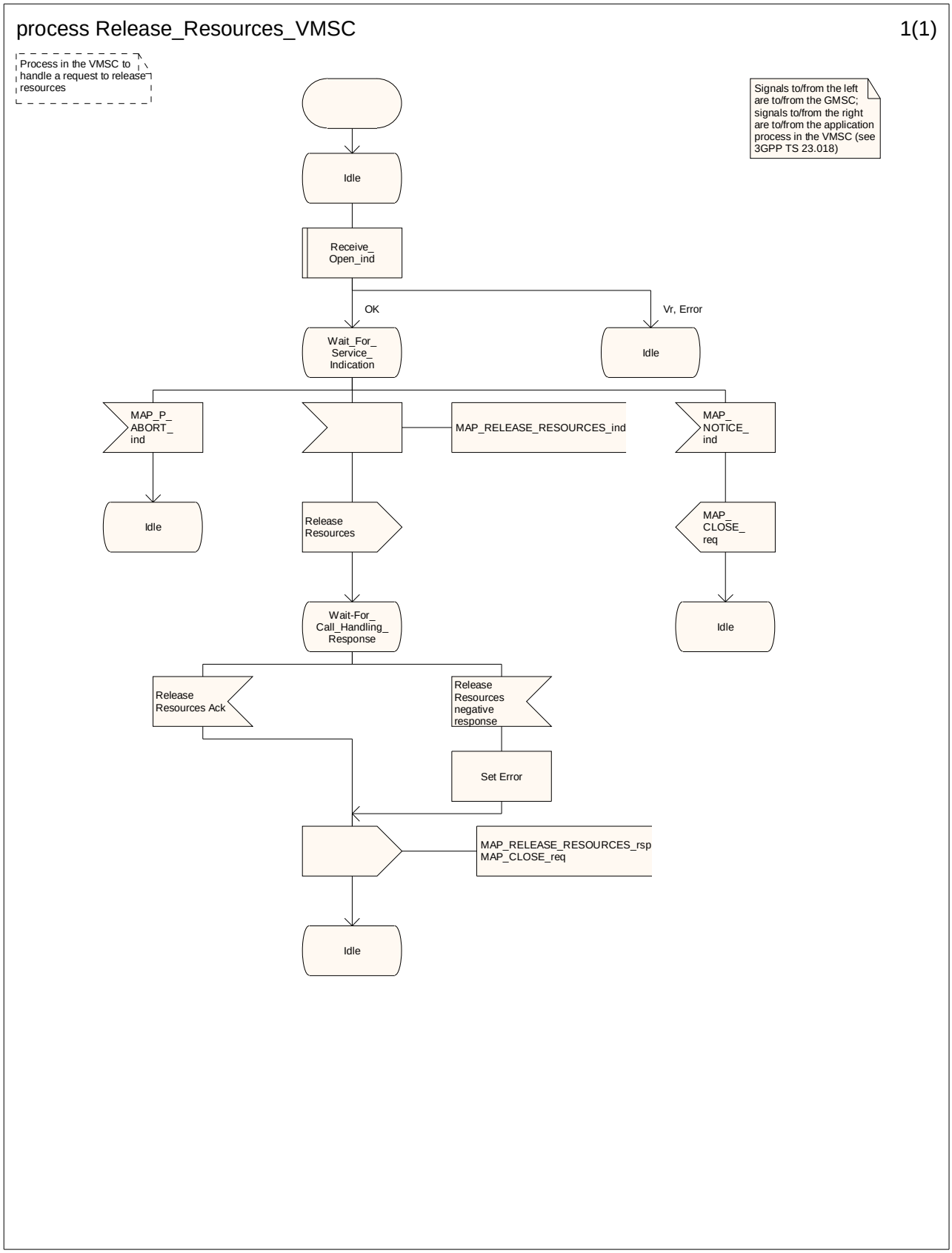


Figure 21.10/3: Process Release Resources_VMSC

22 Supplementary services procedures

22.1 Supplementary service co-ordinator processes

22.1.1 Supplementary service co-ordinator process for the MSC

The co-ordinator process in the MSC to handle a CM connection request with CM service type Supplementary service activation is shown in figure 22.1/1. The process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Process_Access_Request_MSC see subclause 25.4.1.

22.1.2 Void

22.1.3 Functional supplementary service co-ordinator process for the HLR

The MAP co-ordinator process in the HLR to handle a dialogue opened with the networkFunctionalSS application context is shown in figure 22.1/3. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

22.1.4 Call completion supplementary service co-ordinator process for the HLR

The MAP co-ordinator process in the HLR to handle a dialogue opened with the callCompletion application context is shown in figure 22.1/4. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

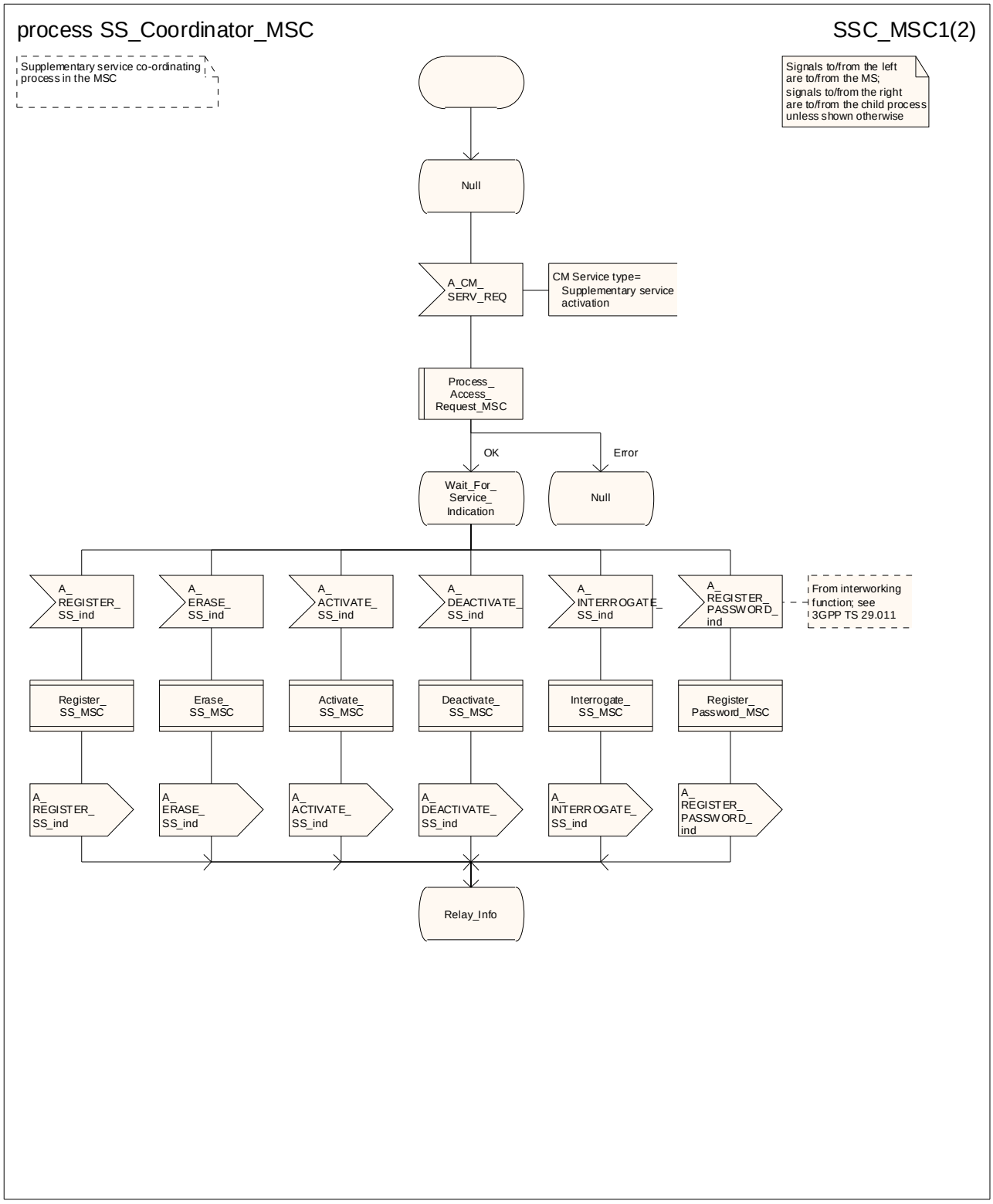


Figure 22.1/1 (sheet 1 of 2): Process SS_Coordinator_MSC

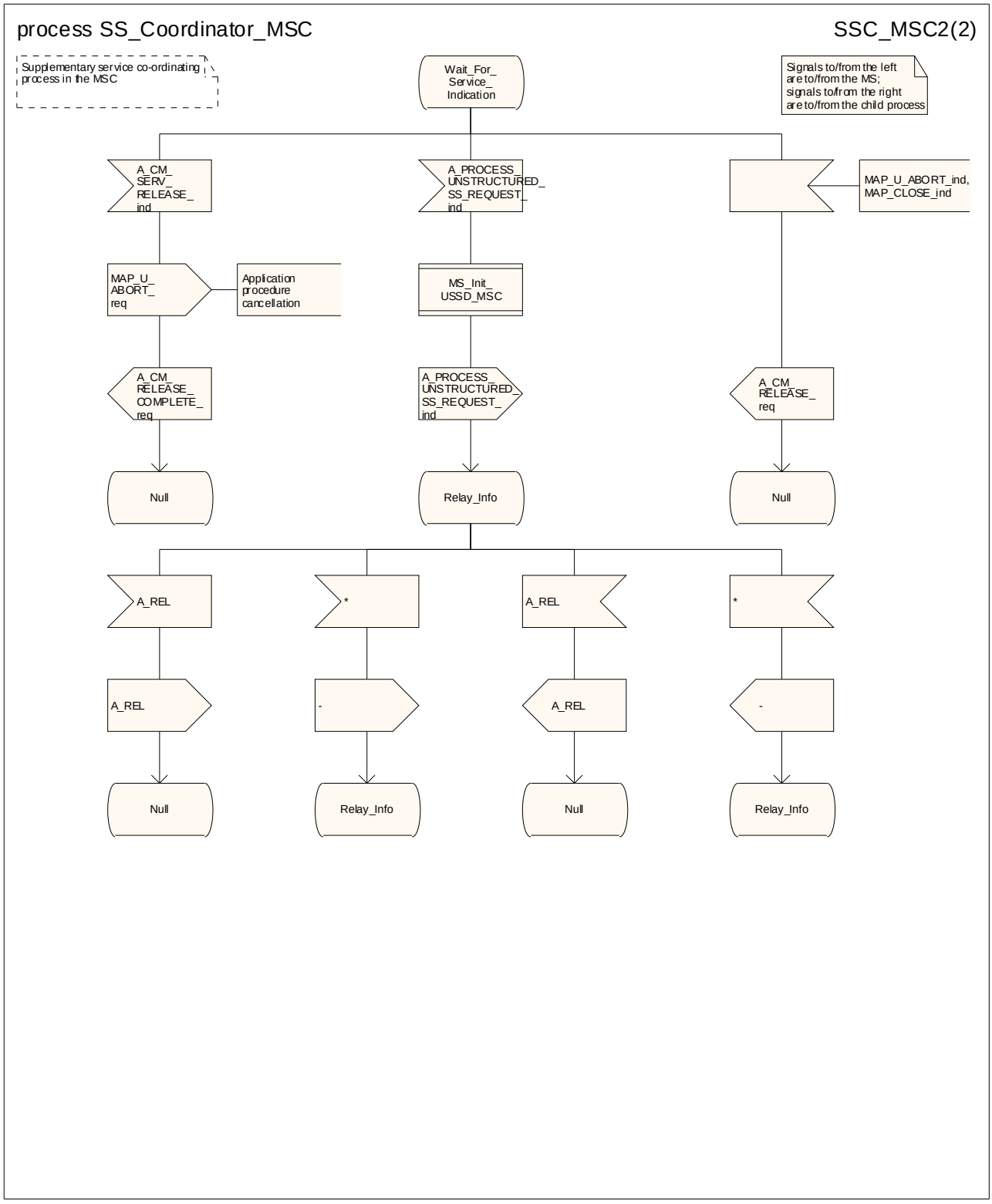


Figure 22.1/1 (sheet 2 of 2): Process SS_Coordinator_MSC

Figure 22.1/2 void

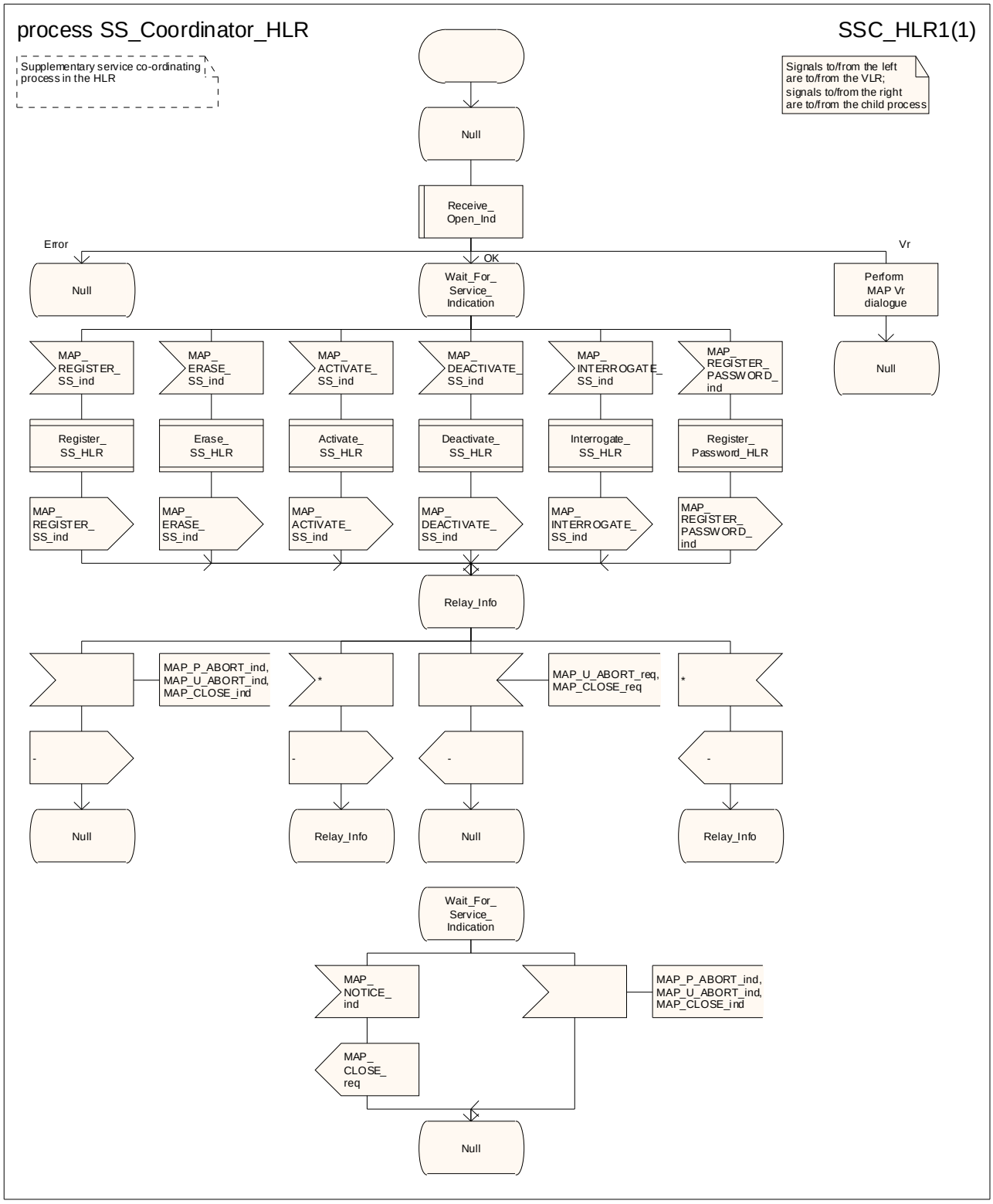


Figure 22.1/3: Process SS_Coordinator_HLR

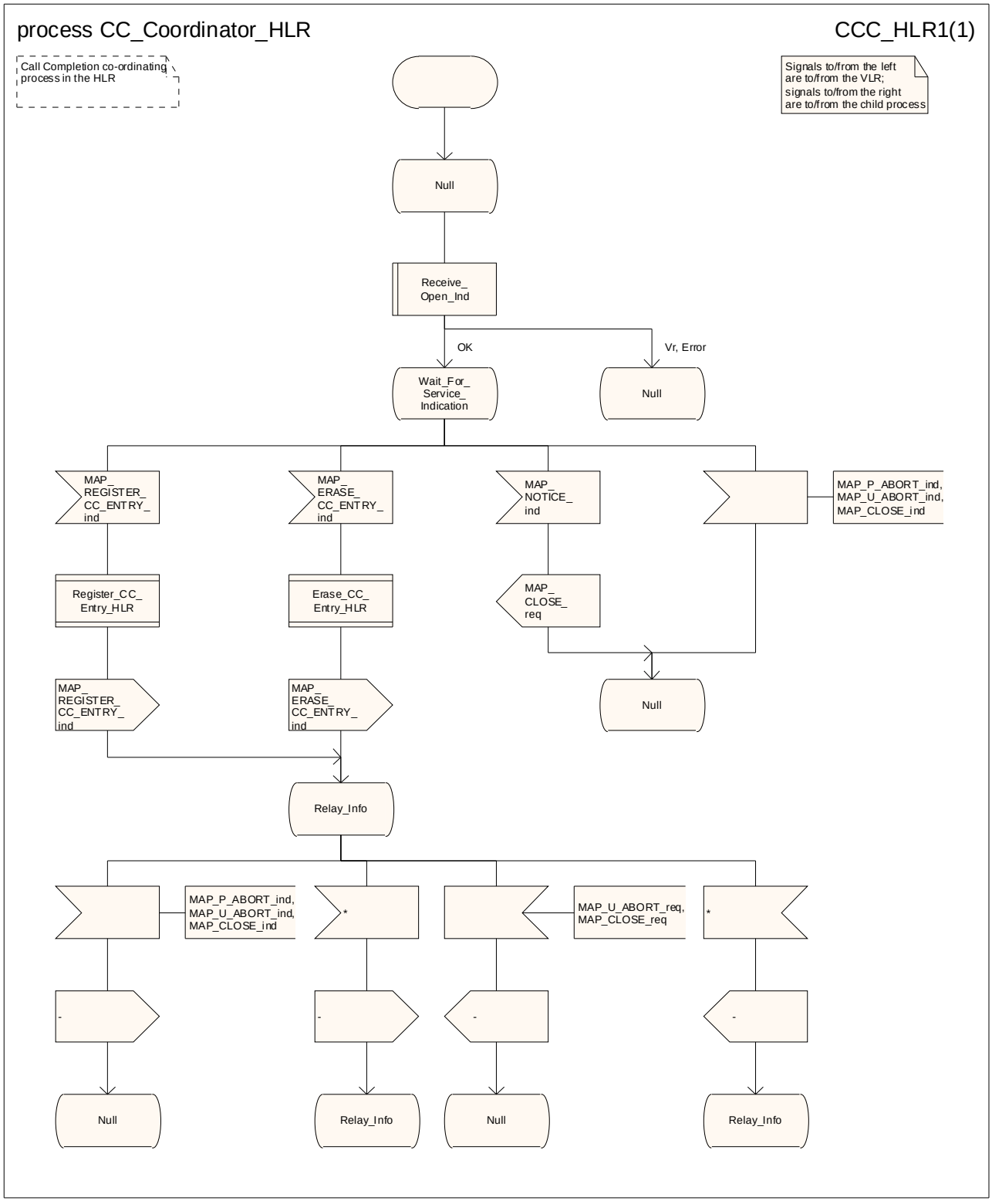


Figure 22.1/4: Process CC_Coordinator_HLR

22.2 Registration procedure

22.2.1 General

The registration procedure is used to register data related to a supplementary service in the HLR. The registration procedure is a fully transparent communication between the MS and the HLR, except that some services may be invoked as a result of the procedure, as described in the clauses below.

The registration procedure is shown in figure 22.2.1/1.

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

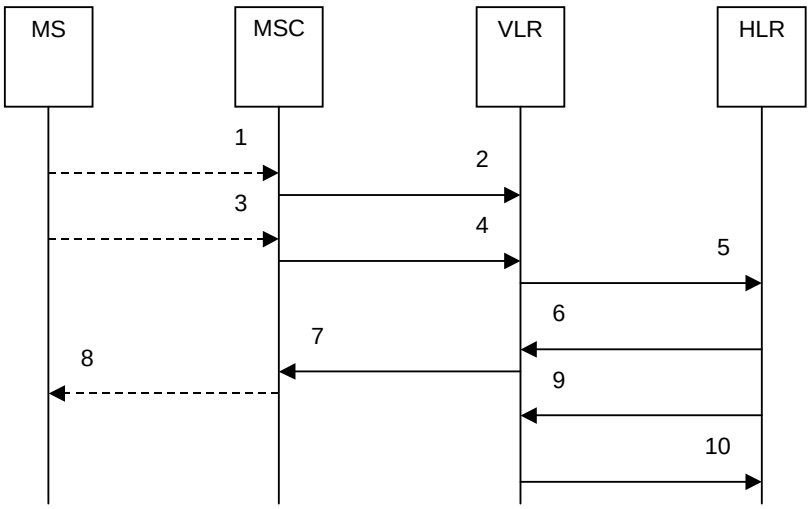
MAP_CHECK_IMEI (see clauses 8 and 25);

MAP_READY_FOR_SM (see clauses 12 and 25);

MAP_INSERT_SUBSCRIBER_DATA (see clauses 8 and 25);

The following service is certainly used:

MAP_REGISTER_SS (defined in clause 11).



1) A_CM_SERV_REQ (Note 1)
2) MAP_PROCESS_ACCESS_REQUEST (Note 2)
3) A_REGISTER_SS (Note 1)
4) MAP_REGISTER_SS_req/ind
5) MAP_REGISTER_SS_req/ind
6) MAP_REGISTER_SS_rsp/cnf
7) MAP_REGISTER_SS_rsp/cnf
8) A_REGISTER_SS ack (Note 1)
9) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 3)
10) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 3)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35], 3GPP TS 24.010 [36], 3GPP TS 24.08x and 3GPP TS 24.09x. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For details of the Process Access Request procedure, refer to subclause 25.4 in the present document.
NOTE 3: Services printed in *italics* are optional.

Figure 22.2.1/1: Message flow for supplementary service registration

22.2.2 Procedure in the MSC

The A_REGISTER_SS service indication received by the MAP process in the MSC contains the SS-Code and any parameters that are related to the supplementary service.

The MAP user transfers the received information to the VLR in the MAP_REGISTER_SS request without checking the contents of the service indication. Rules for the mapping are described in 3GPP TS 29.011 [59].

The information in the MAP_REGISTER_SS confirm from the VLR is relayed to the MS in the A_REGISTER_SS response message as described in 3GPP TS 24.08x, 3GPP TS 24.08x and 3GPP TS 29.011.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

The registration process in the MSC is shown in figure 22.2.2/1.

22.2.3 Procedure in the VLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Process_Access_Request_VLR see subclause 25.4.2.

The MAP process in the VLR transfers the information received in the MAP_REGISTER_SS indication to the HLR in the MAP_REGISTER_SS request without checking the contents. The MAP_OPEN request includes the IMSI of the subscriber as the destination reference and the VLR number as the originating reference.

If the MAP_REGISTER_SS confirm is properly formed and contains a result or a user error, the MAP process in the VLR shall transfer the information contained in this primitive to the MSC in the MAP_REGISTER_SS response.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

The registration process in the VLR is shown in figure 22.2.3/1.

22.2.4 Procedure in the HLR

The MAP process invokes a macro and a process not defined in this clause; the definitions of the macro and process can be found as follows:

Check_Indication see subclause 25.2.1;

Insert_Subs_Data_Stand_Alone_HLR see subclause 25.7.3.

The supplementary service request shall be processed according to 3GPP TS 23.011 [22] and the 23.08x and 23.09x-series of technical specifications. This handling may lead to a successful result, a partially successful result or an error being returned.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]):

The registration process in the HLR is shown in figure 22.2.4/1.

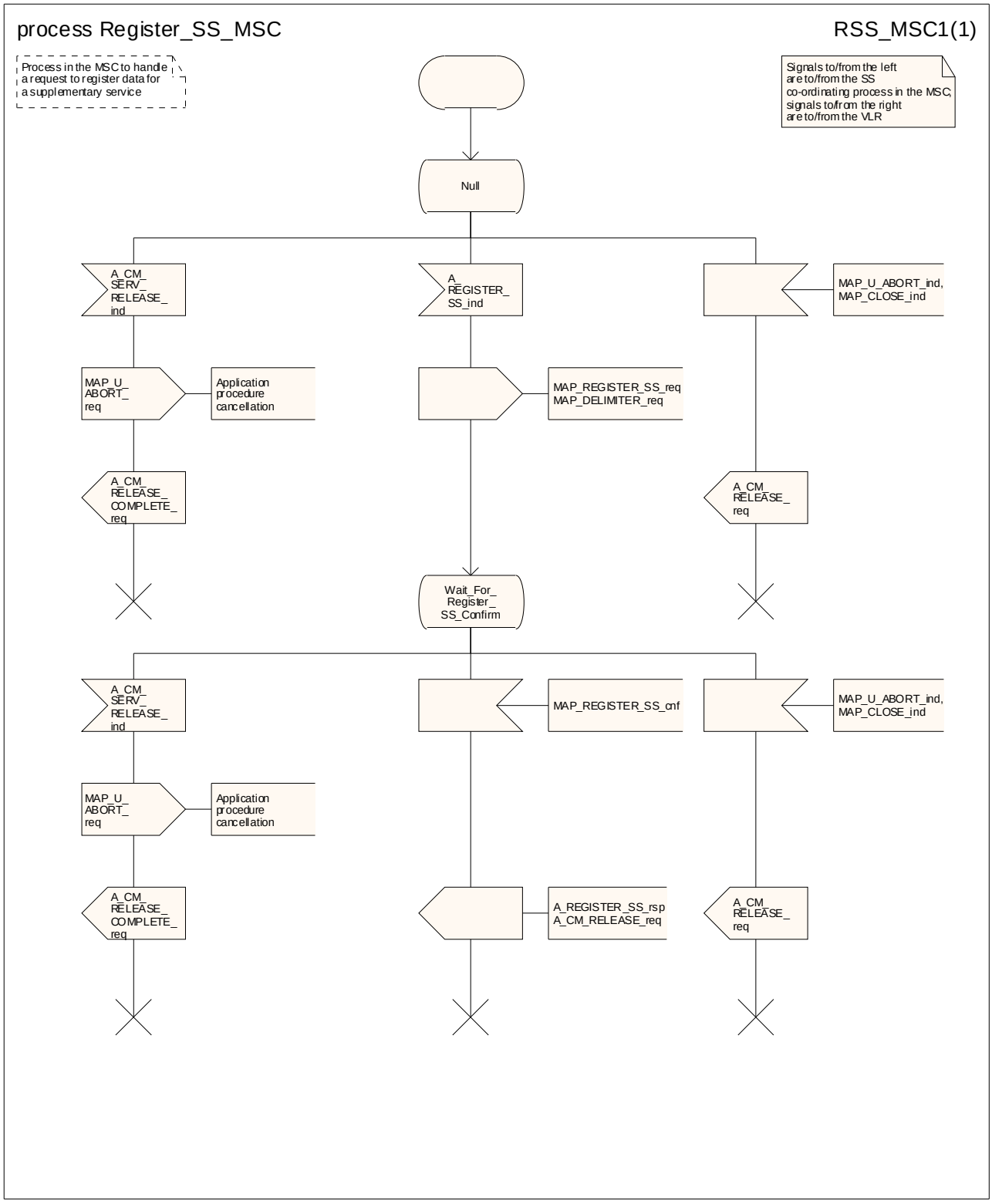


Figure 22.2.2/1: Process Register_SS_MSC

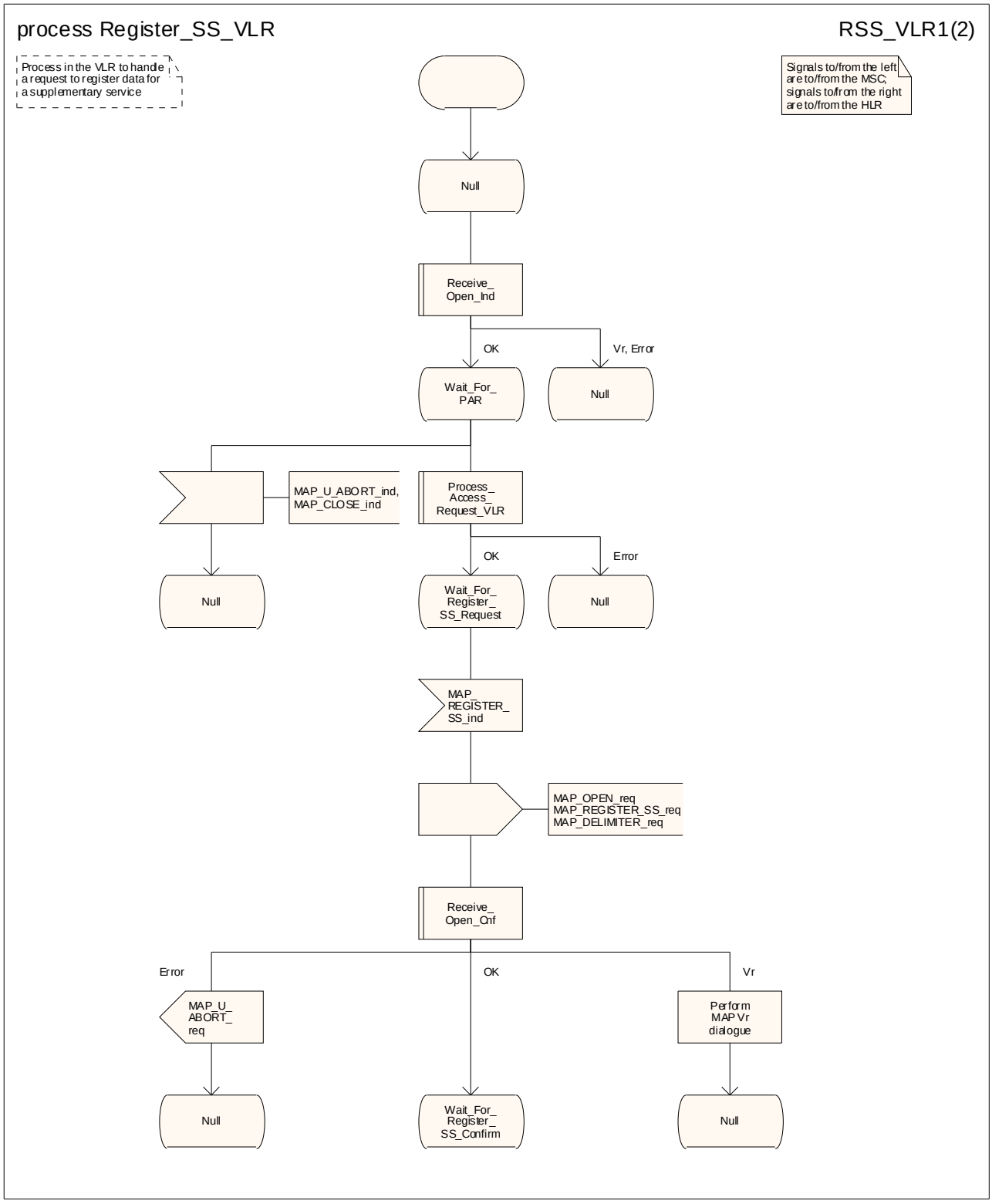


Figure 22.2.3/1 (sheet 1 of 2): Process Register_SS_VLR

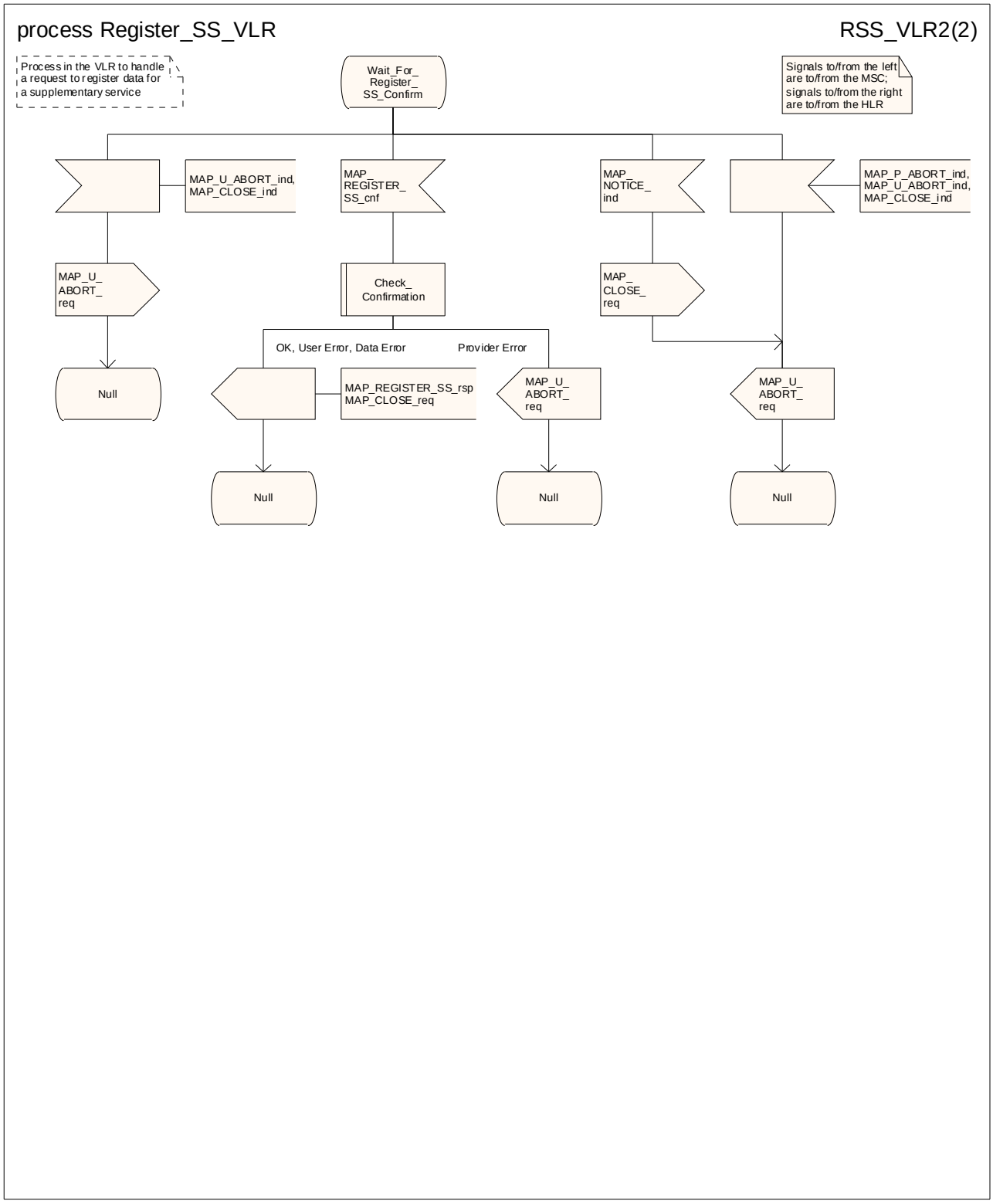


Figure 22.2.3/1 (sheet 2 of 2): Process Register_SS_VLR

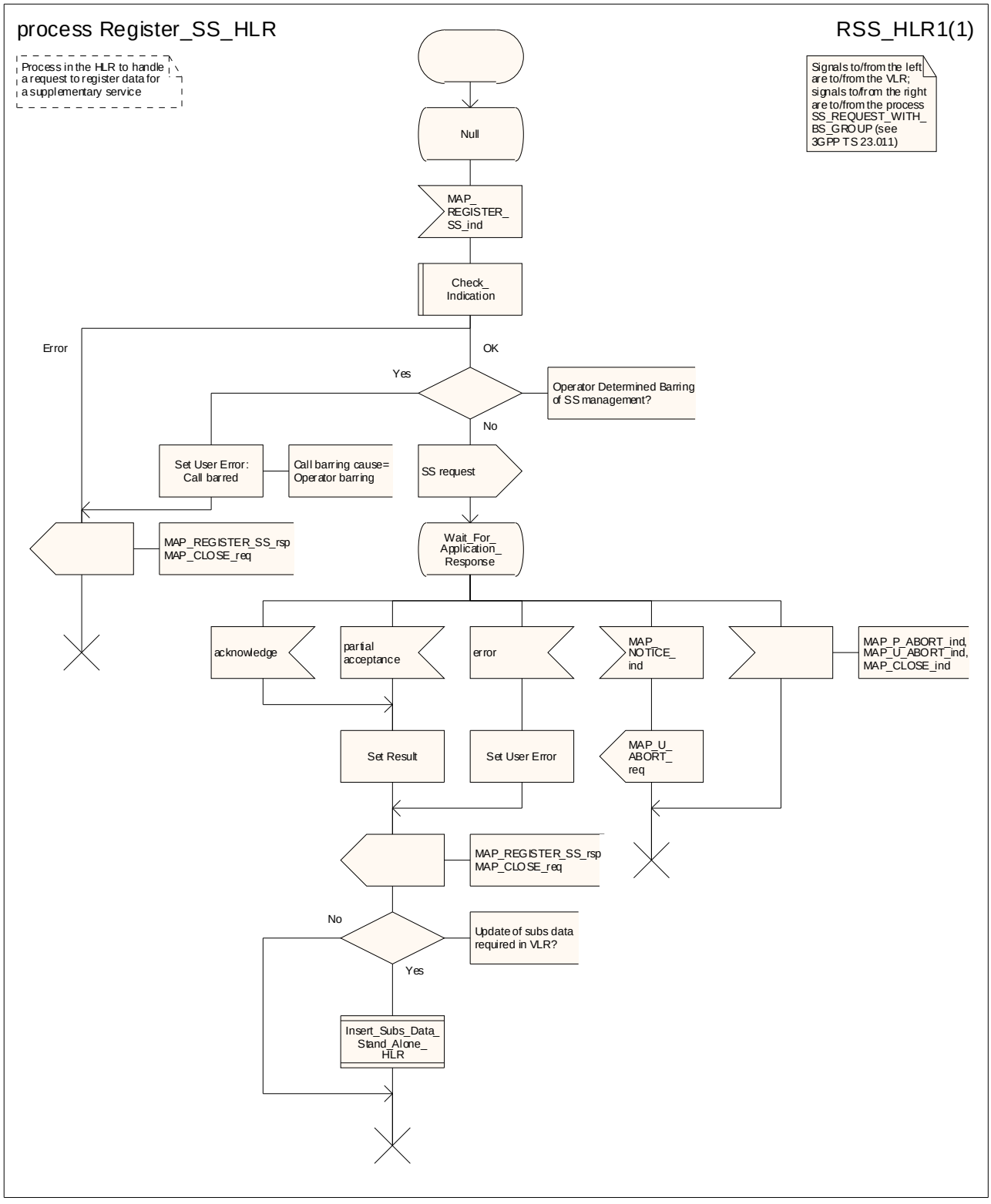


Figure 22.2.4/1: Process Register_SS_HLR

22.3 Erasure procedure

22.3.1 General

The erasure procedure is used to erase data related to a supplementary service in the HLR. The erasure procedure is a fully transparent communication between the MS and the HLR, except that some services may be invoked as a result of the procedure, as described in the clauses below.

The erasure procedure is shown in figure 22.3.1/1.

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

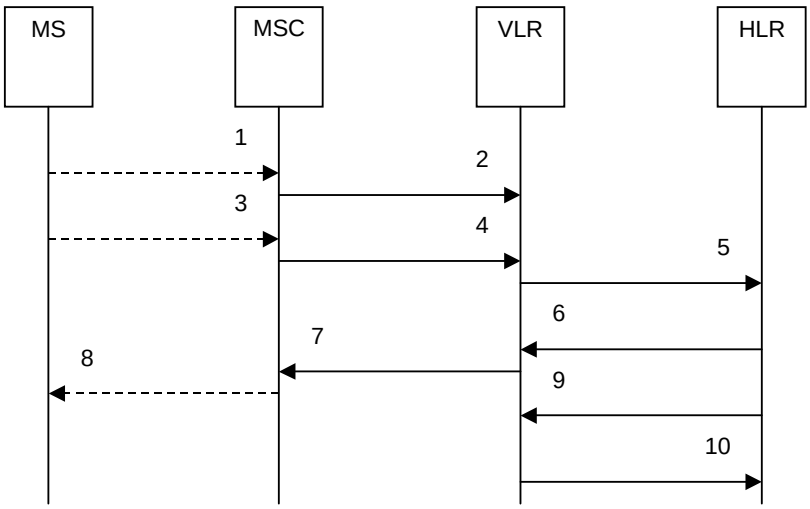
MAP_CHECK_IMEI (see clauses 8 and 25);

MAP_READY_FOR_SM (see clauses 12 and 25);

MAP_INSERT_SUBSCRIBER_DATA (see clauses 8 and 25);

The following service is certainly used:

MAP_ERASE_SS (defined in clause 11).



1) A_CM_SERV_REQ (Note 1)
2) MAP_PROCESS_ACCESS_REQUEST (Note 2)
3) A_ERASE_SS (Note 1)
4) MAP_ERASE_SS_req/ind
5) MAP_ERASE_SS_req/ind
6) MAP_ERASE_SS_rsp/cnf
7) MAP_ERASE_SS_rsp/cnf
8) A_ERASE_SS ack (Note 1)
9) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 3)
10) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 3)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35], 3GPP TS 24.010 [36], 3GPP TS 24.08x and 3GPP TS 24.09x. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For details of the Process Access Request procedure, refer to subclause 25.4 in the present document.
NOTE 3: Services printed in *italics* are optional.

Figure 22.3.1/1: Message flow for supplementary service erasure

22.3.2 Procedure in the MSC

The MSC procedure for erasure is identical to that specified for registration in subclause 22.2.2. The text and diagrams in subclause 22.2.2 apply with all references to registration changed to erasure.

22.3.3 Procedure in the VLR

The VLR procedure for erasure is identical to that specified for registration in subclause 22.2.3. The text and diagrams in subclause 22.2.3 apply with all references to registration changed to erasure.

22.3.4 Procedure in the HLR

The HLR procedure for erasure is identical to that specified for registration in subclause 22.2.4. The text and diagrams in subclause 22.2.4 apply with all references to registration changed to erasure.

22.4 Activation procedure

22.4.1 General

The activation procedure is used to activate a supplementary service in the HLR. The activation procedure is a fully transparent communication between the MS and the HLR, except that some services may be invoked as a result of the procedure, as described in the clauses below.

The activation procedure is shown in figure 22.4.1/1.

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

MAP_CHECK_IMEI (see clauses 8 and 25);

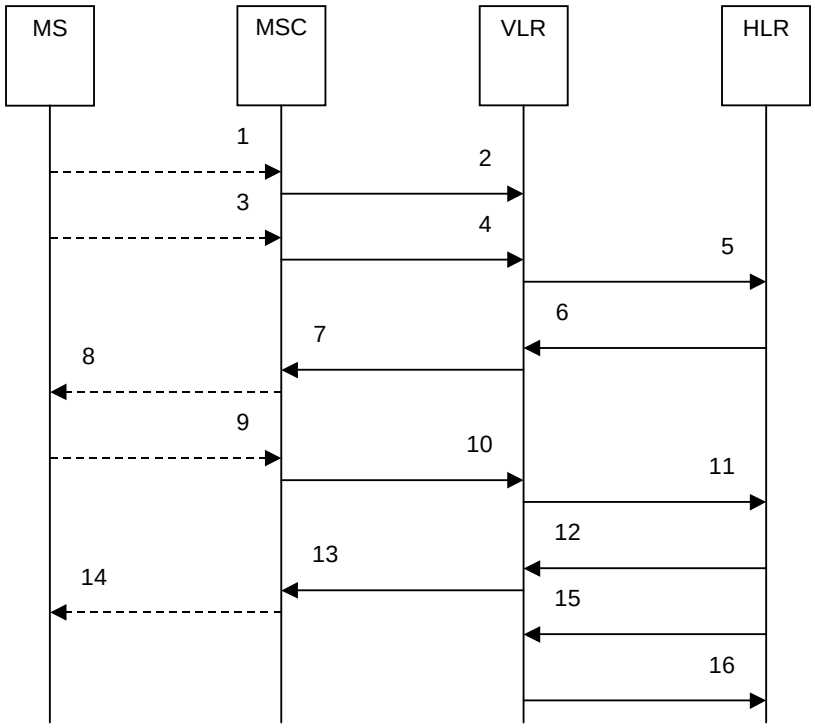
MAP_READY_FOR_SM (see clauses 12 and 25);

MAP_GET_PASSWORD (defined in clause 11);

MAP_INSERT_SUBSCRIBER_DATA (see clauses 8 and 25);

The following service is certainly used:

MAP_ACTIVATE_SS (defined in clause 11).



1) A_CM_SERV_REQ (Note 1)
2) MAP_PROCESS_ACCESS_REQUEST (Note 2)
3) A_ACTIVATE_SS (Note 1)
4) MAP_ACTIVATE_SS_req/ind
5) MAP_ACTIVATE_SS_req/ind
6) *MAP_GET_PASSWORD_req/ind* (Note 3)
7) *MAP_GET_PASSWORD_req/ind* (Note 3)
8) *A_GET_PASSWORD* (Note 1, Note 3)
9) *A_GET_PASSWORD ack* (Note 1, Note 3)
10) *MAP_GET_PASSWORD_rsp/cnf* (Note 3)
11) *MAP_GET_PASSWORD_rsp/cnf* (Note 3)
12) MAP_ACTIVATE_SS_rsp/cnf
13) MAP_ACTIVATE_SS_rsp/cnf
14) A_ACTIVATE_SS ack (Note 1)
15) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 3)
16) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 3)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35], 3GPP TS 24.010 [36], 3GPP TS 24.08x and 3GPP TS 24.09x. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For details of the Process Access Request procedure, refer to subclause 25.4 of this document.
NOTE 3: Services printed in *italics* are optional.

Figure 22.4.1/1: Message flow for supplementary service activation

22.4.2 Procedure in the MSC

The A_ACTIVATE_SS service indication received by the MAP user in the MSC contains the SS-Code and any parameters related to the supplementary service.

The MSC transfers the received information to the VLR in the MAP_ACTIVATE_SS request without checking the contents of the service indication. Rules for the mapping are described in 3GPP TS 29.011 [59].

The information in the MAP_ACTIVATE_SS confirm from the VLR is relayed to the MS in the A_ACTIVATE_SS response message, as described in TS 24.08x, 3GPP TS 24.08x and 3GPP TS 29.011.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

The activation process in the MSC is shown in figure 22.4.2/1.

22.4.3 Procedure in the VLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Process_Access_Request_VLR see subclause 25.4.2.

The MAP process in the VLR transfers the information received in the MAP_ACTIVATE_SS indication to the HLR in the MAP_ACTIVATE_SS request without checking the contents. The MAP_OPEN request includes the IMSI of the subscriber as the destination reference and the VLR number as the originating reference.

If the MAP_REGISTER_SS confirm is properly formed and contains a result or a user error, the MAP process in the VLR shall transfer the information contained in this primitive to the MSC in the MAP_ACTIVATE_SS response.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

The activation process in the VLR is shown in figure 22.4.3/1.

22.4.4 Procedure in the HLR

The MAP process invokes a macro and a process not defined in this clause; the definitions of the macro and process can be found as follows:

Check_Indication see subclause 25.2.1;

Insert_Subs_Data_Stand_Alone_HLR see subclause 25.7.3.

The supplementary service request shall be processed according to 3GPP TS 23.011 [22] and the 23.08x and 23.09x-series of technical specifications. This handling may lead to a successful result, a partially successful result or an error being returned.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]):

The activation process in the HLR is shown in figure 22.4.4/1.

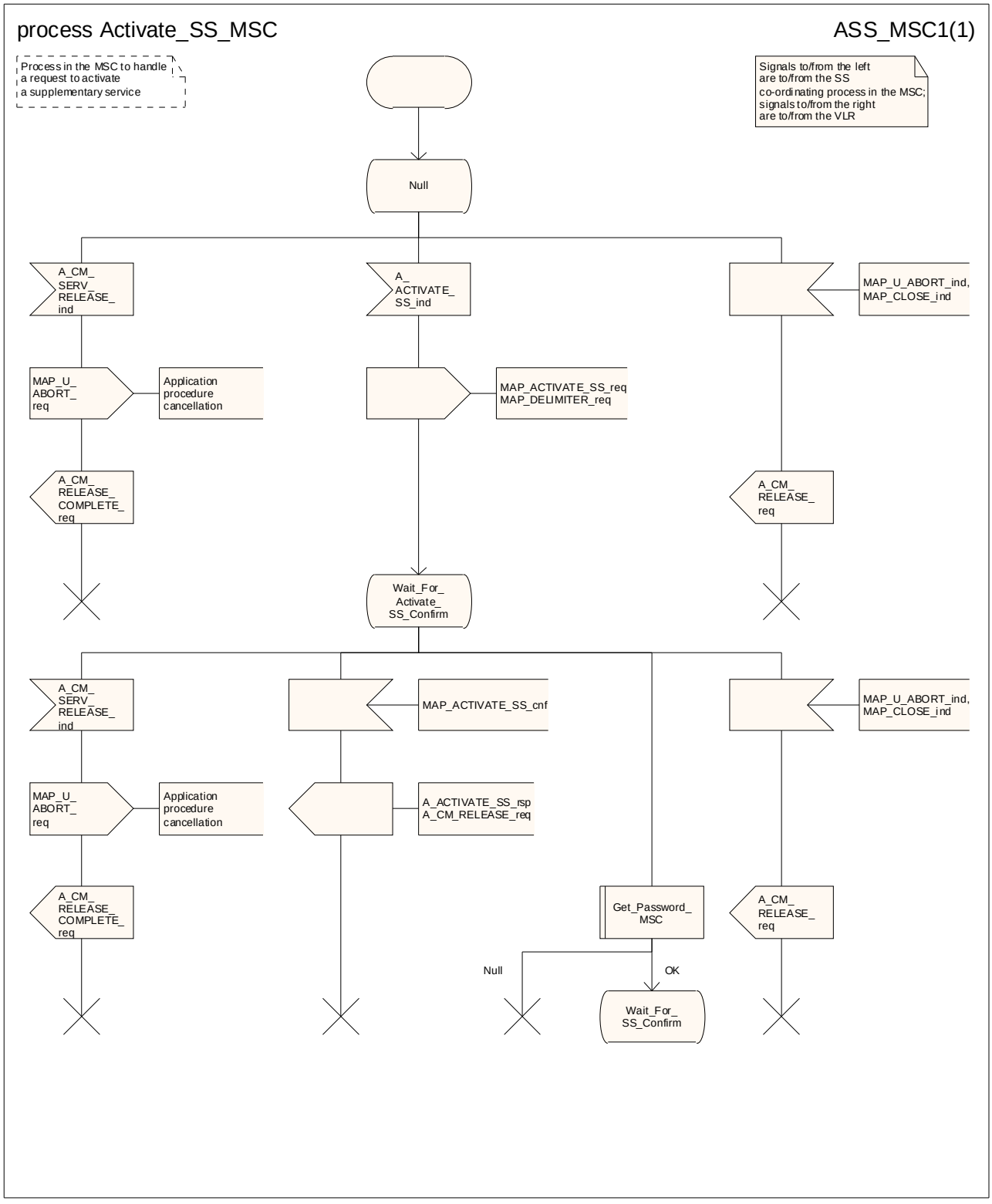


Figure 22.4.2/1: Process Activate_SS_MSC

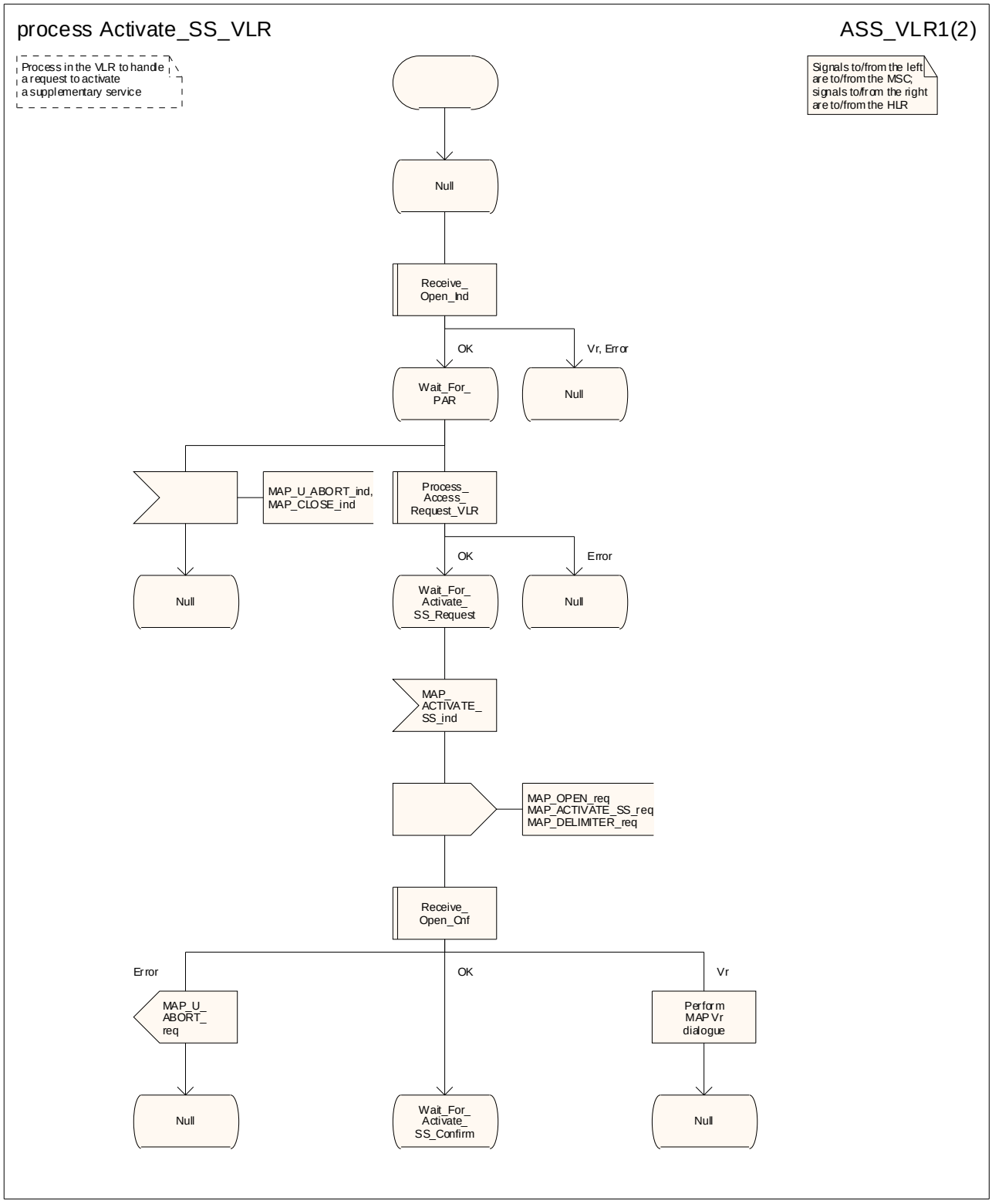


Figure 22.4.3/1 (sheet 1 of 2): Process Activate_SS_VLR

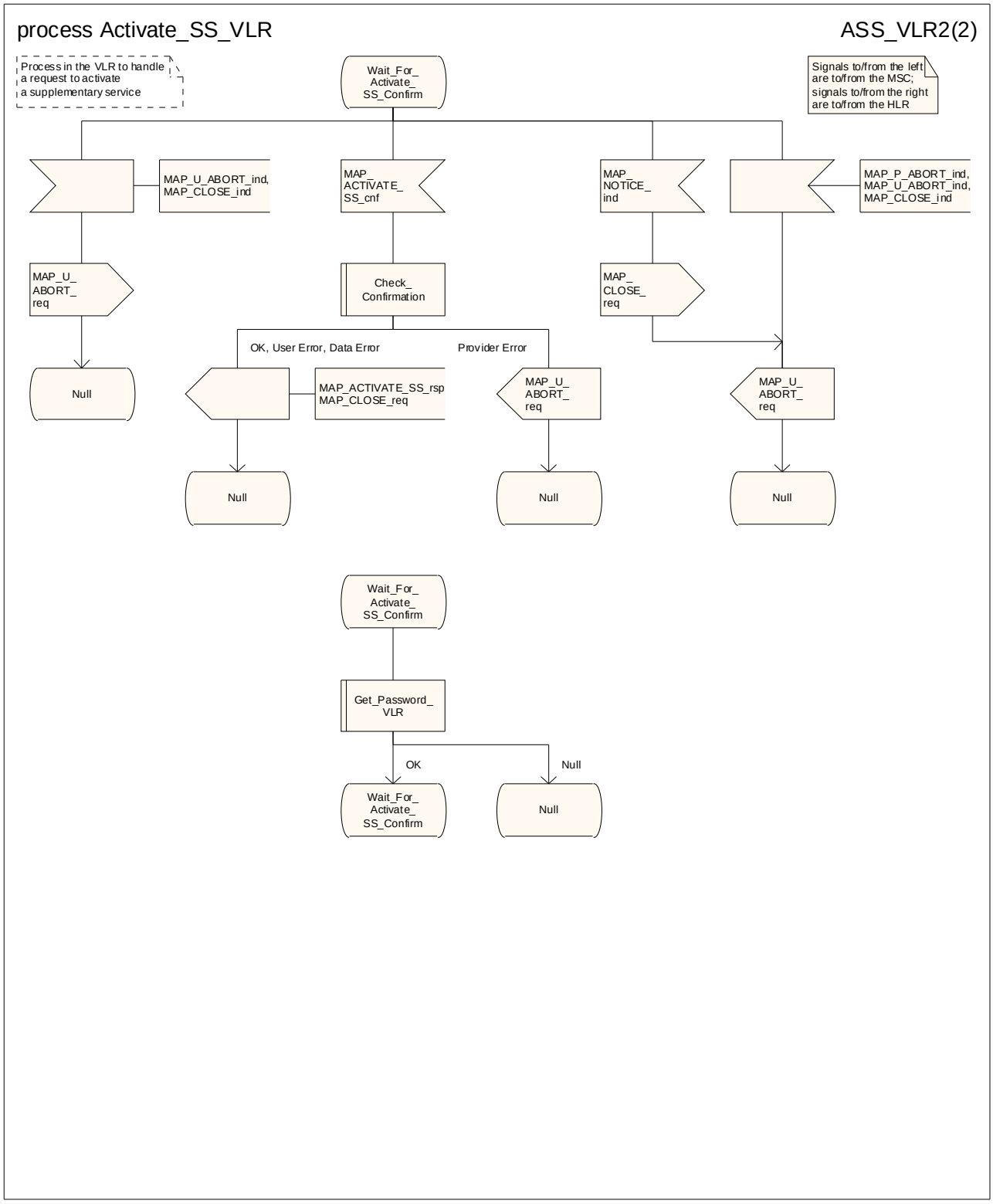


Figure 22.4.3/1 (sheet 2 of 2): Process Activate_SS_VLR

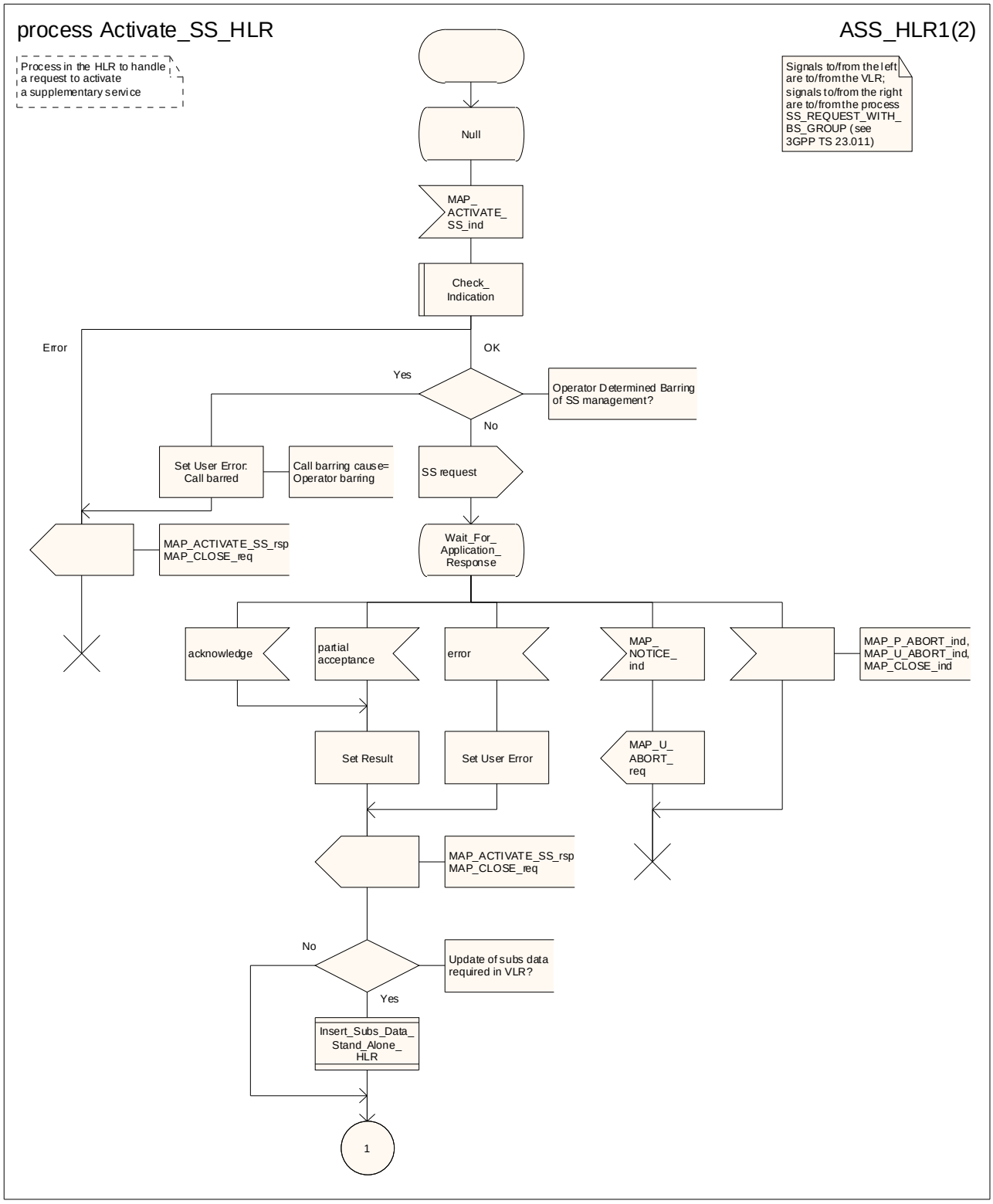


Figure 22.4.4/1 (sheet 1 of 2): Process Activate_SS_HLR

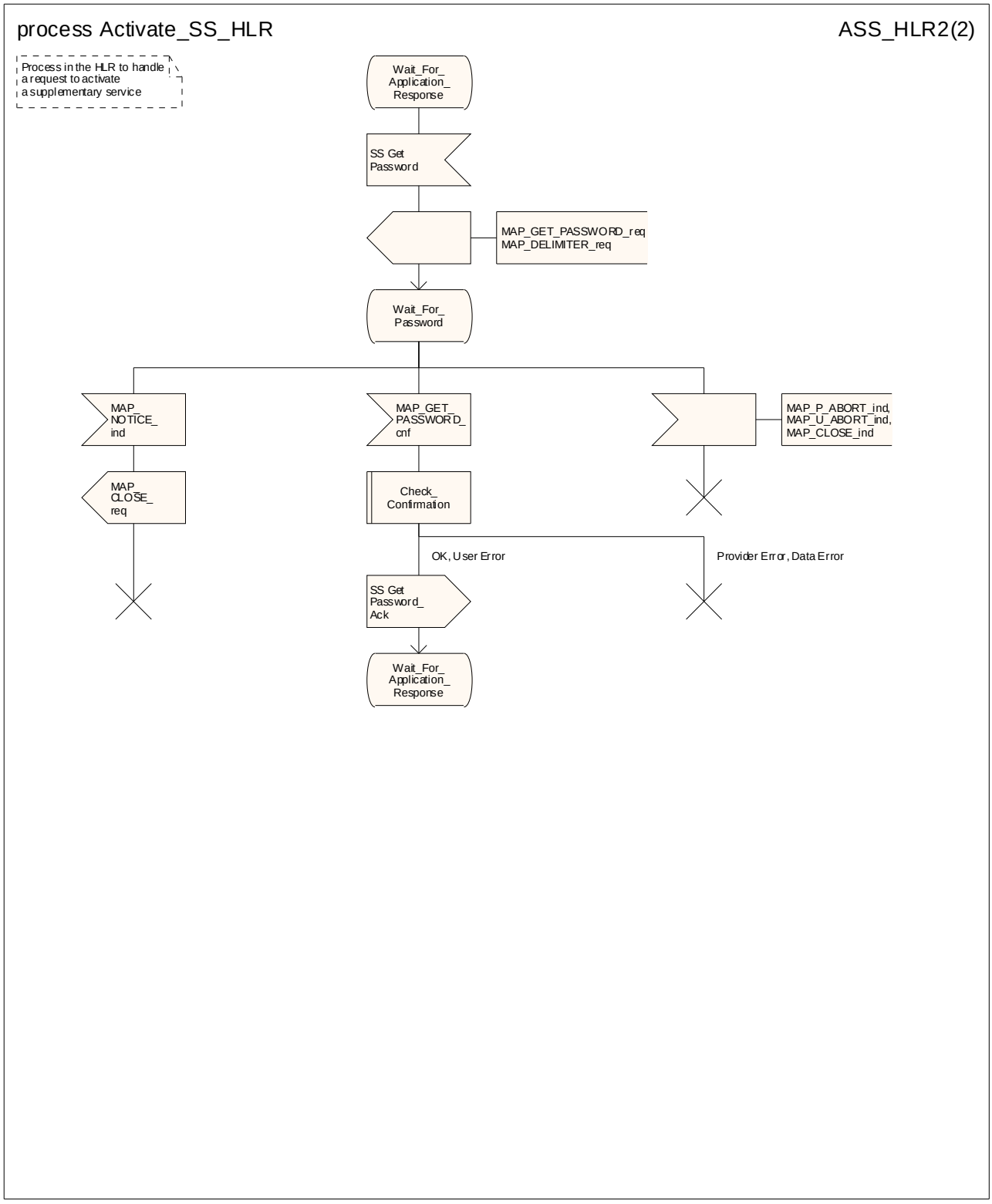


Figure 22.4.4/1 (sheet 2 of 2): Process Activate_SS_HLR

22.5 Deactivation procedure

22.5.1 General

The deactivation procedure is used to deactivate a supplementary service in the HLR. The deactivation procedure is a fully transparent communication between the MS and the HLR, except that some services may be invoked as a result of the procedure, as described in the clauses below.

The deactivation procedure is shown in figure 22.5.1/1.

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

MAP_CHECK_IMEI (see clauses 8 and 25);

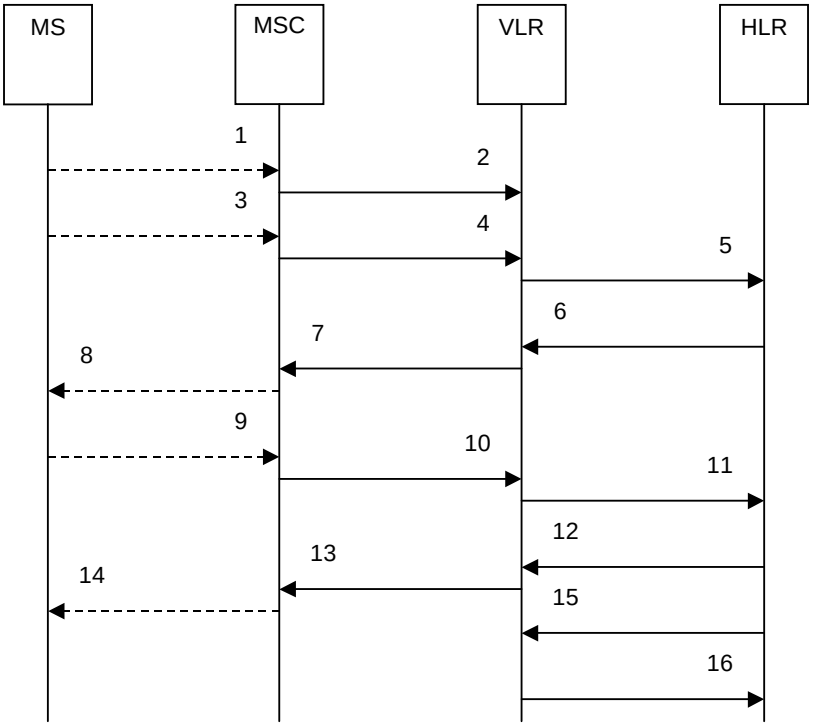
MAP_READY_FOR_SM (see clauses 12 and 25);

MAP_GET_PASSWORD (defined in clause 11);

MAP_INSERT_SUBSCRIBER_DATA (see clauses 8 and 25);

The following service is certainly used:

MAP_DEACTIVATE_SS (defined in clause 11).



1) A_CM_SERV_REQ (Note 1)
2) MAP_PROCESS_ACCESS_REQUEST (Note 2)

3) A_DEACTIVATE_SS (Note 1)
4) MAP_DEACTIVATE_SS_req/ind
5) MAP_DEACTIVATE_SS_req/ind
6) *MAP_GET_PASSWORD_req/ind* (Note 3)
7) *MAP_GET_PASSWORD_req/ind* (Note 3)
8) *A_GET_PASSWORD* (Note 1, Note 3)
9) *A_GET_PASSWORD ack* (Note 1, Note 3)
10) *MAP_GET_PASSWORD_rsp/cnf* (Note 3)
11) *MAP_GET_PASSWORD_rsp/cnf* (Note 3)
12) MAP_DEACTIVATE_SS_rsp/cnf
13) MAP_DEACTIVATE_SS_rsp/cnf
14) A_DEACTIVATE_SS ack (Note 1)
15) *MAP_INSERT_SUBSCRIBER_DATA_req/ind* (Note 3)
16) *MAP_INSERT_SUBSCRIBER_DATA_rsp/cnf* (Note 3)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35], 3GPP TS 24.010 [36], 3GPP TS 24.08x and 3GPP TS 24.09x. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For details of the Process Access Request procedure, refer to subclause 25.4 in the present document.
NOTE 3: Services printed in *italics* are optional.

Figure 22.5.1/1: Message flow for supplementary service deactivation

22.5.2 Procedure in the MSC

The MSC procedure for deactivation is identical to that specified for activation in subclause 22.4.2. The text and diagrams in subclause 22.4.2 apply with all references to activation changed to deactivation.

22.5.3 Procedures in the VLR

The VLR procedure for deactivation is identical to that specified for activation in subclause 22.4.3. The text and diagrams in subclause 22.4.3 apply with all references to activation changed to deactivation.

22.5.4 Procedures in the HLR

The HLR procedure for deactivation is identical to that specified for activation in subclause 22.4.4. The text and diagrams in subclause 22.4.4 apply with all references to activation changed to deactivation.

22.6 Interrogation procedure

22.6.1 General

The interrogation procedure is used to retrieve information related to a supplementary service from the VLR or the HLR. It is the VLR which decides whether an interrogation request should be forwarded to the HLR or not. Some non-supplementary service related services may be invoked as a result of the procedure, as described in the clauses below.

The interrogation procedure is shown in figure 22.6.1/1.

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

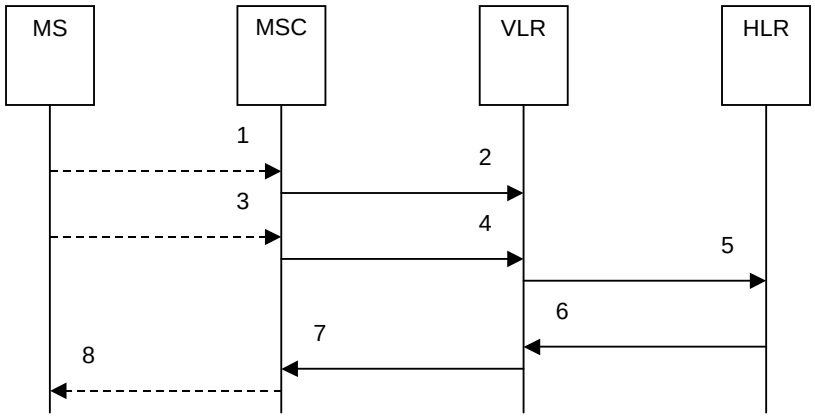
MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

MAP_CHECK_IMEI (see clauses 8 and 25);

MAP_READY_FOR_SM (see clauses 12 and 25);

The following service is certainly used:

MAP_INTERROGATE_SS (defined in clause 11).



1) A_CM_SERV_REQ (Note 1)
2) MAP_PROCESS_ACCESS_REQUEST (Note 2)
3) A_INTERROGATE_SS (Note 1)
4) MAP_INTERROGATE_SS_req/ind
5) *MAP_INTERROGATE_SS_req/ind*
6) *MAP_INTERROGATE_SS_rsp/cnf*
7) MAP_INTERROGATE_SS_rsp/cnf
8) A_INTERROGATE_SS ack (Note 1)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35], 3GPP TS 24.010 [36], 3GPP TS 24.08x and 3GPP TS 24.09x. Services shown in dotted lines indicate the trigger provided by the signalling on the radio path, and the signalling triggered on the radio path.
NOTE 2: For details of the Process Access Request procedure, refer to subclause 25.4 in the present document.
NOTE 3: Services printed in *italics* are optional.

Figure 22.6.1/1: Message flow for supplementary service interrogation

22.6.2 Procedure in the MSC

The MSC procedures for interrogation are identical to those specified for registration in subclause 22.2.2. The text and diagrams in subclause 22.2.2 apply with all references to registration changed to interrogation.

22.6.3 Procedures in the VLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Process_Access_Request_VLR see subclause 25.4.2.

The interrogation is answered either by the VLR or by the HLR, depending on the service interrogated.

1) Interrogation to be handled by the VLR

The supplementary service request shall be processed according to 3GPP TS 23.011 [22] and the 23.08x and 23.09x-series of technical specifications. This handling may lead to a successful result, a partially successful result or an error being returned.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

2) Interrogation to be handled by the HLR

If the interrogation is to be handled by the HLR, the MAP process in the VLR transfers the information received in the MAP_INTERROGATE_SS indication to the HLR in the MAP_INTERROGATE_SS request without checking the contents of the service indication. The MAP_OPEN request includes the IMSI of the subscriber as the destination reference and the VLR number as the originating reference.

If the MAP_INTERROGATE_SS confirm is properly formed and contains a result or a user error, the MAP process in the VLR shall transfer the information contained in this primitive to the MSC in the MAP_INTERROGATE_SS response.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

The Interrogation process in the VLR is shown in figure 22.6.3/1.

22.6.4 Procedure in the HLR

The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

The HLR acts as follows:

The interrogation is answered either by the VLR or by the HLR, depending on the service interrogated.

1) Interrogation to be handled by the VLR

If the interrogation procedure should have been answered by the VLR, then the HLR assumes that the VLR does not support the interrogated supplementary service, and returns the SS Not Available error to the VLR.

2) Interrogation to be handled by HLR

The supplementary service request shall be processed according to 3GPP TS 23.011 [22] and the 23.08x and 23.09x-series of technical specifications. This handling may lead to either a successful result or an error being returned.

For call independent SS operations, each message shall contain only a single component. Messages which contain more than one component will be stopped at the air interface (as specified in 3GPP TS 29.011 [59]).

The Interrogation process in the HLR is shown in figure 22.6.4/1.

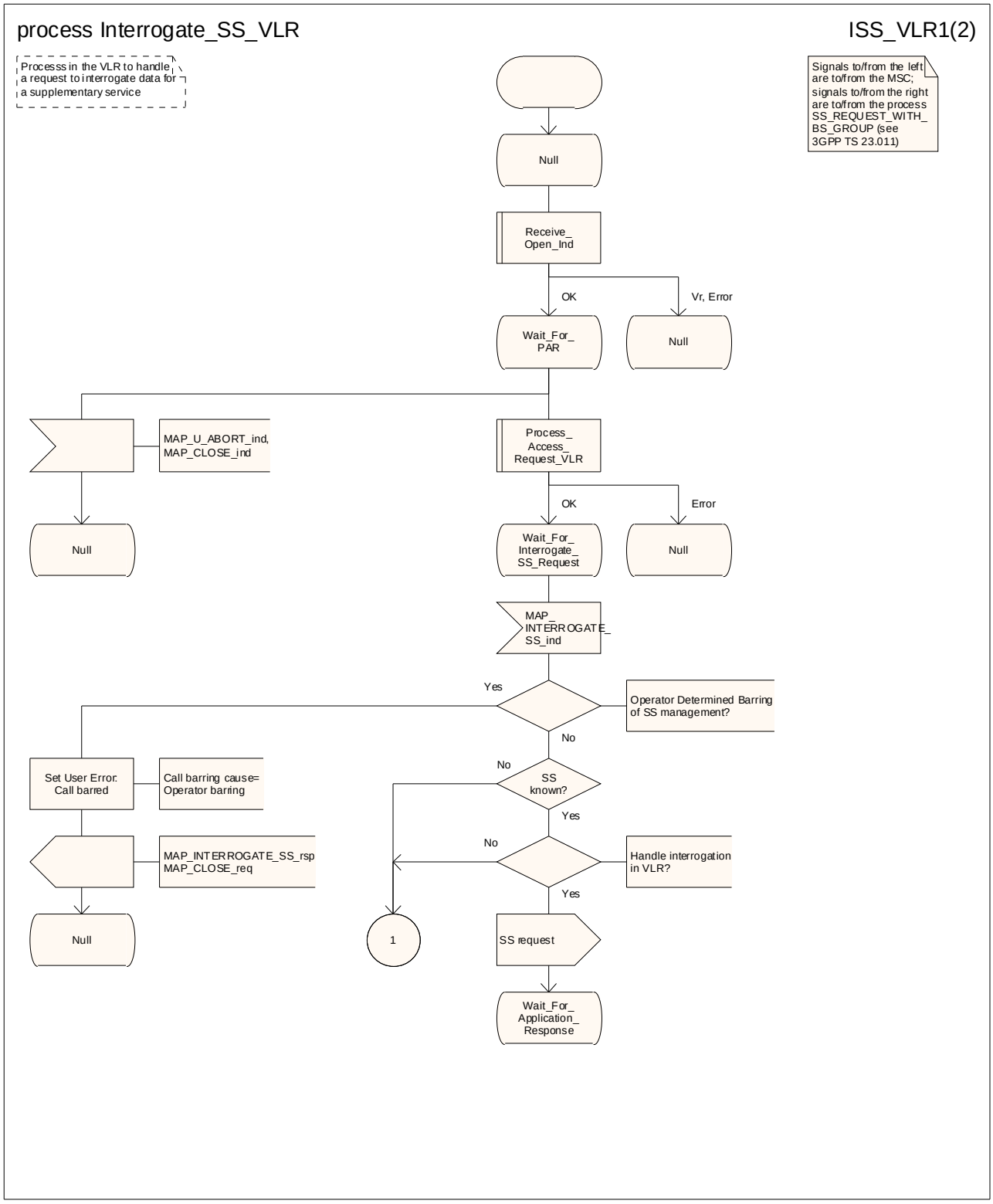


Figure 22.6.3/1 (sheet 1 of 2): Process Interrogate_SS_VLR

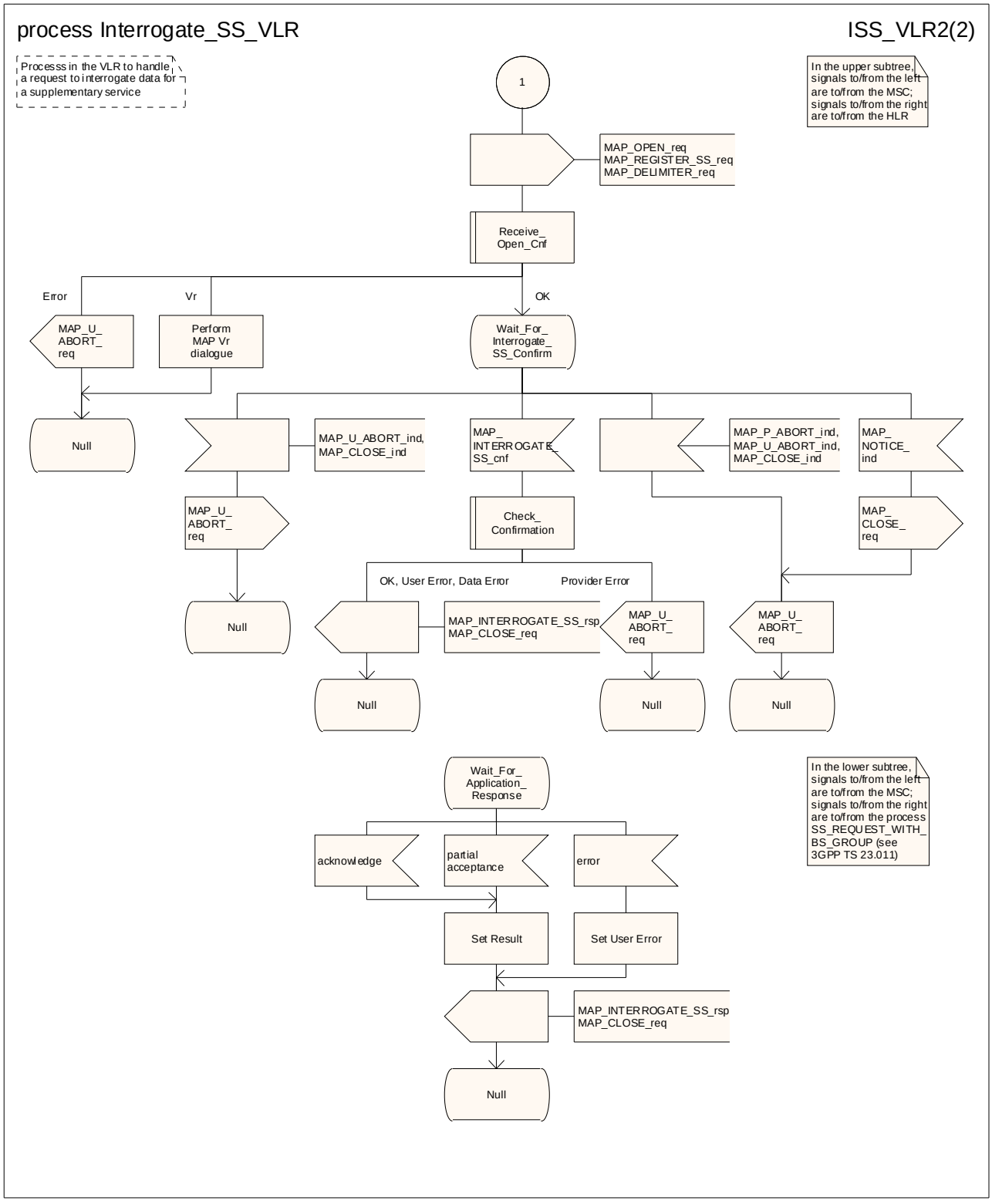


Figure 22.6.3/1 (sheet 2 of 2): Process Interrogate_SS_VLR

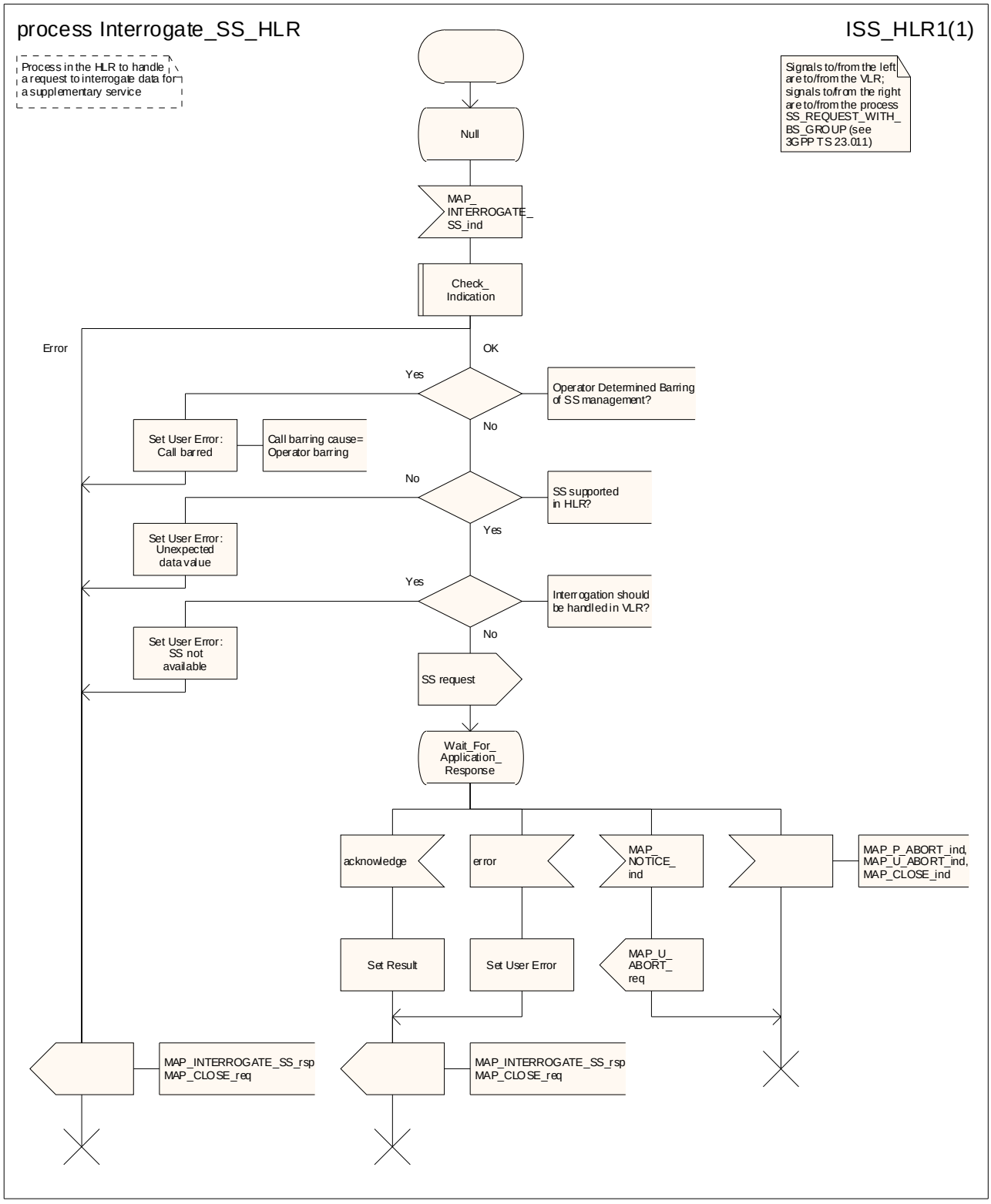


Figure 22.6.4/1: Process Interrogate_SS_HLR

22.7 Void

Figure 22.7.2/1 void

Figure 22.7.3/1 void

22.8 Password registration procedure

22.8.1 General

The password registration procedure is used to register a password in the HLR. The password registration procedure is a fully transparent communication between the MS and the HLR, except that some services may be invoked as a result of the procedure, as described below.

The password registration procedure is shown in figure 22.8.1/1.

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

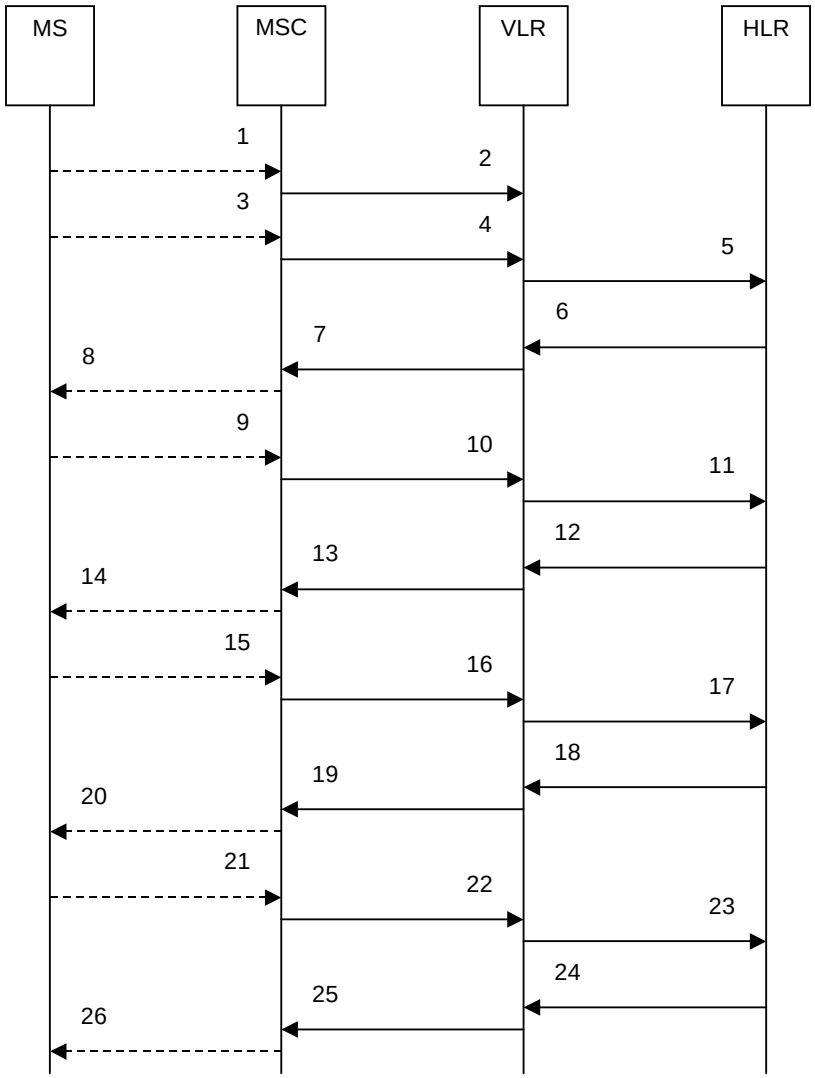
MAP_CHECK_IMEI (see clauses 8 and 25);

MAP_READY_FOR_SM (see clauses 12 and 25);

The following services are certainly used:

MAP_REGISTER_PASSWORD (defined in clause 11);

MAP_GET_PASSWORD (defined in clause 11).



1) A_CM_SERV_REQ (Note 1)
2) MAP_PROCESS_ACCESS_REQUEST (Note 2)
3) A_REGISTER_PASSWORD (Note 1)
4) MAP_REGISTER_PASSWORD_req/ind
5) MAP_REGISTER_PASSWORD_req/ind
6) MAP_GET_PASSWORD_req/ind (Note 3)
7) MAP_GET_PASSWORD_req/ind (Note 3)
8) A_GET_PASSWORD (Note 1, Note 3)
9) A_GET_PASSWORD ack (Note 1, Note 3)
10) MAP_GET_PASSWORD_rsp/cnf (Note 3)
11) MAP_GET_PASSWORD_rsp/cnf (Note 3)
12) MAP_GET_PASSWORD_req/ind (Note 3)
13) MAP_GET_PASSWORD_req/ind (Note 3)
14) A_GET_PASSWORD (Note 1, Note 3)
15) A_GET_PASSWORD ack (Note 1, Note 3)
16) MAP_GET_PASSWORD_rsp/cnf (Note 3)
17) MAP_GET_PASSWORD_rsp/cnf (Note 3)
18) MAP_GET_PASSWORD_req/ind (Note 3)
19) MAP_GET_PASSWORD_req/ind (Note 3)
20) A_GET_PASSWORD (Note 1, Note 3)
21) A_GET_PASSWORD ack (Note 1, Note 3)
22) MAP_GET_PASSWORD_rsp/cnf (Note 3)
23) MAP_GET_PASSWORD_rsp/cnf (Note 3)
24) MAP_REGISTER_PASSWORD_rsp/cnf
25) MAP_REGISTER_PASSWORD_rsp/cnf
26) A_REGISTER_PASSWORD (Note 1)

NOTE 1: For details of the procedure on the radio path, see 3GPP TS 24.008 [35], 3GPP TS 24.010 [36], 3GPP TS 24.08x and 3GPP TS 24.09x. Services shown in dotted lines are triggers/ triggered signalling on the radio path.
NOTE 2: For details of the Process Access Request procedure, refer to subclause 25.4 in the present document.
NOTE 3: The use of each of the three MAP_GET_PASSWORD operations is described in subclause 22.8.4.

Figure 22.8.1/1: Message flow for supplementary service password registration

22.8.2 Procedure in the MSC

The password registration procedure in the MSC is identical to that for activation specified in subclause 22.4.2. All the text and diagrams in subclause 22.4.2 apply with all references to activation changed to password registration.

22.8.3 Procedure in the VLR

The password registration procedure in the VLR is identical to that for activation specified in subclause 22.4.3. All the text and diagrams in subclause 22.4.3 apply with all references to activation changed to password registration.

22.8.4 Procedure in the HLR

The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

The HLR shall process the MAP_REGISTER_PASSWORD indication as specified in 3GPP TS 23.011 [22]. During the handling of password registration, the password procedure is initiated (as specified in 3GPP TS 23.011 [22]) This involves the sending of MAP_GET_PASSWORD requests to the VLR.

The password registration process in the HLR is shown in figure 22.8.4/1.

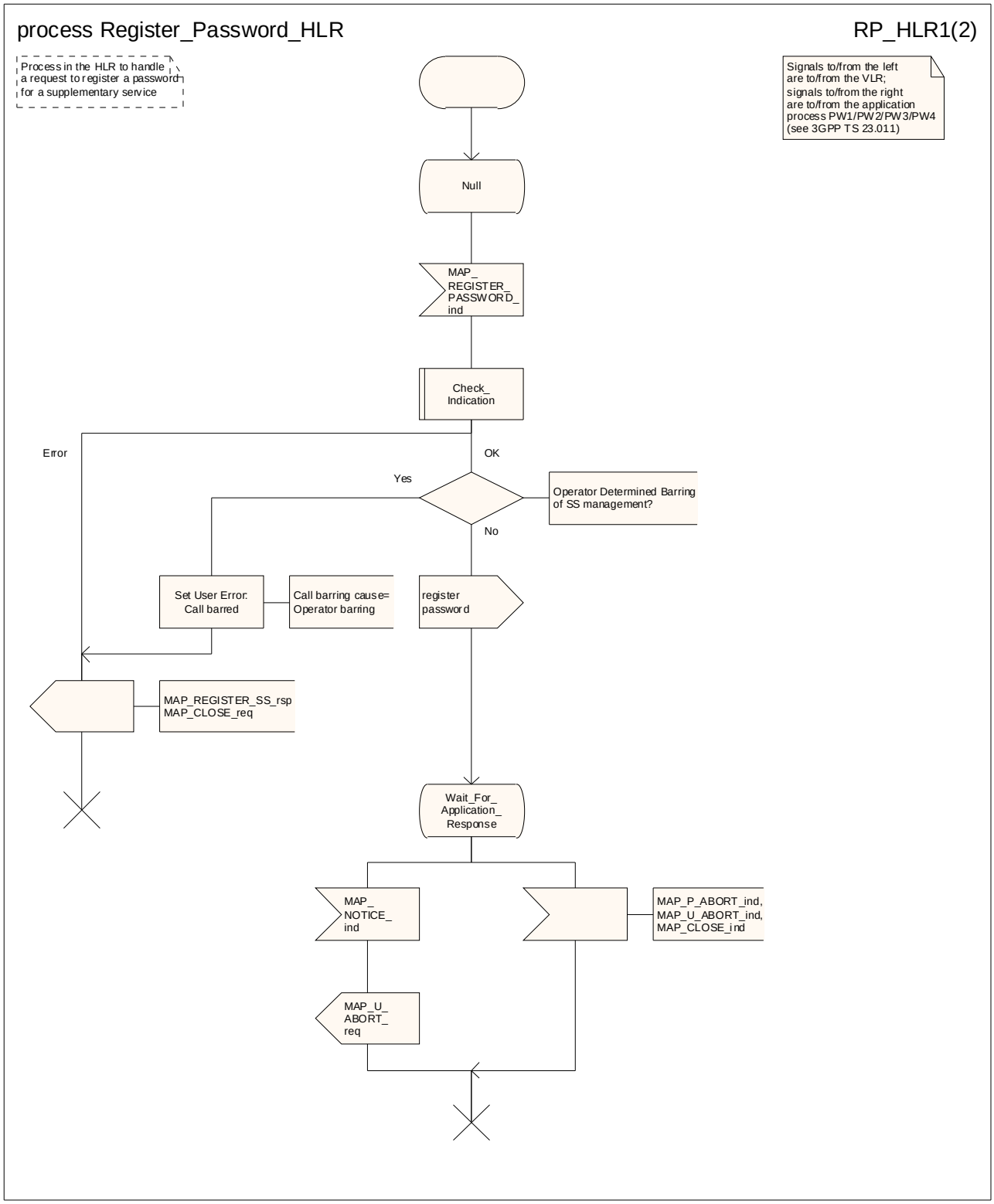


Figure 22.8.4/1 (sheet 1 of 2): Process Register_PW_HLR

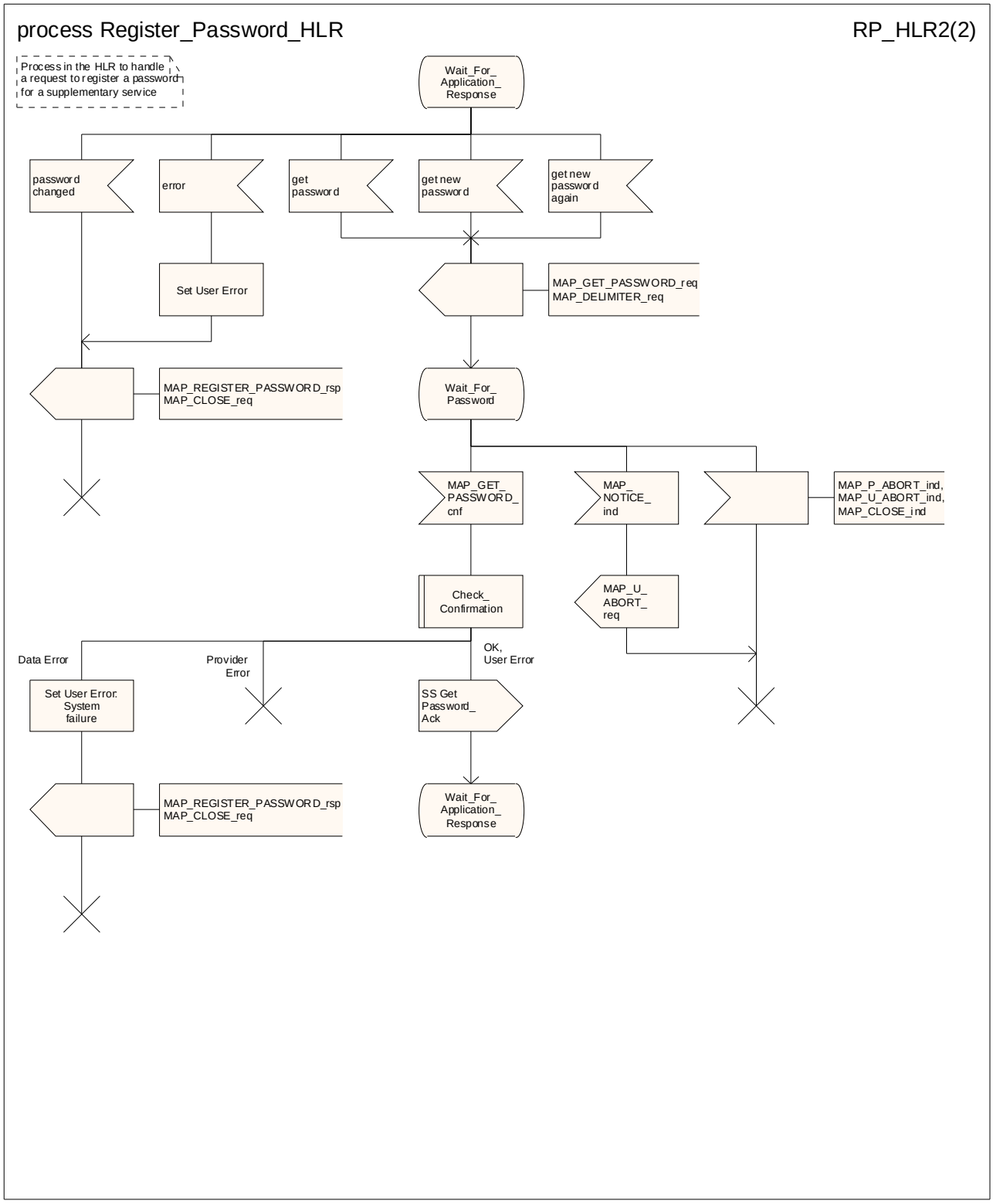


Figure 22.8.4/1 (sheet 2 of 2): Process Register_PW_HLR

22.9 Mobile Initiated USSD procedure

22.9.1 General

The procedure supports supplementary service signalling procedures which allow PLMN specific services to be introduced.

The message flow for the procedure can be found in 3GPP TS 23.090 [34].

The following services may be used:

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_TRACE_SUBSCRIBER_ACTIVITY (see clauses 9 and 25);

MAP_PROVIDE_IMSI (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

MAP_CHECK_IMEI (see clauses 8 and 25);

MAP_READY_FOR_SM (see clauses 12 and 25);

MAP_UNSTRUCTURED_SS_REQUEST (defined in clause 11);

MAP_UNSTRUCTURED_SS_NOTIFY (defined in clause 11).

The following service is certainly used:

MAP_PROCESS_UNSTRUCTURED_SS_REQUEST (defined in clause 11).

22.9.2 Procedure in the MSC

The process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Confirmation see subclause 25.2.2.

The A_PROCESS_UNSTRUCTURED_SS_REQUEST from the MS contains information input by the user; the message may be fed to an application contained locally in the MSC or to the VLR. The rules for determining this are specified in 3GPP TS 23.090 [34].

1) Message Destined for the VLR

If the message is destined for the VLR then the MSC shall transfer the message to the VLR using the mapping specified in detail in 3GPP TS 29.011 [59].

2) Message Destined for the Local Application

If the message is destined for the local USSD application then the MSC shall transfer the information contained in the message to the application.

The process in the MSC is shown in figure 22.9.2/1.

22.9.3 Procedure in the VLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2;

Process_Access_Request_VLR see subclause 25.4.2.

The MAP_PROCESS_UNSTRUCTURED_SS_REQUEST from the MSC contains information input by the user; the message may be fed to an application contained locally in the VLR or to the HLR. The rules for determining this are specified in 3GPP TS 23.090 [34].

1) Message Destined for the HLR

If the message is destined for the HLR then the VLR shall transfer the message transparently to the HLR.

2) Message Destined for the Local Application

If the message is destined for the local USSD application then the VLR shall transfer the information contained in the message to the application.

The process in the VLR is shown in figure 22.9.3/1.

22.9.4 Procedure in the HLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The MAP_PROCESS_UNSTRUCTURED_SS_REQUEST from the VLR contains information input by the user. If the alphabet used for the message is understood then the message shall be fed to an application contained locally in the HLR or to the gsmSCF or to a secondary HLR where the USSD application is located.

1) Message Destined for the Local Application

If the message is destined for the local USSD application then the HLR shall transfer the information contained in the message to the local application.

2) Message Destined for the gsmSCF or the secondary HLR

If the message is destined for the gsmSCF or the secondary HLR then the primary HLR shall transfer the message transparently to the next node.

The process in the primary HLR is shown in figure 22.9.4/1.

22.9.5 Procedures in the gsmSCF/secondary HLR

The MAP process invokes a macro not defined in this clause; the definition of this macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

The process in the gsmSCF or secondary HLR is shown in figure 22.9.5/1.

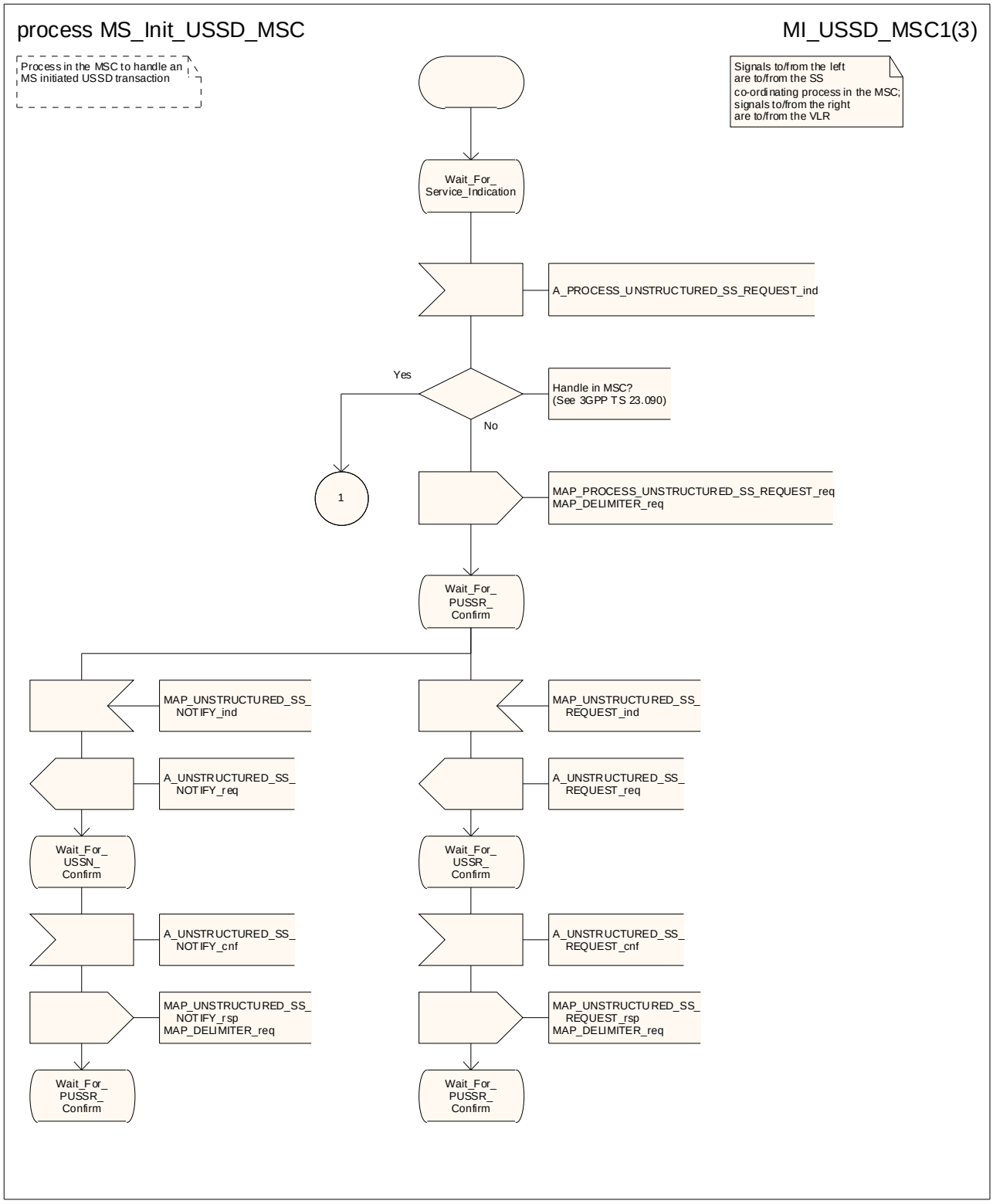


Figure 22.9.2/1 (sheet 1 of 3): Process MS_Init_USSD_MSC

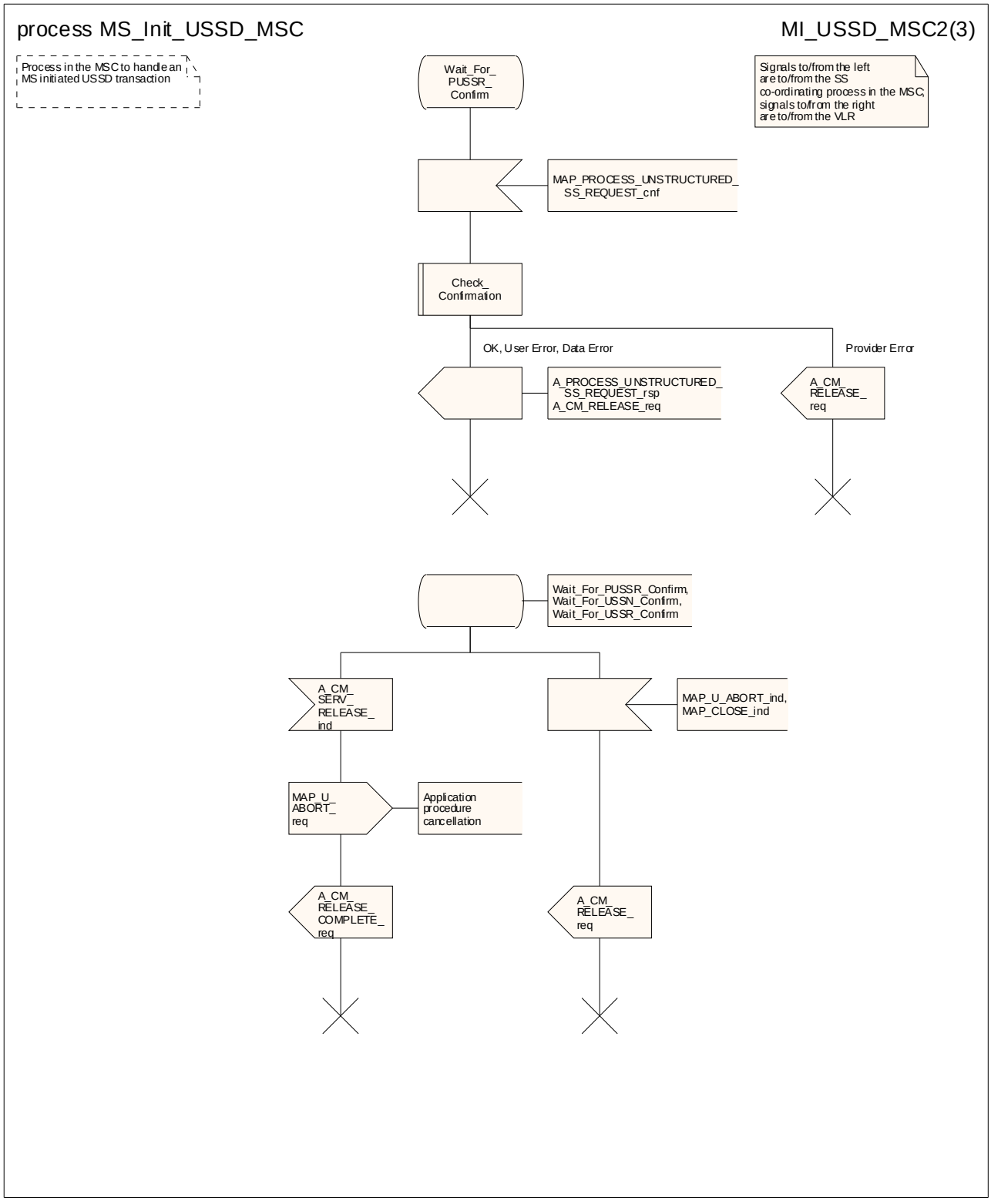


Figure 22.9.2/1 (sheet 2 of 3): Process MS_Init_USSD_MSC

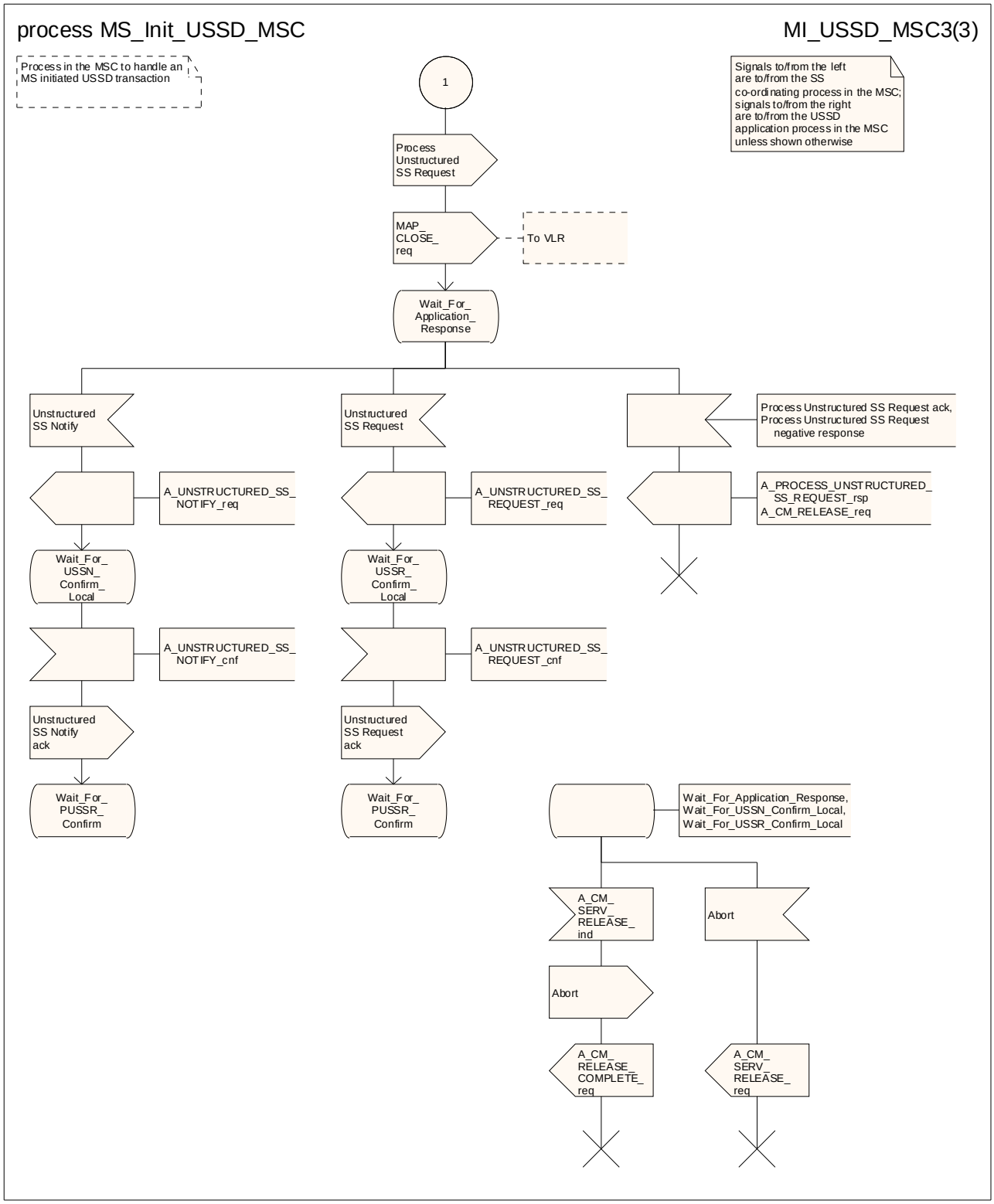


Figure 22.9.2/1 (sheet 3 of 3): Process MS_Init_USSD_MSC

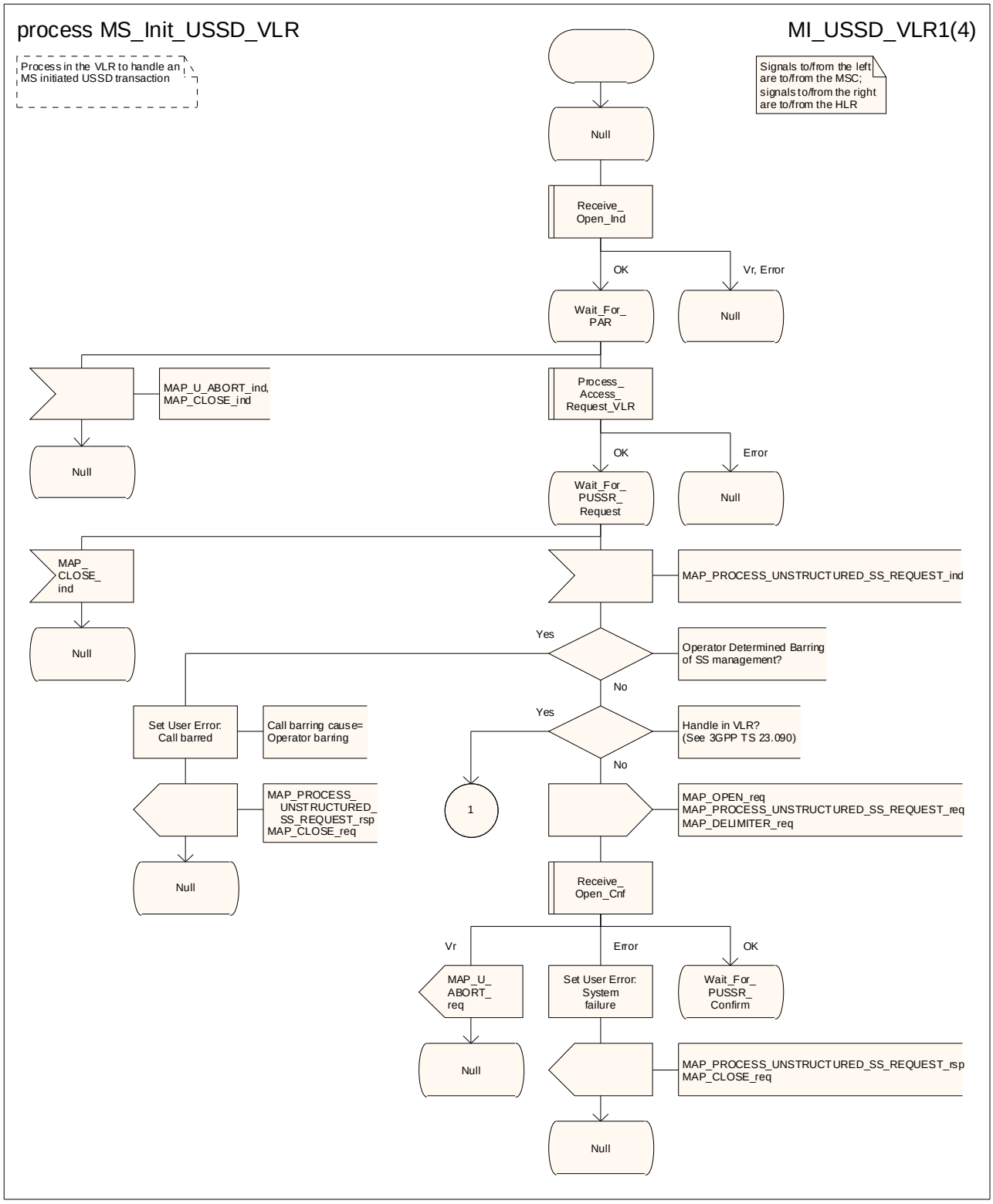


Figure 22.9.3/1 (sheet 1 of 4): Process MS_Init_USSD_VLR

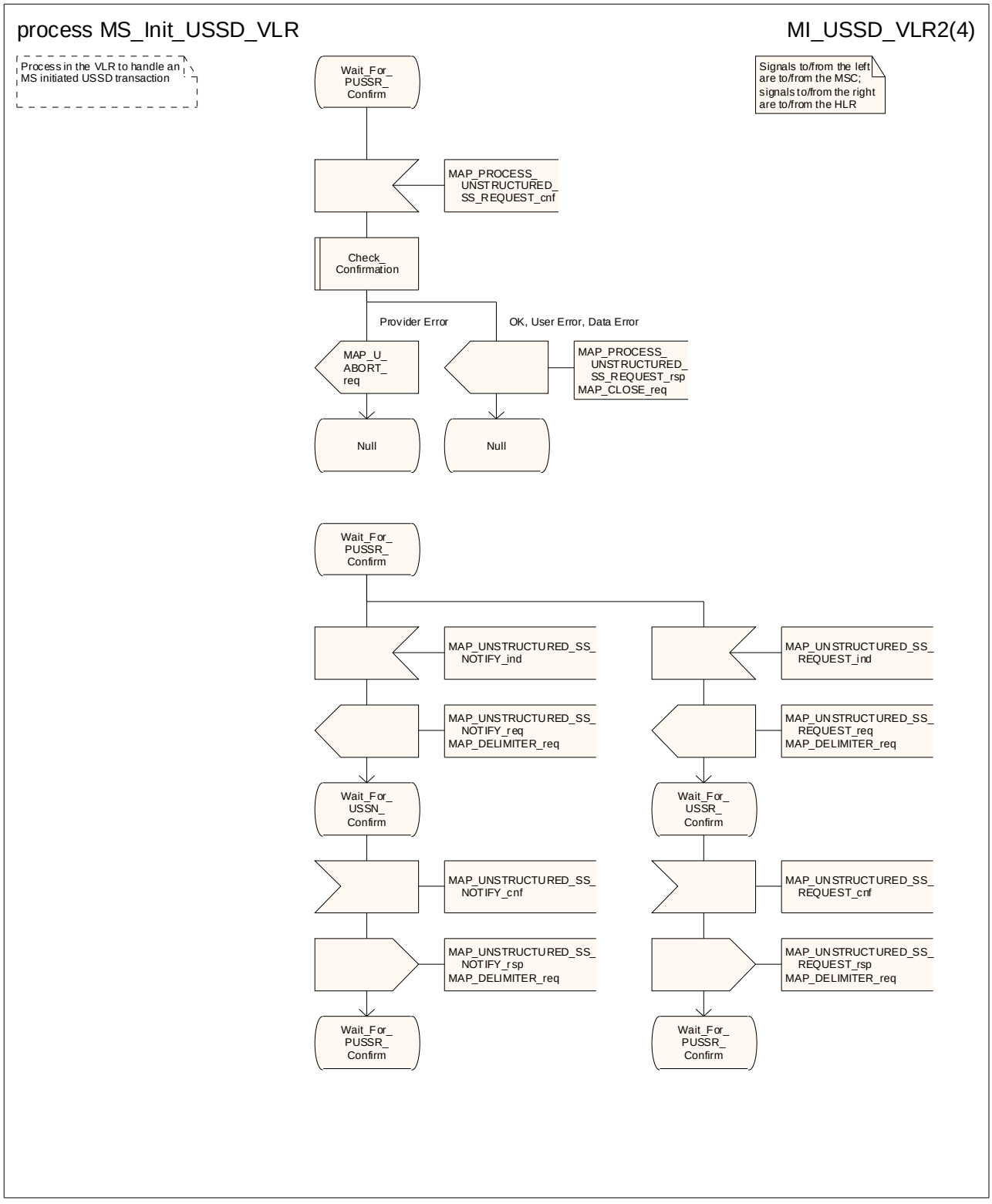


Figure 22.9.3/1 (sheet 2 of 4): Process MS_Init_USSD_VLR

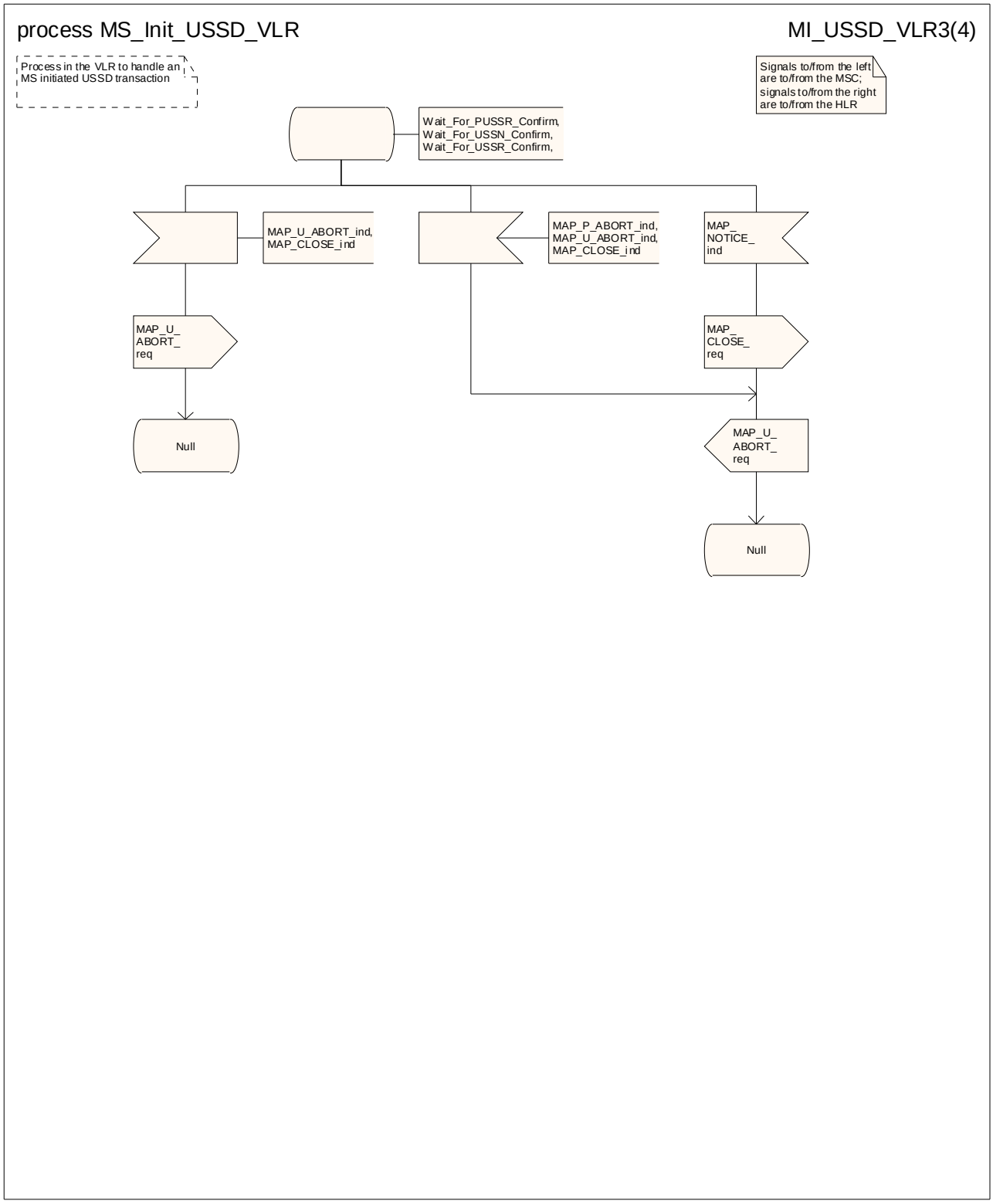


Figure 22.9.3/1 (sheet 3 of 4): Process_MS_Init_USSD_VLR

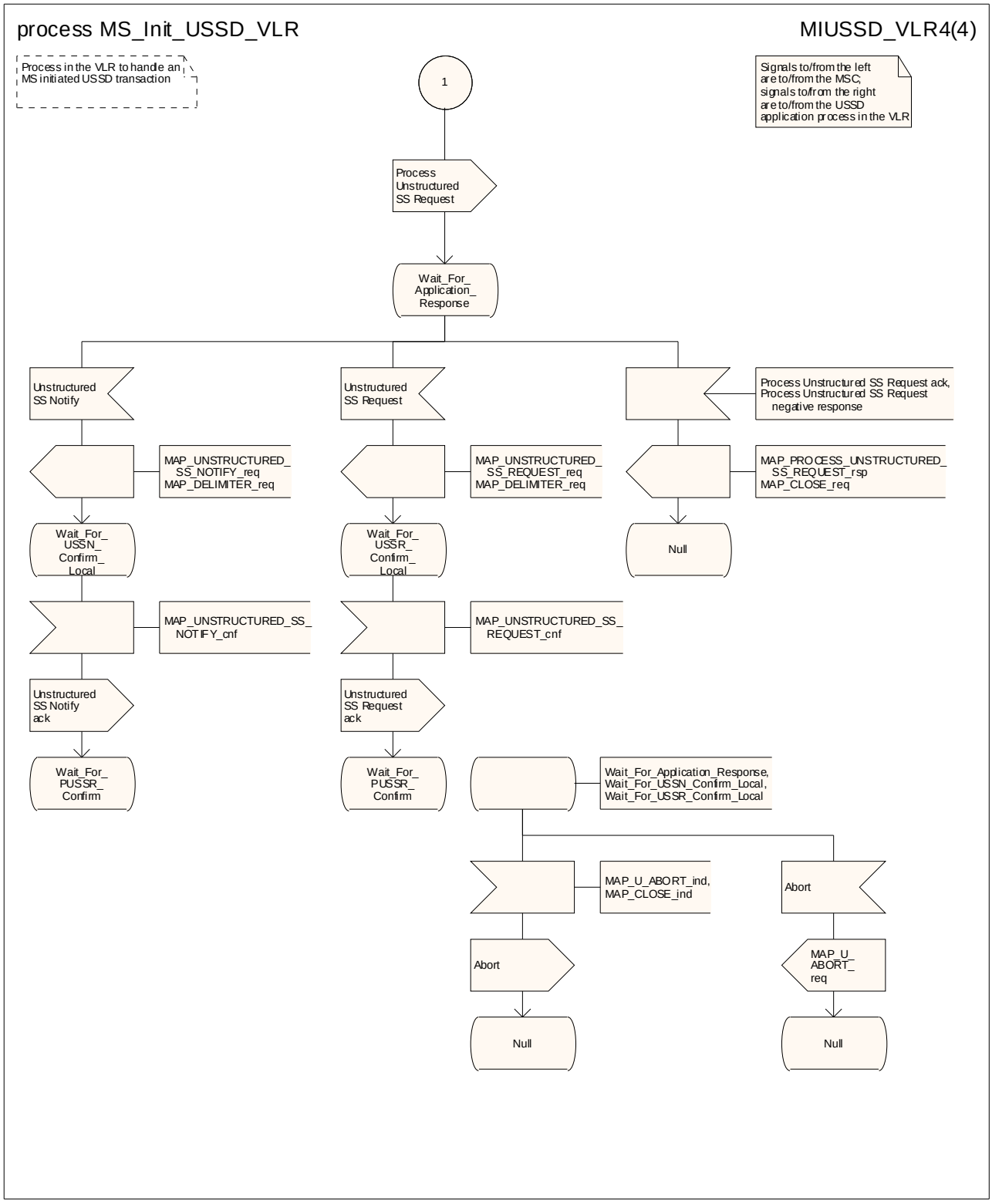


Figure 22.9.3/1 (sheet 4 of 4): Process_MS_Init_USSD_VLR

Figure 22.9.3/2 void

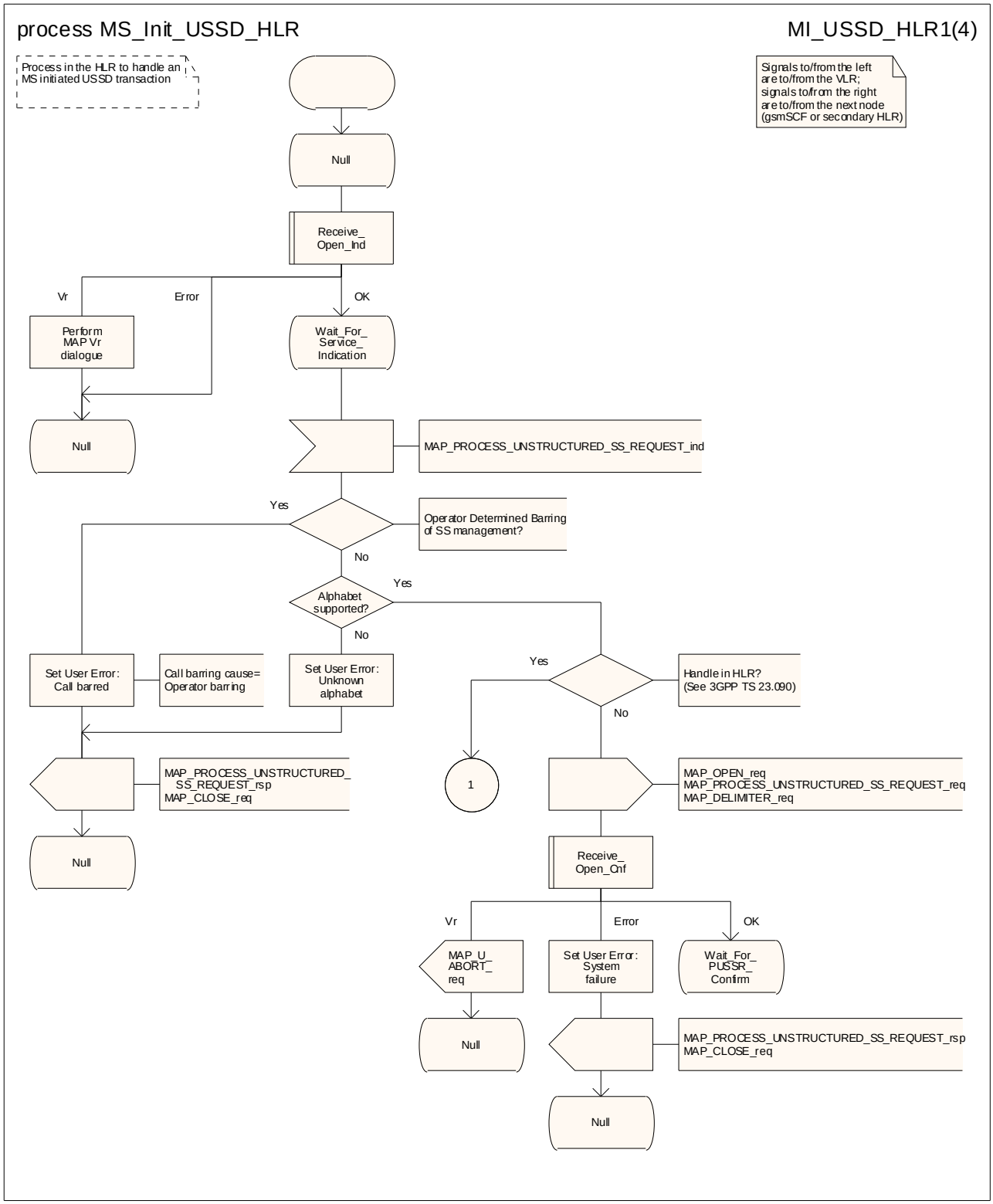


Figure 22.9.4/1 (sheet 1 of 4): Process MS_Init_USSD_HLR

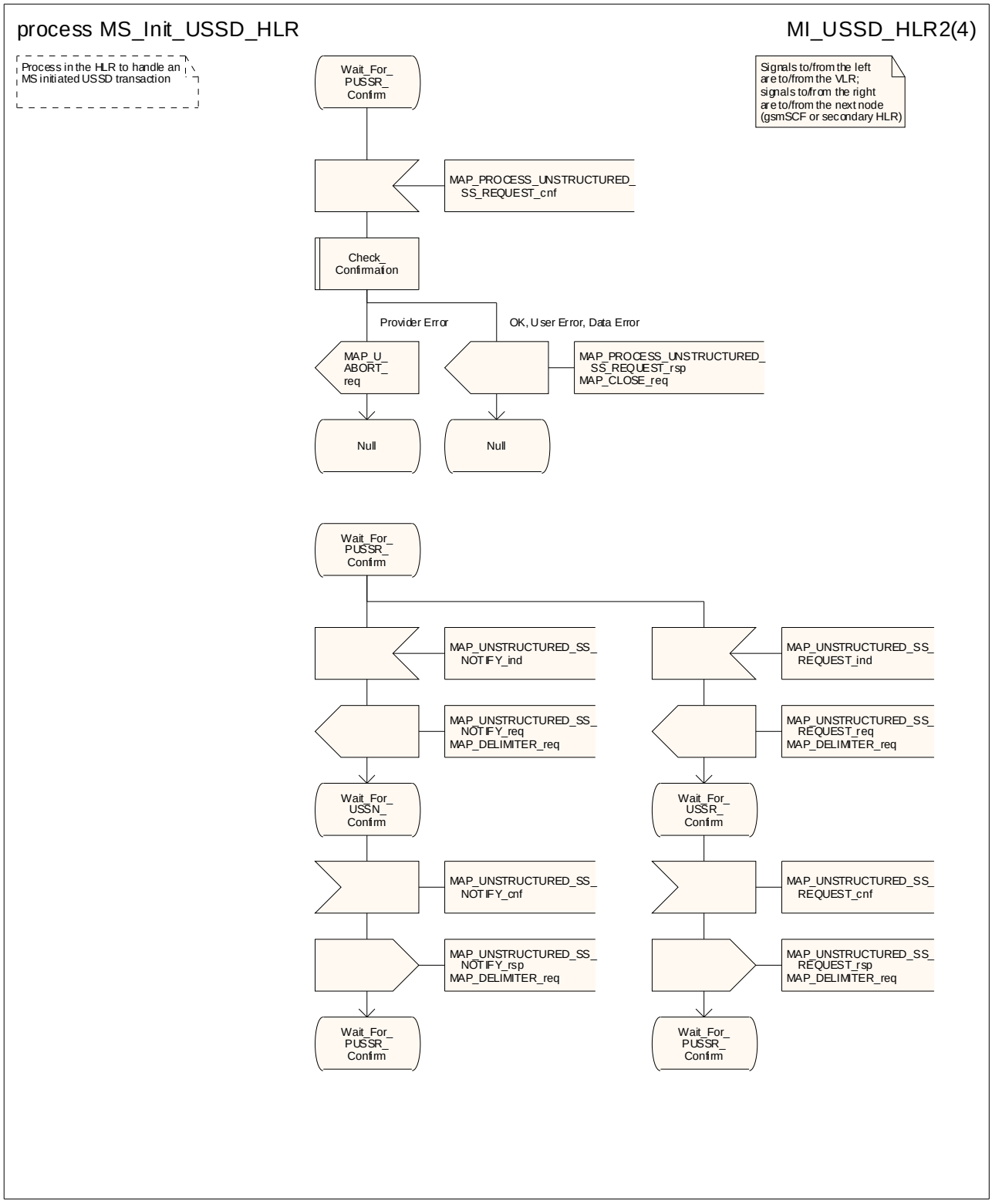


Figure 22.9.4/1 (sheet 2 of 4): Process MS_Init_USSD_HLR

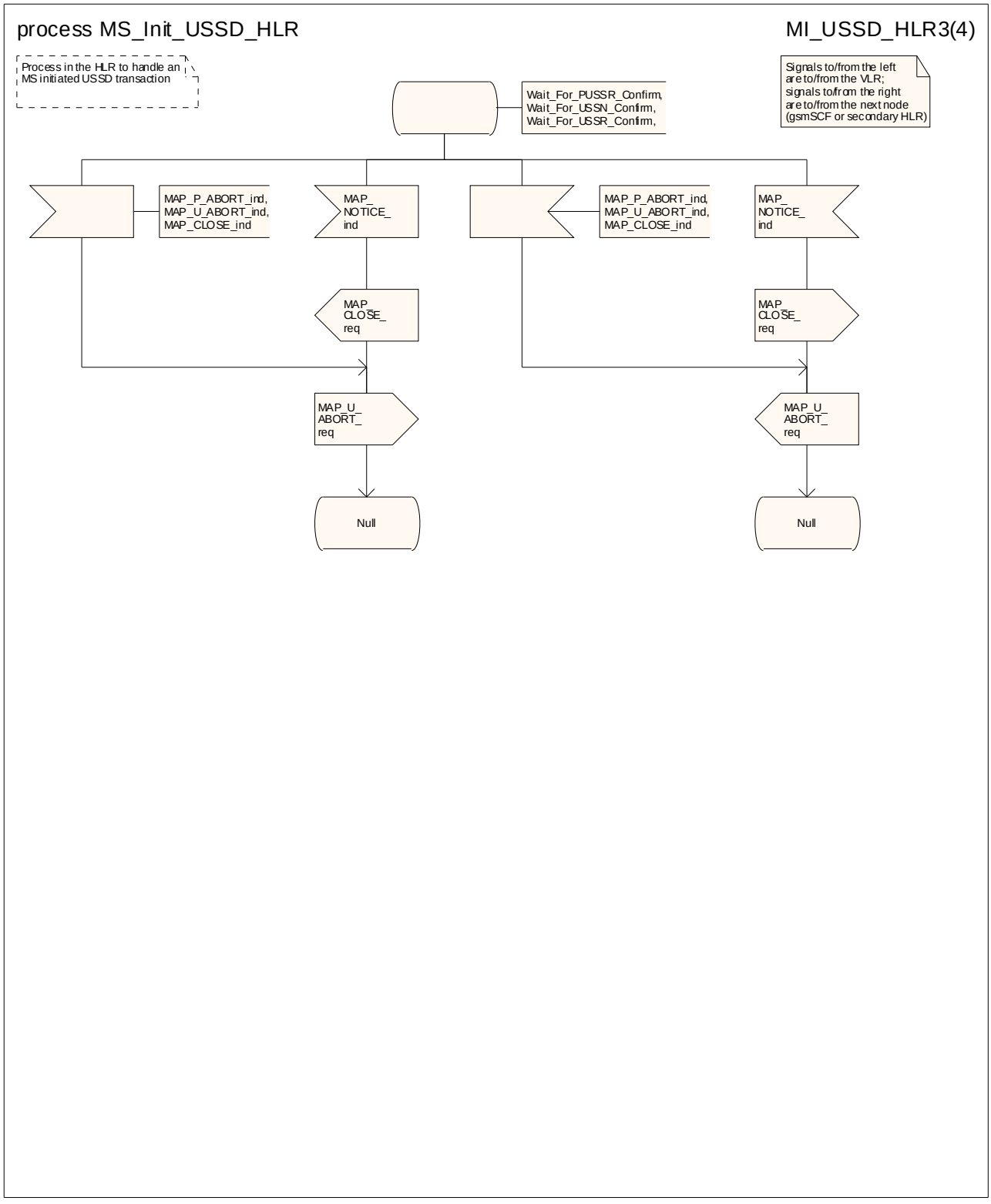


Figure 22.9.4/1 (sheet 3 of 4): Process MS_Init_USSD_HLR

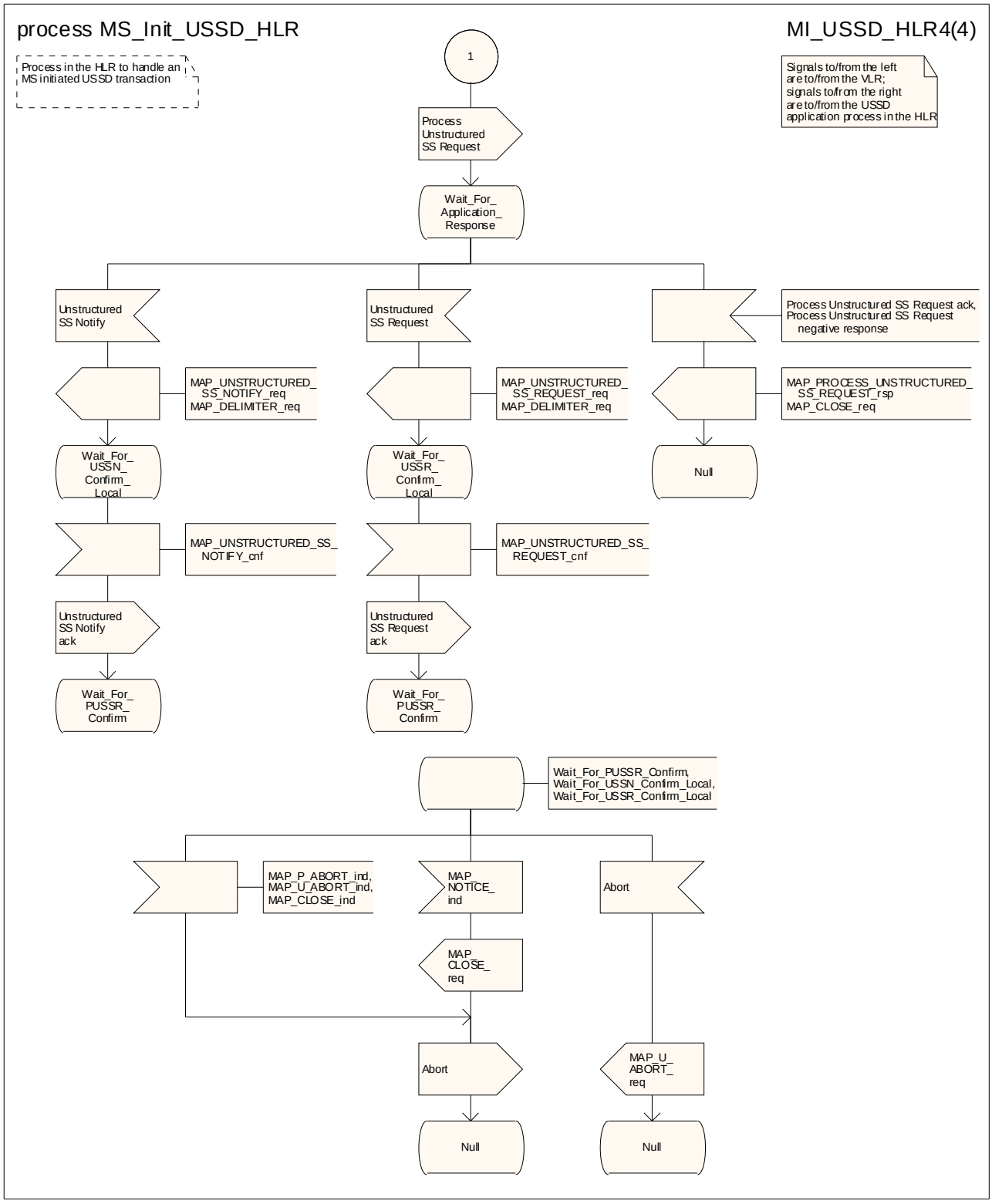


Figure 22.9.4/1 (sheet 4 of 4): Process MS_Init_USSD_HLR

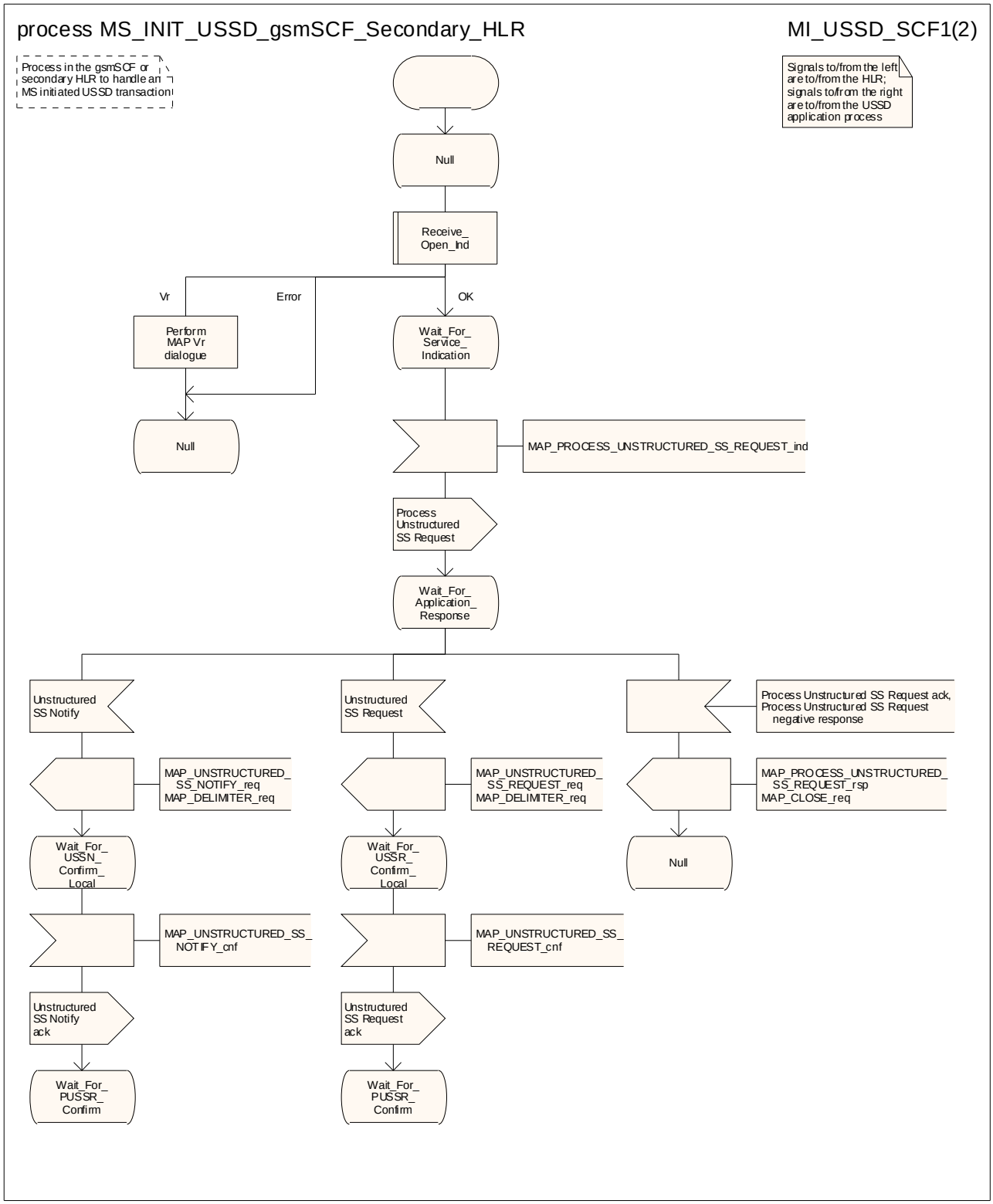


Figure 22.9.5/1 (sheet 1 of 2): Process MS_Init_USSD_gsmSCF_Secondary_HLR

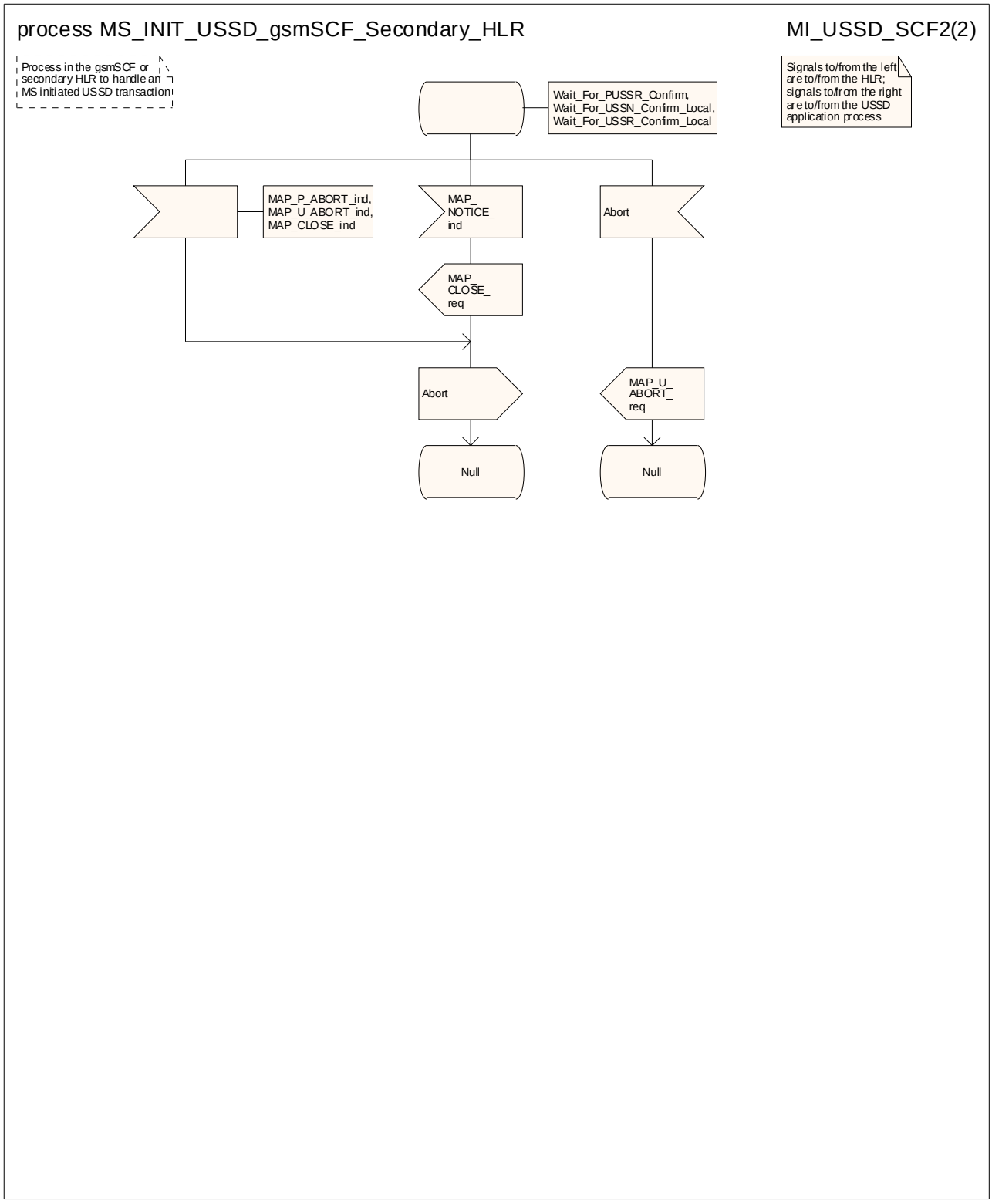


Figure 22.9.5/1 (sheet 2 of 2): Process MS_Init_USSD_gsmSCF_Secondary_HLR

22.10 Network initiated USSD procedure

22.10.1 General

The procedure supports supplementary service signalling procedures which allow PLMN specific services to be introduced.

The message flow for the procedure can be found in 3GPP TS 23.090 [34].

The following services may be used:

MAP_PAGE (see clauses 8 and 25);

MAP_SEARCH_FOR_MOBILE_SUBSCRIBER (see clauses 8 and 25);

MAP_PROCESS_ACCESS_REQUEST (see clauses 8 and 25);

MAP_AUTHENTICATE (see clauses 8 and 25);

MAP_SET_CIPHERING_MODE (see clauses 8 and 25);

MAP_FORWARD_NEW_TMSI (see clauses 8 and 25);

MAP_READY_FOR_SM (see clauses 12 and 25).

At least one of the following services will certainly be used, and both may be used:

MAP_UNSTRUCTURED_SS_REQUEST (defined in clause 11);

MAP_UNSTRUCTURED_SS_NOTIFY (defined in clause 11).

22.10.2 Procedure in the MSC

The process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Page_MSC see subclause 25.3.1;

Search_For_MS_MSC see subclause 25.3.2;

Process_Access_Request_MSC see subclause 25.4.1.

The process in the MSC is shown in figure 22.10.2/1.

22.10.3 Procedure in the VLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2.

The process in the VLR is shown in figure 22.10.3/1.

MSC Initiated USSD

If a USSD application in the MSC wishes to use the network initiated USSD procedure, and a connection to the MS does not exist then the MSC opens a dialogue with the VLR. This dialogue leads to the VLR performing page or search using the macro Start_USSD_VLR.

Macro Start_USSD_VLR

The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Confirmation see subclause 25.2.1;

Process_Access_Request_VLR see subclause 25.4.2.

The macro is shown in figure 22.10.3/2.

22.10.4 Procedure in the HLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2.

The process in the primary HLR is shown in figures 22.10.4/1 and 22.10.4/2.

22.10.5 Procedure in the gsmSCF or secondary HLR

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The procedure in the gsmSCF or secondary HLR is shown in figure 22.10.5/1.

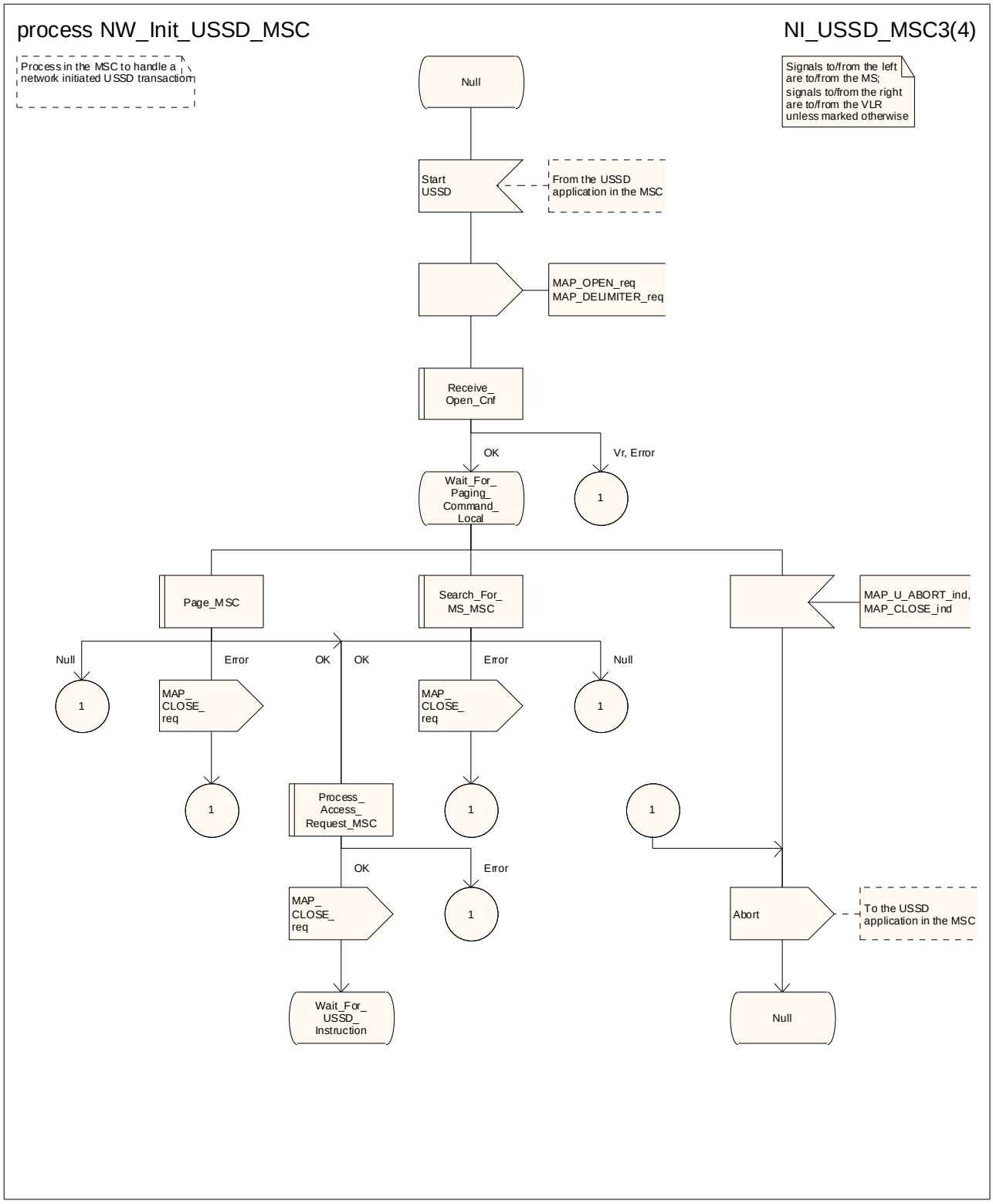


Figure 22.10.2/1 (sheet 1 of 4): Process NW_Init_USSD_MSC

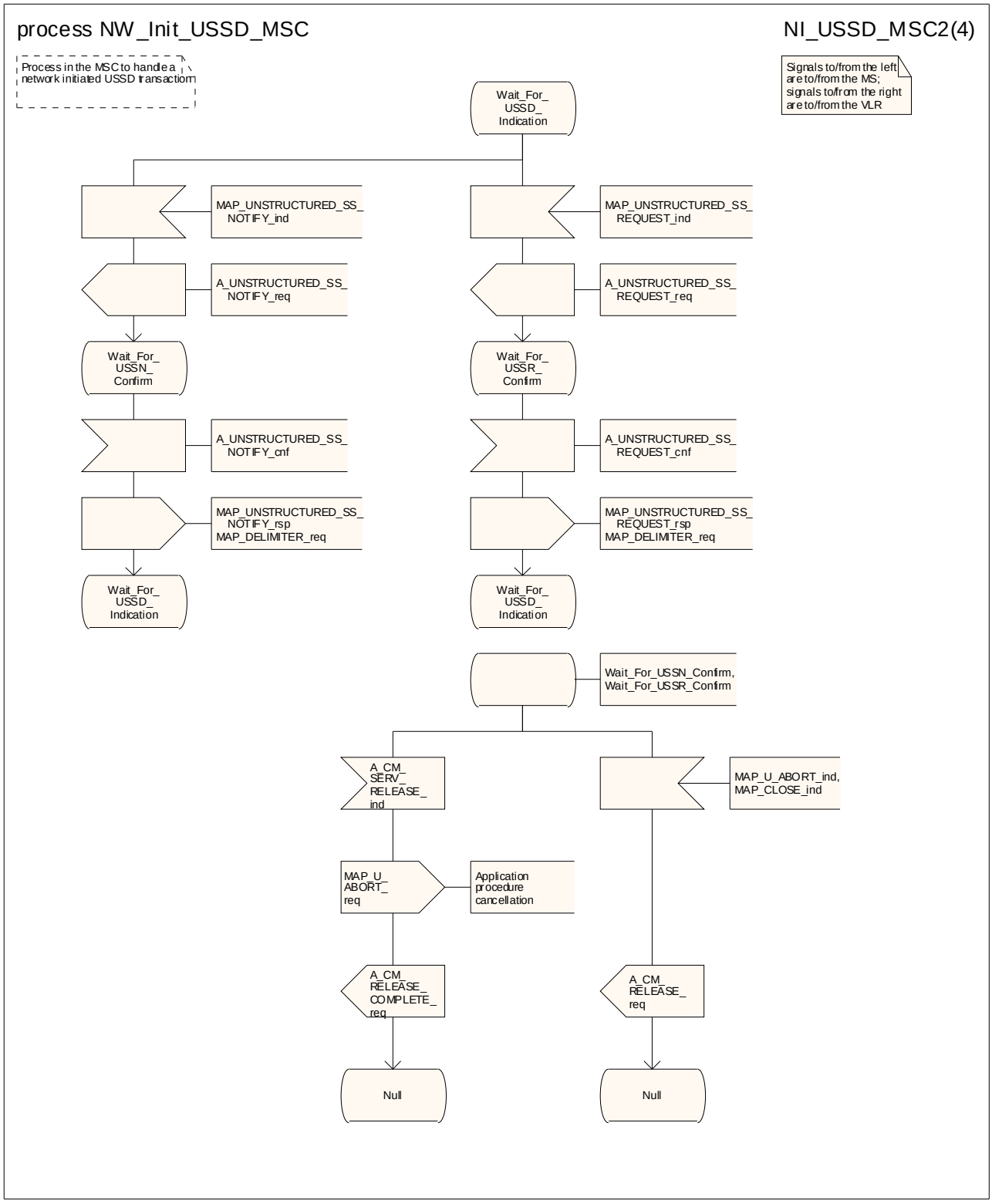


Figure 22.10.2/1 (sheet 2 of 4): Process NW_Init_USSD_MSC

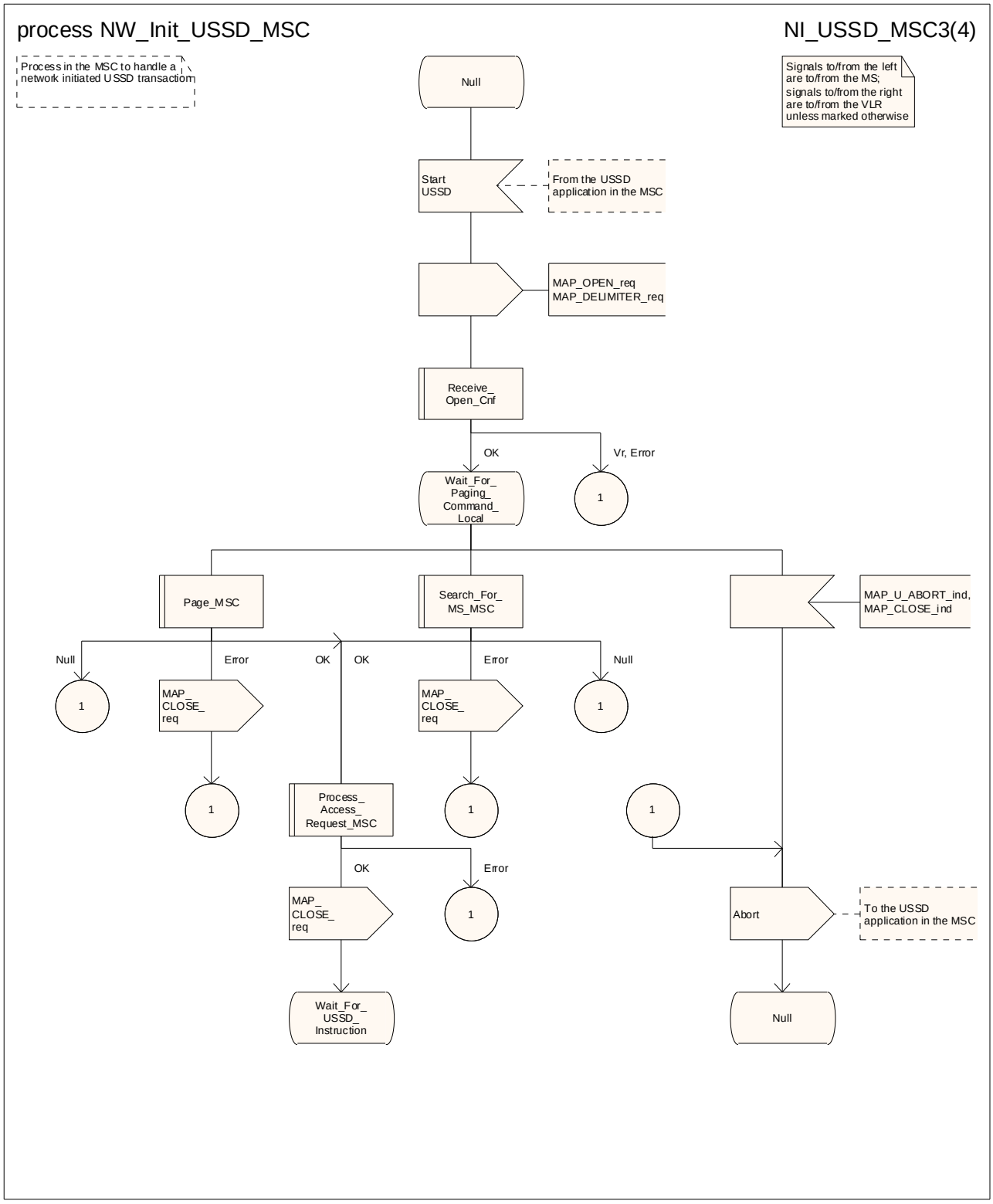


Figure 22.10.2/1 (sheet 3 of 4): Process NW_Init_USSD_MSC

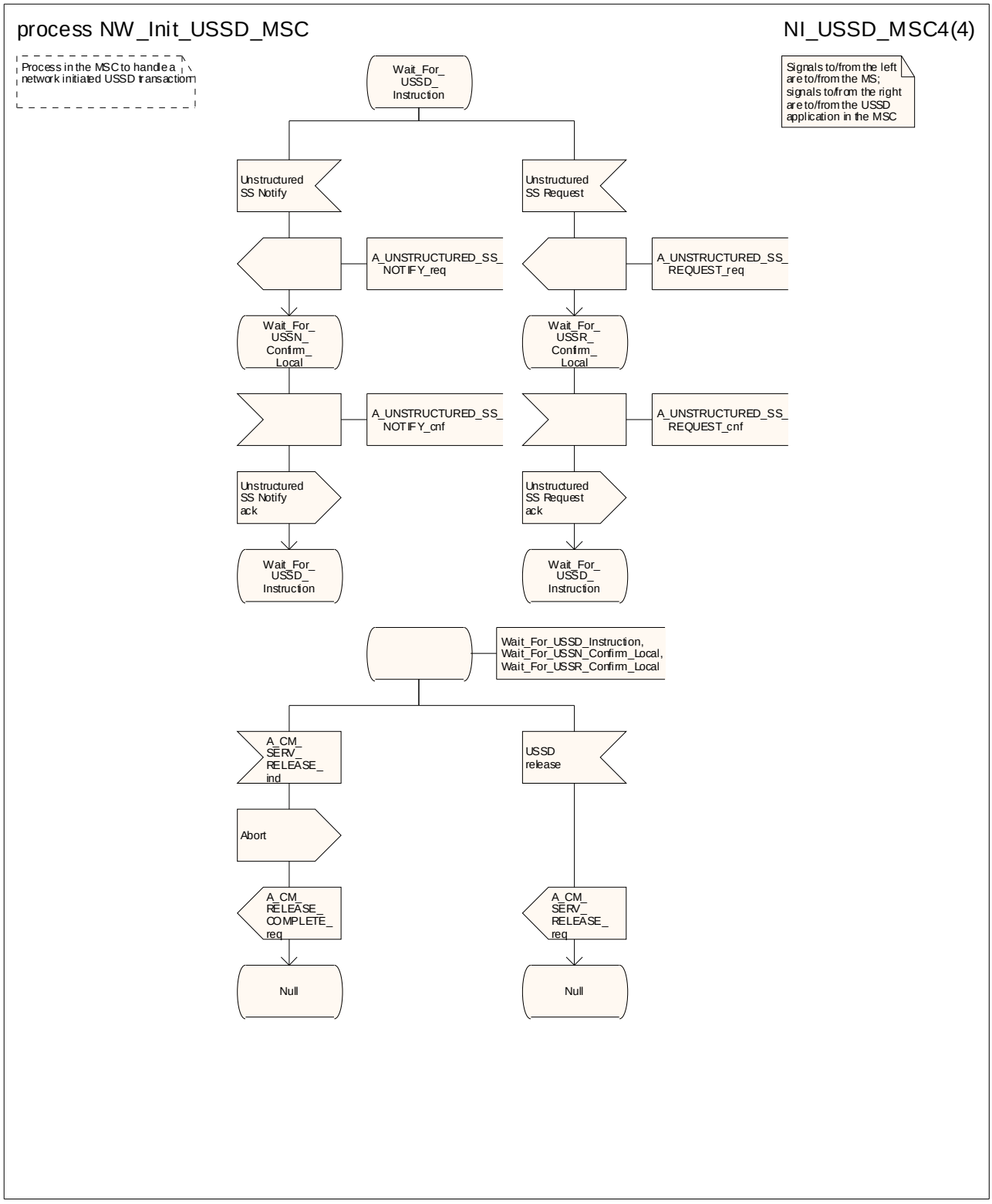


Figure 22.10.2/1 (sheet 4 of 4): Process NW_Init_USSD_MSC

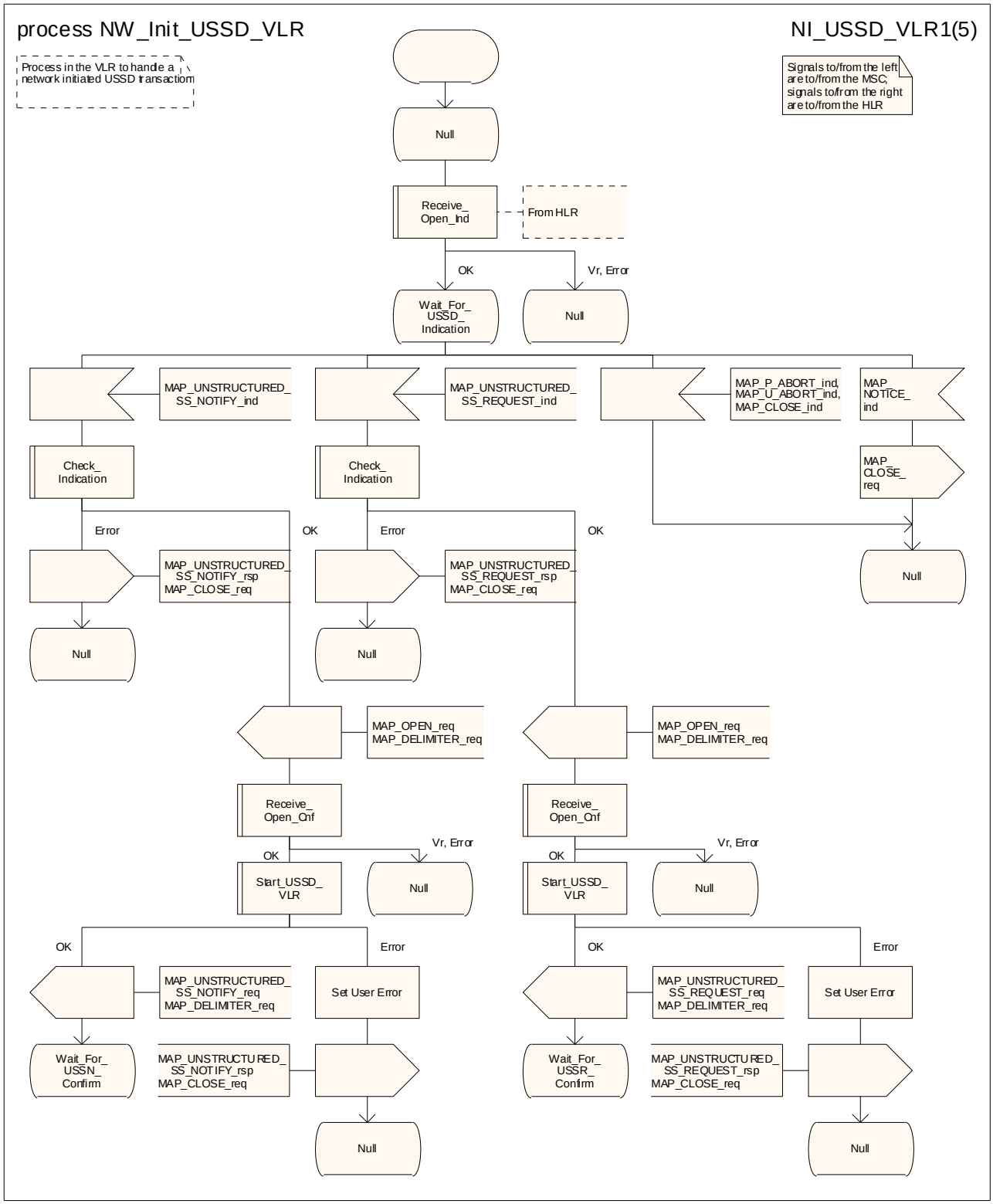


Figure 22.10.3/1 (sheet 1 of 5): Process NW_Init_USSD_VLR

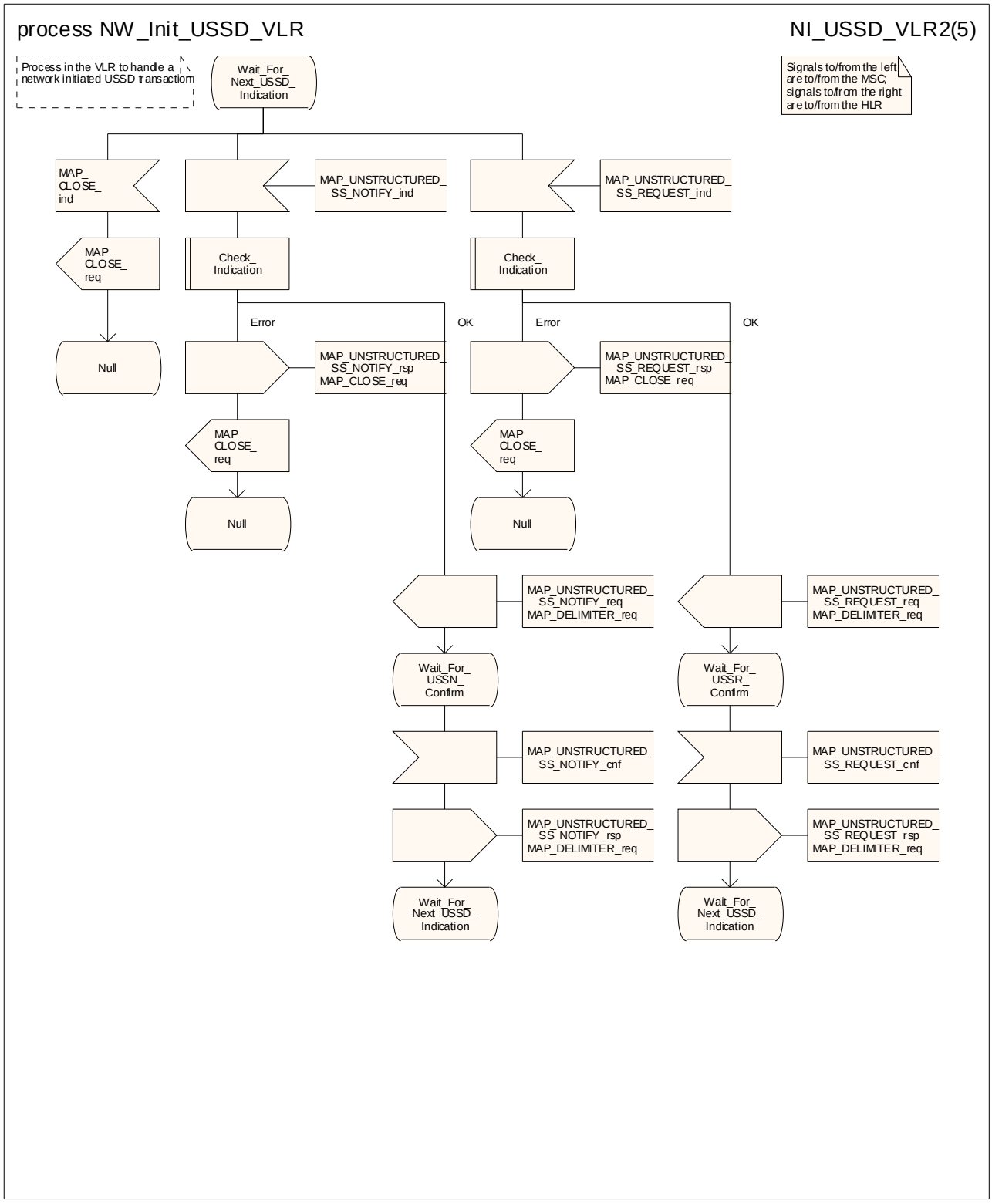


Figure 22.10.3/1 (sheet 2 of 5): Process NW_Init_USSD_VLR

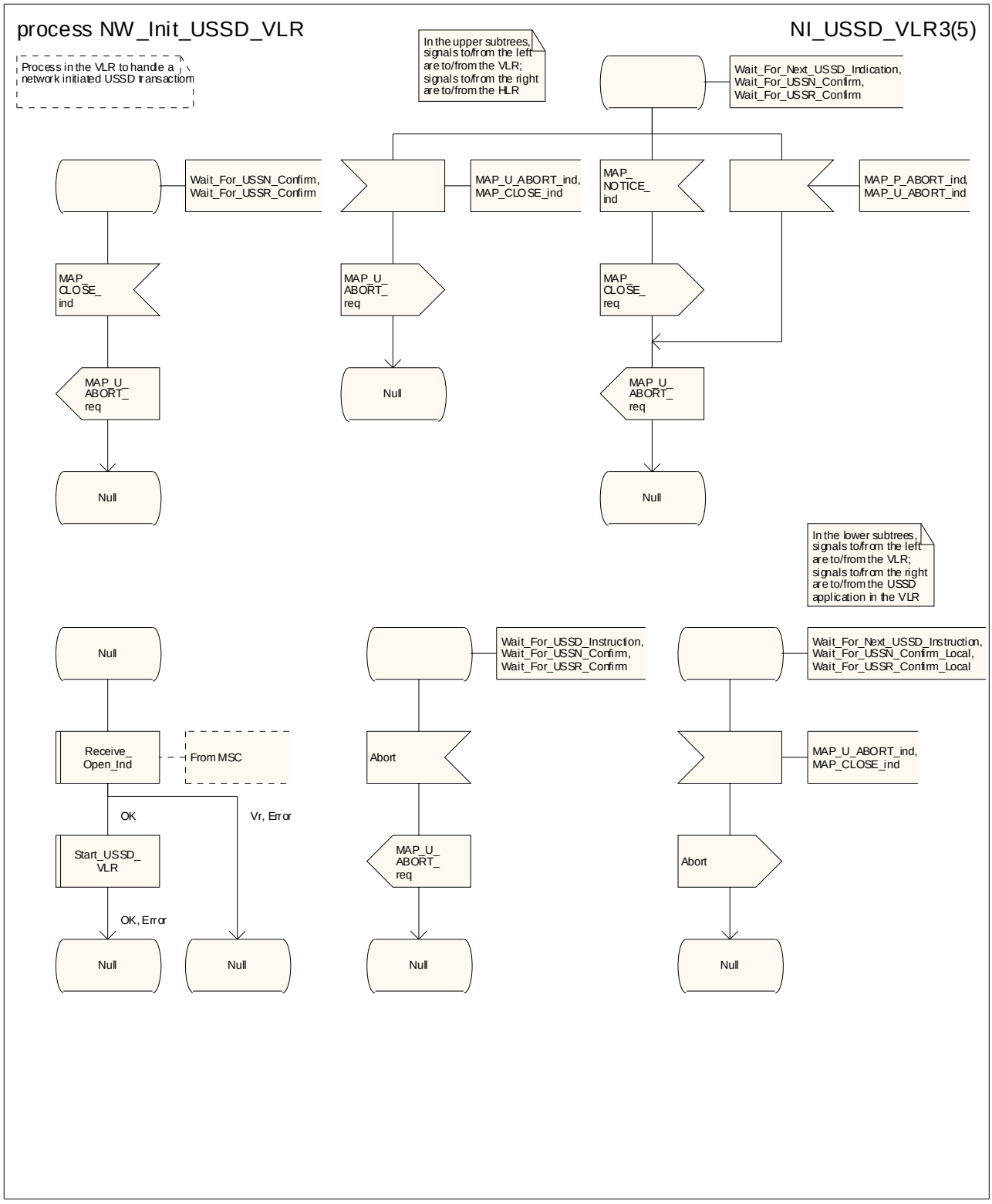


Figure 22.10.3/1 (sheet 3 of 5): Process NW_Init_USSD_VLR

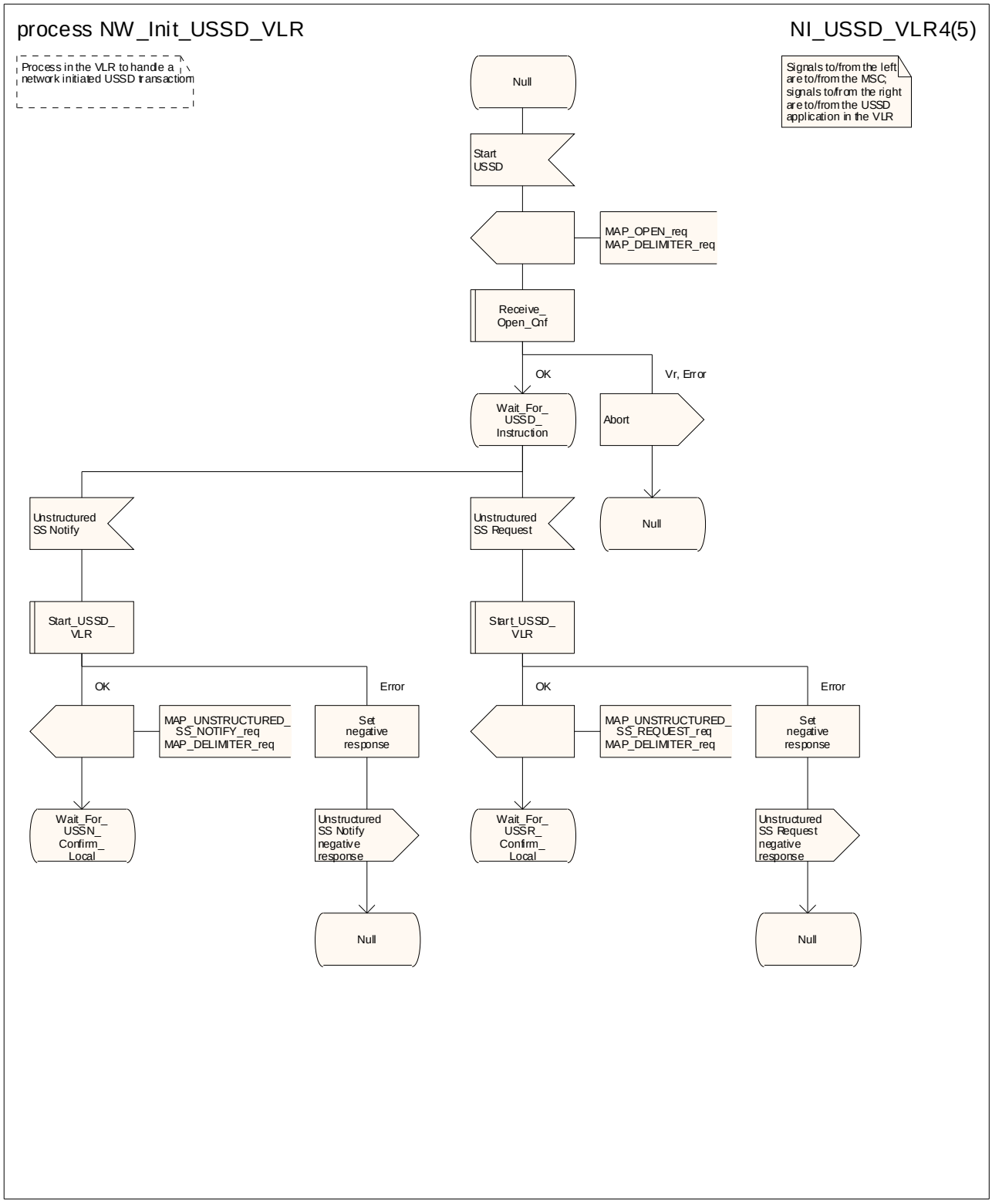


Figure 22.10.3/1 (sheet 4 of 5): Process NW_Init_USSD_VLR

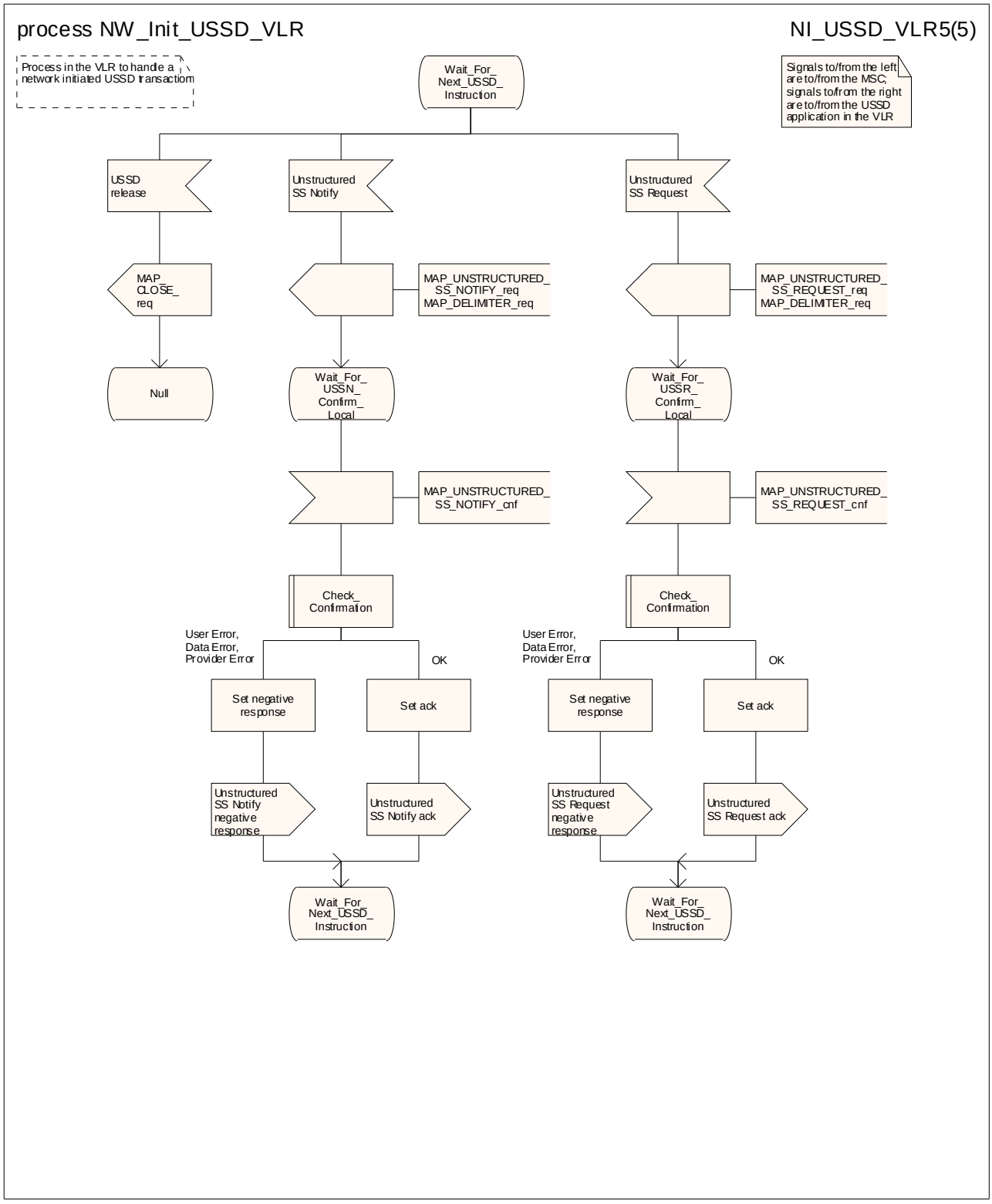


Figure 22.10.3/1 (sheet 5 of 5): Process NW_Init_USSD_VLR

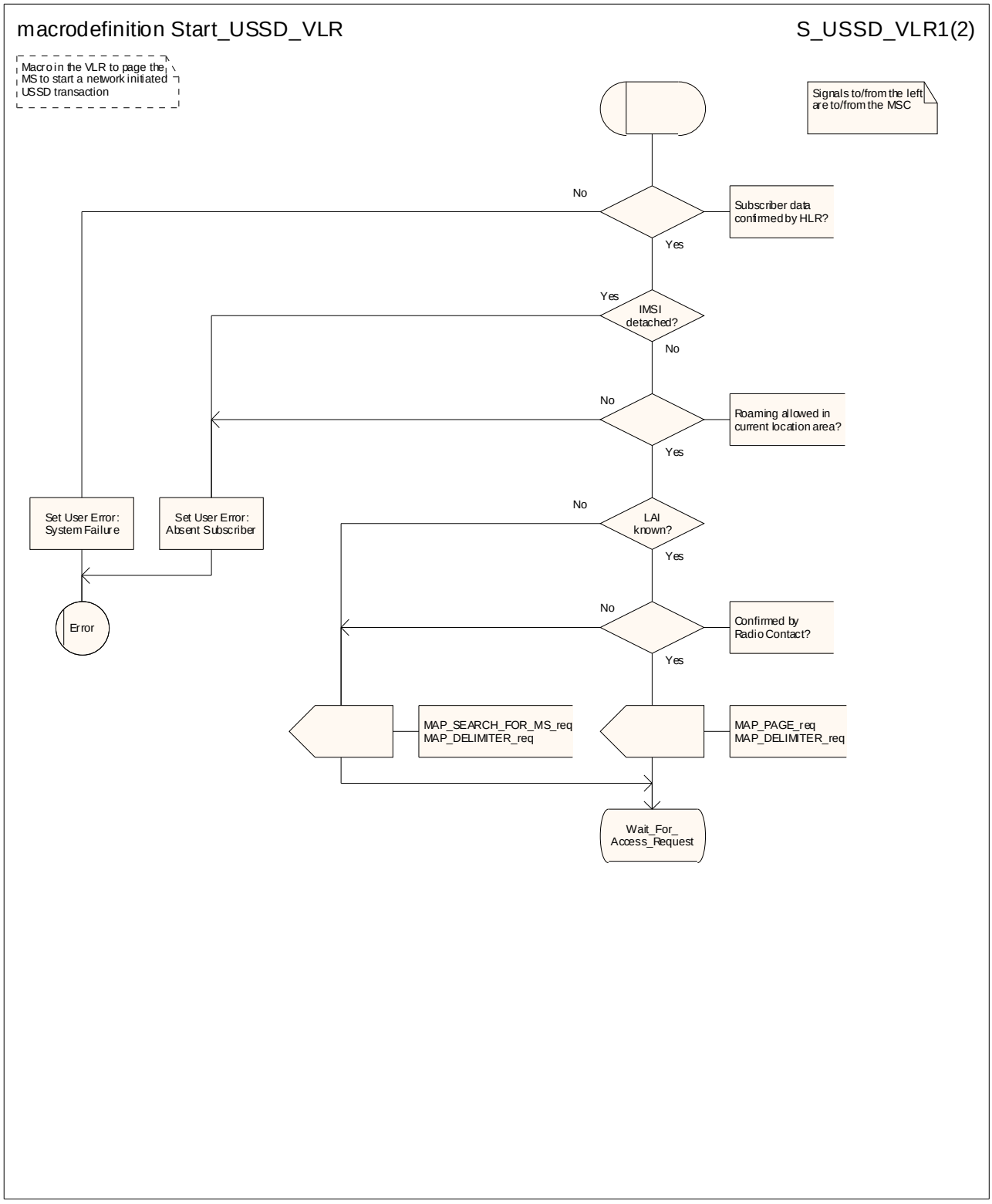


Figure 22.10.3/2 (sheet 1 of 2): Macro Start_USSD_VLR

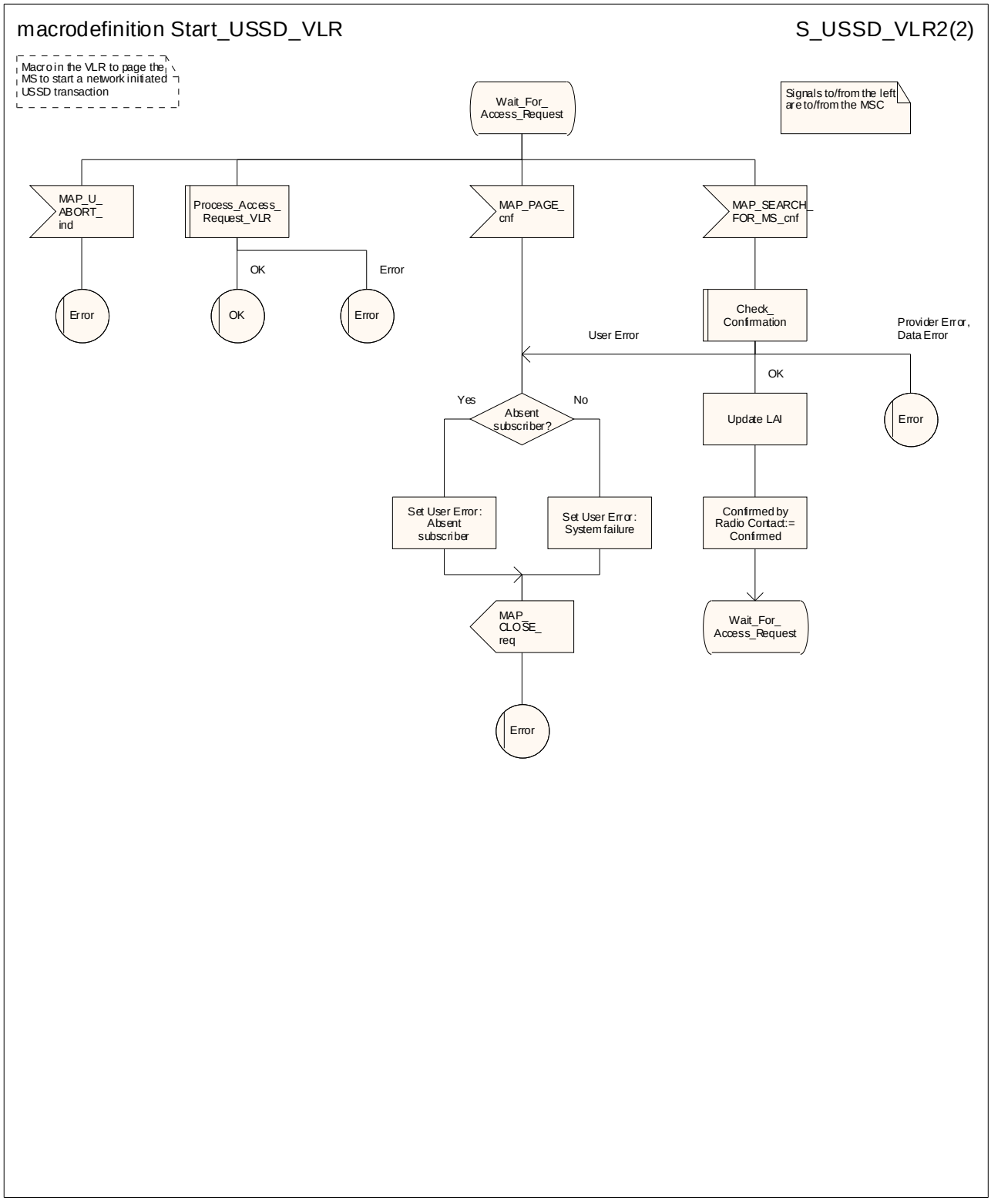


Figure 22.10.3/2 (sheet 2 of 2): Macro Start_USSD_VLR

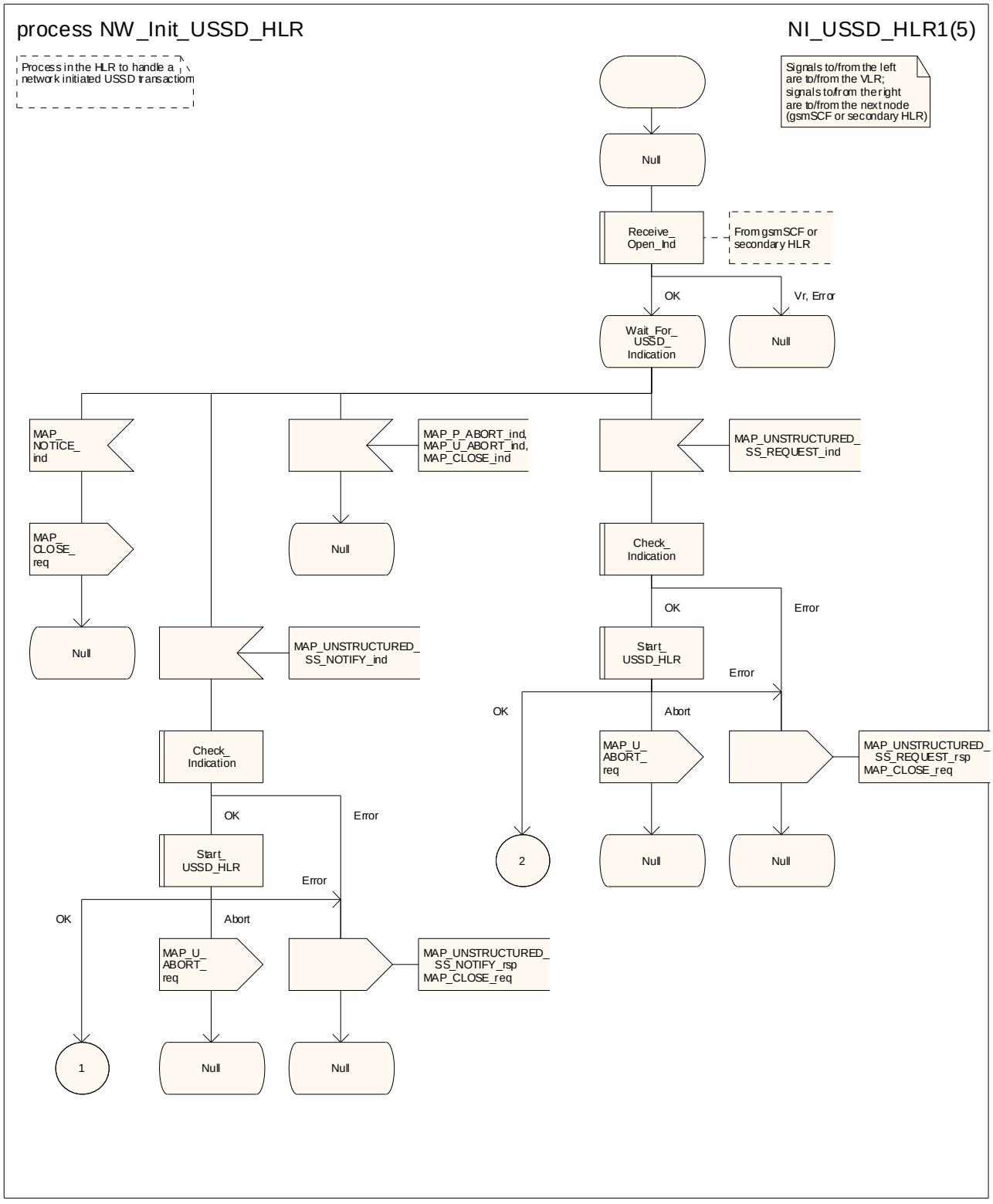


Figure 22.10.4/1 (sheet 1 of 5): Process NW_Init_USSD_HLR

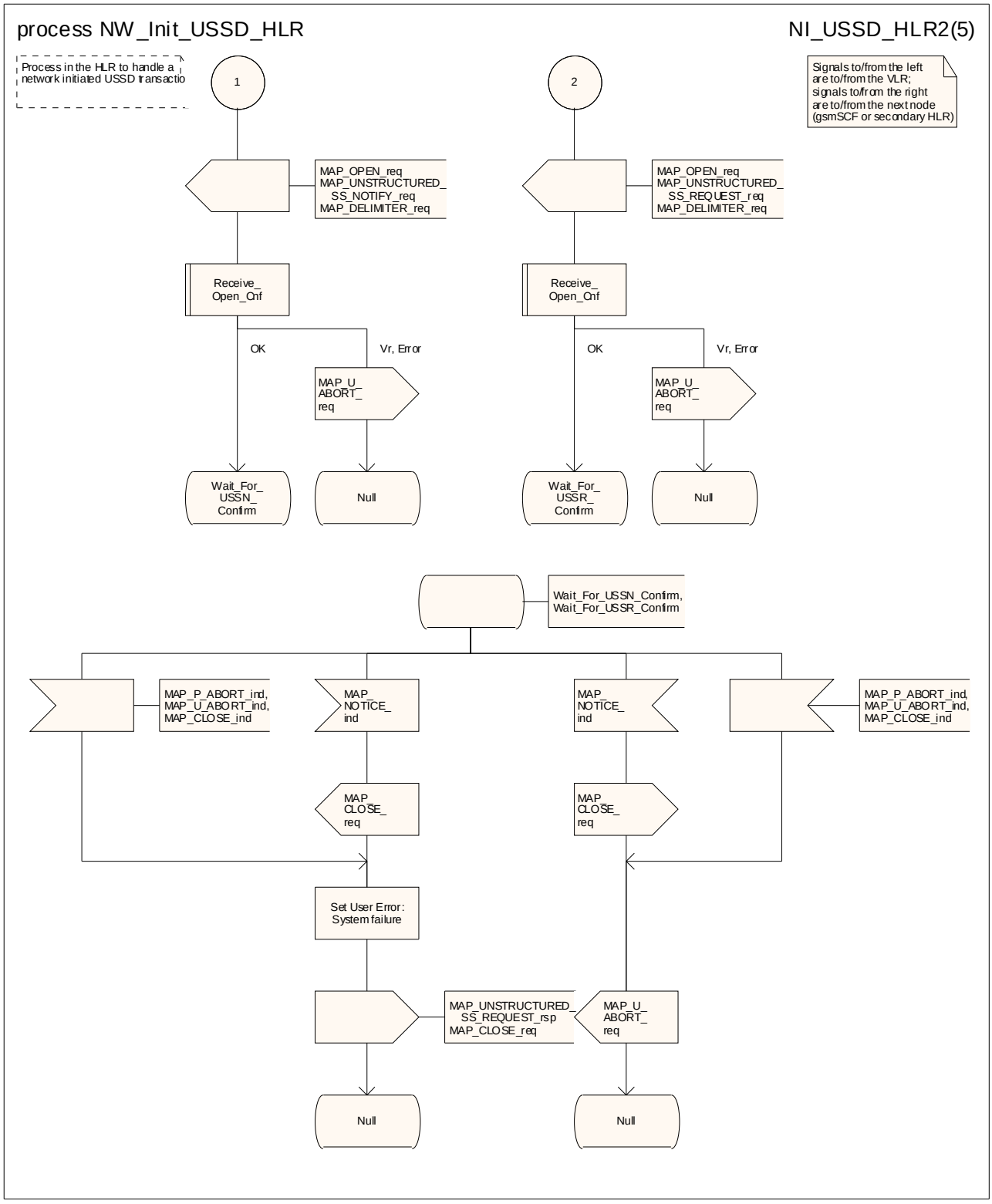


Figure 22.10.4/1 (sheet 2 of 5): Process NW_Init_USSD_HLR

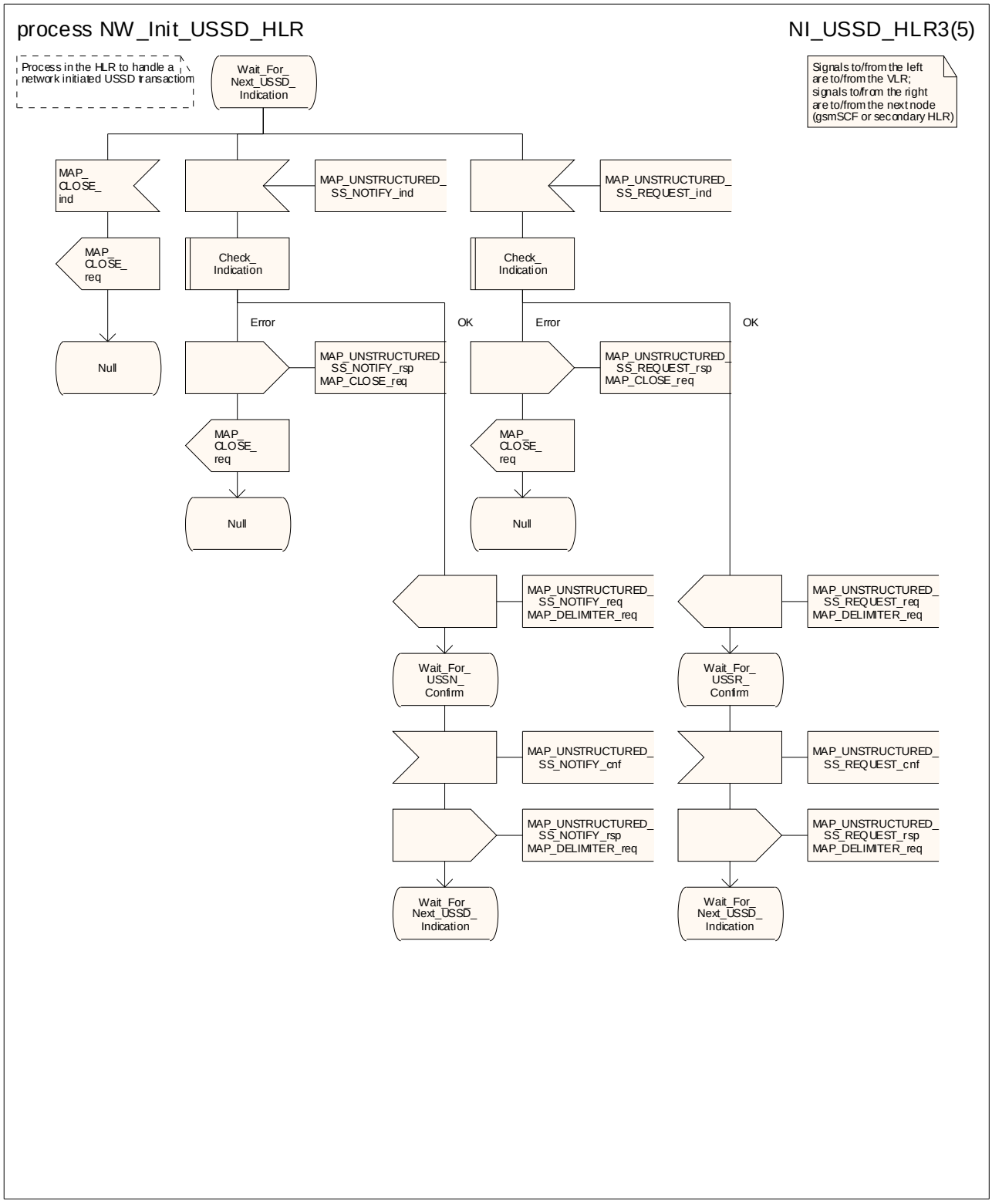


Figure 22.10.4/1 (sheet 3 of 5): Process NW_Init_USSD_HLR

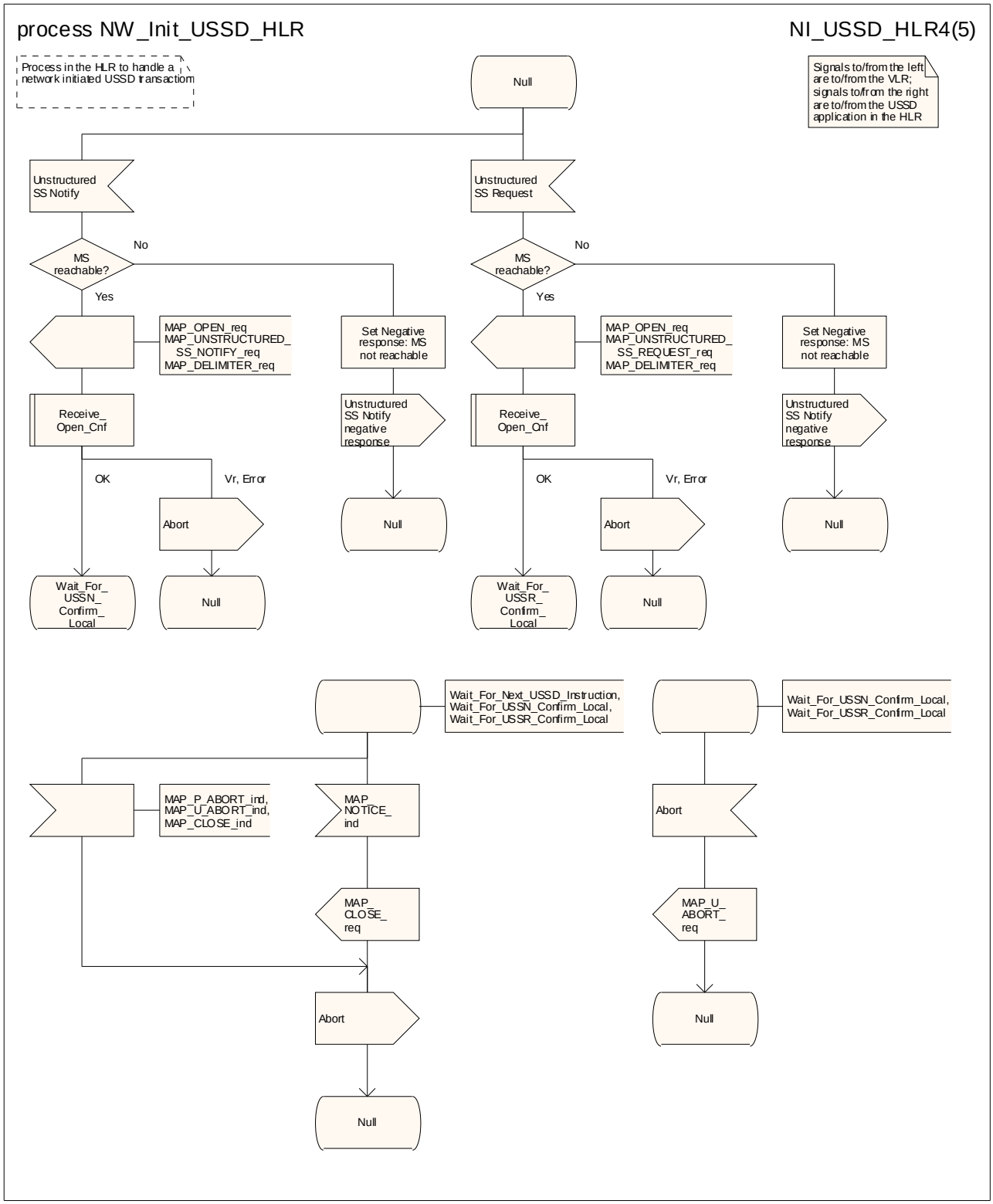


Figure 22.10.4/1 (sheet 4 of 5): Process NW_Init_USSD_HLR

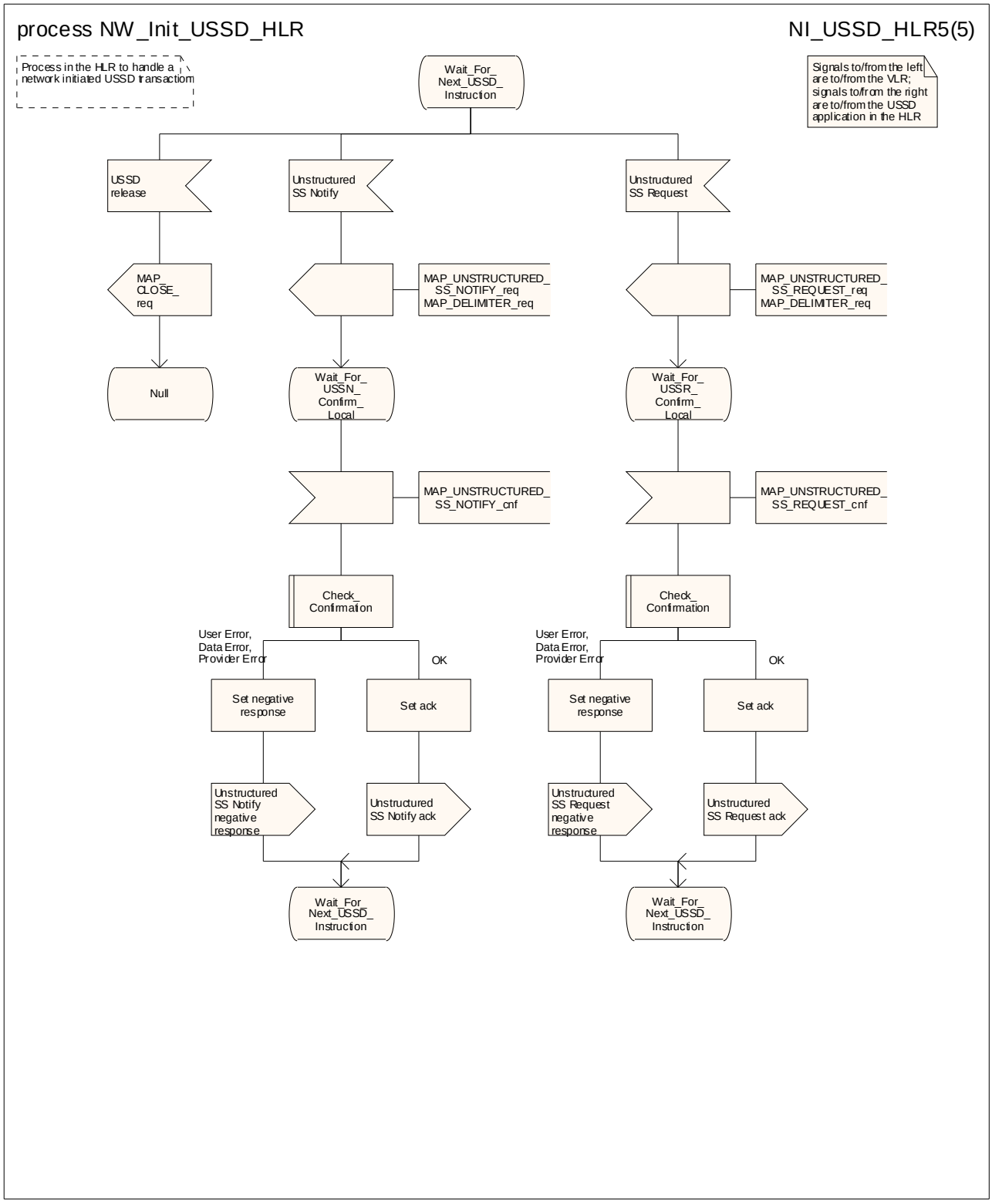


Figure 22.10.4/1 (sheet 5 of 5): Process NW_Init_USSD_HLR

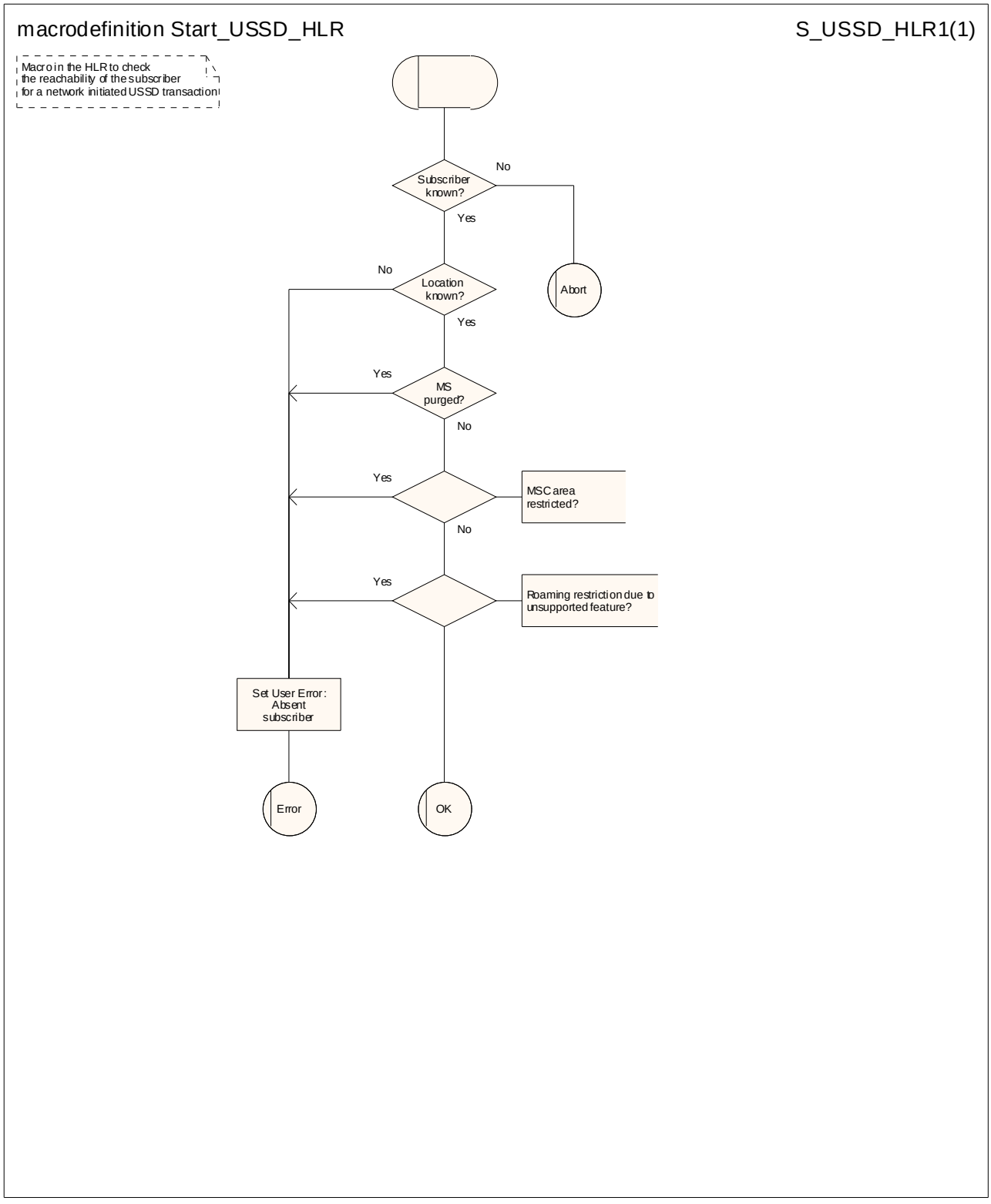


Figure 22.10.4/2: Macro Start_USSD_HLR

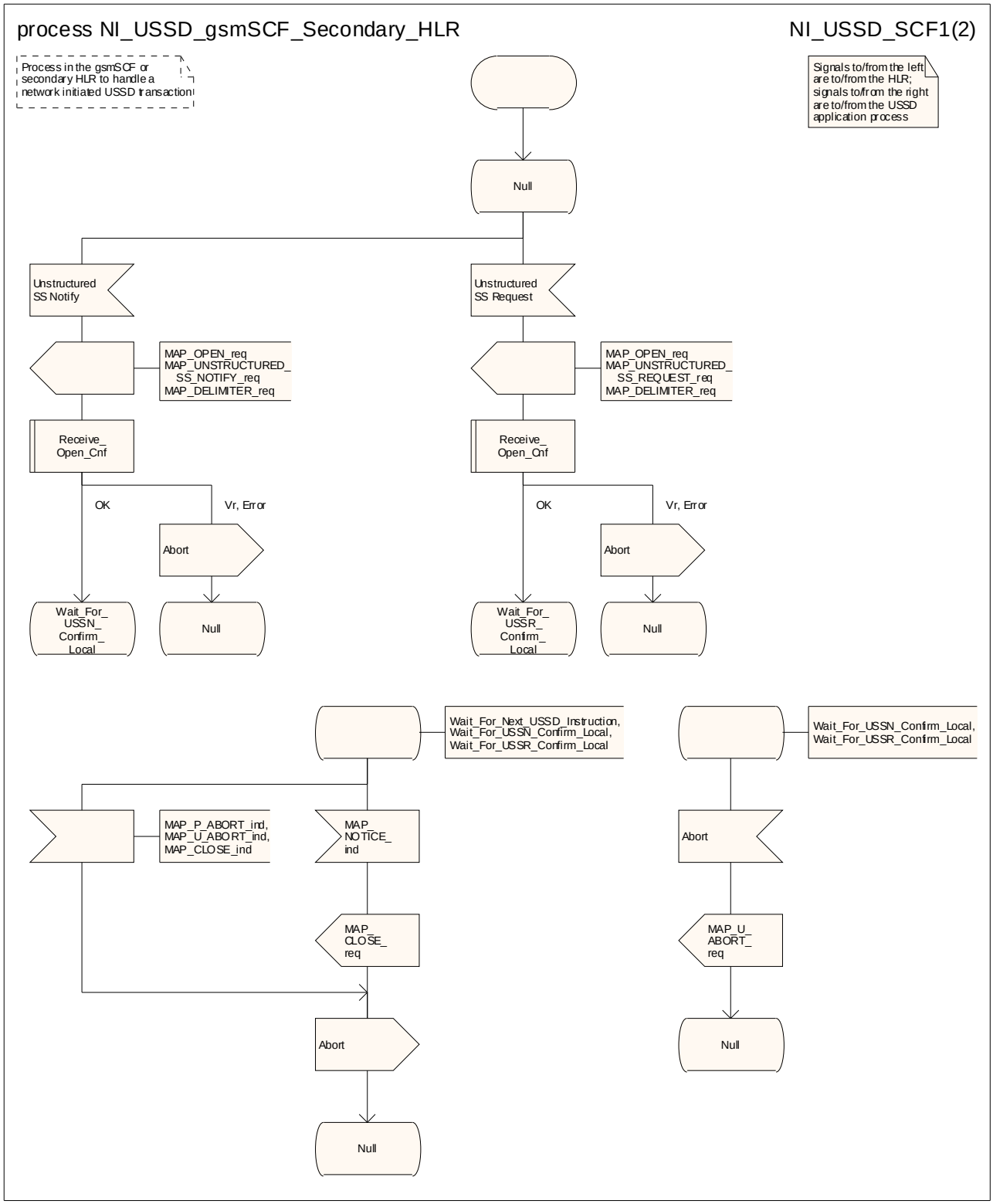


Figure 22.10.5/1 (sheet 1 of 2): Process NW_Init_USSD_gsmSCF_secondary_HLR

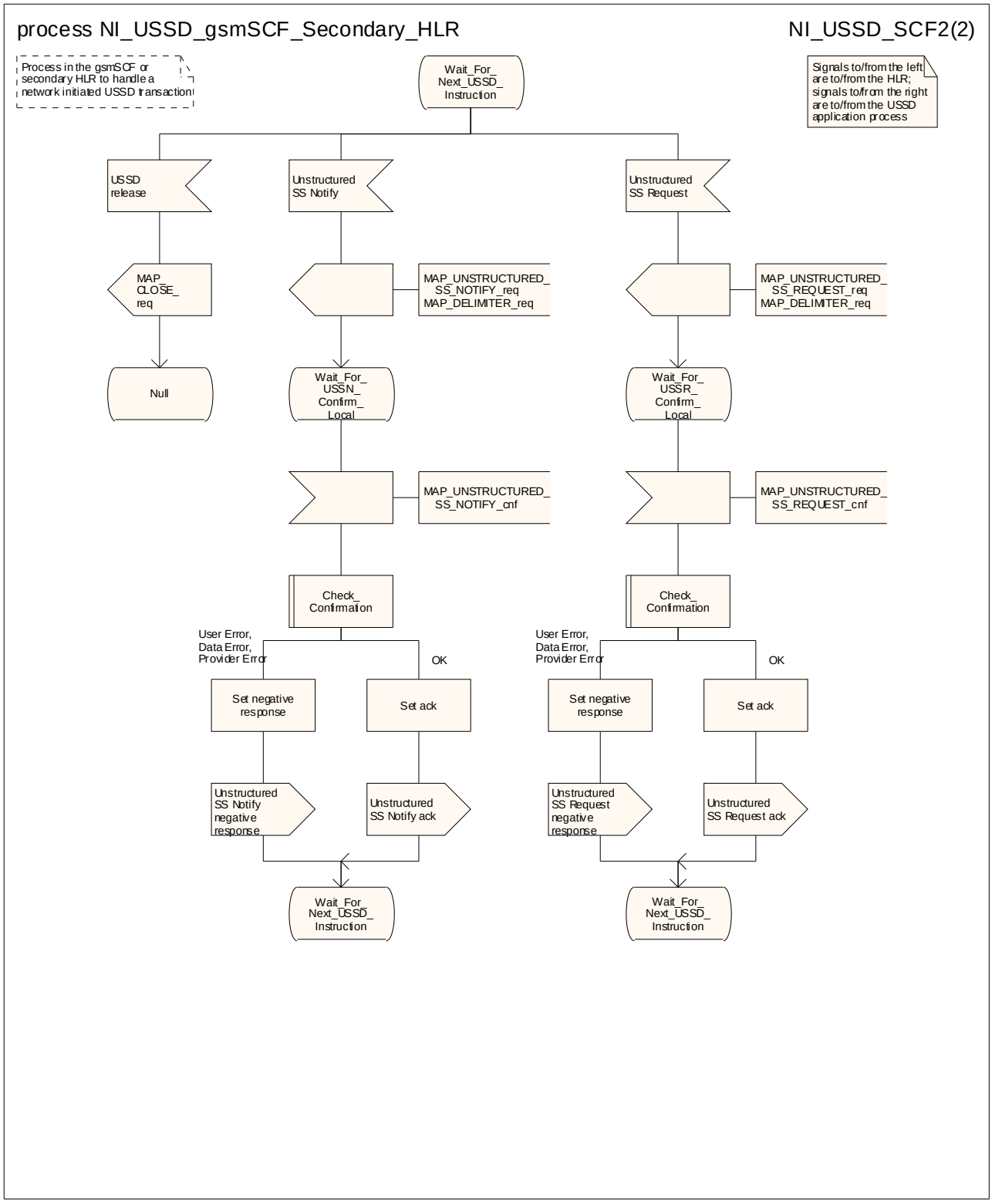


Figure 22.10.5/1 (sheet 2 of 2): Process NW_Init_USSD_gsmSCF_Secondary_HLR

22.11 Common macros for clause 22

The following macros are used for the description of more than one of the supplementary service processes described in clause 22.

22.11.1 SS Password handling macros

Macro Get_Password_MSC

This macro is used by the MSC to relay a request for password from the VLR to the MS, and to relay a response from the MS back to the VLR. The macro is shown in figure 22.11.1/1.

Macro Get_Password_VLR

This macro is used by the VLR to relay a request for password from the HLR to the MSC, and to relay a response from the MSC back to the HLR. The macro invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

The macro is shown in figure 22.11.1/2.

22.11.2 Void

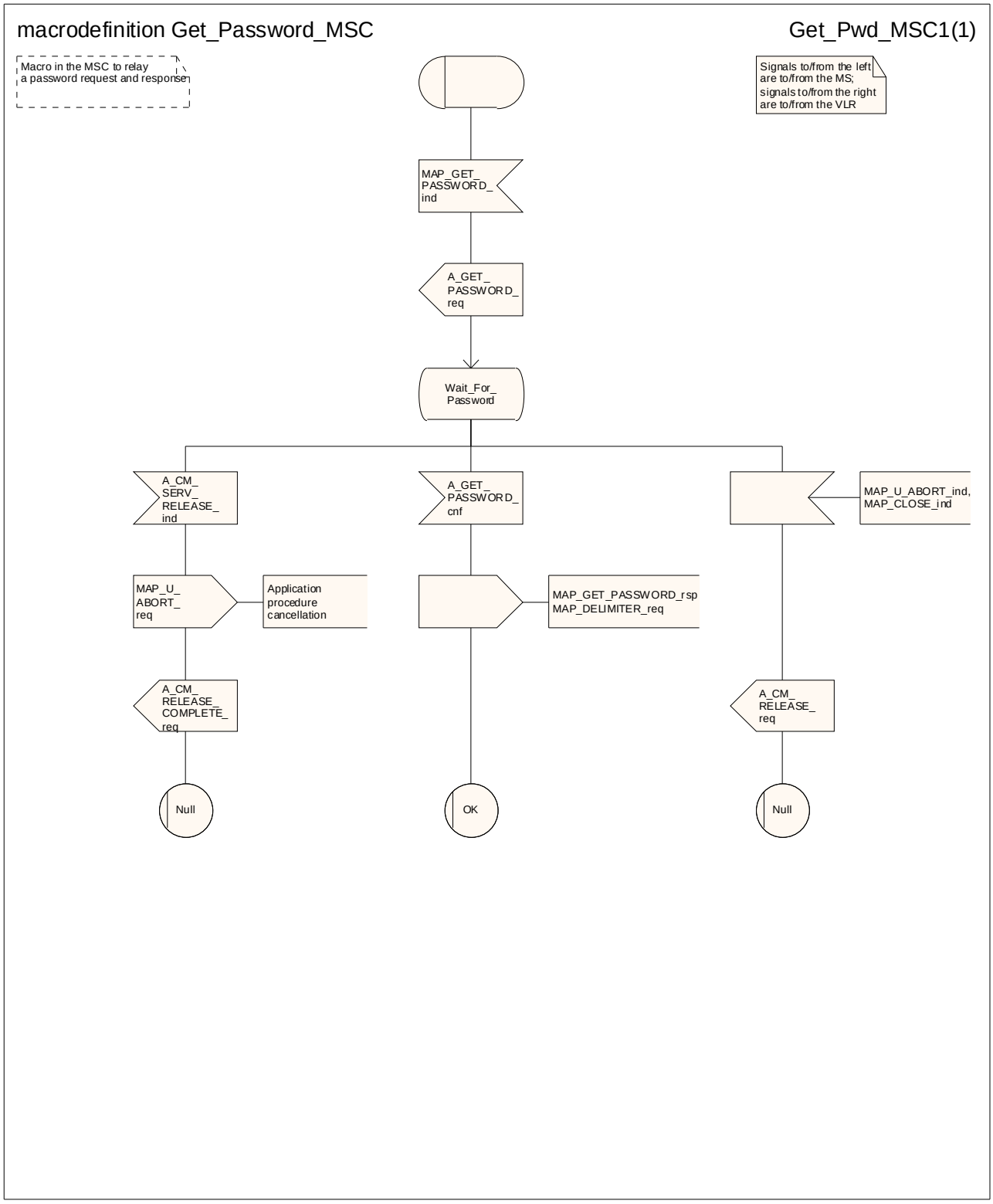


Figure 22.11.1/1: Macro Get_Password_MSC

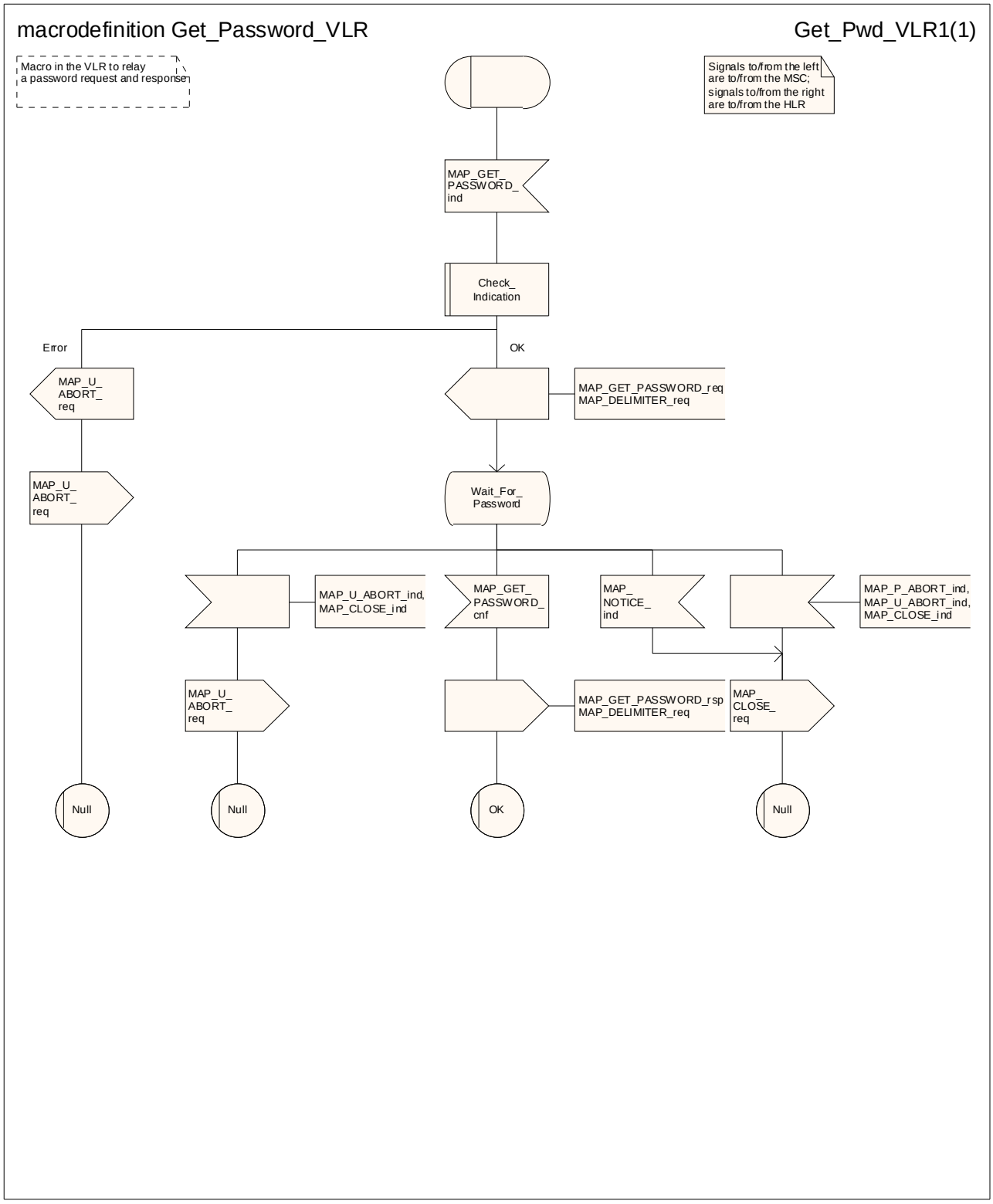


Figure 22.11.1/2: Macro Get_Password_VLR

Figure 22.11.2/1 void

Figure 22.11.2/2 void

Figure 22.11.2/3 void

Figure 22.11.2/4 void

Figure 22.11.2/5 void

22.12 Supplementary Service Invocation Notification procedure

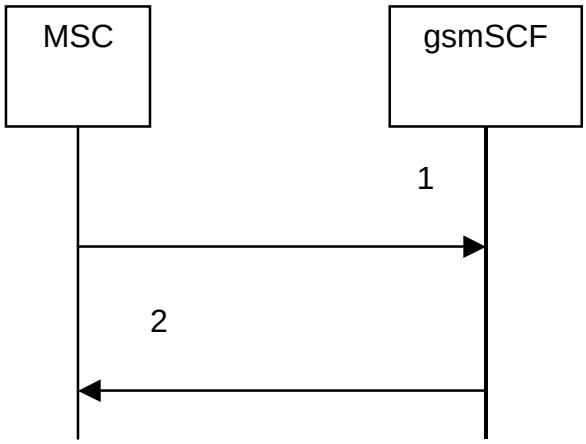
22.12.1 General

The Supplementary Service Invocation Notification procedure is used to notify a gsmSCF about the invocation of a GSM Supplementary Service.

The supplementary service invocation notification procedure is shown in figure 22.12.1/1.

The following service is certainly used:

MAP_SS_INVOCATION_NOTIFY (defined in clause 11).



1) MAP_SS_INVOCATION_NOTIFY_req/ind
2) MAP_SS_INVOCATION_NOTIFY_rsp/cnf

Figure 22.12.1/1: Message flow for supplementary service invocation notification

22.12.2 Procedure in the MSC

The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The supplementary service invocation notification process in the MSC is shown in figure 22.12.2/1.

22.12.3 Procedure in the gsmSCF

The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

The supplementary service invocation notification process in thegsmSCF is shown in figure 22.12.3/1.

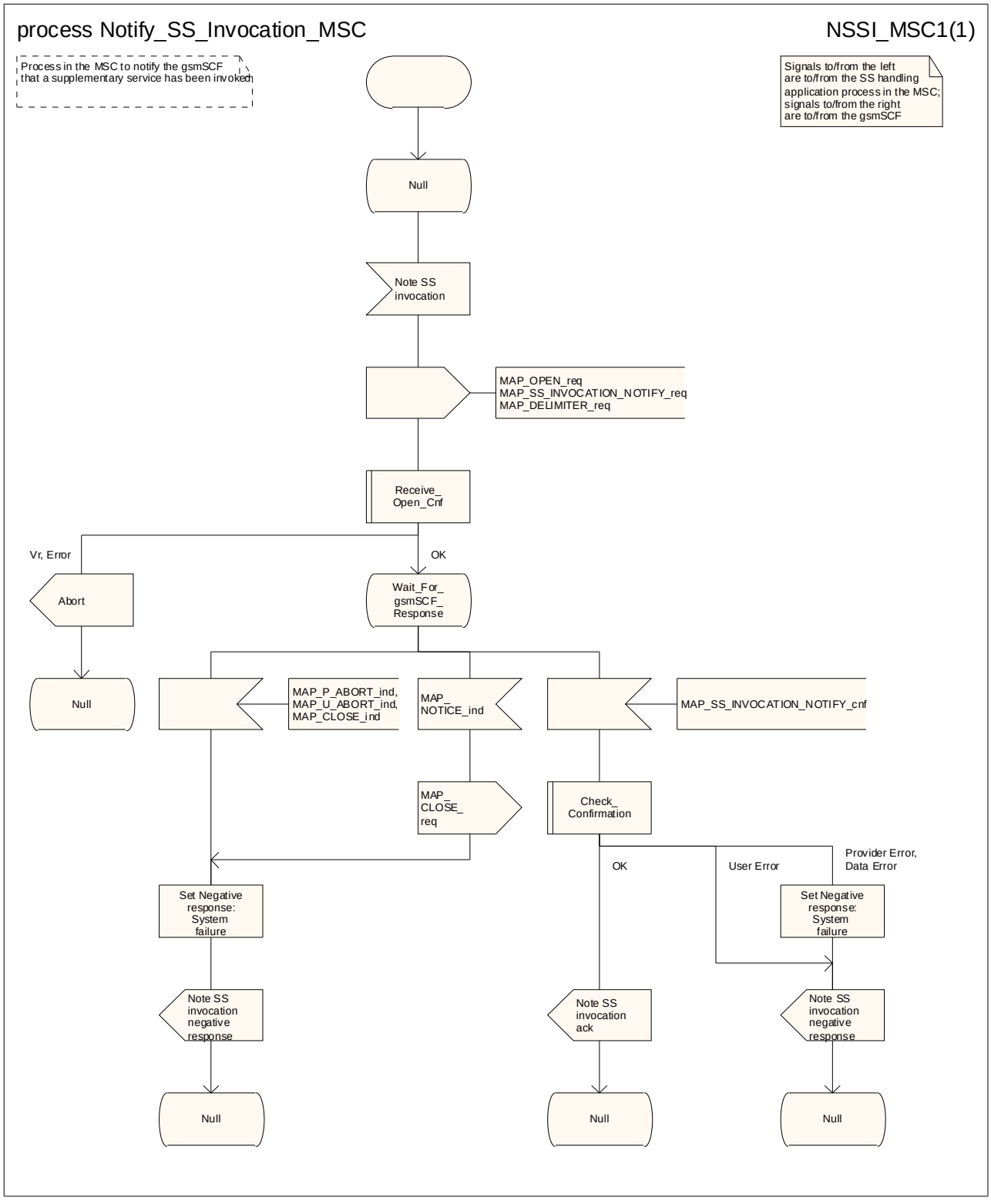


Figure 22.12.2/1: Process Notify_SS_Invocation_MSC

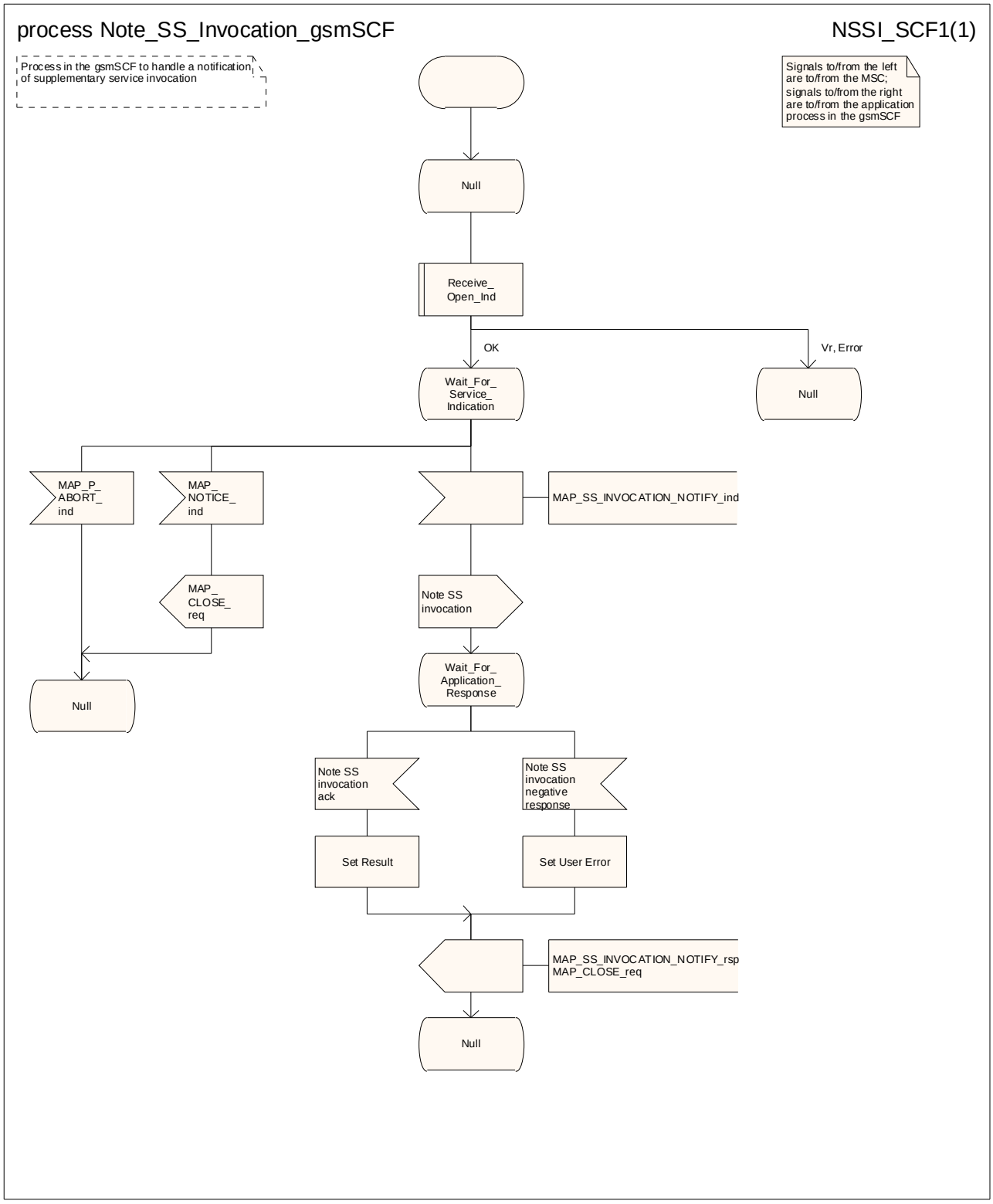


Figure 22.12.3/1: Process Note_SS_Invocation_gsmSCF

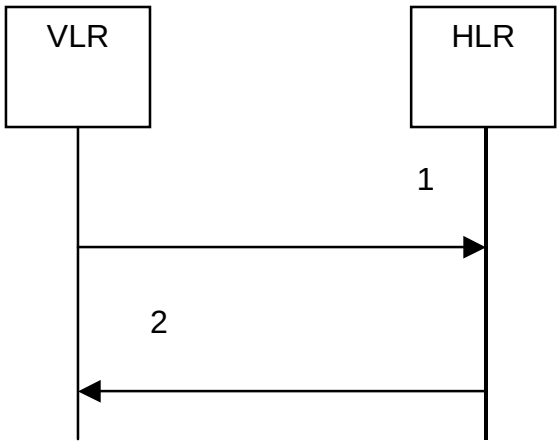
22.13 Activation of a CCBS request

22.13.1 General

The message flow to activate a CCBS request is shown in figure 22.13.1/1.

The following service is certainly used:

MAP_REGISTER_CC_ENTRY (defined in clause 11).



1) MAP_REGISTER_CC_ENTRY_req/ind
2) MAP_REGISTER_CC_ENTRY_rsp/cnf

Figure 22.13.1/1: Message flow to activate a CCBS request

22.13.2 Procedure in the VLR

The MAP process in the VLR to activate a CCBS request is shown in figure 22.13.2/1. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

22.13.3 Procedure in the HLR

The MAP process in the HLR to activate a CCBS request is shown in figure 22.13.2/1.

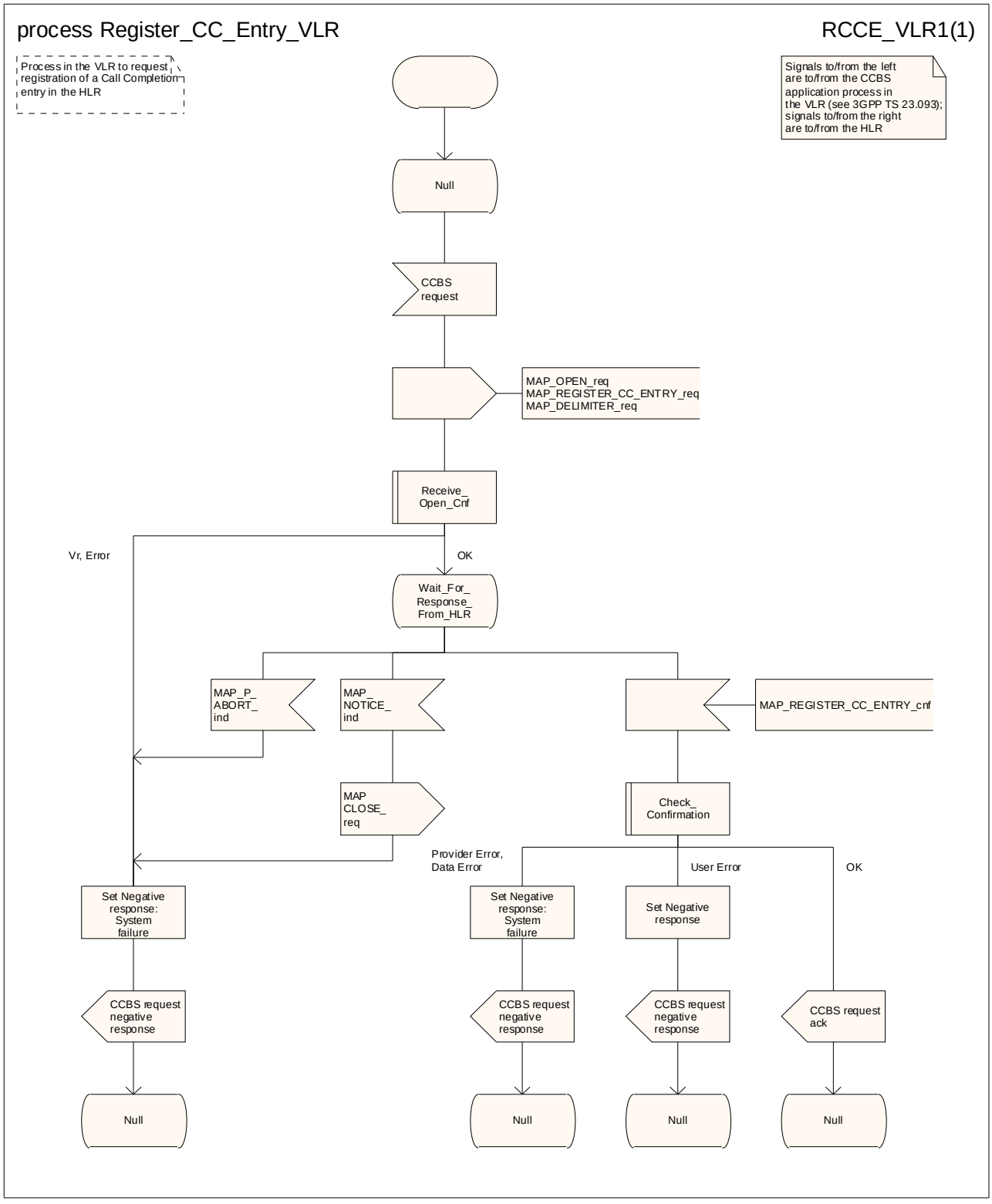


Figure 22.13.2/1: Process Register_CC_Entry_VLR

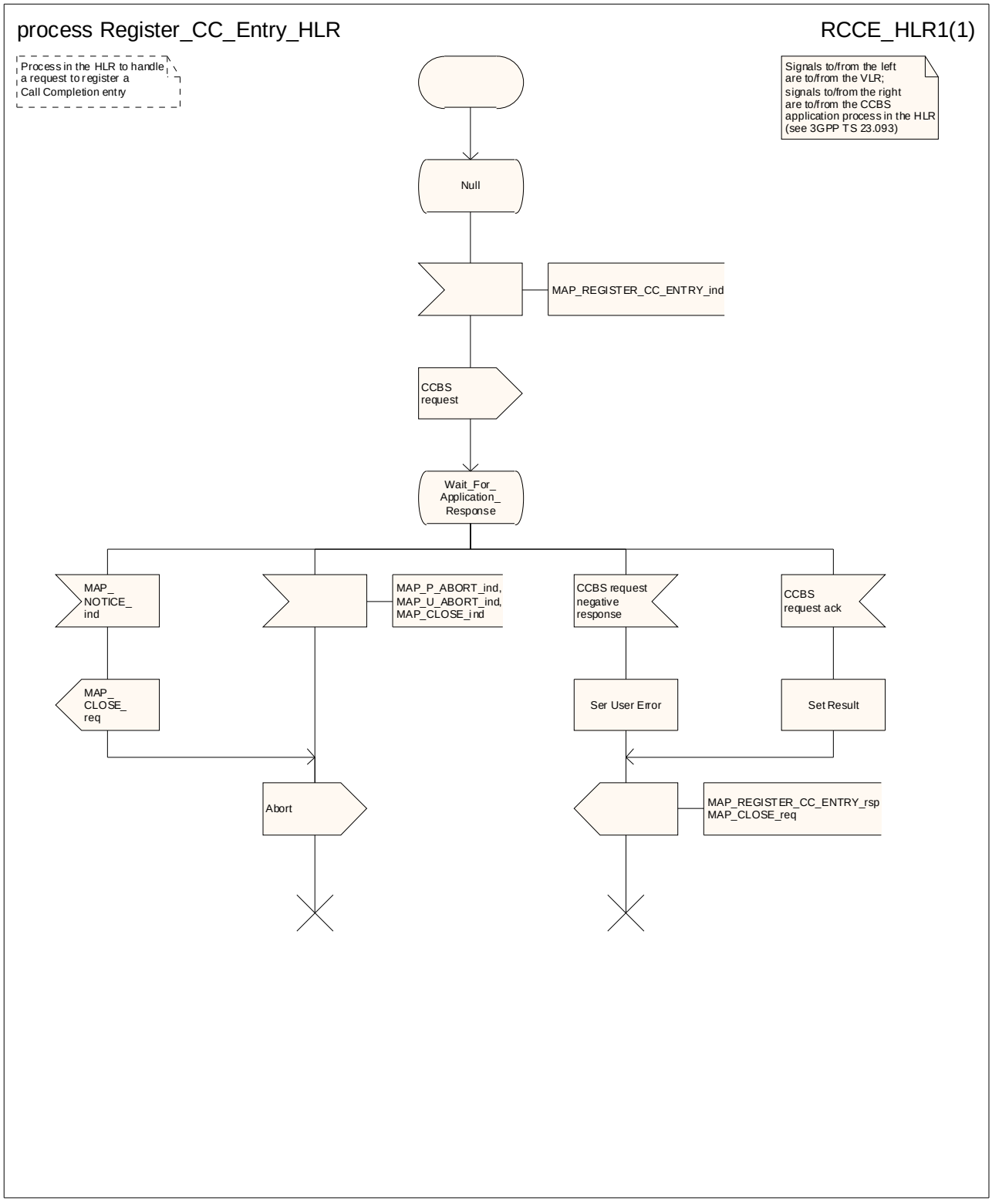


Figure 22.13.3/1: Process Register_CC_Entry_HLR

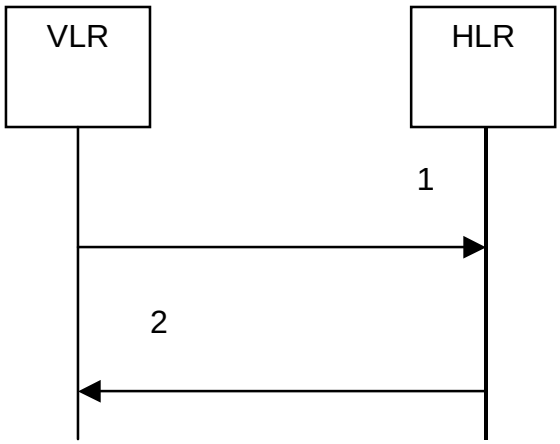
22.14 Deactivation of a CCBS request

22.14.1 General

The message flow to deactivate a CCBS request is shown in figure 22.14.1/1.

The following service is certainly used:

MAP_ERASE_CC_ENTRY (defined in clause 11).



1) MAP_ERASE_CC_ENTRY_req/ind
2) MAP_ERASE_CC_ENTRY_rsp/cnf

Figure 22.14.1/1: Message flow to deactivate a CCBS request

22.14.2 Procedure in the VLR

The MAP process in the VLR to deactivate a CCBS request is shown in figure 22.14.2/1. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

22.14.3 Procedure in the HLR

The MAP process in the HLR to deactivate a CCBS request is shown in figure 22.14.2/1.

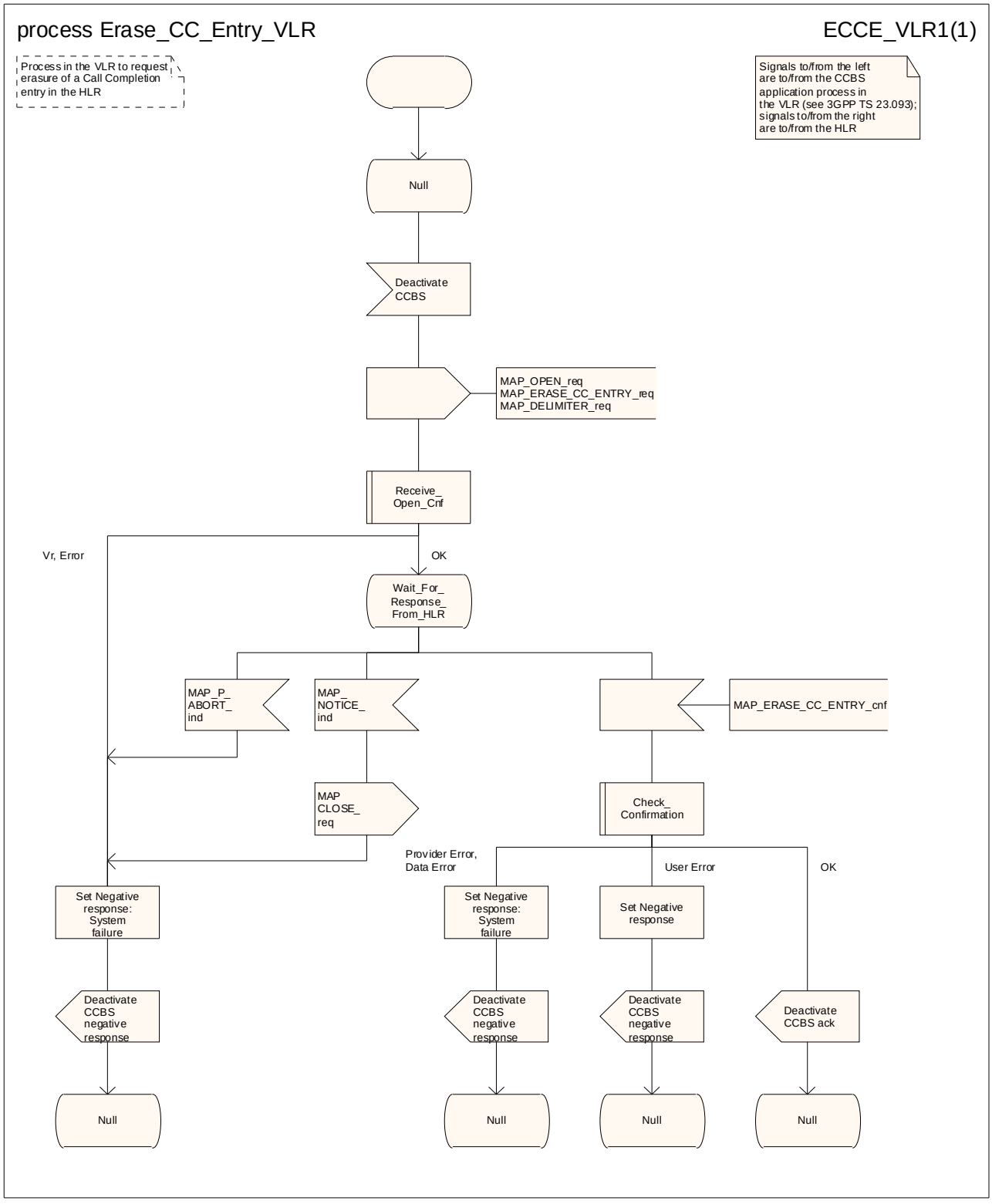


Figure 22.14.2/1: Process Erase_CC_Entry_VLR

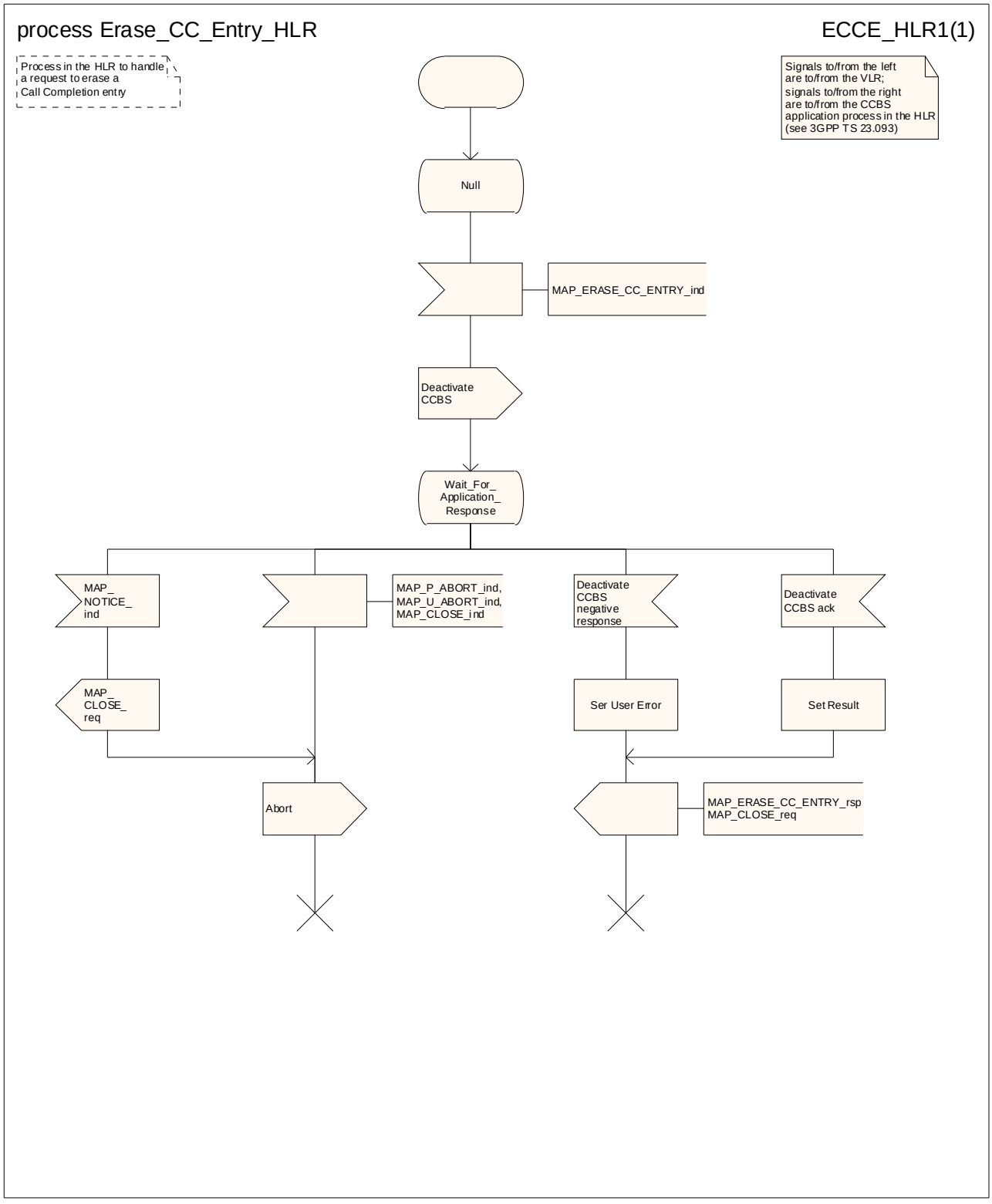


Figure 22.14.3/1: Process Erase_CC_Entry_HLR

23 Short message service procedures

23.1 General

The short message service procedures are used to control both mobile originated and mobile terminated short message transfer.

Four procedures exist for short message services:

- mobile originated short message service transfer;
- mobile terminated short message service transfer;
- short message alert procedure;
- short message delivery status report procedure.

The following application context refers to a complex MAP user consisting of several processes:

- shortMessageGatewayContext.

This application context needs a co-ordinating process in the HLR. Additionally a co-ordinating process needed for the mobile originated situation in the MSC, because the A_CM_SERV_REQ message does not distinguish between mobile originated short message transfer and the short message alert procedures.

NOTE: the A_CM_SERV_REQ message is not used for SMS over GPRS. The modelling is based on the assumption that the SGSN will trigger the appropriate process, according to whether an RP_MO_DATA or an RP_SM_MEMORY_AVAILABLE is received over the LLC layer.

When the "SMS in MME" architecture option is supported, the subclause 23 and its associated figures applies where HSS replaces HLR and where MME is handled as a MSC, except the procedures between the serving MSC and the VLR which, here, are not applicable to the MME.

NOTE: MSC and MME cannot be both registered as serving nodes for MT SMS at a given time (cf 3GPP TS 23.272 [2])

23.1.1 Mobile originated short message service Co-ordinator for the MSC

The process starts when the MSC receives an A_CM_SERV_REQ message (see 3GPP TS 24.008 [35]), with a CM service type indicating short message service, from the A-interface. The process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Process_Access_Request_MSC see subclause 25.4.1.

If the macro Process_Access_Request_MSC takes the "OK" exit (which means that the MSC has sent an A_CM_SERVICE_ACCEPT to the MS), , the MS initiates mobile originated short message transfer or sends an indication that it has memory available for more short messages.

The SMS Co-ordinator process in the MSC is shown in figure 23.1/1.

23.1.2 Short message Gateway Co-ordinator for the HLR

The process starts when the HLR receives a MAP_OPEN indication using when the application context shortMessageGatewayContext. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

The SM Gateway Co-ordinator process in the HLR is shown in figure 23.1/2.

If the Receive_Open_Ind macro takes the Vr exit then HLR shall perform the MAP dialogue as specified for the appropriate application context version. Depending on the subscriber data, handling at the MAP user application level may be performed as specified in subclauses 23.3.2 and 23.5.2 of the present document:

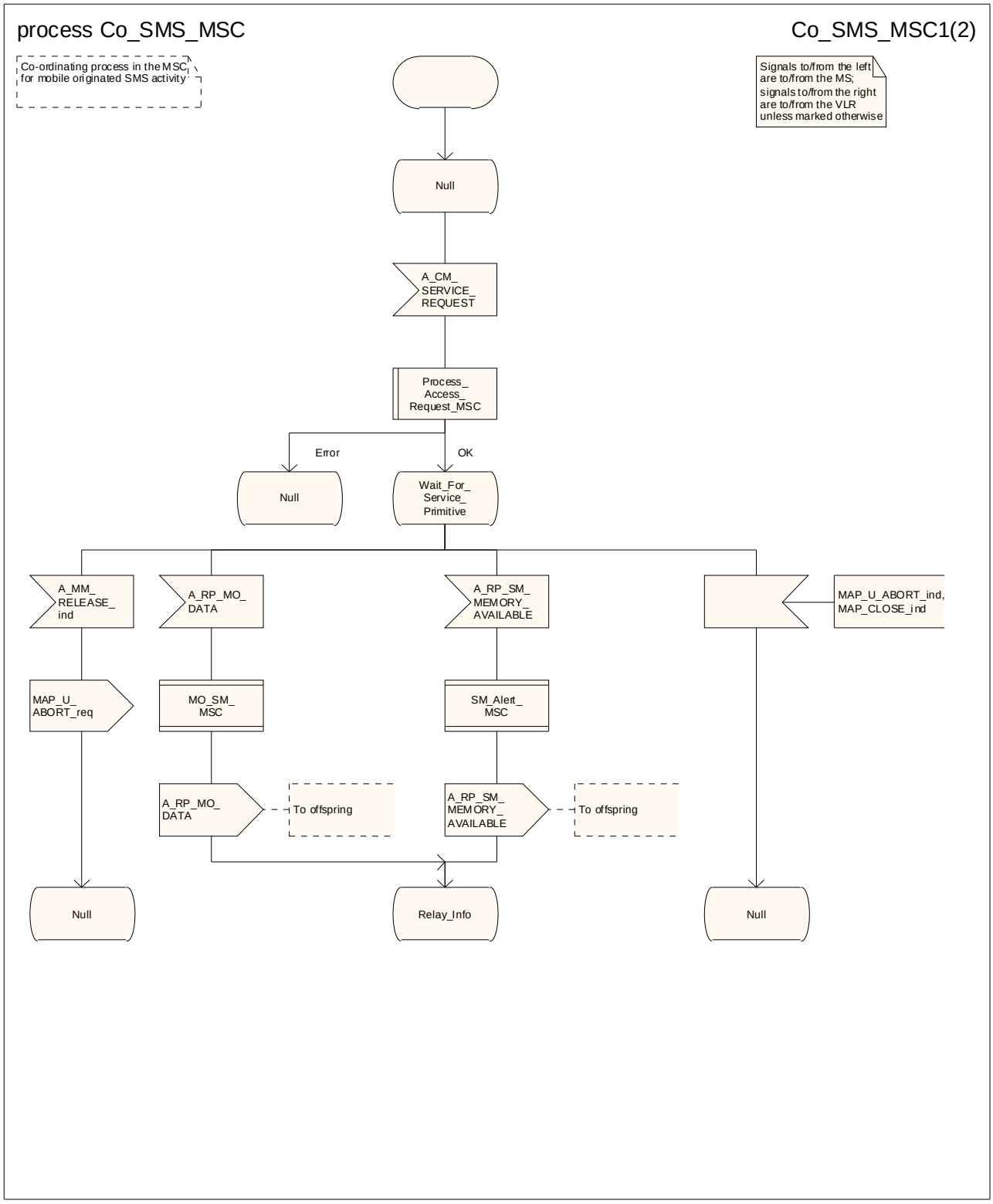


Figure 23.1/1 (sheet 1 of 2): Process Co_SMS_MSC

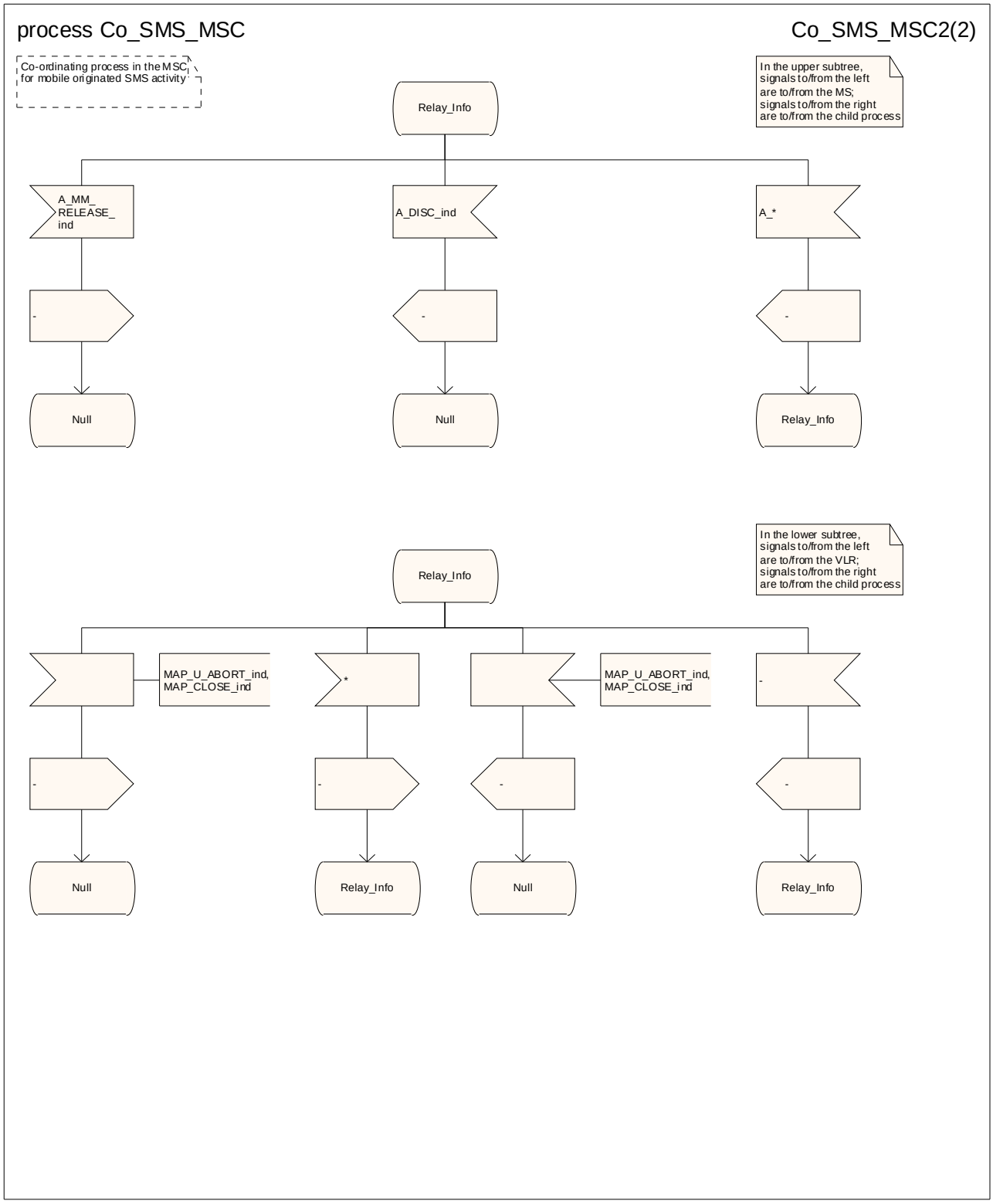


Figure 23.1/1 (sheet 2 of 2): Process Co_SMS_MSC

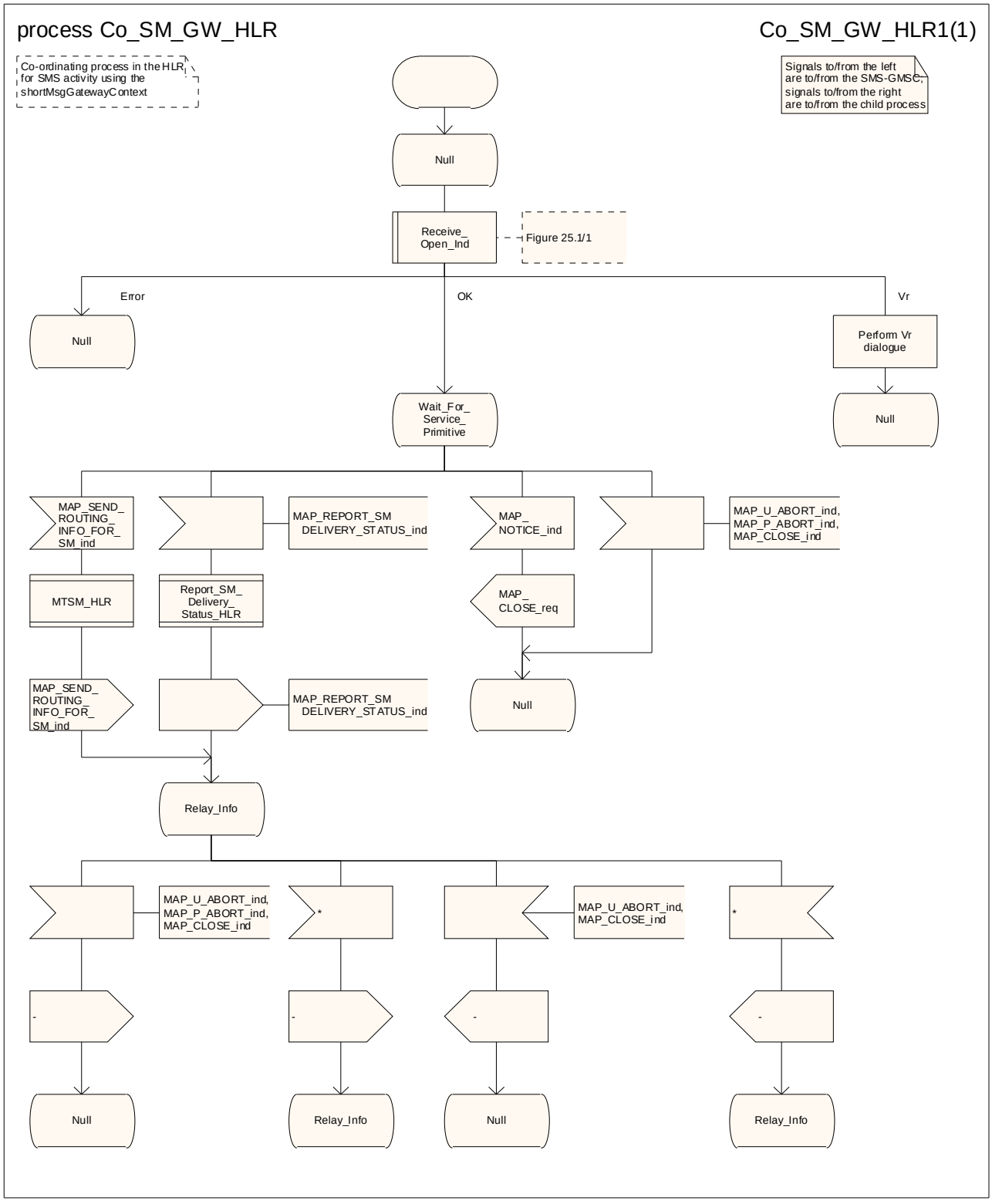
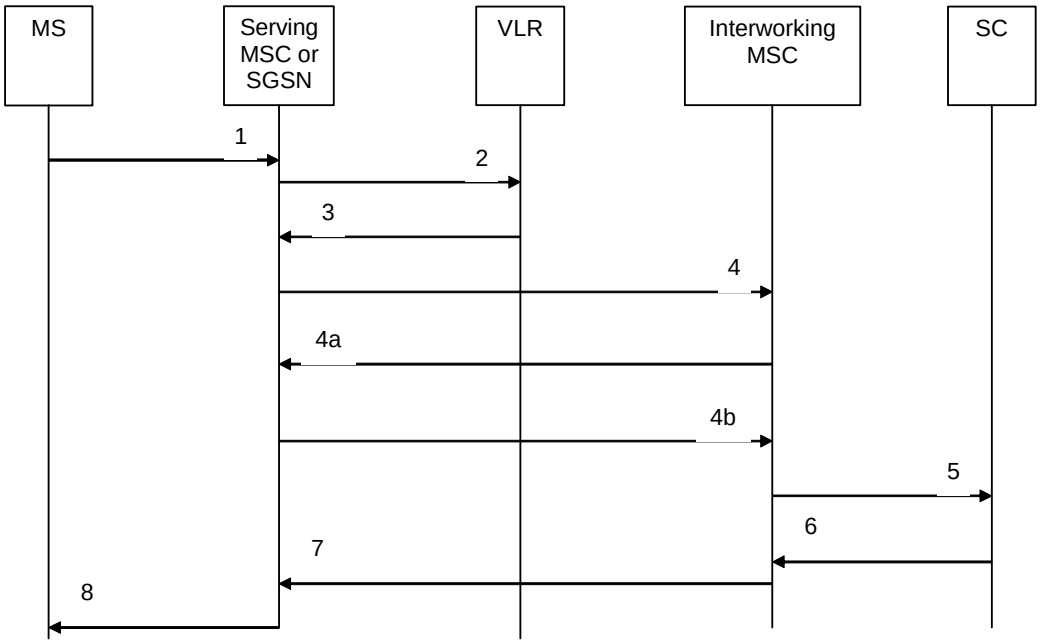


Figure 23.1/2: Process Co_SM_Gateway_HLR

23.2 The mobile originated short message transfer procedure

The mobile originated short message service procedure is used to forward a short message from a mobile subscriber to a Service Centre. The message flow for the mobile originated short message service procedure is shown in figure 23.2/1.



1) Short Message (3GPP TS 24.011 [37]).
2) MAP_SEND_INFO_FOR_MO_SMS (*).
3) MAP_SEND_INFO_FOR_MO_SMS_ACK (*).
4) TCAP BEGIN (**)
4a) TCAP CONTINUE (**)
4b) MAP_MO_FORWARD_SHORT_MESSAGE.
5) Short message (3GPP TS 23.040).
6) Short message Acknowledgement (3GPP TS 23.040).
7) MAP_MO_FORWARD_SHORT_MESSAGE_ACK.
8) Short Message Acknowledgement (3GPP TS 24.011 [37]).
(*) Messages 2) and 3) are not used by the SGSN.
(**) If
a)
the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MO_FORWARD_SHORT_MESSAGE request in a single TC message

and
b) the Interworking MSC operator and the serving node (MSC or SGSN) operator agreed not to use the TCAP handshake countermeasure against SMS fraud for messages exchanged between their networks (see 3GPP TS 33.204 [34a])

then
the TCAP handshake may be omitted.

Figure 23.2/1: Mobile originated short message transfer

In addition the following MAP services are used:

MAP_PROCESS_ACCESS_REQUEST	(see subclause 8.3); (*)
MAP_AUTHENTICATE	(see subclause 8.5); (*)
MAP_SET_CIPHERING_MODE	(see subclause 8.6); (*)
MAP_PROVIDE_IMSI	(see subclause 8.9); (*)

MAP_CHECK_IMEI (see subclause 8.7);

MAP_FORWARD_NEW_TMSI (see subclause 8.9); (*)

MAP_TRACE_SUBSCRIBER_ACTIVITY (see subclause 9.1); (*)

MAP_READY_FOR_SM (see subclause 12.4).

(*) These services are not used by the SGSN.

23.2.1 Procedure in the serving MSC

Any CAMEL-specific handling defined in this subclause is omitted if the MSC does not support CAMEL control of MO SMS, or if the subscriber does not have a subscription for CAMEL control of MO SMS.

The process starts when the MSC receives a short message from the MS. The process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2.

Sheet 1: If the MSC is integrated with the SMS-IWMSC, it communicates directly with the Short Message Service Centre (SMSC) using one of the protocols described in 3GPP TS 23.039 [25a]; otherwise it communicates with the SMS-IWMSC using MAP.

Sheet 3: If the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MO_FORWARD_SHORT_MESSAGE request in a single TC message, the test "Message segmentation needed" takes the "No" exit; otherwise the test takes the "Yes" exit.

Sheet 3:The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the serving MSC's operator and the SMS-IWMSC's operator (see 3GPP TS 33.204 [34a]).

The mobile originated short message service process in the MSC is shown in figure 23.2/2.

23.2.2 Procedure in the VLR

Any CAMEL-specific handling defined in this subclause is omitted if the VLR does not support CAMEL control of MO SMS.

The process starts when the VLR receives a dialogue opening request followed by a MAP_PROCESS_ACCESS_REQUEST including a CM service type Short Message Service. The process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1;

Process_Access_Request_VLR see subclause 25.4.2.

The mobile originated short message transfer process in the VLR is shown in figure 23.2/3.

23.2.3 Procedure in the SGSN

Any CAMEL-specific handling defined in this subclause is omitted if the SGSN does not support CAMEL control of MO SMS, or if the subscriber does not have a subscription for CAMEL control of MO SMS.

The process starts when the SGSN receives a short message received from the MS over the Gb interface. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Sheet 2: If the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MO_FORWARD_SHORT_MESSAGE request in a single TC message, the test "Message segmentation needed" takes the "No" exit; otherwise the test takes the "Yes" exit.

Sheet 2:The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the serving SGSN's operator and the SMS-IWMSC's operator (see 3GPP TS 33.204 [34a]).

The mobile originated short message service process in the SGSN is shown in figure 23.2/4.

23.2.4 Procedure in the SMS Interworking MSC (SMS-IWMSC)

This procedure applies only when the SMS-IWMSC is not integrated with the serving MSC or SGSN.

The process starts when the SMS-IWMSC receives a dialogue opening request with the application context shortMsgMO-RelayContext. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

Sheet 1:The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the SMS-IWMSC's operator and the serving node's operator (see 3GPP TS 33.204 [34a]).

The mobile originated short message service transfer process in the SMS-IWMSC is shown in figure 23.2/5.

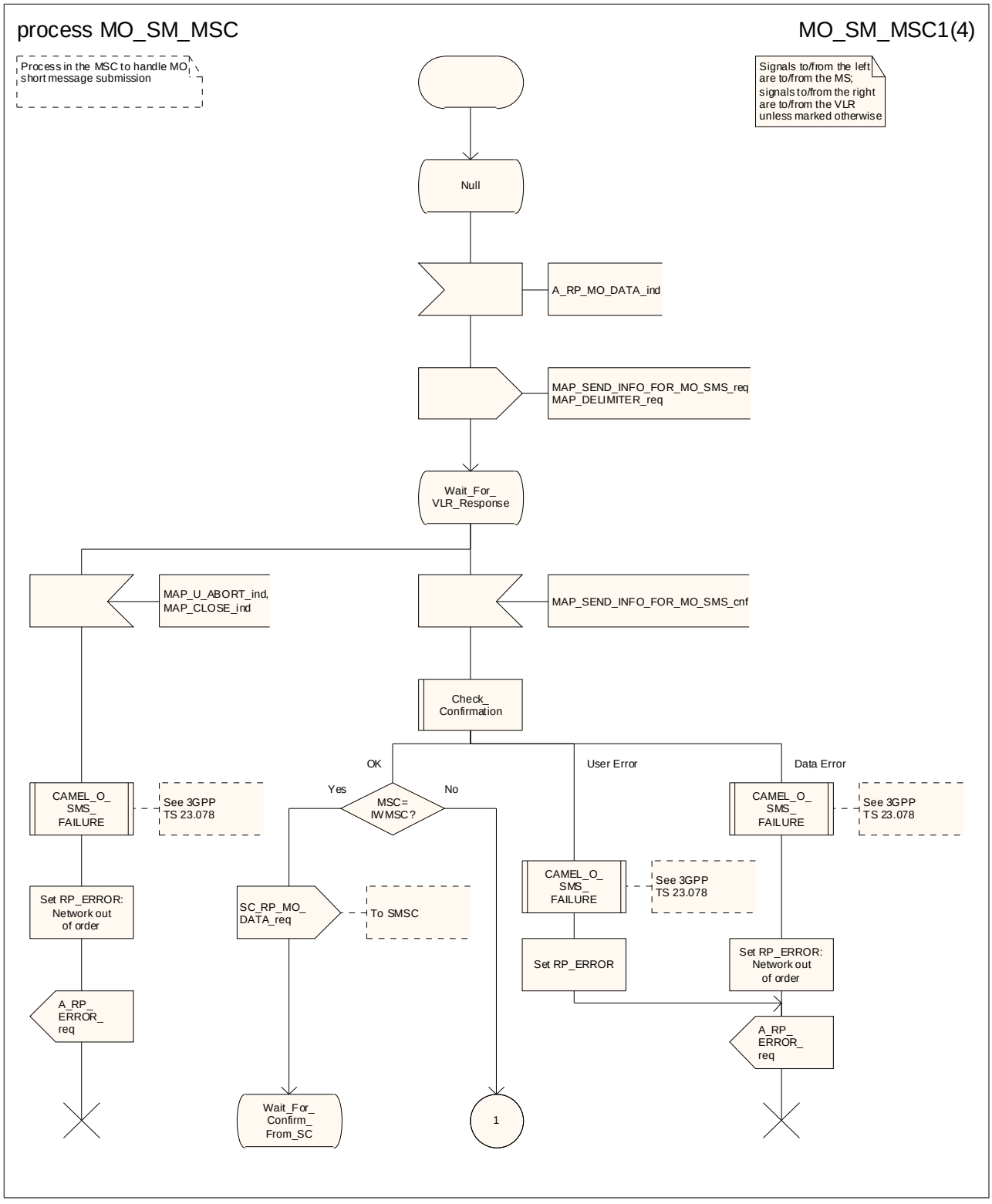


Figure 23.2/2 (sheet 1 of 4): Process MO_SM_MSC

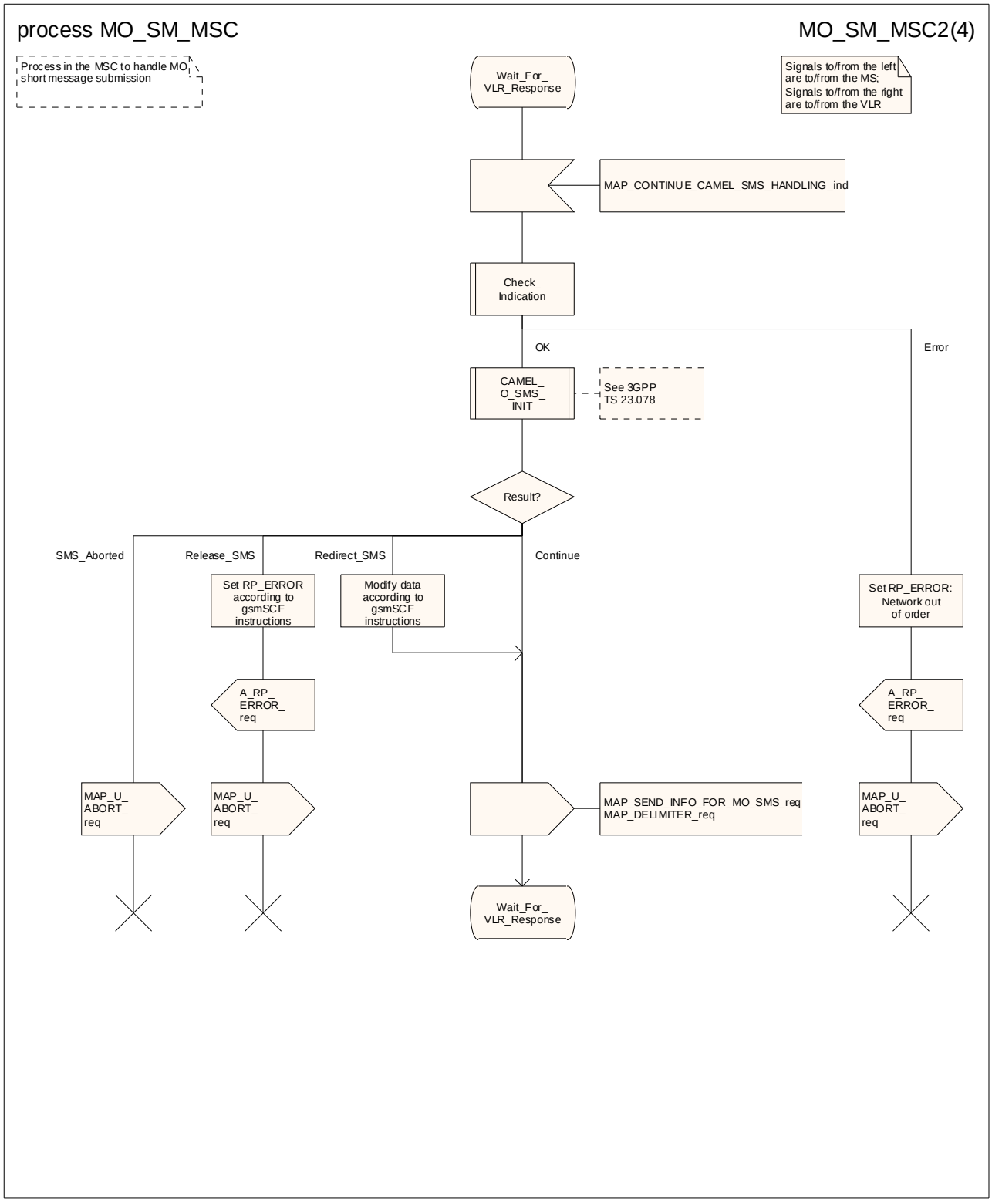


Figure 23.2/2 (sheet 2 of 4): Process MO_SM_MSC

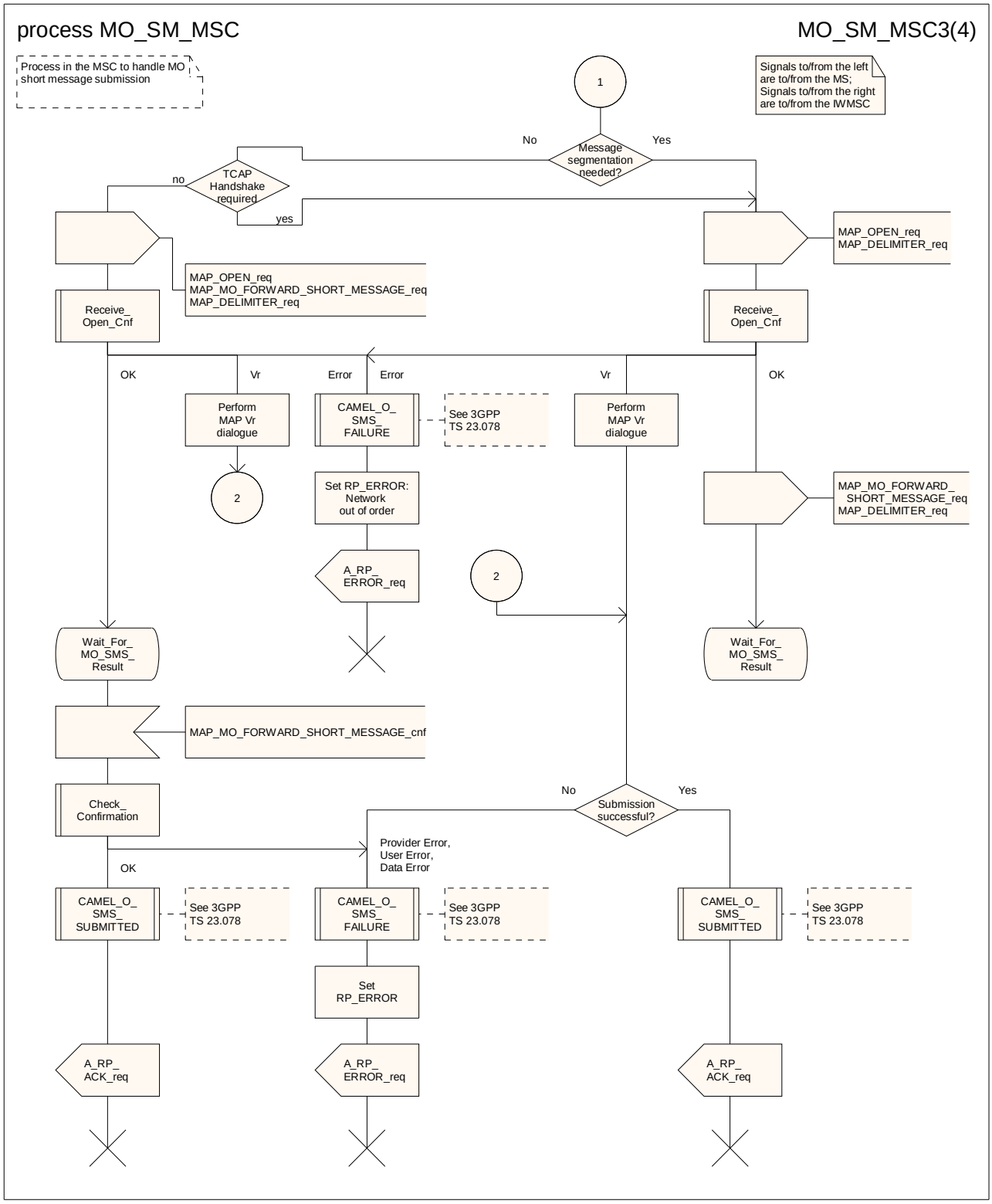


Figure 23.2/2 (sheet 3 of 4): Process MO_SM_MSC

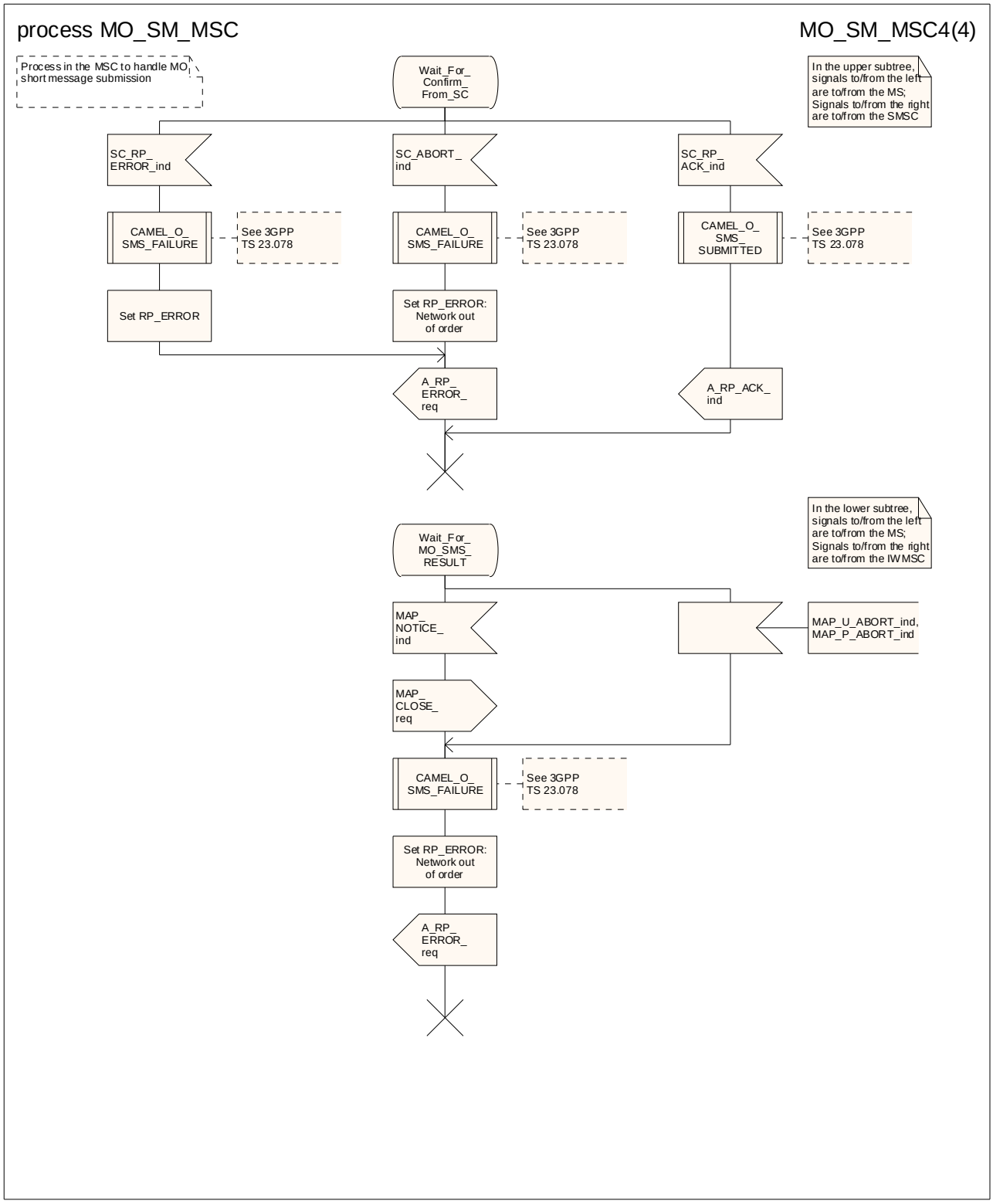


Figure 23.2/2 (sheet 4 of 4): Process MO_SM_MSC

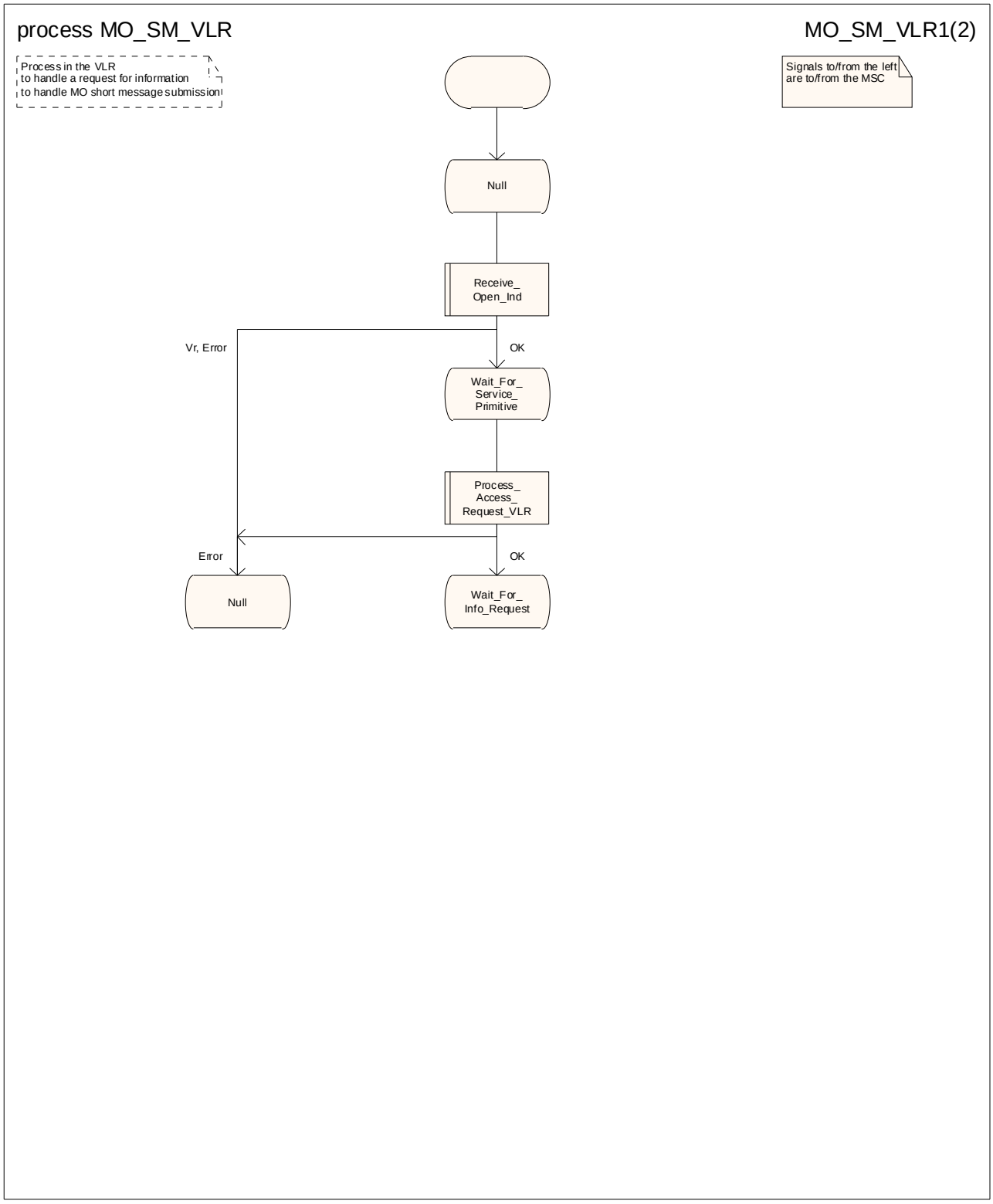


Figure 23.2/3 (sheet 1 of 2): Process MOSM_VLR

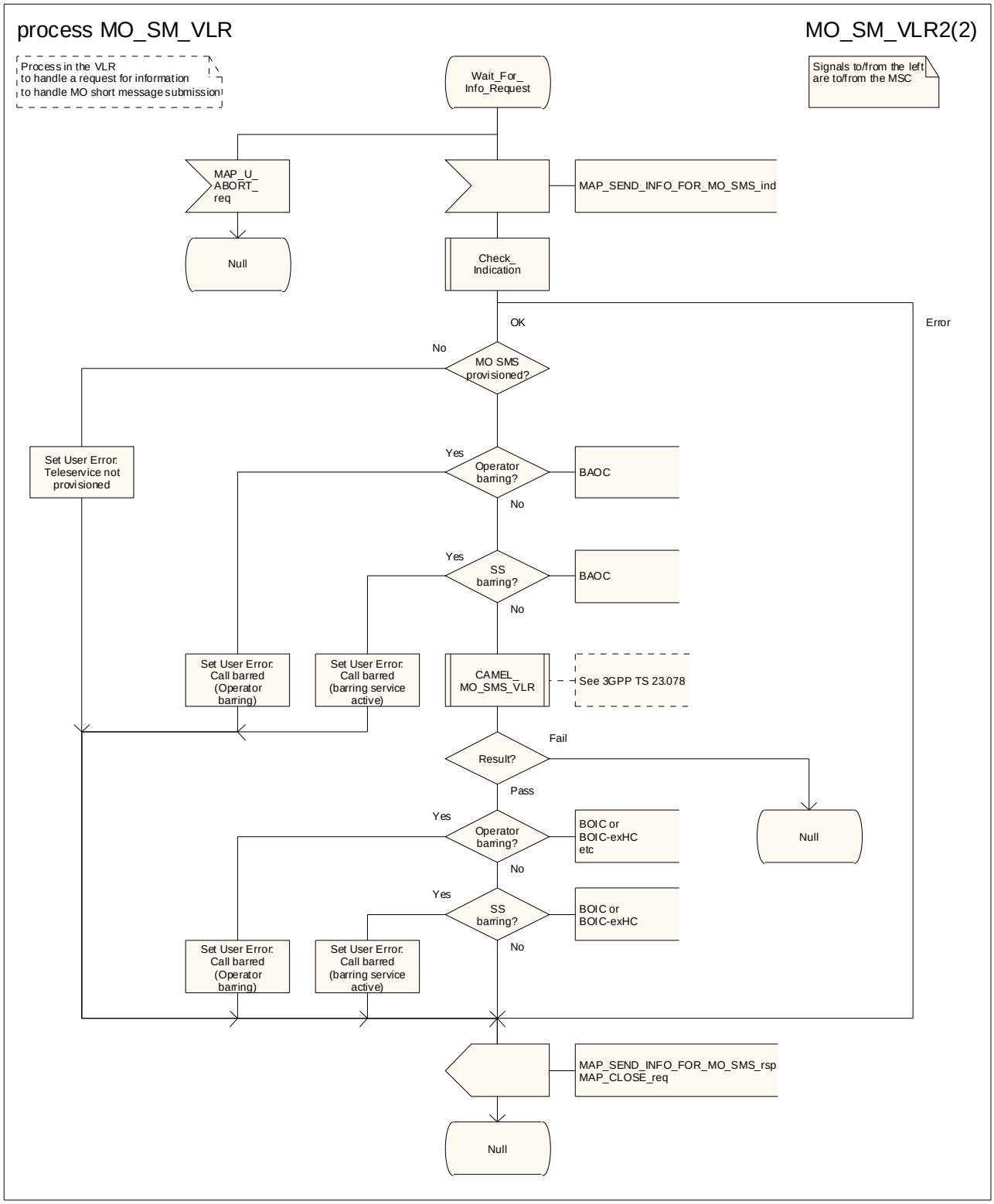


Figure 23.2/3 (sheet 2 of 2): Process MO_SM_VLR

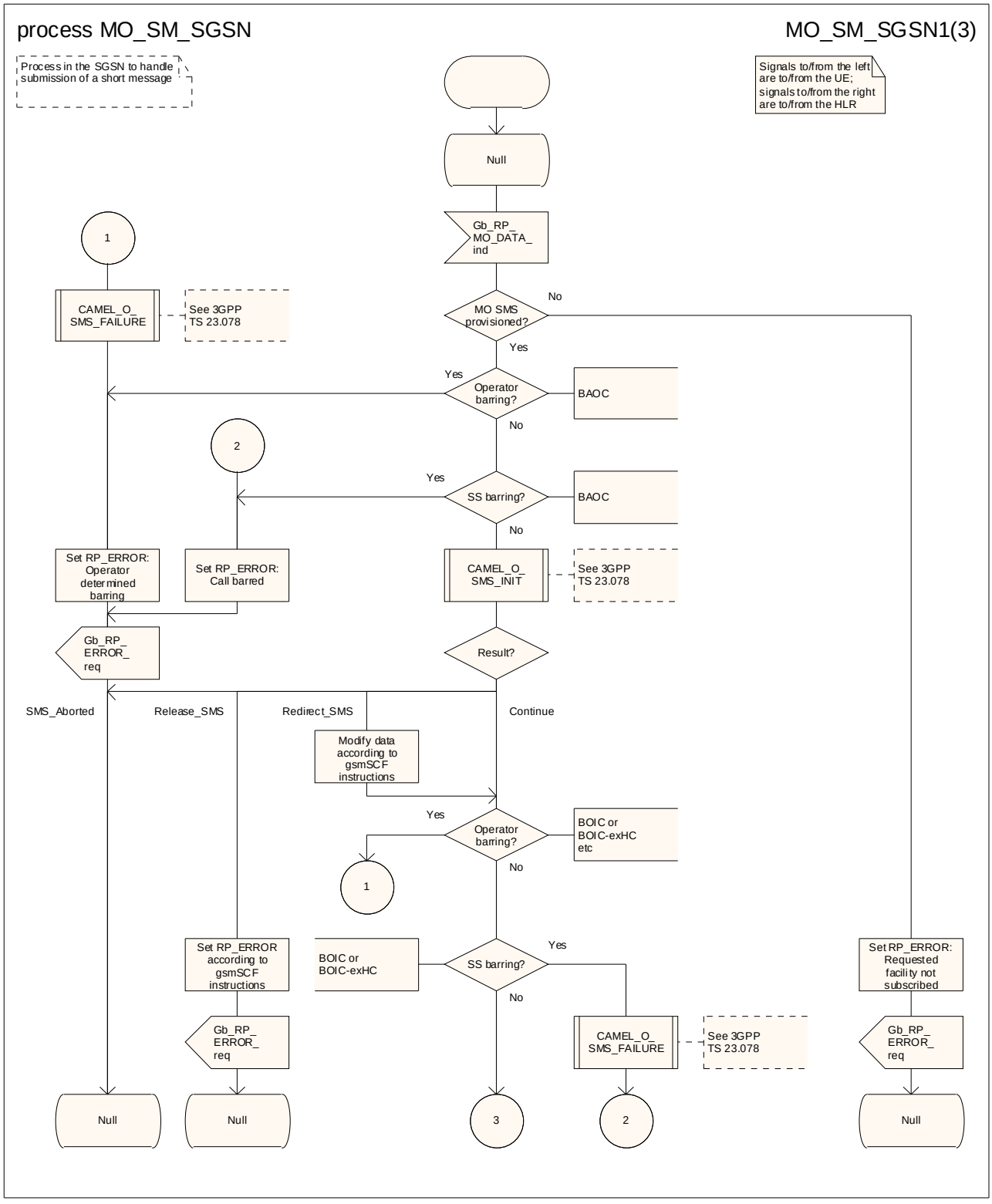


Figure 23.2/4 (sheet 1 of 3): Process MO_SM_SGSN

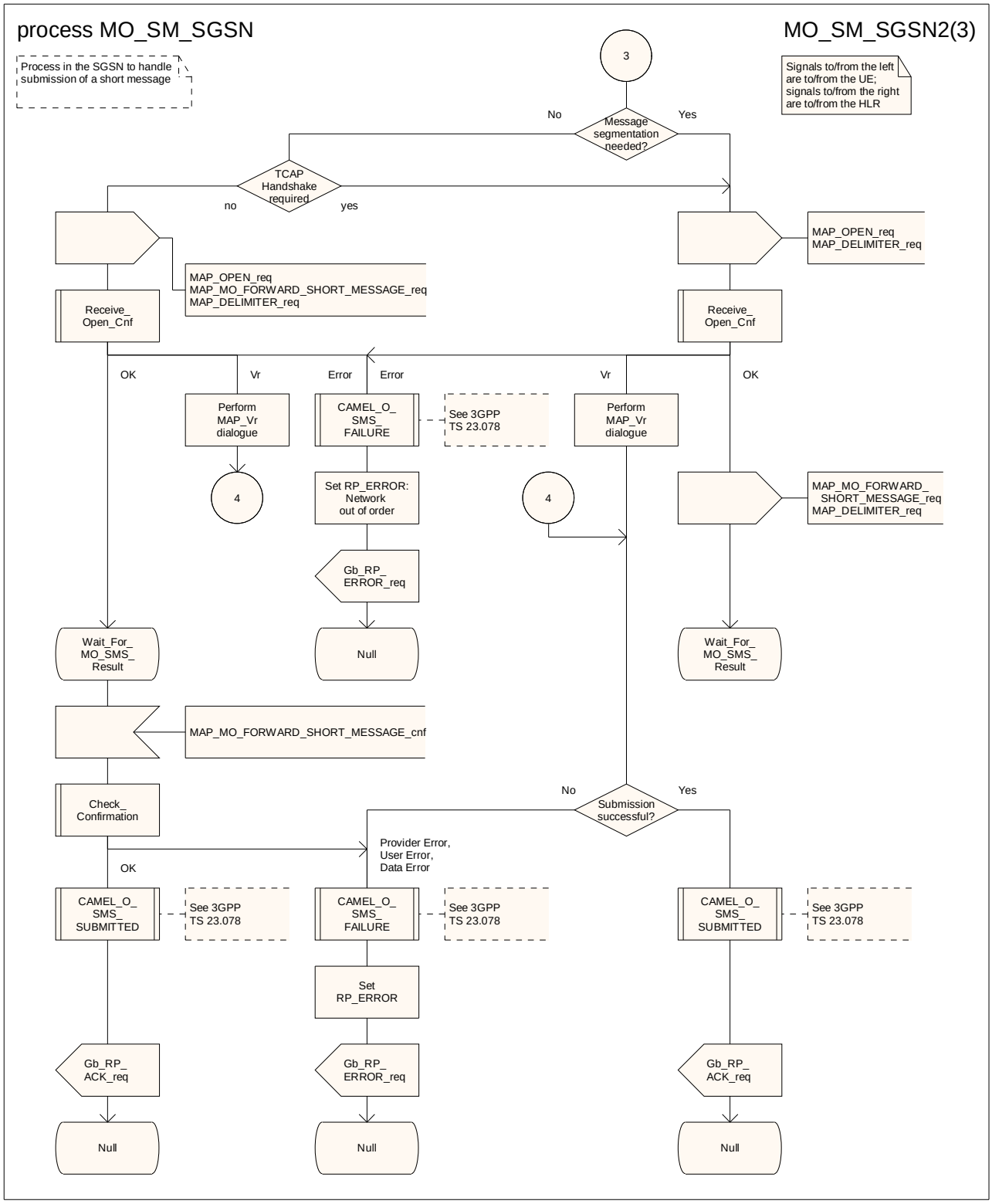


Figure 23.2/4 (sheet 2 of 3): Process MO_SM_SGSN

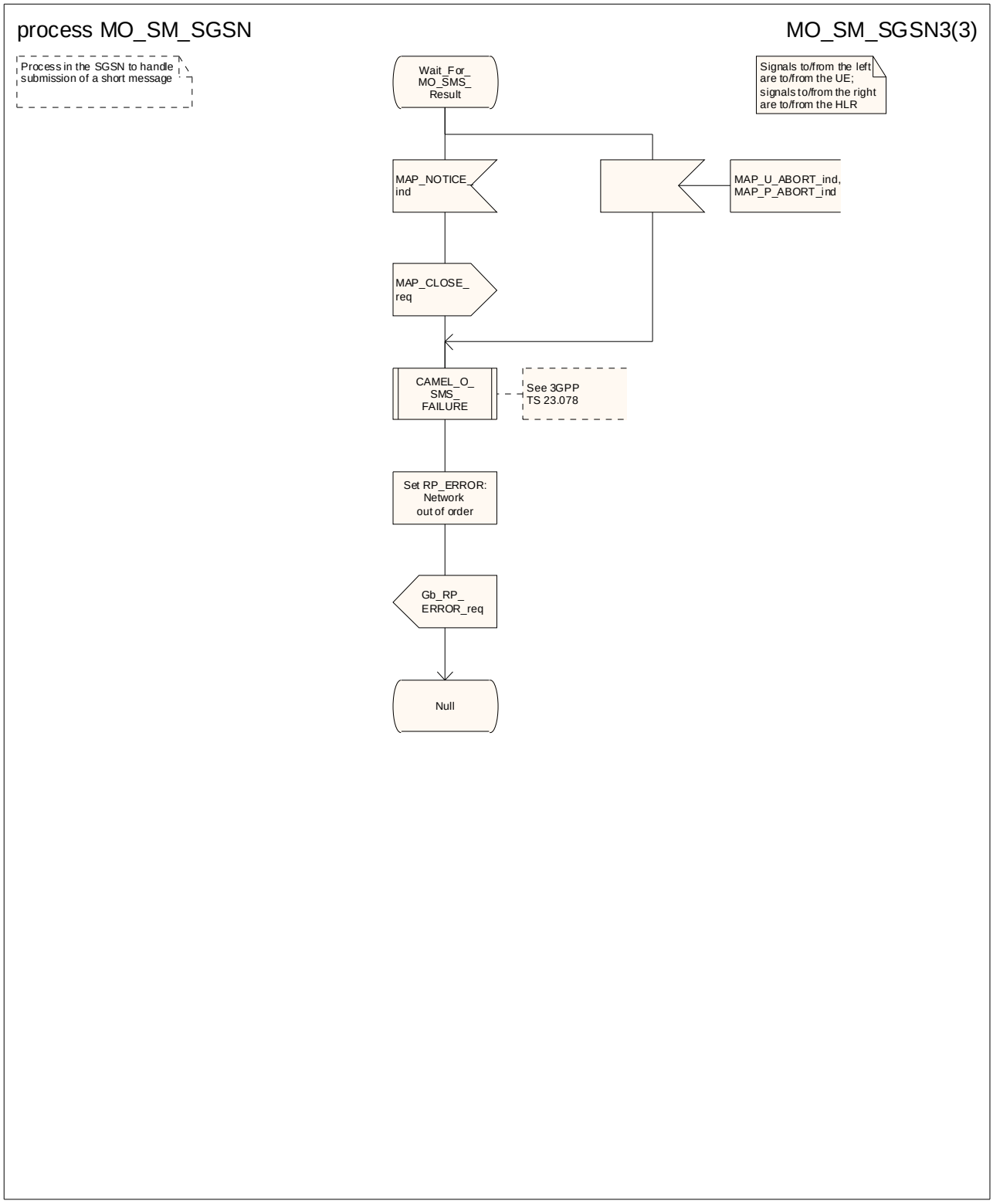


Figure 23.2/4 (sheet 3 of 3): Process MO_SM_SGSN

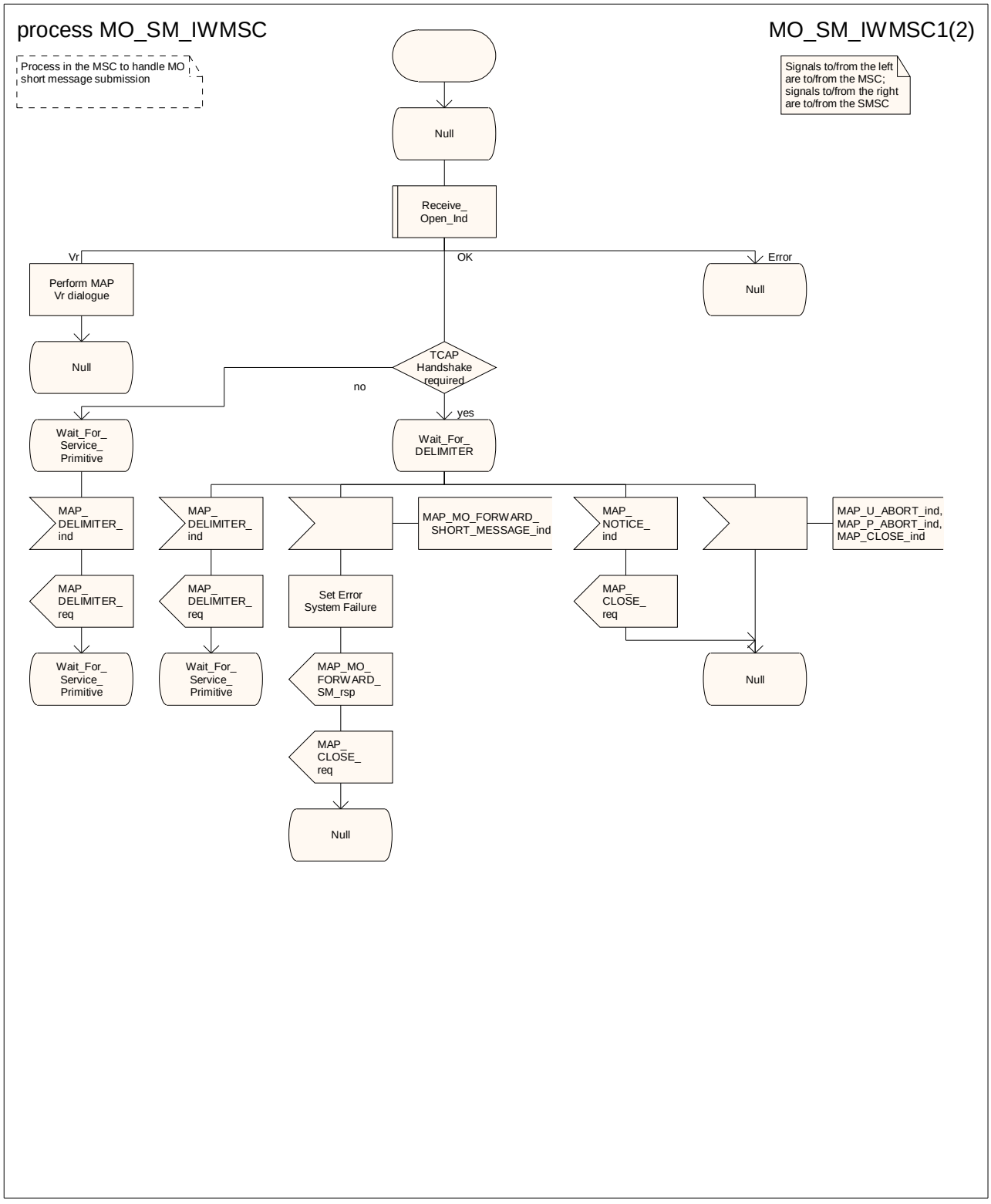


Figure 23.2/5 (sheet 1 of 2): Process MO_SM_IWMSC

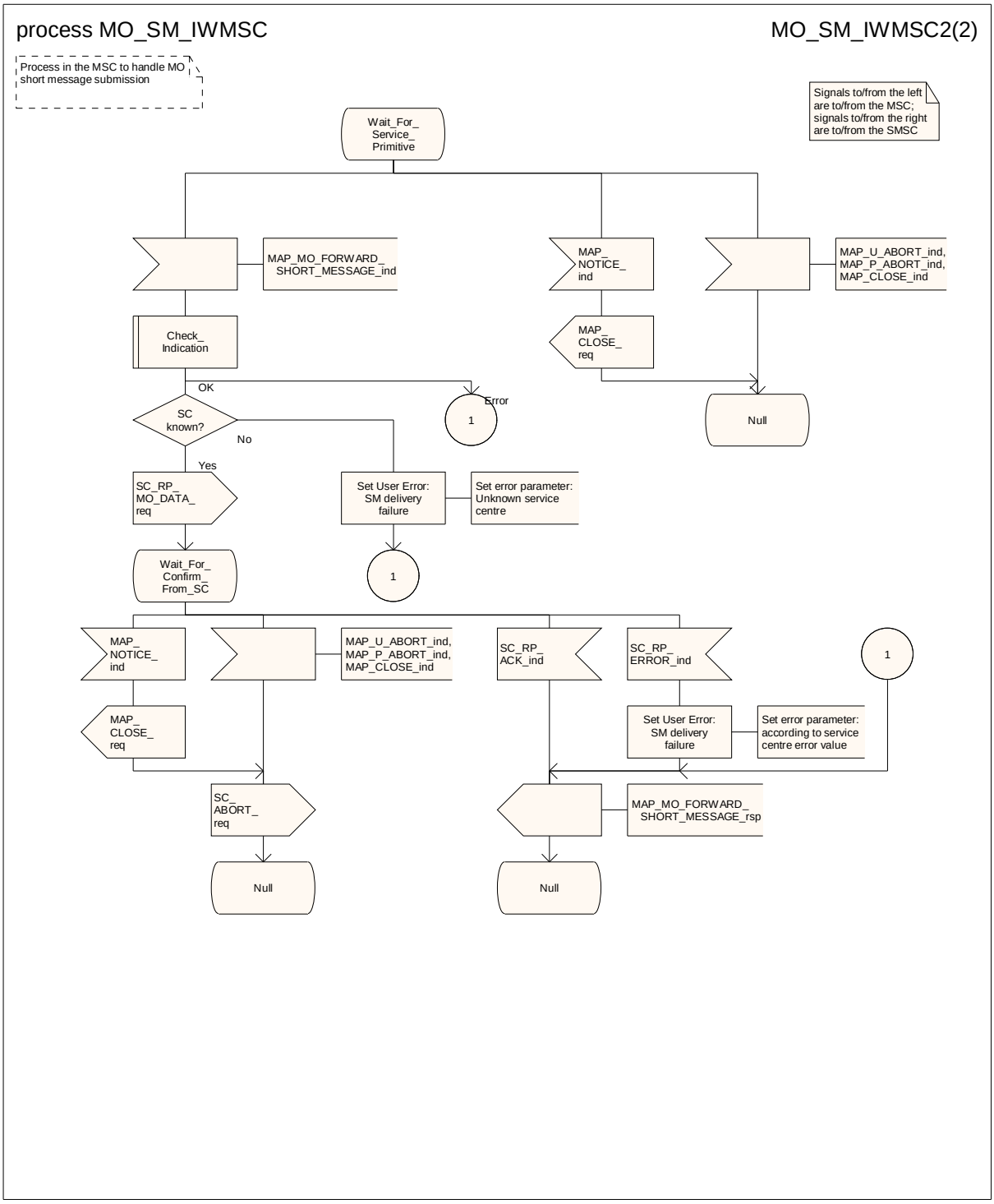


Figure 23.2/5 (sheet 2 of 2): Process MO_SM_IWMSC

23.3 The mobile terminated short message transfer procedure

The mobile terminated short message transfer procedure is used for forwarding a short message or several short messages from a Service Centre to a mobile subscriber. The message flow for the mobile terminated short message procedure for a single short message transfer is shown in figure 23.3/1.

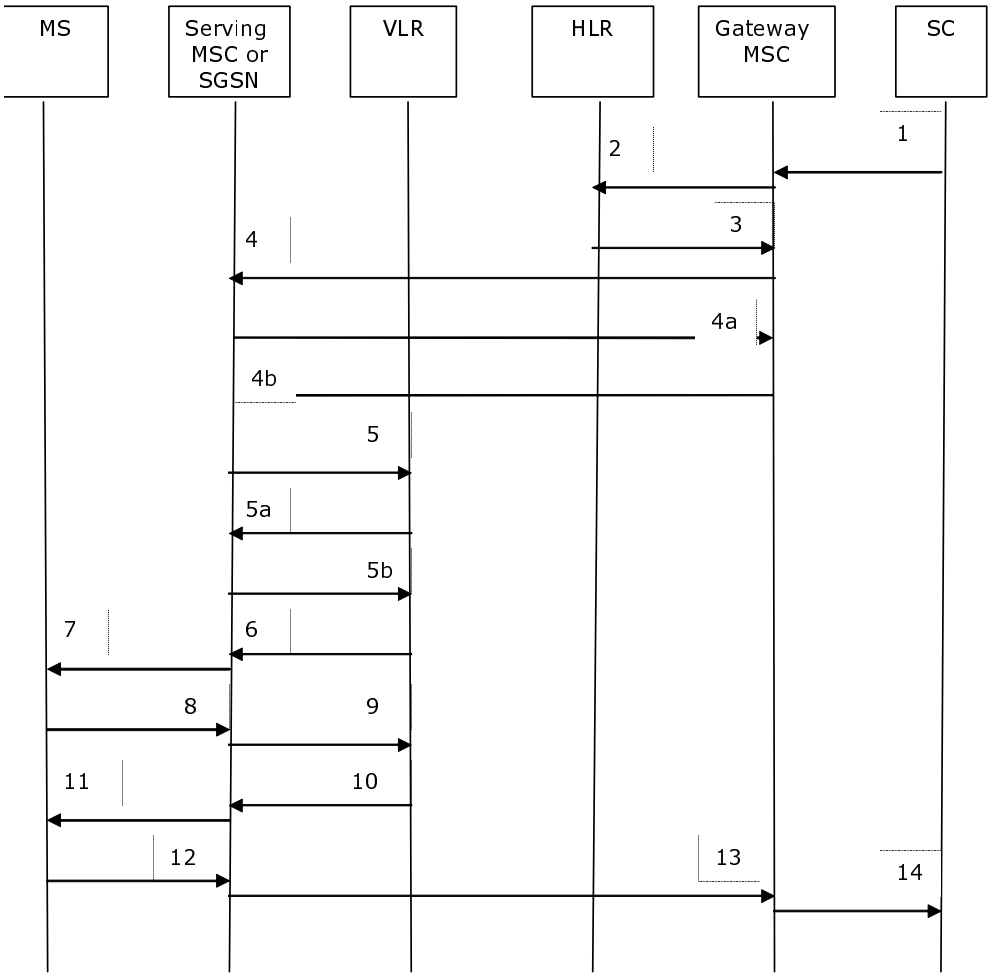


Figure 23.3/1: Mobile terminated short message service procedures

1) Short Message (3GPP TS 23.040).
2) MAP_SEND_ROUTING_INFO_FOR_SM.
3) MAP_SEND_ROUTING_INFO_FOR_SM_ACK.
4) TCAP BEGIN (***)
4a) TCAP CONTINUE (***)
4b) MAP_MT_FORWARD_SHORT_MESSAGE.
The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
5) MAP_SEND_INFO_FOR_MT_SMS (*).
5a) MAP_CONTINUE_CAMEL_SMS_HANDLING (*)(**)
5b) MAP_SEND_INFO_FOR_MT_SMS (*)(**)
6) MAP_PAGE/MAP_SEARCH_FOR_MOBILE_SUBSCRIBER (*).
7) Page (3GPP TS 24.008 [35]).
8) Page response (3GPP TS 24.008 [35]).
9) MAP_PROCESS_ACCESS_REQUEST_ACK and MAP_SEARCH_FOR_MOBILE_SUBSCRIBER_ACK (*).
10) MAP_SEND_INFO_FOR_MT_SMS_ACK (*).
11) Short Message (3GPP TS 24.011 [37]).
12) Short Message Acknowledgement (3GPP TS 24.011 [37]).
13) MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
14) Short Message Acknowledgement (3GPP TS 23.040).

(*) Messages 5), 5a), 5b), 6), 9), and 10) are not used by the SGSN.
(**) These messages are used only for a subscriber provisioned with MT-SMS-CSI in the VLR.
(***) If
a)
- the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MT_FORWARD_SHORT_MESSAGE request in a single TC message,

and
b) the SMS Gateway MSC operator and the serving node (MSC or SGSN) operator agreed not to use the TCAP handshake countermeasure against SMS fraud for messages exchanged between their networks (see 3GPP TS 33.204 [34a])
then
the TCAP handshake may be omitted.

The message flow for the mobile terminated short message procedure for multiple short message transfer is shown in figure 23.3/2.

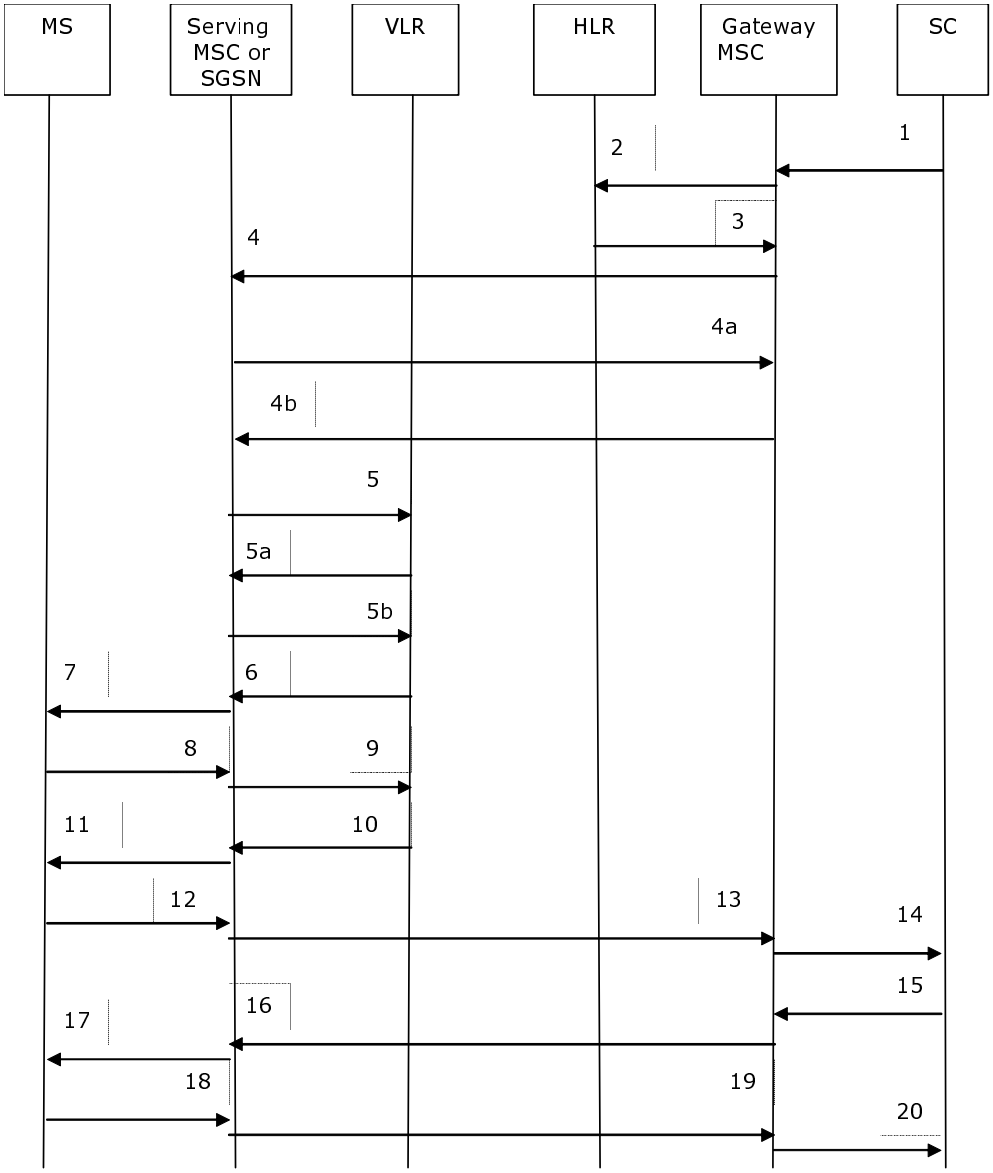


Figure 23.3/2: Mobile terminated short message procedure for multiple short message transfer

1) Short Message (3GPP TS 23.040 [26]).
2) MAP_SEND_ROUTING_INFO_FOR_SM.
3) MAP_SEND_ROUTING_INFO_FOR_SM_ACK.
4) TCAP BEGIN (***)
4a TCAP CONTINUE (***)
4b) MAP_MT_FORWARD_SHORT_MESSAGE (note 1).
The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
5) MAP_SEND_INFO_FOR_MT_SMS (*).
5a) MAP_CONTINUE_CAMEL_SMS_HANDLING (*)(**)
5b) MAP_SEND_INFO_FOR_MT_SMS (*)(**)
6) MAP_PAGE/MAP_SEARCH_FOR_MOBILE_SUBSCRIBER (*).
7) Page (*3GPP TS 48.008 [49]*).
8) Page response (3GPP TS 24.008 [35]).
9) MAP_PROCESS_ACCESS_REQUEST_ACK and MAP_SEARCH_FOR_MOBILE_SUBSCRIBER_ACK (*).
10) MAP_SEND_INFO_FOR_MT_SMS_ACK (*).
11) Short Message (3GPP TS 24.011 [37]).
12) Short Message Acknowledgement (3GPP TS 24.011 [37]).
13) MAP_MT_FORWARD_SHORT_MESSAGE_ACK.

14) Short Message Acknowledgement (3GPP TS 23.040 [26]).
15) Short Message (3GPP TS 23.040 [26]).
16) MAP_MT_FORWARD_SHORT_MESSAGE (note 2).
The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
17) Short Message (3GPP TS 24.011 [37]).
18) Short Message Acknowledgement (3GPP TS 24.011 [37]).
19) MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
20) Short Message Acknowledgement (3GPP TS 23.040 [26]).

(*) Messages 5), 5a), 5b) 6), 9), and 10) are not used by the SGSN.
(**) These messages are used only for a subscriber provisioned with MT-SMS-CSI in the VLR.
(***) If
a) the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MT_FORWARD_SHORT_MESSAGE request in a single TC message,
and
b) the SMS Gateway MSC operator and the serving node (MSC or SGSN) operator agreed not to use the TCAP handshake countermeasure against SMS fraud for messages exchanged between their networks (see 3GPP TS 33.204 [34a])
then the TCAP handshake may be omitted.

NOTE 1: The "More Messages To Send" flag is TRUE.
NOTE 2: The "More Messages To Send" flag is FALSE.

In the multiple short message transfer the service MAP_MT_FORWARD_SHORT_MESSAGE can be used several times. However, the short message transfer is always acknowledged to the Service Centre before the next short message is sent.

In addition the following MAP services are used:

MAP_PROCESS_ACCESS_REQUEST	(see subclause 8.3); (*)
MAP_PAGE	(see subclause 8.2); (*)
MAP_SEARCH_FOR_MS	(see subclause 8.2); (*)
MAP_AUTHENTICATE	(see subclause 8.5); (*)
MAP_SET_CIPHERING_MODE	(see subclause 8.6); (*)
MAP_CHECK_IMEI	(see subclause 8.7);
MAP_FORWARD_NEW_TMSI	(see subclause 8.9); (*)
MAP_REPORT_SM_DELIVERY_STATUS	(see subclause 12.3);
MAP_INFORM_SERVICE_CENTRE	(see subclause 12.6);
MAP_TRACE_SUBSCRIBER_ACTIVITY	(see subclause 9.1); (*)
MAP_READY_FOR_SM	(see subclause 12.4).

(*) These services are not used by the SGSN.

A message flow example for the mobile terminated short message procedure for a single short message transfer in an environment that makes use of an SMS Router for MT-short-message-transfer is shown in figure 23.3/2a.

NOTE: This message flow can be applied only if no IP-SM-GW deployed in the same network.

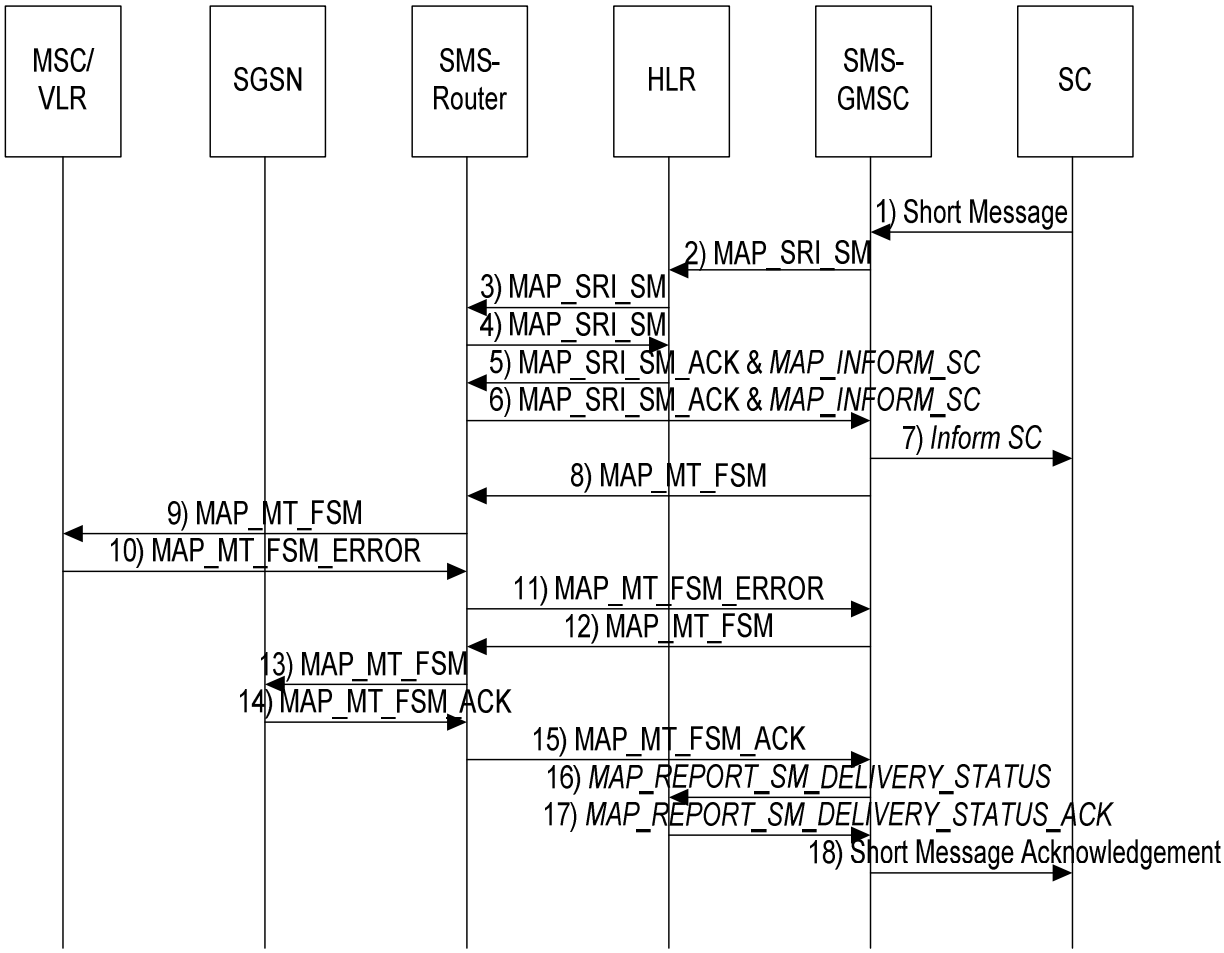


Figure 23.3/2a Mobile terminated short message procedure with SMS Router

1) Short Message (3GPP TS 23.040 [26])
2) & 3) MAP_SEND_ROUTING_INFO_FOR_SM
NOTE: The HLR relays the message MAP_SEND_ROUTING_INFO_FOR_SM received from the SMS-GMSC to the SMS Router on SCCP level. How this is done is implementation specific.

4) MAP_SEND_ROUTING_INFO_FOR_SM
5) MAP_SEND_ROUTING_INFO_FOR_SM_ACK and conditionally MAP_INFORM_SERVICE_CENTRE
6) MAP_SEND_ROUTING_INFO_FOR_SM_ACK and conditionally MAP_INFORM_SERVICE_CENTRE
7) Conditionally: Inform Service Centre (3GPP TS 23.040 [26])
8) MAP_MT_FORWARD_SHORT_MESSAGE
The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
NOTE: In this example the SMS-GMSC decides to attempt delivery via MSC. Therefore the SCCP called party SSN shall be set to SSN for MSC.

9) MAP_MT_FORWARD_SHORT_MESSAGE
The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
10) MAP_MT_FORWARD_SHORT_MESSAGE_ERROR
NOTE: In this example delivery via the MSC is unsuccessful e.g. due to IMSI detached

11) MAP_MT_FORWARD_SHORT_MESSAGE_ERROR
12) MAP_MT_FORWARD_SHORT_MESSAGE
NOTE: In this example the SMS-GMSC decides to retry delivery via the SGSN. Therefore the SCCP called party SSN shall be set to the SSN for SGSN.

13) MAP_MT_FORWARD_SHORT_MESSAGE
The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
14) MAP_MT_FORWARD_SHORT_MESSAGE_ACK
NOTE: In this example delivery via SGSN is successful

15) MAP_MT_FORWARD_SHORT_MESSAGE_ACK
16) Conditionally: MAP_REPORT_SM_DELIVERY_STATUS
NOTE: In this example unsuccessful delivery via MSC and successful delivery via SGSN is reported

17) MAP_REPORT_SM_DELIVERY_STATUS_Ack
18) Short Message Acknowledgement (3GPP TS 23.040 [26]).

A message flow example for the mobile terminated short message procedure for a single short message transfer in an environment that makes use of an IP-SM-GW (see 3GPP TS 23.204 [134]) for MT-short-message-transfer is shown in figure 23.3/2b.

NOTE: SMS Routers can apply this message flow as well.

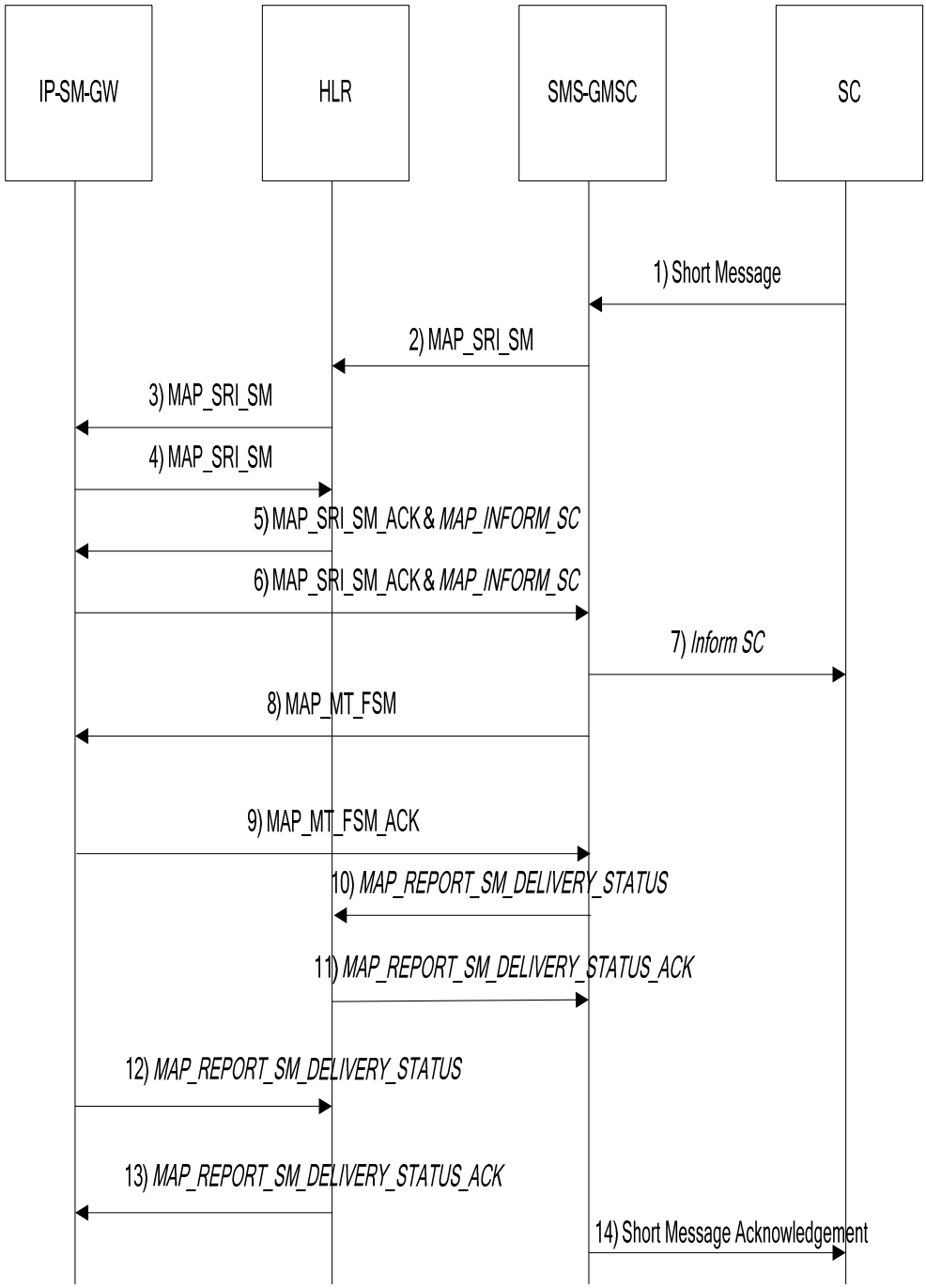


Figure 23.3/2b Mobile terminated short message procedure with IP-SM-GW

1) Short Message (3GPP TS 23.040 [26])
2) MAP_SEND_ROUTING_INFO_FOR_SM
the message is forwarded to the IP-SM-GW assigned to the recipient of the SM
the message may contain IP-SM-GW Guidance support indication
3) MAP_SEND_ROUTING_INFO_FOR_SM

4) MAP_SEND_ROUTING_INFO_FOR_SM
since the message is received from an IP-SM-GW, it is not forwarded to an IP-SM-GW
5) MAP_SEND_ROUTING_INFO_FOR_SM_ACK and conditionally MAP_INFORM_SERVICE_CENTRE
6) MAP_SEND_ROUTING_INFO_FOR_SM_ACK and conditionally MAP_INFORM_SERVICE_CENTRE
The IP-SM-GW returns its own address within the network node number parameter
The message may include IP-SM-GW Guidance
7) Conditionally: Inform Service Centre (3GPP TS 23.040 [26])
8) MAP_MT_FORWARD_SHORT_MESSAGE
If the IP-SM-GW-Guidance support indicator was present in message 2 and IP-SM-GW-Guidance was present in message 6, message 8 shall contain the used timer value for supervision of MAP_MT_FORWARD_SHORT_MESSAGE_ACK; the used timer should be identical to the recommended value received in message 6 to ensure that the IP-SM-GW can attempt delivery to multiple domains if necessary and shall not be lower than the minimum value received in message 6 to ensure that an IP-SM-GW can attempt delivery to at least one domain.
Otherwise, The message should include the timer value used at the SMS-GMSC for the supervision of the MAP_MT_FORWARD_SHORT_MESSAGE_ACK.
9) MAP_MT_FORWARD_SHORT_MESSAGE_ACK
10) Conditionally: MAP_REPORT_SM_DELIVERY_STATUS
NOTE: As an IP-SM-GW is deployed the message is acknowledged ignoring its content

11) MAP_REPORT_SM_DELIVERY_STATUS_Ack
12) Conditionally: MAP_REPORT_SM_DELIVERY_STATUS
since the message is received from an IP-SM-GW, it is processed
13) MAP_REPORT_SM_DELIVERY_STATUS_Ack
NOTE: Step 12 and 13 is independent of steps 10, 11, and 14. They can run in parallel.

14) Short Message Acknowledgement (3GPP TS 23.040 [26]).

23.3.1 Procedure in the SMS-GMSC

Any CAMEL-specific handling described in this subclause is omitted if the SMS-GMSC does not support CAMEL. CAMEL-specific handling is invoked only if the SMS-GMSC is integrated with the VMSC.

The process starts when the SMS-GMSC receives an SC_RP_MT_DATA indication from a Service Centre. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

Process MT_SM_GMSC sheet 1: If the MAP_SEND_ROUTING_INFO_FOR_SM confirmation included an LMSI, it shall be included in the sm-RP-DA information field of the first MAP_MT_FORWARD_SHORT_MESSAGE request sent to the serving MSC. In this case, the IMSI shall be included in the Destination Reference of the MAP_OPEN request. The SMS-GMSC shall not send an LMSI to an SGSN. If the SMS-GMSC does not send an LMSI to the serving node, the sm-RP-DA information field in the first MAP_MT_FORWARD_SHORT_MESSAGE request sent to the serving MSC or SGSN shall contain the IMSI, and the Destination Reference in the MAP_OPEN request shall not be present. The parameter SM_RP_OA shall contain the Service Centre address.

Process MT_SM_GMSC sheet 1: The indication of which number belongs to the SGSN and which to the MSC, received from the HLR in the MAP_SEND_ROUTING_INFO_FOR_SM confirm (see subclause 23.3.2) will enable the SMS-GMSC to map the causes received from one or both serving nodes into the appropriate causes for non GPRS, GPRS or both, and send them to the SC and the HLR.

Process MT_SM_GMSC sheet 2: The SMS-GMSC maps "Unexpected data value" and "System failure" MAP errors from the serving node to a "System failure" RP_ERROR error cause. The mapping between other MAP error causes and the RP_ERROR error cause is given in 3GPP TS 23.040 [26] and 3GPP TS 24.011 [37].

Process MT_SM_GMSC sheet 2: If the SMS-GMSC receives both MSC and SGSN numbers from the HLR as routeing information, it may choose which serving node to use for the first delivery attempt.

Process MT_SM_GMSC sheet 2: If the SMS-GMSC makes two delivery attempts, it may report the result of each delivery attempt to the HLR according to the conditions described below.

Procedure MT_SM_Delivery_Attempt_GMSC sheet 1: if the macro MT_SM_Transfer_MSC takes the Error exit, the SMS-GMSC maps the MAP User Error to the corresponding SC_RP error, as defined in 3GPP TS 23.040 [26].

Procedure MT_SM_Delivery_Attempt_GMSC sheet 3: The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the GMSC's operator and the serving node's operator (see 3GPP TS 33.204 [34a]).

Procedure MT_SM_Delivery_Attempt_GMSC sheet 1, sheet 2, sheet 4, sheet 5: The SMS-GMSC invokes the macro Report_SM_Delivery_Stat_GMSC if:

- the reason received from the serving node for failure to deliver the message is absent subscriber_SM, unidentified subscriber or SM delivery failure with error cause "MS memory capacity exceeded", and the SC address is not yet included in the MWD set, or

- the reason received from the serving node for failure to deliver the message is absent subscriber_SM, unidentified subscriber or SM delivery failure with error cause MS memory capacity exceeded, and the corresponding flag in the HLR (as indicated in the information received in the MAP_INFORM_ SERVICE_CENTRE) is not set, or

- the reason received from the serving node (MSC or SGSN) for failure to deliver the message is absent subscriber_SM and the absent subscriber diagnostic is different from the absent subscriber diagnostic received in the MAP_INFORM_ SERVICE_CENTRE.

Procedure MT_SM_Delivery_Attempt_GMSC sheet 1, sheet 2, sheet 4, sheet 5: If absent subscriber diagnostic information (see 3GPP TS 23.040 [26]) is included with the absent subscriber_SM error indication then the SMS-GMSC relays this information to the HLR using the MAP_REPORT_SM_DELIVERY_STATUS service.

Procedure MT_SM_Delivery_Attempt_GMSC sheet 1, sheet 4: The More Messages To Send flag is set to TRUE or FALSE according to the information received from the Service Centre.

Procedure MT_SM_Delivery_Attempt_GMSC sheet 3: If the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MT_FORWARD_SHORT_MESSAGE request in a single TC message, the test "Message segmentation needed" takes the "No" exit; otherwise the test takes the "Yes" exit.

The mobile terminated short message transfer process in the SMS-GMSC is shown in figure 23.3/3. The procedure MT_SM_Delivery_Attempt_GMSC is shown in figure 23.3/4. The macro MT_SM_Transfer_MSC is shown in figure 23.3/7.

The SMS-GMSC may include the Maximum Retransmission Time IE in the MT Forward Short Message Request to indicate that it is capable to retransmit the Short Message until the indicated maximum retransmission time, if the following conditions are fulfilled:

- the destination user pertains to the PLMN of the SMS-GMSC; and

- if an SMS Router is used for MT SMS sent to destination users pertaining to the PLMN of the SMS-GMSC, the SMS Router is known to support the Alert Service Centre procedure specified in subclause 12.5.

The SMS-GMSC shall include its E.164 number in the SMS-GMSC address and, if available its Diameter Identity in the SMS-GMSC Diameter Address in the request if it also includes the Maximum-Retransmission-Time IE.

If subsequently, the SMS-GMSC receives an Alert Service Centre request from an MME (via an IWF), SGSN or MSC, the SMS-GMSC shall retransmit pending MT SMS(s) for the destination user identified by the User Identifier Alert, to the same serving node if the SMS-GMSC Alert Event indicates that the MS is available for MT SMS, or to the new serving node if the SMS-GMSC Alert Event indicates that the MS has moved under the coverage of another MME, SGSN or MSC. In the latter case, if neither New MSC Number, nor New SGSN Number, nor New SGSN Diameter Address, nor New MME Number, nor New MME Diameter Address are received in the Alert Service Centre request, the SMS-GMSC shall initiate a Send Routing Info for SM procedure to retrieve the new serving node 's address from the HLR.

23.3.2 Procedure in the HLR

The process starts when the HLR receives a MAP_SEND_ROUTING_INFO_FOR_SM indication from the SMS-GMSC. If an SMS Router is deployed, the HLR receives MAP_SEND_ROUTING_INFO_FOR_SM from the SMS Router (step 4 in figure 23.3/2a); relaying a message received from the SMS-GMSC to the SMS Router on SCCP level (steps 2 and 3 in figure 23.3/2a) is done by implementation specific means and is not shown in figure 23.3/5.

The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.1.

Sheet 1: The decision box "Relay to IP-SM-GW" takes the "yes" exit if an IP-SM-GW is deployed and the message was not received from an IP-SM-GW.

Sheet 3: If the SMS-GMSC does not support GPRS functionality, it uses the protocol defined in the Release 96 version of this specification. The parameter "msc-Number" in "RoutingInfoForSM-Res" in the Release 96 version of the protocol definition corresponds to the parameter "networkNode-Number" in "RoutingInfoForSM-Res" in the Release 97 (and later) version of the protocol definition; therefore if the HLR populates the parameter "networkNode-Number" with the SGSN number, the Release 96 SMS-GMSC will interpret the SGSN number as an MSC number. If the HLR populates the "gprsNodeIndicator" parameter in the MAP_SEND_ROUTING_INFO_FOR_SM response, a Release 96 SMS-GMSC will silently discard the parameter.

Sheet 5: If the HLR received a LMSI from the VLR at location updating, it shall include the LMSI in the MAP_SEND_ROUTING_INFO_FOR_SM response only if the MAP_SEND_ROUTING_INFO_FOR_SM response also includes the MSC number.

The mobile terminated short message transfer process in the HLR is shown in figure 23.3/5.

23.3.3 Procedure in the Serving MSC

Any CAMEL-specific handling defined in this subclause is omitted if the MSC does not support CAMEL control of MT SMS, or if the subscriber does not have a subscription for CAMEL control of MT SMS.

The process starts when the MSC receives a dialogue opening request with the application context shortMsgMT-RelayContext. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

The mobile terminated short message transfer process in the serving MSC is shown in figure 23.3/6

Procedure MT_SM_VMSC sheet 1: The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the Serving MSC's operator and the SMS Gateway MSC's operator (see 3GPP TS 33.204 [34a]).

The macro MT_SM_Transfer_MSC may be invoked either in a stand-alone serving MSC or in a serving MSC which is integrated with the SMS-GMSC. It is used to transfer the first MT short message of a possible sequence of messages. The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Confirmation see subclause 25.2.2.

Page_MSC see subclause 25.3.1;

Search_for_MS_MSC see subclause 25.3.2;

Process_Access_Request_MSCsee subclause 25.4.1;

Trace_Subscriber_Activity_MSC see subclause 25.9.1.

The macro MT_SM_Transfer_MSC is shown in figure 23.3/7. The macro Check_Subscr_Identity_For_MT_SMS is shown in figure 23.3/8.

3GPP TS 29.118 [152] specifies the extensions applicable to the MT SMS procedure over SGs for a UE using extended idle mode DRX (as defined in 3GPP TS 23.682 [148]).

If the MS is using a power saving mechanism such as extended idle mode DRX (see 3GPP TS 23.682 [148]), and if the MT Forward Short Message Request includes the Maximum Retransmission Time IE, an MSC using Deployment Option 2 (see subclause 8.2.4a.1 of 3GPP TS 23.272 [143] and 3GPP TS 23.040 [6]) may return an MT Forward Short Message Response with the User Error set to Absent Subscriber_SM and with the Requested-Retransmission-Time IE requesting the SMS-GMSC to retransmit the Short Message at a later time prior to the Maximum Retransmission Time. In that case, the MSC shall store (if not already done) the SMS-GMSC address and, if available, the SMS-GMSC Diameter Identity received in the request and shall not set the MNRF flag.

NOTE: This mechanism does not cause additional signalling at the HSS to retransmit the Short Message.

An MSC using Deployment Option 2 (see subclause 8.2.4a.1 of 3GPP TS 23.272 [143] and 3GPP TS 23.040 [6]) shall initiate the MAP Service Center Alert procedure to alert the SMS-GMSC when the UE, for which one or more MT SMS have been requested to be retransmitted at a later time, becomes available for MT SMS delivery or moves under the coverage of another MSC prior to the requested SM retransmission time. The MSC shall then delete the stored SMS-GMSC address after the Alert Service Centre procedure is completed.

23.3.4 Procedure in the VLR

Any CAMEL-specific handling defined in this subclause is omitted if the VLR does not support CAMEL control of MT SMS.

The process starts when the VLR receives a dialogue opening request from the MSC. The process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2;

Process_Access_Request_VLR see subclause 25.4.2.

The mobile terminated short message transfer process in the VLR is shown in figure 23.3/9.

If the VLR has no IMSI record, or if the record is marked "Subscriber Data Not Confirmed by HLR", the VLR may perform the data restoration procedure as specified in subclause 4.2.2 in 3GPP TS 23.007 [19].

23.3.5 Procedure in the SGSN

Any CAMEL-specific handling defined in this subclause is omitted if the SGSN does not support CAMEL control of MT SMS, or if the subscriber does not have a subscription for CAMEL control of MT SMS.

The process starts when the SGSN receives a dialogue opening request with the application context shortMsgMT-RelayContext. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

The mobile terminated short message transfer process in the SGSN is shown in figure 23.3/10.

Procedure MT_SM_SGSN sheet 1: The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the Serving SGSN's operator and the SMS Gateway MSC's operator (see 3GPP TS 33.204 [34a]).

The macro MT_SM_Transfer_SGSN is used to transfer the first MT short message of a possible sequence of messages. It is shown in figure 23.3/11.

If the MS is using extended idle mode DRX (as defined in 3GPP TS 23.682 [148]) and the MS is expected to respond to paging shortly or within the time frame indicated by the SM-Delivery-Timer and SM-Delivery-Start-Time IEs, the SGSN should page the MS and attempt to deliver the short message to the MS.

If the MS is using extended idle mode DRX (as defined in 3GPP TS 23.682 [148]) and the MS is expected to not respond to paging shortly or within the time frame indicated by the SM-Delivery-Timer and SM-Delivery-Start-Time IEs, the SGSN may behave as specified in figure 23.3/11 for an MS that is not reachachable, i.e. set the MNRG flag and return an Absent Subscriber Error to the SMS-GMSC, while still paging the MS.

NOTE 1: This mechanism is not intended for MSs which are known to wake up shortly (e.g. within the next 10 seconds) as enough time needs to elapse, between the sending of the MT Forward Short Message Response and the subsequent Ready For SM procedure towards the HLR when the MS becomes reachable, for the Report SM Delivery Status procedure to take place beforehand from the SMS-GMSC to the HLR.

If the MS is using a power saving mechanism such as extended idle mode DRX (see 3GPP TS 23.682 [148]), and if the MT Forward Short Message Request includes the Maximum Retransmission Time IE, the SGSN may return an MT Forward Short Message Response with the User Error set to Absent Subscriber_SM and with the Requested-Retransmission-Time IE requesting the SMS-GMSC to retransmit the Short Message at a later time prior to the Maximum Retransmission Time. In that case, the SGSN shall store (if not already done) the SMS-GMSC address and, if available, the SMS-GMSC Diameter Identity received in the request and shall not set the MNRG flag.

NOTE 2: This mechanism does not cause additional signalling at the HSS to retransmit the Short Message.

The SGSN shall initiate the MAP Service Center Alert procedure to alert the SMS-GMSC when the UE, for which one or more MT SMS have been requested to be retransmitted at a later time, becomes available for MT SMS delivery or moves under the coverage of another MME or SGSN prior to the requested SM retransmission time. The SGSN shall then delete the stored SMS-GMSC address after the Alert Service Centre procedure is completed.

The macro Check_Subscr_Identity_For_MT_SMS is shown in figure 23.3/8. The page and search procedures are shown in figures 23.3/12 and 23.3/13.

23.3.6 Procedure in the SMS Router

If SMS Router is deployed together with IP-SM-GW, then mobile terminated short message transfer process for IP-SM-GW applies as described in subclause 23.3.7.

The mobile terminated short message transfer process in the SMS Router is shown in figure 23.3/14.

Procedure MT_SM_SMS_ROUTER sheet 2: Allocated MT Correlation IDs have a limited lifetime, managed by Timer T1. The value of Timer T1 shall be operator configurable (its value being dependant on such factors as subscriber base, network size, number of roaming/SMS-interworking partners, average and peak SMS traffic load, etc.).

The value of the timer T1 shall cover at least the time indicated by the SM Delivery Start Time and SM Delivery Timer and, if the SMS Router support the Alert Service Centre procedure specified in subclause 12.5, the Maximum Retransmission Time signalled in the MAP-MT-FORWARD-SHORT-MESSAGE.

Procedure MT_SM_SMS_ROUTER sheet 2: MAP parameters to be stored against the MT Correlation ID are IMSI, networkNode-Number, gprsNodeIndicator, and additional-Number (if and as received within MAP_SEND_ROUTING_INFO_FOR_SHORT_MESSAGE_cnf), and optionally MSISDN as received within MAP_SEND_ROUTING_INFO_FOR_SHORT_MESSAGE_ind from the SMS-GMSC (and relayed by the HLR)). The SMS Router may also store the GT, or just the CC and NDC parts of the GT, of the SMS-GMSC from which the MAP_SEND_ROUTING_INFO_FOR_SHORT_MESSAGE_ind was received.

Procedure MT_SM_SMS_ROUTER sheet 3: The SCCP called party SSN received with Open_ind is used to decide whether the new dialogue is opend with the MSC or with the SGSN.

The detail of replacing RP-OA is described in TS23.040 [26].

Procedure MT_SM_SMS_ROUTER sheet 4: The decision box "Retry expected" takes the "Yes" exit if two addresses were received from the HLR, the first delivery attempt was unsuccessful, and the second attempt has not yet been made.

Procedure MT_SM_SMS_ROUTER sheet 4: The task "Release MT Correlation ID" includes deleting of data stored against the MT Correlation ID.

If the MT Forward Short Message Request includes the Maximum-Retransmission-Time IE, the SMS Router shall store the SMS-GMSC address and, if available, the SMS-GMSC Diameter Identity received in the request and replace it by its SMS Router address (E.164 number) and, if available, its SMS Router Diameter Identity when forwarding the request to the MME (via an IFW), SGSN or MSC.

If the MT Forward Short Message Response includes the Requested-Retransmission-Time IE, the SMS Router shall include the User Identifier Alert IE when forwarding the response to the SMS-GMSC.

NOTE: The User Identifier Alert is further used in the Alert Service Centre procedure specified in subclause 12.5 to enable the SMS-GMSC to identify and retransmit all pending MT SMS messages towards the destination user.

When receiving an Alert Service Centre request from an MME (via an IWF), SGSN or MSC, the SMS-Router shall replace the IMSI received in the IMSI IE by the User Identifier Alert previously sent in the MT Forward Short Message Response, and forward that request to the SMS-GMSC.

23.3.7 Procedure in the IP-SM-GW

Process MT_SM_IPSMGW sheet 3: After unsuccessful delivery via the S-CSCF the IP-SM-GW may retry delivery via MSC and/or SGSN if MSC address and/or SGSN address are available (unless the reported error was "memory capacity exceeded" in which case a retry shall not be done). If the retry is successful, a positive response is returned to the SMS-GMSC. If the retry is unsuccessful, an error indication is returned to the SMS-GMSC as follows:
If one of the error indications received from S-CSCF, MSC, or SGSN is AbsentsSubscriberSM or UnidentifiedSubscriber, this error shall be returned to the SMS-GMSC.

Process MT_SM_IPSMGW sheet 3: The IP-SM-GW invokes the macro Report_SM_Delivery_Stat_IPSMGW if:

- the reason for failure to deliver the message is absent subscriber_SM, unidentified subscriber or SM delivery failure with error cause "MS memory capacity exceeded", and the SC address is not yet included in the MWD set, or
- the reason for failure to deliver the message is absent subscriber_SM, unidentified subscriber or SM delivery failure with error cause MS memory capacity exceeded, and the corresponding flag in the HLR (as indicated in the information received in the MAP_INFORM_SERVICE_CENTRE) is not set, or
- the reason for failure to deliver the message is absent subscriber_SM and the absent subscriber diagnostic is different from the absent subscriber diagnostic received in the MAP_INFORM_SERVICE_CENTRE.

The mobile terminated short message transfer process in the IP-SM-GW is shown in figure 23.3/15.

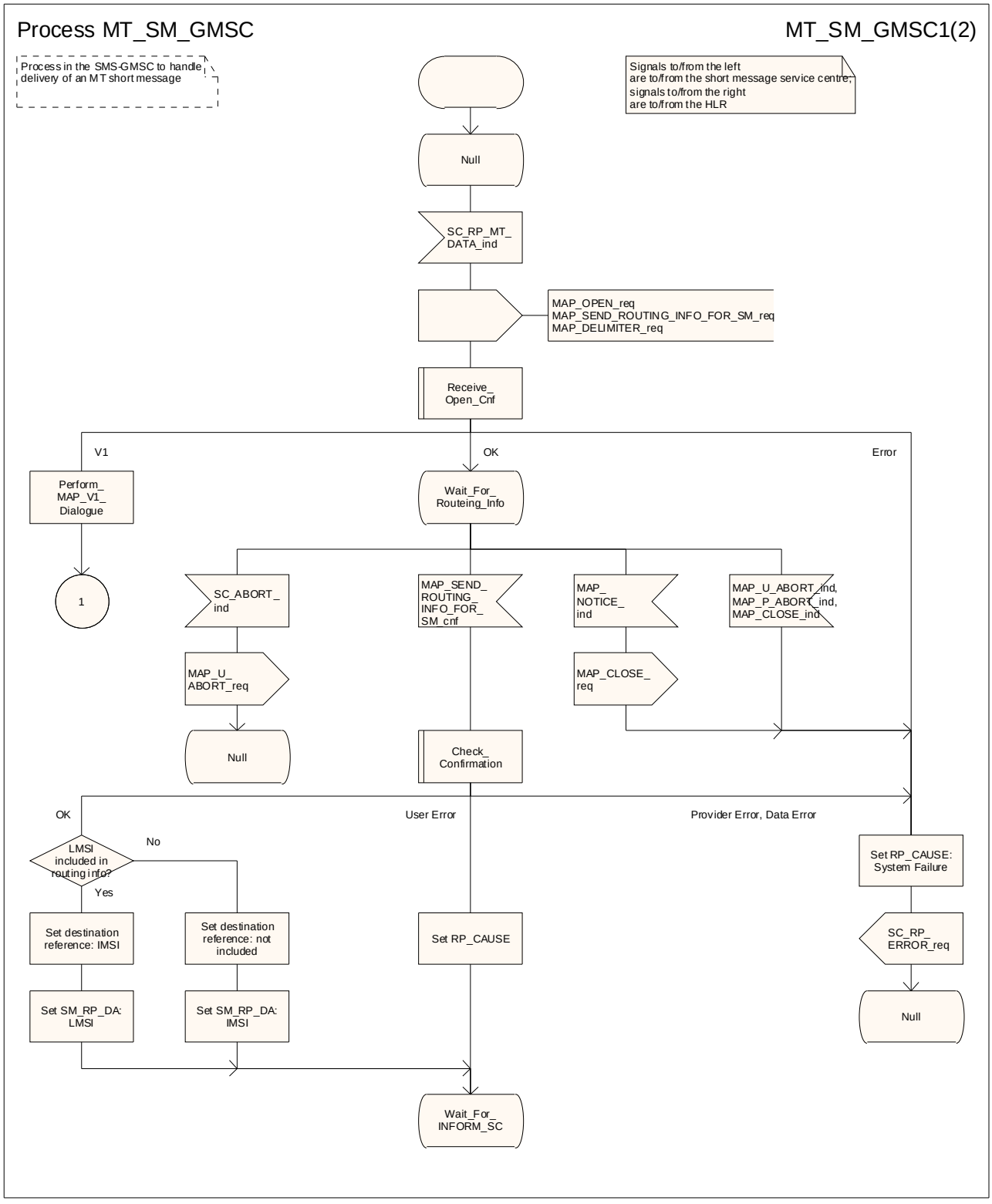


Figure 23.3/3 (sheet 1 of 2): Process MT_SM_GMSC

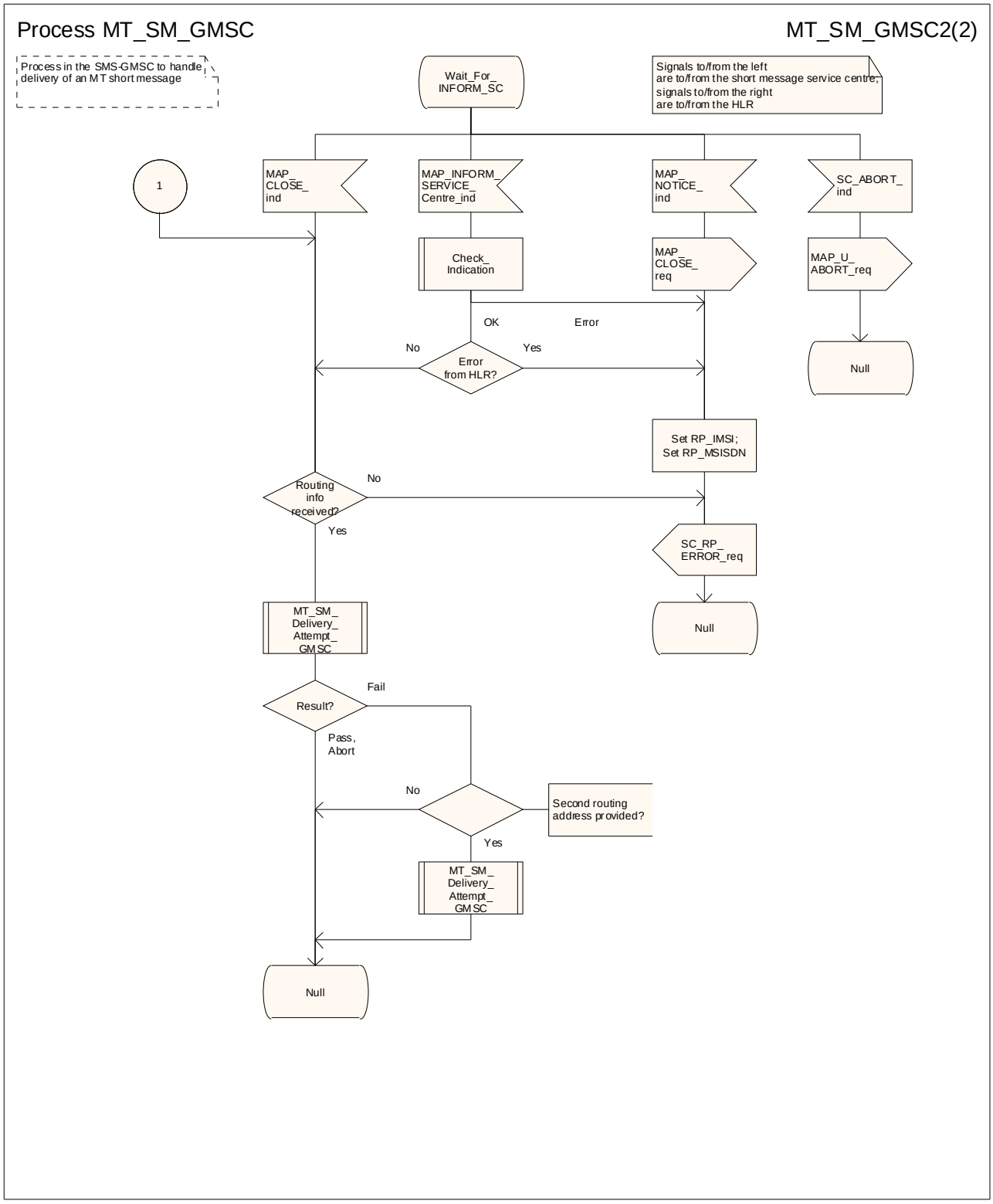


Figure 23.3/3 (sheet 2 of 2): Process MT_SM_GMSC

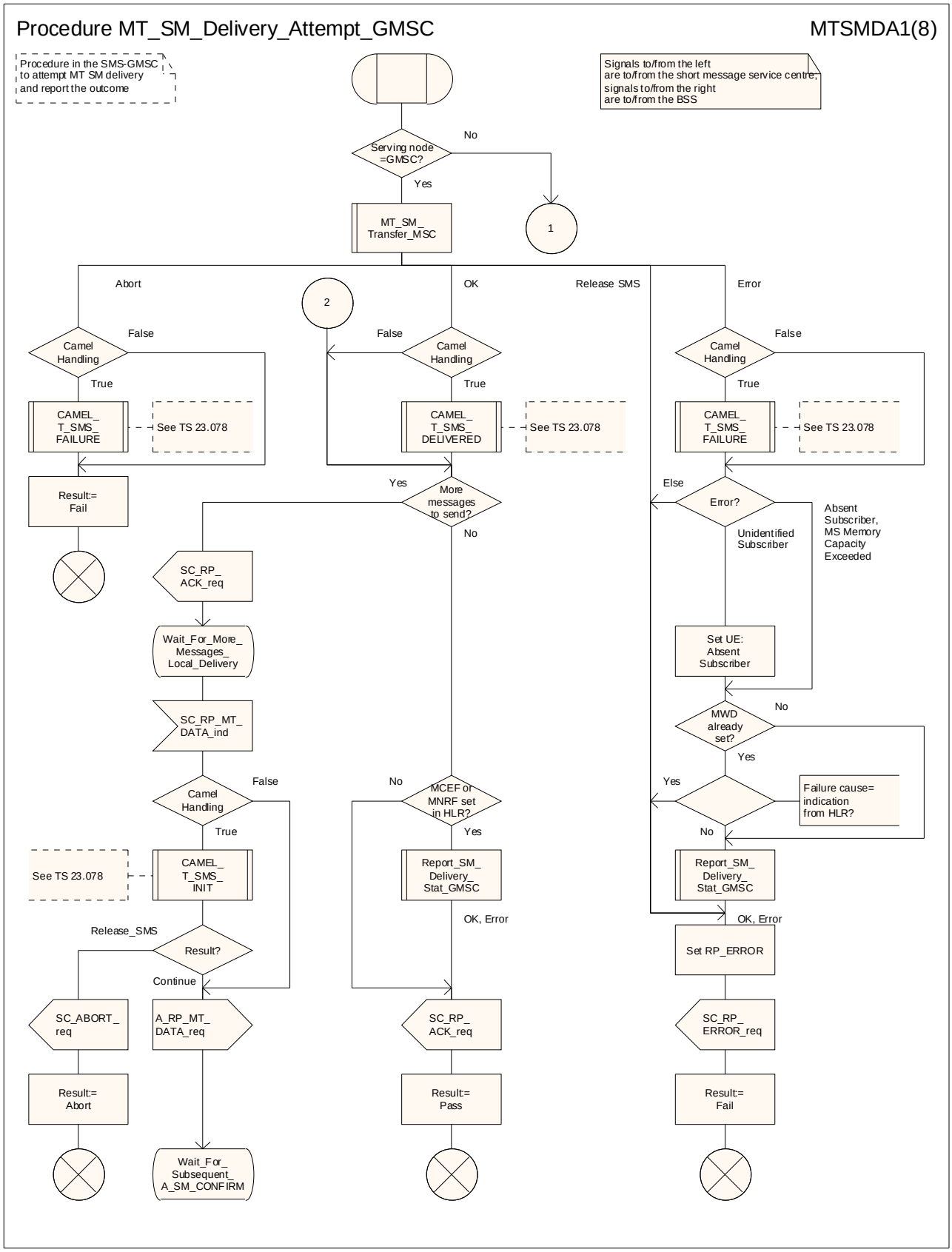


Figure 23.3/4 (sheet 1 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

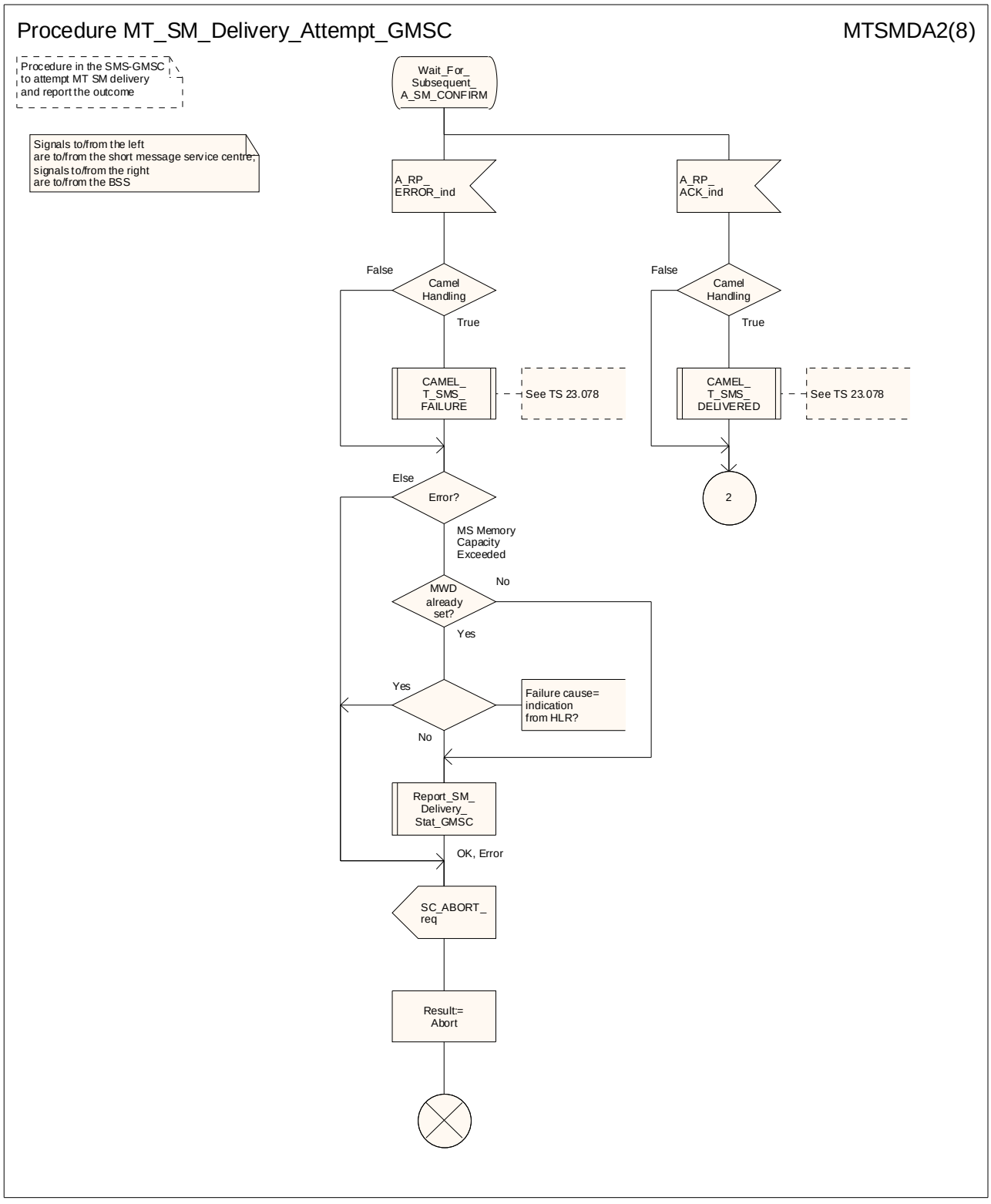


Figure 23.3/4 (sheet 2 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

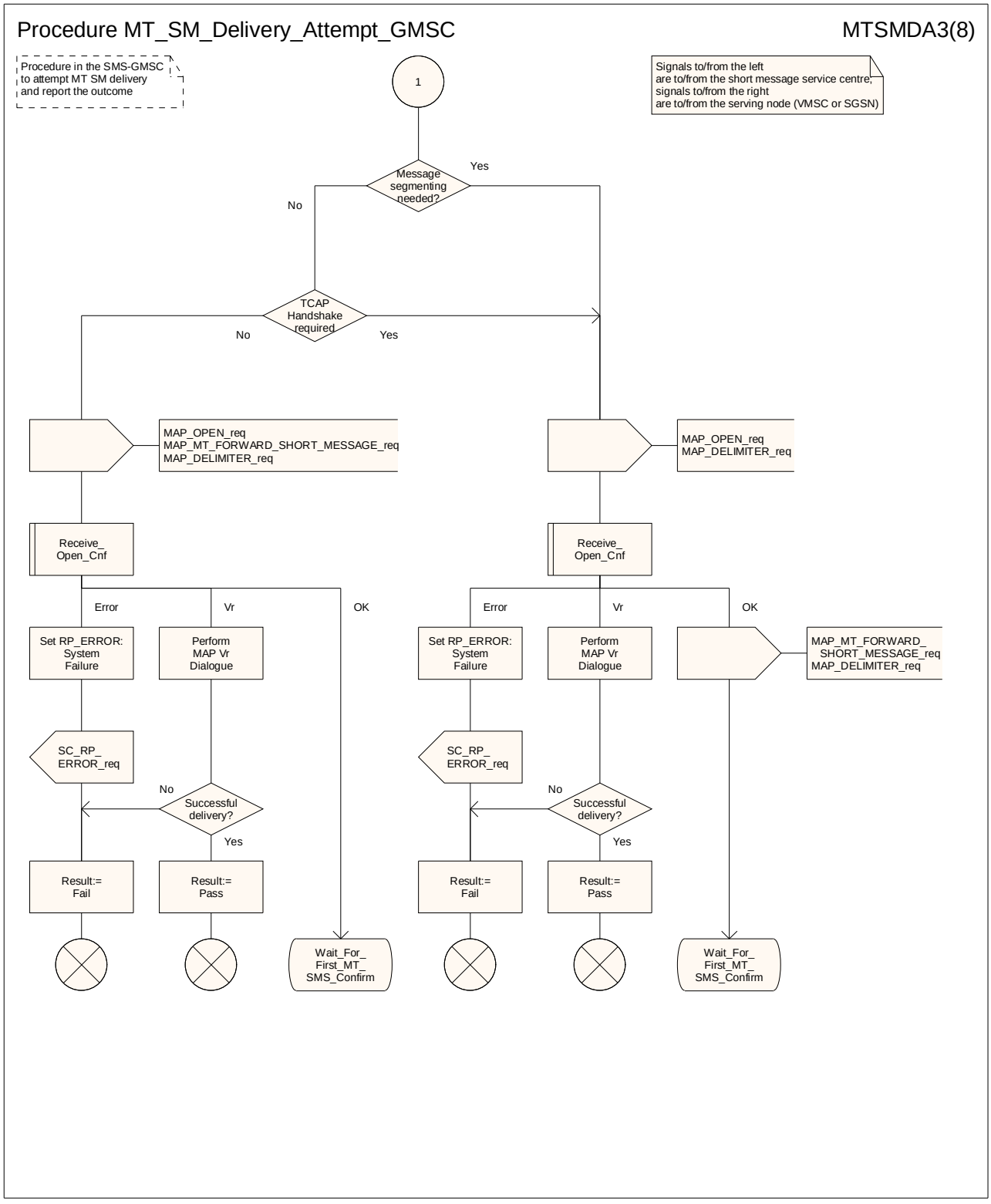


Figure 23.3/4 (sheet 3 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

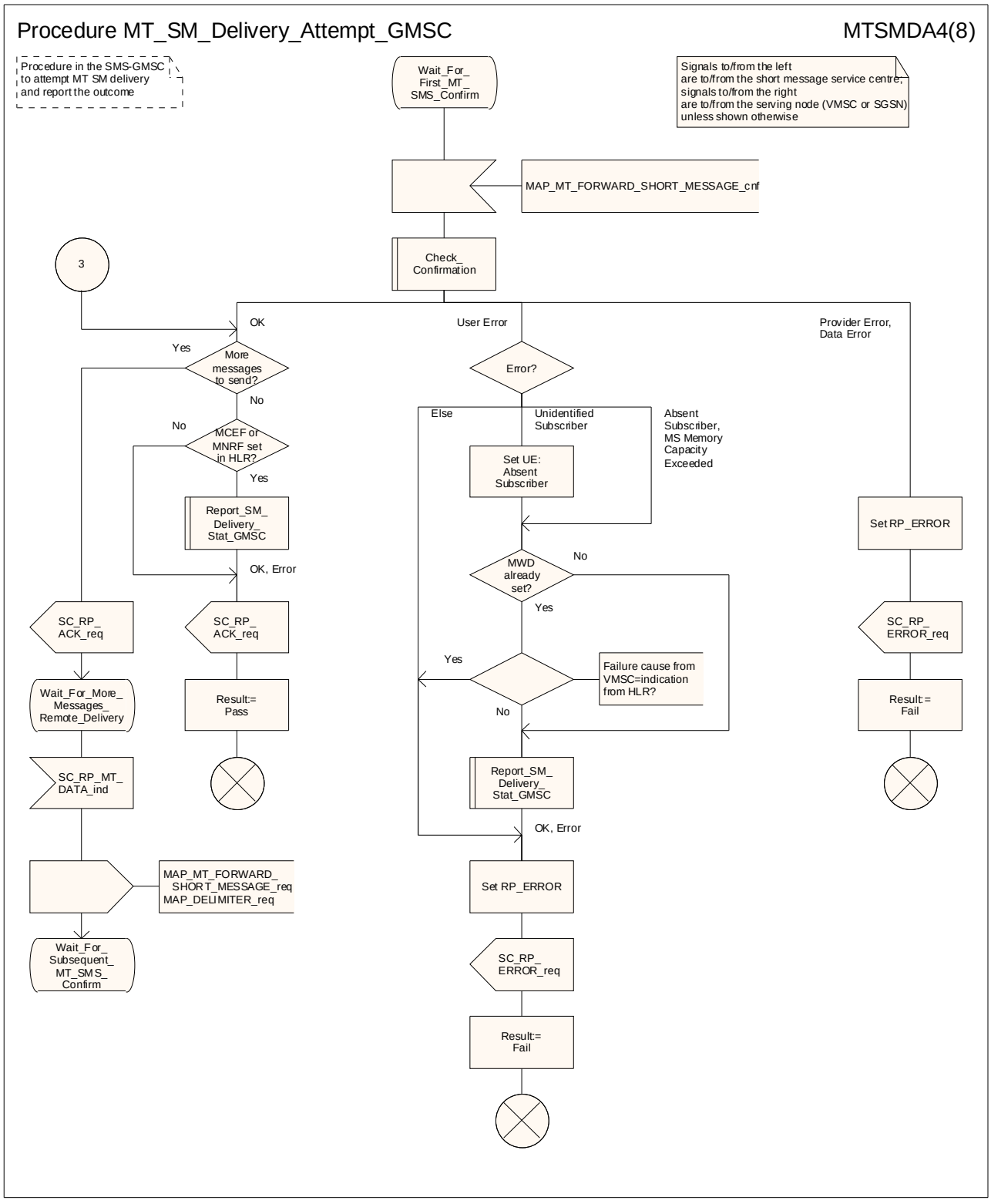


Figure 23.3/4 (sheet 4 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

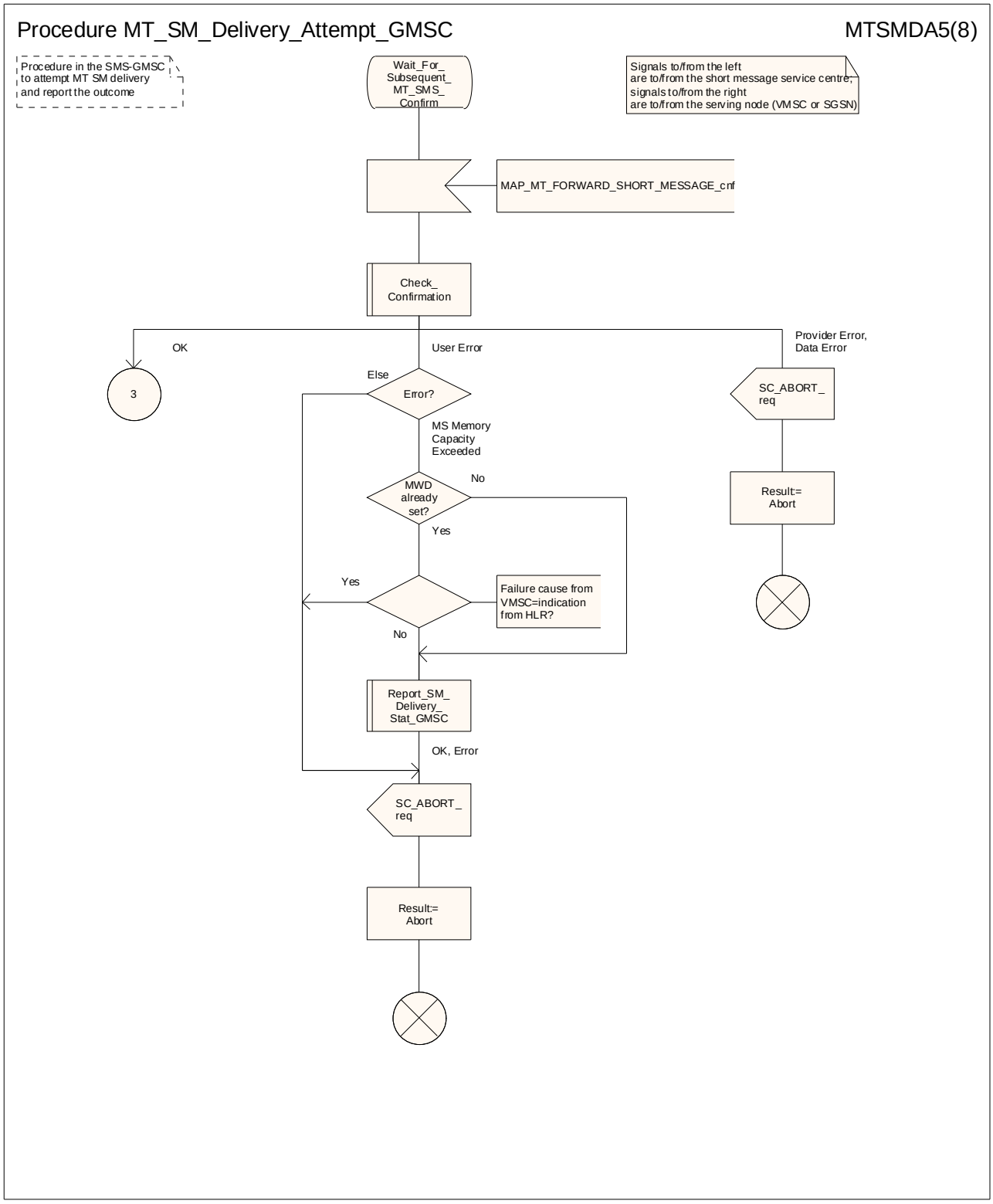


Figure 23.3/4 (sheet 5 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

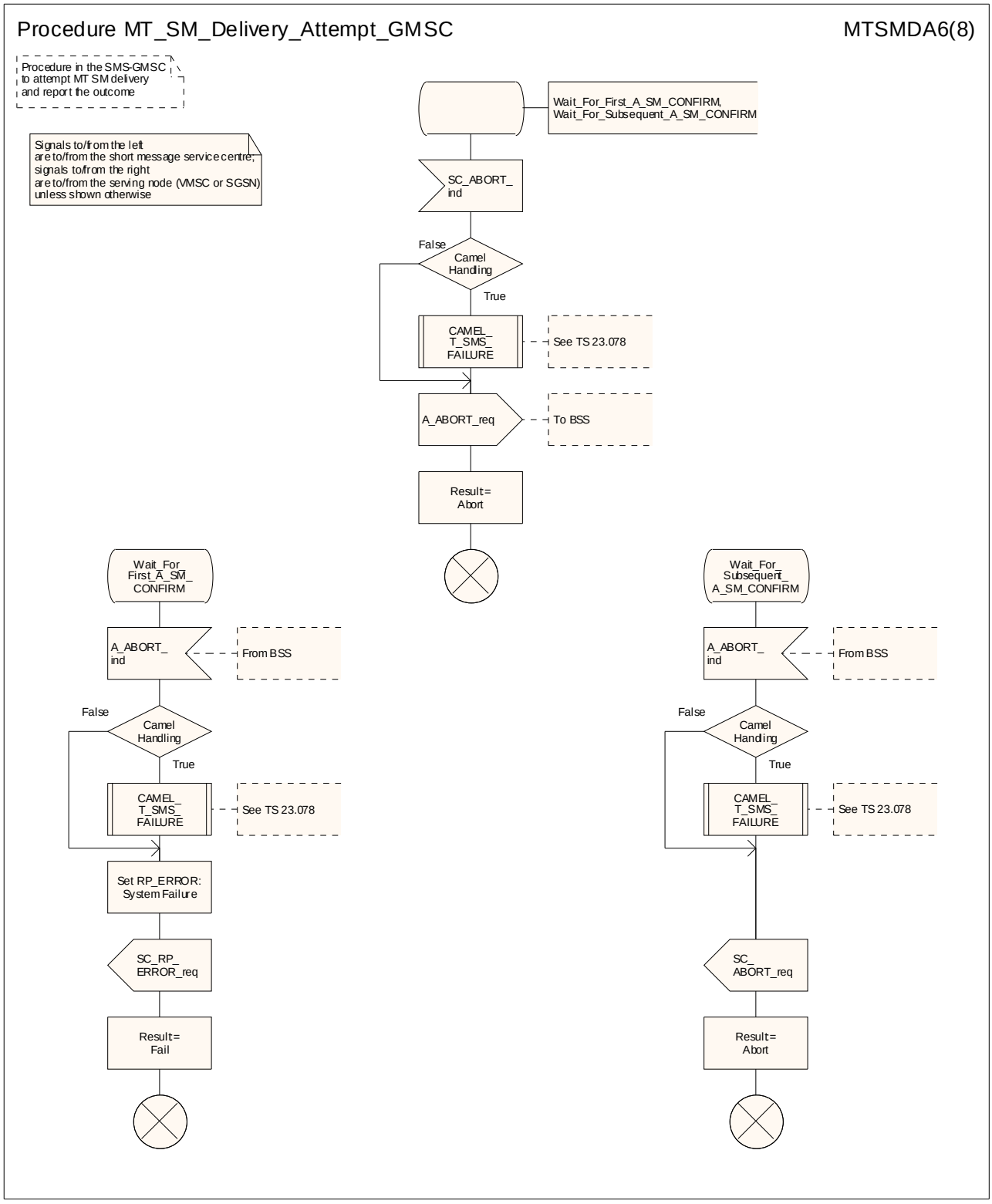


Figure 23.3/4 (sheet 6 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

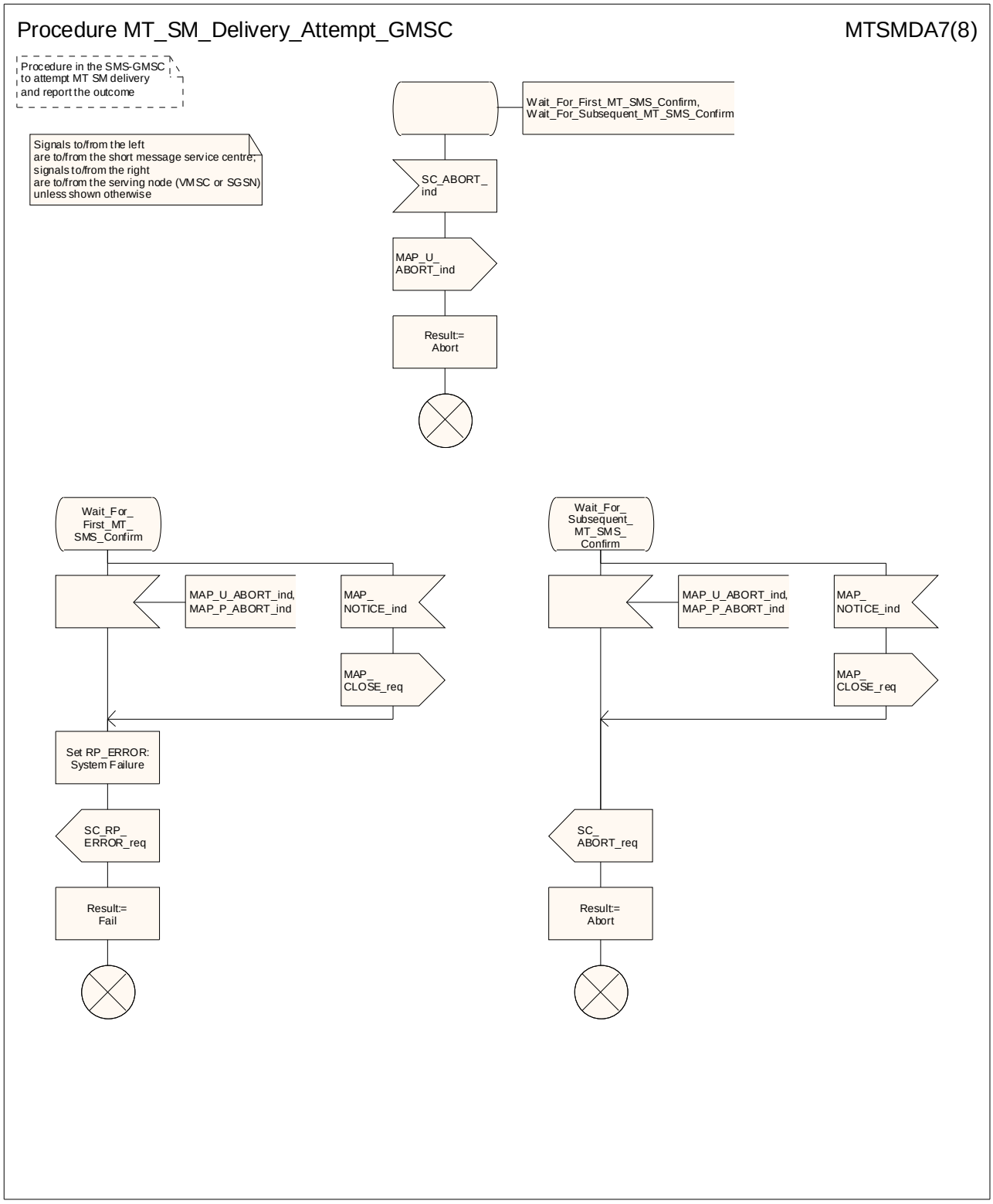


Figure 23.3/4 (sheet 7 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

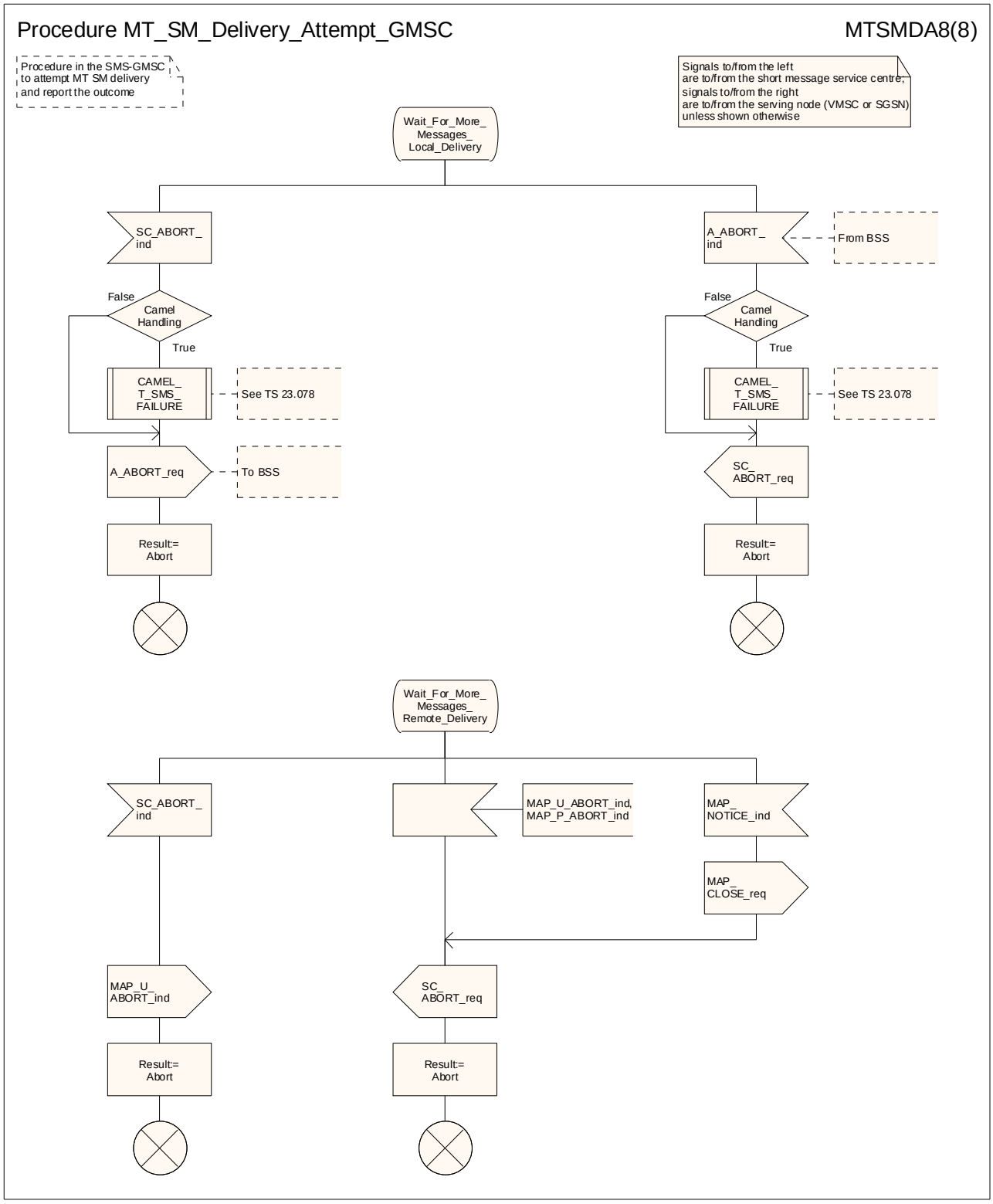


Figure 23.3/4 (sheet 8 of 8): Procedure MT_SM_Delivery_Attempt_GMSC

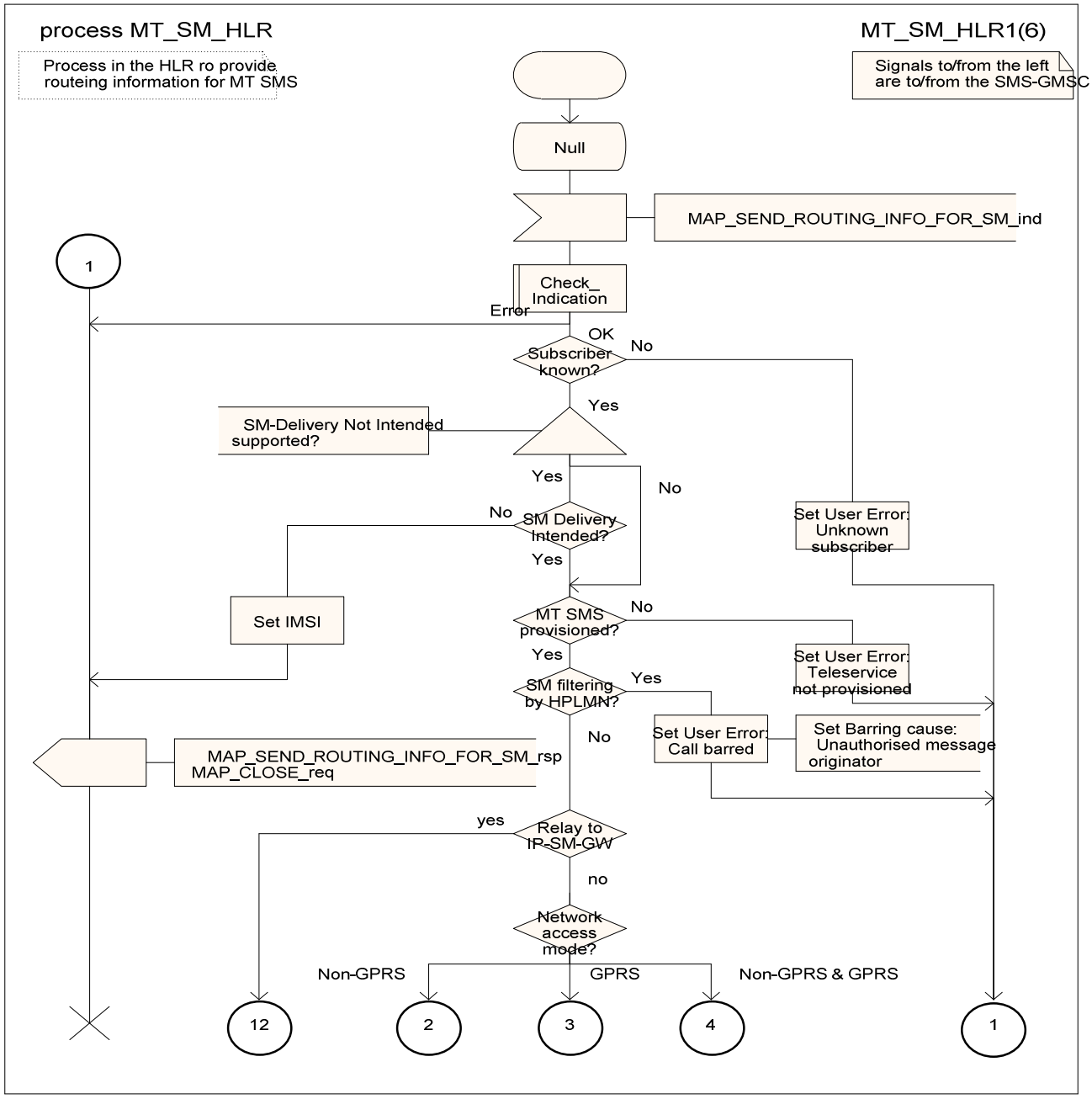


Figure 23.3/5 (sheet 1 of 6): Process MT_SM_HLR

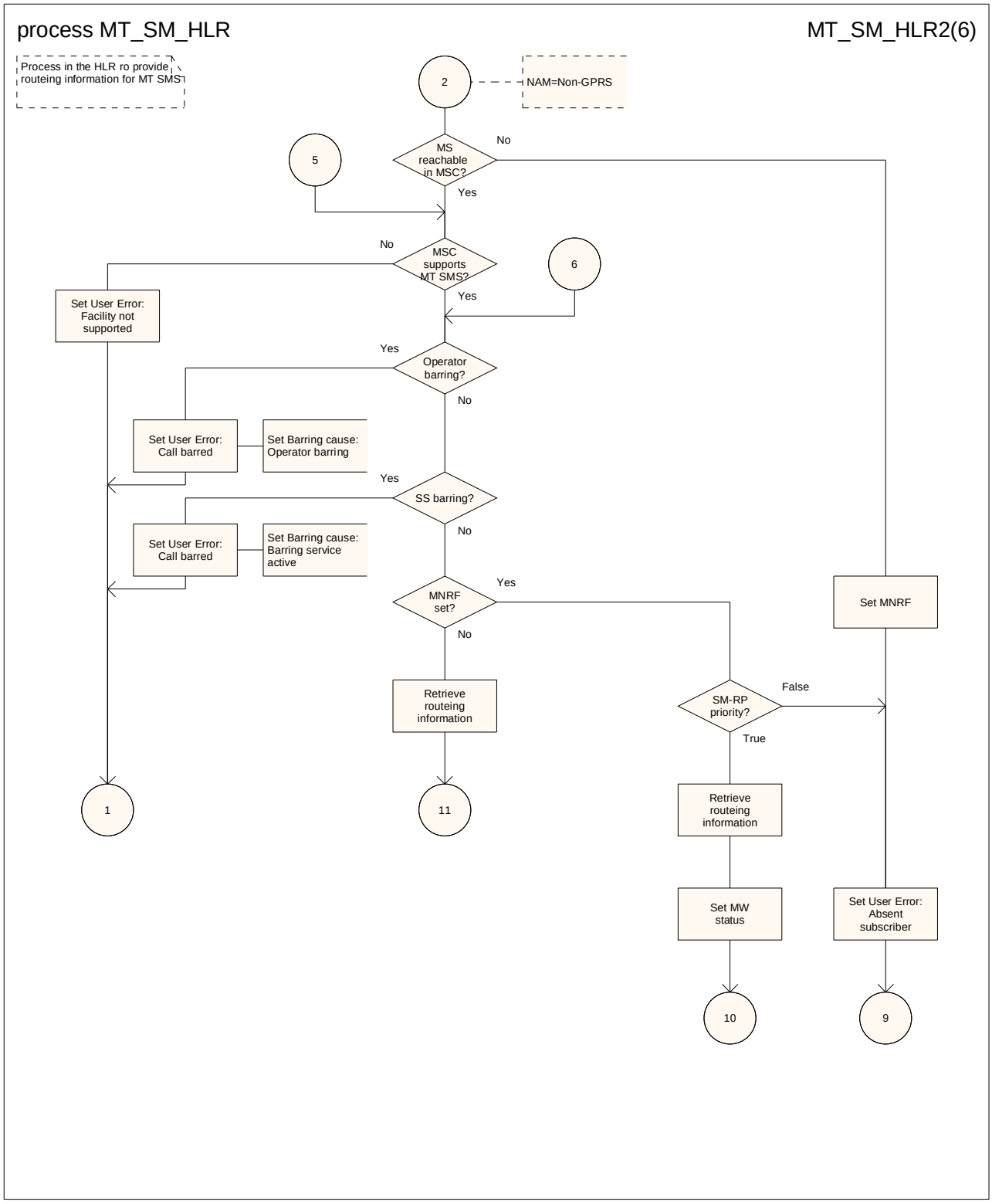


Figure 23.3/5 (sheet 2 of 6): Process MT_SM_HLR

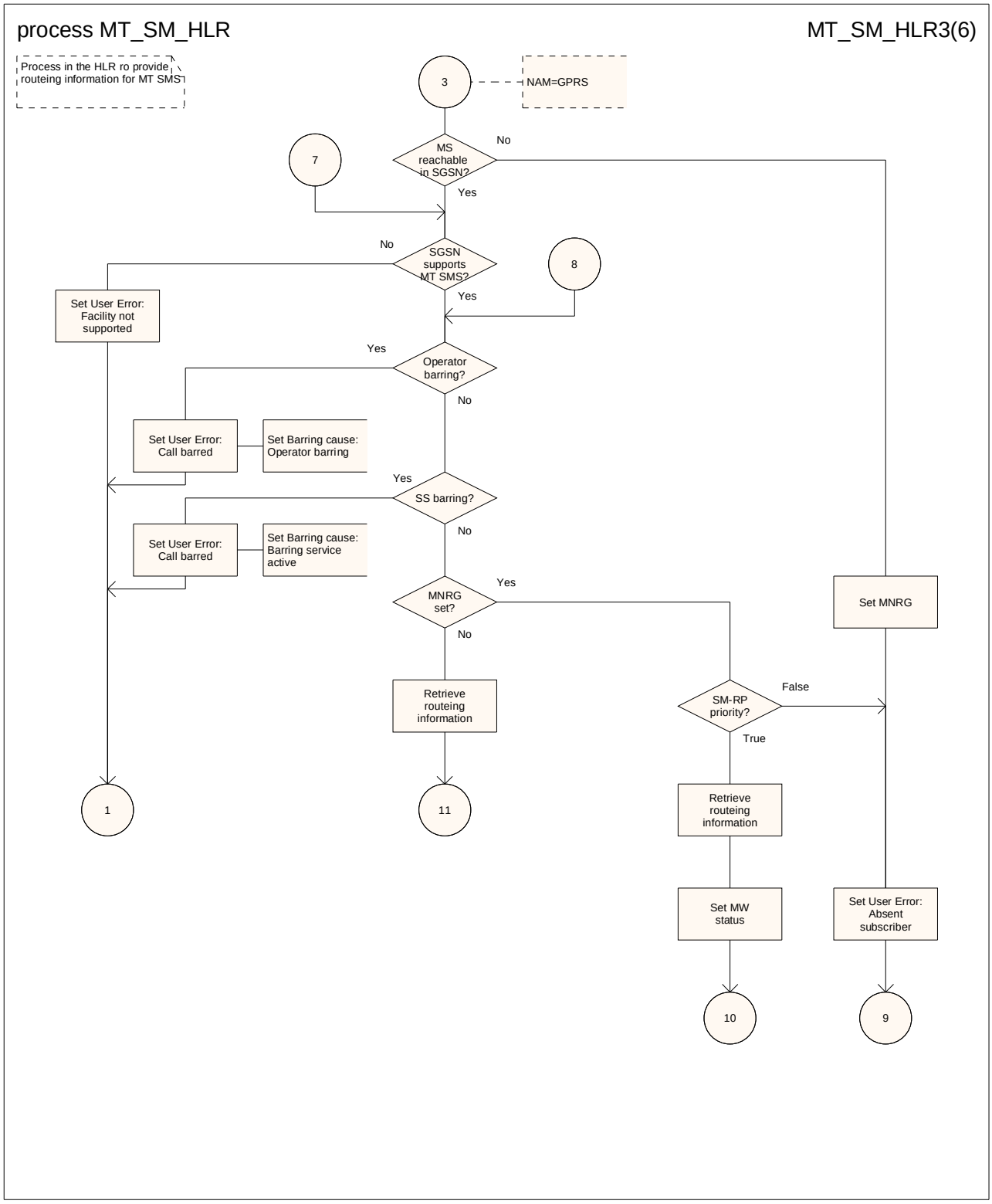


Figure 23.3/5 (sheet 3 of 6): Process MT_SM_HLR

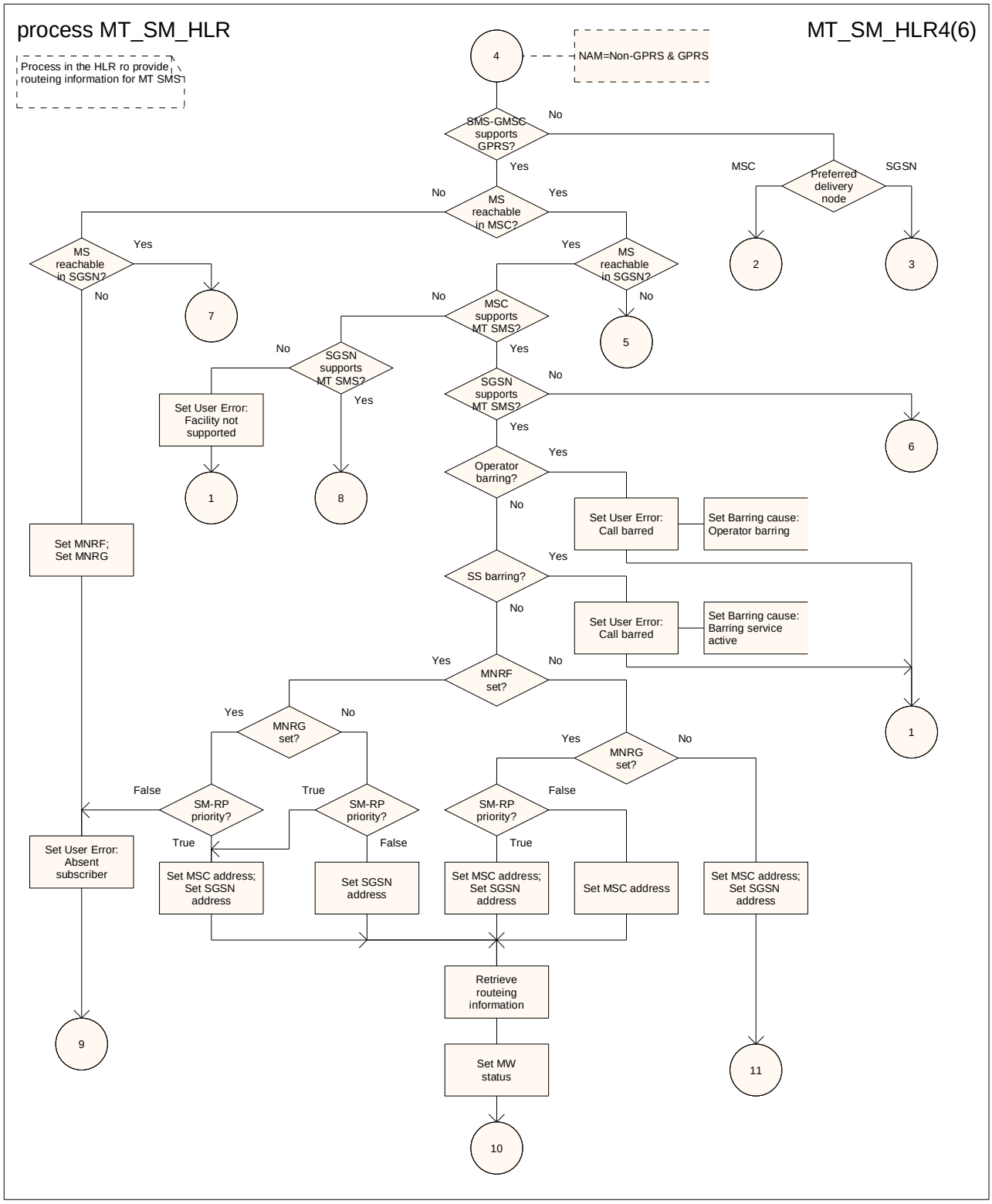


Figure 23.3/5 (sheet 4 of 6): Process MT_SM_HLR

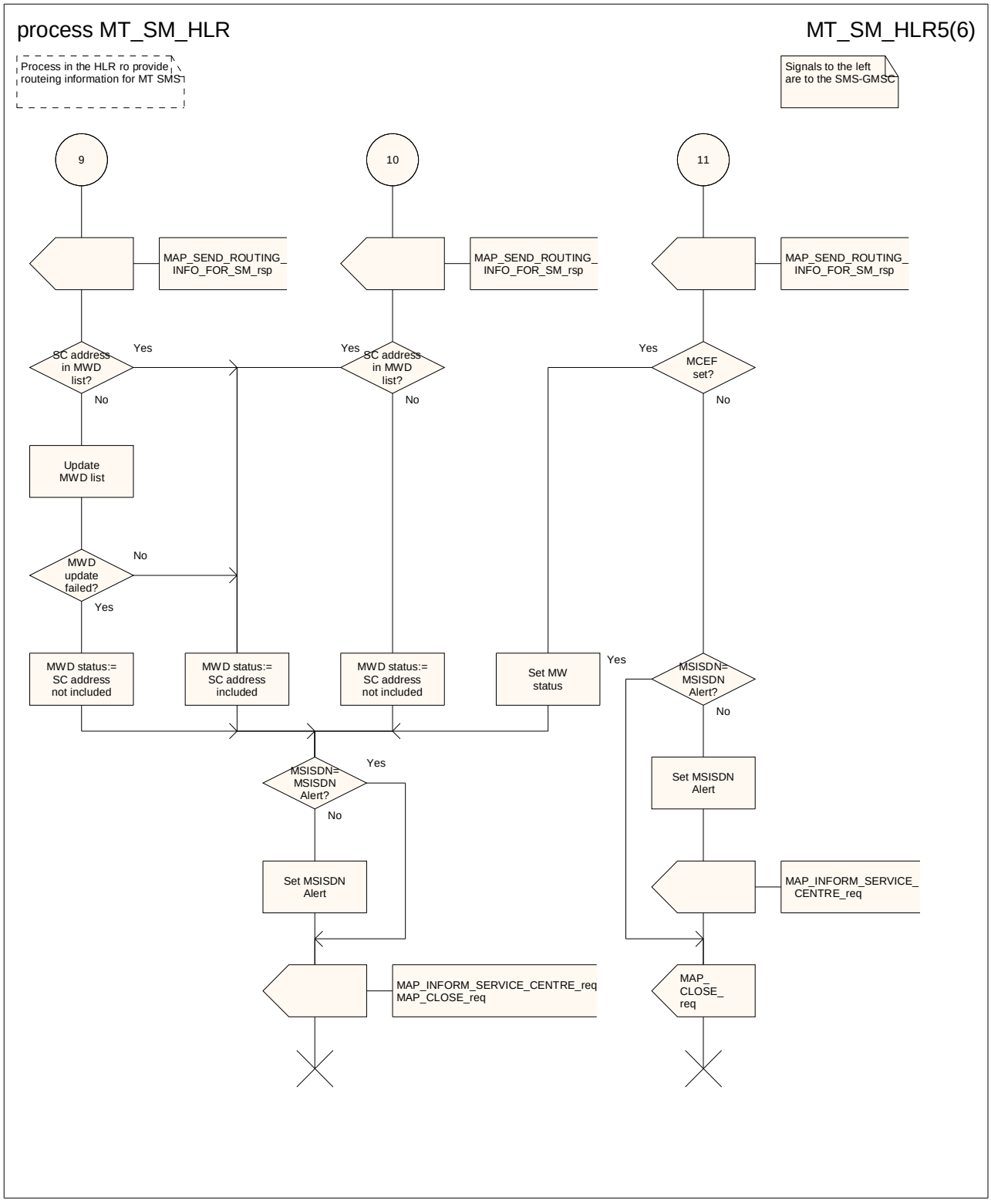


Figure 23.3/5 (sheet 5 of 6): Process MT_SM_HLR

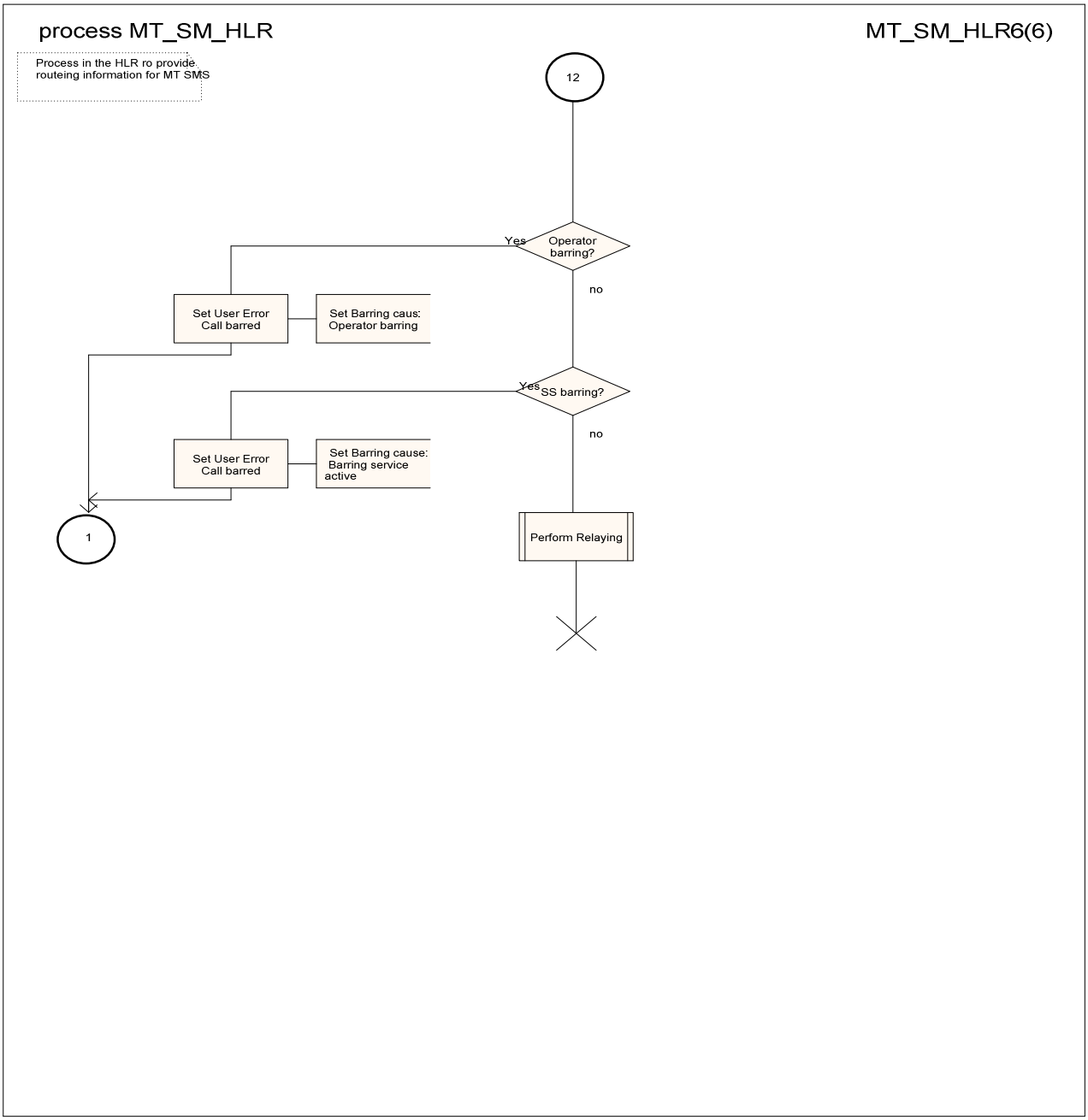


Figure 23.3/5 (sheet 6 of 6): Process MT_SM_HLR

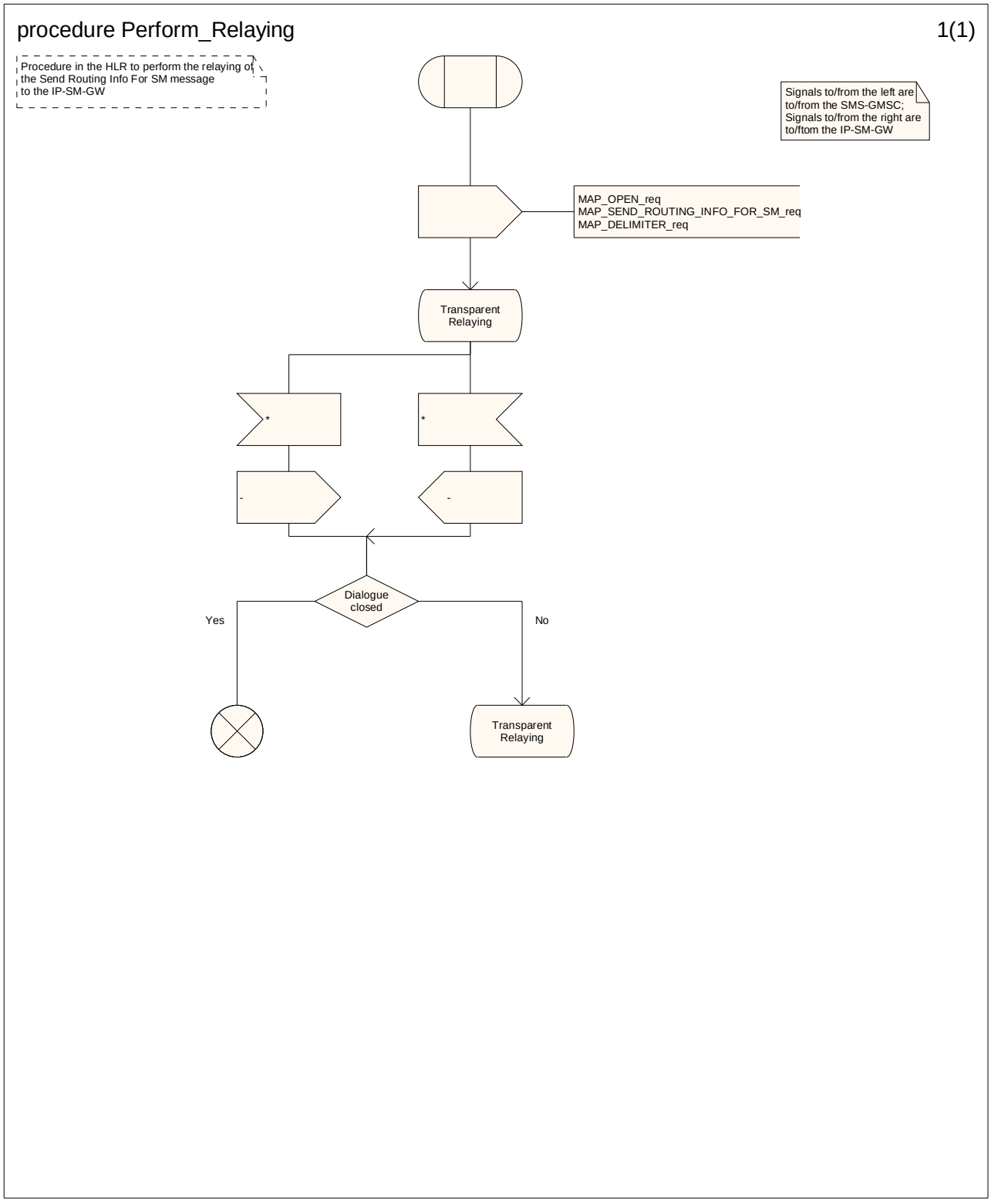


Figure 23.3/5a (sheet 1 of 1): Procedure Perform_Relaying

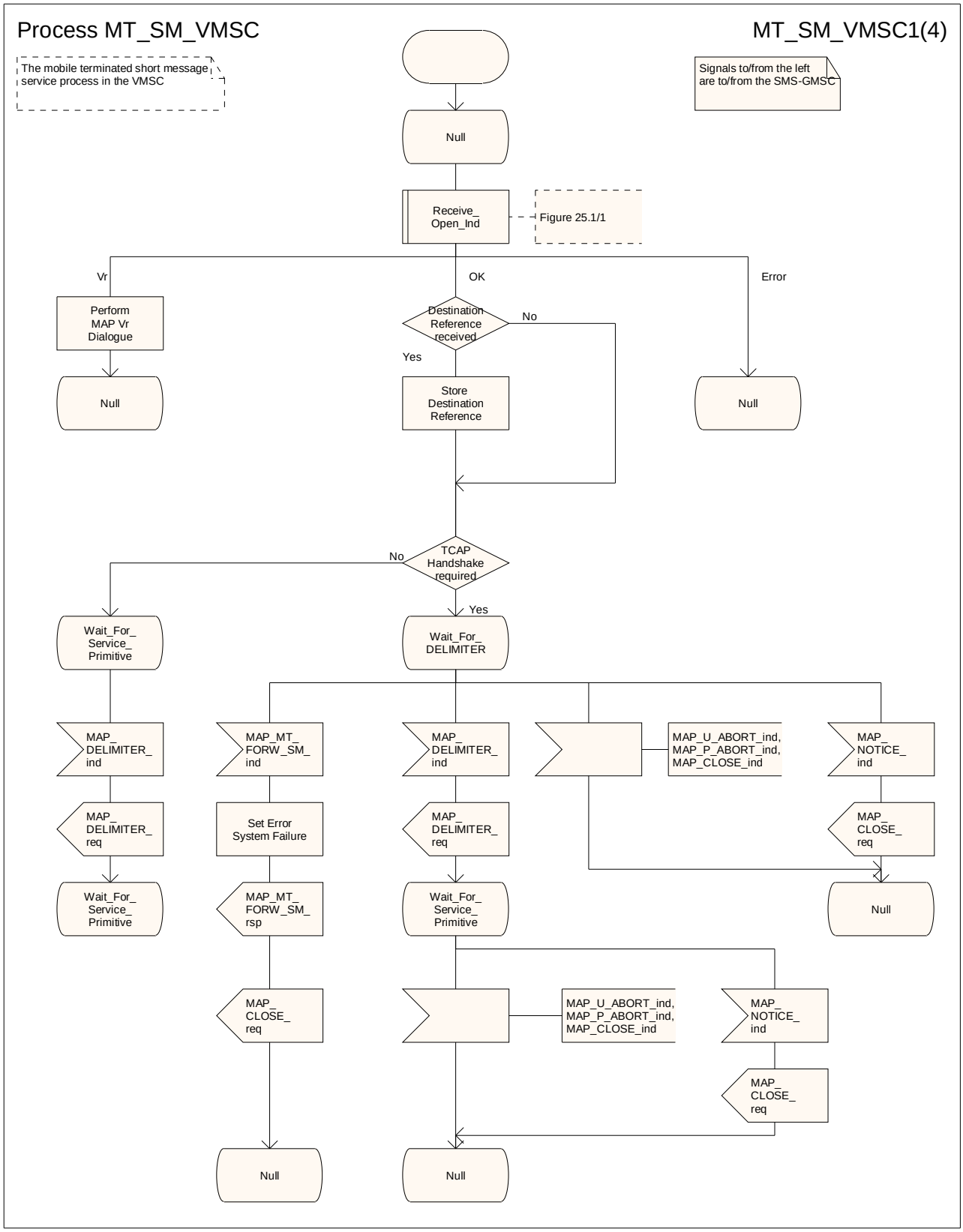


Figure 23.3/6 (sheet 1 of 4): Procedure MT_SM_VMSC

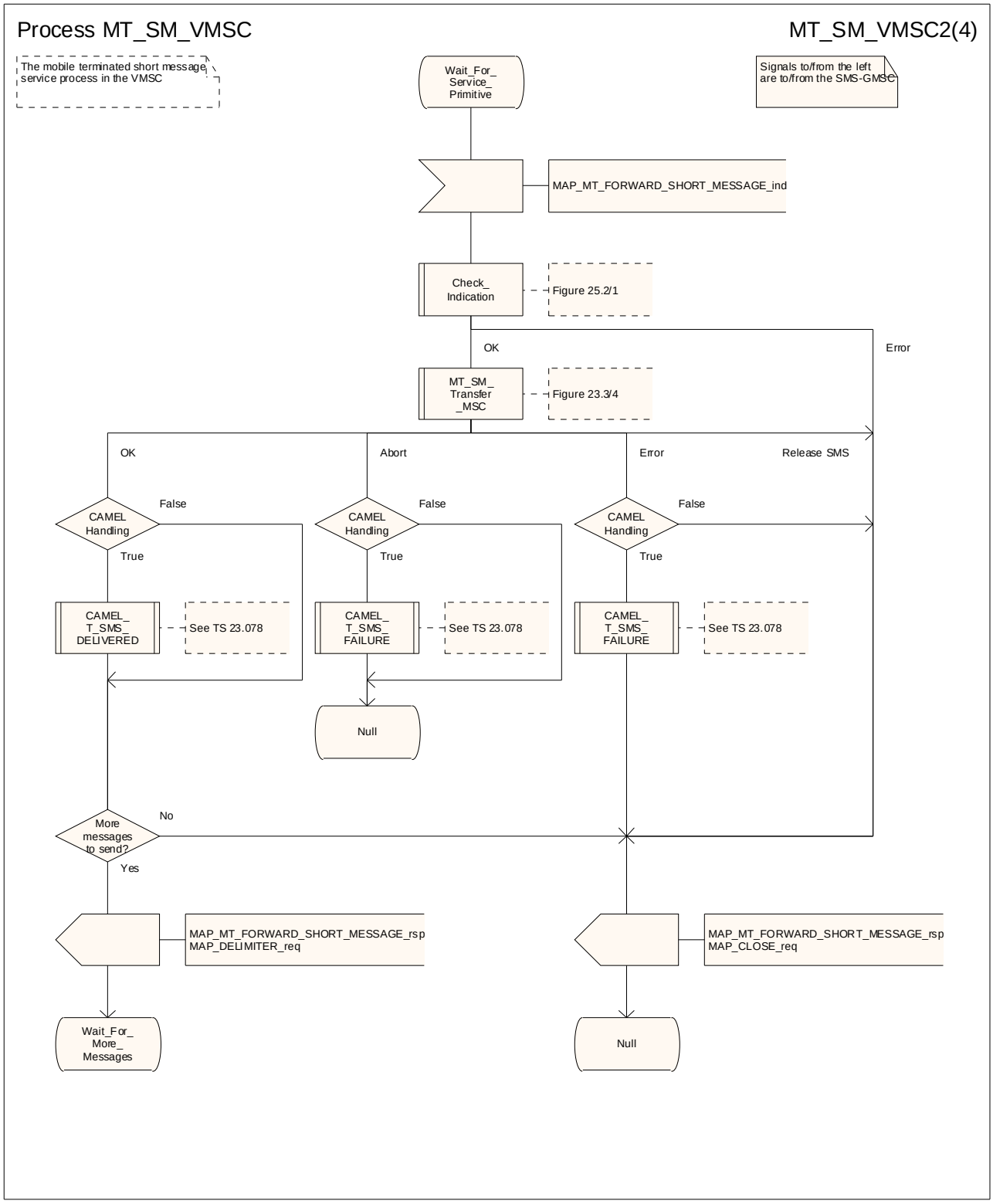


Figure 23.3/6 (sheet 2 of 4): Procedure MT_SM_VMSC

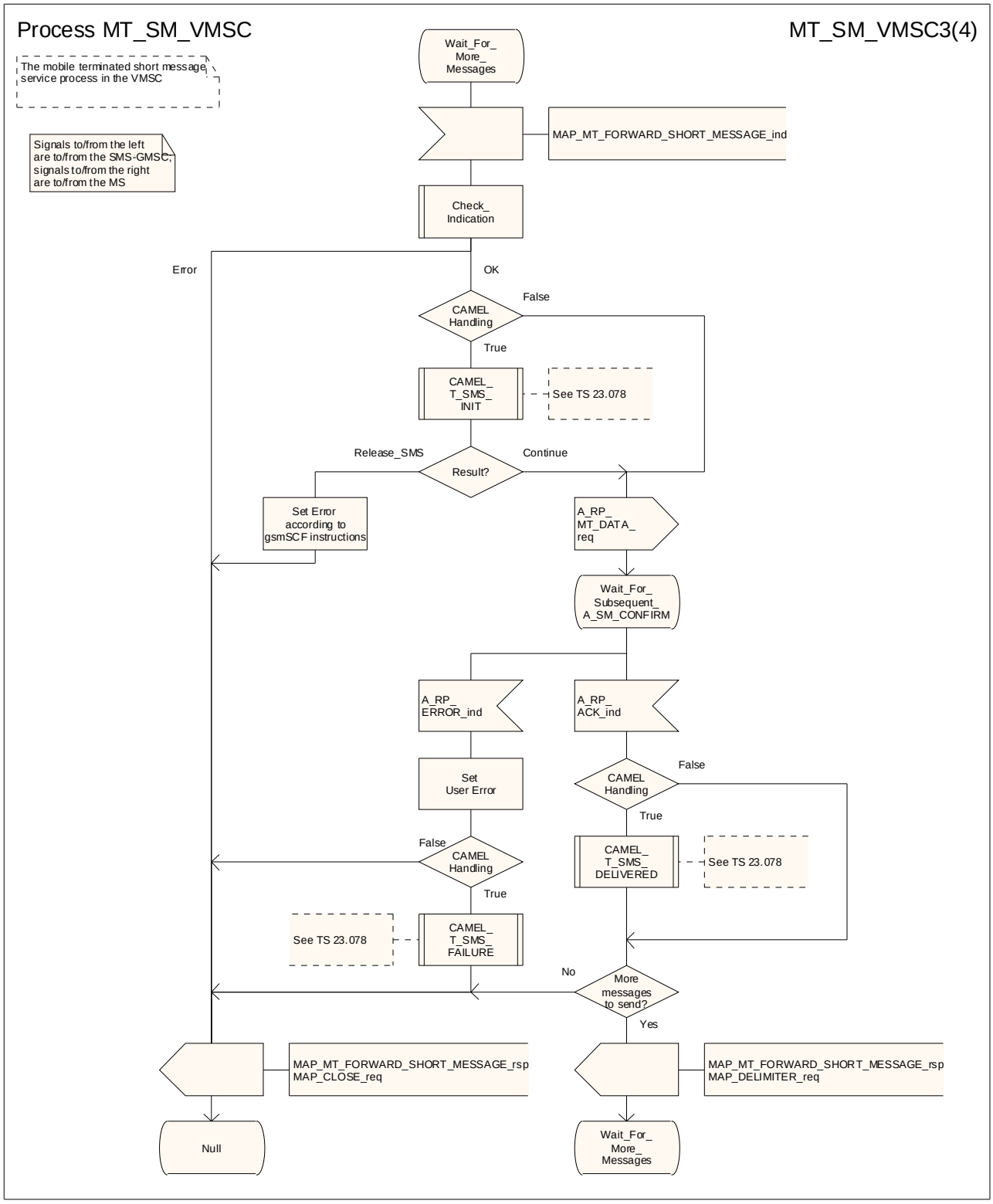


Figure 23.3/6 (sheet 3 of 4): Procedure MT_SM_VMSC

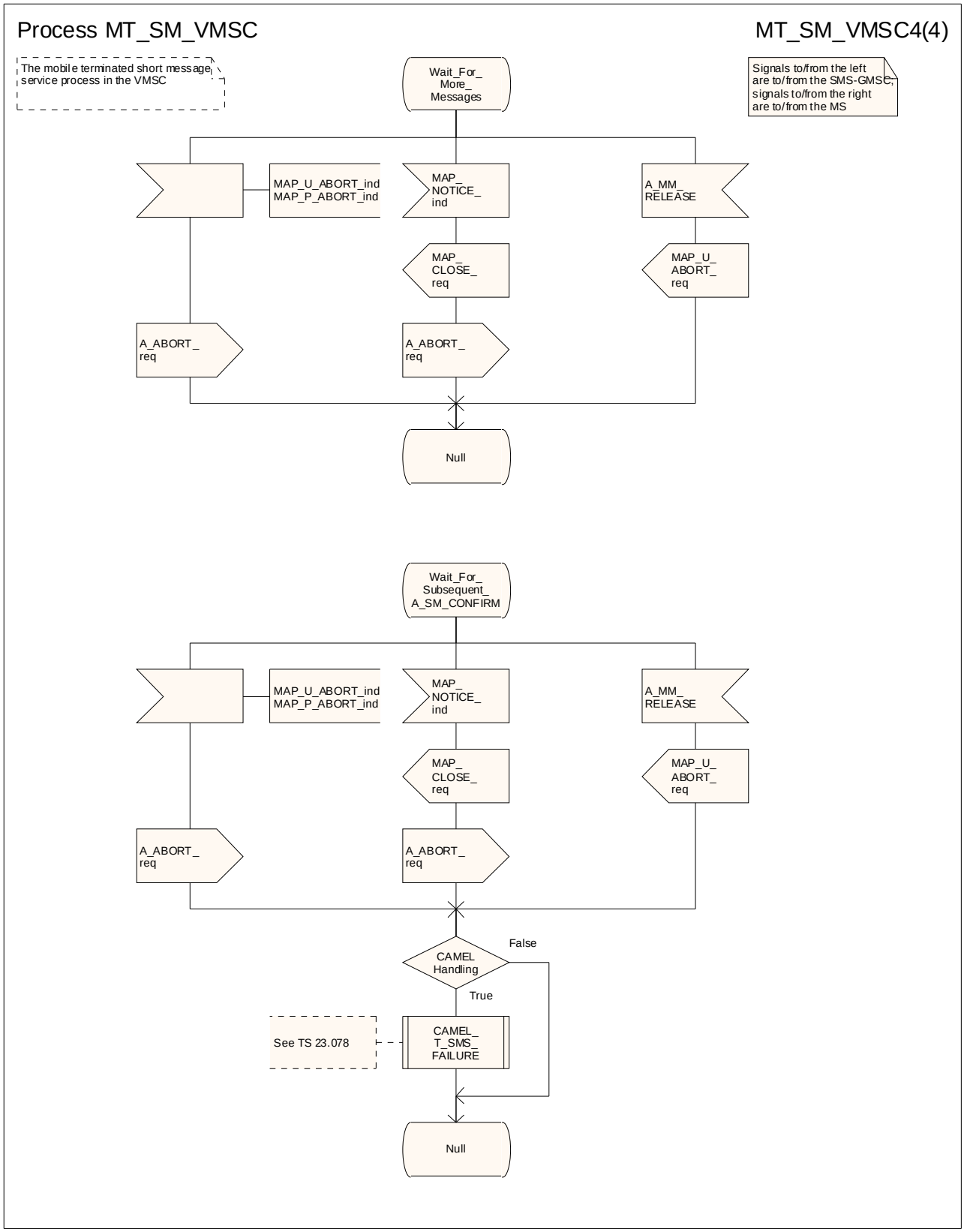


Figure 23.3/6 (sheet 4 of 4): Procedure MT_SM_VMSC

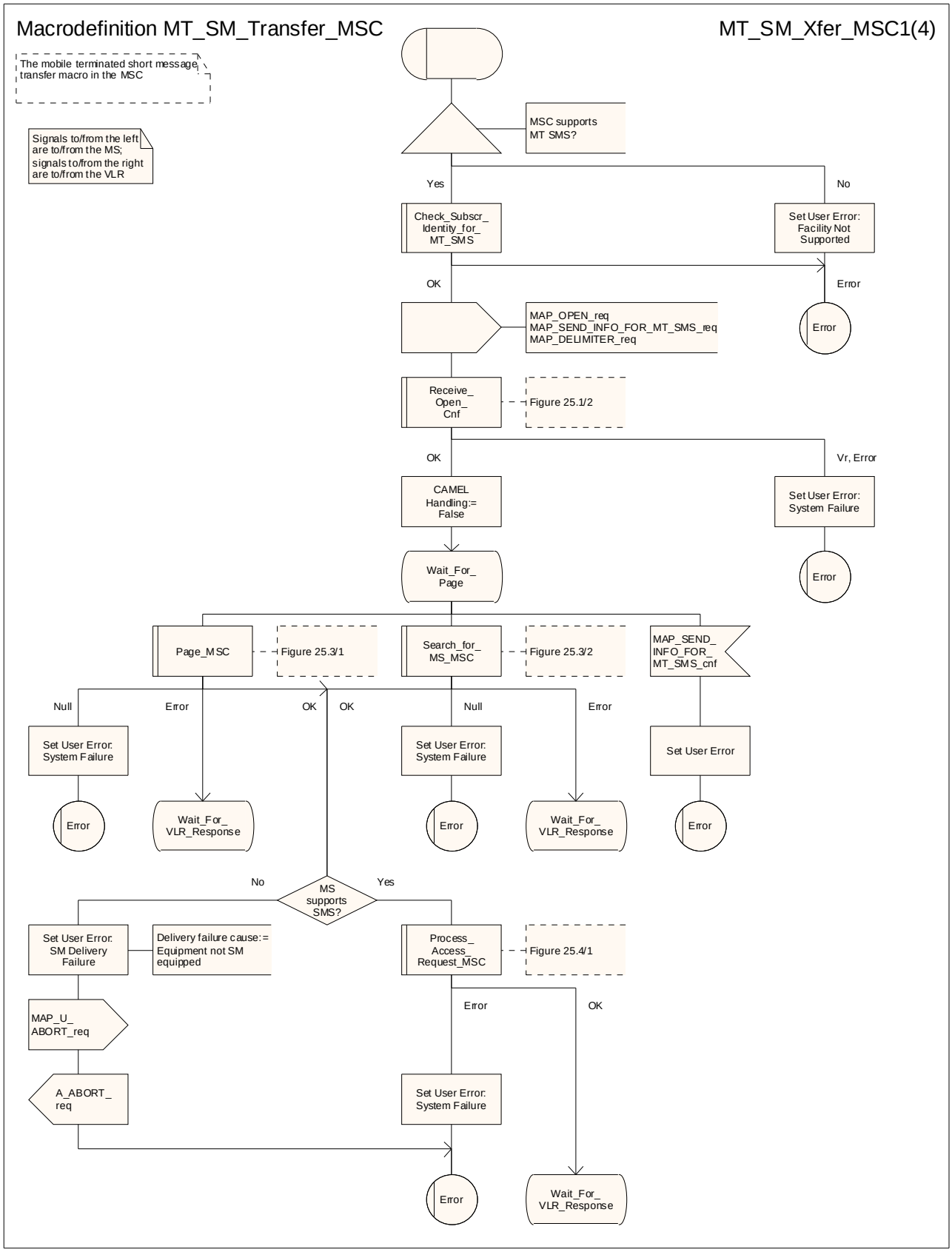


Figure 23.3/7 (sheet 1 of 4): Macro MT_SM_Transfer_MSC

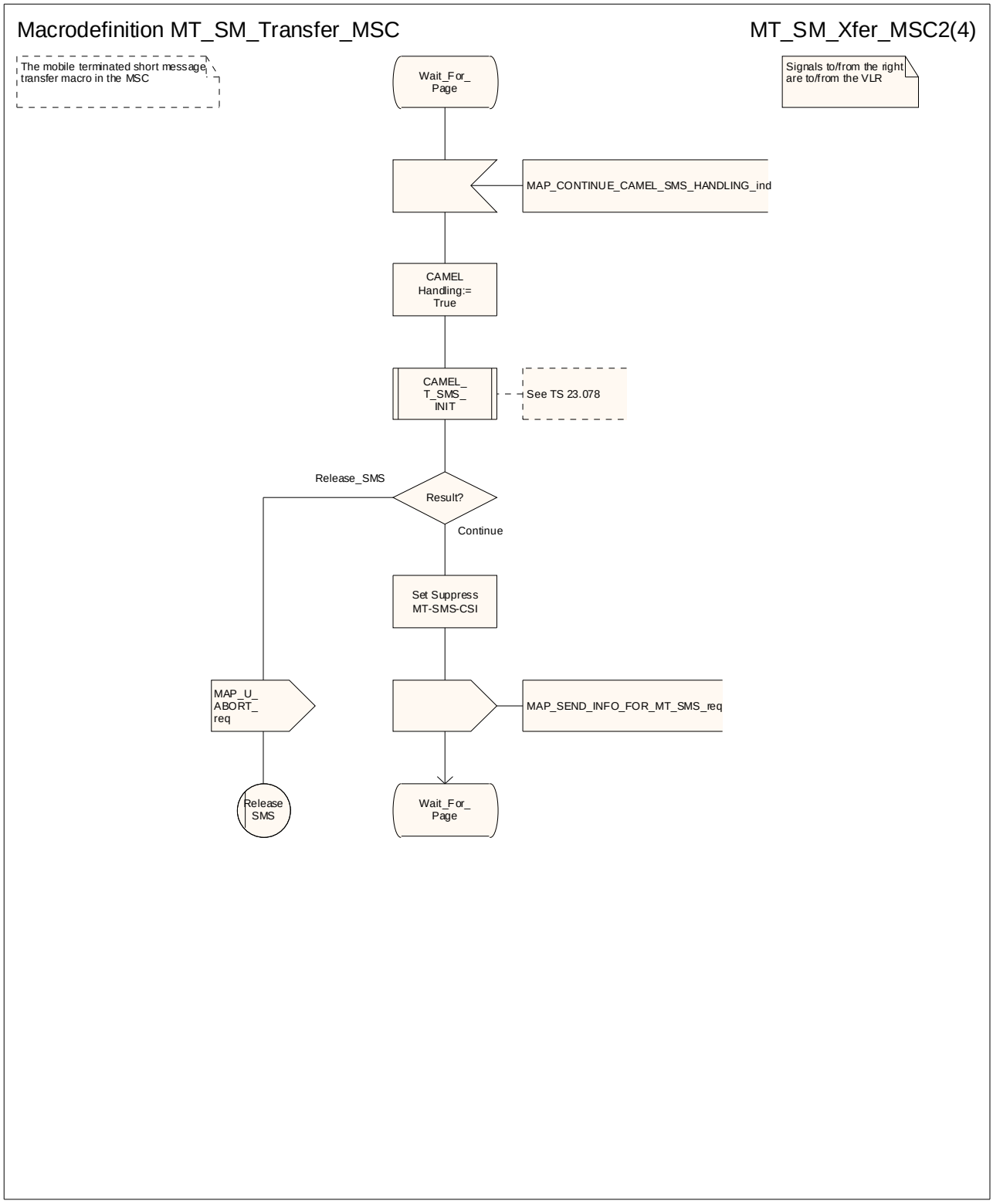


Figure 23.3/7 (sheet 2 of 4): Macro MT_SM_Transfer_MSC

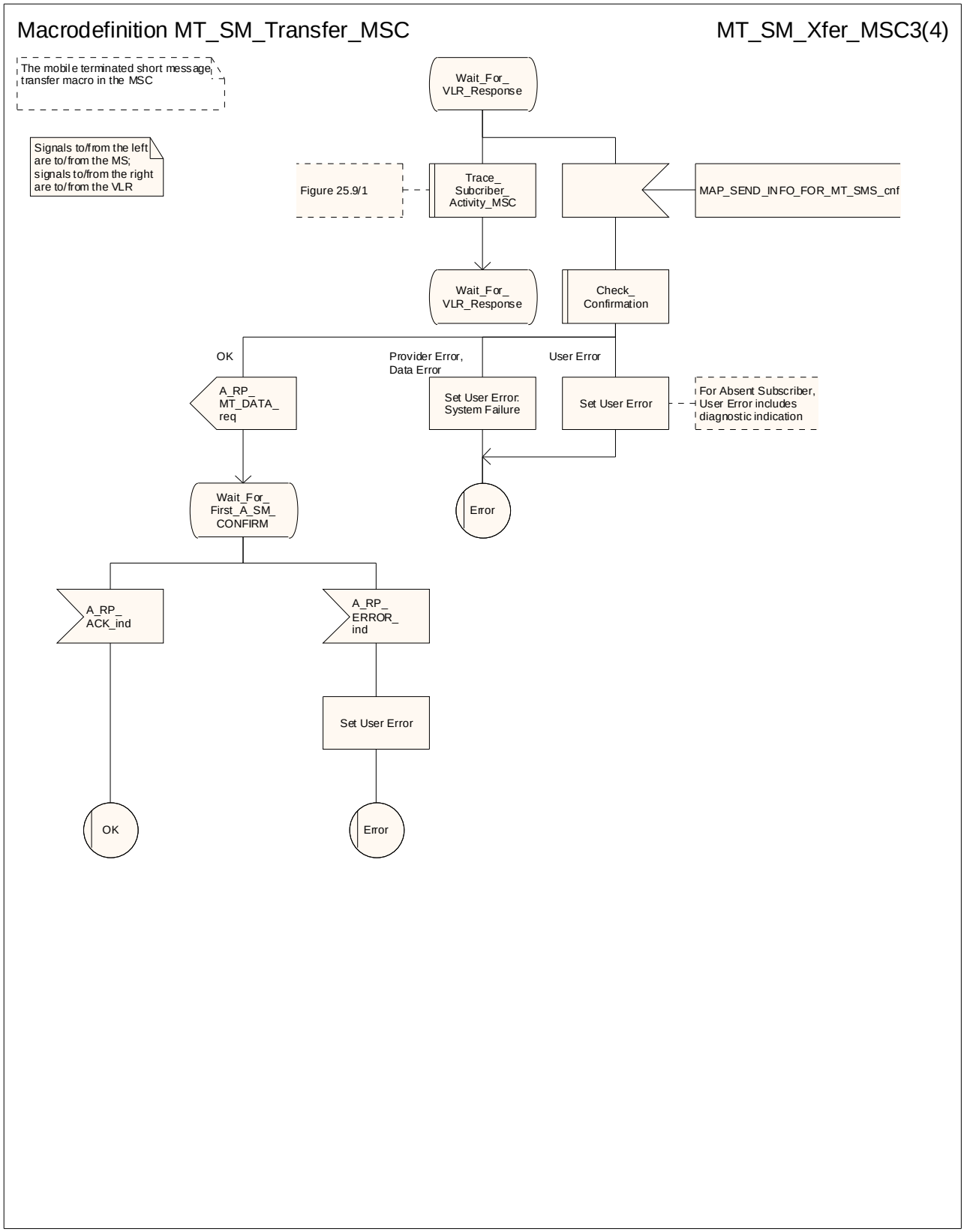


Figure 23.3/7 (sheet 3 of 4): Macro MT_SM_Transfer_MSC

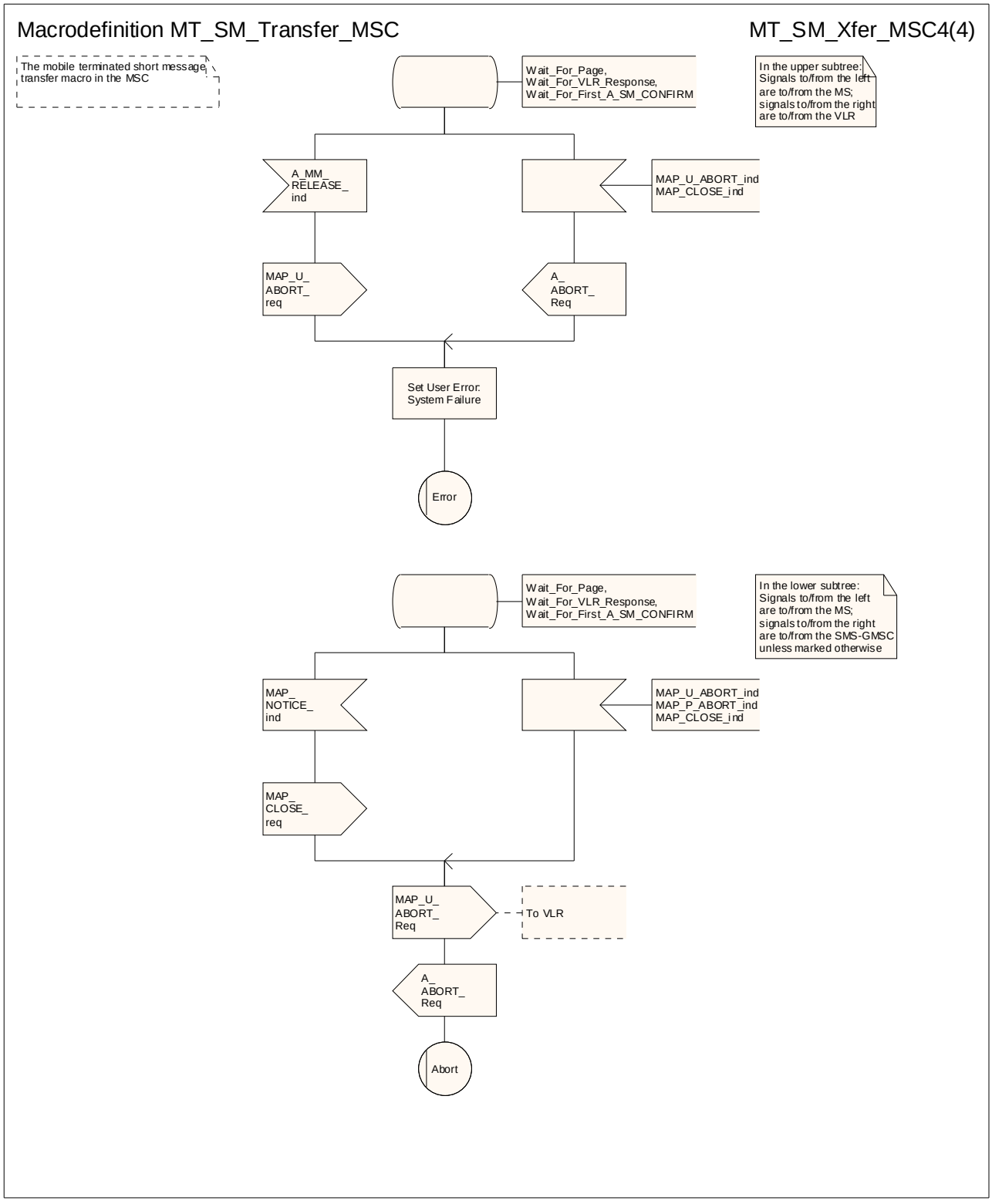


Figure 23.3/7 (sheet 4 of 4): Macro MT_SM_Transfer_MSC

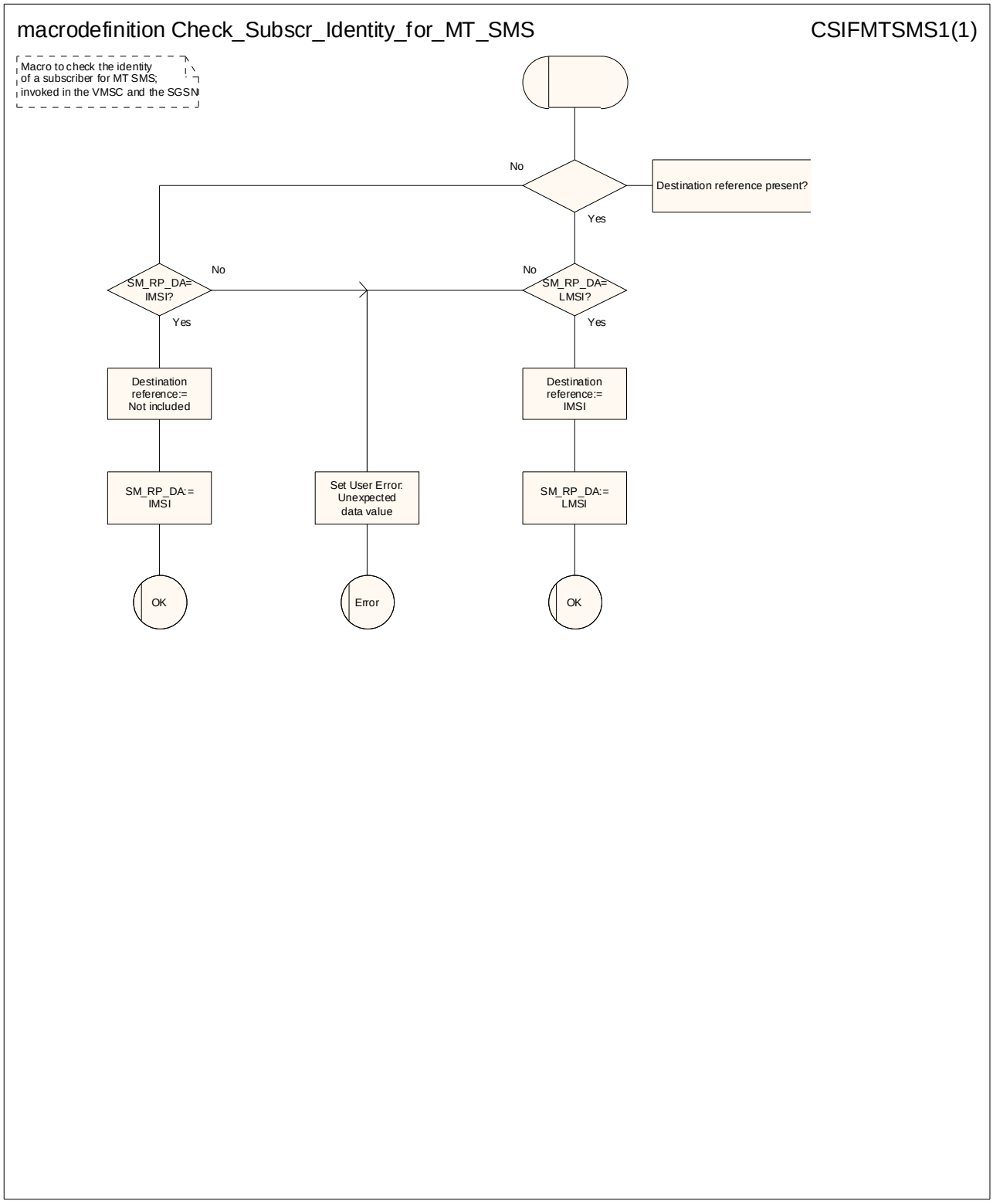


Figure 23.3/8: Macro Check_Subscr_Identity_For_MT_SMS

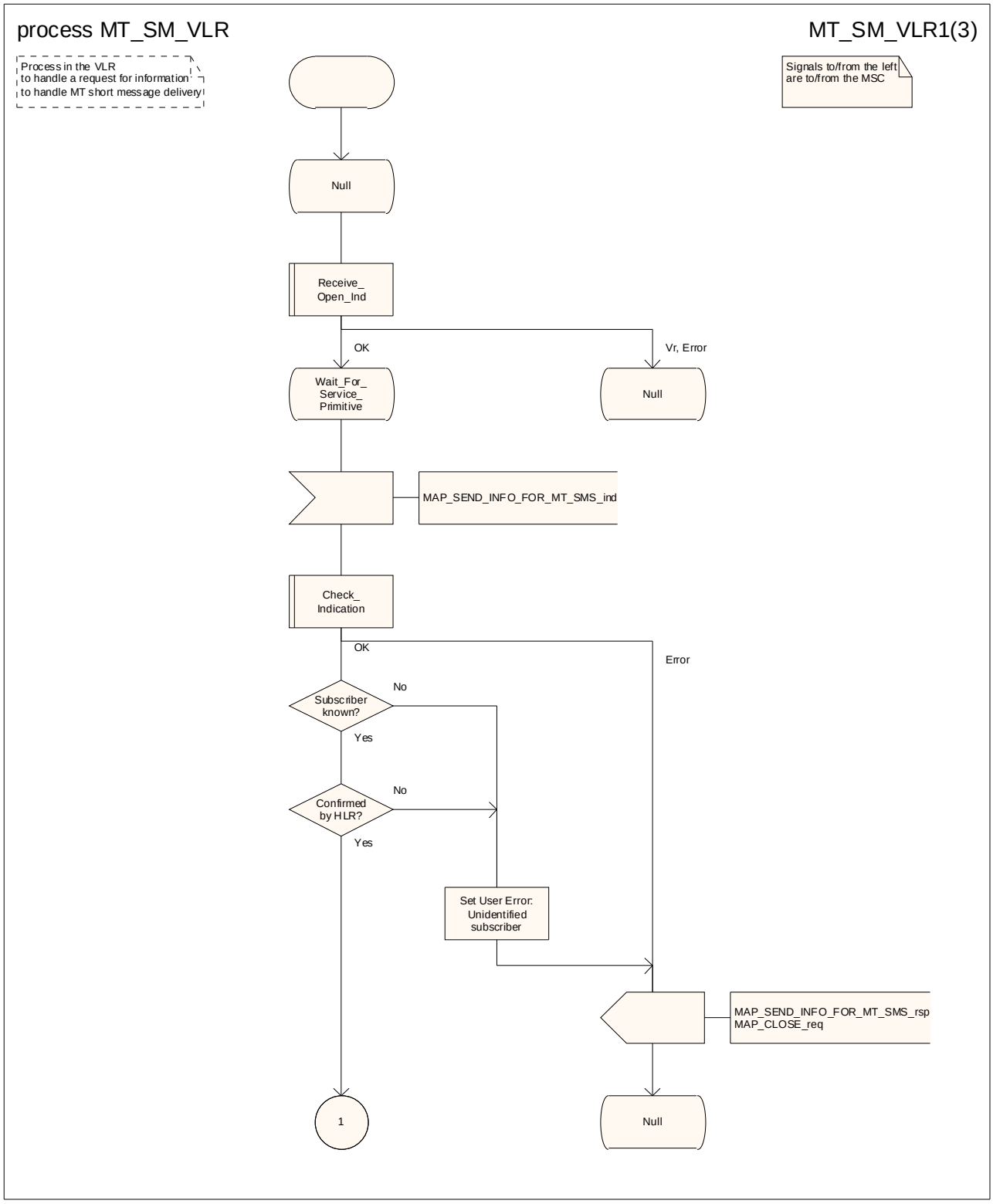


Figure 23.3/9 (sheet 1 of 3): Process MT_SM_VLR

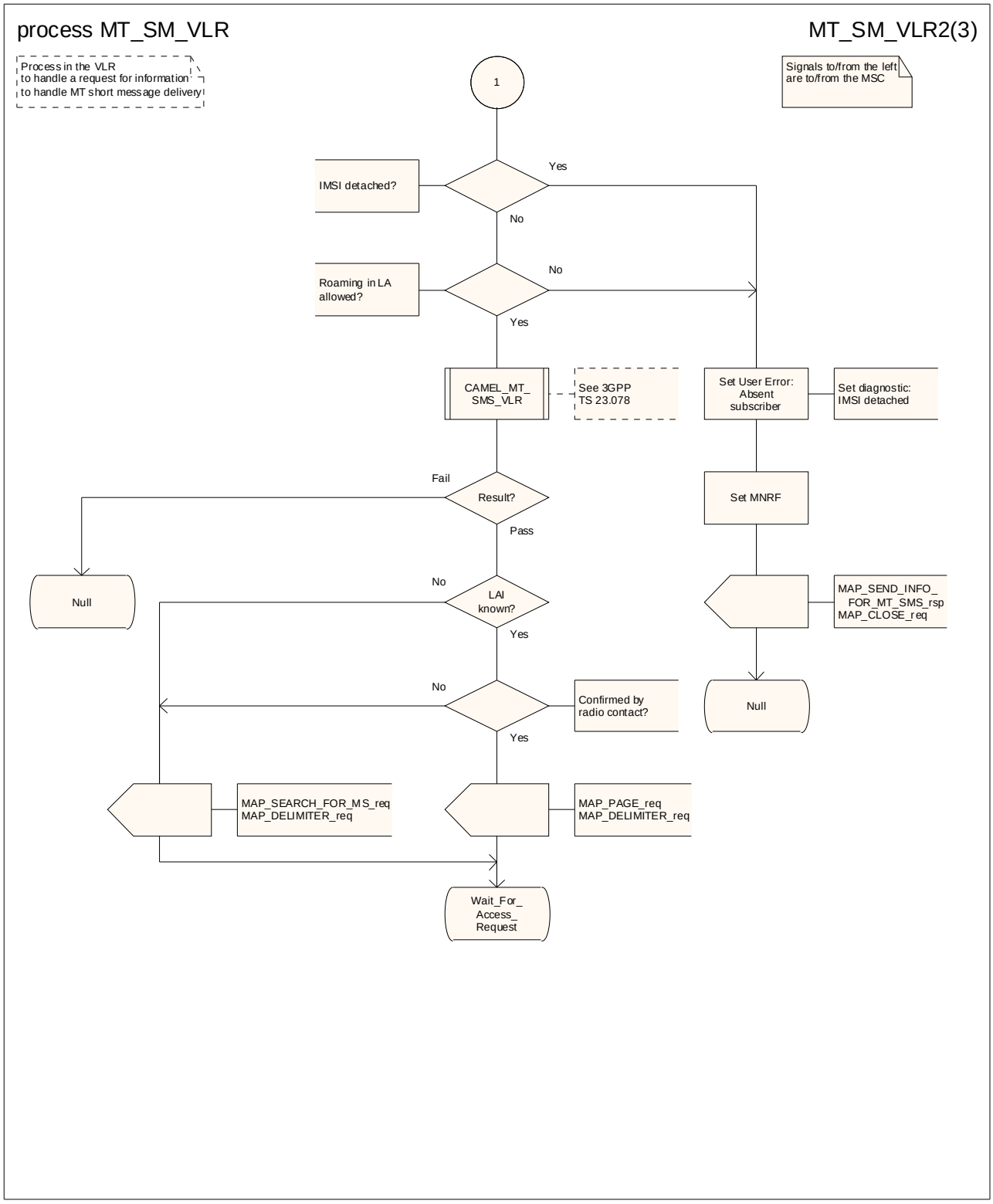


Figure 23.3/9 (sheet 2 of 3): Process MT_SM_VLR

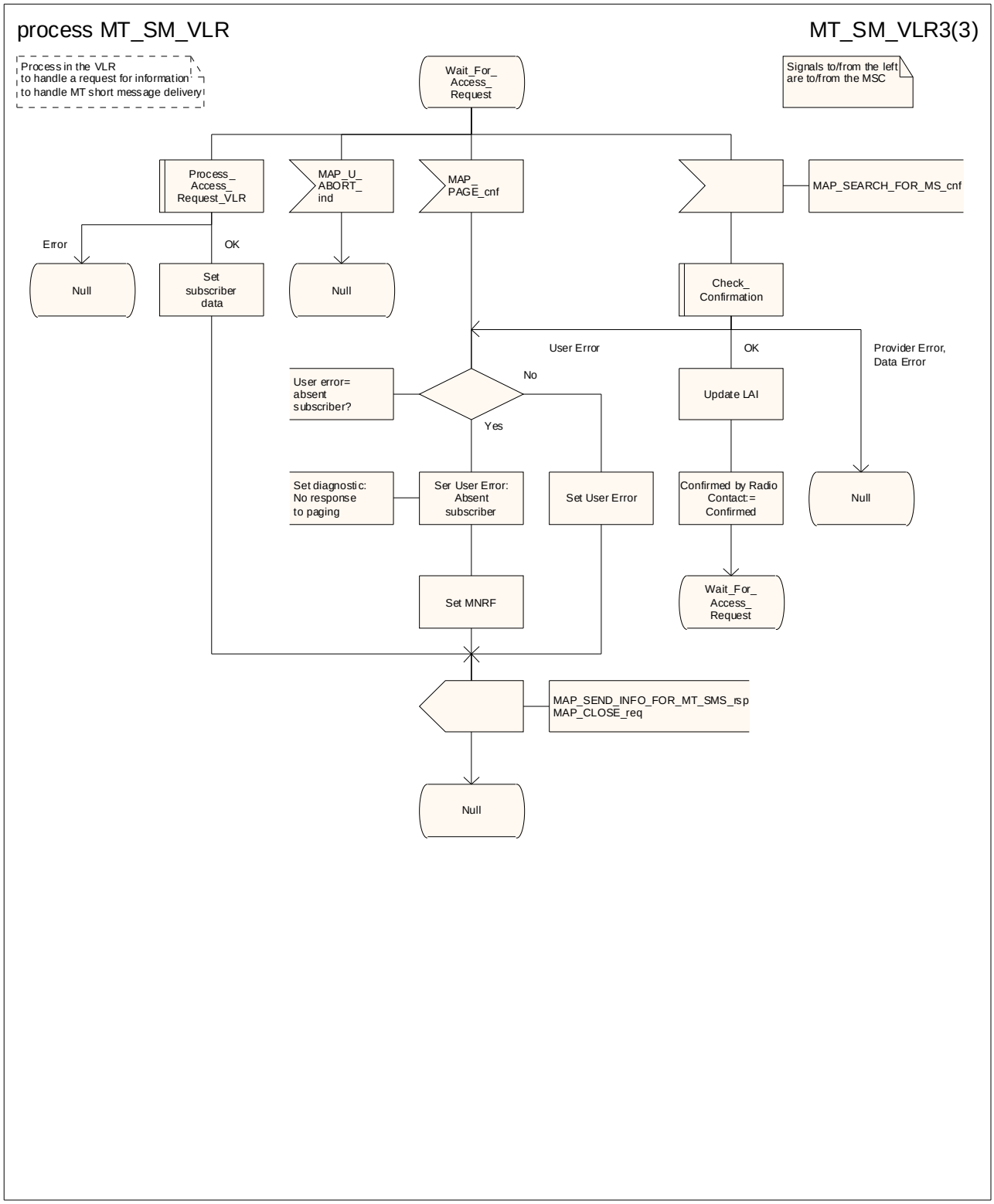


Figure 23.3/9 (sheet 3 of 3): Process MT_SM_VLR

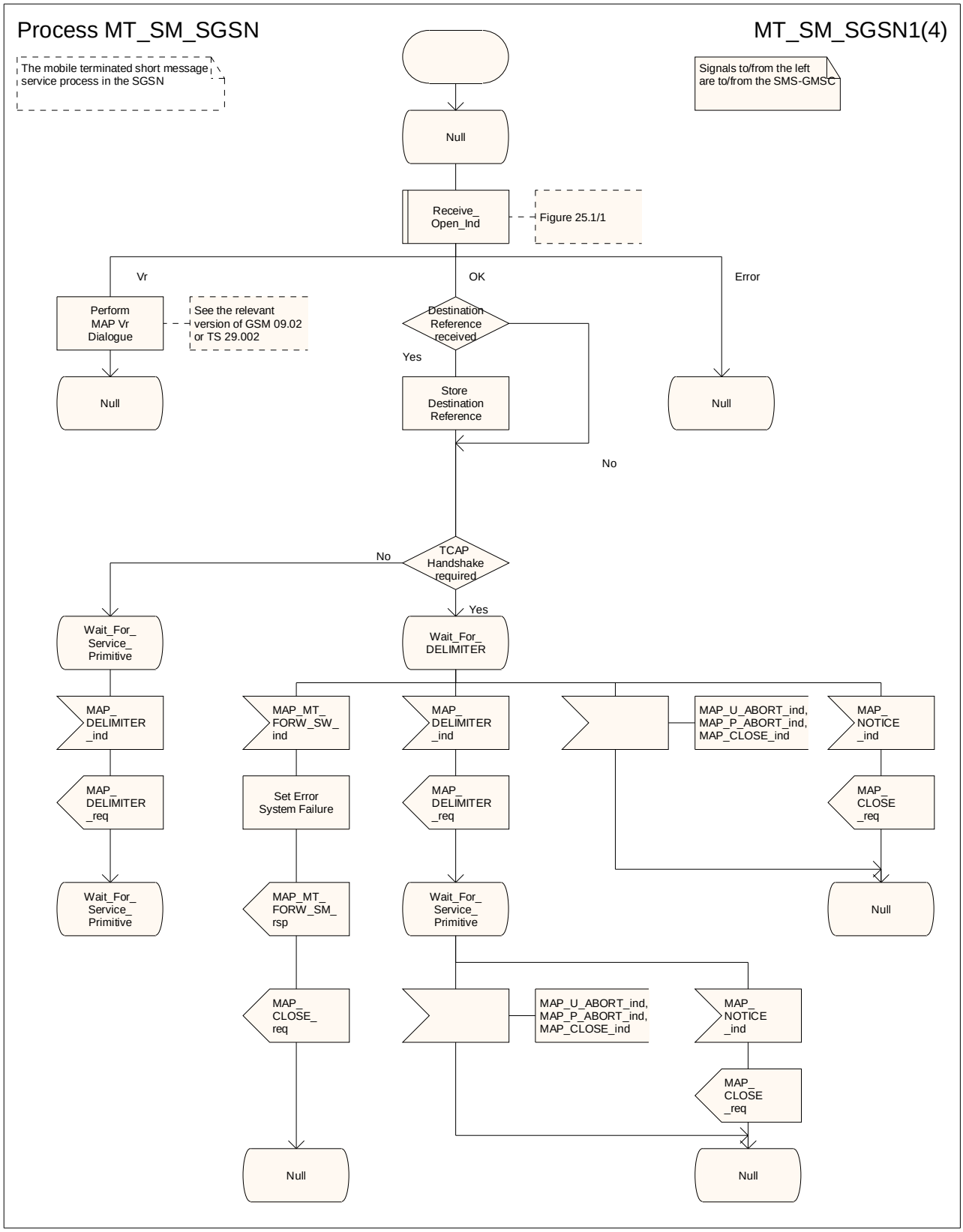


Figure 23.3/10 (sheet 1 of 4): Process MT_SM_SGSN

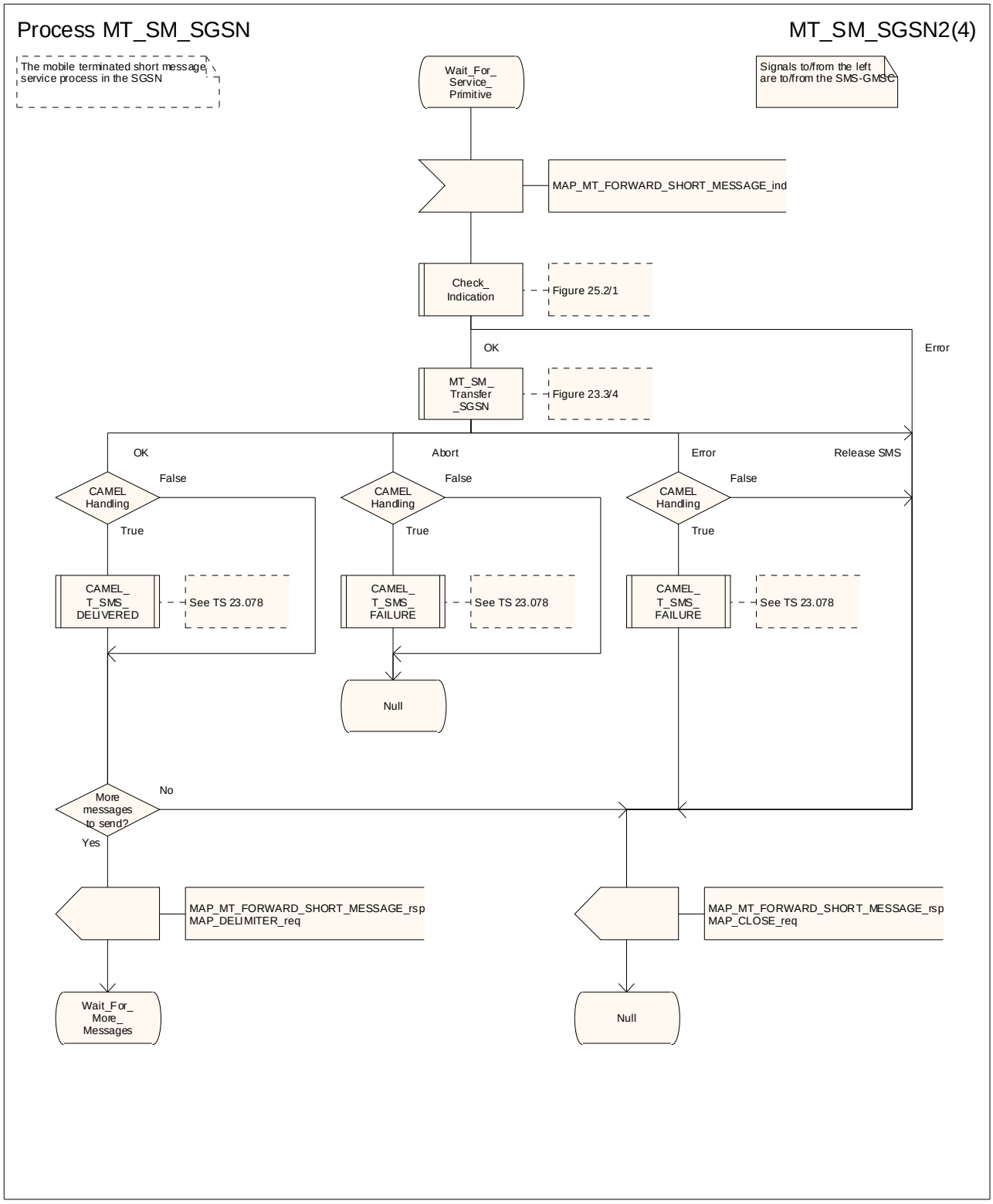


Figure 23.3/10 (sheet 2 of 4): Process MT_SM_ SGSN

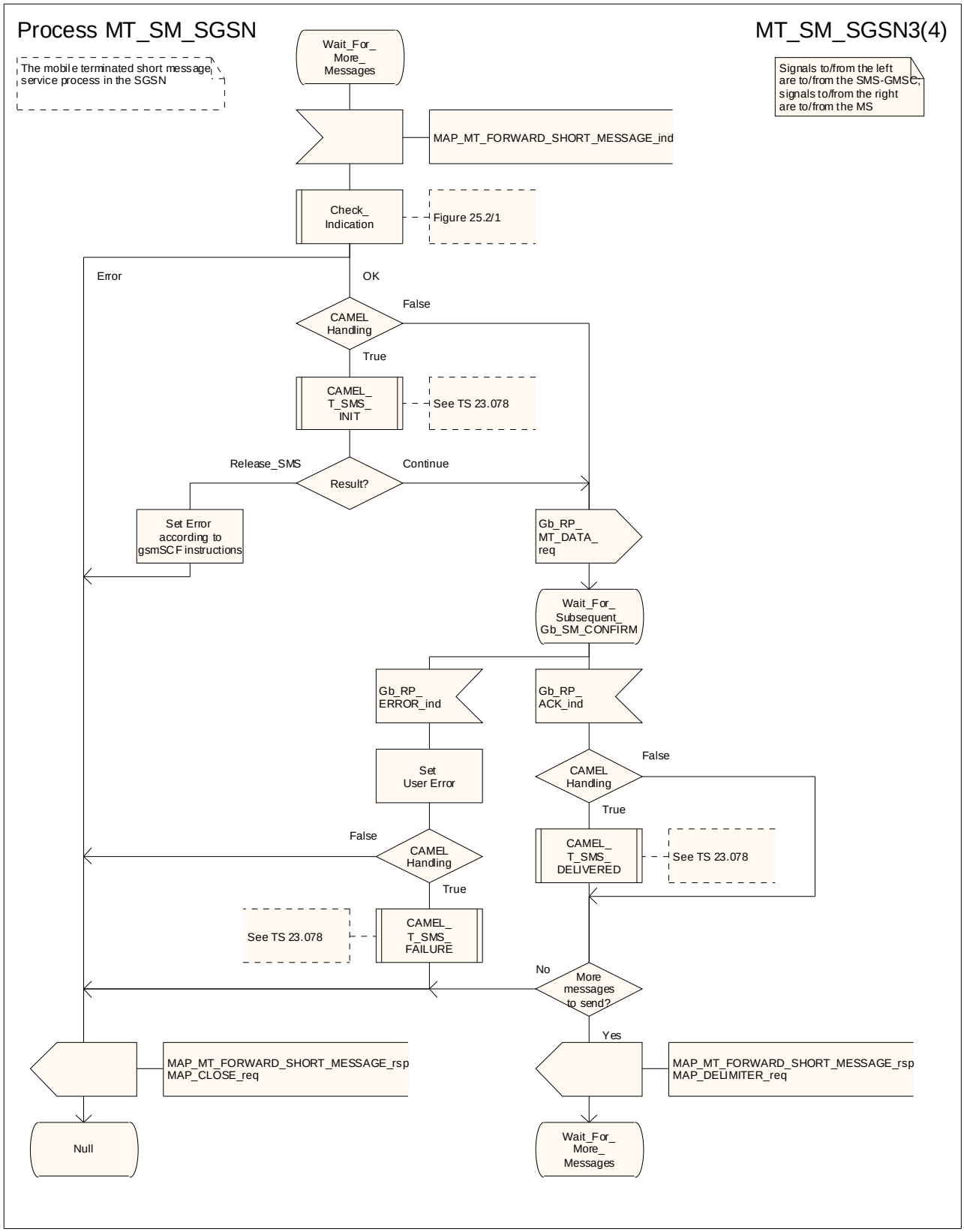


Figure 23.3/10 (sheet 3 of 4): Process MT_SM_ SGSN

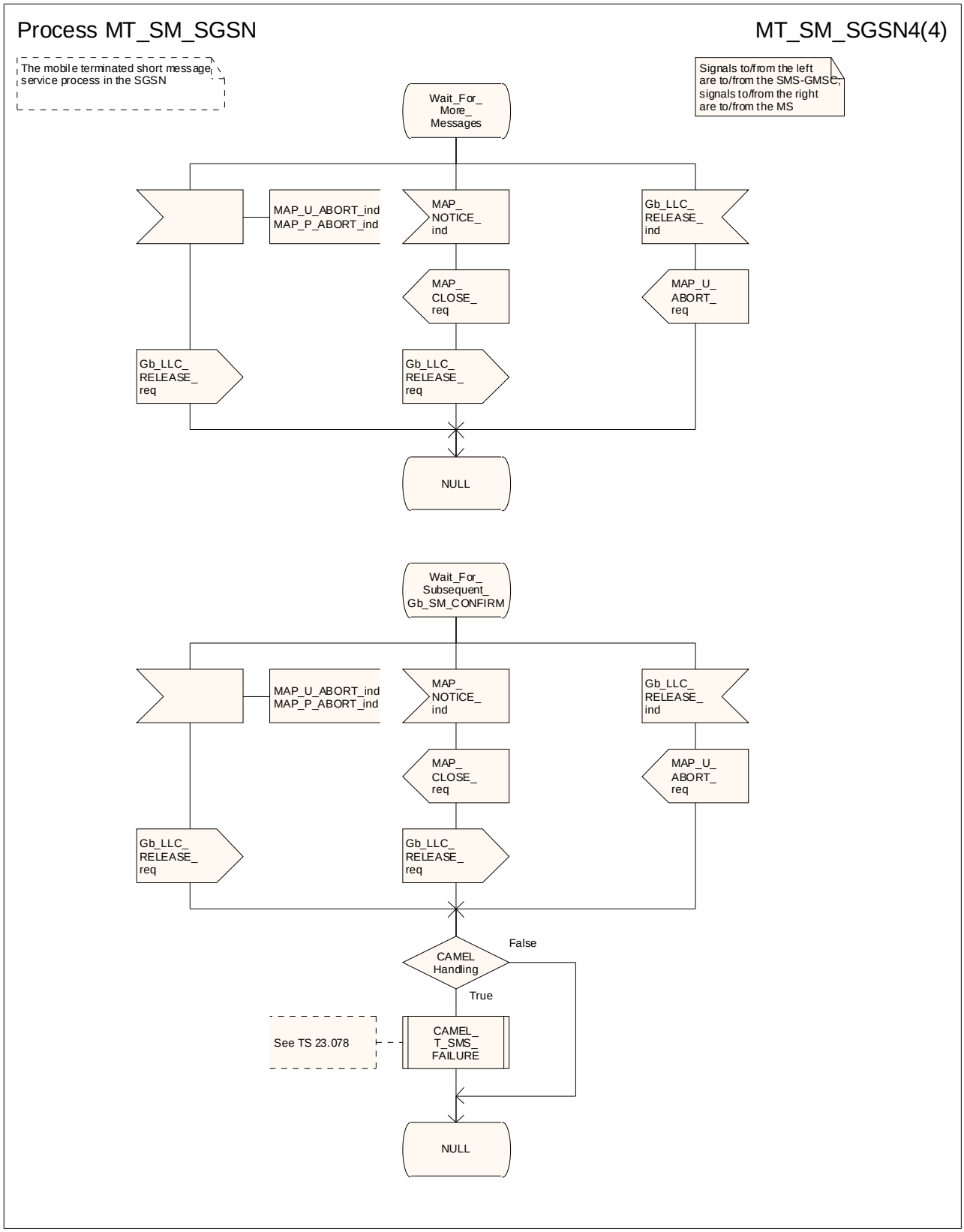


Figure 23.3/10 (sheet 4 of 4): Process MT_SM_ SGSN

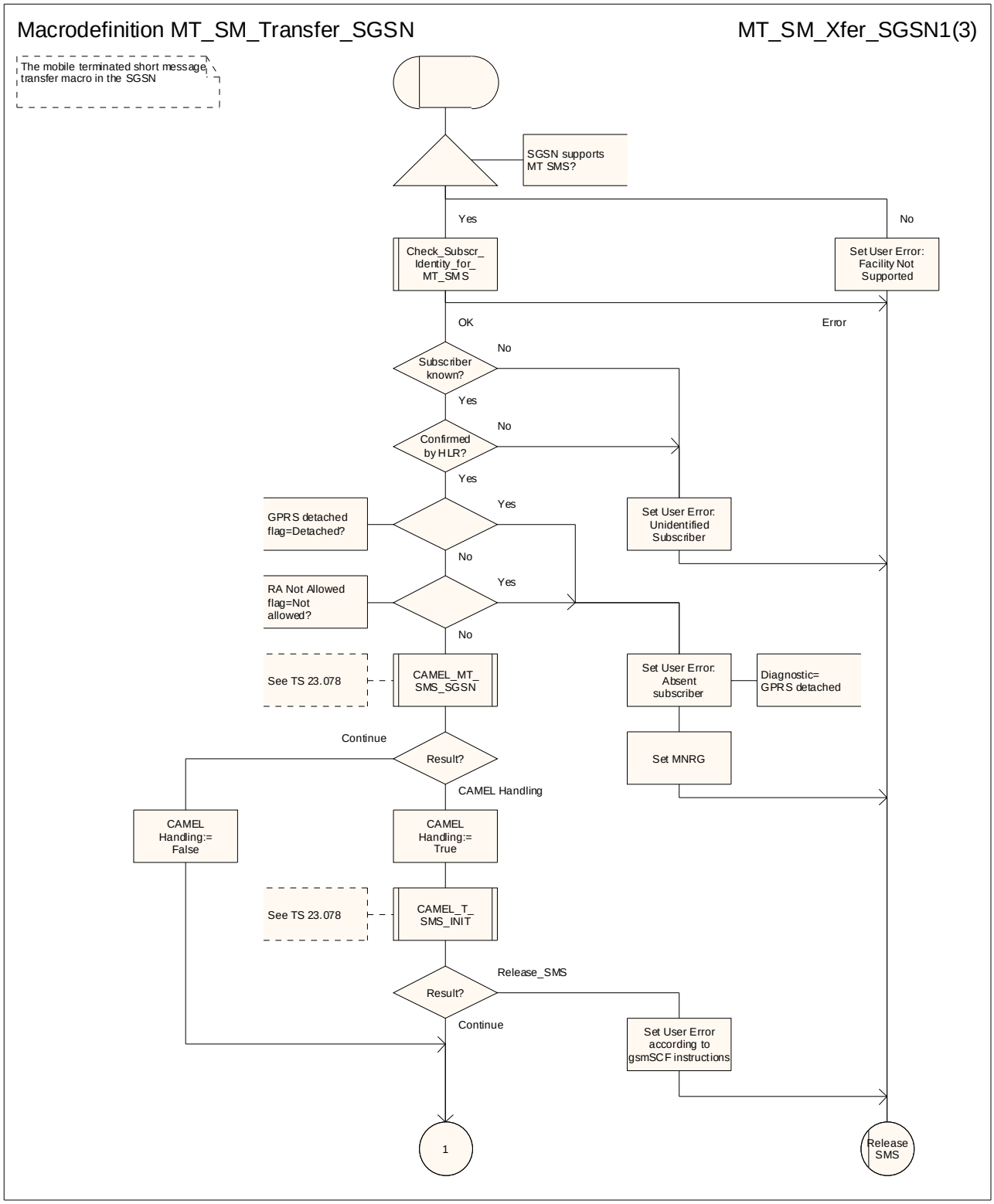


Figure 23.3/11 (sheet 1 of 3): Macro MT_SM_TRANSFER_SGSN

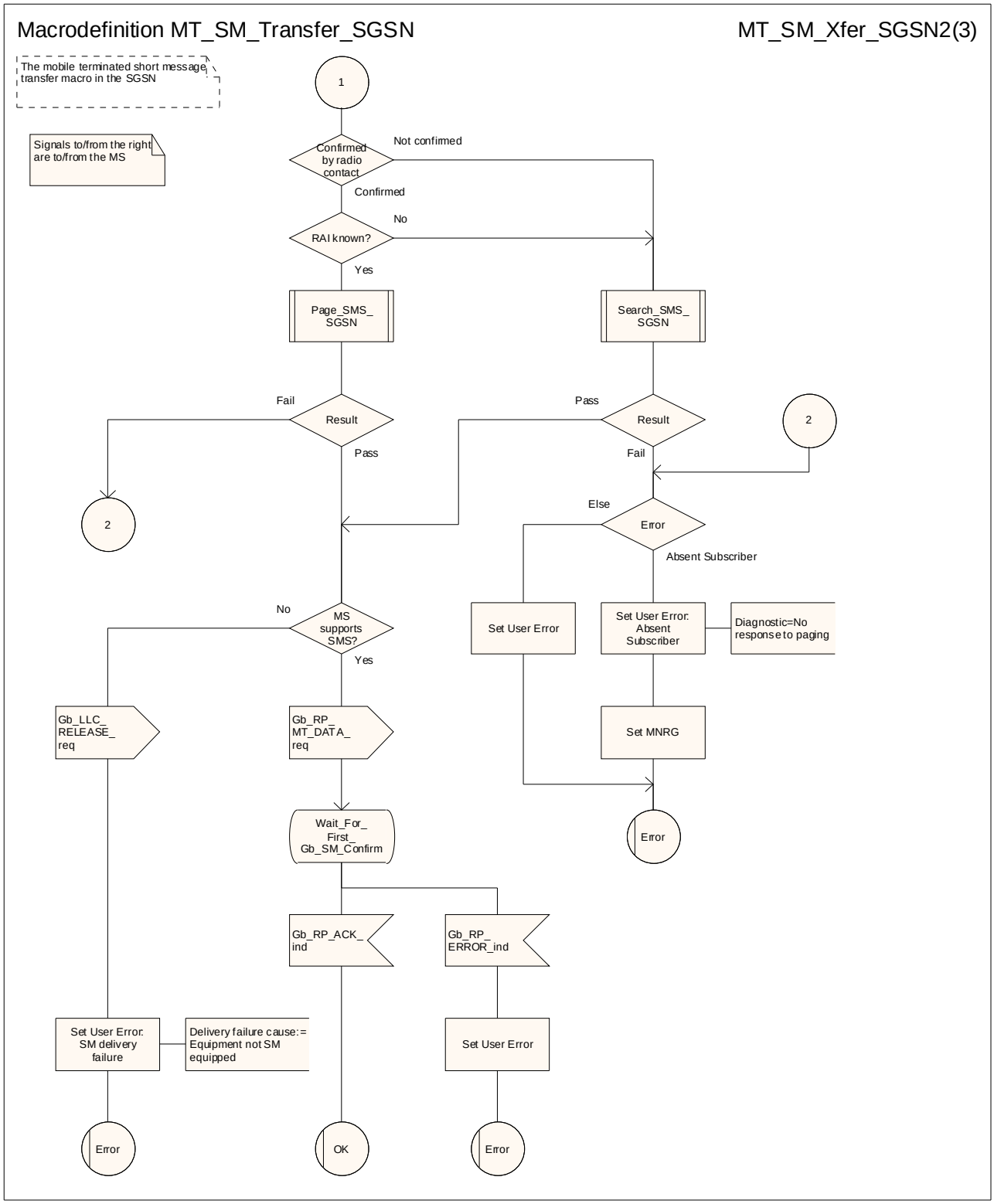


Figure 23.3/11 (sheet 2 of 3): Macro MT_SM_TRANSFER_SGSN

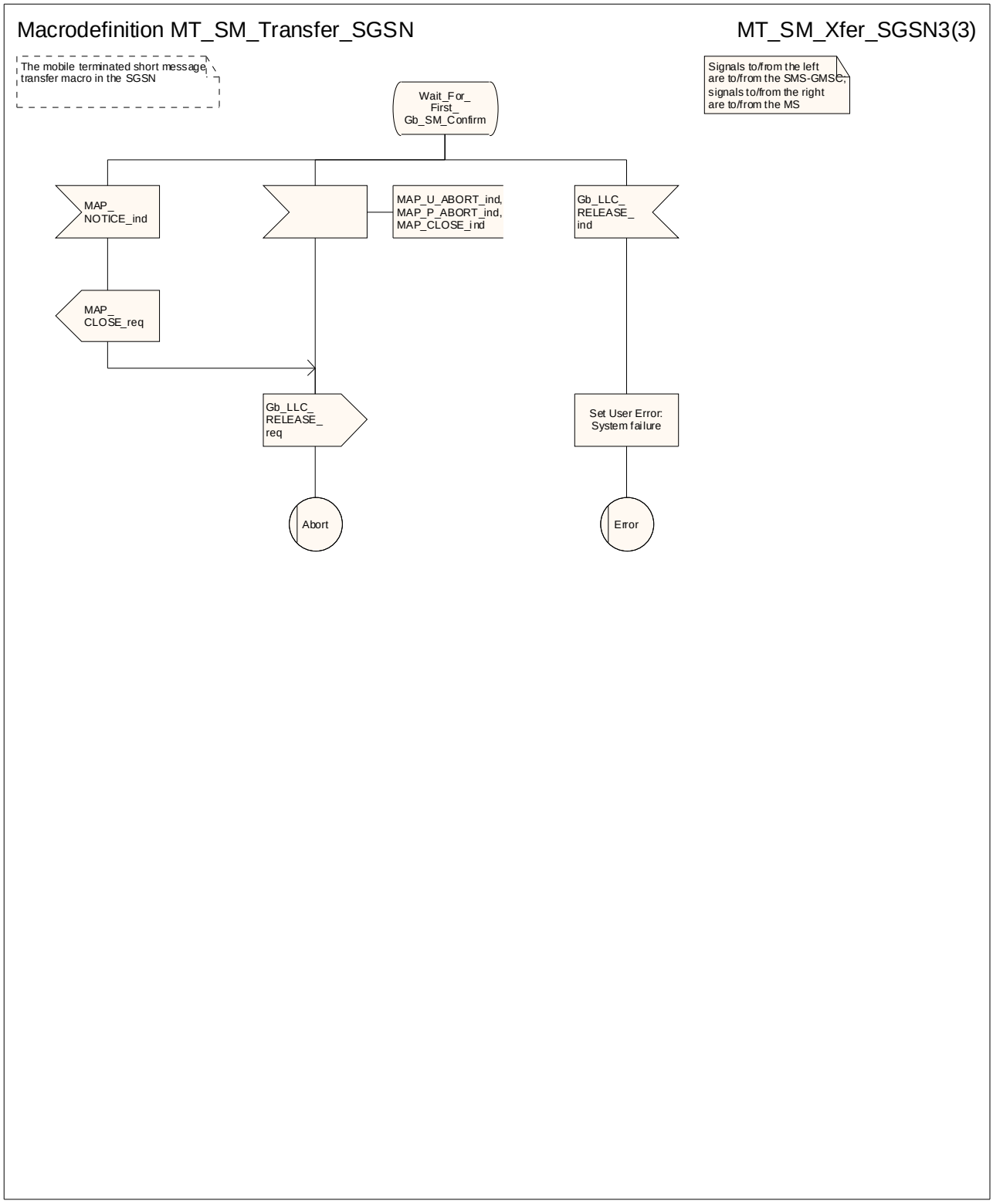


Figure 23.3/11 (sheet 3 of 3): Macro MT_SM_TRANSFER_SGSN

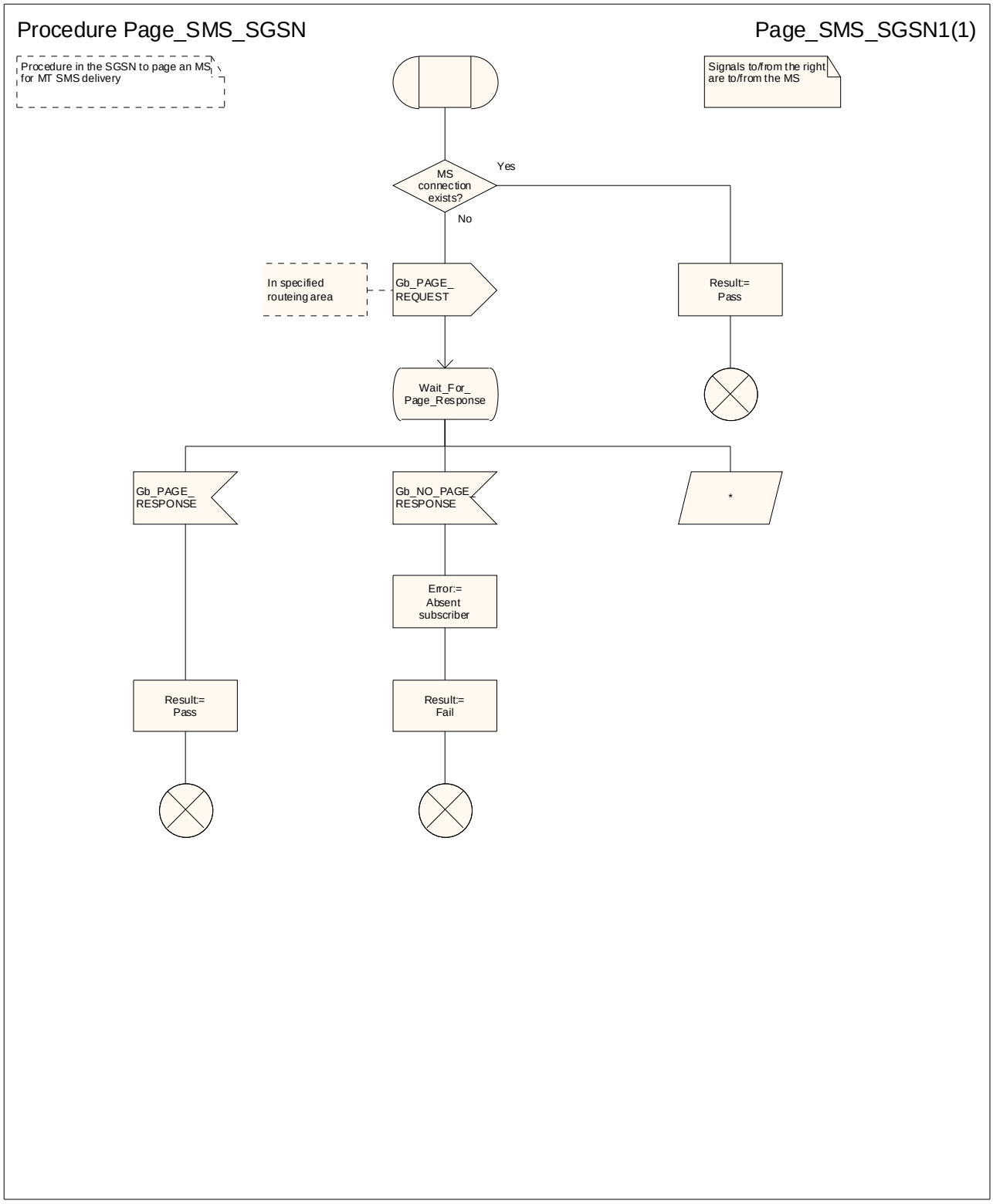


Figure 23.3/12 (sheet 1 of 1): Procedure Page_SMS_SGSN

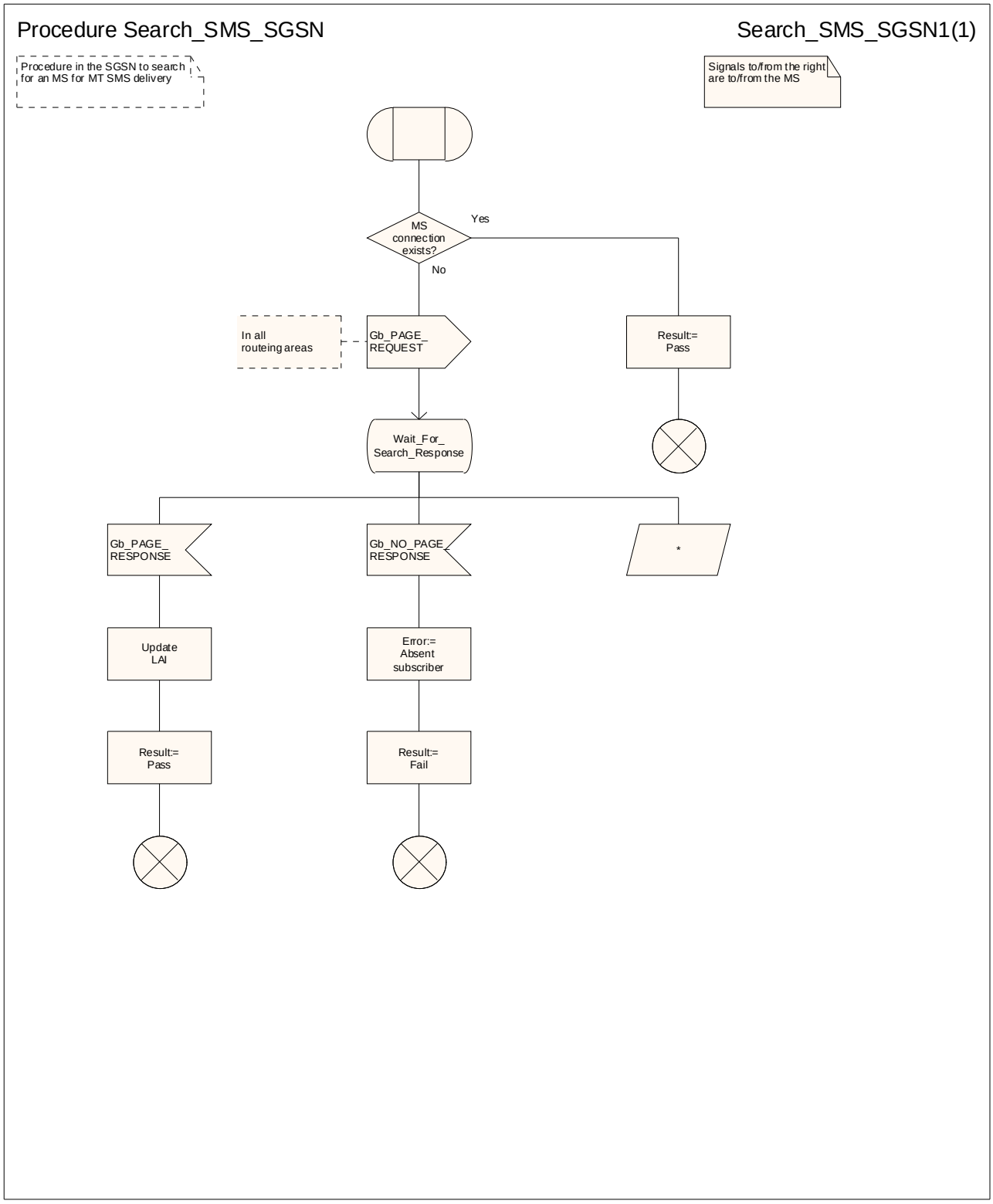


Figure 23.3/13 (sheet 1 of 1): Procedure Search_SMS_SGSN

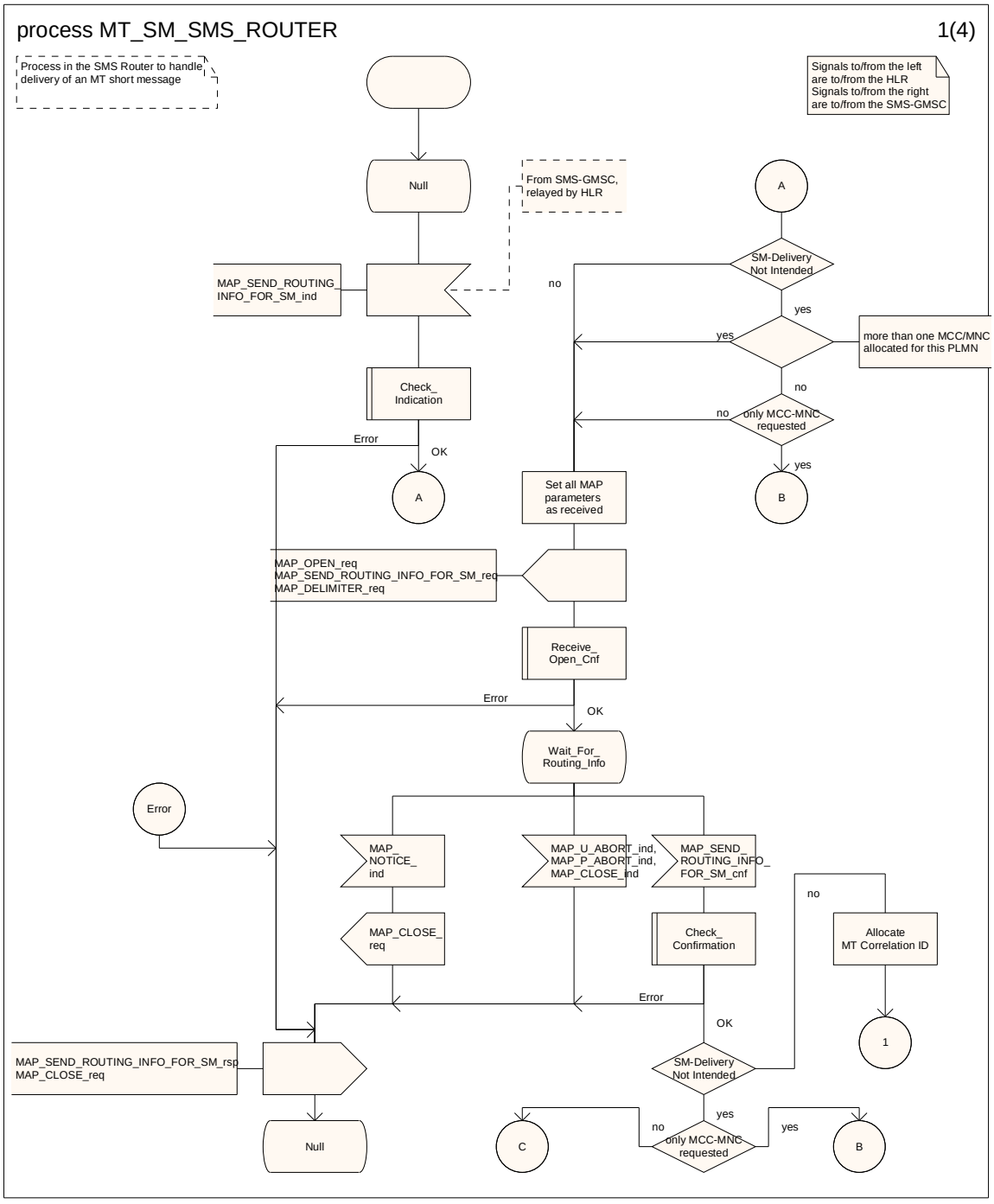


Figure 23.3/14 (sheet 1 of 4): Process MT_SM_ SMS_ROUTER

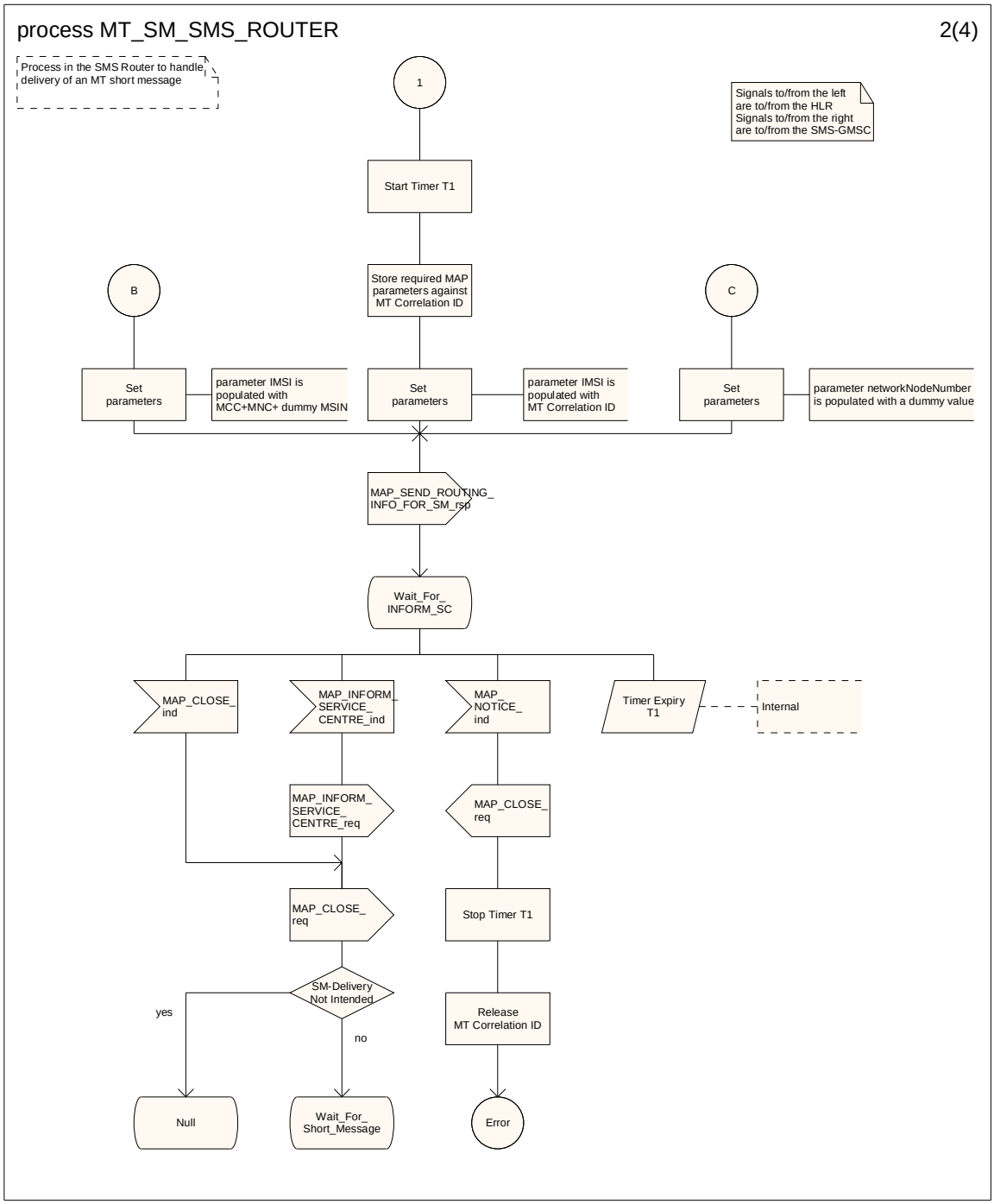


Figure 23.3/14 (sheet 2 of 4): Process MT_SM_ SMS_ROUTER

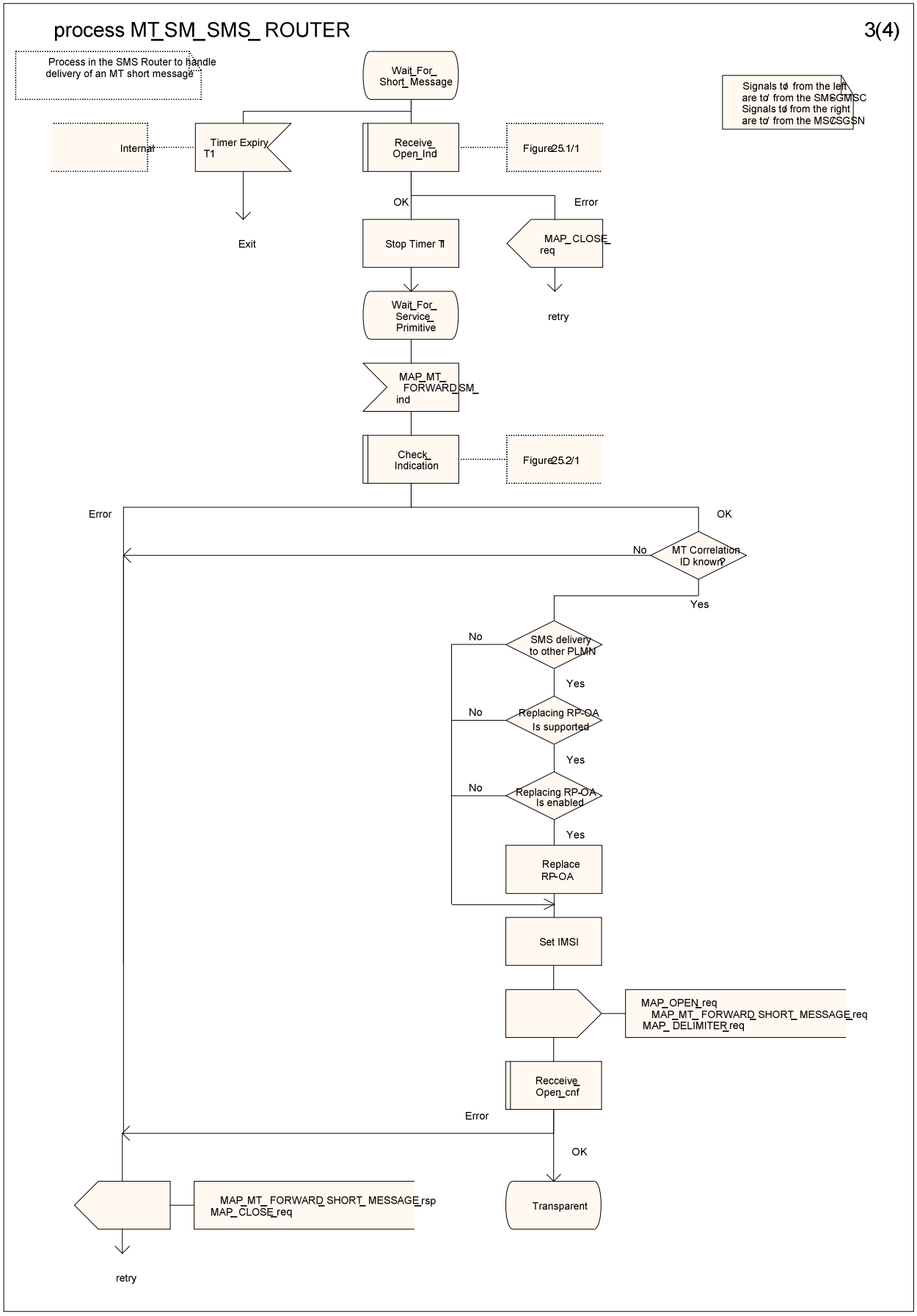


Figure 23.3/14 (sheet 3 of 4): Process MT_SM_ SMS_ROUTER

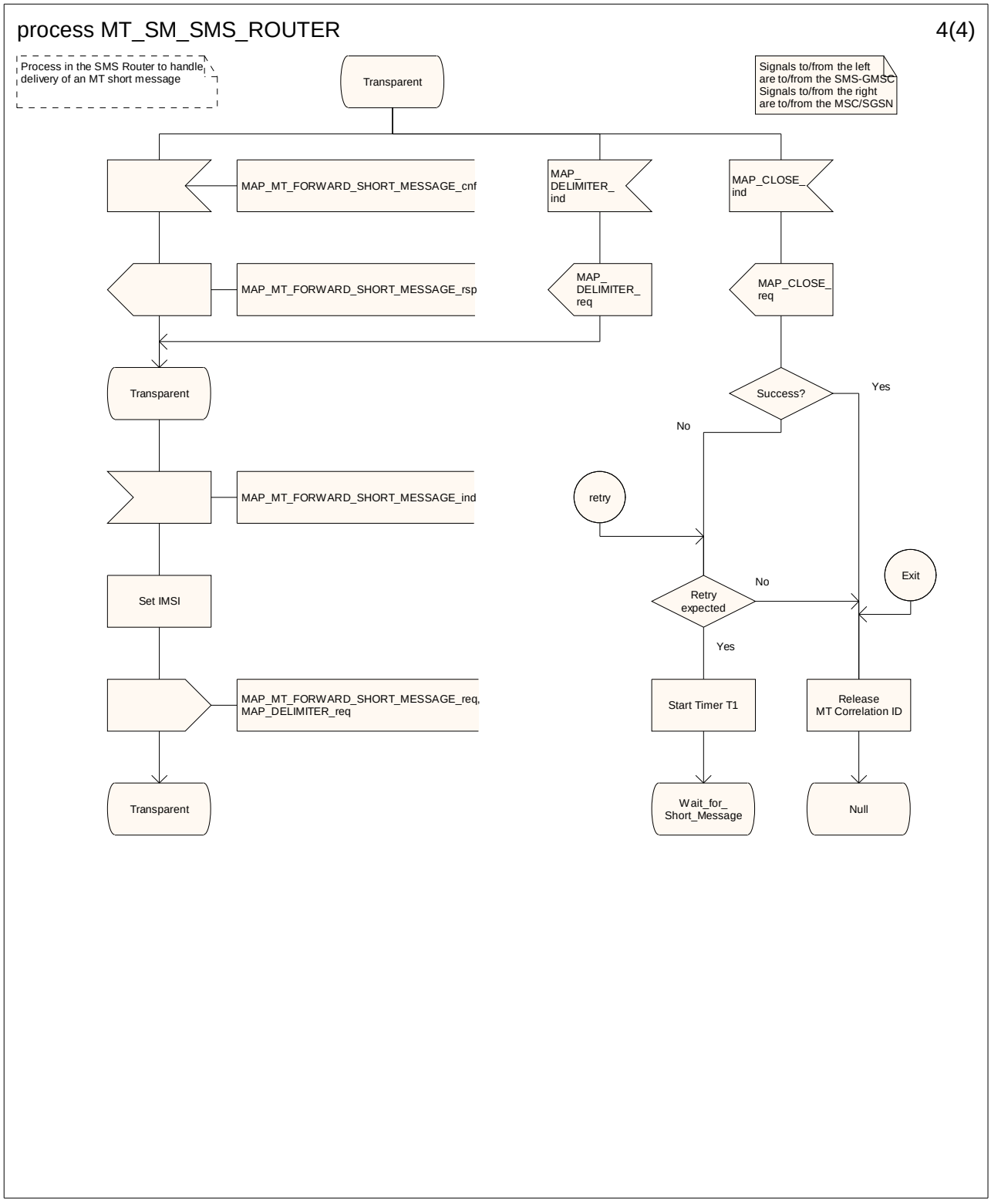


Figure 23.3/14 (sheet 4 of 4): Process MT_SM_ SMS_ROUTER

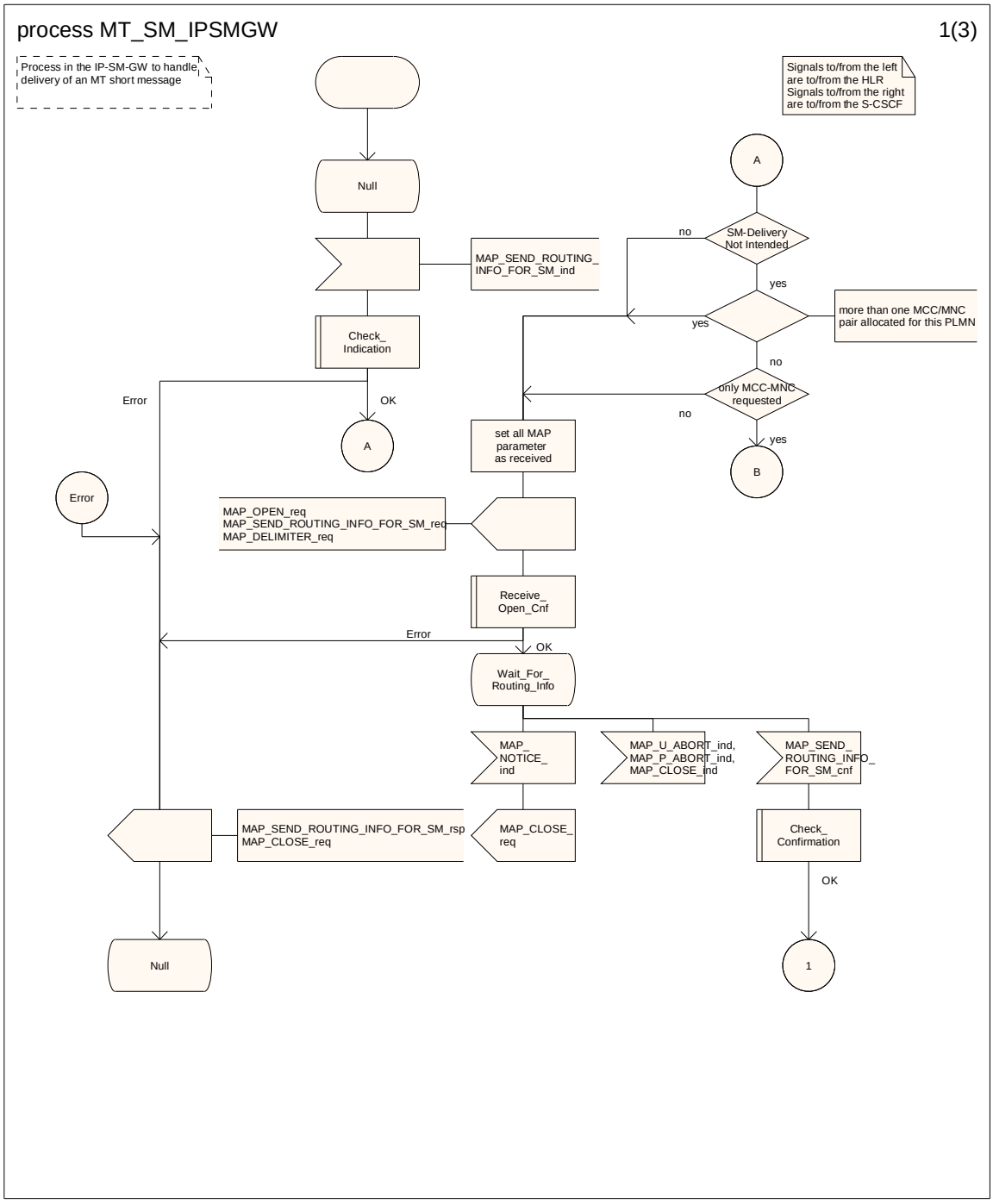


Figure 23.3/15 (sheet 1 of 3): Process MT_SM_ IPSMGW

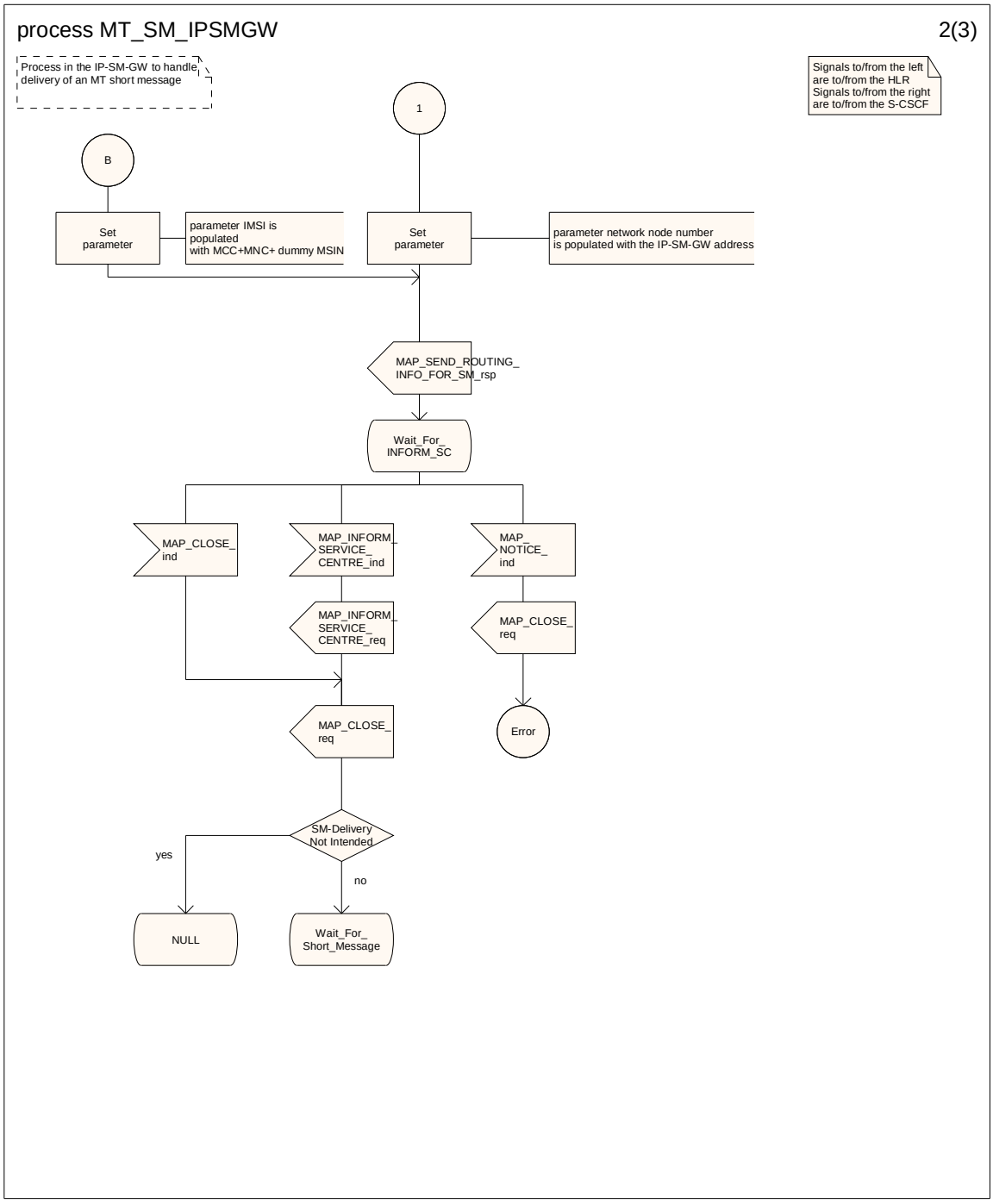


Figure 23.3/15 (sheet 2 of 3): Process MT_SM_ IPSMGW

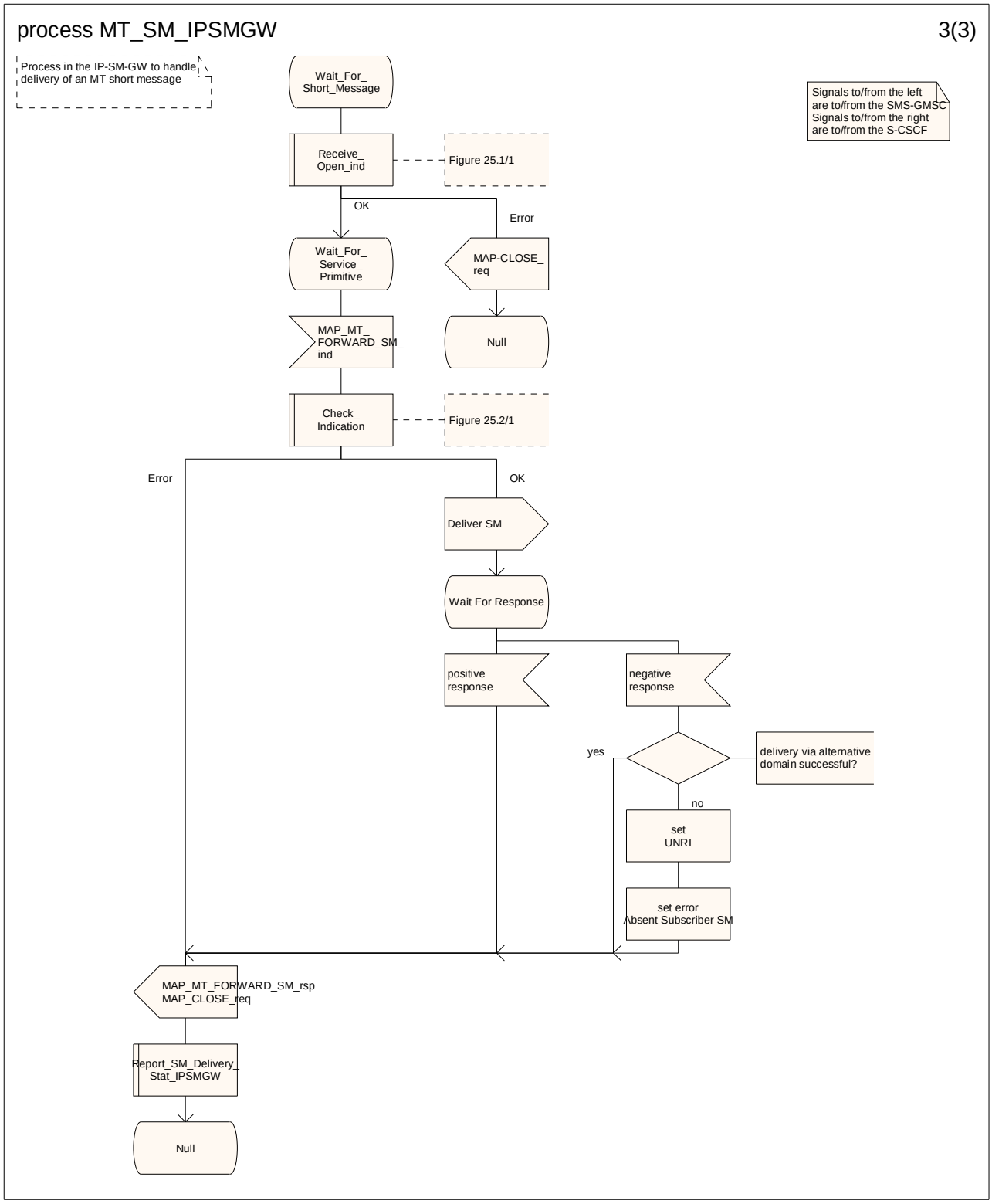
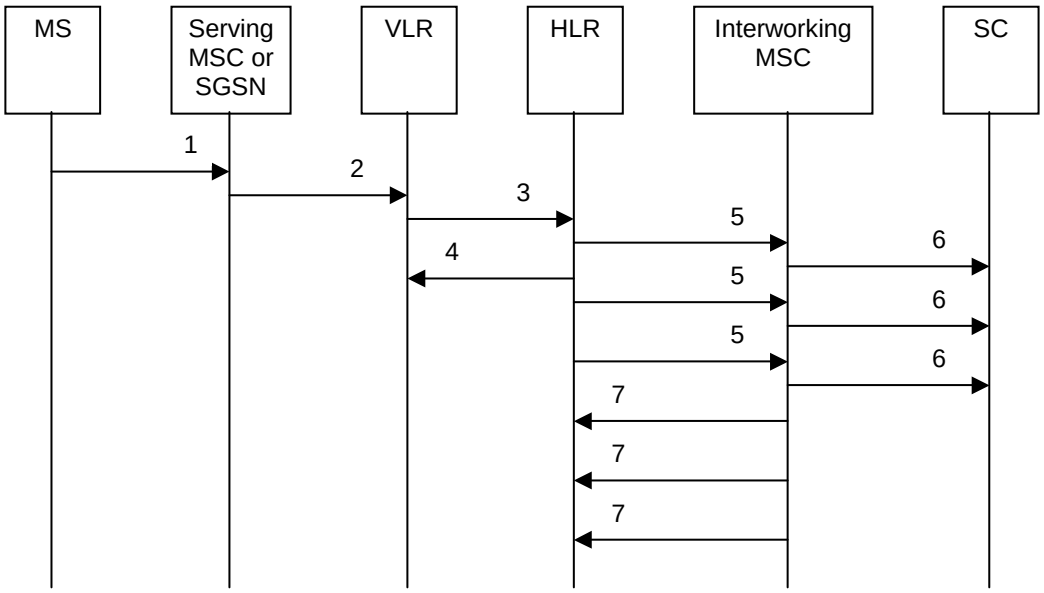


Figure 23.3/15 (sheet 3 of 3): Process MT_SM_ IPSMGW

23.4 The Short Message Alert procedure

The Short Message Alert procedure is used to alert the Service Centre when the mobile subscriber is active after a short message transfer has failed because the mobile subscriber is not reachable, or when the MS has indicated that it has memory capacity to accept a short message.

The message flow for the Short Message Alert procedure for the case when the mobile subscriber was not reachable is shown in figure 23.4/1.

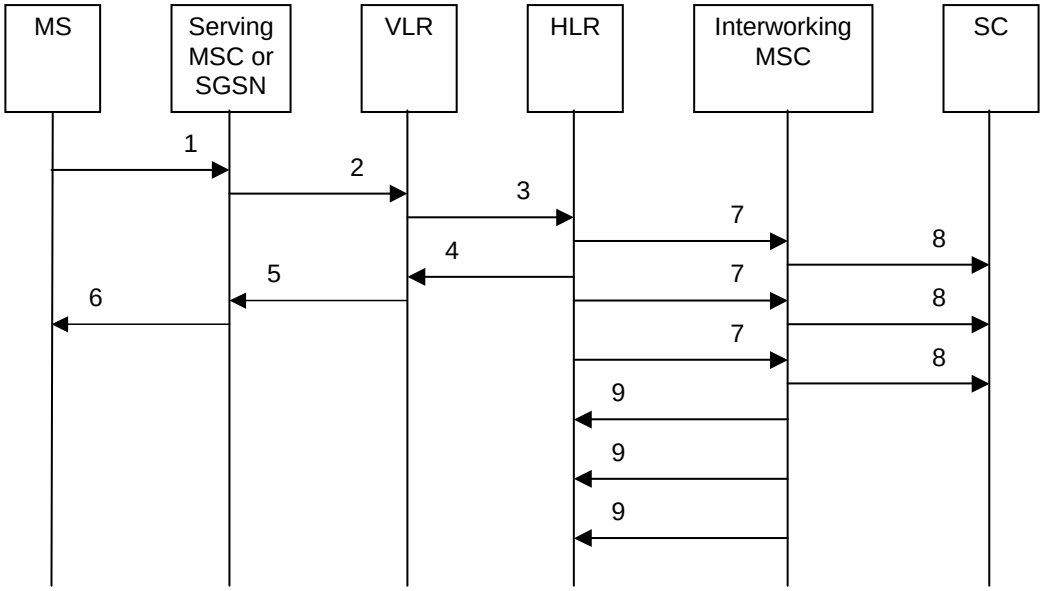


1) CM Service Request (**), Page response or Location Updating (3GPP TS 24.008 [35]).
2) MAP_PROCESS_ACCESS_REQUEST / MAP_UPDATE_LOCATION_AREA (**).
3) MAP_READY_FOR_SM (Mobile Present) / MAP_UPDATE_LOCATION / Supplementary Service Control Request (*).
4) MAP_READY_FOR_SM_ACK (*).
5) MAP_ALERT_SERVICE_CENTRE (notes 1 and 2).
6) Alert Service Centre (3GPP TS 23.040).
7) MAP_ALERT_SERVICE_CENTRE_ACK.

NOTE 1: To all Service Centres in the Message Waiting List.
NOTE 2: The HLR initiates the MAP_ALERT_SERVICE_CENTRE service only if the MS Memory Capacity Exceeded flag is clear.
(*) For GPRS, messages 3) and 4) are sent/received by the SGSN.
(**) These messages are not used by the SGSN.

Figure 23.4/1: Short message alert procedure (Mobile is present)

The message flow for the Short Message Alert procedure for the case where the MS indicates that it has memory capacity to accept one or more short messages is shown in figure 23.4/2.



1) SM memory capacity available (3GPP TS 24.011 [37]).
2) MAP_READY_FOR_SM (Memory Available) (*).
3) MAP_READY_FOR_SM (Memory Available) (**).
4) MAP_READY_FOR_SM_ACK (**).
5) MAP_READY_FOR_SM_ACK (*).
6) SM memory capacity available (Acknowledge) (3GPP TS 24.011 [37]).
7) MAP_ALERT_SERVICE_CENTRE (note).
8) Alert Service Centre (3GPP TS 23.040).
9) MAP_ALERT_SERVICE_CENTRE_ACK.

NOTE: To all Service Centres in the Message Waiting List.
(*) Messages 2) and 5) are not used by the SGSN.
(**) For GPRS, messages 3) and 4) are sent/received by the SGSN.

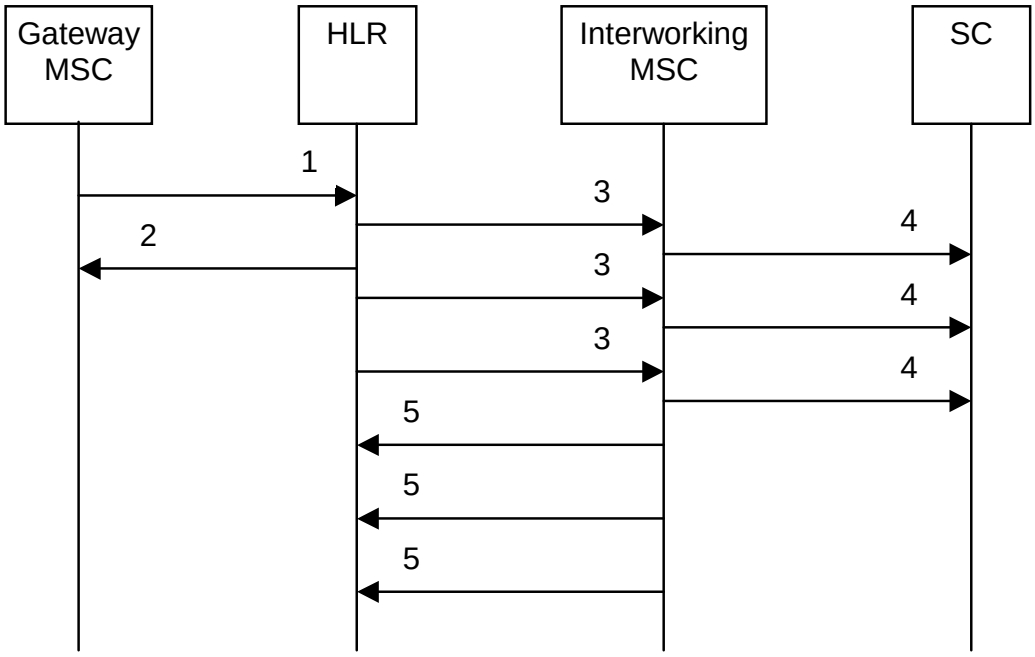
Figure 23.4/2: Short message alert procedure (MS memory capacity available)

In addition the following MAP services are used in the MS memory available case:

MAP_PROCESS_ACCESS_REQUEST	(see subclause 8.3); (*)
MAP_AUTHENTICATE	(see subclause 8.5); (*)
MAP_SET_CIPHERING_MODE	(see subclause 8.6); (*)
MAP_PROVIDE_IMSI	(see subclause 8.9); (*)
MAP_CHECK_IMEI	(see subclause 8.7);
MAP_FORWARD_NEW_TMSI	(see subclause 8.9); (*)
MAP_TRACE_SUBSCRIBER_ACTIVITY	(see subclause 9.1). (*)

(*) These services are not used by the SGSN.

The Short Message Alert procedure when the MS indicates successful transfer after polling is shown in figure 23.4/3.



1) MAP_REPORT_SM_DELIVERY_STATUS (Successful Transfer).
2) MAP_REPORT_SM_DELIVERY_STATUS_ACK.
3) MAP_ALERT_SERVICE_CENTRE (note).
4) Alert Service Centre (3GPP TS 23.040).
5) MAP_ALERT_SERVICE_CENTRE_ACK.

NOTE: To all Service Centres in the Message Waiting List.

Figure 23.4/3: Short message alert procedure (Successful transfer after polling)

23.4.1 Procedure in the Serving MSC – the MS has memory available

The process starts when the MSC receives a notification from the MS that it has memory available. The process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Confirmation see subclause 25.2.2.

The short message alert process in the MSC for the MS memory capacity available case is shown in figure 23.4/4.

23.4.2 Procedures in the VLR

23.4.2.1 The Mobile Subscriber is present

If the VLR successfully handles a MAP_PROCESS_ACCESS_REQUEST indication or a MAP_UPDATE_LOCATION_AREA indication while the MS Not Reachable Flag (MNRF) is set, the VLR sends a MAP_READY_FOR_SM request to the HLR. The Alert Reason is set to indicate that the mobile subscriber is present for non GPRS. If authentication fails during the handling of a MAP_PROCESS_ACCESS_REQUEST indication or a MAP_UPDATE_LOCATION_AREA indication, the VLR shall not send a MAP_READY_FOR_SM request to the HLR. The process in the VLR is described in detail in subclause 25.10.1.

23.4.2.2 The MS has memory available

The process starts when the VLR receives dialogue opening request followed by a MAP_PROCESS_ACCESS_REQUEST indication including a CM service type Short Message Service. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Indication see subclause 25.2.1;

Check_Confirmation see subclause 25.2.2.

The short message alert process in the VLR for the MS memory capacity available case is shown in figure 23.4/5.

23.4.3 Procedures in the SGSN

23.4.3.1 The Mobile Subscriber is present

If the SGSN successfully handles a Page response, Attach request or Routing Area Update request message (3GPP TS 24.008 [35]), while the MS Not Reachable for GPRS (MNRG) flag is set, the SGSN sends a MAP_READY_FOR_SM request to the HLR. The Alert Reason is set to indicate that the mobile subscriber is present for GPRS. If authentication fails during the handling of a Page response, Attach request or Routing Area Update request, the SGSN shall not send a MAP_READY_FOR_SM request to the HLR

The process in the SGSN is described in detail in subclause 25.10.23.

23.4.3.2 The Mobile Equipment has memory available

The process starts when the SGSN receives an RP_SM_MEMORY_AVAILABLE indication from the MS. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The short message alert procedure in the SGSN for the MS memory capacity available case is shown in figure 23.4/6.

23.4.4 Procedure in the HLR

The process starts when the HLR receives a dialogue opening request using the application context mwdMngtContext. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1;

Alert_Service_Centre_HLR see subclause 25.10.3.

Sheet 1: If the dialogue opening request is from an SGSN, version 2 and version 1 of the application context are not applicable.

The short message alert process in the HLR is shown in figure 23.4/7.

23.4.5 Procedure in the SMS Interworking MSC

The process starts when the SMS-IWMSC receives a dialogue opening request using the application context shortMsgAlertContext. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

The short message alert process in the SMS-IWMSC is shown in figure 23.4/8.

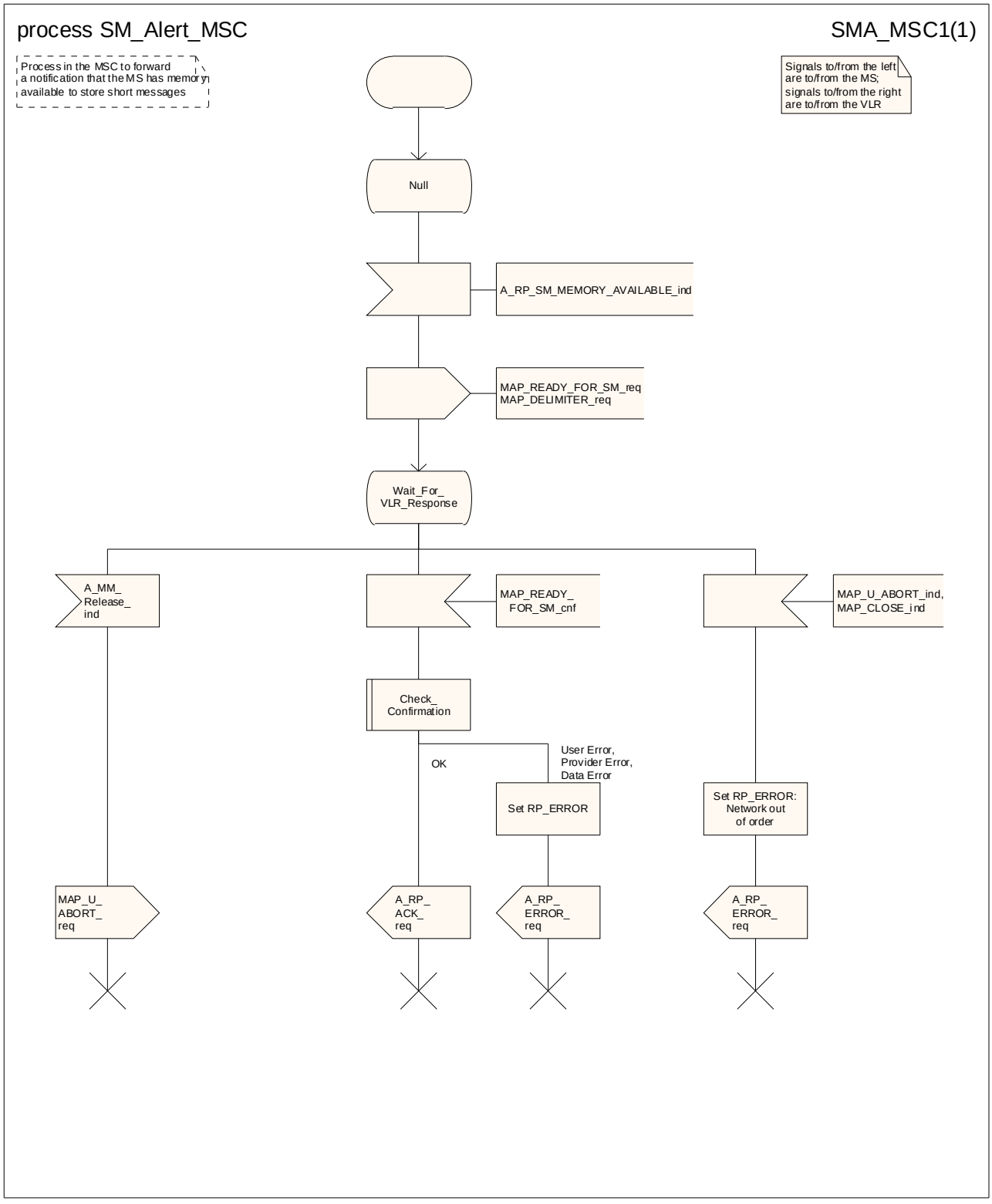


Figure 23.4/4: Procedure SM_Alert_MSC

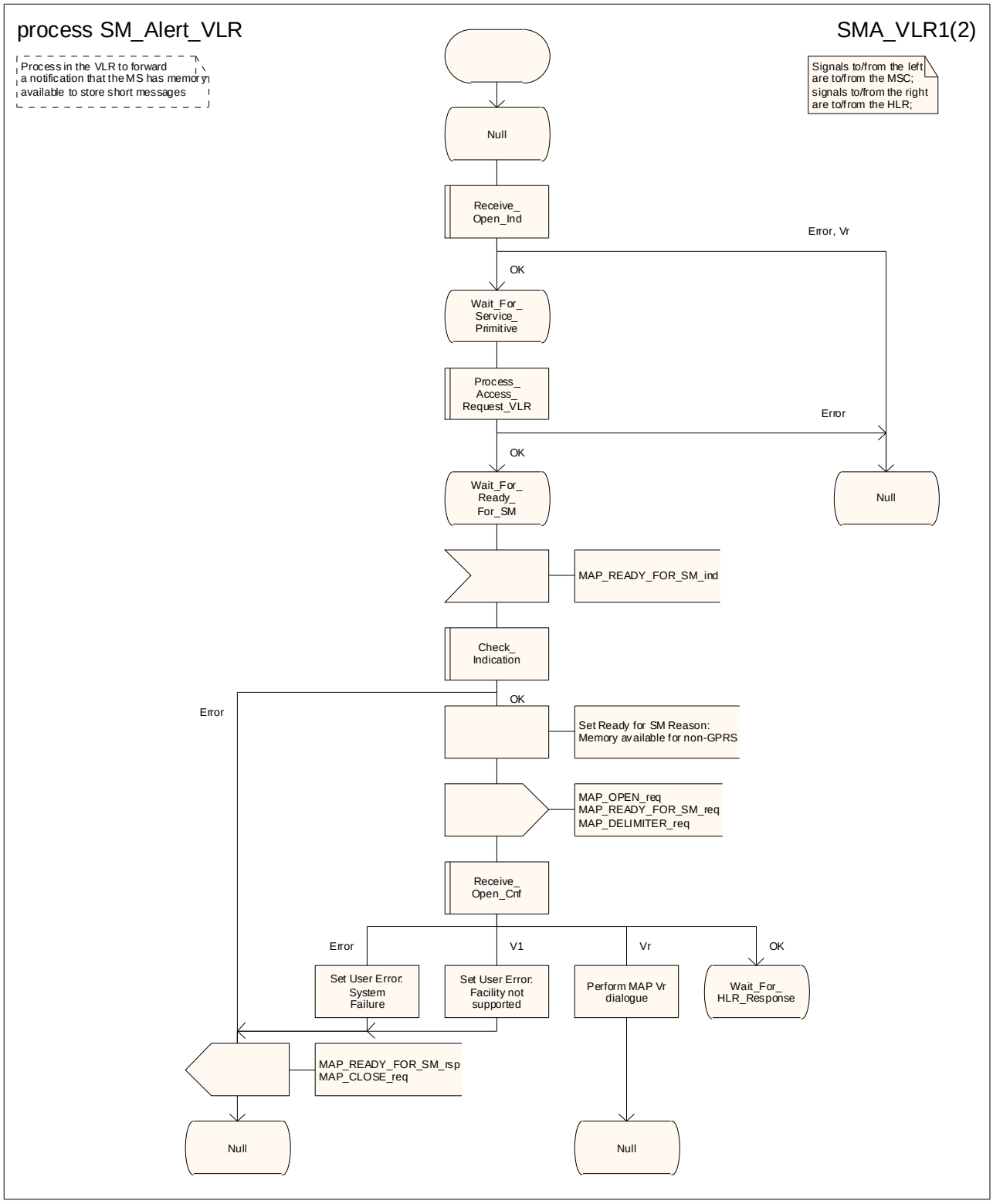


Figure 23.4/5 (sheet 1 of 2): Procedure SM_Alert_VLR

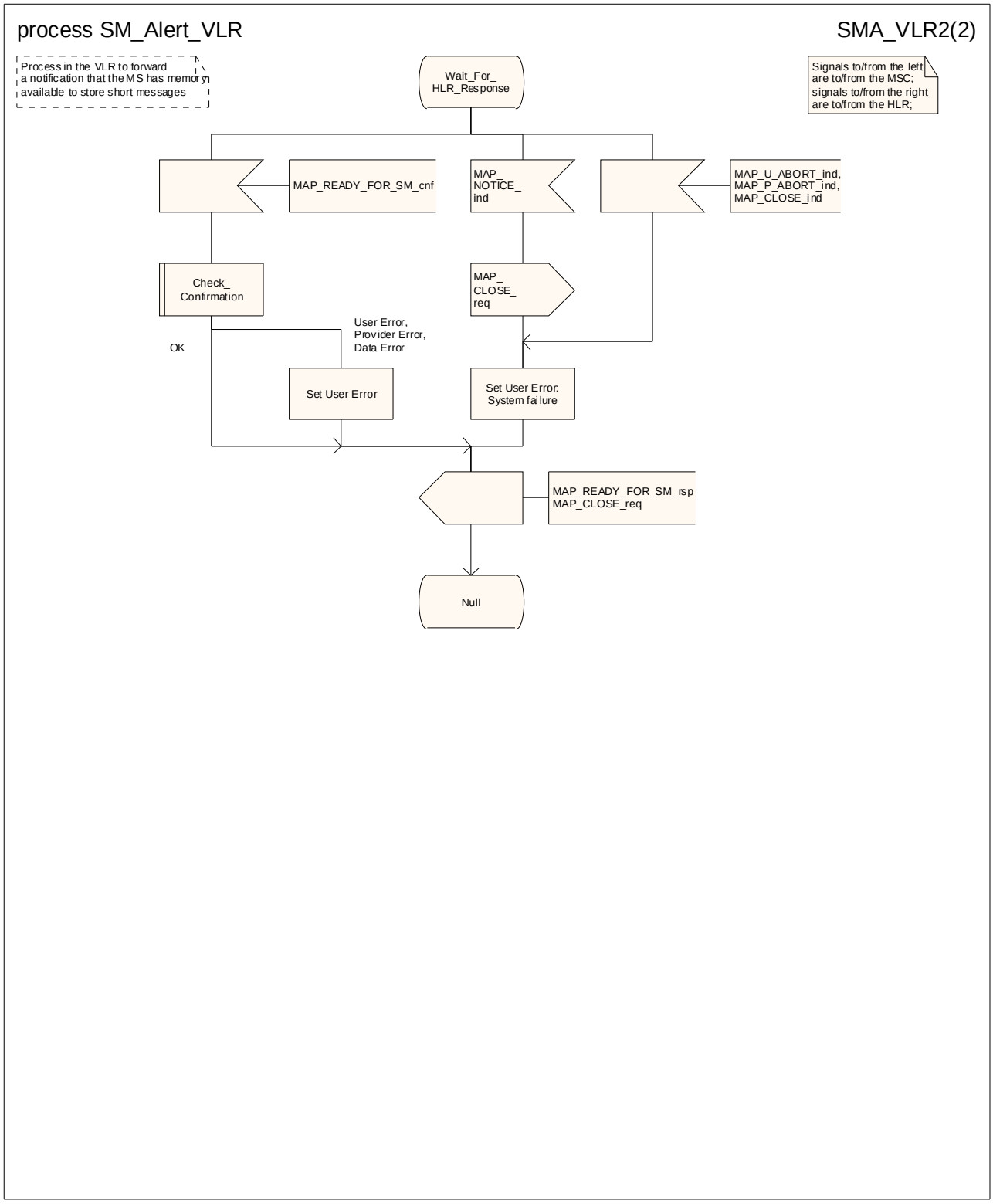


Figure 23.4/5 (sheet 2 of 2): Procedure SM_Alert_VLR

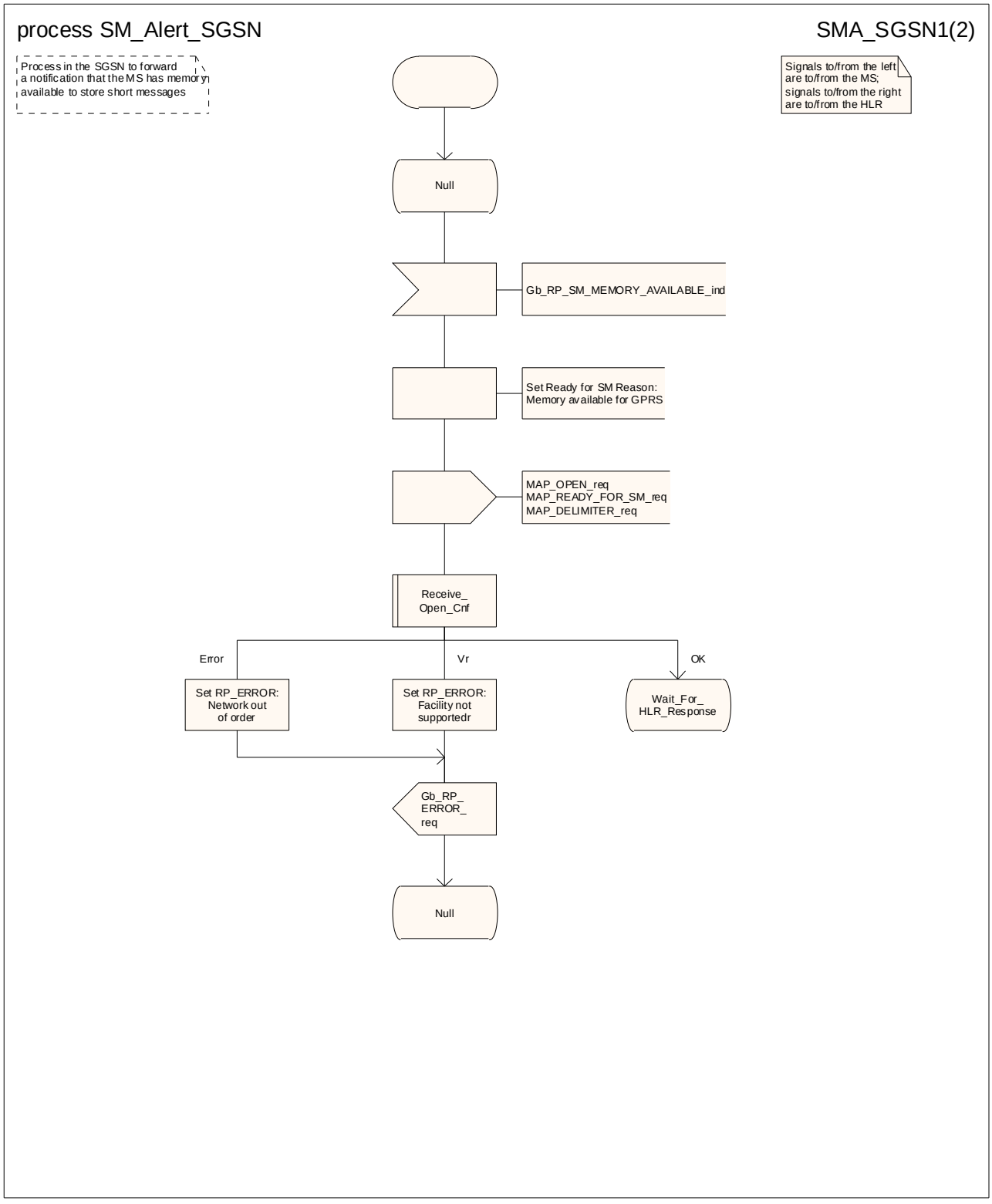


Figure 23.4/6 (sheet 1 of 2): Process SM_Alert_SGSN

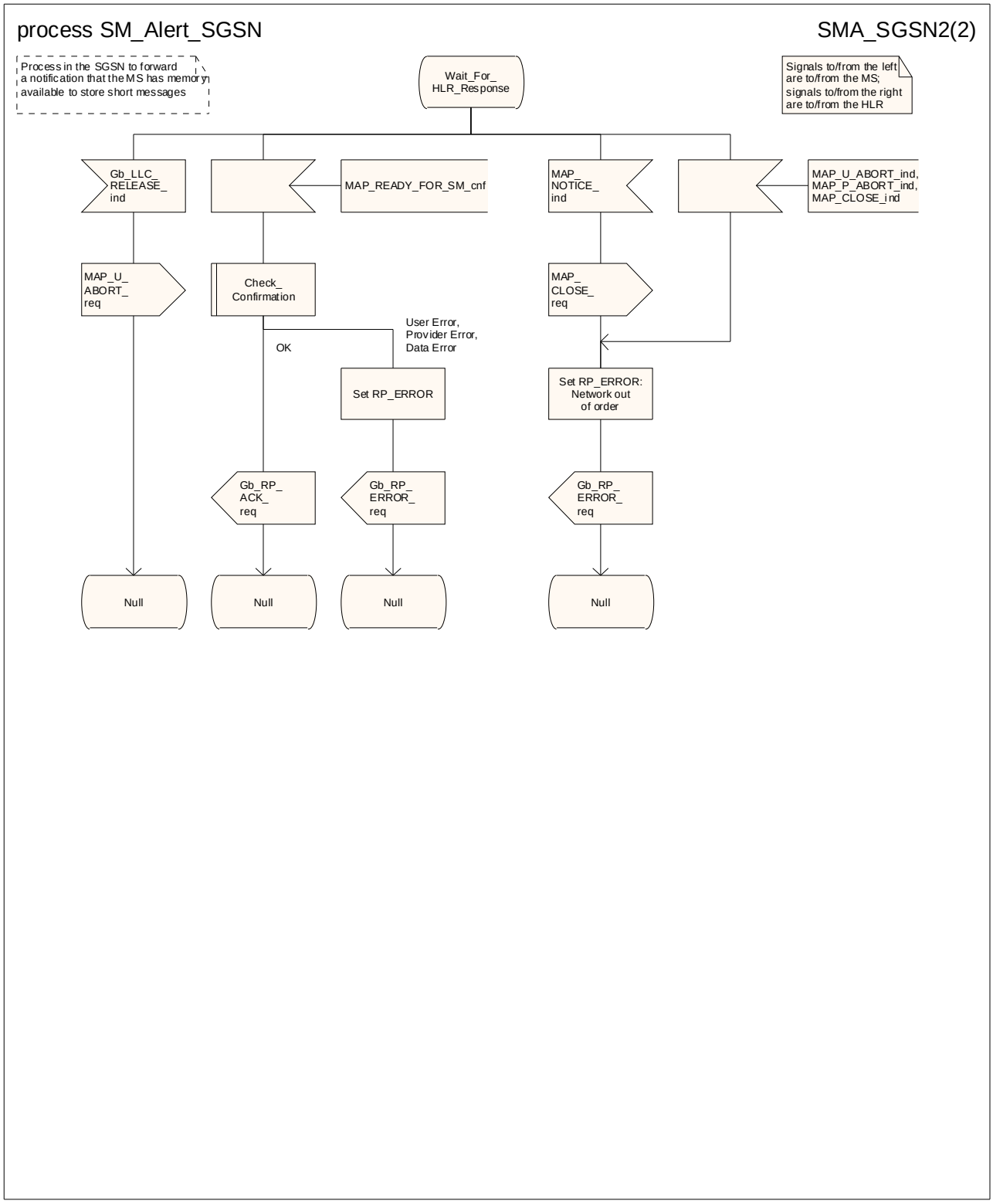


Figure 23.4/6 (sheet 2 of 2): Process SM_Alert_SGSN

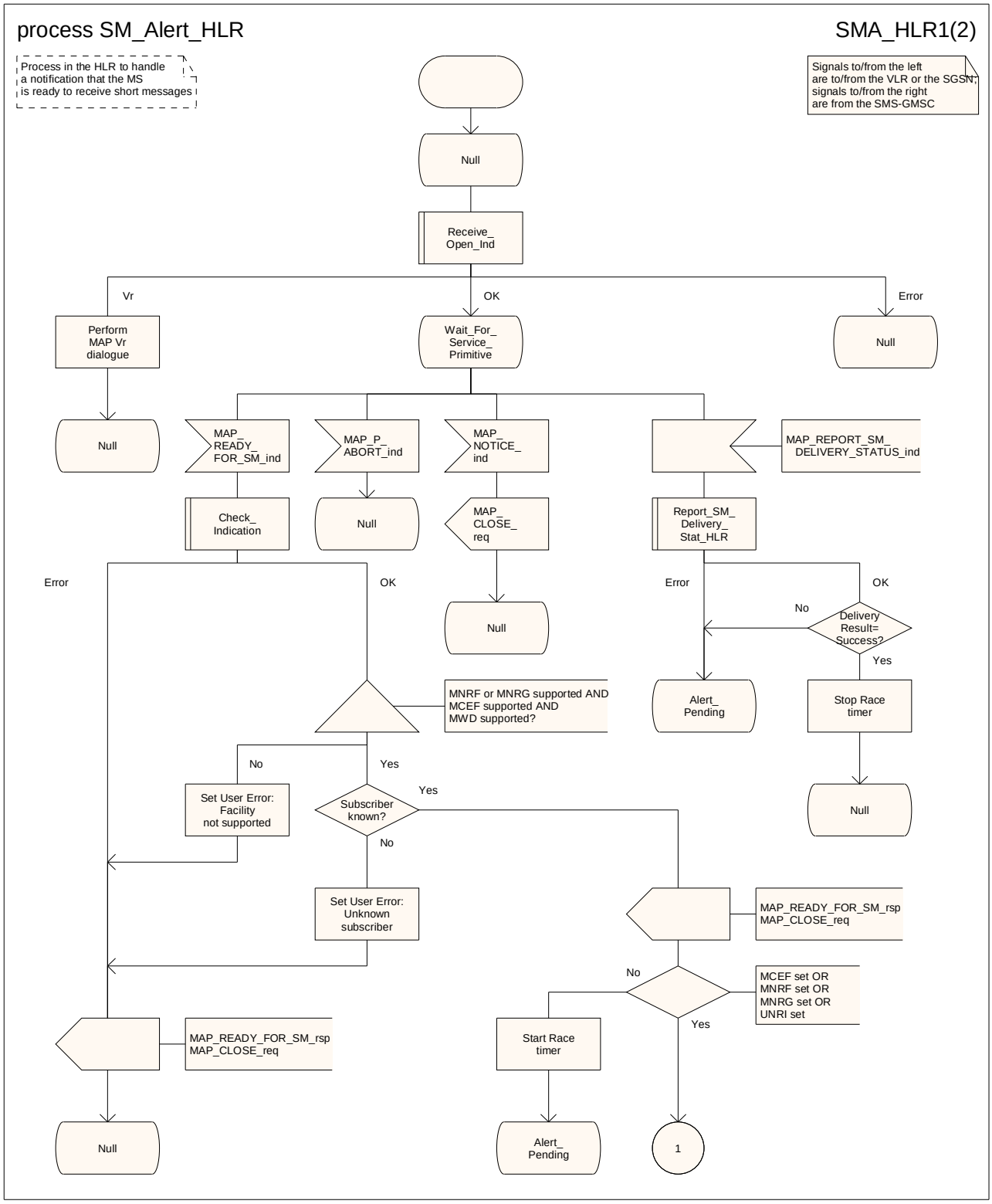


Figure 23.4/7 (sheet 1 of 2): Process SM_Alert_HLR

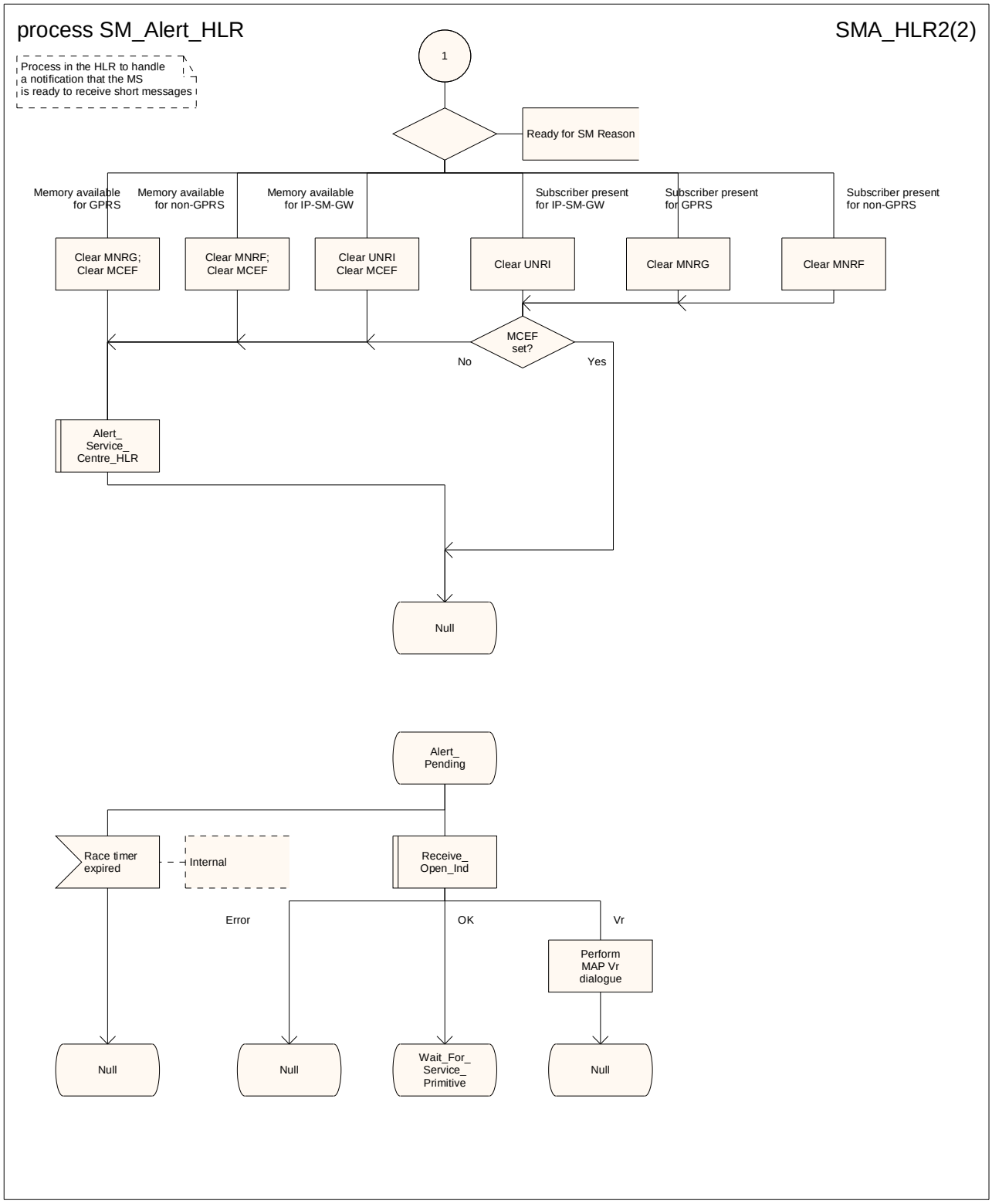


Figure 23.4/7 (sheet 2 of 2): Process SM_Alert_HLR

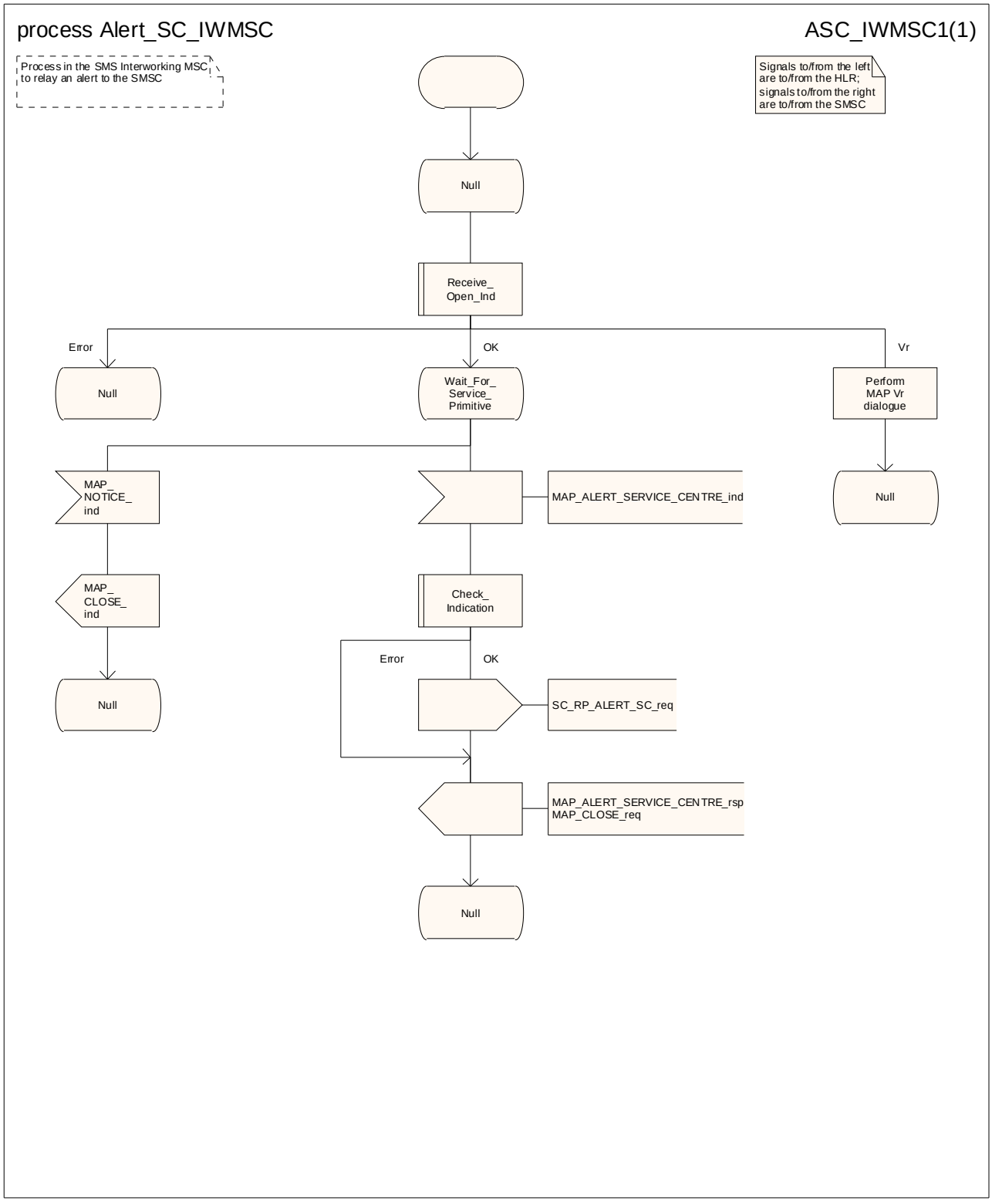


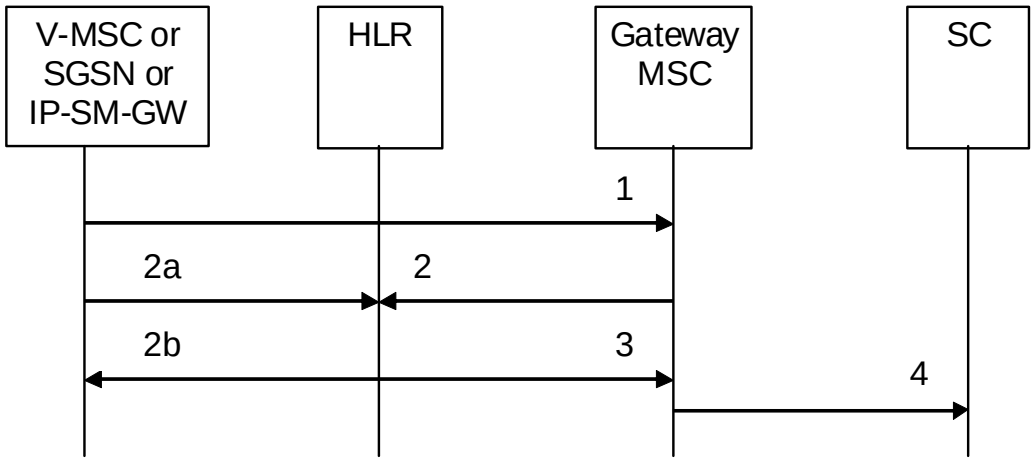
Figure 23.4/8: Process Alert_SC_IWMSC

23.5 The SM delivery status report procedure

The SM delivery status report procedure is used:

- to set the Service Centre address into the message waiting list in the HLR after short message delivery has failed because the subscriber is absent or unidentified or the memory capacity is exceeded. The procedure sets:
  - the Memory Capacity Exceeded Flag (MCEF) in the HLR if the MS memory does not have room for more messages;
  - and/or the MS Not Reachable Flag for non-GPRS if there is no record for the subscriber in the VLR or the subscriber does not respond to paging for delivery via the MSC;
  - and/or the MS Not Reachable for GPRS (MNRG) flag if there is no record for the subscriber in the SGSN or the subscriber does not respond to paging for delivery via the SGSN;
  - and/or the UE Not Reachable for IP (UNRI) flag if delivery via the IMS was not successful.
- to report to the HLRthat delivery has succeeded. The conditions for report of a successful delivery are described in subclause 23.3.1.

The message flow for the SM delivery status report procedure is shown in figure 23.5/1.



1) MAP_MT_FORWARD_SHORT_MESSAGE_ACK/_NACK (Absent subscriber_SM, unidentified subscriber or memory capacity exceeded).
2) MAP_REPORT_SM_DELIVERY_STATUS. (The HLR ignores the content of this message when an IP-SM-GW is deployed)
2a) MAP_REPORT_SM_DELIVERY_STATUS (sent only by IP-SM-GW)
2b) MAP-REPORT_SM_DELIVERY_STATUS_ACK.
3) MAP_REPORT_SM_DELIVERY_STATUS_ACK.
4) Short Message Negative Acknowledgement (3GPP TS 23.040).

Figure 23.5/1: Short message delivery status report procedure

23.5.1 Procedure in the SMS-GMSC

The conditions for the GMSC to invoke the short message delivery status report procedure are specified in subclause 23.3.1.

The short message delivery status report macro in the SMS-GMSC is shown in figure 23.5/2.

23.5.2 Procedure in the HLR

When the HLR receives a MAP_REPORT_SM_DELIVERY_STATUS indication while an IP-SM-GW is deployed in the network and the message is not received from an IP-SM-GW, it ignores the information received in the message; otherwise it acts as described in subclause 23.6, macro Report_SM_Delivery_Stat_HLR.

The short message delivery status report process in the HLR is shown in figure 23.5/3.

23.5.3 Procedure in the IP-SM-GW

The conditions for the IP-SM-GW and for SMS Router, if deployed with IP-SM-GW, to invoke the short message delivery status report procedure are specified in subclause 23.3.7.

The short message delivery status report macro in the IP-SM-GW is shown in figure 23.5/4.

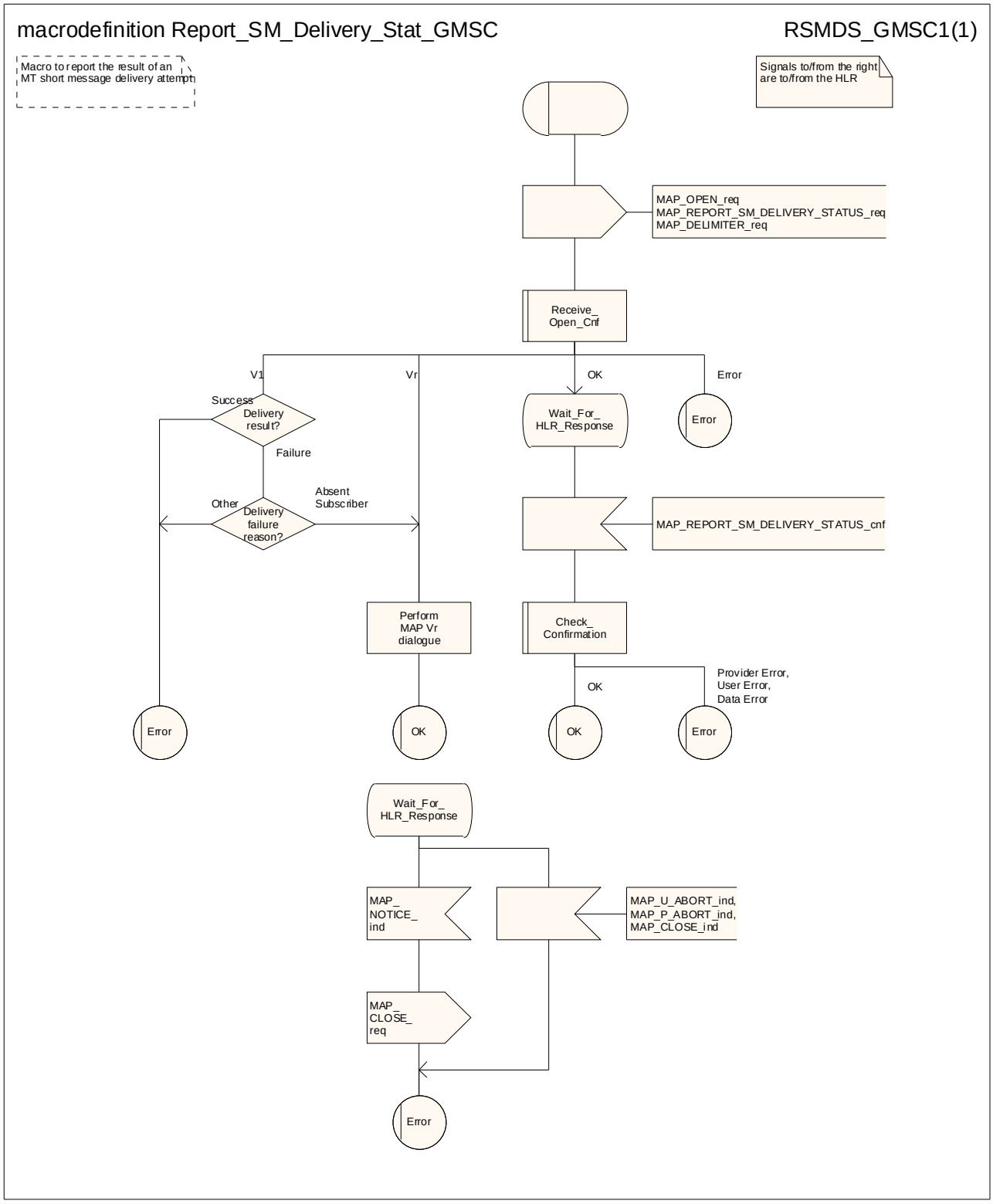


Figure 23.5/2: Macro Report_SM_Delivery_Stat_GMSC

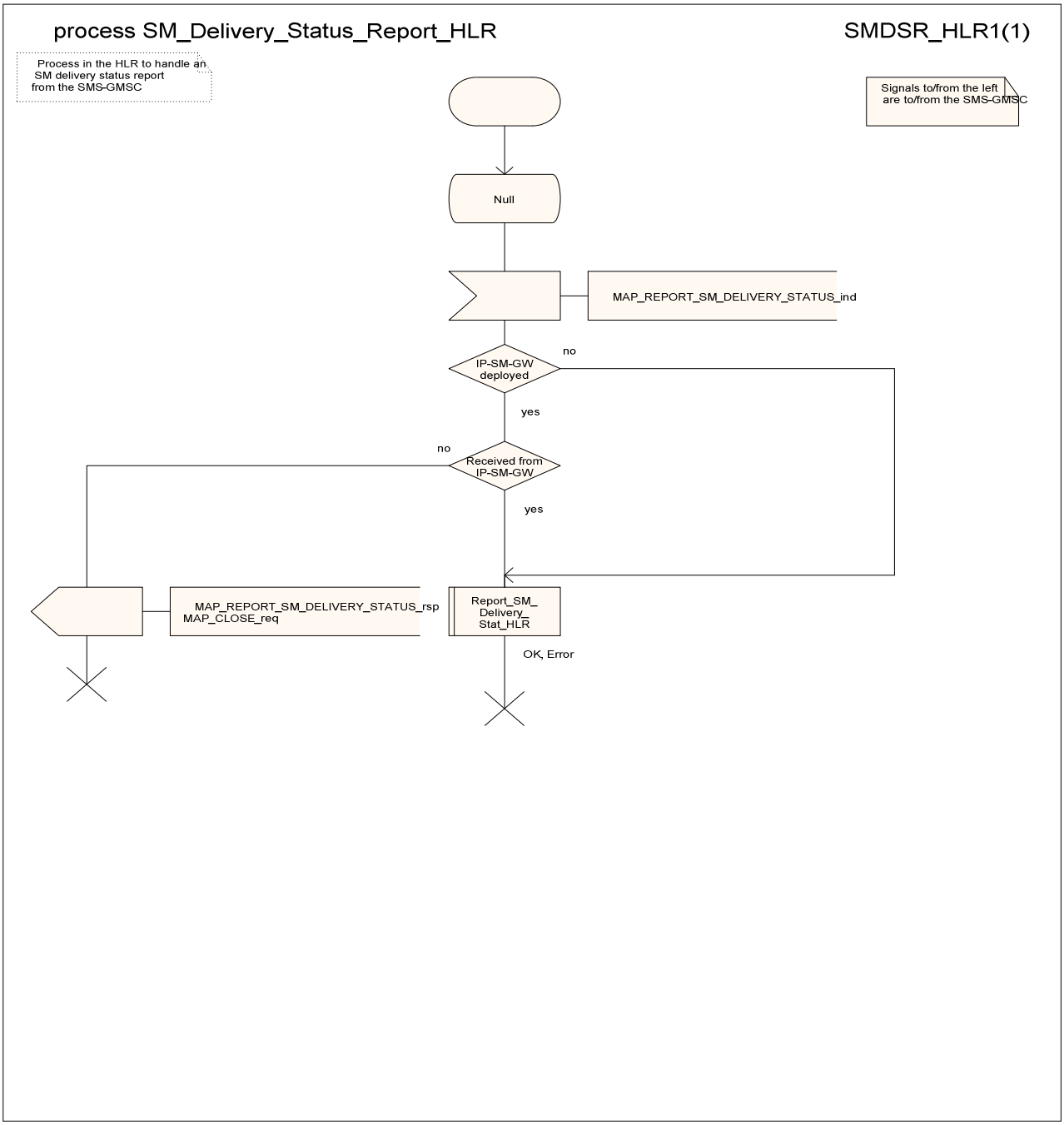


Figure 23.5/3: Process SM_Delivery_Status_Report_HLR

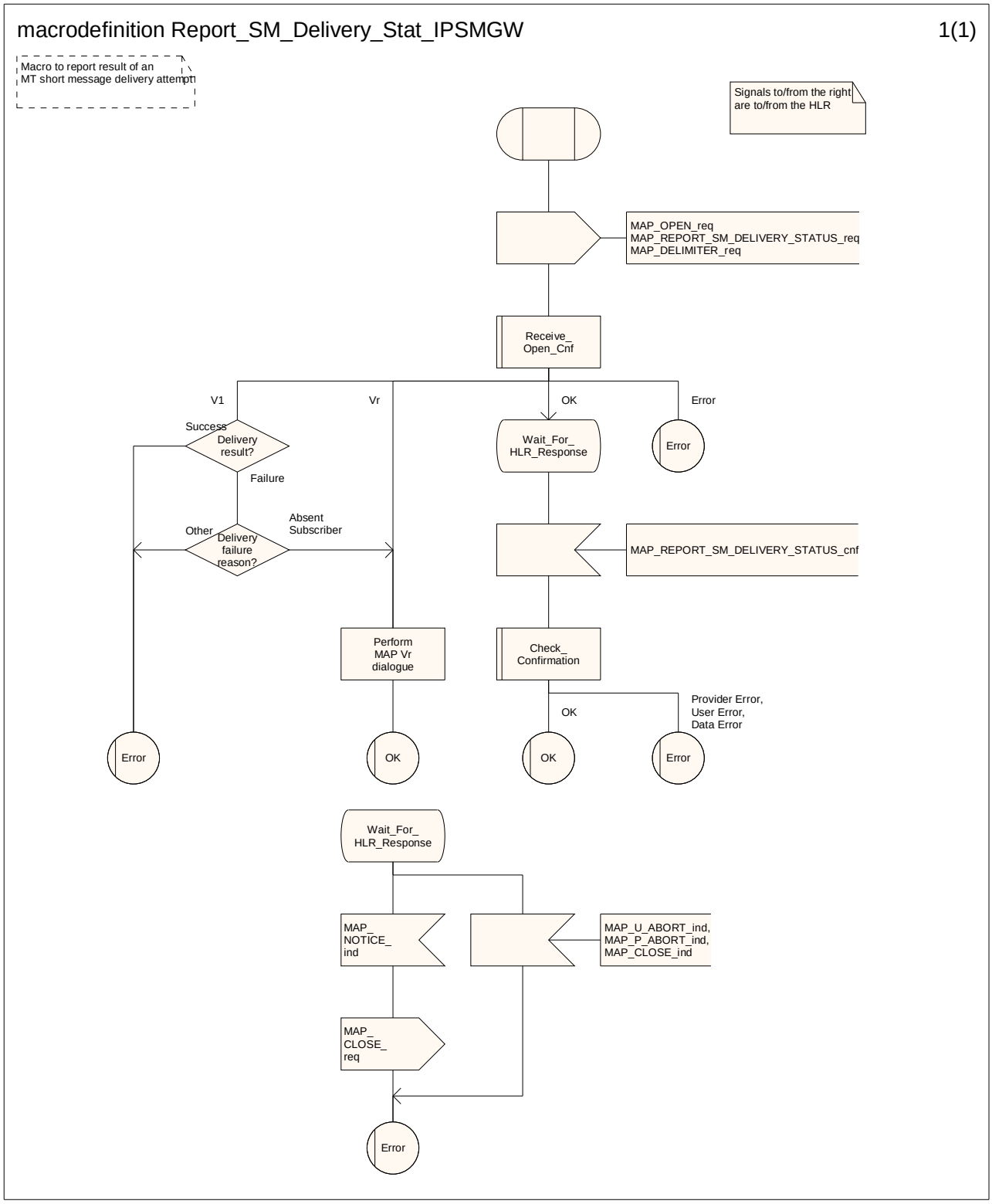


Figure 23.5/4: Macro Report_SM_Delivery_Stat_IPSMGW

23.6 The macro Report_SM_Delivery_Stat_HLR

This macro is invoked when the HLR receives a MAP_REPORT_SM_DELIVERY_STATUS indication from the SMS-GMSC. The macro invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Check_Indication see subclause 25.2.1;

Alert_Service_Centre_HLR see subclause 25.10.3.

Sheet 1: If the MAP_REPORT_SM_DELIVERY_STATUS indication did not include the GPRS support indicator, the HLR deduces the domain for which the delivery report applies as follows:

- if the subscriber is a GPRS-only subscriber, the report applies for GPRS;
- if the subscriber is a non-GPRS-only subscriber, the report applies for non-GPRS;
- if the subscriber is a GPRS and non-GPRS subscriber and the subscription option for MT SMS delivery when the SMS-GMSC does not support GPRS is set to "Delivery via the SGSN", the report applies for GPRS;
- if the subscriber is a GPRS and non-GPRS subscriber and the subscription option for MT SMS delivery when the SMS-GMSC does not support GPRS is set to "Delivery via the MSC", the report applies for non-GPRS;

The short message delivery status report macro in the HLR is shown in figure 23.6/1.

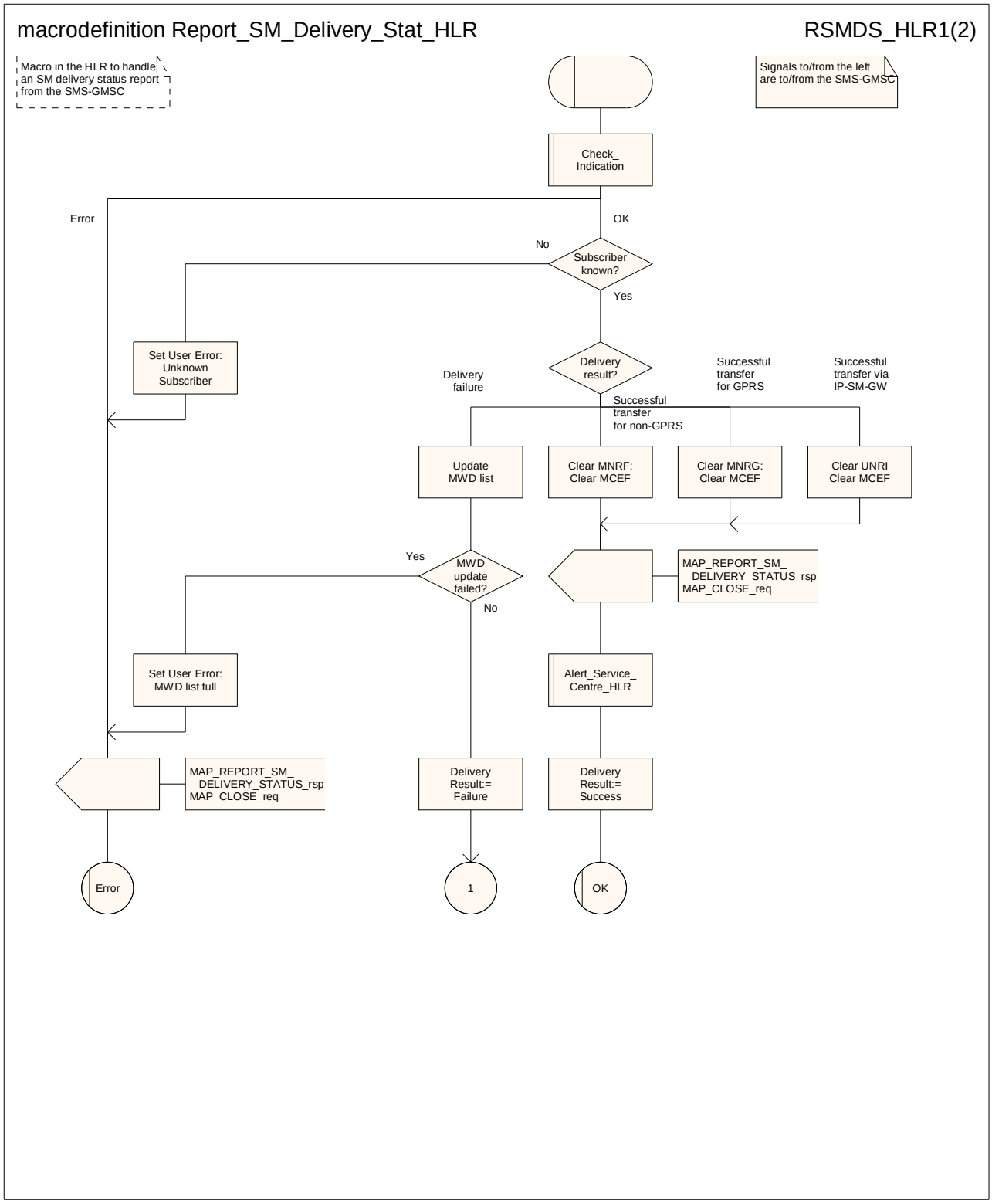


Figure 23.6/1 (sheet 1 of 2): Macro Report_SM_Delivery_Stat_HLR

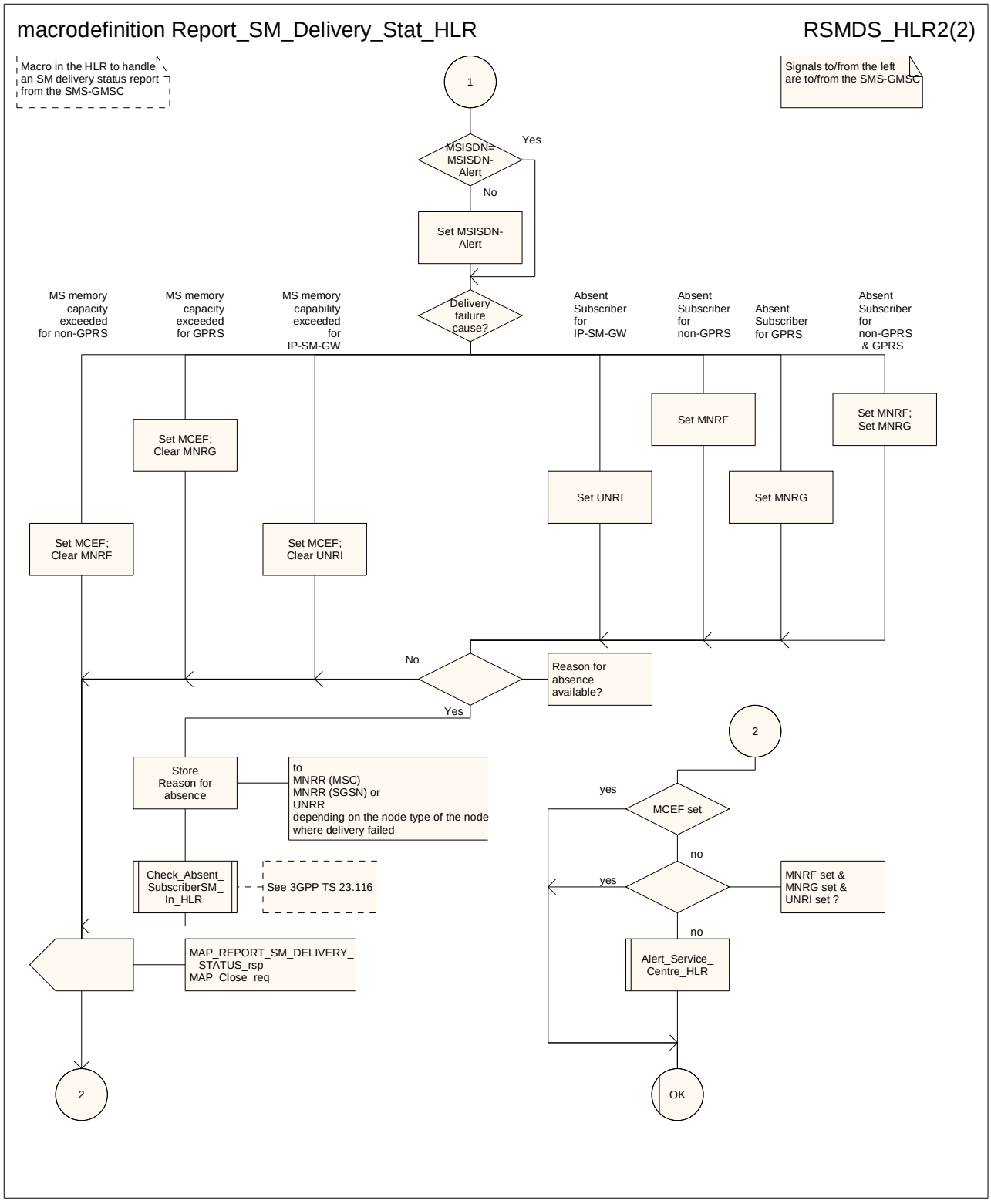


Figure 23.6/1 (sheet 2 of 2): Macro Report_SM_Delivery_Stat_HLR

23.7 The mobile terminated short message transfer procedure for VGCS

The mobile terminated short message transfer for VGCS procedure is used for forwarding a short message from a Service Centre to the group call anchor MSC. The message flow for the mobile terminated short message transfer procedure for VGCS is shown in figure 23.7/1.

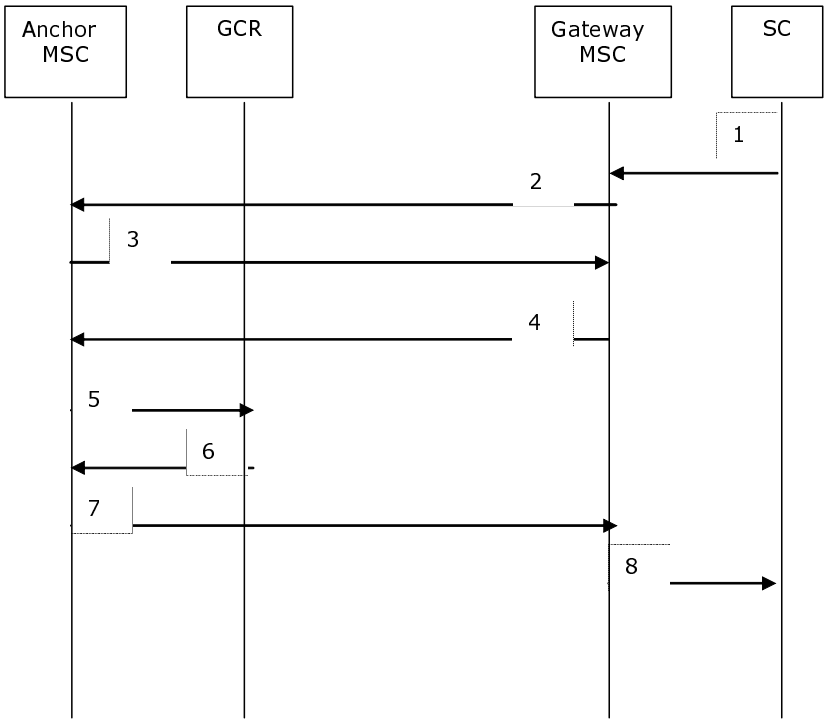


Figure 23.7/1: Mobile terminated short message for VGCS service procedures

1) Short Message (3GPP TS 23.040).
2) TCAP BEGIN (*)
3) TCAP CONTINUE (*)
4) MAP_MT_FORWARD_SM_FOR_VGCS.
5) GCR_SMS_INTERROGATION (3GPP TS 43.068).
6) GCR_SMS_INTERROGATION_ACK (3GPP TS 43.068).
7) MAP_MT_FORWARD_SM_FOR_VGCS_ACK.
8) Short Message Acknowledgement (3GPP TS 23.040).

(*) If
a)
- the capacity of a message signal unit in the lower layers of the protocol is enough to carry the content of the MAP_OPEN request and the content of the MAP_MT_FORWARD_SM_FOR_VGCS request in a single TC message,

and
b) the SMS Gateway MSC operator and the serving node (Anchor-MSC) operator agreed not to use the TCAP handshake countermeasure against SMS fraud for messages exchanged between their networks (see 3GPP TS 33.204 [34a])
then
the TCAP handshake may be omitted.

23.7.1 Procedure in the SMS-GMSC

The process starts when the SMS-GMSC receives an SC_RP_MT_DATA indication from a Service Centre. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The mobile terminated short message transfer for VGCS process in the SMS-GMSC is shown in figure 23.7/2.

23.7.2 Procedure in the Anchor MSC

The process starts when the MSC receives a dialogue opening request with the application context shortMsgMT-Relay-VGCS-Context. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

The mobile terminated short message transfer for VGCS process in the Anchor MSC is shown in figure 23.7/3

Procedure MT_SM_VGCS_GMSC sheet 1: The decision box "TCAP Handshake required" takes the "yes" or "no" exit depending on agreements between the Serving MSC's operator and the SMS Gateway MSC's operator (see 3GPP TS 33.204 [34a]).

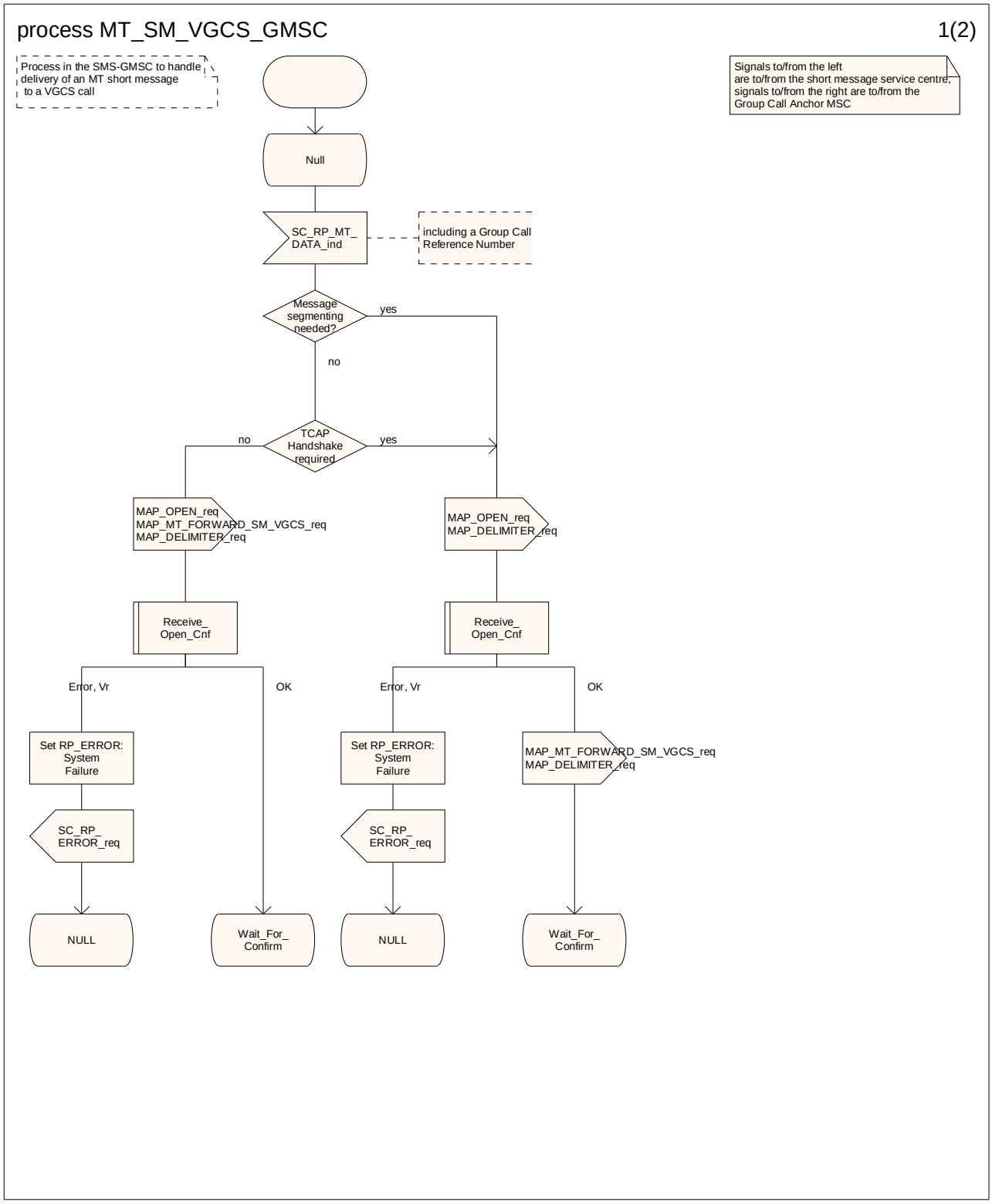


Figure 23.7/2 (sheet 1 of 2): Process MT_SM_VGCS_GMSC

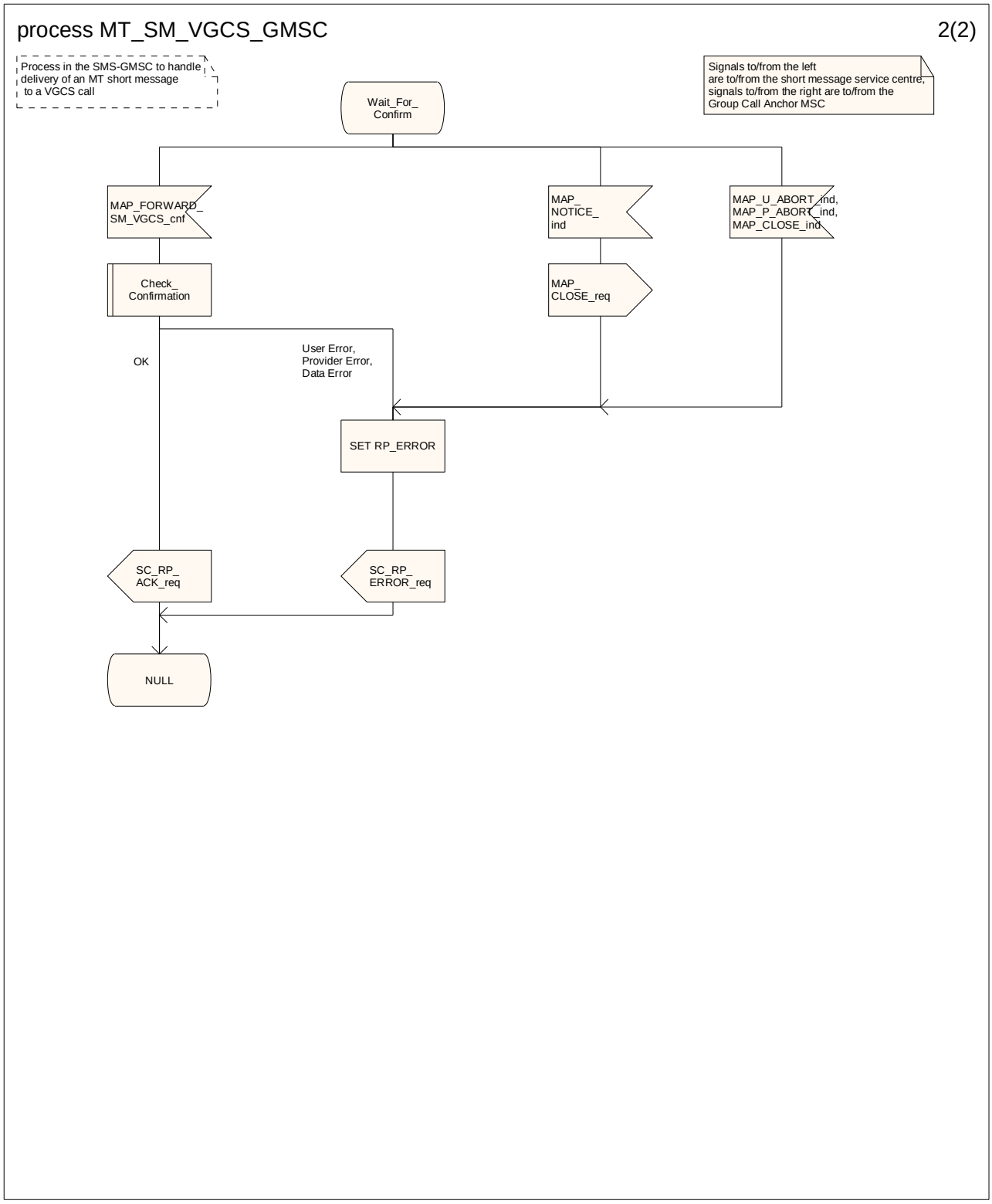


Figure 23.7/2 (sheet 2 of 2): Process MT_SM_VGCS_GMSC

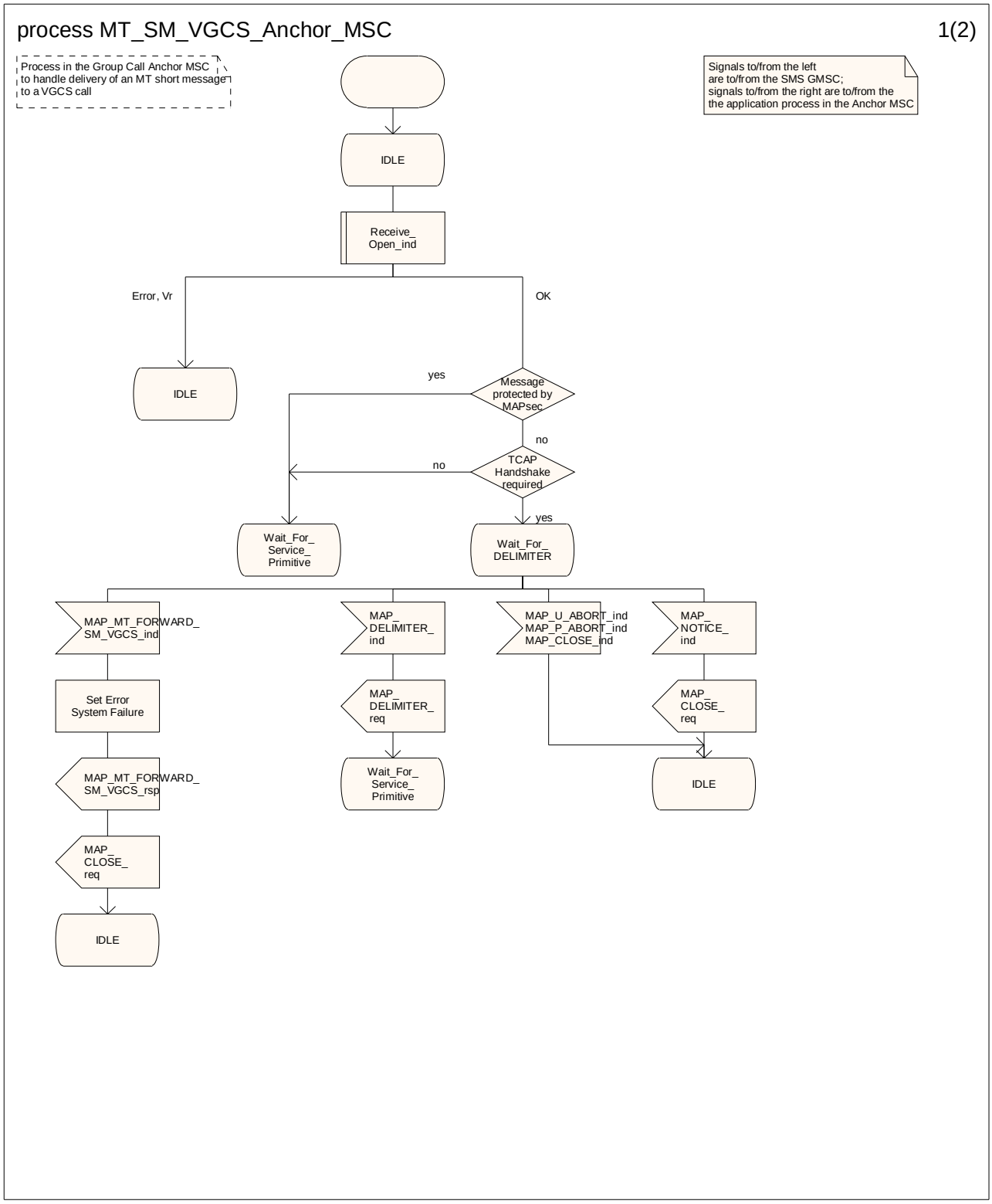


Figure 23.7/3 (sheet 1 of 2): Process MT_SM_VGCS_Anchor MSC

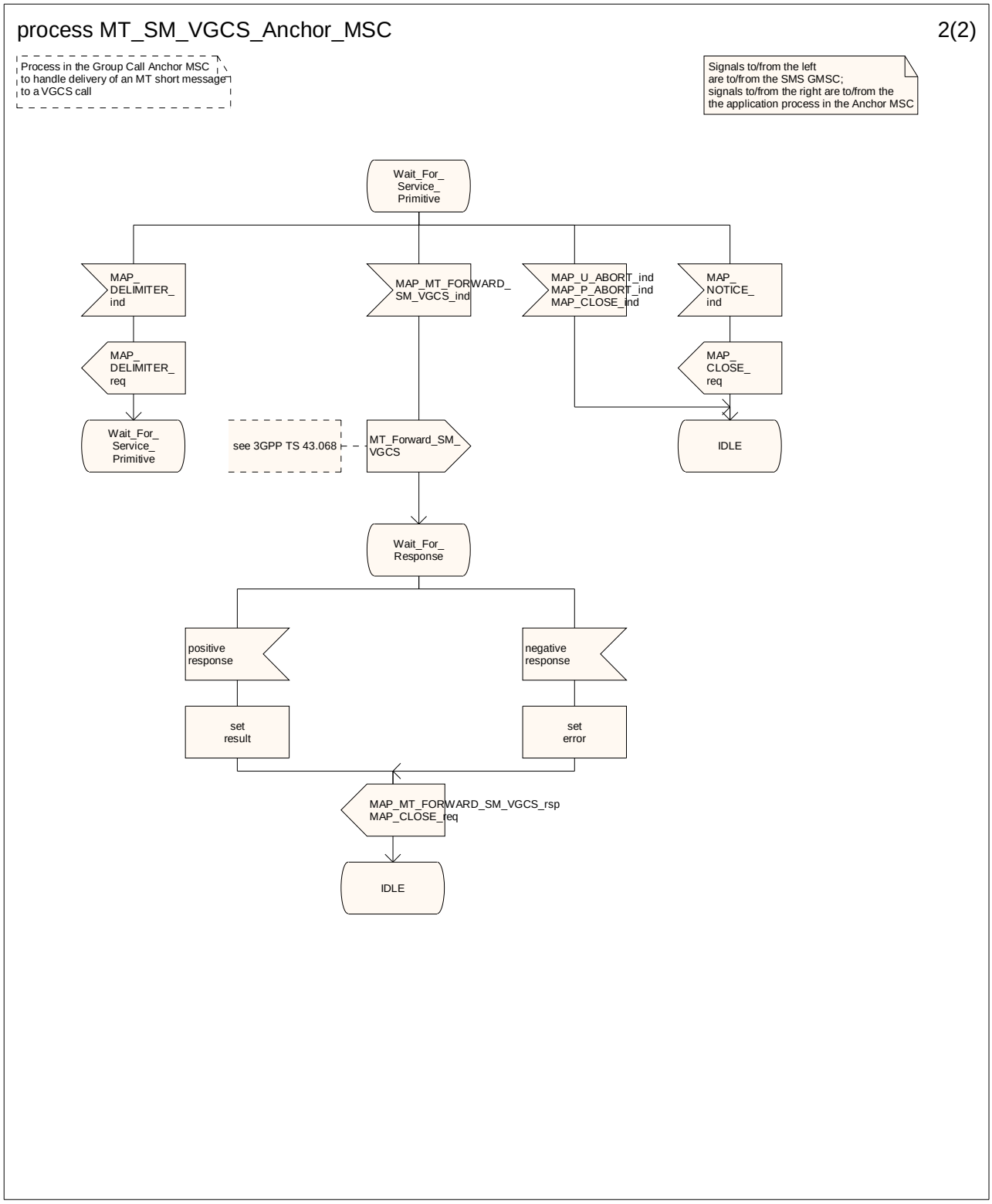


Figure 23.7/3 (sheet 2 of 2): Process MT_SM_VGCS_Anchor MSC

24 GPRS process description

The MAP GPRS procedures are used for the Network Requested PDP-Context Activation procedures.

The stage 2 specification for General Packet Radio Service (GPRS) is in 3GPP TS 23.060 [104].

24.1 Procedure for retrieval of routeing information for GPRS

24.1.1 Process in the GGSN

The MAP process in the GGSN to request routeing information for a network requested PDP context activation is shown in figure 24.1/2. The MAP process invokes macros not defined in this clause; the definition of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24.1.2 Process in the HLR

The MAP process in the HLR to provide routing information for a network-requested PDP context activation is shown in figure 24.1/1. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

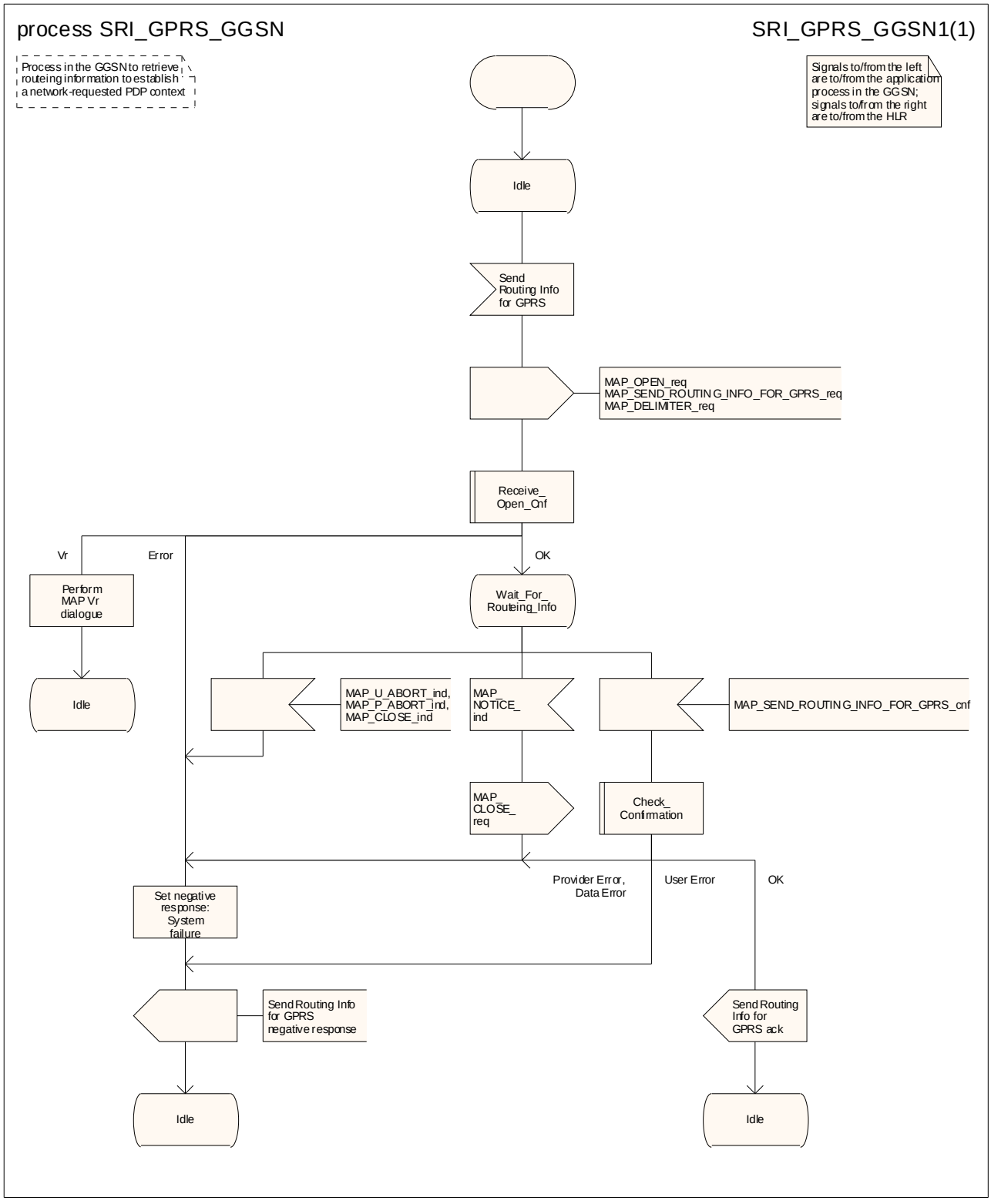


Figure 24.1/1: Process SRI_GPRS_GGSN

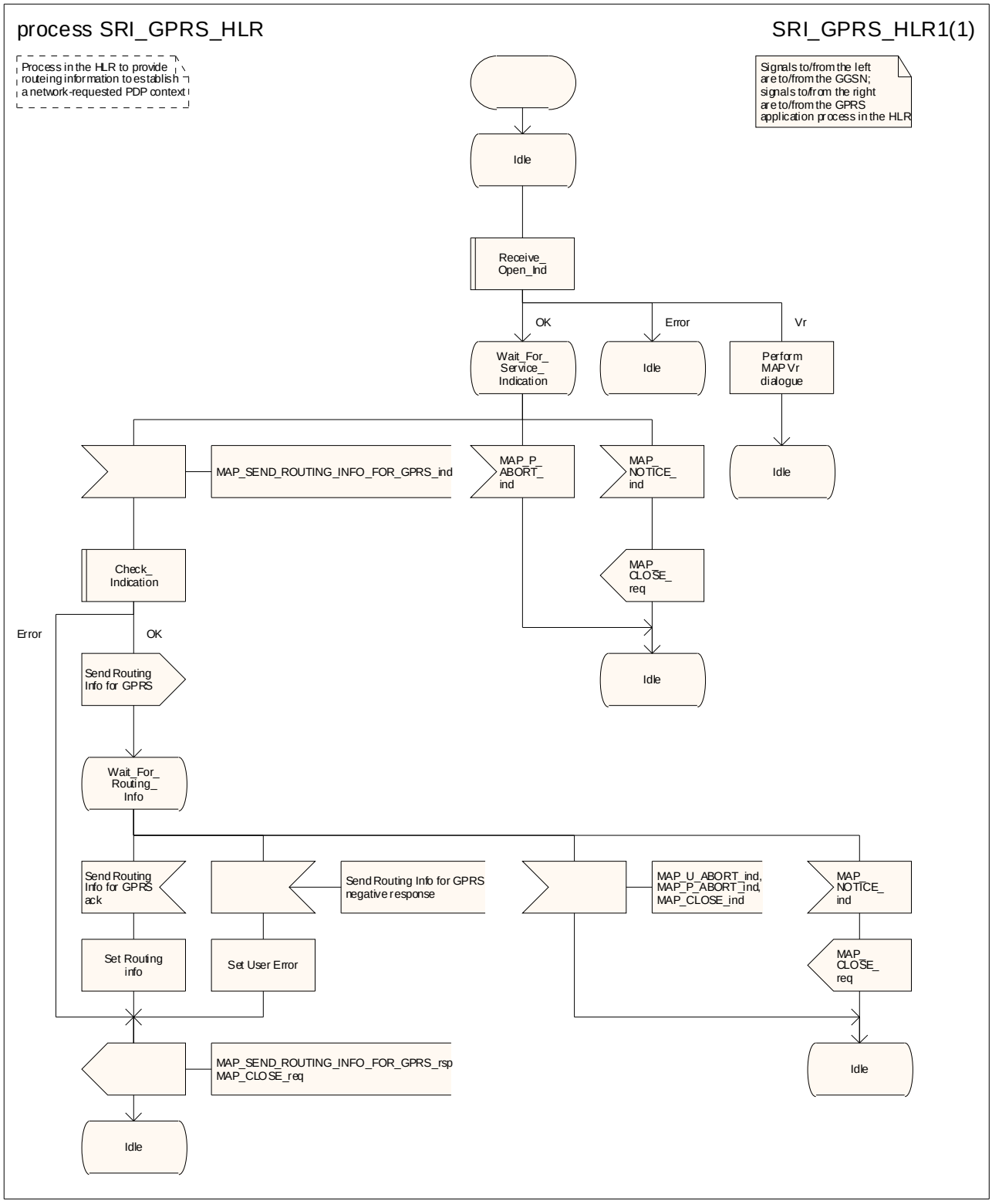


Figure 24.1/2: Process SRI_GPRS_HLR

24.2 Procedure for reporting failure to establish a network requested PDP context

24.2.1 Process in the GGSN

The MAP process in the GGSN to report the failure to establish a network requested PDP context is shown in figure 24.2/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24.2.2 Process in the HLR

The MAP process in the HLR to handle a notification from the GGSN that a network requested PDP context could not be established is shown in figure 24.2/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check Indication see subclause 25.2.1.

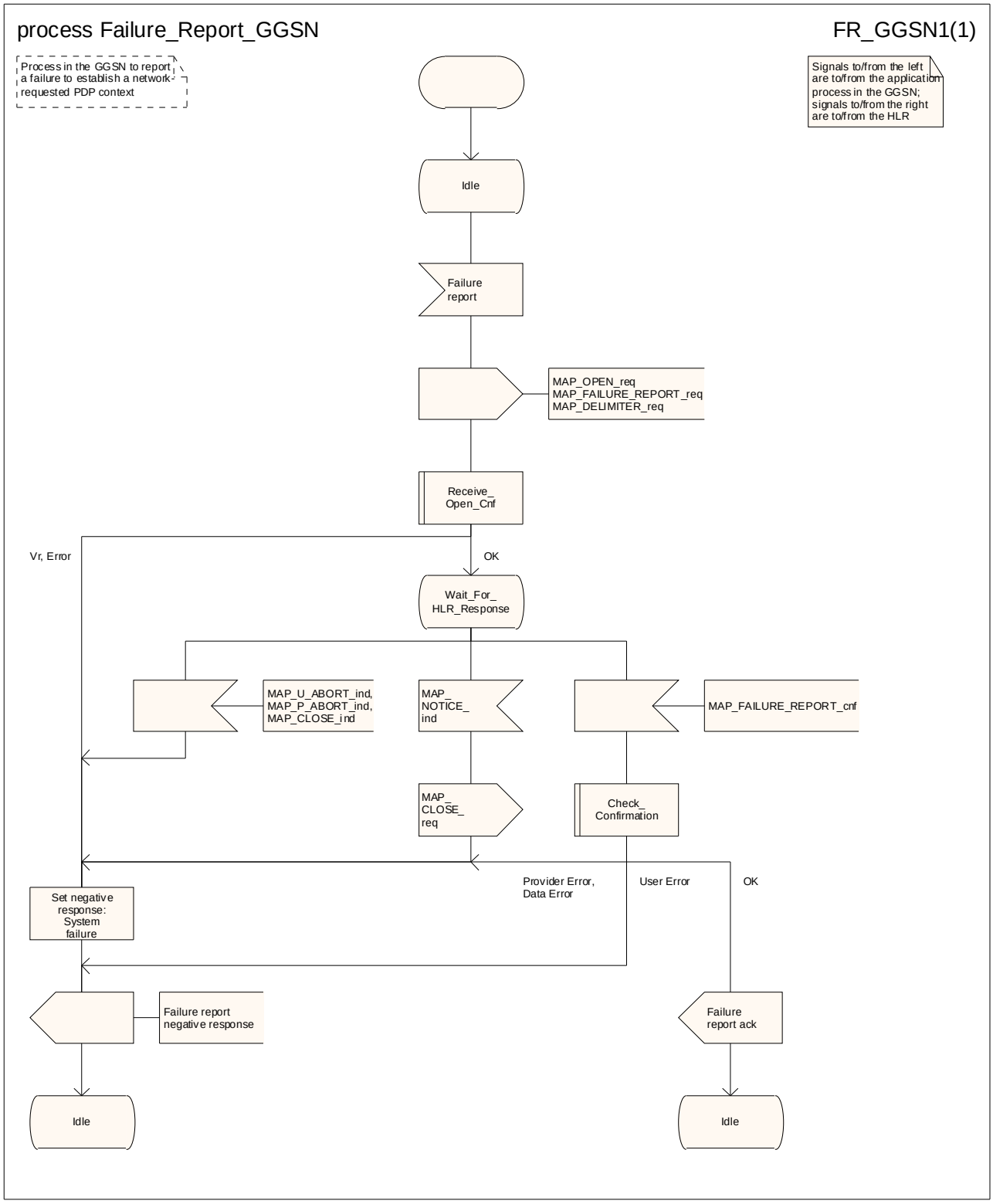


Figure 24.2/1: Process Failure_Report_GGSN

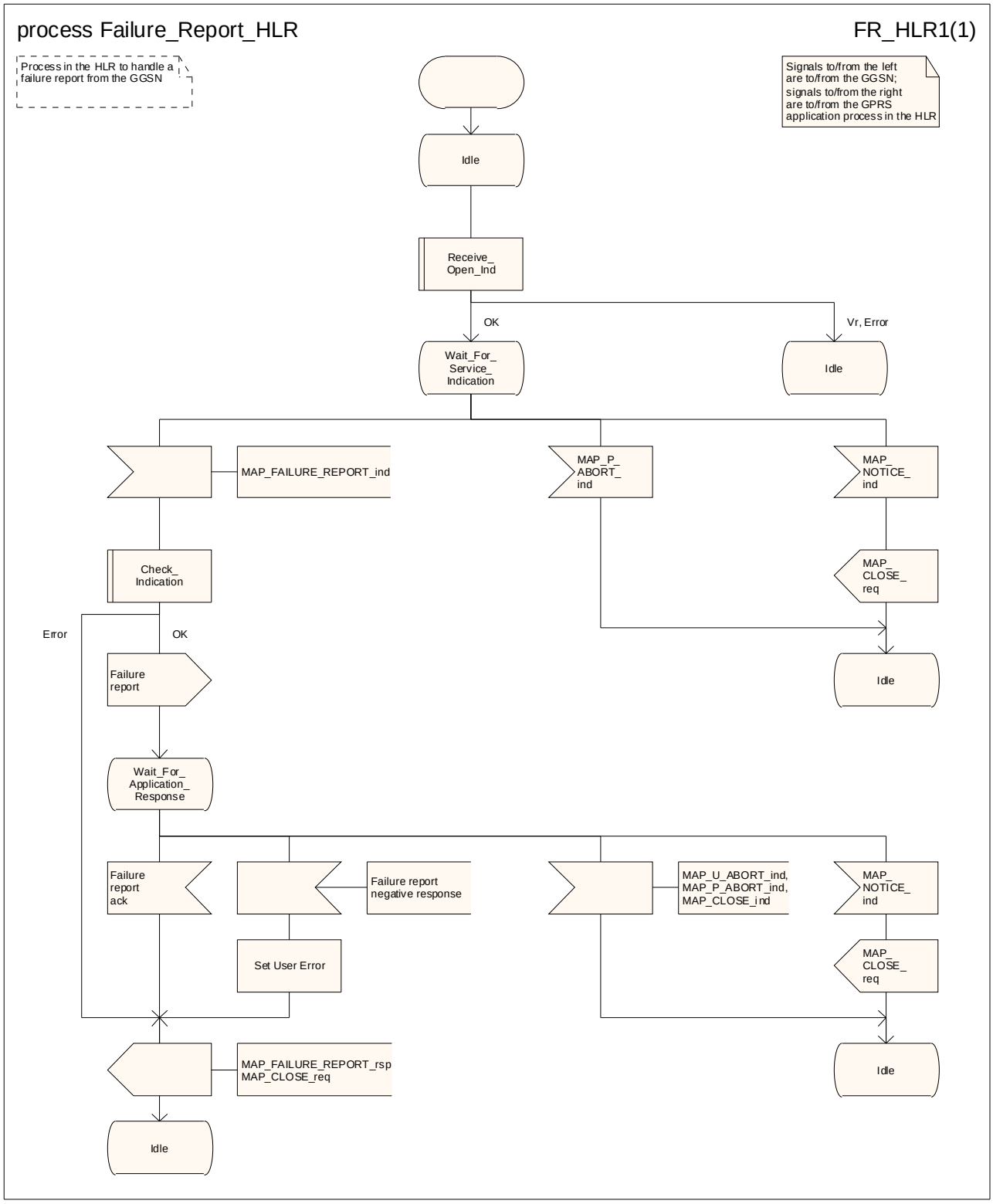


Figure 24.2/2: Process Failure_Report_HLR

24.3 Procedure for reporting that an MS has become reachable for GPRS

24.3.1 Process in the HLR

The MAP process in the HLR to report that an MS is reachable for GPRS is shown in figure 24.3/1. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24.3.2 Process in the GGSN for Note Ms Present For Gprs

The MAP process in the GGSN to handle a notification that the subscriber is present for GPRS again is shown in figure 24.3/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

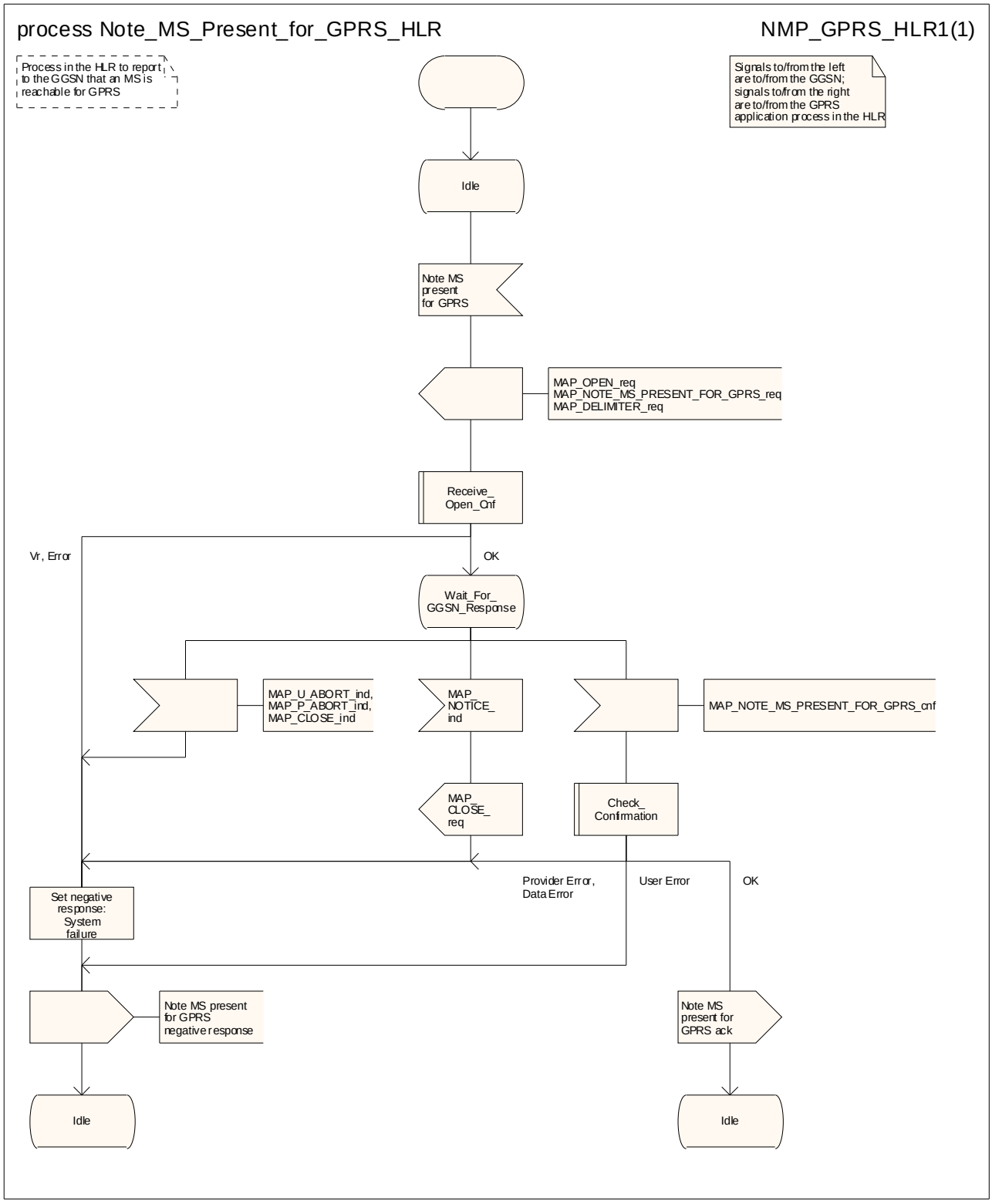


Figure 24.3/1: Process Note_MS_Present_For_GPRS_HLR

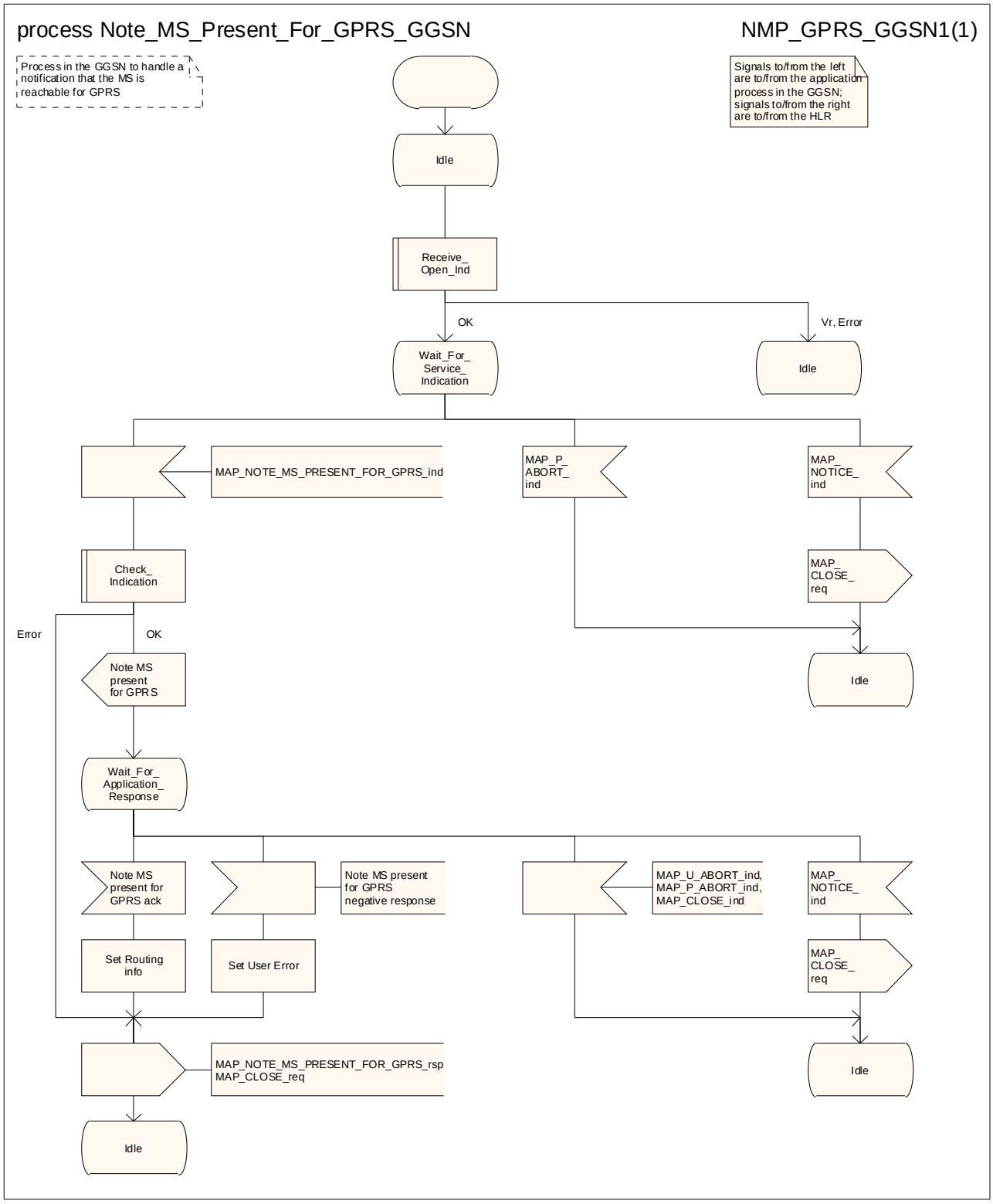


Figure 24.3/2: Process Note_MS_Present_For_GPRS_GGSN

24A CSE interrogation and control of subscriber data

24A.1 General

The MAP procedures for interrogation and control of subscriber data are used to allow the CSE:

- to retrieve subscriber data from the HLR;
- to modify subscriber data in the HLR;
- to receive notification from the HLR when there is a change in subscriber data;
- to request information about the location of a subscriber from the HLR or the GMLC;
- to request information about the state of a subscriber from the HLR.

The following application context refers to a complex MAP user consisting of several processes:

- **anyTimeInfoHandlingContext**

This application context needs a co-ordinating process in the HLR.

The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

The Any Time Info Handling Co-ordinator process in the HLR is shown in figure 24A.1/1.

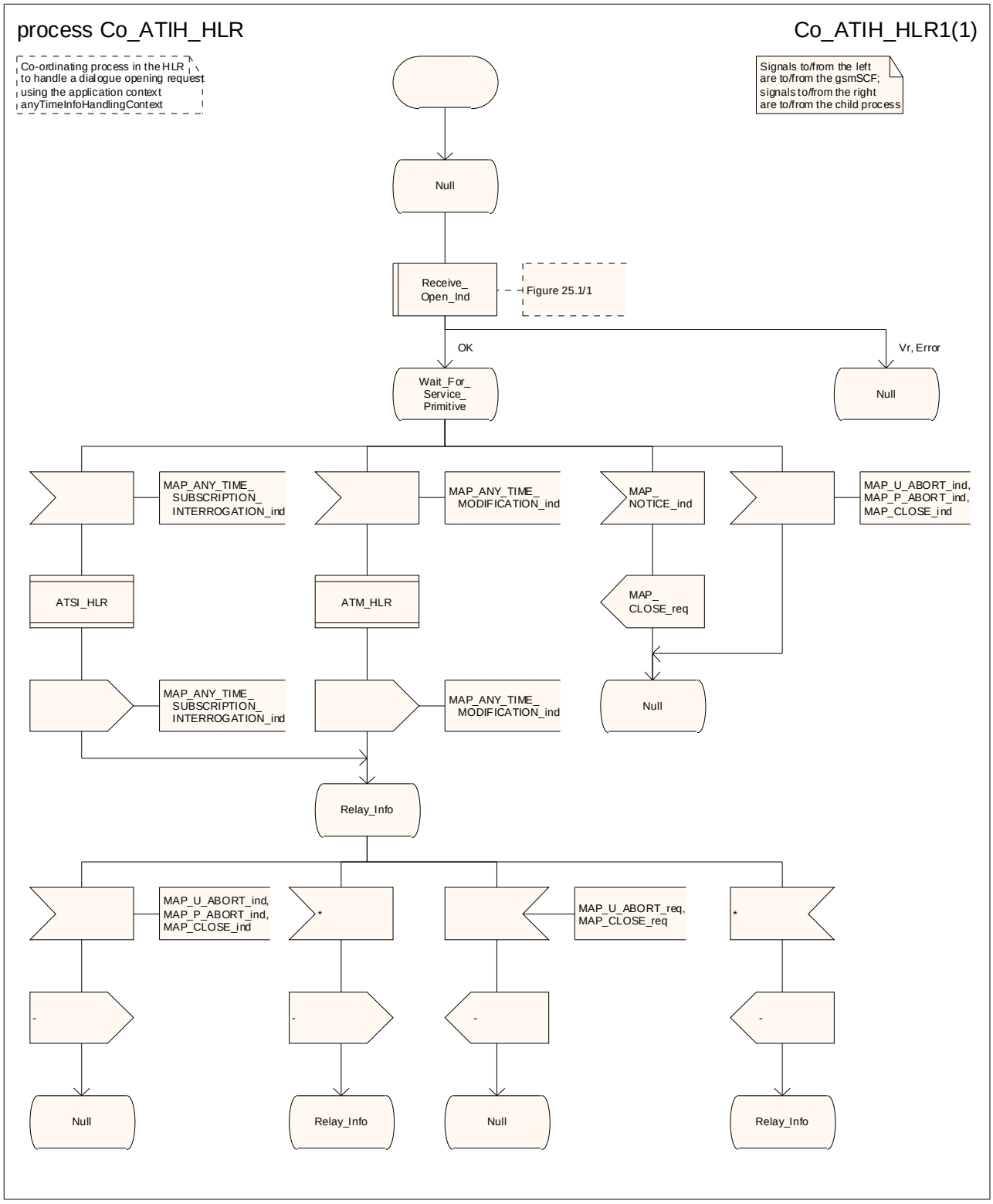


Figure 24A.1/1: Process Co_ATIH_HLR

24A.2 Any Time Subscription Interrogation procedure

24A.2.1 General

The message flow for successful retrieval of subscription information related to an any time subscription interrogation from the CAMEL server are shown in figure 24A.1/1. In an IP Multimedia Core Network, an IM-SSF can take on the role of a gsmSCF for this procedure (see 3GPP TS 23.278 [125]).

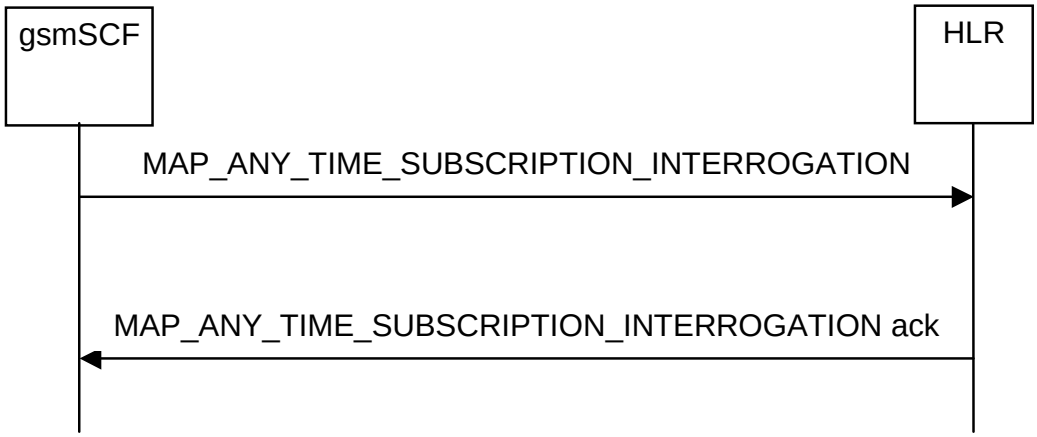


Figure 24A.2/1: Message flow for any time subscription interrogation

The following MAP service is used to retrieve requested information:

MAP_ANY_TIME_SUBSCRIPTION_INTERROGATION see subclause 8.11.3.

24A.2.2 Process in the gsmSCF

The MAP process in the gsmSCF to obtain subscription information in response to a request from the application process in the gsmSCF is shown in figure 24A.2/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2

24A.2.3 Process in the HLR

The MAP process in the HLR to provide subscription information in response to an interrogation from the CAMEL server is shown in figure 24A.2/3. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Check_Indication see subclause 25.2.2

If the MAP_ANY_TIME_SUBSCRIPTION_INTERROGATION service response cannot be carried in a single TC-Result component, it is carried in one or more TC-Result-NL components (each sent in a TC-CONTINUE), followed by a TC-Result-L component in a TC-END message.

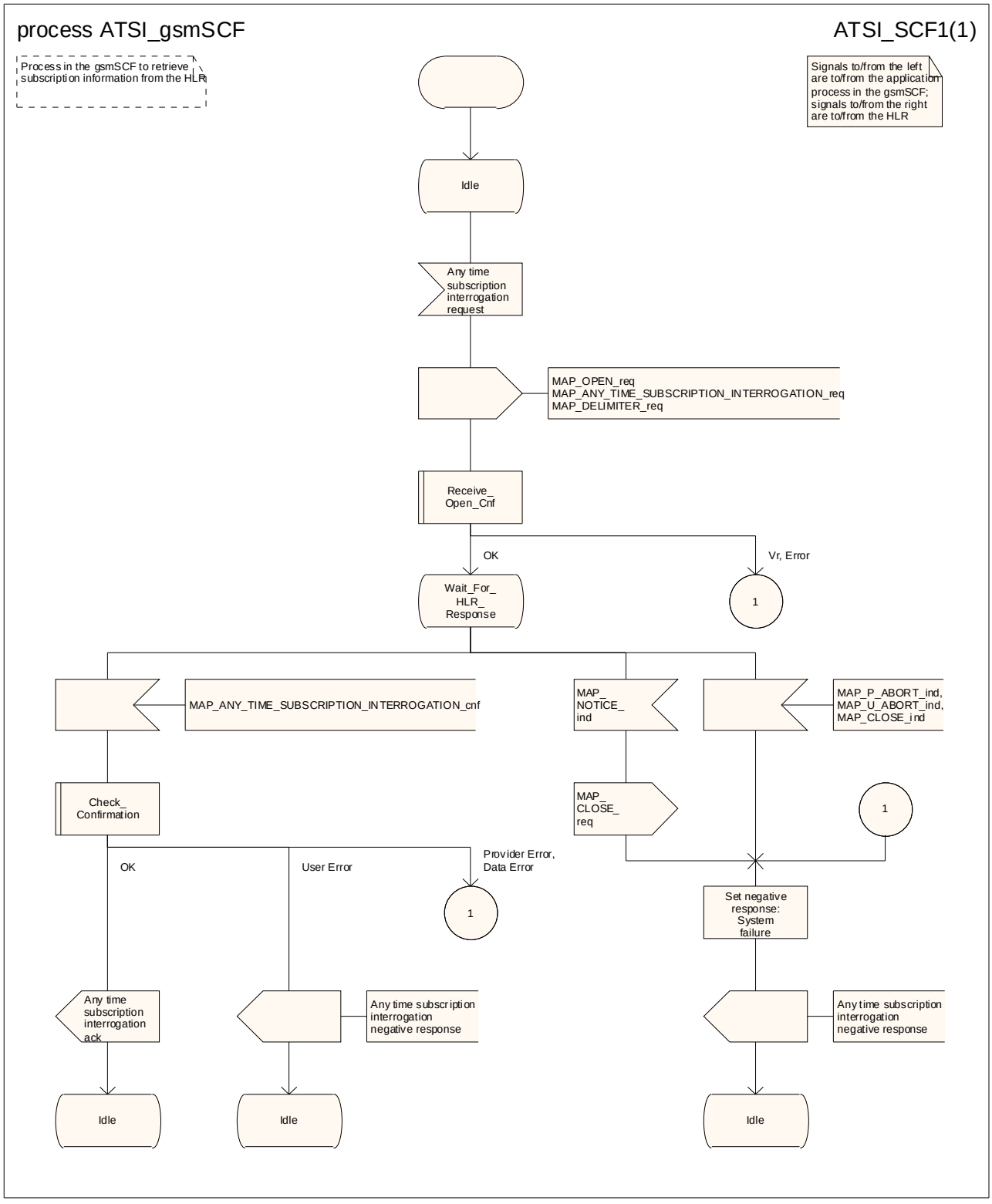


Figure 24A.2/2: Process ATSI_gsmSCF

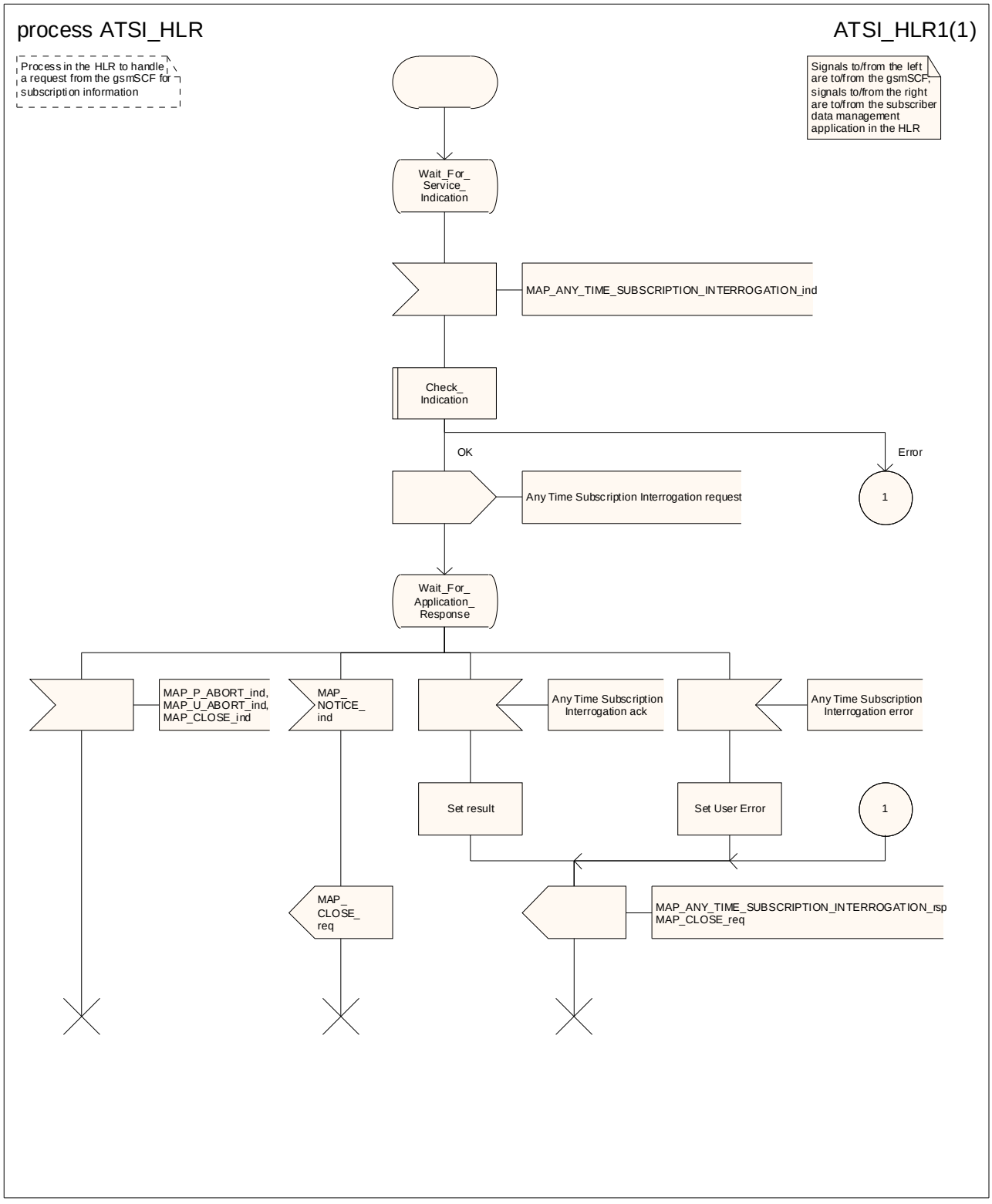


Figure 24A.2/3: Process ATSI_HLR

24A.3 Any Time Modification procedure

24A.3.1 General

The message flow for successful modification of subscription information related to an any time modification request from the CAMEL server is shown in figure 24A.3/1

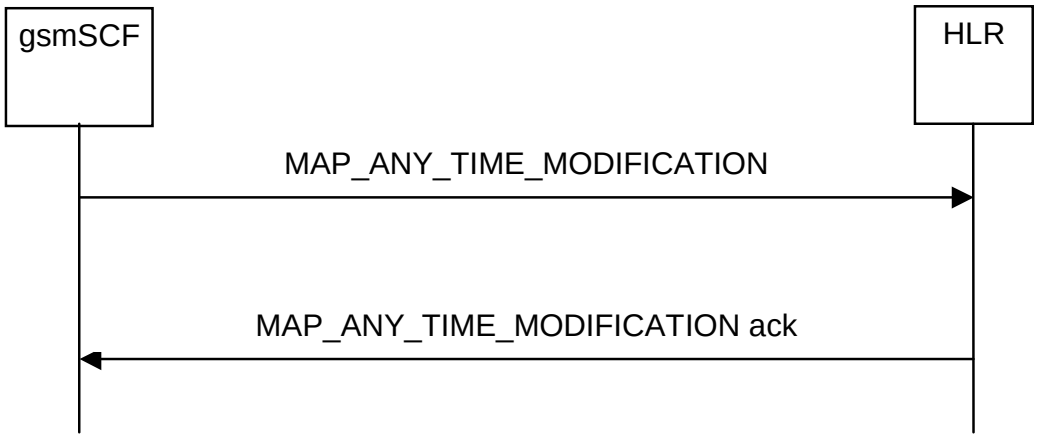


Figure 24A.3/1: Message flow for any time modification

The following MAP service is used to modify subscription information:

MAP_ANY_TIME_MODIFICATION see subclause 8.11.4.

24A.3.2 Process in the gsmSCF

The MAP process in the gsmSCF to modify subscription information in response to a request from the application process in the gsmSCF is shown in figure 24A.3/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2

24A.3.3 Process in the HLR

The MAP process in the HLR to modify subscriber information in response to a modification request from the CAMEL server is shown in figure 24A.3/3. The MAP process invokes a macro and a process not defined in this clause; the definitions of these can be found as follows:

Check_Indication see subclause 25.2.2;

Insert_Subs_Data_Stand_Alone_HLR see subclause 25.7.3;

If the macro takes the OK exit, the MAP process waits for a service indication.

If the MAP_ANY_TIME_MODIFICATION service response cannot be carried in a single TC-Result component, it is carried in one or more TC-Result-NL components (each sent in a TC-CONTINUE), followed by a TC-Result-L component in a TC-END message.

If the serving node (VLR or SGSN) is to be updated after the modification, the MAP process creates an instance of the appropriate process (Insert_Subs_Data_Stand_Alone_HLR for VLR update, Insert_GPRS_Subs_Data_Stand_Alone_HLR for SGSN update).

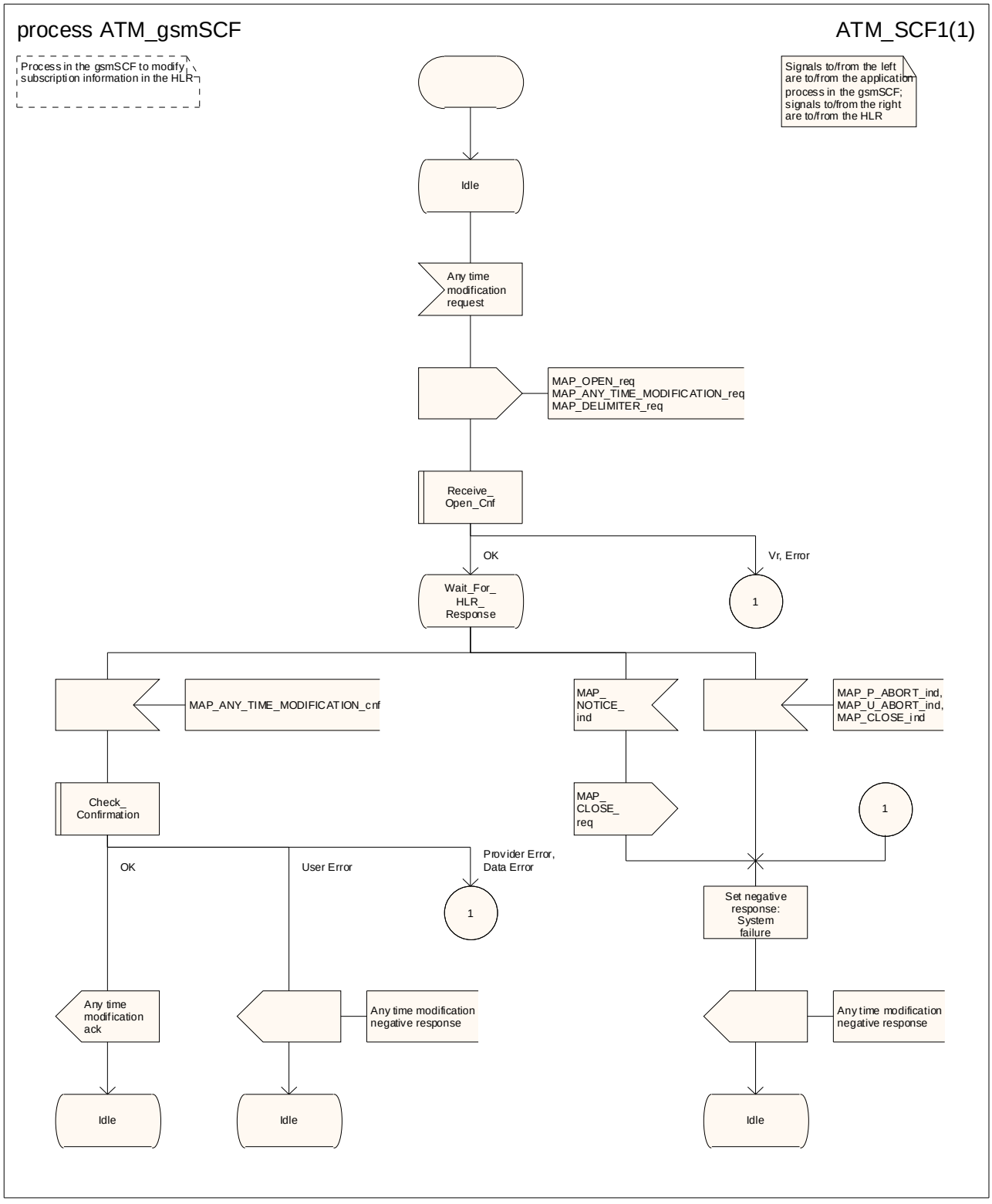


Figure 24A.3/2: Process ATM_gsmSCF

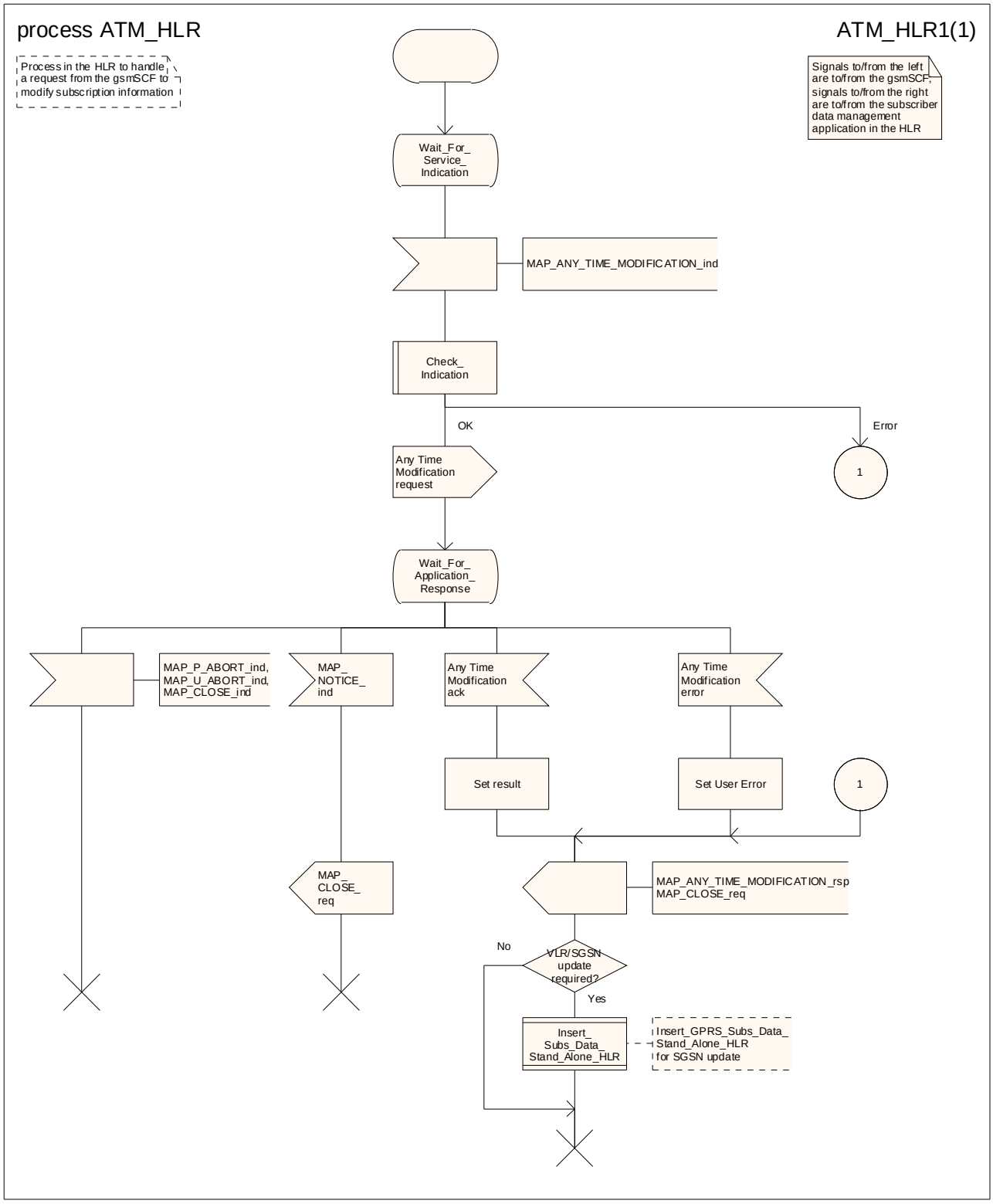


Figure 24A.3/3: Process ATM_HLR

24A.4 Subscriber Data Modification Notification procedure

24A.4.1 General

The Subscriber Data Modification Notification procedure is used to notify a gsmSCF about the modification of subscriber data. In an IP Multimedia Core Network, an IM-SSF can take on the role of a gsmSCF for this procedure.

The stage 2 specification for Subscriber Data Modification Notification is in 3GPP TS 23.078 [98] and 3GPP TS 23.278 [125]. The interworking between the MAP signalling procedures and the Subscriber Data Modification Notification procedures for each entity (HLR, gsmSCF) is shown by the transfer of signals between these processes.

The following services are used:

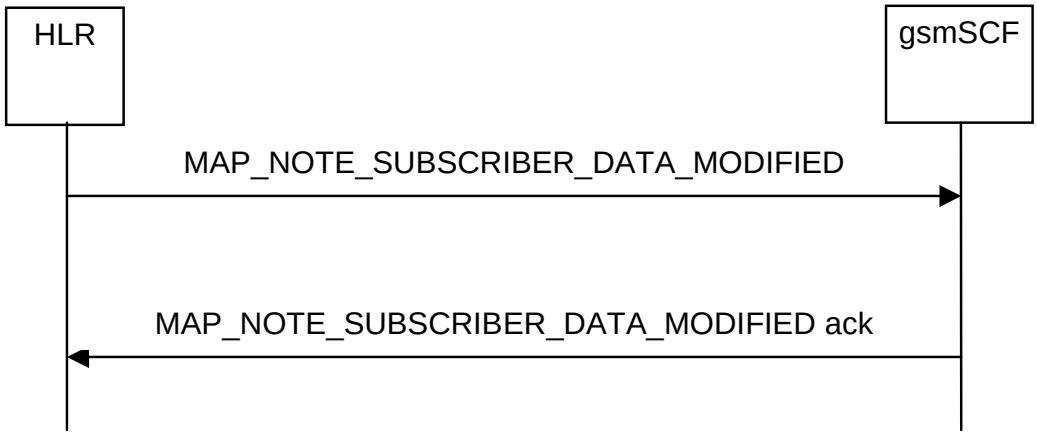


Figure 24A.4/1: Message flow for subscriber data modification notification

The following MAP service is used to send the notification to the gsmSCF:

MAP_NOTE_SUBSCRIBER_DATA_MODIFIED see subclause 8.11.5.

24A.4.2 Process in the HLR

The MAP process in the HLR to send modified data to the gsmSCF is shown in figure 24A.4/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

If the required information cannot be carried in a single MAP_NOTE_SUBSCRIBER_DATA_MODIFIED service request, the HLR segments the information into two or more requests. The "All Information Sent" parameter is omitted from each request except the last.

Sheet 2: If the MAP_NOTE_SUBSCRIBER_DATA_MODIFIED service request contained the "All Information Sent" parameter, the test "All information sent" takes the "Yes" exit.

24A.4.3 Process in the gsmSCF

The MAP process in the gsmSCF to handle a notification to the gsmSCF of change of subscriber data is shown in figure 24A.4/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1

If the MAP_NOTE_SUBSCRIBER_DATA_MODIFIED service indication contained the "All Information Sent" parameter, the test "All information sent" takes the "Yes" exit.

If the test "All information sent" takes the "No" exit, the MAP process stores the data received in the MAP_NOTE_SUBSCRIBER_DATA_MODIFIED service indication. If the test "All information sent" takes the "Yes" exit, the MAP process assembles the data received in all the MAP_NOTE_SUBSCRIBER_DATA_MODIFIED service indications received in the dialogue and sends the assembled data to the application process in the gsmSCF.

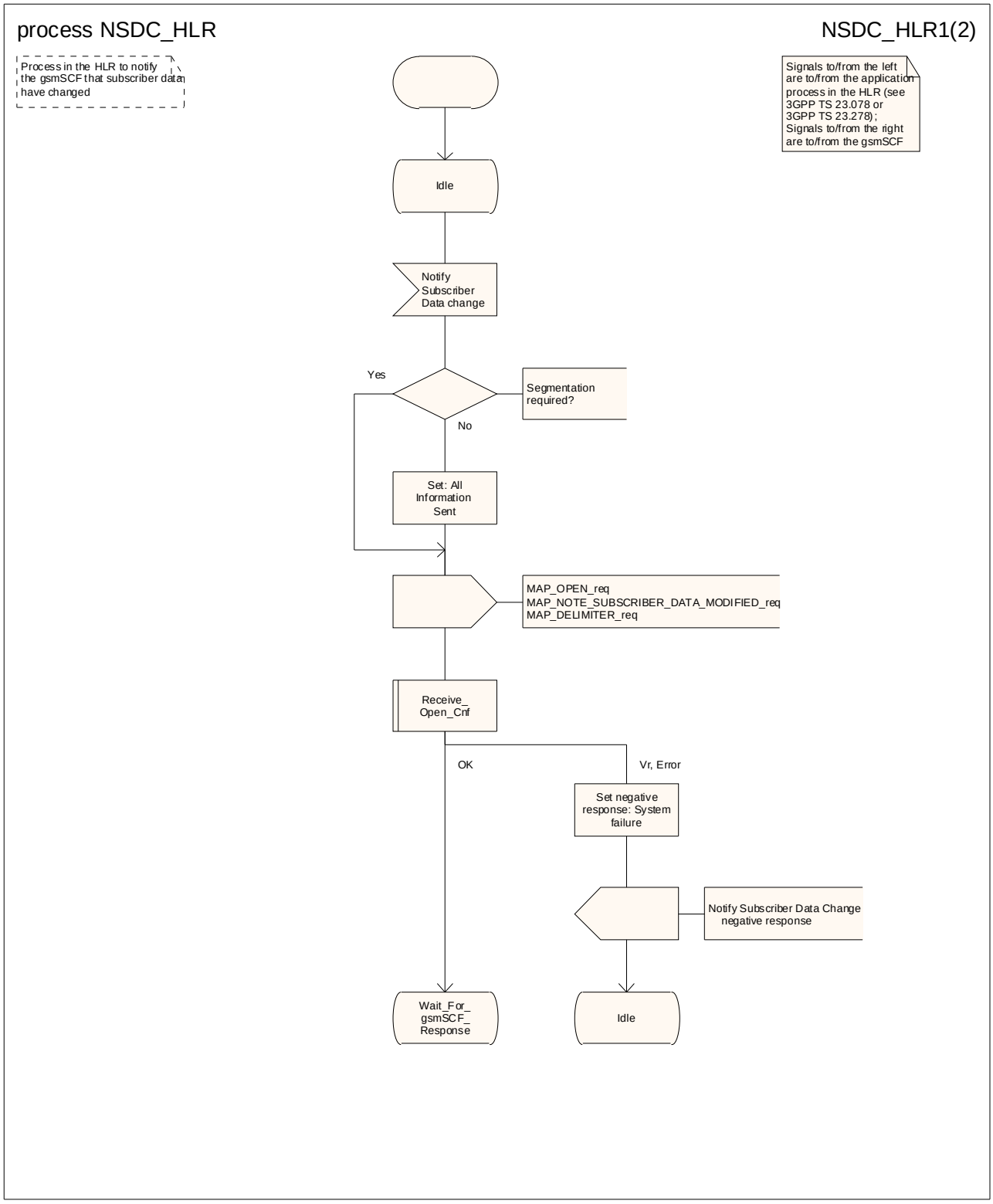


Figure 24A.4/2 (sheet 1 of 2): Process NSDC_HLR

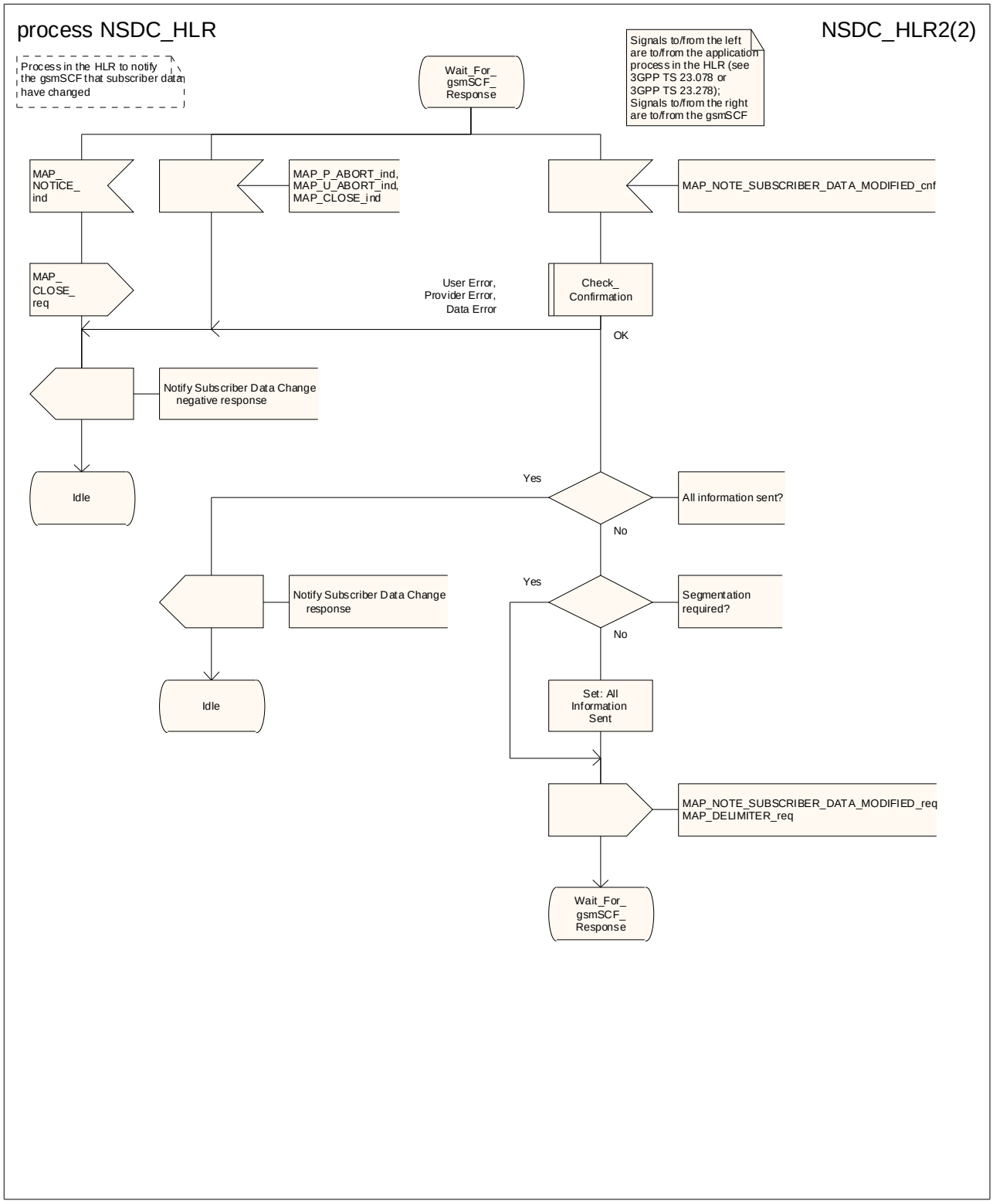


Figure 24A.4/2 (sheet 2 of 2): Process NSDC_HLR

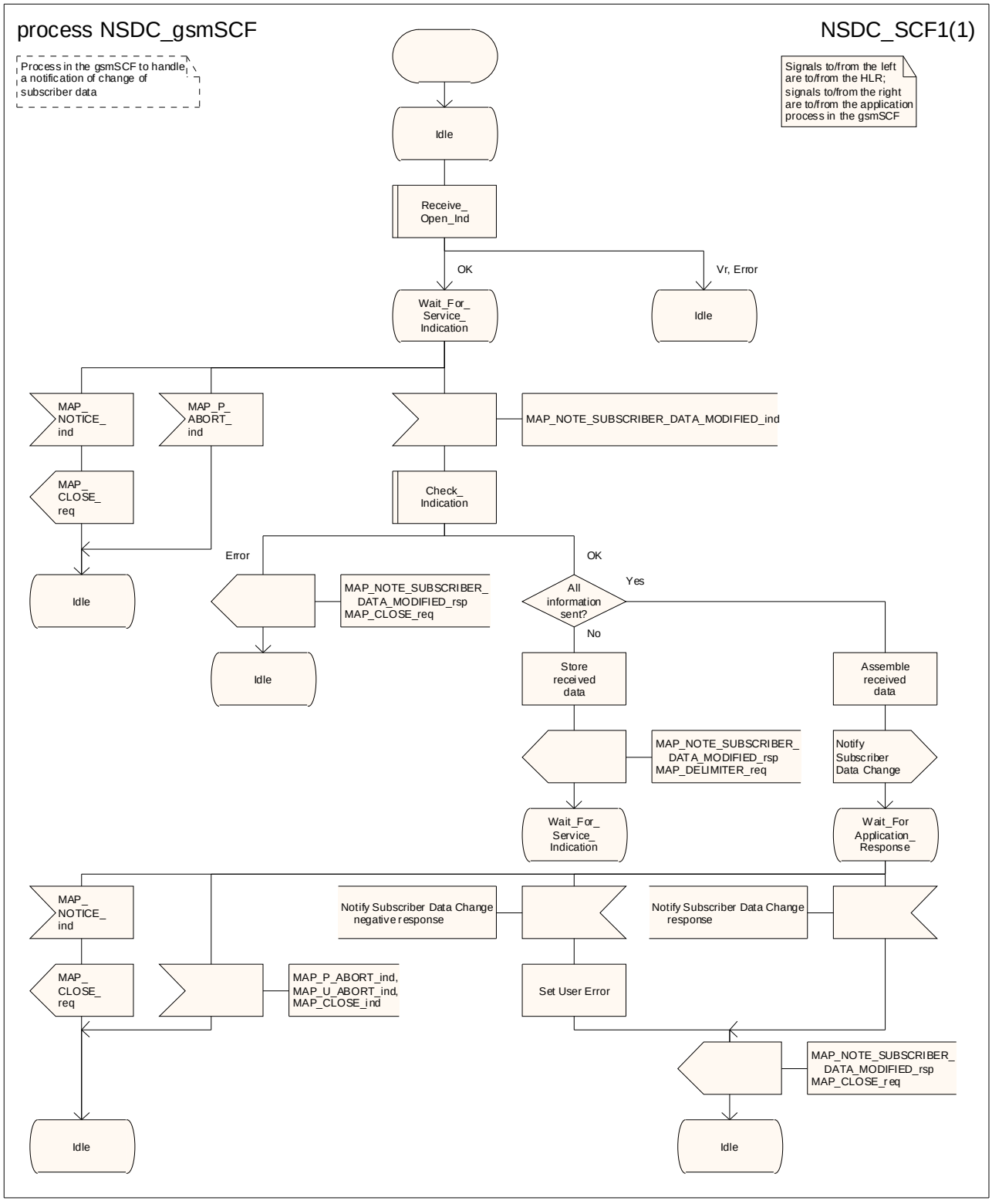
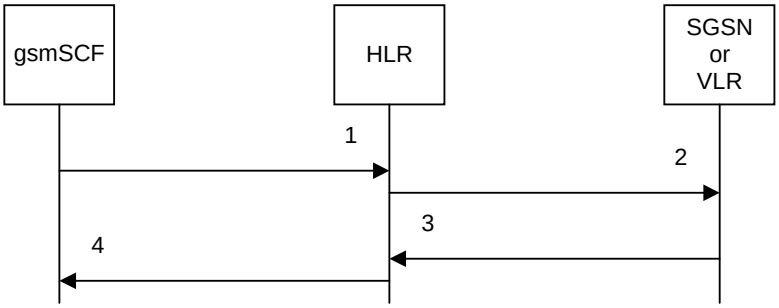


Figure 24A.4/3: Process NSDC_gsmSCF

24A.5 Any Time Interrogation procedure

24A.5.1 General

The message flows for successful retrieval of subscriber information related to an any time interrogation from the CAMEL server are shown in figure 24A.5/1 for interrogation directed to an HLR and figure 24A.5/2 for interrogation directed to a GMLC.



1) MAP_ANY_TIME_INTERROGATION_req/ind
2) MAP_PROVIDE_SUBSCRIBER_INFO_req/ind
3) MAP_PROVIDE_SUBSCRIBER_INFO_rsp/cnf
4) MAP_ANY_TIME_INTERROGATION_rsp/cnf

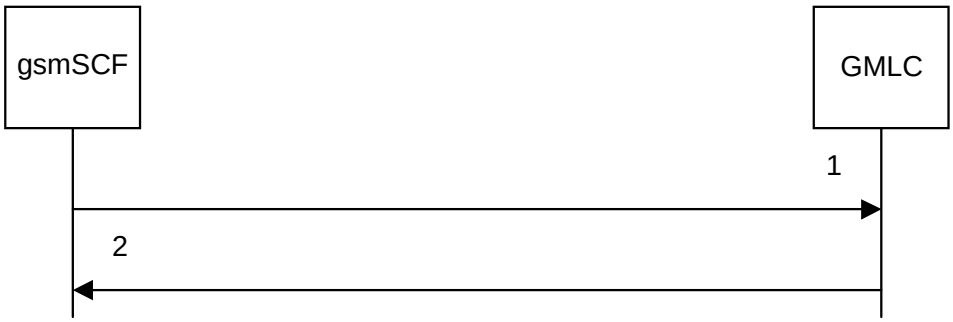
Figure 24A.5/1: Message flow for any time interrogation (gsmSCF to HLR)

The following MAP services are used to retrieve information about the status and/or location of a subscriber:

MAP_ANY_TIME_INTERROGATION see subclause 8.11.1;

MAP_PROVIDE_SUBSCRIBER_INFO see subclause 8.11.2.

The HLR sends the MAP_PROVIDE_SUBSCRIBER_INFO request to the SGSN or the VLR, according to the domain for which the gsmSCF requested the information.



1) MAP_ANY_TIME_INTERROGATION_req/ind
2) MAP_ANY_TIME_INTERROGATION_rsp/cnf

Figure 24A.5/2: Message flow for any time interrogation (gsmSCF to GMLC)

The following MAP service is used to retrieve location information from a GMLC:

MAP_ANY_TIME_INTERROGATION see subclause 8.11.1;

In addition, the GMLC may use MAP Services specific to Location Services.

24A.5.2 Procedures in the gsmSCF

The process in the gsmSCF to request information about the location and/or state of a subscriber from the HLR is shown in figure 24A.5/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

The process in the gsmSCF to request location information from the GMLC is shown in figure 24A.5/4. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24A.5.3 Procedure in the HLR

The MAP process in the HLR to provide subscriber information in response to an interrogation from the CAMEL server is shown in figure 24A.5/5. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24A.5.4 Procedure in the GMLC

The MAP process in the GMLC to provide location information in response to a request from the gsmSCF is shown in figure 24A.5/6. The MAP process invokes a macro not defined in this clause; the definition of this macro can be found as follows:

Receive_Open_Ind see subclause 25.1.1.

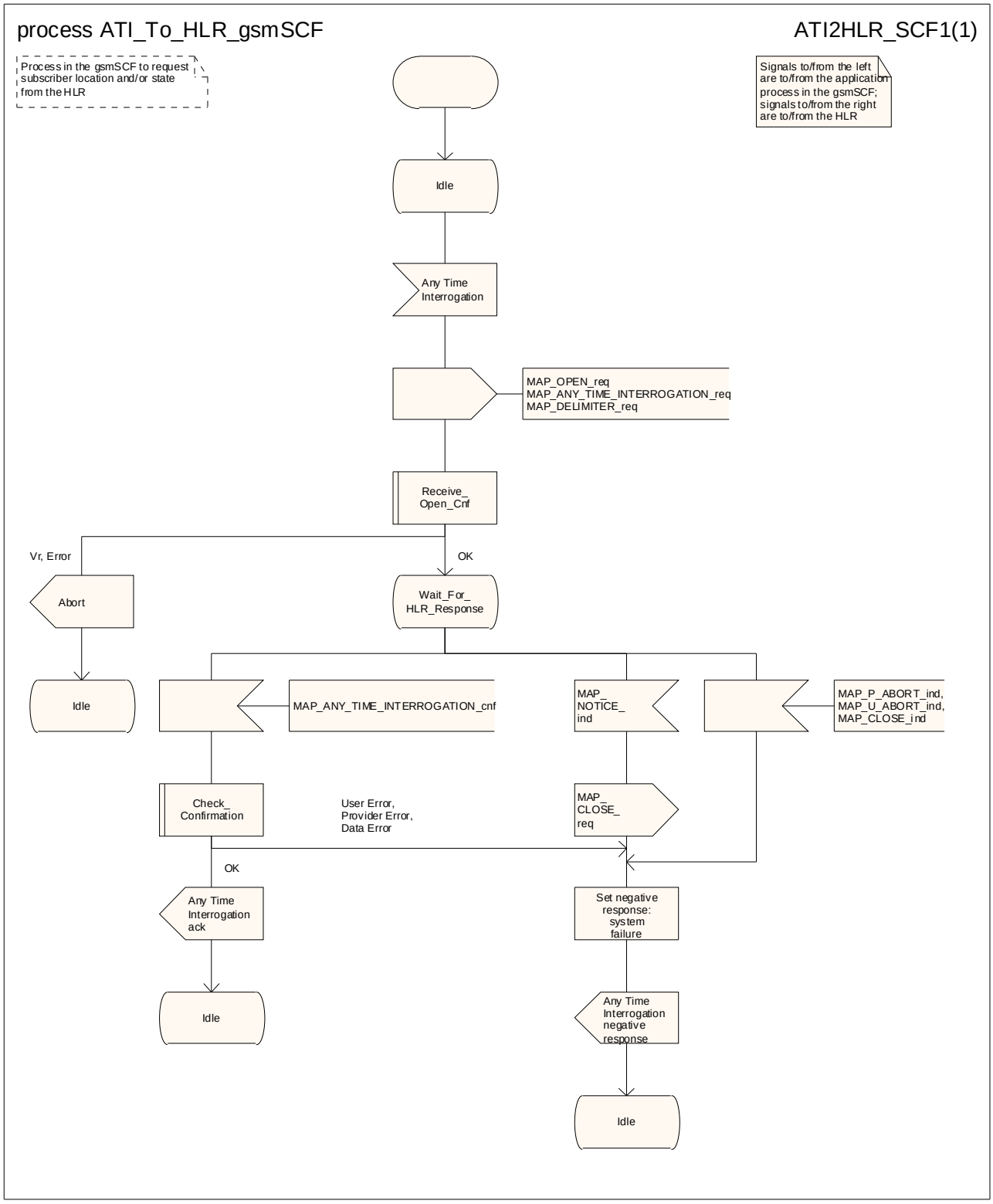


Figure 24A.5/3: Process ATI_To_HLR_gsmSCF

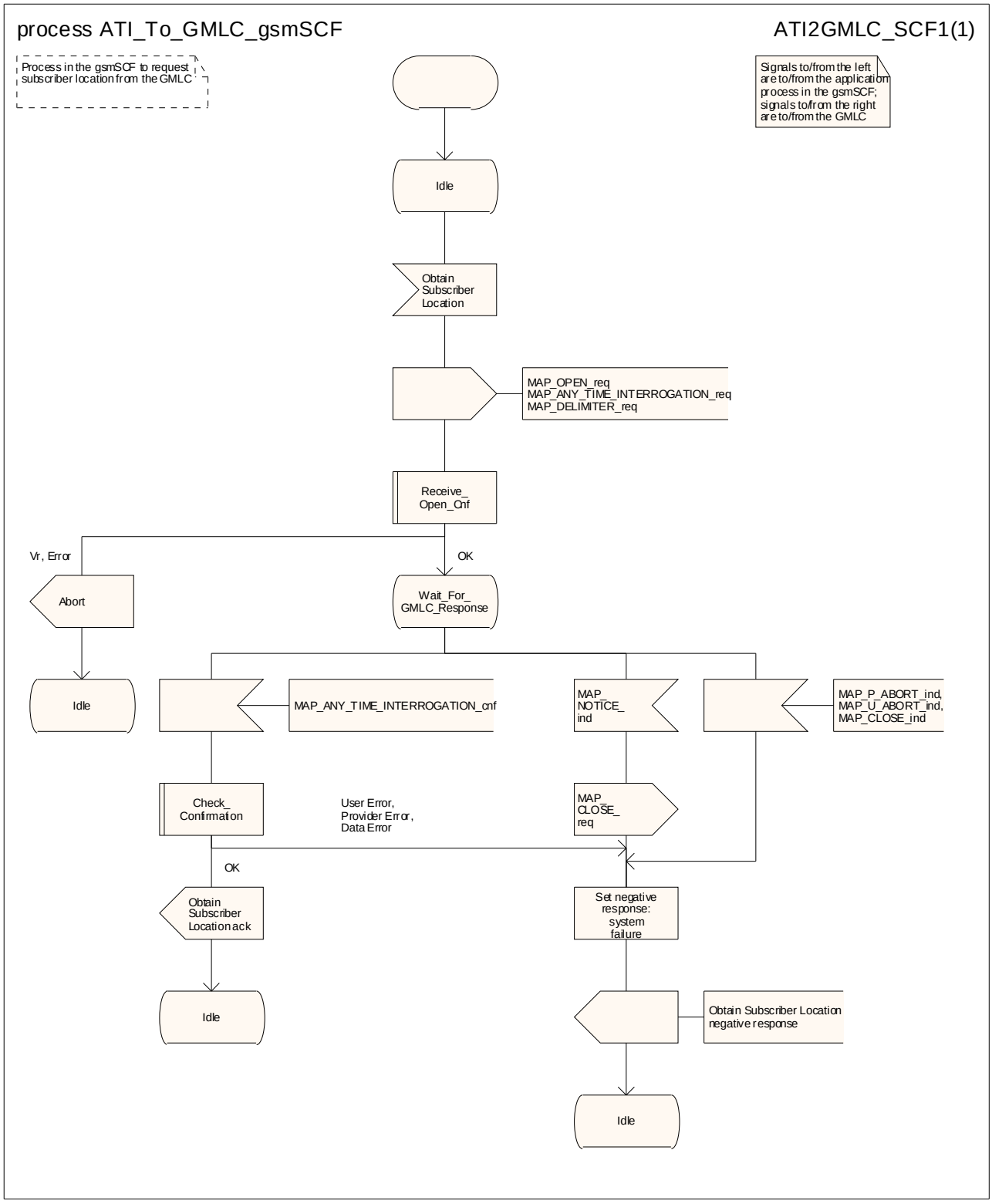


Figure 24A.5/4: Process ATI_To_GMLC_gsmSCF

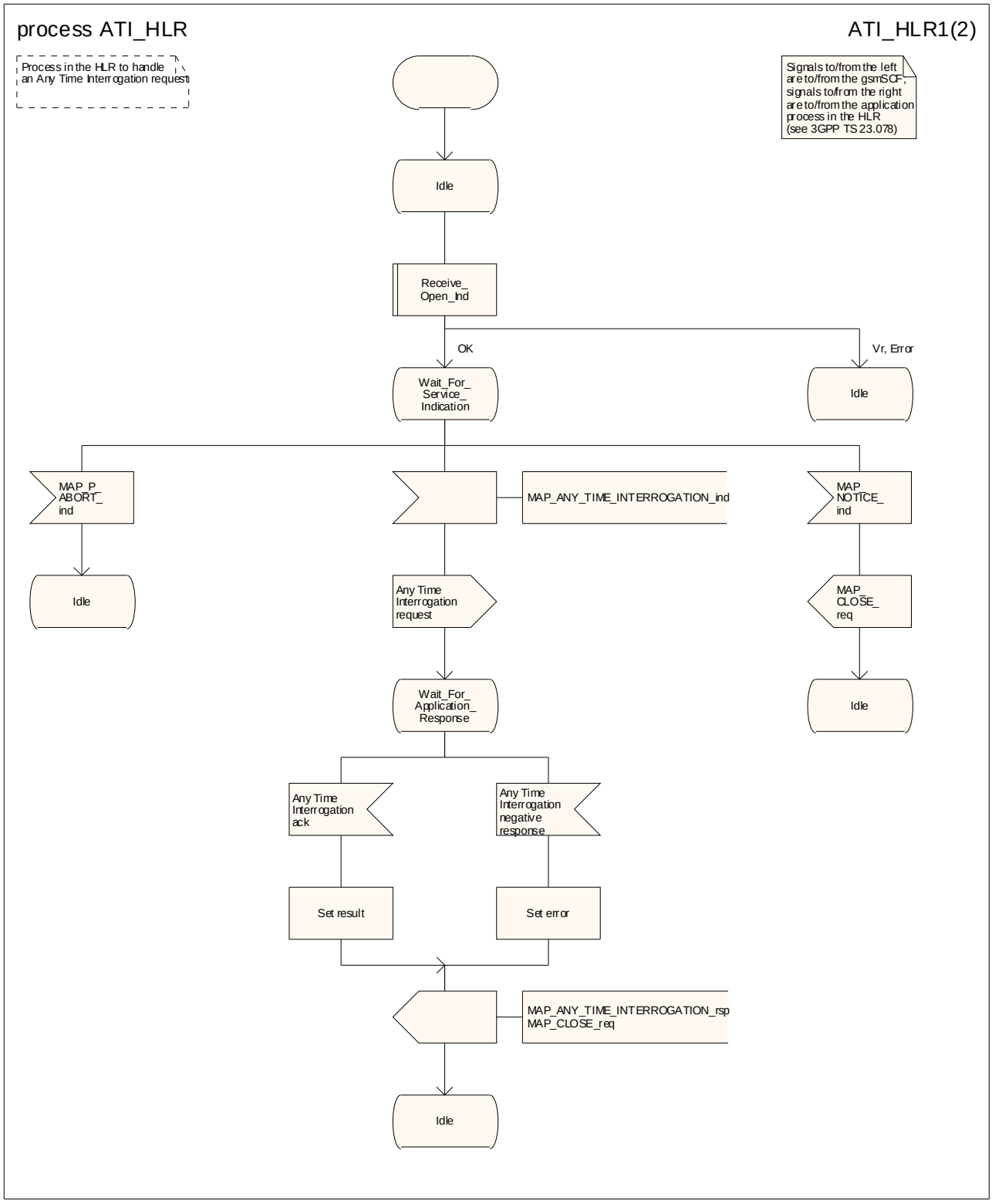


Figure 24A.5/5 (sheet 1 of 2): Process ATI_HLR

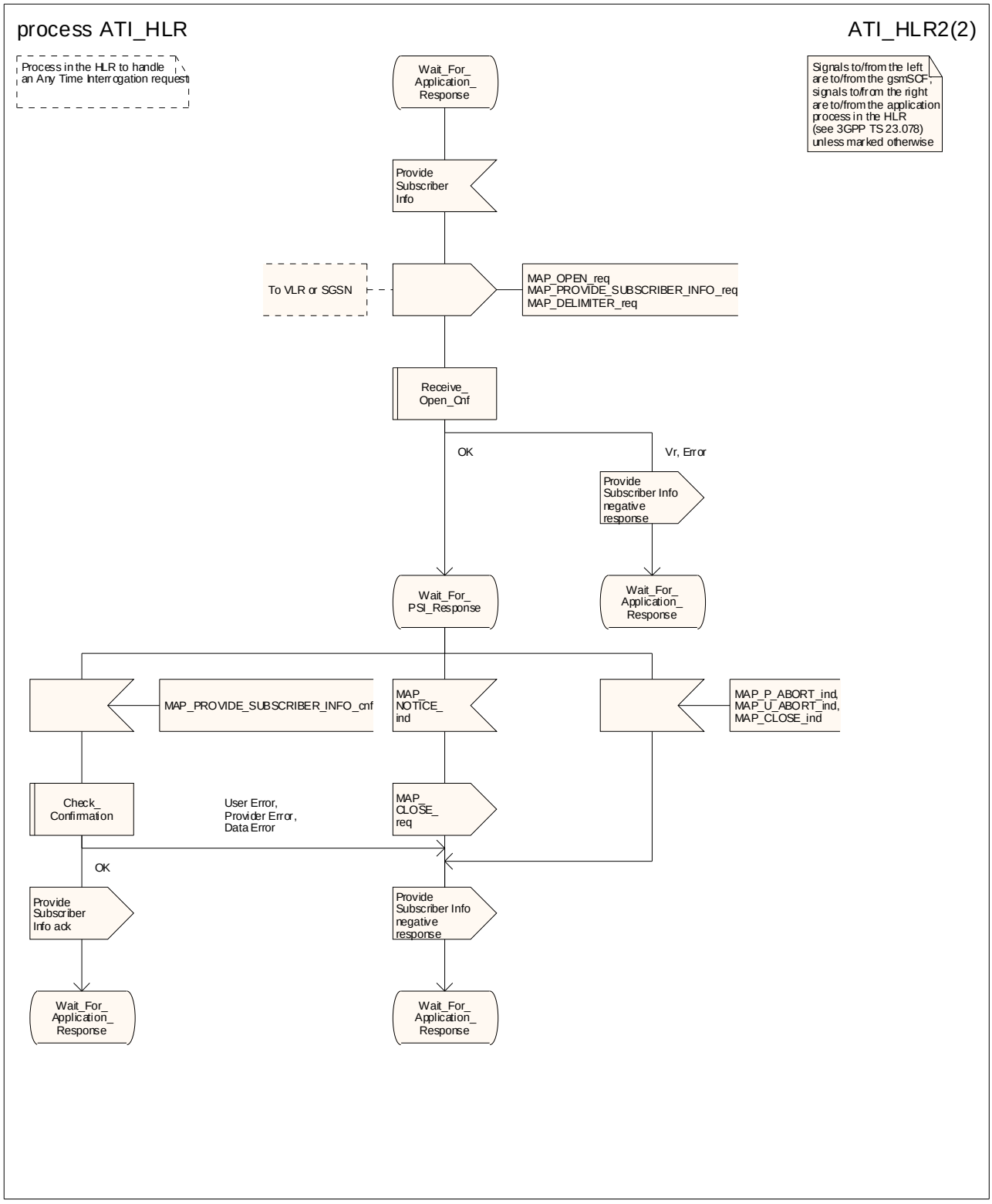


Figure 24A.5/5 (sheet 2 of 2): Process ATI_HLR

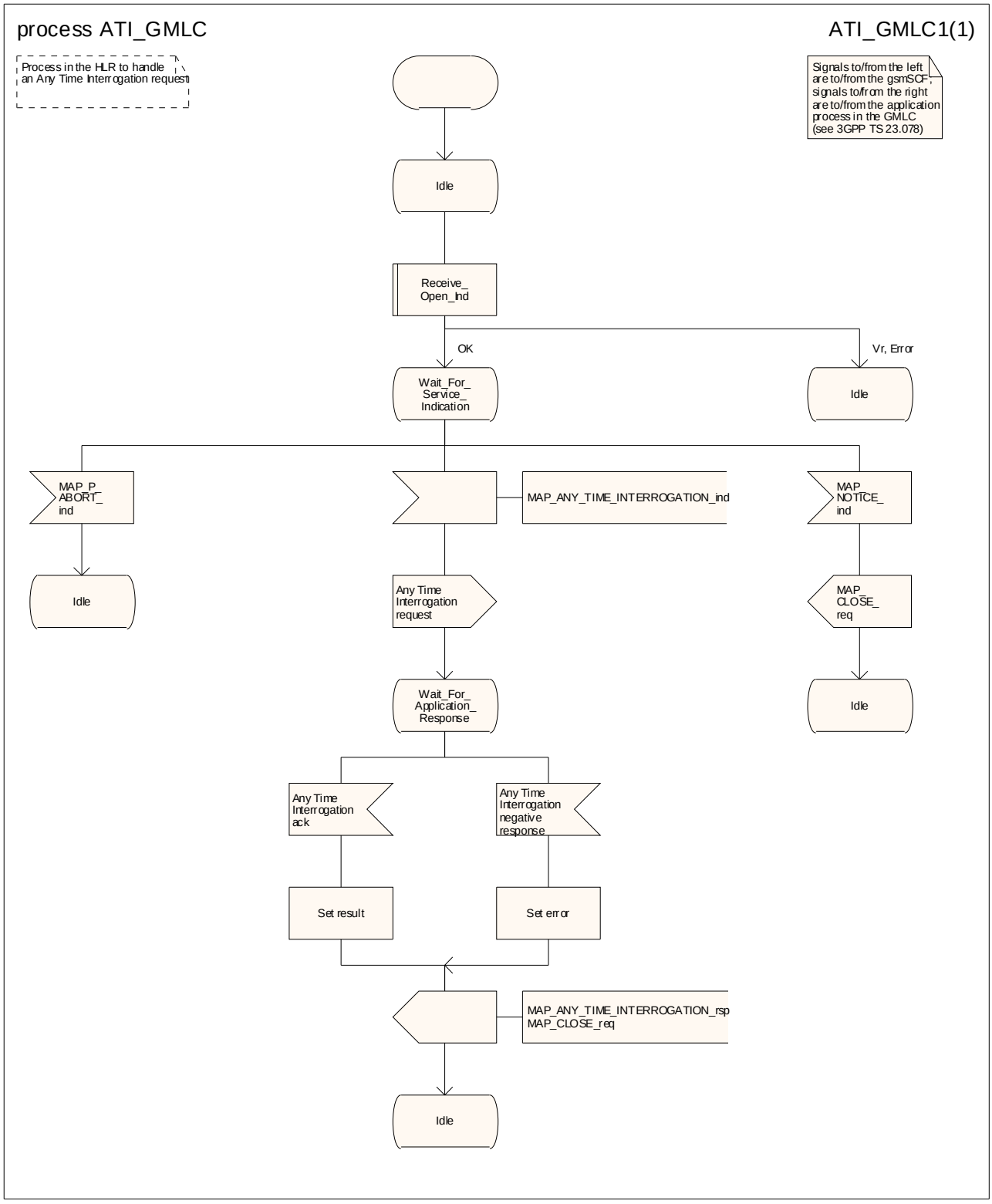


Figure 24A.5/6: Process ATI_GMLC

24B Location Services process description

24B.1 Routeing information retrieval procedure for LCS

24B.1.1 General

The message flow for successful retrieval of routeing information related to location services is shown in figure 24B.1/1.

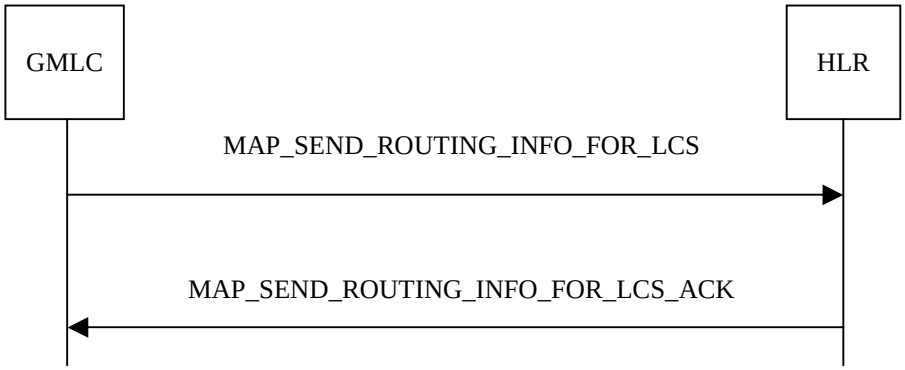


Figure 24B.1/1: Message flow for retrieval of routeing information for LCS

The following MAP service is used to retrieve routeing information:

MAP_SEND_ROUTING_INFO_FOR_LCS see subclause 13A.1.

24B.1.2 Process in the GMLC

The MAP process in the GMLC to request routeing information for LCS is shown in figure 24B.1/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24B.1.3 Process in the HLR

The MAP process in the HLR to handle a request for routeing information for LCS is shown in figure 24B.1/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

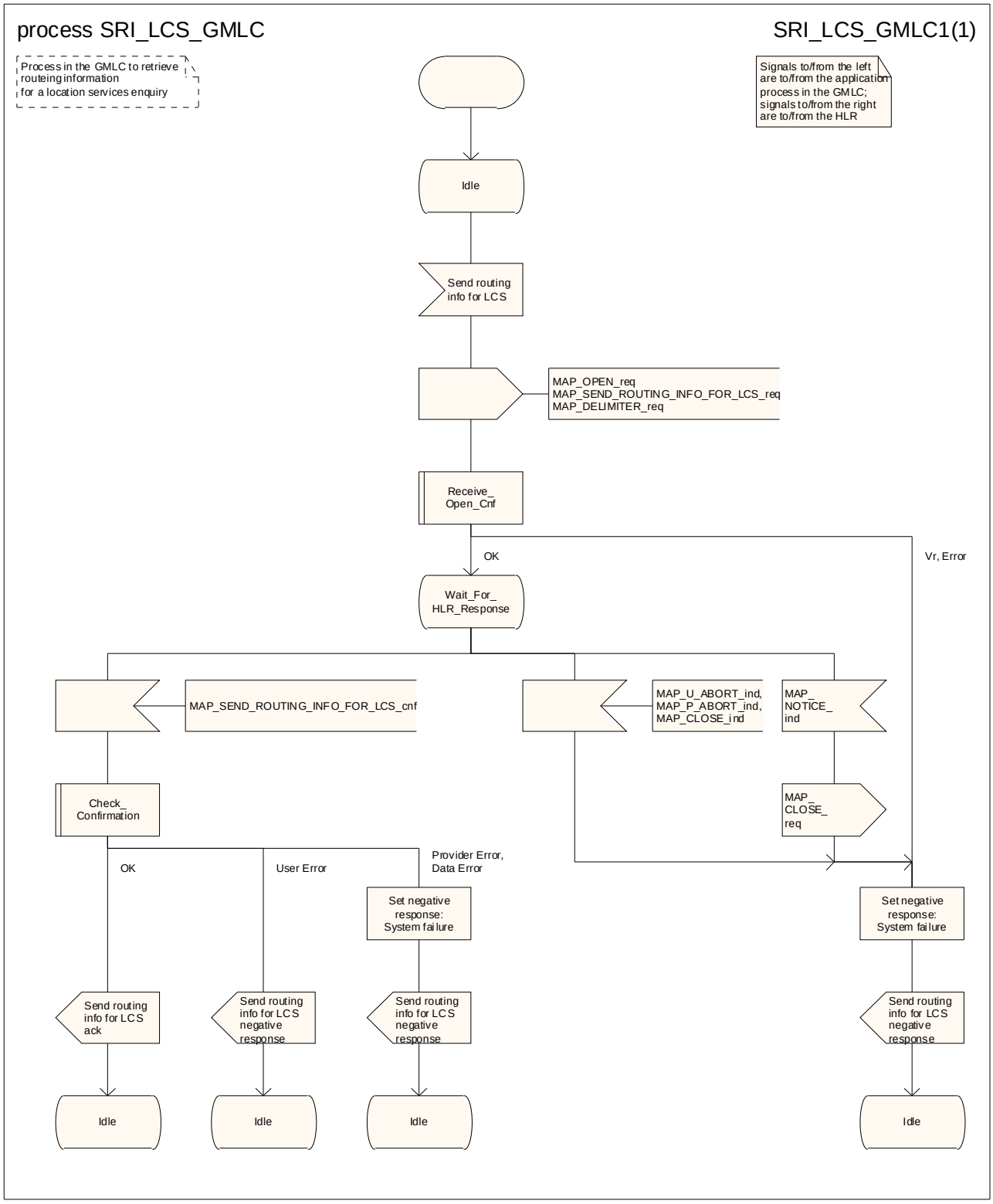


Figure 24B.1/2: Process SRI_LCS_GMLC

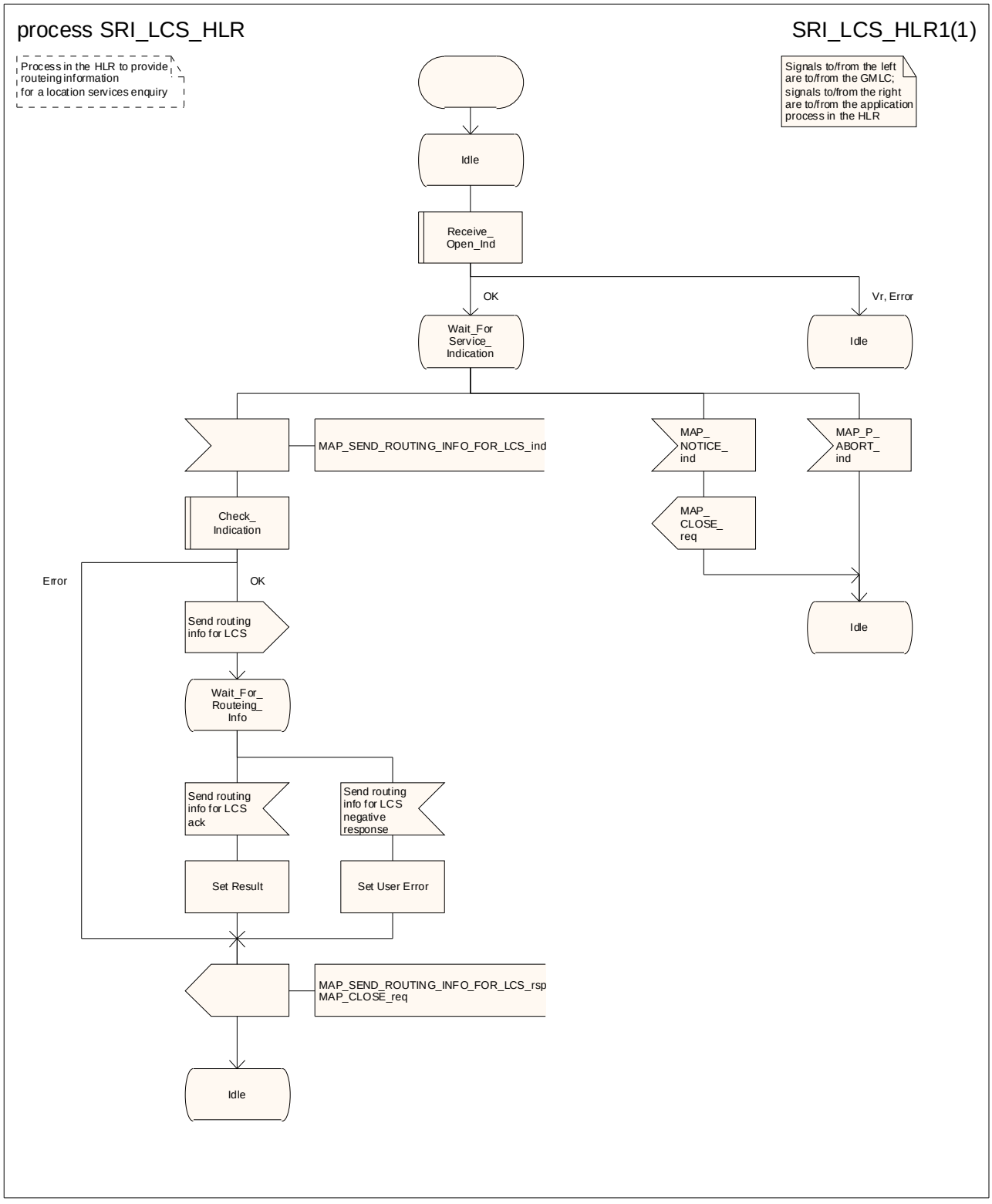


Figure 24B.1/3: Process SRI_LCS_HLR

24B.2 Provide Subscriber Location procedure

24B.2.1 General

The message flow for successful retrieval of the location information of a target MS related to location services is shown in figure 24B.1/1.

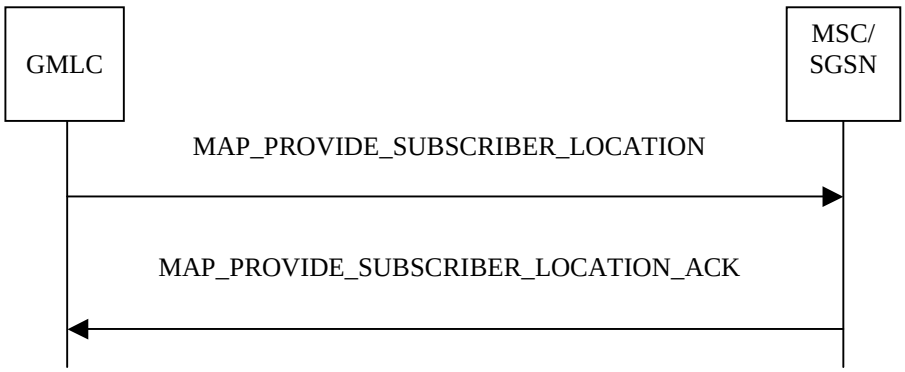


Figure 24B.2/1: Message flow for retrieval of location information

The following MAP service is used to retrieve location information:

MAP_PROVIDE_SUBSCRIBER_LOCATION see subclause 13A.2.

24B.2.2 Process in the GMLC

The MAP process in the GMLC to request location information from an MSC or an SGSN is shown in figure 24B.2/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24B.2.3 Process in the MSC

The MAP process in the MSC to handle a request for location information from a GMLC is shown in figure 24B.2/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

24B.2.4 Process in the SGSN

The MAP process in the SGSN to handle a request for location information from a GMLC is shown in figure 24B.2/4. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

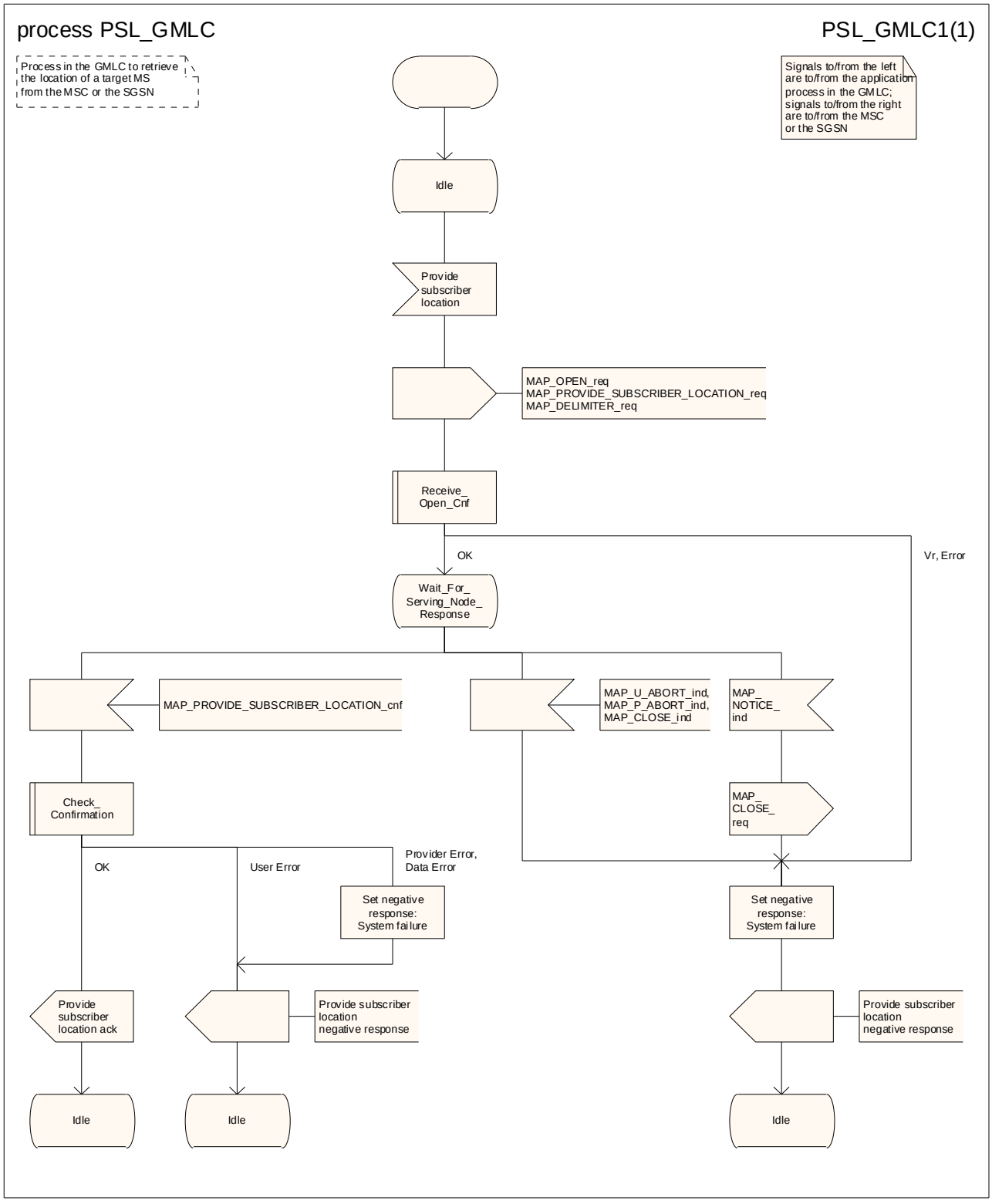


Figure 24B.2/2: Process PSL_GMLC

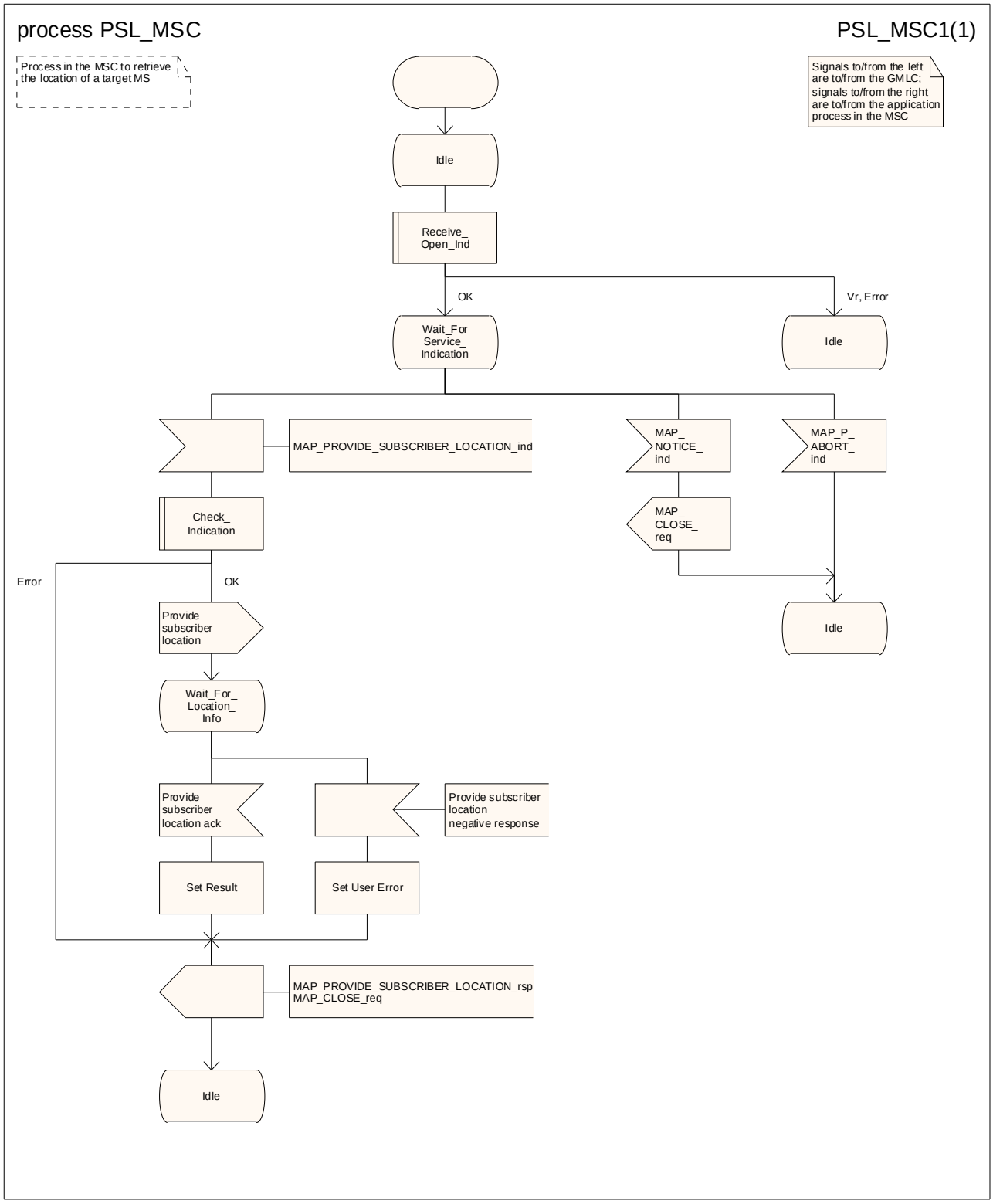


Figure 24B.2/3: Process PSL_MSC

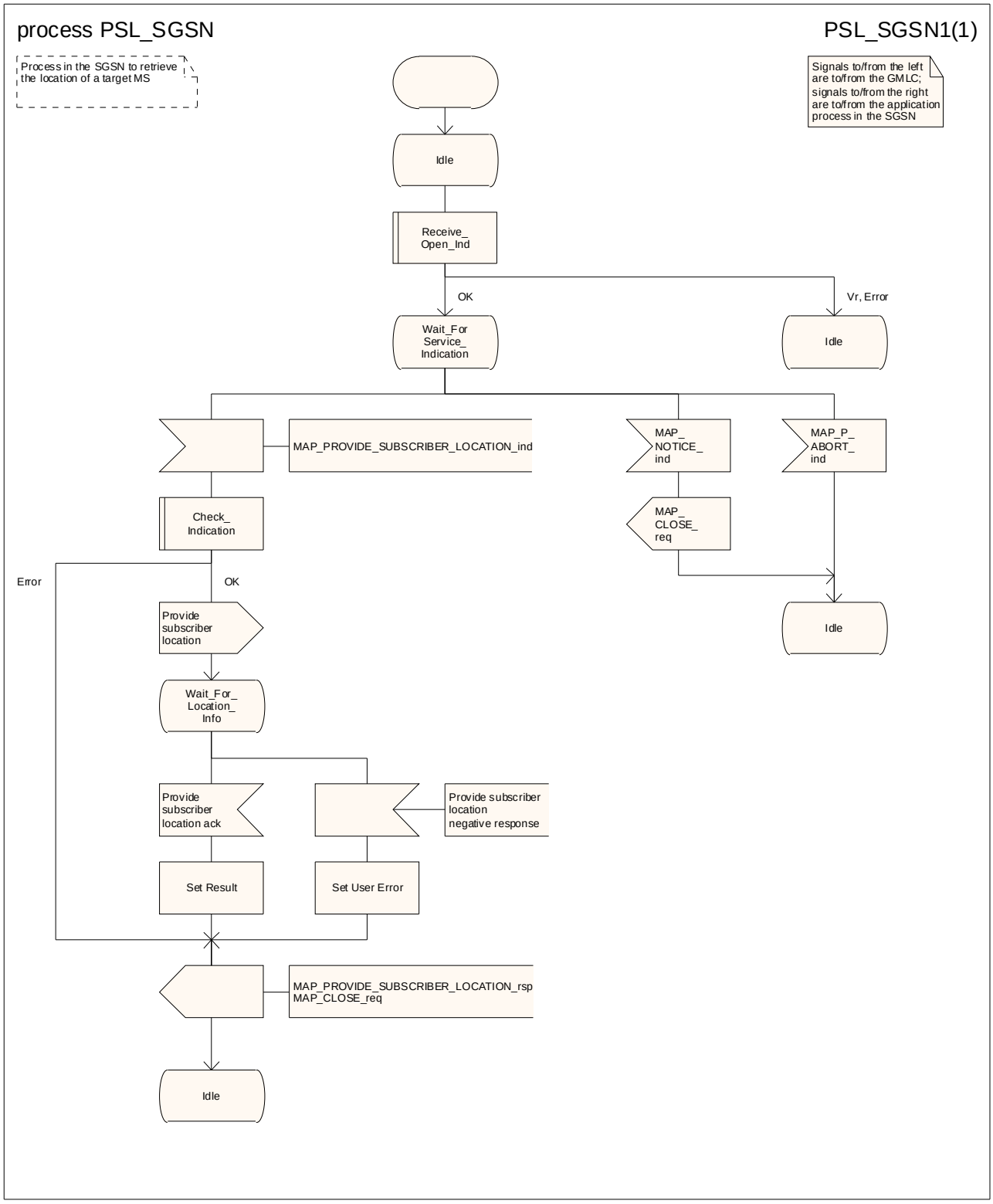


Figure 24B.2/4: Process PSL_SGSN

24B.3 Subscriber Location Report procedure

24B.3.1 General

The message flow for successful report of the location information of a target MS related to location services is shown in figure 24B.3/1.

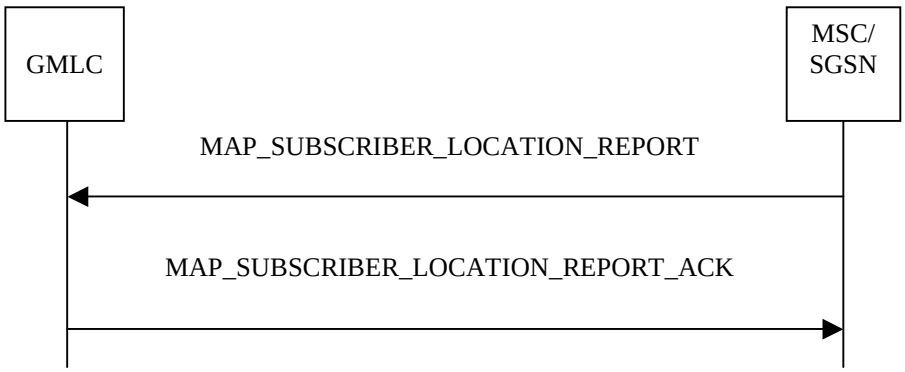


Figure 24B.3/1: Message flow for report of the location information

The following MAP services are used to report location information:

MAP_SUBSCRIBER_LOCATION_REPORT see subclause 13A.3.

24B.3.2 Process in the MSC

The MAP process in the MSC to send a subscriber location report to the GMLC is shown in figure 24B.3/2. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24B.3.3 Process in the SGSN

The MAP process in the SGSN to send a subscriber location report to the GMLC is shown in figure 24B.3/3. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Cnf see subclause 25.1.2;

Check_Confirmation see subclause 25.2.2.

24B.3.4 Process in the GMLC

The MAP process in the GMLC to handle a subscriber location report is shown in figure 24B.3/4. The MAP process invokes macros not defined in this clause; the definitions of these macros can be found as follows:

Receive_Open_Ind see subclause 25.1.1;

Check_Indication see subclause 25.2.1.

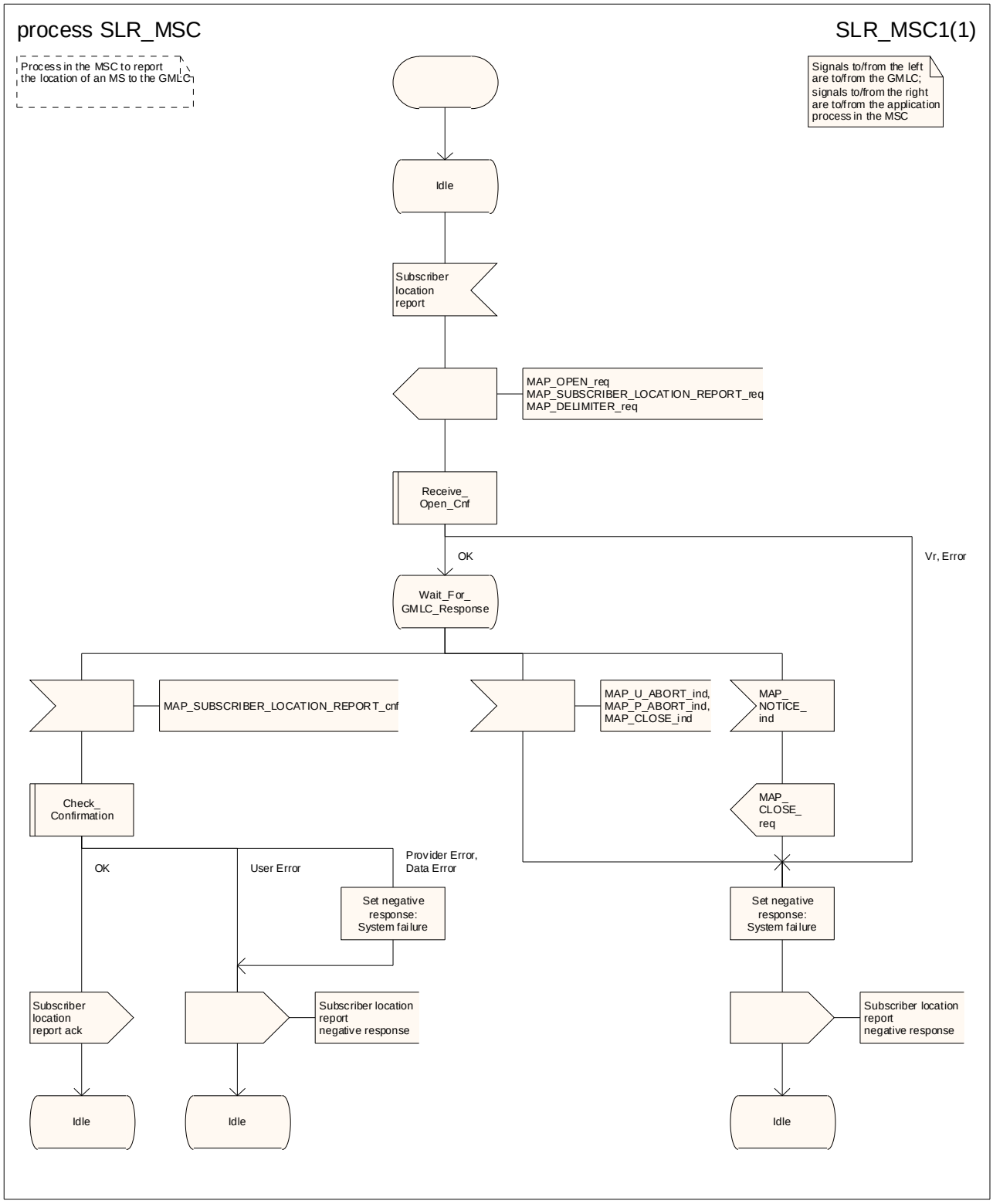


Figure 24B.3/2: Process SLR_MSC

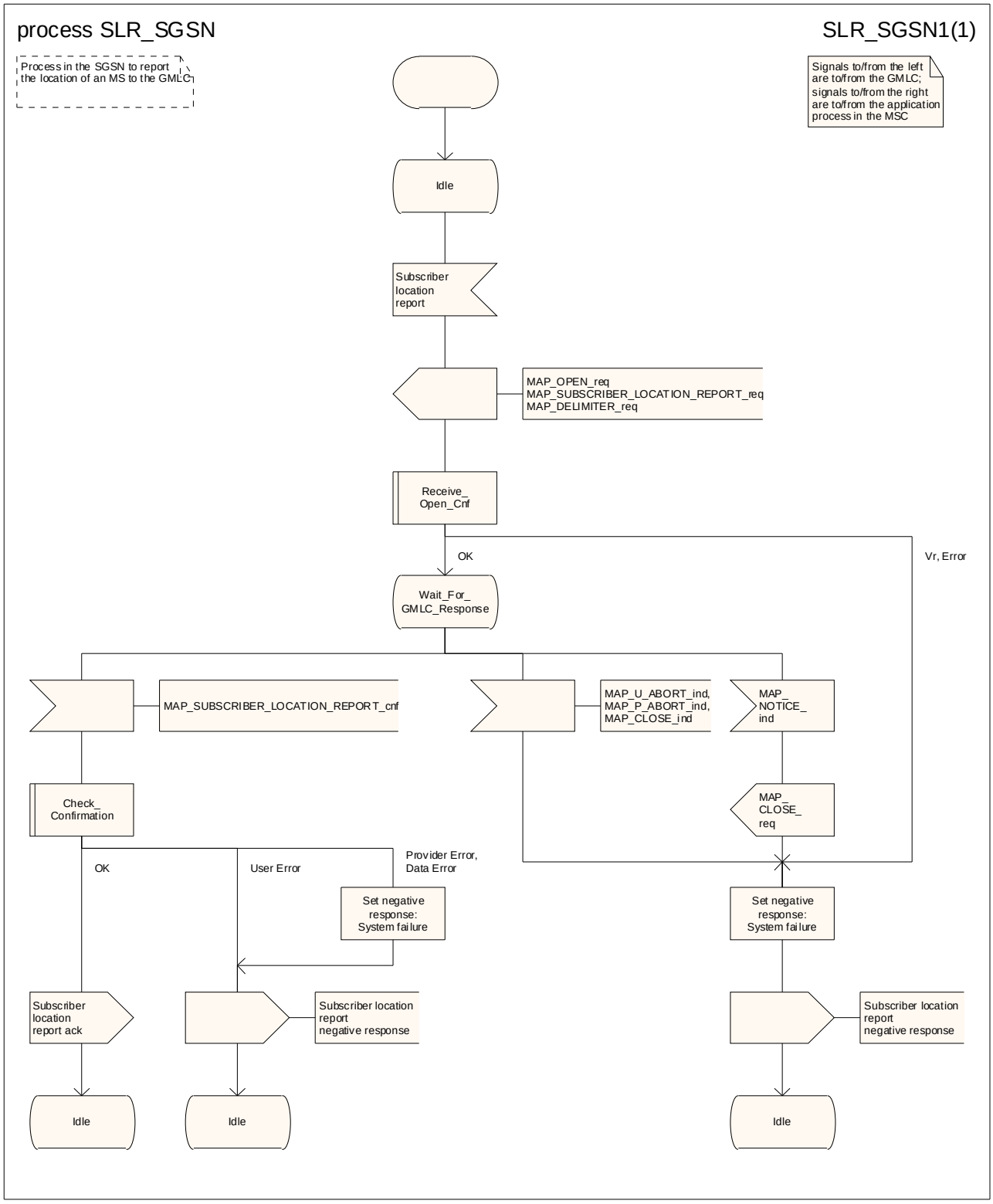


Figure 24B.3/3: Process SLR_SGSN

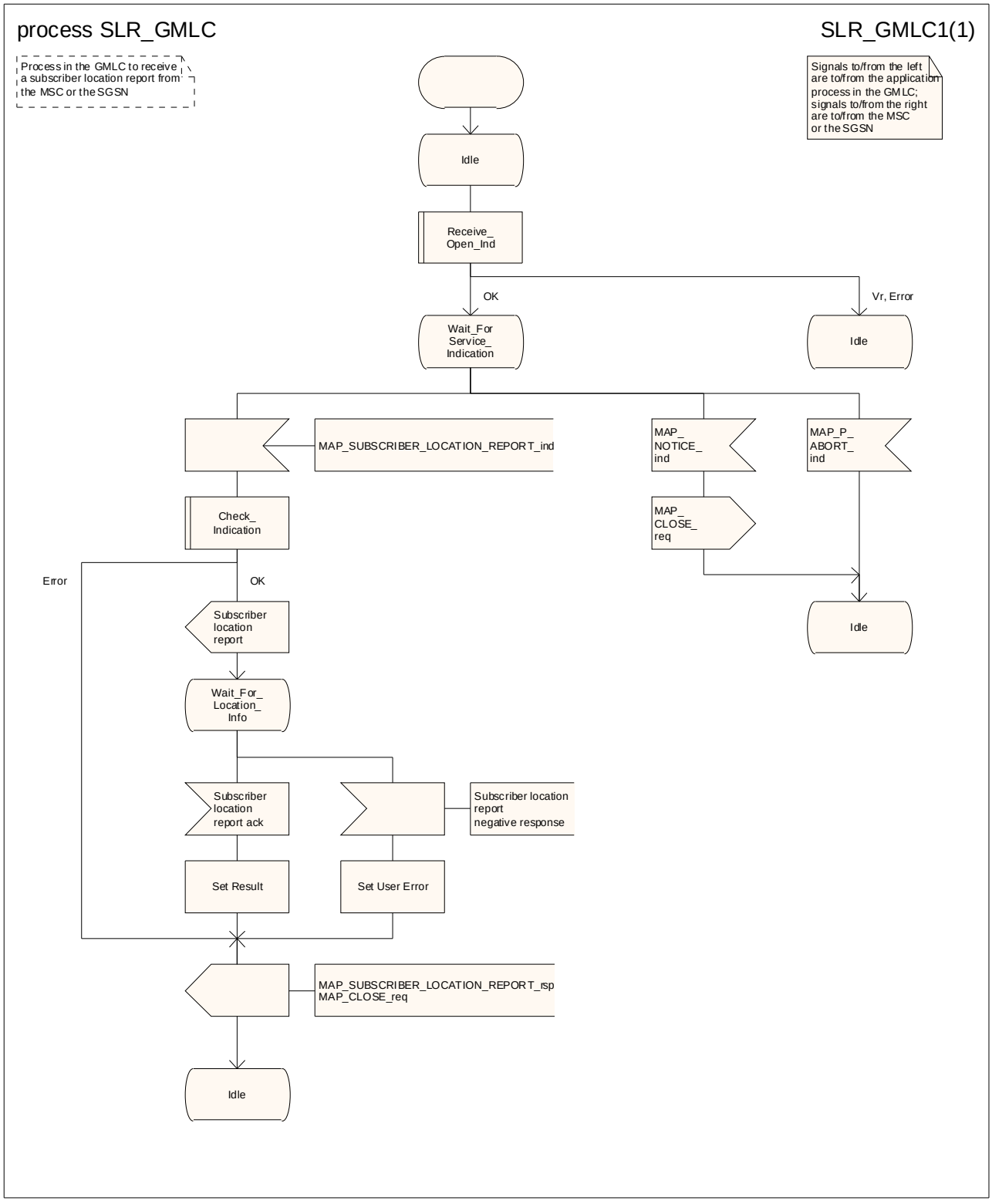


Figure 24B.3/4: Process SLR_GMLC

25 General macro description

25.1 MAP_OPEN handling macros

25.1.1 Macro Receive_Open_Ind

This macro is used by a MAP service-user procedure when a peer entity requests opening of a dialogue.

25.1.2 Macro Receive_Open_Cnf

This macro is used by a user procedure after it has requested opening of a dialogue towards a peer entity.

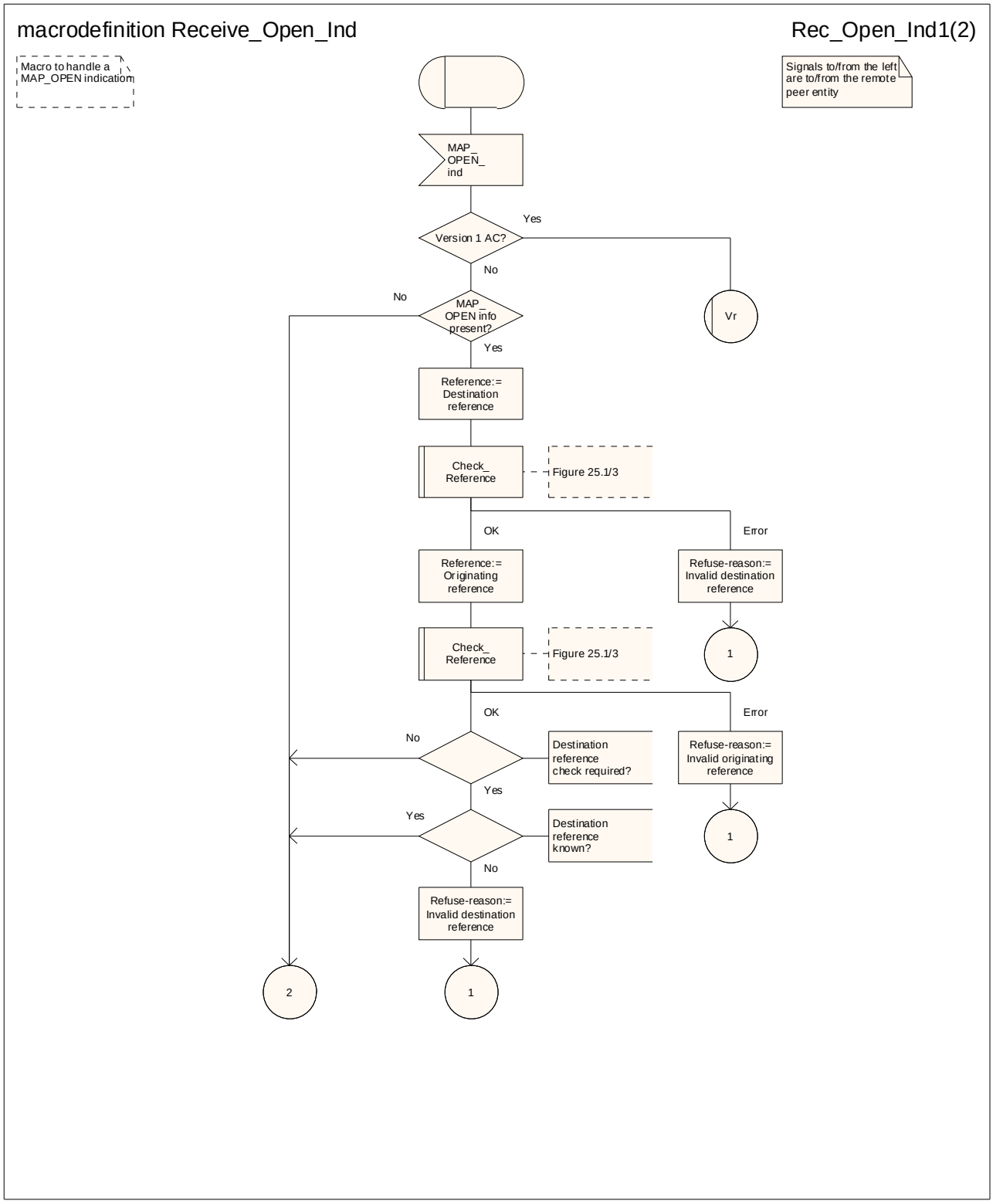


Figure 25.1/1 (sheet 1 of 2): Macro Receive_Open_Ind

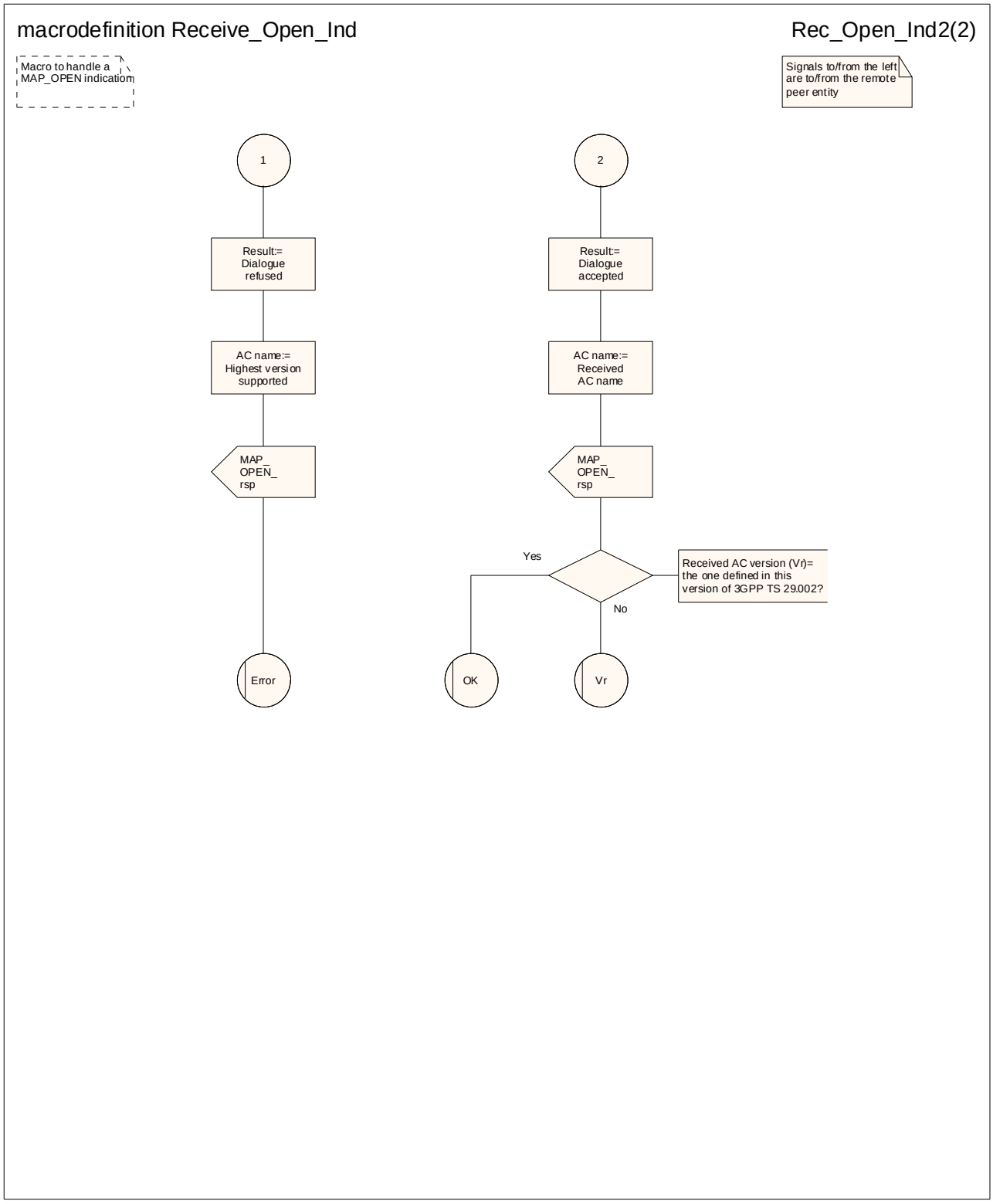


Figure 25.1/1 (sheet 2 of 2): Macro Receive_Open_Ind

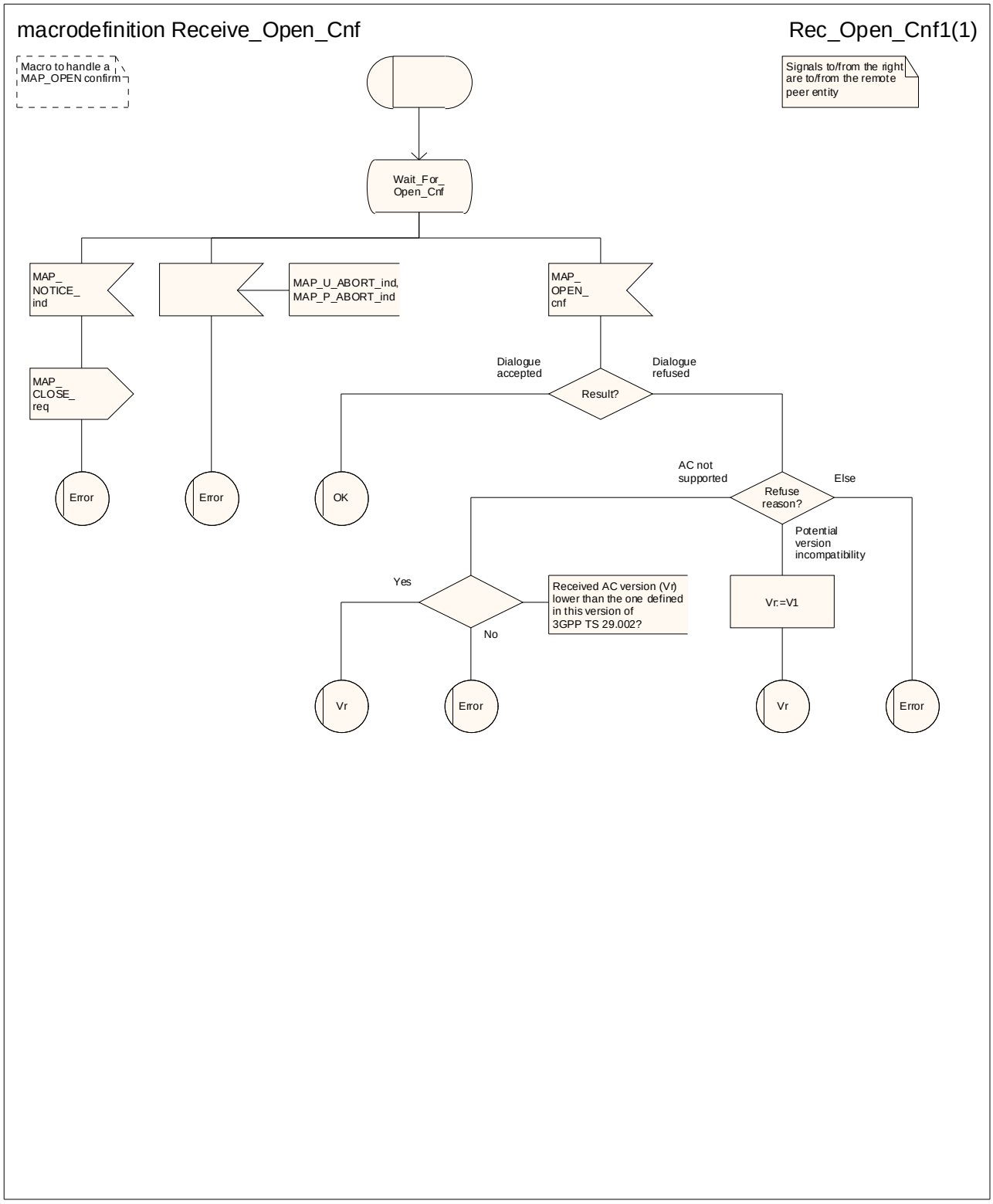


Figure 25.1/2: Macro Receive_Open_Cnf

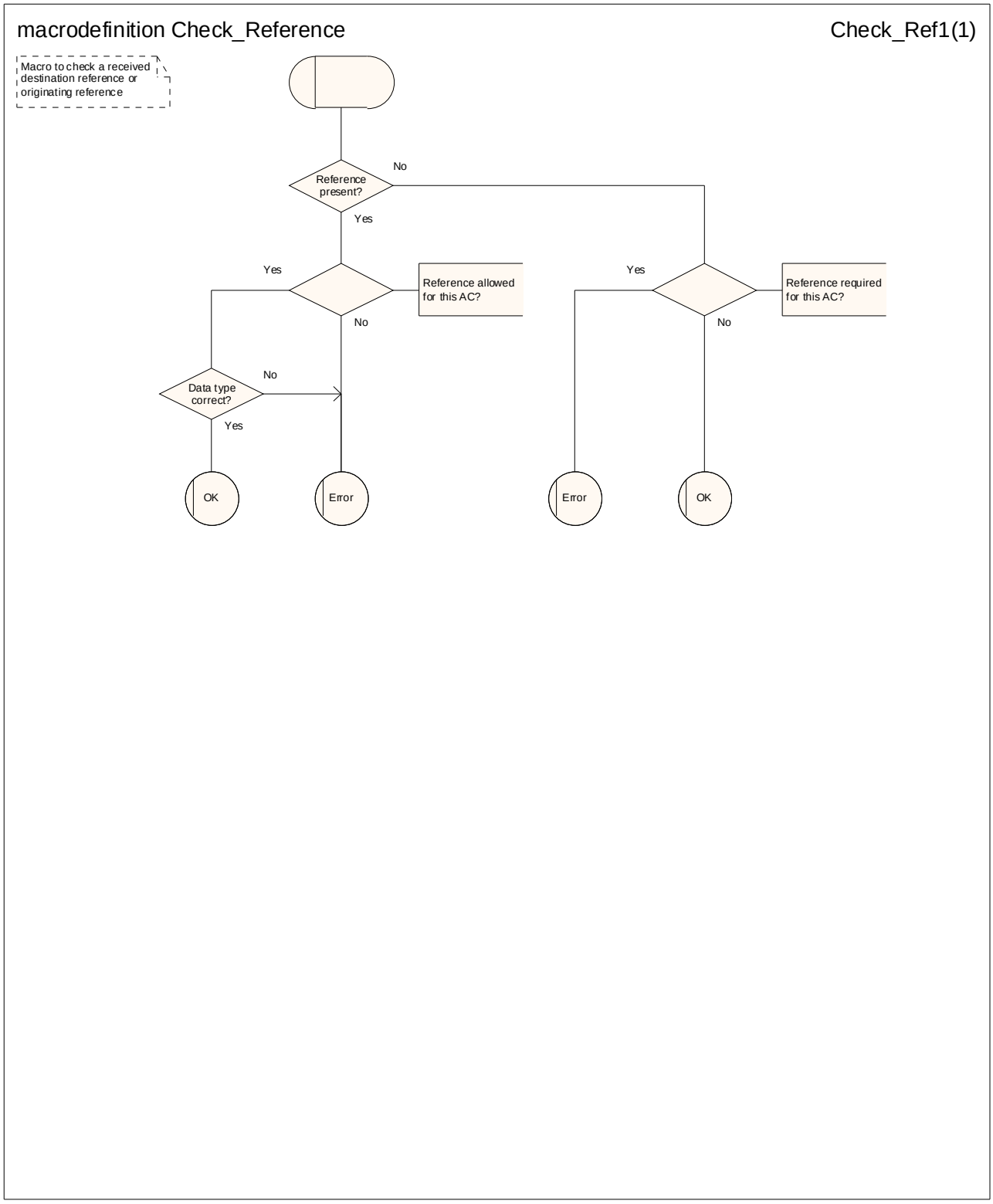


Figure 25.1/3: Macro Check_Reference

25.2 Macros to check the content of indication and confirmation primitives

25.2.1 Macro Check_Indication

This macro checks that an indication includes all the parameters required by the application, no more and no less, and that the parameters are all within the correct range. It does not handle syntax checking; that is part of the function of the MAP protocol machine.

25.2.2 Macro Check_Confirmation

This macro checks whether a confirmation contains an error or a result, and if it contains a result whether the result is correctly formed.

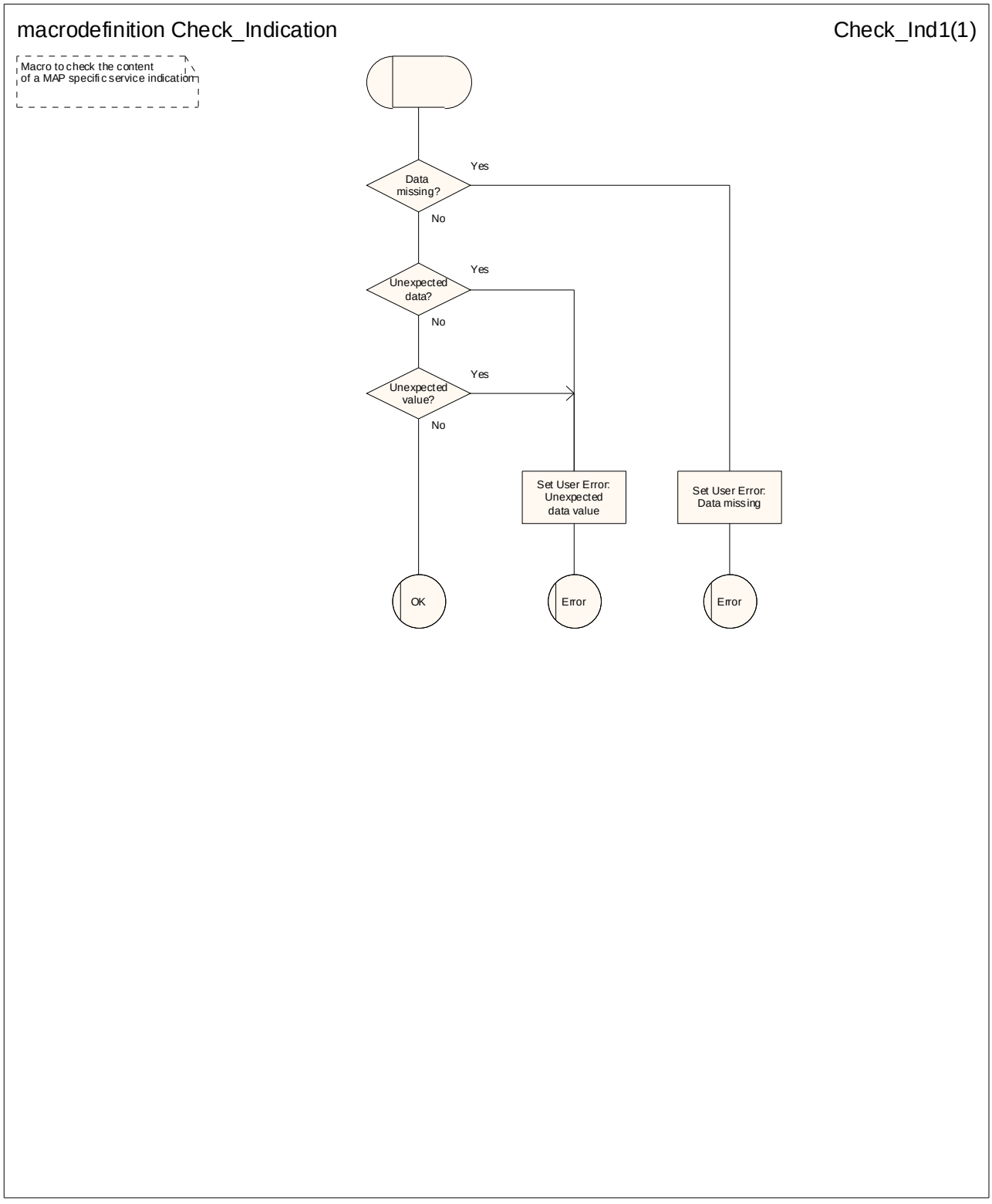


Figure 25.2/1: Macro Check_Indication

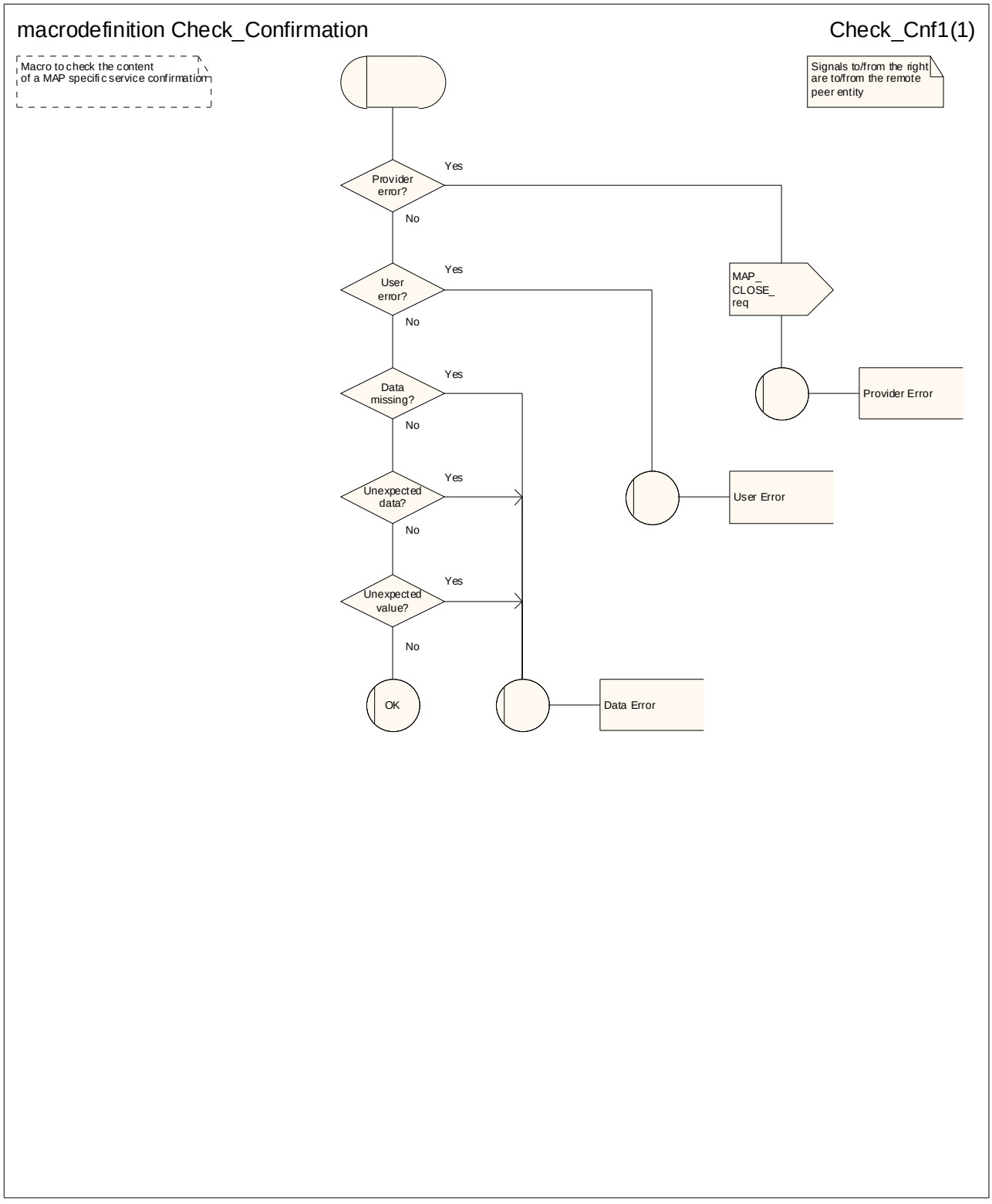


Figure 25.2/2: Macro Check_Confirmation

25.3 The page and search macros

25.3.1 Macro PAGE_MSC

This macro is called if an unstructured SS notification, a network-initiated unstructured SS request or a mobile terminating short message is to be delivered to the MS and the current location area identity of the MS is known in the VLR.

If an MM-connection over the radio link already exists for the given IMSI, the MSC sets the access connection status according to the characteristics of the existing connection (i.e. RR-connection established, ciphering mode on/off, MM-connection existing and authenticated or not).

If the MSC pages the MS and the VLR provided the TMSI, the MSC uses it to identify the MS at the radio interface; otherwise the MSC uses the IMSI. The MSC also uses the IMSI to determine the page group (see 3GPP TS 24.008 [35]).

If the MS responds with a channel request containing an establishment cause which is not "answer to paging" the MSC sends a MAP_PAGE response primitive with user error Busy Subscriber. This gives priority to the mobile originating request. Alternatively, as an implementation option, the MSC may treat this as a response to paging, which gives priority to the mobile terminating request.

If the paging is for MT SMS delivery and the VLR aborts the transaction before the MSC receives a response from the MS, the MSC aborts the transaction with the SMS-GMSC.

25.3.2 Macro Search_For_MS_MSC

This macro is called if an unstructured SS notification, a network-initiated unstructured SS request or a mobile terminating short message is to be delivered to the MS and the current location area identity of the MS is not known in VLR.

If an MM-connection over the radio link already exists for the given IMSI, the MSC returns a MAP_SEARCH_FOR_MS response containing the IMSI and current location area identification of the called MS to the VLR and sets the access connection status according to the characteristics of the existing connection (i.e. RR-connection established, ciphering mode on/off, MM-connection existing and authenticated or not).

If the MSC pages the MS, the MSC uses the IMSI to identify the subscriber and the page group (see 3GPP TS 24.008 [35]).

If the MS responds with a channel request containing an establishment cause which is not "answer to paging" the MSC sends a MAP_SEARCH_FOR_MS response with user error Busy Subscriber. This gives priority to the mobile originating request. Alternatively, as an implementation option, the MSC may treat this as a response to paging, which gives priority to the mobile terminating request.

If the paging is for MT SMS delivery and the VLR aborts the transaction before the MSC receives a response from the MS, the MSC aborts the transaction with the SMS-GMSC.

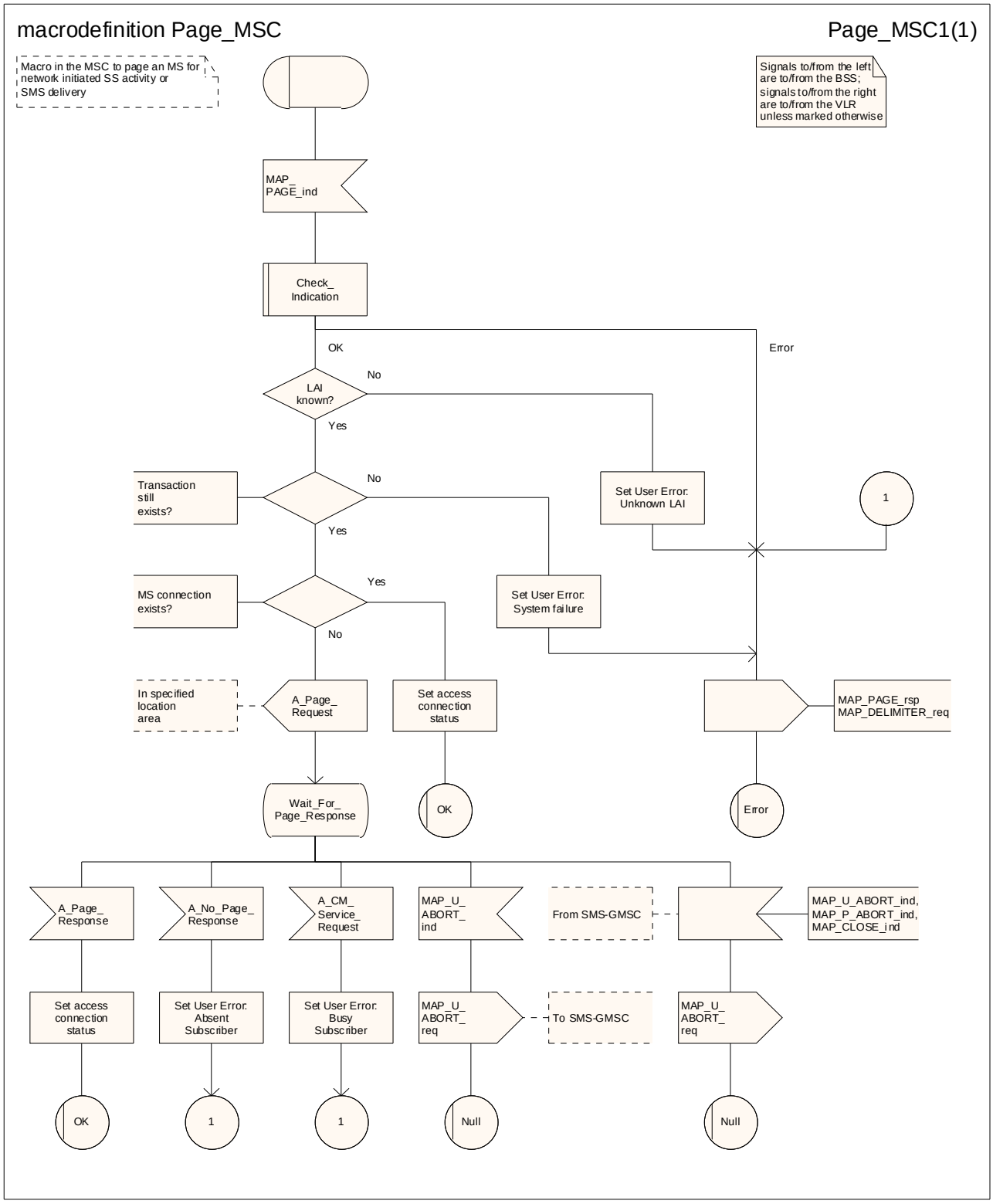


Figure 25.3/1: Macro Page_MSC

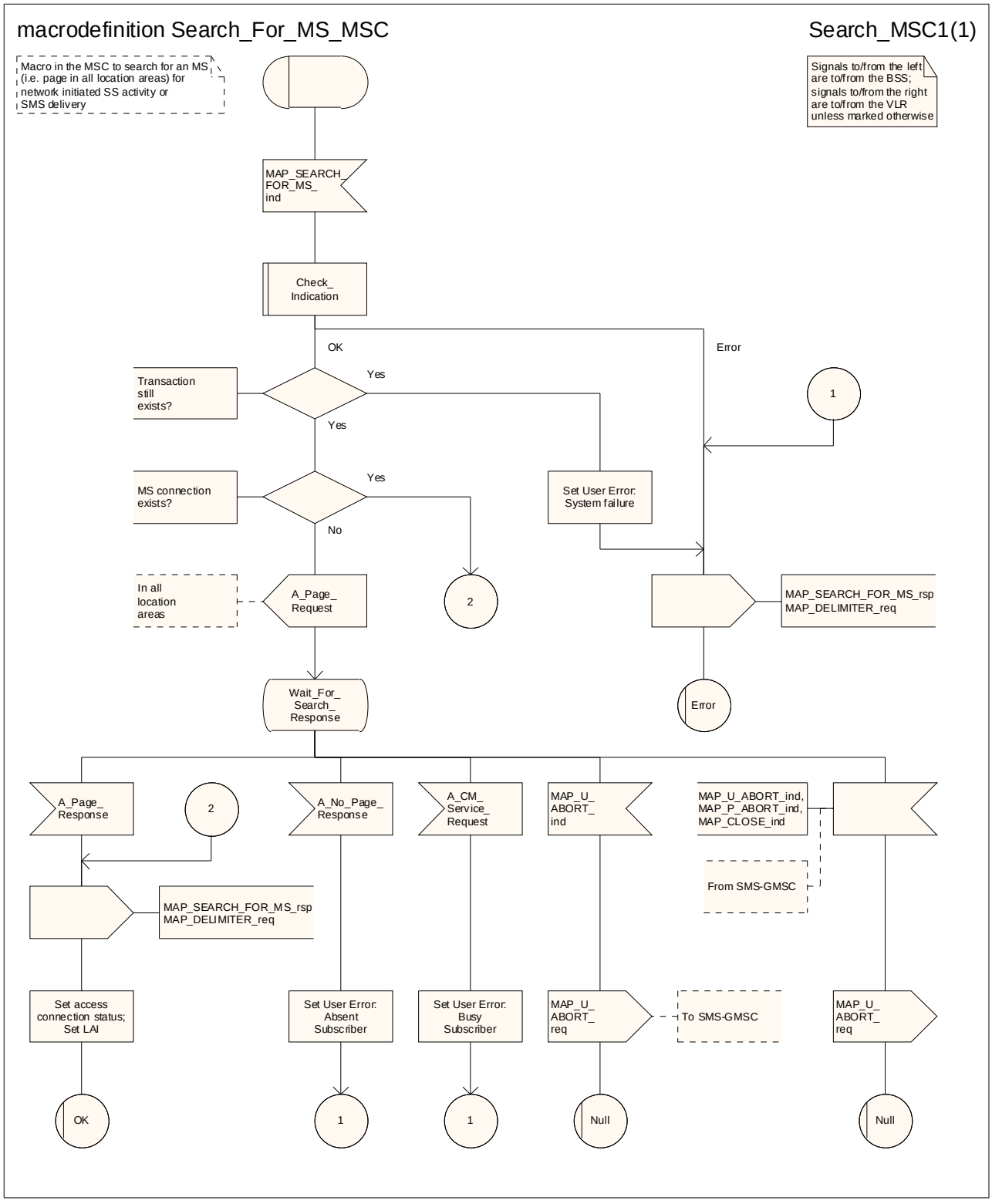


Figure 25.3/2: Macro Search_for_MS_MSC

25.4 Macros for handling an Access Request

These macros are invoked when a MS accesses the network, e.g. to submit an MO short message or when responding to paging. The macros handle identification and authentication of the mobile subscriber as well as invocation of security related features (see 3GPP TS 42.009 [6]).

25.4.1 Macro Process_Access_Request_MSC

Sheet 1: The MAP_PROCESS_ACCESS_REQUEST request includes the following parameters:

- the received subscriber identification (IMSI, TMSI);
- the CM service type, indicating the type of request;
- the status of the access connection, i.e. whether a connection to this MS already exists and if so, whether it is already authenticated and ciphered;
- the current location area id of the MS; and
- the CKSN received from the MS.

Sheet 2, sheet 3: If the MSC receives an A_SETUP indication while it is waiting for further instructions from the VLR or for the acknowledgment of TMSI reallocation from the MS, the MSC saves the setup request for processing after control has returned from the macro Process_Access_Request_MSC to the calling process.

Sheet 3: When the MSC is waiting for a possible instruction to allocate a new TMSI, a MAP_DELIMITER indication indicates that TMSI reallocation is not required.

Sheet 3: If the MS sends a TMSI reallocation failure in response to the TMSI reallocation command, the MSC takes the OK exit; the VLR treats the lack of response as a provider error (see macro Process_Access_Request_VLR).

25.4.2 Macro Process_Access_Request_VLR

Sheet 3: If the MSC does not send a positive response to the MAP_FORWARD_NEW_TMSI request, this is treated as a MAP_FORWARD_NEW_TMSI confirmation containing a provider error. The Macro takes the Error exit. If TMSI reallocation does not succeed, the old TMSI is frozen, to prevent it from being reallocated. In this case, both old and new TMSIs are regarded as valid.

25.4.3 Macro Obtain_Identity_VLR

This macro is invoked by the macro Process_Access_Request_VLR if the subscriber's identity is not known in the VLR.

It is an operator option to allow or prevent retrieval of the IMSI without encryption.

25.4.4 Process Update_Location_Child_VLR

This process is started when the subscriber successfully accesses the network, e.g. for mobile originated short message submission, response to paging or supplementary services handling.

The procedure Notify_gsmSCF is specified in 3GPP TS 23.078.

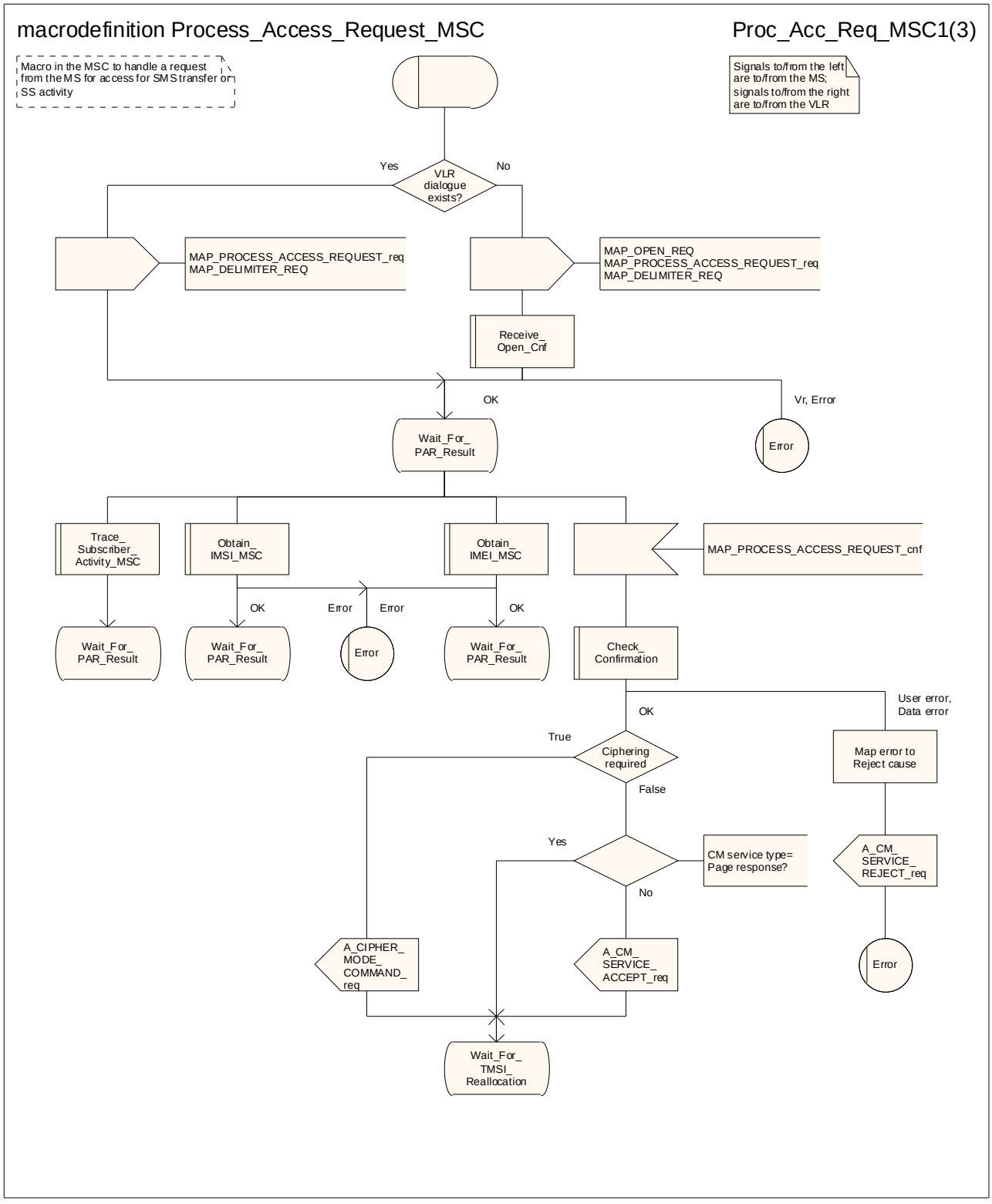


Figure 25.4/1 (sheet 1 of 3): Macro Process_Access_Request_MSC

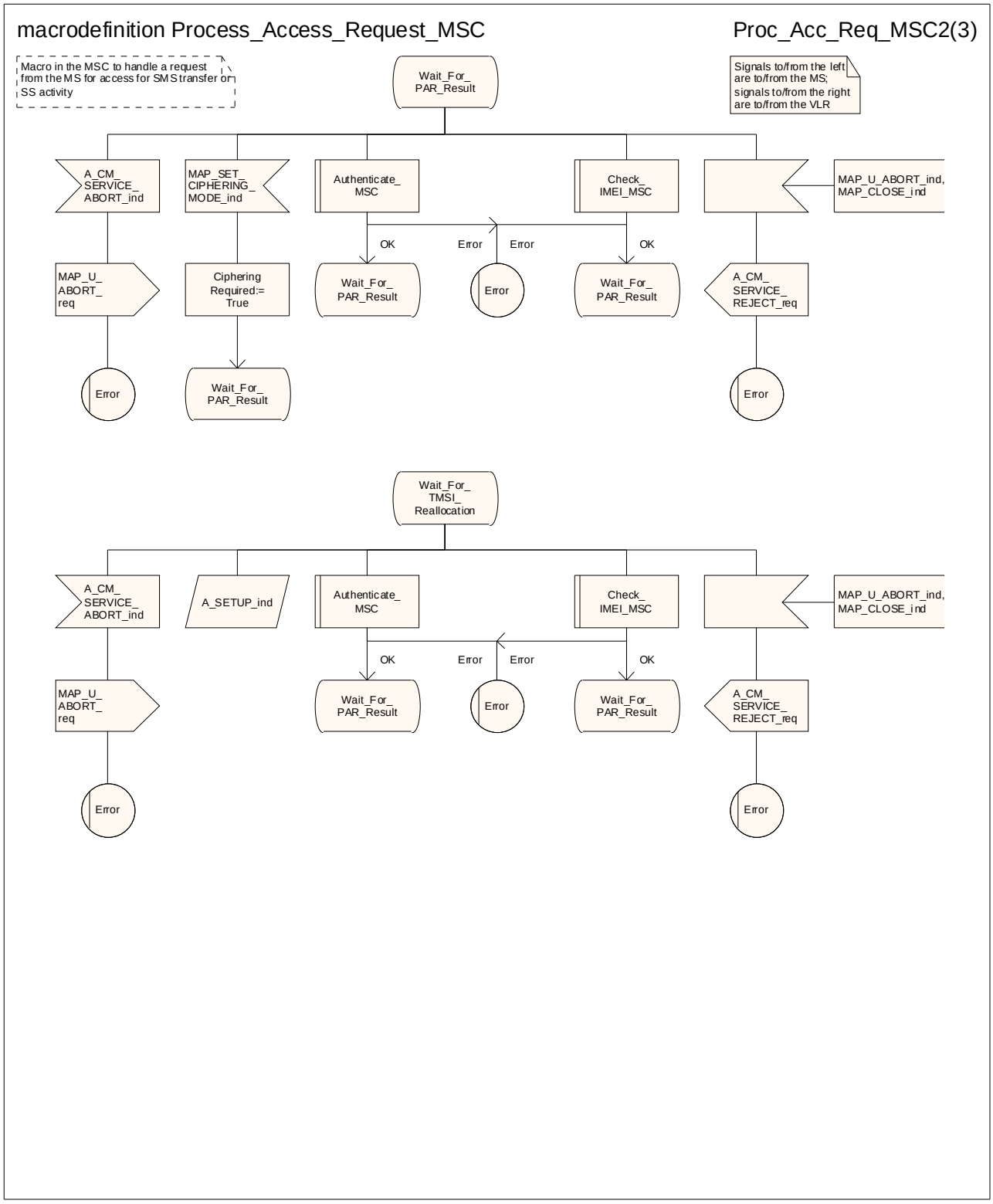


Figure 25.4/1 (sheet 2 of 3): Macro Process_Access_Request_MSC

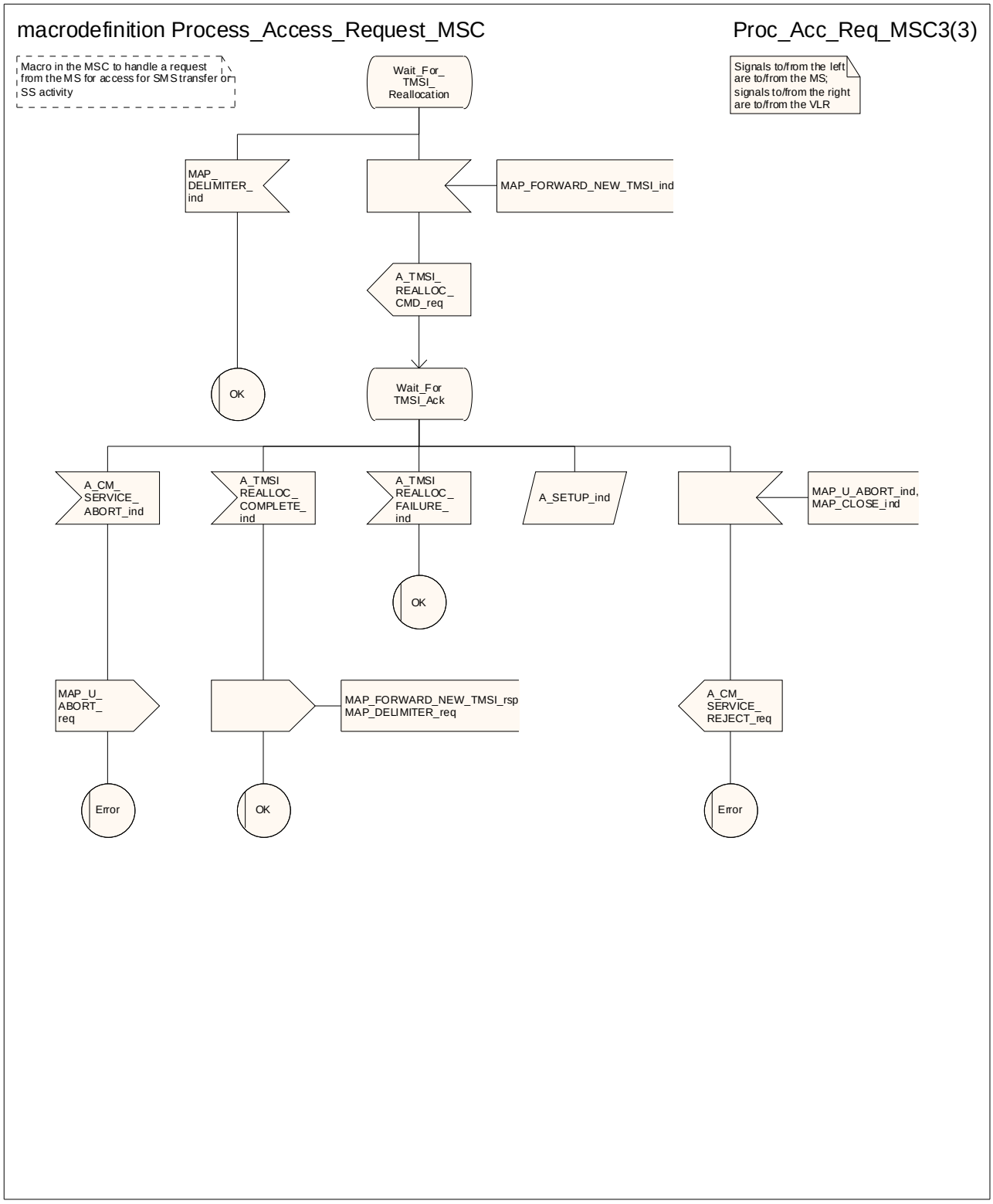


Figure 25.4/1 (sheet 3 of 3): Macro Process_Access_Request_MSC

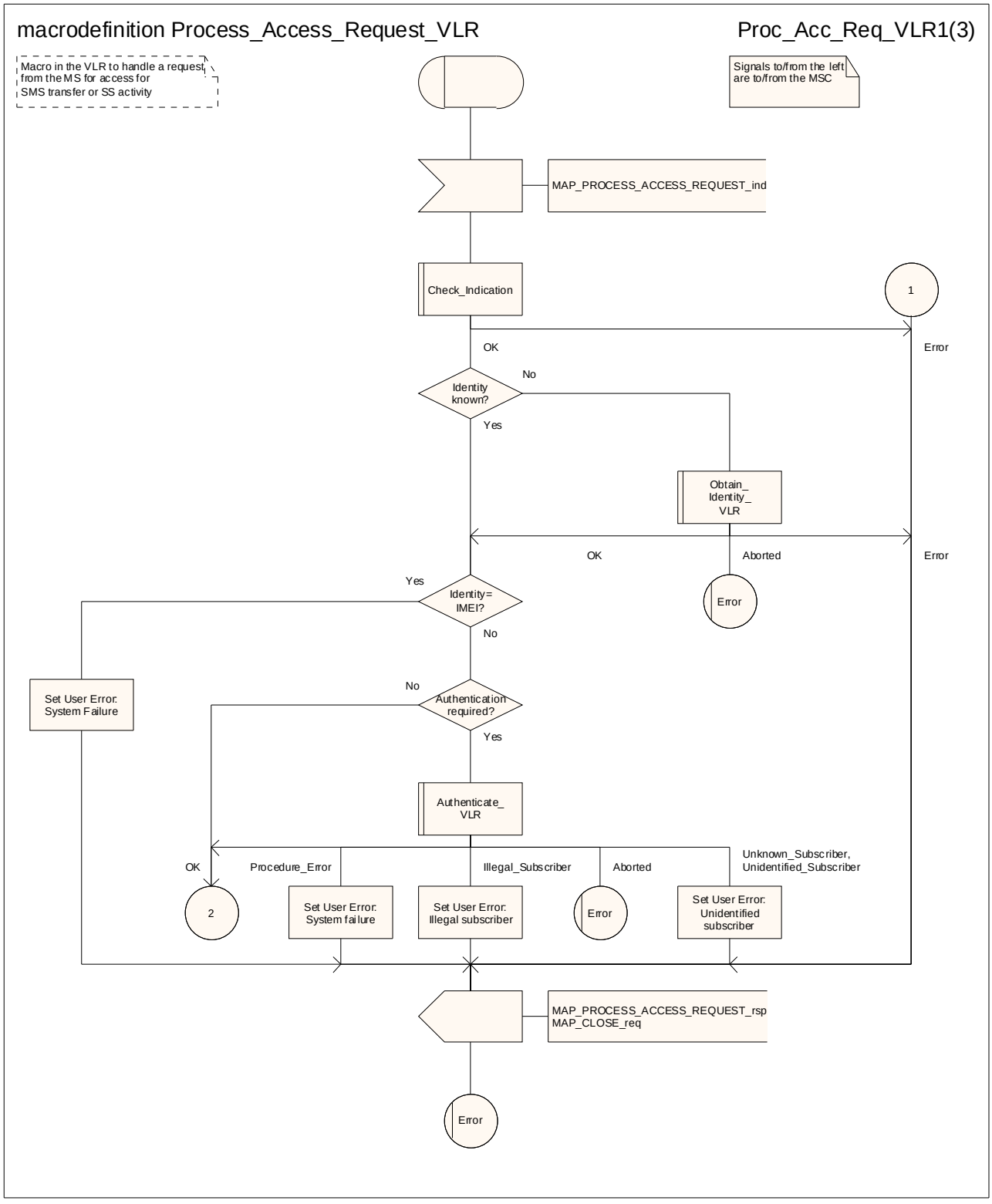


Figure 25.4/2 (sheet 1 of 3): Macro Process_Access_Request_VLR

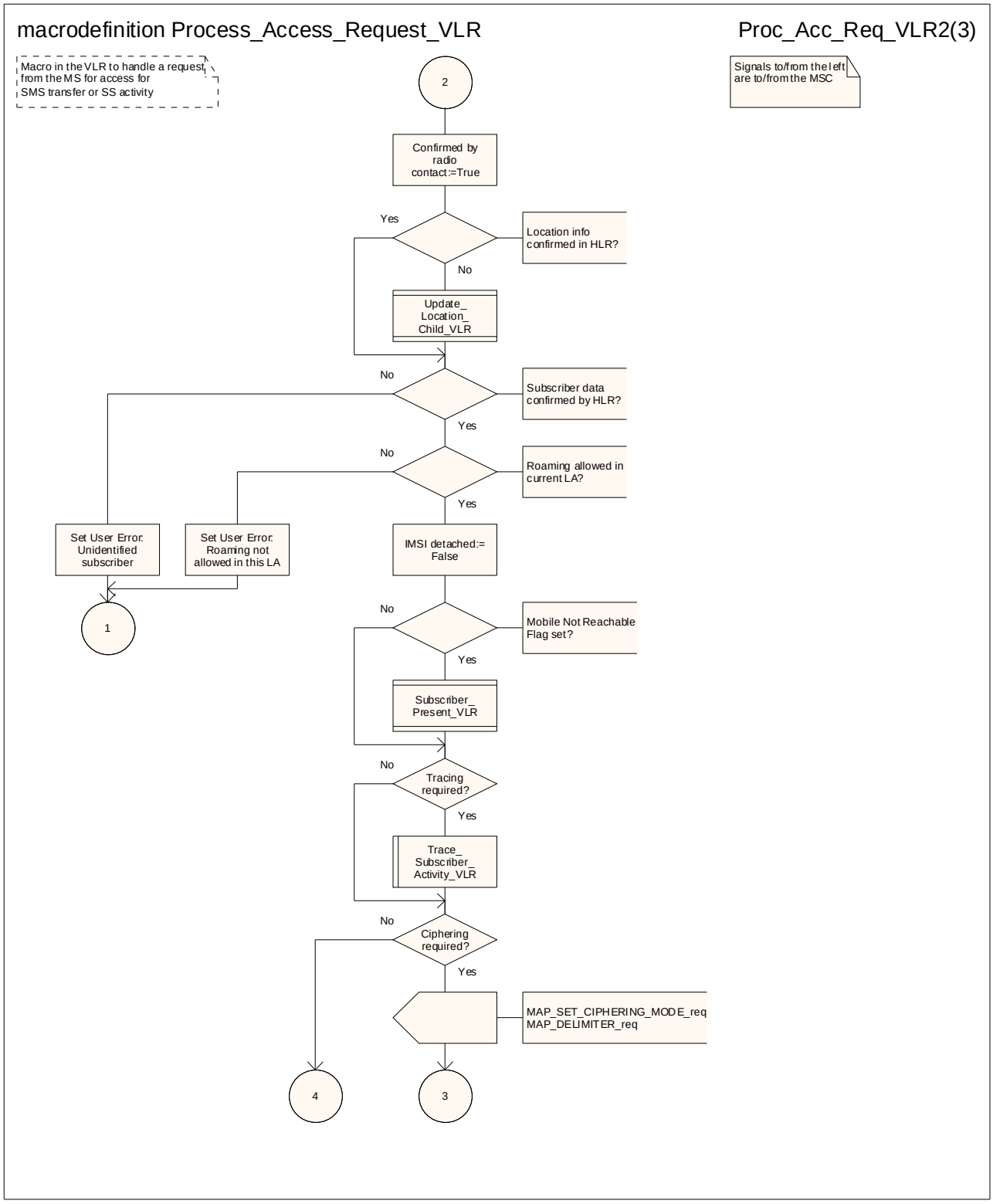


Figure 25.4/2 (sheet 2 of 3): Macro Process_Access_Request_VLR

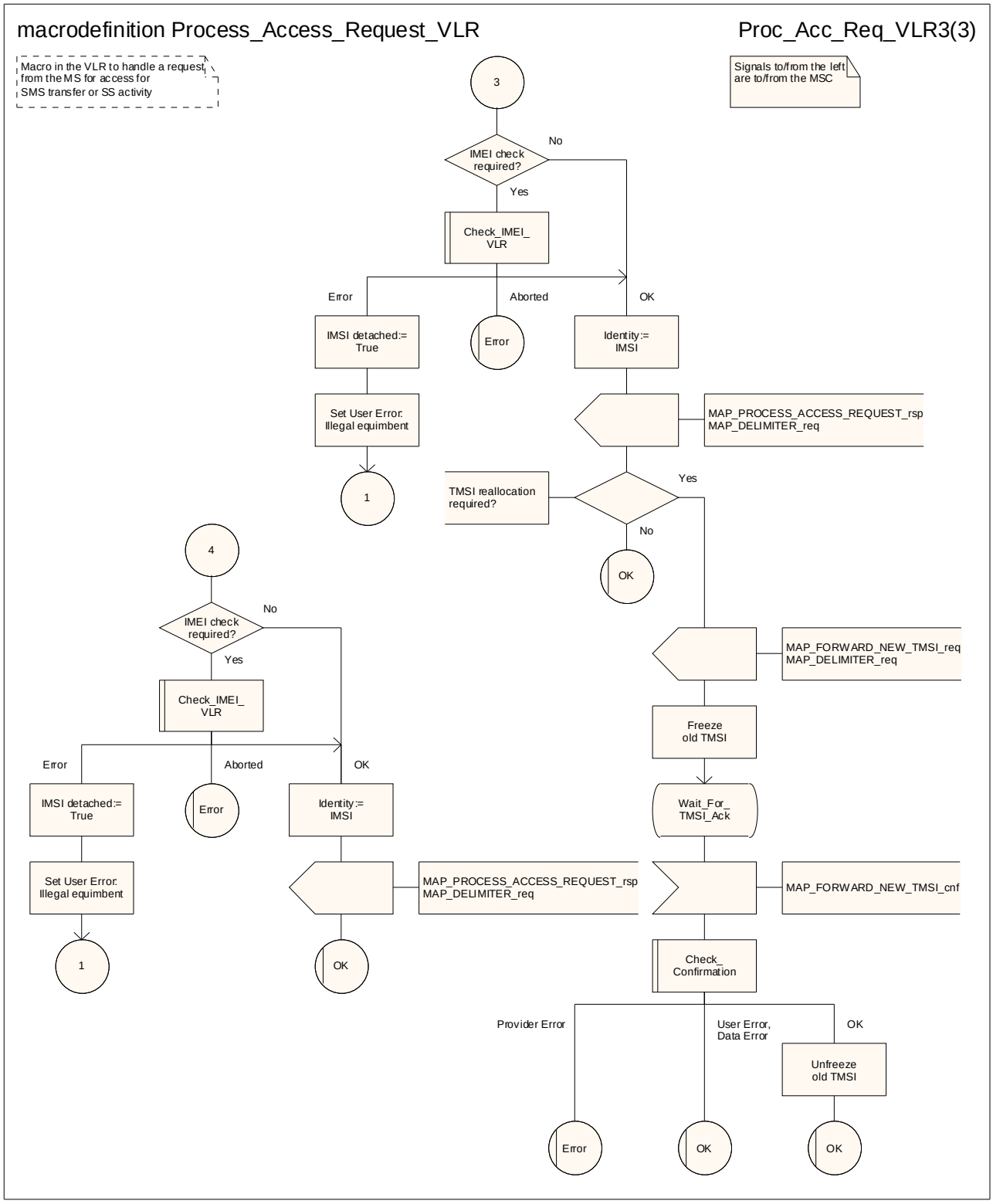


Figure 25.4/2 (sheet 3 of 3): Macro Process_Access_Request_VLR

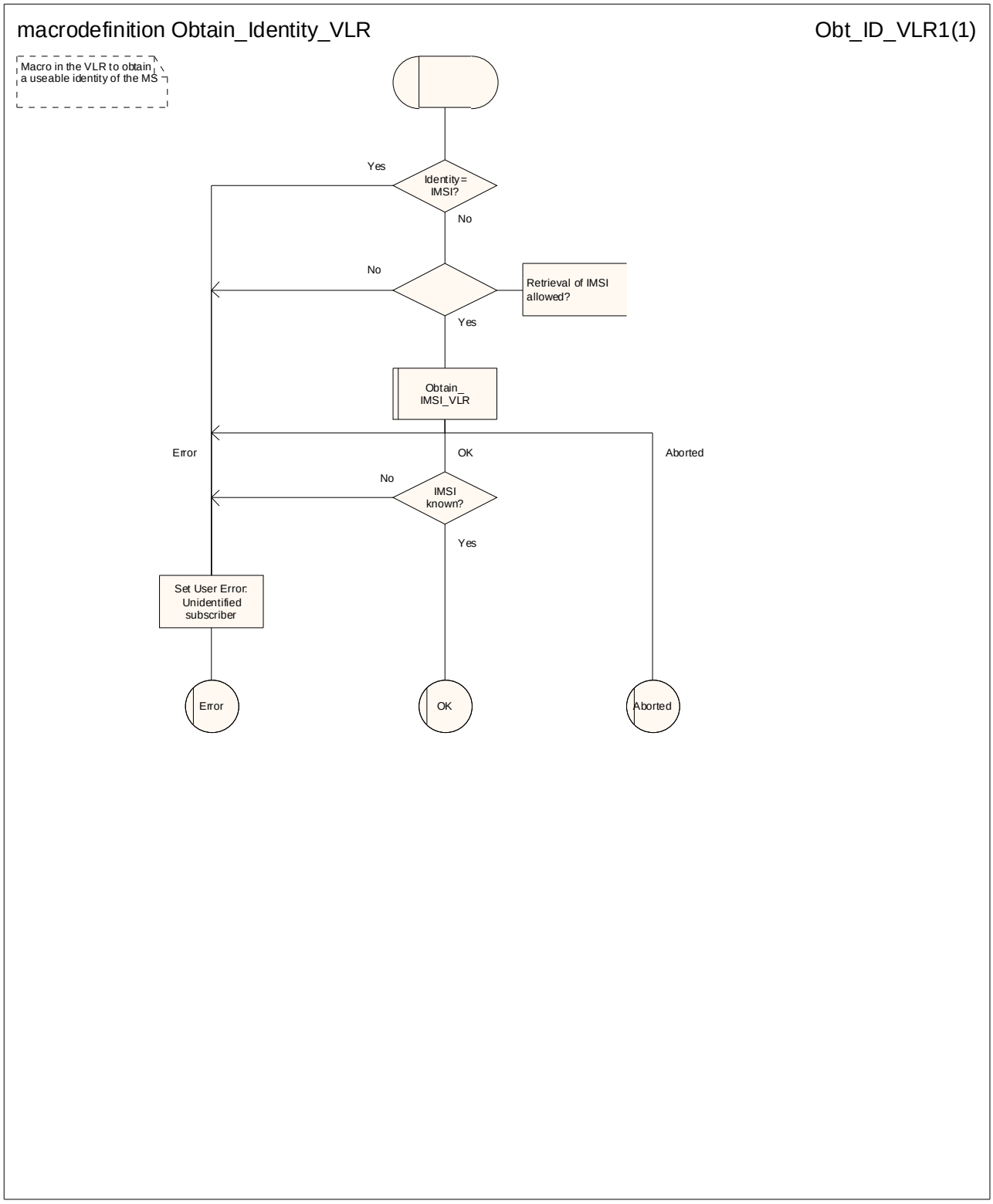


Figure 25.4/3: Macro Obtain_Identity_VLR

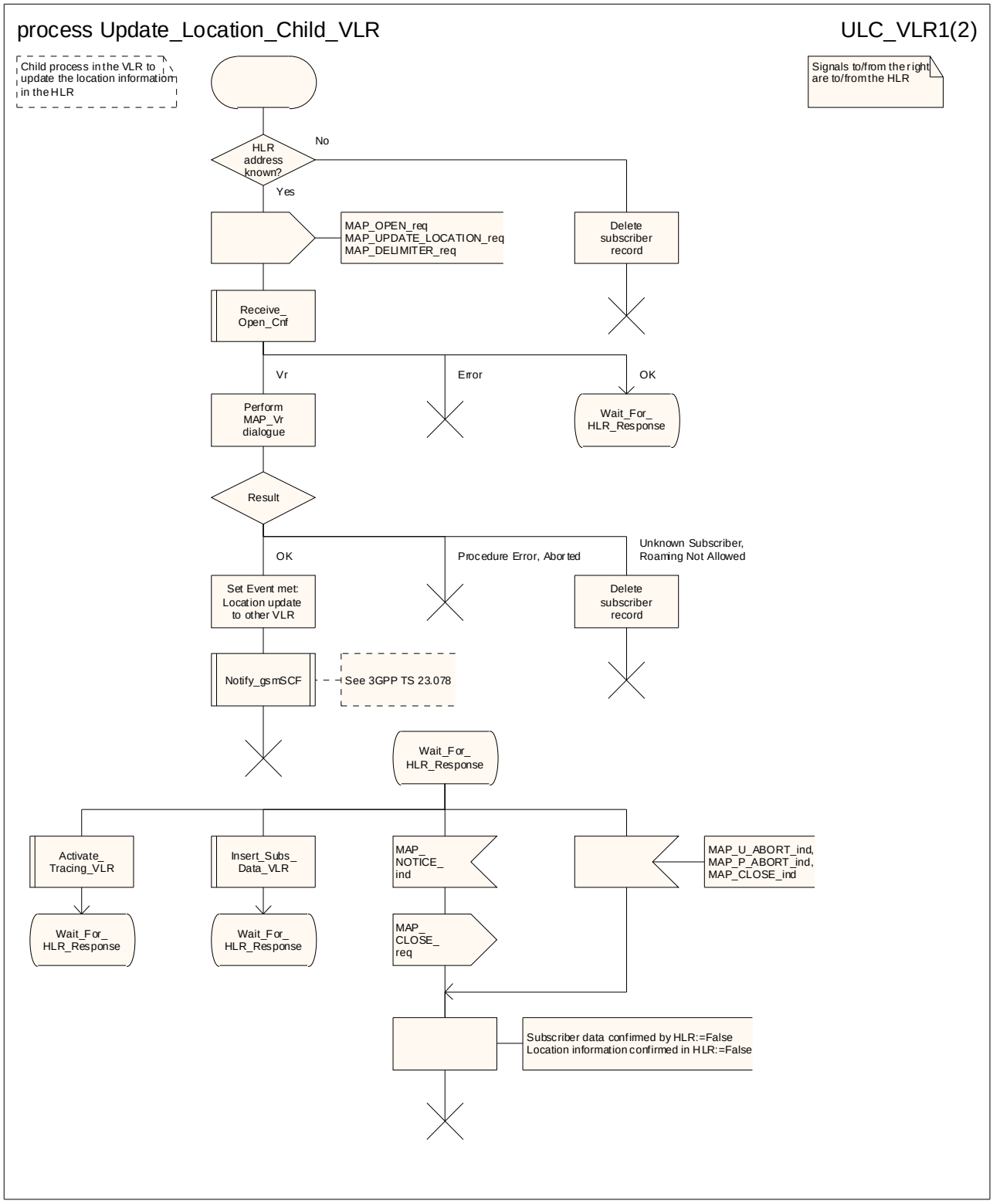


Figure 25.4/4 (sheet 1 of 2): Process Update_Location_Child_VLR

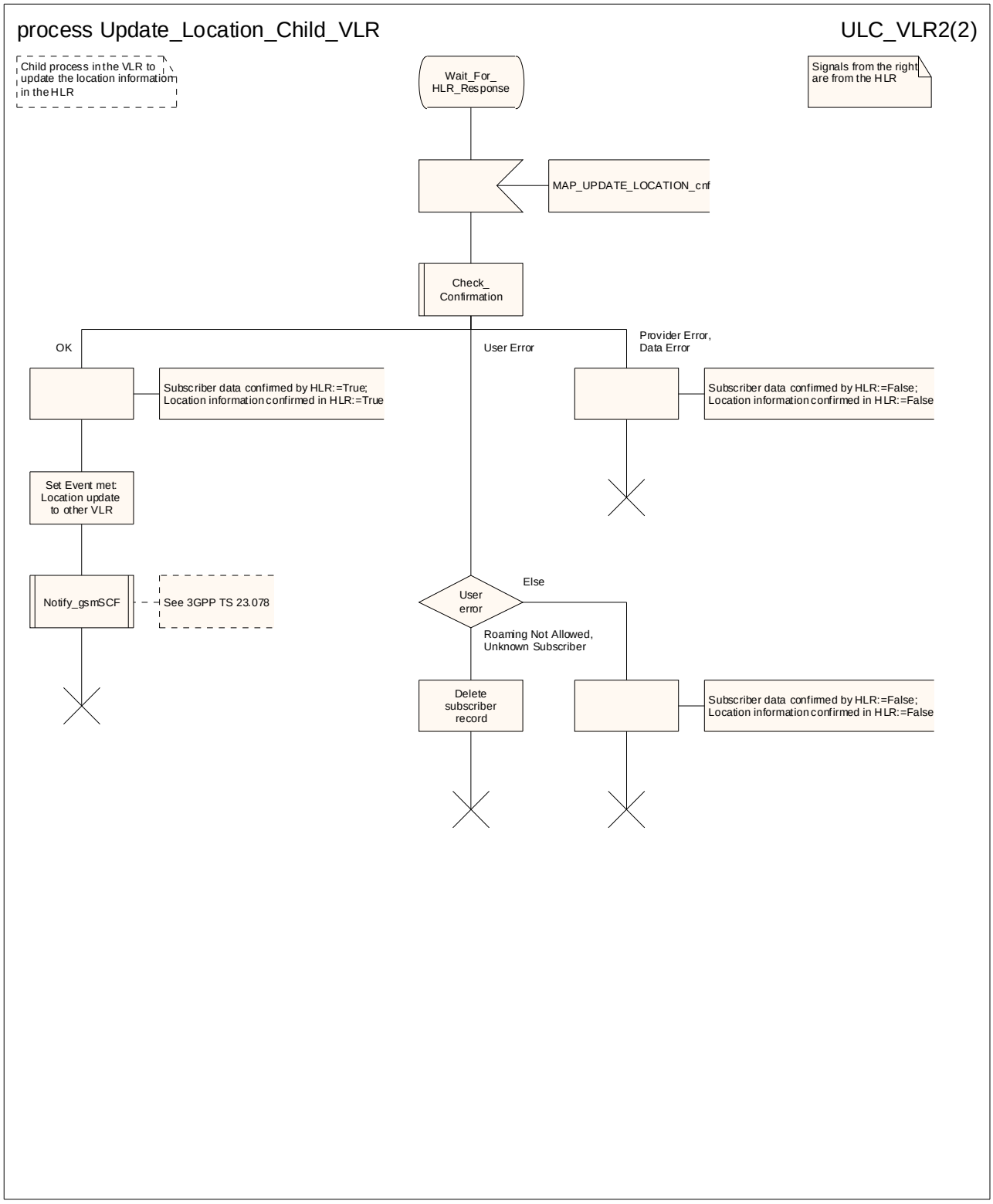


Figure 25.4/4 (sheet 2 of 2): Process Update_Location_Child_VLR

25.5 Authentication macros and processes

The following macros are used in the network in order to enable authentication of a mobile subscriber.

25.5.1 Macro Authenticate_MSC

This macro is used by the MSC to relay a request for authentication transparently from the VLR to the MS, wait for a response from the MS and relay the response from the MS back to the VLR.

25.5.2 Macro Authenticate_VLR

This macro is used by the VLR to control the authentication of a subscriber.

Sheet 1: The test "Received SRES=Expected SRES" indicates:

- a comparison of the Signed RESult received from the MS with the Signed RESult received from the HLR, if GSM authentication is used (see 3GPP TS 43.020 [24]), or
- a comparison of the RESult received from the MS with the expected RESult received from the HLR, if UMTS authentication is used (see 3GPP TS 33.102).

25.5.3 Macro Obtain_Authent_Params_VLR

This macro is used by the VLR to request authentication vectors from the HLR.

Sheet 1, sheet 2, sheet 3: It is an operator option whether to allow the re-use of old authentication triplets.

Sheet 2, sheet 3: Old UMTS quintuplets shall not be re-used.

Sheet 2: if the VLR requests more authentication vectors in the same dialogue, the subsequent MAP_SEND_AUTHENTIFICATION_INFO request has no parameters.

25.5.4 Process Obtain_Authentication_Sets_VLR

This process is initiated by the VLR to fetch authentication vectors from a subscriber's HLR independently of any other processing.

25.5.5 Process Obtain_Authent_Sets_SGSN

The procedure for authentication when the serving node is an SGSN is described in 3GPP TS 23.060 [104] and 3GPP TS 24.008 [35].

This Process is used by the SGSN to request authentication vectors from the HLR.

Sheet 1, sheet 2: It is an operator option whether to allow the re-use of old authentication triplets.

Sheet 2: Old UMTS quintuplets shall not be re-used.

25.5.6 Process Obtain_Authent_Sets_HLR

This process is used to provide authentication vectors (triplets or quintuplets) in response to a request from a VLR or an SGSN.

Upon receipt of an authentication information request for a UMTS subscriber, the HLR shall return authentication quintuplets. If the user is a GSM subscriber, the HLR shall return authentication triplets.

25.5.7 Authentication Failure Reporting

25.5.7.1 General

The Authentication Failure Report procedure is used to notify an HLR about the occurrence of an authentication failure in the SGSN or VLR.

The message flows for this procedure are shown in figures 25.5/7& 25.5/8.

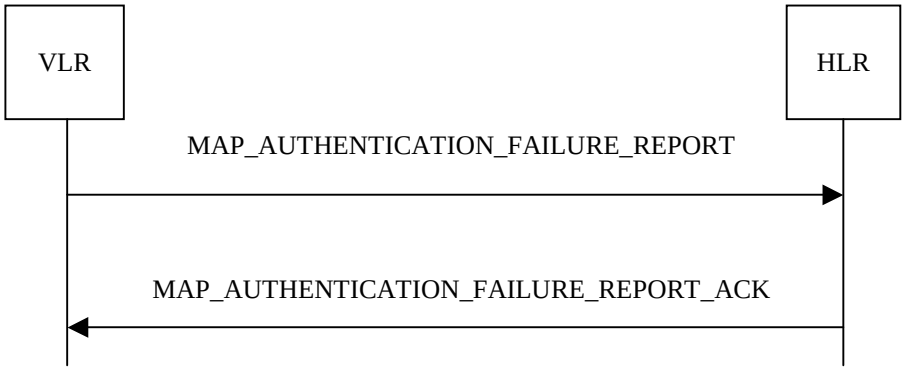


Figure 25.5/7: Message Flow for Authentication Failure Report – VLR to HLR

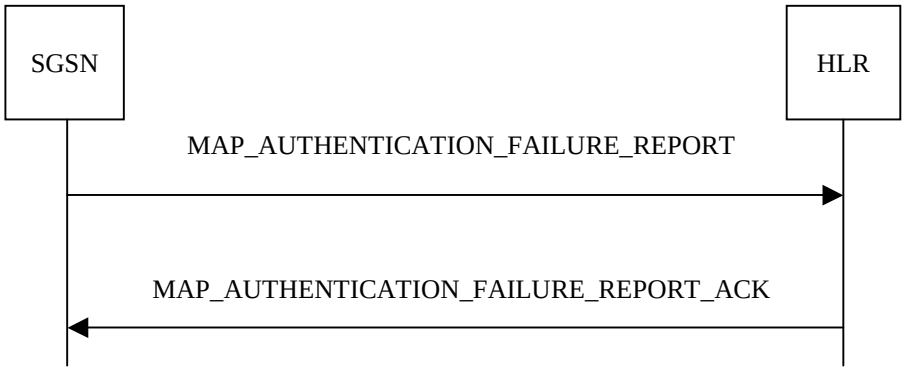


Figure 25.5/8: Message Flow for Authentication Failure Report – SGSN to HLR

25.5.7.2 Process in the VLR

25.5.7.3 Process in the SGSN

25.5.7.4 Process in the HLR

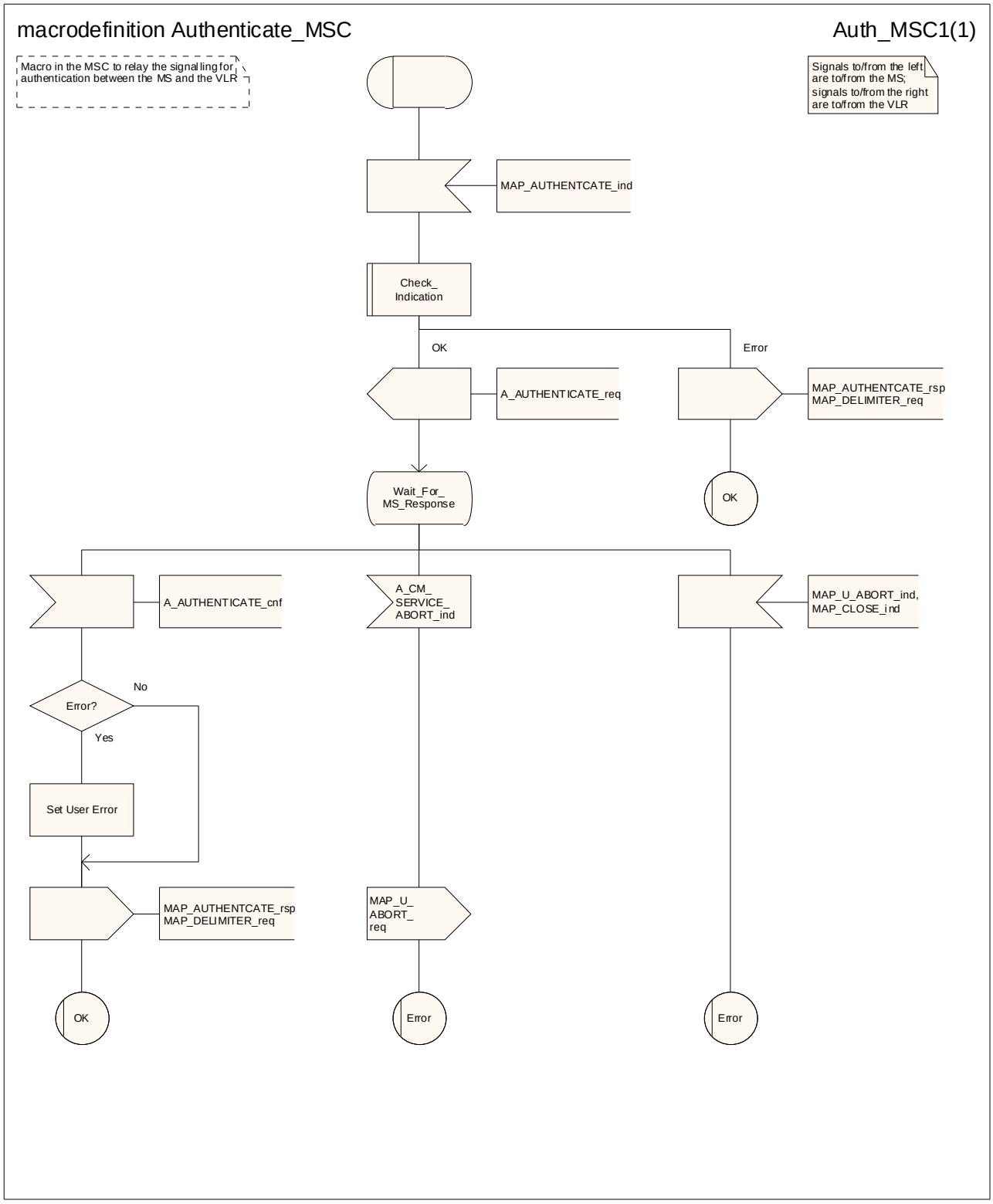


Figure 25.5/1: Macro Authenticate_MSC

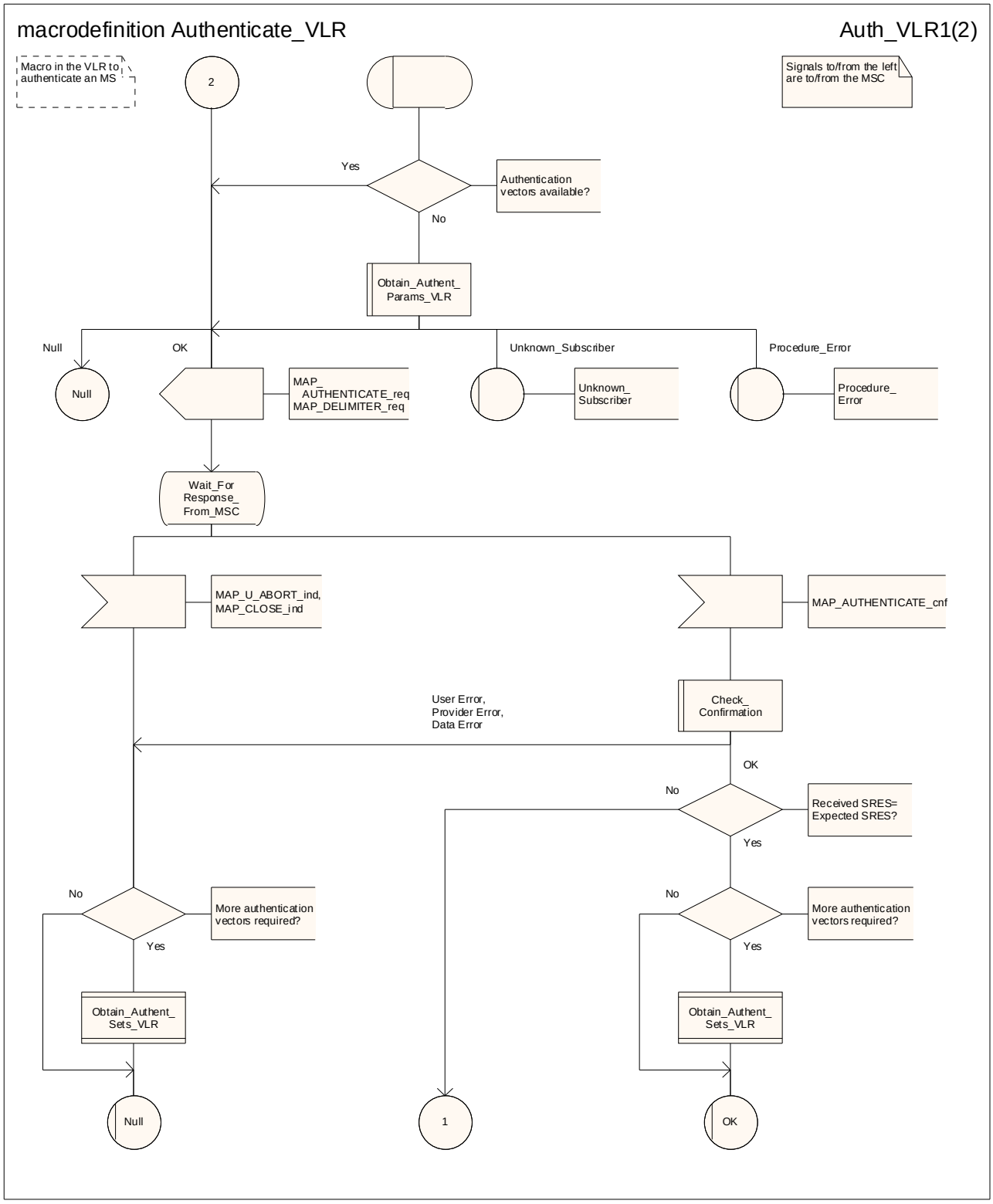


Figure 25.5/2 (sheet 1 of 2): Macro Authenticate_VLR

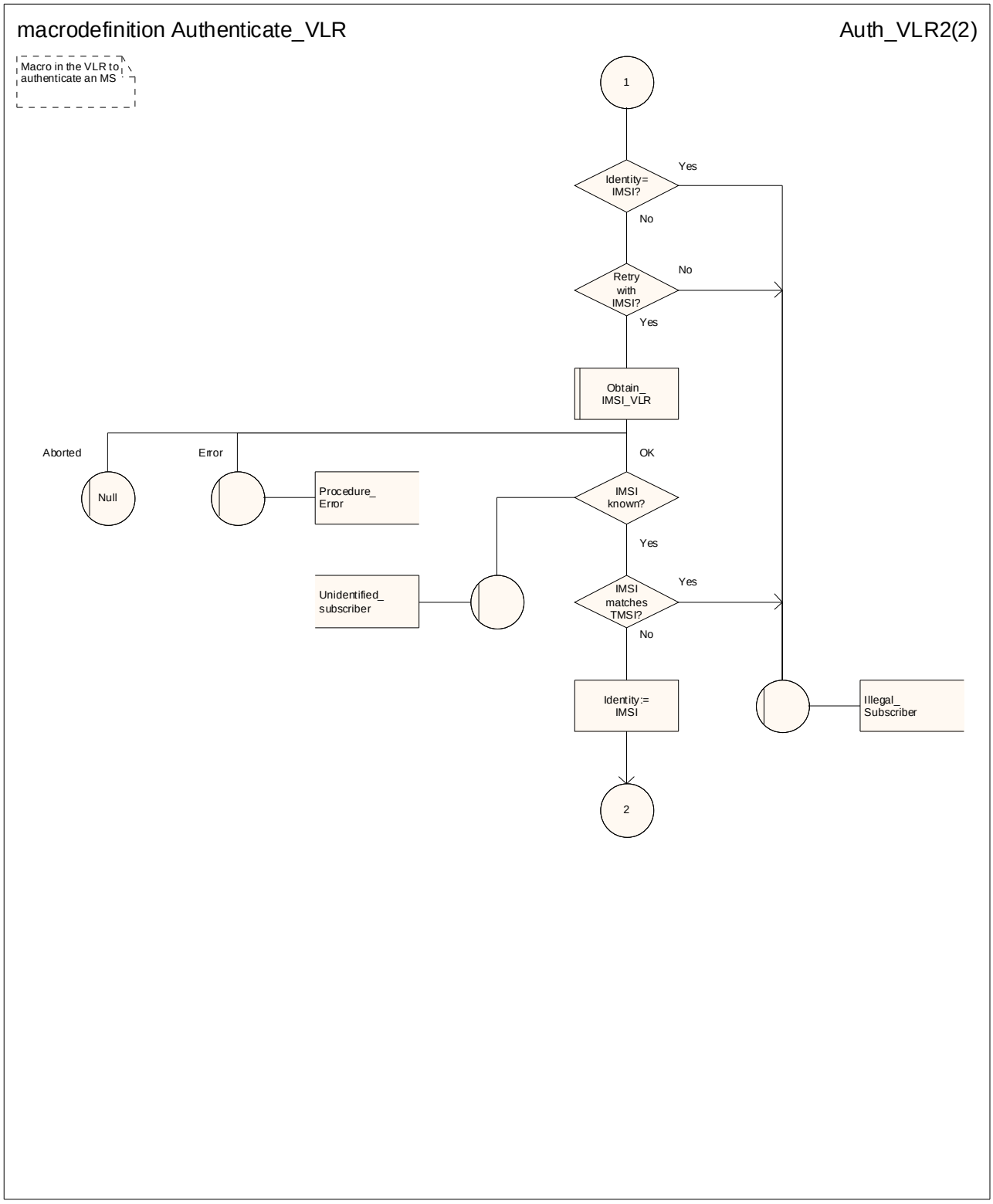


Figure 25.5/2 (sheet 2 of 2): Macro Authenticate_VLR

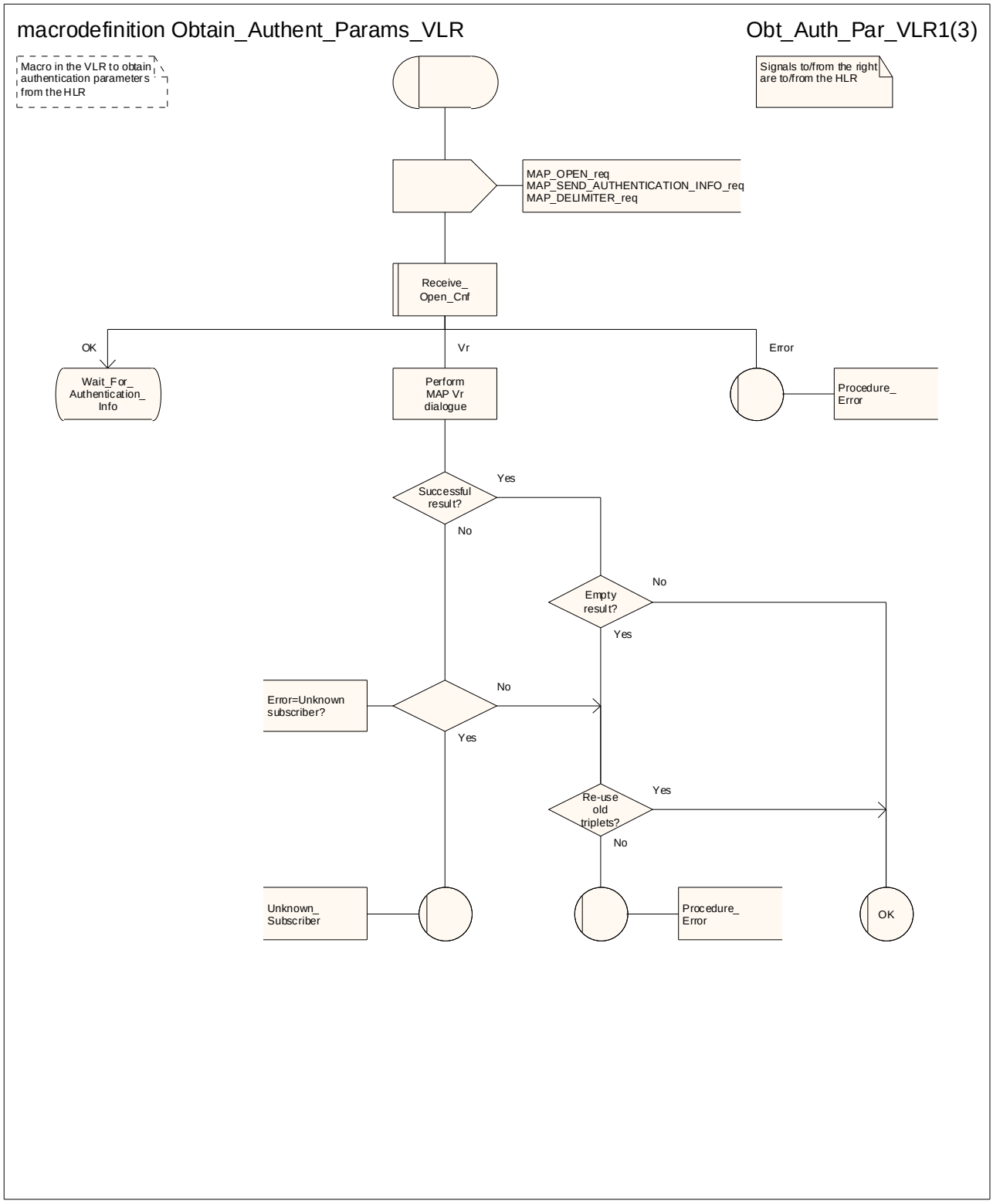


Figure 25.5/3 (sheet 1 of 3): Macro Obtain_Authent_Params_VLR

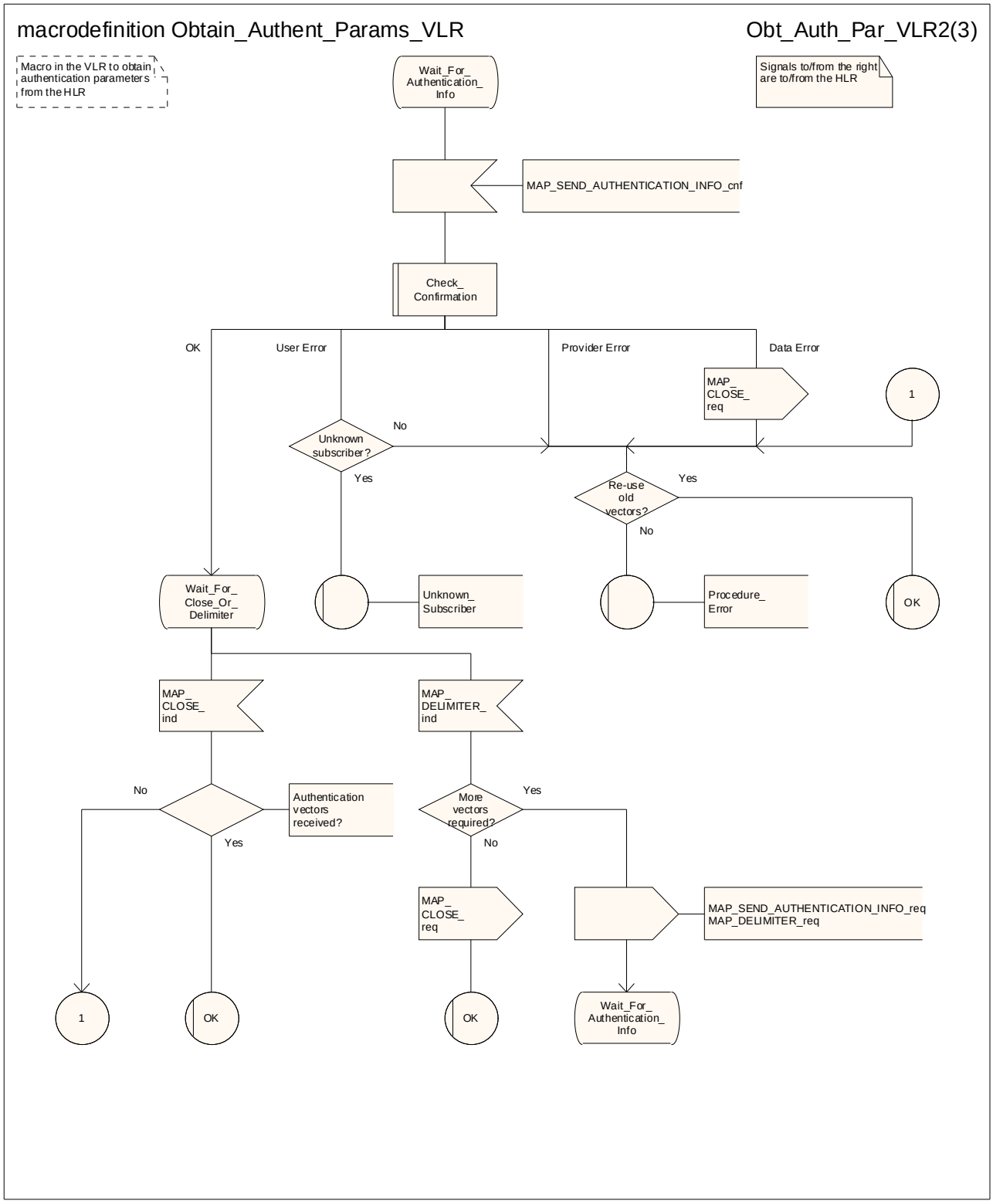


Figure 25.5/3 (sheet 2 of 3): Macro Obtain_Authent_Params_VLR

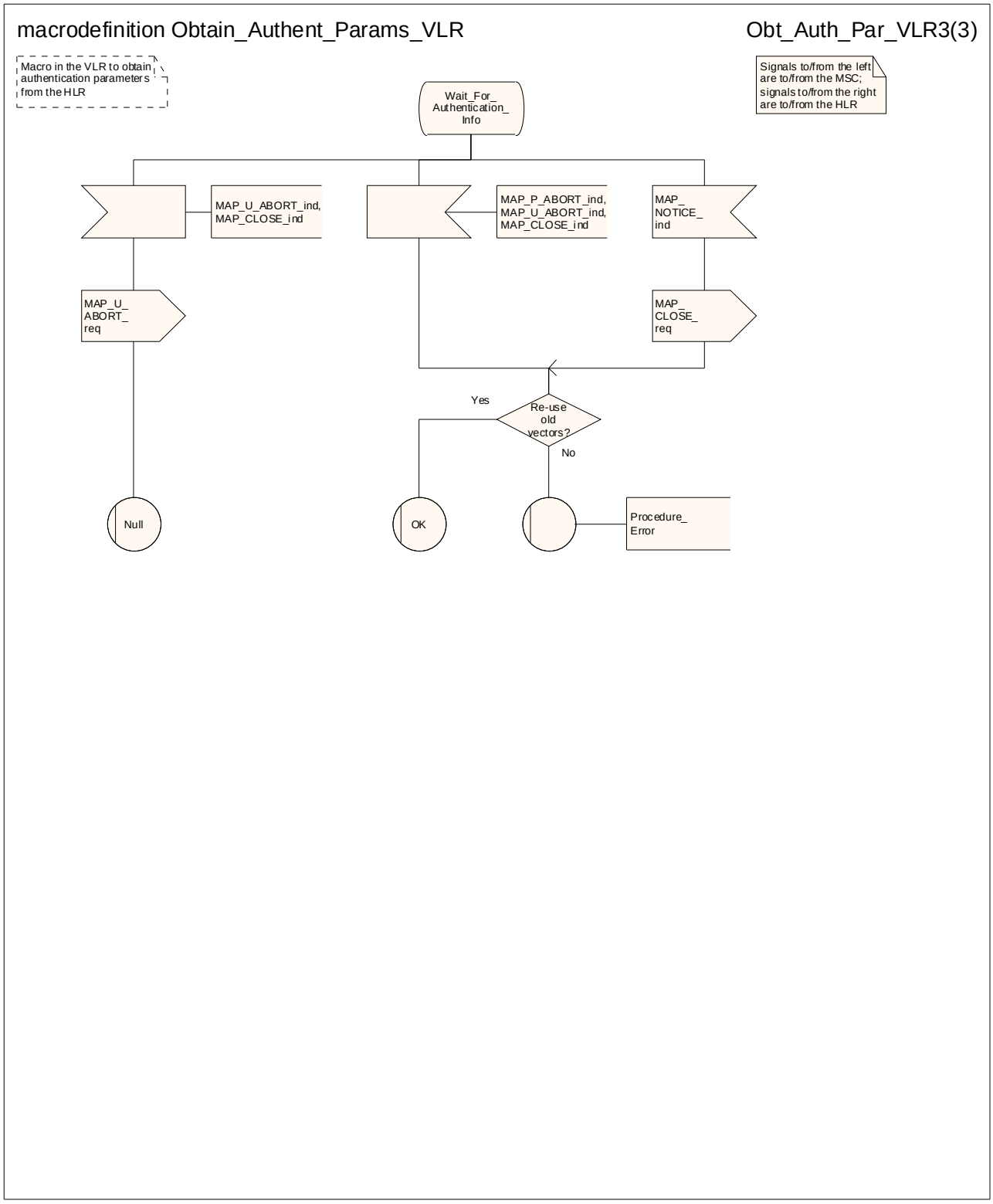


Figure 25.5/3 (sheet 3 of 3): Macro Obtain_Authent_Params_VLR

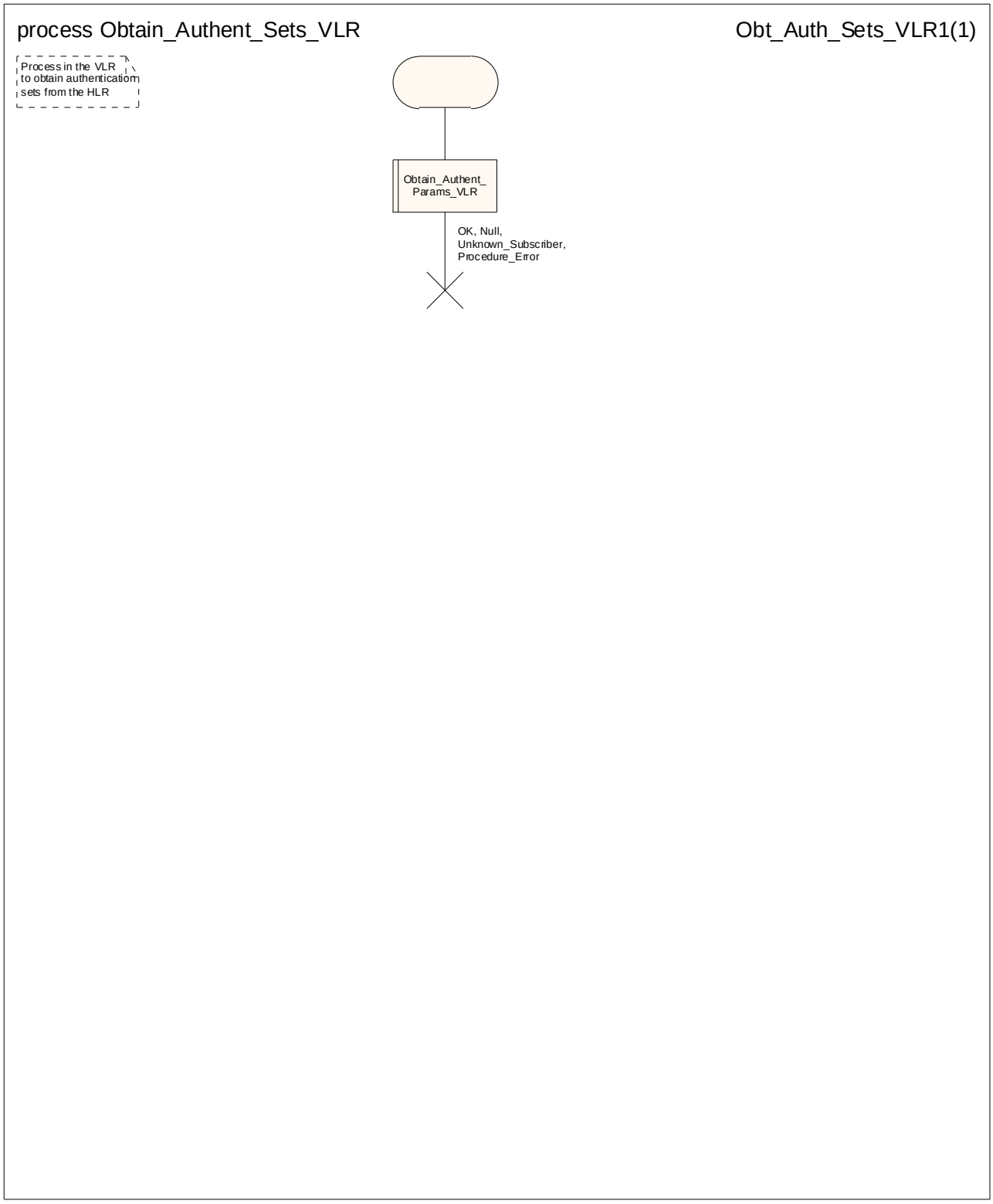


Figure 25.5/4: Process Obtain_Authent_Sets_VLR

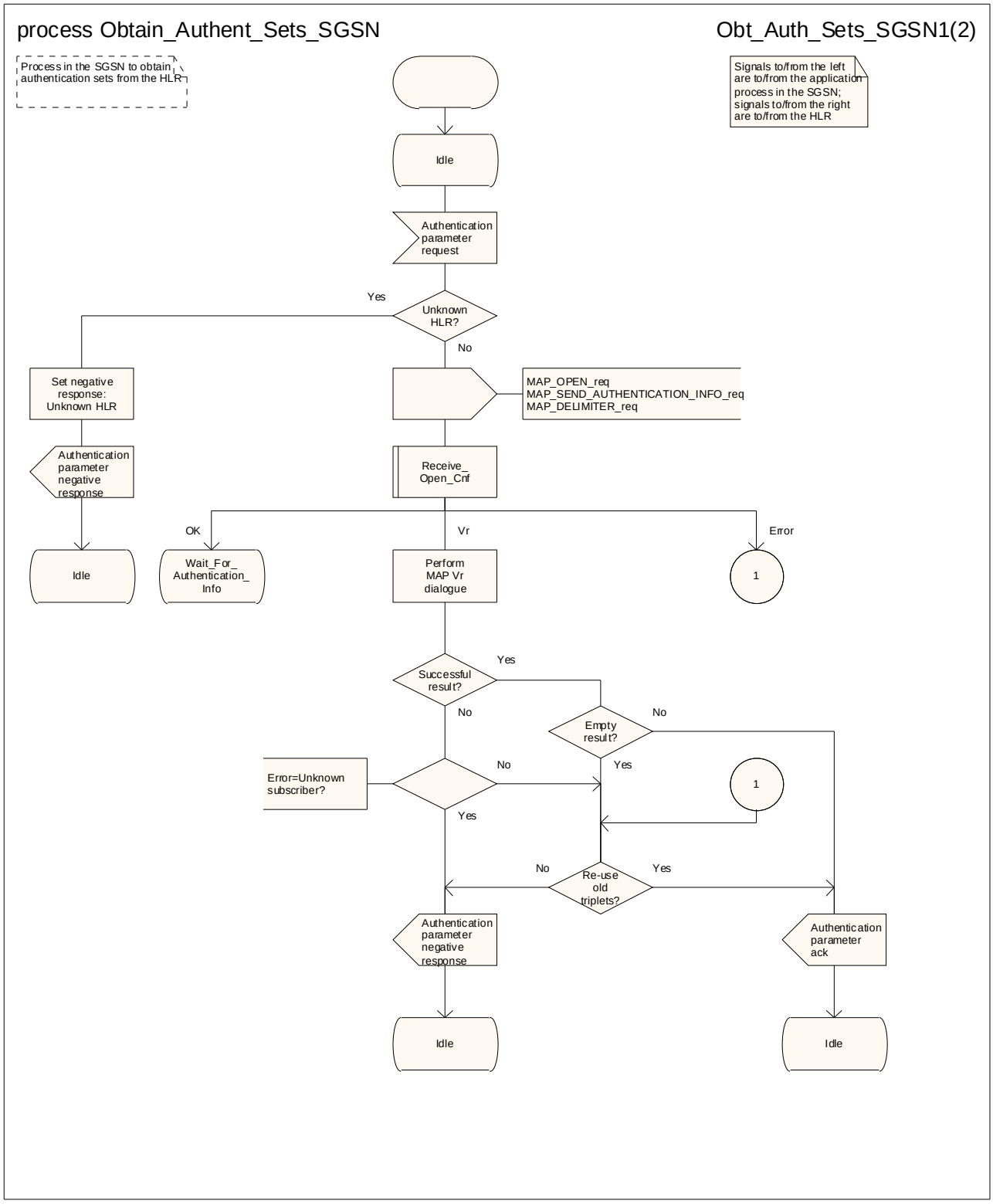


Figure 25.5/5 (sheet 1 of 2): Process Obtain_Authent_Sets_SGSN

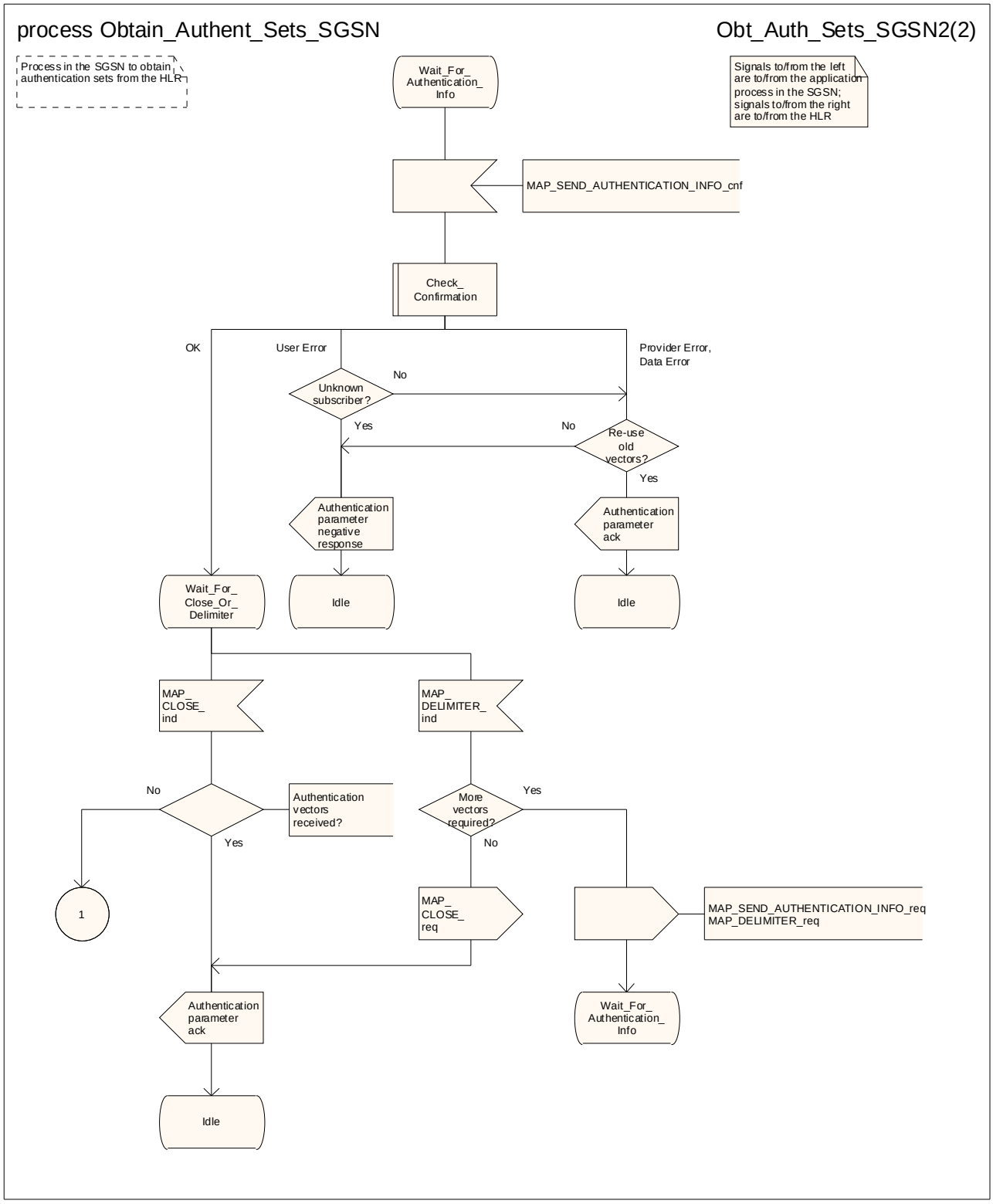


Figure 25.5/5 (sheet 2 of 2): Process Obtain_Authent_Sets_SGSN

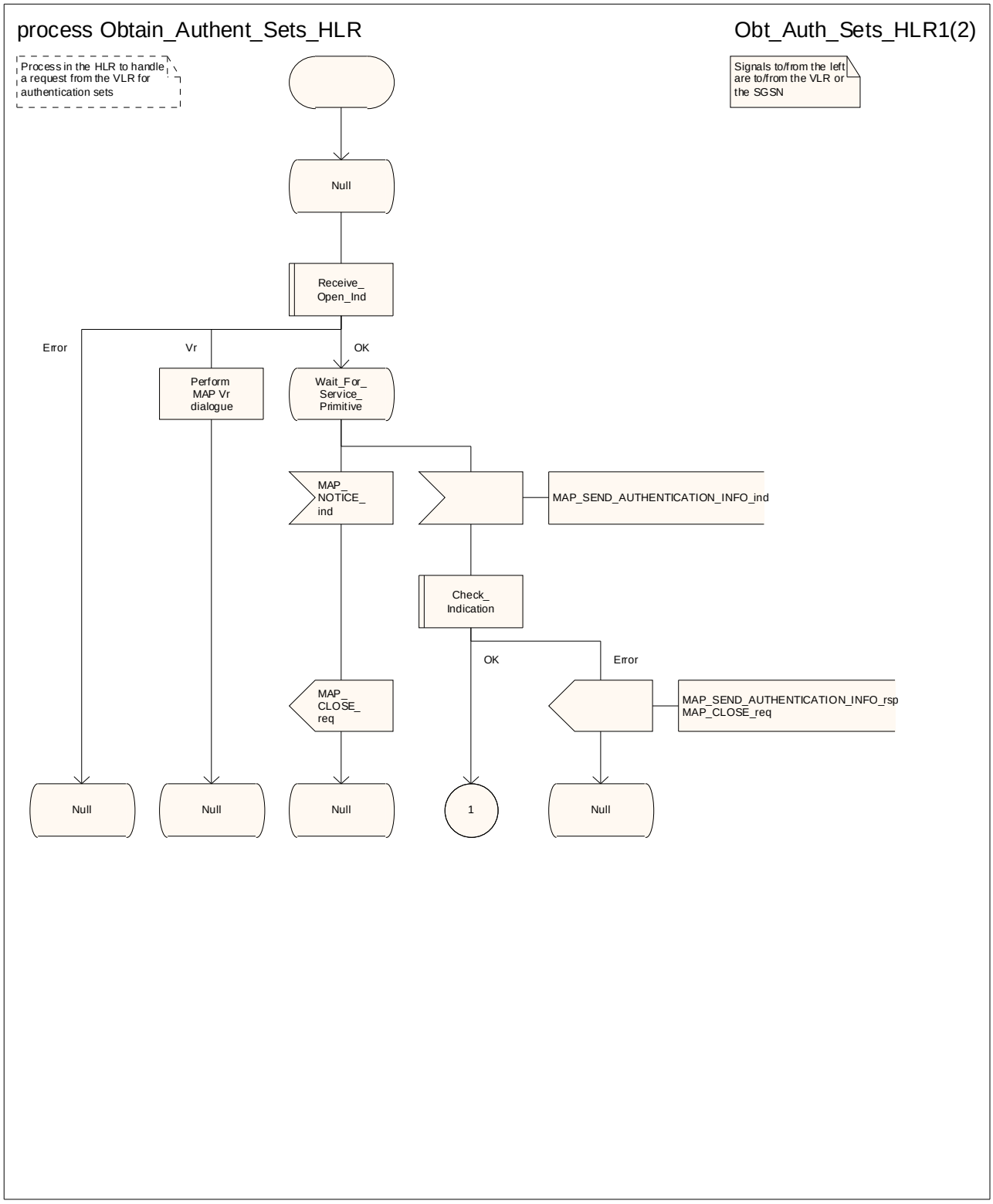


Figure 25.5/6 (sheet 1 of 2): Process Obtain_Authent_Sets_HLR

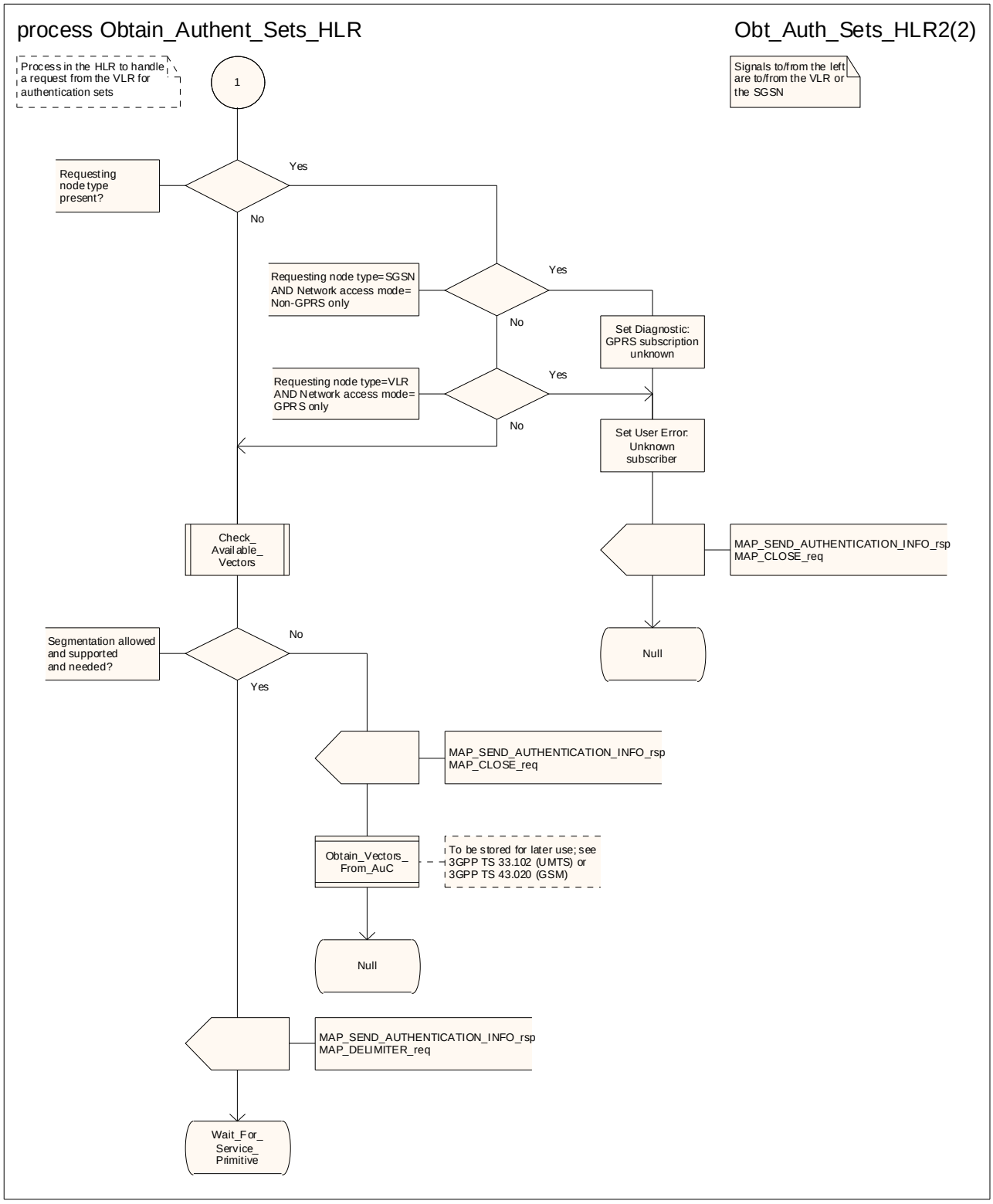


Figure 25.5/6 (sheet 2 of 2): Process Obtain_Authent_Sets_HLR

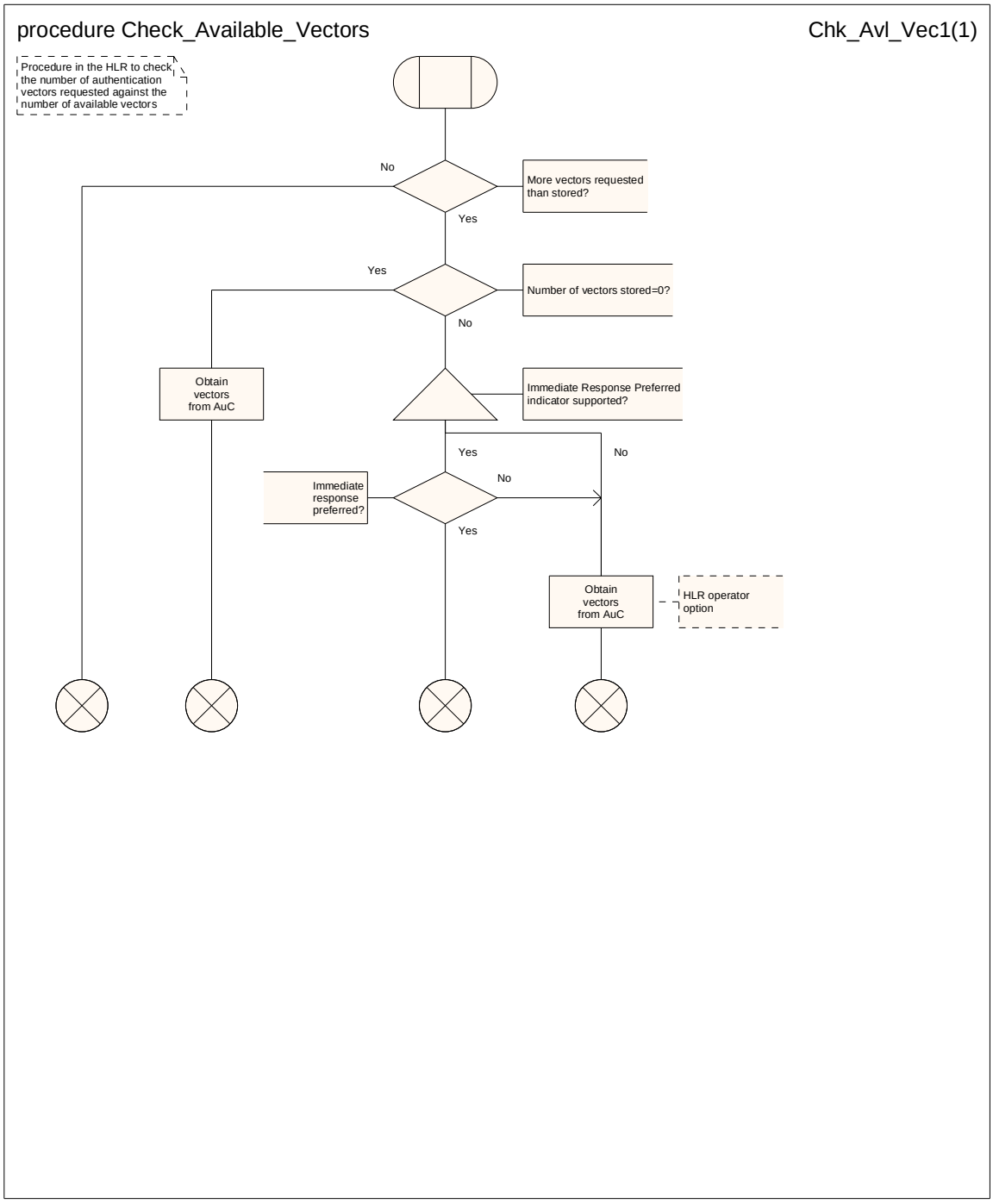


Figure 25.5/7: Procedure Check_Available_Vectors

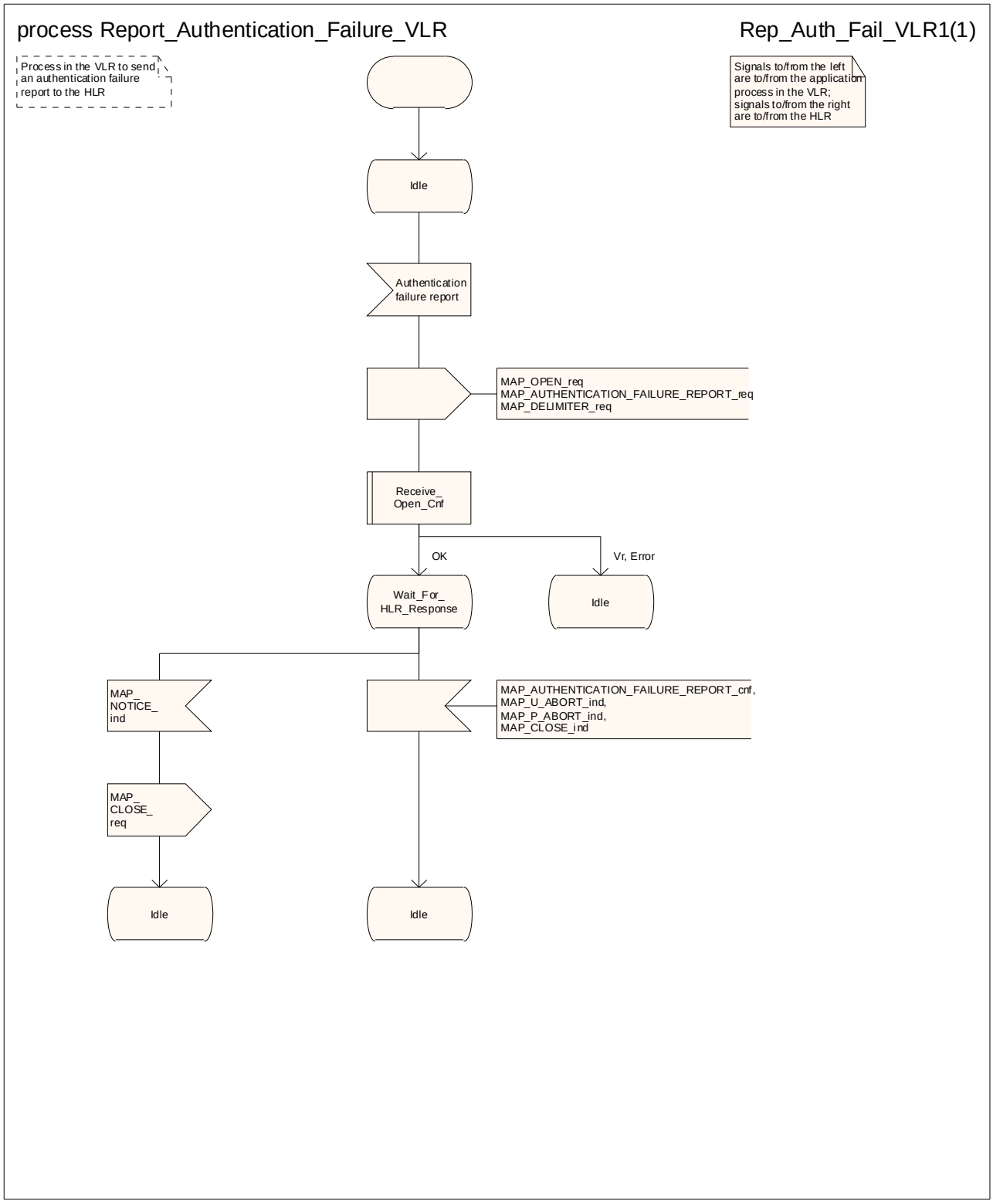


Figure 25.5/9: Process Report_Authentication_Failure_VLR

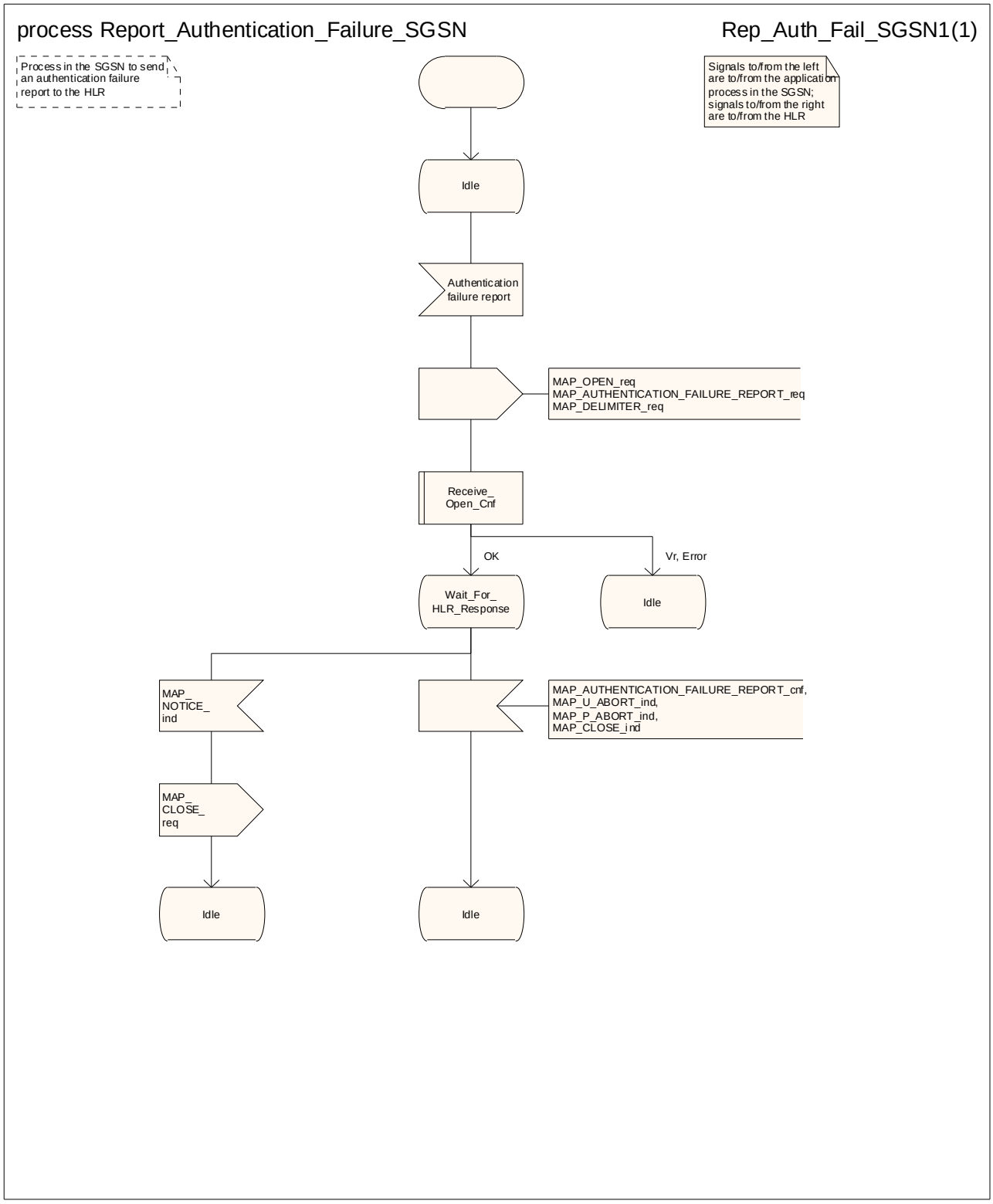


Figure 25.5/10: Process Report_Authentication_Failure_SGSN

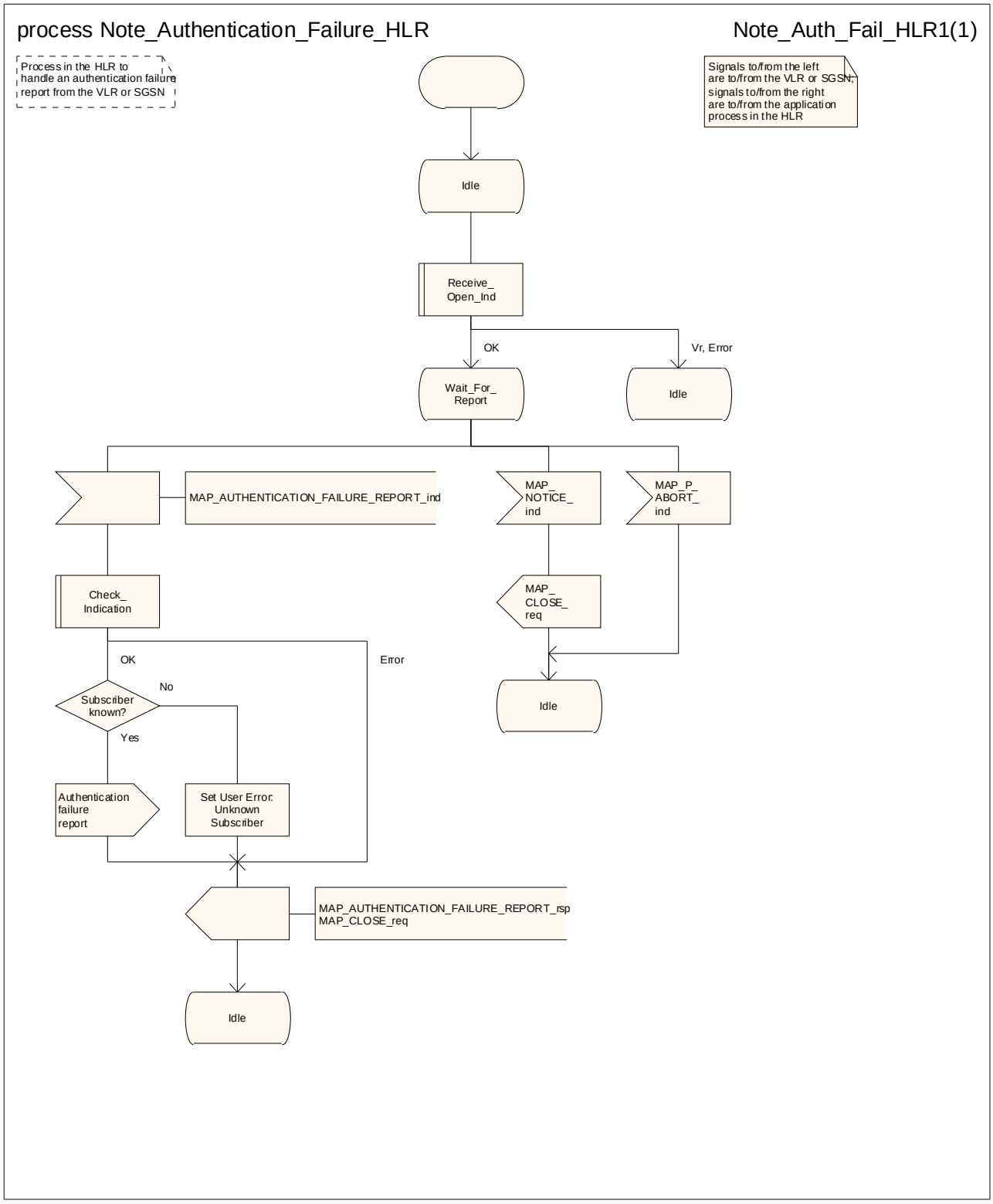


Figure 25.5/11: Process Note_Authentication_Failure_HLR

25.6 IMEI Handling Macros

The following macros are used in the network in order to enable handling and checking of the mobile equipment identity.

25.6.1 Macro Check_IMEI_MSC

This macro is used by the MSC to receive a request from the VLR, relay it to the EIR, and pass the result from the EIR back to the VLR.

Sheet 1: If the dialogue with the EIR drops back to a previous protocol version and the EIR returned an error, the MSC relays the error to the VLR in the MAP_CHECK_IMEI response. If the dialogue with the EIR failed, or the EIR returned a badly formed result, the MSC sends a System Failure error to the VLR in the MAP_CHECK_IMEI response.

25.6.2 Macro Check_IMEI_VLR

This macro is used by the VLR to control the check of a mobile equipment's IMEI. It may also be used to request the BMUEF from the EIR.

25.6.3 Process Check_IMEI_SGSN

This process is used by the SGSN to control the check of a mobile equipment's IMEI. It may also be used to request the BMUEF from the EIR.

25.6.4 Process Check_IMEI_EIR

This process is used by the EIR to obtain the status of a mobile equipment, upon request from the MSC or from the SGSN. It may also be used to obtain the BMUEF.

25.6.5 Macro Obtain_IMEI_MSC

This macro is used by the MSC to respond to a request from the VLR to provide the IMEI.

25.6.6 Macro Obtain_IMEI_VLR

This macro is used by the VLR to obtain the IMEI from the MSC.

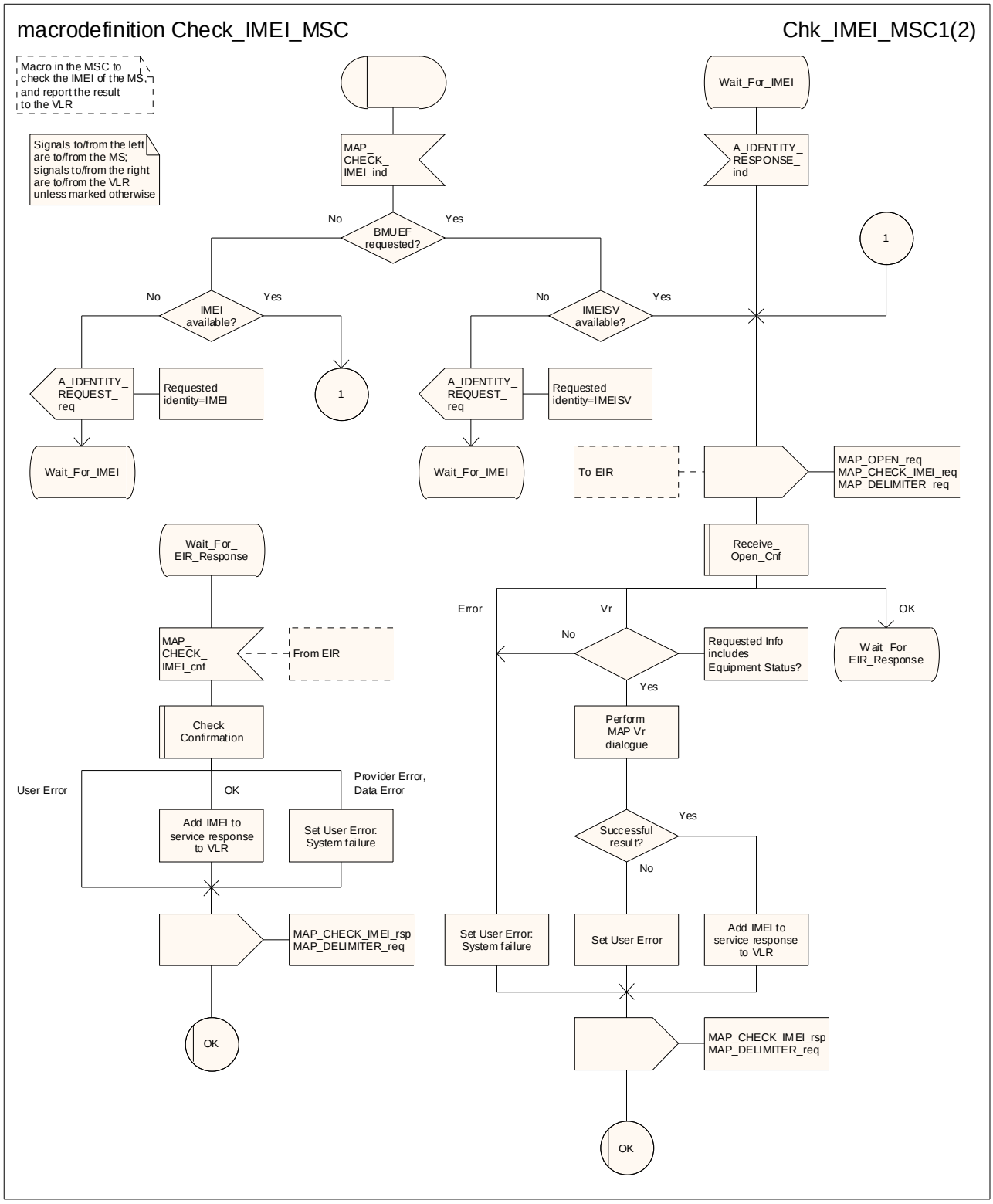


Figure 25.6/1 (sheet 1 of 2): Macro Check_IMEI_MSC

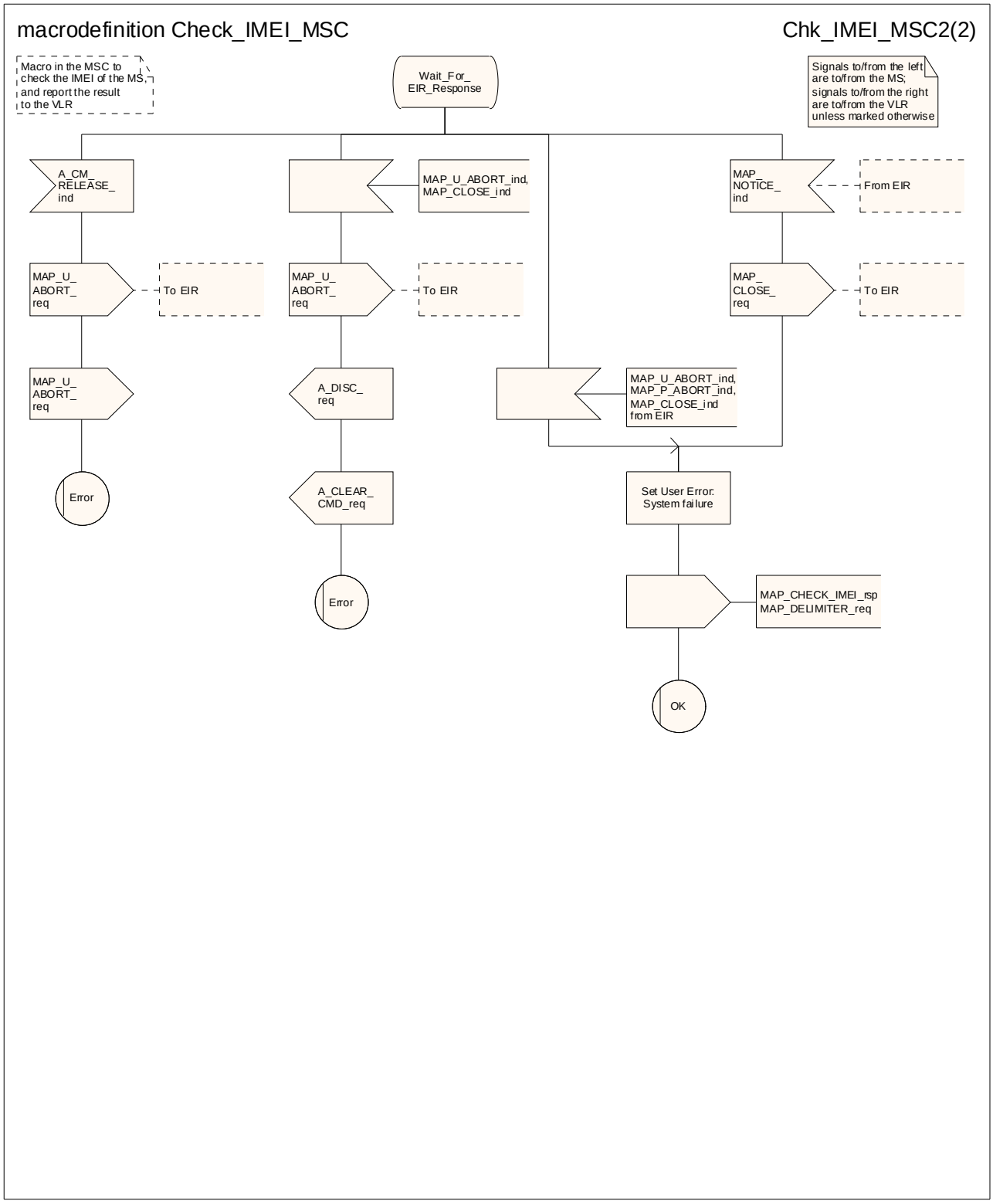


Figure 25.6/1 (sheet 2 of 2): Macro Check_IMEI_MSC

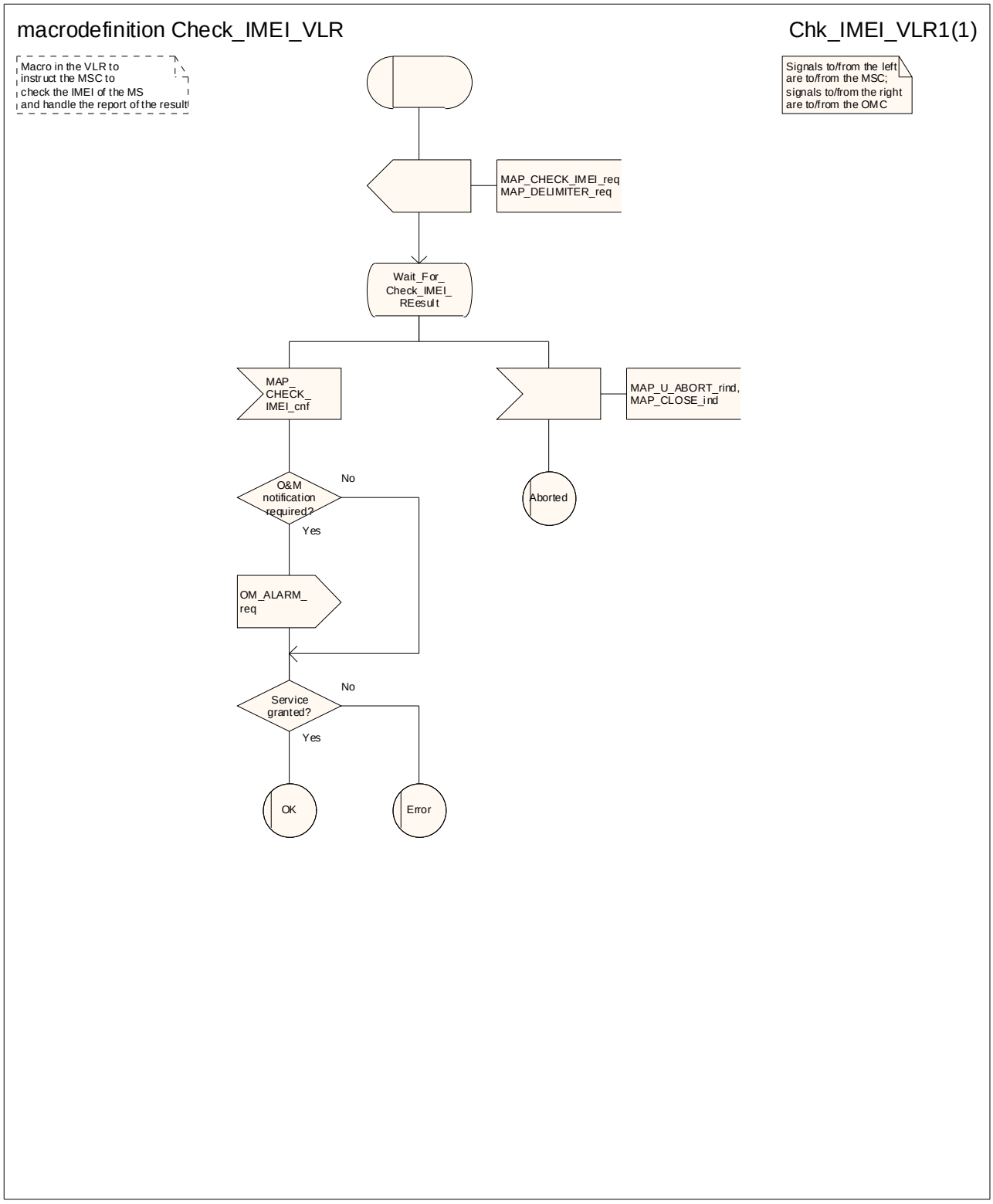


Figure 25.6/2: Macro Check_IMEI_VLR

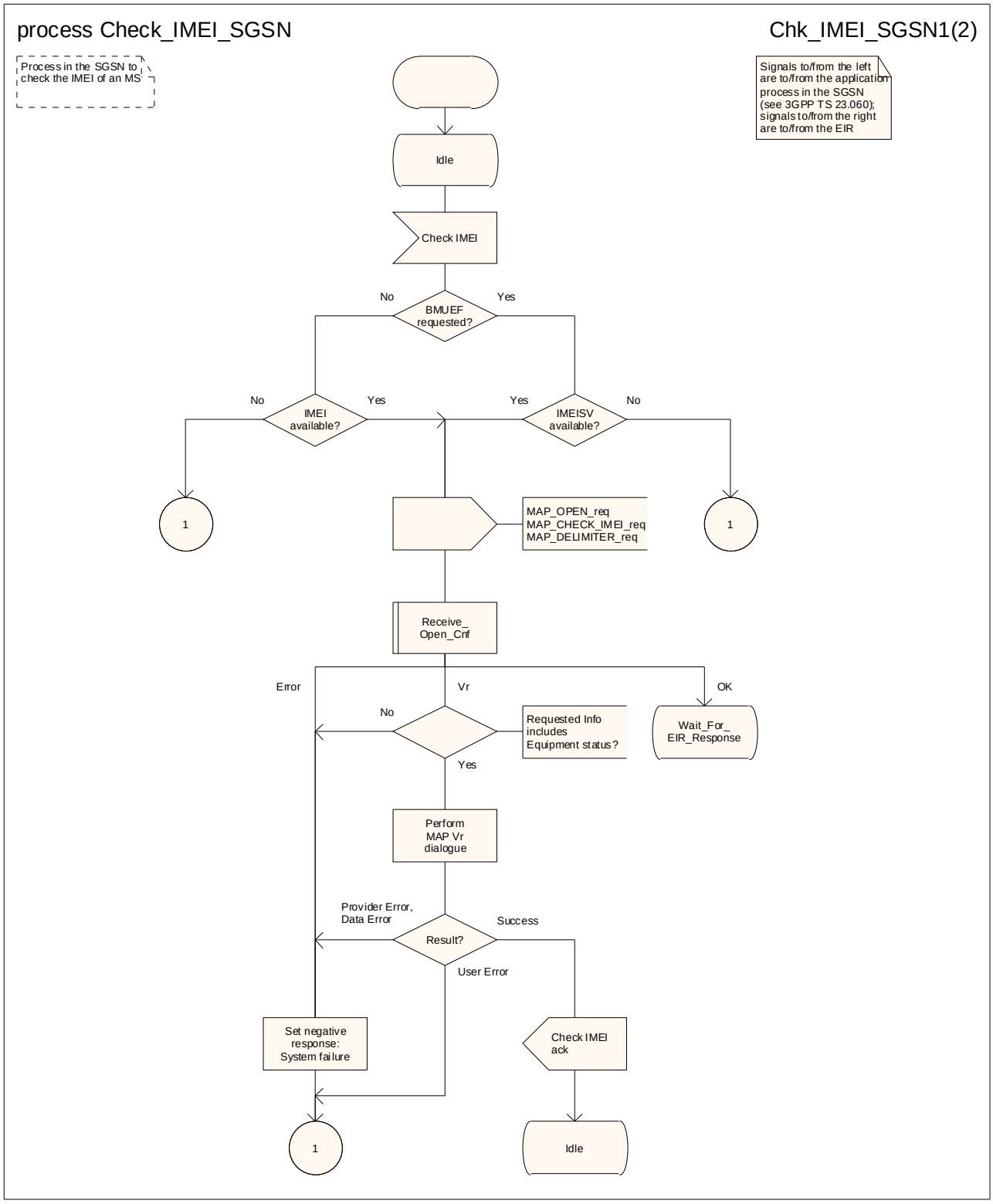


Figure 25.6/3 (sheet 1 of 2): Process Check_IMEI_SGSN

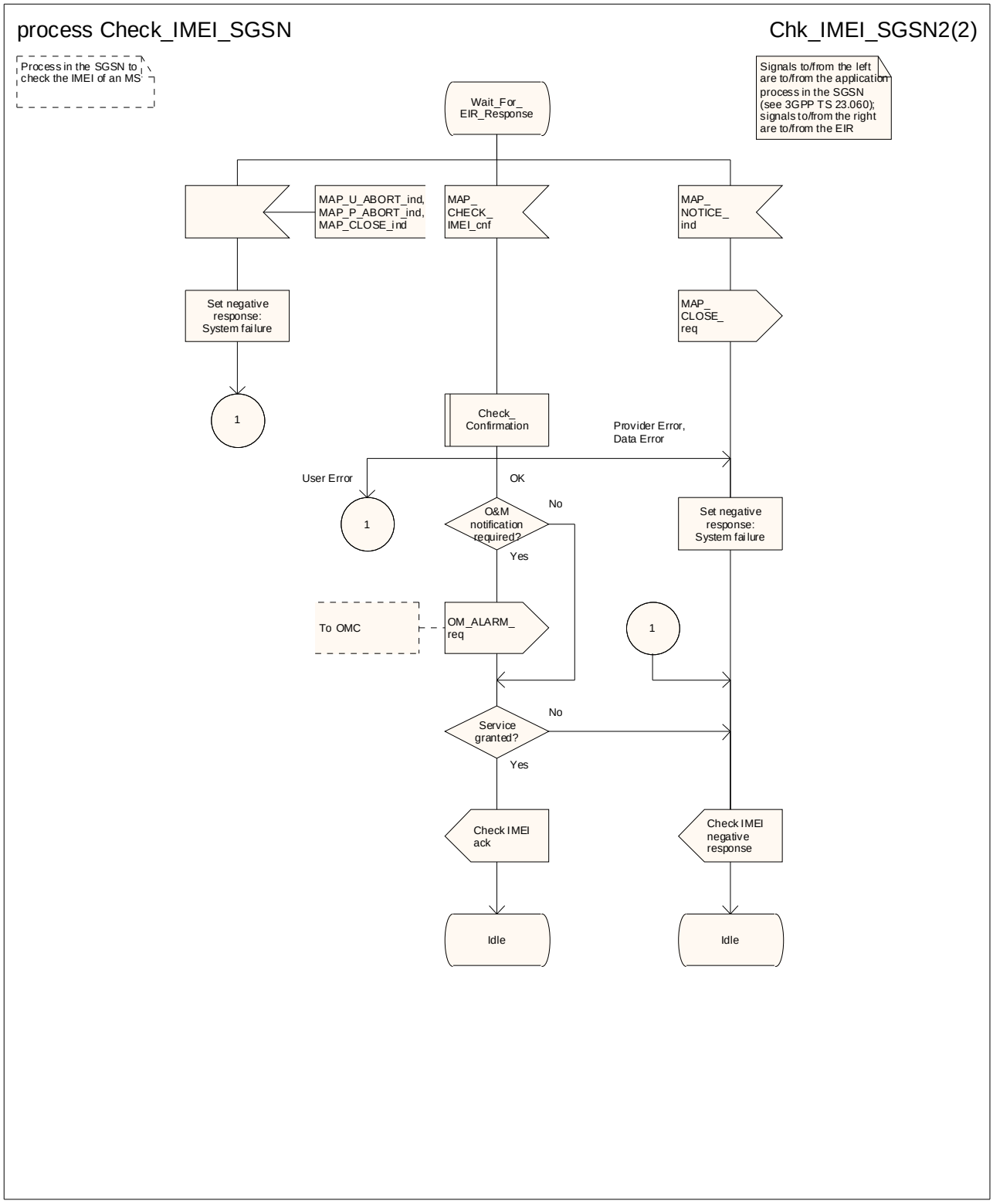


Figure 25.6/3 (sheet 2 of 2): Process Check_IMEI_SGSN

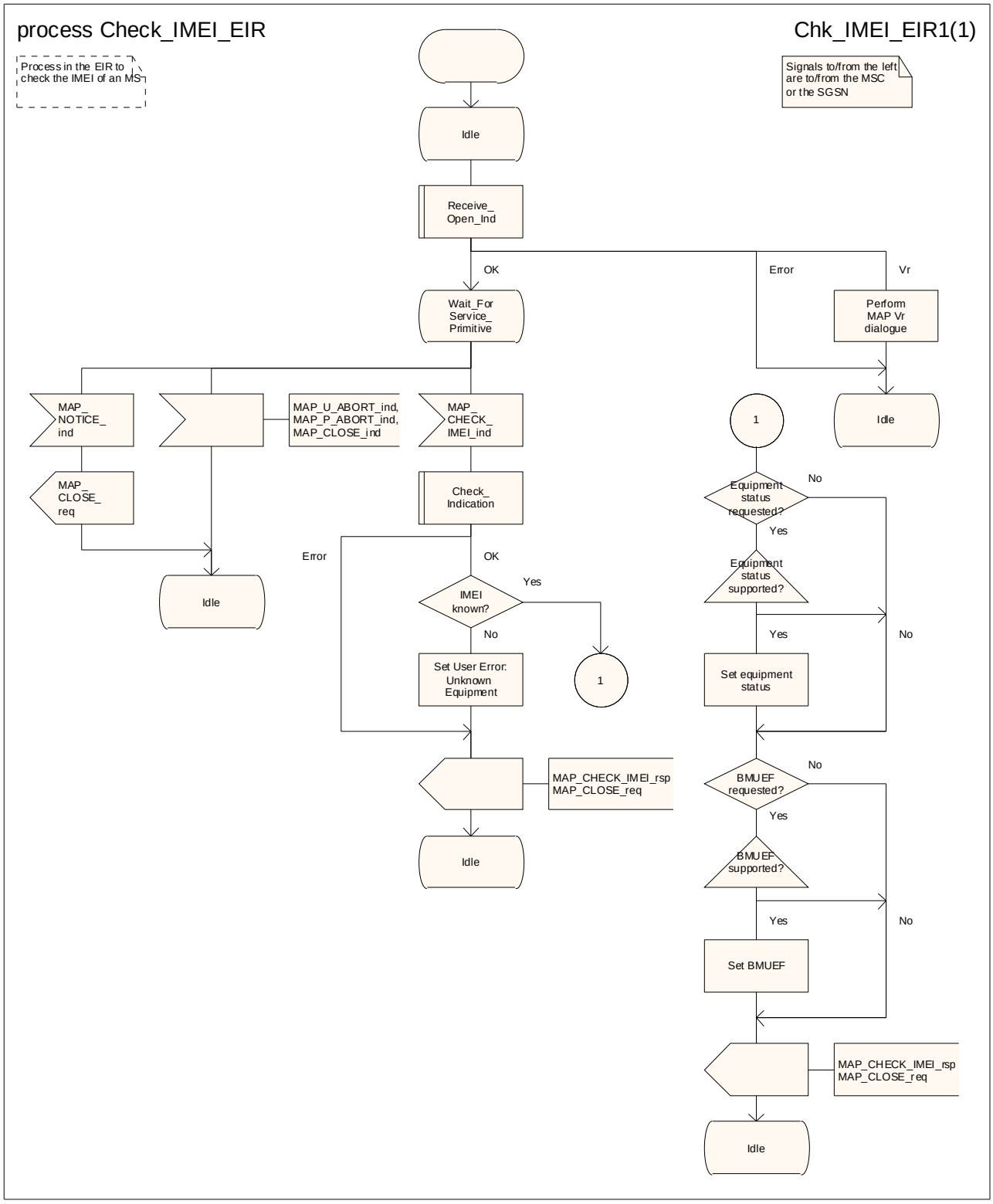


Figure 25.6/4: Process Check_IMEI_EIR

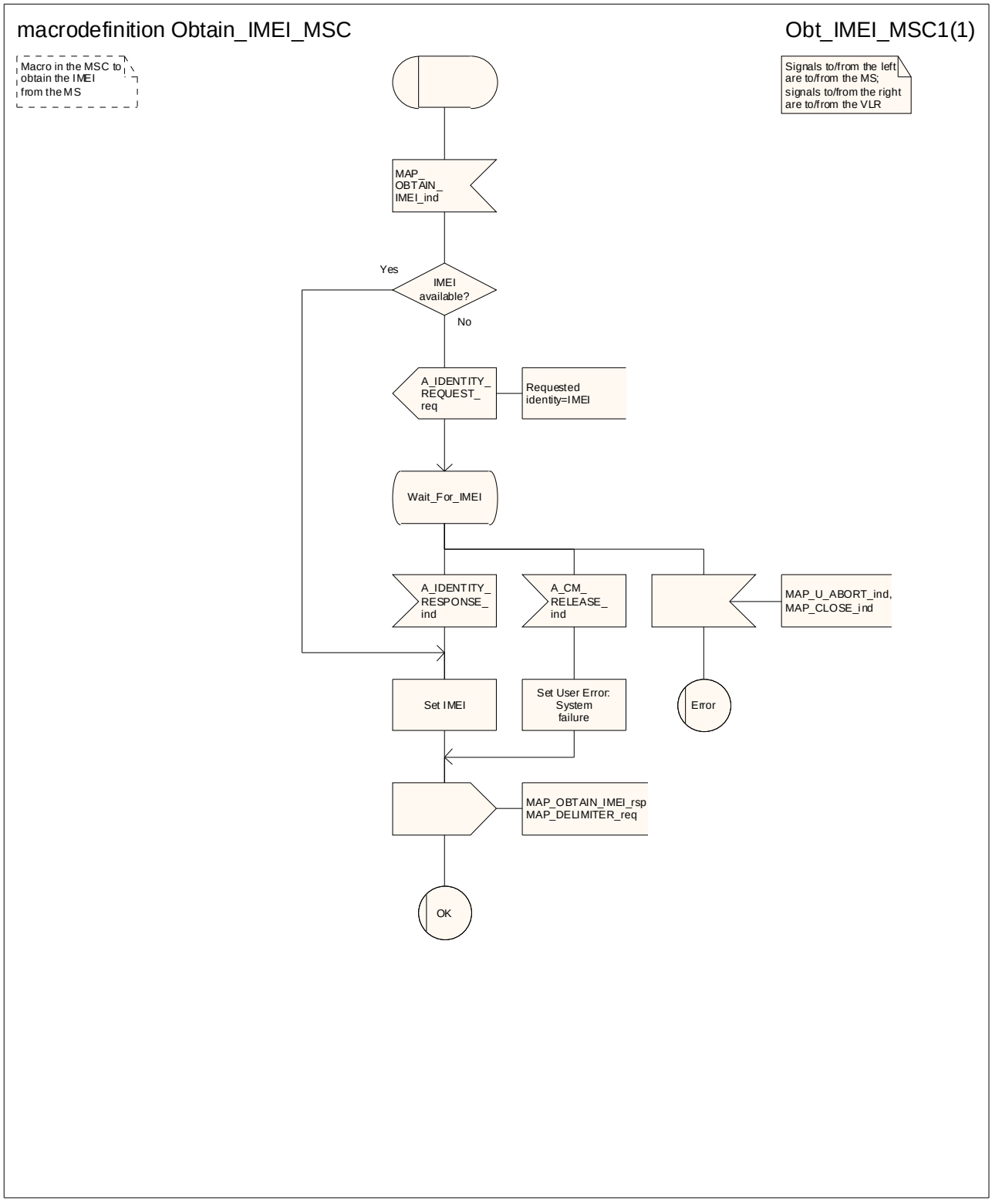


Figure 25.6/5: Macro Obtain_IMEI_MSC

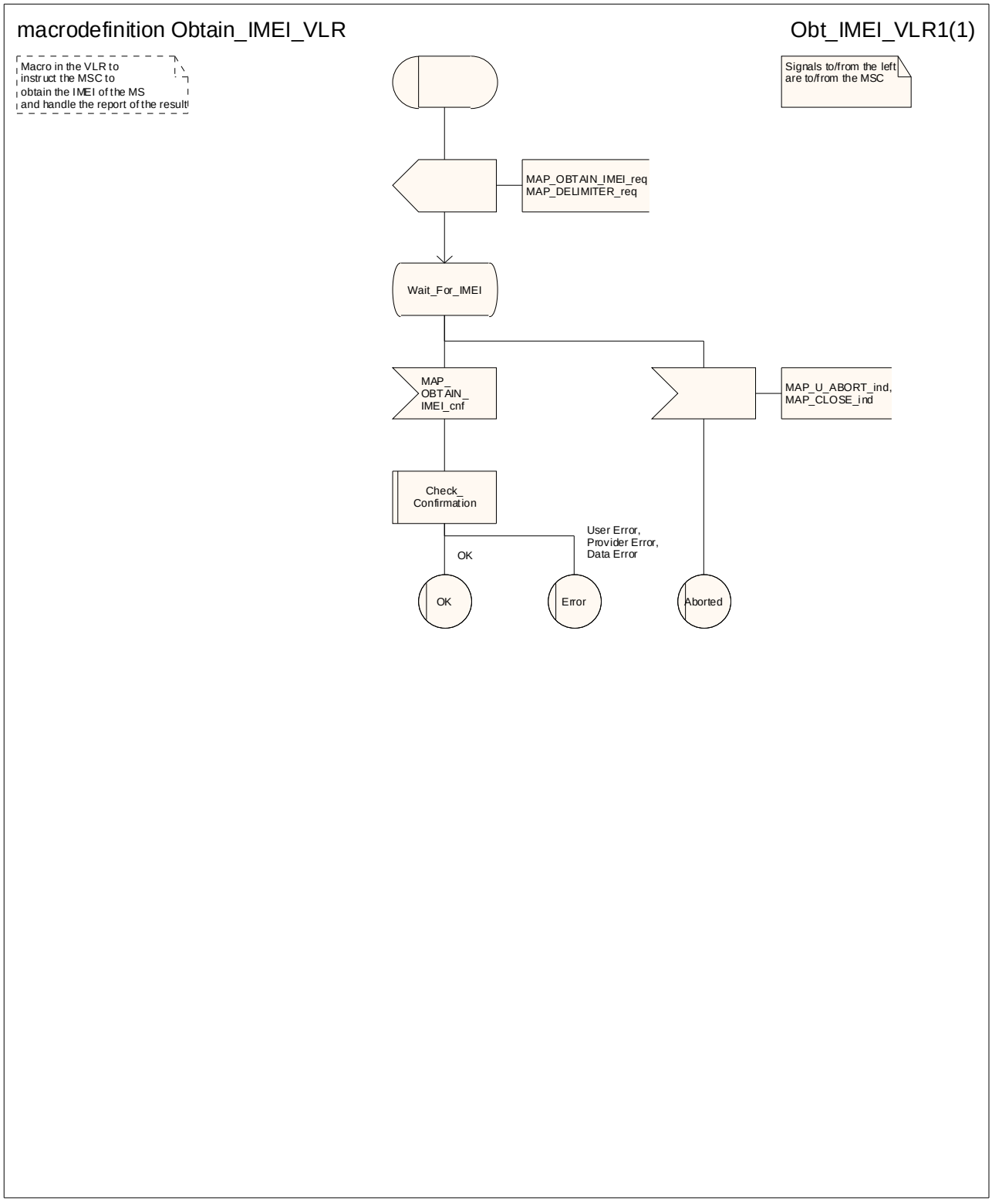


Figure 25.6/6: Macro Obtain_IMEI_VLR

25.7 Insert Subscriber Data macros and processes

25.7.1 Macro Insert_Subs_Data_VLR

This macro is used by any procedure in the VLR that triggers the reception of subscriber data (e.g. Update Location, Update VCSG Location or Restore Data).

25.7.2 Macro Insert_Subs_Data_SGSN

This macro is used by any procedure that triggers the reception of subscriber data (e.g. Update GPRS Location or Update VCSG Location).

25.7.3 Process Insert_Subs_Data_Stand_Alone_HLR

This process is used by HLR to transfer subscriber data to the VLR in a stand alone mode, i.e. in a separate dialogue. This is done whenever a change of subscriber data is performed either by the operator or by the subscriber and this change has to be reported to the VLR.

Sheet 1: The HLR may wait for each MAP_INSERT_SUBSCRIBER_DATA request to be acknowledged before it sends the next request, or it may handle the requests and the confirmations in parallel.

Sheet 1, sheet 2: If the VLR has indicated that it does not support a service or feature (e.g. Closed User Group or Advice Of Charge Charging Level) which the HLR operator regards as essential for the subscriber, the macro Wait_for_Insert_Subs_Data_Cnf takes the Replace_Service exit; the HLR sets the Roaming Restriction Due To Unsupported Feature flag to roaming restricted and sends Roaming Restriction Due To Unsupported Feature in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

Sheet 1, sheet 2: If the HLR operator does not regard the unsupported service or feature as essential for the subscriber but the macro Wait_for_Insert_Subs_Data_Cnf takes the Replace_Service exit, the HLR sends the data for a replacement service in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

Sheet 2: It is an operator option whether to repeat the download of subscriber data if the VLR returns an error response. The number of repeat attempts and the interval between them is also an operator option, depending on the error response from the VLR.

If subscriber data for CAMEL Phase 2 or later services are sent to a VLR which does not support the appropriate phase of CAMEL, the service behaviour may be unpredictable or incorrect. The HLR should therefore ensure that at the conclusion of a stand alone Insert Subscriber data procedure the data in the VLR do not require a capability that the VLR does not have. Possible mechanisms to ensure this are described in 3GPP TS 23.078 [98].

The HLR should send a Forwarded-to number which is not in E.164 international format to the VLR only when the HLR has ascertained that the VLR supports CAMEL Phase 2 or later. Thus, the ISD message containing the Forwarded-to number which is not in E.164 international format shall be sent to the VLR only if the HLR previously received confirmation from the VLR at Location Update that CAMEL Phase 2 or later is supported.

25.7.4 Process Insert_GPRS_Subs_Data_Stand_Alone_HLR

This process is used by the HLR to transfer subscriber data from the HLR to the SGSN in a stand alone mode, i.e. in a separate dialogue. This is done whenever a change of subscriber data is performed either by the operator or by the subscriber and this change has to be reported to the SGSN.

Sheet 1: The HLR may wait for each MAP_INSERT_SUBSCRIBER_DATA request to be acknowledged before it sends the next request, or it may handle the requests and the confirmations in parallel.

Sheet 1, sheet 2: If the SGSN has indicated that it does not support a service or feature which the HLR operator regards as essential for the subscriber, the macro Wait_for_Insert_GPRS_Subs_Data_Cnf takes the Replace_Service exit; the HLR sets the Roaming Restricted In SGSN Due To Unsupported Feature flag to roaming restricted and sends Roaming Restricted In SGSN Due To Unsupported Feature in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

Sheet 1, sheet 2: If the HLR operator does not regard the unsupported service or feature as essential for the subscriber but the macro Wait_for_Insert_GPRS_Subs_Data_Cnf takes the Replace_Service exit, the HLR sends the data for a replacement service in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

Sheet 2: It is an operator option whether to repeat the download of subscriber data if the SGSN returns an error response. The number of repeat attempts and the interval between them is also an operator option, depending on the error response from the SGSN.

25.7.5 Macro Wait_for_Insert_Subs_Data_Cnf

This macro is used by any process or macro that describes the handling in the HLR of the transfer of subscriber data to the VLR (e.g. Update Location or Restore Data).

25.7.6 Macro Wait_for_Insert_GPRS_Subs_Data_Cnf

This macro is used by any process or macro that describes the handling in the HLR of the transfer of subscriber data to the SGSN (e.g. Update GPRS Location).

25.7.7 Process Send_Insert_Subs_Data_HLR

This process is used by any process or macro in the HLR where a MAP_INSERT_SUBSCRIBER_DATA request is sent to the VLR or to the SGSN.

25.7.8 Process Insert_VCSG_Subs_Data_Stand_Alone_CSS

This process is used by the CSS to transfer CSG subscriber data from the CSS to the VLR or the SGSN in a stand alone mode, i.e. in a separate dialogue. This is done whenever a change of CSG subscriber data in the CSS is performed either by the operator or by the subscriber and this change has to be reported to the VLR or the SGSN.

Sheet 1: The CSS may wait for each MAP_INSERT_SUBSCRIBER_DATA request to be acknowledged before it sends the next request, or it may handle the requests and the confirmations in parallel.

Sheet 1, sheet 2: If the VLR or the SGSN has indicated that it does not support a service or feature which the CSS operator regards as essential for the subscriber, the macro Wait_for_Insert_VCSG_Subs_Data_Cnf takes the Replace_Service exit.

Sheet 1, sheet 2: If the CSS operator does not regard the unsupported service or feature as essential for the subscriber but the macro Wait_for_Insert_VCSG_Subs_Data_Cnf takes the Replace_Service exit, the CSS sends the data for a replacement service in a subsequent MAP_INSERT_SUBSCRIBER_DATA request.

Sheet 2: It is an operator option whether to repeat the download of CSG subscriber data if the VLR or theSGSN returns an error response. The number of repeat attempts and the interval between them is also an operator option, depending on the error response from the VLR or the SGSN.

25.7.9 Macro Wait_for_Insert_VCSG_Subs_Data_Cnf

This macro is used by any process or macro that describes the handling in the CSS of the transfer of CSG subscriber data to the VLR or to the SGSN (e.g. Update VCSG Location).

25.7.10 Process Send_Insert_VCSG_Subs_Data_CSS

This process is used by any process or macro in the CSS where a MAP_INSERT_SUBSCRIBER_DATA request is sent to the VLR or to the SGSN.

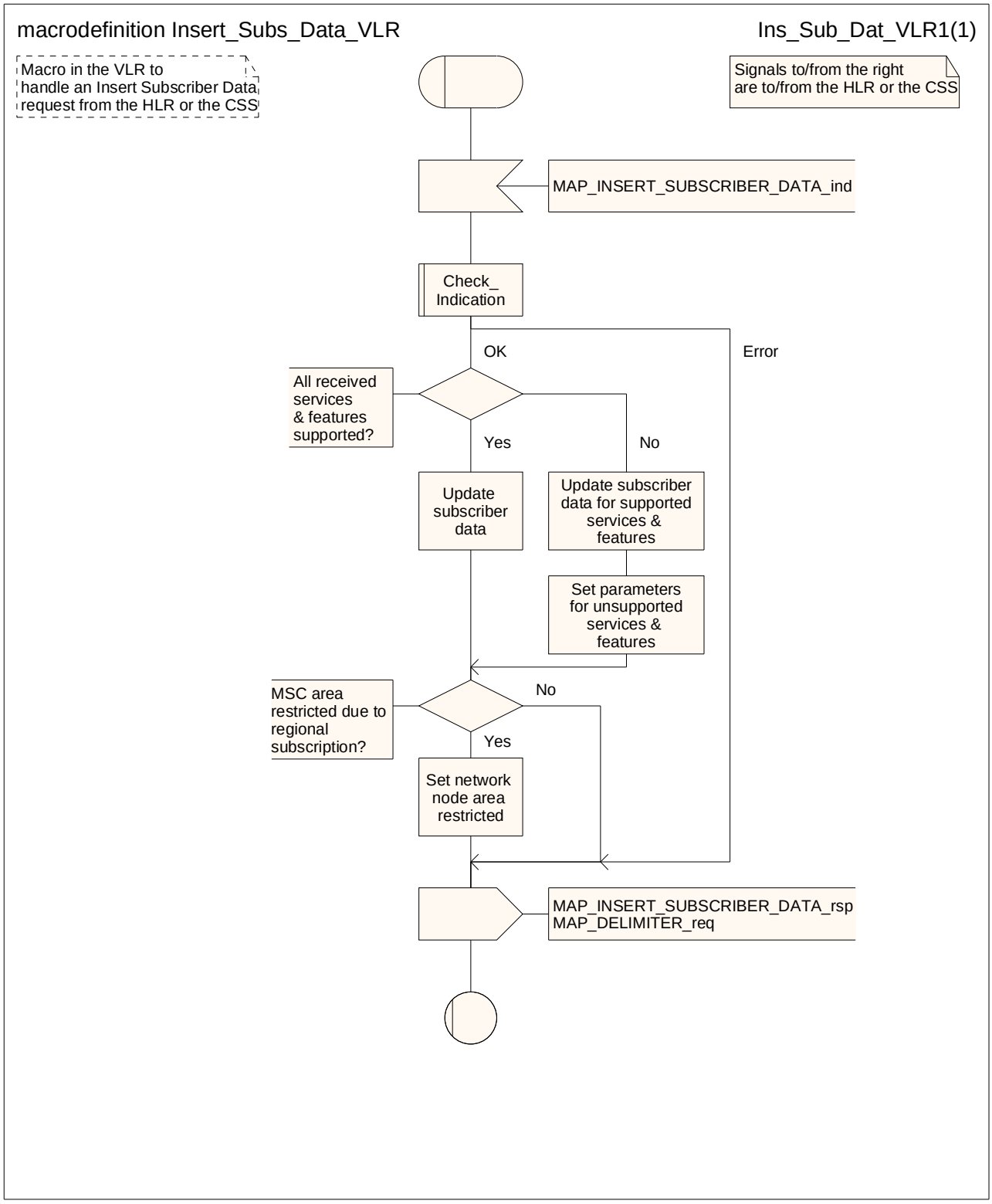


Figure 25.7/1: Macro Insert_Subs_Data_VLR

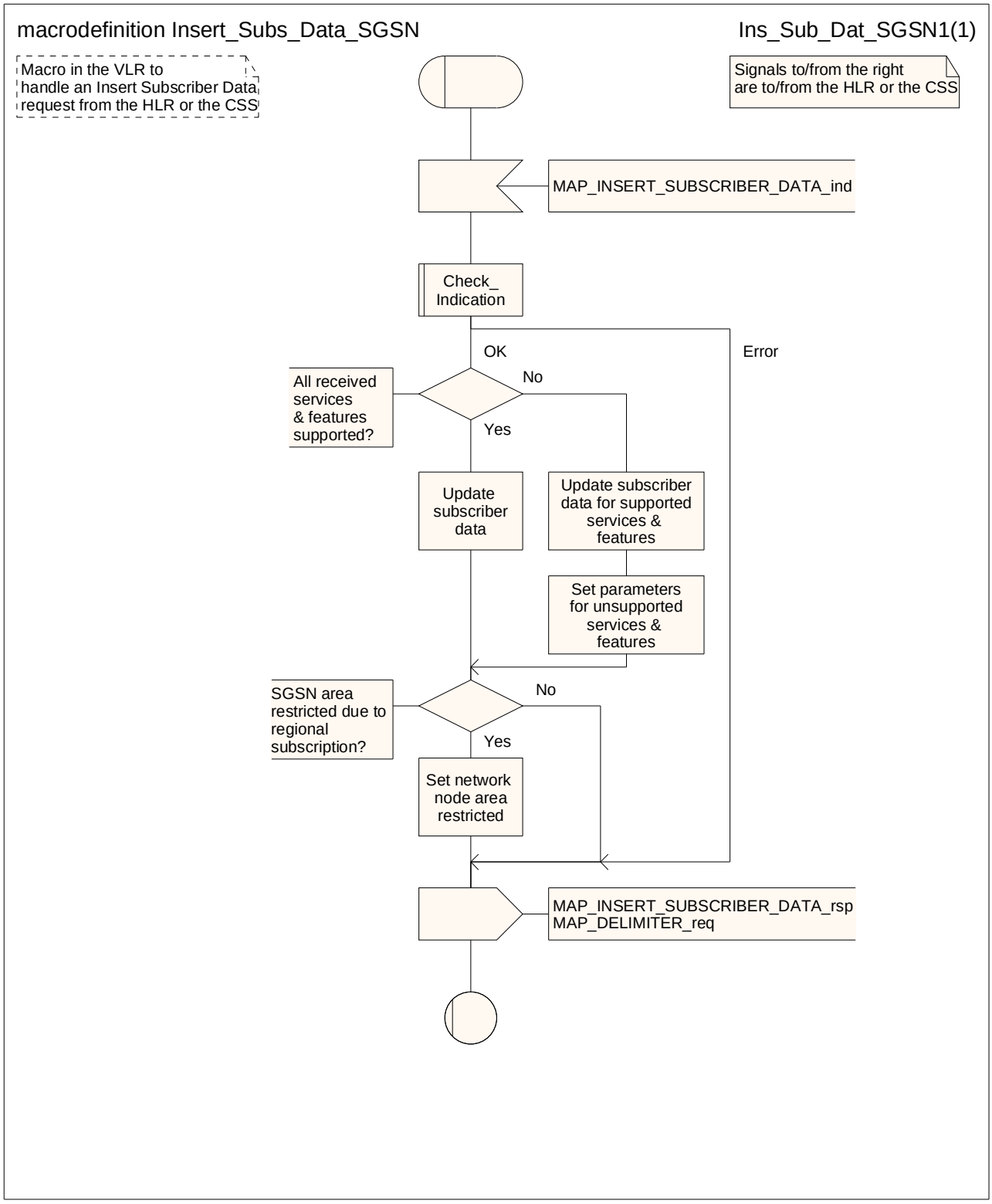


Figure 25.7/2: Macro Insert_Subs_Data_SGSN

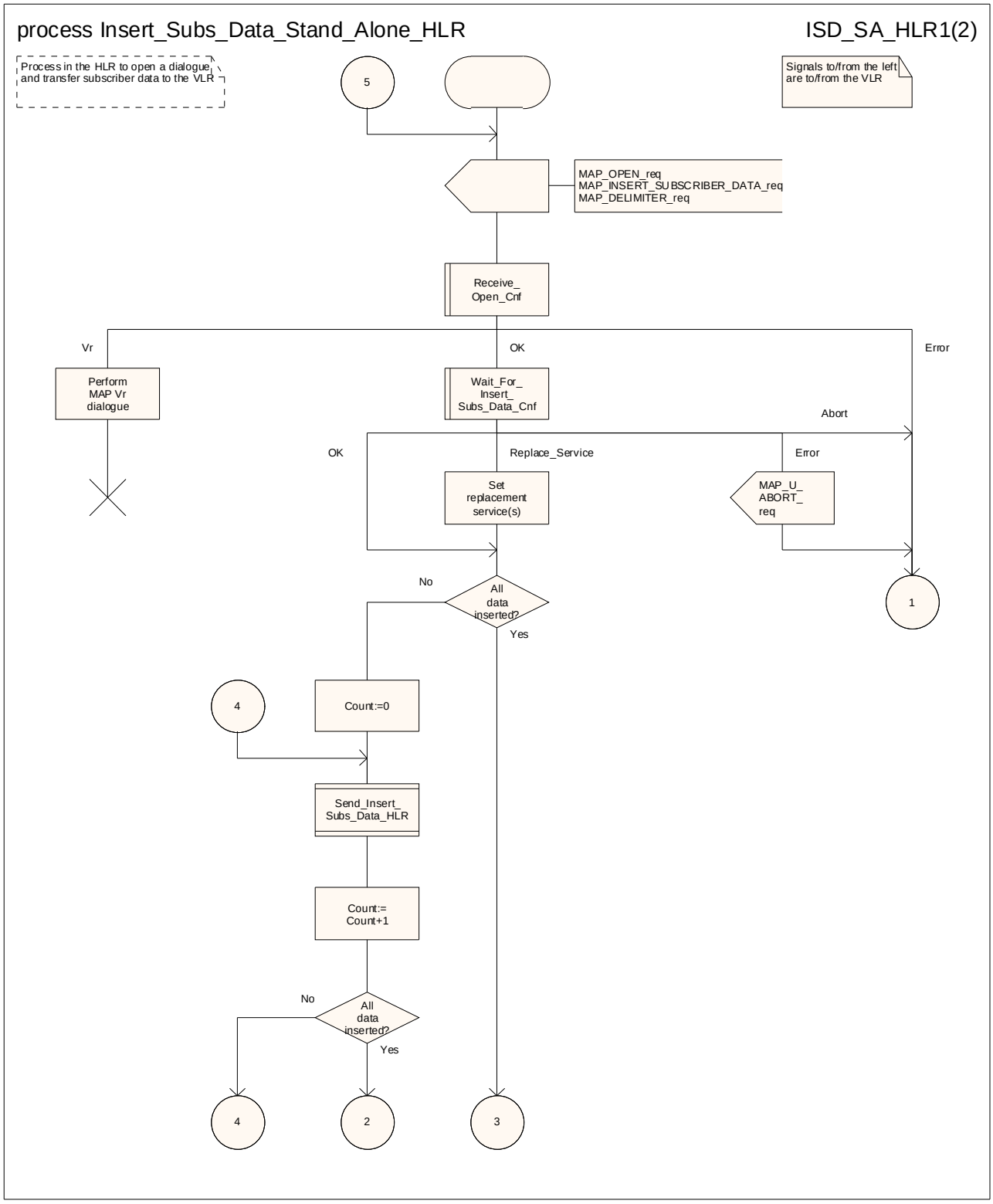


Figure 25.7/3 (sheet 1 of 2): Process Insert_Subs_Data_Stand_Alone_HLR

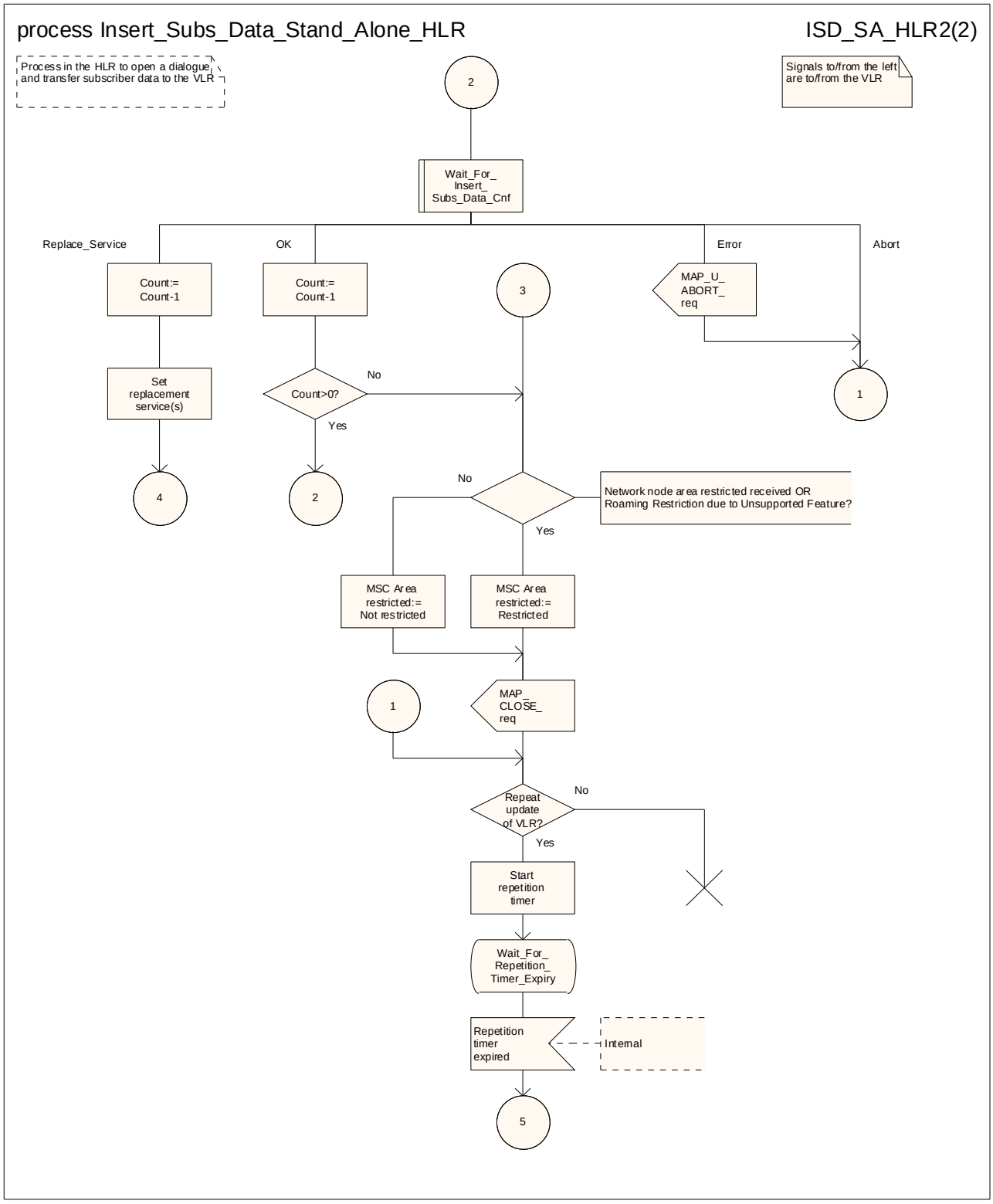


Figure 25.7/3 (sheet 2 of 2): Process Insert_Subs_Data_Stand_Alone_HLR

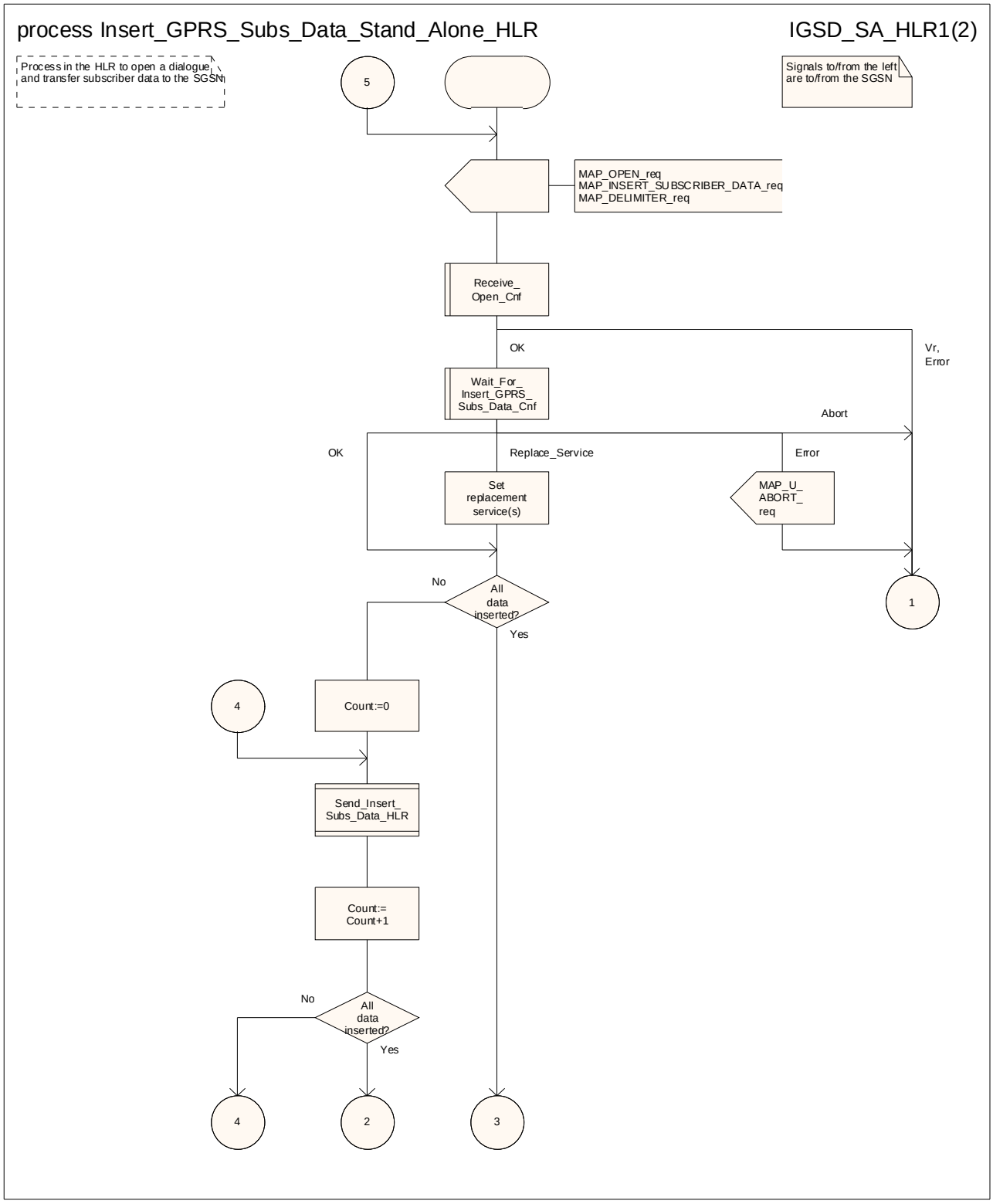


Figure 25.7/4 (sheet 1 of 2): Process Insert_GPRS_Subs_Data_Stand_Alone_HLR

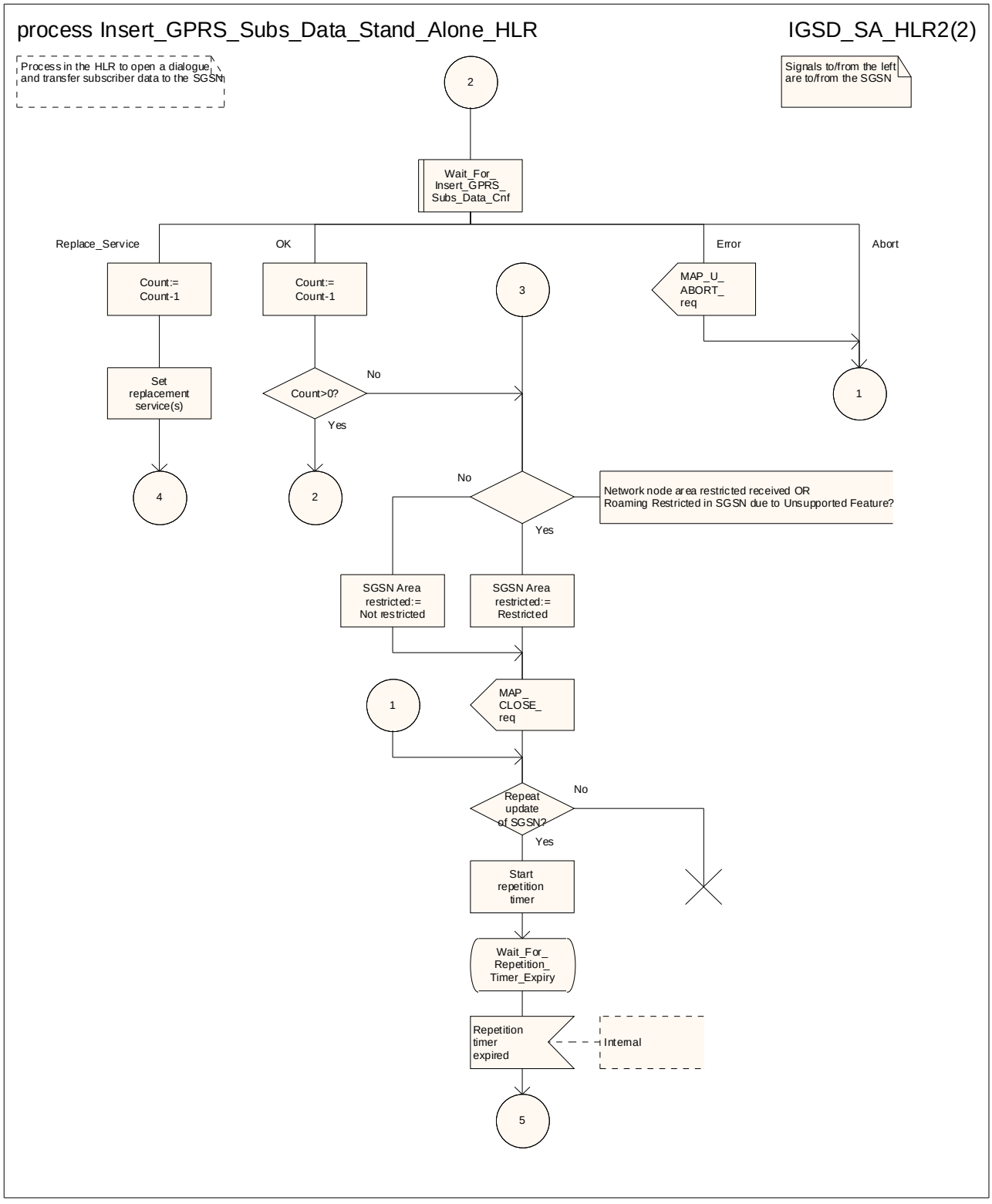


Figure 25.7/4 (sheet 2 of 2): Process Insert_GPRS_Subs_Data_Stand_Alone_HLR

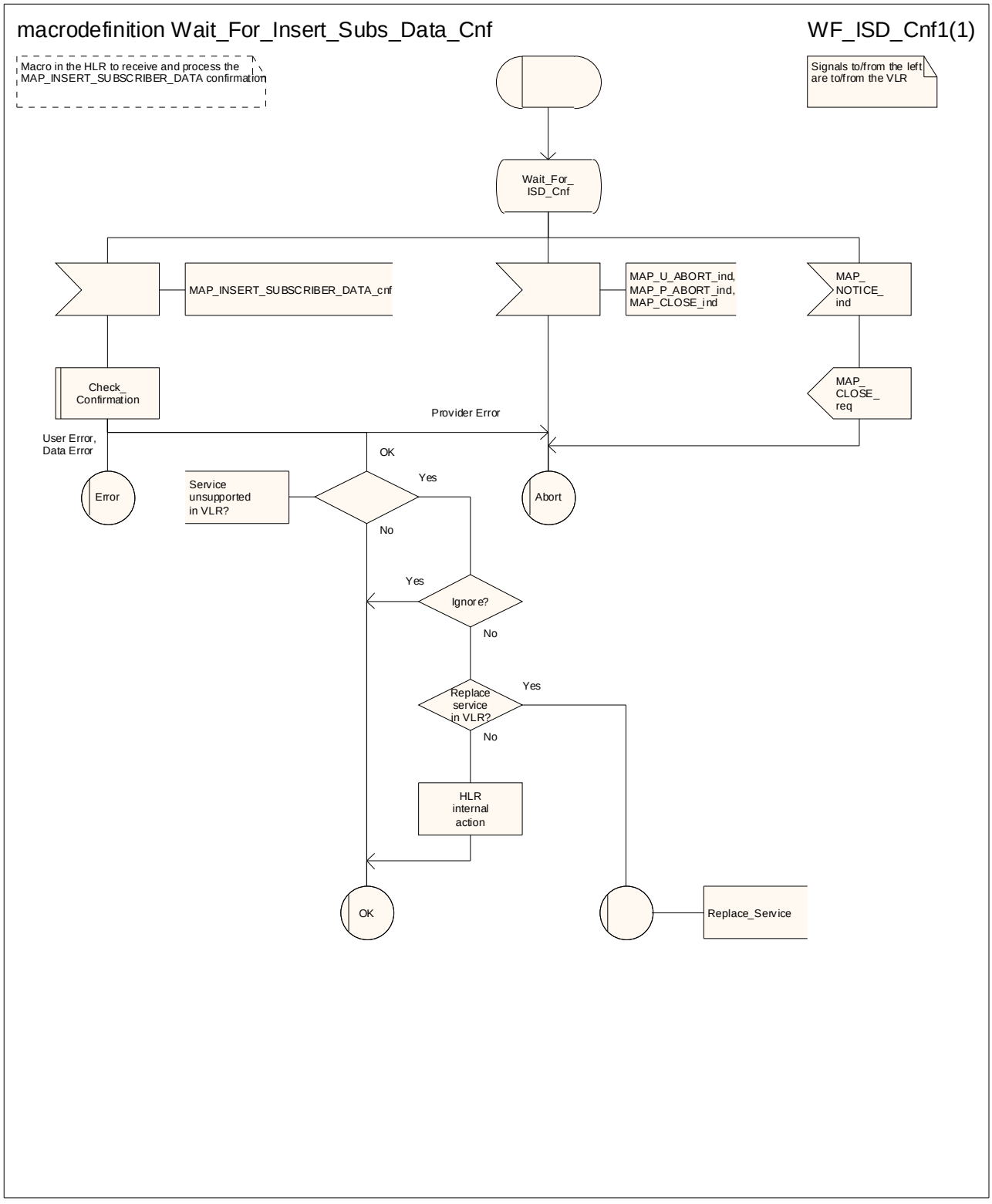


Figure 25.7/5: Macro Wait_for_Insert_Subs_Data_Cnf

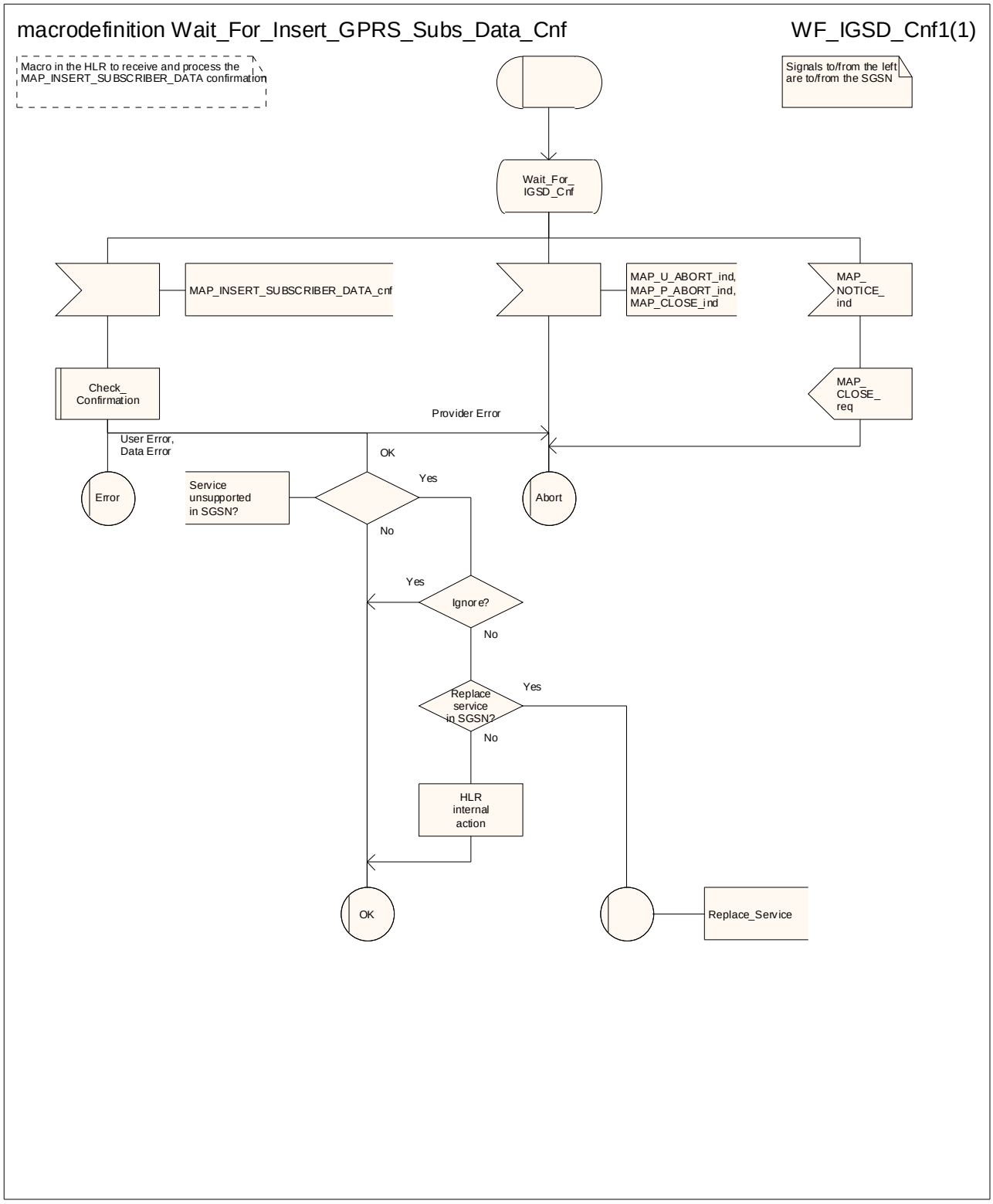


Figure 25.7/6: Macro Wait_for_Insert_GPRS_Subs_Data_Cnf

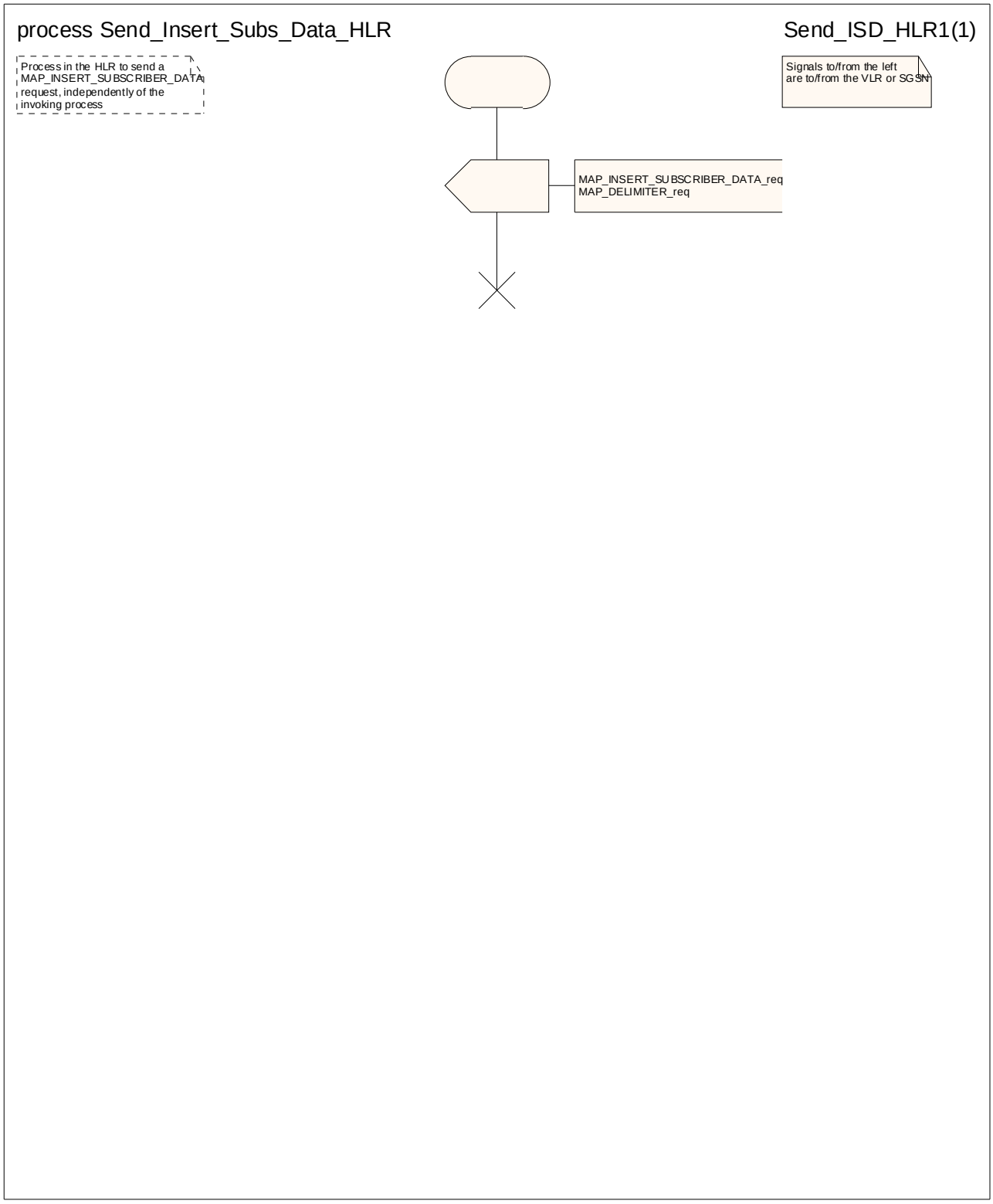


Figure 25.7/7: Process Send_Insert_Subs_Data_HLR

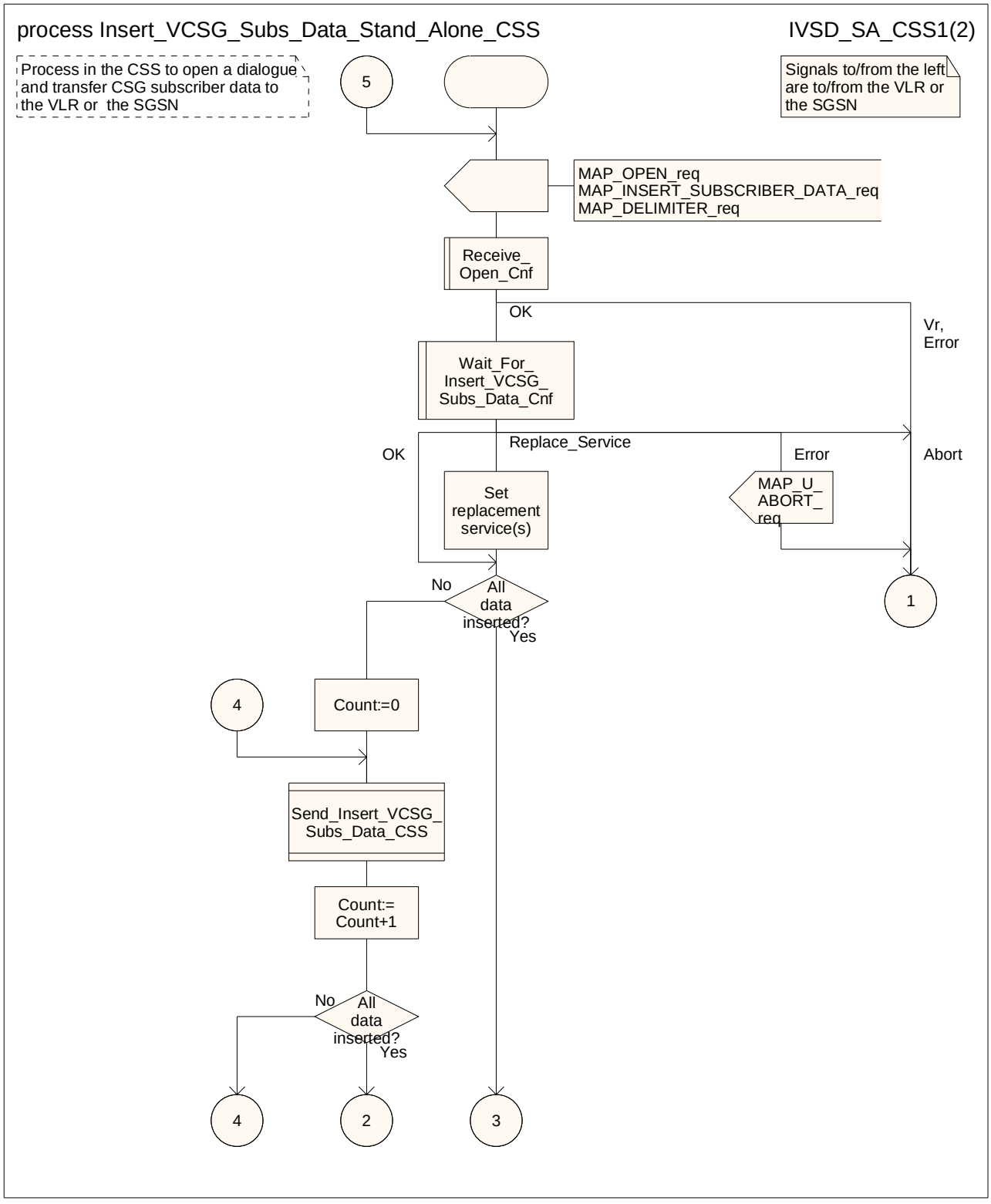


Figure 25.7/8 (sheet 1 of 2): Process Insert_VCSG_Subs_Data_Stand_Alone_CSS

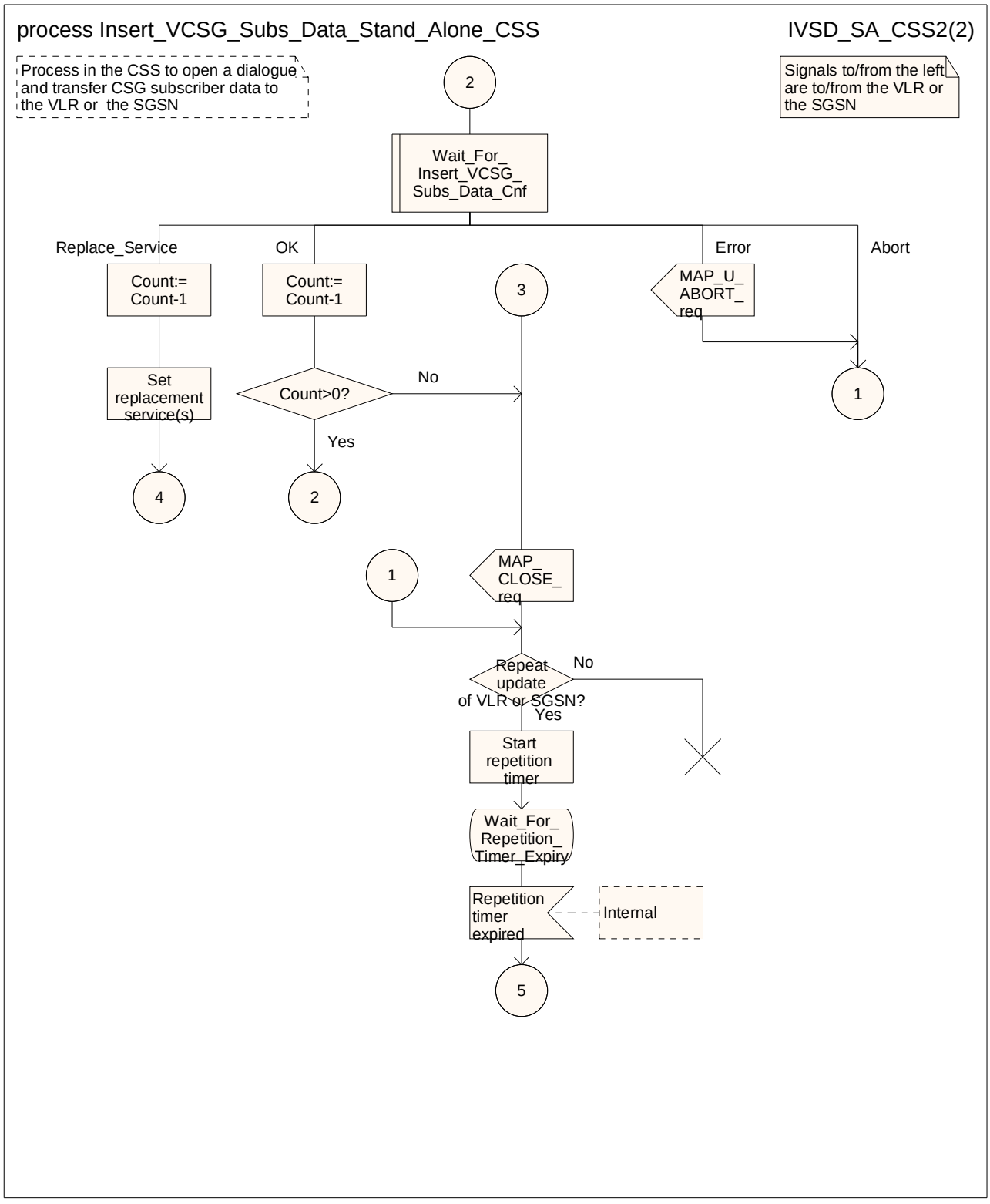


Figure 25.7/8 (sheet 2 of 2): Process Insert_VCSG_Subs_Data_Stand_Alone_CSS

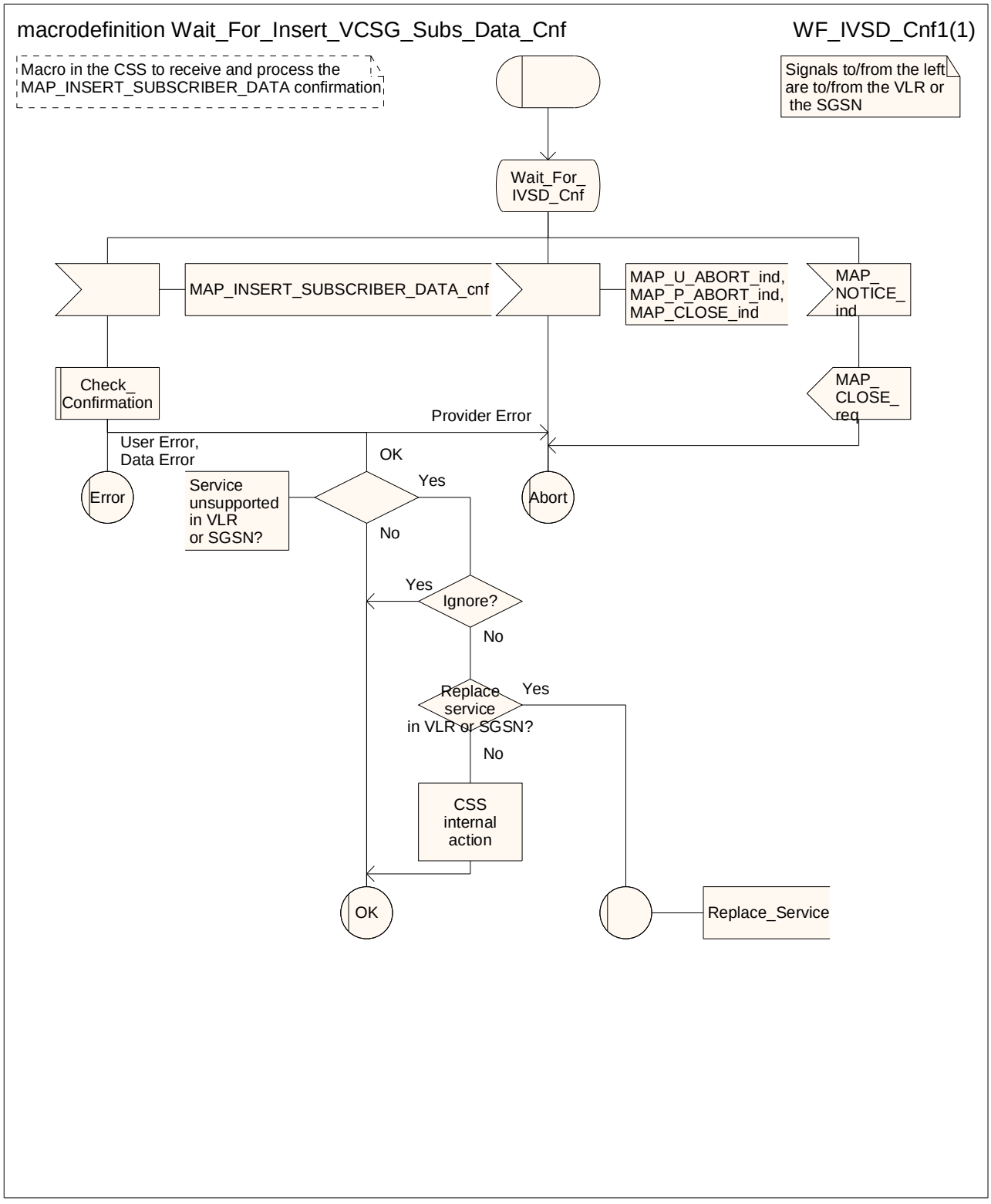


Figure 25.7/9: Macro Wait_for_Insert_VCSG_Subs_Data_Cnf

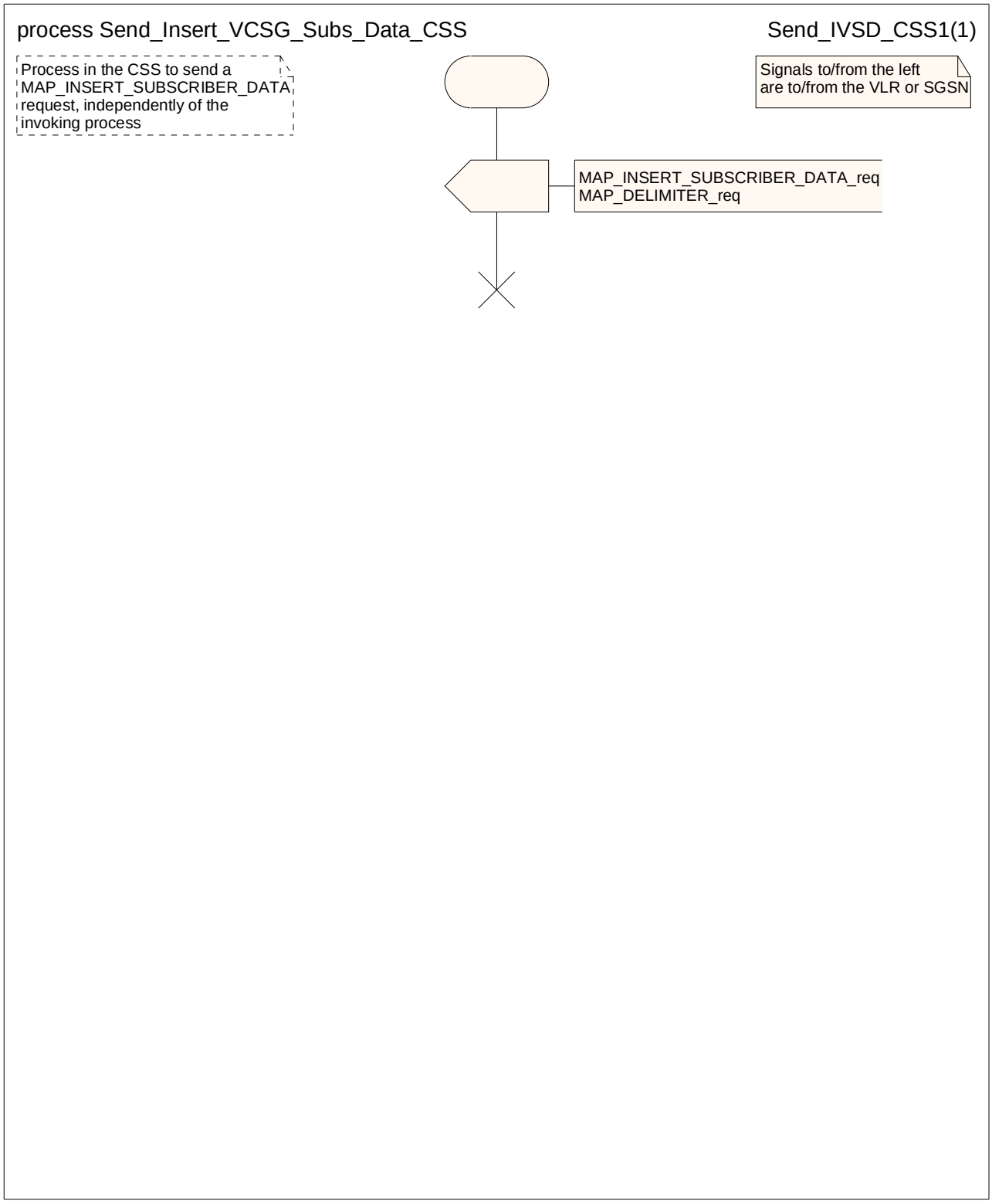


Figure 25.7/10: Process Send_Insert_VCSG_Subs_Data_CSS

25.8 Request IMSI Macros

25.8.1 Macro Obtain_IMSI_MSC

This macro describes the handling of the request received from the VLR to provide the IMSI of a subscriber (e.g. at Location Updating).

25.8.2 Macro Obtain_IMSI_VLR

This macro describes the way VLR requests the MSC the IMSI of a subscriber (e.g. at Location Updating).

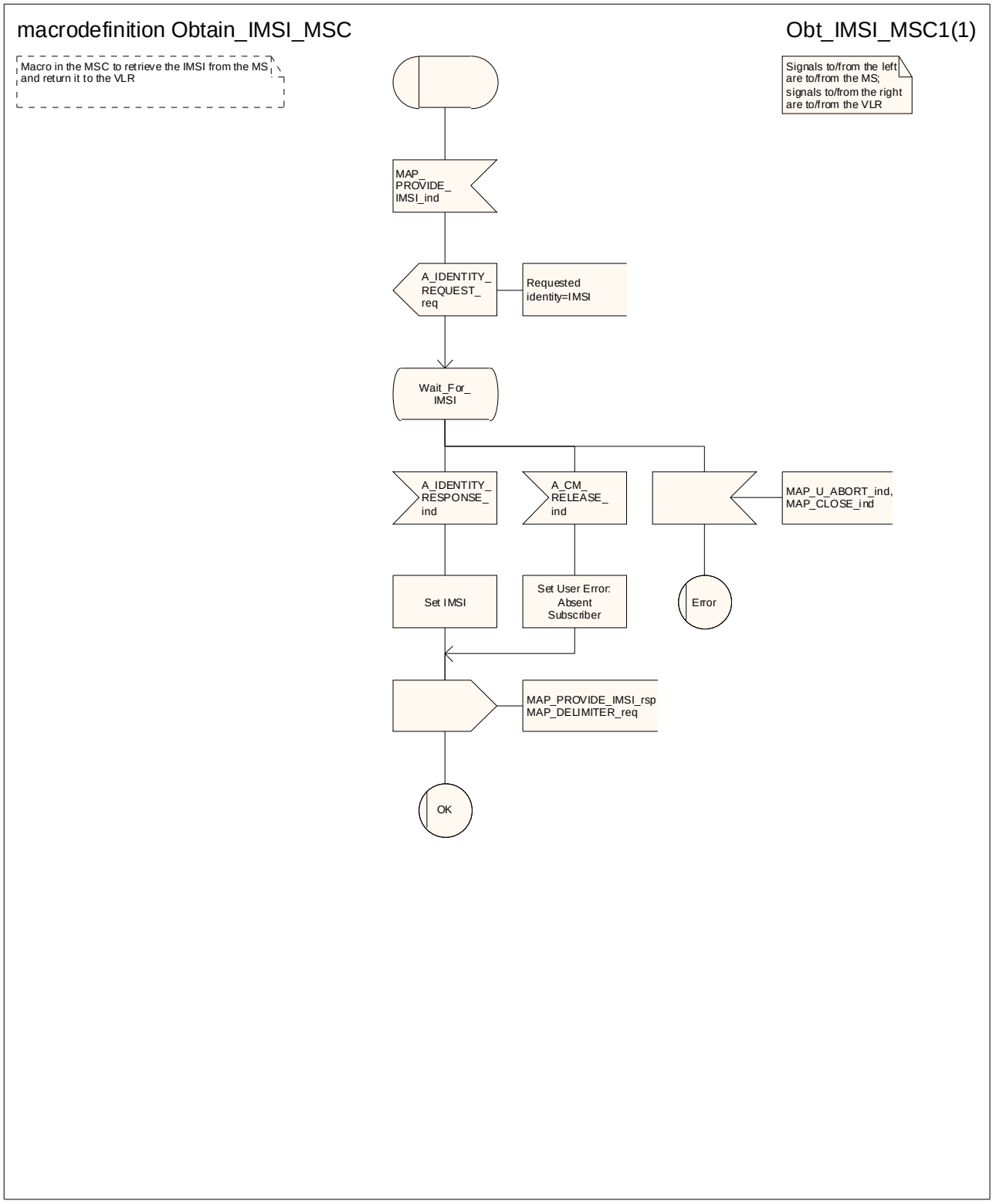


Figure 25.8/1: Macro Obtain_IMSI_MSC

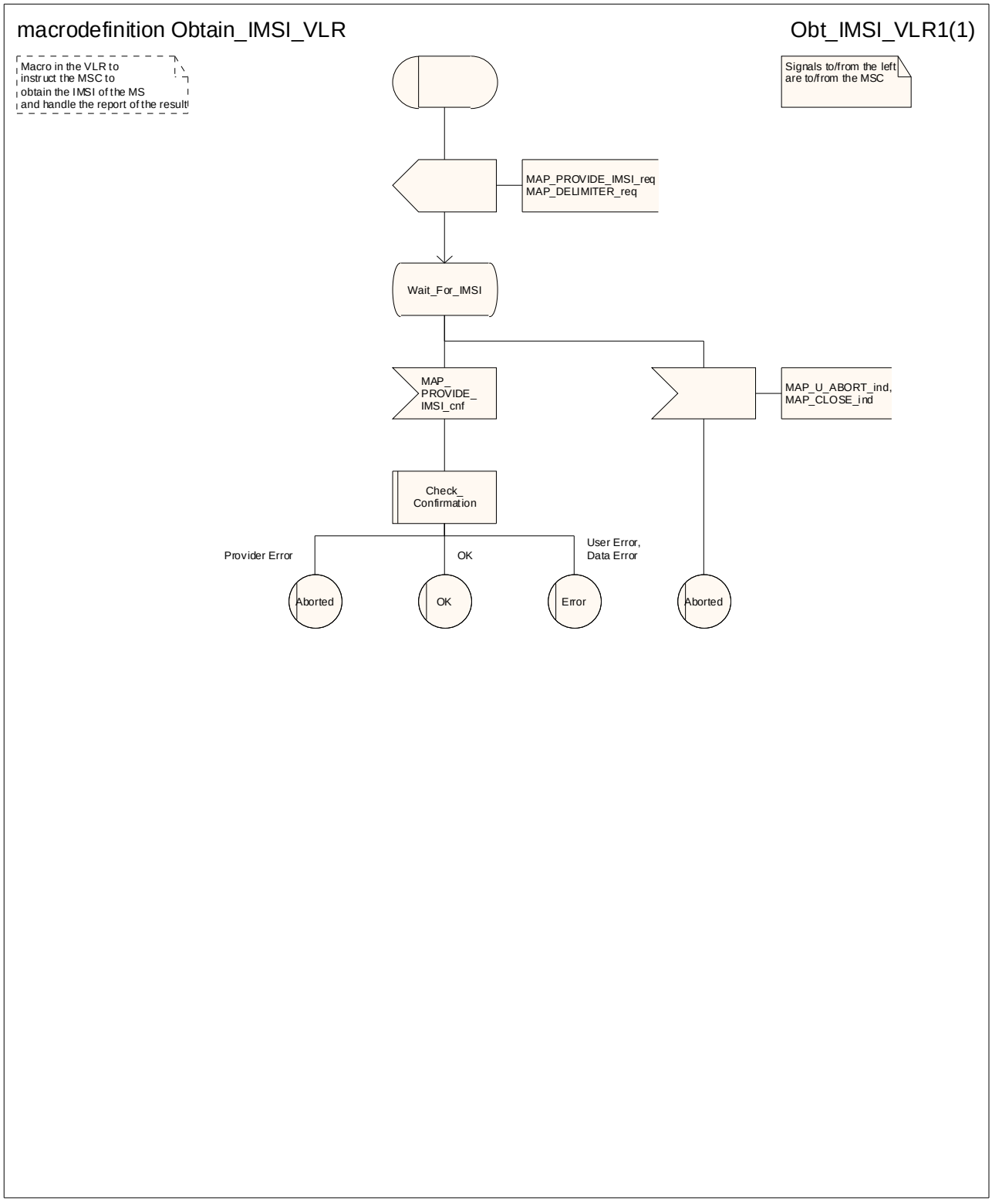


Figure 25.8/2: Macro Obtain_IMSI_VLR

25.9 Tracing macros

25.9.1 Macro Trace_Subscriber_Activity_MSC

This macro shows the handling in the MSC for a request from the VLR to trace the activity of a subscriber.

25.9.2 Macro Trace_Subscriber_Activity_VLR

This macro is called during the handling of subscriber activity in the VLR to activate tracing if necessary.

25.9.3 Macro Trace_Subscriber_Activity_SGSN

This macro is called during the handling of subscriber activity in the SGSN to activate tracing if necessary.

25.9.4 Macro Activate_Tracing_VLR

This macro shows the handling in the VLR for a request from the HLR to activate tracing for a subscriber.

25.9.5 Macro Activate_Tracing_SGSN

This macro shows the handling in the SGSN for a request from the HLR to activate tracing for a subscriber.

25.9.6 Macro Control_Tracing_With_VLR_HLR

This macro shows the handling in the HLR to activate tracing in the VLR if it is required during a dialogue between the VLR and the HLR

25.9.7 Macro Control_Tracing_With_SGSN_HLR

This macro shows the handling in the HLR to activate tracing in the SGSN if it is required during a dialogue between the SGSN and the HLR

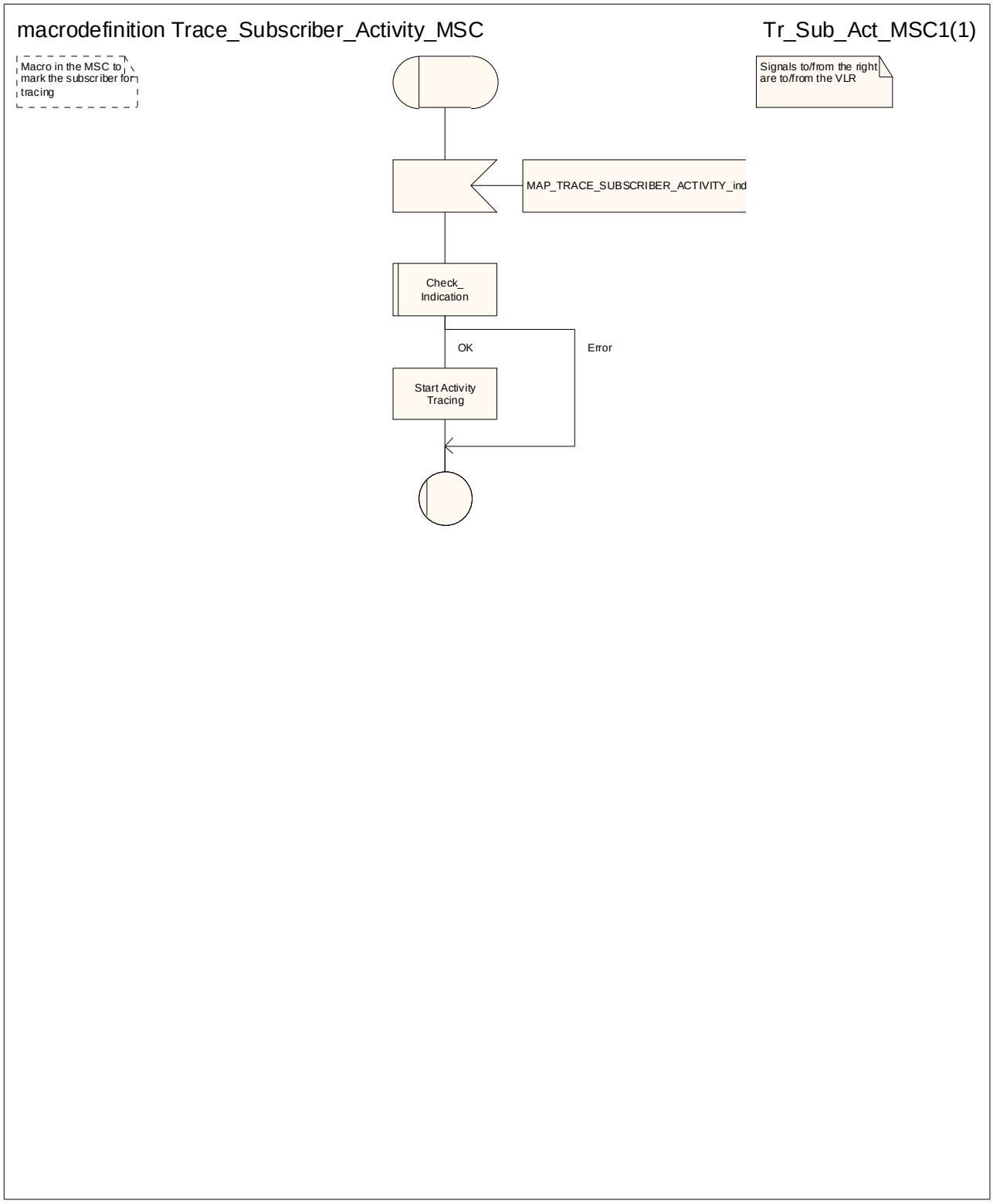


Figure 25.9/1: Macro Trace_Subscriber_Activity_MSC

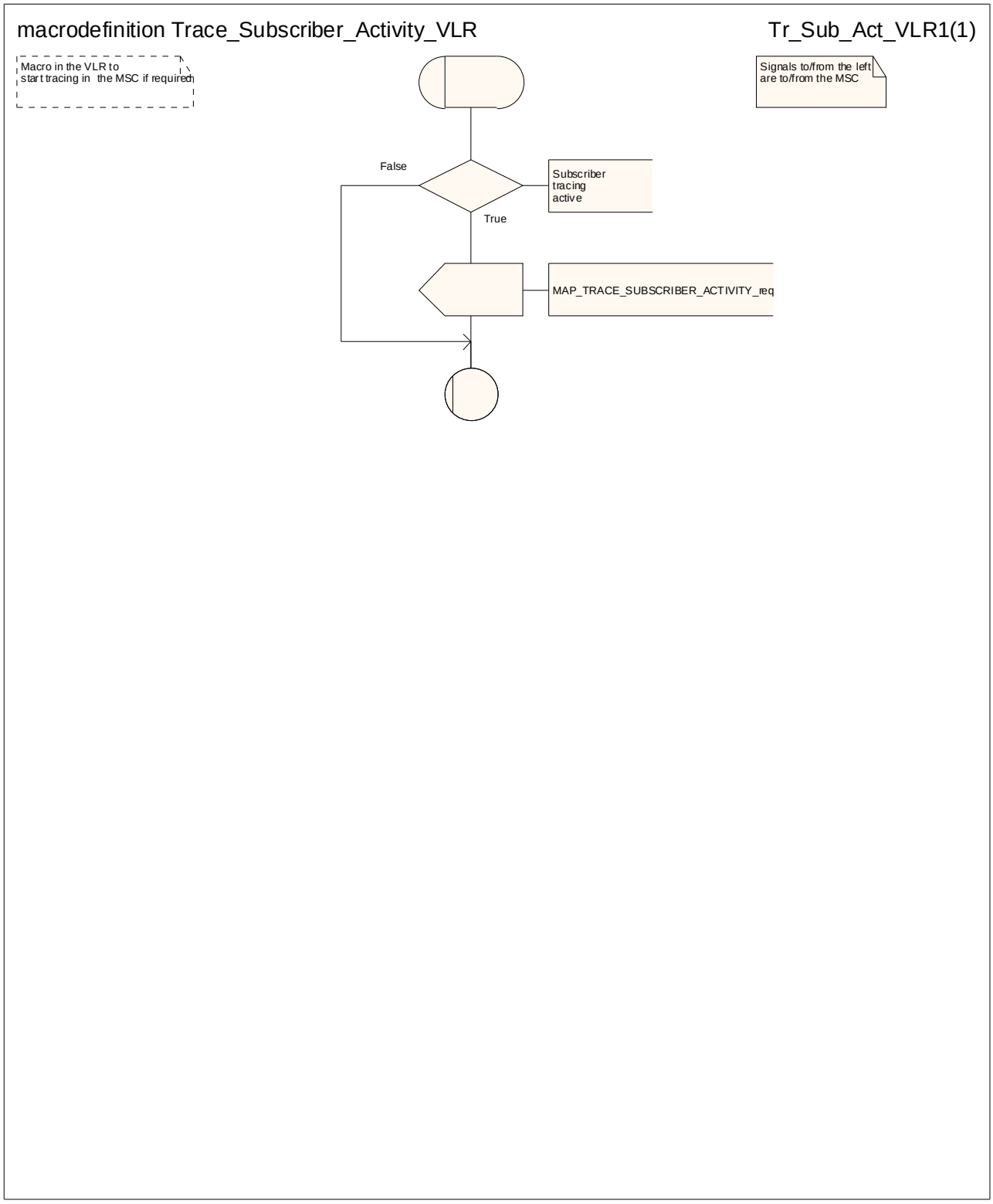


Figure 25.9/2: Macro Trace_Subscriber_Activity_VLR

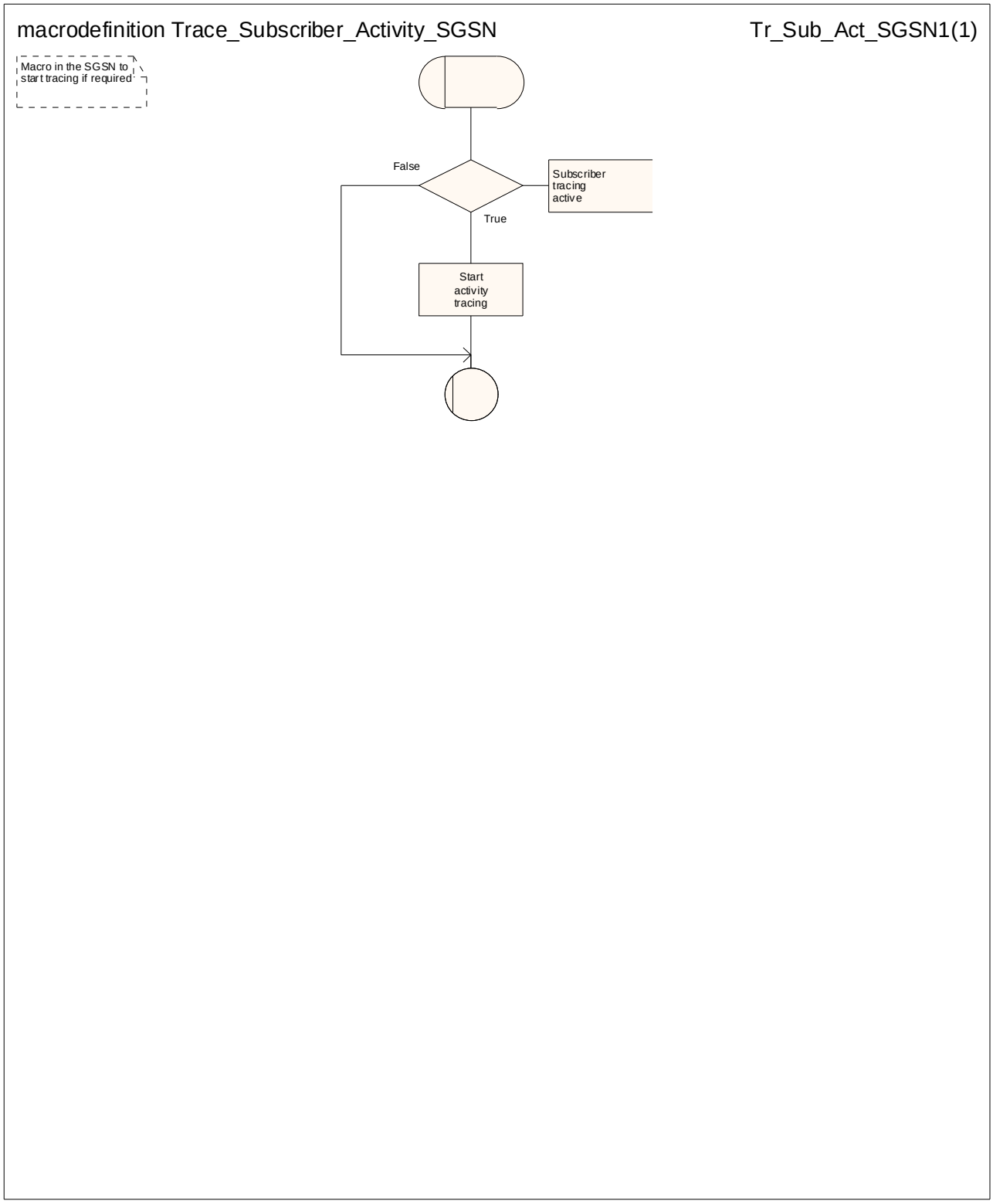


Figure 25.9/3: Macro Trace_Subscriber_Activity_SGSN

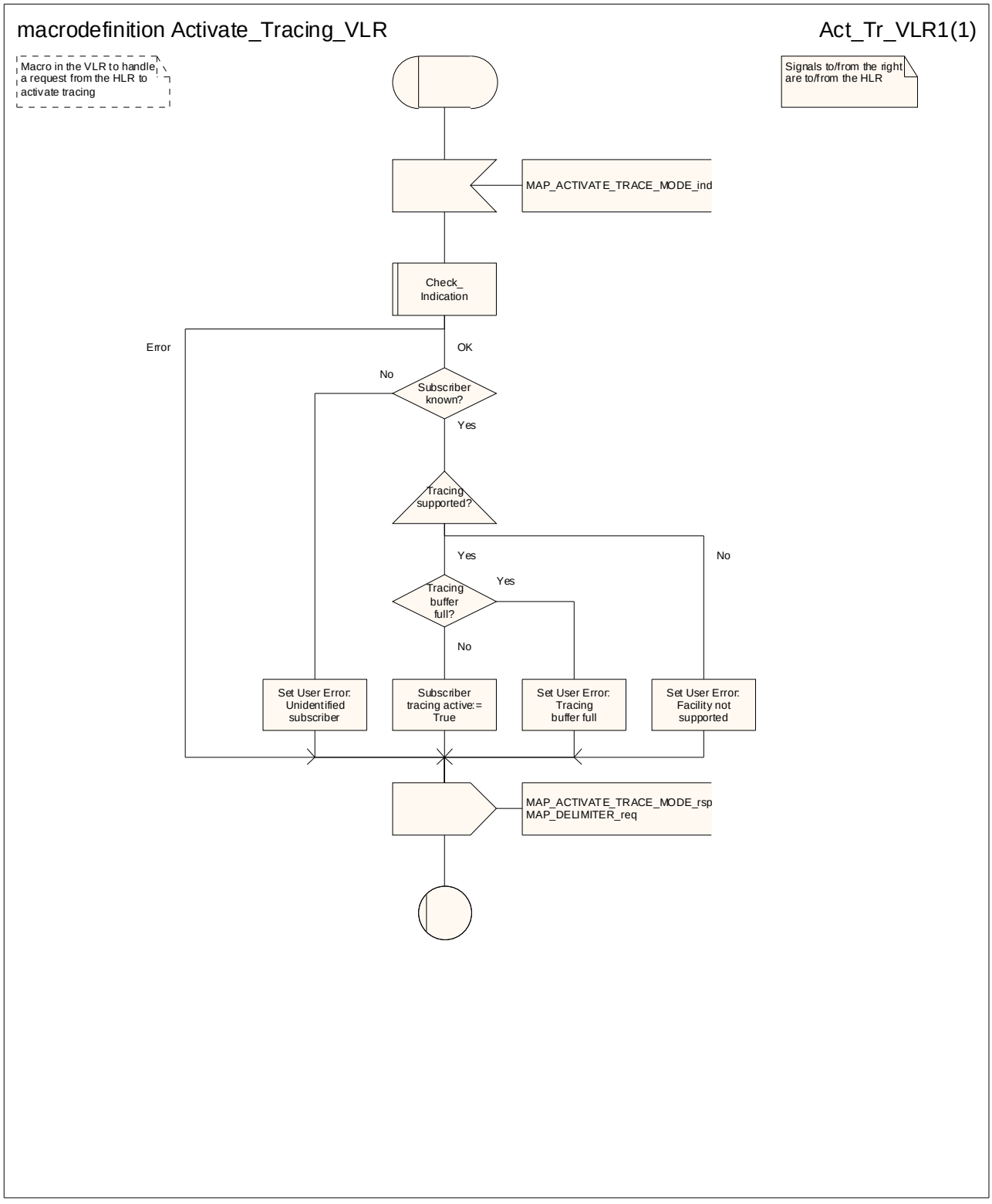


Figure 25.9/4: Macro Activate_Tracing_VLR

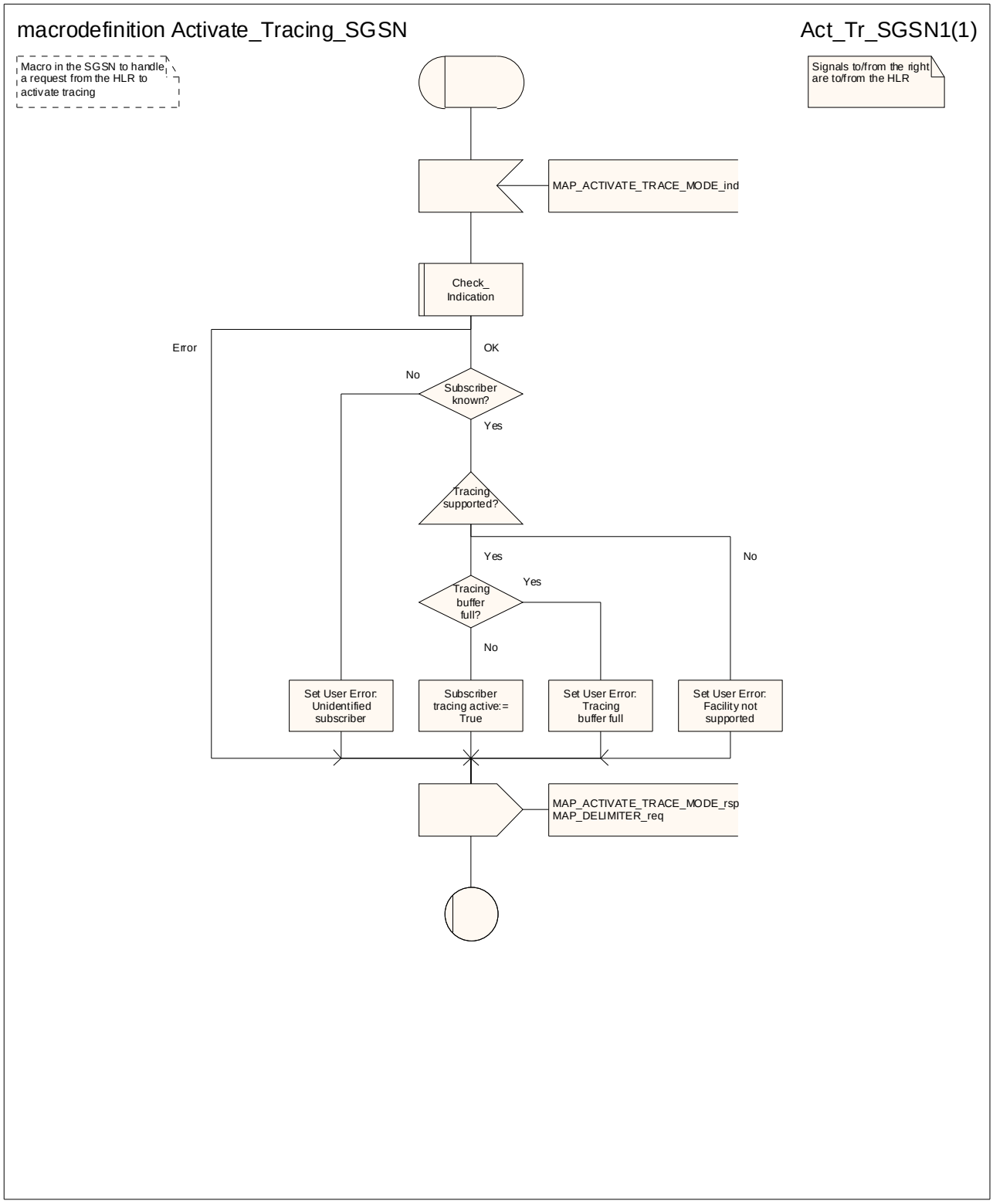


Figure 25.9/5: Macro Activate_Tracing_SGSN

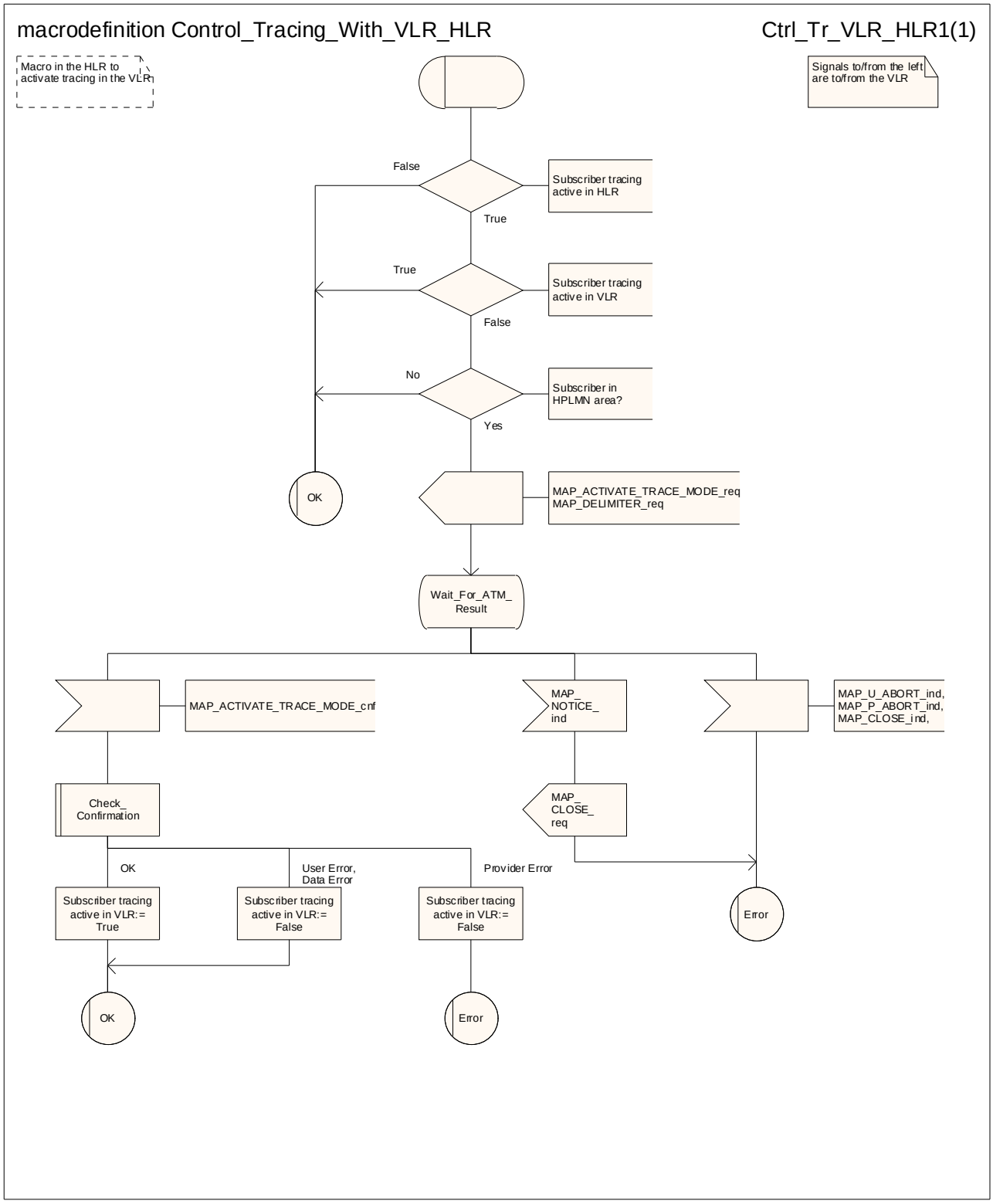


Figure 25.9/6: Macro Control_Tracing_With_VLR_HLR

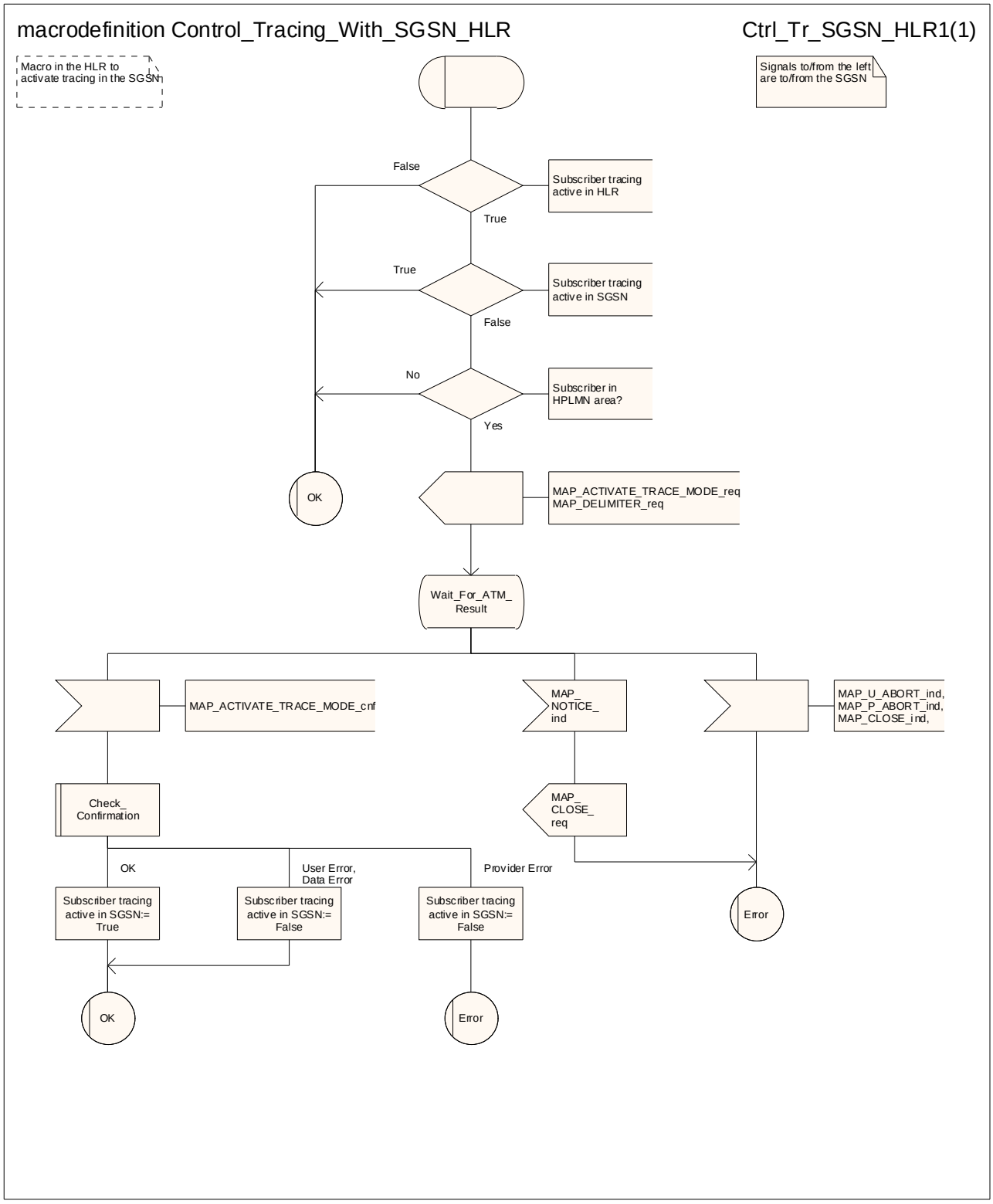


Figure 25.9/7: Macro Control_Tracing_With_SGSN_HLR

25.10 Short Message Alert procedures

25.10.1 Process Subscriber_Present_VLR

The VLR invokes the process Subscriber_Present_VLR when the mobile subscriber becomes active. The general description of the short message alert procedures is in subclause 23.4 of the present document.

25.10.2 Process SubscriberPresent_SGSN

The SGSN invokes the process Subscriber_Present_SGSN when it receives a Page response, a GPRS Attach request or a Routing area update request message (3GPP TS 24.008 [35]). The general description of the short message alert procedures is in subclause 23.4 of the present document.

25.10.3 Macro Alert_Service_Centre_HLR

The HLR invokes the macro Alert_Service_Centre_HLR when Service Centre(s) are to be alerted.

25.10.4 Process Alert_SC_HLR

It is an operator option to resend the MAP_ALERT_SERVICE_CENTRE request to the SMS-IWMSC if the alert is unsuccessful. The number of repeat attempts and the interval between them is also an operator option. The service centre address should be purged from the MWD list if the alert is consistently unsuccessful.

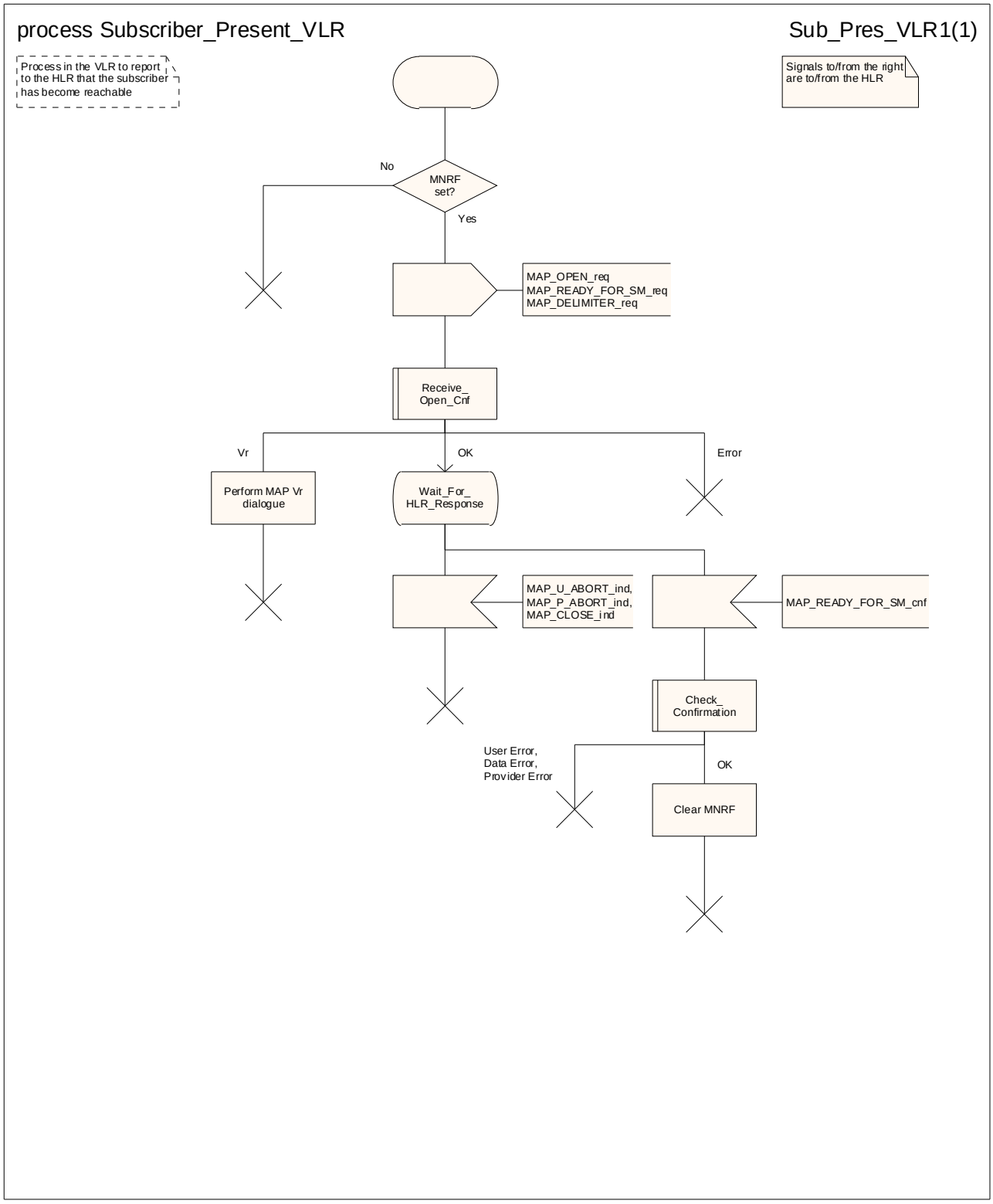


Figure 25.10/1: Process Subscriber_Present_VLR

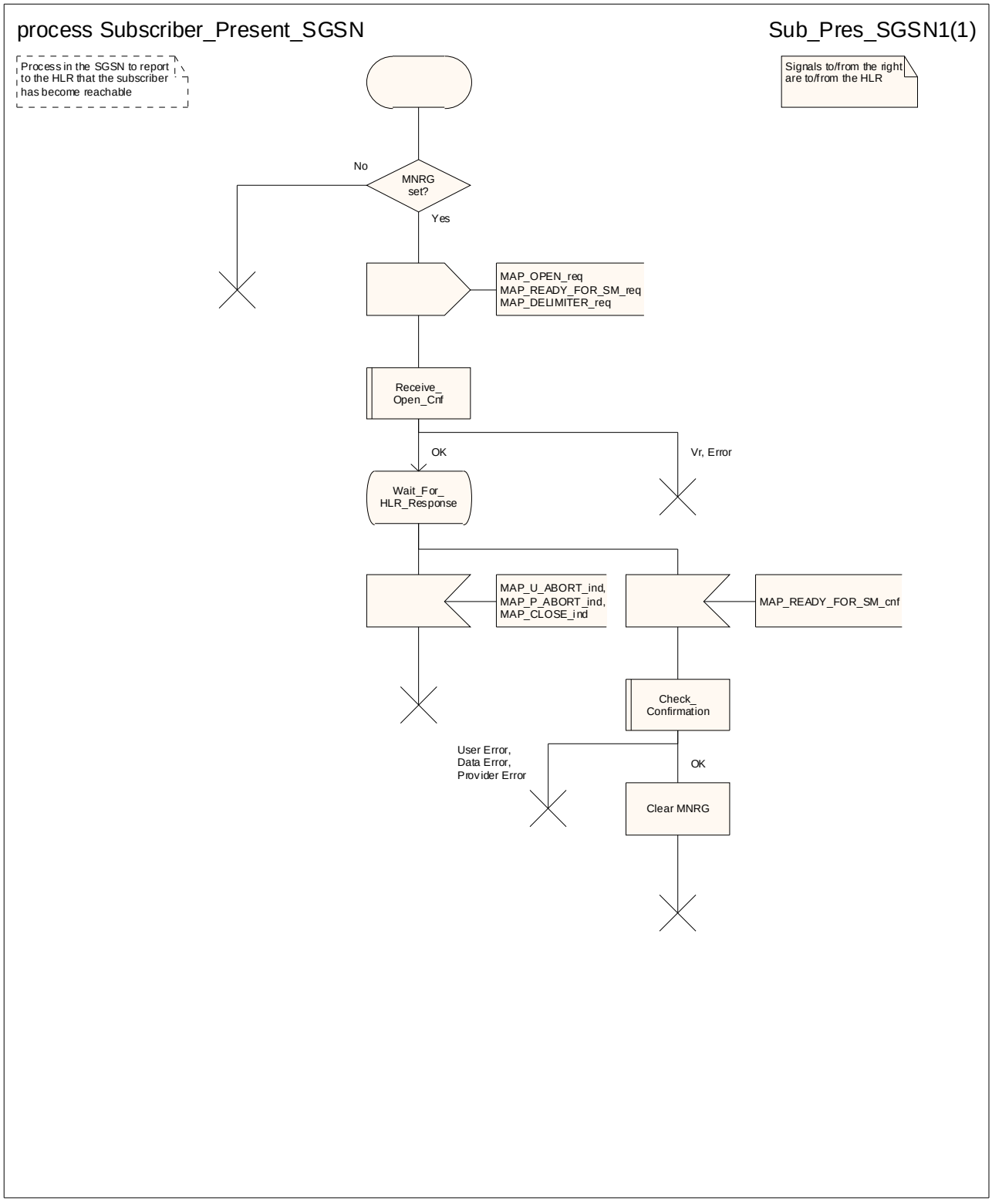


Figure 25.10/2: Process Subscriber_Present_SGSN

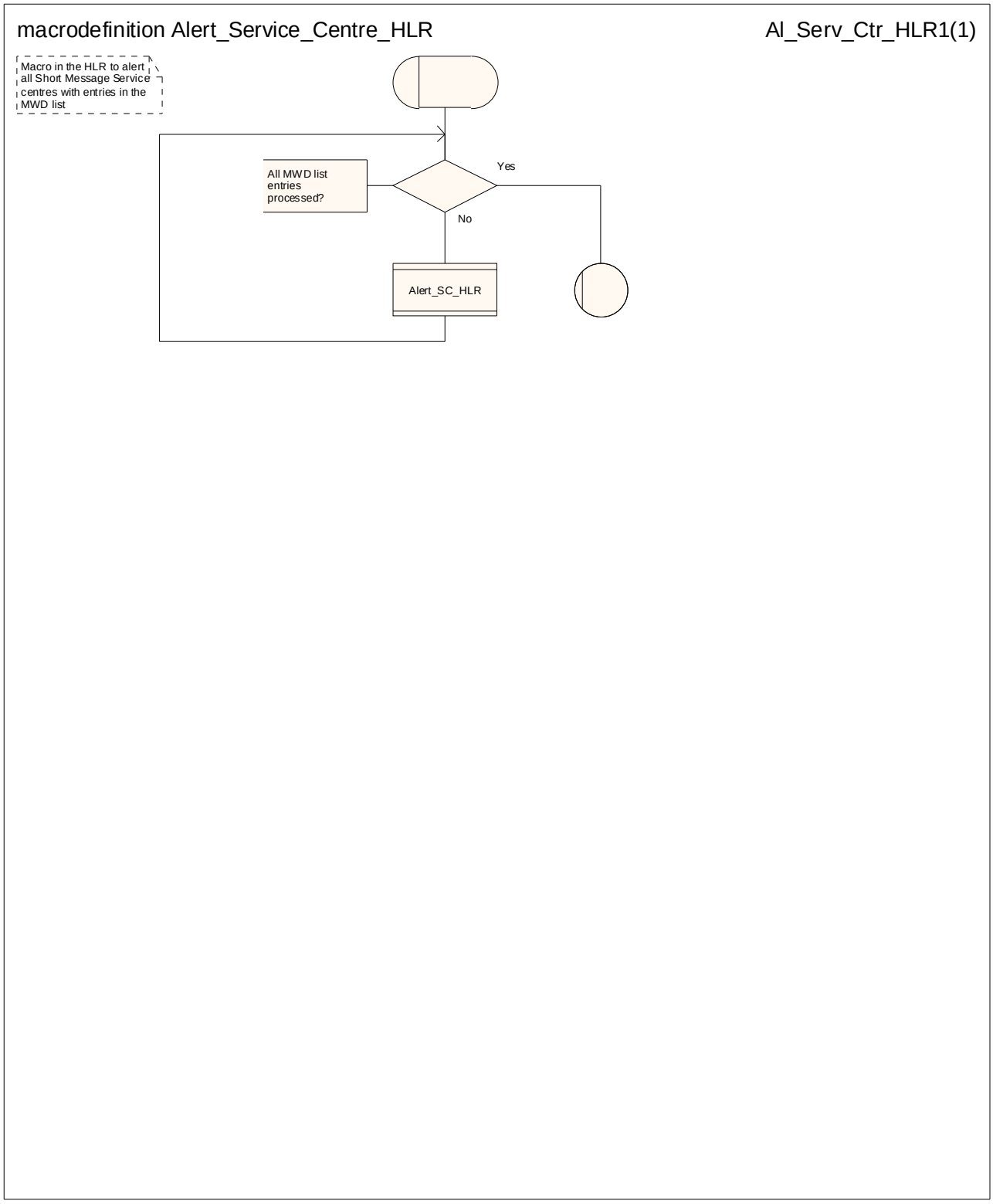


Figure 25.10/3: Macro Alert_Service_Centre_HLR

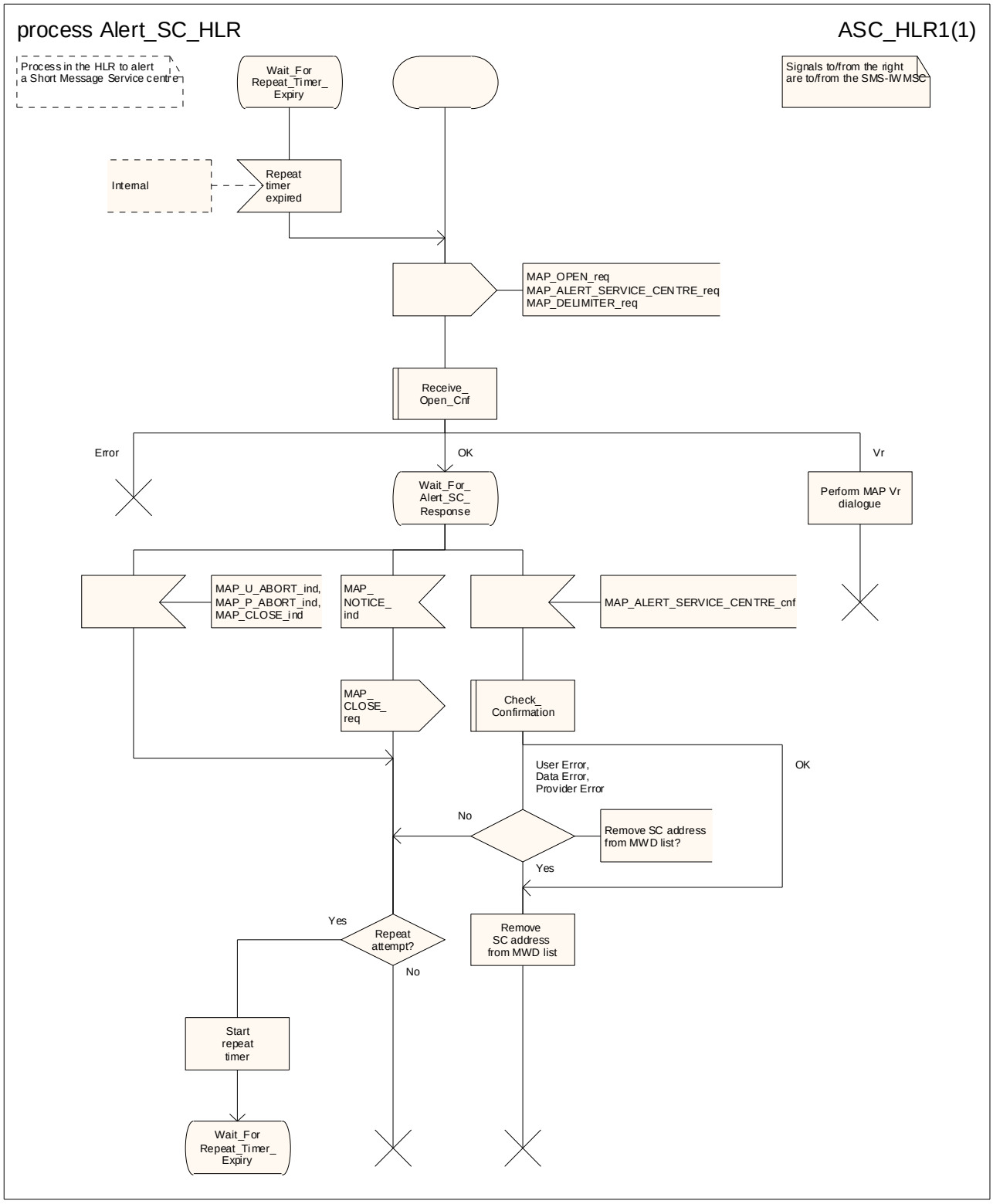


Figure 25.10/4: Process Alert_SC_HLR

Annex A (informative): ASN.1 Cross-reference listing and fully expanded sources

The ASN.1 Cross-reference listing and the fully expanded ASN.1 sources of the MAP protocol are provided for information at http://www.3gpp.org/ftp/Specs/archive/29_series/29.002/ASN.1/

Annex B (informative): Void

Annex C (informative): Message Segmentation Mechanisms

Various segmentation mechanisms are in use to overcome the problem where a MAP parameter carried in an Invoke, Result (or Error) component is too long to fit into a single SCCP UDT message. These mechanisms are:

C.1 SCCP segmentation

Instead of one UDT message several XUDT messages are used according to

> Signalling Connection Control Part, Signalling System no. 7 ITU-T recommendation (07/96) Q.711 to Q.716 ('White Book SCCP').

This mechanism may be used for all MAP messages. If no segmentation mechanism at the TCAP or MAP level is available, this is the only remaining possibility.

This mechanism has no impact on the MAP provider level and above; the MAP provider sees the parameter as being sent in a single segment.

It should be noted that not all SCCP transit nodes (world wide) currently support the transfer of XUDT messages. Therefore XUDT messages may be lost without notice, depending on the route the message takes. The routes which successive messages take between two end points can differ because of load balancing. It is therefore recommended that this mechanism is used only for:

a) messages which do not cross PLMN boundaries (when the PLMN operator ensures that all SCCP transit nodes within his PLMN support White Book SCCP)

b) messages with low priority i.e. loss of the message does not result in serious misoperation.

It should be noted that the decision whether or not a message crosses PLMN boundaries needs to be taken at the MAP application level; it is therefore based on the message's operation code rather than on the SCCP called party address, i.e. only messages which never cross PLMN boundaries due to the type of message (SendIdentification, SendRoutingInfo without OR, AnyTimeInterrogation, ...) can be regarded as not crossing PLMN boundaries.

C.2 TCAP segmentation

At the TCAP level the following segmentation mechanisms are available:

C.2.1 Empty Begin

In a dialogue with AC version >1 the first forward message (Begin) must contain a Dialogue Portion. Instead of sending the Dialogue Portion and the Component Portion in the first forward message, an empty Begin (i.e. without a Component Portion) is sent, followed (after successful dialogue establishment) by a Continue message which can carry a longer Component Portion since no Dialogue Portion is present in the second forward message.

C.2.2 Empty Continue

In a dialogue with AC version >1 the first backward message (Continue / End) must contain a Dialogue Portion. Instead of sending the Dialogue Portion and the Component Portion in the first backward message, an empty Continue (i.e. without a Component Portion) is sent, followed by a Continue/End message which can carry a longer Component Portion since no Dialogue Portion is present in the second backward message.

C.2.3 TC-Result-NL

A Result component may be segmented into one or several Result-Not-Last components followed by a Result-Last component. As specified in subclause 15.6.3, the MAP user parameter shall be split so that each segment is compatible with the type defined for the parameter of the result of the associated operation.

Note that this segmentation mechanism runs the risk that the message carrying the Result-Last component arrives before the message carrying a Result-Not-Last component which results in failure. The use of SCCP class 1 "Sequence guaranteed", which raises the chance of in sequence delivery, is recommended.

C.3 MAP Segmentation

At the MAP level the following segmentation mechanisms are available:

C.3.1 Invoke without explicit indication

An Invoke component may be segmented into several Invoke components. These may be sent in burst mode (in which case SCCP class 1 is recommended) or in acknowledged mode. The receiving node does not get an indication of whether or not more segments will be received, so it must not close the dialogue. The MAP user parameter shall be split so that each segment is compatible with the type defined for the parameter of the invoke of the associated operation.

C.3.2 Invoke with explicit indication

An Invoke component may be segmented into several Invoke components sent in acknowledged mode. Each component contains at the MAP level an indication of whether or not subsequent components will follow. The receiving node terminates the dialogue when the last component is received. The MAP user parameter shall be split so that each segment is compatible with the type defined for the parameter of the invoke of the associated operation.

C.3.3 Result

A Result (last) component may be segmented into several Result (last) components sent in acknowledged mode where a new (empty) Invoke component serves as an acknowledgment. The last segment is not acknowledged. The MAP user parameter shall be split so that each segment is compatible with the type defined for the parameter of the result of the associated operation.

The following tables show the applicability of the mechanisms described above:

AC Version 4:

Parameter	SCCP-segmentation	Empty Begin	Empty Continue	TC-Result-NL	Invoke without indication	Invoke with indication	Result
ResumeCallHandlingArg	allowed	not allowed	n.a.	n.a.	not allowed	recommended	n.a.

AC Version 3:

Parameter	SCCP-segmentation	Empty Begin	Empty Continue	TC-Result-NL	Invoke without indication	Invoke with indication	Result
InsertSubscriberDataArg	risky	not allowed	n.a.	n.a.	recommended	n.a.	n.a.
SendIdentificationRes	allowed	n.a.	not allowed	not allowed	n.a.	n.a.	recommended
PrepareHO-Arg	allowed	not allowed	n.a.	n.a.	not allowed	n.a.	n.a.
PrepareHO-Res	allowed	n.a.	recommended	not recommended	n.a.	n.a.	not allowed
ProcessAccessSignalling-Arg	allowed	n.a.	n.a.	n.a.	not allowed	n.a.	n.a.
ForwardAccessSignalling-Arg	allowed	n.a.	n.a.	n.a.	not allowed	n.a.	n.a.
PrepareSubsequentHO-Arg	allowed	n.a.	n.a.	n.a.	not allowed	n.a.	n.a.
PrepareSubsequentHO-Res	allowed	n.a.	n.a	not recommended	n.a.	n.a.	not allowed
SendAuthenticationInfoRes	risky	n.a.	not allowed	not allowed	n.a.	n.a.	recommended
ProvideSubscriberInfoRes	allowed	n.a.	not allowed	not recommended	n.a.	n.a.	not allowed
AnyTimeInterrogationRes	allowed	n.a.	not allowed	not recommended	n.a.	n.a.	not allowed
AnyTimeModificationRes	allowed	n.a.	not allowed	recommended	n.a.	n.a.	not allowed
AnyTimeSubscriptionInterrogationRes	allowed	n.a.	not allowed	recommended	n.a.	n.a.	not allowed
noteSubscriberDataModifiedArg	allowed	not allowed	n.a.	n.a.	not allowed	recommended	n.a.
SendRoutingInfoRes	allowed	n.a.	not allowed	recommended	n.a.	n.a.	not allowed
MO-ForwardSM-Arg	risky	recommended	n.a.	n.a.	not allowed	n.a.	n.a.
MT-ForwardSM-Arg	risky	recommended	n.a.	n.a.	not allowed	n.a.	n.a.

AC Version 2:

Parameter	SCCP-segmentation	Empty Begin	Empty Continue	TC-Result-NL	Invoke without indication	Invoke with indication	Result
InsertSubscriberDataArg	risky	not allowed	not allowed	n.a.	recommended	n.a.	n.a.
SendIdentificationRes	allowed	n.a.	not allowed	not recommended	n.a.	n.a.	not allowed
SendAuthenticationInfoRes	risky	n.a.	not allowed	not recommended	n.a.	n.a.	not allowed
ForwardSM-Arg	risky	recommended	n.a.	n.a.	not allowed	n.a.	n.a.
PrepareHO-Res	allowed	n.a.	recommended	not recommended	n.a.	n.a.	not allowed

AC Version 1:

Parameter	SCCP-segmentation	Empty Begin	Empty Continue	TC-Result-NL	Invoke without indication	Invoke with indication	Result
InsertSubscriberDataArg	risky	n.a.	n.a.	n.a.	recommended	n.a.	n.a.
SentParameterList	risky	n.a.	n.a.	recommended	n.a.	n.a.	not allowed

In the tables above the keywords "recommended", "allowed", "risky", "not recommended", "not allowed" and "n.a." are used as follows:

"**recommended**"
indicates that the normative part of this specification explicitly specifies the use of this mechanism for the parameter in question;

"**allowed**"
indicates that the normative part of this specification allows the use of this mechanism for the sending node and mandates support of this mechanism for the receiving node;

"**risky**"
indicates that the mechanism is "allowed".However, the use of this mechanism for the parameter in question may result in serious misoperation because SCCP transit nodes are not guaranteed to support XUDT messages.

"**not recommended**"
indicates that the normative part of this specification does not explicitly specify the use of this mechanism for the parameter in question.

"**not allowed**"
indicates that the normative part of this specification implicitly prohibits the use of this mechanism for the parameter in question.

"**n.a.**"
indicates that the mechanism is not applicable for the parameter in question.

Annex D (informative): Void

Annex E (informative): Change History

Date	TSG #	TSG Doc.	CR	Rev	Subject/Comment	New
	04	N2-99227	A002	3	Use of E interface	3.1.0
	04	N2-99578	A003		Introduction of TIF-CSI for Call Deflection	3.1.0
	04	N2-99233	A004		Clarification in ASN.1 encoding of O-CSI and T-CSI	3.1.0
	04	N2-99269	A005		Introduction of MSISDN in USSD operation	3.1.0
	04	N2-99650	A006		Modification of the O-CSI ASN.1 structure	3.1.0
	04	N2-99250	A007		Adding of MAP_DELIMITER_req to the Status report operation	3.1.0
	04	N2-99628	A008		Correction to the Purge MS "Detailed procedure in the HLR"	3.1.0
	04	N2-99677	A009		Adding of MNP-indicator to the SRI ack	3.1.0
	04	N2-99228	A010		New subscription options for call forwarding	3.1.0
	04	N2-99585	A011		Adding the support of ANSI SCCP which is required in North America (World Zone 1)	3.1.0
	04	N2-99515	A012		Introduction of 3-digit MNCs correction	3.1.0
	04	N2-99520	A013		Export of NAEA-CIC	3.1.0
	04	N2-99548	A014		Clarification to text to identify how the LSA data relevant in the current VPLMN can be determined	3.1.0
	04	3C99-468	A015		Alignment with 04.80	3.1.0
	04	N2-99519	A016		VBS data	3.1.0
	04	N2-99461	A017		Introduction of Data Missing error to the Resume Call Handling	3.1.0
	04	N2-99583	A018		Removal of 3-digit MNCs	3.1.0
	04	N2-99676	A019		Corrections of mapping from MAP service to TC service	3.1.0
	04	3C99-206	A020		Introduction of UUS service to Resume Call Handling	3.1.0
	05	N2-99906	021		Clarification on VLR CAMEL Subscription Info	3.2.0
	05	N2-99908	022		Clarification on DestinationNumberCriteria	3.2.0
	05	N2-99910	023		Removal of TDP-Criteria from RCH	3.2.0
	05	N2-99934	025		Various corrections related to GGSN-HLR Interface.	3.2.0
	05	N2-99936	034		Update Location handling for GPRS-only subscription	3.2.0
	05	N2-99938	035		Correction of OP & AC definitions for NoteMS-PresentForGPRS	3.2.0
	05	N2-99952	036		Removal of redundant information from RCH	3.2.0
	05	N2-99956	026		OR capability IE in PRN	3.2.0
	05	N2-99964	024	1	GMSC-CAMEL phase 2 support IE in PRN	3.2.0
	05	N2-99A19	028		Alignment of 29.002 with 02.67	3.2.0
	05	N2-99A45	029	1	Non-CAMEL IST implementation	3.2.0
	05	N2-99B57	027	2	Addition of the information elements and the ASN.1 definitions for Pre-paging	3.2.0
	05	N2-99C27	042		Clarification on 'Supported CAMEL Phases' in ISD ack	3.2.0
	05	N2-99C78	044		Editing error correction on VLR capabilities	3.2.0
	05	N2-99D06	043	1	Addition of exception handling to the CancellationType	3.2.0
	05	N2-99D33	046		Clarification of LR-REJECT cause corresponding to RoamingRestrictionDueTo UnsupportedFeature	3.2.0
	05	N2-99D35	047		Clarification of returning the MSISDN in SRIack	3.2.0
	06	N2-99G06	033	3	Introduction of the Super-Charger Concept in TS 29.002	3.3.0
	06	N2-99G18	032	2	Introduction of White Book SCCP in MAP	3.3.0
	06	N2-99G50	070		Addition of GGSN number for the SRIforGPRS	3.3.0
	06	N2-99J88	075	1	Introduction of Follow Me	3.3.0
	06	N2-99K12	077		Use of SSN for GPRS	3.3.0
	06	N2-99K24	069		Correction of the USSD procedure in the HLR.	3.3.0
	06	N2-99K52	060	1	MAP Impacts for Location Services (LCS)	3.3.0
	06	N2-99K58	045	4	Authentication Enhancements	3.3.0
	06	N2-99K60	050	5	QoS-Subscribed field modification	3.3.0
	06	N2-99L20	073	1	Introduction of CAMEL Phase 3 in 3GPP TS 29.002	3.3.0
	06	N2-99J52	074		Restructuring of MAP Location Management Procedures for the Circuit Switched Domain	3.3.0
	06	N2-99J92	068		Update of SDLs to support Super-Charger	3.3.0
					New version created to fix a CR implementation error	3.3.1
	07	N2B000436	048	5	Introduction of Multicall	3.4.0
	07	N2B000319	059	1	Alternative solution for ALR	3.4.0
	07	N2B000461	063	4	MNP Database Mismatch	3.4.0
	07	N2B000375	066	5	Addition of the FTN-AddressString	3.4.0
	07	N2B000456	079	4	Correction of SS Invocation Notification for CCBS	3.4.0
	07	N2A000023	080		Corrections to ATSI, ATM, NCSD	3.4.0
	07	N2B000046	083		Privacy notification/verification for call related privacy class	3.4.0

	07	N2B000142	084	2	Addition of CS Allocation/retention priority	3.4.0
	07	N2B000144	086	1	Editorial cleanup of 29.002	3.4.0
	07	N2B000100	087		Correction of LSA information	3.4.0
	07	N2B000067	089		Security interworking between release 99 and pre-99 MSC/VLRs	3.4.0
	07	N2B000113	090	1	Improving GPRS charging efficiency	3.4.0
	07	N2B000120	094	2	QoS-Subscribed field enhancements	3.4.0
	07	N2B000322	095	1	RANAP support on the E-interface	3.4.0
	07	N2B000191	099		UMTS Authentication	3.4.0
	07	N2B000466	100	5	Support of 3G Handover, including Multicall	3.4.0
	07	N2B000372	101	1	Introduction of Service Area Identification	3.4.0
	07	N2B000380	102	2	Clarification on Authentication Info Retrieval	3.4.0
	07	N2B000330	103	1	Addition of UMTS security to MAP B interface	3.4.0
	07	N2B000244	104		Re-Synchronisation Info	3.4.0
	07	N2B000324	105	1	Introduction of additional service parameters for inter-system handover	3.4.0
	07	N2B000281	107		Removal of architectural information from clause 4	3.4.0
	07	N2-000454	110	1	Introduction of Authentication Failure Report	3.4.0
	07	N2B000357	111		Use of MAP private extensions to implement region-specific requirements	3.4.0
	07	N2B000470	112		Prioritisation of MAP application context related to VGCS/VBS	3.4.0
	07	N2B000472	113		Correction of SS-Codes for LCS	3.4.0
	08	N4-000098	115	1	Minor corrections to CAMEL3 NSDC/ATM/ATSI information flows	3.5.0
	08	N4-000094	117	1	Using DSD to delete CCBS-B from the subscriber	3.5.0
	08	N4-000089	118	1	Indication in PRN of support of Long FTNs	3.5.0
	08	N4-000073	120	1	QoS-Subscribed field enhancements	3.5.0
	08	N4-000050	121		Correction of introduction of additional service parameters for inter-system handover	3.5.0
	08	N4-000100	122	2	Proposed information flow on NSDC	3.5.0
	08	N4-000321	124	3	CAMEL Subscription Info	3.5.0
	08	N4-000068	125		Clarification to GMLC List definition	3.5.0
	08	N4-000320	127	1	Optionality of parameters in d-csi and in sms-csi	3.5.0
	08	N4-000209	130		Version 3 tags for handover messages	3.5.0
	08	N4-000211	132		Correction of version handling at dialogue establishment	3.5.0
	08	N4-000357	133	1	Various corrections and/or cleanup to 29.002	3.5.0
	08	N4-000217	134		Correction of errors in Figure 25.1/1: Macro Receive_Open_Ind	3.5.0
	08	N4-000326	135	1	Addition of charging characteristics per PDP context	3.5.0
	08	N4-000264	138		Clarification of SAI-ack segmentation procedure	3.5.0
	08	N4-000392	139	1	Indication of unsupported position method	3.5.0
	08	N4-000276	141		Clarification for ReportSM-DeliveryStatus operation	3.5.0
	08	N4-000349	142	1	Addition of a parameter in the subsequent Handover from UMTS to GSM with Multicall	3.5.0
	08	N4-000278	143		Editorial correction to MSC-A handover SDLs	3.5.0
	08	N4-000378	144	1	Use of NAM parameter with MAP-INSERT-SUBSCRIBER-DATA service between HLR and SGSN	3.5.0
	08	N4-000293	145		Addition of state attributes in Forward group call signalling	3.5.0
	08	N4-000294	146		New user error 'target cell outside group call area' in MAP Prepare Handover message	3.5.0
	08	N4-000374	149		Correction to the description of MAP-MO-Forward-Short-Message service	3.5.0
	08	N4-000407	148	4	Changes to MAP for secure transport of MAP messages	4.0.0
	08				Version 4.0.1 created to allow inclusion of automatic update of Annexes A and B and of section 17	4.0.1
	09	N4-000543	152	1	Clarifications for secure MAP transport	4.1.0
	09	N4-000539	153	1	Generalization of version handling text in clause 18.2.4	4.1.0
	09	N4-000491	158		Deletion of informative Annexe C	4.1.0
	09	N4-000540	159		Aligning 29.002 with 25.413 (UTRAN Iu Interface RANAP Signalling)	4.1.0
	09	N4-000541	160		AUTS and AUTN parameter length	4.1.0
	09	N4-000744	161	2	Clarification on Authentication Failure Report ack	4.1.0
	09	N4-000666	163	1	Correction on Location Information	4.1.0
	09	N4-000777	174	2	Optionality of parameters in GPRS-CSI	4.1.0
	09	N4-000788	176	1	Correction to QoS indication	4.1.0
	09	N4-000747	178	1	Clarification of use of Radio Resource Information	4.1.0
	09	N4-000750	180	2	Correction to MSC-A handover SDLs	4.1.0
	09	N4-000736	182		Removal of LSAIdentity from NoteMM-EventArg	4.1.0
	09	N4-000772	184		LCS Support for CAMEL Phase 3	4.1.0
	09	N4-000751	186	1	Correction to MSC-A handover SDLs	4.1.0
	09	N4-000779	188		Clarification for segmentation of D-CSI and SMS-CSI	4.1.0
	10	N4-000912	166	3	Corrections and clarifications for USSD procedures on the HLR - gsmSCF interface	4.2.0
	10	N4-000908	191	1	Corrections of ISD data structure for CAMEL phase 3	4.2.0
	10	N4-001069	193	2	USSD Corrections for Follow Me	4.2.0
	10	N4-001071	196	1	GSM to 3G Handover: MAP parameter Target Cell ID	4.2.0
	10	N4-000921	198		ASN.1 description of targetCellId	4.2.0
	10	N4-001073	200	1	IMSI in MAP_PREPARE_HANDOVER	4.2.0
	10	N4-001076	208	1	Alignment of the Target RNC-ID	4.2.0
	10	N4-001089	211	1	Export of GSN-Address data type	4.2.0

	10	N4-001095	212		Transport of long RANAP messages on MAP-E interface	4.2.0
	-	-	-	-	Automatic update of annexes A and B	4.2.1
	11	N4-010036	206	1	Correction to LCS application context	4.3.0
	11	N4-010276	215	2	Add parameters to ISD and SRI for GPRS to handle ODB for PS	4.3.0
	11	N4-010033	217		Correction to maximum number of RAB's	4.3.0
	11	N4-010198	222	2	PS domain support for LCS Release 4	4.3.0
	11	N4-010058	224		Failure of Update GPRS Location when HLR is not reachable	4.3.0
	11	N4-010287	231	1	Extension of call related privacy class for LCS Release 4	4.3.0
	11	N4-010375	232	2	Maximum number of LCS Clients	4.3.0
	11	N4-010261	234		MAP over IP according to SIGTRAN	4.3.0
	11	N4-010465	236	1	Requesting node type in authentication set request	4.3.0
	11	N4-010360	246		Adding EXPORT definition for LSAIdentity	4.3.0
	11	N4-010361	247		Removing duplicate parameters from ss-CSI	4.3.0
	11	N4-010362	248		Correction to description of SS-CSI in HLR to VLR information flow	4.3.0
	11	N4-010365	250		GSM to UMTS handover: addition of MAP parameter RNC ID	4.3.0
	11	N4-010393	252		Clarification of the use of multicall bearer information	4.3.0
	11	N4-010428	258		Adding EXPORT definition for GeographicalInformation	4.3.0
	11	N4-010446	260		Failure of Authentication Parameter GPRS when HLR is not reachable	4.3.0
	11	N4-010484	262	1	Correction to D-CSI	4.3.0
	12	N4-010728	239	4	Addition of selected UMTS algorithm indication to the handover procedures	4.4.0
	12	N4-010730	241	4	Addition of allowed GSM algorithms indication to the handover procedures	4.4.0
	12	N4-010733	244	4	Addition of allowed UMTS algorithm indication to the handover procedures	4.4.0
	12	N4-010735	245	4	Addition of selected GSM algorithm indication to the handover procedures	4.4.0
	12	N4-010739	254	2	Addition of radio resource list to the handover procedures	4.4.0
	12	NP-010247	256	3	Addition of GSM channel type and GSM chosen channel indications to handover procedures	4.4.0
	12	N4-010787	264	3	Add support in MAP for all shapes defined in 23.032	4.4.0
	12	N4-010633	270	1	Correction to description of RNCId parameter	4.4.0
	12	N4-010635	272	1	Correction to Encryption Information and Integrity Protection parameters	4.4.0
	12	N4-010767	279	3	Essential drawbacks on services due to introduction of Super-Charger function	4.4.0
	12	N4-010741	283	1	Introduction of selected Rab-id to the Process Access Signalling operation	4.4.0
	12	N4-010673	285		Mistake in the definition of Authentication Failure Report Application Context	4.4.0
	12	N4-010551	266		Add support in MAP for Ellipsoid Point	4.4.0
	12	N4-010778	168	5	Security Header modification	4.4.0
	12	N4-010785	267	3	Additional Parameters in Authentication Failure Report	4.4.0
	12	N4-010783	268	3	MS presence notification procedure for LCS	4.4.0
	12	N4-010790	289	2	Component level granularity of protection	4.4.0
					Corrupted headers fixed	4.4.1
	13	N4-010840	290		Clarifications on long forwarded-to numbers	4.5.0
	13	N4-010929	291	1	Corrections for Deferred MT-LR	4.5.0
	13	N4-010930	292	2	Clarifications on SupportedLCS-CapabilitySets	4.5.0
	13	N4-010958	295	2	Corrections on the introduction of LCS for PS domain	4.5.0
	13	N4-010970	302	2	Additional SGSN related values to Access Type	4.5.0
	13	N4-010976	306		Addition of data type definitions to EXPORT statements for the usage in CAP	4.5.0
	13	N4-011017	307	2	Minimum MAP application context for intersystem MSC handover from GSM to UMTS	4.5.0
	13	N4-011019	309	2	Minimum MAP application context for intersystem MSC handover from UMTS to GSM	4.5.0
	13	N4-010845	277	1	Correction on the SDL of NW initiated USSD operations	4.5.0
	13				Editorial Clean up	4.5.0
	14	N4-011031	313		Clarification on LCS parameters in MAP	4.6.0
	14	N4-011043	314		Handling of linked operations in the MAP protocol machine	4.6.0
	14	N4-011285	316		Corrections on the SDL diagrams for LCS	4.6.0
	14	N4-011198	318	1	Indication of deletion of CSI in Notify Subscriber Data Change	4.6.0
	14	N4-011074	320		Correct length of Add-GeographicalInformation	4.6.0
	14	N4-011091	322		Clarify encoding of RNC Id	4.6.0
	14	N4-011094	324		Clarify encoding of RANAP parameters in MAP	4.6.0
	14	N4-011097	325		Clarifications on long forwarded-to numbers	4.6.0
	14	N4-011227	331	1	Clarification of methodology for maintaining data consistency in Supercharger	4.6.0
	14	N4-011173	334		Addition of RAB ID to Prepare Handover procedure	4.6.0
	14	N4-011175	336		Correction to the Allowed GSM Algorithms parameter	4.6.0
	14	N4-011177	337	1	Correction of references	4.6.0
	14	N4-011190	339		CUG-Info is not exported from 29.002	4.6.0
	14	N4-011209	341		Clarification on NSCD when data is withdrawn	4.6.0
	14	N4-011211	343		Clarification of sending CAMEL information in stand alone ISD case	4.6.0
	14	N4-011262	344		Correction of the priority for "SRI for LCS"	4.6.0
	14	N4-011273	347		ASN.1 correction	4.6.0
	14	N4-011437	349	2	Handling of MNRR in the HLR & SMS-GMSC	4.6.0
	14	N4-011433	354	1	Minimum MAP application context for G2G inter-MSC handover	4.6.0

	14	N4-011439	359	2	Alignment of parameter lengths with those prescribed in 08.08	4.6.0
	14	N4-011423	360	1	Aligning the security header elements with TS33.200	4.6.0
	14	N4-011394	364		Syntax error in the ATM result and ATSI result	4.6.0
	14	N4-011381	355	1	LCS Capability Handling for UE's	5.0.0
	15	N4-020300	368	4	Collective CAMEL Phase 4 CR	5.1.0
	15	N4-020013	373		Inclusion of complete ODB data in ATSI and NSDC	5.1.0
	15	N4-020266	381	2	Introduction of the "Requestor ID"	5.1.0
	15	N4-020068	386		Correction to AC version of gprsLocationInfoRetrievalContext	5.1.0
	15	N4-020248	390	1	Incomplete description of Restore Data parameters	5.1.0
	15	N4-020183	403		Clarification on CODEC-Info	5.1.0
	15	N4-020250	407	1	ODB alignment	5.1.0
	16	N4-020530	428	2	LCS: error handling if shape not supported by GMLC	5.2.0
	16	N4-020622	453		Addition of Radio Resource List to the Forward Access Signalling operation	5.2.0
	16	N4-020641	460		Clarification on Resume Call Handling	5.2.0
	16	N4-020746	440	2	Clarification on SendAuthenticationInfo	5.2.0
	16	N4-020750	446	1	Addition of Service Handover parameters to MAP Handover messages	5.2.0
	16	N4-020318	398		Check of NAM and Requesting Node Type on receipt of SendAuthenticationInfo	5.2.0
	16	N4-020333	410		Handling the MNRR flag in the HLR & SMS-GMSC	5.2.0
	16	N4-020499	420	1	Clarfication of introducing Session related and unrelated class	5.2.0
	16	N4-020511	430	1	Corrections on the introduction of LCS for PS domain	5.2.0
	16	N4-020743	448	1	Corrections in SS-code chapter	5.2.0
	16	N4-020408	423		Clarification of handling of MT-SMS-TPDU-Type and SMS-TDP	5.2.0
	16	N4-020410	425		Clarify conditions to trigger restart of MTLR-Deferred procedure	5.2.0
	16	N4-020468	414	1	Corrections to the handling of Any Time Interrogation and Provide Subscriber Info	5.2.0
	16	N4-020476	435	1	Change PS-connected in PS-PDPactive	5.2.0
	16	N4-020483	422	1	Triggering of gsmSCF for MT-SMS-CSI	5.2.0
	16	N4-020485	408	2	Transferring the MS classmark & IMEI to the gsmSCF	5.2.0
	16	N4-020543	441		Correction of Object Identifiers for ASN.1 modules	5.2.0
	16	N4-020608	450		Enhancement to LCS in the PS domain	5.2.0
	16	N4-020623	454		Addition of Location Information GPRS to Note MM Event operation	5.2.0
	16	N4-020703	421	4	LCS: Codeword and Service Type	5.2.0
	16	N4-020756	436	2	Splitting of CAMEL phase 4	5.2.0
	17	N4-021001	437	3	Compatible upgrade to ASN.1:1997 of 29.002	5.3.0
	17	NP-020399	462	2	Introduction of GERAN classmark	5.3.0
	17	N4-020841	465		Clarification on Call Deflection	5.3.0
	17	N4-021040	470	1	Correction to the usage of "Roaming not allowed" error	5.3.0
	17	N4-021041	471	1	Clarifications on Send Identification	5.3.0
	17	N4-021094	479	2	Handling of partial implementations of CAMEL phase 4	5.3.0
	17	N4-021047	480		Removal of ChargingNotification feature	5.3.0
	17	N4-020810	481		CR29.002-443 (rel5) on extensions to ATM for CAMEL control of IMS	5.3.0
	17	N4-020809	482		CR to 29.002 for the support of the MAP Si interface	5.3.0
	18	N4-021290	499		Correction to segmentation of O-CSI and T-CSI	5.4.0
	18	N4-021418	508		ODB correction	5.4.0
	18	N4-021563	511	1	Addtion of reference number to deferred location request procedure	5.4.0
	18	N4-021573	516	2	Correction to the Service Handover parameters	5.4.0
	18	N4-021299	442	3	Description of MT SM delivery via two serving nodes	5.4.0
	18	N4-021294	474	2	Correction of handling of MT-SMS in the SGSN	5.4.0
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	18	N4-021153	486		Correction of IMEI check for SGSN	5.4.0
	18	N4-021467	489	5	Available codecs list and selected codec indication	5.4.0
	18	N4-021194	490		Clarification of the use of Requested CAMEL Subscription Info parameters	5.4.0
	18	N4-021252	495		Correction to RCH – adding O-CSI trigger criteria	5.4.0
	18	N4-021264	496		Additional MM-Code for MG-CSI	5.4.0
	18	N4-021296	497	1	Additional handling of partial implementations of CAMEL phase 4	5.4.0
	18	N4-021383	512		Correcion of Codeword Handling	5.4.0
	18	N4-021443	513		Reference to TS 23.078 in TS 29.002 regarding handling of VMSC address is missing	5.4.0
	18	N4-021524	521	1	Editorial clean-up	5.4.0
	18	N4-021531	522		Introduction of the CHOICE element "netDetNotReachable" for PS-SubscriberState	5.4.0
	18	N4-021260	491	1	Addition of LCS Format Indicator to LCS Client ID	6.0.0
	18	N4-021504	517	2	Addition of V-GMLC Address to the Update Location and Update GPRS Location requests	6.0.0
	18	N4-021567	518	3	Addition of V-GMLC and H-GMLC Addresses to the Send Routing Info for LCS response	6.0.0
	18-	N4-021506	519	2	Addition of PPR Address to the Send Routing Info for LCS response	6.0.0
	19	N4-030234	509	3	Introduction of Call Barring for SMS in PS domain	6.1.0
	19	N4-030325	524	3	Clean-up of SMS procedures chapter	6.1.0
	19	NP-030068	545	2	Correction to interactions between CAMEL control of MO SMS and barring	6.1.0
	19	N4-030061	526		Incrementing ASN.1 module versions	6.1.0
	19	N4-030063	528		LCS diagnostic alignment	6.1.0

	19	N4-030054	529		Addition of LCS Capability Set 4	6.1.0
	19	N4-030301	533	1	Correction to the definitions of Radio Resource List and BSSMAP Service Handover List	6.1.0
	19	N4-030305	541	2	Handover of Group Calls where MSC-B has bearer established	6.1.0
	19	N4-030287	551	1	Change of SS-Code List description for Insert Subscriber Data	6.1.0
	19	N4-030289	559	1	Missing of "Continue Monitoring message" in SDL 21.7_3.2	6.1.0
	19	N4-030297	563	1	Alignment of TS 29.002 with TS 23.107 regarding QoS subscribed data	6.1.0
	19	N4-030222	566	1	Introduction of MSC Number as a new parameter in MAP-SEND-IDENTIFICATION operation	6.1.0
	20	N4-030692	536	2	Additional SGSN Related Access Type – Detach	6.2.0
	20	N4-030658	568	4	Addition of Positioning Data IE to Provide Subscriber Location and Send Location Report	6.2.0
	20	N4-030638	574	1	Provision of SDL diagrams and removal of redundant text in chapter 25	6.2.0
	20	N4-030713	595	2	Removal of redundant text from 29.002 Chapter 23	6.2.0
	20	N4-030439	599		LCS Client external ID	6.2.0
	20	N4-030682	607	1	Provision of SDL diagrams and removal of redundant text in chapter 22	6.2.0
	20	N4-030608	608	1	Addition of LCS capability sets to MAP_SRI_for_LCS response	6.2.0
	20	N4-030647	612	1	Enhancement of the CheckIMEI operation to retrieve the BMUEF	6.2.0
	20	N4-030678	619	1	Correction to naming of PRN parameter	6.2.0
	20	N4-030609	624	1	Addition of Privacy Check Related Action to Provide Subscriber Location request	6.2.0
	20	N4-030642	610	1	Transfer of UE-specific behaviour bitmap at handover	6.2.0
	20	N4-030601	633		Missing SMSs over MSC even if the MS is capable of such sending	6.2.0
	21	N4-031043	584	2	Correction to MAP Process Secure_MAP_DSM SDLs	6.3.0
	21	N4-031053	664	1	Correction of encoding description of Group-Id	6.3.0
	21	N4-030828	657		Reduce maximum length of "LCS Requestor ID" and "LCS Codeword".	6.3.0
	21	N4-030922	647	1	UESBI -IU format	6.3.0
	21	N4-031069	616	3	Incorrect Charging with MNP	6.3.0
	21	N4-031057	660	2	Notification of the 2nd BSG in case of Late CF with OR	6.3.0
	21	N4-031059	614	3	HLR Interrogation for SCUDIF calls	6.3.0
	21	N4-030785	644		Removal of tables in section 7.6	6.3.0
	21	N4-030806	649		Correction of References	6.3.0
	21	N4-030815	648		Correction of wrong AC name in the table in 17.1.6	6.3.0
	21	N4-030824	654		New LCS Service Types	6.3.0
	21	N4-030951	671		SS-Barring Category	6.3.0
	21	N4-031006	650	1	Add SGSN, GGSN, GMLC, gsmSCF, NPLR and AuC to network resource parameter	6.3.0
	21	N4-0301038	645	1	Introduction of North American Interim Location Based Routing of Emergency Call	6.3.0
	21	N4-031065	674		Positioning Data for UTRAN LCS	6.3.0
	21	N4-030953	637	1	Provision of SDL diagrams and removal of redundant text in chapter 19	6.3.0
	21	N4-030745	639		Provision of SDL diagrams and removal of redundant text in chapter 20	6.3.0
	21	N4-030747	641		Provision of SDL diagrams and removal of redundant text in chapter 21	6.3.0
	21	N4-030748	642		Removal of SIWF description	6.3.0
	21	N4-030749	643		Deletion of redundant Annex D	6.3.0
	22	N4-031098	677		Enhancements for the Partial Implementation for "Change of position procedure armed with criteria"	6.4.0
	22	N4-031135	687		Collective CR for Rel-6 Enhanced Dialled Services	6.4.0
	22	N4-031274	648	2	Message Segmentation Mechanisms	6.4.0
	22	N4-031315	703		Addition of requestingPLMN-ID to Send Authentication Info Request	6.4.0
	22	N4-031372	680	2	Addition of CGI to LCS procedures	6.4.0
	22	N4-031373	696	2	Include v-gmlc parameter in RESTORE DATA MAP message	6.4.0
	22	N4-031365	702	2	Deferred MT-LR Area Event	6.4.0
	22	N4-031132	686		More spare bits for CAMEL4 enhancements	6.4.0
	22	N4-031163	692		Clarification on D-CSI segmentation	6.4.0
	22	N4-031342	676	2	MNP correction for prepaid charging	6.4.0
	22	N4-031338	695	1	Remove reduntant option for retrieval of routeing information in figure 21.2.3	6.4.0
	22	N4-031108	679		Modification of description for conditions on inclusion of Positioning Data	6.4.0
	22	N4-031317	689	2	HSDPA impacts to MAP	6.4.0
	22	NP-030533	704		EXPORT data types to CAP (Change of position armed with criteria)	6.4.0
	23	N4-040310	668	3	Codec Modification/ Mid-Call Codec Negotiation after Inter-MSC Relocation	6.5.0
	23	N4-040193	670	2	Correction of Inter-MSC SRSN Relocation procedure	6.5.0
	23	N4-040249	701	3	Introduction of Presence Stage 3 (Ph, Pc and Pg) to the MAP interface	6.5.0
	23	N4-040333	708	2	Correction to Insert Subscriber Data message for LCS SS	6.5.0
	23	N4-040328	709	1	SCCP segmentation for Inter PLMN MAP message	6.5.0
	23	N4-040327	711	2	Inclusion of UTRAN Positioning Data parameter	6.5.0
	23	N4-040284	717	1	Include administrative restriction subscription parameter	6.5.0
	23	N4-040340	720	2	Add new Unavailability cause for SCUDIF	6.5.0
	23	N4-040171	721		CR implemented by fault	6.5.0
	23	N4-040182	724		Removal of R-GMLC Address	6.5.0
	23	N4-040322	725	1	MO-LR Service Identity support	6.5.0
	23	N4-040267	726	1	CAMEL4 SCUDIF notification during active call for prepay	6.5.0

	24	N4-040520	731		Introduction of North American Interim Location Based Routing of Emergency Call	6.6.0
	24	N4-040585	735		Modify IMEI parameter usage definition in MAP-PSL and MAP-SLR	6.6.0
	24	N4-040600	736		Addition of SAI-Present indication to the LCS procedures	6.6.0
	24	N4-040601	737		Clarification on the use of MSISDN parameter for Follow Me functionality	6.6.0
	24	N4-04732	734	1	Add Additional V-GMLC parameter in MAP-SRI-INFO-FOR-LCS	6.6.0
	24	N4-040736	718	6	Addition of IMEISV to Update Location Procedure for ADD function	6.6.0
	25	N4-040929	739		Export of UU-Data data type	6.7.0
	25	N4-041021	743		Wrong SDL flow page implemented	6.7.0
	25	N4-041128	732	2	Pre-Paging Resource Optimization	6.7.0
	26	N4-041272	747		Incorrect Implementation of CR 731	6.8.0
	26	N4-041477	752		Correction to the service response parameters of ATI	6.8.0
	26	N4-041662	746	1	Introducing VGCS/VBS ciphering	6.8.0
	26	N4-041683	757	2	Clarification about returning authentication data for a subscriber (GSM or UMTS)	6.8.0
	26	N4-041684	748	1	LCS Capability Handling for UE's	6.8.0
	26	N4-041685	753	1	Enable NA-ESRD Provision from a GMLC for E911 Location in North America	6.8.0
	26	N4-041641	740	2	SMS Fraud countermeasures	6.8.0
	27	N4-050212	749	1	Management Based Activation Impacts	6.9.0
	27	N4-050369	761	1	Addition of LAI to SendIdentification Request	6.9.0
	27	N4-050430	760	1	Subscribed Charging Characteristics	6.9.0
	27	N4-050444	759	1	Addition of TCAP-Handshake for MO-ForwardSM	6.9.0
	27	N4-050446	745	2	Introduction of Hop Counter for Send Identification	6.9.0
	27	N4-050463	738	8	Rel-6 trace management additions to trace activation and deactivation procedures	6.9.0
	27	N4-050467	763	2	Pseudonym indicator support in MO-LR	6.9.0
	28	C4-050737	769	1	Correction to Trace parameters to allow trace at the BM-SC	6.10.0
	28	C4-050832	770	6	Full RANAP support of network initiated SCUDIF	6.10.0
	28	C4-050895	766	2	Clarification on the use of Access Restriction Data parameter	6.10.0
	28	C4-050784	765	1	Addition of CollectInformation procedure to OfferedCAMEL4Functionalities	7.0.0
	29	C4-051013	771		ASN.1 module version update	7.1.0
	29	C4-051295	776	2	Enabling the Providing of Velocity	7.1.0
	29	C4-051333	772	1	Support of talker priorities and talker identity presentation	7.1.0
	29	C4-051334	773	1	Delivery of SMS to voice group call	7.1.0
	29	C4-051368	777	2	CS data Mobile Terminating calls from PSTN	7.1.0
	29	C4-051336	780	2	Correction on misalignment with stage 2 for Location Services	7.1.0
	30	C4-051775	783	2	Addition of UMTS Trace parameters to handover procedure	7.2.0
	31	C4-060320	794	1	Addition of UMTS Trace parameters to handover procedure	7.3.0
	31	C4-060295	790	1	Removal of MAPsec material	7.3.0
	31	C4-060315	787	1	addition of "supported RAT types indicator" during location/routing area update	7.3.0
	31	C4-060378	792	1	Addition of Periodic Location Feature Support	7.3.0
	31	C4-060434	781	3	New LocationType for the notification based on current location of target UE	7.3.0
	31	C4-060318	788	1	SMS Relay Application Context Names for Version 1	7.3.0
	31	C4-060041	789		Precision on segmentation of MAP GPRSSubscriptionData parameter	7.3.0
	31	C4-060250	801		Improvements to VGCS Call Establishment	7.3.0
	31	C4-060011	786		Addition of Authentication Domains in MAP Send Authentication Info	7.3.0
	32	C4-060813	0808	2	List of MSISDNs and Basic Service Code for MAP Any Time Subscription Interrogation.	7.4.0
	32	C4-060499	0803		Correction of LCS parameter for emergency call usage	7.4.0
	32	C4-060680	0814		SSN for FFN	7.4.0
	32	C4-060706	0817		Removal of MAPsec material	7.4.0
	33	CP-060522	0818	1	Removal of ASN.1 Expanded Source	7.5.0
	33	C4-061047	0805	1	Interoperability between VBS/VGCS and RANflex	7.5.0
	34	CP-060741	0795	1	Support of SMS over IP networks	7.6.0
	34	C4-061800	0828	1	Extension of Group ID	7.6.0
	34	C4-061633	0829		Addition of Teleservice Code to SendGroupCallInfo	7.6.0
	34	C4-061775	0834		Accuracy Fulfillment Indicator parameter to MAP SLR for deferred MT-LR	7.6.0
	34	C4-060693	0832	2	Optional Suppress Terminating Services Bit String in SRI	7.6.0
	34	C4-061632	0807	2	Introduction of sending application-specific data to group call members	8.0.0
	35	C4-070140	0843		ASN.1 module version update	8.1.0
	35	C4-070097	0837		Corrections to RAB Configuration Indicator and Iu-Selected codec	8.1.0
	35	C4-070229	0840		Addition of capability to route MT-SMs via the HPLMN of the receiving MS	8.1.0
	36	C4-070388	0849		Mobile Termination whilst the MS is moving to another MSC	8.2.0
	36	C4-070394	0842	2	Addition of SMS over IP functionality	8.2.0
	36	CP-070476	0859		Detailed procedure in the IP-SM-GW	8.2.0
	37	C4-071055	0862		QoS Extension	8.3.0
	37	C4-071072	0863		Talker Channel Parameter	8.3.0
	37	C4-071266	0869		LMSI For MT-SMS	8.3.0
	37	C4-071281	0873	1	NPI for the call forwarding to number	8.3.0
	37	C4-071285	0864	1	Limit on number of concurrent MT-LR location requests	8.3.0

	37	C4-071383	0868	2	Corrections to SMS over IP handling	8.3.0
	38	C4-071724	0876		TCRT: Clarification on coding of Notification Data	8.4.0
	38	C4-071815	0879		Removal of CCBS_Call_Report_Ack and Event_Report_Ack	8.4.0
	38	C4-071855	0881		Restriction on the use of ccbs-A SS indication	8.4.0
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	38	C4-071997	0875	1	Behaviour of the IP-SM-GW for SM Delivery Status Report	8.4.0
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	39	C4-080148	0883		SDL correction for procedure Check_Available_Vectors	8.5.0
	39	C4-080532	0886	2	HLR involvement in SMS Router Optimization	8.5.0
	40	C4-081277	0888	1	Extension of Group ID	8.6.0
	40	C4-080647	0882	1	Paging optimization with A/Iu flex	8.6.0
	41	C4-081730	0890		Addition of IMS Centralized Service subscription information	8.7.0
	41	C4-082447	0891	1	eMLPP Priority in MAP SRI, PRN and PSI request	8.7.0
	41	C4-082335	0892	1	Gr+ enhancements for EPS	8.7.0
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	42	C4-082758	0896		RAT Frequency Selection Priority	8.8.0
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	42	C4-083247	0903		Access Restriction	8.8.0
	42	CP-080706	0906	1	Access Restriction Data Handling	8.8.0
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	42				SDL files added in Zip-file	8.8.1
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	44	C4-091377	0921	1	Subscription Data Clarification for MAP Interface	8.10.0
	44	C4-091429	0920	1	Trace	8.10.0
	44	C4-091435	0922	1	Supported Features	8.10.0
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	45	C4-092254	0925	2	Cancel Location for Initial Attach	8.11.0
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	49	C4-101809	0976	1	Addition of SIPTO permissions in PS subscription data	10.0.0

	49	C4-102250	0957	4	Prevention of Timeout in IP-SM-GW	10.0.0
	49	CP-100608	0979	2	Addition of SS codes to the ATSI and ATM procedures	10.0.0
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	50	C4-102699	0988	1	Removal of MAP Update GPRS Location message during detach or last PDN connection deactivation via 3GPP access	10.1.0
	50	C4-102737	0993	1	URRP for SGSN	10.1.0
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	50	C4-103157	0997	2	Correction to ATM for call waiting	10.1.0
	50	C4-103314	1002	2	Periodic TAU/RAU timer in HSS subscription	10.1.0
	50	C4-102614	0991		ASN.1 module version upgrade	10.1.0
	50	C4-102687	0987	1	Addition of MPS Priority in Subscription Data	10.1.0
	50	C4-102809	0986	1	Addition of LIPA permission in Subscription Data	10.1.0
	51	C4-110389	1006	2	UE SRVCC Capability Support in MAP Message	10.2.0
	51	C4-110292	1008	1	Use of recovered MME Name / SGSN Name in MSC/VLR	10.2.0
	51	C4-110133	1009		Zone Code Propagation at Handover	10.2.0
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	51	C4-110759	1018	1	Mobile Terminating Roaming Forwarding	10.2.0
	51	C4-110778	1007	2	Minimization of Drive Tests (MDT)	10.2.0
	51	C4-110793	1015	1	Introduction of LCLS functionality in TS 29.002	10.2.0
	51	C4-110958	1017	2	Enhancements of T-ADS data retrieval via MAP ATI	10.2.0
	52	C4-111112	1024		Correction on Subscriber Data Withdrawal	10.3.0
	52	C4-111611	1030	2	Missing MME Name in EPS Location Information	10.3.0
	52	C4-111534	1020	1	MDT user consent	10.3.0
	52	C4-111567	1021	1	SC Address in IP-SM-GW Register Response	10.3.0
	52	C4-111402	1025	1	Periodic LAU timer in HSS subscription	10.3.0
	52	C4-111414	1026	1	New LMSI handling for MTRF	10.3.0
	52	C4-111416	1029	1	Addition of VMSC Address in PRN Ack	10.3.0
	53	C4-112067	1047		Use of UE-Reachability by SGSN	10.4.0
	53	C4-112081	1042	1	APN-AMBR for GPRS	10.4.0
	53	C4-112096	1033	1	MTRF and Super Charger interactions	10.4.0
	53	C4-111986	1043		ASN.1 exports for 32.298	11.0.0
	53	C4-112033	1032	1	Addition of Anonymous Call Rejection in the CS domain	11.0.0
	53	C4-112209	1037	2	Add vSRVCC updates to the Gr interface	11.0.0
	53	C4-112222	1040	2	MAP-READY-FOR-SM for IP-SM-GW	11.0.0
	54	C4-112930	1053	1	Provide Subscriber Information handling for UE under LTE	11.1.0
	54	C4-112988	1063		PDN-Type	11.1.0
	54	C4-112990	1056	1	Equivalent PLMN CSG Subscription Request	11.1.0
	54	C4-113037	1055	2	LCLS negotiation MAP update	11.1.0
	54	C4-112464	1039	4	Cancellation type initial attach	11.1.0
	55	C4-120406	1064	1	Initial Attach Indication in MAP_CANCEL_LOCATION	11.2.0
	55	C4-120420	1073	1	Removal of Subscribed Periodic TAU/RAU timer in HSS subscription	11.2.0
	55	C4-120528	1072	2	User Unknown Error in Provide Subscriber Info MAP operation	11.2.0
	55	C4-120224	1066		UE reachability	11.2.0
	55	C4-120322	1070		TC-RT: Introduction of group IDs with prefix	11.2.0
	55	C4-120416	1067	1	CSG subscription data propagation	11.2.0
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	56	C4-120707	1078		Editorial corrections to TS 29.002	11.3.0
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	57	C4-121813	1090	2	SRI for SM and MME Diameter address	11.4.0
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	58	C4-122660	1106	4	Clarification on EPS Info	11.5.0
	58	C4-122497	1108	1	Pdp Type	11.5.0
	58	C4-122164	1093	1	Trace Depth extension	11.5.0
	58	C4-121847	1094		AC version for Reset	11.5.0
	58	C4-122469	1095	4	MSISDN-less UEs	11.5.0
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	59	C4-130293	1119	1	Registration for SMS Request for SMS in SGSN	11.6.0
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	59	C4-130335	1121	1	Providing Diameter identity of the SGSN to the GMLC over Lh interface	12.0.0
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	59	C4-130256	1114		Mobile Terminating call pending flag in MAP Send Identification response	12.0.0
	60	C4-131046	1147	2	Subscribed RFSP index for Gn SGSNs	12.1.0
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	60	C4-130859	1131	1	Clarification on RNC ID value	12.1.0
	60	C4-130979	1129	1	Storing Last known Location Information of purged UE in HSS	12.1.0
	60	C4-131040	1135	2	Maximum value for subscribed periodic timers	12.1.0
	61	C4-131488	1158	2	Addition of Missing Supported Features	12.2.0
	61	C4-131261	1160		ASN.1 module version update	12.2.0
	61	C4-131540	1149	1	Returning to former LTE PLMN after CSFB	12.2.0
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	61	C4-131445	1153	1	GERAN Iu Mode	12.2.0
	61	C4-131478	1150	2	CancelLocation requesting reattach	12.2.0
	61	C4-131529	1161		Enforcing access restriction during I-RAT RAU/TAU procedures	12.2.0
	61	C4-131370	1130	2	SMS for IMS UE to IMS UE without MSISDN	12.2.0
	62	C4-131758	1162	1	Addition of a reference to TS 23.018 for MTRF optimal routing	12.3.0
	62	C4-131759	1163	1	Removal of Editor's Notes for Single-shot SM	12.3.0
	62	C4-131764	1165	1	Clarification on Serving Node for SMS	12.3.0
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	62	C4-132125	1169	1	Update of Homogeneous Support of IMS Voice Over PS Sessions	12.3.0
	62	C4-132202	1171	1	Time Zone retrieval from a Gn/Gp-SGSN	12.3.0
	63	C4-140243	1174	1	Addition of SGSN CAMEL Capability to SupportedFeatures	12.4.0
	64	C4-140515	1177		CS to PS SRVCC	12.5.0
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	65	C4-141526	1181	1	P-CSCF Restoration Indication	12.6.0
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History

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V16.0.0	November 2020	Publication